

ECOLOGY ENVIRONMENTAL SCIENCE AND CONSERVATION

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PREFACE

This book addresses a wide range of subjects and presents a view of ecology and environmental science, somewhat different from that held by many ecologists. The authors feel that the traditional ecology is really a natural, integral part of environmental science. This book is an attempt to demonstrate the existence of a deep underlying core of principles to which all aspects of environmental science can contribute, and reflects the recent strong shift of emphasis in ecology to examination of processes. An effort has been made to break down artificial barriers between pure and applied science, and in that many applied topics are described in detail. As apparent, many practical problems in environmental science are reaching crisis dimensions for humans, and the attention of our young generation must be directed to them. Most of the global environmental problems require a thorough understanding of ecological principles for resolution. Understanding the basic ecological principles is important for the sustainable use of resources and to evolve strategies for the mitigation of environmental problems at local, regional and global levels. We intended to show how the basic science of ecology can be applied to a fundamental understanding of current position of human beings in the global environmental, and socio-economic scenario. This book is designed in such a way that it can cater to the needs of basic and multidisciplinary courses in college biology, for university students, and advanced courses offered to students in ecology and environmental science. The present book will fill the gap for a basic book on ecology and environment science for undergraduate and graduate students in India and other developing countries based on examples from India, tropical countries, and the global literature.

The introductory chapter deals with the scope and importance of ecology and environmental science, evolution of ecological thought, the linkages between ecology, environment and resource conservation, and environmental sustainability. The basic concepts of the physical and chemical environment, the effects of environmental factors (light, temperature, water, soil, disturbance and fire) on organisms, and the adaptational features of organisms are elaborated. The chapter on soils highlights the role of soil in ecosystem services, and that of belowground biodiversity in ecosystem regulation. The processes and patterns in biogeography relate to speciation, extinction, and the characteristic biota of different geographical regions, with special reference to biogeographical zones in India. The species population and species interactions mainly deal with the population dynamics, life-history strategies of organisms, the role of competition, predation, and parasitism in population regulation, population genetics, and the significance of positive and negative species interaction at the interface of population and community ecology. The discussion on biological community includes the emerging concepts of the nature and organization of communities, species diversity, keystone species, ecological niche and community stability. The approaches to vegetation analysis are outlined to facilitate analysis of structure and dynamics of plant communities. The chapter on community dynamics describes the dynamic nature of community, the kinds, patterns, mechanisms, the recent models of succession, and the changes in ecosystem attributes during succession.

The concept of ecosystem is a central dogma in ecology and the concerned chapter deals with the ecosystem structure, functioning and processes as seen in the context of ecosystem regulation and explores the scope for translating ecosystem paradigms into practice for nature conservation. Productivity, energy flow, decomposition processes, and the mineral-nutrient cycling are the central themes of ecosystem ecology. The global biogeochemical cycles deal with carbon, nitrogen, phosphorus, sulfur and hydrological cycles as influenced by human activities. The nutrient cycling chapter provides a mechanistic understanding of nutrient transformation and conserving

mechanisms at the ecosystem level. The description of terrestrial biomass and aquatic ecosystems gives an insight into the biophysical basis for the spatial distribution of organisms. In view of the increasing human impacts on natural systems, the urgent need for conserving different biomes, freshwater and marine ecosystems is emphasised.

The interaction of human beings with the environment is now an important theme in ecology. Therefore, an attempt has been made to understand socio-economic and ecological dimensions of environmental pollution, ecotoxicology, and biological invasions, global environmental change, natural resources, biodiversity conservation, ecological restoration, and environmental impact assessment. The discussion on climate change touches various aspects of greenhouse gas emissions, impact of climate change, climate change adaptation and mitigation strategies. Biodiversity is essential for human survival and economic well-being, and plays a prominent role in modulating ecosystem function and stability. The magnitude of biodiversity, ecosystem services, gradients of biodiversity, biodiversity hotspots, extinction of species and various biodiversity conservation strategies have been described with latest data. Ecological restoration of degraded ecosystems emphasizes linkages between restoration and human well-being, strategies of restoration and landscape restoration. The processes and application of remote sensing and geographical information system have a focus on the geospatial aspects of the environment, ecosystem analysis and natural resources. Ecological economics and sustainable development highlight the need to integrate ecology with social sciences with greater emphasis on environmental policy, planning and implementation for maintaining ecosystem services and improving human well-being. The growth of human population and problems of urbanization are also discussed.

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Part – I

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CHAPTER

1

Ecology and Environmental Sustainability

Although ‘Ecology’ is a relatively young branch of biological sciences, as a philosophy or view point its roots lie in antiquity. In fact, thinking about ecology is indeed as old as the first humans inhabiting the earth. The very survival of humans depended on resources drawn from the nature and the understanding of their surrounding environment. With the freed upper limbs, humans used fire, and stones in order to compete with the forces of nature, and this paved the way not only for human survival but also for his disproportionate dominance in the biosphere. Early humans depended on natural resources (mostly biological) largely to satisfy their hunger, and developed a basis for identification and assessing distribution of resources within a specified territory (small spatial scale). This was later replaced by innovative ways of resource utilization with the advent of human civilization. Cultivation of plants and domestication of animals became a practice, and the knowledge base on nature and its components widened. With the progress of time, humans came to utilize such varied resources as fossil fuels and minerals from deep profiles of earth and oceans, timber from the highest mountains and deepest part of virgin forests, and fish from polar caps to deep oceans of equatorial regions and biodiversity from different types of ecosystems. Thus the capacity to utilize the resource, which enabled the primitive man to survive, gradually attained the dimensions, which, in modern times, tend to threaten the life support systems-air, water, land and living forms and the very survival of human beings on the planet earth.

Over the years, the scope of our scientific understanding and technical skills have widened significantly. We are moving in a direction wherein barriers of disciplines are becoming porous to accommodate and address societal issues and concerns. In this context, being a cross-cutting interdisciplinary science, ecology holds a key position. Broadly speaking, ecology is the study of structure and function of the ‘nature’, while environmental science deals with the application of scientific/ technological tools for solving the environmental problems faced particularly by humans. When one speaks of ‘nature’, the contemporary ecologists agree that ‘nature’ is heterogeneous, its patterns and dynamics are scale-dependent, and diversity and complexity are its inherent characters. Most of the front line global environmental problems, be it biodiversity loss or climate change, habitat fragmentation or population growth, soil degradation or pollution increase, etc., fall within the gambit of ecological studies and require a thorough understanding of ecological principles for resolution. The ecologists therefore, are required to interpret the human-induced environmental changes and postulate varied scenarios for alternative policy options for legislative and implementable administrative decisions. Ecology has grown by internalizing the different disciplines of natural and, more recently, social sciences (Misra, 1985), and with the passage of time, it is evolving as a science of Human Sustenance and Sustainability. Therefore, understanding the ecological principles is important for the sustainable use of resources and to evolve strategies for the mitigation of environmental problems at local, regional and global levels.

1.1 DEFINING ECOLOGY AND ENVIRONMENTAL SCIENCE

In nature, organisms can survive only in appropriate environments, interact with each other and are influenced by the whole complex of environmental factors. Ecology is the branch of biology that deals with the study of interactions between organisms and their environment. In 1859, Geoffrey Saint Hilaire (a French Zoologist) used the term *Ethology* to refer to the study of relationships between the organisms and the environment; this term was ignored for a long time but it has been resurrected in recent years as an important part of ecology dealing with the field of animal behaviour. A German biologist, Reiter in 1866 coined the term *Oekologie* derived from the Greek “oikos” meaning “home”, and “logos”, a “discussion or study”. A year later in 1867, Ernst Haeckel, a leading proponent of Darwinian thought in Germany introduced this term in a book relating animal morphology to Darwin’s theory of evolution by natural selection, and claimed that the theory of evolution explained and formed the groundwork of ecology. As a recognized distinct field of biology, the science of ecology thus dates from about 1900, but only in the 1960s this term became part of the general vocabulary.

The term *Oekologie*, as introduced by Haeckel, meant the study of the relationship of the animal to its organic as well as its inorganic environment. Literally ecology means the study of organism ‘at home’. The ‘home’ includes the entire physical environment as well as other organisms that are there. Thus ecology is defined as the study of the interrelationships between organisms and their environment.

Odum (1971) cites one of the elaborate definitions for ecology namely, “the totality of patterns of relations between organisms and their environment”. The basic concept of ecology is that “everything depends on everything else”. Ecology deals with numerous and varied components of nature, which can be categorized variously (climate, soil, litter lying over soil, plants, animals, production, decomposition, diversity, dominance, etc.) and the linkages existing between them (e.g., how diversity is related to production or litter lying over soil to production). This is why an inter-disciplinary team (incorporating biologists, chemists, geologists, statisticians, mathematicians, geographers, socio-economists, anthropologists etc.) is often required for modern ecological research.

Ecology is a unifying discipline where techniques and knowledge of several other disciplines have been internalized for testing and developing concepts. Most sciences, particularly biological sciences, could be represented as a trunk from which sprout a number of branches (cytology, histology, physiology, etc.), which are further subdivided into smaller and increasingly specialized ramifications (molecular biology, neurophysiology, etc.). Ecology can be shown as a mass of roots all tending to form a common trunk: first botany, zoology, climatology, the soil sciences and physical geography, then biochemistry and molecular biology and perhaps advance mathematics and finally sociology, human geography, psychology and even economics. It is evident that one cannot build up a common trunk by collecting roots haphazardly; it can only be erected from intricately woven root mass. The various disciplines that form ecology have to be in a harmonized state with respect to approaches and interactions in order to build up ecological concepts. Ecology, thus is a multidisciplinary and holistic science, in which attempt is made to grasp a natural phenomenon in its totality by studying interactions between all its variables, and forcing functions. Odum (1971) emphasized that ecology can be appropriately defined as the study of the structure and function of nature.

It is important to distinguish ecology from environmental science. Ecology deals with the interrelationships of all organisms with their environment. Environmental science is an applied science with an emphasis on the analysis of human impacts on the physical, chemical and biological environment of Earth. Environmental science is a multidisciplinary field that integrates ecology,

geology, climatology, environmental economics, political science and philosophy. Environmental science provides an integrated, quantitative, and interdisciplinary approach to the study of environmental systems. Taking the example of the global warming phenomena, physicists create computer models of atmospheric circulation and infra-red radiation transmission, chemists examine the inventory of atmospheric chemicals and their reactions, biologists analyse the plant and animal contributions to carbon dioxide fluxes, and specialists such as meteorologists and oceanographers add additional breadth in understanding the atmospheric dynamics. However, ecology must be at the core of environmental science programme in view of increasing environmental problems and an urgent need for conserving natural resources (Singh, 1991a, b).

1.2 BIOLOGICAL SCALES: LEVELS OF ORGANISATION

In ecology we study the distribution and abundance of organisms and their interactions with the physical environment comprising lithosphere, hydrosphere and the atmosphere as well as the influences of one organism on the other (Fig.1.1). Having emphasized the holistic nature of ecology, we can now introduce the basic units, unique to ecology. One system of classifying units of biology is the hierarchical levels expressing the concept of biological scales (Allen and Starr, 1982). Biology is a wide field of enquiry from genes to the ecosystems. The whole area of ecology is seen as forming a biological spectrum, with many levels of organization. In the ascending level of complexity, these levels include the genes, cells, organs, individual organism, population, biological community, ecosystem, landscape, biome and the biosphere. Each level of organization has abiotic and biotic components, forming a functional biological system characterized by the exchange of matter and energy.

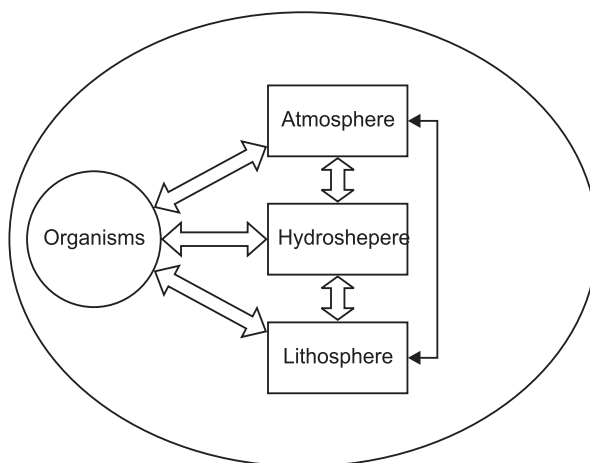


Fig. 1.1. The organisms interact with the physical environment comprising the atmosphere, hydrosphere and lithosphere.

The concept of levels of organization provides a framework for dealing with complex biological systems. For example, the basic unit of the study of genetics is the gene, that of cytology is the cell and that of morphology is the organ. In ecology, we focus on organisms, populations, biological community, ecosystem and the global processes occurring in the biosphere (Fig. 1.2). One level of organization integrates with the other levels and there are no sharp lines or breaks in the functional sense among various levels. For example, an organism in the presence of other individuals of the same species is referred to as population. Similarly, a biological community is sustained only due to the cycling of nutrients and flow of energy in the ecosystem. O'Neil et al. (1986) recognized that levels beyond that of organisms could arise during the course of energy dissipation in ecosystems. Thus, the idea of scale is central to hierarchy theory in ecology (Schneider, 2001).

An ecologist normally studies at three levels, viz., population (group of individuals of any one kind of organism), community (all the populations occupying an area) and ecological system or ecosystem (where community and abiotic environment function together as a system). However, a basic and unique unit to ecologist is the ecosystem, in the confines of which one can conveniently study the integrated functions of nature (the interacting plants and animals and the abiotic environment). When we deal with different levels, it is too tempting to ask which level is more important for human's understanding of problems and solutions of nature. The answer can be

BIOSPHERE

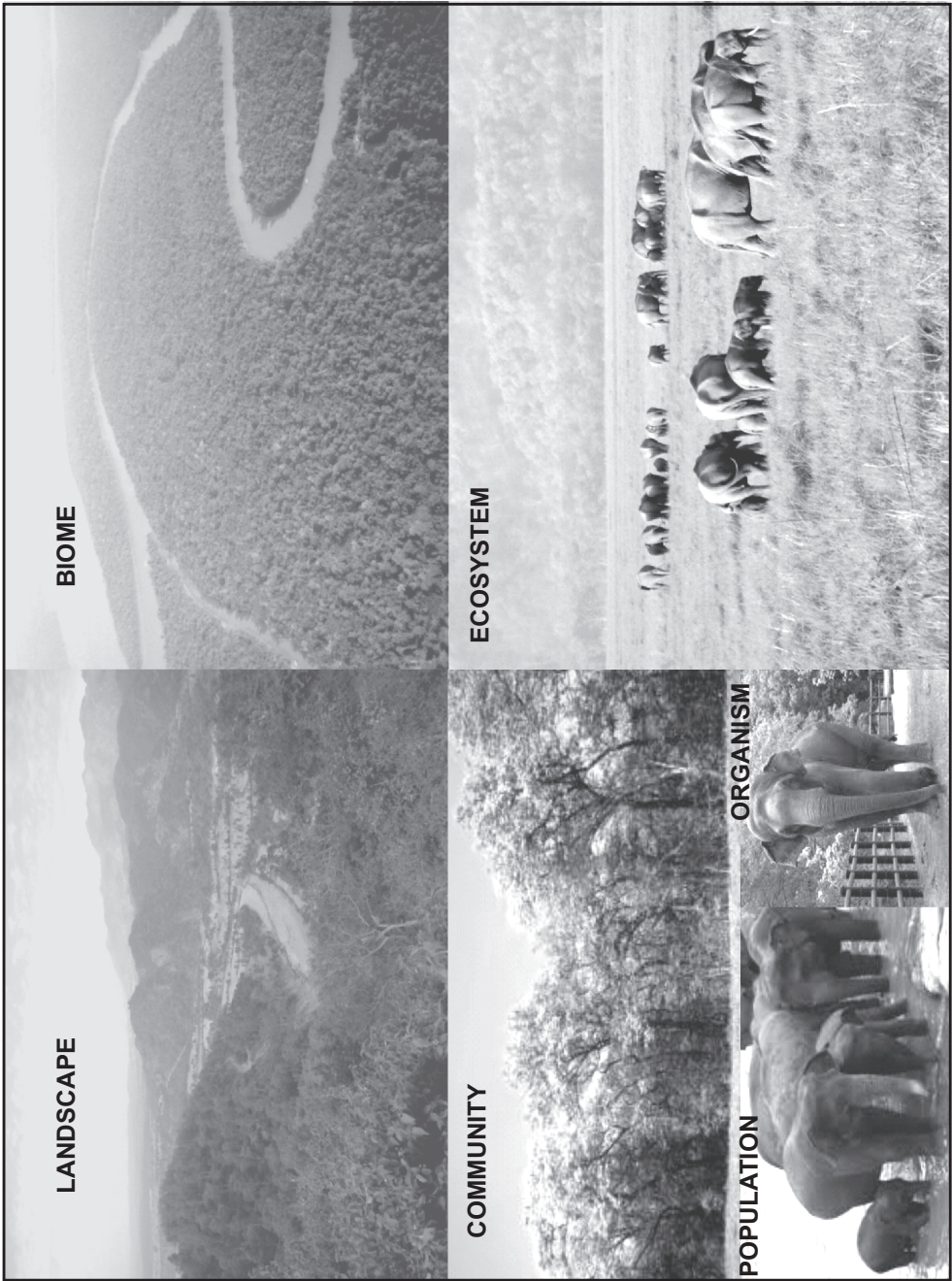


Fig. 1.2. Biological scales: the levels of biological organization from species to biosphere.

that all are equally important as the findings of one level aid in the study of another. For example, understanding of growth and metabolism at cellular level helps assessing the same at organism level, and that developed at the latter level aids in understanding at population level, and so on. However, one will have to assess the growth and metabolism at each of the levels for the findings of one level will never fully explain those of another level. Because an organism is more than the sum of cells, a population is more than the sum of individuals it consists of, and a community is more than the sum of populations of an area. For example, it is possible that a plant species population when grown in garden without individuals of other species may exhibit peak production at 20% soil moisture content, but when the population of this species is grown in a community, i.e., along with individuals of several other species its peak production may occur at 15% moisture level because of the competitive effects of other species. Similarly, we can know the transport mechanism of photosynthate only at cellular or organism level, but the productivity of species in interaction with other species can be only known at community level.

Organisms form a basic unit of study in ecology. At the level of organism, we intend to understand the form, physiology and behavior, distribution and adaptations in relation to the environmental conditions. Similar organisms having the potential for interbreeding and producing fertile offspring constitute what we call a species.

Population is a group of individuals (plants or animals) of the same species of plants or animals inhabiting a given area. An example of population is all individuals of elephants of an area. Many populations may share the same habitat. A population has group characteristics like density, natality (birth rate), mortality (death rate) and age structure. We also study how two populations interact, a predator with its prey or a parasite with its host? Competition, mutualism, parasitism and predation are various types of interactions between species.

The *biological community* is an assemblage of populations of plants, animals, bacteria and fungi that live in an area and interact with each other. A biological community has a distinct species composition, structure and environmental relations as characterized by distinct functional attribute.

The concept of *ecosystem* emphasizes the integration of living and nonliving components such that flow of energy and cycling of nutrients lead to clearly defined trophic structure and ecological diversity. The ecosystem studies help to illustrate phenomena responsible for biological productivity at a level where numerous organisms interact among themselves and with the soil and water components of the ecosystem.

A *landscape* is a unit of land with a natural boundary having a mosaic of patches. These patches generally represent different ecosystems. A *biome* is a large regional unit characterized by a major vegetation type and associated fauna found in a specific climatic zone. Some examples of terrestrial biomes are desert, temperate deciduous forest, tropical rain forest, etc. Aquatic systems are mainly lakes, rivers and marine ecosystems.

All the earth's terrestrial biomes and aquatic systems operating on the surface of the earth on a global scale constitute the *biosphere*. In fact, we may visualize biosphere as a thin film of living organisms on earth. The biosphere consists of the lower atmosphere, land and the oceans, rivers and lakes, where living beings are found.

It is now realized that the Earth itself is a system wherein the interlinked physical, chemical and biological processes cycle materials and energy in complex dynamic ways, providing the conditions necessary for life on the planet. The Earth is perceived as a single, materially closed system with sun as a primary external source of energy, within which the biosphere is an active, essential component.

1.3 SPATIAL AND TEMPORAL SCALES

Environment is the sum total of all biotic (living) and abiotic (non-living) factors that surround and potentially influence an organism. Some components of the environment contribute resources, while others act as regulatory factors. The different components of the environment are interlinked and interdependent. Interactions among the various components of the environment are systematic and governed by the laws of conservation of energy and matter. Environment consists of numerous subsystems and each subsystem represents interacting living and non-living components. The environment can be understood at both large and small scales including regional and global climatic patterns and the microclimate. Some ecological phenomena can only be understood at very large scales. For example the relationship between climate and the distribution of vegetation or the relationship between elevated greenhouse gases in the atmosphere (e.g., carbon dioxide, methane and nitrous oxide and global climate change).

Most organisms interact with their environment at many spatial and time scales. A single bacterium in soil, for example, interacts with air and water within a fraction of a cubic centimeter. On the other hand, the trees in the forest interact with a large volume of air, water and soil. The environment varies from place to place due to variations in climate, soil type and topography. The activities of organisms also influence the hydrosphere, the lower atmosphere and the near surface part of the lithosphere through the exchange of matter and energy. Most organisms have to cope with the external environment over a variety of time scales that may range from few minutes to days, seasons or over a much longer period of geological time scale. For example, phytoplankton populations may change very rapidly with the change in light conditions in aquatic systems. On the other hand, the variations in lithosphere occur very slowly over a long period of time at a particular spatial scale. In general, the scales of patterns and processes in time and space are correlated; the longer the time scale the greater may be the area affected. While many global environmental change issues require time-scales of decades to a century, the basic understanding of the Earth system dynamics consider much longer time-scales.

1.4 EVOLUTION OF ECOLOGICAL THOUGHT

The European dominance in 15-17th century brought a major shift in man-nature relationship. The progress in the science and technological sphere during this period resulted into a qualitative change in human ability to use natural resources. Exploitation of resources started exceeding biological necessities, and this phenomenon continued as science and technology advanced. Thought processes on assessment and distribution of resources, their use potential and economic value led to the enrichment of knowledge base on complex systems of nature and its components, thereby laying the foundation of modern ecology.

The ecological thought and development has been immensely influenced by the natural history. The developments in the field of ecology have been gradual from natural history to biogeography to ecosystem ecology and more recently global ecology with an emphasis on global change, biodiversity conservation, restoration ecology and ecological sustainability (Fig. 1.3). Ecology has now evolved into a modern observational, experimental, and theoretical field describing the diversity of the biophysical environment and the processes regulating the functioning of the natural systems (see Starzomski et al., 2004).

Ecology as a science has reached a fair level of maturity with significant researches in both basic and applied fields and the development of environmental educational programmes. The ecological principles are being applied in environmental management and resource conservation. A brief account of history and development in the field of ecology is given below.

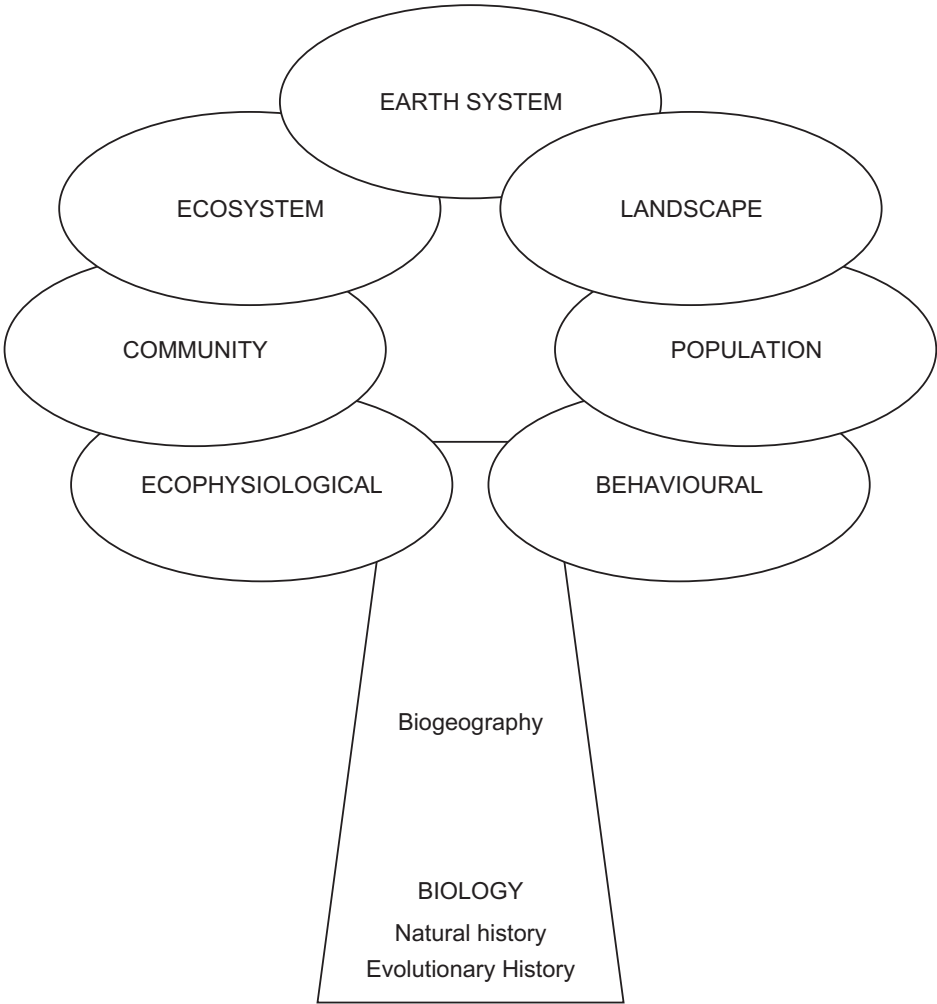


Fig. 1.3. Ecology has developed as a multidisciplinary and holistic science comprising various emerging disciplines of study from biogeography, natural history, biology and evolutionary biology during the last three centuries.

Biogeography and Ecology

Initially, ecology started with descriptive accounts with a focus on biogeographical patterns with a strong taxonomic leaning (Fig.1.3). This biogeographical approach of ecology was strengthened by the contributions of early European (e.g., Braun-Blanquet, 1932; Wulff, 1943) and American ecologists (e.g., Cain, 1944; Kuchler, 1947). Later, Danserau (1957), Fosberg et al. (1961), among others, patronized this approach for tropical regions. In Indian context, the ecological literature accumulated in the form of floristic accounts following the establishment of Botanical Survey of India (BSI) in 1889. This approach with certain refinements and additions of geographical patterns has remained in the mainstream (Chatterjee, 1939; Legris and Meher-Homji, 1968). Monumental book ‘*Silviculture of Indian Trees*’ by Troup (1921) included basic understanding of biology and regeneration of trees in relation to the environmental conditions. H.G. Champion (1936) used the available climatic and forest floristic data to propose the first ever classification of Indian Forests.

Population and Evolutionary Ecology

The study of population dynamics received impetus in the early 19th century after Thomas Malthus (an English economist) called attention to the conflict between expanding human population and limited supply of resources particularly food. The logistic equation of population growth discovered in by Verhulst (1838) was rediscovered and exploited by Pearl and Reed (1920). The new biometric or statistical methods were incorporated into ecological theory and practice. Carrying capacity of the environment, mechanism of population regulation, *r*- and *K*- selection strategies, etc began to be investigated. Theoretical and experimental studies in population ecology also led F. Gause to formulate the competitive exclusion principle which stated that two species that compete for a common and limiting resource cannot coexist and one will be eliminated in time. It led to the assumption that species that coexisted in nature inhabited somewhat different niches. The niche concept which was developed by J. Grinnell in 1917 was fully expanded and formalized by E. Hutchinson, between 1944 and 1959, as a region of an *n*-dimensional hyperspace. The population biology of plants by J.L. Harper provided impetus to the study of plant demography.

There is continuing interest in interpreting ecological phenomena in terms of evolution. Quantitative genetics is being increasingly used, employing techniques of molecular biology, to study ecologically relevant problems.

Experimental and Dynamic Ecology

The beginning of the 20th century also saw the emergence of experimental and dynamic ecology (e.g., Clements, 1916). The idea of the unity of organisms and environment was continued to be pursued after Mobius and Forbes. Karl Friederich in 1927 proposed the concept of *holocoenotic environment* which states that it is impossible to isolate the importance of single environmental factors because the factors are interdependent and synergistic. The concept of 'ecosystem' (a structural and functional entity of biotic communities and their environment) was introduced by Tansley (1935). Charles Elton (1927), an English ecologist, developed the concept of ecological niches, pyramid of numbers and food chains.

Lindeman (1942) elaborated the "trophic-dynamic" aspect of ecology, with emphasis on energy flow. (Lindeman died at the age of 27, and his article on trophic-dynamic aspect was published posthumously). Soon the laws of the thermodynamics were invoked to explain the energy flow in ecosystems. Traditional distinction between plant and animal ecology was reduced, and interdisciplinary research teams to tackle the complexities of ecosystems began to be formed. E.P. Odum's *Fundamentals of Ecology*, first published in 1953, provided a solid framework for ecosystem theory and revolutionized the education as well as research in ecology. Energy flow, mineral-nutrient cycling, population biology and ecosystem dynamics thus became the central themes in ecology.

Mathematical and Statistical Ecology

The mathematical and statistical approaches in ecology with strong experimental background started dominating by the second half of the 20th century. This was the take-off stage of quantitative ecology. The studies made by Goodall (1963), Gounot (1969), Whittaker (1970), etc., strengthened this approach. The experimental and mathematical considerations of ecology provided ability to explain a number of unresolved issues of relative abundance and species area relationships. In 1950s and 1960s, in addition to the availability of complex and sophisticated instrumentation, there was a great increase in the use of quantitative techniques in collecting and analysing data. Contributions of Greig-Smith, Pielou, Patil, etc. firmly established the discipline of statistical ecology. Simple empirical methods of numerical analysis were rapidly supplemented by more elaborate mathematical techniques such as principal components, detrended correspondence, and discriminate analyses which are yielding new insights into species distribution and vegetation organisation through ordination or gradient analysis.

Ecosystem Ecology

During 1950s, there was a widespread concern about rapidly burgeoning resource-related and environmental problems facing mankind. This led to the development of the International Biological Programme in 1960 by ICSU, which had a dramatic effect on ecology. This programme made substantial progress in advancing understanding of ecosystems, particularly the operating processes: photosynthesis and productivity, water and mineral-nutrient cycling, competition, predation, decomposition processes and the role of detritus in the ecosystems. The programme revolutionized the research and teaching world-wide.

Some of the important contributions of the IBP were (i) the development of valuable methodologies which were usually adapted throughout the world in similar studies; (ii) it made possible for specialists to begin to understand the functioning of at least the simpler ecosystems, such as tundra, lakes, deserts, deciduous and coniferous forests and temperate grasslands; and (iii) for the first time tropical rain forests came under the per view of sophisticated ecological studies. The latter highlighted the limitations of techniques developed earlier in tackling as diverse and complex ecosystems as tropical rainforests, and drew attention to the fragility of these systems. The IBP (1964-1974) helped in promoting the ecosystem concept [see Coleman (2010) for details]; the emphasis was, however, on production at primary, secondary and tertiary levels in terrestrial and aquatic ecosystems. In fact, ecosystem concept follows top down thermodynamic approach, wherein fundamental entities are ecosystems, the components of which are multi-scaled interacting processes, such as photosynthesis, energy transfer from one trophic level to the next and nutrient cycling (Allen and Starr, 1982; Allen and Hoekstra, 1992).

The IBP also generated interest in establishing Long-term Ecology Experiments. In fact, of the first six long-term ecological research sites (LTER sites) established in 1980, four of them originated from previous IBP projects (Coleman, 2010). Long-term studies are designed to capture the effects of the environment and biotic interactions (including human impact) in on-going ecological processes. LTERs are also required to validate mechanistic and predictive models of global change and ecological results (Pickett, 1991). Even to obtain knowledge on vegetation succession, long-term permanent plot monitoring is an appropriate research tool. According to Bakker et al. (1996), the relevance of the permanent plot observations lies in: (1) separating trends from fluctuations; (2) the possibility to extrapolate beyond the time range of available observations (now often 20 to 40 years); (3) improving extrapolation to other sites (which have of course their own characteristics); (4) predicting vegetation change at future combinations of environmental factors (scenario analyses); (5) testing ecological models which are often based on assumptions not derived from solid field studies. Since the founding of International LTER's in 1980, global long-term ecological research programmes have expanded rapidly, reflecting the increased appreciation of the importance of long-term research in assessing and resolving complex environmental issues.

In 1971, UNESCO launched an interdisciplinary programme of research, the Man and the Biosphere Programme (MAB), which emphasizes an ecological approach to the study of interrelationships between man and the environment. It is meant to focus on the general study of the structure and functioning of the biosphere and its ecological divisions, the changes brought about by man in the biosphere and its resources, and on the overall effects of these changes upon the human species itself. Thus man and nature interactions became a central theme in ecology. The ecology began to be recognized as a problem-solving science. One of the major outcomes of MAB programme was the establishment of Biosphere Reserves across the world. The sustainable biosphere initiative was proposed by the ecological society of America during 1970 to study the ecology of global change, conservation of biological diversity and sustainable development strategies. The IUCN Commission on Ecosystem Management has adopted ecosystem approach for implementing action oriented programmes for biodiversity conservation, sustainable human livelihoods, empowerment of local people and maintaining ecosystem productivity (IUCN, 2002).

Conservation Biology

Conservation biology is the scientific study of the nature and status of Earth's biodiversity with the aim of protecting species, their habitats, and ecosystems from excessive rates of extinction and the erosion of biotic interactions (Sahney and Benton, 2008; Soulé, 1985). Conservation biology, a new stage in the application of science to conservation problems, addresses the biology of species, communities, and ecosystems that are perturbed, either directly or indirectly, by human activities or other agents. Its goal is to provide principles and tools for preserving biological diversity. Conservation biology is tied closely to ecology in researching the dispersal, migration, demographics, effective population size, inbreeding depression, and minimum population viability of rare or endangered species (Soulé, 1985).

The International Union for the Conservation of Nature (IUCN) has organized a global assortment of scientists and research stations across the planet to monitor the changing state of nature in an effort to tackle the extinction crisis. The IUCN provides annual updates on the status of species conservation through its Red List (www.iucnredlist.org) The IUCN Red List serves as an international conservation tool to identify those species most in need of conservation attention and by providing a global index on the status of biodiversity (Vié et al., 2009).

Recent research provides increasing evidence that biodiversity crisis is a potential threat to ecosystem processes and services (Loreau et al., 2002; Hooper et al. 2005; Balvanera et al., 2006). Biodiversity and ecosystem multifunctional studies have shown that high species diversity is essential for multiple ecosystem functions because different species promote different functions (Hector and Bagchi, 2007; Isbell et al., 2011). Studies have explored the relationship between biodiversity and ecosystem functioning keeping in view the concerns of increasing loss of biodiversity in different types of terrestrial ecosystems. Especially in grassland ecosystems an increased understanding of the functional role of biodiversity has been achieved. Large-scaled experiments carried out (Hector et al., 1999; Tilman et al. 2001a; Spehn et al., 2005) showed significant positive impacts of plant diversity and composition on ecosystem processes such as biomass production, nutrient use, decomposition. High biodiversity is essential to maintain ecosystem services and functions under changing environmental conditions (Isbell et al., 2011).

Predictive Ecology

The amount of data in different fields continues to increase and at the same time rapid advances have been made in computer technology. Van Dyne, Patten and Olson in 1966 laid the foundations of systems ecology, which uses the techniques and concepts of systems analysis derived substantially from engineering. R. Margalef had already introduced information theory into ecology in 1958, and the concepts of information, entropy, and cybernetics were pursued by ecologists. Theoretical concepts relating to species diversity began to be formulated. Thus 1960s was the decade in which theoretical-mathematical ecology grew, allowing predictions about natural and man-made ecological changes. According to Starzomski et al. (2004), some important research areas using advances in computer technology include network theory; the resilience, robustness, and stability of ecosystems; species/system response to climate change, and macroecology. Macroecology has emerged as an important field in ecology due to the application of GIS programmes and remote sensing technologies, development of allometry theory (Enquist and Niklas, 2001, 2002), species area relationships (Sagar et al., 2003) and application of network theory (Dunne et al., 2002 a, b). In a series of recently published papers, Enquist and Niklas (2001, 2002) have discussed the importance of allometric theory to interrelate attributes of organisms, community and ecosystem across temporal and spatial scales. The effect of recent climatic changes on species distribution and phenology, range boundary shifts and the physiology of the organisms have been analyzed through metadata analysis (see Parmesan and Yohe, 2003; Root et al., 2003; Badeck et al., 2004).

Enhancing predictive capacity has remained one of the most important future endeavours for ecologists. There is need to develop and use models to formulate testable hypotheses, forecast change, inform research priorities, and integrate information from monitoring systems, observation networks, and experiments. Mechanistic, quantitative simulation modelling, should be a core component of long-term ecological research. These models need to address the system level linkages between drivers, feedbacks, ecosystem services, economic valuation and human wellbeing indicators.

Human Dimensions in Ecology

Until recently, the ecosystems were considered to be producer controlled. In spite of the clear understanding of the close link of humans with natural ecosystems and their capacity to alter/modify the properties of the ecosystem, they were considered external to the ecosystems. This school of thought is referred to as compositionlists who tend to think that people are a case apart from nature (Callicott et al., 1999). Therefore, any man-made alteration/modification of nature is unnatural. With the establishment of two central axioms (i.e., nature is heterogeneous and scale matters) in contemporary ecology, concepts of landscape ecology emerged. These considerations deal with the spatially explicit relationships among patches in landscape mosaics and with the effects of spatial patterns on ecological processes (Wiens, 1999). With the advent of landscape ecology in the 1970s and the realization that human activities have environmental effects over large areas than local ecosystems (Forman and Gordon, 1986), the human dimensions started to be a central theme of ecological studies. Thus, it is important to recognize that ecology has interface with earth system science, social science and contemporary environmental problems (Fig. 1.4).

Articulation of human dimensions in ecology shifted the emphasis towards predator controlled or consumer controlled ecosystems. Disturbances are being seen as a driving force in the evolution of ecological systems. More researches are being focused on human impacted ecosystems (Ramakrishnan, 1992a, b). Recent studies that contribute to improving the quality of

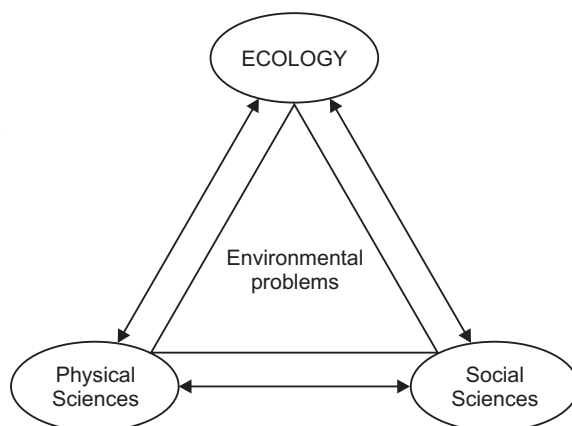


Fig 1.4. Ecology has linkages with physical and social sciences to formulate management strategies for environmental problems at different spatial and temporal scales.

life have created a definite niche. This includes sustainability, equity and coexistence with other species and components of our heritage in a biologically and culturally diverse world (Huntley et al., 1991). The applied fields like, conservation biology, restoration ecology, landscape ecology and ecotoxicology are receiving increasing attention. In conservation biology, we address the issues of distribution and utilization of natural resources, environmental pollution, global climate change as well as the management of habitats and species. Ecotoxicology encompasses the study of toxic effects of chemicals and contaminants such as hazardous wastes, industrial chemicals, heavy metals, pesticides and fertilizers on the organism, population, community or ecosystems including human beings.

The process level understanding of decomposition, nutrient cycling and regulation of ecosystem functions through top down and bottom up controls are being put to practice to improve productivity, profitability and more importantly, the sustainability of wide ranging terrestrial and aquatic systems. These considerations usually advocate two basic concepts-sustainable yield and sustainable

development. The genesis of these concepts can be traced to the conservation philosophies, which dominated the first three quarters of the 20th century i.e. (i) resource conservation (resourceism-natural is valued only to the extent that it is humanly useful) and (ii) wilderness (preservationism) (Salwasser, 1990; Callicott et al., 1999).

Concept of sustained yield is largely based on harvestable surplus that so much and no more can be received continuously. However, the theoretical models of sustained yield are complicated by such biological variables as the growth rates and optimum harvest sizes and ages of targeted organism (Larkin, 1977). The concept of sustainable yield received considerable opposition from conservation biologists particularly in view of the fact that most species are not harvestable resources and many of the species face danger of genetic impoverishment, local extirpation, and global extinction not due to over harvesting but because their habitats are being polluted and destroyed (Ehrlich, 1988).

1.5 ENVIRONMENTAL SUSTAINABILITY

In a very short history of Earth, humans and their activities are having significant impact on the natural dynamics of the Earth system. The main cause of environmental changes is ever increasing human population. During the 20th century, global human population has increased by 3.7 fold (to 6 billion people) and consumption of resources has increased twenty-fold world over. In the past 50 years the world has experienced an unprecedented increase in population growth. The global human population is estimated to be 7.064 billion in 2012 by the United States Census Bureau (US Census Bureau, 2013). According to the United Nations, world population reached 6 billion at the end of 1999. In just 12 years, another 1 billion people have been added to the Earth, with global population reaching 7 billion in October 2011. Demographers project that the global population will reach eight billion by 2030, and will likely be around nine billion by 2050. It is expected that global population will be more than 10 billion by 2100 (UN, 2011), Fig. 1.5. Given current trends in climate change, food production, and energy supplies confronting human society, population growth is critical to the future sustainability of the planet.

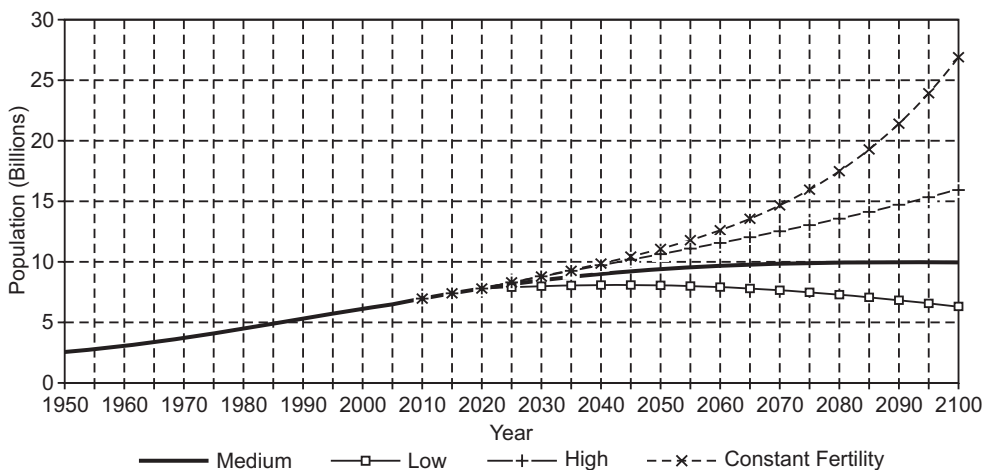


Fig. 1.5. Population of the world, 1950-2100, according to different projections and variants. According to the medium fertility scenario, the total population will stabilize at 10 billion around the end of the century. (source : Population Division of the Department of Economic and Social Affairs of the United Nations Secretariat, 2011). World Population Prospects: The 2010 Revision. New York: United Nations. esa.un.org/wpp/

The various human activities such as food production, industrial development, international commerce, energy production, and urbanization are transforming the earth system at various scales ranging from local to global. The collapse of some ocean fisheries, the destructive influence of agricultural and animal husbandry practices on unsuitable lands, and the destruction of tropical rain

forests are all caused, directly or indirectly, by population growth pressure. Human population and the economic wealth of people (particularly in the developed countries) have significantly increased degradation of natural resource.

During the last two decades, the major global environment changes that are being realized and are occurring at ever-increasing pace have been the focus of research under the Millennium Ecosystems Assessment, launched in June 2001 by the UN. The Millennium Ecosystems Assessment was followed by PAGE (Pilot Analysis of Global Ecosystems), which assessed five major ecosystem types including agro ecosystems, forests, freshwater, grasslands and coastal and marine ecosystems (WRI/UNDP/UNEP/World Bank, 2000). The trends of ecosystem health in terms of production, water quality, water quantity, and biodiversity and carbon storage are shown in Fig. 1.6. It is evident that biodiversity has shown decline in all the major ecosystems.

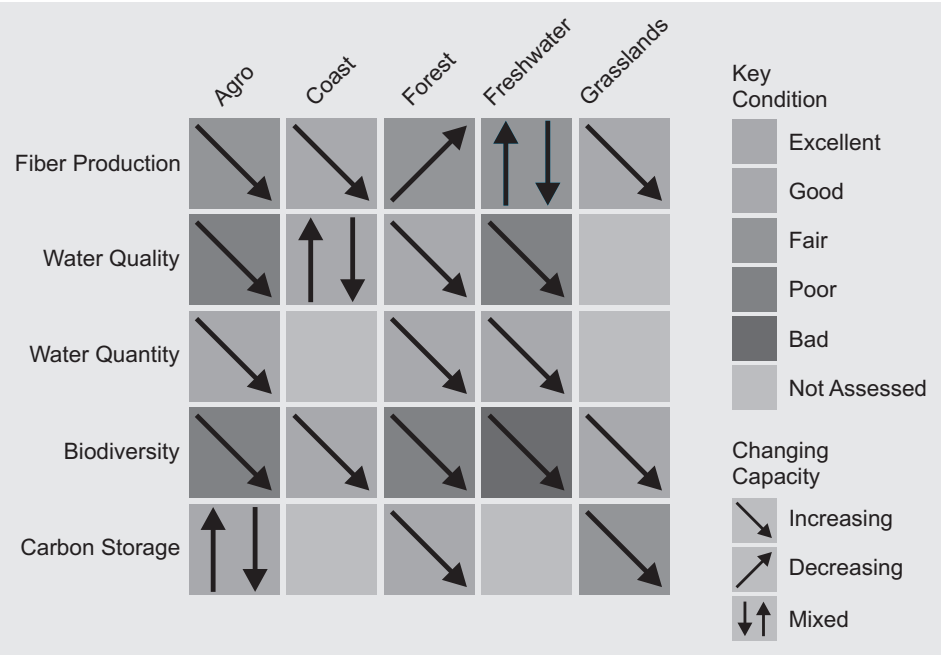


Fig. 1.6. Environmental scoreboard to show the trends of change in major ecosystem conditions and changing capacity (Reprinted, with kind permission, from *A Guide to World Resources 2000-2001 : People and Ecosystems: the Fraying Web of Life*. © 2000 World Resources Institute). www.wri.org.

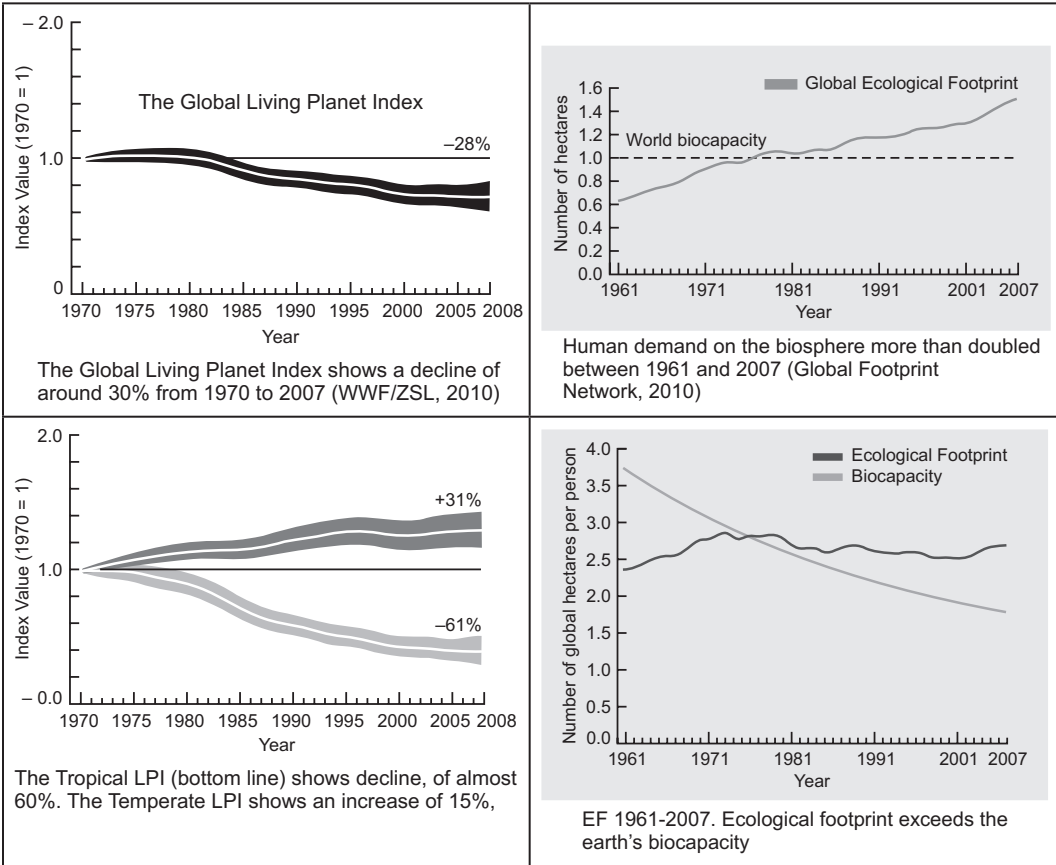
International Geosphere-Biosphere Programme (IGBP) is one of the world's leading scientific research programmes studying changes to the global environment by addressing the interactive physical, chemical and biological processes that define Earth-system dynamics, changes that are occurring in these processes and the role of human activities in these changes.

Ecosystem Health and Biological Integrity

A healthy ecosystem is able to provide flow of essential services to the society without which human civilization would cease to thrive. Unfortunately, these undervalued services are being impaired and destroyed. The Living Planet Index is a measure of the natural wealth of Earth's ecosystems, which has declined by about 28% globally between 1970 and 2008 (Box 1.1 and Box 1.2). The Living Planet Index, reflecting changes in the state of the planet's biodiversity, using trends in the size of 9,014 populations of 2,688 mammal, bird, reptile, amphibian and fish

species from different biomes and regions (WWF/ZSL, 2012), has shown a 28% global decline in biodiversity health since 1970 (Box 1.1). The tropical Living Planet Index declined by more than 61% from 1970 to 2008, while the temperate Living Planet Index increased by 31% over the same period (WWF/ZSL 2012, Box 1.1). However, recent average population increases of major groups of animals may not be indicative of better state of temperate ecosystems than tropical ecosystems.

Box 1.1.Recent information on LPI and EF (based on WWF/ZSL, 2012)



Box 1.2 Living Planet Index (LPI) and Ecological Footprint (EF) (based on WWF/ZSL, 2012)

Two synoptic indices to acquire general picture of environmental conditions

LPI is a measure of the natural wealth of earth's forest, freshwater, and oceanic/coastal ecosystems

EF is a measure of area of biologically productive land or sea required to produce food, material and energy for human population.

The global terrestrial Living Planet Index declined by 25 % between 1970 and 2008 based on 1,432 species of birds, mammals, amphibians, and reptiles from a broad range of temperate and tropical habitats, including forests, grasslands and drylands (Table 1.1). The marine Living Planet Index declined by more than 20% based on 675 species of fish, seabirds, marine turtles and marine mammals found in temperate and tropical marine pelagic, coastal and reef ecosystems between 1970 and 2008 (Table 1.1). The freshwater Living Planet Index declined more than for any other biome. The index includes 737 species of fish, birds, reptiles,

amphibians and mammals found in temperate and tropical freshwater lakes, rivers and wetlands. Overall, the global freshwater index declined by 37% between 1970 and 2008 (Table 1.1).

Table 1.1. The Change in Living Planet Index from 1970 to 2008

Index	# of species monitored	% change since 1970	Notes
Terrestrial	1432	-25%	Global
Freshwater	737	-37%	Global
Marine	675	-22%	Global

(Source : WWF/ ZSL, 2012)

The Ecological Footprint (the area required to produce food, materials and energy, increased by 50% between 1970 and 1997, and has far exceeded the existing biological capacity in most Asian countries (Wu and Overton, 2002). The Ecological Footprint tracks humanity's demands on the biosphere by comparing humanity's consumption against the Earth's regenerative capacity, or biocapacity (Box 1.1). For 2008, humanity's total ecological footprint has been estimated at 1.5 planet Earths, that is, humanity uses ecological services 1.5 times as quickly as Earth can renew them. By the 2030s we may need the equivalent of two Earths to maintain our current lifestyles (WWF/ZSL, 2012). The human influence on the earth system is so pervasive that the present era is called anthropocene. Nature is finite and we have almost reached a critical point beyond which ecological decline would lead to disasters.

Main causes of disruption of ecosystems and ecosystem services are:

Land degradation: Over 40% of vegetal land surface of earth has been directly disturbed (Daily, 1995) and its natural production capacity has been reduced, diverted or destroyed (Vitousek et al., 1986). Habitat degradation and the loss of biodiversity are threatening the livelihoods of more than 1 billion people living in dry and sub-humid lands globally.

Biodiversity loss and ecosystem services: The over-exploitation of ecosystems is evident at local to global scales with profound negative impacts on biological diversity and livelihood opportunities of the people. Current extinction rates caused by human activities are orders of magnitude higher than natural background levels (Pimm et al., 1995, see Singh, 2002). The loss of biological diversity is having impacts on the local rights of people along with their cultural diversity.

Ecosystem services are the processes and conditions of natural system that support human activity and well-being. The ecosystem services that benefit society are air quality, climate regulation carbon sequestration, water purification, pollination, and prevention of erosion and have been valued in the range of 16 to 54 trillion (10^{12}) US dollars per year (Constanza et al., 1997). The Millennium Ecosystem Assessment defined four categories of ecosystem services, i.e., provisioning services, regulating services, cultural services and supporting services that contribute to human well-being (MEA, 2005a). 'The degradation of ecosystem services represents loss of a capital asset; loss of ecosystem services are seldom, if ever, brought into the balance sheet of GDP. When loss through unsustainable use is factored into GDP, many of the countries that show positive GDP growth are actually experiencing net loss of capital, with dire consequences for future growth' (MEA, 2005a).

The ecosystem services approach can save many ecosystems with high biodiversity and willingness of society to protect their biodiversity. TEEB (The Economics of Ecosystems and Biodiversity) is a major global study to draw attention to the tangible benefits of biodiversity, and highlights the growing costs of biodiversity loss and ecosystem degradation (TEEB, 2010). The Convention on Biological Diversity, the Millennium Development Goals, and other international agreements clearly indicate relationship between biodiversity conservation, poverty alleviation, and human well-being (MEA, 2005a; Sachs et al., 2009; Turner et al., 2012).

Deforestation and fragmentation: Forests covered about 50% of the earth's land area 8000 years ago, as opposed to 30% today (Ball, 2001). Forest plays an important role in maintaining ecological processes as well as in providing livelihood and supporting economic growth (FAO, 2010). More than 1.6 billion people rely on forests and non-timber forest products for their livelihoods.; wood and other goods removed from the forests have been valued at 122 billion US dollars in 2005 (FAO, 2010a). The forests are the home to 78% of the land biodiversity, and are among the most biodiverse terrestrial ecosystems (FAO, 2010a; IUCN, 2010). Tropical regions lost 15.2 million hectares of forests per year during the 1990s and forest degradation was most extensive in Southeast Asia (0.42% per year) (FAO, 2001a). Forest fragmentation replaces large areas of native forest by other ecosystems leaving isolated forest patches, with deleterious consequences for most of the native forest biota (Murcia, 1995). With an increasing number of habitats becoming fragmented, we are faced with the task of conserving a patchy world. There are two seemingly contrasting options: 'preserve optimal habitat quality wherever it can be found' as opposed to 'preserve a large number of well-connected patches' (Roslin, 2002).

Intensification of agriculture: The emergence of agriculture was one of the most fundamental transformations of the human–environment interaction (Kinzig, 2002). Land conversion from natural ecosystems to simplified agroecosystems results in a substantial reduction in biodiversity and a decrease in habitat for displaced wildlife and plant communities (Cassman et al., 2003). Historically, humans have increased agricultural output mainly by bringing more land into production (Lambin et al., 2003). However, increase in the world's food production over the last decades was mostly due to agricultural intensification which is defined as higher levels of inputs and increased output of cultivated or reared products per unit area and time (Tilman, 1999). This signifies the use of high-yielding crop varieties, fertilization, irrigation, and pesticides on land already under agriculture (Matson et al., 1977). Water used for irrigation accounts for 80% of all freshwater consumption (Rosegrant et al., 2002). Nitrogen (N) fertilizer applied to agricultural land comprises more than 50% of the global reactive N load attributable to human activities (Smil, 1999). The high input agriculture has a number of negative environmental consequences such as the use and release of limiting resources, release of pesticides and conversion of natural ecosystems to agriculture, which influence ecosystem functioning.

Environmental pollution: Heavy use of coal, high-sulphur fuels and N fertilizers and growth in vehicular traffic have increased the atmospheric pollution load. The rising levels of total suspended particles, respirable particles and nanoparticles are posing huge risk to human health. The contamination of air, water and soil is accelerating the accumulation of toxic metals in the human food chain.

Biological invasion: The frequency of biological invasions has increased. Presently the alien invasive species constitute a global challenge of great magnitude by posing great threat to biological diversity, harming human health and causing enormous economic losses (Vitousek et al., 1997b; McNeely et al., 2001). The concept of biological integrity [i.e., native species diversity in naturally structured biotic communities (Angermeier and Karr, 1994)] is receiving increasing attention because biodiversity may be artificially increased by introducing exotic species into vacant or under-exploited habitats, thereby disturbing the integrity of a biotic community. Disturbance in biological integrity can result in competitive exclusion of sensitive and endemic species (Callicott et al., 1999). Interestingly, plant invasions are most pervasive in areas of high plant endemism posing a still greater threat to native biodiversity (Stockwell et al., 2003). Invasive species can threaten

biological diversity by reducing genetic variation and eroding gene pools, through the extinction of endemic species, and by altering habitat and ecosystem functioning. Biological invasions are a pervasive global change challenging the conservation of biodiversity and natural resources (MEA 2005a; TEEB, 2010). The cost of damage caused by invasive species is estimated as 1.5 trillion dollars per annum in environment and economic damage-close to 5% of global GDP (GISP, Global Strategy 2008 – 2010, www.diversitas-international.org).

Global Warming and Ecological Considerations

Among the most obvious consequence of global warming is the change in altitudinal and latitudinal distribution of organisms and their assemblages. While affecting the distribution, phenology and physiology of the organisms at individual level (IPCC, 2001; Walther et al., 2002). These shifts merit consideration with regard to changes in both structural and functional properties of the ecosystems. In the Indian context, Himalayan Mountains are supposed to be the most sensitive to these effects. Rapid glacial melt can cause serious flood damage in heavily populated regions, and as the glaciers recede, large regions that rely on the glacial run-off for water supply could experience severe water shortage (McDowell, 2002). On a much wider perspective, the expected rise in sea level will have profound effects on human sustenance and sustainability of coastal habitats around the world including India (IPCC, 2001; IPCC, 2007).

Sustainable Development and Ecological Sustainability

As defined in the Brundtland Commission Report (WCED, 1987), sustainable development is meeting the needs of the present without compromising the ability of future generations to meet their own needs (Fig. 1.7). This was further strengthened at the United Nations Conference on Environment and Development held at Rio de Janeiro in 1992, and the World Summit on Sustainable Development held at Johannesburg in 2002. The sustainable development is more commonly understood to mean economic development that does not appreciably harm the natural environment (WRI, 1992). Another sustain-rooted conservation concept-ecological sustainability focuses on conserving the biota of ecosystems that are inhabited by humans and are economically exploited. In simple words, ecological sustainability is meeting human needs without compromising the ecosystem health (Callicott and Mumford, 1997). The concept advocates that proposed developmental projects deemed to be sustainable, in addition to economic criterion, should meet analogous ecological criteria, i.e., they should be ecologically sustainable and economically sound (Callicott et al., 1999).

The ecological interpretation of sustainability and the emerging conservation concept- ecosystem health have both anthropocentric and ecocentric value dimensions (Callicott and Mumford, 1997). Such realizations have called upon two separate disciplines (i.e., ecologists and social scientists

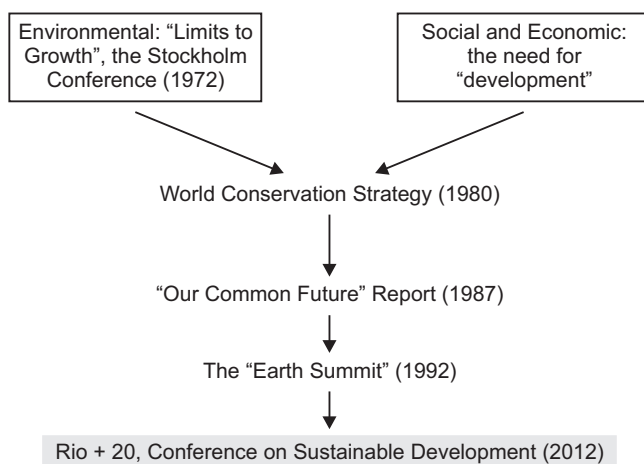


Fig. 1.7. Historical roots of sustainable development

who have traditionally worked separately) to integrate their approaches in a complementary manner (Huntley et al., 1991) to address the issues of physical, biological and socio-economic components of ecosystems. Ultimately, this must address the question of most fundamental importance: that of the earth's human population supporting capacity. The complexity intertwined in this question, calls for considering varied socio-economic scenarios, implied living standards, and potential per capita material and energy use scenarios. Considerations also focus on level of agricultural inputs, alternate use of potentially productive lands, industrial inputs, and energy subsidies and fluxes (Huntley et al., 1991). Herein lies the motivation to develop sustainable economic alternatives through maintenance of ecosystem health. Ecosystems are healthy when the linked ecological processes that compose them occur normally - that is, as they occurred historically (Costanza et al., 1992).

As human interaction with the natural environment is increasing immensely and day by day, we need to refocus ecology towards research that ensures a future in which natural systems and the humans they include coexist on a more sustainable planet. Sustainability science is a field of research which deals with the interactions between natural and social systems, and with how those interactions affect the challenges of sustainability. It has emerged in the 21st century as a new academic discipline (Kates et al., 2001). The core of this discipline is the concept of sustainability which means meeting the needs of present and future generations while substantially reducing poverty and conserving the planet's life support systems. Thus, sustainability science addresses the fundamental character of interactions between nature and society, and society's capacity to guide those interactions along sustainable trajectories (Kates et al., 2001). Sustainability science, therefore, seeks real world solutions by breaking down artificial disciplinary gaps between the natural and social sciences through the creation of new knowledge and its practical application to decision making (Clark and Dickson, 2003; Palmer et al., 2005).

Sustainable Development Goals

At the Rio+20 United Nations Conference in June, 2012 held in Brazil, the world's governments agreed to produce a set of sustainable development goals with a global reach (Glaser, 2012), and integrated into the follow up of the Millennium Development Goals (Griggs et al., 2013). The idea of sustainable development goals was put forward by Columbia and Guatemala in 2011, received widespread support at the Rio+20 UN conference. The sustainable development goals are an important idea that could help to move the world to a sustainable trajectory (Sachs, 2012). The Rio+20 outcome document proposes that sustainable development goals must be "action oriented, concise, easy to communicate, limited in number, inspirational, global in nature and universally applicable to all countries, respecting national policies and priorities" (Griggs et al., 2013). In view of increasing consumption and population increase, definitions of sustainable development must also include the security of people and the planet (Griggs et al., 2013), Fig. 1.8. Adopting the planetary boundaries works, using recent scientific studies, and existing international processes, a list of sustainability must haves for human prosperity have been suggested as shown in Fig. 1.8 (Griggs et al., 2013). However, long-term research initiatives are needed to refine targets and provide sustainable solutions for human well-being.

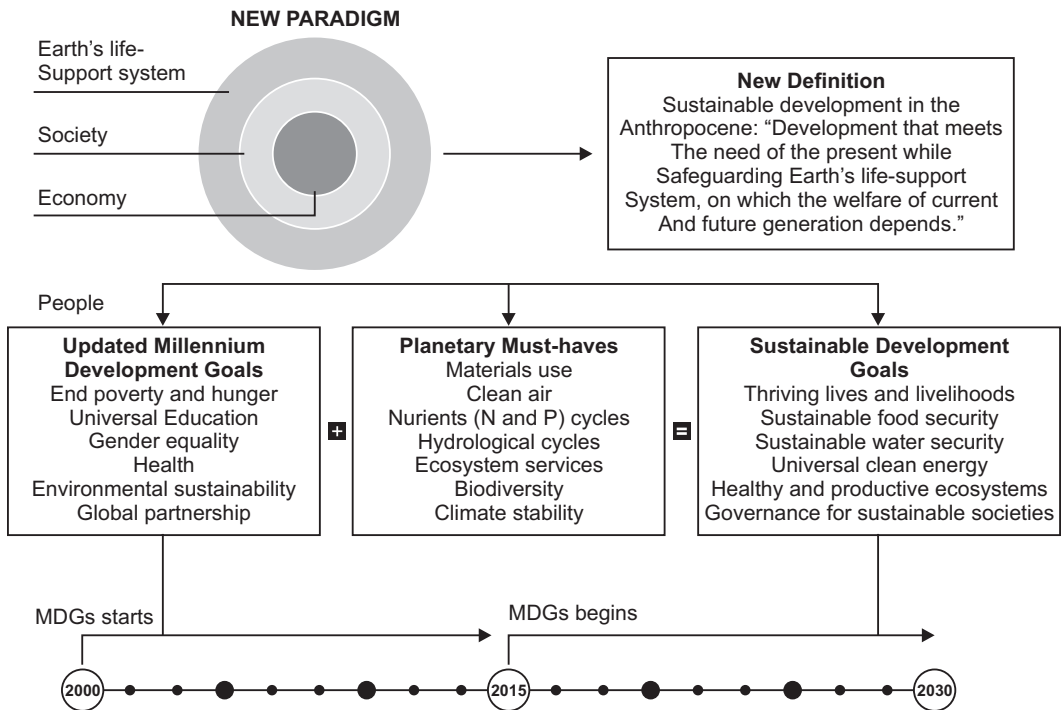


Fig.1.8. The Millennium Developmental Goals (MDGs) combined with global environmental targets indicating six sustainable development goals with provisional targets for 2030. (adapted from Griggs et al., 2013).

Ecoinformatics

Understanding and sustaining the environment and natural resources depends on improving our capacity to access data on ecological, earth science and human dimensions at different spatial-temporal scales. This requires mining of the data from various sources for new knowledge and obtaining new insight for decision making and effective people's participation. Therefore information technology network and infrastructure should facilitate discovery, access, integration, interpretation and analysis of widely distributed ecological information. Ecological information can provide conceptual and practical tools for understanding, generation, processing, and dissemination of ecological data and information. For example, Sugden and Pennisi (2000) emphasized the importance of digitizing biodiversity data, sharing of information across data bases, delivering the data to the desktop and continued monitoring of biodiversity. Ecoinformatics is framework that helps scientists to generate new knowledge by using innovative tools and approaches to integrate and analyze large amounts of data at scales from genes to biosphere (Michener and Jones, 2011). The US National Ecological Observatory Network (NEON) and Global Lake Ecological Observatory Network (GLEON) provide research platforms that enable scientists to examine phenomena across diverse ecosystem types and collecting diverse environmental observations (Michener and Jones 2011 and references there in).

Remote sensing technologies and Geographical Information Systems are being used to understand large-scale patterns and spatial relationships between components of the landscape (Bawa et al., 2002), resource management and biodiversity conservation. Global Biodiversity Information Facility (GBIF), the European Network for Biodiversity Information (ENBI), Inter-American Biodiversity Information Network (IABIN) are some of the biodiversity related databases which provide open access. In India, Jeeva Sampada, Indian Bioresources Information Network (IBIN) and Biodiversity Information System (BIS) are good starts in this direction. See Roy, Roy and Karnataka (2012) for details on web-enabled, Biodiversity Information System.

Planetary Boundaries

Planetary boundaries is the central concept in an Earth system framework proposed by a group of Earth system and environmental scientists led by Johan Rockström from the Stockholm Resilience Centre and Will Steffen from the Australian National University (Box 1.3; Fig. 1.9).

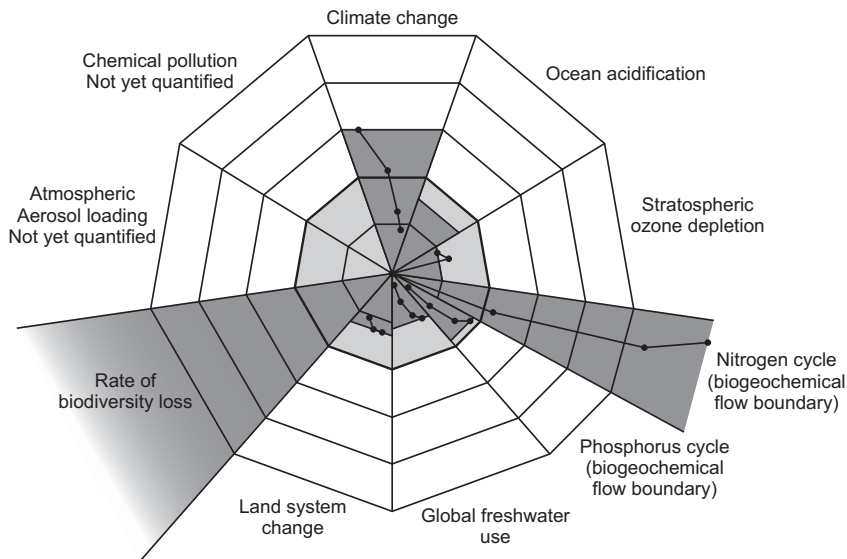


Fig. 1.9. Nine components of Earth systems that show signs of global environmental change driven by human activities. (Adapted from Rockström et al., 2009b. Planetary boundaries: exploring the safe operating space for humanity. Ecology and Society 14(2): 32.) [online] URL: <http://www.ecologyandsociety.org/vol14/iss2/art32/>

Box 1.3. Nine components of Earth systems that show signs of global environmental change driven by human activities

In 2009, a group of 28 internationally renowned scientists identified and quantified a set of **nine planetary boundaries** within which humanity can continue to develop and thrive for generations to come (Rockström et al., 2009a, b). Crossing these boundaries could generate abrupt or irreversible environmental changes. Respecting the boundaries reduces the risks to human society of crossing these thresholds.

The scientists first identified the Earth System processes and potential biophysical thresholds, which, if crossed, could generate unacceptable environmental change for humanity. They then proposed the boundaries that should be respected in order to reduce the risk of crossing these thresholds.

The **Nine boundaries identified** were: climate change, stratospheric ozone, land use change, freshwater use, biological diversity, ocean acidification, nitrogen and phosphorus inputs to the biosphere and oceans, aerosol loading, chemical pollution (Figure 1.8). The study suggests that climate change, biological diversity and nitrogen input to the biosphere may already have crossed planetary boundaries. In addition, it emphasizes that the boundaries are strongly connected — crossing one boundary may seriously threaten the ability to stay within safe levels of the others.

Source : Rockström et al. (2009a, b)

In 2009, the group proposed a framework of “planetary boundaries” designed to define a “safe operating space for humanity” for the international community, including governments at all levels, international organizations, civil society, the scientific community and the private sector, as a precondition for sustainable development (Rockström et al., 2009a, b). This framework is based on scientific research that indicates that since the Industrial Revolution, human actions have gradually become the main driver of global environmental change. The scientists assert that once human activity has passed certain thresholds or tipping points, defined as “planetary boundaries”, there is a risk of “irreversible and abrupt environmental change”.

1.6 ENVIRONMENTAL ETHICS

Ecology studies the complex physical relationships between living and non-living systems. Economics analyzes the nature of commercial transactions in human society. Ethics is the systematic examination of our rights, our responsibilities and the consequences of our conduct. Environmental ethics is a new sub-discipline of philosophy that deals with the ethical problems surrounding environmental protection (Yang, 2006). It aims to provide ethical justification and moral motivation for the cause of global environmental protection. Environmental ethics extends the scope of ethical concerns beyond one’s community and nation to include not only all people everywhere, but also animals and the whole of nature – the biosphere – both now and beyond the imminent future to include future generations.

There are many overlapping concerns and areas of consensus among environmental ethics, environmental politics, environmental economics, environmental sciences and environmental literature. The distinctive perspectives and methodologies of these disciplines provide important inspiration for environmental ethics, and environmental ethics offers value foundations for these disciplines. Three normative principles of environmental ethics are the Principles of environmental justice, Principle of intergenerational equality, Principle of respect for nature (Yang, 2000).

Environmental ethics is a diversified discourse (Yang, 2006). In modern society, it comprises four schools of thought: enlightened (or weak) anthropocentrism, animal liberation/rights theory, biocentrism and eco-centrism (which include the land ethic, deep ecology and the theory of nature’s value). When an environmental ethic constrains our use of resources, economics evolves into ecological economics, which is based on an understanding of society’s dependence on the natural world. By appreciating the ethical significance of environmental resources as well as their economic worth, we will be better able to assess their value and communicate their benefits to the public.

Environmental Degradation and Ethics

The 1960s and 1970s witnessed an ecological crisis brought about by industrial civilization. This crisis was composed of environmental pollution (such as air pollution, water pollution, soil pollution, toxic chemical pollution, and solid waste pollution), resource shortages (such as shortages of energy, cultivated land, minerals and fresh water) and ecological imbalances. Rachel Carson’s *Silent Spring* (Carson, 1962) revealed the life-threatening nature of chemical pesticides and questioned the dominating concept of conquering nature. Rachel Carson’s was an American marine biologist and conservationist. The year 2012 marked the 50th anniversary of publication of Rachel Carson’s seminal book, *Silent Spring*, considered being the most significant paving the way for the modern environmental awareness and action movement. Rachel Carson’s environmental ethic provides guidelines for the challenges we face in the 21st century. Her message distilled from all of her writing can be summarized in four basic principles, i.e., (i) live in harmony with nature, (ii) preserve and learn from natural places, (iii) minimize the effects of synthetic chemicals on the natural systems of the world, (iv) to consider the implications of all human actions on the global web of life.

Climate Change and Ethics

Global warming is the most severe challenge confronting humanity in the twenty-first century. Global environmental change gives rise to ethical challenges that need to be grasped within a framework of critical and forward-looking thinking. Environmental challenges often tend to be framed as costs to be borne and technologies to be discovered, leaving behind the opportunities and co-benefits associated with serious engagement with their ethical dimensions. The impact of climate change on health is one of the most significant measures of harm associated with our warming planet. Protecting health is therefore one of the most important motivations for climate action. Climate change is affecting human health in multiple ways: both direct – through extreme weather events, food and water insecurity and infectious diseases – and indirect – through economic instability, migration and as a driver of conflict.

The recently concluded UN Climate change conference (COP18.CMP8) in Doha, Qatar, in November 2012 has clearly emphasized that issues of ethics and justice must not be ignored in the search for a global solution to climate change. Climate change is a civilization challenging ethical issue and need to give ethical considerations in policy formation (Brown, 2012). Global environmental change gives rise to ethical challenges that need to be grasped within a framework of critical and forward-looking thinking (Brown, 2012).

Biodiversity and Ethics

Environmental ethics can help people to understand clearly and comprehensively the economic as well as non-economic values of biodiversity. An integrated ethical framework based on the theoretical research, animal, and environmental ethics traditions can provide practical guidelines for biodiversity conservation. The ecosystem services approach can save many ecosystems with high biodiversity and willingness of society to protect their biodiversity. Agricultural biodiversity is under serious threat in many parts of the world despite extensive conservation efforts. Ethical considerations regarding the collecting, research, and use of agricultural biodiversity are currently topics of great concern. Moral values, norms and traditions will be needed to reshape our relationship with the living world of which we are part.

Public Interest in Environment

During the last four decades, a number of international programmes and events have increased the public awareness of problems and need for conserving environmental resources. As stated earlier, Rachel Carson in 1962 published *Silent Spring*, a monumental work to create public awareness of environmental pollution. On April 22, 1970, the first annual Earth Day was celebrated to create awareness about environmental quality. The International Union for Conservation of Nature and Natural Resources (IUCN), the United Nations Environment Programme (UNEP), and the World Wide Fund for Nature (WWF) published “*Caring for the Earth: a strategy for Sustainable Living*” (IUCN/ UNEP/WWF, 1991). In June 1989, the world conference on changing atmosphere in Toronto led to the formation of the Intergovernmental Panel on Climate Change (IPCC); now a consortium of over 2000 scientists and policy analysts from over 100 nations. The IPCC has produced three major reports on the various aspects of the climate change; the most recent one is Third Assessment Report. The Convention on Biological Diversity and Kyoto Protocol are some important policy instruments for greater environmental protection on a global scale.

1.7 ENVIRONMENTAL SUSTAINABILITY ISSUES IN INDIA

In spite of strong traditions of nature preservation and its due recognition in the Indian Constitution (Article 48 a) - “the State shall Endeavour to protect and improve the environment and to safeguard forests and wildlife in the country” - the real orientation towards contemporary environmental issues began in the aftermath of the 1972 Stockholm Conference on Human

Environment. In this conference, the then Prime Minister of India, Mrs. Indira Gandhi forcefully emphasized that the removal of poverty should form an integral part of an environmental strategy for the world. Since then, integration of poverty alleviation with environmental protection became a reality in India. Between Stockholm Conference and the World Summit on Sustainable Development, India has established a fairly strong organizational structure and legal and policy framework for the protection of environment, focusing on poverty alleviation and natural resource conservation.

In spite of these and other initiatives, India still needs to make considerable strides (Singh 1991a and 1991b, also see Box 1.4). Burgeoning mass of data gathered through research has largely remained un-translated into management practices. For India, a formidable issue is the available land/water mass (2.5% of the world) for a rapidly increasing population (16% of the world). This creates enormous pressure on natural resources and human development as reflected by poverty and inequality, quality of water resources and environmental health risks (MoEF, 2002a, 2002b). The Human Development Index (HDI) is 0.611 for India, the life expectancy is 63.6 yr in comparison to an average of 64.5yr for other developing countries. Still we have nearly 27% of population under poverty line and over 40% of city dwellers do not have proper shelter.

With roughly 23% of forest area (forest + tree cover), including nearly 42% degraded forests, we still are far behind the national goal of 33% of total area under forests set by the National Forest Policy, 1988. Rapid deterioration of land resources is yet another major concern. Due to increasing demand of land by ever growing population, nearly 50% of the country's land is degraded at varying levels. The problems associated with water scarcity (both qualitative and quantitative) are aggravating with every passing day. India is among 17 countries which are most water scarce, and in 2025 will not have enough water to maintain 1990 levels of per capita food production from irrigated agriculture and meet industry, household and environmental needs (Seckler et al., 1999).

Environmental health risks arise due to poverty, unsafe drinking water, sanitation in rural and urban areas, air quality, indoor air pollution, unsafe solid waste management, and agro-industrial pollution. The quality of air in most of the urban India has deteriorated to an alarming level. The faulty disposal of municipal, hazardous and medical wastes causes many types of diseases. Chemical contamination of harvestable portions of crops and vegetables, particularly by heavy metals, from sources such as industries, vehicles, pesticides, contaminated irrigation water are matters of growing concern (Agrawal, 2003). Likewise, the rapid developmental activities and increasing population density are putting the coastal zones of India under severe stress. The wetland systems are now among threatened categories. Similar is the case of Indian Mountains where the pressures on natural systems are increasing at an alarming rate.

In India, although legislation has evolved on wild-life conservation, water and air pollution, biodiversity protection, and environmental assessment for development projects (Sivaramakrishnan, 2011), we continue to face enormous challenges to address pollution, and its consequences for human health, declining soil and water ecosystem services, deforestation and biodiversity loss, climate change, and human-wildlife conflicts (Singh and Bagchi, 2013). In India, as in other parts of the world, different types of human-wildlife conflicts are increasing because of loss of natural habitats and land-use change. In India, human-wildlife conflicts particularly relate to mortality and morbidity of humans due to wild-life, and wildlife damage to crops and other property (Singh and Bagchi, 2013 and references therein). Annually conflicts related mortality is estimated at 400 people and 100 elephants, and 500,000 families affected by crop damage (Ministry of Environment and Forests, http://moef.nic.in/downloads/public-information/ETF_REP_ORF_FINAL.pdf).

Government of India has launched eight Missions as part of the National Action Plan on Climate Change (NAPCC) in specific areas, i.e., solar energy, enhanced energy efficiency, sustainable habitat, water, sustaining the Himalayan eco-system, green India, sustainable agriculture and strategic knowledge for climate change. The Rio+20 Summit recognized the importance of greening the economy as an important tool in achieving sustainable development and poverty eradication in India. From both economic as well as ecological point of view, maintenance of ecosystem health

emerges as the most important issue to be addressed. Considering the diversity of ecosystems and their sensitivity to human or climate-induced perturbations, we need to develop capabilities to maintain the integrity of these ecosystems.

1.8 ECOLOGY RESEARCH IN INDIA

The mainstream Ecology was established in India around Professor R. Misra and his students, notably P.S.Ramakrishnan, J.S.Singh, K.P.Singh and Brij Gopal at Banaras Hindu University, Varanasi. Other early ecologists in India were F.R.Bharucha (Royal Institute of Science, Bombay) who introduced the methodology of Zurich-Montpellier school of vegetation analysis in India, and G.S.Puri, whose focus was on forest ecology, and who together with R.Misra founded the International Society for Tropical Ecology and the journal, Tropical Ecology. Prof. Misra's (1937-1970s) position in Indian ecology is like that of Justus von Liebig in the chemistry of the nineteenth century world. Like Liebig in chemistry, Prof.Misra taught most of the important ecologists in India (to name a few, S.C.Pandeya, L. P.Mall, M. C. Joshi), and he was 'scientific grandfather' to many more in different universities and institutions in India and abroad. Professor J.S Singh has been one of the leading ecologists in India, with a focus of his research on ecosystem structure and functions of tropical and Himalayan Ecosystems. The studies by Madhav Gadgil, P.S.Ramakrishnan, and K.G. Saxena, and by ecologists at ATREE incorporated a fair amount of social component in ecological research.

In recent years, research on animal behaviour and wildlife conservation (Centre for Ecological Sciences, and Indian Institute of Science, Bengaluru) has greatly enriched social dimensions in ecology of India. Like all other sciences, ecology in India, developed around a few isolated individuals having touching devotion to the ideal of the scholar, each one showing an extraordinary vision and intellectual vigour. In all, Indian ecologists have made significant contribution in the fields of phytogeography, population ecology, aquatic and wetland ecosystems, marine and mangrove ecosystems, savanna, grassland, agro-and forest ecosystems, tree phenology, tree water relations and ecosystem services, environmental pollution, ecological restoration, application of remote sensing for ecosystem analysis, and biodiversity conservation (see Tripathi, 1985; Singh and Singh, 1987; Singh, 1993; Singh and Gupta, 1992; Singh and Singh, 1995; Singh, 2002; Gopal and Zutshi, 2001, Ganeshaiah, 2002, Singh 2011). A summary of the efforts of these ecologists is listed in Box 1.5.

Box 1.4 India's Major Environmental Issues

Population: India is one of the oldest civilizations in the world, with rich cultural heritage. Geographically, it accounts for only 2.4% of the world's total surface area. India supports and sustains a 16.7% of the world population. India has seen its population explode from 300 million in 1947 to approximately 1.22 billion in 2012. This rapidly growing population has placed a strain on India's infrastructure and environment. There is currently ~ 28% urban population, 20 to 40% urban population in slums – with environmental and health implications ; undernourished population of ~ 231 million.

Land degradation: Area covered by land degradation is 45% of total land area due to water erosion, wind erosion, soil acidity, alkalinity and salinity. (i) during the last 25 years, Indian soils have been experiencing a net negative nutrient balance of 8-10 million tonnes per annum; (ii) about 70% of the total cropped area experiences nutrient depletion of $>50 \text{ kg ha}^{-1} \text{ yr}^{-1}$; (iii) the annual depletion of major nutrients (N,P,K) is to the tune of 5.8 million tonnes.

Pollution: Increasing trend of RSPM and SPM levels in most cities, the primary drivers being Vehicular pollution and Industrialisation. Suspended particulate matter (SPM) considerably exceed the WHO standards as well as National Ambient Air Quality Standards set by the Indian Central pollution Control Board; (iii) India's energy-related carbon emissions have grown nine-fold over the past four decades.

Water: There are Quantitative limits to Water Use because consuming 75% utilisable resource. (i) the water requirement of major water consuming industries such as agriculture, refineries, petrochemicals, fertilizers and chemicals has grown 40 time. The projections indicate that water demand in the agriculture sector will increase from 46 Mham in 1990 to about 85.5 Mham in 2025. In domestic and industrial sector 3-fold and

2-fold increase by 2025 is expected; (ii) The over exploitation without commensurate recharge has resulted in fall in ground water table. Further, leaching of pollutants from pesticides and fertilizers into the aquifers has resulted in quality deterioration of groundwater. A 1994 survey of groundwater quality at 138 sampling locations in 22 industrialized zones revealed that water in all 22 zones was not fit for drinking.

Biodiversity and Forest cover: With 2.4% global land area has 8% biodiversity wealth; 10% wild flora and fauna in threatened list. According to the India State of Forest Report, 2011, released by the Forest Survey of India (FSI, 2011) on February 7, 2012, the total forest cover in the country is now at 6,92,027 sq km. which is 21.05 per cent of the total geographical area of the country.

Sources: <http://envfor.nic.in/nfap>; www.exhibitionsindia.org; www.eia.doe.gov/emeu; Down to Earth Jan 31, 2003 p 31.; MOEF, 2009a. State of Environment Report 2009, India, envfor.nic.in/downloads/home/home-SoE-Report-2009.pdf. India State of Forest Report, 2011 (FSI, 2011)

Box 1.5. A synoptic view of contributions of Indian ecologists

University/Institution ; leading ecologists	Major research areas and contributions
Botany department, Banaras Hindu University (BHU), Varanasi; from mid 1950s- continues; R.Misra, K.C.Misra, J.S.Singh, K.P.Singh; others were D.N.Rao, R.S. Ambasht, Madhoolika Agrawal, A.S.Raghubanshi.	One of the early leading groups in production ecology in the world, the time of which coincided with IBP; fine root production in trees, functioning of tropical dry forests particularly nutrient cycling, role of nutrient immobilization by litter microbes in dry tropical forests, diversity patterns; restoration of mine spoils, autecology, ecology of invasive alien species, air and water pollution. The Ecology Workbook developed from here played a key role in expanding ecology in India.
French Institute, Pondicherry; During most of the later half of the 20th century; V.M. Meher-Homji, P. Legris, F.Blasco, P. Pascal.	Extensive vegetation cartography and bioclimate mapping; ecological nature of Shola grasslands, classification of vegetation of Western Ghats.
Sagar University, Sagar; 1960s- S.C. Pandeya, L.P. Mall, G.P. Mishra	Full paper on ecology was developed for the first time in India by R.Misra in 1955-56. Forests and grasslands of Central India
Saurashtra University, Rajkot; 1969-1988, S.C. Pandeya	Semi-arid grasslands and croplands.
North Eastern Hill University, Shillong; from 1970s- onwards, peak during 1980s-1990s; P.S.Ramakrishnan, R.S.Tripathi, H.N. Pandey, S.K.Barik.	Ecology of various aspects of shifting cultivation, population ecology, invasive alien species, and species regeneration, tree phenology, traditional knowledge system.
Kashmir University, Srinagar; 1960s-1990sV. Kaul, D.P.Zutshi on lakes; currently, invasive species, Zafar A. Reshi.	Lake limnology, lake classification on the basis of thermal stratification, macrophytic and plankton communities. Invasive alien species is the current research area.
CAZRI, Jodhpur and IGRI, Jhansi; 1970s, K.A.Shankarmarayan, P.M.Dabadghao, P.S.Pathak, V. Shankar.	Grazing land ecology, agroforestry. The book, "Grass Cover of India", written by K.A.Shankarmarayan, and P.M.Dabadghao.
Kumaun University Nainital; 1977-onwards; J.S.Singh (1977-1985), S.P. Singh (Until 2005), valuable inputs from D.B.Zobal of Oregon State University in tree water relations research.	Vegetation analysis; structure and functioning of forest ecosystems along an extraordinarily wide altitudinal range; characterization of oak and pine forest ecosystems, linkage between forest and agriculture; phenology, diversity, and regeneration of trees; resorption of nutrients from senescing leaves; tree water relations; mycorrhizal ecology, incorporation of Himalayan forest ecosystem services in national accounting. The research of the centre resulted in a major book on Himalayan forests and the establishment of Govind Ballabh Pant Institute of Himalayan Environment and Development.

Vikram University, Ujjain; Early 1960-1970; L.P. Mall.	Grassland vegetation, mangrove vegetation, and other stressed plants.
Kurukshetra University, Kurukshetra; 1970s-onwards; J.S.Singh, P.S. Yadava(now at Manipur University, Imphal), S.R.Gupta.	Productivity, Soil respiration, litter decomposition, and carbon dynamics in early successional grasslands; ecological rehabilitation of salt-affected soils, forest ecosystems of Siwaliks, the role agroforestry systems in carbon sequestration; diversity of AM fungi.
JNU, Delhi; 1980s- onwards; P.S.Ramakrishnan, C.K.Varshney, Brij Gopal, K.G.Saxena.	Indigenous ecological knowledge systems, air pollution, Environmental flows, ecosystem services.
Delhi University, Delhi; 1990s onwards; C.R.Babu,Inderjit Singh, M.K.Pandit,K.S.Rao.	Ecological restoration using legumes microbes, and soil invertebrates; the mechanism of invasion of exotics; molecular basis of invasion capacity and biodiversity.
Jaipur University, Jaipur; 1980s and 1990s; Brij Gopal.	Macrophytic vegetation productivity and decomposition.
Centre for Ecological Sciences (CES), Indian institute of Science (IISC) Bengaluru; Since 1973; Madhav Gadgil, Raghvendra Gadakar, R.Sukumar.	Promoting human dimensions in ecology; Social biology of Ropalidia ; the research was converted into a major book; Human-elephant conflict; dynamics of tropical forests.
Indian Institute of Remote Sensing (IRS), Dehradun 1990s- onwards; P.S.Roy, S.P.S.Kushwaha.	Development of geospatial database on vegetation of large areas using satellite data with focus on physiognomy, species composition, biodiversity, fragmentation.
University of Agricultural Sciences, Bengaluru 1990s-onwards; K.N. Ganeshaiah, R.UmaShankaar.	Biodiversity data base.
ATREE (AshokaTrust for Research in Ecology and Environment, Bengaluru); 1990s onwards; K.S.Bawa, Jagdish Krishnaswamy, and several others.	NTFPs (Non Timber Forest Products), vegetation, climate change, Invasive alien species, biodiversity.
Wildlife Institute of India, Dehradun; 1990s-onwards; H.S.Panwar, R. Chellam, A.J.T. Johnsingh, G.S.Rawat.	Wildlife-focused areas and high altitude vegetation.
Pondicherry University, Pondicherry; 1990s onwards; S.B. Chaphekar,N.Parthasarathy.	Tropical forests of Eastern Ghats-biodiversity, species composition, Lianas.

Science-based NGOs have also contributed substantially to our ecological knowledge. For example, the Ashoka Trust for Research in Ecology and the Environment (ATREE), the Nature Conservation Foundation, Mysore, and Bombay Natural History Society (BNHS). BNHS has conducted commendable long term ecological studies in the wetlands of Bharatpur, Bhitarkanika, and Point Calimere with particular reference to birds. The Centre for Wildlife Studies, Bengaluru, has studied the prey - predator relationships and habitat partitioning in coexisting predators in Bandipur - Nagarhole National Park. For ecological and environmental challenges facing India, there is a need for a landscape approach, total system perspective and integrative approach, long-term data and experiments on ecosystem properties, understanding of sustainability parameters, ecology education, communicating ecology to general public, and integration of social and ecological sciences (Singh, 2011; Singh and Bagchi, 2013).

CHAPTER

2

Solar Radiation and Temperature

The physical environment determines where organisms can live, make use of resources, and allow populations to grow. The physical environment includes climate which shows long-term trends in temperature and precipitation. The physical environment also includes radiation from the sun, which mainly drives the climate system as well as biological production. Wind is an important factor in climate through its influence on global air circulation pattern. Thus, understanding the physical environment is the key to understanding ecological phenomena.

Sun is the only important source of energy; it supports life on earth, and energizes most of the atmospheric processes. Solar radiation is the driving force for the functioning of ecosystems. Though only wavelength bands between 400 nm and 700 nm of solar radiation are absorbed by chlorophyll of plants in photosynthesis, the entire electromagnetic spectrum of the solar radiation influences life on earth and is an important constituent of environment of most organisms. Organisms at the surface of earth are influenced by the downward flow of solar radiation and thermal radiation from the nearby surfaces. Both components contribute to temperature and other conditions of existence.

2.1 SOLAR RADIATION

Electromagnetic spectrum

Energy is emitted from the sun in the form of electromagnetic radiation. Different forms of electromagnetic energy can be distinguished by their wavelengths. The collection of all possible wavelengths of electromagnetic energy is known as electromagnetic spectrum that comprises gamma rays, X-rays, ultraviolet light (< 400 nm), visible light (400 nm to 700 nm), infrared radiations (> 740 nm); the microwaves (radar waves), and radiowaves (greater than 100000 nm) (Fig. 2.1.). The visible range, i.e., the energy used in photosynthesis is known as photosynthetically active radiation (PAR). The long-wave infrared radiation is the 'heating' part of sunlight. Radiant energy reaching the surface of earth on a clear day is about 10% ultraviolet, 45% visible and 45 per cent infrared.

The sun at 5800°C radiates a tremendous amount of energy than does the surface of the earth, which radiates at an average temperature of 15°C . The sun emits radiation strongly in the visible region whereas earth emits radiation in the infrared regions (relatively long wave radiation of about 10000 nm). The infrared wavelength bands for which there is little or no absorption by the greenhouse gases (8000 to 12000 nm) constitute the atmospheric windows. In the region of atmospheric windows, natural greenhouse gases do not absorb infrared radiation. Chlorofluorocarbons (CFC), a family of greenhouse gases, strongly absorbs the radiation in these regions of atmospheric window.

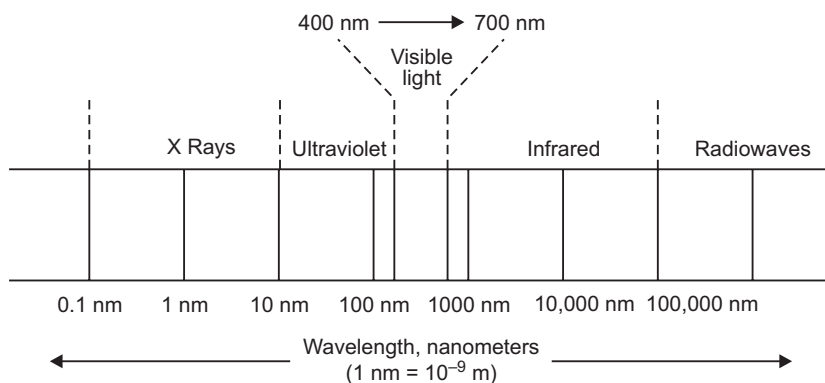


Fig 2.1. The Electromagnetic spectrum of solar radiation.

The ultraviolet radiation is of short wave length ranging from 100 nm to 400 nm, which is of high intensity. Because of ozone layer in the stratosphere, only a little ultraviolet radiation reaches the earth's surface. 99% of the UV radiation is screened out by the ozone layer present in the stratosphere, extending from 15 to 45 km above earth's surface. The ozone shield, developed early in the history of biosphere, made it possible for primitive life to emerge from the waters and evolve in to higher life forms. On the basis of wave length, there are three types of ultraviolet radiation, UV-A, UV-B and UV-C.

UV-A radiation (from 320 to 400nm) can cause some damage to living cells; is not affected by the stratospheric ozone and transmitted to the surface of the earth.

UV -B radiation has a wave length from 280 to 320 nm and has high intensity. Ozone layer absorbs some radiation in the 220 to 320 nm band. UV-B radiation causes damage to living organisms.

UV-C radiation has shortest wavelength from 100 to 280 nm, is of very high intensity and is lethal to oranisms. It can dissociate molecular oxygen into atomic oxygen. Atomic oxygen combines with other O₂ molecules to produce O₃. UV-C is strongly dissipated by the ozone present in the stratosphere, only negligible amounts can reach the earth's surface.

The Fate of Solar Radiation Incident on the Ground

Solar energy comes at the rate of about 5 million kcal m⁻²yr⁻¹. This large flow is reduced exponentially as it passes through the clouds, water vapour and other gases of the atmosphere. The amount of energy reaching the top of Earth's atmosphere each second on a surface area of one square metre facing the sun during daytime is about 1,370 Watts, and the amount of energy per square metre per second averaged over the entire planet is one-quarter of this, i.e., 342 Watts (Fig. 2.2). About 30% of the sunlight that reaches the top of the atmosphere is reflected back by clouds, atmospheric particles, or bright ground surfaces like sea ice and snow to space. Roughly two-thirds of this reflectivity is due to clouds and small particles in the atmosphere known as 'aerosols'. This energy plays no role in Earth's climate system. About 23 percent of incoming solar energy is absorbed in the atmosphere by water vapour, dust, and ozone; about 23% of the solar energy flow is used to recycle water. Around 46% of incoming solar radiation is converted to heat. A small fraction of energy (0.2%) is used in driving the wind, waves and currents. Only a very small fraction of solar radiation reaches the autotrophic layer of ecosystems and only about 0.8% of this is converted to organic matter by photosynthesis. Vegetation absorbs the blue and red wavelengths, which are the most useful for photosynthesis. The green light and the near infrared light are reflected by vegetation. Therefore, these spectral bands are used in aerial and satellite remote sensing and photography to reveal patterns of natural vegetation.

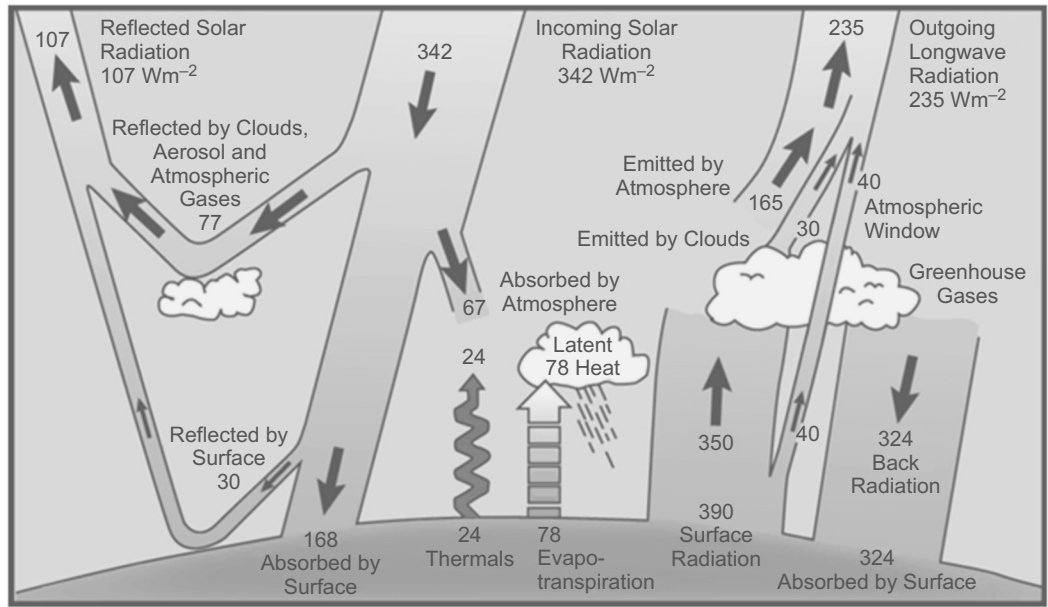


Fig. 2.2. Estimate of the Earth's annual and global mean energy balance. Over the long term, the amount of incoming solar radiation absorbed by the Earth and atmosphere is balanced by the Earth and atmosphere releasing the same amount of outgoing longwave radiation. About half of the incoming solar radiation is absorbed by the Earth's surface. This energy is transferred to the atmosphere by warming the air in contact with the surface (thermals), by evapotranspiration and by longwave radiation that is absorbed by clouds and greenhouse gases. The atmosphere in turn radiates longwave energy back to Earth as well as out to space. (source: Kiehl and Trenberth, 1997).

Distribution of Solar Radiation

Solar radiation generally decreases from regions around equator (between equator and 30°N and 30°S latitudes) to Polar Regions (Table 2.1). However, within a latitudinal belt solar radiation can vary widely because of cloud cover, e.g., within the tropical latitudes the rainforest areas receive far less solar radiation than do the deserts (Table 2.1). In tropical rainforest areas the sky remains cloudy on most of the days of the year while cloudy days are fewer in the desert.

Table 2.1. Broad pattern of distribution of solar radiation in relation to variations in latitude and ecosystems (k cal cm⁻² yr⁻¹). (from Landsberg et al., 1966)

Latitude effect		Ecosystem effect	
Latitude	Solar radiation	Ecosystem (0-30°N and S lat.)	Solar radiation
0-30°	220	Tropical rain forest	100
30-60°	180	Other forests	160-200
>60°	100	Desert	200

Solar radiation is generally greater in high altitudes, compared to low altitudes. According to one report, irradiance increases by about 1% per 100 m rise in altitude due to the decline in atmospheric scattering and absorption of solar beam, as atmospheric pressure and air density decrease toward higher altitudes. However, where cloudiness increases toward higher altitudes, such

as in the Himalaya, Scandinavia or Scotland, the resultant decrease in sunshine (Fig.2.3) may reduce solar radiation.

The angle at which rays from the sun strike earth's surface is called angle of incidence. When the sun is directly overhead, it has the angle of incidence of 90°C . A ray striking the surface at a slant has the angle of incidence smaller than 90°C . For rays striking earth at either pole, the angle of incidence is zero. The larger the angle, the more concentrated the energy, resulting in greater heating. It is obvious that during the year as a whole, the insolation received by high altitude regions is less intense compared to that of tropical areas.

Latitudinal variation also depends on the duration of sunlight. In tropical regions, there are equal numbers of hours between sunrise and sun set. The light and dark period is of 12 hours each every day of the year. In middle and high latitudes, there are pronounced seasonal variations in day length. In summer there are long hours of day light, where as winter light is received for less duration.

Vegetation itself is a modifier of solar radiation. Thus, light decreases exponentially as it moves downward into a plant canopy, depending upon the amount of foliage. Tropical forests have a very low transmittance to the forest floor, as low as 0.2% of the mean daily PAR at the top of the canopy, while in temperate deciduous forests it is about 3-10% in the growing season and as high as 50-70% during winter periods. In a temperate deciduous forest during the leafless period of spring, up to 60% of full solar radiation may reach the forest floor. The resultant rise in temperature on the ground often leads to increased decomposition and nutrient release from the floor litter. As trees leaf out and leaf area index approaches 8 or so during the summer, only less than 10% of full sun light may reach the ground of a deciduous forest. This may lead to establishment of different herbs in the forest undergrowth during different seasons. The spring herbs are adapted to sun and the summer herbs to shade. An evergreen forest does not experience such a seasonal variation in light, litter decomposition and herb growth.

Apart from the leaf area, orientation of leaves affects penetration of solar radiation through crown. Vertical orientation of leaves permits markedly greater light penetration than the horizontal leaf orientation. In areas of high intensities of solar radiation, the canopy species may have vertically oriented leaves and the under canopy species living in shade may have horizontally oriented leaves, to maximize light interception and absorption.

Radiation Measurement

Radiation is measured with radiometers, which integrate all incoming wavelengths and measure in energy units. Another instrument called pyranometer measures direct and diffuse sunlight. Its sensitivity is for 360-2500 nm, which includes some ultraviolet light ($<450\text{ nm}$), all visible light and considerable infrared light ($>700\text{ nm}$). If needed, filters can be used to restrict measurement of only visible light or part of the visible light spectrum. While on a clear day, sunlight accounts for approximately 85% of the energy received, on a cloudy day diffused, indirect sunlight may account for up to 100%. A shading device is used to measure diffused light separately. If one has to measure only direct solar radiation, *pyrheliometer* is used.

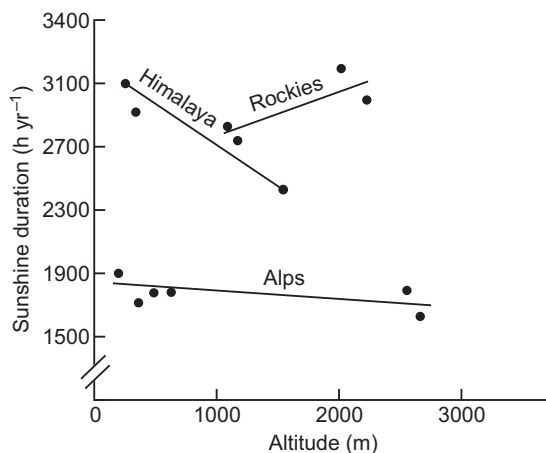


Fig. 2.3. Sunshine duration for three contrasting mountain ranges. It sharply declines in the Himalaya with elevation, as cloudiness increases (adapted from Muller, 1982).

2.2 EFFECTS OF LIGHT

Photosynthesis

The photosynthetic characteristics of plants that are of ecological significance and the pathways of carbon dioxide fixation have been discussed in detail in chapter 14.

Photoperiodism

Plants are adapted to day-length of the area and day-length of the season. Some plants require light period longer than a critical minimum (long-day plants) within a 24 hr cycle, some less than a critical maximum (short day plants) for flowering to commence and certain others do not respond to photoperiodism (day-neutral). This determines distribution of plants. Light adaptations are genetically fixed, e.g., plants of southern grassland communities in America are adapted to long photoperiods and those of northern grasslands are adapted to short photoperiods. Northern species show growth inhibition in longer photoperiods. Majority of the long-day plants have originated in the temperate region, while most of short-day and day-neutral plants have tropical origin. The examples of short-day plants are Chrysanthemum, coffee, cotton, rice, maize and sugarcane, whereas long-day plants are barley, wheat, radish, spinach and lettuce.

Light Sensitive Seeds

Exposure to red light causes germination and to far-red light stops germination. Exposure to light, for example, is essential for germination of *Verbascum*, *Lactuca*, *Anagallis* etc., and light exposure inhibits germination in *Vanilla*, *Primula*, etc. Control of germination by light is important for disturbed fields and under canopy environments. Several weeds are able to germinate as soon as soil is disturbed by ploughing. Occurrence of canopy gaps can permit germination of light sensitive, shade avoiding species.

Heliophytes and *sciophytes* are referred to as the plants adapted to high and low light environments, respectively. C_4 plants typically have far higher light saturation levels than C_3 plants, though exceptions do occur (see chapter 14). Individuals of the same species or even leaves of the same plant growing in different light conditions, show morphological and physiological differences. Shade grown species have larger and thinner leaves borne on shoots with longer internodes. They have larger but less dense mesophyll cells than sun leaves. These adaptations increase utilization efficiency of light, increase light availability or reduce reflectance of light. Generally, shade leaves perform better than sun leaves at low solar radiation, but their light saturation point is much lower than that of the sun leaves (Fig. 2.4).

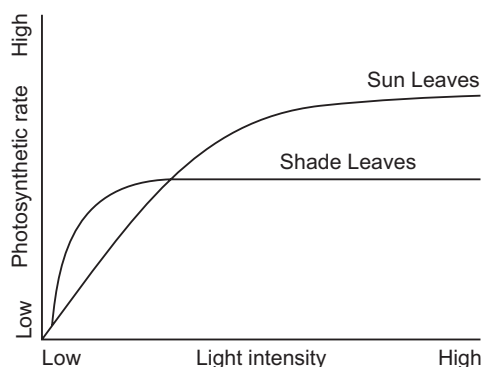


Fig. 2.4. Rate of photosynthesis in response to light in sun and shade leaves. Curves are based on hypothetical values assumed to show general trends.

Light Stress

Because all plant life depends on solar energy, it is difficult to believe that light can become a stress factor. But light can become a cause of stress to plants when light intensity increases abruptly and when high light intensity combines with another stress factor, such as water stress. Subsequent to a gap formation in the forest canopy (as a result of tree fall), light intensity may increase suddenly,

exceeding the tolerance limits of under canopy plants developed in the shade. Exposure of the shade-grown plants to high photon flux densities causes light-dependent damage to photosynthetic apparatus, or photo inhibition (Osmond et al., 1987). Insufficient light may hold several plants indefinitely in vegetative condition.

The light stress becomes more severe when a plant is under the stress of another factor, and vice versa. Bjorkman and Powles (1984) have clearly shown that the loss of photosynthetic activity following water stress in sun-grown *Nerium oleander* is light dependent (Fig. 2.5a).

An interesting example of interaction among stress factors is found in a pasture legume (*Macroptilium atropurpureum*) (Fig. 2.5b). In this legume, leaf movement is an effective means of avoiding high light stress and high temperature damage linked with water stress. The sun-avoiding leaf movement, called *paraheliotropic movement*, reduces the photoinhibition caused by high light intensity whether or not the legume is water stressed. When not water stressed, leaves move in a way that light absorption is maximum. The damage to photosynthetic apparatus due to chilling usually occurs when it is followed by a clear day of bright light. Apart from light quantity, the light quality also affects plants. Increased UV radiation at high elevation and far-red enrichment in shaded habitats may affect plants.

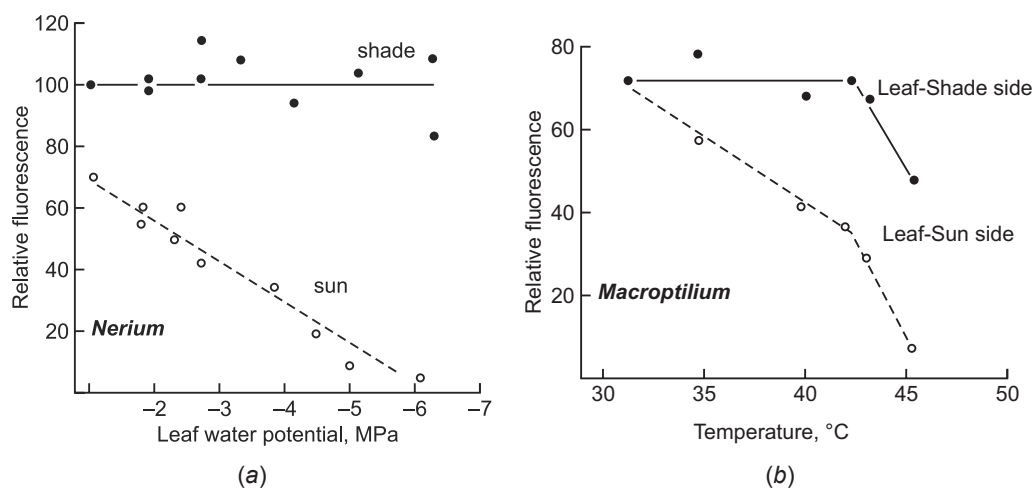


Fig. 2.5. Light-dependent aggravation of stress effects on photosynthesis shown by decline in chlorophyll fluorescence. (a) *Nerium oleander*, shows no evidence of damage to the primary photochemistry at very low water potentials (high water stress) provided leaves remain in the shade. (b) In mildly water-stressed *Macroptilium* high temperatures precipitate damage to primary photochemistry only on the upper, sun side of the leaf. There is no change in fluorescence on the shade side, until the thermal tolerance limit is exceeded. (developed from Bjorkman and Powles, 1984; Ludlow and Bjorkman, 1984).

Effect of Light on Aquatic Systems

Water absorbs or scatters enough light to limit the depth of the sunlit zone in aquatic systems. For example, in pure seawater, the energy content of light in the visible part of the spectrum decreases to 50% of the surface value at a depth of 10 metres. It further drops to less than 7% with increase in depth to 100 metres. Moreover, water absorbs longer wavelengths more strongly than shorter ones. Most of the infrared radiation disappears within the topmost metre of water. Short wavelengths (violet and blue) tend to scatter when they strike water molecules and cannot penetrate deeply. Green light predominates with increasing depth due to the absorption and scattering of light by water.

The absorption of light by water determines the depth at which aquatic photosynthetic organisms can exist. The narrow zone close to the surface (euphotic zone), has sufficient light for photosynthesis. The lower limit of the euphotic zone, where the assimilation of energy by photosynthesis just balances the release of energy by respiration is known as *compensation point*, which may be about 100 metres in some very clear water lakes and oceans.

Life under water is often controlled by the availability of light, being a limiting factor for plants in deep water in the oceans and deep lakes. Aquatic systems have varying levels of light based on the amount of light penetration through the water. The presence of light determines where producers and consumers live in water. For example, the phytoplanktons live in the illuminated surface layer of water, whereas benthos live in the sediments of a lake.

2.3 TEMPERATURE

Heat is a form of kinetic energy, which is constantly transferred from a relatively warm body to a cold body. The means of transfer are the same as of the light, viz., radiation, convection and conduction. The unit of heat measurement is gram calorie, the quantity of energy which raises the temperature of 1 g water by 1°C (to be more exact, from 15°C to 16°C). The term temperature is used to refer to a particular level or degree of molecular activity. While temperature is a qualitative term, heat is quantitative. Suppose, there are two buckets of water, one containing 5 litre water and the other 10 litre water, and both have the same temperature. Now if we were to raise their temperature by the same degree, say by 2°C, the water of the second bucket would require twice as much energy as the first.

Factors Influencing Temperature

Exposure to solar radiation, called *insolation* ($\text{g cal m}^{-2} \text{ hr}^{-1}$) varies in space (spatially) and in time (temporally). Since temperature depends on insolation, it too varies spatially and temporally. These variations are primarily influenced by latitude, altitude and distance from water bodies. Variations in temperature are also influenced by topography, cloud cover, vegetation and slope aspect, but they can be regarded only as contributing factors.

Latitude: Latitude is of the basic significance as far as spatial variation in temperature is concerned. It is the tilt of the earth by 23.5° which makes our seasons. When the earth's polar axis is perpendicular to the radius between the earth and the sun, day and night are virtually equal in length and sunlight shines from poles to poles. This happens twice in a year, once on 21 March and then on 23 September and called *vernal* and *fall equinoxes*, respectively. Alternatively, when the earth's polar axis is in a plane parallel to the earth-sun radius, one pole will be in 24 hours darkness and the other in 24 hours sunlight. In the northern hemisphere we have our longest day on 21 June and the shortest on 22 December, referred to as summer and winter solstice, respectively. The day length also varies with latitude, e.g. on the winter solstice the day-length at 23.5°S latitude is 12 hours while at 66.5°S latitude the day length is 24 hr. Comparison of solar radiations at different latitudes in different months or seasons indicates that we have least day-to-day difference at the lowest latitude and most difference at the highest latitude (Fig. 2.6). For example, the difference between the monthly mean maximum temperature in May (warmest month) and the monthly mean minimum in January (coldest month) is only 19°C for Madras, located at 13°N and 34 °C for Delhi located at 28°N (Table 2.2). This large difference is also due to the other factors, such as proximity to ocean but difference in latitude is primary. Temperature differences related to latitude have several implications for agriculture. For example, wheat is generally sown earlier in areas around Delhi than those around Varanasi. Since Delhi's latitude is higher, its winter is also cooler than that at Varanasi. Thus the optimal temperatures cool enough for wheat sowing in winters is experienced at earlier dates in Delhi than in Varanasi having warmer winters.

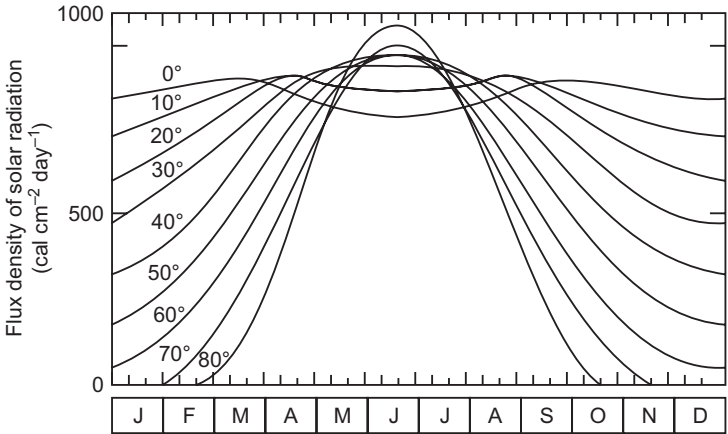


Fig. 2.6. Daily total solar radiation (undepleted) received on a horizontal surface as a function of latitude and time of year (from Gates, 1962).

Table 2.2. Temperature differences in response to differences in latitude and altitude. See that the difference in mean monthly maximum (May) and mean monthly minimum (January) for Delhi located at 28°N lat. is twice as much as for Madras located at 13°N latitude.

Station	Latitude	Altitude	Temperature °C		
			Max.	Min.	Difference
Madras	13°N	Plain	38.5	19.5	19.0
Delhi	28°N	Plain	40.4	6.3	34.1
Mukteshwar	29°N	Mountain (1938 m)	22.9	1.9	21.0

Altitude: In general, at different sites located at the same latitude, temperature would be lower at the site located in higher altitude than that at lower. Air temperatures (mean annual) decline by about 0.4-0.7°C per 100 m rise in altitude, called lapse rate in different mountains (Fig. 2.7). The average lapse rate (normal vertical gradient of temperature) is 6.5°C per 1000 m elevation. The fall in temperature with altitude occurs because an increased parcel of air expands as the atmospheric pressure declines toward higher altitudes. Expansion of air parcel extracts heat from the air itself (an endothermic process), causing it to cool. Because of decline in temperature, rates of evaporation and transpiration decline with altitude, especially in areas where cloud cover increases and consequently solar radiation decreases towards higher altitude. Crop performances have been measured along latitudinal gradients. In a grass crop in southern Scotland, floral development was delayed by 4 days and yield fell by about 10% per 100 m rise in altitude.

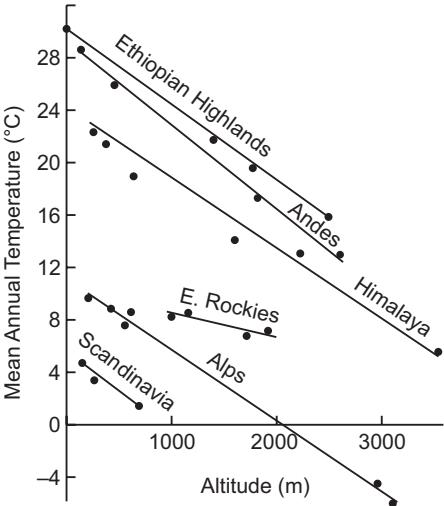


Fig. 2.7. Trends in mean annual temperature at meteorological stations located on six mountain ranges (from Muller, 1982).

Climatological data of high altitude tree-line stations from all over the world indicate that trees are restricted to those areas where the mean summer temperature is greater than 10°C (Grace 1989).

Distance from Water: Since the specific heat of water is high (about five times of soil), temperature of a large water body varies more slowly than that of the land of comparable size. During summer, when water evaporates, the heat demand of the evaporation keeps nearby land areas cooler than the distant areas. Since more heat loss is required to cool water than the land, nearby areas are relatively warmer than lands away from the water body in winter. Thus, the differences between summer and winter temperatures is less in the nearby areas close to a big water body than in the distant areas of land. This is also one of the reasons for relatively narrower temperature differences in the climate of Madras located close to the Indian Ocean than in the continental climate of Delhi

In most plants of temperate regions, growth occurs during the frost-free warm seasons. Therefore, it is important to record the dates of the last killing frost during spring and the first killing frost during autumn. For the north America, Miller and Thompson (1975) recorded that the former occurred at earlier dates and the latter at the later dates in coastal areas with maritime (or oceanic) climate than in interior parts of the land having continental climate (Table 2.4). Obviously, the growth periods are longer for the areas of maritime climate than for those of continental climate.

Table 2.3. Average dates of last killing frost in spring (I) and the last dates of first killing frost in autumn (II) in the USA (developed from Miller and Thomson, 1975)

	Maritime climate		Continental climate	
	I	II	I	II
Southernmost sites	Feb. 10	Nov. 20	Mar. 30	Oct. 20
Middle sites	Mar. 10	Oct. 30	Apr. 30	Oct. 10
Northernmost sites	Apr. 10	Oct. 30	May 10	Sep. 10

Effect of proximity to water also affects diurnal range of temperature, at the maritime stations (e.g., Mumbai) the diurnal variations being far narrower compared with more continental stations (e.g., Chanda) (Fig. 2.8).

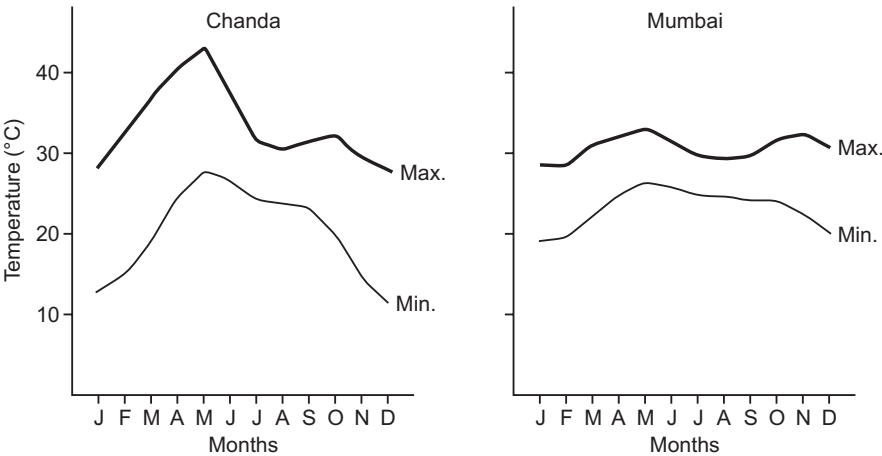


Fig. 2.8. Mean monthly maximum and minimum temperatures for two stations of Maharashtra, India, Chanda is located far from ocean (continental climate), and Mumbai is located at the sea coast (maritime climate). See the difference in maximum and minimum temperatures are larger for Chanda than for Mumbai.

The significance of these contrasts between land and water body is that a continental climate shows greater seasonal extremes of temperature (hotter in summer, colder in winter) than a maritime climate. The worldwide temperature pattern is broadly controlled by latitude, altitude, land water contrasts and surface ocean currents. The coldest places are over the landmasses in high latitudes (such as Antarctica, Siberia, and sub arctic portions of Canada). The warmest areas are localised in parts of Australia, Africa and South America. Average annual temperatures are highest in equatorial regions.

Temperature Inversions: Two sites in mountains with the same elevation may have different temperatures because of topographical variations. The more dissected the topography, the more complex is the pattern of temperature variations in a mountainous area. It is possible that during the night, the soil may release heat to overlaying layers of air so rapidly that it becomes relatively cooler. This may lead to transfer of heat through conduction from the warmer air layers to soil, resulting in a situation when air layers close to soil become cooler than those lying above them. Thus a temperature inversion is created, for normally layers close to ground are warmer than the higher layers. Temperature inversions are of brief duration and occur near the earth's surface. Temperature inversions influence weather and climate. Inversion inhibits movement of air and can contribute significantly to air pollution by limiting the natural upward dispersal of urban-industrial pollutants.

The cold air is denser than the warm air, and may accumulate in ravines and valleys at night, resulting in a situation when bottom is cooler than the slopes above them. The situation is reversed in the day time, as warmed air moves back upslope. In a situation of very slow air movement, the cold air accumulated in a small basin may remain there for long time, developing into a frost pocket.

Compared to broad open valleys, the narrow mountain valleys are cooler. The latter receive comparatively less sunlight, and consequently their walls reflect less heat into the narrow spaces. In contrast, in a broad valley heat is accumulated due to the release from the wide sides.

Cloud Cover: Temperature variations in the tropics are as little as 2°C on cloudy days, but can vary as much as 9°C on sunny days (Barbour et al., 1980). The sites of humid areas having high precipitation and cloud cover on most days receive far less solar radiation than the desert sites (Table 2.4) because water vapour is opaque to certain wavelengths of solar radiation. Then water vapours in atmosphere hold radiated heat energy, thus much of the heat radiation which normally escapes is held by clouds. The net result is, while the maximum temperature is lower on a cloudy day than on a clear day, whereas the minimum temperature which is lower on a clear day than on a cloudy day. Farmers are familiar with the fact that frosts are much more likely on clear nights.

Plant cover: Normally in day time, shaded ground under a forest cover is distinctly cooler than open places, provided there is no free air movement. This occurs, firstly because the soil heating is checked by shading and soil absorbs energy from the warmer air above it, cooling the forest air. Secondly, as photosynthesis proceeds, vegetation transpires and quantity of heat required to raise temperature in a humid condition is increased. However, since the crown cover impedes the reradiation loss, nights are warmer in forests than in nearby open areas. Thus in a heavily forested area temperature oscillations are far less than in clearings. For example, in a virgin forest, Larsen (1922) recorded maximum and minimum air temperatures, 25.9 and 7.4°C, respectively, compared with 30.0 and 3.9°C in adjacent clearing. The mean soil temperatures of soil in the forest and clear areas were 12.8 and 17.0°C, respectively.

The above example is useful to give an idea of concept of the microclimate. It is apparent that the plants and insects occupying the shaded microsites have different environmental conditions than that of the region as a whole. In deserts the burrowing animals breathe far more cooler and humid air in their burrow than in open in daytime. In other words, the microclimate in the burrow is distinctly cooler and humid than in open, where in daytime sand is rapidly heated by sunlight.

Slope aspect: In mountainous areas of northern hemisphere, north-facing slopes (also called north aspect) are distinctly cooler and humid than south-facing slopes (south aspect). Because of the inclination of the sun, a south-facing slope receives greater amount of insolation than a north facing slope (Fig. 2.9). This causes variations in temperature and humidity conditions of the aspects, and these effects can be seen on plant growth. The more music north aspect generally has greater tree cover and biomass. Individuals of the same species leaf out generally earlier on the warmer, south aspect.

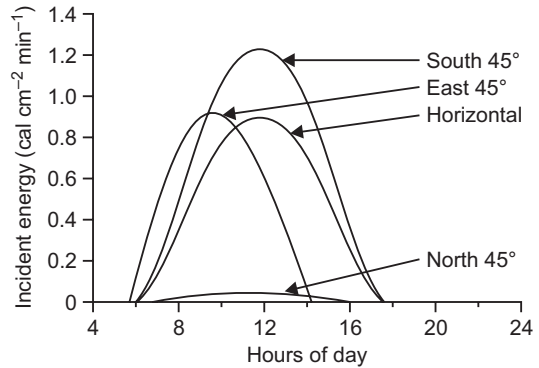


Fig. 2.9. The amount of solar radiation incident upon sloping surfaces in different hours on 21 March (vernal equinox) at 40°N latitude.

Thermal Stratification in Aquatic Systems

Temperate zone lakes respond quickly to change in temperature because of the changing seasons (see Figure in chapter 18). During winter, a typical temperate lake has an inverted temperature profile, i.e., water is at freezing temperature on the surface (0°C), whereas in the lower layer the temperature is about 4°C.

In early spring, the sun warms the lake surface gradually, and the sun-warmed surface water tends to sink into the cooler layers immediately below. This vertical mixing distributes heat throughout the water column from the surface to the bottom, resulting in a uniform temperature profile and free mixing of water in the whole water body. Winds cause deep vertical movement of water, or spring overturn, in early spring. During this process, nutrients are brought to the surface from the bottom sediments and oxygen from the surface to the depths, resulting in bloom of phytoplankton growth.

During summer, the surface layers of water gain heat faster than deeper layers, creating a zone of rapid temperature change at intermediate depth (thermocline). The depth of the thermocline is affected by winds, depth and turbidity of the lake. The thermocline demarcates an upper layer of warm water called the *epilimnion* and a deeper layer of cold water called the *hypolimnion*.

During autumn, the surface layers of the lake cool more rapidly than the deeper layers, become denser than the underlying water. This vertical mixing, called fall overturn, persists into late fall, until the temperature at the lake surface drops below 4°C and winter stratification occurs. Fall overturn speeds the movement of oxygen to deep waters and pushes nutrients to the surface. In lakes where the hypolimnion becomes warm by midsummer, deep vertical mixing may take place in late summer, while temperatures remain favourable for plant growth. The resulting infusion of nutrients into surface waters often causes an overshoot in the population of phytoplankton, the fall (autumn) bloom.

Thermal stratification develops only weakly if at all, in lakes at high and low latitudes. In arctic and sub arctic regions, too little heat enters lakes to establish a thermocline and turbulent mixing of water. Thus, the water column tends to warm uniformly. In the tropics, the lack of a pronounced seasonal temperature cycle reduces the sharpness of thermal stratification because the sunshine and constant high air temperatures warm water uniformly.

Ocean Currents and El Nino

Various kinds of ocean movements are categorised as currents. It is the air blowing over the surface of water that causes the major surface currents. The heat energy stored in the ocean has a significant effect on atmospheric circulation. The broad scale surface currents make up the general circulation of the oceans. The major components of oceanic circulation are closely related to major components of atmospheric circulation. The general circulation of the ocean is a pole ward flow of warm water along the western edge of each ocean basin. There is an equator ward movement of cool water from high latitude along the eastern margin of each ocean basin.

The variations in the marine environment are caused by the global climate patterns that govern all life on earth due to variations in temperature, rainfall, and humidity in different regions of the globe. The earth's major climate zones are formed due to unequal heating of the atmosphere, land masses and the oceans and the rotation of the earth on its axis, major ocean currents and the trade winds.

El Nino, also known as the *El Nino southern Oscillation (ENSO)*, is an occasional phenomenon that causes major changes in temperature and precipitation on a global scale. *El Nino* usually first appears during the Christmas season in December and is the Spanish term for the 'Christ Child'; hence the name. It primarily affects the rainfall pattern off the west coast of South America, and may result in large scale global climatic changes periodically. During the *El Nino*, there is a wind shift in the equatorial Pacific Ocean; the normal east to west winds reverse direction and drag the large volume of water eastwards in the pacific ocean towards off the west coast of South America; the normally cold water becomes warmer. This top layer of warm water prevents cold nutrient rich deeper water to come on the surface, preventing the *upwelling*. As a result, the growth of microscopic algae that normally flourishes in the nutrient rich areas is inhibited. Related changes in ocean temperature and wind result in dry conditions in the Indonesia-Australian region and heavy rains on the west coast of South America (Fig. 2.10).

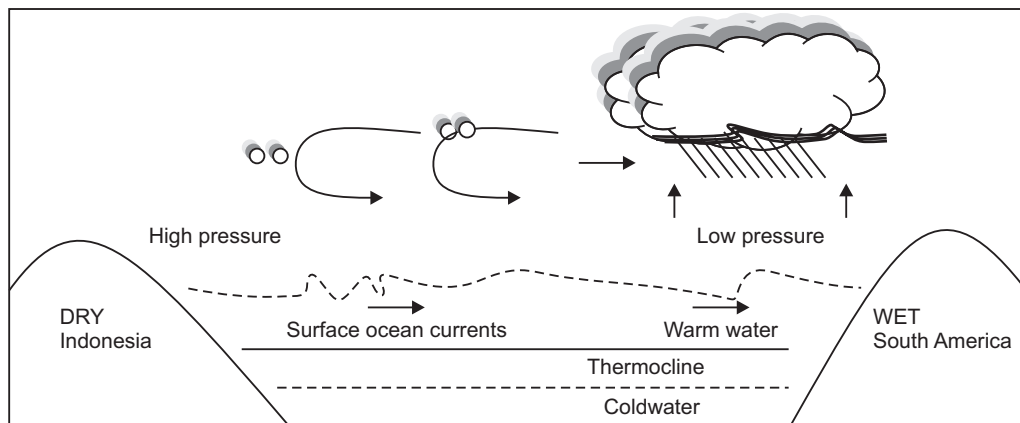


Fig. 2.10. The phenomenon of *El Nino-Southern Oscillations* showing major changes in temperature and precipitation on a global scale.

The exact incidence pattern of *El Nino* is not known, but it seems to occur once every seven to ten years. This phenomenon illustrates the interdependence of organisms, their susceptibilities to environmental changes and various interactions that link organisms, atmosphere, land and the ocean systems.

2.4 EFFECT OF TEMPERATURE

Biological Activity of Organisms

Activities of organisms occur within a very narrow range of temperature, 8° to 50°C . For example, the normal activity of the housefly occurs only between 15 and 23°C , and death occurs within a few minutes when temperature rises to 46.5°C , or drops to 5°C (Fig. 2.11). In tundra, where temperatures are very low few species occur - one can easily count all the plant species, while in tropical wet areas where temperatures are favourable most of the year one may fail to count easily even the plant species of one hectare area.

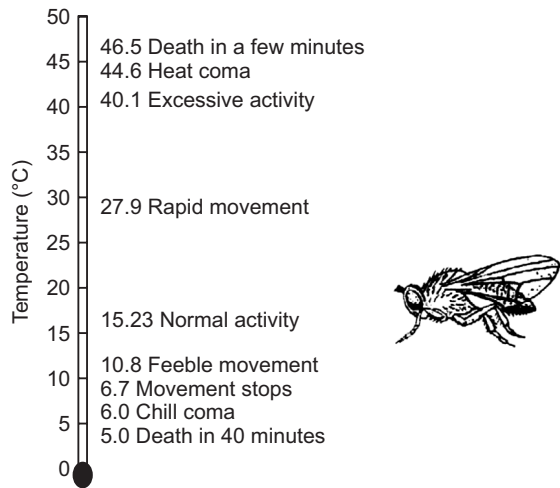


Fig. 2.11. Range of tolerance for air temperature in the housefly

Thermoregulation and Homeostasis

Temperature is important in controlling the rate of processes inside an organism, and its activities. The relation of body temperature to the external environmental temperature differs among organisms. Most organisms are ectotherms (cold blooded animals), their body temperature tends to match with the environmental temperature in which they live. Animals belonging to groups such as amphibians, most fish and most reptiles behave as ectotherms. Virtually, all invertebrates are ectotherms. Many active ectotherms such as frogs and snakes control their body temperature by moving around or by seeking shade; some animals are nocturnal and they feed during the night.

Endotherms (warm blooded animals) regulate their body temperature by physiological means and maintain a more or less constant internal temperature even when external environmental temperature fluctuates. The warm blooded animals are mainly birds and mammals. The Pigmy white toothed shrew (*Suncus etruscus*) found in Africa is the world's smallest mammal, weighs only 2 gm as an adult and is an endotherm. Endotherms have physiological mechanisms for keeping body temperature constant or within tolerance limits. You may recall that the maintenance of relatively constant internal environment under varying external environment is called *homeostasis*. For example, humans maintain homeostasis by negative feedback responses through physiological and behavioural mechanisms. The *negative feedback* is a special type of systems response to maintain the system to work within tolerance limits (lower and upper limits of temperature range) of the system (Fig. 2.12). The negative feedback is also known as corrective feedback as it resists change due to external conditions.

The average body temperature of human body is 37°C and it varies within a range of 36°C to 41°C . The hypothalamus acts like a thermostat and maintains body temperature through feedback information to the brain. Hypothalamus in brain senses body temperature and sends neural and hormonal signals to the body organs. Figure 2.12 shows that body temperature can be regulated by the rate of heat production and the rate of heat loss or gain with respect to the environment. The body experiences a decrease in metabolism at high temperature, whereas increase in metabolism at low temperature. At low temperature, thermoregulation is achieved through shivering and non shivering heat production as regulated by the activity of nervous system and the endocrine systems. At high temperature, hypothalamus initiates a series of responses that slow down heat production and increase the loss of heat from the body. The skin starts sweating and this produces a cooling

effect by evaporation. In some mammals like dog and cat (which do not sweat), heat loss through the lungs and via the tongue is increased by rapid panting, which maintains body temperature. The negative feedback acts as a on and off switch to the hypothalamus for achieving the control of body temperature (Fig. 2.12).

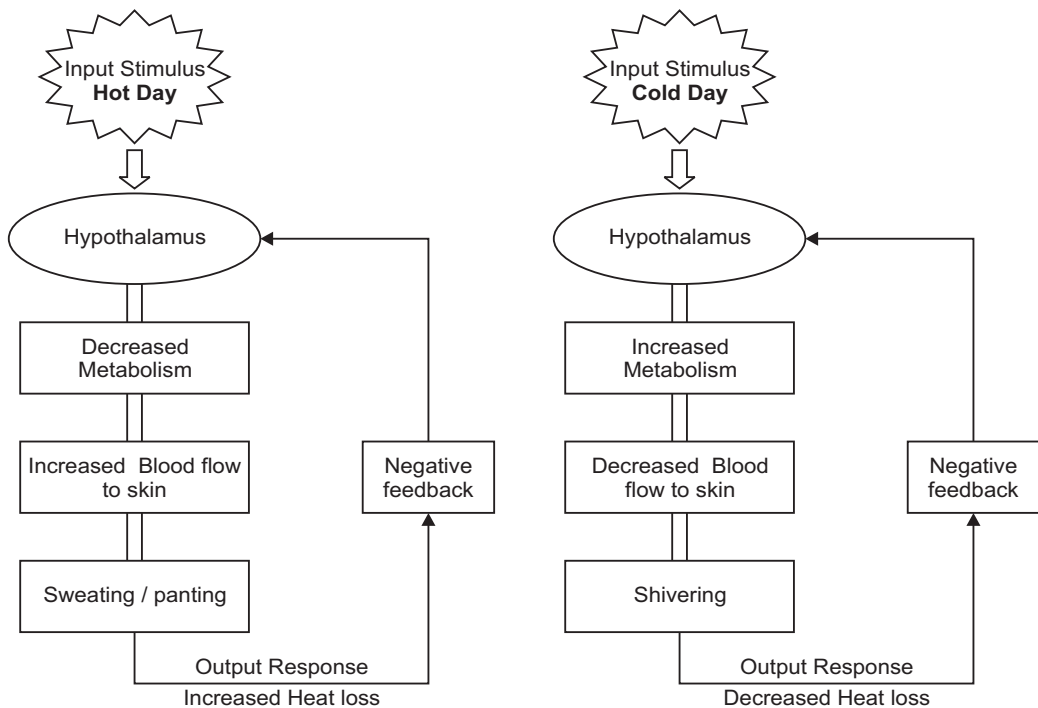


Fig. 2.12. The homeostatic control of body temperature in endotherms by negative feedback control.

Maintaining a constant body temperature in the case of endotherm animals provides them a metabolic advantage over other organisms. As we know, biochemical pathways and the enzyme activity is often maximum at a temperature near 37°C in endotherms. Therefore, these animals are able to remain active under cold conditions by exhibiting different types of adaptations to minimize the loss of heat.

Thermoperiodism

Several species show their best growth when the day and night temperatures are quite different. For example, loblolly pine shows far more growth when the day and night temperatures are 30° and 17°C, respectively, i.e., when thermo period is of 13°C than when grown on constant temperature or temperatures when difference is less than 13°C. A positive response to a temperature regime having a marked diurnal variation is called *thermoperiodism*. It is not necessary that all species show thermoperiodism. For instance, tree species of coastal areas where diurnal temperature differences are limited, plants may grow at a constant temperature as well as they do under a temperature regime showing diurnal differences. Species showing positive response to thermoperiodism generally are of continental climate having a marked difference in day and night temperatures. Thermoperiodism is also often recorded for seed germination.

Plant Adaptations to Temperature

The temperature of plant organs is closely related to the temperature of their immediate environment. If the air is cooler than the plant organ, energy moves away from the organ, and its direction is reversed when the air is relatively warmer. Much of this difference in temperature is caused by the cooling effect of evaporation of water. Recall that the vaporization of one gram of water requires about 536 cal of energy, called its *latent heat*. The heat energy demand of vaporization cools the leaf. The higher the wind velocity the higher the rate of evaporation and transfer of heat from the plant organ, such as leaf.

The leaf width affects the leaf temperature. At the same air temperature and wind velocity, the larger leaves (more width) would lose less heat than the smaller leaves. Thus they would get warmer in a given time than smaller leaves. A leaf is surrounded by a layer of still air (unstirred air), called *boundary layer*, the thickness of which increases as leaf width increases. The thicker the boundary layers around the leaf, the lesser the heat exchange between leaf and its environment, resulting in temperature rise of leaf. This is one of the principal reasons for the common occurrence of plants with small leaves in deserts.

Plant Height in Relation to Temperature: The aerodynamic factor is strongly related to the height of vegetation. As a consequence, temperatures of vegetation of great heights, such as trees of a forest, are never very different from the general climate, while those of dwarf vegetation may be totally different from it (Table 2.4). Leaves of dwarf vegetation in bright sunshine are considerably warmer than the air, so much so that near-lethal high temperatures have been observed in cold, alpine belts on days with bright sunshine and low wind speeds.

Table 2.4. Shortwave radiation ($\text{J cm}^{-2} \text{ yr}^{-1}$) for three major ecosystem types in the United States (from Webb et al., 1983)

Ecosystem type	Number of sites	Precipitation (mm yr^{-1})	Range of shortwave radiation ($\text{J cm}^{-2} \text{ yr}^{-1}$)
Forest	5	639-2592*	344850*-549670
Grassland	1	241	591470
Desert	4	204-394	648160-789650

*The unusually high precipitation, 2592 mm and low radiation (344850 J) is for Andrew forest of giant *Pseudotsuga menziesii*.

In cold climate, woody plants become stunted or assume prostrate habit, so that the leaves remain close to the ground, the warmest micro site for plants. Along an elevation gradient, as an example, the air temperature at the highest point may be lower by 30°C than at the lowest elevation, but leaf temperature may not vary at all with the elevation rise. Certain desert annuals which have leaves in rosettes on the soil surface during the cool spring, assume a basket shape by raising their leaves up during the warm summer season. Firstly, the basket shape reduces the leaf surface area to absorb solar radiation, and secondly it raises leaves to relatively cooler environment, allowing easier heat transfer to air.

Leaf Reflectance: Increasing the reflectivity by developing light coloured leaves to check the temperature rise in the condition of continuously high solar radiation is an important adaptation of desert plants. Mooney et al. (1977) found that in the desert holly (*Atriplex hymenyletra*, of Chenopodiaceae) the leaves produced during the cool season are nearly twice the size of summer leaves. In this species, young leaves are covered with hydrated salt bladders which collapse as summer progresses. The salt solution spreads over the leaf surface and gives white appearance, thus increasing the reflectance. In this way leaf temperature is not allowed to rise beyond a critical level.

Another adaptation for increasing reflectance is to reduce water content of tissues, for both are inversely correlated. The desert holly and other desert plants also keep the leaf angle steep, thereby reducing the heat load. The steep angle also allows greater interception of light during the early morning and evening hours, when relative humidity is relatively more adequate for photosynthesis. Many desert plants shed their leaves or replace them with spines to limit the increase of tissue temperature.

Flower-level Adaptation: Temperatures of flowers may be especially important to the completion of the life-cycle. Studies show that in some species petals focus the solar radiation to the gynoecium, or inflorescence peduncles show *heliotropic movements*, both leading to raised temperatures, suitable for seed development. It may be significant that the spring forest herbs (when temperatures are low) are often dwarf cushion forming, and have flowers on short peduncles, close to the ground, while the tall summer herbs of forests have long peduncles, that raise flowers in cooler air.

Dormancy, Vernalization and Summorization

Dormancy: It is a period of inactivity for seeds or other plant organs, which is broken (i.e., activity re-starts) only when certain requirements are met. In temperate regions vegetative buds of trees, shrubs and other plants remain dormant and would not resume growth until the spring when temperature or/and photoperiod requirements are met. Similarly, seeds may not germinate even if all the conditions required for germination, such as moisture, oxygen and photoperiod are supplied because such seeds are in dormant state. Plants adopt such a strategy to escape unfavourable periods-times of short photoperiod, lack of heat or sunlight or water. For example, in trees and shrubs of temperate regions dormancy is usually initiated by short photoperiod in interaction with cold temperatures. However, hormones play an important role in this. These plants resume growth as soon as photoperiods and temperatures begin to steadily rise in spring. Dormancy is a very important adaptation mechanism having a precise relationship with seasonal rhythm. Trees often do not start leafing out during winters if per chance there is a unreasonable temporary rise of temperature, for leaves may be killed subsequently by the cold.

Vernalization: Refers to cold exposure experienced by a plant species. Some plants require to be exposed to a period of winter chilling before flower initiation can occur. For instance, the flowering of winter rye (*Secale cereale*) and of some other cereal species occurs in 7 weeks when sown before the beginning of winters so that they are exposed to cold temperatures, but occurs in 14 weeks when sown after winters and, thus, are not exposed to cold temperatures.

Table 2.5. Influence of height on temperatures of apical meristems near a timberline. Temperatures represent approximately the maximum between 9-12 noon on a day of sunshine (derived from Grace, 1989)

Plant	Plant height (m)	Temperature (°C)
<i>Pinus sylvestris</i>	2 – 3	13
Dwarf shrub	0.1	19

The seeds of most desert annuals are subject to heat after falling on the ground. Possibly, high temperatures are required for seed ripening. This heat pre-treatment, called *Summorization* (summor = summer) promotes seed germination. An experiment indicated that the percentage of seed germination was higher in desert annuals when seeds were stored at 50°C than when they were stored at 20°C or were untreated (Capon and Van Asdall, 1966) (Table 2.5).

Temperature and Plant Distribution

Distribution patterns of plants are probably determined by failure of a part of the life-cycle caused by extremes of temperature, or for that matter by extremes of any climatic and environmental factors. Extreme values of temperature, or other specific variables often occur once or twice in a century and affect distributional changes of organisms. Cold hardiness of woody species determines their elevational distribution. Species of *Juniperus* are the hardiest (resisting more than -23°C) among the Himalayan woody species and they go up to the highest elevations, 4000 m or more. Trends in the absolute minimum temperature (observed over 30 years) in relation to elevations are shown in Fig. 2.13. Sometimes unusual combinations of specific variables may develop to disrupt the life cycle patterns in plants. For example, some C_3 species may fail to survive under a given temperature regime even if it were suitable for their photosynthesis and growth. The growth of pollen tube may require higher temperatures than other plant activities including photosynthesis or cell division. Life cycles of such species are not completed because of insufficient pollen growth and they fail to survive.

Some species require winter to accumulate sufficient amounts of carbohydrate reserves for normal shoot growth in spring. For example, accumulation of photosynthate during winter when plants were dormant caused doubling of their dry weight in *Picea sitchensis* in a nursery at south Scotland. Such species fail to compete with other species when temperatures are not low enough. There is a genetic basis for temperature tolerance. For example, Bhardwaj et al. (2013) have shown that *Caragana jubata* occurring in cold desert of Himalaya has an exceptional capability to express genes encoding chaperones and those associated with growth and development. Some of these were constitutively expressed, whereas others were strongly up-regulated within 5 min to 3 h of exposure to low temperature. Genes encoding six groups of late embryogenesis abundance (LEA) proteins and those involved in growth and development exhibited up-regulation in response to low temperature. The species exhibited a quick acclimation response to low temperature with concomitant expression of the relevant genes. Photosynthetic acclimation of the species to low temperature was an important trait for functional advantage in the cold environment.

Environmental stresses, low temperature in combination with drought in relation to major vegetation types are shown in Fig. 2.14. Tropical moist forests experience almost no stress, while arctic alpine deserts bear stresses, both of low temperature and drought. Other vegetation types are exposed to intermediate levels of disturbances. The vegetation types also differ in regard to their exposure to other stresses. For example, stress of shade on the forest floor plants is very high in case of tropical moist forests, and that radiation stress is high in deserts of arctic and alpine regions.

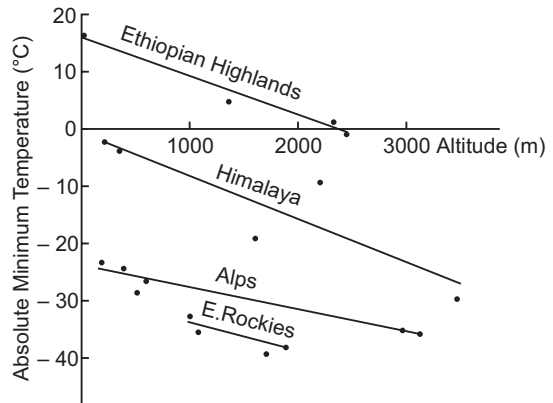


Fig. 2.13. Trends in the absolute minimum temperature (observed over 30 years) (Muller, 1982).

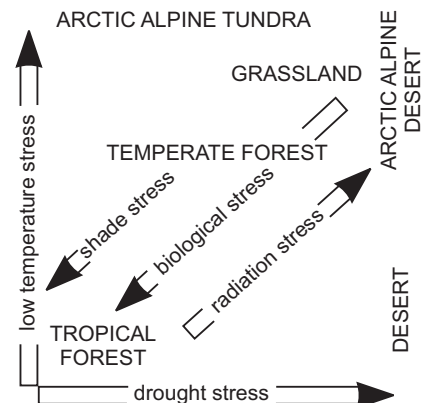


Fig. 2.14. Schematic gradients of environmental stress in relation to major vegetation types. The X axis is a gradient of decreasing temperature, and the Y axis represents decreasing water availability.

Measurement of Temperature

Ordinary mercury thermometer may be used for instantaneous temperature measurements. A variety of electrical devices are now used, particularly to measure temperature of environmental parts where it is difficult to use ordinary mercury thermometer, such as deep waters and plant tissues. Platinum resistance thermometer, thermocouples (bimetallic functions) and thermistors (metallic oxide, thermally sensitive resistors) are few such devices.

While recording air temperature with a mercury thermometer, make sure that the bulb is protected from direct sunlight in order to prevent heat absorption. This can be done by keeping the thermometer in a well ventilated box. It needs to be large, so that distance between thermometer and side walls is at least 30 cm.

Maximum-minimum thermometer is used to study the environment over a period of time. It consists of a closed 'U' tube containing a column of mercury with an expansion bulb filled with fluid at one end (Fig. 2.15). In each arm above the mercury column a steel pin is located. It will not slip down on its own. The rise in temperature expands the fluid of bulb, which in turn pushes the mercury down the limb A and up on the other limb B. The rise of the mercury in B limb pushes up the steel pin. When temperature declines, the whole process is reversed - fluid of bulb contracts, pushing up the mercury and pin in A limb. In the B limb, mercury shows simultaneous fall but the steel pin does not leave the place where it was pushed indicating the maximum temperature. The steel pin in limb A also stays, indicating the minimum temperature.

After recording maximum and minimum temperatures steel pins are brought back to the top of mercury column in each limb with the help of a magnet. Now the thermometer is ready for next day temperature measurement.

At thousands of meteorological stations throughout the world, temperature data are recorded on a regular basis. The data provide important information useful in weather reports and forecasts as well as for determining long-term climatic changes.

Climographs

It is the graphic system of the climatic diagram depiction (climograph) of monthly temperature and precipitation for a specific weather station (Walter, 1979). In the climatic diagram one division on the vertical scale is equal to 10°C temperature and 20 mm of rain. It describes simultaneously the seasonal or monthly changes in temperature and rainfall.

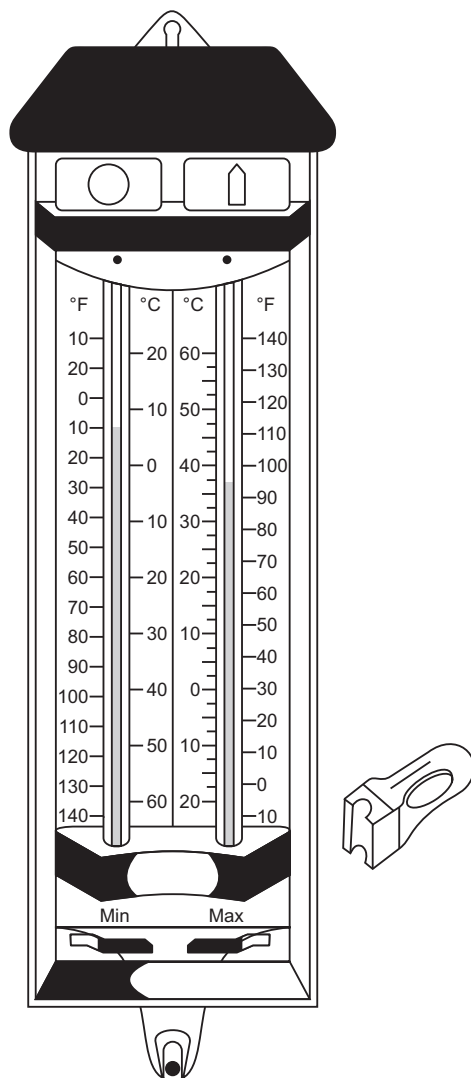


Fig. 2.15. A diagrammatic view of the minimum and maximum temperature.

In a glance, one can have an idea of period of drought and humid period. We note immediately that the seasons in Mumbai bring marked change in rainfall, but only limited change in temperature (Fig. 2.16). Some other information, such as annual rainfall, mean annual temperature and months with mean daily minimum below 0°C are given (Fig. 2.16). The climatic diagrams permit ecologically meaningful comparisons of climate between localities. Different localities within the same climate zone have similar climates, regardless of their geographic position. For example, the tropical climates of Brasilia (Brazil), Salisbury (Zimbabwe) and Darwin (Australia) all have year-round warm temperatures and summer rainfall. All these areas support deciduous forest vegetation grading into savannas, where precipitation is low. The climographs are a useful tool in the general study of world climatic classification. The world biome types also follow the patterns of temperature and precipitation on earth.

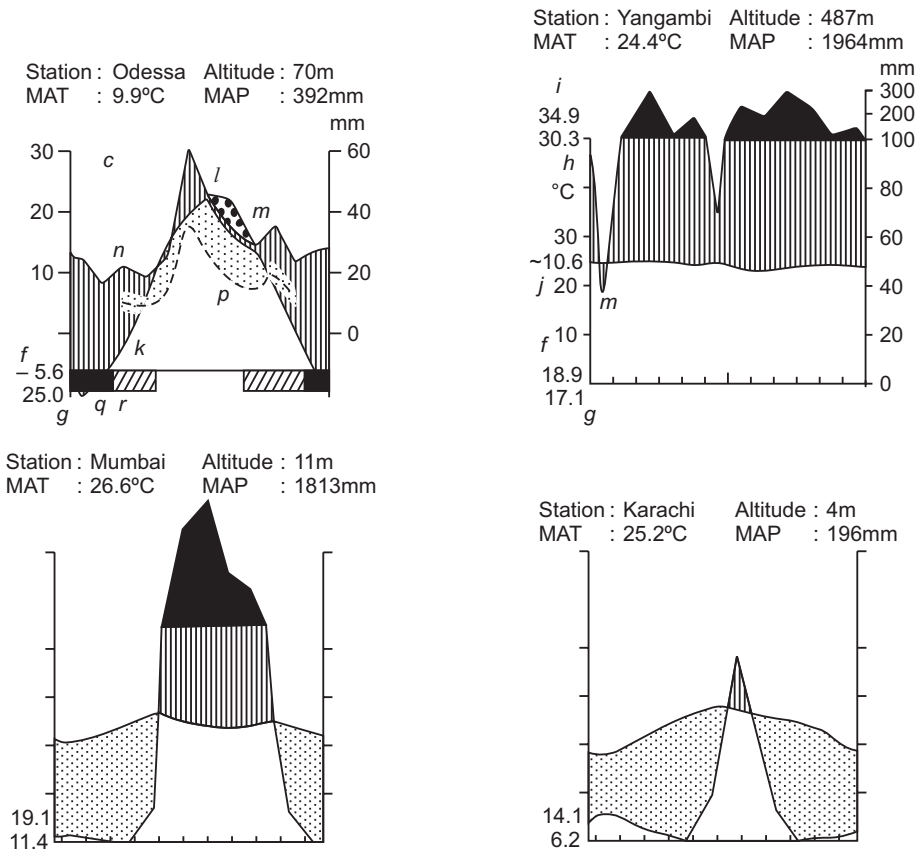


Fig. 2.16. Key to climatic diagrams. Horizontal axis months (January to December, warm season in the centre of diagram). Vertical axis (1 division = 10°C or 20 mm precipitation). MAT = Mean annual temperature (°C), MAP = Mean annual precipitation (mm) (f) = mean daily temperature minimum of the coldest month; (g) = absolute minimum temperature (lowest recorded); (h) = mean daily temperature maximum of the warmest month; (i) = absolute maximum temperature (highest recorded); (j) = mean daily temperature fluctuation; (k) = curve of mean monthly temperature (1 division = 10°C); (l) = curve of mean monthly precipitation (1 division = 20 mm, i.e., 10°C = 20 mm); (m) = period of relative drought (dotted); (n) = corresponding relatively humid season (vertical shading); (o) = mean monthly precipitation above 100 mm (black areas, per humid season); (p) = supplementary precipitation curve, reduced to 10°C = 30 mm. Yangambi shows warm and wet climate throughout the year. Mumbai shows a typical monsoon climate, conditions being per humid during monsoon months and dry during most months of the remaining year. Karachi shows drought condition during most of the year (redrawn from Walter, 1979).

CHAPTER

3

Water Relations

3.1 WATER AND PLANT FUNCTIONING

Water is important to plants as a constituent, solvent and reactant and because it produces turgor pressure in plant cells. Water constitutes 80 to 95% of fresh weight of non-woody tissues of plants like leaves and young shoots. The protoplasm may die with the removal of water; its presence is necessary for protoplasm structure. Water is strongly attracted to proteins and other cell components and provides the supporting structure for non-woody plants. Water loss from them may cause wilting. A reduction in water content of 15-20% of the value of full hydration may result in stoppage of almost all vital processes of plants. It is an extremely good solvent in which chemical reactions of the cell take place. Water acts as a reagent or reactant, being used, or formed in chemical reactions occurring in the cell.

Water uptake by a cell causes a hydraulic pressure on its walls and on surrounding cells. Pressure on the cell wall is necessary for plant growth and plant movement. Water serves as a transport system, movement of substances occurring through water in and around plant cells. Some unique properties of water are given in the Box 3.1.

BOX 3.1. Properties of Water

Water has (a) a high specific heat, thus absorbs or loses a lot of heat with small temperature changes, and (b) a high latent heat, taking a lot of energy with evaporation or releasing a lot of heat with freezing. These properties help in stabilizing temperatures of plants and nearby environment. Water remains liquid over much of temperatures that occur in biosphere. It expands on freezing. Water molecules have great attraction for one another, which results in a strong cohesive force. Water has highest heat capacity of all liquids (except ammonia), which helps in balancing temperature changes. It is almost transparent to visible light but absorbs infrared radiation.

At global scale net primary productivity and precipitation are positively correlated (Leith, 1975). Water loss via transpiration is associated with photosynthesis as the stomata which allow atmospheric CO₂ to enter the leaf, also provide exit points for water vapour.

3.2 WATER SOURCES

Water occurs in the atmosphere in three physical states, viz., solid (snow, hail, sleet and ice), liquid (rain, water droplets) and gas (water vapour). The gaseous state of water regulates the atmospheric moisture dynamics. When there is enough moisture in the atmosphere, it condenses to form haze, fog, cloud, rain or snow. Precipitation has both short- and long-term effects on the plants and landscape as a whole. The survival of plants depends on the presence and absence of

precipitation. There is a continuous interchange of moisture between the earth and the atmosphere constituting the hydrological cycle. The movement of water through the hydrological cycle has a major influence on rainfall distribution and temperature modification. Water cycle plays a vital role in creating habitable climate and moderating temperature. Plants play an important role in the hydrological cycle through the process of transpiration. In tropical forests, as much as 75% of annual precipitation may be returned to the atmosphere by the plants.

Precipitation

In terrestrial habitats precipitation is the only source of water for growth of most plants. Forms of precipitation are rain, snow, dew and fog. Of these, rain water is of the greatest use for plants, and dew and fog are of only minor importance.

Rain and Snow

Water droplets formed when atmospheric vapour cools to dew-point and condenses around condensation nuclei are called *clouds*. There is no condensation layer between clouds and the earth. When water droplets become sufficiently large, they fall as rain under the influence of gravity. Water arriving at the surface of earth as ice crystals, small pellets or flakes is called *snowfall*. One of the necessary conditions for snow fall is that condensation occurs below freezing temperatures. Apart from the amount (here rainfall + snowfall), seasonal pattern of precipitation and its kind (rainfall or snow) affect the vegetation. Amount and distribution of precipitation in time and space depend on distance from ocean, temperature and moisture content of atmosphere, latitude, elevation, topography, etc. The distribution of terrestrial vegetation, the life forms that constitute communities and productivity the communities realize, are closely related with the distribution of rainfall in space and time.

Mountain ranges affect the rainfall pattern markedly. Water-laden air deposits rain on mountain slopes of a side of the summit, as it gets cooler with increasing elevations and loses capacity to hold water. Areas of such a *windward* side often receive heavy rainfall. Depleted of water, the air mass becomes warm as it descends from summit to valleys on the other side of mountain, referred to as *leeward side* (Fig. 3.1). Consequently, rainfall on the leeward side is scanty. The rain shadow is the leeward slope and the area beyond as far as drying influences extend.

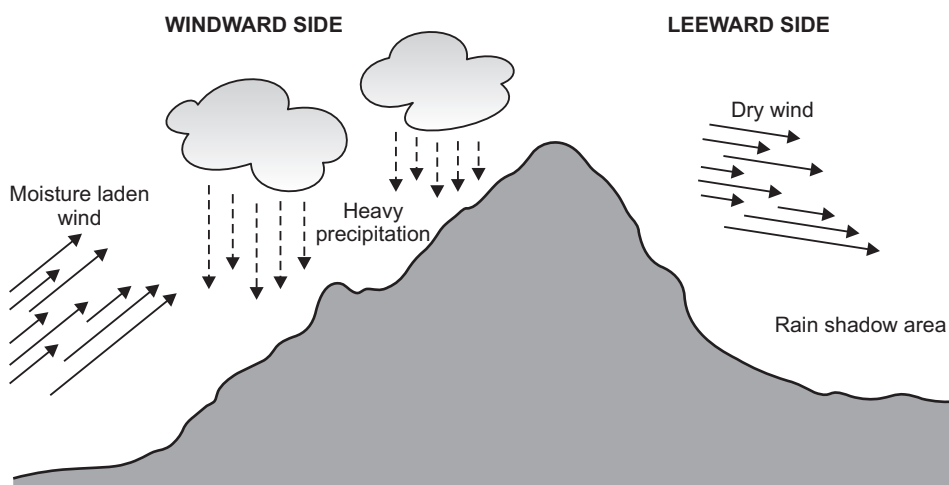


Fig. 3.1. A diagrammatic representation of windward and leeward sides of a mountain range, the former receives heavy precipitation while the latter is dry.

The Himalayan mountains rose in phases, and it was during the later phases of rise (during the Miocene) that the monsoon pattern of rainfall was established in India. The outer mountain ranges of the Himalaya receive usually 150-250 cm rainfall and support mesic forests, while the inner valleys are dry, favouring different flora. On the northern side of the main Himalayan ranges is located the Tibet plateau, one of the world's largest *rain shadow* areas, receiving less than 10 cm annual rainfall. Similarly, in the Western Ghats of India, rainfall is heavy on slopes facing the ocean (windward side), while the Deccan plateau (leeward side) is dry. Consequently, the Western Ghats have tropical rain forests and the Deccan plateau have thorn forests. Mountains can also change the seasonal pattern of rainfall. For example, more winter rainfall in Kashmir valley than in most of other areas, and consequently more Mediterranean taxa in the flora of Kashmir valley, are attributed to the rise of mountains during the Palaeocene.

Latitude also influences rainfall. Rainfall is potentially highest in equatorial region and lowest in the polar region of the earth. The polar regions are too cold to hold much moisture in air, and cold air tends to settle rather than to rise and expand. There is a contrast between precipitation distribution between continental margins and interiors. Coastal regions usually receive more precipitation than the interior regions, coastal regions being closer to the water source. Air masses accumulate water vapour from several sources; of these oceans are the most important. In this, salt particles form condensation nuclei and cloud formation results. That is why; generally areas near oceans receive heavy rainfall.

The seasonal distribution of precipitation is important for vegetation. The same amount of rainfall received in summer season would lead to the development of more mesic vegetation than when it is received during winter season when temperature becomes limiting to plant growth. For example, an annual rainfall of 70-80 cm supports deciduous forest when received during summer season, but a drought-enduring schlerophyllous vegetation when received during winter season (e.g., Mediterranean region).

In some deserts, the average annual rainfall is sufficient to support more mesic plants than actually found occurring. But rainfall in these areas is very unpredictable and gaps between effective precipitations may be up to several years. As a result, vegetation is extremely xerophytic. Whether vegetation itself affects rainfall is difficult to assess, but there are evidences to suggest that the large-scale deforestation in Amazon region has affected the rainfall pattern of the nearby regions.

Precipitation-effectiveness ratio is used to determine how effective is rainfall (P in mm) temperature (T in °C):

$$PE = P / (T + 10)$$

This is applied to annual means, and to increase its usefulness, monthly evaluations are developed as follows:

$$PE = (P \times 12) / (T + 10)$$

A given amount of precipitation is more effective if it is spread over many days than if it falls in a day. A simple way of expressing this for monthly or annual means is (Conrad and Pollak 1950):

$$\frac{\text{Number of rainy days} \times P \text{ in mm}}{T^{\circ}\text{C} + 10}$$

Snowfall has significant ecological influence in a large part of the world, encompassing temperate zone, tundra, and sub alpine and alpine belts of mountains of both temperate and tropical zones of the earth. Perennial rivers of India and for that matter all over the tropical zone, generally owe their origin to snowfall in the high mountain ranges. During summer much of the water flow in the Ganga and other perennial rivers of Northern India are because of the snowmelt in the high ranges of the Himalaya.

Snowmelt is usually gradual. Therefore, the resultant release of water and consequent runoff is slower than in the case of an intense rainstorm. In the alpine belt of Himalaya, herbs start growing well before the commencement of the monsoon rains, as snow-melt water supplies soil moisture, while herbs of lower altitude and plains start growing only when monsoon sets in. Apart from supplying soil moisture for plants, snow protects plants from extreme cold, as temperatures under snow seldom fall much below 0°C.

Dew and Fog

Importance of dew and fog to plants is a matter of much debate. It is suggested that leaves can absorb dew and fog, as the wet cuticle is slightly penetrable. *Dew* is condensation of water on surfaces of objects, which are cooled to *dew point* due to loss of heat by radiation. The dew point is the temperature at which the air becomes saturated, which varies with moisture content of air. A clear sky, which allows rapid reradiation, low wind speed and high vapour pressure, favour dew formation. No dew is formed under a closed forest canopy, as reradiation is reduced. The dew does not enter soil, but by reducing the difference between leaf and air water potentials it may reduce water requirement of plants. Stone (1963) found a favourable effect of dew on *Pinus ponderosa* seedlings subjected to drought. Depending upon morphological features, species differ in amounts of dew formation. At same time periods, plants of one species may be full of dew, and those of others, may have little dew.

Water droplets formed around condensation nuclei are called *fog* when in contact with the earth. Generally, fog is formed when (1) warm, moist air passes over cold water or land as in the coastal areas of sea, e.g., Pacific Northwest in U.S.A., (2) warm, moist air ascends a mountain range, e.g., tropical cloud forest areas and high mountain areas in eastern Himalaya; and (3) such an air is cooled by rapid reradiation from soil, e.g., inland valleys. According to Lovett et al. (1982), fog and clouds account for 46% of total water input to subalpine forests in the northern Appalachian Mountains.

Dust, pollen and other particulate matter and salt particles, (because of which condensation nuclei are formed), greatly enhance fog formation. Obstacles in environment cause the fog to precipitate. That is why fog gauges are the rain gauges having arrangement for a structure (a kind of obstacle) that extends above the opening, causing droplets to enter the rain gauges.

The giant coast redwood forests of northern California owe their existence to fog. In a study on these forests, 123 to 388 cm (depending on the species) fog precipitation was recorded in 28 summer days (Azevedo and Morgan, 1974). However, how much plants are able to use fog and dew is not yet confirmed. Then, stomata are usually closed at the time of deposition of dew. Slatyer (1967) concluded that absorption of surface water is so minute that, at the best, dew may only enhance recovery of turgor pressure of wilted plants in night and delay the beginning of stress during the day.

Other forms of Precipitation

The deposition of water vapour on surfaces cooled to subfreezing temperatures constitutes frost. When water droplets fall from a warm air mass through a layer of freezing or below freezing air, small pellets of ice mixed with highly cooled rain are formed which freeze on striking any surface- this is called sleet. Rapid and repeated ascent of water vapour through areas of super cooled water droplets in the cloud masses results in the formation of ice pellets called *hail*.

3.3 HUMIDITY AND VAPOUR PRESSURE

Understanding *evapotranspiration* is basic for determining the interchange of energy and water between plants and environment. It is the process by which water moves from the soil to atmosphere and constitutes evaporation (vapour from soil and plant surfaces) and transpiration (water movement through plants). When water changes into vapour, a large amount of energy

in the form of heat is required. Similarly, a large amount of heat is released when water freezes and changes into solid form. Because of the high value of latent heat of water, it takes most of the solar energy incident on an unit area on a clear summer-day to evaporate water from that unit area. Similarly, the release of heat from freezing water has warming influence on the environment, which is why crops of irrigated crop fields escape frost damage while those of dry crop fields are damaged.

Now we define three terms, viz., saturation vapour pressure, relative humidity and saturation deficit.

Saturation vapour pressure: is the maximum possible partial pressure of water that can be held in the air at a given temperature. Air is rarely saturated, so usually the level below saturation is given. Water vapour content of atmosphere can be measured by psychrometers, which have two ventilated thermometers, viz., one that measures air temperature is dry, and the other that measures the reduction of temperature resulting from evaporation.

The invisible water vapour content of air can be expressed as *relative humidity*. This is not the actual moisture content of a volume of air, but is the amount as a percentage of the maximum quantity that the air can hold at prevailing temperature. Relative humidity is defined as a constant over a range of air temperatures and vapour pressures, being meaningful, therefore, only when temperature is given. However, saturation deficit is more meaningful measure of the evaporative power of the air. It is the difference between the actual vapour pressure and the saturation vapour pressure (100% relative humidity) at the same temperature and can be explained as follows:

	At 20°C	At 30°C
Saturation vapour pressure, i.e., at 100% relative humidity (mbar)	22.5	40.5
Vapour pressure at 50% relative Humidity (mbar)	11.2	17.4
Saturation deficit	11.3	23.1

Thus, at 50% relative humidity the potential evaporation is much greater at 30°C than at 20°C. Because of the higher evaporative power of air at higher temperatures higher humidity levels are required than at lower temperatures for plant growth. The surface area of mesophyll cells of leaf in touch with internal air is about 7-80 times the external leaf area (Lambers et al. 1998), which causes the air inside the leaf to be saturated with water vapour (100% relative humidity). Since atmospheric air has generally much lower humidity, there is huge gradient in water vapour concentration from inside to outside leaf, resulting in a massive water loss.

3.4 WATER POTENTIAL AND WATER RELATIONS OF CELLS

Plant water status is a broad term used to indicate general status of plants in relation to water. It has been difficult to find an appropriate expression to describe status of plant water. Plant water potential along with its various components and water content in plant tissues relative to that present at full saturation, i.e., relative water content (RWC) are useful for studying plant water status.

Water potential (ψ_w) is a measure of the free energy of water of plant or soil in comparison with the free energy of pure and free water. All the energy associated with matter, such as wood, leaf and water in a system may not be available to do work. The energy that is available to do work in a system may be called free energy. The amount of free energy of a mole of substance is defined as the chemical potential which is a measure of the ability of a substance to do work. Water potential is generally measured in pressure units (megapascals, MPa = 10 bars*). By definition, water potential

at atmospheric pressure and at a temperature of 298 K of pure water is zero. But in plants and soil it is less than zero, like -2 MPa, and -3 MPa (value of -2 MPa is greater than -3 MPa, and water stress at -2 MPa is less than that at -3 MPa). The more negative the ψ in soil, the less available is water to plants.

Movement of water into and through a plant occurs from higher to lower ψ_w , e.g., from 0.5 MPa to -1.5 MPa. Diffusion, which is the net movement of a given substance from one area to another due to random motions of particles (molecules, atoms, ions, etc.) which it is composed of, occurs from the site of higher chemical potential of substances to that of lower chemical potential of that substance. Interaction of water with solute particles results in decrease of its (water's) free energy. Therefore, diffusion of water takes place from less concentrated solution to more concentrated solution (water in the latter is less free than in the former).

Water potential ψ_w consists of several components, i.e., matric potential ψ_m , osmotic potential or solute potential ψ_s , hydrostatic pressure ψ_p and gravitational pressure which refers to the effect of gravity on water potential ψ_g . However, in any plant part water potential is mainly algebraic sum of osmotic potential and hydrostatic pressure (the other component is generally ignored when describing transport of water between cells). Gravitational potential can be substantial when vertical distances are greater than 5 - 10 m. The matric potential determines the interaction of solid surfaces such as cell walls and colloids with water; generally values are small and difficult to determine and its impact on water potential is generally ignored.

The matric potential ψ_m and osmotic potential ψ_s , (also written as ψ_π) values are always negative; the hydrostatic pressure potential ψ_p can be negative as well as positive. Therefore, water potential values are always negative, with the exception of a fully turgid cell where positive value of P may balance the negative ψ_s and ψ_m (Kramer, 1969). ψ_m refers to the force with which water is adsorbed onto surfaces by various substrates (adsorbing colloids to which water molecules are attracted or adhered include macro-molecules, such as of proteins in cells and soil particles in rooting zone) and is important in soil. As for plants ψ_m is poorly understood, and its effects may be lumped with ψ_s , except for seeds.

Osmotic or solute potential ψ_s always measures the reduction in free energy of water resulting from the presence of solutes. It can be called chemical potential of water due to the presence of dissolved materials. The addition of solutes in cells decreases the free energy of water.

Pressure potential ψ_p or the *hydrostatic pressure* of water is the physical pressure exerted on water in the system. A living plant cell is an osmotic system having a differentially permeable membrane (Box 3.2), e.g., some molecules or ions can diffuse faster than others through it. The membranes are generally much more permeable to water than to solutes. Water would move (diffuse) from high concentration of water, that is water without solutes to low concentration of water, i.e., solution (contains solutes, therefore water concentration is lower). Water diffusion causes pressure to build on the walls of the cell having solution, or sap. Pressure must be applied to prevent the movement of water into the solution compartment. In case of the plants cells the rigid cell wall supplies the opposite force to check diffusion of water into it, and is thereby responsible for the increase of pressure inside the cell. Adding pressure in living cells by cell wall results in positive, pressure often referred to as turgor pressure. However, water in the dead xylem vessels of rapidly transpiring plant is under tension, i.e., in negative pressure.

* $1 \text{ bar} = 10^5 \text{ Pascals (Pa)} = 0.1 \text{ Megapascals (MPa)} = 0.987 \text{ Atmospheres} = 14.5 \text{ pounds per square inch}$

Box 3.2.

Differentially permeable membranes are necessary for life. They occur in the protoplast inside the cell wall (plasma membrane) and in the vacuole, as well as in many cell organelles: plastids, mitochondria, endoplasmic reticulum, nucleus, golgi apparatus, sphaerosomes, etc. The membrane controls what substances enter and leave the cell and vacuole and from distinct compartments within the cell, which differ in their contents. Much of the control over reactions, which occur in cells, is due to the compartmentalization of different substrates and enzymes. Permeability of these membranes in the cell may help control reaction rates by controlling the movement of substances from one compartment to another.

The ψ_w of a cell is a result of the balance between ψ_s and ψ_p ; thus

$$\psi_w = \psi_s + \psi_p$$

If the cells $\psi_p = 0.5$ MPa and $\psi_s = -1.5$ MPa, then $\psi_w = -1$ MPa

Effect of ψ_p and ψ_s in relation to water movement can be explained by following examples: consider two cells A and B in contact with each other, with following MPa values:

Cell	ψ_p	ψ_s	ψ_w
A	+1.0	-2.2	-1.2
B	+1.3	-2.0	-0.7

Water movement would occur from B to A.

Consider another two cells, M and N in contact with each other. Now we want to know the value of ψ_s for cell N (indicated below as x) which would not allow water movement between the two cells. The values of the other water potential components of the cells, are as following:

Cell	ψ_p	ψ_s	ψ_w
M	+1.0	-1.8	-0.8
N	+0.5	x	-0.8

For no movement ψ_s of N cell $= -0.8 = 0.5 + x$, i.e.,

$$x = -1.3$$

(since ψ_s is always negative, the value is -1.3 MPa).

However, the above relationships are not so simple. As water moves into a cell its wall may stretch and some cells may stretch more than others. The change in ψ_p would depend on how much wall stretches and consequently how much volume increases. Increase in cell volume would reduce ψ_p . Another influence of moving water is dilution of cell solution, resulting in higher ψ_s . The change in ψ_s can be calculated if volume (V) of change is known as follows:

$$\psi_{s1} V_1 = \psi_{s2} V_2$$

Where ψ_{s1} , V_1 and ψ_{s2} , V_2 indicate original and changed values, respectively. Suppose a cell's $\psi_s = -1.5$ MPa and the volume changes by 100%, or two times, The n:

$$\psi_{s2} = -1.5 \times 1 = \psi_{s2} \times 2$$

$$\psi_{s2} = \frac{1.5 \times 1}{2} = 0.75 \text{ MPa}$$

Thus, the increase in cell volume, or dilution of cell solution due to the moving water increases ψ_π . The effect of increase in solute concentration is just opposite.

Water potential also affects many plant processes other than water movement. Most processes slow down as ψ_w declines. Decline in rate of leaf production is one of the first major responses in the field. Growth is more sensitive than many other processes. Some effects are primarily due to reduced pressure, others result from stomatal closure and still others from hormonal changes induced by low ψ_w . Trees with low ψ_w are often more subject to disease and insect damage than well watered trees with high ψ_w .

The ability to maintain adequate turgor as water potential declines, called *turgor maintenance* is considered an important adaptation to water deficits. Thus a key property of the species is how its ψ_p changes as it loses water. Changes in ψ_p as water is lost from plants depend on changes in ψ_s and elastic properties of the cell wall. At a given water potential ψ_w , reducing ψ_s increases ψ_p , or say improves turgor maintenance. A tissue having low ψ_s will have high ψ_p at any value of ψ_w , and the osmotic potential at full turgor sets the upper limit to the turgor which can be developed. These properties change seasonally in plant tissues as plants adapt to changing environmental conditions.

Relative Water Content

In recent years, questions have been raised concerning the value of leaf water potential as a measure of plant water status in relation to physiological processes and relationship of turgor to growth and stomatal conductance. Proposals have been made to express plant water status in terms of relative water content (RWC). The RWC of plant tissue is determined as:

$$\text{RWC} = (\text{Fresh weight} - \text{dry weight}) / (\text{Turgid weight} - \text{dry weight}) \times 100$$

Leaf discs or needles are floated on water (distilled) until they become fully turgid to determine turgid weight. 1-RWC gives the value of relative water deficit (RWD). The RWC of transpiring leaves has been reported to range from 85 to 95 % when water uptake by roots closely follows water loss by leaves.

3.5 PRESSURE CHAMBER METHOD FOR MEASURING WATER POTENTIAL AND PRESSURE VOLUME CURVE

The pressure chamber technique devised in its modern form by Scholander et al. (1965) is the most widely used method for measuring osmotic potential., as it is reliable, simple to use, and it does not require fine regulation of temperature (see Ritchie and Hinckley, 1975).

A leaf or twig is inserted through a rubber stopper, which is used to seal the pressure chamber. The leaf or shoot remains mostly in the chamber, except for the petiole of leaf, or cut-end of the shoot, which protrudes through the rubber stopper (Fig. 3.2). Using a nitrogen cylinder, the atmospheric pressure in the chamber is slowly increased until sap (fluid) appears at the cut end. At the point of sap appearance, the hydrostatic pressure inside the xylem elements is exactly balanced by the applied external pressure. This is the ψ_w of that leaf or shoot. Following precautions must be taken:

- (i) The leaf or twig should be covered with a plastic sheet at the time of sample collection to reduce water loss by transpiration.
- (ii) Pressure should be increased slowly, usually at the rate of 0.25 to 0.35 bars (0.025-0.035 MPa) per second (Ritchie and Hinckley, 1975).
- (iii) The twig or petiole should be cut only once and by a sharp blade or razor.
- (iv) The cut end should extend only 1.5 to 2.5 cm from the stopper. Pressure-volume curves (P-V curves) enable us to separate ψ_s (osmotic potential) and ψ_p (pressure potential). The theory behind P-V curves depends on a relationship between ψ_s and the solutes in the cell sap, known as van't Hoff equation:

$$\psi_s = R T N_s / V$$

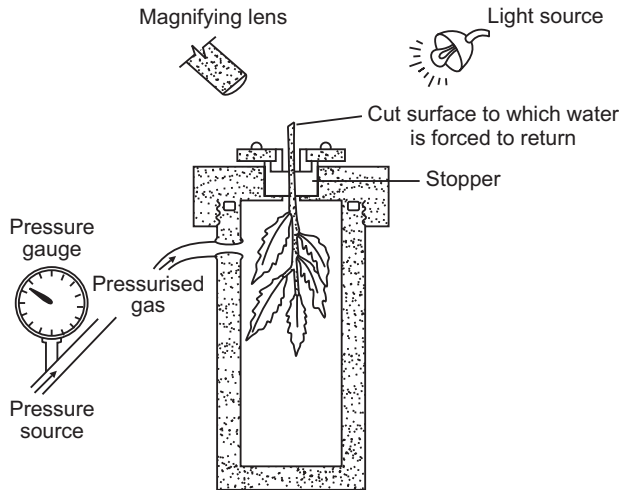


Fig. 3.2. A diagrammatic representation of a pressure chamber used to measure plant water potential.

Where ψ_s is a coefficient relating to the type of solutes, R is the universal gas constant, T is Kelvin temperature, N_s is the number of moles of solute, and V is the volume of water in the cell. When water is squeezed out of cells (as by applying pressure to leaf or twig in pressure chamber), the volume of water in the cells decreases (while R , T and N_s all remain constant). Therefore, ψ_s changes linearly with $1/V$ and $1/\psi_s$ is linearly related to V . As for total water potential, it drops with decrease in water volume as the turgor in the cells also decreases (Fig. 3.3). With the continued loss of water from cell the turgor falls to zero. This is the point of plasmolysis. Turgor will not decrease below zero, therefore, after the turgor loss the water potential of the cells is a function of only ψ_s . Hence, after turgor loss water potential is also linearly related to $1/V$. To collect data for P-V curve, one way is to place the leaf or twig sample in a pressure chamber, apply pressure, and collect all expressed water (cell sap) from the cut-end to compute volume. This exercise is repeated several times for increasingly higher pressures. In another way, samples (twig or leaf) are dried passively and water loss is measured by weighing samples before placing them in the pressure chamber to determine ψ_w .

To construct P-V curves in practice, $1/\psi_w$ is plotted against the RWC or RWD (which is $1-\text{RWC}$). Generally, $1/\psi_w$ drops sharply with increasing RWD until plasmolysis ($\psi_p = 0$), then becomes linear. Now, this line is extrapolated to vertical axis (ordinate), to calculate $1/\psi_s$ at full turgor and $1/\psi_w$ at zero turgor. The relationship between turgor potential and RWD can also be determined from this curve. ψ_s is calculated for several RWD values, then ψ_p is calculated as $\psi_p = \psi_s + \psi_w$.

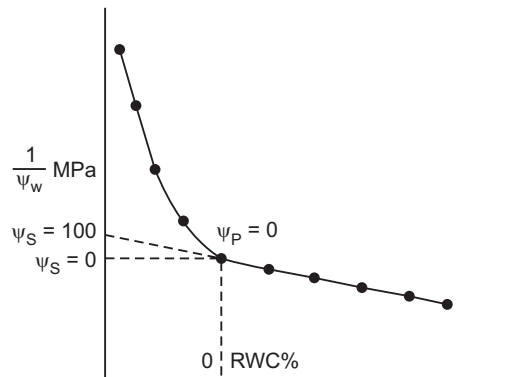


Fig. 3.3. A pressure-volume curve showing the relationship of water potential to relative water content, and the derivation of the osmotic potential at full and zero turgor and RWC at zero turgor.

3.6 WATER IN SOIL

Water comes into the soil by percolation of rainfall and snowmelt as well as by upward movement of ground water by capillary action. There are two important factors of soil water with regard to its availability to plants, viz., amount of water stored in soil and its relationship with soil water potential. The water storage largely depends on pore size. Pores larger than 30 μm , such as in sandy soils hold little water as water easily drains out. In contrast, pores smaller than 0.2 μm hold water so firmly to surrounding soil particles that water hardly drains out and is scarcely available to plants. Therefore, water that plants can easily tap occurs when soil pore size ranges between 0.2 and 30 μm in diameter.

Soil water potential determines the availability of water for plant growth and it is influenced by physical properties of soil. When soil is irrigated, the water fills pore spaces and drains down through pores under the influence of pull of gravity, and this is referred to as gravitational water.

The amount of water retained in the soil when gravitational water has left soil, is called the *field capacity* (FC), which is the water content of saturated soil. This is the upper limit of water available to plants, and is constant for a given soil and differs from one soil to another, depending on soil texture, soil structure, organic content, and is close to 0.01-0.03 MPa. The lower limit of water in soil that plant can access is called *permanent wilting point* (PWP). At this point, plants wilt permanently and remain wilted even when brought to cool, humid and dark chamber (but addition of water may enable plants to revive). At PWP, soil water potential equals to or is less than the minimum osmotic potential of plants so that water cannot be taken up from the soil. The PWP for mesophytic plants is often around -1.5 MPa, but can range from -1 to -8 MPa. The amount of water between FC and PWP is called *available water* or *capillary water* (for capillary and matric actions retain water against the gravitational pull). Plants can remove most of the available water, but not all water. In unsaturated soils, water potential is largely determined by matric potential, but in soils such as saline soils, osmotic potential is predominant. At given water potential, clay soil would hold far more soil water content than the light soils, such as sandy, loam and sand (Table 3.1).

Irrespective of soil type, as soil moisture declines, the water film surrounding soil particles becomes thinner, and the water that is left is held more tightly, giving a more negative matric potential. In soil hydrostatic pressure is generally negligible, and matric potential is the most important component of soil water potential. However, in saline water, osmotic potential is also an important component of water potential.

Table 3.1. Pore size distribution and soil water content in three soil types based on texture (from Lambers et al., 1998)

	Soil		
	Sand	Loam	Clay
Pore space (% of total)			
>30 μm	75	18	6
0.2-30 μm	22	48	40
>0.2 μm	3	34	53
Water content (% of volume)			
Field capacity	10	20	40
Permanent wilting point	5	10	20

Measurement of Soil Water Content and Potential

The measurement of soil water content can be made on soil samples of known weight or volume dried in an oven at 105 °C to constant weight. Soil sample is collected with an auger. Water content is shown as percentage of dry weight. However, to make this value useful, both PWP and FC should be known. When weight percentage of soil is multiplied by its bulk density, it gives water content in terms of volume.

Soil water content can also be measured indirectly by neutron probes which consist of a source of neutrons and a detector. These can make frequent and continuous measurements of soil water content at several depths without undue soil disturbance. A neutron probe makes use of the following facts: (i) hydrogen atoms have a high capacity to slow down and scatter fast neutrons, and (ii) in most soils, water is the principal source of hydrogen atoms (Kramer and Boyer, 1995). In this method water content is measured by counting slow neutrons around a fast neutron source.

There are several methods to measure soil water potential. However, the use of thermocouple psychrometer is now quite common. It measures the total soil water potential. In one method, soil samples are removed and placed in psychrometer (in sample cups). In the first and the last cups, KCl solution of known osmotic potential at 20°C (0.1 and 0.9 m KCl solutions, respectively) it placed. The instrument has a micro-voltmeter attached to it, which gives micro-volt and temperature readings for the samples. A graph is prepared taking osmotic potentials on x-axis and microvolt reading on y-axis. Two fixed points are marked taking known osmotic potential values for 0.1 and 0.9 m KCl solutions and microvolt readings (read from the micro-volt meter for the solutions) and connecting the two points by a straight line. The line is extrapolated in both directions. The values of soil water potential (osmotic potential) are read from this graph after marking the micro-volt reading on the y-axis for different soil samples. The points marked on the y-axis (micro volts) are then connected to the extrapolated line connecting the two fixed points by straight lines. Perpendiculars are dropped from these points on to the x-axis and the osmotic potential (water potential) read.

Movement of Soil Water

Soil organic matter is hydrophilic (with the exception of the one that is rich in plant- derived waxy compounds) and, therefore causes water retention to increase. For example, the water holding capacity is more than doubled when organic matter content is raised from 0.2 to 5.4% (Lambers et. al., 1998).

Availability of water within soil depends partly on how rapidly and how far water moves upward from the water table toward an absorbing root, or downward into the soil during a rainstorm through a process called *infiltration*. In soils of similar texture and structure, infiltration is faster in dry soil than in wet soil. Through a saturated soil the rate of water flow v , is:

$$v = -k \frac{\Delta}{D} \psi_{\text{soil}}$$

where k is *hydraulic conductivity*, i.e., the capacity of soil to transport water (may be measured as cm/second); $\Delta\psi_{\text{soil}}$ indicates changes in ψ soil, and D is the soil depth. The hydraulic conductivity drops as soil water drops. Compared to fine textured soil (rich in clay), this drop is sharper in coarse-textured soil (sandy soil) where continuity of water is easily broken. Clay holds a much greater amount of water than sand at a given soil water potential, consequently in a fine textured soil, continuity of the water is not broken even at a low soil water potential.

Lateral movement of water through the soil to the zones of absorption by roots depends on the water potential gradient and the soil hydraulic conductivity; soil water potential must be higher than root water potential for absorption to occur.

Water infiltration into the soil is a critical factor and becomes more critical on steep hill slopes where rainfall is also intense. Infiltration is faster in sandy soil than in fine textured soils, where it is restricted with increasing proportion of clay particles. However, where organic compounds derived from plants form a “skin” (a thin impenetrable layer) over sandy particles, infiltration stops, and water droplets bead up on soil surface. Thus, the soil becomes “water repellent” or water phobic.

3.7 WATER IN CELLS

At a plant cell level, adjustment of water potential occurs through changes in hydrostatic pressure ψ_p as well as osmotic potential ψ_s . However, only through adjustment of ψ_s plants can modify ψ_w within hours. In long term, however plants can also adjust ψ_w by modifying cell wall elasticity. As for dead cells, they have very dilute cell solutes, and ψ_w can change largely through changes in hydrostatic pressure.

Osmotic adjustment

Reduction in ψ_s of cells as a result of accumulation of osmotically active compounds is called *Osmotic adjustment*. As a consequence of increased solute concentration, cells develop a higher turgor when fully hydrated and lose turgor (turgor loss point) at more negative water potential. Thus, it enables plants to acquire water from drying soils and maintain some growth where none would occur otherwise. The Himalayan oaks and their common undercanopy associate, *Machilus duthei* reduce their water potential by more than 1 MPa through osmotic adjustment, (Fig. 3.4), thereby maintain growth throughout winters (Singh et al. 2006) Solutes which do not have a negative effect on cell metabolism are called *compatible solutes*. They are synthesized in cytoplasm to maintain enzyme functions. In contrast, osmotic solutes in the cell vacuole (which are often inorganic ions and organic acids) reduce the activity of enzymes of cytoplasm. Sugars, glycerols, amino acids such as glycine betaine, proline, sugar alcohols like mannitols and other low molecular weight metabolites are compatible solutes. Marine algae like *Dunaliella* and *Oochromonas* can withstand even saturated solutions of NaCl by producing compatible solutes (glycerol or galactosyl glycerol) in cytoplasm (Kramer and Boyer, 1995). These solutes are produced from reserves, largely starch and they return to starch when plants are not under stress.

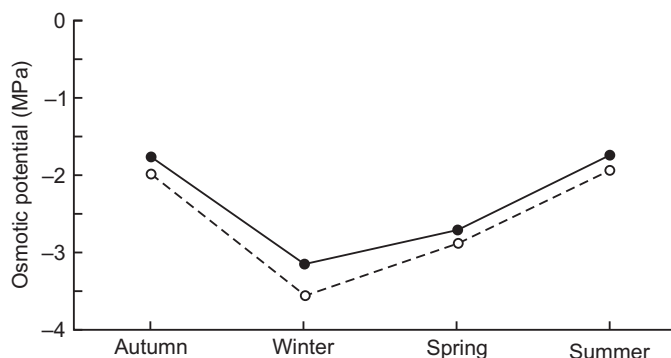


Fig. 3.4. Seasonal course of osmotic potential at full turgor (solid line) and osmotic potential at zero turgor (dotted line) in *M. dutheie*, a common understorey species of the central Himalayan oak forests (Tewari, 1998).

Osmotic adjustment can enable growing tissues to extract water from mature tissues of a plant (Matyssek et al., 1991). In that case water can move backward from mature shoots to growing roots. Since the ability of solutes largely depends on photosynthesis, plants cannot employ osmotic adjustment for long in the face of continued water limitation.

Plants differ in the elasticity of their cell walls, which depends on chemical interactions between the various cell wall components. It is an important character as it determines the extent to which a cell can decrease its volume. This in turn, determines the extent to which water potential of cell can decline until the turgor loss point is reached. Cell walls of CAM and other such plants are highly elastic; therefore, hold more water at full turgor. Hence, their volume can decrease more before the turgor loss point is reached.

The *bulk elastic modulus*, ϵ (measured in MPa) is reverse of cell elasticity and is the amount of pressure by which a change in cell volume brings about a change in turgor at a certain initial volume (Tyree and Jarvis, 1982).

$$\Delta P = \epsilon (\Delta V/V) \text{ or } \epsilon = \Delta P/(\Delta V/V)$$

Where ΔP is change in turgor pressure (MPa), and $\Delta V/V$ is the relative volume change. A large number of measurements of ϵ have been made in giant algal cells (e.g., *Nitella*) where values may range up to 100 MPa. In higher plant cells, the ϵ values range from 1 to 10 MPa (Dainty, 1976). The bulk elasticity modulus is generally greater for species with thick walled cells than those with thin walled cells. Among the Central Himalayan species, conifers have clearly lower ϵ than sclerphyllous oaks and other broad-leaved tree species (Table 3.2).

Table 3.2. Elasticity modulus (ϵ) of the Central Himalayan trees by life form (Singh et al., 2006). Values (MPa) are mean of four seasons (rainy, autumn, winter and summer). The mean of broad-leaved evergreen group was based on 7 species (including three oaks, deciduous 3 species and conifers 2 species)

	Means of seasons
Broad leaved evergreen	10.43-24.17
Deciduous	10.53-14.85
Conifers	3.73-6.25

From Höfler diagrams that are used to relate turgor pressure, osmotic potential and water potential to relative water content, ϵ can be derived. Figure 3.5 shows such a relationship for two Mediterranean sclerphyllous tree species. Of the two species, *Laurus nobilis* (a laurel) shows a much greater change in turgor for a change in volume than *Olea oleaster* (olive).

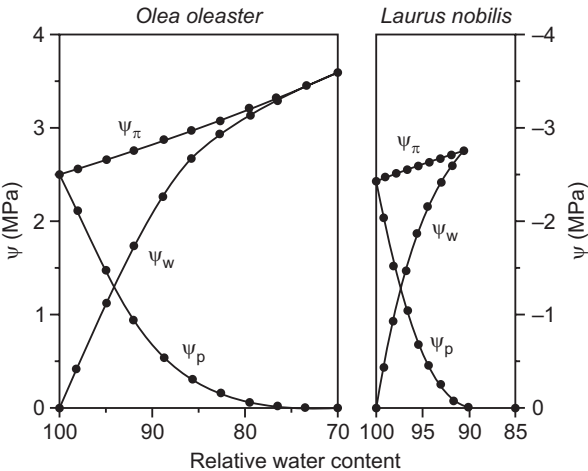


Fig. 3.5. Höfler diagrams, relating turgor pressure (ψ_p) and osmotic potential (ψ_{π}), to relative water content for leaves of two Mediterranean tree species. The slope of ψ_p with relative water content is used to calculate the bulk elasticity modulus (ϵ). The bulk elasticity modulus (ϵ) is the initial slope of ψ_p with relative water content (Lo Gullo and Salleo, 1988).

3.8 SOIL-PLANT-ATMOSPHERE WATER MOVEMENT

As to how water moves from soil through the plants and into the atmosphere, of the several hypotheses suggested, two are important, viz., cohesion and multforce hypothesis (Barbour et al., 1999). According to cohesion hypothesis, water movement is driven by transpiration through the path of soil-plant-atmosphere called soil, plant and atmosphere continuum (SPAC). However, there are several evidences to suggest that the transpirational pull alone may not account for this water movement. For example, plants can obtain transpirational water not only from the soil but also from their own tissues. According to this, each tissue has ability to lose water in a transpiring plant called capacitance; it is measured as the change in tissue water content per unit change in water potential. This tissue water content includes both apoplastic (cell wall water) and symplastic (protoplasm water). Desert succulents are known to have a large capacitance. Furthermore, some water may move from the roots and surrounding tissues under the influence of osmotic force, or water movement may not be strictly pressure driven. Similarly, reverse osmosis which enables water movement from sieve tubes to xylem may also contribute to water movement.

Evapotranspiration

Evaporation of water occurs from soil surfaces and those of the plants. The amount of moisture that evaporates from soil is small as compared to that comes from plants. The process by which plants give up moisture through their leaves is called *transpiration*. The combined process of water loss from both soil and plants is called evapotranspiration. The wet or dry condition of a soil depends on the rate of evapotranspiration and precipitation. Evaporation from the soil is the direct pathway for the movement of water vapour to the atmosphere. Soil evaporation rates are high after irrigation or rainfall as the soil is wet. The soil evaporation rates decline with the drying of soil. The rate of evaporation from a water body is regulated by temperature of the water and the air and the degree of windiness. Higher temperature and greater windiness cause more evaporation. A typical pattern of soil water potential in monsoon climate is shown in Fig. 3.6. It keeps on declining after the monsoon.

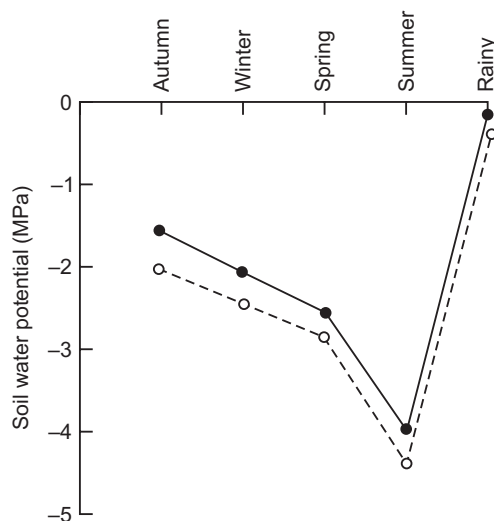


Fig. 3.6. Seasonal changes in soil water potential at 10 and 60 cm soil depth (solid and dotted lines, respectively) in an oak forest of Nainital, Uttaranchal, India (Tewari, 1998). This is the typical seasonal pattern in a climate with monsoon pattern of rainfall.

The water status of a terrestrial plant is determined by a dynamic equilibrium between the source of supply, the soil and sinks, the atmosphere and/or use of water in leaf growth. Transpiration involves two processes: (i) evaporation from cell walls into intercellular spaces, and (ii) diffusion from intercellular spaces to air. The water moves along a potential energy gradient, which decreases from the soil to the leaf surface and then to the atmosphere. However, difference between leaf and air is far steeper than between soil and leaf. Even among plant parts there should be difference in water potential if water has to move through the plant. Thus, the order of water potential needs to be like this for water movement through the plant need to be : soil > root > stem > leaf > air. A common set of water potential measurements

along the gradient might be -0.1 MPa, -1 MPa, -1.5 MPa and -100 MPa, respectively in soil, stem, leaf and atmosphere (Slatyer, 1967). Thus, the final loss of water through transpiration (from leaf to atmosphere) is associated with the steepest part of the water potential gradient.

Water Uptake by Roots

Kramer and Boyer (1995) identify two mechanisms of water absorption from soil, viz., osmotic absorption in slowly transpiring plants, and passive absorption in rapidly transpiring plants. The first occurs in plants growing in moist, well aerated soils, and it leads to creation of root pressure, exudation and guttation. Here, roots function as osmometers. In rapidly transpiring plants, much of water is pulled in by matric forces created in the evaporative surface of cells and is transmitted to roots through the cohesive water columns of the xylem cells, where the decreased water potential causes inflow from the soil.

Water enters the root primarily through root hairs, because they are much more permeable than older roots and their surface area is far larger. Water enters roots probably through a region few centimetres behind root tips, where root hairs are most abundant (Kramer and Boyer, 1995). Little water enters both through the meristematic part of roots because of a lack of functional xylem and older parts where suberization and lignification of tissues (hypodermis and endodermis) reduce water permeability. Root hairs also maintain contact between root and soil particles, and in this mycorrhizae play an important role. Water finds least resistance until it reaches endodermis in roots, where it has to move through the cell membrane instead of cell wall because the casparian strip makes it impermeable and attaches cell membrane to cell wall.

Plants continuously explore soil with the establishment of root system, by extending new roots. That is why root productivity and biomass are quite high (up to 60-80% of total biomass in some communities). Root density, or root length per unit volume of soil and root architecture may have a significant influence on water uptake. Within soil mass, roots have an extraordinary ability to grow away from dry pockets and toward wet ones (hydrotropic) and root caps are likely sites of *hydrosensing* (Takahashi and Scott, 1993).

Hydraulic lift: In some species water can move from deep moist soils to dry surface soils through their roots and is called *hydraulic lift*. The water lifted, thus, to dry surface soil is often utilised by shallow rooted neighbouring species. For example, Dawson (1993) showed that 3-60% of water transpired by shallow rooted herbs came from the water lifted by the deep rooted sugar maple (*Acer saccharum*) from deep soils. This upward water movement occurs during night. This kind of facilitation by a species to its neighbouring species is common in the Great basin desert of Western North America (Richard and Caldwell, 1987), where shallow rooted *Agropyron desertorum* received 20-50% of its water use from the water hydraulically lifted by deep rooted neighbour sage brush (*Artemesia tridentata*) (see Lambers et al., 1998).

Water in leaf: A decrease in soil water availability below critical level may lead to a reduced cytokinin production and its transport to leaves (Davies et al., 1986). The lowered cytokinin production may result in partial closure of stomata (Blackman and Davies, 1985).

The water potential of a plant part depends on how fast it is losing water in transpiration and how fast its supply from soil can be replaced.

A simple equation for transpiration, T , can be written as:

$$T = \frac{C_{ias} - C_{air}}{\sum r_i}$$

Where C = concentration of water vapour,

i_{as} = intercellular spaces of leaf,

a_{ir} = air around leaf, and

r_i = is resistance to diffusion ;

Σ before r_i indicate that there are resistances due to several factors.

Air and leaf temperature, absolute humidity of air and resistances of stomata and leaf boundary layer affect the diffusion of water from leaf to air.

The *leaf resistance* to gaseous diffusion (resistance to water vapour diffusing outwards or CO_2 inwards) is measured in second cm^{-1} . Alternatively, diffusion can be expressed as *conductance*, the reciprocal of resistance, with units, cm second^{-1} . Porometers measure this resistance or conductance. Most field measurements are made by diffusion prometers, which measure diffusion of water vapour out of leaf. A chamber or cup is pressed onto the leaf. Water vapour diffuses from the leaf through stomata into cup. This increases relative humidity (RH). The rate of RH rise in the cup is a measure of the leaf surface conductance.

The stomata, which are located at the steepest portion of the water potential gradient, have also the greatest controlling power on the water flux from plant to the air, for they have ability to open and close.

Transpiration is directly proportional to vapour pressure difference between leaf and air and inversely proportional to resistance in the diffusion pathway. The vapour pressure difference is influenced by water content of air. Air in intercellular space is usually saturated and vapour pressure in saturated intercellular air depends on leaf temperature. Solar radiation elevates leaf temperature above air temperature and wind action brings down leaf temperature to the level of air temperature.

There is an external layer of still air along leaf surface through which water must diffuse. This is called *boundary layer*. The resistance r_a caused by this still layer of air to water diffusion increases with increase in leaf size as the thickness of air increases. Wind action reduces thickness of boundary layer, so it affects transpiration by reducing r_a .

Leaf resistance has two components, viz., r -cuticle (r_c) and r -stomata (r_s). While r -cuticle is very high and essentially invariable, r -stomata which depend on stomatal density and opening size, varies widely from $< 0.4 \text{ sec cm}^{-1}$ in fully open stomata to infinity in closing stomata, and is thus the most important plant mechanism for controlling water loss. Stomata are pores in the leaf; their size is controlled by guard cells surrounding each pore. Thus, the total leaf resistance is equal to sum of r_a , r_c and r_s . Generally, stomata close in the dark, in high CO_2 , in low humidity and with extreme temperatures.

Stomata on leaves, stalk, and stems, regulate the water loss from plants as whole. At global scale, $32 \times 10^{15} \text{ kg yr}^{-1}$ of water vapour pass through stomata, which is double the water vapour content of atmosphere ($15 \times 10^{15} \text{ kg yr}^{-1}$; Hetherington and Woodward, 2003). The opening and closing of stomata are regulated by differences in guard cell turgidity. This involves the changes in osmotic potential in guard cells caused by movement of K^+ into and out of cells. The entry of potassium ions into guard cells from the surrounding cells increases the turgidity of guard cells and opening of stomata. When plants are under water stress the production of abscissic acid (ABA) is stimulated, which inhibits the proton pump that drives K^+ into cells. As a consequence K^+ diffuses out from the guard cell, which in turn causes stomatal closure.

The dumb bell-shaped stomata (Fig. 3.7) of the grasses with smaller pores are more efficient in regulating pores than the kidney-shaped stomata, with larger pores.

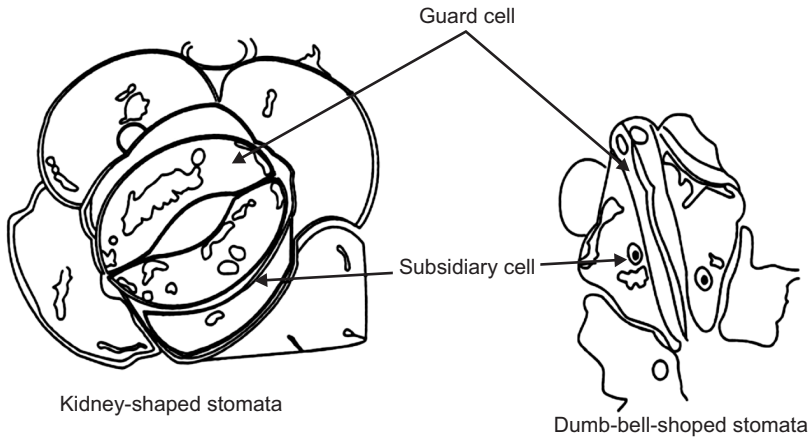


Fig. 3.7. Two stomata types: (i) kidney-shaped found in dicotyledonous plants and (ii) dumb bell-shaped associated with grasses and crops of grass family.

These are faster to close and open; and demonstrate lower potential for hydraulic dysfunction under drought (Hetherington and Woodward 2003). The spread and diversification of grasses during the increase in aridity, 30-45 million years ago would have been made possible partly because of their dumb bell- stomata. On the whole, the size of stomata range from 10-80 μm and they occur at densities between 5 and 1000 per mm^2 of epidermis. There is an inverse correlation between stomatal density and guard cell length (Fig. 3.8).

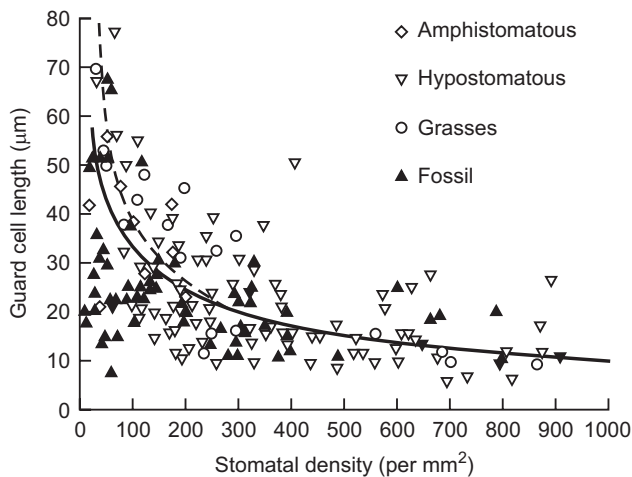


Fig. 3.8. Relationship between guard cell length and stomatal density. The solid curve is log-normal, $y = -28.75 + 162 x^{-0.2086}$, $r^2 = 0.5$. The dashed line shows equal stomatal conductance with variable stomatal density. Samples included amphistomatous (stomata on both leaf surfaces), hypostomatous (stomata confined to lower leaf surface), both of living and fossil species. (reproduced from Hetherington and Woodward 2003. Based on data from Willmer and Fricker, 1996; Beerling and Woodward, 1997; Sugden 1985; Tanner and Kapos, 1982; Knapp, 1993).

3.9 TEMPORAL CHANGES IN WATER POTENTIAL

Diurnal patterns

For a day, soil moisture remains relatively invariant, while transpiration is low at night and high during the day. There is a time lag before the water supply from soil can replace the transpiration water loss. As transpiration decreases at night, water uptake can replace water deficit and leaf water potential can reach close to soil water potential. The daily change in plant water potential from predawn to mid-day can be considerable in a rapidly transpiring plant growing in a favourable environment (Fig. 3.9).

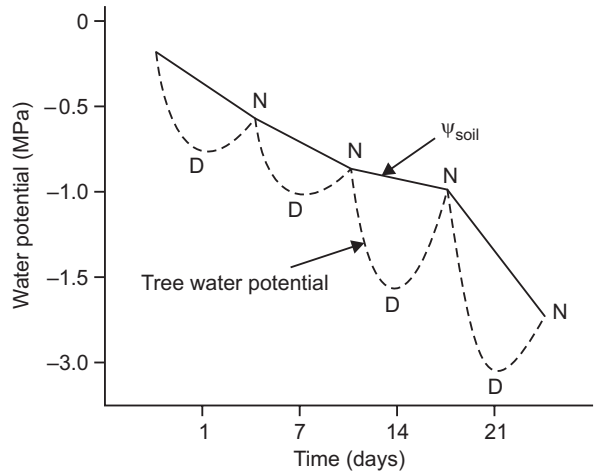


Fig. 3.9. Typical diurnal changes in twig water potential (broken line) and soil water potential (solid line) of banj oak seedlings rooted in soil allowed to dry from water potential near zero to a water potential at which wilting occurs. D indicates day time and N the night time water potential.

Seasonal pattern

In trees, the shoot ψ_s is low during winters and is followed by rapid increase in spring, the time of bud swell (Fig. 3.4). The ψ_s is highest in subsequent summer months when shoots elongate vigorously. Following shoot elongation, ψ_s again drops. The low winter osmotic potential in evergreen species is due to increased sugar concentration. During the bud swell and shoot elongation sugar of leaf in evergreen species is transported to new shoots, and decline in sugar causes increase in ψ_s . During shoot elongation, elasticity is high as shoots have mainly cells without secondary thickenings. Elasticity is low during mid summer, by which time cells develop secondary thickenings. Plants also show seasonal changes in daily reduction of water potential (predawn to mid-day). In Himalayan trees the daily reduction increases from monsoon to winter season (Fig. 3.10).

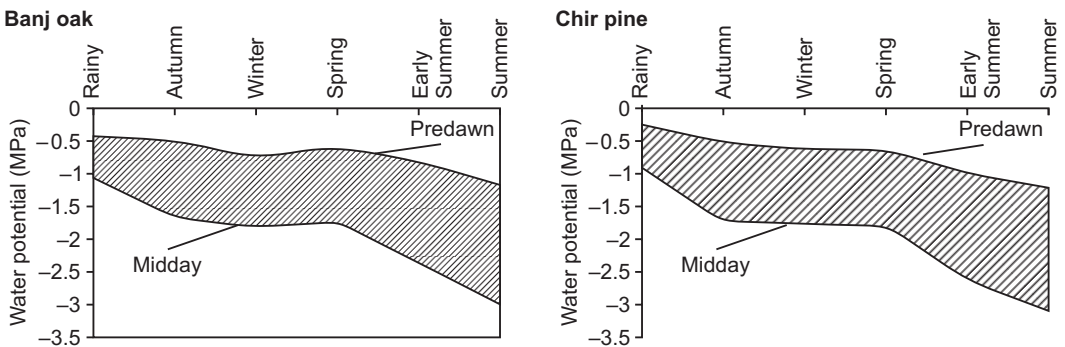


Fig. 3.10. Daily change in tree water potential (from predawn to midday) across different seasons in two major forest species of central Himalaya. Banj oak is *Quercus leucotrichophora* and chir pine is *Pinus roxburghii* (derived from Tewari, 1998).

3.10 WATER FLOW AND STORAGE IN STEMS

Within the stems of vascular plants, water flows through vessels and tracheids of xylem. The water flow rate in xylem is a function of the radius of xylem elements and the viscosity of the xylem solution (Nobel, 1983). The hydraulic conductance is proportional to the fourth power of the diameter of the lumen of vessels and tracheids. For a given hydrostatic pressure gradient, water flow in vessels which are wider, is greater than in tracheids. In addition, tracheids have pits, which impose resistance to water flow. Thus generally, water flow is more rapid in angiosperms than gymnosperms, which have only tracheids. Generally, tracheids have diameter upto 30 μm , vessels in the range of 20-700 μm . Partly because of this, gymnosperms fail to compete with angiosperms in tropical humid forests where soil water is less limiting. Among the angiosperms, ring porous species have wider vessels than the diffuse porous species. But wider the conducting element, the more vulnerable it is to *embolism* or *cavitation* which is the obstruction of conducting elements due to the filling with air bubble or water vapour, rather than water. That is why embolism is found maximum in ring porous species and minimum in conifers. The empty air occurs when water stress is high or when tension is caused by freezing and thawing. Cavitated conducting tissue can refill by dissolution of air bubble, when xylem pressure becomes less negative, and this recovery occurs more readily in narrow conducting tissues than in broader ones. That is why deserts and colder regions

have preponderance of plants with narrow conduits. Species greatly differ in their vulnerability to cavitation or embolism (Fig. 3.11). Species with narrow conduits also cavitate at more negative water potential than the species with wider conduits. Apart from diameter of conducting tissues, exposure to low water potential in their natural environments determines their vulnerability to cavitation. For example, *Cassipourea elliptica* which generally experiences xylem water potential above -1.5 MPa in field condition is more vulnerable to cavitation than *Rhizophora mangle*, which is frequently exposed to -4 MPa or lower water potential in its habitat. Furthermore, species from habitats which are exposed to a very low water potential for many weeks at a time, operate with a large safety margin. For example, creosote (*Larrea tridentata*) would completely cavitate only where water potential reaches below -16 MPa, though the minimum water potential it encounters in its habitat is much less negative (about -9 MPa).

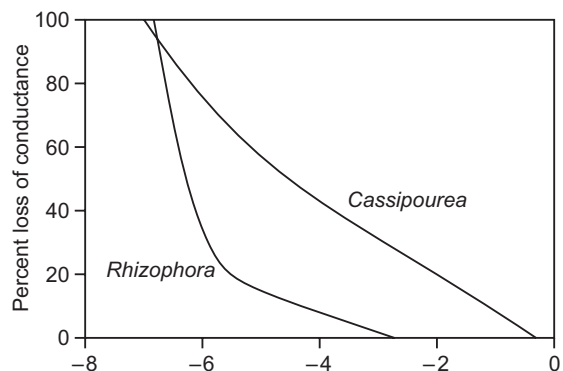


Fig. 3.11. Difference in vulnerability of two species of different habitats to embolism, measured as the loss of hydraulic conductance (vertical axis) in relation to water potential (horizontal axis). *Rhizophora mangle*, which is regularly exposed to below -4 MPa xylem water potentials in its natural habitat is less vulnerable to embolism than *Cassipourea elliptica*, which has generally above -1.5 MPa xylem water potentials in its natural habitat. However, both have a safety margin, i.e., they lose 100% conductance at more negative xylem water potential than the values they experience in their habitats (derived from Tyree and Sperry, 1989).

The surface area for transpiration is coupled with the amount of tissues through which water flows. It is indicated by the linear relationship between the canopy leaf area and sapwood (outer wood, light in colour and having living cells, in contrast to heartwood which is entirely dead) area of

stem. Species of dry environments generally support a markedly smaller leaf area per unit sapwood area (Table 3.3) than those from mesic environment. Since species of dry habitats have narrower vessels and tracheids they need a larger sapwood area to achieved a given water transport capacity.

Table 3.3. Ratio of total leaf area (sum of area of all leaves, A_l) to sapwood area (A_s) of conifer trees from mesic and xeric environments (Margolis et al., 1995; Lambers et al., 1998)

	Environment	
	Mesic	Xeric
Range of $A_l : A_s$ (m^2/m^2)	1600 – 7500	800 – 3000
Average	5000 ± 500	1800 ± 200
Genera to which species belong, species number in parenthesis	<i>Abies</i> (4), <i>Picea</i> (3), <i>Tsuga</i> (2), <i>Pseudotsuga</i> (1), <i>Larix</i> (1)	<i>Pinus</i> (6), <i>Juniperus</i> (2)

Water in dead cells of xylem of transpiring plant has a negative hydrostatic pressure (under a tension) due to interaction of water molecules with the capillaries in the cell walls of vessels. Capillary effect results from the attraction between water and hydrophilic surfaces at an air-water interface. Capillary effect causes large negative values of matric potential in soil and hydrostatic pressure in xylem of transpiring plants. According to the cohesion theory of Dixon (1914), water column in the stem is sucked upward to replace water lost in transpiration. This is made possible due to the cohesion among water molecules from hydrogen bonding.

Water flow in the stem of trees generally lags behind canopy transpiration by some time. The morning water supply for transpiration comes from sapwood rather than from soil. Thus, the sapwoods of stem, branches and twigs store water. In some conifers with large stature, sapwood water storage may be as large as 300 t ha^{-1} , equivalent to the amount transpired by a forest over a period of 5-10 days. In tropical dry forests stem water storage enables trees to flower and produce leaves during the dry period (Borchert, 1994).

Water Use Efficiency

Water use efficiency refers to the ratio between productivity or carbon gain in photosynthesis and water loss in transpiration. The former is called *water use efficiency of productivity* and the latter, *photosynthetic water use efficiency* (A/E) (Lambers et al., 1998), and both are positively correlated. In case of the latter, one can also consider the ratio of leaf conductance for CO_2 and water vapour (gc/gw), in place of ratio between rates of photosynthesis and transpiration. The A/E ranges between (mmol mol^{-1}) 4-20 for CAM, 4-12 for C_4 , 2-11 for woody C_3 , 2-5 for herbaceous C_3 , and 0.3-2.5 for hemiparasitic C_3 species (Lambers et al., 1998).

3.11 ADAPTATIONS TO DROUGHT

Mechanism of plant adaptation to drought can be divided into two types, viz., drought avoidance, which includes desert annuals and drought deciduous species which avoid drought by remaining dormant until water is available, and drought tolerance that include evergreen shrubs and other plants that can tolerate the state of dehydration in the absence water and “resurrect” upon water availability.

Drought avoidance

It results from either a reduction of water loss or maintenance of water uptake. In both cases, plants maintain high water potential in drought.

Leaf-drop during drought: Generally, the proportion of drought- deciduous species (shedding leaves during dry season) increases with decreasing rainfall. Leaf drop, by checking transpiration water loss, enables species to avoid severe water deficits. Certain North American cactus species are known to produce and lose leaves six times in a year in response to the period of rain and of the absence of water, respectively. Even evergreen broad-leaved species like Himalayan oak (*Quercus floribunda*) shed a fraction of their foliage during a drought period to restrict water loss. This is more pronounced in seedlings and saplings than in trees (Singh unpubl). In desert habitats deciduous plants (e.g., *Acacia* and *Prosopis*) with compound leaves are common. Such plants save transpirational losses by also shedding rachises bearing compound leaves. The drought-deciduous species have advantage of having high rates of photosynthesis and water conductance, but that are offset by cost of producing new leaves in each new growth period.

Desiccation avoidance by desert annuals: Many desert annual plants have little or no physiological tolerance to drought that is why they ensure that their life cycle is completed when soil moisture is enough. Seeds of these species have water – soluble germination inhibitors that are leached out only when rainfall is sufficiently large ensuring seed germination only on the onset of rainy period. These ephemeral plants generally have high rate of photosynthesis, stomatal conductance and growth rates, enabling them to quickly complete life cycle while desert soils still have moisture.

Phreatophytes: Another way to avoid desiccation in deserts is to produce extremely deep roots having access to water table, e.g., *Prosopis*, *Tamarix* known as *phreatophytes*. These plants also have little capacity to restrict water loss, and generally have high rates of photosynthesis and transpiration.

Drought Tolerance

Evergreen shrubs and trees particularly of a Mediterranean or monsoon-type climate with a highly seasonal rainfall pattern are exposed to long drought periods. These woody species with scleromorphic leaves have typically low relative water content. They can withstand comparatively low water potential before they close stomata or reach turgor loss point because of greater solute concentration that enables them to lower osmotic potential. Their vessels are more resistant to cavitation than those of drought-sensitive plants. Some Mediterranean shrubs are characterized by presence of vasicentric tracheids, which act as subsidiary conducting system for water transport in xylem, allowing continued conductance when vessels have cavitated.

Desiccation resistant species have generally lower leaf area per unit sapwood area, than desiccation-sensitive species. There are several plants which combine characters of both drought -tolerating and drought- avoiding plants.

Resurrection plants: These represent an extreme case of desiccation tolerance and include many mosses and ferns and some flowering plants including woody species, like *Myrothamnus flabellifolius*. These plants can almost fully restore their physiological activities even after their protoplasm has dried out, and that is why they are called resurrection plants are. Within a day, following a shower their seemingly dead tissues regain their turgor. The resurrection plants that lose chlorophyll and break down their chloroplast upon drying are called *poikilochlorophyllous*, and those that retain some or all of chlorophyll and chloroplast ultra structure are called *homoichlorophyllous*.

Leaf traits checking water loss: Leaves employ several mechanisms to avoid drought.

(i) **Leaf rolling:** Many grasses roll their leaves as a result of the loss of turgor in the bulliform cells which are large epidermal cells with thin and elastic walls (Fig. 3.12). These cells lose more volume than surrounding cells with decreasing relative water content, resulting in leaf wilting. Rolling of leaves may reduce area exposed to sun rays and transpiration by as much as about 70%.

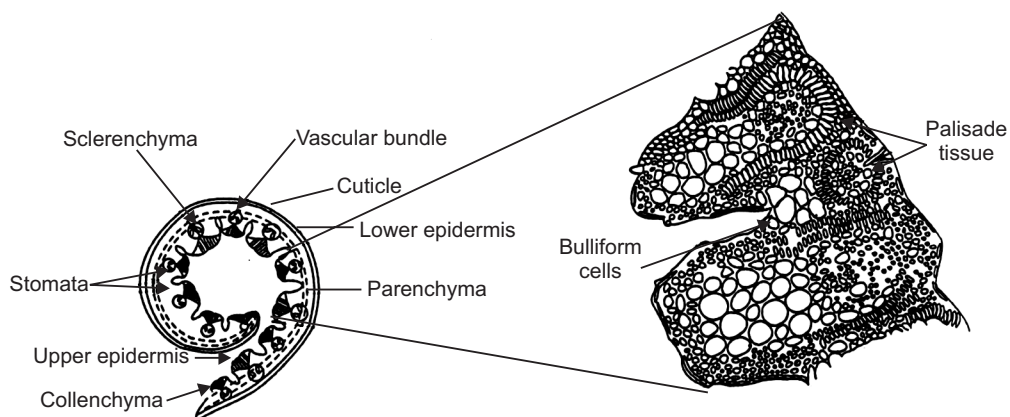


Fig. 3.12. Cross-section of *Ammophila arenaria* leaf; the blade is rolled inward hiding the sunken stomata; cuticle is thick and sclerenchymatous hypodermis is well developed. Location of bulliform cells is shown in the magnified portion of the leaf.

(ii) **Stomatal closure:** Stomatal closure enables species to avoid severe water deficits of tissues during dry seasons. Plants employ various mechanisms to close stomata during drought. Some plants such as barrel cactus, may close their stomata for 210 days, allowing the water potential to fall by only -0.5 MPa against the fall of -9.0 MPa in soil water potential. In some *Acacia* species of Australia, stomata do not close, but the cuticle become very thick and closes over the stomata.

(iii) **Leaf adjustment to sun rays:** Leaf movements (heliotropism) may reduce radiation load and leaf temperature. In Fabaceae (e.g., *Mimosa*, *Glycine max*) light stimulus giving rise to leaf movement is perceived in the pulvinus (at leaf joint), while in Malvaceae it is observed in leaf lamina. In common bean, air temperature is the stimulator of leaf movement.

In some evergreen species of California Chaparral, leaf angle changes seasonally to avoid water loss (Comstock and Mahall, 1985). When there is a water stress, leaves are vertically inclined (leaf surface being parallel to light), while during the periods of better water supply they are horizontally inclined (rays falling perpendicular to leaf surface).

A similar adjustment to light is realised diurnally by some herbaceous species of semi-arid and arid regions (Ehleringer and Forseth, 1980). Diurnal movement of leaves and resultant orientations not only enables leaves to reduce transpiration but also minimizes the likelihood of photo-inhibitory damage to leaves (Ludlow and Bjorkman, 1984).

(iv) **Decrease in leaf absorptivity:** Decrease in leaf absorptivity is made possible by the presence of structures such as hairs, salt glands and waxes on leaf surface. Such structures can substantially lower leaf temperatures and transpirational water losses.

(v) **Water storage in leaves:** Leaves of succulent plants store water, often in specialized tissues, such as *hydrenchyma* (e.g., *Pepromia* an epiphyte) or multiple epidermis and tissues lying just under epidermis, which may account for upto 60 % of leaf volume. *Hydrenchyma* is suggested to function as a reservoir for water loss through transpiration, allowing *chlorenchyma* to continue to remain photosynthetically active (Lambers et. al., 1998).

Table 3.4. A list of characters that distinguish desiccation-tolerant evergreen species from desiccation-sensitive evergreen species. All descriptions about desiccation-tolerant species are in comparison to desiccation-sensitive species (Lambers et al., 1998).

Characters	Desiccation-tolerant species (in comparison to sensitive species)
Maximum rate of transpiration	Lower
Photosynthesis Rate	Lower
Conductance	Lower
Specific leaf area	Lower
Leaf size	Smaller
Leaf life span	Longer
Leaf area per unit sap wood area	Lower
Potential growth rate	Lower
Allocation to root	Higher
Solute (compatible) concentration in leaf	Higher
Water potential at turgor loss point	Higher
Stomatal regulation	Lower
Safety margin for cavitation	Anisohydric*
	Larger

* Desiccation sensitive species have both iso- and anisohydric stomata regulation.

To conclude, the characteristics that enable plants to tolerate desiccation are interdependent (Table 3.4). The characters include, e.g., small leaf size, long leaf life span, high root mass ratio and low specific leaf mass. They need to be understood at biophysical, physiological and molecular levels. Of several groups of plants, resurrection plants offer most remarkable examples of how plants cope with extreme water scarcity. They are peculiar in their extraordinary ability to cope with water shortage. They can be used as source of genes to be used to improve crop performance in dry habitats.

3.12 XEROPHYTES AND HYDROPHYTES

With regard to water relationships, plants are generally classified as xerophytes, hydrophytes and mesophytes, referring to plants of dry, wet and moist habitats, respectively.

Xerophytes

Several morphological adaptations have been associated with xerophytes in response to water deficit which is the characteristic feature of their habitats. The plants that are structurally adapted to withstand drought are xerophytes.

Generally, xeromorphic leaves are small and compact, with the high ratio of volume to surface. They have thick blades, small sunken and dense stomata on lower surface, dense covering of hair (pubescence), a thick cuticle, heavily lignified epidermal cells, strongly developed palisade mesophyll and less inter-cellular spaces. A cross section of *Nerium* leaf (Figs. 3) shows several of these xeromorphic characters. The strongly developed sclerechyma is likely to reduce the injurious effect of wilting (Stalfelt, 1956). Occurrence of stomata in cavities (sunken stomata), crypts (e.g., *Nerium* leaf) or in grooves lined with epidermal hair is quite common. Some succulent xerophytes contain water-storage tissues and sparse stomata.

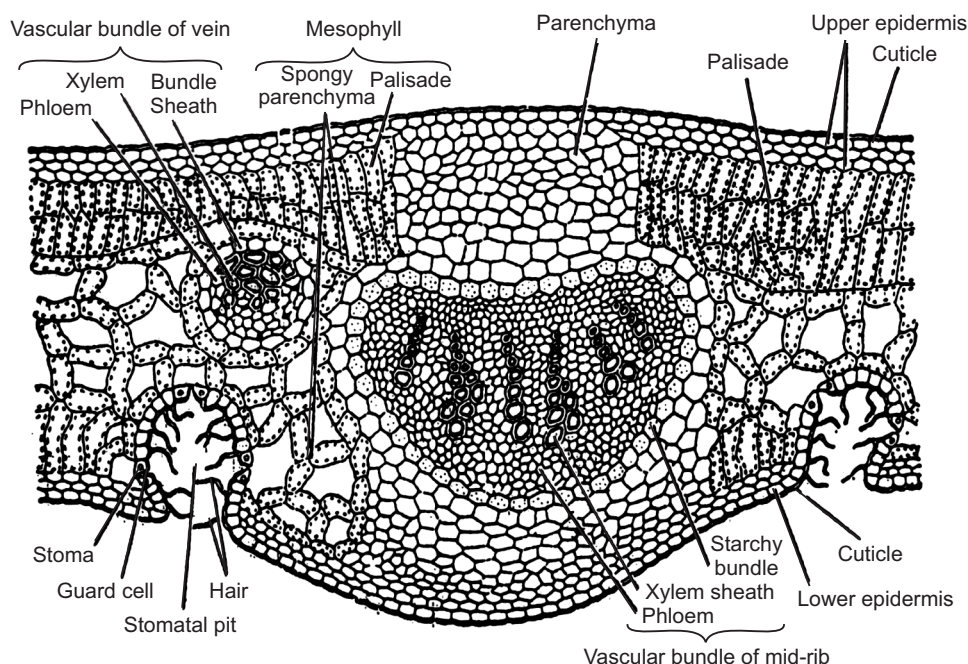


Fig. 3.13. A cross-section of *Nerium* leaf showing several characters such as stomata in pits and covered with hair, thick cuticle and well developed palisade tissue.

Some of the xeromorphic characters might be an adaptation to cold, strong light and nutrient deficiency, apart from moisture deficiency. Such conditions are often experienced in deserts and other dry areas. In many types of dryland shrubs, leaves are replaced by thorns, which almost have no transpiration.

Plant adaptations vary from species to species growing at same site under same level of dry conditions, e.g., creosote bush (*Larrea tridentata*), the most common shrub in the warm North American deserts, has mesomorphic leaf characteristics - no pubescence, stomata distributed equally on both surfaces (Barbour et al., 1974).

Leaf characters may also depend on adaptations shown in other plant parts. For example, the rocky heaths of the Mediterranean shores abound in low shrubs with shallow roots, therefore cannot utilize deep soil water. These plants possess large leaves during rainy season and replace them with smaller leaves in dry summer (Oppenheimer, 1960). As described earlier, some xerophytes roll their leaves to restrict water loss.

Hair have been considered to decrease transpiration, but there are also evidences to indicate that transpiration was reduced with the removal of hair. It is possible that hair of xerophytic plants serve other purposes, such as prevention of insect attack and cooling of leaf by increasing total radiating surface.

Apart from adaptations reflected in leaves, plants show adaptations in terms of water storage in roots (e.g., *Asparagus*) and stems (succulents), deep penetration of roots so that deep water-soil is made available. For example, *Robinia pseudoacacia* is known to have 8-9 m deep roots, whereas roots of other species on that site penetrate less than 2 m depth. The succulent plants like *Opuntia* develop fleshy organs, accumulating large amounts of water during the short wet season of the year. Succulents have generally shallow roots (e.g., *Opuntia*, *Euphorbia*, *Agave* and cacti) as they can store water in aerial parts.

Hydrophytes

Hydrophytes (hydric = wet; phytes = plants) are the plants growing in lakes, ponds, rivers, oceans, swamps, bogs, etc., with at least some plant parts in water for substantially long periods. Hydrophytes can be divided into following four categories, primarily from the point of view of the proportion of plant parts submerged under water (Fig. 3.13) as follows:

- (i) **Emergent hydrophytes** are usually rooted in water with leaves and stems fully or partly exposed to air. In a way they represent a transitional stage between aquatic and terrestrial plants.
- (ii) **Free-floating** are the plants floating freely on the water surface having no contact with soil or mud of water body; upper surface of leaves are usually fully exposed to sun and air (e.g., *Eichhornia*, *Lemna*, *Pistia*).
- (iii) **Rooted, with floating leaves** are as (i) and (ii) except that roots are fixed in soil or mud (e.g., *Nelumbo*, *Nymphaea*, *Marsilea*).
- (iv) **Submerged** plants generally growing below the water surface, with roots (e.g., *Myriophyllum*, *Vallisneria*), or without roots, floating in water (e.g., *Ceratophyllum*).

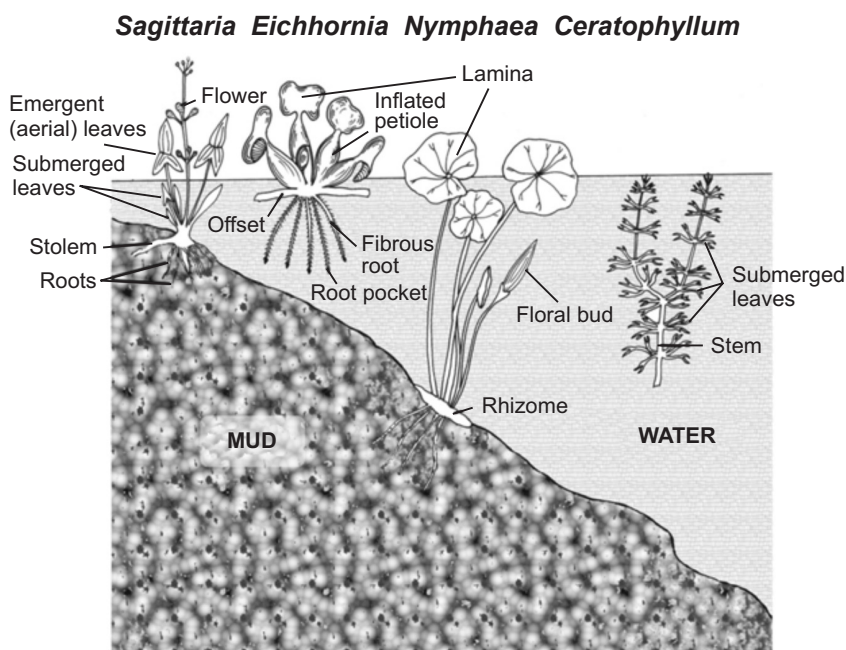


Fig. 3.14. Examples of hydrophytes, *Sagittaria* (emergent), *Eichhornia* (free-flowering), *Nymphaea* (rooted floating leaved) and *Ceratophyllum* (submerged).

Plants living in water have not to develop a large root system as water availability is not a problem. Therefore, hydrophytes do not require too many stomata or a thick cuticle to regulate the water loss. Then, mechanical tissues are not required to keep plant body erect. Adaptation features which are generally found in hydrophytes to deal with the aquatic environment are listed below:

- (1) Sponginess of the tissues is a characteristic shared by most hydrophytes. This is because of the presence of large air chambers or cavities (lacunae), resulting from disruption of group of cells or enlargement of intercellular spaces (Fig. 3.14). Parenchyma containing cavities are called *lacunar tissues*, and periderm containing intercellular spaces is called *aerenchyma*. Some species like, *Eichhornia* develop inflated petioles (Fig. 3.14), or other such structures

having air-filled tissues which keep plants floating on the water surface. The thin and spongy leaves also maximize the gaseous exchanges. In submerged plants, air cavities store the O_2 released in photosynthesis during the day. This is used up in respiration in night when CO_2 accumulates. Oxygen in water is only one-twentieth or less than that in atmosphere. In emergent and rooted, floating leaved hydrophytes, cavities generally form a continuous system of air passage ways by means of which submerged organs (e.g., corn, lower part of leaves) can exchange gases with air via stomata in the emerged organs (e.g., leaves).

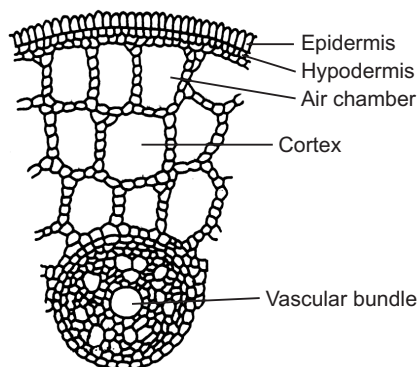


Fig. 3.15. Transverse section of a submerged hydrophyte, showing the presence of large air chambers and reduced vascular tissues.

- (2) Roots may be entirely absent (e.g., free-floating, *Wolffia*, submerged, *Ceratophyllum*) or they are poorly developed, without root caps, root hairs and branches, or with few of them.
- (3) Cuticle in leaves is either absent, as in submerged plants like *Potamogeton*, or is poorly developed. Similarly, stomata are absent or are non-functional in some submerged plants, they are present only on the upper leaf surface in floating plants and mainly in emergents their presence can be seen on both leaf surfaces. To cope with low light availability, leaves may be finely dissected and chloroplasts may be confined to outer cell layers.
- (4) Tissue differentiation is poor, including those of vascular bundles, and mechanical tissues, such as sclerenchyma are generally altogether absent. To quote some examples, epidermis of stem is usually single-layered (Fig. 3.14), mesophyll in leaves is undifferentiated and may be single layered in some submerged species (in emergents such a differentiation may occur); vascular tissues are markedly reduced, they are thin-walled, often with no clear-cut differentiation of xylem and phloem. Lack of lignification of cell wall is common.
- (5) Leaves in submerged species are thin, linear and finely dissected (Fig. 3.13). The finely dissected leaf was thought to be a useful adaptation to high speed currents which may tear the entire leaves into pieces. However, more careful examinations suggest that the finely dissected leaves in submerged plants are an adaptation to limitation of CO_2 in the water (Givnish, 1987). The CO_2 diffusion in water is several hundred times lower than in atmosphere. The smaller the leaf the thinner the boundary layer (the still layer along the leaf), and hence more the conductance. Thus, finely dissected leaves in submerged plants are selected so that leaf resistance to gaseous exchange in the water environment is minimal. Several of them have the capability to use bicarbonate ions, can absorb CO_2 from sediments by roots and can fix carbon during night.
- (6) Most of the hydrophytes are characterised by a rapid vegetative growth and have the potential to reproduce by sexual and asexual means. They can simultaneously reproduce by rhizomes, tubers and turions as well as seeds produced from sexual reproduction. This gives opportunity for creation of new genotypes. A typical example is *Potamogeton pectinatus*.

Hygrophytes

These moisture loving plants need frequent soaking with water, as in the case of mosses and ferns. Hygrophytic trees like *Cypress* have wide flanking tree trunks near the ground to provide a firm footing in the wet environment. Many hygrophytic plants which can grow in standing and moving water have generally weak stems to withstand the ebb and flow of water.

CHAPTER

4

Soil and Belowground Interactions

Soil can be defined as a medium of plant growth representing a mixture of minerals and organic matter, capable of supplying all the essential nutrients and water. The word soil comes from Latin word *solum*, meaning floor (groundfloor). The soil upon the planet earth is not uniform. It is tens of metres deep in the Gangetic plains of India, but only a few centimetres deep on many mountain slopes. There is a foundation called bedrock, above which is the regolith consisting of unconsolidated, weathered (i.e., broken down or disintegrated) rock material and finally soil, the highly weathered component. It is the uppermost layer of the earth's crust consisting of loose products of the disintegration of rocks mixed with organic matter. It is the ground where living and non-living components of the world meet.

Most soils are products of the interaction of a living community and geological substrate, under the influence of climate, and control ecosystem processes based on their physical, chemical and biological properties. The soil is not only a complex biochemical material on which plants grow, but it represents a dynamic ecological system characterised by the presence of diverse and interacting populations of micro organisms (Singer and Munns, 1999). The soil biota is composed of a diverse community of soil microbes and invertebrates which have a major influence on nutrient cycles and biological transformations.

Soil is a vital component in the functioning of terrestrial ecosystems, provides a habitat for diverse and interacting populations of soil organisms, accounts for decomposition processes, and is a critical link in carbon sequestration for mitigating climate change. The soils in terrestrial ecosystems support millions of genetically distinct prokaryote organisms (bacteria, archaea), and eukaryotic species across many taxonomic groups. Soil microbial communities occur at a broad range of scales, from soil macroaggregates to plant rhizosphere, to field plot and to the ecosystem (Tiedje, 1995). The functioning of terrestrial ecosystem depends on soil biodiversity as many of plant interactions take place below-ground. The detritus food webs are largely based in the soil systems as the organic matter available to the microbes constitutes the base of food webs. The detritus food webs are linked to energy flow, nutrient cycling and productivity, particularly, in terrestrial ecosystems.

4.1 CONTRIBUTION OF SOIL TO ECOSYSTEM SERVICES

Soils and soil biodiversity form the basis of terrestrial production systems and contribute to ecosystem services, such as control of greenhouse gases, pests and invasive species, and retention of nutrients. Healthy soil provides a wide range of ecosystem goods and services that play a crucial role in sustaining biological diversity of planet earth. The soil functions and processes that benefit human society are the bases of these ecosystem services. The soil functions such as decomposition of organic materials, soil nutrient cycling, and detoxification of soil contaminants, plant productivity and regulation of plant-soil water relationships that benefit humankind are some of the important soil

ecosystem services (UNESCO and SCOPE 2007, Palm et al., 2007). The maintenance of fertility by soil biota is one of the most vital ecological services.

Soils deliver all four kinds of ecosystem services recognised by MEA (2005a), viz., provisioning, regulating, cultural and supporting; these services are regulated by the physical, chemical and biological properties of the soil (Palm et al., 2007). There has been a large increase in provisioning services of the soils due to large increases in the production of food crops, livestock, timber, and fuel wood (MEA, 2005a). However, the large increases in production from the land systems have caused degradation of soils as well as impacted the regulatory and supporting services of soils (MEA, 2005a). The ability of soils to deliver the ecosystem services depends directly on soil's regulatory services such as (i) filtering and detoxifying water, (ii) soil biodiversity, (iii) decomposition of organic materials, (iv) regulation of fluxes of greenhouse gases to and from the atmosphere, (v) plant-soil nutrient cycles. The relationship between provisioning services, soil processes and soil properties are shown in Table 4.1. The processes mediated by the soil biota such as, waste recycling, soil formation, nitrogen fixation, bioremediation of chemicals, biotechnology, biocontrol of pests, and pollination by organisms having edaphic phase in their life cycle, contribute to the ecosystem services.

Table 4.1. Relationship between provisioning ecosystem services, soil processes and soil properties (adapted from Palm et al., 2007)

Provisioning services	Soil processes	Soil property
Physical support for plants	Soil formation	Depth; *State factors of soil formation, clay mineralogy
Provision of nutrients	Mineral weathering	Primary minerals
Provision of nutrients	Litter decomposition	Soil biota and soil Texture
Provision of nutrients	Soil organic Matter mineralization	Soil organic matter quantity and quality ; Soil Texture
Provision of water	Infiltration	Macroporosity-aggregation, texture, soil organic matter, soil biota
Provision of water	Storage of water in soil	Aggregation, bulk density, depth, Texture, mineralogy, soil organic matter

* State Factors are the major factors that influence soil formation including climate, living organisms, topography or relief, rock type or parent material and time. The combined influence of these soil-forming factors determines the properties of a soil and their degree of expression.

4.2 SOIL COMPOSITION AND STRUCTURE

Soils are made up of four components; the two solid substances in soils are small mineral particles and organic matter. The small mineral particles come from weathering of rocks. The organic matter is of plant and animal origin, getting added into the soil both from dead and living organisms. In an average soil, about 45% of the volume is mineral particles, and about 5% is organic matter. The other 50% is soil air and soil water, the amount of air or water varies greatly depending on how wet or dry the soil is at the time. Soil air has a higher concentration of carbon dioxide and lower concentration of oxygen than the atmosphere because of the respiration by soil organisms and plant roots. Quartz (SiO_2) is the most common mineral found in the soil with the micas and feldspars varying with the degree of weathering. Most of the soil organic matter is made up of remains of

organic residues in various stages of decomposition and a highly decomposed fraction (humus). A very small amount of the soil organic matter consists of microorganisms including bacteria and fungi, collectively known as soil microbial biomass. The soil microbial biomass is the living part of soil organic matter comprising all soil organisms with a volume $< 5 \times 10^3 \mu\text{m}^3$. Living plant tissue, such as roots, is not included. The soil microbes play an important role in nutrient cycling, decomposition of plant residues and soil nutrient conservation. For example, some microbes can take up nitrogen gas from the atmosphere and convert it biologically into available forms that the plants can use.

The mode of arrangement of the soil particles is termed soil structure. Soil structure is a property of soils that has a major influence on plants and other living organisms in the soil. The soil particles are bound to form compound particles, the binding of soil particles or aggregation results in the formation of structural subunits in the soil body. Aggregates formed by natural processes are known as peds. Aggregates are secondary units composed of many soil particles, organic substances, iron oxides, carbohydrates, clays and silica. The soil structure depends on the shape of peds and can be described with terms such as granular, crumb, platy, blocky, sub-angular blocky, prismatic and columnar (Fig. 4.1). The structure less soil is termed single grain (e.g., dune sand or massive large uniform mass of cohesive soil).

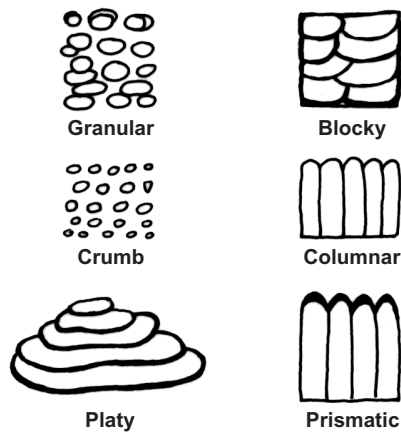


Fig. 4.1. Soil structural types and their location in the soil profile

Granular - spherical peds found in surface mineral horizon;

Platy - Horizontal plate like structures found in E horizon;

Angular Blocky - In B horizon of humid regions;

Subangular Blocky - In B horizon of of humic region soils

Prismatic - Columnar peds in B horizon of arid and semiarid soils;

Columnar - Column like peds in B Horizon of arid and semiarid soils.

4.3 SOIL ORIGIN AND DEVELOPMENT

Soil is derived from rocks through a number of sequential or simultaneous steps. Soil- forming rocks are of three major types:

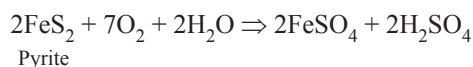
1. Primary rocks, formed due to solidification of molten magma and are referred to as igneous, e.g., granite, basalt.
2. Sedimentary rocks, the mechanical and chemical break down and redistribution of the primary rocks give rise to sedimentary rocks, e.g., sandstone, shale, limestone.
3. Metamorphic rocks, formed due to the influence of heat and /or pressure on the above types of rocks, e.g., slate, marble, Gneiss.

The predominant soil-forming minerals in these rocks are alumino-silicate minerals (Feldspars, micas) which yield fine-grained mineral matter, and silicas (or quartz) which persists in the soil as sand and gravel.

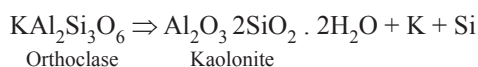
Soil formation occurs in two phases: (1) weathering of rocks and (2) conversion of raw material into a new body through the action of biosphere.

Chemical processes which decompose the complex substances of the rocks and minerals into

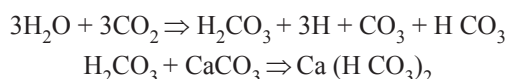
simpler ones, constitute chemical weathering. These processes are oxidation, hydrolysis, carbonation and soluviation. Oxidation involves taking up of oxygen by a mineral compound resulting in a product which is usually soluble in water. For example, atmospheric oxidation of pyrites in presence of water vapour:



Some mineral compounds combine with water and form hydrates, increasing their bulk and becoming soft, such that they disintegrate at a touch. For example, concentration of sandstone is disrupted when hematite (Fe_2O_3) is hydrated to limonite ($\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$). Hydrolysis represents the reaction of dissociated H^+ and OH^- ions of water with the mineral ions. For example, silicate is broken down to release K and Si :



Carbonation refers to the combination of carbonate (CO_3^{2-}) or bicarbonate (HCO_3^-) ions with a mineral. For example, CO_2 in rainwater forms carbonic acid which converts insoluble calcium carbonate in soluble calcium bicarbonate:



Soluviation refers to the solvent action of water which removes the mobile components of minerals, and products of chemical weathering into soil solution. Chemical weathering is thus strongly influenced by the atmosphere, hydrosphere and biosphere.

Weathering is the process of disintegration and decomposition of rocks and minerals, and occurs as a result of combinations of physical, chemical and biological effects. Changes in temperature and expansion of ice in the cracks of rocks that break them are examples of physical factors involved in weathering. Change of minerals to different forms by exposure to water, oxygen and weak acids and other compounds represent the effects of chemical processes. Splitting of cracked rocks by plant roots is a biological cause of rock weathering.

The rocks expand due to heat and contract due to cold. Since rocks are mixtures of minerals, all parts do not expand or cool at the same rate, causing the rocks to disintegrate under the influence of changes in temperature. Rock is also a poor conductor of heat, and therefore outer and inner layers respond differently to temperature and this difference causes cracking and spreading of rocks. The fissures and crevices thus formed (also by gravity tension and tectonic activity) are filled with water. In cold regions the water freezes and expands, causing enormous hydrostatic pressure (more than $10,000 \text{ kg cm}^{-2}$). This force widens the cracks and causes break down of rocks; the process is called frost shatter. If the crevice is formed alround, it leads to exfoliation, the peeling-off of rock layers in a fashion similar to that of the scales around onion bulbs. Rock weathering is also promoted by erosive processes of water and wind. Glaciers (the moving ice) transport enormous amounts of material and disintegrate substrates to smaller pieces.

Plant roots exert an impressive amount of force when they have entered fissures. Plants also exude chemicals which erode or disintegrate rocks.

During chemical weathering of the mineral particles, stronger bases are removed into solution and are washed out of the surface layers through leaching. From the soil solution some ions are leached out (e.g., Cl, SO_4 , Ca, Na, K, etc.), some are precipitated (e.g., Ca, Fe,) while some are bound to clay particles through ion exchange (e.g. Na, K, Ca, etc.). With the development of vegetation, their death and decay, organic matter is added into the soil which helps the particles in aggregation. Dispersion, migration and aggregation ultimately result in profile development (Fig. 4.2). Soil genesis

(or pedogenesis) is the process in which soil develops from the parent material through the process of weathering of the parent material, transformation of primary and secondary soil minerals, and addition of organic matter and formation of soil horizons.

Mode of Origin

Depending upon the mode of origin, the soil and its mineral matrix can be categorized as follows:

Residual

It results from the disintegration of rock in place. The surface layer becomes most weathered because it is most exposed to atmosphere. From surface to deeper soil levels, the weathering effect decreases. Accordingly, the size of material particles increases with increasing depth.

Transported

The materials are transported from the place of origin. They can be transported by the force of (i) gravity (colluvial), (ii) running water (alluvial), (iii) glaciers (glacial), and (iv) wind (eolian).

Colluvial: pull of gravity dislodges fragments from steep rocky slopes or cliffs from time to time which may accumulate below as talus. The material which forms talus is characteristically coarse consisting of large pieces of rocks. Since water percolates rapidly, talus fails to retain water and allows only plants with deep tap roots to establish.

In mountains a slow creeping of the wet soil with stones down an incline is a common feature during rains. Mudflows (river-like movement of saturated layers of materials particularly when rock surface is smooth) and landslides (sudden mass removals) are other common factors of transportation of material by gravity. There is no stratification in a colluvial deposit. Particles of all sizes including many sizes of rock fragments and soils are mixed. Colluvial deposits are confined to mountainous parts, but can occupy large areas.

Alluvial: This refers to the material deposited by running water. The so called “Bhabar” belt along the foothills of central Himalaya is the talus, built up partly as a result of rock fall from the hill slopes. However, the contribution of rivers to the formation of talus is considerable, for they have transported much of the coarse particles from the mountain ranges. Most of the material deposited by rivers tend to be rounded and smoothed by the action of running water, in contrast to rough and uneven surfaces of rock fragments that result from gravity pull. Formation of distinct strata each containing materials of a particular size class, depending upon the speed of water that transported them, is another distinguishing feature of the alluvial deposits

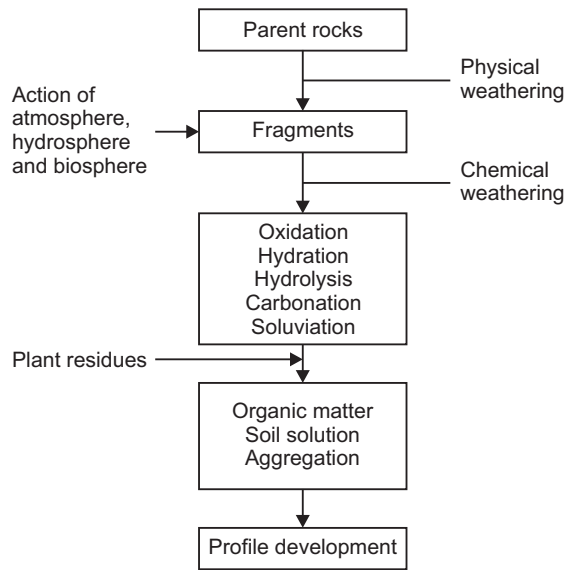


Fig. 4.2. Summary of soil-forming processes.

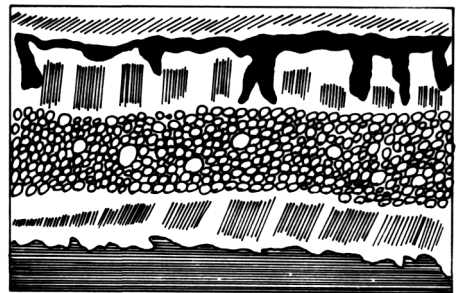


Fig. 4.3. Alluvial deposits. Note the sorting effect on running water on different texture of the alluvium. The stones have round and smooth edges.

(Fig. 4.3). The swift water deposits larger particles and the slowly flowing water lays down finer particles. The India's fertile Gangetic plain is the product of deposition of fine particles by calm water, as the swift flowing mountain rivers slow down in their flow.

Glacial: In the glacial deposits, called till, particles of all sizes mix up thoroughly. The edges of rock fragments are sharp, differing markedly from the rounded edges of the rocks in the alluvial deposits (Fig. 4.4). Within one glacial deposit there is no stratification; but different strata are formed when several deposits are laid one after another.

The glacial deposits are generally flat plains with many depressions, lacking surface drainage. In much of Canada soil development has occurred on the parent material deposited by glaciers. In India, glacial deposits of limited sizes occur in the high Himalayan ranges.

Eolian: Eolian or wind transported materials can be seen as dune or loess. Their formation, however, involves action of other forces along with that of wind. The particles are relatively finer. Dunes may occur (i) along the shores of oceans and lakes as a result of water currents; (ii) may form along river valleys where water deposits sand upon floodplains and sand is then dried out and moved by wind; and (iii) weathering of sandstone and other rocks may produce sand in dry areas. Compared to other parent materials of soil, the dunes have more uniform particles and chemical characters.

In addition to above, grey silty sediments that accumulate on bottom of lakes (Lacustrine materials) and marine materials transported by ocean currents are also the parent materials of soil formation.

4.4 SOIL DEVELOPMENT PROCESSES

On the basis of physical, chemical and biological processes prevailing in different regions, there are five major types of soil forming processes: podzolization, gleization, laterization, calcification and salinization, resulting in the formation of different kinds of soils, which have been described as follows:

Podzolization

In humid and cold climates plant decomposition is very slow and releases organic acids. Consequently, on soil surface large amounts of litter mass are accumulated (3 to 7 kg m⁻² in spruce taiga forests as an example). The acidic plant material is called *mor* litter layer and is sharply separated from the underlying soil. The soil pH may be as low as 3.5, fungi predominate and earthworms are absent. The release of organic acids from the gradually decomposing litter contributes to leaching of materials downward into the soil. Beneath the *mor*, the upper leached layer of soil is grey or white like ash. Because of this bleached layer, the soil has been given the name podzol (or podsol), a Russian word for "like ash". Beneath the zone of leaching, a layer of deposits is formed where iron-humus complexes are transported by water (Fig. 4.5). The iron-humus complexes may cement to form a hard iron pan, difficult for plant roots to penetrate. Though traditionally associated with taiga, podzol soils occur in a wide range of climates and communities, and not all taiga communities have podzol.

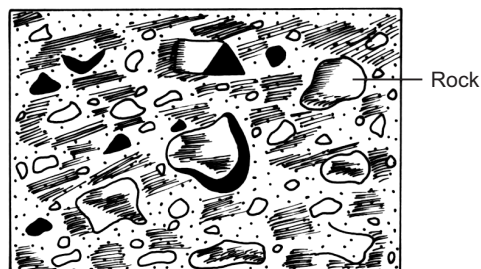


Fig. 4.4. Vertical section through a deposit of glacial till. Particles of all sizes from clay to large rocks are thoroughly mixed. The sharp edges of rocks differ markedly from the round edges of rocks of alluvial deposits.

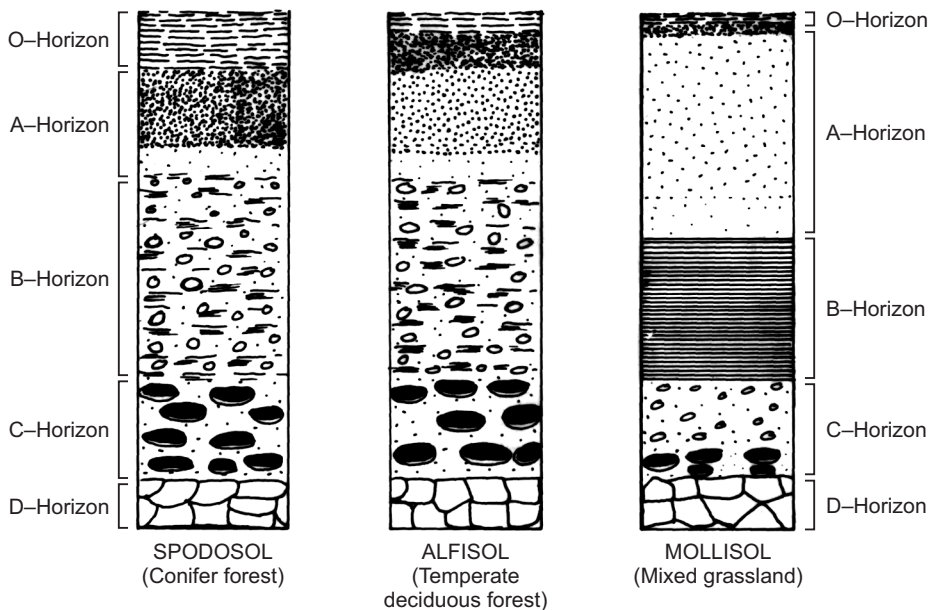


Fig. 4.5. Profiles for three great soil groups. Podzol is for a boreal spruce-fir forest; brown earth is for a temperate deciduous forest; chernozem is for mixed grassland. (derived from Whittaker, 1975).

Podzols are generally infertile soils, the leached layer being poor in plant- available nutrients. The spruce forest on podzol has a large part of nutrients in biomass and litter, and little in soil. Nutrient content in plant tissues is, however, low (Table 4.2), as nutrient availability is poor. These soils are prone to accelerated erosion if the vegetation cover is disturbed by human activities or by fire.

Table 4.2. Content of nutrients in plant tissues of communities developed on different soils (data from Whittaker, 1975)

Forest	Soil	Nutrient content in dry Biomass of above ground total vegetation	
		Ash content (%)	Nitrogen (%)
Taiga spruce forest	Taiga podzol	0.5-0.9	0.2-0.4
Deciduous forest	Forest brown earth	1.0-1.3	0.3-0.5
Tropical rain forest	Rain forest latosol (red soil)	1.58	0.57
Temperate grassland	Chernozem (black earth)	2.5-3.5	1.0
Desert	Sierozem (grey desert soil)	4-7	0.3-0.5
Salt desert	Solonchak	10-14	Low
Bog	Bog soil, peat	–	Low

Mildly podzolized Forest Brown Earth develops under deciduous forests of temperate regions. In this case surface litter layer is thin ($1-2\text{ kg m}^{-2}$) because of rapid decomposition. Climates are relatively warm, and nutrient-rich thin leaves of deciduous forests are less resistant to decomposition. The surface litter layer grades continuously from freshly fallen leaves, through

fragments to particles and humus colloids mixed with other soil materials. Such a litter layer, with thorough mixing of the organic matter downward into the soil is called mull (Fig. 4.5). Present in abundance, earthworms play significant role in soil development.

Though upper soil layers are rich in humus, they are poor in calcium and some other nutrients because of the downward transport due to the leaching. However, the effect of leaching is far less than in podzols. The soils are generally mildly acidic and rich in silica. The plant tissues have higher nutrient concentration, and the total nutrient stock of mature deciduous forest may be twice the spruce forest's nutrient stock, though both may have similar biomass values. The turnover of nutrient in the deciduous forest is rapid, and the larger part of ecosystem nutrient is in the soil and smaller fraction in biomass.

Laterization

The abundant rainfall and high temperatures lead to the formation of rain forest latosol or tropical red forest soil. In the latosol the situation is just opposite to that of podzol. The litter decomposes very rapidly and leaching is primarily by inorganic acids (the weak carbonic acid formed by CO_2 and dilute nitric acid formed by NO_2 in water). Consequently, silica is transported downward, while the oxides of iron and aluminium remain along with clay in upper soil layers. This forms a red leached layer several metres deep, and restricts the availability of nutrients released from the parent materials to plant community. The leached layer is mildly acidic, nutrient-poor and extremely deficient in exchangeable bases.

The soil materials dominated by oxides of iron and aluminium, which harden into a rock like substance when exposed to air, are called "laterite". Such soils are referred to as lateritic soils.

Though having a greater quantity of litterfall than temperate forests, in tropical rain forest litter forms a thin layer on soil (0.1 to 0.6 kg m^{-2}) because of rapid decomposition. The turnover of nutrients is markedly rapid and enables the plants to maintain high nutrient levels in tissues even on nutrient-poor soils. When forest is cleared for agriculture or other human activities, most of the base nutrients are rapidly lost. As a result, the soils become poor in nutrients, leaving crusts of iron and aluminium compounds. The latosols are not good for agriculture.

Calcification

Calcification is the dominant process in arid and semiarid climates, as exemplified by dry prairies of North America, the steppes of Eurasia, savannas and steppes of the subtropics.

Since in dry condition, evapotranspiration exceeds precipitation (in contrast to forests, described above, where precipitation exceeds evapotranspiration), the rainwater is drawn back toward surface subsequent to movement into soil by capillary action and root transport. The rainfall is insufficient to leach much of the calcium from the soil profile and therefore these soils have fairly high percentage base saturation. The soils are known as Chernozem or black earth soils. In these conditions the humic acids are neutralized and clay minerals are stable. There is a zone of accumulation of CaCO_3 at a depth corresponding to the average penetration of rainwater.

Presence of numerous roots and animals in soil promotes the mixing up of organic matter, which occurs in large amounts (8-10% of soil mass). Though deeper soils differ from the surface soils in having low organic contents, contrary to forest soils, distinct zones of leaching and accumulation are not observed in black earths.

The high organic content and calcium carbonate content give a loamy and granular structure to the chernozem. All these make a favourable condition for agriculture. The soil is fertile because it is not leached out, but the water balance is such that soil is moist enough for crop cultivation. The soils have abundance of humus giving a dark colour to the soil and it can retain both nutrients

and soil moisture. In prairie and steppe, the grasslands that the chernozem supports, aboveground biomass is small (far smaller than in forests). In them litter mass may be larger and may store more nutrients than the aboveground biomass. The plant tissue is more nutrient-rich than in the forests (Table 4.2).

Sierozem is the grey desert soil. Here the plant cover is sparse and litter cover is negligible. Thus the modifying effect of plants on soil is much less, organic content in soil being less than 1%. The weathering of parent material is very slow in the dry climate. Because of dry condition, soil water moves upward and evaporation concentrates many of the soluble materials near the soil surface. Thus this process is reverse to that of leaching observed in humid areas under forest covers. The soil is basic in reaction. The calcium carbonate, which appears as nodules in the chernozem forms a continuous rock like-layer (caliche) in sierozem.

Because woody plants (shrubs) dominate the desert vegetation, fractions of plants that turnover through litter is smaller than in grasslands but larger than in forests. Nutrient content of biomass and litter is high, the content of sodium and chlorine being especially high in some species. The desert is an ecosystem of low productivity and biomass, provided with a superabundance of some nutrients (Whittaker, 1975). Only a small part of nutrients is circulated even though tissues may be highly concentrated with them.

Salinization (Solonchak, the salt desert soil)

In arid and semiarid areas, there are areas of inadequate drainage. The evaporative water leaves behind various salts in or on the surface of soil. The process of salt concentration in soil surface just described is so intense in desert basins (adjacent to mountains) that landscape can be mainly occupied by a salt lake or dry plain or salt flat. The concentration of salt at surface may produce a whitish or grey crust on the soil. Plants are absent or they are salt-adapted shrubs, including succulents with very high mineral ratios (10-14%). Solonchaks, because their development depends on local drainage, can also occur locally in semiarid desert, grassland climates from tropics to arctic. The salts are mostly the chlorides and sulphates of calcium and sodium, which are toxic to many plants and soil organisms. Therefore, only salt tolerant grasses and shrubs grow under these conditions.

Gleization

This process commonly occurs in water logged areas of the cool climates and results in the formation of peat and bog soil. Bogs occur in extremely cold and humid climates (tundra and taiga belts). They have deep deposits of organic soils having compacted and relatively undecomposed plant remains. Peat represents an enlarged form of mor layer of a podzol. Roots of plant have no contact with the weathered material because of the thick layer of organic matter. The soil is acidic and poor in nutrients. In response, plant's nutrient content is also very low. Only water-tolerant vegetation can grow in gley soils.

Factors of soil formation

Pedogenesis is the process of soil formation, that generally leads to the formation of soil horizons. Soil formation is dependent on climate (cl), organisms (o), topography or relief (r), rock type or parent material (p) and time (t), which can be expressed according to Jenny (1941) as:

$$S = f(\text{cl, o, r, p, t} \dots)$$

The equation has been kept open for further recognition of factors.

In this equation, *f* is a constant factor; the organisms include all organisms, particularly vegetation and soil organisms. Both the amount of litter (which depends on primary productivity) and its quality (how rapidly or slowly litter decomposes) affect the soil formation. Needless to

say, climate influences both the productivity and concomitantly litter input, and the rate of litter decomposition. Soil organisms and litter interactions influence soil porosity; soil rich in organic matter and organisms like earthworm are porous. Climate (temperature and precipitation) and vegetation are the most important factors. Relationship between the type of vegetation and climate on soil development is quite close. For example, podzol soils (soils with a conspicuous ash grey upper horizon) develop beneath coniferous forests in cool, moist climates, while chernozem soils ("black earth") develop under temperate grasslands.

Soils may owe their chemical characters mainly to the parent material. For example, acid soils are derived from substrates such as sands, slates, etc., while ultra basic rocks, rich in silicates of iron and magnesium give soils with poor fertility (e.g. serpentine soil). It may, however be noted that different parent materials may give rise to the 'same' kind of soil, whenever the climate and biosphere are the same, and when these factors are dissimilar same type of parent material may give rise to different soil type.

Effect of time or age on soil development is reflected in degree of weathering and transport of materials, and extent of biological influences. In hilly regions or mountain topography normal geological erosion removes weathered materials from slope and hinders profile development. Slope exposure (south-facing slopes are warmer than north-facing slopes) also influences soil formation particularly through temperature and vegetation. Soil development occurs more slowly on impenetrable bedrock compared with loose and weather able material, such as glacial till.

Soil-forming factors may affect local plant diversity

There is a global gradient in soil development, the lower latitude soils tend to be older and more strongly weathered as compared with that of higher-latitude soils (Sanchez, 1976). The most species-rich plant communities occur on ancient, strongly weathered soils, whereas those on recently developed soils tend to be less diverse (Laliberte et al., 2013). The most strongly weathered soils in the world (oxisols) occur only in the tropics. These soils support highly diverse lowland tropical forest. For example, lowland tropical rainforest in Yasuni can contain up to 644 tree species ha^{-1} (Valencia et al., 2004); the soils in the area being strongly weathered, very acidic, kaolinitic, high exchangeable aluminium concentration, and very low base saturation. The tropical rain forests in the Lambir Hills National Park (Sarawak, Malaysia) contain 610 tree species ha^{-1} , the greater diversity being on infertile ultisols (Lee et al., 2002) the global pattern of increasing plant diversity with pedogenesis is reproduced at local scales along well defined soil-age gradients from around the world. Laliberte' et al. (2013) have given a conceptual model of mechanisms by which pedogenesis (the process of soil development) might derive local plant diversity (Fig. 4.6). These mechanisms include (i) nutrient availability and stoichiometry, (ii) diversity of N and P forms, (iii) soil spatial heterogeneity, (iv) belowground heterotrophs, (v) aboveground heterotrophs, (vi) species pool, (vii) belowground herbivores (Figure 4.6). Laliberte' et al. (2013) have suggested that chronosequences can be used to quantify the relative importance of different mechanisms affecting plant species diversity depending upon environmental conditions. However, there is need for much empirical research with a focus on multivariate controls over plant diversity, particularly in relation to belowground mechanisms. A greater understanding of mechanisms would help to understand why some ecosystems with ancient, strongly weathered soils exhibit high plant diversity, whereas younger and more fertile ecosystems are often dominated by a fewer species (Laliberte' et al., 2013).

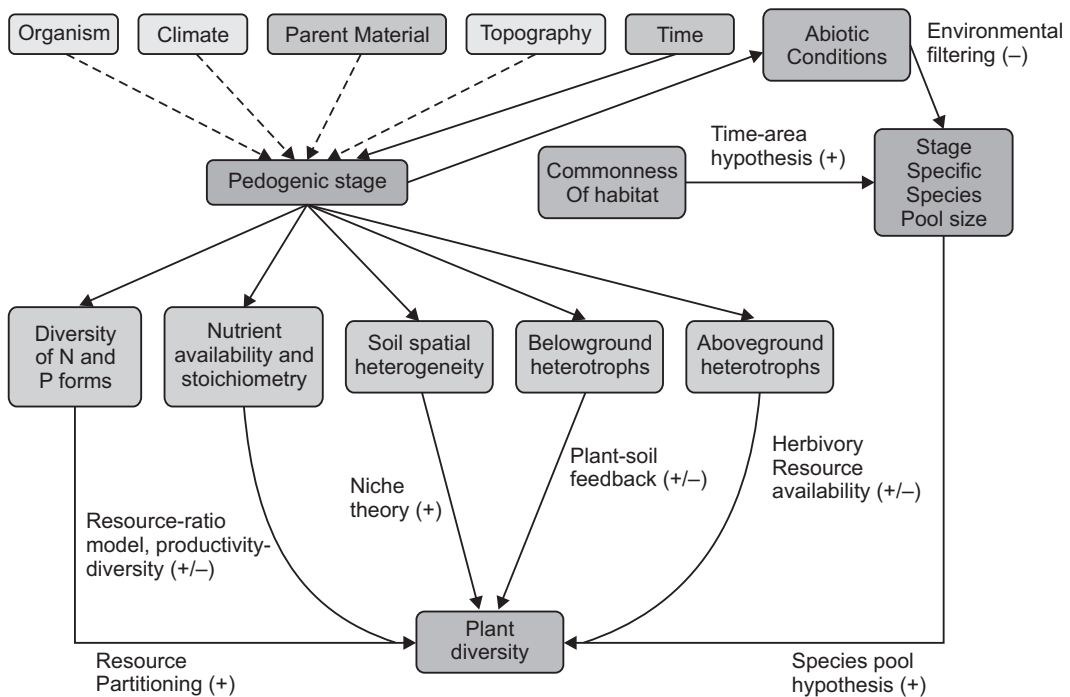


Fig. 4.6. General conceptual model representing how soil-forming factors can drive local plant diversity. Relevant theories and hypothesized effects (+, positive; -, negative; +/-, nonlinear or context dependent) are shown next to the arrows. N, nitrogen; P, phosphorus. (adapted from Laliberte' et al., 2013).

4.5 SOIL HORIZONS AND PROFILES

If a pit or trench is dug into soil, especially deep soils, we can see characteristic layers or horizons differing in colour, texture, structure, thickness, chemical composition and biological properties. These superimposed layers or horizons of soil are collectively designated as soil profile (Fig. 4.7).

The differentiation of horizons is brought about largely through the action of rain water, which leaches materials from surface layers and deposits them at different depths. A typical soil consists of the following horizons:

O horizon It consists of plant litter on the mineral soil surface. It is divisible into O_1 , having recently fallen litter (fresh litter) with little decomposition and O_2 , having partly decomposed organic debris, derived from litter (particles having lost identities, matted together).

A horizon It is referred to as *eluviation or extraction zone* or zone of leaching, which indicates downward movement of materials. It is represented by mineral material that contains most of humus in the profile. This horizon easily loses soluble salts, clay, Fe, Al and organic colloids by leaching but is enriched by silica. Its subdivisions are:

A_1 = This zone has maximum humus with dark colour and granular structure;

A_2 = The zone of maximum eluviation, light colour and structure less due to the loss of clay, Fe, Al and organic colloids;

A_3 = Transitional zone to B horizon, but more like A than B horizon.

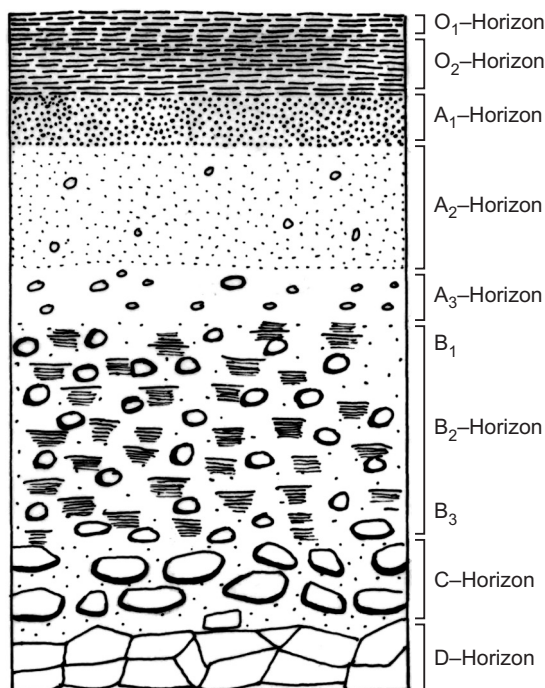


Fig. 4.7. Diagrammatic representation of a soil profile (vertical section of the soil) showing soil horizon nomenclature and symbols for mineral soils are as follows:

O₁ - largely undecomposed organic debris (leaves, etc.); O₂ - Partly decomposed organic debris; A₁ - Zone of maximum humus accumulation (usually dark coloured); A₂ - Zone of maximum leaching (usually light coloured); A₃ - Portion of A horizon transitional to B; B₁ - Portion of B horizon transitional to A; B₂ - Zone of maximum deposits (clay, iron, alluvium and/or humus); B₃ - Portion of B horizon transitional to C; C - Mineral horizon usually unconsolidated but sometimes reconsolidated (below the principal root zone); D - Consolidated bedrock.

B horizon It is commonly called the subsoil, and represents the zone of illuviation or deposition. This layer receives and accumulates the soluble salts, clay minerals and oxides of aluminium and iron carried by percolation waters from the A horizon. The intense colour is due to sesquioxide coating of mineral grains. This zone is of heavy texture, greater density and higher clay content than A horizon. In the B horizon, a hard pan might form in some soils.

C horizon This is the mineral horizon below the B layer but excluding true bed rock; this is formed under conditions little affected by pedogenic processes. The A and B horizon may or may not have been formed from the material of C horizon. Frequently, however, the C horizon contains parent material weathered to some degree.

D horizon It consists of unweathered bedrock and may not be the material from which A,B and C layers are formed.

O horizon may be absent in deserts, as plant cover is scarce. A young soil may have only A and C horizons. This implies absence of downward translocation (eluviation) of clay from A horizon. It is characteristic to the chernozem, the black earth soil of steppe region. The formation of an illuvial B horizon is normally an indication of a mature soil.

Granular structure is generally found in A horizon. Platy structure is not very common, can be found in the lower horizons in some forest soils. The blocky, prismatic and columnar structure is found in B horizon. Angular and subangular blocky structures are found in soils of humid regions. Columnar and prismatic structures are common in soils of arid and semiarid regions.

Soils in grassland, forest and desert biomes differ markedly in their colour, clay, organic matter content, and depth. In desert soils, the top soil has little or no humus, whereas the sub soil is a mixture of sand, clay, minerals and salts. In the grassland systems, the plant roots penetrate deep in the ground, forming a dense sod that prevents erosion, and improves soil organic matter besides retaining moisture. In temperate forest soils the top horizon is a rich mixture of humus and inorganic soil components. The tropical rain forests are characterised by dense clay subsoil and nutrient-poor shallow soils because of heavy rainfall, and high temperatures.

4.6 SOIL PHYSICAL, CHEMICAL AND BIOLOGICAL PROPERTIES

Soil texture

Soil texture is defined by proportions of sand, silt and clay. From sand to clay, the size of particles decreases. Mineral particles in soil vary greatly in size. Those between 0.2 to 2 mm are called coarse sand, 0.02 to 0.2 mm fine sand, 0.002 to 0.02 mm silt and those less than 0.002 mm clay. Fine textured soils have a predominance of clay and silt particles, and coarse textured soils are predominated by sand particles. Particles greater than 2 mm diameter, called gravels, though occur everywhere, are not considered for defining soil texture; for describing texture, we do not consider also other constituents of soil, such as organic matter which may constitute up to more than 5% of soil. Sand provides skeletal support of soil and finer particles, like clay helps to store nutrients and to bind particles together into aggregates. Larger particles are inert, while clay particles with colloidal properties are highly active physico-chemically.

Soils in which clay, silt and sand are well represented are called loams. The use of soil texture triangle (Fig. 4.8) enables one to determine textural class of soil. For example, a soil with 70% silt, 40% sand and 20% clay is silt loam. General terms used to describe soil texture classes are given in Table 4.3.

Table 4.3. General terms u/sed by the US Department of Agriculture to describe texture classes of soil.

Common names	Texture	Basic soil textural class names
Sandy soils	Coarse	Sand, loamy sand
	Moderately	Sandy loam, fine sandy loam
Loamy soils	Medium Silt loam, silt	Very fine sandy loam, loam,
	Moderately Fine	Clay loam, sandy clay loam, silty clay loam
Clayey soils	Fine	Sandy clay, silty clay, clay

An estimate of texture can be made by mixing a small amount of soil with little water. When the soil paste is handled, one feels grainy when it is mainly sand, silky when it is silt, and when a flexible ribbon is formed between thumb and finger it is mainly clay. Soil texture can be roughly determined by assessing the combinations of these properties.

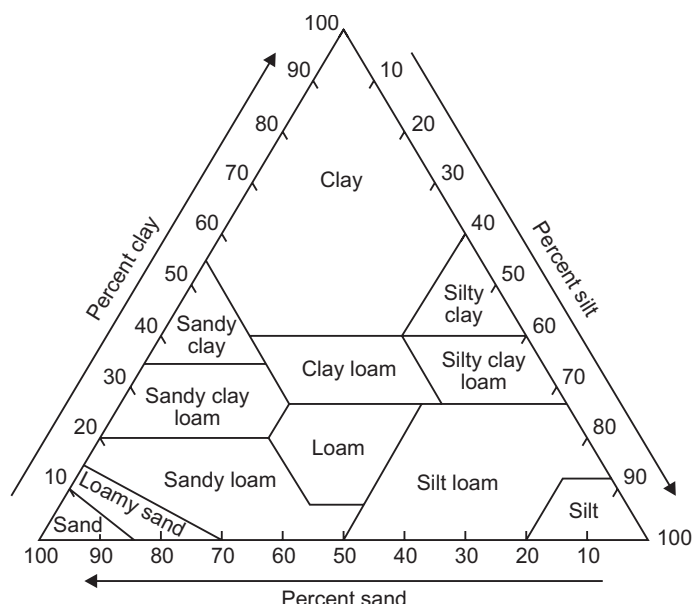


Fig. 4.8. A soil texture chart showing the percentages of clay (below 0.002 mm) silt (0.002 to 0.05 mm), and sand (0.05 to 2.0 mm) in the basic soil textural classes.

A more precise determination of soil texture is made with the help of sieve plates. Coarse particles (larger than 2 mm) are separated out. The sample then is treated to remove organic matter. Usually hydrogen peroxide is used to oxidise organic matter; its effect is also the removal of clay particles from organic matter as their bonds are ruptured. The sample is then dried and weighed. The remaining materials are flushed and shaken through a series of sieve plates which allow particles of different sizes to pass through, e.g. particles of 1 mm, 0.5 mm, 0.25 mm, 0.1 mm and 0.05 mm sizes. The fraction passing through the sieves contains a mixture of silt and clay, and is separated with the help of hydrometer or pipette method of analysis, the latter being based on differential rates of settling of silt and clay.

Soil texture influences physical properties and nutrient content of soil. The physical properties influenced are (i) porosity, the space in soil that is not occupied by soil particles; and (ii) permeability which indicates how easily liquid, gases and roots move through soil. In general, soils with fine texture have greater porosity – 40-60% (of total space occupied by soil) for clay and silt, compared to 35-50% for sandy soils. However, much of the space in a fine textured soil is occupied by very minute pores, through which movement of water is slow as water has strong adhesion and cohesion capacity. Molecules of **hygroscopic water** form a film around each clay particle, and easily bridge the gap between particles because of the strong cohesion between water molecules. In short, permeability of fine textured soil is relatively lower than that of the coarse textured soil. Consequently, the water holding capacity of fine textured soil is greater than that of coarse textured soil.

Clay particles serve as an important reservoir of plant nutrients, because (i) for a given volume their surface area is much larger than that of coarse particles and (ii) they have negative electrical charge. The relationship between small size of particle and large surface area can be explained as following. Suppose there is a block with sides of length 1 mm, its surface area would be 6 mm². Now it is cut into blocks of 0.1 mm sides. This results in 1000 such bodies, with a sum of surface area, 60 mm². Remember, it is the exposed surface area to which nutrients can be attached. In case the smaller blocks have 0.001 mm sides, the exposed surface area would be 6000 mm².

The positively charged atoms (cations) of Ca, Mg, K, Zn, Fe, etc. are attracted and chemically bonded to the negatively charged clay particles. This process is called **adsorption**. Without adsorption nutrients would be easily leached out. However, negatively charged clay particles have poor ability to retain negatively charged NO_3 particles (anions, one of the major forms of N taken up by roots), which are easily leached out with rainwater.

Soil minerals

On the basis of their origin, soil minerals are classified as primary and secondary minerals. Primary minerals, for example, quartz (SiO_2), Muscovite or mica ($\text{KAl}_3\text{Si}_3\text{O}_{10}(\text{OH})_2$) and microcline (KAlSi_3O_8) are prominent in sand and silt fractions. Secondary minerals including hematite (Fe_2O_3), dolomite ($\text{Ca Mg}(\text{CO}_3)_2$), calcite (CaCO_3), and gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) are formed from soluble products during the weathering of rocks. Clay minerals (aluminosilicates) are the most common secondary minerals. In arid and semiarid regions, carbonates and sulphates accumulate in B-horizon.

Particle and bulk density

The density of the mineral particles is the mass of particles per volume of soil (g cm^{-3}). It is also called specific gravity of soil particles. The density of soil particles varies from 2.60 to 2.75 g cm^{-3} , the average particle density being 2.63 g cm^{-3} . Bulk density is the measure of the mass of undisturbed soil per volume and is represented as g cm^{-3} . The bulk density is lower than particle density due to presence of pore space (the proportion of soil body not occupied by particles) and humus. The bulk density of the soil is a variable property due to the effects of weathering, cultivation and effects of animal activity, and increases with compaction. The bulk density of a soil is inversely related to its porosity and generally ranges between 1 and 1.6 g cm^{-3} . Fine textured soils, such as clay and silt loams have lower bulk density compared to coarse-textured sandy soils which have lower pore space than silt or clay soils.

Soil water infiltration, percolation, water holding capacity

Movement of water into soil is called infiltration, and the downward movement of water within the soil is called *percolation*. Pore space in soil is the conduit that allows water to infiltrate and percolate. It also serves as the storage compartment for water. Infiltration rates can be near zero for very clayey and compacted soils, and high for sandy and well aggregated soils. Low infiltration rates lead to ponding on nearly level ground and runoff on sloping ground. Organic matter, especially crop residue and decaying roots, promotes aggregation so that larger soil pores develop, allowing water to infiltrate more readily. Percolation also varies with soil texture and structure. This is the mechanism by which water reaches the subsoil and rooting zone of plants. It also refers to the movement of water below the root zone. Water that percolates deep in the soil may reach the water table or groundwater aquifer. If the percolating water carries chemicals such as nitrates or pesticides, these water reservoirs may become contaminated.

Infiltration and percolation describe the manner by which water moves into and through soil. Water held in a soil is described by the term **water content**. Water content can be quantified on both a gravimetric (g water/g soil) and volumetric (ml water/ml soil) basis. The volumetric expression of water content is used most often. Since 1 gram of water is equal to 1 milliliter of water, we can easily determine the weight of water and immediately know its volume.

Field capacity The maximum amount of water a soil can hold before it is saturated is known as “field capacity”. Field capacity is the amount of water held in the soil after excess gravitational water has drained away and after the rate of downward movement has materially decreased. It is generally considered as the water content retained in the soil at -0.33 bar water potential (suction). *Field capacity* is the soil water content after the soil has been saturated and allowed to drain

freely for about 24 to 48 hours. Free drainage occurs because of the force of gravity pulling on the water. The redistribution of draining water in a soil profile is a continuous process, which may be influenced by many factors including soil texture, type of clay present, organic matter and the rate of evapotranspiration. *Gravitational water* refers to the amount of water held by the soil between saturation and field capacity.

Permanent wilting point is defined as the water content at which the leaves of a growing plant reach a stage of wilting from which they do not recover (usually considered water content at -15 bar water potential (suction). At the permanent wilting point, a plant will wilt and not recover. Plant- available water is the difference between the water content at field capacity and permanent wilting point. Unavailable water is the soil water content that is strongly attached to soil particles and aggregates, and cannot be extracted by plants. This water is held as films coating soil particles.

Water holding capacity of soil is the specific ability of a particular type of soil to hold water against the force of gravity. On the other hand, Field capacity is the point where the soil water holding capacity has reached its maximum for the entire field. Different types of soils have difference water holding capacities, for example a sand soil has a lower capacity to hold water when compared to a clay soil. The nature of the soil, composition of the soil, amount of organic component and size of the soil particles determine its ability to retain water.

Soil pH

Soil pH (a symbol used as quantitative measure of hydrogen ion concentration; hydrogen ion is a positively charged hydrogen atom; latin *pondus hydrogenii*, *potentia hydrogenii*, or potential hydrogen) is defined as the decimal logarithm of the reciprocal of the hydrogen ion activity. It generally varies from 4 to 8, but in exceptional cases values can be higher or lower than this range. In salt brush scrub of California or in saline lands in the Indo-Gangetic plain pH can reach 9-10. On the other hand, ancient soils of humid regions under coniferous forests, such as the pygmy cypress (*Cupressus pygmaea*) the soil pH can be as low as 2.8 to 3.9. This strongly acidic soil is called black lock podzol.

The effect of pH on plant growth is generally indirect, but significant. But toxic effect of metals like aluminium and manganese, which become quite soluble at low pH (less than 6.5), is direct. The most significant direct effect of pH is on the availability of nutrients (Table 4.4). Major nutrients like nitrogen and phosphorus are most available at and around neutral pH (6 to 8). Iron and boron are most available at strongly acidic to moderately acidic pH (4 to 6), while availability of molybdenum increases with increasing pH.

Table 4.4. Soil pH values at which nutrients availability is greatest.

Nutrient	pH	Nature
Iron	4-6	Strongly acidic
Manganese	5-6.5	
Boron	5-6	Slightly acidic
Copper and zinc	5-6.5	
Nitrogen	6-8	Slightly acidic to slightly
Phosphorus	6.5-7.5	Alkaline
Calcium	7-8.5	Slightly alkaline
Magnesium	7.5-8.5	
Potassium	6-7.5; 8.5-10	Slightly acidic to strongly alkaline
Sulphur	6-10	Acidic to strongly alkaline
Molybdenum	7.5-10	Slightly alkaline to strongly alkaline

Cation Exchange Capacity

Cation exchange capacity (CEC) is a measure of the number of negatively charged sites on soil particles and humus that attract exchangeable cations, the positively charged ions (e.g. Ca, Mg, K, NH_4 , Na) that can be replaced by other such ions in the soil solution. Clay and humus contents increase the CEC of soil. It is also influenced by amorphous colloids (called allophane). Generally, the higher the cation exchange capacity, the more fertile the soil. The cation exchange capacity of soils depends on the amounts and kinds of clay and organic matter present in the soil; CEC increases with the increase in soil organic matter.

The CEC of a soil is expressed in terms of milligram equivalents per 100 grams of soil and is written as meq/100g. The CEC of organic matter ranges from 200 to 400meq/100g. Where soils are highly weathered and organic matter levels are low, CEC values are low. Where less weathering has occurred and organic matter levels are usually higher, CEC values can be quite high. Clay soils with high CEC can retain large amounts of cations against potential loss by leaching, and sandy soils with low CEC retain small quantities of cations. Therefore, the amount of clay and organic matter content greatly influence the CEC of soils. The measurement of CEC is important to decide about the timing and application of fertilizers in agricultural soils.

Sodic or alkaline soils contain excessive amounts of sodium (Na) on the soil CEC sites. Soils are sodic if Na saturation exceeds 15% of the cation exchange capacity. Sodic soils have soil pH exceeding 8.5 and exchangeable sodium more than 15%. Excess sodium disperses the soil structure and limits the movement of air and water due to poor physical properties. Sodic soils can be reclaimed by replacing the Na on CEC complex by calcium. Gypsum (calcium sulphate) is the most common treatment, but elemental sulphur can be used if the soil is calcareous. Soil salinity refers to the amount of soluble salts in soil and expressed in term of the electrical conductivity of a saturated extract of soil. If the electrical conductivity exceeds four decisiemens per metre (4 dsm^{-1}), the soil is saline. The problems of soil salinity and sodicity are particularly acute in semi-arid regions.

Anion exchange

Anions are negatively charged ions found in the soil, which are NO_3^- , SO_4^{2-} , HCO_3^- , and OH^- . Anions tend to leach from the soil rapidly than the cations. However, phosphorus is retained against leaching as it forms insoluble precipitates with compounds containing iron, aluminium and calcium.

Soil organic matter, Microbial biomass, Aggregates and Humus

The carbon fixed by the plants is the primary source of organic matter of the soil; the inputs are both from aboveground and belowground parts of the plants. Soil organic matter forms a highly heterogeneous mixture of organic materials in the soil along a continuum from freshly fallen litter to highly decomposed organic materials and thus consists of materials in various stages of decomposition from recent inputs of organic residues (plant animal and microbial origin) to highly refractory materials. Organic matter influences soil physical, chemical, and biological properties. Soil structure, supply of nutrient elements, cation exchange capacity and pH buffering are affected by the amount of soil organic matter. Soil organic matter is a highly complex mixture of carbon compounds which contains nitrogen, sulphur and phosphorus. Organic matter is also a source of carbon energy for microorganisms.

Models of Soil Organic matter

Soil organic carbon simulation models have been used to predict the effect of management practices and climate change on the fluxes and stocks of soil organic carbon. In the models, soil organic matter is commonly divided into several pools depending upon resource quality, turnover time and

functional pools (Jenkinson et al., 1987; Parton et al., 1987; Davidson and Janssens, 2006). Different carbon pools existing in the soil have different rates of turnover, ranging from one year to few years to decades or more than 1000 years (stable fraction) as influenced by the biochemical composition of litter (Jenkinson et al., 1987, Parton et al., 1987). These models have been successfully used to simulate changes in total soil organic matter. The two best-known models of soil carbon dynamics are the CENTURY (Parton et al., 1987) and ROTH-C (Jenkinson et al., 1987; Jenkinson, 1990). A simplified structure of the ROTH-C and CENTURY models of soil organic matter dynamics have been adapted from Davidson and Janssens (2006) in Fig. 4.9. These models compartmentalize soil carbon into 5–7 conceptual pools, including 2–4 pools of decomposable plant material near the soil surface (litter layer) and three pools of carbon in the mineral soil, with mean residence time ranging from years to millennia. Roth-C only models soil processes, with plant residue carbon as the input.

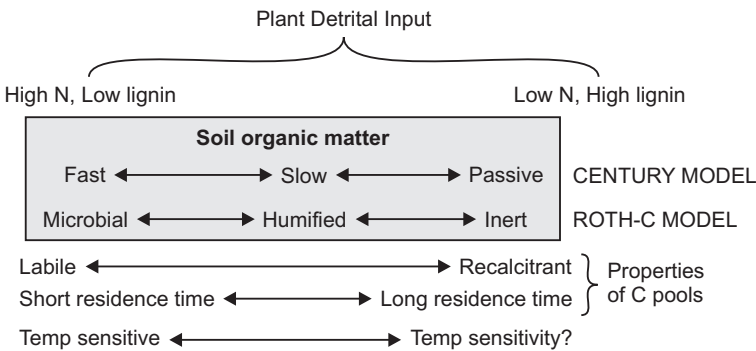


Fig. 4.9. Conceptual Pools of soil Organic Matter. Three discrete soil carbon pools in the mineral soil along a continuum of decomposability and mean residence time (MRT) in the soil in CENTURY model and Rothamsted-Carbon (Roth-C) model.
(based on Davidson and Janssens (2006) NATURE | Vol. 440 | 9 March 2006)

CENTURY is an ecosystem model that recognizes three carbon pools in the mineral soil, i.e., fast, slow and passive pools (Parton et al., 1987) (Fig. 4.9). The CENTURY model has been used extensively to simulate the long-term (10–100 years), response of ecosystems to changes in climate, atmospheric CO₂ levels, and agricultural management practices.

Soil microbial biomass. The soil microbial biomass is a labile pool of organic matter and comprises 1-3% of total soil organic matter (Jenkinson and Ladd, 1981). It is a source and sink of the plant nutrients (Jenkinson and Ladd, 1981; Singh et al., 1989) which regulates the functioning of soil system (Fig. 4.10). A major supply of nitrogen for plant growth comes from organic matter through the microbial mediated mineralization and immobilization turnover. The dominating influence of plant species on soil microbial biomass has been found to be associated with litter quality and quantity as well as carbon inputs to soil. The soil microbial biomass can be used for assessing changes in soil organic matter caused by soil management (Powelson et al., 1987), tillage practices, bioamelioration of sodic soils by silvopastoral systems (Kaur et al., 2002a), and land management.

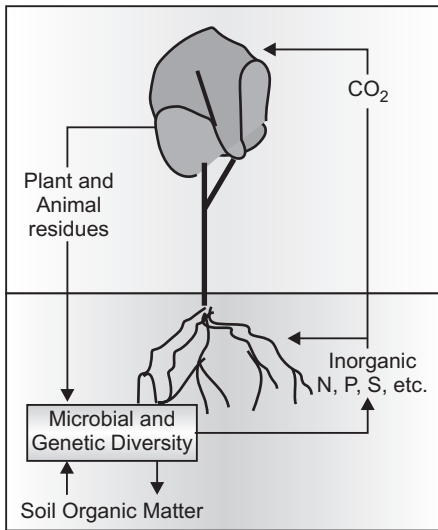


Fig. 4.10. The Soil Microbial biomass concept.

Soil aggregates. Organic matter is an important component of aggregates. Fungal hyphae and fine roots bind soil particles together to form aggregates larger than 1 mm. Medium sized aggregates (20 to 250 μm) are more stable because of binding by humus, iron, aluminium oxides and clay particles. Smaller aggregates may be more stable, though contain same binding agents. On the basis of structure, the soils are classified as weak, moderate and strong. Soil structure has a large influence on the infiltration of water and tilth of soil. For arable cropping, a soil should have small aggregates or crumbs. This allows seedlings to emerge readily or provide a supply of water and oxygen to the roots. The most stable aggregates are formed in neutral and calcareous soils (containing calcium carbonate under grasslands) or in some tropical soils in which primary mineral particles are cemented together by iron oxides.

Humus. Humus is an intermediate product of decomposition. This finely divided black coloured organic matter decomposes slowly, for instance, in temperate regions at the rate of 3% per year. Nearly 90% of humus is due to the microbial tissues and microbial activities on litter; however, it also includes root exudates, earthworm excreta, etc. This slowly decomposing material is a reliable source of nutrients for plant growth. Because the humus particles are very small, they can coat soil particles to the point that the soil appears black, even if total organic matter is as little as 5%.

Biodiversity of soil organisms

The soil biota consists of soil fauna, bacteria, fungi, actinomycetes and algae. The soil organisms are important regulators of nutrient cycling, decomposition, soil organic matter formation and the maintenance of soil structure. They contribute significantly to the structure of soil organic matter, which arises from four sources: humus, vegetation (both live and dead), micro organisms and soil fauna. From the standpoint of size, soil fauna can be divided into three types: (i) macro fauna e.g., oligochaetes, large larvae and adult insects, isopods, myriapods, mollusks, etc.; (ii) mesofauna - e.g., microarthropods, nematodes; and (iii) micro fauna - e.g., protozoans (Fig. 4.11).

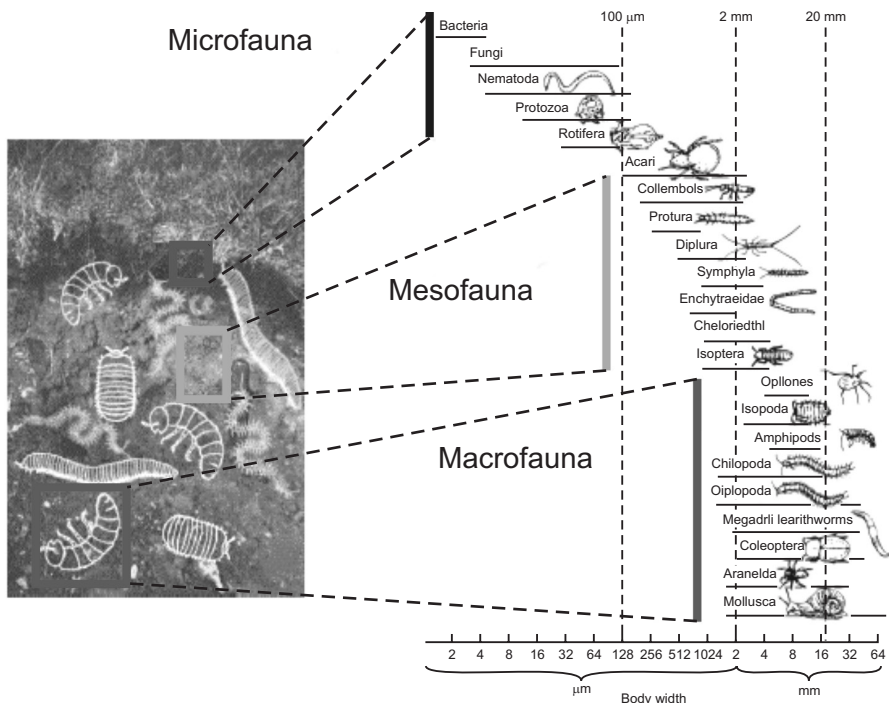


Fig. 4.11. Soil biodiversity: the biological engine of the earth. (Soil profile adapted from <http://soilbio.nerc.ac.uk>, soil fauna size classes based on Swift et al., 1979).

Their trophic categories can be herbivores, saprophytes or carnivores. With the exception of tundra, where micro arthropods (mesofauna) are dominant component, macro fauna account for over 90% soil zoomass in forests and grasslands (Zlotin, 1985). In arid regions, their importance declines (48% to 56%), nevertheless they continue to be the greatest component.

Earthworms are major decomposers of dead and decomposing organic matter in most temperate soils and many tropical soils. Nematodes (mesofauna) which are insignificant component in the soils of forests, share up to 32% zoomass in some deserts. Among the three trophic groups, saprophytes (feeding on dead matter) are by far the largest group, accounting for 82%-90% of soil zoomass, except in dry conditions where herbivores (phytophages) also become important (Table 4.5). Being primarily hygrophilous, saprophytes become less predominant in dry regions. Comparison between invertebrates and vertebrates suggests the absolute predominance of the former in all soils.

Table 4.5. Structure of the soil zoomass (%) in certain ecosystems (from Zlotin 1985, based on several studies). (a) includes two forest types; (b) includes a mountain moist meadow, a meadow steppe and a true steppe; (c) includes a dry steppe and a mountain dry steppe; (d) includes a semi-desert, a desert and a mountain semi-desert and a mountain desert.

Ecosystem	Size-systematic group			Trophic group		
	Macrofauna	Microarthropods	Nematodes	Saprophages	Phytophages	Zoophages
Moist region						
Arctic tundra	7	78	15	97	1	2
Mountain tundra-like cushion plant formation	25	52	23	82	14	4
Forest (a)	94.5	3.5	2	95	2.5	2.5
Steppes (b)	91	4	5	90	16	4
Dry regions						
Dry steppes (c)	56	25	19	59	32	6
Deserts (d)	48	20	32	44.5	48	7.5

Soil fauna constitute far lower biomass than the bacteria (Table 4.6). They represent only 0.1% of the soil carbon and 2-3% of the total soil biomass (Rosswall and Paustian, 1984). Obviously, microbes (bacteria and fungi) account for most of the soil biomass (Table 4.6). However, the role of soil fauna in association with bacteria, fungi and actinomycetes may exert high and low intensity effects on the ecosystem functioning both directly or indirectly. The high intensity effects may regulate plant growth and production, e.g., feeding on plants, dissemination of pathogens, and pollination of plants. Direct or indirect influences of lower intensities include effects on humification and mineralization of organic matter, mineral cycling, interactions with soil micro organisms, impacts on soil structure and its physico-chemical conditions, which become visible after long periods of time through their cumulative influences. The standing crop of soil fauna shows highest values in deciduous forest and low values in the grassland and agroecosystem (Table 4.7). The structure of the humified horizon in soils is largely due to the invertebrates, earthworms and others. As an example, lumbricids digest and mix a layer of soil (0-50 cm) in 200 to 300 years. In meadow steppes lumbricids in association with ants and burrowing mammals, dig up 0-30 cm soil layer in 100 to 150 years, the total depth transformed being over 2 m. In tropical ecosystems, earthworms other than those of the family lumbricidae occur (Lumbricidae account for 10% of the earthworm species of the world), though their relationships with the soil are same as that of the

lumbricids. In lowland tropical forests earthworms constitute 60% of the total macrodecomposer biomass, compared to 24% in deciduous forest and 78% in tropical highland forest (Lee, 1983). Earthworm's feeding and casting activities improve and modify the physical properties of soil and increase the availability of nutrient elements like N, P, K, Ca and Mg. Casts are generally richer in organic matter and nutrients than surrounding soil, their pH is higher and their C : N ratio is lower. But the greatest importance of the burrowing activity of earthworm is that by enhancing infiltration and reducing erosion.

Table 4.6. Range of biomass (g dry wt m⁻²) of fauna protozoans and bacteria and in soils of different ecosystems (based on temperate climatic zone). Each value is based on several samples.

	Agroecosystem	Natural ecosystem
Soil fauna	1.3-10.1	1.6-45.0
Protozoans	0.2-3.0	—
Bacteria	166-415	243-5,300

Table 4.7. Standing crop (g dry wt m⁻²) of soil fauna in soil (without protozoa) in various ecosystems of the temperate climatic zone (from Ryszkowski, 1985).

Ecosystem	Biomass* (g dry wt m ⁻²)
Deciduous forest	18.0-45.0
Mixed deciduous coniferous forest	9.0-12.0
Coniferous forest	5.4-9.0
Grassland	1.3-1.7
Agro ecosystem	1.3-1.7

Several groups of arthropods help in decomposition of plant residues and mixing of fragmented litter into the soil. For example, millipedes are saprophytic as they consume dead organic materials. Ants and termites can mix large volumes of soils through their tunneling effects and forming galleries (in the case of termites).

Functional Diversity of Soil Microorganisms

The soil microorganisms including bacteria, fungi and actinomycetes show great taxonomic and functional diversity. Spatial heterogeneity in soil microbial communities occurs at a broad range of scales from soil particles (soil macroaggregates), to plant rhizosphere, to field plots and to the ecosystem (Tiedje, 1994). The existence of the functional groups of soil organisms provides a useful means of describing the spatial heterogeneity in ecological systems. Soil microbial populations impact many soil processes and therefore ecosystem functions and productivity. The key functional groups of soil biota that influence ecosystem functions include N-fixing bacteria and mycorrhizal fungi. The functional diversity of the soil organisms also includes a wide range of bacteria, fungi, predators and pathogens as well as photo- and litho-autotrophic microorganisms.

Soil microorganisms have also major roles in determining global warming, ground water quality, plant and animal health and organic matter cycling. A greater understanding of soil microbial diversity can benefit land management decisions, monitoring and predicting environmental change, restoration biology of higher organisms and for understanding biological interactions (Kennedy and Gewin, 1997). The metabolic, phenotypic and genotypic profiles of soil microbial communities

* One exceptional value was 32.2 g dry wt m⁻² for a meadow steppe of USSR and in short grass prairie the values can be as low as about 0.8.

can provide a holistic representation of the community structure. It has been estimated that only 1 to 5% of all microorganisms have been classified and named. The tools of molecular biology have been used to unlock the secrets of soil microbial biodiversity. The studies have shown that soil contains more microorganisms that can be cultured by traditional methods. The molecular methods in combination with stable isotope probing provide a cultivation-independent means of linking taxonomic diversity of microbes with functional processes (Radajewski et al., 2000). Using stable isotope probing method, these workers have identified methanol utilizing microorganisms that comprised 3-protobacterial and acidobacterium lineages involved in carbon and nitrogen cycling in the soil (Radajewski et al., 2000). A quantitative analysis of the interaction between microbes, plant residues and disturbance at various scales could be useful to understand the functioning of the ecosystems to ensure sustainability as well as exploiting microbial processes for economic gains.

Bacteria are tiny, one-celled organisms – generally 1 μm wide and somewhat longer in length. A teaspoon of productive soil generally contains between 100 million and 1 billion bacteria. Bacteria fall into four functional groups, i.e., decomposers, mutualists, pathogens and chemoautotrophs. Most are decomposers that consume simple carbon compounds and convert energy in soil organic matter into forms useful to the rest of the organisms in the soil food web. Decomposers are especially important in immobilizing nutrients in their cells, thus preventing the loss of nutrients, such as nitrogen, from the rooting zone. *Mutualists* such as nitrogen fixing bacteria form partnerships with plants. Bacterial pathogens include *Xymomonas* and *Erwinia* species, and species of *Agrobacterium* that cause gall formation in plants. The *chemoautotrophs*, obtain their energy from compounds of nitrogen, sulfur, iron or hydrogen instead of from carbon compounds; these bacteria play an important role in nitrogen cycling and degradation of pollutants.

Fungi are microscopic cells that usually grow as long threads or strands called hyphae, which push their way between soil particles, roots, and rocks. Hyphae sometimes group into masses called mycelium or thick, cord-like “*rhizomorphs*” that look like roots. Fungal fruiting structures (mushrooms) are made of hyphal strands, spores, and some special structures like gills on which spores form. Fungi play an important role in water dynamics, nutrient cycling, and disease suppression. Soil fungi can be grouped into three general functional groups including decomposers; *mutualists* (the mycorrhizal fungi that colonize plant roots), and pathogens and parasites.

Soil algae can enrich the soil in organic matter. Blue green algae are capable of fixing atmospheric nitrogen. Lichens are symbiotic association of an algae and a fungus. These constitute the first colonizers of the bare rock and can play an important role in weathering of rocks through the production of organic acids.

Microbial communities play a major role in soil fertility through the decomposition of crop residues, livestock manure and soil organic matter. These microbes are often affected by variations in rainfall and temperature patterns caused by climate change. Recent advances in the use of stable isotopes like carbon-13, nitrogen-15 and oxygen-18 as biomarkers to characterize microbial communities and their interactions with soil nutrient and organic matter processes, known as stable isotope probing (SIP), are important for soil–water–nutrient management. SIP helps to understand the interactions between soil microbial communities, and their specific functions in soil carbon sequestration, soil organic matter stabilization, soil fertility and soil resilience, as well as the soil productive capacity for sustainable intensification of cropping and livestock production.

4.7 MYCORRHIZAE

Mycorrhizae (= fungus-root), comprising the mycelia of fungi, which live mutualistically with the roots of plants, play a significant role in mineral cycling as well as in growth of plants in most ecosystems. The mycorrhizal fungi interact with root tissues to form a kind of composite

“organs” called “fungus root”. The mycorrhizae increase the ability of plants to extract minerals from soil, especially phosphorus. The plants, however, have to bear the cost by providing some of photosynthate in return. Roots and the mutualistic symbiosis they form with mycorrhizae fungi utilize a substantial portion of net primary production (NPP) in the growth of new roots-mycorrhizae and in the senescence of old roots mycorrhizae. These are a major source of nutrient return to soil, possibly greater than the leaf litter. In a Douglas fir ecosystem, 70% of NPP is invested in root-mycorrhiza processes; and root-mycorrhiza throughput accounts for 83-87% of N, P and K, plus 45-57% of calcium and magnesium returned to the soil by trees (Fogel and Hunt, 1983). The hyphae that cover ectomycorrhizal roots in the Douglas fir ecosystem, for instance, account for 50% of soil fungal biomass and 40% of the fungal biomass returned to soil (Fogel and Hunt, 1979). In forest soils, certain species of ectomycorrhizal fungi form mat like structures at the interface of humus layer and upper mineral layer. Within the fungal mats, calcium oxalate crystals are in abundance, which enhance decomposition rates and nutrient availability. Mycorrhizae take primarily the following forms:

Ectotrophic mycorrhizae : In this, fungal mycelia form a thick (20-40 μm) mantle on outer sheath around the root, with hyphae traversing the rhizosphere (1-2 mm) zone of soil surrounding growing roots) to colonise soil particles and organic substrates. Intracellular penetration of hyphae does not occur, hyphae can, however, penetrate between the epidermal cells and then the sub epidermal cortical cells, forming a “*Harting net*”. Most of the ectomycorrhizal fungi belong to Basidiomycetes including Agaricales such as *Amanita*, *Boletus* and *Trichoderma*. Some Ascomycetes such as “Tuber the truffle”, the fungus with under ground fruiting body, also form ectomycorrhizal sheaths. About 3% of plant species, distributed across 130 genera of woody plants are ectomycorrhizal.

Endotrophic mycorrhizae: These fungi are members of *Phycomycetes* in which hyphae penetrate the cortical cells of plant root (Fig. 4.12). Among these, arbuscular (AM) endomycorrhizae are most wide-spread, occurring in association with about 50% of all vascular plant species, including grasses, shrubs, legumes and many hardwood species. They are also common in trees of tropics and southern hemisphere. *Glomus*, *Gigaspora* and *Sclerocystis* are the four genera forming associations with host of plant species. Their colonization of the roots of various groups of plants is very high, whereas the level of host specificity is very low.

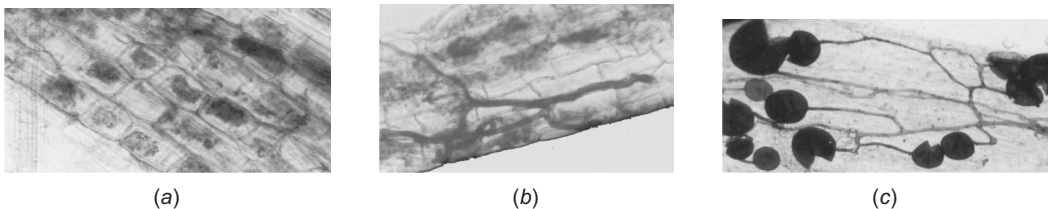


Fig. 4.12. AM fungal infection in wheat roots under Zero-tillage : (a) An arbuscule with mycorrhizal hypha in a root cortical cell; (b) the penetration of infective AM hypha in cortical cells; (c) mycorrhizal hyphae with globose vesicles. (source: Neelam, 2006; Gupta et al., 2009)

Ericaceous type of endomycorrhizae: These are found within the family Ericaceae (e.g. *Rhododendron* and others such as heathers and bilberry). The endomycorrhizae assume ericoid form and are commonly found in nutrient poor, acidic soils of cold regions. The Ericaceae plants are able to grow in such harsh conditions largely because of ericoid endomycorrhizae. These plants do not have root hairs, but root ends are very fine. These fungi are members of *Ascomycetes* or *Deuteromycetes* and are not host specific. They can grow on soil over a long period. These fungi are also able to tolerate up to 100 ppm of heavy metals, this allows the host to grow in soils having high concentrations of cadmium, zinc and copper. Ericaceous mycorrhizae also include ectendotrophic forms, which combine features of both ecto- and endomycorrhizae and ectendotrophic monotropoid form, involving the achlorophyllous, saprophytic higher plant genus, *Monotropa*.

Orchidaceous mycorrhizae: This is another form of endomycorrhizae, developed by all species of Orchidaceae (orchid) family worldwide. In this, the septate hyphae invade the very young embryo from the soil. After the seed coat has been ruptured, the hyphae penetrate intracellularly into epidermal cells. Subsequently, cortical cells are colonized by both inter and intercellular hyphae. Unlike most other mycorrhizae, the fungus in this case supplies organic carbon compounds to the host plant from the surrounding soil or from neighboring plants, also infecting them. The orchid seeds are very small, having little food reserve. Therefore, they can germinate only when mycorrhizal association is established.

The role of lichens and ectomycorrhizal fungi in soil formation

Recent studies have shown the role of lichenous fungi and ectomycorrhizal fungi in mobilization of essential nutrients from insoluble mineral sources and in soil formation (Landeweert et al., 2001). Lichenous fungi and ectomycorrhizal fungi get organic carbon from their host and can cause weathering of minerals. The mycobiont (fungal component of a lichen or mycorrhizal symbiosis) present in lichens produce a range of organic acids, including oxalic acid, which helps in mobilization of elements from the rocks. The organic acids have been found to enhance Biotite (mica type of mineral containing potassium, iron and magnesium) dissolution under the lichen thallus.

Ectomycorrhizal fungi are commonly found in the roots of trees in Pinaceae and Fagaceae (temperate forests) and trees belonging to Myrtaceae and Dipterocarpaceae (found in subtropical regions). About 6000 fungal species may form ectomycorrhizal associations with different tree species. They can either mobilise phosphorus, potassium, calcium and magnesium from solid mineral substances through organic acid excretion, depending upon species specificity of different ectomycorrhizal fungi. In forest soil, some species of ectomycorrhizal fungi (*Hysterangium* spp., *Hydnella* spp. and *Gautieria* spp.) form mat like structures at the interface of the surface humus layer and upper mineral soil. These fungal mats cover several m² of forest floor and the mineral soil within the mass of hyphae is often weathered. The ectomycorrhizal fungi establishing links of plants to rocks may have implications in soil formation and nutrient cycling (Landeweert et al., 2000) because of their role in the following processes :

- Essential plant nutrients can be released from mineral weathering through organic acid excretion.
- Can mine nutrient sources in minerals deep in the rocks that are not accessible to roots.
- Excretion of oxalates by ectomycorrhizal fungi can reduce the effect of acid rain.
- Contribute to soil formation via their effects on mineral weathering and podzolization

4.8 SOIL - A SPECIAL TYPE OF ECOSYSTEM

An ecosystem is regarded as a discrete entity that has predictable properties and behaviour and that exchanges energy and materials with the outside. The soil ecosystem is a special representative of such a system (Golley, 1985). It consists of the subsystems, biota, the physical substrate, air and water and the linkages include flows of energy elements. Elliott (1987) developed a simplified conceptual model of a tropical ecosystem with emphasis on soil and related processes. The soil solution (X_1) is the centre of system processes (Fig. 4.13). It may exchange or come into equilibrium with the exchange sites and/or the mineral soil ($X_1 - X_2$; $X_2 - X_1$), and receive dissolved organic matter (X_3) from microbes (X_5), roots and associated organisms (X_7) and soil fauna (X_6). It is the donor for uptake by microbes and plants. Nutrient losses can occur, through leaching (see solution sink) or denitrification (atmospheric sink) due to disturbance. These losses are a result of the accumulation of inorganic nitrogen in the soil solution which makes it vulnerable to biotic and abiotic losses other than plant uptake. The undisturbed ecosystems have mechanisms that reduce these losses through immobilization by plants and mycorrhizae (both retain the nutrients in their body until death and decay), and microbes and fauna.

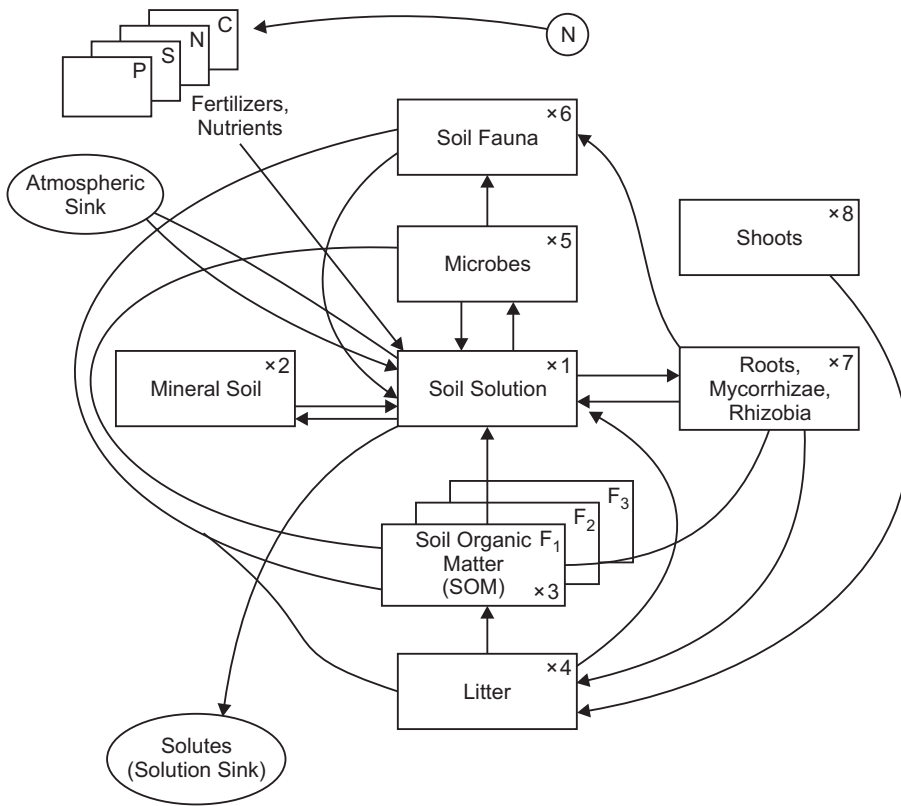


Fig. 4.13. A simple conceptual model of a tropical ecosystem with emphasis on soil biological components and nutrient availability. The soil solution is the hub of the soil system (after Elliott, 1987).

4.9 SOIL CLASSIFICATION

Naming of soils, their classification and concepts of soil science were founded on the plains of Russia mainly by V.V. Dokuchayev and K.D. Glinka during 1870s. They identified broad relationships of soil profile characteristics to climate and vegetation. The broadly defined units of this approach have been called “soil types” in Russia, and great soil groups, as per the international terminology. These great soil groups are broadly defined types, having characteristic soil profiles, developed on level terrain corresponding to prevailing climax vegetation. They may be referred to as *zonal soils*. However, development of soil in the same climate may be greatly affected by the kind of parent material, or by conditions of drainage. These are called “non-zonal” soils; they are also accepted as great groups.

Because soils differ and resemble in so many directions, their classification is difficult. The most accepted system of a classification is that of the United States System of Soil Taxonomy, developed by the U.S. Department of Agriculture. The system is hierarchical, and begins with 10 upper-level classes (termed orders), and the remaining levels of the hierarchy (from higher to lower) are suborders, “great groups”, subgroups, families, and series. Most of the formative elements in the names of soil orders are derived from Latin or Greek words appropriate to that soil order, and

the soil order name is formed with the suffix *sol*, from the Latin *solum* (soil): e.g., Aridisol derived from, *aridus* (Latin), meaning, dry; Spodosol derived from, *spodos* (Greek), meaning wood ash; Histosol, derived from *histos* (Greek) meaning tissues; Inceptisol, derived from *inceptum* (Latin), meaning soft; Ultisol, derived from *ultimus* (Latin) meaning last. Oxisols are defined as soils with an oxic horizon [i.e., containing <10% weatherable minerals in the fine sand fraction (50–200 mm) and a low cation-exchange capacity] or a kandic horizon (clay enriched and low cation-exchange capacity) with >40% clay in the surface soil. Oxisols represent soils that have undergone extreme weathering, are low in nutrients, and occur predominantly in stable regions of the lowland tropics.

There are eleven orders in soil Taxonomy, 10 are arranged in the hierarchy on the basis of mineral alteration and profile development. The 11th order (Histosols), is an organic soil and outside the concept of hierarchy.

An example of hierarchical classification is given below:

Categories	Example of taxa within categories
Order	Entisol
Suborder	Orthent
Great group	Cryorthent
Subgroup	Typic cryorthent
Family	Loamy-skeletal, carbonatic typic cryorthent
Series	Swift creek

A brief description of major soil groups along with suitable examples is included in Table 4.8.

Table 4.8. Major soil orders with brief descriptions, examples and percent of land area in the world (from Brady, 1990; Barbour et al., 1995).

Order	Diagnostic features	Examples	% land area	Typical vegetation
Alfisols	Soil with high clay translocation and deposition; high medium to high base status	Gray-brown podzols	14.7	Deciduous forest soil
Andisols (<i>An</i> , dark)	Volcanic soils with minimum profile development	Volcanic soils	-	Many types
Aridisols (<i>aridus</i> , dry)	Low in organic matter; light in colour soils in dry climates, salts and carbonates accumulate	Desert soils	19.2	Desert scrubs
Entisols (recent)	Immature soils; little or no horizon development; recently formed soils	Sand dunes and rocky soils	12.5	Many types
Histosols (<i>Histos</i> , tissue)	Organic soils; peat and fen soils	Peat	0.8	Marsh
Inceptisols (<i>inceptum</i> , beginning)	Young soils, usually wet, with weakly developed profiles	Tundra and alluvial soils	15.8	Cool grasslands
Mollisols (<i>mollis</i> , soft)	Temperate grassland soils organic rich surface horizons; high base status	Prairie soils, Chernozems	9.0	Grasslands
Oxisols (<i>oxide</i>)	Deeply weathered soils of the tropics	Lateritic	9.2	Tropical rainforest
Spodosols (<i>spodos</i> , wood ash)	Soils with surface accumulation of organic matter and aluminium / iron oxide; acidic	Podzols	5.4	Coniferous forest
Ultisols (<i>ultimus</i> , last)	Soils with clay translocation and deposition; low base status	Red-yellow podzol	5.6	Forest
Vertisols (<i>verto</i> , turn)	Swelling clay soils	Grumusols	1.8	Many types

4.10 SOIL GROUPS IN RELATION TO CLIMATE

Whittaker (1975) identifies two ecoclines along which patterns of relationships between soils can be identified (Fig. 4.14). (i) The first is tropic-arctic series in humid climates. This series leads from tropical and subtropical latosols through warm-temperate red-yellow forest soils and mid-temperate brown earths to taiga podzols and tundra gleys and, as non-zonal extreme members, bog peat soils. Along this series there (i) is a shift from inorganic acid leaching with iron and aluminium oxides left as residual material (see tropical rain forest) to humid acid (organic acid) leaching with silica left as a residual material. (ii) The mass of surface litter increases. (iii) The overall nutrients of soil and soil pH increase from latosol (tropical rain forest) to brown earth (temperate deciduous forest and then from the latter to podzol (taiga). (iv) soil nitrogen content increases northward but may be lower in podzol. (v) Nutrient stock in the community, and the fraction of that stock turned over per year decreases northward. (vi) productivity and litter fall decrease northward, and the mineral concentrations of litter also decrease. There is consequently a marked decline in the return of nutrients to soil (in terms of litter fall and root mortality) from tropical forest ($128 \text{ g m}^{-2} \text{ yr}^{-1}$) to taiga forest ($9 \text{ g m}^{-2} \text{ yr}^{-1}$).

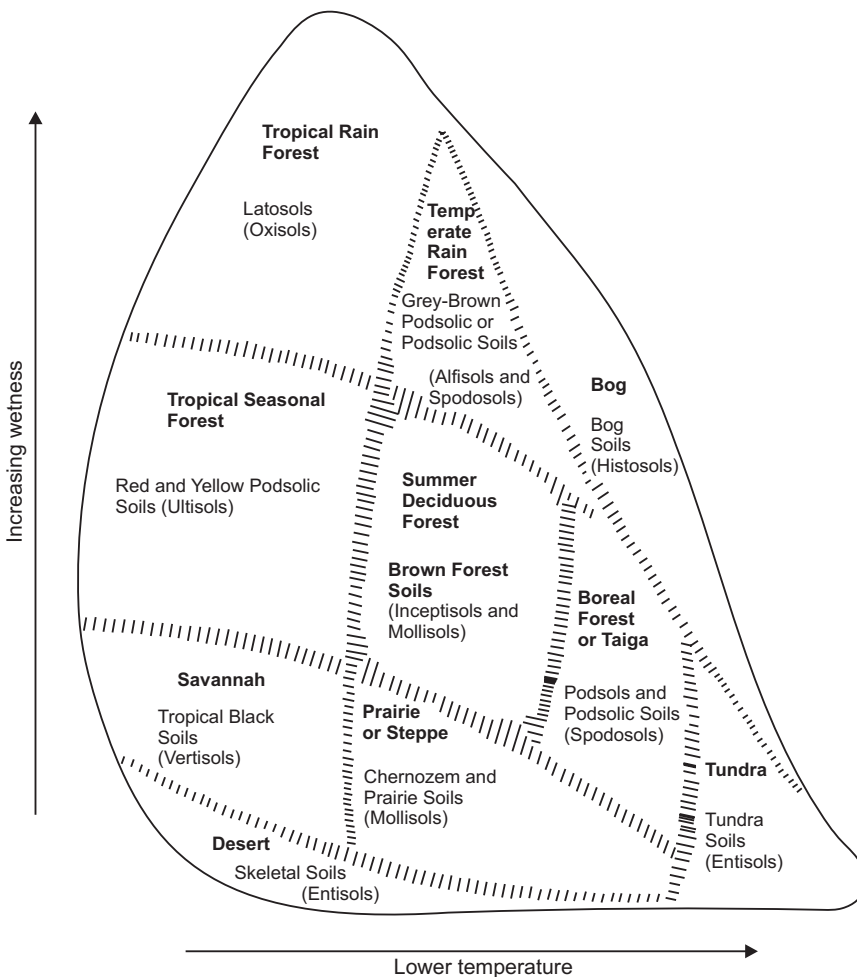


Fig. 4.14. The geographical distribution of soil types in relation to climatic, temperature and wetness regimes. (after Muckenhirn et al., 1949)

The second series extends from forest brown earths (temperate deciduous forest) through chernozems (prairie soils) to sierozems and desert soils. Obviously, this gradient is along the decreasing moisture in the temperate regions. The broad patterns are as following:

1. The weathering and clay content tend to decrease.
2. The pattern of nutrient transport with water shifts from predominantly downward leaching in the forest, with somewhat a balance between upward and downward movements in the chernozem to predominance of upward movement and evaporation, with solutes left near soil surface, in desert soils.
3. Along the gradient, overall nutrient content and pH of soil increase, but nitrogen content decreases from grassland (chernozem) to desert.
4. Community biomass decreases along the series, but soil organic content and darkness of soil colour increase from brown forest soils to black chernozem, then decrease from the latter to the desert soil.
5. From forest to desert the nutrient stock in vegetation decreases, but relative nutrient content per unit of plant tissue increases.
6. The return of nutrients to soil by litter fall and root mortality is highest in the middle of the gradient, on prairie (nitrogen return $\text{g m}^{-2} \text{yr}^{-1}$ being 7, 12 and 3.5 for deciduous forest, grassland and semi desert).

4.11 SOIL TYPES OF INDIA

The soils vary in their characteristics, and properties. They are classified so as to establish the interrelationships between their characteristics. The major soil types of India are shown in Fig. 4.15. A brief description of major soil groups of India along with their modern nomenclature is given in Table 4.9. These soil groups have been extensively studied for their distribution and agricultural importance and briefly described as follows based on Biswas et al. (1985) and ICAR (2009):

Red Soils

The red soils of India, including red loams and yellow earths, extend over a large part of Tamil Nadu, Mysore, South-East Maharashtra and a tract along the eastern part of Madhya Pradesh to Chotanagpur and Orissa. In the north and north-east these extend into and include great part of the Santhal Parganas of Bihar; Birbhum, Bankura and Midnapur districts of West Bengal; Khasi, Jaintia, Garo and Naga Hills areas of Assam Meghalaya, Nagaland and Sikkim; Mirzapur, Jhansi, Banda and Hamirpur Districts of Uttar Pradesh; Baghelkhand Division of Madhya Pradesh and Aravallis and the eastern half of Rajasthan. Mostly, these soils occur in sub-humid to humid areas with some exception in Southern India. Due to tropical weathering, the soils are highly weathered and dominated by low activity clays. The main features of these soils are

1. Light in texture and porous and friable nature.
2. The absence of lime (*kankar*) and free carbonates.
3. These soils are generally neutral to acid in reaction, deficient in nitrogen, phosphorus, humus and perhaps lime.
4. The soils differ greatly in depth and fertility, and produce a large variety of crops under rainfed or irrigated conditions.
5. The soils have undergone excessive weathering and very low amount of decomposable mineral hornblende.
6. The red soils are rich in kaolinite type of clay mineral.

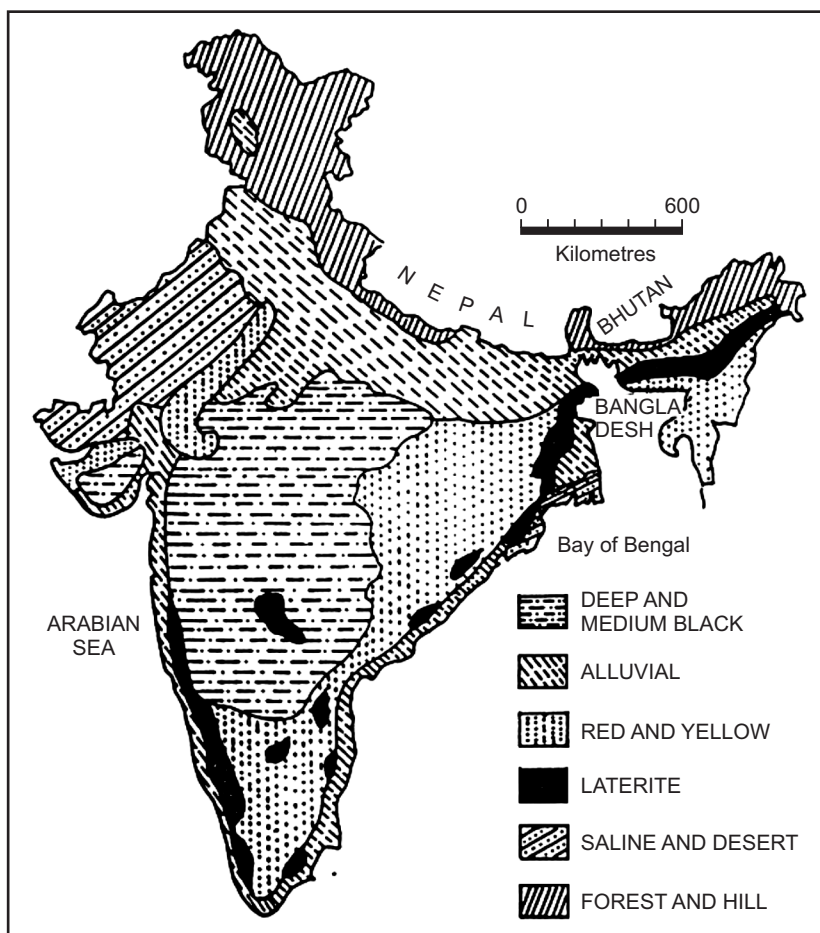


Fig. 4.15. The major Soil Types in India. (based on Biswas et al., 1985)

Table 4.9. Major Soil Types of India (based on Biswas et al., 1985; ICAR, 2009)

Soil groups	Soil orders	Main properties
Red soil	Alfisol, Inceptisol and Ultisol	The soils have a wide distribution in almost all the states. The red colour is due to diffusion of iron in the soil profile.
Laterites	Alfisol, Ultisol and Oxisol	These soils are composed of a mixture of hydrated oxides of aluminium and iron with small amounts of manganese oxide.
Black soil	Vertisol, Inceptisol and Entisol	These soils are dark in colour and contain a high proportion of calcium and magnesium carbonates and have a high degree of fertility
Alluvial soils	Entisol, Inceptisol and Alfisol	These are the largest and agriculturally most important group of soils, include calcareous alluvium, deltaic alluvium and coastal alluvium.
Desert soils	Entisol and Aridisol	These soils occur mostly in dry areas and have high content of quartz.
Forest and Hill soils	Alfisol	Soils high in organic matter.

Laterites

Laterite is a highly weathered material rich in secondary oxides of iron, aluminium or both. It may contain large amounts of quartz and kaolinites. The laterite is specially well-developed on the summits of Hills of Karnataka, Kerala, Central India, Madhya Pradesh, the Rajmahal Hills, the Eastern Ghats, in certain plains of Orissa, Maharashtra, Malabar and Assam. These are found to develop under fair amount of rainfall, and alternating wet and dry periods. Both the high-level and low-level laterites occur in Tamil Nadu. They are both *in situ* and sedimentary formations, and are found all along the West Coast and also in some parts of the East Coast, where the rainfall is heavy and humid climate prevails. On higher levels these soils are exceedingly thin and gravelly, but on lower levels and in the valleys they range from heavy loam to clays and produce good crops, particularly rice. In the laterites on lower elevations paddy is grown, while tea, cinchona, rubber and coffee are grown on those situated on high elevations. The main features of these soils are

1. The soils are rich in nutrients including organic matter.
2. The pH is generally low, particularly of the soils under tea (pH 3.5 – 4.0) and at higher elevation.
3. Soils are characterized by a compact to vesicular mass in the subsoils horizons composed essentially of mixture of the hydrated oxides of aluminium and iron.
4. Soils are deficient in potash, phosphorus and lime.
5. The colour of these soils is highly variable ; mostly reddish, reddish brown and yellowish brown.

Black soils

These soils cover a large area of 72.9 million ha (22.2%) throughout the southern half of the peninsula, the Deccan Plateau, greater part of Maharashtra State, western parts of Madhya Pradesh and Andhra Pradesh, and some parts of Tamil Nadu State, including the districts of Ramnad and Tinnevely. The typical black soils or *regur* (black cotton soil) occur in a large number of physiographic regions, each within a zone having its own combination of soils.

The main features of the black soils are:

1. The soil depth varies one to two or several feet deep.
2. The soil texture is loamy to clayey.
3. The soils crack heavily in summer, the cracks reaching up to more than three or four feet in depth, especially in the case of heavy clays.
4. The soils contain lime *kankar* and free carbonates (mostly CaCO_3) mixed with the soil at some depths.
5. These soils are often rich in montmorillonitic and beidllitic group of minerals, and are usually suitable for the cotton cultivation.
6. The soils are generally deficient in nitrogen, phosphorus and organic matter; potash and lime are usually sufficient.
7. The black soils are very heavy, contain 65-80 per cent of finer fractions, have high pH (8.5 – 9.0) and are rich in lime (5-7 per cent).
8. They have low permeability, high values of hygroscopic coefficient, pore-space, maximum water-holding capacity and true specific gravity.

Depending upon depth, black soils are classified as shallow, medium, and deep. The shallow black soils are entisols, medium black soils are predominantly inceptisols, and deep black soils are vertisols.

Alluvial soils

Various types of alluvium are classed as alluvial, e.g., calcareous soils, deltaic alluvium, and coastal soils. In these soils, there is wide variation; the main feature of these soils is that they are derived from the deposition by numerous tributaries of the Indus, the Ganga, and the Brahmaputra. The alluvial soils occur mainly in the southern, north-western and north-eastern parts of India: the Punjab, Uttar Pradesh, Bihar, West Bengal, parts of Assam, Orissa, and coastal regions of Southern India including the deltaic alluvium. These soils are the most fertile amongst the Indian soils. The whole of the Indo-Gangetic plain is, in this alluvial area.

These soils are very deep, deeper than 90 m at some places, and deficient in nitrogen and humus. They support a variety of crops, including rice, wheat and sugarcane. They may be subdivided into two broad groups, the old and the new; both are geological groupings. The former, locally called *bangar*, represents reddish-brown, sandy loams with increasing content of clay in the lower horizons; the latter, known as *khaddar*, represents the fairly coarse sand on the *chars* and banks of the river to the soils of very fine texture in the low-lying marshy tracts. The old alluvium reddish in colour, is deficient in nitrogen and humus, and occasionally in phosphoric acid.

The immature soil near the rivers is calcareous and light brown in colour with salt impregnation. On higher situations it becomes brown to deep brown in colour and is non-calcareous. *Kankar* beds are found in the soil. Most of the alluvial soils in Uttar Pradesh and Bihar are of the above pattern.

Desertsoils

A large part of the arid region in Rajasthan and part of Haryana, lying between the Indus and the Aravallis, is affected by desert conditions of recent geological origin. This part is covered under a mantle of blown sand which inhibits the growth of soils. Some of the desert soils contain high percentage of soluble salts, high pH, varying percentage of calcium carbonate and poor organic matter, the limiting factor being mainly water. The soils could be reclaimed if proper facilities for irrigation are available.

Forest and hill soils

Nearly 22% per cent of the total area of India is under forests. The formation of forest soils is mainly governed by the characteristic deposition of organic matter derived from the forest growth. Broadly two types of soil-formation may be recognized (1) soils formed under acid conditions with presence of acid humus and low base status; and (2) soils formed under slightly acid or neutral condition with high base status which is favourable for the formation of brown earths.

The soils of the hilly districts of Assam are of fine texture and reveal high content of organic matter and nitrogen, perhaps due to the virgin nature. Their chemical and mechanical composition show great variations.

In Uttar Pradesh the sub-Himalayan tract comprises three regions, viz., *bhabar* area immediately below the hills, *terai* and the plains. Four major groups, i.e., red loams, brown forest soils, podsoles and transitional podsoles have been observed in the Himalayan tract; of these, brown forest soils and podsoles are predominant. The *terai* area is characterized by excessive soil moisture and luxuriant vegetation. The soils are clay loam, loam and sandy loam. The loam may have high content or minor quantities of lime, or lack it altogether.

In the Himachal Pradesh, the typical soil profiles under deodar, spruce, blue pine and *chir* are related to the podsoles but have significant differences, probably due to the relatively high calcium content of the first layer. These differences are brought out in the high degree of saturation of various horizons.

Problem soils

Saline and alkali soils: In India, 9.38 million hectare land is salt affected out of which 5.5 million hectare land is saline and 3.88 million hectare is alkali/sodic soils (IAB, 2000). The distribution of saline and alkaline soils is extensive throughout India in all climatic zones. Many parts of the North, especially Bihar, Uttar Pradesh, Punjab, Haryana and Rajasthan, give rise to saline and alkaline efflorescence in the same way as the soils capping the Upper Tertiary rocks. There are many yet undecomposed mineral fragments in these alluvial clays and silts, which, on weathering, liberate sodium, magnesium and calcium salts. Large areas, once fertile and populous, have become impregnated with these salts locally known as *reh* or *kalkar* with highly deleterious effect on their suitability for cultivation.

Acid soils: In India about 6.5 million hectares of land area is covered by strongly acid soils ($\text{pH} < 5.5$). The acid soil areas occur in and around Bhawali (UP), Palampur, Kangra (HP), Baster and Jagdalpur (MP), Jorhat and Titabar (Assam), Ratnagiri (Maharashtra) and large areas in Ooty (Tamil Nadu) and Kutanad (Kerala). The acid soils suffer due to lack of calcium and magnesium and in some cases due to aluminium and iron toxicity.

Peaty and marshy: Peaty soils originate in humid regions as a result of accumulation of large amounts of organic matter in the soil. They may also contain considerable amounts of soluble salts. They are found in Kerala. They are generally submerged under water during the monsoon. After the rains, these are put under paddy cultivation. Soils are black, heavy and highly acidic, with pH as low as 3.9, and contain 10-40 per cent of organic matter. Their acidity is due to the decomposition of organic matter under anaerobic conditions, and no nitrification is possible. Sometimes the soils contain ferrous and aluminium sulphates. Marshy soils of this type are found in the coastal tracts of Orissa, the Sunderbans and some other places in West Bengal, central portion of North Bihar, Almora District of Uttar Pradesh, and south-east coast of Chennai.

CHAPTER

5

Fire Ecology and Disturbance

FIRE ECOLOGY

Almost all lands on earth have been affected by lightning caused fires and other natural fires caused by sparks from falling rocks, by volcanic activity or natural combustion. Human evolution is closely linked to the use of fire ever since the origin of humans in the savannas and woodlands of Africa (Schule, 1990). The prehistoric humans used fires to flush out game and to improve visibility within thickly vegetated habitats. Humans now employ control-burning as a common management practice for promoting the growth of economically important plant species and for various types of land use changes. The occurrence of natural wildfires (natural fire regimes) is a major environmental factor in the development and existence of many vegetation types of the world. Among the ecosystems most influenced by fire are the xeric shrub communities of the Mediterranean-type ecosystems (named as chaparral, fynbos, maqui and matorral), African savannas and tropical grasslands, the Brazilian Cerrado and prairie grasslands. Fire is a key factor in vegetation dynamics of prairies and stand structure of the Canadian Boreal forests. It is estimated that fire-prone ecosystems cover about 40% of the land surface (Chapin et al., 2002). In India, nearly 80% of fire incidence in forests, grasslands and agricultural ecosystems are human induced for facilitating grazing, shifting cultivation, and collection of minor forest products by local people. Fire is also a source of atmospheric pollution. On a global scale, forest fires in some years may account for a significant amount of CO₂ and the tropospheric ozone (Andreae, 1991).

5.1 TYPES OF FIRE

The different types of fires are as follows:

Surface fires

The surface fires are of natural occurrence in many types of ecosystems. The surface fire sweeps over the ground surface rapidly (Fig. 5.1 *a*). The flames consume litter, herbs, shrubs and seedlings, but in adult trees only lower parts of stems are burnt, that too usually superficially. The underground parts, such as root stocks and tubers are not harmed. The ground temperature in surface fires may vary from 90° to 120°C.

Crown fires

As the name indicates, crown fires occur in the crowns of trees and shrubs usually ignited by a surface fire; flames travel from one tree crown to another (Fig. 5.1 *b*). Lightning that hits tree crown and surface fires intensified by heavy accumulation of litter and coarse woody materials and dry windy weather lead to crown fires. They are common in coniferous forests and Chaparral type shrub lands. They are fastest spreading of fires and may consume the entire plant community and everything on ground and below it. It burns the surface litter and the organic matter in the A1 soil

horizon, and various types of organisms and regenerative structures.

Ground fires

They are flameless and subterranean fires, consuming organic matter beneath the surface litter of forest floor. Ground fires take place in peat bogs and boreal forests where organic matter of hundreds of years is accumulated. Though of very infrequent occurrence, ground fires can be very destructive, consuming most of the belowground parts and structures, such as, roots, tubers, rhizomes and all organic material of soil itself (Fig. 5.1 c). An ecosystem destroyed by ground fires may take tens to thousands of years to recover.

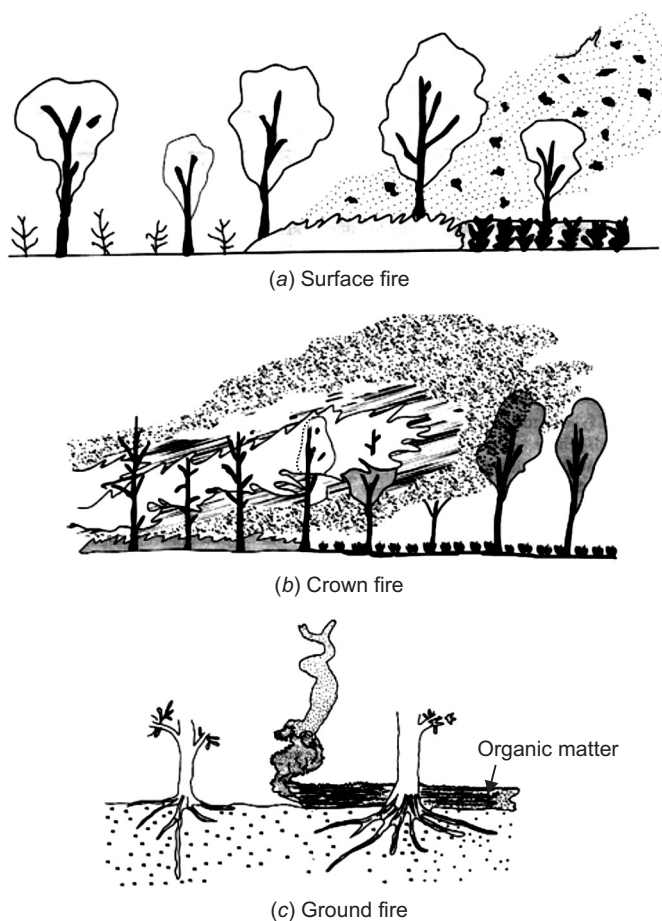


Fig. 5.1. Three kinds of fire. Ground fires occur when the organic matter layer on the ground is very thick.

5.2 MAIN TYPES OF VEGETATION FIRES

Both wildfires and land use fires can directly or indirectly cause immediate damages or have long-term environmental or humanitarian consequences. Wildfires are common in all vegetation zones under different climatic conditions. In the temperate and northern boreal forests wildfires occur regularly during the dry northern summers in North America, Eurasia and Mediterranean region.

The equatorial rain forests are usually too moist to allow the propagation of wildfires.

However, extreme droughts in association with forest exploitation periodically create conditions of flammability, fuel availability and fire spreads in the equatorial rain forests. Such events regularly occur in the forests of tropical South Asia in association with cyclic climate variability caused by the El Nino-Southern Oscillation (ENSO) phenomenon. The largest areas affected by uncontrolled wildfires in tropical forests are in the seasonal forest biomes (deciduous and semi-deciduous forests, sometimes also referred to as “monsoon” forests). Here, the fire occurs in short return intervals of 1 to 3 years. The tropical submontane coniferous forests (pine forests) are also subjected to regular fires.

In the last few years, catastrophic fires have devastated large areas of forests in Central America, Indonesia and Mexico, largely caused by agricultural burning. In the rain forests of South-East Asia and the tropical montane forests, lowland rain forests in Amazon, frequent fires may affect the ecosystem balance. Wildfires under extreme weather conditions are threatening the existence of both natural forests and forest plantations. The large scale fire incidences are having detrimental impacts on biodiversity, economies and human health and safety.

Savannas typically consist of a more or less continuous layer of grass with interspersed trees and shrubs, and cover large area of the land worldwide. The surface fuels in these ecosystems which are dominated by grasses and leaves shed during the dry season, are burned at intervals which may range from one to four years. The area of savannas potentially subjected to fire each year is up to several hundred millions of hectares, releasing about three times as much gas and particle emissions to the atmosphere as deforestation burning.

Two types of forest clearing for agricultural use are common, predominantly in the tropics, viz., shifting agriculture (also called slash and burn agriculture), where the land is allowed to return to forest vegetation after a relatively short period of use, and permanent removal of forest to be converted to grazing or crop lands. In shifting agriculture, which is practised by several hundred million people worldwide, the cleared areas are used for agriculture for a few years until yields decline, and then are abandoned and new areas cleared. The generally observed shortening of shifting agriculture cycles is increasingly associated with site degradation and makes this traditional land use technique one of the leading causes of global tropical deforestation.

Burning of agricultural residues

A substantial amount of agricultural residues, e.g., straw and stalks is disposed off by burning and it is a major source of atmospheric pollution, mainly in the tropics.

5.3 FIRE BEHAVIOUR

The manner in which fuels ignite, flames develop and wildfire spreads is collectively referred to as fire behaviour. In general terms, fires require a continuous and dry layer of “fuels” on the surface of the ground, air and sufficient heat in order to continue spreading (Fig. 5.2). The word fuel refers to live and dead vegetation that can contribute to combustion. Fuel components include items such as downed dead material, litter, duff, herbaceous vegetation, live foliage or live stems. Fine fuels may be conifer needles, hardwood leaves, grass, lichens, moss or small shrubs. The moisture content of the fuels is a very important factor in determining fire behaviour. A high moisture level will slow the fire because much of the heat energy is used to eliminate this moisture in the fuels. Live trees usually contain a great deal of moisture, dead trees very little. The effect of fuel size and arrangement is also important; larger fuels require more energy to ignite and burn, than finer fuels.

Slower the rate of decomposition of litter and higher the C : N ratio, the greater would be the inflammability of litter. The

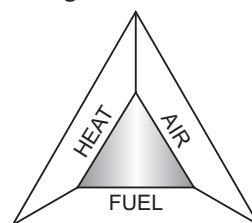


Fig. 5.2. Fire triangle to show basic requirements for the fires to occur.

higher surface to volume ratio for leaf litter (less compacted litter) means presence of a greater amount of oxygen. Thus, a loose litter catches fire more easily than the compacted litter having oxygen in lower amounts. Fuel composition affects fire intensity, oils and resins increase the heat yield of the reaction, whereas high content of minerals can reduce flammability. Many species of fire dependent vegetation have highly flammable foliage, e.g., the *Eucalyptus* trees of Australia with their high oil content burn rapidly.

The impact of fire depends on its intensity (I), which depends on the heat yield H (K cal g^{-1} dry matter), fuel availability W ($\text{g dry matter m}^{-2}$) and rate of fire spread R (m sec^{-1}) (Wakimoto, 1977). This can be expressed as follows:

$$I = H \times W \times R$$

The fire behaviour is influenced by weather condition, topography and fuel properties and amount of fuel. Weather is the state of the atmosphere and its changing nature. Fire weather includes all those elements of weather that affect fires, such as wind, temperature, moisture, humidity and precipitation. Climate and vegetative condition combine to make fire season, a term which is generally used in relation to wildfires. The humid condition would check the fire intensity, and warm, dry and windy conditions would intensify it. Topographic features such as streams and lakes can create fire breaks and influence the spread of fire across the landscape. The intensity of fire is also affected by topography, e.g., a fire ignited on the top of slope is likely to spread slowly as it burns downhill, whereas a fire at the bottom of slope spreads rapidly uphill.

Oxygen is an essential requirement and must be available in sufficient quantities to sustain combustion. Again, weather plays an important role, as wind can bring a fresh supply of oxygen to the fire and also push the fire toward a new fuel source. An external source of heat is usually required to start a fire. In forest fires, this can be a natural source, such as a lightning strike, or it can be caused by people, with sources ranging from campfires, to spark from machinery. The local cattle grazers often set ablaze grazing areas in the hope of getting new shoots. The head loaders destroy vegetation to create pathway through the forests. The encroachers set fire to forest in order to clear the land for agricultural purpose. The careless tourists throwing away lighted matches and cigarette butts start fire.

5.4 FIRE REGIMES

Fire regime refers to the nature of fire occurring over long periods of time that characterizes an ecosystem. Fire regimes are the patterns of wildland fires that include factors like frequency, extent, intensity, type, and season. Regimes vary by ecosystem because of differences in composition and structure. Humans have altered many aspects of natural fire regimes over time. Altered fire regimes, too much or too little of fire at the wrong time and wrong place indicate that ecosystem is out of balance. Altered fire regimes can have devastating impacts on natural resources including water, soil organic matter, and flora and fauna. Understanding natural fire regimes should lead to the most appropriate resource management policies.

Fire regime may be described considering frequency (time interval) of fire, its scale and intensity; fire intensity being fire severity. Heinselman's (1981) classification of natural fire, which is widely accepted, is given below:

Class 0 – No or very little natural fires.

Class 1 – Infrequent light surface fires, more than 25 years return interval.

Class 2 – Frequent light surface fires, 1-25 yr return interval .

Class 3 – Infrequent severe surface fires.

Class 4 – Short return interval crown fires.

Class 5 – Long return interval crown fires.

Class 6 – Very long return interval crown fires.

The fire regimes on the basis of historical records, fire return interval and severity are given in Table 5.1.

Table 5.1. The Five historical natural fires regime groups (Hardy et al., 1988).

Fire Regime Groups	Frequency (Fire return interval)	Severity
I	< 35 years	Low severity
II	< 35 years	Stand replacement fire
III	35-100 + years	Mixed severity fire
IV	35-100 + years	Stand replacement fire
V	> 200 years	Stand replacement fire

Regional fire potential strongly depends on regional climate through the interaction of climatic conditions, vegetation characteristics and fire (Fig. 5.3). Any kind of climate change affects the ignition and spread of fire. The physical and chemical properties of fuel affect flammability, which in turn varies with climate. Increase in the frequency of fire and severity of drought and extension of length of the fire season results in more severe fires. Wind is a major factor determining fire behaviour. The composition of plant community affects the spread of fire as some species are more flammable than others.

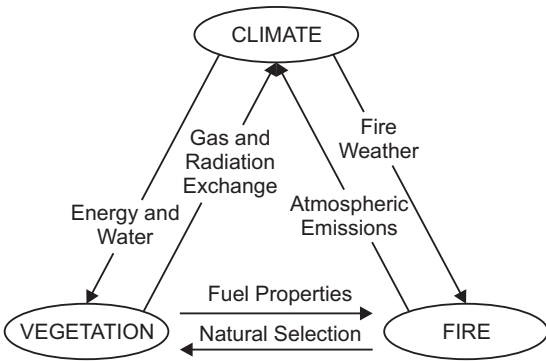


Fig. 5.3. Climate, vegetation, and fire are dynamically coupled. Any changed in one factor will lead to changes in the other two (modified from Brown et al., 2000).

The vulnerability of the forests to fire varies from place to place depending upon the type of vegetation and the climate. In India, normal and peak fire seasons usually occur during January to June. In the plains of Northern and Central India, most of the forest fires occur between February and June. In the Himalayan region, fires are common in May and June. In the hills of northern India most of the fires occur during April and June. In the southern part of the country, fire season extends from January to May.

5.5 EFFECT OF FIRE ON AIR

Complete combustion of biomass results in the production of CO₂ and H₂O. Combustion is rarely complete; a variety of chemicals such as carbon monoxide, methane, nitrous oxide, ammonia, trace hydrocarbons and trace gases (including volatile organic compounds) are produced during vegetation fire (Crutzen and Goldammer, 1993). A number of studies have shown that due to large scale fires, ozone, carbon monoxide and other trace gases are released into the atmosphere causing changes in atmospheric chemistry (see Andreae and Merlet, 2001). Burning of forests also destroys an important sink for atmospheric carbon. Fire also has an impact on local and regional atmospheric conditions as well as global atmospheric chemistry.

Particulate fire emissions are composed of organic carbon and black carbon. Due to large scale fires, large amount of fine particulate matter (smoke aerosols) is formed, which affects radiation budget and the microphysical properties of the atmosphere (Crutzen and Andreae, 1990). During

1997 episode of fire, large areas in South East Asia were affected by smoke-haze pollution due to burning of vegetation in Indonesian islands, Kalimantan and Sumatra (Heil and Goldammer, 2001).

5.6 EFFECT OF FIRE ON SOIL

Fire affects physical, chemical and biotic properties of soil. In general, soil temperatures are raised, some soil nutrients and organic matter are lost, and soil biota are changed. Some of these changes are given in Table 5.2.

Table 5.2. Comparison of certain soil properties of a forest in western Washington before and immediately after a severe fire (from Issac and Hopkins, 1937). Soil values are averages of 0-7 cm and 7-15 cm depths. All values in percentages are in relation to dry weight of soil.

Property	Before fire		Immediately after fire	
	Ao	Soil	Ao	Soil
Organic matter (%)	88.5	4.7	9.7	3.3
Total N (%)	0.9	0.1	0.3	0.1
C:N ratio	57	26.5	17	18
Field capacity (%)	190	59	60	52.5
pH	4.9	4.9	7.6	5.9

Organic matter and nutrients

Nitrogen and potassium compounds are volatilized, released and lost by distillation under the influence of burning. Nitrogen loss is little up to 200°C, but at 700°C, 60% may be lost. On the other hand, a moderate rise in temperature helps the release of nutrients like calcium, sodium and magnesium, thereby enhancing their recycling into biota (De Bano et al., 1977). Thus, though fire reduces the soil nutrient pool, it can increase nutrient availability to plants. Some of the loss of nitrogen is compensated by a better growth of nitrogen-fixing plants which often prevail in the post-burn stage (e.g., *Alnus*, *Robinia*, *Lespedeza*, and *Desmodium*). Substantial amounts of sodium, potassium, iron, zinc are also volatilized at high temperature. The loss of organic matter results in loss of the cation exchange capacity of soil and capacity to hold nutrients.

Soil moisture

Fires lead to decline in soil moisture as a consequence of the destruction of organic matter and resultant increase in soil bulk density. Water infiltration through the burnt soil declines as soil pores are reduced, resulting in increases of soil run-off and in a drier soil. Furthermore, the ash resulting from fire and the formation of a charred crust plug the micropores.

The presence of hydrophobic substances prevents the movement of water into the soil and this condition is called nonwettability. As plants decompose, the hydrophobic substances accumulate on the soil surface in the years between burns. It is a characteristic of the forests of *Pinus ponderosa*,

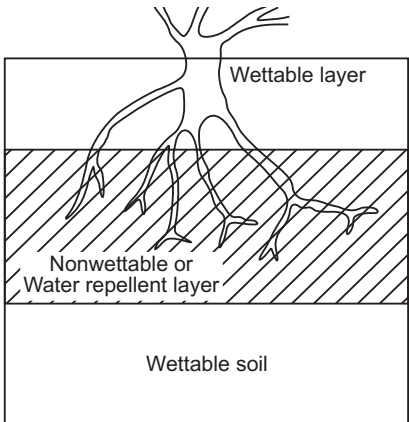


Fig. 5.4. Establishment of water repellent layer in soil column after fire burn.

Sequoiadendron giganteum, and *Abies concolor* and chaparral community. Depending on several factors a nonwettable layer, referred to as water repellent zone is established at a depth in soil column (Fig. 5.4), restricting water infiltration into deeper soils.

Fire can also influence the hydrological cycle by transferring heat into the soil. Intensive fire affects water holding capacity and erosivity in the Mediterranean–type ecosystems.

Soil pH

Subsequent to burning, soil pH increases because of the consumption of litter which is acidic and other factors, such as a greater loss of elements that form anions (e.g., N, P and Cl) than of those elements which form cations (e.g., Ca, K and Mg), and conversion of the soluble oxides of alkali earths into carbonates.

Soil temperature

Fire causes both the immediate temperature rise as well as long-term temperature increases. The immediate rise of soil temperature that follows burning improves conditions for plant growth and nutrient release in cold areas. Fire also causes a long-term rise in soil temperature. First, by darkening soil surface fire promotes absorption of sunlight, and second, by consuming the surface organic layer, it promotes transfer of heat to the mineral soil. However, being a good insulator, soil does not allow temperature to rise even just below the surface (Table 5.3). Soil moisture plays most significant role in restricting the rise of temperature, which can be above 100°C only when all soil water has evaporated. The evaporation requires large amount of heat. The ash resulting from fire acts as a good insulator.

Table 5.3. Typical temperatures (°C) at different soil depths during different intensities of burn in chaparral (from De Bano et al., 1977). Temperature values are the maximum values which generally reached after 5 minutes of burn at soil surface (0 cm) after 10-15 min. at 2.5 cm soil depth and after about 20 minutes at 5 cm soil depth.

Fire intensity	Soil depth (cm)		
	0	2.5	5
Intense burn	690	200	80
Moderate burn	410	150	70
Light burn	240	100	60

Nutrient losses by soil erosion and mass movement after fire

One of the most adverse effects of fire on soil subsystem is the loss that results after burning due to soil erosion. In three watersheds studied in the Cascade Range of Washington, Helvey et al., (1985) have shown that after a wildfire the annual sediment yields increased 14-28 fold of the pre-fire level. Losses of sediment-borne nutrients during five post-fire years averaged 14 times greater for available P, 26 times greater for Ca and Mg, and 38 times for K than average values during the 4 years prior to fire. However, in managed fires these losses are much smaller.

Soil biota

Fire affects soil biota, both by directly killing them, and by reducing their food base. The increase in pH after burning is expected to favour bacterial growth relative to fungal growth (Ahlgren, 1974). In many situations, the initial decline in microflora is followed by increases in their

populations, as fire by raising pH and nutrient release improves conditions for their growth. Grass growing in burned areas is often more lush and green and contains more protein than in unburned areas, possibly because of an enhanced population of microbes that help increase in nutrient availability to the plant. Frequent burning, however, may reduce microbial populations. Among the Central Himalayan forests, populations of bacteria, fungi, actinomycetes and microarthropods are markedly smaller in frequently burned chir pine (*Pinus roxburghii*) forest than in others (Table 5.4).

Table 5.4. Comparison of decomposers associated with leaf litters of chir pine (*Pinus roxburghii*) and broadleaf forests in Central Himalaya (after Singh and Singh, 1992).

Population ^a	Chir pine forest	Several broadleaf forests
Fungi (10^4 g^{-1})	21.02	25.93-32.42
Bacteria (10^6 g^{-1})	31.18	35.23-51.95
Actinomycetes (10^6 g^{-1})	3.21	4.81-7.78
Microarthropods (individuals m^{-2})	1171	2259-7413
Community coefficient of fungi (%) ^b	21.4	28.0-33.3

a = Average of 12 months;

b = Average of coefficient values between fungal communities of a given leaf litter and leaf litters of the other forests occurring between 300-2200 m elevations.

5.7 EFFECT OF FIRE ON PLANTS AND VEGETATION

Fires are commonly very patchy, killing most trees and other plants in some areas and few or none in others. Recurrent fires favour herbaceous plants over woody plants. Thick barked and root sprouting woody plants are favoured over less fire adapted plants. At a landscape level a mosaic of vegetation in different stages of succession results.

Fire and vegetation interaction

Wright and Heinzelman (1973) (see in Pyne et al., 1996) have summarised the effect of fire on plant species and communities.

1. Fire alters seed beds by removing litter and humus and creating bare soil. The bare soil patches may promote establishment of species with small seeds, such as giant sequoias and Douglas fir.
2. Fire temporarily reduces competition for light, moisture and nutrients, thereby selectively favouring some species.
3. Fire frequency (return interval of fire) influences community composition and successional stages.
4. Fire may trigger the release of seeds in species like lodgepole pine, jack pine, paper birch and *Eucalyptus*.
5. Fire may stimulate flowering and fruiting of some shrubs and herbs.
6. Fire stimulates vegetative reproduction of some woody and herbaceous species when overstorey is reduced (sprouting from root collar is shown by some oaks, maples and alders, and root suckering is common in aspen).
7. Fire creates patchy condition, as it generally eliminates only a part of community.
8. Fire frequently regulates susceptibility of forests to blow downs.
9. In prairie grasslands, fire is essential, almost annually, for the normal functioning of the ecosystem. Fire prevents the invasion of woody plants.
10. The Californian chaparral depends on a fire initiated cycle for nutrient regeneration and

reducing the amount of plant debris and allelochemicals.

11. Indirect effects occur due to fire. For example, sudden and drastic modification of habitat is the major short-term effect. Increase in sediment input is detrimental to the stream's biotic life. In less intense fire, diversity of habitat is increased. Long-term effects include increase in animals and plants of early successional nature.

Plant adaptations to fire

In fire prone ecosystems such as chaparral, lodge pole pine and *Eucalyptus* forests in Australia, plants exhibit adaptive traits that enable them to cope with fire.

Enhanced seed setting and reproduction: Enhanced reproduction due to better seed setting, as in bermuda grass (*Cynodon dactylon*), following a fire and fire-induced flowering are associated with fire adaptation. These characters may give competitive advantage to species over other species which do not possess such characters.

Serotiny: In the condition called serotiny (meaning late to open) the closed cones remain on the tree until the heat of a fire opens them (e.g., at 45-50°C in the lodge pole pine and at 200°C for knob cone pine). Fire promotes the opening of cones by burning resin of cones. In conifers with serotinous cones (e.g., *Pinus banksiana*, *P. contorta*, *P. muricata*, *P. attenuata*, *Cupressus macrocarpa*, *Sequoiadendron giganteum*) viable seeds are retained for several decades, and seeds accumulated over several years are dispersed when forest is burnt, allowing species to colonise the burnt site (Biswell, 1989).

Improved seed germination: Certain species of fire-adapted grasslands have hard seed coats requiring scarification to germinate (e.g., legumes, *Astragalus* and *Trifolium*). Fire ruptures or splits seed coat, making it permeable to oxygen and water, which in turn enables seeds to germinate. Grassland seeds are also known to tolerate high temperatures. The seeds of *Bromus mollis* can survive more than 200°C for two minutes heating (Daubenmire, 1968a) and some others up to 370°C for short periods (Biswell, 1974), and some *Avena* seeds germinate even after being charred by fire. Heat shock-stimulated germination is widespread in families like Fabaceae, Rhamnaceae, Convolvulaceae and Sterculiaceae. There are species which show higher percentage germination when exposed even to fire-generated smoke (see Khurana and Singh, 2001).

Long seed viability: In fire adapted communities like chaparral seeds remain dormant in soil and litter for decades and centuries between periods of two fire incidents. For example, seeds of snowbrush (*Ceanothus velutinus*) may remain viable in forest litter for up to 575 years (Zavitkovski and Newton, 1968). They germinate as fire consumes litter.

Fire-induced growth: The longleaf pine seedlings remain at the "grass stage" for several years. This is a seedling stage characterised by little height growth, more allocation of photosynthate to roots than stem, presence of the terminal bud close to the ground, protected from fire damage by the densely packed needles low in oxygen and rich in tissue water content (Fig. 5.5). When burned, the carbohydrates stored in roots are mobilised for height growth enough to lift the terminal bud beyond heat of the next fire.

Bud protection and resprouting: Resprouting of perennial buds protected at the soil surface is an important mechanism of survival of hemicryptophytic plants in grasslands experiencing frequent burning. For example, *Aristida stricta* and *Sporobolus curtisii*

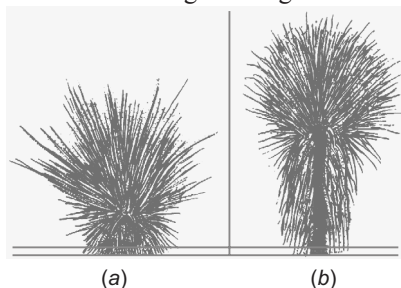


Fig. 5.5. Effect of fire on height growth on long-leaf pine seedlings. (a) Grass stage of seedling, (b) Tall seedling resulting from fire-induced growth (derived from Pyne et al., 1996).

have buds buried 40 mm or so below the soil. In the chir pine (*Pinus roxburghii*) forest of the Central Himalaya periodic fire has promoted growth of grasses, but have eliminated almost entire shrub flora. However, many shrubs and trees of chaparral communities have buds at the base which may survive fire. Their dormant buds resprout subsequent to the burning of shoots.

The sprouting may be from the stem, called ‘*epicormic*’, or may be from basal burl or ‘*lignotuber*’, which are conspicuous turnip-shaped or tabular swellings of the axis (reaching up to 4 m across) entirely or mostly below the ground surface, which bear hidden buds having capacity to produce new shoots quickly because of large food reserves (Daubenmire, 1959a). The stem buds (epicormic) have bark protection and lignotubers soil protection. The common chamise (*Adenostoma fasciculatum*) and *Eucalyptus*, for example, bear lignotubers and species of *Arctostaphylos* bear stem buds.

Fire-resistant bark : Of all the protective structures of trees, the bark which is an insulating structure is the most important. The thicker the bark, particularly at the base, the more resistant is a species to fire. It is important how rapidly bark is thickened, for young trees are more vulnerable to fire than old trees. Lodge pole pine, a fire resistant species, does not have significantly thicker bark at old age than the other pines, but when young its bark is markedly thicker than other pines. Relationship between bark thickness and fire-resistance is seen also in the Central Himalayan species.

Other adaptations: High crowns, open stands, large buds and deep roots are also found to be related to fire resistance, e.g., the western larch (*Larix occidentalis*) which is an extremely fire-resistant species has unusually thick bark at the base and high crowns, and forms fairly open stands. A large bud is more fire-resistant than a small bud, as relatively more heat would be required to raise their temperature to a lethal level.

Fire resilience in plant community

The species composition of community is determined by the successive birth and death of individuals in a community, determining vegetation physiognomy and structure in the temperate and boreal ecosystems (Brown et al., 2000). The severity of fire, which depends upon the quantity and quality of biomass present and weather conditions at the time of fire, has a major influence on plant survivorship and regeneration (Fig 5.6). Short fire cycles favour species that endure fire by juvenile sprouting or evade fire by storing seeds in the soil, invading from offsite and having short life cycles.

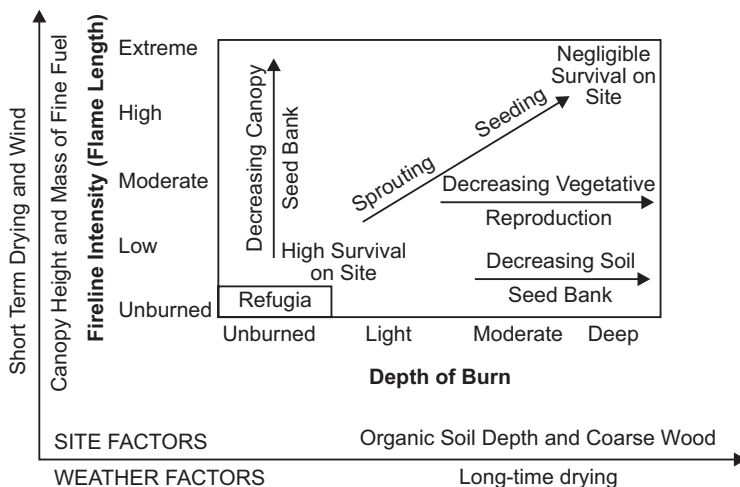


Fig. 5.6. Post fire vegetation recovery varies with fire frequency (redrawn from Ryan and Noste, 1985).

The Intermediate fire cycle favours species that resist fire when mature or evade fire by storing seeds in the canopy; sprouting and offsite colonization does occur. Long fire cycles favour species that typically avoid fire. These species exhibit low resistance to fire injury and regenerate predominantly by seed. If the period of fire return interval is less than the time of sexual maturity, then the species is not able to complete the life cycle.

Ability to bear seeds at an early age (precocious seed production) and resprouting of buds after burn may give advantage to species over others to colonise a site in the post-burn stage. Some species of communities that are frequently burnt possess fire-enhancing traits, such as fire-branching, dense canopies with a high proportion of dead wood, high volatile content and loose and flaky bark. These highly flammable species are suggested to have competitive advantage over the less flammable species (damp species) because of the greater success of their offspring in the post-burnt stage.

Effects of fire on forest ecosystems in India

In India, forest fires are significant and one of the increasing contributory factors in the degradation of existing forest resources (Saigal, 1999). It is estimated that proportion of forest areas, prone to forest fires annually ranges from 33% in some of the states to over 90% in others (MoEF 1991). In India there are very few cases of fire due to natural causes. The majority of the forest fires in the country are human caused. It is widely acknowledged that most of these fires are caused by the people deliberately and have a close relationship to their socio economic conditions. India has about 1.7 million hectares of productive coniferous forests with valuable timber and pulpwood species such as fir (*Abies* spp.), spruce (*Picea smithiana*), deodar (*Cedrus deodara*), kail and chir pine (*Pinus wallichiana* and *P. roxburghii*). People set surface fire in pine forests so that the litter layer is destroyed and grazing is facilitated. In forested regions of India, the deciduous forests rich in bio diversity also get damaged by forest fires.

A characteristic feature of the Nilgiris biosphere is the occurrence of sholas (montane forests) above 1500 m. They are found in patches in hollows and sheltered folds surrounded by rolling downs in the Anamalais, Nilgiri and Palni hills and the high ranges of Kerala and Karnataka. Meher-Homji (1967) opines that two adverse factors, viz., frost during the cold season, and fire during the drier months, prevent the spread of the sholas in landscape dominated by the grasslands. In the Nilgiri biosphere reserve, forest fires affect the natural regeneration (Srivastava, 2002). The recurrent fire set by grazers during summer has adversely affected the floristic composition and biodiversity. Forest fires either natural or man-made also increase erosion and alter the physical and chemical properties of the soil. The soluble ash is washed away in the next rain. Fires can also make trees more susceptible to insect attack. Severe burns such as those caused by wildfire can result in nearly complete destruction of organic matter and bring about changes in the physical, chemical and biological properties of the upper layers of mineral soil

5.8 EFFECT ON FAUNA

Animals with limited mobility get severe injury during fire, for example, mortality of ground nesting birds having eggs or nestlings. Soil fauna that lay eggs or have immature individuals in litter are vulnerable to fire. Surface insects may escape by flying or moving down tunnels or getting protection in bark. Rodents that nest underground may survive as soil is a good insulator. Large mammals and birds are able to move away.

Fires affect animals mainly through effects on their habitat, which influences faunal populations and communities (Smith and Kapler, 2000). Fires often cause short-term increases in wildlife foods that contribute to increases in populations of herbivores. The effect of fire on small

carnivore mammalian population could be both positive and negative. Large carnivore and omnivore species are characterized by large home range; their populations change very little in response to fire. Many animal-fire studies have shown that animal communities usually reorganize in response to fire; causing increases in some species and decreases in another species.

5.9 PRACTICAL VALUE OF FIRE

Fire control has favourable effects on ecosystems, where they are a natural phenomenon. For example, most grasslands, and some types of forests (some *Eucalyptus* forests in Australia and pine forests of South eastern United States) are adapted to low intensity fires at frequent intervals. Their vegetation changes due to suppression of fire because of influx of fire-intolerant species and accumulation of large amount of fuel. Any fire that gets started in these systems is very intense. The fire in Yellowstone National Park in 1988 occurred due to suppression of fire for many years.

Knowledge of these fire-dependent communities can be used profitably for the ecosystem management. Some of the management uses are given below:

(1) To promote desired species

In a situation where an economically useful species are fire-resistant and their associates of lesser economic value are fire-susceptible, periodic fire can be used to promote the growth of desired species. An original sagebrush-grass community of Western North America comprises a dense cover of perennial grasses among which are scattered shrubs, particularly, *Artemisia tridentata* (sagebrush). As grazing pressure increases the palatable grasses are reduced and the unpalatable sagebrush increases in number. In contrast to grazing, fire kills the shrub, sagebrush but not grasses which resprout from protected buds.

In Central Himalaya, the foresters have been using fire for more than a century to promote chir pine (*Pinus roxburghii*) at the cost of banj oak (*Quercus leucotrichophora*), because of its commercial value.

(2) To maintain species diversity and diverse landuse

Bell et al. (1984) have shown that periodical burning is necessary to check the loss of many endangered species in south Western Australia where over two-thirds of the 4,000 or so angiosperms are endemic and nearly 2,000 species are listed as poorly known, possibly rare. Based on the study of Kwongan (a fire-dependent shrub vegetation of high species diversity) they have given a recommendation schedule of burning frequency for the maintenance of diversity. For example, vegetation close to crop fields is suggested to be burnt in shorter time intervals so that the shrubs do not invade crop fields, while that away from them could be burnt after more than 20 years, so that shrubs are able to grow and flower and serve the purpose of bee keepers.

In longleaf pine (*Pinus palustris*)-wiregrass (*Aristida stricta*) ecosystem of the Southern U.S.A., lightning directly strikes into ground (not crown), consuming the highly flammable dense cover of wiregrass and pine needles. This leads to the periodical establishment of a rich herbaceous flora which supports a fauna unique to the region. Here, the lightning striking ground is the keystone factor (Rebecca et al., 1992). On the basis of dynamic global-vegetation model, Bond et al. (2005) have shown that a majority of warm season grasslands and savannas, especially in South America and Africa, have the potential climate for formation of forests. In the forested regions of both tropical and temperate regions, the reduction of trees by fire has facilitated the spread of grassland systems and resulted in the evolution of the most biodiverse ecosystems in the world. Fires have been the most important factor for the origin of C₄ grass-dominated floras between 6 million and 8 million years ago in the tropics, which quickly spread around the globe (Bond et al., 2005).

(3) To improve forage quality

Browsers like deer, moose and others depend heavily on browse (leafy twigs) of shrubs and tree seedlings. When old, they can grow so tall that their leaves and twigs become out of reach and also less nutritious. Burning is required not only to bring the shrubs and tree seedlings within the reach of browsers, but also to induce the re-sprouts yielding young and nutritious leafy twigs. Similarly, grasslands are burnt to obtain young, succulent and nutritious herbage.

(4) To remove organic debris as preventive measure

The greater the accumulation of organic matter (fuel) on the ground, the more devastating a fire is. In order to prevent the disaster caused by severe fires frequent controlled burning is applied to keep the forest fuel in check. This practice may save some species which are not adapted to severe fires.

(5) To improve seedling growth

In accumulated litter, roots of germinated seeds may remain suspended and die because of the lack of root contact with soil. The removal of litter by fire may expose mineral soil, making it easy for the root of seedlings to come in contact with soil.

(6) To control insects and diseases

Periodic burning of ground litter may remove harmful insects and fungi.

Restoration of Wildlife habitat

The Kirtland's Warbler (*Dendroica kirtlandii*) is dependent on fire-maintained landscapes in its breeding grounds and on portions of its winter range. The world's population of this endangered species has ranged from 167 to 905 pairs in the last 30 years. It breeds only in Northern Michigan and winters almost exclusively in the Bahamas Archipelago. Kirtland's warbler habitat requirements are quite specific; nest only in stands of young jack pine (5-20 feet tall) and winter in fire-maintained tropical scrub communities of Caribbean pine (*Pinus caribaea*). The suppression of fire in fire-dependent jack pine forests resulted in near extinction of Kirtland's warblers because of the shortage of young pine stands (Probst and Weinrich, 1993). The U.S. Forest service began a fire management programme to restore the Kirtland's Warbler habitat needs.

5.10 DETECTING FOREST FIRE DISTURBANCE IN TROPICAL FORESTS

Fire plays a significant role in tropical ecosystems and is a major evolutionary force shaping the structure and composition of forests worldwide (Bond and Keeley, 2005). The geographical extent of many tropical rainforests is limited by fire as well as edaphic and climatic factors (Bowman, 2000). Most tropical rainforest trees are poorly adapted to fire stress, and even low-intensity forest wildfires can lead to the mortality of more than 40% of stems 10 cm and above in diameter at breast height (Barlow and Peres, 2008). The Amazonian forest is a large, ecologically intact area of ancient (or primary) forest. However, the Brazilian Amazon is becoming increasingly fragmented due to human activities, building of roads within the forest, and selective logging of the mahogany (*Swietenia macrophylla* King) trees, that are widely scattered in the forest. This makes the remaining forest vulnerable to increased fires and to climatic change. An increase in the severity and frequency of forest fires is a significant negative outcome of industrial logging in tropical forests, along with the logging operations themselves (Nepstad et al., 1999).

Satellite-based deforestation monitoring is an essential tool in studies of human effects on tropical forests because it documents the most extreme form of landuse, over large areas, and at low cost (Nepstad et al., 1999). Amazon deforestation has been measured by remote sensing for three decades (Peres et al., 2006). However, selective logging has been mostly invisible to satellites.

Asner et al. (2005) developed a large-scale, high-resolution, automated remote-sensing analysis of selective logging in the top five timber-producing states of the Brazilian Amazon. Their analysis of 480 Landsat ETMC scenes, which detected small logging gaps (10.5 ha) and associated understorey disturbance caused along skid trails and logging decks, shows that an area of 12075–19823 km² was selectively logged each year, which is 60 to 123% of previously reported deforestation area. Up to 1200 square kilometers per year of logging were observed on conservation lands. Each year, 27 million to 50 million cubic meters of wood were extracted, and a gross flux of ~0.1 billion metric tons of carbon could be released to the atmosphere by logging (Asner et al., 2005).

Souza et al. (2005) field tested another image-processing technique that relies on a gap fraction index (the Normalized Difference Fraction Index, NDFI) combined with the green and non-photosynthetic vegetation, soil and shade fractions derived from SMA to enhance the detection of log landings and skid trails, the spatial signature of selective logging (De Wasseige and Defourny 2004). An assessment of this method based on aerial digital videography provided a high level of mapping classification accuracy (94%) of forest canopy damage. In their study area of 30000 km², 65% was classified as intact forest, 22% as non-forest and 13% as canopy areas that had been damaged by logging and burning (Souza et al., 2005). Thus, remote sensing techniques could play an important role for preventing fires from spreading further into new Amazonian forest frontiers as well as for climate change mitigation.

5.11 MEGA-FIRES

Very large wildland fires are a part of the planet's history (Pyne, 1995), but the onset of high-impact mega-fires signals a disturbing new reality. Large-scale (e.g., >10⁵ ha) fires of high intensity that influence climate through emissions of CO₂, and production of aerosol particles have been referred to as mega-fires (Adams, 2013). Global warming, xericness and forest management leading to increased fuel-load have exacerbated the problem of mega-fires. Most of the mega-fires are human-initiated either accidentally or deliberately. The mega-fires are high-impact wildfires that adversely affect forested ecosystems already under strain in many places around the world. They are a global phenomenon, tied to internationally significant issues like public safety, deforestation, forest degradation, black carbon emissions, and carbon sequestration (Williams, 2013). The “dramatic increase in wildfire size, frequency and related environmental impacts” in 1997 and 1998 for countries such as Indonesia, Brazil, Mexico, Canada, USA, France, Turkey, Greece and Italy as well as the Russian Federation and China's northeastern Inner Mongolia have been described (Levine et al., 1999; Adams, 2013). The Yellowstone and Canyon Creek have been cited for the ‘never-before-seen’ massive wildfires.

Tropical forest fires at scales of >10⁶ ha produce truly global impacts by increasing global atmospheric concentrations of CO₂ (Page et al., 2002); smoke plumes and hazes affect whole regions across many countries. In South East Asia alone, tens of millions of people were exposed to high levels of fire-produced gases and particulates for weeks at a time. The wildfire effects on climate are: (i) cooling of the earth's atmosphere from more numerous and smaller droplets and increasing cloud albedo (Adams, 2013); (ii) increased absorption of radiation and atmospheric and surface temperatures as a result of black carbon (e.g., ash and soot) of tropical origin released in the atmosphere and deposited on surfaces (Ramanathan and Carmichael, 2008); (iii) increased albedo and biophysical cooling due to the nature of re-growth after high intensity fires (Beck et al., 2011). Adams (2013) has discussed mega-fires with emphasis on fire-related ‘tipping points’ (leading to abrupt changes) in major biomes and their consequences for dominant woody species. The effects of large scale (>10⁵ ha), high intensity fires on carbon and water as two of the most important ecosystem services provided by forests, and mitigation of mega-fires and their impacts on vegetation type (and biodiversity), carbon and water with special reference to Southern Australia, Western USA have been analysed (Adams, 2013). It is argued that for managing the mega-fires, it is important to incorporate them into land management, policy and planning at state, regional and national levels.

DISTURBANCE

Disturbance is a temporary change in average environmental conditions that causes a pronounced change in natural ecosystems; it is a physical disruption of the structure of a system which may lead to alteration in the functioning of the system. Disturbance has been recognized as a structuring force in ecological communities. Disturbances can be abiotic (e.g., drought, fire, wind and geomorphological events), or biotic (e.g., insects, disease or grazing) or directly caused by humans (e.g., logging, burning, introduction of exotic species, and mining). White and Pickett (1985) define disturbance as a relatively discrete event in time that causes abrupt change in ecosystems, community, or population structure and changes resources, substrate availability, or the physical environment. In forest ecosystems, both natural and human induced disturbances influence their composition, structure and functional processes (Dale et al., 2001). Disturbance plays a key role in ecosystem and landscape dynamics; specifically, in initiating secondary succession and maintaining ecosystems in a constant state of flux. Disturbance frequently leads to killing, displacement or damaging one or more individuals or colonies which creates an opportunity for new individuals to become established. After disturbance, ecosystems undergo *succession*, a directional change in ecosystem structure and functioning resulting from biotically driven changes in resource supply. Natural disturbances play a key role in ecosystem dynamics and are important factors for sustainable forest ecosystem management (Seidl et al., 2011).

5.12 DISTURBANCE REGIMES

The impact of disturbance on ecosystem processes depends on its severity, frequency, type, size, timing, and intensity. The intensity, size and frequency of disturbance are often correlated. In general, communities subject to disturbances recover slowly to their original state through the process of succession (Fig. 5.7). The severe disturbance impacting a community at the same time or in quick succession can push the community to altered state (Fig. 5.7) (Paine et al., 1998). In natural communities, small disturbances of low intensity are much more frequent than large intense disturbances. These characteristics differ depending upon the type of community and their distribution.

A combination of frequency, size, and type of disturbances refer to the disturbance regime or disturbance pattern. The disturbance regime is often a fundamental characteristic of forests associated with processes ranging from carbon and water cycling to the interactions among the flora and fauna (Vitousek and Denslow, 1986; Denslow, 1987; Asner, 2013). The following descriptors are used for disturbance regimes:

Type: the identity of the event

Area/Size: The areal extent of the disturbances, including the size of disturbance patches; the area per event per time period, or the total area per disturbance per time period.

Spatial distribution: The spatial distribution of the disturbance refers to the distribution of events relative to topography, soils.

Frequency: The mean number of disturbance events per time period within a specified area. This is perhaps one of the most commonly reported attributes of a disturbance regime.

Magnitude: There are two aspects of magnitude: *Intensity* refers to the magnitude in physical force of the event (such as fire temperature, wind velocity, etc.) per unit area and time; *Severity* refers to the magnitude of impact on organism, community, or ecosystem (i.e. the amount of damage caused by a disturbance event).

Turnover or rotation period: The mean time required to disturb the entire area, and fire return interval.

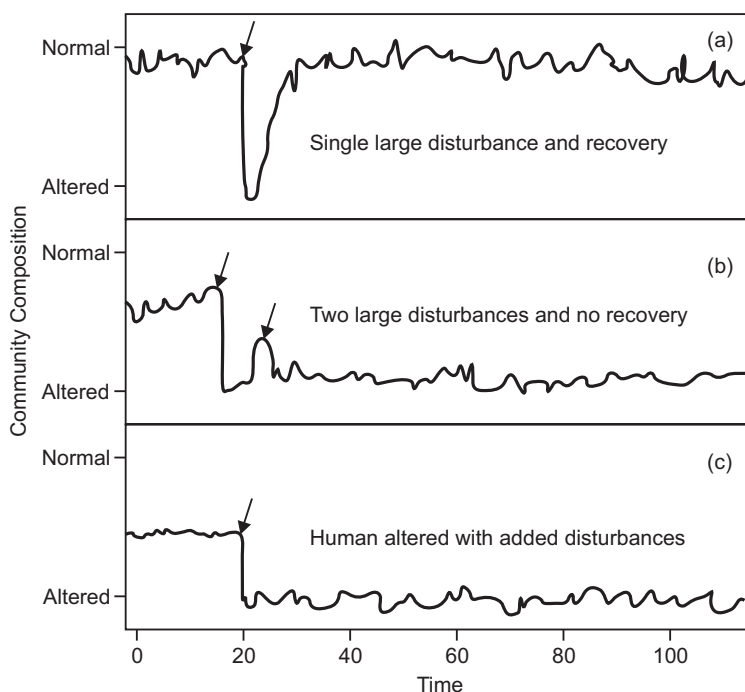


Fig. 5.7. Schematic illustration of the effects of disturbance on ecological communities. (a) A normal community is subjected to a single large disturbance subsequently it recovers. (b) A normal community undergoes a second (or multiple) disturbance(s) before recovery from the first is completed; the combined effects lead to long-term alteration in community state; (c) Major disturbance occurs in the community already altered by human activity or disease; again the combination of stresses leads to long-term alteration of community state. (modified from Paine et al., 1998).

Synergism: The effects of a disturbance event on the occurrence of other disturbances. For example, there may be a synergistic relationship between insect infestations in certain forest types and the occurrence of fire.

Disturbances can be transient or sudden and persistent. Transient disturbances are repeated interruptions in time or space, also called chronic disturbances such as repeated lopping of trees. Sudden and persistent disturbances represent a sudden change of conditions in which the change is then maintained.

5.13 AGENTS OF DISTURBANCE

The disturbance can be caused by physical processes, such as fires, Ice storms, floods, drought, high winds, landslides, large waves, desiccation stress, etc. or by biological processes, such as predation, disease outbreaks, grazing, invasion, etc. The agents can also be categorise as natural (e.g., lightning-induced fire, volcanic eruptions, landslides, drought, etc.) or anthropogenic (e.g., deforestation, grazing, introduction of exotic species, habitat destruction, shifting cultivation, etc.). Disturbance may propagate from outside of the system or may occur within the system. For example, invasive species from road sides may slowly migrate and establish in the system of interest.

5.14 EFFECTS OF DISTURBANCE

No population, community or ecosystem is immune to disturbance. The effect may vary, however, from system to system depending upon the type and magnitude of disturbance and the nature of the target community/ecosystem. Modest disturbances have small or short-lived effects, whereas large perturbations may shift the ecosystem from one state to another, for example conversion of forest into savanna or grassland. Further, each ecosystem has a threshold, that is, up to a point it can tolerate or absorb a given disturbance and maintain its integrity, beyond the threshold it may collapse.

Some disturbances can be beneficial also. For example, a grazed grassland exhibits rapid cycling of carbon and nutrients due to treading and excretion by large mammals, which prevent the accumulation of standing dead litter and return nutrients to the soil in plant-available form. Light grazing can increase primary production because it encourages the formation of new tissues which have higher photosynthetic rates than the old tissues. On the other hand, rapid deforestation results into release of NO_3 into nearby streams; thus on the one hand, N capital of the forest ecosystem declines and on the other, nutrient enrichment in the water bodies leads to eutrophication.

A major initial response of disturbance is the rapid change in population sizes of many species, usually dropping sharply as a result of death or damage of individuals caused directly by the disturbance. This leads to the formation of gaps whose size depends upon the force of disturbance. For example, in a tropical rainforest, a wind-blown tree creates a small gap, while multiple treefall creates large gap. The second response to disturbance which follows quickly is another drastic change in population sizes of several surviving species, their numbers increase. For example, some individuals who were light limited grow quickly due to opening of the canopy. The species which were previously absent can also immigrate to the gap to colonise the newly opened area. Smaller gaps are more transient than the larger gaps. The importance of gap dynamics is discussed below under the heading Disturbance and Gap dynamics.

Another major effect of disturbance which has drawn considerable interest is that on species richness/diversity. It has been postulated that at high levels of disturbance species richness is reduced to the species best able to tolerate the habitat conditions so produced. At low levels of disturbance, on the other hand, the late-successional/climax species may come to dominate to a degree that most other species are lost. This is discussed below under the heading Intermediate Disturbance Hypothesis.

Disturbance and Gap dynamics

Gap phase succession occurs in small gaps created by the death of one or a few plants. Many tropical wet forests or intertidal communities, for example, are mosaics of gaps of different ages. The disturbance regimes of tropical forests are largely defined by a process known as gap phase dynamics (Brokaw, 1985). Slow-growing shade-tolerant species survive well in closed-canopy forest (Connell, 1978). Disturbances such as hurricanes and tree-falls from lightning or wind events create a mosaic of forest patches of different microhabitats at varying stages of succession (Denslow, 1987). On the basis of several studies, it has been shown that openings in the forest canopy are important for the establishment and growth of trees in tropical rainforests in different regions of the world (Denslow, 1987). The gaps created in communities are colonised by light harvesting species, which are replaced by slow growing species over a period of time.

The Intermediate Disturbance Hypothesis

The 'intermediate disturbance hypothesis' (IDH) was originally proposed by Connell (1978) as an explanation for species diversity in plants and sessile animals (e.g., corals and trees). The IDH predicts that diversity will reach its maximum at intermediate levels of disturbance, while

remaining low at high and low levels of disturbance (Fig. 5.8). At high disturbance frequency, species diversity is predicted to be low, because only “weedy” species that quickly colonize and reach maturity are able to survive. In contrast, at low disturbance frequency, species diversity is expected to be low, because competitively dominant species exclude competitively inferior species (Fig. 5.8). Only at intermediate levels of disturbance do a mix of colonizers and competitors co-exist. This hypothesis applies regardless of whether the disturbances are biological (e.g., predation in predator-prey relationships) or physical (e.g., wave action functioning as a disturbance agent in inter tidal zone communities).

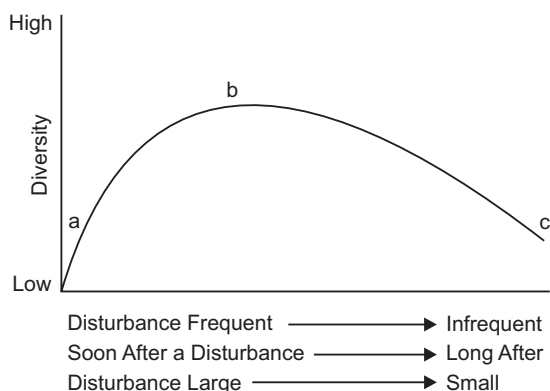


Fig. 5.8. The intermediate disturbance hypothesis (IDH). (a) Species diversity is low at low disturbance frequency because of competitive exclusion. (b) Species diversity is higher at intermediate disturbance frequency due to a mix of good colonizer and good competitor species. (c) Species diversity is low at high disturbance frequency because only good colonizers or highly tolerant species can persist. (based on Connell, 1978).

The IDH has the following postulates :

1. Species richness will be greatest in communities experiencing some intermediate level of disturbance.
2. This relationship is based on the fact that disturbances alter the availability of resources (physical, chemical, and biological) and are the source of multiple levels of environmental heterogeneity.
3. At low and high disturbance levels, environments tend toward homogenization and competitive relationships lead to selection for fewer species that are best adapted to the predominant environment that is produced.
4. At intermediate disturbance levels, maximum environmental heterogeneity is maintained, which allows for maximum niche differentiation and maximum opportunities for coexistence of competing species, as well as predators and their prey.

Marine macroalgal communities provide a classic example of the IDH, several studies illustrate a hump-shaped relationship between physical or biological disturbance and algal species diversity (Lubchenco, 1978; Sousa, 1979; Carpenter, 1981; Sousa, 2001). The species diversity of algae growing on boulders in the marine intertidal zone was studied by Sousa (1979). These boulders get overturned due to the physical force of wave action, and algae growing on the boulders get crushed and/or smothered as a result. Because the frequency of boulder movement is inversely proportional to boulder size (i.e., small boulders overturn frequently, whereas large boulders do not), boulder size can be used as a proxy for disturbance frequency. By experimentally stabilizing small boulders, Sousa (1979) demonstrated that the physical disturbance caused by boulders overturning

with different frequency leads to variation in species diversity, rather than some other difference between boulders of different sizes. According to this study :

1. Species diversity is low on small, frequently disturbed boulders, consisting primarily of fast-growing colonizer species;
2. Large boulders are dominated by competitively superior algae, leading to low diversity;
3. Diversity is maximized on intermediate boulders, with a mix of colonizers, competitive dominants, and species that lie in between these two extremes.

Disturbance significantly affects diversity in a majority of experimental manipulations (~70%) which attempt to examine the relationship between disturbance and species diversity (Hughes et al., 2007). However, this relationship can take a variety of shapes other than the unimodal pattern predicted by the IDH (Hughes et al., 2007). The intermediate disturbance hypothesis has implications for conservation planning. Areas with intermediate levels of disturbances are likely to contain the greatest diversity of species and thus serve as a logical focus for biodiversity conservation. Conservationists have used this concept not only as the basis for prioritizing sites for conservation protection, but also as the basis for managing conservation lands through active management.

Criticism of IDH

The IDH states that diversity of competing species is, or should be expected to be, maximized at intermediate frequencies and/or intensities of disturbance or environmental change. The IDH has, however, been refuted on both empirical and theoretical grounds by Fox (2013) who has argued that IDH should be abandoned. According to Fox (2013), a review of over 100 published diversity–disturbance relationships found that the predicted peak of diversity at intermediate disturbance levels rarely occurs (< 20% of studies; [Mackey and Currie, 2001]). A more recent review reports the same result as discussed above (Hughes et al., 2007). Even tightly controlled and highly-replicated microcosm and mesocosm experiments generally fail to find the predicted peak in diversity at intermediate disturbance or environmental fluctuation levels (Cadotte, 2007; Volle, 2010; Hall, 2012). Hall (2012) has argued that the humped diversity–disturbance relationships have no special empirical significance, being just one among many possibilities.

CHAPTER

6

Adaptations, Ecotypes and Genecology

The environment in which an organism exists provides numerous selection pressures on the population of species in response to environmental conditions. The changes in soil conditions, light, temperature and mineral nutrients can lead to changes in the form, physiology and behaviour of organisms. The phenotype (the appearance of organism as we see) results from the interaction of genotype (plant's genetic make up) and the environment. The special characteristics of plants and animals that enable them to be successful under prevailing set of environmental conditions are called adaptations. The organisms in natural world exhibit various types of morphological, physiological and behavioural adaptations. These special features have evolved over a long period of time, through the process of natural selection. The ultimate aim is to seek food and space, the organism's needs for its survival. The adaptive traits enable many organisms to live and thrive in different types of ecosystems and habitats. These traits also describe the characteristics of an organism that increase its fitness as well as distribution. The study of the gene frequency of a species in relation to its population distribution within a particular environment refers to genecology.

6.1 TOLERANCE RANGES

Under natural conditions, each organism possesses a range of tolerance to variations in its physical and chemical environment. The organisms respond to variations in environmental conditions in terms of their growth, reproduction and distribution. Any of the physical or chemical components of the environment that may inhibit the growth of living organisms, through either its lack or excess, is said to be a limiting condition or limiting factor. For example, low temperature limits plant growth at higher elevation, water availability limits plant growth in deserts, and changing levels of salinity affect organisms in estuaries, whereas low phosphorus availability limits the growth of phytoplankton in deep lakes.

The Liebig's law of minimum

The Liebig's law of minimum states that "growth of plants is dependent on the amount of food stuff which is presented to it in minimum quantity". Limiting factors most often affecting terrestrial organisms are moisture and temperature; those affecting aquatic plants and animals are mainly oxygen and light. The law of the minimum and the law of limiting factors operate within steady state conditions, when inflows balance outflows of energy and materials.

The Law of Tolerance

However, not only too little of something is a limiting factor, but also too much may limit the growth and distribution of an organism. The concept of the effect of maximum as well as minimum has been incorporated in to the law of tolerance by Victor E. Shelford (1913). According to this

law, for most environmental factors, such as temperature, light and moisture have an ecological minimum ('too little') and an ecological maximum ('too much'). The range between these two conditions represents limits of tolerance.

The tolerance curve is a measure of organism's fitness and survival. A bell shaped curve results when measures of survival or fitness are plotted against the environmental gradient (Fig. 6.1). Those environmental conditions under which an organism grows and reproduces maximally comprise the optimum range. The organisms show wide distribution due to wide ranges of tolerance for all factors and restricted distribution if the tolerance range for one or more than one factor is narrow. At limiting levels, reproduction of an organism is most critical, whereas factor interaction influences the fitness of an organism.

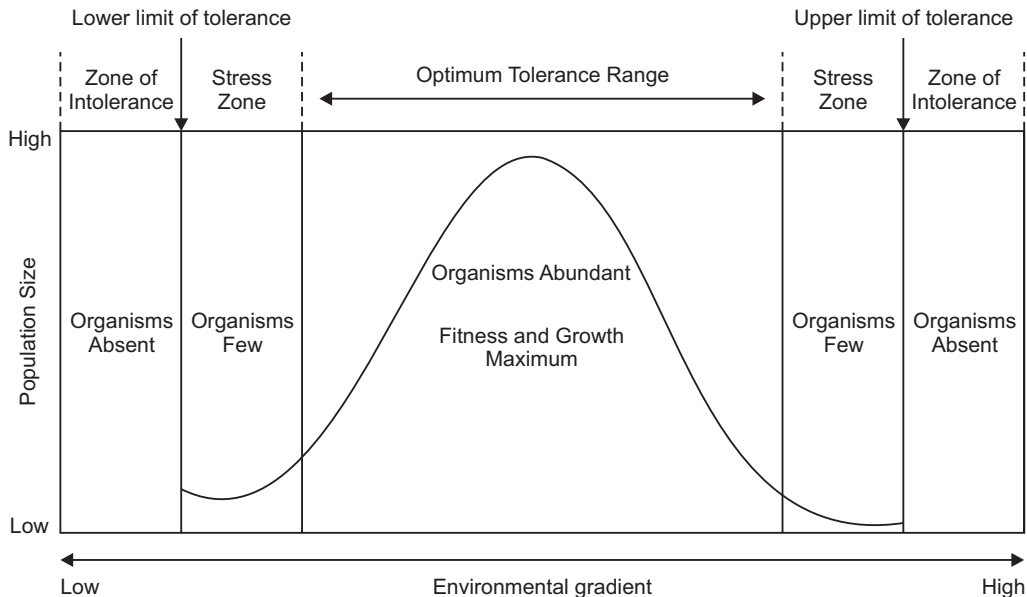


Fig. 6.1. The bell shaped curve shows the response of an organism to a range of single environmental variable (Temperature, sunlight or nutrient concentration). The organisms are abundant in the central optimum range showing greatest fitness, growth and survival. In the zone of stress, only a few organism survive and not able to reproduce. Organisms are absent from the zone of intolerance.

The prefix “steno” (narrow) and “eury” (wide) have often been used for comparing the relative degree of tolerance of an organism to a given environmental factor (Fig. 6.2). For example, a stenothermal organism will mean that the organism can tolerate only a narrow range of temperature. With narrow range of tolerance, organisms are concentrated at either end of the curve; having the optimum, minimum and maximum in a very narrow range. The organisms which are confined towards low temperature are called oligo thermic and those towards higher temperature are polythermal. Even a small change in temperature can have marked effect on the stenothermic organisms. Euthermal organisms have a wide range of tolerance for temperature. Similarly, stenohydric and euhydric refer to tolerance range of an organism to water. Euryhaline would describe a species with a wide range of tolerance for salinity and stenohaline a narrow salinity tolerance.

The red maple tree is found over a large area in south western United States due to broad ranges of tolerance to environmental variations as it grows under varying conditions of moisture and temperature. The coastal redwood tree (*Sequoia sempervirens*) is limited in its distribution to a narrow range in California in the areas receiving summer fog. In estuaries (a transition zone between

freshwater and marine ecosystem), different species of amphipods have narrow tolerance range to salinity and are restricted to different regions of the estuary due to changing levels of salinity.

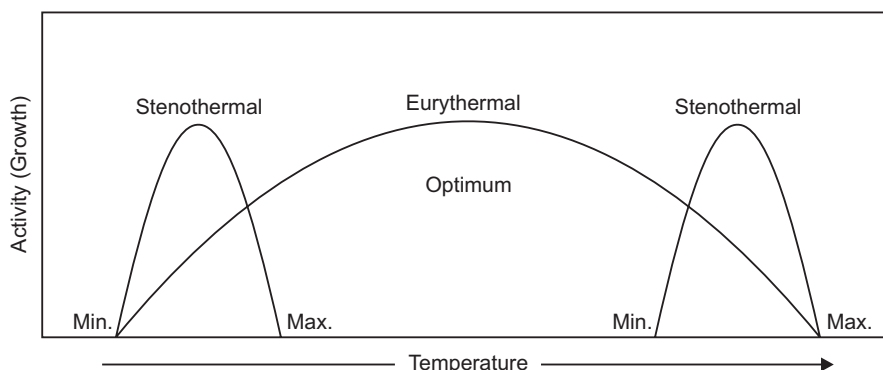


Fig. 6.2. Comparison of the relative limits of tolerance of stenothermal and eurythermal organisms (redrawn from Odum, 1983).

6.2 ACCLIMATION

The tolerance limits and optimal range of distribution may vary seasonally in some species populations. If some environmental factor shifts beyond the tolerance range of an organism, the organism can come to the resting stage or migrate or it can acclimate. The gradual, physiological adjustment to slowly changing environmental conditions is known as acclimation. Acclimation is a shift in the physiological tolerance of an individual and is a strategy to adapt to seasonal and other persistent variation in environmental conditions. During acclimation, an organism alters its homeostasis so as to accommodate shifts in its external environment. A period of acclimation before abiotic stress (e.g., cold temperature) may help in stress tolerance.

For example, the range of tolerance of a fish, an insect or an oak tree may vary seasonally, depending upon the temperature variations over the seasons. This is due to the fact that these organisms are able to exist in low temperature range in winter as well as under higher temperature during the summer. A well studied example of acclimation is the creosote bush (*Larrea divaricata*), found in the low altitude hot deserts of Southwestern United States. It is a C_3 plant and can shift its thermal optimum from winter to summer by acclimatization to maintain high rates of photosynthesis (Mooney et al., 1976). Acclimatization provides an advantage to an organism to exist in changeable environments as well as to overcome stress conditions.

6.3 STRATEGIES OF ADAPTATIONS IN PLANTS

The plants have special traits that help them to enlarge their tolerance limits to light regimes, dry conditions, high temperature, water-saturated conditions and saline environments. In plants, flowers have evolved special structures to ensure pollination by insets or other animals. The plants have developed various mechanisms to deal with stress conditions of the environment which are described as follows:

Adaptations to light regime

Individual plants as well as plant communities adapt to different light intensities by becoming shade tolerant (*sciophytes*) or sun adapted (*heliophytes*). Heliophytes are adapted to high intensity of light, and have higher temperature optima for photosynthesis as well as have high rates of respiration.

The shade adapted plants generally have low photosynthetic and respiratory, and metabolic activity. The plants such as ferns and several herbaceous plants, growing on the groundfloor under the dense canopy of forests, are shade tolerant plants. Thus adaptations in sun and shade conditions enable different plants to exist under different habitat conditions.

Adaptations to water scarcity and heat

Desert plants are adapted to cope with dry conditions of soil and high temperature. The plants evading the dry conditions are known as ephemerals. Plants of hot deserts are adapted to cope with dry conditions of soil and high temperature. Deserts provide many striking examples of how plants are adapted to hot and dry conditions. Many xerophytes can lay dormant for many years without perishing. With the onset of rainfall, these plants quickly pass through various growth phases, germination, flowering, fruiting and seed dispersal in only a few days. The plants become dormant again with the advent of dry conditions.

Plants adapt to aridity through the reproductive cycle. In desert areas of Rajasthan, many annual plants (e.g., *Crotalaria burhia*, *Solanum xanthocarpum*, *Tephrosia purpurea*) germinate from seeds and complete their life cycle quickly during the rainy season. With the onset of dry conditions, the plants undergo dormancy again and survive in the form of seeds. Various morphological adaptations in xerophytes to low soil water and high temperature have been discussed in Chapter 3.

The effect of heat on plants depends on the duration of the heat exposure. Heat stress is produced if leaves absorb solar radiation without heat loss. Hydrated and actively growing tissues of most plants cannot survive extended exposure to temperatures $>45^{\circ}\text{C}$; dehydrated material, e.g., seeds and pollen, can tolerate temperatures even in excess of 100°C .

Photosynthesis and respiration are inhibited by high temperature; photosynthesis is more sensitive to high temperature than respiration. Temperature compensation point is defined as the temperature at which CO_2 release in respiration equals to that of CO_2 fixed during photosynthesis. Carbon reserves are depleted at temperatures above the compensation point, i.e., stress effect. Plants that are native to environments of extreme high temperature (e.g., deserts) can tolerate much higher temperatures. For example; *Opuntia* (cactus) and many desert CAM plants can tolerate high temperatures. Transpirational cooling (both mass water flow and vaporization) is the principal mode by which plants cool leaf surfaces and dissipate heat, whereas in CAM plants stomata remain closed during the day. Many xerophytes may accumulate proline (an amino acid) in the cells to maintain osmotic and water potential in their leaves.

Heat Shock is a condition when there is a sudden increase in temperature ranging from greater than 5 to 15°C . Heat shock causes protein denaturation, (manifested as protein misfolding) that leads to cellular dysfunction. Plants and other organisms, both prokaryotic and eukaryotic, respond to heat shock by the production of heat shock proteins (HSPs), during progressive high temperature increase. Many HSPs are chaperones that facilitate proper protein folding (preventing aggregation of proteins into inactive complexes) and protein targeting because a misfolded protein is degraded rapidly in the cell. Plants produce large number of low molecular mass HSPs (sm HSPs) that function to facilitate protein folding and membrane fluidity during high temperature stress episodes. The low molecular mass HSPs are rapidly produced in response to heat shock or other environmental stress and are transferred to different plant compartments like chloroplasts, mitochondria, endoplasmic reticulum, cytosol and membranes imparting thermo tolerance to plants (see Buchanan et al., 2000).

Adaptation to low temperature

Resistance to cold may result from stress avoidance which means the plant excludes the stress by morphological properties or from tolerance which means that the stress is effective but the plant survives. Chilling injury occurs at temperatures that are substantially below the cardinal optimal but are above the freezing point of water. Chilling tolerance capacity is an adaptation that is associated with ecological fitness to low temperature. Plant tissues that are dehydrated organs, e.g., seeds, are very tolerant to freezing. Low temperature stress causes damage to foliage, flowers and fruits and usually manifested in discolouration. Loss of membrane function (fluidity) is the primary cause of chilling injury which leads to inhibition of photosynthesis, reduced carbohydrate translocation, lower respiration, inhibition of protein synthesis and protein degradation and formation of ice crystals in plants. Cold resistance is based either on tolerance to extra cellular ice formation or cold avoidance through super cooling. Water in xylem tissues of many temperate deciduous trees, fruit trees and forest species may remain in liquid form at very low temperature through the process of super cooling, enabling the plants to escape light frost damage.

In the plants growing at low temperatures (e.g., birches, alders, quacking aspen and willow) and native to the boreal forests, extracellular freezing occurs. The cells are extremely tolerant of the dehydration of their protoplasm and cold acclimation is very prominent, a response to low but non freezing temperatures that occurs before freezing. Acclimation through hardening and direct exposure to freezing temperatures enables the plants to become tolerant to extreme low temperatures. Though the process of freeze tolerance is not well understood, the possible mechanisms imparting freeze tolerance could be stabilization of membranes, accumulation of sugars, other osmolytes, antifreeze proteins or multiple change in gene expression (see Buchanan et al., 2000). Sugars and cold induced proteins appear to have cryoprotective functions, stabilizing other proteins and membranes. Sugars also contribute to the colligative properties of water (were negative osmotic potential) lowering the freezing point. The antifreeze proteins limit the formation of extracellular and intracellular ice crystals. The cold acclimation proteins in cold hardened plants protect the thylakoid membranes from freezing damage.

Adaptations in aquatic environments

The plants which remain permanently immersed in water are hydrophytes. They may be submerged or partly submerged and show the presence of aerenchyma (large air spaces) in the leaves and petioles. Aerenchyma tissue helps to store oxygen produced during photosynthesis and permits its free diffusion to other parts, including roots which may be in anaerobic soil. These tissues also impart buoyancy to the plants. The adaptations of hydrophytes have been described in Chapter 3.

Adaptations in saline environments

Halophytes are plants of saline environments, which are adapted to high concentration of salt in soil or water. Tidal marshes and coastal dunes, mangroves and saline soils support halophytes. In wet and marshy conditions of tropical deltas and along ocean edges mangroves are found. In mangroves, some species can excrete salts through the salt glands on the leaves. Some plants can exclude salts from the roots by pumping excess salts back into the soil. For coping with conditions of high salt concentration and osmotic potential, many mangrove plants have high levels of organic solutes such as proline, glycinebetaine (amino acids), mannitol and sorbitol.

Mangroves have developed adaptations, e.g., pneumatophores, prop and stilt roots, and vivipary (seeds germinate while on the tree), etc. to cope with high concentration of salts.

In *Avicennia*, presence of pneumatophores (the respiratory roots) helps to take up oxygen and transport it to the main roots. The pneumatophores are root extensions that grow out of water and up in the air. These roots have lenticels and spongy tissues that allow oxygen to diffuse through them. In red mangroves (*Rhizophora*), seeds germinate while on the tree (vivipary), grow into a seedling and then drop into water. The waves transport the seedling to shallow waters so that they can penetrate the mud. Prop and stilt roots in many species of mangroves gives support to the plants.

Spartina alternifolia is a dominant plant of temperate coastal salt marshes. It can withstand salinity by selectively concentrating sodium chloride in its cells as compared to sea water. Seagrass (*Zostera marina*) is a marine angiosperm that is found submerged under water in tropical coastal environment. *Halobacterium salinarum* (a halophilic bacterium accumulates large amounts of salts into cells and survives only in salty environments. *Dunaliella* species (green and halophytic algae found in hyper saline lakes) can tolerate saline conditions by accumulating glycerol in the cells, which helps in osmoregulation.

The halophytic plants under hot and dry conditions may become succulent and dilute the ion concentration of salts with water they store in cells of stems and leaves.

Adaptations to oligotrophic soils

The oligotrophic soils are nutrient poor soils which develop in old and geologically stable areas, such as found in much of tropical rain forests. These soils have a poor nutrient retention capacity due to intense weathering and high rates of leaching from the surface soil. In nutrient poor soils, nutrient retention is high in vegetation (e.g., in tropical rain forests). In many plants growing on nutrient poor soils, nutrients are retained in the root mats, a mutualistic association of roots with the mycorrhizae. Mycorrhizae promote plant growth in nutrient poor soils. Ectomycorrhizae are widespread, especially among roots of trees and shrubs in regions of the temperate zones with poor soils. Endomycorrhizae are found in most of the species of vascular plants, which help in mineral nutrition of plants particularly in nutrient poor soils. The adaptations of plants to oligotrophic soils are discussed in chapter 17 in detail.

6.4 STRATEGIES OF ADAPTATIONS IN ANIMALS

As with plants, animals also adjust to different environmental conditions to survive and flourish. Carnivores and herbivore animals have adaptations to eat a certain kind of food. Some animals have adaptations to avoid being eaten by the predators; others have behavioural adaptations to attract a mate. The males of some animals (particularly the plumage of the birds) have bright coloration, which gives advantage in sexual selection and mate attraction. However, a majority of animal adaptations to environmental variations and stress conditions are physiological and behavioural (Tables 6.1 and 6.2). Camouflage (to blend with surroundings) in some animals is a common example of adaptation. Some insects, reptiles and mammals have markings on their bodies which make it difficult to distinguish them from shadows and branches or from other members of the group. In very cold or dry environments, animals are not able to function normally. Under such conditions, the species that are not capable of migration, shift to a physiological dormant state. Hibernation is spending winter in dormant condition, whereas aestivation is spending the dry-hot period in an inactive state; examples are included in Table 6.1.

Table 6.1. Behavioural strategies of adaptations in animals.

Type	Examples	Processes and activities
Hibernation	Northern ground squirrels	True hibernators go into sleep during winter; body temperature drops, breathing and heart beat becomes slow.
Aestivation	Ground squirrels in south-west deserts	It avoids heat by spending dry-hot period in a torpid state into burrows
Cryptic appearance (camouflaged)	Leaf like grasshopper (<i>Arantia rectifolia</i>)	The grasshopper resembles the complete leaf or appears to be a part of leaf
	The praying mantis (<i>Phyllocrania paradoxa</i>)	Mimics a dead leaf and resembles back-ground vegetation
Batesian mimicry	Monarch butterfly and the mimic Viceroy butterfly	Monarch butterfly (contains toxins in the body) and mimicked by viceroy butterfly (contains no toxins)
	The Vasp (<i>Vespula vulgaris</i>) and two mimics, the Wasp Beetle, <i>Clytus arietis</i> , and the Hoverfly, <i>Syrphus ribesii</i>	In all three body is dark brown or black with yellow bands
Mullerian mimicry	Monarch butterfly and the mimic Queen butterfly	Both butterfly species look similar and are also distasteful
Echolocation	Horseshoe bat, dolphins	These animals produce high frequency sounds. They detect the presence of the echoes produced from the objects, on the same principle of sonar.
Auditory location system	Barn owls	The facial structure and the positional asymmetry of the ear of owl help to locate and catch a freely moving mouse in total darkness.
Electromagnetic sensing	Dogfish	Capable of detecting prey buried in the sand.
	Electric ray	It produces electric shocks capable of stunning the prey

Adaptations in preys

In nature, the various kinds of preys have specialized adaptations to avoid the predators. Warning signals, cryptic appearance and mimicry are usually associated with the prey species. In the cryptic appearance strategy, the basic idea is not to be conspicuous. The prey often remains concealed, sheltered or camouflaged. It is commonly found in all groups of animals. Warning coloration is well developed in the order *Hymenoptera*, including bees and wasps. The bright coloured insects often have bad taste and emit bad smell or have powerful stings. Concealing form and coloration enables a species to avoid a natural predator by camouflage (e.g., walking sticks).

The poison dart frogs (*Phyllobates bicolor*; *Dendrobates pumilio*) of the tropical rain forests of South America are brightly coloured. Their colouration (called *aposematic colouration*) is an adaptation for diurnal foraging in which predators can easily recognise and avoid these poisonous amphibians. These species of frogs contain highly toxic neurotoxin alkaloids in their skin. It blocks neuromuscular transmission, resulting in muscle and respiratory paralyses and even death of the predators. Native people in South America use the toxin on blowgun darts.

Mimicry is the resemblance of one organism with another. The organism imitated is model, whereas the imitator is the mimic. Batesian and Mullerian mimicry are two common examples of mimicry. In Batesian mimicry (named after the English naturalist in 1862), the potential predators confuse the mimic with the models and avoid both. For example, non-poisonous snakes resemble the poisonous ones, good tasting butterflies resemble the noxious ones and harmless flies resemble bees and wasps. In *Batesian Mimicry*, the mimic is defenseless and the model has a defense. A harmless moth (*Aegeria*) is a *Batesian mimic* as it is incapable of stinging another animal, but resembles the yellow jacket wasp (*Vespula*). In this type of mimicry, the anti-predatory markings on species (the model) are imitated by another species, the mimic, which has advantage. The monarch butterfly (contains poison, toxic to predator) is mimicked by the viceroy butterfly (contains no poison).

Related concept is Mullerian mimicry in which all the members of a species complex are distasteful, dangerous or toxic. It involves both unpalatable models and unpalatable mimics. All converge on a similar appearance so that predators avoid them. Mullerian mimicry is the process when the mimic shares the same defensive mechanism as the model. The yellow jacket wasp and bumble bee (*Bombus*) are *Mullerian mimics*. Both the organisms have bright yellow and black colours and use powerful stings as a defensive mechanism. In this mimicry, one unpalatable species mimics another. It is good for both the species. The caterpillars of monarch butterfly (found in North America) take up the toxic compounds while feeding on the milk weed plant. These butterflies can tolerate these compounds and retain them even after the caterpillars metamorphose into adult butterflies. Any predator tasting the monarch butterfly gets sick. The similar looking queen butterfly also tastes bad to the birds. Therefore, it is an example of Mullerian mimicry.

Adaptations in predators

There are also mechanisms used by predators to increase their hunting efficiency and the capture of the prey. The adaptations in predators are group hunting, cryptic appearance and mimicry, visual accuracy, auditory accuracy, echolocation and sensing of electromagnetic and thermal radiation signals. Cryptic and concealing colouration enables the predator to blend into background or can deceive prey by resembling the host or prey. *Echolocation* is a method used by bats while they emit high frequency sound waves during their flight to catch their prey. In some predators eyes are adapted for night vision due to the presence of the slit shaped eye pupils. The sensing of the infrared radiation (heat) by rattle snakes using pit organs (located between the eyes and the nose) helps them to catch their prey.

Migration

Many organisms that fly or swim undertake extensive migration; a long distance or short distance movement of animals from one region to another (Table 6.2) that allow organisms to take advantage of resources in different locations. Migratory behaviour is found in many groups of animals including insects (monarch butterfly, locust), fishes (salmons and eels), reptiles (sea turtles), birds (arctic tern, waterfowl, and song birds) and mammals (whales, seals and ungulates). Activities of migration exhibited by various types of animals are given in Table 6.2. The best known migrations are those of birds that winter in tropics and breed in colder regions of northern hemisphere. For example, Arctic terns make a trip of thousands of kilometres between their North Atlantic breeding grounds to Antarctica every year. Millions of waterfowl and song birds migrate thousands of kilometers from their southern wintering grounds to breeding grounds in arctic tundra. In arid regions, seasonal migrations occur in response to change in moisture. In Africa, wild beasts migrate long distances, following a geographical pattern of seasonal rainfall and availability of fresh vegetation. Locust migrate in search of new feeding grounds from the food depleted areas in large numbers in the arid regions.

Table 6.2. Migration as a Strategy of adaptation in animals.

Type of migration	Examples	Activities
Long distance	Arctic tern	It nests close to north pole in summer, in autumn flies south to Antarctica; each spring returns north pole again.
Short-distance	Many birds and insects	Birds migrate by using sun, moon, stars or magnetic field for direction and navigation.
	Caribou, elk and whales	These animals migrate in search of food each winter to warmer places.
Periodic migration	Locust	Large populations migrate due to change in environmental conditions from arid regions.

Adaptation to water scarcity

Two types of adaptations are prominent in animals living in arid regions i.e. lowering of water loss as much as possible and by adapting to in arid conditions. For example, the kangaroo rat conserves water by excreting solid urine other than liquid. Water is produced from the oxidation of lipids in the diet of kangaroo rat. Kangaroo rats can exist from birth to death without ever drinking water. They survive on moisture ingested in the food. The camels have developed the most remarkable adjustments to desert conditions. They are very economical in water use; can tolerate wide fluctuations in body temperature, which could be as high as 7.0°C on a hot day. They are able to maintain blood stream moisture even during extreme heat stress.

Adaptations to cold

Sessile animals living in very cold inter tidal zones of northern shores (barnacles and mollusks); several insects and spiders resist the effect of cold spells by a process known as cold hardening. The freeze tolerant organisms (able to survive extra cellular freezing) have ice nucleating proteins, which induce ice formation in the extra cellular spaces at very low sub zero temperatures. Some freeze avoiding animals can tolerate environmental temperatures below 0°C by accumulating glycerol or antifreeze proteins that lowers freezing point of their body fluids. A 10% glycerol solution lowers the freezing point of water by about 1.3°C. Such type of antifreeze compounds allows the fish in Antarctica region to remain active in sea water. The antifreeze proteins also act as cryoprotectant in freeze tolerant organisms by inhibiting the damage due to freezing conditions.

8.5 ECOTYPES AND GENECOLOGY

Darwin's theory of natural selection initiated the interest of modern biologists to study variation among organisms in different kinds of environments. The plant biologists conducted thorough studies of variation and to incorporate experimentation in their studies by focusing on plants (Turesson, 1922; Clausen et al., 1940). Some of the experimental studies of phenotypic and genetic variations in plants are as follows:

Phenotypic Plasticity

Phenotypic plasticity is defined as the environmentally induced phenotypic variation resulting in better survival potential of organisms. Phenotypic plasticity can be viewed as a form of the

reaction norm that may allow a given individual to develop a narrow tolerance range adaptive to the environment it is occupying (Gabreil and Lynch, 1992). The phenotypic variation in plant populations occurs because of the effect of environment on plant growth and development. For example, plants growing in low/high light conditions show phenotypic plasticity in biochemical and physiological aspects of photosynthesis, leaf anatomy and morphology, canopy structure and whole plant growth (Givnish, 1988). Phenotypic plasticity in plant structure in patchy environments enables plants to adapt to the heterogeneous environments (Via et al., 1995). Morphological plasticity in growth forms is commonly found in clonal and wetland plants due to strong environmental gradients and changing water levels. Many studies have examined the role of phenotypic plasticity as an individual mode of plant adaptation and the factors regulating plasticity in plant species along environmental gradients (see Givnish, 2002). The phenotypic plasticity in vegetative and reproductive traits in response to nutrient enrichment has been found to occur in an aquatic plant, *Sagittaria latifolia*; the dioecious populations exhibited increase in leaf size, whereas monoecious populations produced more flowers at high nutrient levels (Dorken and Barrett, 2004). Morphological changes also occur in the planktonic animal, such as water fleas (Cladocerans) in response to the presence of predators. The species showing differences in morphology of plants due to varied natural environments are called ecophenes or ecads and the differences are not heritable.

Phenotypic and genetic variations in *Hieracium umbellatum*

The study of genetic variation in plant populations in relation to environment is called genecology (Turesson, 1922). In the pioneering genecological studies, G. Turesson (Swedish botanist) collected seeds of a number of plant species from their natural habitats and grew them under similar environmental conditions in the uniform Experimental Garden in Southern Sweden. In the test garden, many plants maintained their morphological and phenological differences as noted in the field, indicating that species consists of genetically different populations (called ecotypes or ecological races). The ecotype differentiation was found to be prominent in *Hieracium umbellatum* (hawkweed – Asteraceae) which occurred under varying habitat conditions such as woodland, sandy fields, the sand dune, and cliff tops. In the common garden, the field differences in morphology from all the three habitats were maintained, indicating the profound selective advantage of different genotypes in different habitats (Table 6.3). From 1922 onwards, Turesson described ecotypes in more than 50 common European species.

Table 6.3. Some morphological and phenological traits of *Hieracium umbellatam* ecotypes as revealed in the uniform garden developed by Turesson, (Turesson 1922; see Barbour et al., 1980).

Traits	Woodland Ecotype	Field Ecotype	Dune Ecotype
Habit	Erect	Prostrate	Intermediate
Leaves	Broad	Intermediate	Narrow
Pubescence	Absent	Present	Absent
Autumn dormancy	Present	Present	Absent

Phenotypic and genetic variations in *Potentilla glandulosa* and *Achillea millefolium*

Clausen, Keck and Hiesey at the Carnegie Institution, Washington and Stanford University in California studied ecotypic differentiation in *Potentilla glandulosa* (Rosaceae; Fig. 6.3) and *Achillea millefolium* (Asteraceae) using common garden experiments and reciprocal transplant set along a transect from coastal conditions at Stanford to the alpine climate of the Timberline in Sierra Nevada mountains (Clausen et al., 1940). Responses of four ecotypes of *Potentilla glandulosa* to the three contrasting environments at Stanford, Mather, and Timberline is shown in Fig. 6.4. When the *Potentilla* plants from Sierra Nevada, the Coastal range and other locations were brought together in a uniform garden; the plants retained the difference in their morphology. There were four distinct ecotypes in *Potentilla glandulosa* differing in morphology and physiology, and adaptations to the range of environmental conditions along the climatic transect (Table 6.4). The hypothesis of these workers that ecotypic variants of *Potentilla glandulosa* differed genetically received support from a series of crossing experiments.



Fig. 6.3. A close-up of *Potentilla glandulosa*.
(from Molles, 2008)

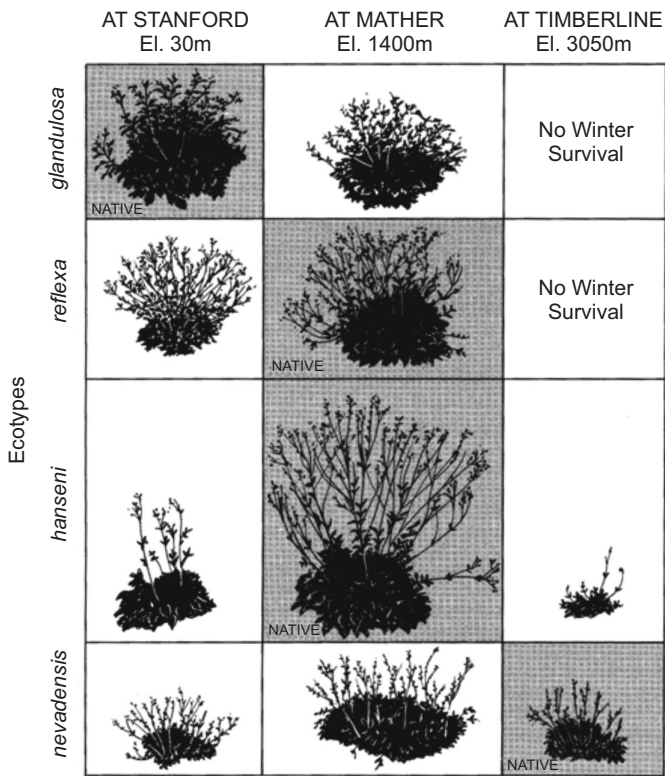


Fig. 6.4. Responses of four ecotypes of *Potentilla glandulosa* to the three contrasting environments at Stanford, Mather, and Timberline (after Clausen et al., 1940). El = Elevation

Table 6.4. Genecological classification of *Potentilla glandulosa* (on the basis of study of Clausen et al., 1948; see Barbour et al., 1980).

Cenospecies	Ecospecies	Ecotypes	Environment and plant characteristics
<i>Potentilla</i>	<i>glandulosa</i>	<i>nevadensis</i> 1500-3500 m elevation	Timberline form (alpine, subalpine) short form, winter dormancy
		<i>hanseni</i> 250-2200 m elevation	Mid elevations (wet meadow), Winter dormant, summer flowering
		<i>reflexa</i> 250-2200 m elevation	Mid elevation (dry slope), Winter flowering in lowland forest, summer flowering in Ponderosa pine forest
		<i>typica</i> 30 m elevation	Coastal range, winter and summer flowering

Similar studies were carried out on ecoypic differentiation in *Achillea millefolium* complex;eleven ecotypes were identified along the altitudinal gradient in the low elevation species (*Achillea borealis*; n=27) and the high elevation species (*A. lanulosa*; n=18). In the uniform garden in Stanford, the populations of *Achillea lanulosa* showed significant differences in the height of plants, the texture of leaves, the number of flower heads and the number of flowers per head. The expression of phenotype and the environmental factors were significantly correlated; the tallest ecotypes were from the lowest altitude, whereas plant height decreased with increasing altitude (Fig. 6.5).

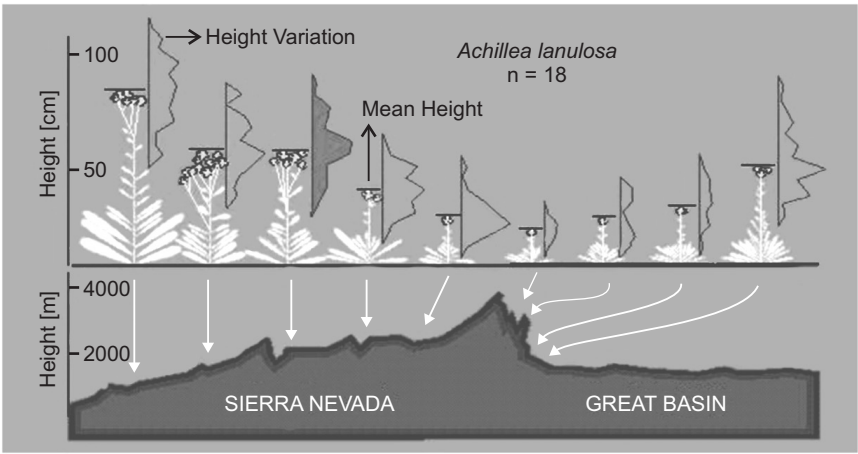


Fig. 6.5.Ecotypic differentiation in *Achillea lanulosa* from a transect across *California* along an altitudinal gradient from the coast to timber line in Sierra Nevada mountains to the Great Basin Desert in USA. (based on Clausen et al., 1948).

On the basis of classical studies of Turesson (1922) and Clausen et al. (1940), the ecotypes are defined as “mosaic of populations in which environmentally induced variations in morphology and physiology are genetically fixed and are heritable”. Some important attributes of ecotypes are as follows:

1. Ecotypes are discrete entity because they have clear differences in morphology, physiology and phenology in relation to particular environmental conditions.
2. The ecotypes are potentially interfertile with other ecotypes of the same species. But ecotypes are prevented from interbreeding due to ecological barriers.

3. The environmentally induced variations in ecotypes are heritable and have adaptive value to a particular habitat.
4. The ecotypic differentiation generally occurs over a wide range of environmental conditions.
5. On the basis of ecotypic differentiation, a genecological classification has been given, where a *cenospecies* is equivalent of a genus with a few species, whereas *ecospecies* is a biotic unit having ecotypes (Table 6.4).
6. Random genetic variants within an ecotype are called biotypes

Clausen, Keck, and Hiesey used differences in growth form of *P. glandulosa* plants grown in common gardens to infer genetic differences among populations of this species.

Ecotypic Variations in *Potentilla nivea* complex

More recently, Hansen et al. (2000) combined molecular and morphological techniques to explore genetic variation in populations of *Potentilla* species living on Spitsbergen Island in the high Arctic. Spitsbergen is the largest island in the Svalbard archipelago, a far northern part of the country of Norway. These workers studied *Potentilla nivea* complex comprised of three species: *P. chamissonis*, *P. insularis*, and *P. nivea*. Genetic analyses were done using the randomly amplified polymorphic DNA, or RAPD, method. The results of this study demonstrate the utility of joining morphological information with genetic information. The RAPD method identified three genetically distinct groups of plants within the *P. nivea* complex, which we can call “RAPD phenotypes.” Significantly, most of the genetic variance within the *P. nivea* complex was due to variation between the proposed species, while much less was due to variation within each of the proposed species. It has been shown that the separation of plants achieved by Hansen et al. (2000), which was based on combined genetic and morphological data, corresponded precisely to the three previously proposed species: *P. chamissonis*, *P. insularis*, and *P. nivea*. These results support the continued recognition of these taxa.

The wide spread occurrence of ecotypic differentiation in plants

A number of studies have shown evolutionary divergence in ecophysiological and life-history traits including photoperiod, temperature responses, photosynthetic rates, water–use efficiency and leaf area. (see Hiesey and Milner, 1965; Grant, 1981; McGraw and Antonovics, 1983). The natural population of plants exhibit ecotypes closely adapted to their immediate environment. Ecotypic differentiation has been widely studied at morphological, physiological and genetic level in high and low altitude plants (Clausen et al., 1948; Vaartaja, 1959; Mooney and Billings, 1961), plants of metal contaminated soils (Bradshaw, 1959; McNeilly and Bradshaw, 1968; Antonovics et al., 1971) and grasses.

Ecotypic differentiation at physiological and metabolic level has been found in the case of perennial herb, *Oxyria digyna* having a circumpolar – boreal distribution in Arctic Tundra (Mooney and Billings, 1961). The alpine and arctic ecotypes have been established on the basis of photosynthetic response to temperature. The alpine ecotypes showed higher temperature optima (38°C) for photosynthesis as compared to the arctic ecotype (20°). The sun and shade ecotypes have been examined in *Solidago virgaurea* on the basis of rate of photosynthesis and the enzyme activity (Bjorkman, 1968). The concentration and activity of Rubisco (ribulose 1-5 biphosphate carboxylase) was two to five times greater in sun ecotypes as compared to shade ecotypes; the rate of photosynthesis also being higher in the case of sun ecotypes. Photoperiod has a crucial effect on the growth and development of woody plants especially in the northern conditions at high latitudes. The

photoperiod changes from very short to very long days in these latitudes. Photoperiodic ecotypes are common in trees in the north-south range in northern hemisphere, which may be considered evolutionary adaptations to climate (Vaartaja, 1959).

In alpine areas of Alaska, natural populations of *Dryas octopetala* (Rosaceae) show large site specific variations in the phenotype (McGraw and Antonovics, 1983). The common garden and reciprocal transplant experiments showed that there are mainly two types of ecotypes. i.e., fellfilled (*Dryas octopetala* subsp *octopetala*), and snowbed ecotype (*Dryas octopetala* subsp *alaskensis*). The snowbed ecotype showed greater morphological plasticity accounting for the phenotypic variability between populations. The ecotypes differ in leaf size, leaf pubescence and the timing of flowering. The seedling leaves of fell field ecotype show fine silky hair, whereas the leaves of snow bed seedlings are glabrous. Both the ecotypes differed in flowering time and 90% of the pollen movement occurred between the flowers of the same ecotype (McGraw and Antonovics, 1983).

Ecotypic differentiation over narrow range

Ecotypic differentiation may be found over narrow range of environmental conditions. For example, *Agrostis tenuis*, is a out crossing and wind pollinating grass having more or less continuous distribution in the rolling hills of Wales (England) and grows under a range of soil conditions, altitude, climate and grazing. The populations, in particular, show ecotypic variations in relation to lead and nickel concentrations in soil. According to Bradshaw (1959), there are lead metal tolerant (from metal contaminated soils) and intolerant ecotypes (one those from normal soils). On metal contaminated soils, plants from normal populations could not survive, whereas plants from metal contaminated soils could grow normally (McNeilly and Bradshaw, 1968). The evolution of heavy metal tolerance in contaminated soils has been found in populations of a variety of plant species (Antonovics et al., 1971). The ecotypes of *Thlaspi caerulescens*, a hyperaccumulator plant, showed differences in Cd uptake in both hydroponic and field conditions. The ecotypes have a physiological basis for a specific Cd transport system in the root cells (Lombi et al., 2001).

Silene nutans has high genetic variation at both the ecotypic and population level because of perennial, long-lived life form and the out crossing breeding system of the species (Van Rossum et al., 1997). The plant shows a silicicolous (Si) and a calcicolous (Ca) ecotype with contrasting morphological traits.

Methods to study ecotypes

The ecotypic differentiation in plants has been studied using different methods such as common garden technique, spaced plant experiments, reciprocal transplant experiments and provenance and seed source tests. In common garden technique, individuals from different habitats are grown under standard conditions for comparative analysis of plants (Fig. 6.6 a). The reciprocal transplant experiments are useful for studying population differentiation in wide ranging populations and testing the adaptive nature of population divergence. In these experiments, genotypes from two different environments are grown in their native site as well as in the site of other populations to compare growth and functional processes (Fig. 6.6 b). In spaced plant experiment, the material is transplanted into two or more environments into prepared gardens from which the material was derived. This method was used by Clausen et al. (1940, 1948) for evolutionary studies in *Potentilla glandulosa* and *Achillea millefolium*. The provenance and seed source tests involve the collection of seeds from different populations termed as provenances. The seedlings are raised in a common garden to study the differences among provenances. The gardens used by Clausen Keck, and Hiesey (1940) in transplant experiments with *Potentilla glandulosa* were established at Timberline (3050 m), Mather (1400 m), and Stanford (30 m) sites.

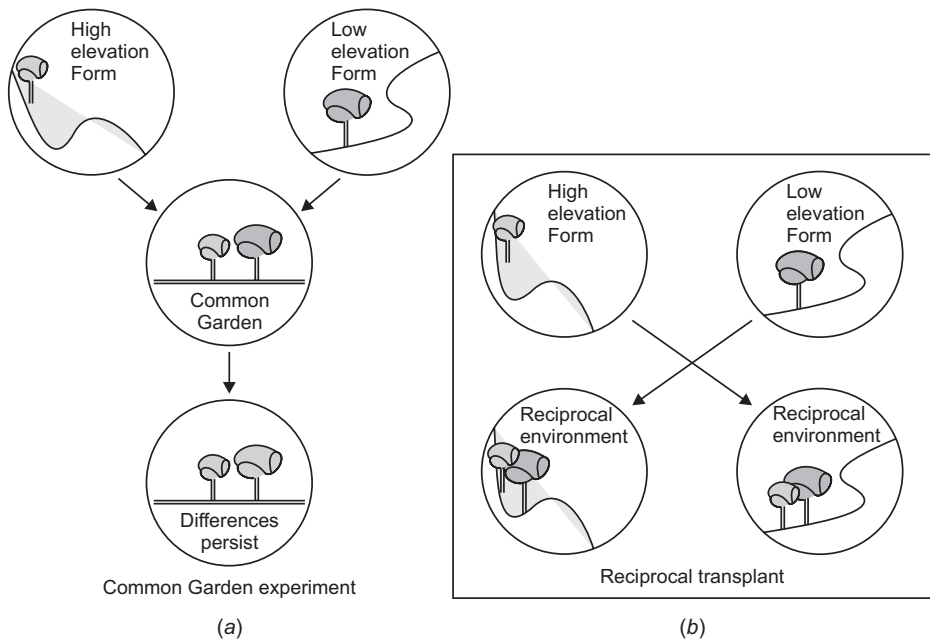


Fig. 6.6. (a) Common garden experiment and (b) Reciprocal transplant experiment for studying ecotypes.

Ecoclines

Langlet (1934) pointed out that most important habitat factor, such as, temperature and rainfall, commonly varied in continuous fashion resulting in graded variations in many widespread species rather than discontinuous variations. This view was supported by Gregor (1946). Thus, the term cline refers to any gradient in measurable characters and variations in a species in response to ecological factors. A series of populations of a species may occur along an environmental gradient, showing small differences in its genetic makeup, which help to adapt to changing conditions. Thus, an ecocline is a gradation in the attributes of a species associated with environmental gradients. J.W. Gregor (1946) found ecoclinical variations in *Plantago maritima* (Plantaginaceae) which grows under varying conditions from water-logged mud in salt-marshes to grasslands on the land ward side. Ecotypes distinguished by quantitative characters and growth habit showed continuous variations corresponding to habitat gradient. The plants growing in high tide zones (salinity level 2.5%) were prostrate, short lived and short seeded near the sea. The plants became progressively more erect with longer leaves, longer seeds and taller upright stalks in coastal grasslands. In this ecotonal area, the plants showed a continuous gradation from one extreme ecotype to the other, the boundaries were not discrete between ecotypes. The presence of ecoclines has also been shown in *Pinus sylvestris* (Langlet, 1959) and many other tree and plant species. Some examples of ecoclinical variations are given in Table 6.5.

Table 6.5. Some examples of clinal variations (source: Briggs and Walters, 1997).

Species	Variation	Reference
<i>Anthoxanthum odoratum</i>	Clines for various characters at mine/pasture boundary	Antonovics and Bradshaw (1970)
<i>Silene latifolia</i>	Clinal variation in seed morphology	Prentice (1986)
<i>Allium schoenoprasum</i>	Longitudinal cline in chromosome banding pattern in Eastern North America	Tardif and Morisset (1991)

CHAPTER

7

Biogeography and Life Zones

Biogeography is the study of the origin of world biota, environmental relationships, and distribution of organisms, both past and present, over the face of earth. There is a long history of biogeography, dating back to seventeenth and eighteenth century as is evident in the works of Linnaeus, Darwin and Wallace, and Alexander von Humboldt and other eminent biogeographers. Charles Darwin gave the dynamic concept of species, suggesting that species originates, migrates to new areas and evolves under the selection pressure of environmental conditions. According to the theory of evolution, a new species evolves in a particular location where a population has been modified by natural selection. We can assume that the new species is adapted to its particular environment and that there are other areas of the world with similar environments where it could live. According to Alexander von Humboldt, the geographic units of vegetation are related to climate irrespective of species composition. There are three fundamental processes of biogeography, i.e., evolution of species, extinction and dispersal. Speciation is the process of formation of new species and it has played a critical role in the evolution of life on earth. Extinction is the process of species becoming permanently eliminated with no more living individuals on earth. Dispersal refers to the movement of organisms away from their place of origin.

7.1 FLORISTIC REALMS

The shapes of the continents and their position in respect to poles and equator have been changing through geological time. The land masses have shown movements under the influence of plate tectonics, and convection currents in the earth's mantle. During the Carbo-Permian, about 280 million years ago, there existed a large hypothetical land mass called, Gondwanaland land consisting of present day Australia, Antarctica, South Africa, South America and India. Its characteristic flora was called Glossopteris flora, because of the abundance of Glossopteris leaves (meaning tongue-shaped leaves). Subsequently, at the end of Mesozoic the continents began to drift apart, becoming increasingly isolated in time. In response to this, there occurred differentiation of six floristic realms on the earth (Fig. 7.1). The time of separation was not same for all land masses, and this affected the degree of floristic differentiation. In the Northern Hemisphere, as for example North America and Greenland separated from the Eurasia only recently during the Pleistocene epoch. As a result, floristic differences in these land masses (North America, Greenland and Eurasia) are so small that they together are considered to represent one floristic realm (literally means kingdom or sphere), (i) the Holarctic. In contrast, much larger floristic differences occur within the tropical floras of the Old and New World, so that two floristic realms are recognised, the (ii) Paleotropical and (iii) Neotropical, respectively. Further, isolation of land masses promoted floristic differentiations. Since the southernmost parts of South America and Africa, and Australia separated into extreme isolation, their flora became conspicuously different from one another. Consequently, they are considered to represent three different realms: (iv) the Antarctic, which comprises southern part of South America

and sub-Antarctic islands, (v) the Australian, which is geographically identical with the continent of Australia, and (vi) the Capensis, which is the outermost south-west corner of Africa. Though smallest in size (Fig. 7.1), the Capensis is an especially rich floristic realm. One should not get the impression that these six floristic realms are discrete entities, sharply differentiated in the floristic content. Far from it, elements from one realm are found deep inside the next realm, often showing mosaic like distribution. As an example, both Paleotropical and Antarctic elements form mosaics in New Zealand.

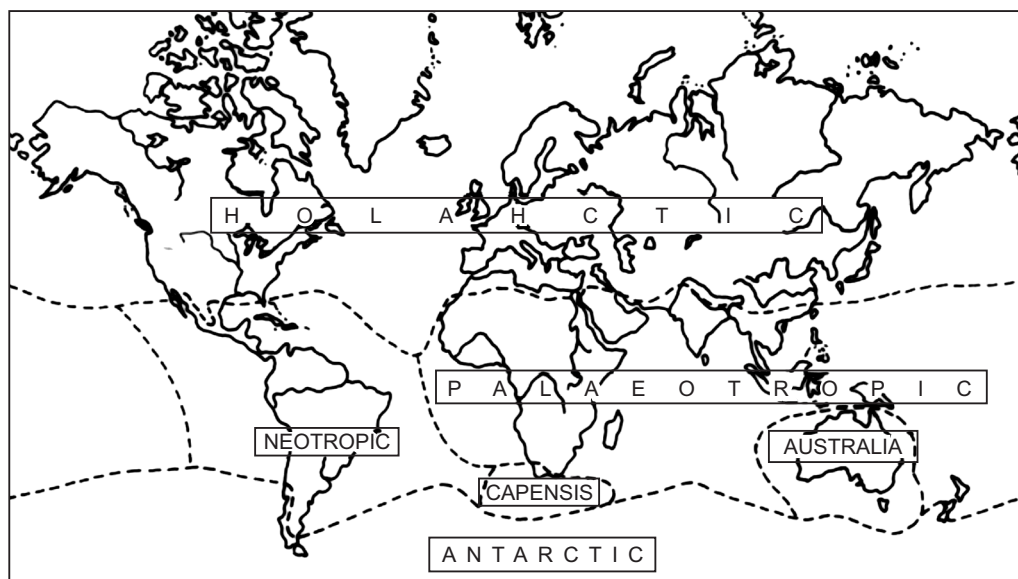


Fig. 7.1. The floristic realms of the earth. In New Zealand and Tasmania, Antarctic, as well as Palearctic and Australian, elements occur. Holarctic includes North America, Eurasia and Greenland. The Greenland separated from Eurasia only recently in the Pleistocene.

Evolution of the floristic realms has some relationship with the evolution of plant groups. Angiosperms or flowering plants which represent the youngest group of plant kingdom are relatively much more differentiated than the older group, gymnosperms. Though the oldest families of flowering plants are known to have existed in the lower Cretaceous, the group realized most development in the Tertiary period, when the landmasses had split into different continents of the present day. It is not possible to deal with the highly differentiated group of angiosperms because of the limitation of space. We cite some examples of differentiation in gymnosperms, with which students are more familiar because of lower number of taxa. Distribution of the family Podocarpaceae and genus *Araucaria* is limited to the Southern Hemisphere, whereas all the over 200 species of Pinaceae and almost all Taxodiaceae occur only in the Northern Hemisphere. The Cupressaceae are unique in having widespread distribution over all continents.

On the whole, the animal realms as suggested by A.R. Wallace (1896) correspond to the floristic realms, with the exception of the Capensis which lacks any typical fauna (Fig. 7.2). As is the case with flora, the Australian fauna is unique. Marsupials prevail, and other mammals are conspicuous by their absence. The floristic and biogeographic realms are now known as biotic provinces as the basic concept has been extended to include all life forms including plants, invertebrates and vertebrates (Udvardy, 1975). A biotic province is a region inhabited by a characteristic set of taxa (species, families and orders) bound by barriers that prevent the spread of distinctive kinds of life to other regions and the immigration of foreign species. In a biotic province, organisms share a common

genetic heritage, but live in a variety of environments, isolated genetically from other regions.

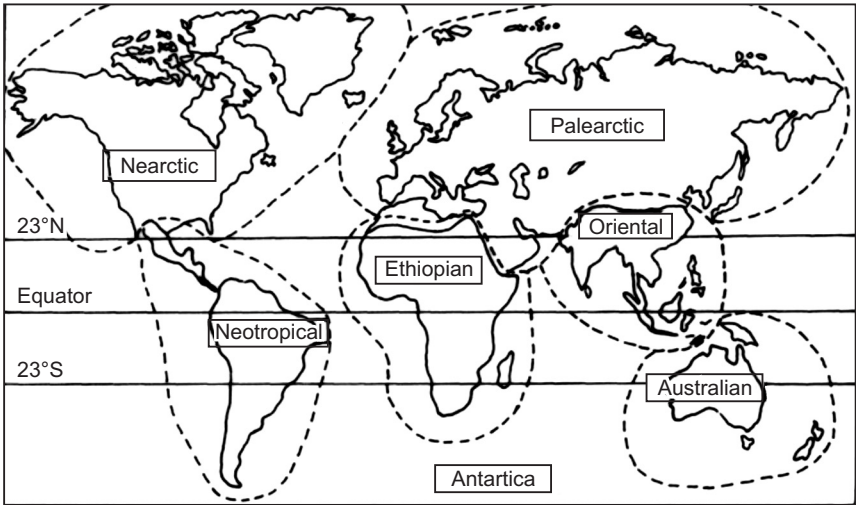


Fig. 7.2. The faunistic realms of earth.

The analytical approach and zoogeographical maps provide a baseline for a wide variety of comparative ecological, bio-geographical, evolutionary, and conservation based studies (Ladle and Whittaker, 2011; Wilson et al., 2006; Holt et al., 2013). Holt et al. (2013) have generated a global map of zoogeographic regions by combining data on the distributions and phylogenetic relationships of world’s amphibians (6110 species), non-pelagic birds (10, 074 species), and non-marine mammals (4853 species), a total of 21,037 species of vertebrates. Pair wise phylogenetic diversity ($p\beta$) matrices have been used to quantify the change in phylogenetic composition among species assemblages across the globe. A total of 20 zoogeographic regions have been identified, which are nested within 11 larger realms (Holt et al., 2013), Fig. 7.3. These 11 broad scale realms include:

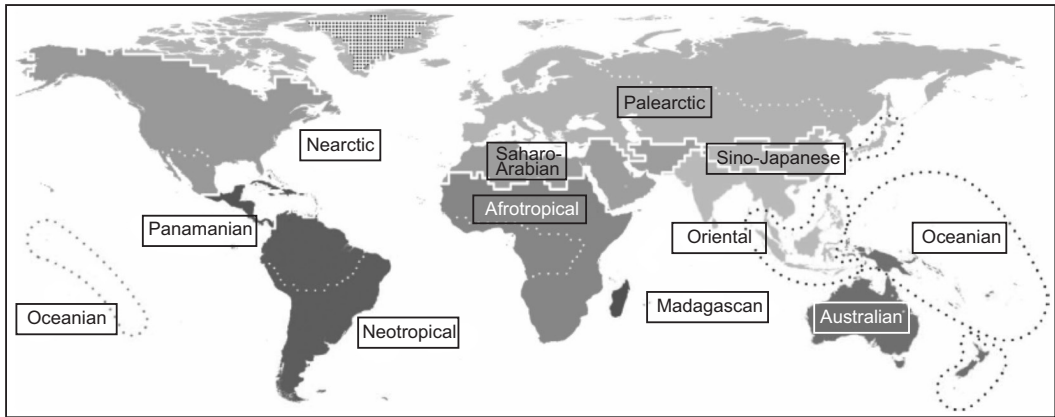


Fig. 7.3. Map of the terrestrial zoogeographic realms and regions of the world. Zoogeographic realms and regions are the product of analytical clustering of phylogenetic turnover of assemblages of species, including 21,037 species of amphibians, nonpelagic birds, and nonmarine mammals worldwide. Dashedlines delineate the 20 zoogeographic regions. Thicklines group these regions into 11 broad-scale realms, which are named. Dotted regions have no species records, and Antarctica is not included in the analyses (adapted from Holt et al., 2013).

Nearctic, Palearctic, Afrotropical, Neotropical, Oriental, Australian, Madagascan, Sino-Japanese, Panamanian, Oceanian, Saharo-Arabian. These evolutionally unique regions harbor radiations of species from several clades, which are restricted to given regions or are found in only a few regions. The maps of geographical realms and regions, given by Holt et al. (2013), give new insight into the distribution of the world's amphibians, birds and non-marine mammals. These maps also reflect major advances made in recent decades regarding our knowledge of the distribution and phylogeny of vertebrates. The maps can also be used to elucidate the forces and historical events responsible for the formation of bio-geographical realms.

7.2 DISPERSAL OF SPECIES

Dispersal is one of the fundamental processes in biogeography, crucial for understanding the distribution of organisms. Dispersal is an ecological process that is part of the life history of every species. Without dispersal, genetic interchange is limited, and species do not have a chance to adapt to a new environment. Dispersal is an important part of the process of evolution. The most basic distinction between different types of dispersal is between organisms that disperse using their own energy and those that use energy from the environment. Dispersal can lead to two main patterns of range expansion. Either a population can slowly expand from the margins of its geographical range or a small number of individuals can disperse to a new location some distance from the current edge of the species range, or a combination of both of these processes can occur. Humans have assisted the dispersal of many organisms to new parts of the world, sometimes leading to problems when they become pests in their new range. Understanding dispersal is also important in attempting to predict how organisms will respond to climate change.

There are many barriers to dispersal, such as oceans, large expanse of land and high Mountains. These barriers are frequently species-specific and therefore affect different species differently.

7.3 SPECIATION

Charles Darwin in his book "Origin of Species", clearly indicated that species are altered by natural selection over time. In a broader sense, speciation includes the diversification of all aspects of phenotype, including the evolution of differences among populations that prevent interbreeding and understanding the evolutionary forces that lead to the formation of new species. Speciation is the fundamental cause of biodiversity and has allowed the exploitation of different ecological resources so that species are able to coexist.

The 'species' literally means the "natural kinds". The evolutionary geneticist Dobzhansky (1937) and systematist, Ernst Mayr (1942) postulated the biological species concept, which states that "species are groups of actually or potentially interbreeding populations which are reproductively isolated from other such groups in nature" (Mayr, 1942). On the basis of biological species concept, speciation has been defined as the evolution of reproductive isolating mechanisms that prevents gene exchange between newly arising species (Turelli et al., 2001). There are many biological incompatibilities that can cause one population to be reproductively isolated from another, thus maintaining these populations as separate species. Recent work indicates that sexual antagonism might be important factor for the reproductive incompatibility.

Reproductive Isolating Mechanisms

The reproductive isolating mechanisms are the property of living organisms that prevent gene flow between organisms. Gene flow, the exchange of genetic information, can occur between members of the same species but not between members of different species. The isolation behaviour could be due to difference in time of flowering or type of pollinator in plants, geographical isolation

or inability to form fertile hybrids. Therefore, species living in close proximity remain distinct and each species is a repository of unique genetic information. They are classified as different species as they do not interchange genes.

Sibling Species

The phylogenetic species concept recognizes that each species is a category that exists as a pattern of recurrence in the natural conditions (Cracfort, 1983 and 1997). These natural kinds are known as “taxa”. A number of morphological and biochemical characteristics are used to separate taxa. They are a result of the evolutionary processes that have caused biological diversity. The term sibling species was introduced by Mayr (1942). The sibling species are defined as morphologically similar or identical natural populations that are reproductively isolated (Mayr, 1963). Sibling species are of great importance in biology as they have been used to test the validity of biological species concept vs. morphological species concept (typological species concept) and to study of speciation. There are numerous examples of sibling species. Such species occur in all the groups of animals but they are common in some groups such as insects. An interesting case of sibling species has been reported in malaria-mosquito complex in Europe which comprises of six sibling species. In *Drosophila* which is a model organism in biology to study genetics, evolution and behaviour (Singh, 2010), a large number of sibling species have been reported. The classical pairs of sibling species are: *D. pseudoobscura* and *D. persimilis*; *D. melanogaster* and *D. simulans*. Recently reported pair of sibling species is: *D. ananassae* and *D. pallidosa*.

In plants also sibling species have been reported. *Gilia transmontana* (small herb of the Mojave Desert and Western North America) is also an example of the sibling species. According to Grant (1965), it is a group of five interrelated diploid and tetraploid species which are *G. clokeyi* (diploid), *G. minor* (diploid), *G. transmontana* (tetraploid), *G. malior* (tetraploid) and *G. aliquanta* (diploid). Under natural conditions, these species grow sympatrically without interbreeding and show interspecific differences in flower size, seeds, the corolla colour and the capsules (Grant, 1965).

Causes of Speciation

Reproductive isolating mechanisms

Under the commonly used ‘*Biological Species Concept*’ (Mayr, 1942), the formation of new species involves the evolution of reproductive barriers to the production of viable offspring either before (pre-zygotic barriers) or soon after (post-zygotic barriers) mating. Thus, new species form when individuals from diverging populations no longer recognize one another as potential mates, or opportunities for mating become limited by differences in habitat use or reproductive schedules, such as difference in time of flowering, or type of pollinator, etc. This prevents gene flow, the exchange of genetic information, between members of the diverging populations. Reproductive isolation is weak in the early stages of speciation, but changes to strong or complete in later stages. One or more of the many types of isolating mechanisms may play a role in the evolution of species along a continuum.

Dobzhansky viewed that both pre and postzygotic isolating mechanisms are inevitable by-product of genetic divergence in allopatry. Prezygotic isolating mechanisms act first, and will always restrict current gene flow more than does postzygotic isolation. New species of polyploid plants, for example, originate entirely through postzygotic isolation. But later these species develop prezygotic mechanisms that prevent the formation of hybrid. Reproductive isolation might be functionally related to adaptive divergence. For example, the presence of different pollinators can lead to floral isolation. Similarly, adaptation to different prey can lead to the evolution of less efficient predators.

Under natural conditions, prezygotic isolation may evolve directly via ecological forces acting differently in different places. Postzygotic isolation can either be extrinsic or intrinsic. In extrinsic isolation, the relative viability and fertility of hybrids varies with the environment in which organisms live. The intrinsic isolation depends on the development problems. The prezygotic and postzygotic isolating mechanisms are summarised in Table 7.1.

Table 7.1. Dobzhansky’s classifications of reproductive isolating mechanisms

Prezygotic isolating mechanisms	
Ecological or habitat selection	The populations are spatially separated in different habitats or host plant use.
Spatial Mating or flowering time	Occurs in different seasons a ecological difference between habitats can affect either male traits or female preferences.
Sexual isolation	Mutual attraction between sexes is weak
Mechanical	Defences in the shape and sizes of sex organs or the flower parts.
Isolation by different pollinators	In flowering plants, related species may be specialized to attract different pollinators.
Gametic isolation	In organisms with external fertilization, female and male Gametes may not be attracted to each other
Postzygotic isolating mechanisms	
Hybrid inviability	Hybrid zygotes have reduced viability
Hybrid sterility	The hybrids of one sex or both sexes fail to produce functional hybrids
Hybrid breakdown	The F ₂ or back cross hybrids have reduced viability or fertility.

Sexual Selection

In many groups of animals, sexual selection may be a potent agent of speciation, which is based on the variations in success to obtain a mate. The effect of sexual selection is clearly evident in the case of rapidly speciating groups of organisms such as cichlid fishes in the rift valley lakes, birds of paradise in New Guinea and *Drosophila* in Hawaiian Islands.

Mechanisms of Speciation

The process of speciation takes a very long time and therefore has not been observed directly. Most of the understanding of speciation is on the basis of observations of patterns in nature and their theoretical interpretation. Allopatry is a major mode of speciation Sympatric speciation can occur through polyploidy, whereas incidences of parapatric speciation are rare. These three types of mechanisms for the formation of new species, i.e., allopatric, sympatric and parapatric speciation are described below.

Allopatric speciation

Allopatric speciation, also known as geographic speciation, is consequence of populations evolving in different geographical regions. Ernst Mayr (1963) proposed that speciation takes place due to geographical isolation. There are two modes of allopatric speciation i.e., the vicariance model (Fig. 7.4) and the peripheral isolation model (localized speciation). In the allopatric speciation

model an ancestral population is divided into two separate populations by a physical barrier (geographic barrier) (Fig. 7.4). Due to this separation, one ancestral gene pool of an inbreeding population is separated into two gene pools. The geographical isolation occurs due to geological and geomorphologic changes such as mountain ranges, continental drift, glaciations, river formation or even dispersion forces (wind). In each isolated gene pool, populations diverge genetically due to change in alleles and gene frequencies under the action of evolutionary agents. Due to accumulation of sufficient genetic changes during the isolation period, the two populations attain sexual incompatibility (Fig. 7.4). The two populations will not exchange genes if they come closer once again due to occasional geographic unification.

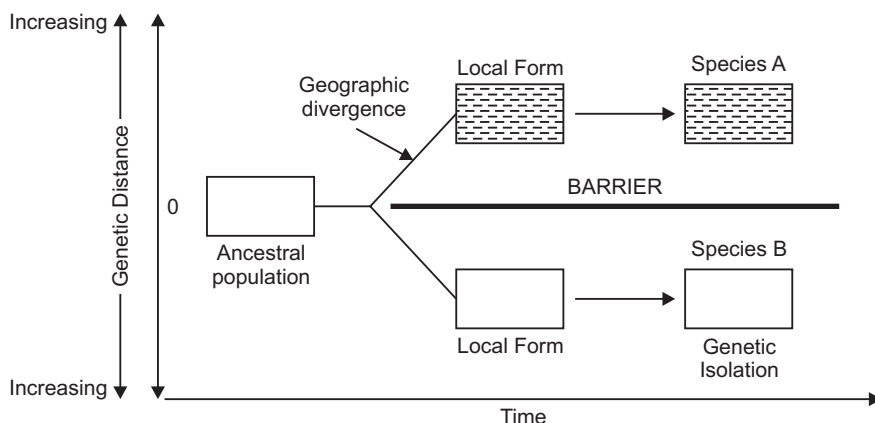


Fig. 7.4. Model of allopatric speciation.

According to the peripheral isolation model, a new species is formed from a strongly differentiated population isolated at the edge of the ancestral range. Some evidences in support of allopatric speciation are as follows:

- Highly fragmented fresh waters have provided opportunities for fish to form geographically isolated populations.
- Differentiation in the European butterfly (*Ernebia epiphnaon*) having different colours and forms; these races probably originated in geographic isolation.
- The extremely high level of speciation is found among picture winged *Drosophila* in Hawaiian islands because new populations are founded by individuals dispersing among the islands. The islands, which vary in age, vary in species distribution of *Drosophila* from older to young islands through founder effect.

Speciation and Adaptive Radiations

Adaptive radiations are the process by which plants, animals, and insects adapt to changed environmental conditions in a way that they form new species in a relatively short period of time. Adaptive radiations happen when one or two ancestral species give rise to several new species within relatively short period of time in a specific geographic area for exploiting the different array of resources available to the new species.

High speciation rates often coincide with low extinction rates when species invade islands that have a few other species. This has been observed particularly on the islands that have few other species. Charles Darwin and his shipmates from the HMS Beagle were the first to collect small songbirds, now known as Darwin's (Galapagos) finches. These birds were later recognized as being closely related to each other but having evolved beaks of distinct shapes and sizes as adaptations for different food sources. As such, they became a classic example of evolutionary processes and continue to provide us with valuable insights into principles of evolutionary change. The finches of the Galapagos Islands (isolated in the Pacific Ocean about 1000 km off the coast of Ecuador), provide a classic example of adaptive radiation—the evolutionary process through which a single lineage gives rise to species occupying diverse environmental niches (Grant, 1986). The finches of the Galapagos Islands, demonstrate the importance of geographic isolation for evolution of species. The different species of Galapagos finches might have originated from one mainland ancestor (blue-black grassquit) (Fig. 7.5). Diversification of habitat, availability of otherwise unoccupied ecological niches and the lack of competition possibly promoted the radiation of the finches. Grant and Grant (2002) while discussing the adaptive radiation of Darwin's finches have refined the model that geographical separation leads to evolutionary divergence. The evolutionary relationships among the various Darwin's finches have become clearer through studies of DNA sequence variation. The molecular techniques confirm that the family tree of Galapagos finches radiated from a single ancestral stock and it was monophyletic. Also, it is now clear that the Galapagos Islands were not static and have changed radically over the 3 million years during which the finches have evolved. The birds' physical appearance and song appear to act as cues that help isolate populations when mating (Grant, 1981; Grant and Grant, 2002).

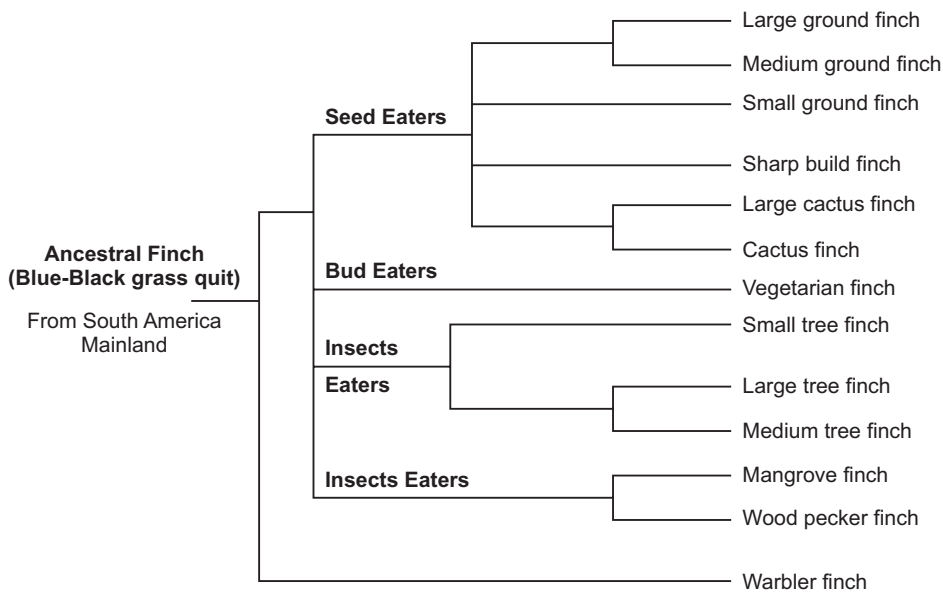


Fig. 7.5. Evolution among Galapagos finches.

The flora and fauna of many islands show many examples of species richness as well as occurrence of many endemics. The most remarkable radiations have occurred in the Hawaiian Islands, which are the most isolated islands from the mainland. More than 90% of species are

endemic and rapid evolution has occurred among many Hawaiian plants and animals. Honeycreepers are a widely known example of adaptive radiation and evolution of new species in the isolated islands, from a common ancestor. About 50 species of the Hawaiian honeycreepers, both living and extinct, are the most numerous element of the indigenous songbird fauna of Hawaiian islands (Lovette et al., 2002). Honeycreeper species occupying even a single Hawaiian island exhibit high diversity of bill morphology (see Lovette et al., 2002). Many specialized honeycreeper have bullfinch-like bills (seed predators), short warbler like bills (insectivores) and hummingbird-like bills (nectivores). Phylogenetic relationships based on mitochondrial-DNA-based genetic distances among honeycreeper and representative crude line finches suggest that the honeycreepers colonized the Hawaiian Islands at the time of formation of the oldest of the present-day main islands (Lovette et al., 2002) (Fig. 7.6). Similarly, *Argyroxiphium*, *Dubautia* and *Wilkesia* (three related genera), belonging to family Asteraceae have possibly originated from a recent common ancestor tarweed (*Madia sativa*) that colonized Hawaii from pacific coast of North America. Chloroplast DNA data show that these species share a relatively recent common ancestor. Thus, evolution of the Hawaiian islands has influenced the radiation of the plants and animals over the years because the islands have changed ecologically over-time facilitating adaptive radiation to occur.

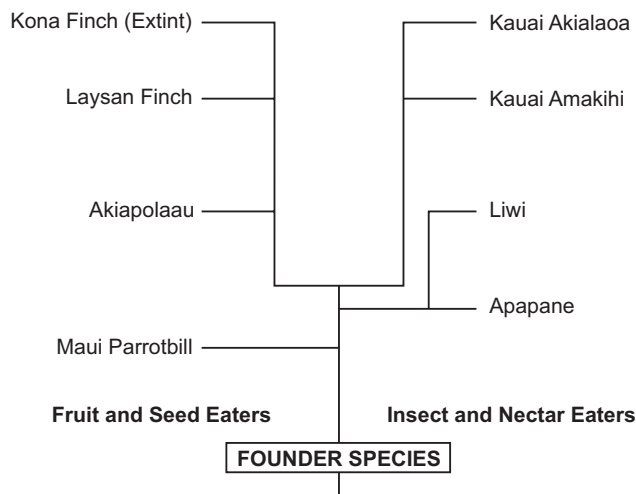


Fig 7.6. Adaptive radiations in a few Hawaiian Honeycreepers. Sixteen species of birds (eight shown in the figure) evolved from a single ancestor to fit different ecological niches by having specialised beaks for its food (derived from Cox et al., 1973).

There is a large diversity of cichlids in Lake Victoria, Lake Malawi and Lake Tanganyika in East Africa. Adaptive radiations have resulted in an amazing diversity of cichlid fishes with several hundreds of different species from independent lineages recognized in each lake. In these lakes, approximately 2000 unique species of cichlids have evolved only during the last 10 million years (see Kocher, 2004). Most of these fishes are endemic and restricted in their distribution to very small areas of the lake in relation to habitat conditions. The differences in trophic morphological characters and colouration patterns enable the fishes to occupy very different ecological niches. The rapid evolution of new species has occurred because of adaptations to local habitat conditions followed by radiation in trophic morphologies in each habitat and diversification of male colour pattern within each lineage (Danley and Kocher, 2001; Kocher, 2004) (Fig. 7.7).

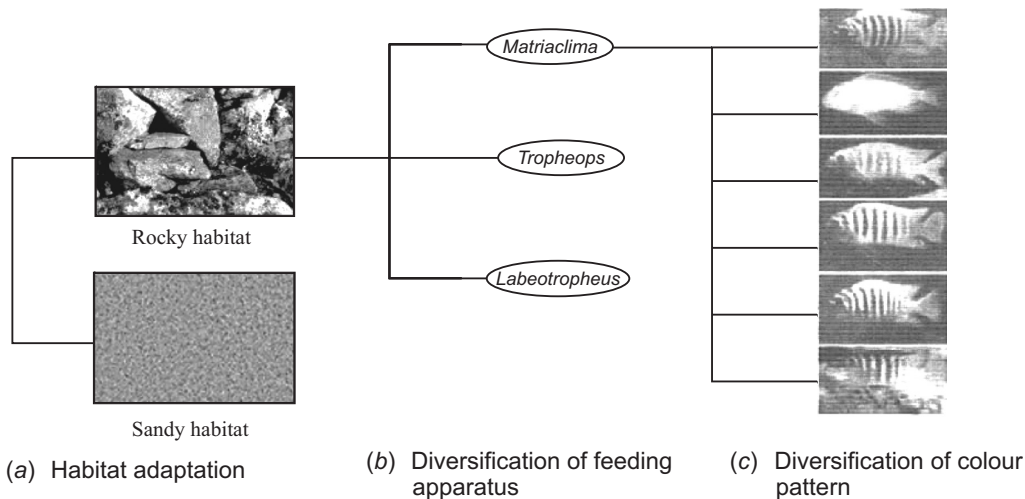


Fig. 7.7. Three stages of adaptive radiation of cichlids in Lake Malawi. (a) The first stage is habitat adaptation to distinct rocky and sandy habitats in the lake. (b) The second stage is diversification of feeding apparatus within each habitat, which is represented by the jaws of *Metriaclima*, *Tropheops* and *Labeotropheus* (top to bottom). (c) The third stage is diversification of male colour pattern within each lineage, which is represented by the genus *Metriaclima* (adapted from Danley and Kocher, 2001 and Kocher, 2004).

Sympatric Speciation

It is the process of differentiation and acquiring reproductive isolation within the same geographical area (Fig. 7.8 a). Sympatric speciation relies on the development of reproductive isolating mechanisms to allow divergence of two sub-populations. The reproductive isolation can be a result of a number of factors including competition for resources, disruptive selection and sexual selection. In sympatric speciation, selection acts against individuals of an intermediate type, through decreased viability and decreased fecundity. Selection enhances traits that promote divergence of the populations. The speciating populations may differ in host preference, food preference and habitat selection. Polyploidy (increase in number of chromosomes) has resulted in the formation of new species in plants. The new individual has a multiple of the original chromosomes and a more vigorous form. The polyploid plant is no longer sexually compatible with the parent population and establishes a distinct population. It is estimated that more than 50% of all flowering plants and 95% of all fern species are polyploids. Most of these polyploids have been formed due to hybridization between two species, followed by self fertilization. It is presumed that new species in wheat have been formed as a result of polyploidy. The sand dune grass, *Spartina townsendii* is a polyploid, whereas *S. anglica* is a diploid. It is known that *Spartina townsendii* is highly successful in coastal marshes as compared to *S. anglica*.

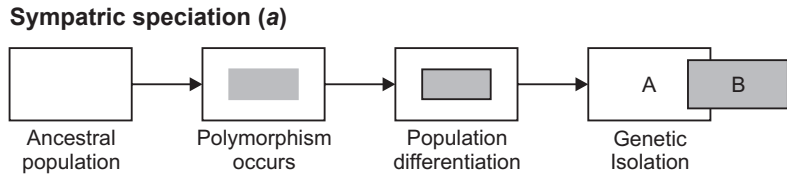


Fig. 7.8. (a) Model of sympatric speciation.

Polyploid speciation may occur after the formation of allopolyploids or autopolyploids; allopolyploids species being more common than autopolyploids. A well studied example of allopolyploid speciation is that of *Tragopogon* species (Asteraceae), that were introduced into North America from Europe in about 1900 (Soltis et al., 1995; Soltis et al., 2004). Hybrids among the three Eurasian diploid species (*T. dubius*, *T. porrifolius* and *T. pratensis*) were observed in 1949 in Eastern Washington. The hybridization among the diploid species leading to the formation of allopolyploids has been studied in great detail using morphological, genetic and molecular techniques (Soltis et al., 1995; Soltis et al., 2004). *Tragopogon mirus* Ownbey and *T. miscellus* Ownbey are allopolyploids that formed repeatedly during the past 80 years following the introduction of three diploids into North America. These allopolyploids have increased in numbers and in geographic range in their region of origin in North America.

The best known example of sympatric speciation is the divergence of *Rhagoletis pomonella*, a true fruit fly (Feder et al., 1988, 1990). This species has recently diverged into two subspecies due to introduction of apple trees in northeastern United States. Initially, the flies have used the hawthorn trees (which are native to the area in North America) to reproduce and to lay their eggs. The introduction of apple provided more nutritious food source for the developing maggots. The samples of fruit flies from apple and hawthorn trees differ genetically in the time of emergence, which is correlated with the fruiting phenology of their hosts. Thus differences between the maturation times of two fruit flies allowed the isolating mechanisms to operate resulting in the divergence of two species. Thus host specificity has taken the place of geographic isolation.

Another recent example of sympatric speciation is that of the cichlid fishes in crater lake (diameter 5 km; maximum depth 200 m) in Nicaragua (Barluenga et al., 2006). Until recently only three species (*Amphilophus citrinellus*, *Amphilophus labiatus* and *Amphilophus zaliosus*) were described from the two large lakes and several crater lakes in Nicaragua. *Amphilophus zaliosus*, the arrow cichlid, occurs in only one of the crater lakes (Lake Apoyo) and based on several lines of genetic evidence and ecological data such as stomach contents analyses and the quantification of morphological differences, it has been shown that this species originated sympatrically in this crater lake (Wilson et al., 2000; Barluenga and Meyer, 2004; Barluenga et al., 2006). It is ecologically distinct from the other endemic species now known exclusively from this lake. Population genetic analysis based on microsatellites also supports an origin of all cichlid species in this lake from a single colonization. Microsatellite-based Bayesian population assignment tests identified two clusters within Lake Apoyo corresponding to *A. citrinellus* and *A. zaliosus* which are clearly distinct from any other *Amphilophus* population in the area (Barluenga et al., 2006). *Amphilophus zaliosus* prefers the open water to the benthos associated environment occupied by other fish within this small crater lake and has a diet associated with open water and the surface rather than the benthic environments of the lake. Barluenga et al. (2006) suggested that sympatric speciation was driven by ecological mechanisms through divergent habitat preferences and resource partitioning, and by assortative mating through behavioural isolation. Thus, mechanisms of ecological speciation as well as sexual selection can act together in sympatry to bring about new species in crater lakes.

There is an extraordinary diversity of crustaceans in the depths of Siberian Lake (Lake Baikal). Sympatric speciation driven by ecological factors may account for the great diversity of crustaceans.

Speciation has been inferred to require thousands or millions of years to be completed. Speciation may be rapid through polyploidy as well as without polyploidy. For example, rapid speciation occurs without polyploidy involving two species in the monkey flower genus *Mimulus* (Phrymaceae) in which a barrier to gene flow may have been caused by floral isolation (Bradshaw et al., 1995). *Mimulus lewisii* and *M. cardinalis* differ strikingly in floral morphology. Flowers of *Mimulus lewisii* are pollinated by bumblebees; their petals are pink with yellow nectar. The flowers of *M. cardinalis* are adapted by humming bird pollination; they are bright red, with a narrow corolla tube and copious nectar. It is interesting to note that natural hybridization between the two species has never been reported, even though they flower at the same time (Bradshaw et al., 1995). In

Mimulus species, reproductive isolation for speciation could have been achieved quite rapidly even without polyploidy (Bradshaw et al., 1995).

Parapatric speciation

Speciation which occurs between adjacent populations within a broadly continuous habitat is called parapatric speciation (Fig. 7.8 b). The existence of narrow hybrid zones shows that selection can dominate gene flow over quite small distances allowing parapatric divergence. Examples are known in sunflower and *Drosophila*. Any factor that reduces gene flow or increases the gradient in selective pressure along small distances can generate conditions favourable for parapatric speciation.

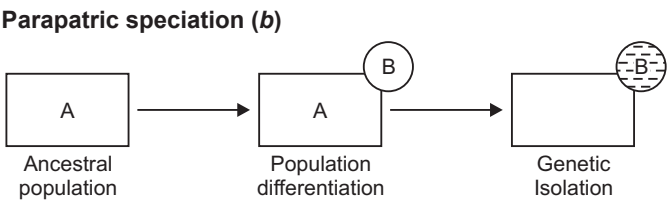


Fig. 7.8. (b) Model of parapatric speciation.

New Insights Needed on Speciation

The Spatial and non-spatial definitions of sympatry, parapatry, and allopatry as given by Mallet et al. (2009), are summarised in Table 7. 2. On the basis of models, it has been suggested that in sympatric speciation, initial gene exchange rate (m) between diverging demes is maximal, so that mating is random (i.e., $m = 0.5$) By contrast, in allopatric speciation, there is no gene exchange rate at all ($m = 0$). All intermediate cases, when migration between diverging (sub) sub populations is neither zero nor maximum are considered parapatric. Thus populations specializing on different resources locally within a defined geographic area will have $m < 0.5$. Recently, evolutionary biologists have defined speciation non-spatially on the basis of panmixia ($m = 0.5$) between a pair of demes before onset of reproductive isolation (Mallet et al., 2009). Panmixia seems to be a suitable starting point in models of speciation.

Table 7.2. Spatial and nonspatial definitions of sympatry, parapatry, and allopatry. (adapted from Mallet et al., 2009)

	Nonspatial, demic	Spatial population genetic
Sympatry	Panmixia $m^* = 0.5$	Where individuals are physically capable of encountering one another with moderately high frequency. Populations may be sympatric if they are ecologically segregated, as long as a fairly high proportion of each population encounters the other along ecotones; and they may be sympatric, yet breed at different seasons
Parapatry	Nonpanmictic $0 < m < 0.5$	Where groups of populations occupy separate but adjoining geographic regions, such that only a small fraction of individuals in each encounters the other. Typically, populations in the abutment zone between two forms will be considered sympatric
Allopatry	Zero gene flow $m = 0$	Where groups of populations are separated by uninhabited space across which dispersal and gene flow occurs at very low frequency

* m is the per generation fraction of individuals exchanged among demes.

The Marie Curie SPECIATION Network (2012) outlines some important themes and questions central to future studies on speciation. According to the workers of this network, the traditional organising principles of speciation do not provide a satisfactory framework by categorizing speciation mechanisms into allopatric, parapatric and sympatric classes. Beyond the geographic context, emphasis must be on the revised consideration of speciation mechanisms, the genetic basis of speciation, and relationship between speciation and diversity. According to these workers, there is clear evidence that ecologically important divergence, perhaps beyond the point of no return, can occur despite gene flow. Further they suggest that new insights into the genetic processes involved in speciation would be especially profitable if they examine the genomics of diverging populations, the genomics of hybrid zones, the genetic architecture of genes and networks underlying isolation, and the sources of variation in gene expression. There is need to focus on evolutionary force, ecological circumstances, and genetic mechanisms underlying speciation. A good understanding of speciation is essential for understanding the dynamics and pattern of biodiversity.

Macroevolutionary Consequence of Speciation

The important consequences of speciation are the evolution of biological diversity and transformation of ancestral lineages. However, the persistence of primitive characters in some of ancient lineages in some cases makes it possible to trace phylogenies.

According to the theory of origin of species by natural selection, new species are produced by gradual changes accumulating over long periods of time. Because of the low probabilities of fossilization, many of the transition forms are missing from the fossil record.

The theory of punctuated equilibrium postulates that new species are formed fairly quickly often during periods of rapid environmental change. There are long periods of time during which speciation does not occur and the periods of equilibrium are punctuated by sudden appearance of new species. These changes occur in small populations at the edge of ancestral species range, when a peripheral population is subjected to strong selective force. Once a new species is formed, it stays there for a long-time.

7.4 EXTINCTION OF SPECIES

Extinction is a natural process affecting various species on earth. Species have disappeared and new ones have evolved to take their place over the long geological history of the earth. It is estimated that 99% of all the species that have lived in the geological past are now extinct. In geological times, large-scale species extinction was almost as common as species origination. Average durations of species were generally less than 10 million years, and biological composition of earth at species level changed drastically several times. The most serious extinction occurred near the end of the Permian period (250 million years ago), removing 52% of the families of marine animals. It was followed by four mass extinctions, including the Cretaceous-Tertiary event 65 million years before the present. However, each of these mass extinctions was spread over million years or less, compared to only 200 years for the anticipated extinction due to human interference. Thus in comparison to mass extinctions of geological past, the man-made mass extinction represents a very severe depletion of biodiversity, which has been termed as sixth mass extinction.

A species ceases to exist because it can not genetically adapt and successfully reproduce under changed environmental conditions. It is useful to distinguish three types of extinction processes.

Natural extinction: With the change in environmental conditions, some species disappear and others, which are more adapted to changed conditions, take their place. This loss of species which occurred in the geological past at a very slow rate is called natural or background extinction.

Mass extinction: There have been several periods in the earth's geological history when large number of species became extinct because of catastrophes. It is abrupt rise in extinction rates above the background level. Mass extinctions occurred in millions of years.

Anthropogenic extinction: An increasing number of species is disappearing from the face of the earth due to human activities. This man-made mass extinction represents a very severe depletion of biodiversity, particularly because it is occurring within a short period of time. The current rate of extinction of species is 1000 to 10000 times higher than the background rate of extinction.

The role of extinction in biodiversity loss will be discussed in chapter 26.

7.5 CONVERGENCE AND ENDEMISM

Different species may resemble in ecological features, and because of that they may form similar vegetation in widely separated geographical regions. Such species, for example, may assume similar life-forms under the influence of extreme climatic conditions. They are termed convergences. In the arid regions of both Africa and the America stem-succulent life-form predominates. While in the former they belong to the family Cactaceae, in the latter species of the genus *Euphorbia* (Euphorbiaceae) prevail. Interestingly, in the arid areas of Australia and India succulents are absent. Australia shows several other convergences, not seen in other continents. The Australian realm is unique also from other standpoints. The two genera, *Eucalyptus* and *Acacia* form most of the Australian forest vegetation. Deciduous forests are common in the Holarctic realm, but are absent in New Zealand. *Endemism* is the ecological state of being unique to a defined geographic location or habitat type; organisms that are indigenous to a place are endemic to it as they are not found elsewhere. For example, the Orange-breasted Sunbird is exclusively found in the fynbos vegetation zone of Southwestern South Africa. Physical, climatic, and biological factors can contribute to endemism.

High level of Endemism, that is the occurrence of species restricted to a region (found nowhere else), is a characteristic feature of islands. This is because of the isolation of islands from the mainland; the greater the distance from the mainland, the larger the proportion of endemic species in the island. The proportion of endemic species in the flora is: Hawaii, 97.6%; New Zealand, 72%; Fiji, 70%; Madagascar, 66%, Galapagos (only dry region), 64%; Canary, 50-55%; islands close to mainland, 12% (Walter, 1979). In comparison, the percentage of endemism is far less in a large subcontinent, like India (30%).

Roberts et al. (2002) have analyzed the centres of endemism for the marine organisms in oceanic systems. They have identified 18 richest multi-taxon centres of endemism, which includes 35.2% of the world's coral reefs, covering only 0.028% of the world's oceans. The 10 richest centres of endemism in the oceans are the oceanic islands (Mauritius, La Reunion in Indian Ocean; Hawaii and Easter Islands in the Pacific Ocean; St Helena and Ascension in the Atlantic). Also the regions of non reversing ocean currents (which move water from tropical to temperate latitude) like East and West Australia, Eastern South Africa and Southern Japan also show high degree of endemism in the oceanic systems. The richest centres of endemism cover 15.8% of world coral reefs, covering only 0.0012% of the oceans and including 44.8 to 54.2% of the restricted range of species (Roberts et al., 2002).

7.6 LIFE ZONES

The effect of climatic variables, especially temperature and precipitation, on the distribution of plant life has long been recognized. Alexander van Humboldt in his pioneering work on plant geography in 1807 found a close relationship between climate and vegetation. In 1889, C. Hart Merriam studied the relationship between mean annual temperature and the distribution of flora and fauna in the western United States. He recognized that similar zones or belts of vegetation occurred with both increasing latitude and increasing elevation; these belts formed Life Zones.

Temperature and precipitation regimes usually determine major vegetation zones. Since latitude affects both the temperature as well as precipitation, latitudinal zones of vegetation are recognised. However, these zones are modified by the altitude in mountain areas (Table 7.3).

Table 7.3. The change in vegetation with altitude at different latitudes.

Latitudinal zonation	Altitudinal Zonation			
	0-1000 m	1000-2000 m	2000-4000 m	4000-6000 m
Tropical (0°-20°)	Tropical	Subtropical	Temperate	Artic-alpine
Subtropical (20°-40°)	Subtropical	Temperate	Artic-alpine	—
Temperate (40°-60°)	Temperate	Artic-alpine	—	—
Arctic and Antarctic (60°-80°)	Artic-alpine	—	—	—

The precipitation-effectiveness index abbreviated as P-E index is a measure of the long-range effectiveness of precipitation in promoting plant growth for a given location:

$$\text{P-E index} = 10 \sum_{m=1}^{12} (\text{P-E index})_n,$$

that is, it is equal to 10 times the sum of the monthly precipitation-evaporation ratios (monthly precipitation amounts divided by monthly evaporation amounts. The P-E Index has been used to estimate precipitation effectiveness for plant growth. Thornthwaite (1931) recognized humidity provinces and their characteristic vegetation types (Table 7.4), using P - E index.

Table 7.4. Humidity provinces of North America (from Thornthwaite, 1931).

Humidity province		Characteristic Vegetation	P-E index
A	Wet	Rain forest	128 and more
B	Humid	Forest	64-127
C	Subhumid	Grassland	32-63
D	Semiarid	Steppe	16-31
E	Arid	Desert	Less than 16

Precipitation evaporation ratio abbreviated as PE ratio; also called precipitation-effectiveness ratio. For a given locality and month, an empirical expression devised for the purpose of classifying climates numerically on the basis of precipitation and evaporation:

$$\text{P-E ratio} = 11.5 \left(\frac{P}{T - 10} \right)^{10/9},$$

where P is the normal monthly precipitation (inches) and T the normal monthly temperature in (degrees Fahrenheit). All temperatures below 28.4 F are given the value of 28.4, and PE ratios greater than 40 are counted as 40.

Holdridge Life Zone Classification

L. R. Holdridge devised a life zone classification of the worlds plant communities based on climate. Holdridge (1967) based his classification of world life zones or plant formations on three major climatic factors: heat, precipitation and moisture (Fig. 7.9). In this he recognized that moisture is dependent upon heat and precipitation. Each life zone refers to a certain type of plant association and also to the ranges of values of temperature and precipitation which produces this association. Some of the major assumptions points of Holdridge (1967) system are as following:

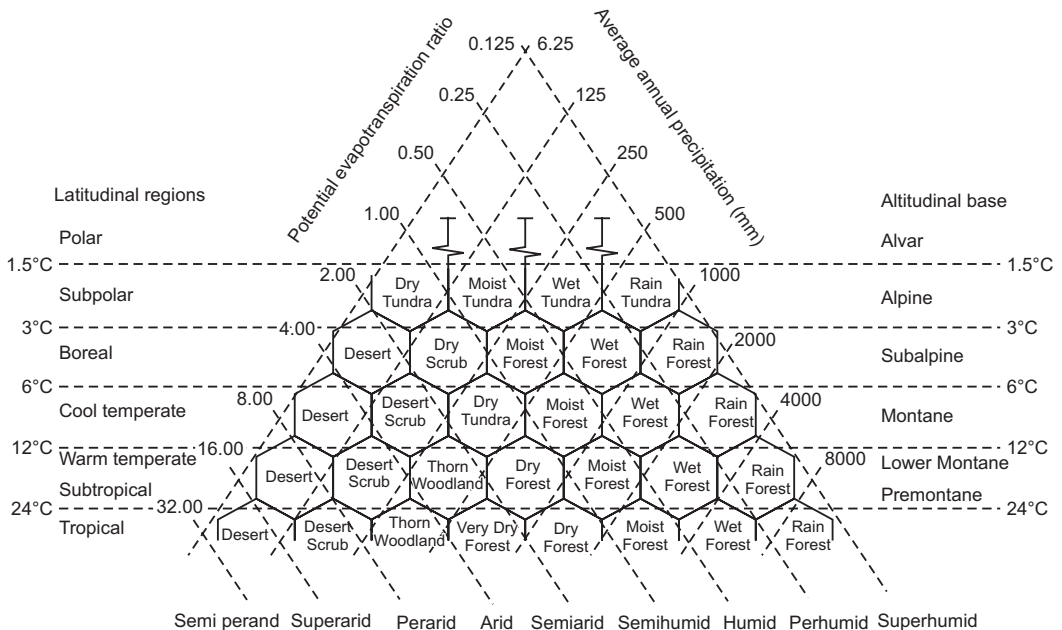


Fig. 7.9. Holdridge classification of world life zones (from Holdridge, 1967).

In this system heat is expressed as mean annual biotemperature. Holdridge defines biotemperatures as those falling in the range in which vegetative growth occurs, i.e., 0 to 30°C. To estimate the mean annual biotemperature, all the daily temperatures that fall between 0°C and 30°C are summed and divided by 365. All temperatures above 30°C are eliminated and those below 0°C are considered 0 for calculation purposes. The precipitation is expressed as annual amount of precipitation in mm.

The influence of temperature, precipitation or potential evapotranspiration on plants is related to the logarithm of the measured value of these variables. The lower limits of temperature are 24, 12, 6, 3, 1.5°C for tropical, subtropical and warm temperate, cool temperate, boreal, subpolar and polar, respectively. Similarly, the values for annual precipitation are 8,000, 4,000, 2,000, 1,000, 500, 250, 125 mm, respectively.

A simple method was developed to obtain potential evapotranspiration (PET). The normal annual biotemperature of a station (in degrees C) multiplied by the factor 58.93 gives PET value.

The world's life zones or plant formations are depicted in consideration of biotemperature, latitude; precipitation and PET ratio (mean annual PET in mm divided by mean annual P in mm). The last parameter is considered as an index of humidity (and moisture availability). Latitudinal regions have their equivalents in altitudinal belts.

This life zone classification better explained the complexities of tropical vegetation. Let us consider Dras, a place in Kashmir as an example of how to use Holdridge's system. Monthly mean temperatures in degrees C from January to December are as following: -15.7, -14.6, -8.6, -4.0, +8.4, +13.7, +16.9, +17.2, +12.8, +5.6, -2.7, -9.1. The sum of these is 74.6 (all negative temperatures were considered 0°C). The total divided by 12, gives a mean annual biotemperature of 6.4°C. The mean annual rainfall of Dras is 649 mm. The juncture of these two parameters falls within the subalpine wet forest (Fig. 7.10). Birch (*Betula utilis*) - fir (*Abies pindow*) forest occurs in this area.

Another example can be from the hot plains of India. Let us consider Varanasi which has following monthly mean temperatures: +16.2, +18.8, +24.8, +30.4, +33.5, +33.1, +30, +28.9, +29.0, +26.3, +20.8, +16.5. This gives a biotemperature of 17.6°C. The mean annual rainfall of Varanasi

is 1,040 mm. According to Holdridge's calculation, Varanasi comes under dry forest life zone. Dry deciduous forest is a characteristic vegetation of the area, though savanna has replaced it under the influence of disturbance. Though Nainital, located at about 2,000 m elevation is much cooler than Varanasi, its biotemperature, 14.0°C is not very different, because temperatures of all months fall between 0 and 30°C and, therefore all of them are considered in calculating biotemperature. While in case of Varanasi, mean temperatures from April to May exceed 30°C and therefore are not considered useful for vegetation. Nainital with the mean annual rainfall of about 2,500 mm comes under wet forest category. Evergreen oak forest (*Quercus floribunda* - *Q. leucotrichophora*) is the characteristic vegetation of Nainital. The above calculations are crude, because the mean annual biotemperature should be calculated by adding up all the mean daily biotemperatures and dividing the sum by 365.

7.7 BIOGEOGRAPHIC REGIONS OF INDIA

Large distinctive units of relatively homogeneous ecology, biome representation, and community and species composition are called biogeographic zones.

Table 7.5. Biogeographic zones and biotic provinces of India.

Biogeographic zone	Biotic provinces
1. Trans-Himalayan	1A Tibetan
2. Himalayan	2A North-West Himalaya
	2B West Himalaya
	2C Central Himalaya
	2D East Himalaya
3. Desert	3A Kutch
	3B Thar
4. Semi-Arid	4A Punjab
	4B Gujarat-Rajwara
5. Western Ghats	5A Malbar coast
	5B Western Ghat Mountains
6. Deccan Peninsula	6A Deccan Plateau South
	6B Central Plateau
	6C Eastern Plateau
	6D Chhotanagpur
7. Gangetic Plain	7A Upper Gangetic Plain
	7B Lower Gangetic Plain
8. North-East India	8A Brahmaputra Valley
	8B Assam hills
9. Islands	9A Andaman Islands
	9B Nicobar Islands
	9C Lakshadweep Islands
10. Coasts	10A West Coast
	10B East Coast

India is divisible into ten biogeographic zones: (1) Trans Himalayan, (2) Himalayan, (3) Indian desert, (4) Semi-Arid, (5) Western Ghats, (6) Deccan Peninsula, (7) Gangetic Plain, (8) Coasts, (9) North-East India, and (10) Islands. (See Rodgers and Panwar, 1988).

The biogeographic zones are divided into secondary units, called biotic province (Table 7.5). For example, the biogeographic zone, Gangetic Plain is divided into two biotic provinces, Upper Gangetic Plain and Lower Gangetic Plain (Fig. 7.10). Barriers to animal dispersal, such as major river valleys may lead to evolution of distinctive species elements in different biotic provinces. Further classification is also made. Within a biotic province subdivisions or regions are recognized and they, in turn are divided into biome, an ecological unit, such as temperate broadleaf forest and tropical dry deciduous forest. Some biomes occur in more than one biogeographic zone. Here we concentrate only on biogeographic zones of India.

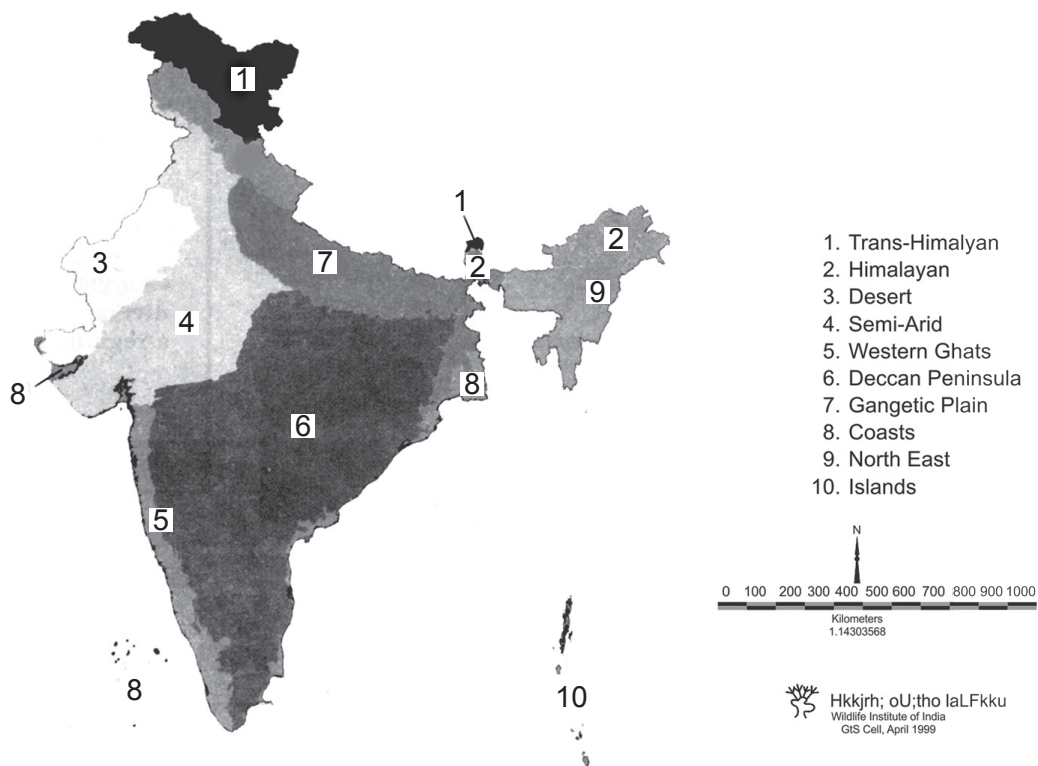


Fig. 7.10. The biogeographical zones of India.

Zone 1: The Trans-Himalayan

This zone (estimated 186,200 km² in India) is far more extensive than the area within India. It covers 2.6 million km² of high altitude cold desert plateaux of Tibet. Virtually this entire zone in India is in Lahul-Spiti district of Himachal Pradesh and in Ladakh. Biologically, the zone is poorly known. Vegetation is sparse alpine steppe, and endemism is high. (Endemic is the species which is unique to a region, it does not occur outside the region of its occurrence). The mountains of this region (including North-West India, Pakistan and Tibet) have the richest wild sheep and goat community in the world with eight distinct species and subspecies.

The flat plateaux have a distinct herbivores community (grazers) of wild yak, ass, a gazelle and four-horned antelope. In association with these grazing animals there has developed an equally distinctive set of carnivores with snow leopard, Tibetan wolf and endemic lesser cat (Pallas' cat), fox and smaller animals e.g., marbled pole cat, Royals pika and Himalayan marmot. The black necked crane is the most distinctive bird of an impressive and distinct avifauna developed in lakes and marshes.

Zone 2: The Himalayan

Importance of this Himalayan zone in terms of natural resources is far more than that is indicated by the proportion of country's surface area it occupies (7% only). More than 300 million populations of the Indo-Gangetic plains are dependent on the Himalayan waters. Unfortunately, the Himalayan Mountains represent one of the most actively degraded areas on earth. The steep slopes, unconsolidated soils and seasonally intense rainfall, all make the Himalayan environment extremely fragile. Excessive livestock density and grazing, cultivation on steep hill slopes and firewood and timber demands of the people have worsened the situation by destroying forests and other natural ecosystems in most of areas below 2000 m elevation. Impact of the degradation on biological resources is severe; consequently there are more endangered species in the Himalaya than anywhere else in India. The elevation range where plants and animals occur is among the widest (over 6000 m) in the world, and annual rainfall varies 10-fold or more. In response to variations in climate and geology vegetation varies widely from tropical rainforest like communities to alpine meadows and scrubs.

Because Himalaya is located at the boundary of so many biogeographical regions, it is one of the richest areas in habitat and species diversity. In this region following five major regions with their special taxa meet:

Region	Examples	
	Plant	Animal
Palearctic	<i>Hippophae</i>	Tibetan Ass
Mediterranean	Deodar (<i>Cedrus deodara</i>)	Ibex
Indo-Chinese	Rhododendrons	Takin
Indo-Malayan	Dipterocarpus	Sun bear, Gibbon
Peninsular India	Euphorbia	Jungle Fowl

Interestingly the Himalayan zone has also evolved a high proportion of its own endemic taxa - e.g., 60 species of balsams and tahr among animals.

The list of endangered species is long, and includes many large mammals. For example, the Sikkim stag may be lost soon from India and the tahr may have gone from Kashmir. Populations of markhor, tahr, and serow are at very low densities. Most animals show wide elevational migrations, moving toward lower slopes during winters and toward higher elevations during summers. Habitat destruction may limit the seasonal migrations, leading to extinction of large mammals.

In lower subtropical belt mixed deciduous forests occupy lowest elevations, they are replaced by chir pine (*Pinus roxburghii*) and then by banj oak (*Quercus leucotrichophora*), around 2000 m elevations. Larger fauna of this belt, viz., samber, muntjac and wild boar have affinity to Peninsula. In temperate areas, below 3500 m several forest types occur: broadleaf mesophyll forests of maples (*Acer* spp.) and other deciduous species, broadleaf sclerophyllous evergreen forest mainly of oaks, and evergreen coniferous forests of deodar (*Cedrus deodara*), blue pine (*P. wallichiana*), spruce

(*Picea*), silver fir (*Abies*), etc. Typical large fauna are musk deer, and serow, with koklas and monal pheasants. The subalpine forests/scrubs have rhododendrons, interspersed with meadows. Here, musk deer, serow and tahr share the lower ranges with beharal and ibex, which are more common in higher elevations. Compared to west, which is rich in brown bear, beharal and tahr, east Himalaya has Mishmi takin as large herbivore. The binturong, red pandas and lesser cats are conspicuous Eastern Himalayan fauna. Among plants Eastern Himalaya has more diversity of oaks, tree ferns, orchids and rhododendrons than the Western Himalaya.

Zone 3: The Indian Desert

The Thar Desert of Western India and Eastern Pakistan is of relatively recent formation. Biogeographically it is the eastwards extension of the Sahara-Arabian desert system which spreads through Iran, Afghanistan and Baluchistan to the Thar area on the India-Pakistan border. Annual rainfall in this desert is less than 400 mm. But the extreme seasonality of rainfall (most of the annual rainfall occurring within couple of rainy season months) and livestock density far greater than the land can support, have made the desert condition very harsh. Within the zone the salt flats of Kutch are quite dissimilar from the typical desert condition Salt marshes and islands in Kutch area, dunes in deserts and southern hills provide variety to desert environment. The forests in this area comprise bushy thorny plants such as *Acacia nilotica*, *Prosopis cineraria*, *Salvadora oleoides* and *Tecomella* spp. Lack of cacti is the special feature of Indian desert. *Prosopis juliflora* and other exotic species are becoming increasingly widespread.

There are wild animal species of interest. The wild ass, a distinct subspecies, is now confined to the Rann of Kutch. The desert fox, Indian desert cat, Honbara bustard, and some sand grouse species are found only in the Thar. The desert cat is found all over Rajasthan and Kutch, but extensive killing for its skin has made it rare and endangered. Flamingoes find breeding sites only in Raun of Kutch within the Indian subcontinent. Furthermore, chinkara, blackbuck, wolf, caracal and great Indian bustard have sizeable populations in this zone.

Zone 4: The Semi-Arid

It is the semi-arid regions with their grasses and shrubs where generally some of the largest biomasses of wildlife occur.

This area of India has strong biological links with Western Asia and Northern Africa. Many plant taxa have African affinity: *Acacia*, *Anogeissus*, *Balanites*, *Capparis*, *Grewia*, etc. Pure *Anogeissus pendula* forest community occurs only in this zone, on gentler slopes of the Aravalli and associated hill ranges.

The larger herbivores are bovidae, e.g., blackbuck, chowsingha, nilgai and gazelle. The carnivores, viz., lions (now restricted to a small area in Gujarat), cheetah (now extinct), caracal, jackal and wolf have analogues in North Africa. Endemism of both plants and animals is low, the Asiatic lion being the notable exception. The Asiatic lion is restricted to single site (Gir National Forest Reserve) in this zone. Pressure of livestock and agriculture has caused local extinction of many wildlife species.

Occupying 15% of India, this region has some of the most productive agricultural areas in Punjab and Haryana, as well as a larger proportion of the total livestock population. One part of this zone surrounds desert zone of Western Gujarat and Rajasthan (in Punjab, Haryana, Delhi, Rajasthan and Gujarat) and the other part consists of the rain-shadow areas behind the Western Ghats (in Maharashtra, Andhra Pradesh, Karnataka and Tamilnadu).

Zone 5: The Western Ghats

This zone extends from Kanyakumari in south to the hills south of the Tapti river in north. The ghats can rise from sea level to 2700 m elevations. The moist evergreen forests are most extensive in the Western Ghats. The zone is exceptionally species-rich. It has 4,000 species of higher plants, which account for 27% of the India's 15,000 species. There are 1,800 endemic species in the region, which constitute about two-third of endemic species of higher plants in India. Of the 4000 species close to 3000 occur in the evergreen forests and about 850 in the moist deciduous forests. The region has 13 of the 29 species of Dipterocarpaceae present in India. These 13 species are found nowhere else. The area is of the great conservational concern. The rainfall is heavy; possibly more than 2000 mm in most areas, but can exceed 5000 mm in some areas. Where soils are good on hill slopes forests have been replaced by tea, coffee, cocoa, rubber, cardamom, Chincona, and other plantations. The sub-tropical forest, commonly called Sholas occur in Palni, Anaimalai and Nilgiris, usually at an altitude of 1800 m.

The region is a great natural resource. It is a source of water for the huge river systems of Peninsular India for hydroelectricity and irrigation schemes. The Western Ghats have viable populations of most of the vertebrate species found in the Peninsular India, plus a number of endemic animals of their own, distributed in all vertebrate groups – fish, amphibians, reptiles, birds, mammals. Some of them show extremely restricted distribution: the Travancore tortoise and cane turtle are endangered taxa. There are 62 mammal genera, one of which, a rodent, is endemic. There are several linkages of biogeographic interest: to the Himalaya (e.g., the tahr), to the North-East India and to Sri Lanka.

Significant species and species groups restricted to Ghats are: (i) primates - Nilgiri Langur, Lion Tailed Macaque, (ii) rodents - *Plataothomys*, an endemic monotypic genus, the spiny dormouse; (iii) squirrels - the grizzled giant squirrel; (iv) carnivores - Malabar Civet, rusty spotted cat; (v) ungulates - Nilgiri tahr (a wild goat), and hornbills. The Malabar grey hornbill is a good indicator of healthy and mature deciduous forest along the Western Ghats.

The Western Ghats have large and potentially viable populations of the other species of conservational significance, which also occur in other zones - e.g., tiger, leopard, dhole, sloth bear, Indian elephant and gaur.

Zone 6: The Deccan Peninsula

Covering 43% of the Indian subcontinent, the Deccan biogeographic zone of Peninsula India is by far the largest in area. Though disturbed in most areas due to man's activities, the zone does contain some more conserved forest areas of India in Madhya Pradesh, Maharashtra and Orissa. The Deccan highlands constitute the principal catchment for a number of South India's main river systems (Narmada, Tapti, Mahanadi, Godavari).

It is the premier area for the deciduous forests, thorn forests and degraded shrublands. Only in small areas evergreen forests occur. Sal (*Shorea robusta*) and teak (*Tectona grandis*), the precious timber species of India along with several other tree species occur in the northern half of the zone. The southern half has generally drier thorn forests with *Acacia*, *Albizia amara* and *Hardwickia* associations. Natural grasslands are rare, but grasslands have developed as a result of fire, cutting and grazing of forests.

Most wildlife species are/were widespread throughout the zone – e.g., chital, sambar, nilgai, chowsingha. Species which have small relict populations include the elephant, wild buffalo (a small area at the junction of Orissa, Maharashtra and Madhya Pradesh) and the hard ground swamp deer. Gharhial is restricted to some rivers. Among carnivores, tiger is absent from many drier areas in the south, and the rusty spotted cat and wolf occur only isolated low-density populations.

Zone 7: The Gangetic Plain

The Gangetic plains are one of the largest zones in India, stretching from the Yamuna river eastwards across Uttar Pradesh, Bihar, West Bengal and coastal plains of Orissa (also in Nepal tarai, Bangladesh and Coastal Burma). The plain is topographically homogenous for hundreds of kilometers, and represents one of the most fertile areas and “robust” agricultural lands in the world. From great aridity in the west (less than 500 mm rainfall), the plain goes to great moistness in the east (more than 2000 mm). Along the Himalayan mountains rainfall is high, consequently rainfall increases from south to north.

There are still a large number of lakes, seasonal swamps and other small wetlands where wetter vegetation types and wildlife of significance have developed. Sal (*Shorea robusta*) forest represents potential vegetation along the Himalaya and mixed dry deciduous forests in plains. Lack of endemism is the characteristic wildlife feature. Either no species were differentiated on the plains or all have been eliminated by now due to expansion of agriculture and high density of human populations. However, centuries ago, the area was rich in wildlife with rhino, elephant, buffalo, swamp deer, etc., and attendant carnivores. Western areas still hold relict populations of nilgai, blackbuck and chinkara. Swamp and hog deer occur only in small pockets. The rhino, Bengal florican and hispid hare remain in low numbers in the tarai. The foothill (of Himalaya) forest have a sambar chital community with goral on steeper slopes.

Zone 8: North-East India

Of all zones in India, perhaps North-East India is the richest in communities, in species and in endemics. However, we know little about distribution and status of several species, *let alone* their biology. About 40% of the total surface area distributed in states of Assam, Manipur, Meghalaya, Mizoram, Nagaland and Tripura is forested.

North-East India is biogeographically very interesting, as it represents the transition zone between the Indian, Indo-Malayan and Indo-Chinese regions as well as the meeting place of the Himalayan Mountains and Peninsula India. After the break up of Gondwana land in the early Tertiary, the northward moving Indian plate (Peninsula Deccan) first touched The Asian landmass in this region. The north-east is thus the biogeographical gateway for much of India's taxa and is of great biogeographic richness and significance. There are transition zones between low and high-land areas, where a great number of species-rich communities occur. Botanically Khasi-Jaintia hills of Meghalaya are known as one of the richest in Asia. Among the animals, the hornbill and the small carnivore communities display a species richness not known anywhere else.

It is only in the north-east the full richness of the large herbivore fauna typical of alluvial grasslands can still be found: rhinoceros, buffalo, elephant, swamp deer, hog deer, pygmy hog and hispid hare. The region represents an important fly-way for waterfowl and other herds seasonally migrating between the warm climate of Indian subcontinent and cold climate of Siberia and China.

Zone 9: The Islands

Two clearly different groups of islands from this zone: (i) The Andaman and Nicobar islands - located in Bay of Bengal, thinly populated with tropical rainforests; and (ii) Lakshadweep islands are located in the Arabian Sea and with little remaining natural vegetation.

The Andaman and Nicobar islands (consisting of 348 islands) are elongated in shape, stretching for 590 km (13°, 45 N to 6°, 45 N lat.). The present description is based on them. The islands are separated by shallow waters from the Burma-India coast, and were connected to the mainland up to recent geological past (the Pleistocene times). They form an extension of the Arakan mountain range of Burma and there are several peaks over 500 m. As expected, the Andaman has biogeographical affinity with Burma. The Nicobars have stronger Indonesian and South-East Asian elements.

The species richness in the islands' tropical rainforests is high (they represent one of the three tropical rainforests areas of India, the other being in the Western Ghats and the north-east). Because of isolation from the mainland their species assemblages are not found elsewhere in India and endemics are high proportionately.

In the Andaman and Nicobar islands, 86% area is legally under forests, which are classified in to mangroves, beach forests, evergreen, semi-evergreen and deciduous forests.. But intensive tree cutting in the past has degraded the evergreen forests and more into less species-rich deciduous forest communities. Because some of the islands are very small forests occur in isolated patches (as many as 57 protected forests being less than 1 km² in area). *Dipterocarpus* spp are the most characteristics of the evergreen forests. In the deciduous and semi evergreen forests, *Terminalias* and *Lagerstroemia* are common species.

Some 2,200 plant species (higher plants) are found in the islands which is 10% of the species of India in only 8,000 km². Over 200 are strict endemics. A further 1,300 species are not found elsewhere in India, but occur in Burma, Malaysia, Indonesia and Polynesia. Having affinities with Indonesia, the Nicobars lack in *Dipterocarpaceae*, and are rich in tree-ferns, palms and orchids

The importance of this zone is in its species richness and endemism of plants and birds (not in mammals). Because of the isolation of islands and their relatively small size mammal fauna is poor. Most species are of rodents and mammals. Mention may be made of some exotic mammals that have naturalised. One is "Andaman Pig", which was introduced centuries ago, and is now regarded as to have attained a sub-specific status. It has bred constantly smaller individuals over the passage of time.

Among the native mammals, the crab-eating or Nicobar macaque and the Nicobar tree shrew are interesting and warrant conservational measures. The Nicobar megapode, a mound building bird that occurs around sandy shores and littoral forests is highly endangered species of the Andaman and Nicobar islands. The Narcodium hornbill, now represented by some 200 birds is endemic to 7 sq km area of the volcanic island, Narcodium. Many more bird species are endemic and are restricted to evergreen forests – e.g., the Nicobar pigeon, Andaman wood pigeon, the Nicobar parakeet, etc. A large proportion of amphibian's reptiles are endemic for the islands. Many species await discovery. One called, water monitor is important with respect to island ecology. The salt water crocodile once widespread, is now very restricted in distribution. The islands have 4 species of marine turtles. The coastal systems have mangrove estuaries, coral reefs and marine cliffs. Fish and coral communities are very rich and await exploration. Dolphins are there and whales have been seen.

The Laccadive and Minicoy group of islands along the western coast form a distinct botanical region. Coconuts, bread fruit, plantains and betel vine grow well in hot and wet areas.

Zone 10: The Coasts

India's long coastline has a diverse set of communities, but is very poorly explored. In brief the features of interest can be listed as follows:

- (i) Mangroves, showing a great variety of communities including estuaries, lagoons and deltas.
- (ii) Sandy beaches, including distinctive plant communities such as *Casuarina-Calophyllum-Pandanus*.
- (iii) Mudflats with a range of successional stages.
- (iv) Raised coral and rocky coastlines.
- (v) Marine angiosperm pastures.

The animals of especial interest are: dungdong, hump-back dolphin of turbid estuarine waters, which is poorly known; estuarine or salt-water crocodile; varied turtles, especially Batagur basker of Sunderbans estuary and *Pelochelys bibroni*, the huge soft-shell estuarine turtle of the Utkal-Bengal coast; avifaunal communities of mangroves, mudflats and lagoons.

Part – II

POPULATION AND COMMUNITY

- ♦ POPULATIONS : CHARACTERISTICS, GROWTH AND REGULATION, DYNAMICS AND GENETICS
- ♦ SPECIES INTERACTIONS
- ♦ COMMUNITY: NATURE, STRUCTURE , ATTRIBUTES AND CONTROL
- ♦ METHODS OF SAMPLING VEGETATION AND DATA ANALYSIS
- ♦ COMMUNITY DYNAMICS: SUCCESSION

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CHAPTER

8

Populations: Characteristics, Growth and Regulation, Dynamics and Genetics

Ecologically, a population is a group of interbreeding individuals of a species, occupying a particular area. For example, all the individuals of oak trees in a forest, a group of snails of the same species on the forest floor, or a pure culture of a bacterium represent their populations. In human demography, a population is a set of humans in a given area like a village, town, state or a country. A population exhibits characteristics which are also shared by the individual organisms of which it consists of, as it undergoes the same life cycle and similar ecological processes at a particular stage of the life cycle. For example, a population has a life history in that it grows, differentiates, and maintains itself as does the individual organism. However, it has characteristics such as birth rate, death rate, age ratio, that are manifested only at group level, i.e., at population level. Although an individual organism is born, it grows, it dies and also has age, it is not possible to describe its birth rate or death rate or age ratio.

Populations can also be defined at various spatial scales, i.e., local, meso - and regional scale. Andrewartha and Birch (1954, 1984) included spatial dimensions in the definition of population and recognised the concept of local populations because individuals of a population interact with conspecifics to respond to a particular environment, allowing immigration and emigration between local populations. Local populations can occupy very small habitat patches like a temporary pool of water or a small field. A set of local populations connected by dispersing individuals is called a metapopulation. Populations can also be considered to occur at the scale of regions, islands, continents or seas; even all the individuals of an entire species can be viewed as a population of that species.

The attributes of a population can be conveniently divided into two categories: (1) those dealing with numerical and structural properties, and (2) those dealing with general genetical properties.

8.1 MATERIAL OF POPULATION STUDIES

Most of the population studies are animal based. Although there are fundamental similarities between the population attributes of animals and plants, the differences are sufficiently great that it may not be appropriate to generalize from one kingdom to the other (Harper, 1977).

Why generally animals have been preferred objects for studying the population attributes and other aspects of population dynamics? After all, plants normally stand at one place and, therefore, are easy to count, whereas animals which move from one place to another have to be trapped, chased, or shot for counting. One confronts altogether different problems in the case of plants as far as counting is concerned, that is, in numerous cases it is difficult to identify an individual,

therefore difficulty arises in regard to ascertaining the unit of population. A count of the number of rabbits, or monkeys or elephants gives nearly precise information on the basis of which approximate predictions of population growth rate or age distributions are possible. In contrast, there are two levels of population structure in plants. One level identifies individuals that are derived from original zygotes or seeds, and are called genets. An individual genet may be a tiny seedling or a clone spreading in fragments over several square metres of land area. Each genet may consist of numerous easily countable units which may be shoots of a tree, the ramets of a clone, the tillers of a grass, or leaves with their buds in annuals. These units can be referred to as modular units. The number of modular units or constructional units per genet may be indicated as n and the number of genets as N . The number of modular units in a plant population will be described by Nn , and changes in N as well as n or both will influence the behaviour of plant populations (Harper, 1977).

An elephant will continue to have four legs whether it is undernourished or overnourished, but an individual plant will show wide changes in the number of a part under varying environmental stress. For example, an individual of wheat growing under favourable water and nutrient conditions may attain 1.5 m height and bear hundreds of seeds, but under conditions of deficiency of nutrients and water it may not grow beyond 50 cm and bear only a few seeds.

Unlike plants, a majority of animals do not lose their parts in their lifetime. Leaves on a plant may have different life cycles, they appear at different times, and consequently their expansion, senescence, and shedding vary in time. Similar is the case with branches and twigs. Thus, units or modules of birth and death can be leaves, branches, twigs, etc., depending upon the kind of plant. In ferns, like *Pteridium aquilinum*, leaves are useful counting units of modules. In certain other vegetatively propagating species such as *Ranunculus repens*, the ramet is the unit of clonal growth, as they will have independent existence after separation from the parent plant. The above examples highlight the fact that in plants the genetic units and physiological units (module) are not the same. In most of the cases, number of animals will represent the number of genotypes, whereas in plants all the individuals, for example, in a marshland of *Typha* sp., may belong to a single genotype (i.e., clone).

Diagrammatic model of plant population

Harper and White (1971) have summarised the sequence of events that determines the success of a seed in producing more seeds (Fig. 8.1). The first step can be recognized as dispersal of seeds from plants which leads to storage of seeds in a dormant state in and above the soil and litter, termed seed bank (I). Depending upon the physical condition that they experience, the seeds germinate and seedlings are recruited (II). More often than not, a great number of seeds of seed bank fail to germinate. The survivors are indicated in the model as branched systems (modules), potentially branching exponentially but restricted within resource constraints (III). The growing plants make their environment different from the earlier ones; this changed environment influences the subsequent recruitments. The growth of plants may culminate in the phase of seed production (IV) (Fig. 8.1 *a*), as it happens in annuals and many biennials and some perennials. They are called monocarpic. In many perennials, the seed production is followed by further growth of plants and this pattern may be repeated several times (Fig. 8.1 *b*). In any case, the seeds produced are dispersed and enter the cycle again.

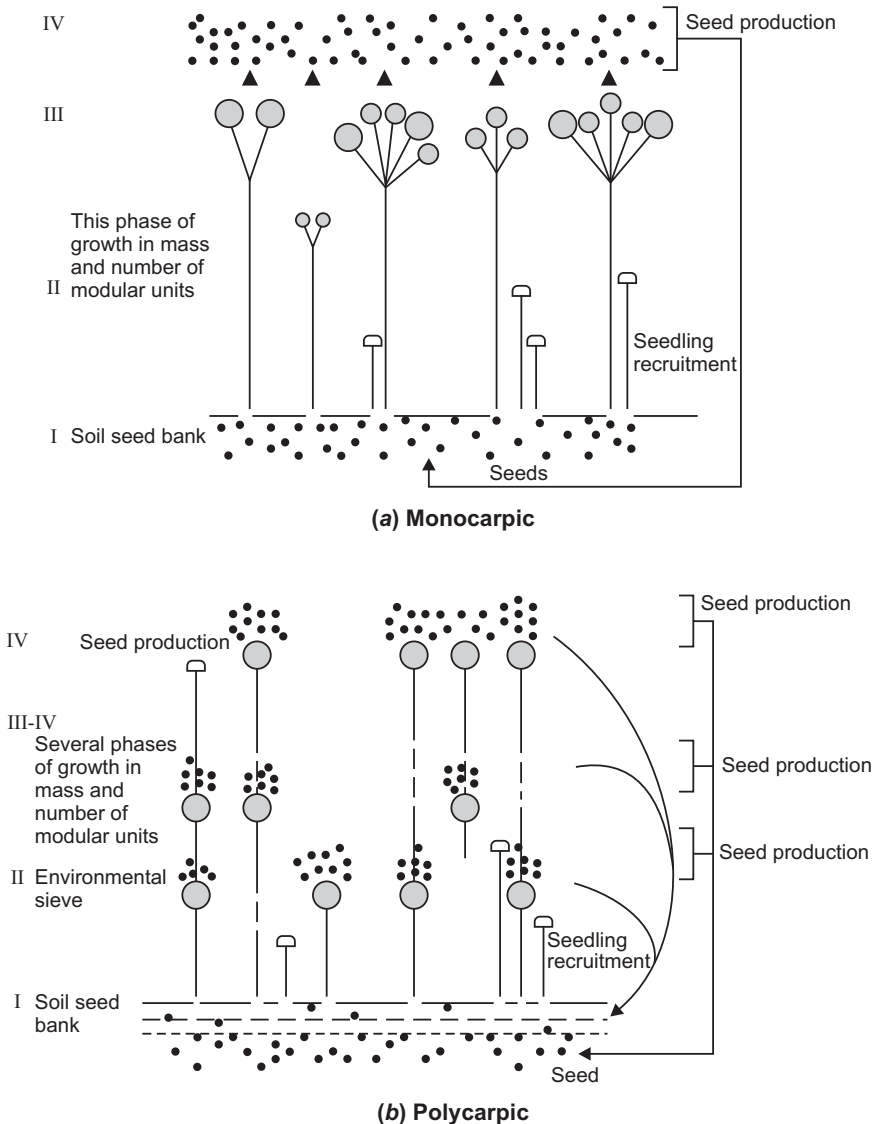


Fig. 8.1. Elements of population dynamics of a monocarpic (a) and polycarpic (b) plants. Apart from various phases of plants, soil seed bank is shown. (derived from Harper and White, 1971).

8.2 CHARACTERISTICS OF POPULATION

The individual populations have group characteristics that can be measured as density, age ratios, natality and mortality. Population size is influenced by natality, mortality, emigration and immigration of individuals (Fig. 8.2 a). Many terrestrial populations are closed populations in which birth rates and death rates are dependent solely on population density, with little emigration and immigration taking place between local population (Fig. 8.2 b). While emphasizing the closedness and persistence of populations, Berryman (2002) defined that a population “is a group of individuals of the same species that live together in an area of sufficient size to permit normal dispersal and/or migration behaviours and in which population changes are the result of birth and death processes”.

Presence of open populations is prevalent in marine environments where dispersal of juveniles from a relatively sedentary adult phase is broad. Most sessile marine species release their highly mobile larvae in open water. The marine benthic system is open if its emigrants (e.g. larvae) can be transported outside the parental range or most of the recruits come from distant places (Fig 8.2 c). For open populations, like in most marine populations, replenishment is largely or exclusively dependent on an exterior supply of juveniles. For example, in the case of coral reef fishes, their populations are dependent on a supply of larvae (from the pelagic water) in open water to replenish their populations. The local production of offspring in sedentary phase has little or no direct role in setting local population size because larval recruitment from elsewhere provides the only substantial input of new individuals. If recruitment fails, the local population will decline to extinction, regardless of local fecundity. Conversely, a different local population could persist as long as recruitment continues, even if the adults produce no viable offspring.

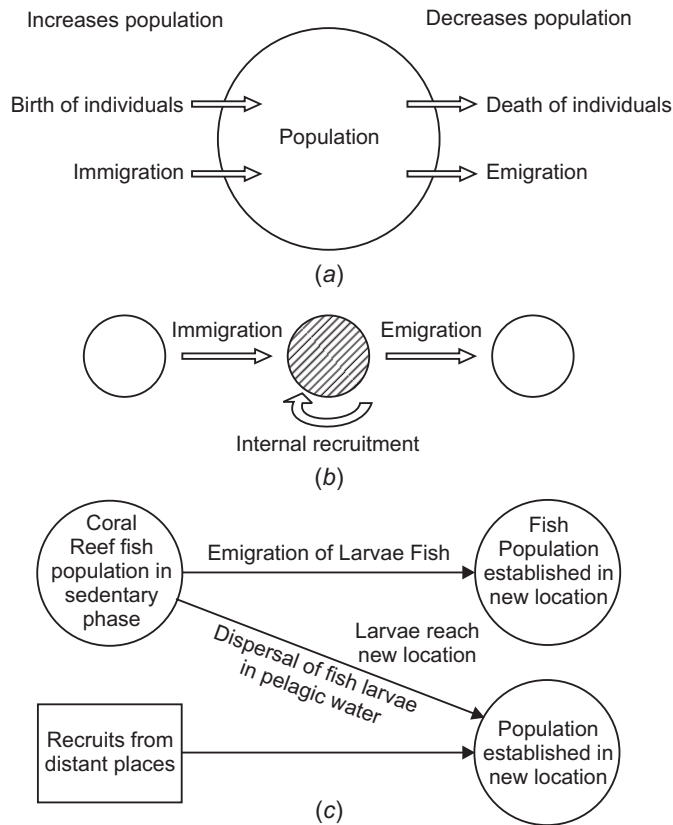


Fig. 8.2. The number of births, loss of individuals, immigration and emigration interact to change the size of a local population (a); emigration and immigration are of low significance for a closed population (b), in open populations of marine systems most of the juveniles are transported out of parental range and most of the recruits come from distant places (c).

In terrestrial populations, the openness of population is due to the dispersal of seeds in the case of plants, and ‘aerial dispersal’ in the case of insects. In marine systems, most of small units of population behave in an open manner (Camus and Lima, 2002). The authors further opine that in strictly biological terms, all population systems can be treated as closed, whether terrestrial or marine, when studied at the proper scale and are open at any smaller scale in an operational term (see Gaylord and Gaines, 2000; Camus and Lima, 2002).

Density

Density of a species is the number of individuals per unit area or per unit volume. For example, the density of oak trees in a forest can be represented as 250 trees per hectare of land area, and that of phytoplankton in a water body as 50,000 cells per litre of water. An investigator may report the number of an orchid per ha of a forest, but the entire area of that hectare of forest is not utilized by the orchid, for it grows epiphytically only on tree trunks and branches. In a pond light is not always sufficient throughout the depths, so living space for an alga, such as *Scenedesmus* or *Microcystis*, may be only the top 1 m of water up to which the intensity of light is sufficient for adequate photosynthesis. Density in terms of the amount of area available as living space is known as ecological density. The former type of density (e.g. number of individuals of orchids per hectare) is termed crude density.

Although density of an organism varies widely in time and space depending upon the size and trophic level, some limit on the upper size of density can be set. The upper limit of the density of herbaceous plants such as soybean (*Glycine soya*) may be about one million individuals per ha, that of the adult trees of common Himalayan oak (*Quercus leucotrichophora*) about 5000 trees per ha, and that of giant adult trees of Californian redwood forests (*Sequoia sempervirens*) 100 trees per ha. Knowledge of such relationships in densities of organisms is of great practical importance in wildlife and vegetation management.

Dispersion

Dispersion indicates how the individuals of a population are distributed in space and time. Evenness in concentration of individuals is highest in regular, lowest in contagious and of intermediate level in random dispersion. Random distribution emphasises that every individual has an equal chance to occur at a place (Fig. 8.3 a). But it can occur only where the environment is uniform, resources are equally available throughout the year, and interaction among members of the population produce no pattern of attraction or avoidance and one individual has no bearing on the distribution of the other individual.

Regular or uniform distribution (Fig. 8.3 b), which implies even-spacing of individuals, may result from intraspecific competition among members of a population. The intraspecific competition involves territoriality among carnivore birds, severe struggle for crown and root space among forest trees and moisture requirement in desert bushes. Auto toxicity (the production of exudates which are toxic to seedlings of the same species) may also bring about a regular distribution.

In contagious, also called clumped or clustered or aggregated distribution, organisms congregate in groups with some scattered individuals outside the groups (Fig. 8.3 c). In this, presence of one individual means there is a high probability of finding other individuals of the same species in immediate neighbourhood. Since environment is never uniform in space, since populations frequently produce attraction and avoidance, and since social behaviour promotes aggregation, plants and animals are mostly contagiously distributed. Susceptibility to interspecific competition also contributes to contagious distribution. A large aggregation of individuals of *Typha* or of a fern is due to vegetative propagation. Large seeds of oak and horse chestnut (*Aesculus indica*) which reach the ground due to gravitational force form clusters of seedlings near the parent tree. The animals may collect and store seeds, burying them in local aggregations from which a dense patch of seedlings may emerge. Many birds eject pellets of a number of undigested seeds; e.g. English robin may eject a pellet containing 20 raspberry seeds, and 5 seeds of other species (Lack, 1966). In such cases, a small cluster of seedlings is ultimately formed. Even small seeds, which are mobile, can reach great distances and produce clumping of varying degrees. The pattern of distribution also changes with the change in the activity of organisms.

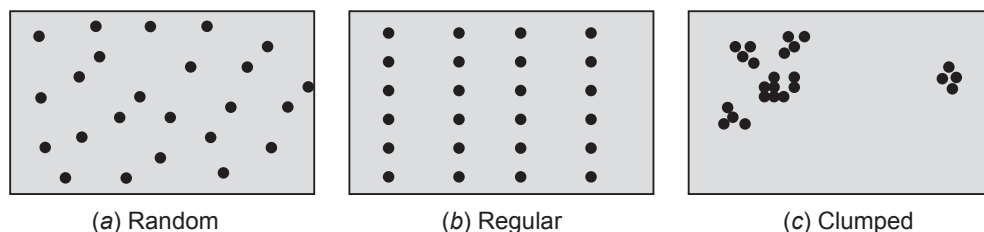


Fig. 8.3. Types of population dispersion in a community. (a) random (apparently irregular) distribution; (b) regular or uniform (negatively contagious) distribution; (c) clumped (contagious) distribution.

Natality

Natality means the production of new individuals of organisms which include the born ones (as in humans), hatched ones (as in birds), those arise after germination (as in plants), or by divisions (as in bacteria), etc. Natality that a population realizes without any constraint is its physiological natality. It is constant for a population. However, in nature, several physical constraints restrict the natality to a much lower level than it is theoretically possible. For this, we use the term *Ecological* or *realized natality* (or just “Natality”), which is the number of individuals produced under an actual or specific environmental condition. Since the level of the physical constraints in nature varies in time as well as in space, the ecological natality is not a constant.

The various expressions that measure the ecological natality are as below :

ΔNn = production of new individuals in the population, or new individuals added to the initial population.

$\frac{\Delta Nn}{\Delta t} = B$ or natality rate per unit time (here, t is time).

$\frac{\Delta Nn}{N\Delta t} = b$ or natality rate per unit population.

N often includes only the reproductive part of the population, e.g. females in the case of higher organisms. The specific natality rate, b , is often defined as age-specific natality rate for different age groups in the population. For example, age-specific rate in wild rabbit is considerably higher in 2-year-old females (1.5 individuals per female per year) than in others. In the case of vegetative propagating plants, N represents the genets which originate from zygote and n the fragmented units (modules), such as tillers in grasses, originating from a genet. It may be pointed out that natality rate is either zero or positive, but never negative.

Mortality

Mortality, which means death of individuals in the population, has a profound effect on the population size. In plants, it begins right from seeds and in animals, right from eggs. Mortality can be represented as the number of individuals dying in a given period, i.e. death per time, or as specific mortality – usually in terms of the proportion of initial population dying within a given time. The former is simply *mortality rate* and the latter is the probability of dying. The complement of mortality rate is the *probability* of living or *surviving*. If mortality rate is expressed as a fraction, M , then survival rate is $1-M$ (i.e., the fraction of population that did not die).

The term *Ecological* or *realized mortality* (i.e., actual mortality) means the loss of individuals due to death under a given ecological condition. Ecological mortality is not a constant, for it is liable to vary in response to variations in environmental conditions and size of population. The theoretical possible mortality under most favourable conditions (ideal) or non-limiting conditions is a constant and is expressed in terms of the minimum.

Life Tables

A life table summarizes information of the death and survival of a population by age. The unit of age could be in hours, days, months and year depending on the life-span of the species or age class (e.g., 0 to 1 year, 1 to 2 years, etc.). The group of individuals born or germinated at the same time is called cohort. The cohort life tables are readily applied to populations of plants and sessile animals as it is easy to measure the marked individuals. Murie (1944) constructed the static life table for Dall mountain sheep (*Ovis dalli*) in Mount Mckinley (now Denali) National Park, Alaska, USA, during the 1930s. On the basis of the horns of the dead animals (as recovered from 608 skeletons of sheep), the age of the individuals was judged for constructing a static life table.

A life table can be constructed with five columns: age class (x), number of individuals dying in an age interval out of, say, 1000 born (d_x), number of individuals surviving at the beginning of the age interval out of, say, 1000 born (l_x), mortality rate per thousand alive individuals at the beginning of the age interval (q_x), and life expectancy (mean life time) of the remaining individuals that attain the age interval (e_x) (Table 8.1). For the estimation of e_x , two other quantities, L_x and T_x are calculated as explained below.

Table 8.1. Life table for the Dall mountain sheep. Based on known age at death of 608 sheep dying before 1937 (both sexes combined). Developed from Deevey (1947). In d_x , l_x and q_x column, the values are number of individuals starting from 1000 born individuals.

x (years)	d_x	l_x	1000 q_x	L_x	T_x	e_x (years)
0-0.5	54	1000	54.0	973	8005	8.0
0.5-1	145	946	153.0	873.5	7032	7.4
1-2	12	801	15.0	795	6158.5	7.7
2-3	13	789	16.5	782.5	5363.5	6.8
3-4	12	776	15.5	770	4581	5.9
4-5	30	764	39.3	749	3811	5.0
5-6	46	734	62.6	711	3062	4.2
6-7	48	688	69.9	664	2351	3.4
7-8	69	640	108.0	605.5	1687	2.6
8-9	132	571	231.0	505	1081.5	1.9
9-10	187	439	426.0	345.5	576.5	1.3
10-11	156	252	619.0	124	231	0.9
11-12	90	96	937.0	51	57	0.6
12-13	3	6	500.0	4.5	6	1.2
13-14	3	3	1000.0	1.5	1.5	0.7

The mortality (d_x) for an age interval is calculated by subtracting l_{x+1} from l_x . For example, in Table 8.1, l_x is 1000 and l_{x+1} is 964 for the age class 0-0.5 yr, so $d_x = 1000 - 946 = 54$, and for the age class 0.5-1 yr $= 946 - 801 = 145$.

The probability of dying or age specific mortality rate (q_x) in terms of initial population at the beginning of the age-interval is determined by dividing the number of individuals that died during the age interval (d_x) by the number of individuals alive at the beginning of the age interval (l_x). For example, for age class 1-2 yr in Table 8.1, it is $= 12/801 = 0.0149 = 15.0$ per 1000 individuals.

The calculation of life expectancy (e_x) at the end of each age-interval needs two further calculations, L_x (the average time period lived) and T_x (the total time period lived). These are calculated as follows.

$$L_x = \frac{l_x + l_{x+1}}{2}$$

For example, for age class 1-2 yr, L_x is $= (801 + 789)/2 = 795$ (Table 8.1). This calculation is made for each age interval. The calculated L_x values are summed from the bottom up to obtain T_x for an age interval. If the values of L_x are calculated from data in Table 8.1, the T_x for age class 10-11 yr is 231 (L_x values from the bottom of table to age class 10-11 yr; $1.5 + 4.5 + 51 + 174 = 231$).

The e_x for each age class is obtained by,

$$e_x = \frac{T_x}{l_x}.$$

For example, life expectancy for the age class 10 to 11 yr will be $231/252 = 0.9$ yr (Table 8.1). Usually, survivorship (l_x) and mortality (d_x) are presented as fractions.

These life tables could be useful to determine the environment effects on population processes in long lived organisms (Murie, 1944; see Deevey, 1947; Ricklefs, 2001). The life expectancy at the end of each age-interval is little difficult to understand. Life expectation at age 0 is the mean natural longevity.

It is easy to obtain data on mortality, survival and age for laboratory organisms and for human beings, but not in the case of organisms living in natural communities. Marking or banding is done for numerous organisms of a population in nature at the time of birth or germination and various techniques which may give correct or approximate estimates are used to determine the age.

Survivorship curves

Based on the l_x column, *survivorship curves* can be derived. The curve is derived by plotting logarithms of the number of survivors, or the l_x column on the vertical axis against time, on the horizontal axis. In Table 8.1, the l_x values are given in numbers starting from 1000 born individuals. However, for survivorship curves, these values are represented as frequency (or proportion) of individuals that survive from birth to age x , where the frequency ranges from 1.0 down to 0. For example, if only 50% of the individuals of a population survive to the age of 1 yr, then $l_{x1} = 0.5$, and if only 10% survive to an age of 2 yrs, $l_{x2} = 0.1$, and so on.

Three basic forms of survivorship curves are shown in Fig. 8.4.

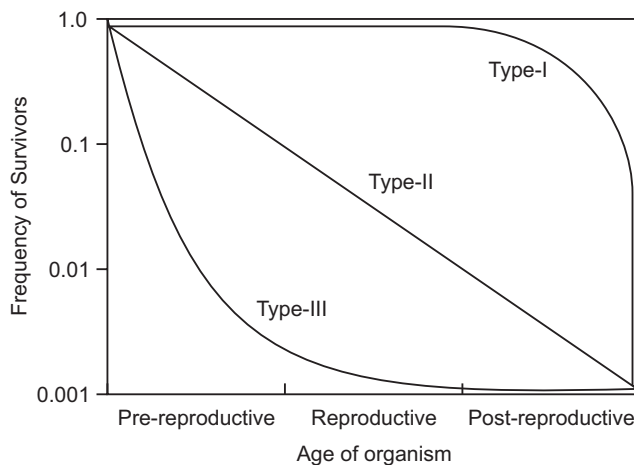


Fig. 8.4. Three basic types of survivorship curves.

(I) The convex curve (type I) indicates remarkably low mortality during most part of the organism's life (i.e. accidental mortality is kept to a minimum), and heavy mortality when the members reach the age of senescence. Such a curve is typical of some grasses, mountain sheep, elephant, whales and humans.

(II) Linear curve (type II), indicating constant mortality rate throughout all age classes (i.e. the probability of death remains about the same at every age), can be exemplified by several perennial plants, *Hydra*, and many rodents, birds and lizards. A fixed fraction of each age group is removed by predators, accidents and other causes.

(III) The concave curve (type III), which indicates excessively high mortality rates in early life, is the most common in nature and is typical of several plants, most insects, many fish, and sea turtle.

The extreme *r*-strategists tend to have type III curves, while the extreme *K*-strategists follow type I curves.

Age structure

Age structure is another population characteristic. Age structure refers to the proportion of individuals of various ages in a population. Age may be expressed in days, months or years or even in categories such as pre-reproductive, reproductive and post-reproductive (Bodenheimer, 1937). The age distribution is conveniently represented by plotting percentage of individuals in the different age classes by the relative width of successive horizontal bars; the youngest class being at the base and the oldest at the top.

A high percentage of individuals in reproductive phase foretell rapid growth in population. In many plants and some insects, post-reproductive period is absent, as individuals die after producing eggs or seeds. Bamboos are known for mass flowering and subsequent mass mortality. In perennial plants, such as trees, although the capacity to produce seeds may decline at older stage, usually they continue to bear seeds until their death. Apart from these, in plants a number of patterns with respect to reproductive phase can be found. For example, the annuals can be of two types: one in which main meristem is used in flowering, and the reproductive event leads to end of the growth and life. In the second category, the potentiality to grow even after seed setting exists because main meristem is not used in reproductive structure.

The age structure (pyramid) varies from population to population, but is likely to fall in one of the three distinct types (Fig. 8.5). (I) A pyramid with a very broad base and tapering strongly toward the top indicates high proportion of individuals in young classes and a decreasing proportion of individuals in succeeding older classes. (II) A more or less bell-shaped pyramid, i.e. having a moderately broad base and also moderate tapering toward top, indicates a moderately high proportion of young to old. (III) The urn-shaped pyramid, having a narrower base and a bulged middle portion, indicates low proportion of young to old. These three pyramids suggest expanding, stable and declining populations, respectively. It is evident that in the first, there are many more individuals in younger classes to replace those of the older classes. The population, if it is of humans or of some higher animal, will expand also because increasingly greater number of individuals will enter the reproductive phase. The third type of pyramid indicates just the opposite, i.e., number of individuals in the young size classes is not adequate to replace the individuals of older classes; hence the population will decline as time proceeds. The second type of pyramid represents an intermediate situation, suggesting that the number of individuals in younger class is only as much as required to replace the number in older classes, hence a stable state of population exists.

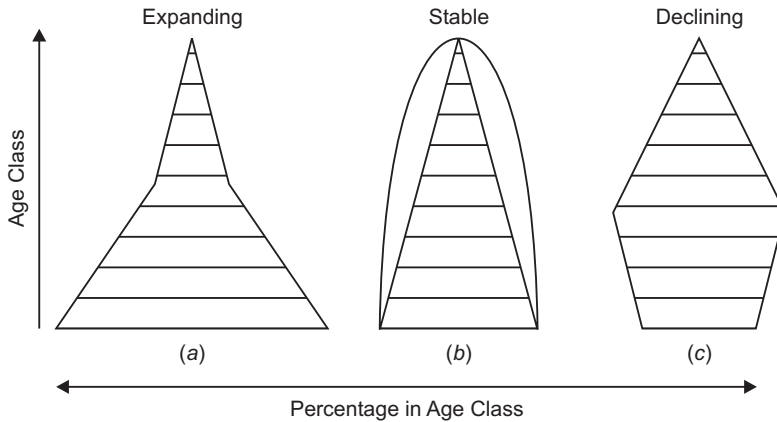


Fig. 8.5. Three types of age pyramids representing a large (a), moderate (b), and small (c) percentage of young individuals in the population. Accordingly, populations are expanding, stable and declining, respectively.

In plant populations, particularly of trees, several workers have tried to determine the size class distribution instead of age class distribution assuming that individuals of lower size classes are younger than those of higher size classes. It is often found convenient to divide a tree population in three major size classes: seedlings (up to 30 cm height); saplings (more than 30 cm height but diameter less than 10 cm), and trees with dbh above 10 cm. However, several diameter intervals may be used. Examples of expanding, stable and declining populations in tree species of a dry tropical forest are given in Fig. 8.6. It is important to recognize that the population structure of the same species may vary in different areas, at some place slowing an expanding population and at other places a declining population. The age or size structure presentations attract management attention. For example, the tree species showing a declining population at a given site needs immediate attention for finding out the causes and possible remedial measures such as reducing biotic stress, augmenting the population through wild-collected or nursery-raised seedlings, conservation, etc.

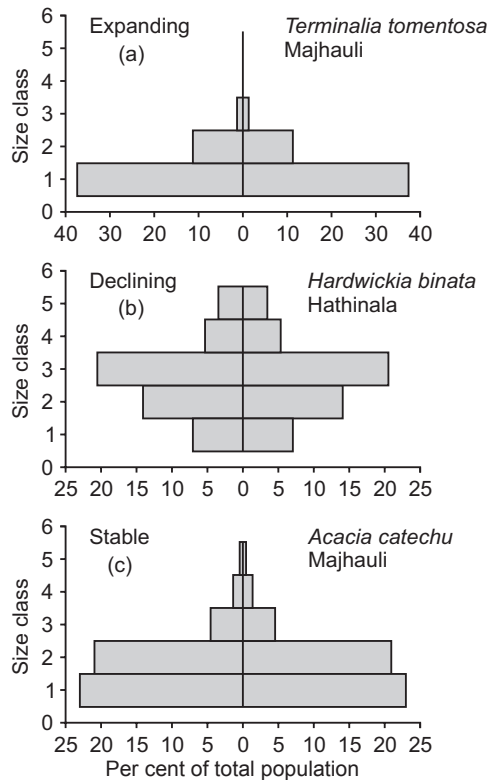


Fig. 8.6. Size class pyramids of three dry tropical tree species showing (a) expanding population; *Terminalia tomentosa* at Majhauri site, (b) declining population; *Hardwickia binata* at Hathinala site and (c) relatively stable population, *Acacia catechu* at Majhauri site (data from Sagar, 2002). Size classes represent increasing diameter.

8.3 POPULATION GROWTH FORMS

Rate of growth

In a simple expression, the growth rate of a population is the number of individuals added to the population per unit time. If N is used to denote the population size (usually as number of individuals), Δ is used to denote a difference in or “a change in”, and t for time, then:

ΔN = the change in population size, i.e., number of individuals.

$\frac{\Delta N}{\Delta t}$ = the average rate of change in the population size in a given interval of time.

The size of this increase, however, will depend on the original size of the population, i.e., it would be proportional to the initial population size. The absolute amount of increase in population size would be much more in population with initial size 100 than in that of 10 individuals. Therefore, in order to compare the growth rate of population of different sizes the increase per individual in time or increase per capita would be required.

The above can be explained by the following hypothetical example: The initial population size of a plant species = 100 individuals. After one year the population size = 300.

N (the initial population size) = 100

ΔN (change in population size) = 200

$\Delta N / \Delta t$ (average rate of change per time) = 200 per yr

$\Delta N / N \Delta t$ (average rate of change per time per individual) = 2 per yr per individual.

$\Delta N / (N \Delta t) \times 100$ (per cent growth rate) = 200% increase per yr.

However, afore-cited expression measures only the average rate of population growth over a period of time, it does not measure the instantaneous rate of change at a given instant in time. dN/dt is the expression used for the rate of change in population size at some instant in time, and $dN/(Ndt)$ = the rate of change in the population size per time per individual at a particular instant. The difference between $\Delta N/\Delta t$ and dN/dt is that in the latter Δt is made infinitesimally small. However, neither (i.e. dN/dt or $dN/(Ndt)$) can be measured directly from the population counts such as given in the afore-cited example. For this, information would be required about the nature of growth curve for a period of time, and growth in populations may not occur at the same rate during different periods of time. Variations result from several factors which vary in time. All the individuals may not produce offspring, and those that produce the offspring produce them in varying numbers. Then, many die, the number of which also varies in time.

Now a stage is set where we can combine the influence of natality and mortality to determine: how population as a whole grows, what will be its response to varying conditions and what is its most possible performance? The population has two basic patterns of population growth designated as exponential growth (J-shaped) and sigmoid growth (S-shaped).

Growth without limit (exponential growth curve):

Growth without limit can occur in an unlimited environment with respect to space and food. Suppose a single individual of *Paramecium*, introduced in a nutrient medium, divides after every 2 hours then we will record 2, 4, 8, 16 and 32 individuals respectively, after 2, 4, 6, 8 and 10 hours. If no deaths occurred during this period we will record 4096 individuals by the end of 24 hours. The population growth over time will depend upon the rates of birth and death; and the number of

birth and death will depend upon the number of individuals in the population. Suppose there exists some average fertility or average birth rate (b) (e.g. one female produces two females in the next generation, two females produce four females and so on, the rate of birth is 2 females per female per generation, and the probability of death per individual (d) (e.g. if one out of ten individuals dies in a given interval of time the death rate is 0.1 individual per individual per unit time), i.e. B (births) and D (deaths) are proportional to the number of individuals in the population at a given time, N :

$$B = bN$$

$$D = dN$$

Then referring to our earlier notations, we can represent the rate of growth as follows:

Where $(b-d)$ is usually written as r ,

$$\frac{dN}{dt} = rN \quad \text{or} \quad \frac{dN}{dt} = (b-d)N$$

The above is the exponential growth equation. The quantity $b-d$ or r is called the *intrinsic growth rate* or the *Malthusian factor*. In population studies usually b_0 and d_0 are used instead of b and d to emphasize that the rates represent the situation when population is very small or is growing very fast as if it were very small. The implication of this equation is that if b is greater than d , the population size (N) will continue to increase faster and faster without limit. If we were to compute the population size after a given interval of time that is at t_1 from the initial population (at t_0):

$$N_{t1} = N_{t0} + (r \times N_{t0}).$$

If we continue the calculations for successive intervals of time and plot the results of our calculations, it would yield an exponential growth curve (Fig. 8.7 a). The exponential growth curve (also called J-curve) becomes a straight line if the log of number of individuals is plotted against time (Fig. 8.7 b). Now r can be easily calculated as :

$$r = \frac{\ln N_{t1} - \ln N_{t0}}{(t_1 - t_0)},$$

where $\ln N_{t0}$ is natural log of number of individuals initially present, i.e., at time t_0 , and $\ln N_{t1}$, is the natural log of number of individuals after an interval $(t_1 - t_0)$ of time, i.e., at time t_1 . The solution of the equation

$$\frac{dN}{dt} = rN \quad \text{is given as:}$$

$$N_t = N_0 e^{rt}$$

where N_t = number of individuals at time t , N_0 = number of individuals initially (at time t_0), $e = 2.71828$ that is the base of natural logarithms and t is the time interval.

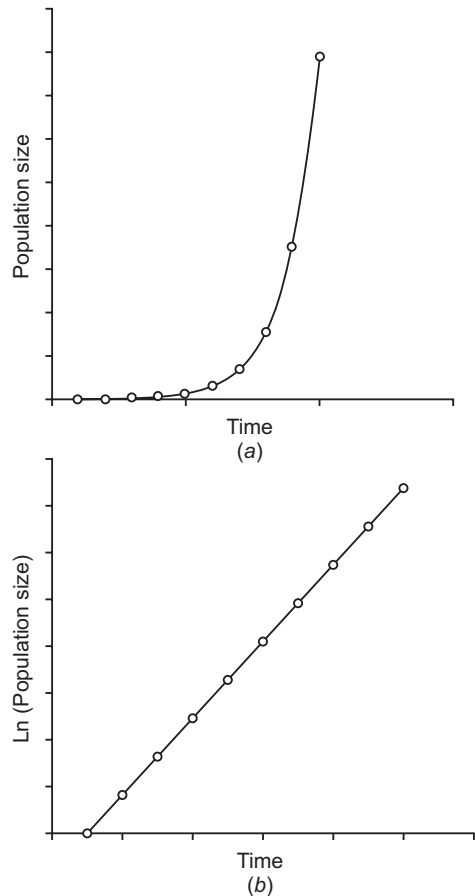


Fig. 8.7. Exponential population growth, for a population in an environment that has not limited its growth. On arithmetic scale, the exponential growth curve is J-shaped (a), while on a semi-logarithmic scale, it is a straight line (b).

So using the initial population size and r , we can calculate the size of population for any subsequent time. The exponential growth model is a characteristic of many organisms in the new environment and could be useful for studying growth of bacteria in laboratory cultures, predicting the yield of insect rearing and the growth of exotic species in the new environments.

In situations where no limit to population growth exists (e.g., no competition, no predation, always surplus food and space) the value of r will be maximum = rm . This (rm) represents the *biotic potential* of a species. In nature, however, the realized r is much lower because of some limits or environmental resistance. The difference between rm and realized r is a measure of environmental resistance. When the exponentially increasing population encounters a limiting resource the population crashes, e.g., growth of algae in a water body may be exponential for a time resulting into algal bloom and then nutrients may become limiting and the population sharply decreases.

Growth within limit (Logistic growth curve)

With rate of division given earlier, and with unlimited food and space we will have more than a billion paramecia within some hours. However, neither food nor space will remain unlimited for long, and unchecked exponential growth will lead to the extinction of the population as food supply is exhausted. Thus, there is a kind of conflict between the intrinsic tendency of population to increase, and the limits the increase encounters.

As shown in Fig. 8.8 a, population growth is eventually checked and stabilizes at a population ceiling, denoted as K . The sum total of environmental limiting factors that prevent the biotic potential to be realized is called environmental resistance, which tends to increase as population attains its maximum. Environmental resistance is the unutilized opportunity for population growth due to both abiotic and biotic factors.

Often as population increases d increases and b decreases till $b = d$. At this time, there would be no net growth in the population or $dN/dt = 0$; and the N at which $dN/dt = 0$ is called the **carrying capacity of the environment** (K). The *carrying capacity* is defined as the maximum population size that a natural ecosystem or environment can support under a given set of environmental conditions. This is the level at which population is theoretically in equilibrium with its environment.

The decrease in b with increasing population is represented as:

$$b = b_0 - K_b N,$$

where K_b is a constant (slope of decrease for birth rate); other notations are as given earlier.

Similarly increasing d is represented as:

$$d = d_0 + K_d N,$$

where K_d is a constant (slope for increase in death rate).

Now:
$$\frac{dN}{dt} = [(b_0 - K_b N) - (d_0 + K_d N)] N$$

This is the logistic growth equation often called Verhulst-Pearl logistic equation. Population can maintain itself (i.e. no further increase or $b = d$) when

$$(b_0 - K_b N) = (d_0 + K_d N).$$

Solving for N ;
$$N = (b_0 - d_0) / K_b + K_d$$

This N will be the population at limit, or $N = K$

Now combining $N = (b_0 - d_0) / K_b + K_d$, and $r = b_0 - d_0$

We have:
$$\frac{dN}{dt} = rN \frac{(K - N)}{K} = rN \left(1 - \frac{N}{K} \right)$$

In this equation, K = carrying capacity and $K-N/K$ = environmental resistance. The maximum growth rate is at $\frac{1}{2} K$. Solution for this equation is $N = K / (1 + e^{a-rt})$ (where a is a constant, rest as in earlier equations).

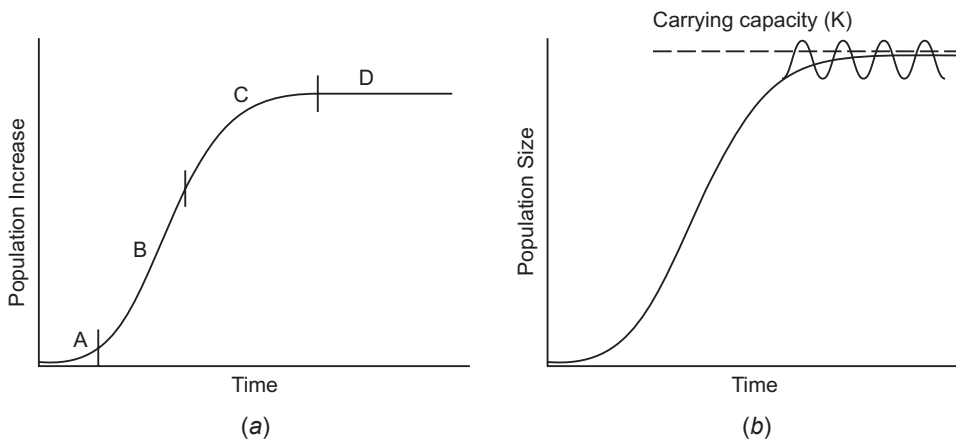


Fig. 8.8. Sigmoid population growth curve for a population in an environment that limits its growth at some point. The curve has four phases (see text), the last phase is an upper asymptote where $dN/dt=0$. The population as it grows faces environmental resistance, growth rate declines and tends to zero as the population size approaches the carrying capacity (b). The population size usually oscillates around the carrying capacity (dotted curve).

See Figure 8.8 a for a plot of this equation, the curve is also called S-curve. This curve shows that as N increases, dN/dt decreases, and when $N=K$, the $dN/dt=0$. The curve can be conveniently divided into 4 phases: (i) positive acceleration phase (a gradual increase in population size), (ii) logarithmic phase (a rapid increase in population size), (iii) negative acceleration phase (rate of increase declines due to environmental resistance), and (iv) upper asymptote (the growth rate tends to be zero) (Fig. 8.8 a). This logistic equation in brief tells us that when N is close to zero (i.e. when population is just starting, dN/dt will be nearer $=rN$; when N is greater than K , population will exceed K , and subsequently, dN/dt will become negative and the population will again level off (Fig. 8.8 b). The general relationship between the exponential and logistic growth curves is represented in Fig. 8.9. Populations growing in a logistic fashion may experience time lags resulting in population fluctuations which include chaotic fluctuations, stable limit cycles, damped oscillations and population overshoots. In chaotic situations, the population fluctuations are sudden and irregular (Fig. 8.10). In certain cases, large fluctuations in populations may lead to sudden extinction as simple processes often lead to complex behaviours. In stable limit cycles, the population fluctuates around the equilibrium level K , rising and falling between an upper and a lower limit (Fig. 8.10). In damped oscillations, the initial fluctuation in population after overshooting the K (irruptive) may diminish over time because the population tends to maintain at k through compensatory birth and death rates. Sometimes, population strongly

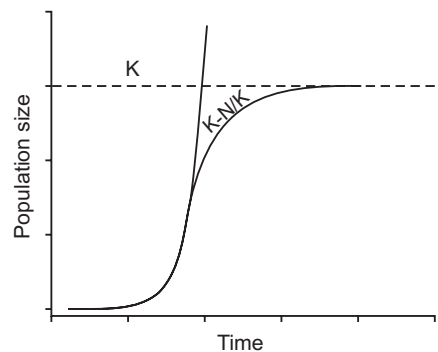


Fig. 8.9. Relationship between J- and S-shaped curves. $K-N/K$ represents the environmental resistance.

overshoots k , and crashes thereafter. It may depress to a low level, then recovers and attains some lower equilibrium as compared to the initial equilibrium level (Fig. 8.10).

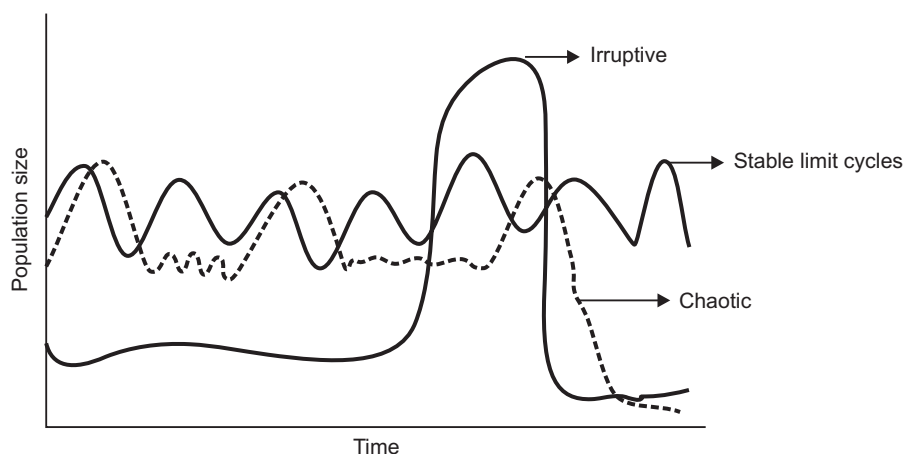


Fig 8.10. Population fluctuations in sigmoid growth curves due to time lags.

8.4 POPULATION REGULATION

Density-dependent v/s density-independent actions

The concept of population regulation was introduced by Verhulst (1838) and the term density-dependence was coined by Smith (1935). Population growth is always, sooner or later, checked, and population eventually attains a form of stability or equilibrium with its resources and environment (Fig. 8.8). Whether or not the controlling action is negative or positive on population growth, if it is independent of the size of population the action or effect is termed *density-independent*, and when the effect is a function of density it is called *density-dependent*. A sudden decline in population growth by a sudden drop in temperature, or in response to severe storms, for example, is largely independent of the density or size of the population. Such cases are clear-cut examples of density-independent effects on population growth. In stressful environments, such as deserts or a disturbed site (dominated by *r*-selected species), population growth is usually affected by unpredictable environmental factors, and the population size is seldom large enough to be affected by density-dependent actions.

In contrast, in favourable environment of moist tropics, and in organisms of longer life span (mainly *K*-selected species) and larger biomass, the role of density-dependent factors, such as, self limiting intraspecific competition and interspecific checks (e.g., competition, parasitism and predation) becomes more pronounced in regard to population regulation. As a population attains a certain density, some density-dependent effects reduce its rate of growth by decreasing the birth rate and by increasing the death rate. In an early study, Harper and McNaughton (1962) have shown that in *Papaver* (cornfield poppies) the chance of a seed producing a plant declines with increasing density.

In plants, response to density results in a decrease in the mean density of individuals and an increase in mean weight of individuals. This relationship is expressed in the $-3/2$ power law, i.e., $w = cd^{-3/2}$. This is also known as Yoda's $-3/2$ Law, describing the reciprocal changes in plant biomass (w) and the number of plants per unit area (d) during the development of a monospecific even-aged plantation with a complete crown closure. The relationship between plant density and plant biomass is called the self-thinning curve, which typically has a slope of $-3/2$ on a log-log plot

(Fig. 8.11). In monoculture plantations, the plants arrive at a point where growth compensates for losses through mortality.

According to Hixon et. al. (2002), both competition and predation may be source of density dependence. Competition for resources is always density-dependent. It can affect both density-dependent fecundity as well as the loss rate by causing density-dependent starvation. Predation including disease, parasitism, parasitoids and herbivory are also density-dependent, but not always. There seems to be a consensus that a combination of density-dependent and density-independent factors is important in controlling population growth. To explain the subject of population regulation and the principles outlined above, we take the example of population dynamics of the Australian psyllid insect *Cardiaspina albitextura*, which feeds on eucalypt leaves (*Eucalyptus blakelyi*), described by Clark (1964). Several factors both density-independent and density-dependent related to weather, predation, parasitism, etc. influence the population dynamics of the insect.

Following phases of the insect population are identifiable (Fig. 8.12). Normally, the population remains stabilized at a low level, which is well below the level that can be reached if all available resources are optimally used (Level 1). In this phase, the population is too small to exploit all the food and space resources. Regulation of the population size at this level results from prevailing weather, density-independent predation and parasitism on nymphal stages, and density-dependent predation by birds on adults. Occasionally, population size can increase above the Level 1 as a result of, say, unusually low temperatures that reduce parasitism of insect individuals. This increase is followed by a rapid, irruptive population growth because parasitism on nymphs is further reduced due to a rapid increase in hyperparasites, and the bird predators cannot increase as rapidly as the insects. The insect population uses up the food supply (*Eucalyptus leaves*), the population growth is halted at Level 2. The insect population then crashes, as the trees are defoliated and as parasitism and predation increase, to below Level 1, and is controlled again by a combination of density-dependent and density-independent factors.

Intraspecific competition, social structure, and territoriality tend to stabilize population densities at a level lower than saturation. Intraspecific competition for resources also acts as a density-dependent mechanism in population regulation through scramble (exploitative) or contest (interference) competition. Each individual taking as much as it can get of some resource is called scramble (exploitative) competition and population is affected by depletion of the living resource. On the other hand, in interference competition, successful individuals divide the resource, whereas unsuccessful ones are denied access to it. There is a direct effect on the population through allelopathy (chemical interference) and territoriality.

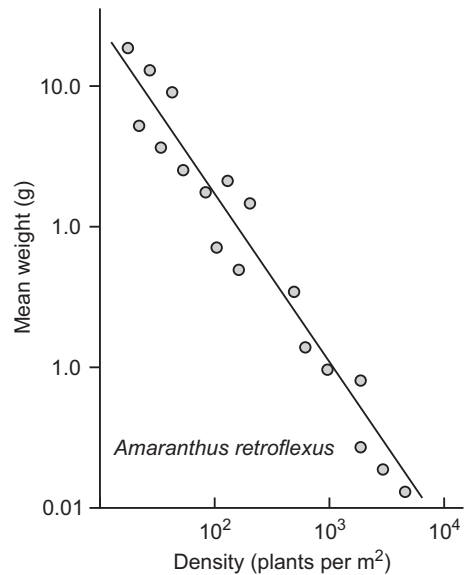


Fig. 8.11. The relationship between plant density and plant weight showing self-thinning ($-3/2$ power law) in *Amaranthus retroflexus*. The slope of relationship is $-3/2$ or -1.5 (based on Harper, 1977).

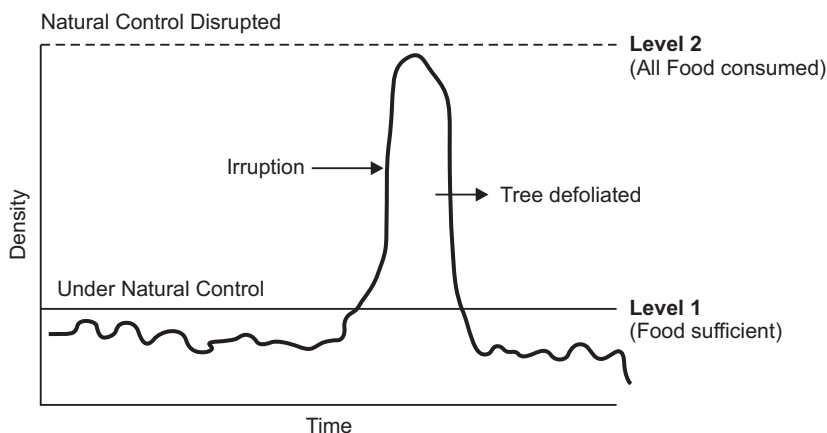


Fig. 8.12. Ecology of a population irruption, as illustrated by population dynamics of the psyllid (*Cardiaspina albiflora*), feeding on leaves of *Eucalyptus blackelyi*. The insect population generally remains at low density (Level 1) by combined density-dependent and density-independent action of predators, parasites and inclement weather. However, occasionally natural control fails to operate and insect population irrupts to a high level density (Level 2), resulting in defoliation of the eucalypt trees (derived from Clark, 1964 and Odum, 1983).

Territoriality is generally a form of intraspecific competition and social behavioural mechanism for spacing the individuals among the population. Selection pressure can also result in interspecific territoriality. Home range is an area in which an animal generally lives, whereas territory is the defended area. It exerts internal environmental resistance on a population by restricting the food supply and partners for mating. For example, birds, mammals, lizards, fishes and many social insects often defend space by aggressive signals. Some primates and ungulates employ olfactory signals, whereas use of some kind of scent or pheromones is widespread among animals and insects to defend their territory. Secretion of toxins into the soil by some plants can also be viewed as a means of territorial spacing. Separation may range from a few millimetres between neighbouring limpets on a rock to several kilometres between herds of antelopes and packs of hunting dogs. From cost-benefit standpoint, there would be a trade-off between greater access to food and time, and energy and the risk of establishing and defending the suitable site. Mating and nesting sites are also decided on the basis of similar trade-offs.

Group behaviour or pecking order is sexual dominance or other behaviour that increases fitness of offspring but reduces their numbers as well as enhances the quality of the individual environment. As already discussed, aggregation is varying degree of clumping due to habitat differences, weather changes, reproductive processes and social attraction in animals. It also has a group value resulting in optimum growth by affecting mortality of propagules.

In the case of some species of social animals and colonial plants, the growth rate is greater above a minimum density. According to Allee's Principle undercrowding as well as overcrowding may be limiting to populations. For example, in species of sea gulls that nest in colonies, the number of young produced per pair is high at an intermediate density of the colony rather than at a very high or low density (Allee, 1958). In summary, density-dependent factors regulate population size at or near carrying capacity by regulating the availability of abiotic and biotic environment and resources. The density-independent factors often reduce populations below the carrying capacities or may cause populations to fluctuate about their equilibrium sizes.

Yadav and Tripathi (1981) have reported on the population flux of a rapidly colonising weed in the north-eastern India to show that despite a substantial natural control, the plant still maintained a

high population growth (Box 8.1). Although a recent introduction to the flora of India, *Chromolaena odorata* (*Eupatorium odoratum*) has become a dominant weed due to its rapid spread during the last few decades (Yadav and Tripathi, 1981). It is an early colonizer in abandoned cultivated fields in the north-eastern region of India and elsewhere at lower altitudes. If the *Chromolaena odorata* population continues to grow at this speed, within a few years it may suppress and eliminate many useful elements of the original flora of the north-eastern region of India (Box 8.1).

Box 8.1

Population flux of *Chromolaena odorata* over a one year period, June 24, 1977 to June 30, 1978

(Based on Yadav and Tripathi, 1981). Values are number of plants/m²

Variable :

(a) No. of plants on 24 June 1977	27
(b) No. of plants on 30 June 1978	36
(c) Net change (b-a)	9
(d) Rate of increase (b/a)	1.33
(e) No. of plants recruited between 24 June 1977 and 30 June 1978	1358
(f) Total no. of plants lost between 24 June 1977 and 30 June 1978	1349
(g) Plants present on 24 June 1977 alive by 30 June 1978	17
(h) Percentage survival of plants in (a) ((g/a) x 100)	62.96
(i) Total no. of plants recorded during the study period	1385
(j) Percentage annual mortality of all individuals	97.4

Population cycles and fluctuations

Gause's experiment with *Paramecium caudatum* and *Didinium nastutum* gives laboratory evidence of population oscillations of interacting predator-prey populations (Gause, 1934). When the populations of the protozoan, *Paramecium caudatum* and its predator, *Didinium nastutum* were placed in a homogeneous culture, the predator ate up all the prey and followed it to extinction. However, if the culture medium had sediment in the bottom, prey in the sediment were immune to attack because *Didinium* would not enter the sediment.

In nature, some populations overshoot or surpass the carrying capacity of the environment, which may result in repeated cycles of population growth and decline. These cycles may be very regular if they depend on a few simple factors such as seasonal variation in light and temperature-dependent bloom of algae in lake. These cycles may be complex if they depend on complex environmental and biotic relationships. For example, the peaks in cyclic populations tend to occur at fairly regular intervals. The prey and predator populations oscillate in a sort of synchrony with each other. Oscillations in the populations of snow-shoe hare and lynx found in Canadian boreal forests show a 9-10 year cycle in population growth and decline, as indicated by the fur trapping records (Fig. 8.13). When the hare population is high, food is plentiful for the lynx. As a result, the lynx reproduction is also successful. The rapid population increase of lynx diminishes the number of hare, until eventually the lynx do not have enough to eat and their population size falls. With the decreasing number of predator, the prey population begins to increase again. However, the predator population shows time delays of 1-2 years, which could be the time required for sexual maturity. This sequence of increase in prey and predator population causes regular oscillation in population

numbers. The delayed density dependence produces cycles in predator and prey abundance (Fig. 8.13). Cycles also appear to be related to habitat. Longer cycles have been observed in forest dwelling species such as snow-shoe hare and muskrat and their predators (lynx and minks). The short cycles of 3-4 years have been observed in tundra dwelling species such as voles and arctic lemmings, arctic foxes and snowy owls. Arctic lemmings are small mouse-like rodents that show a boom-and-bust cycle in their populations; the population cycles of increase and decline are repeated almost every four years. At the peak of their population densities, large number of lemmings is found in the tundra landscape. The following year, only a few lemmings are seen.

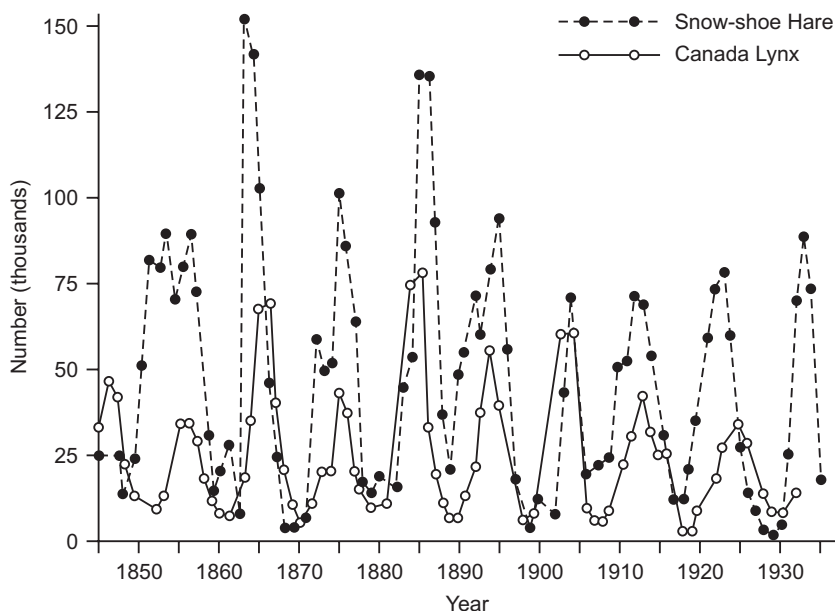


Fig. 8.13. Cyclic variations in the population of snow-shoe hare (prey) and the Canadian lynx (predator) over 9 to 10 year cycle (based on data from MacLulich, 1937).

In irruptive cycles, long periods of stable population size are followed by a sudden population explosion. For example, outbreaks of locusts in deserts or tent caterpillars in northern forests show periodic population irruptions taking place over a period of decade or more. After the overshoot the population decreases and comes at a lower carrying capacity as compared to the initial one as shown in Fig. 8.12.

8.5 LIFE HISTORY STRATEGIES

r- and *K*-selection

The concept of *r*- and *K*-selection was originally given by MacArthur and Wilson (1967) and developed by Pianka (1970) to focus on life-history evolution due to selection pressures on species colonizing an island. In uncrowded, resource-rich environments, natural selection would favour organisms which are able to achieve rapid population growth. On the other hand, in crowded and not so resource-rich conditions natural selection would favour organisms which are competitive and maintain population growth near the carrying capacity. The former are termed *r*-strategists and the latter *K*-strategists. In the *r*-strategists, the population remains on the ascending part of the logistic growth curve (i.e., $dN/dt = rN$), while in the *K*-strategists, population is maintained near the upper asymptote of the logistic growth curve (i.e., when $N = K$ or close to it). The concept of *r*- and *K*-selection is further explained in Figure 8.14 with the help of two hypothetical populations.

Population 1 has lower reproductive output and a higher competitive ability; therefore Population 1 is *K*-selected. On the other hand, population 2 is *r*-selected as it is characterized by higher reproductive output and lower competitive ability. At low population density and high resource availability, population 2 has higher growth rate than population 1, and at high density and low resource availability population 1 has a higher growth rate than population 2.

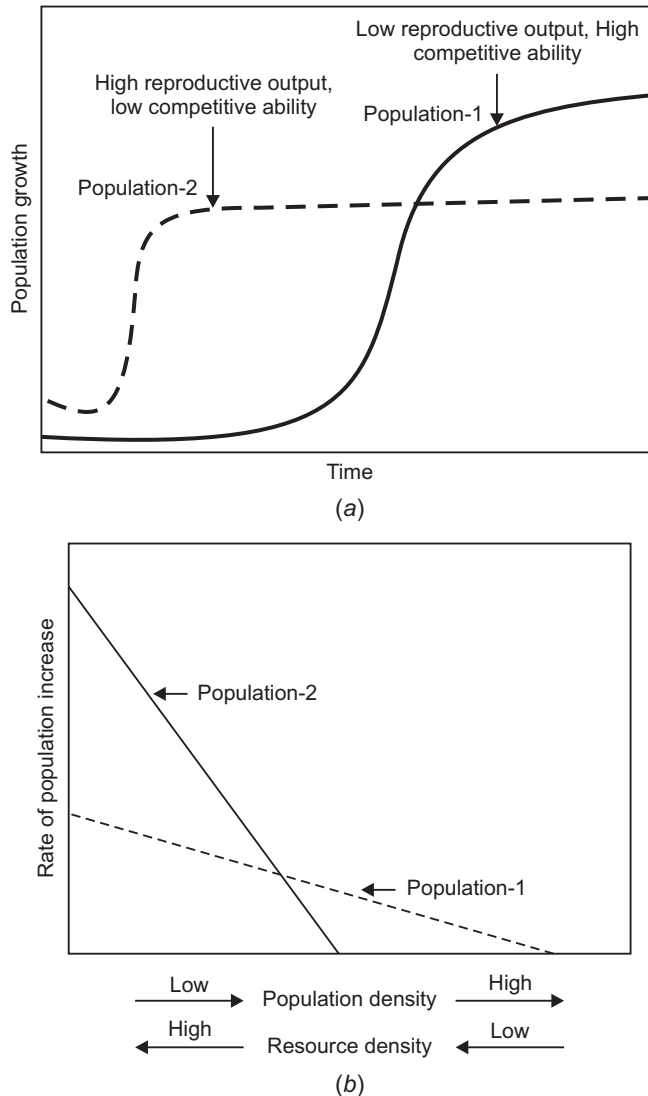


Fig. 8.14. The model of *r*- and *K*-selection in two populations growing under uncrowded (population 2) and crowded condition (population 1). (a) population size with time, and (b) growth rates (based on MacArthur and Wilson, 1967; MacArthur, 1972).

Relatively unstable populations (of *r*-strategists) are selected for wide dispersal and rapid growth in new environments, such as eroded land where they are less affected by competition as few organisms exist. This type of selection is called *r*-selection in contrast to the *K*-selection which involves selection toward tolerance of competition or toward specialization in the community or both (Table 8.2). The former is characteristic of populations of pioneer communities and the latter of populations of mature communities (late successional). In higher plants, *K*-selection may result in

larger individuals, such as shrubs or trees which could, for example, reduce light availability to other plants that germinate close by, and therefore have higher competitive ability. In animals, *K*-selection may lead to greater specialization to avoid interference with competitors or an increased tendency to occupy and defend territories.

Table 8.2. Some traits attributed to *r*- and *K*-selection (modified and developed from Pianka, 1970).

Trait type	<i>r</i> -selection	<i>K</i> -selection
Climate	Variable, unpredictable	Fairly constant, predictable
Habitat	Harsh physically, more vacant space	Less harsh, less vacant space
Population size	Variable	Fairly constant, near <i>K</i>
Competition	Variable, less important, lax	Keen, important
Successional status	Pioneer	Later successional stage
Mortality	Density-independent often catastrophic	Density-dependent
Development time	Rapid	Slow
Energy to reproductive structure	High	Low
Seed number	High	Low
Seed size	Small	Large
Seedling growth	Photosynthate dependent	Partially seed reserve dependent
Dispersal mechanism	Well developed	Less developed
Survivorship Curve	Usually type III	Type I
Size of individuals	Small	Large
Selection favours	Rapid development, reproduction at early stage of life span, reproductive effort single or few	Slower development, delayed reproduction but repeated in life span
Life span	Short	Long
Seed life span	Long	Short
Specialization	Low	High
Overall needs	Greater net productivity	Greater efficiency of resource use

The *r*-strategists are opportunistic in as much as they are able to make immediate use of any uncrowded environment. They develop strategies such as large amount of small seeds which can reach great distances through suitable dispersal mechanisms, well developed dormancy which enables them to persist in unpredictable environments for quite long time. The last strategy enables these species to utilize temporary, unpredictable environments, since their seeds are present in the soil; these species fill any vacant space that is created by the removal of the existing vegetation. *K*-strategists, which are part of mature stable and predictable environments, are competitive species with long-lived individuals, such as of Oak (*Quercus*), Sal (*Shorea robusta*) and others. Because of their high longevity, the populations remain stable for a fairly long period. Pianka (1970) points out that the *r*-selection favours the genotypes with the highest possible intrinsic rate of increase, while *K*-selection favours those which can use the environment with greatest efficiency.

Certain species, however, may exhibit features common to both *r*- and *K*-selections. The *r*- and *K*-selected species occupy the end points of a continuum from *r*-selection to *K*-selection. In fact, *r*- and *K*-selections are relative terms useful in comparing organisms. Among *r*-species, in general, selection favours the genotypes which confer highest possible intrinsic rate of increase, rapid development of population size, reproduction at early stage of life span. Among *K*-species, selection is for opposite characteristics favouring such genotypes that are able to maintain densest populations near the carrying capacity of the environment. In current understanding, many factors interact to shape life histories, which may include density dependence, age-specific mortality, resource limitation and environmental fluctuations as components of selection.

Ruderal, Competitive and Stress-tolerance Strategies

Grime (1979) expanded the *r*- and *K*-selected system and recognized three categories of plants, i.e., ruderals (R), competitors (C) and stress tolerants (S). Grime visualizes two groups of environmental factors, the stress factors and the disturbance factors. The stress factors are those which reduce biomass production, e.g. low light, low moisture, etc. The disturbance factors remove part or whole of biomass, e.g. grazing. Four combinations of these groups are possible: (i) high stress, high disturbance, (ii) low stress, high disturbance, (iii) low stress, low disturbance, and (iv) high stress, low disturbance. The first combination is too harsh to permit colonisation. The combination (ii) is suitable for ruderal species, combination (iii) for competitive species, and combination (iv) for stress tolerants. The ruderal species are similar to *r*-selected species, which colonise disturbed patches of habitats. The R species exhibit rapid growth, early maturation and have high reproductive rates. Competitive species are found in habitats that are rich in resources, relatively undisturbed, and with dense populations. Competitors tend to allocate carbon to resource gaining structures including roots, leaves and stems, and are long-lived. S species have also long life spans and occur in stressed environments such as deserts, arctic tundra, peat bogs and serpentine soils, and grow slowly. The traits of competitor, stress tolerant and ruderal species are compared in Table 8.3.

Table 8.3. Characteristics of plant life history strategies based on competition, stress and disturbance (based on Grime, 1979).

Trait type	Competitive (C)	Stress Tolerant (S)	Ruderal (R)
Life form	Herbs, shrubs or trees	Lichens, herbs, shrubs or trees	Herbs, usually annual
Shoot morphology	Dense canopy	Variable	Small
Leaves	Deciduous	Evergreen	Deciduous
Longevity	Variable	Very long	Very short
Flowering	Annual polycarpic	Intermediate polycarpic	Annual monocarpic
Reproductive maturity	Delayed	Delayed	Early
Reproductive effort	Small	Small	Large
Perennation	Vegetative spread, buds and seed	Vegetative spread by leaves and roots	Seeds
Growth rate	Rapid	Slow	Rapid
Stress response	Rapid	Slow	Reproduces and dies
Resource allocation to seed production	Small	Small	Large

Under natural conditions, plants grow in habitats with combination of disturbance, competition and stress (Grime, 1979). Therefore, plants show mixed strategies such as competitor-ruderal (C-R), competitor-stress tolerator (C-S) and stress tolerator-ruderal (S-R). A model in triangular form shows three main and the mixed strategies of plants (Fig. 8.15). The three axes of the triangle represent per cent stress, per cent disturbance and per cent importance of competition and it is divided into areas within the triangle to show the mixed strategies of plants. For example, annual herbs would occur in areas of high disturbance, low competition and low stress. Biennial herbs are found in areas of high competition and low stress, whereas lichens occur in high stress, low competition and high disturbance areas. The competitive-ruderal (C-R) live in environments of low stress and moderate disturbance, stress tolerant-ruderals (S-R) live in harsh environments, and C-R-S are found in habitats with reduced competition due to combined effects of stress and disturbance.

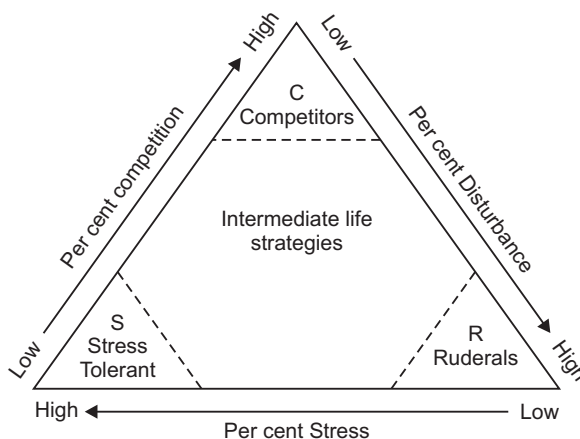


Fig. 8.15. Expected plant life history strategies as regulated by stress, disturbance and competition. Competitors occupy the upper part, stress tolerators the lower left-hand part, and ruderals the lower right-hand part of the triangle. Species within the triangle are intermediate in their response (based on Grime, 1979).

8.6. POPULATION GENETICS AND NATURAL SELECTION

The theory of evolution by natural selection, the unifying concept of modern biology, was given by Charles Darwin on the basis of his observations on plants and animals in the Galápagos Islands, lying 1,000 km off the Pacific coast of Ecuador. Variations exist within all populations of organisms, partly because of random mutations that cause changes in the genome of an individual organism, and due to wide ranging variations in the environmental conditions. Random genetic changes (genetic drift, founder events) in different parts of a population's range can cause geographic variation in allele frequencies among sub populations. *Population genetics* is the study of allele frequency distribution and change under the influence of natural selection, genetic drift, mutation and migration. It attempts to explain the phenomena of adaptation and speciation. This section discusses mutations as source of genetic variations, the Hardy-Weinberg equilibrium, genetic drift, and natural selection.

Mutations and Recombination as a Source of Genetic Variation

Genes, contained in chromosomes within cells, are inherited from one generation to the next. Genes are made up of complex compounds called deoxyribonucleic acid (DNA). A mutation is a change in the sequence of an organism's DNA. Mutations can be caused by high-energy sources such as radiation (X-rays or gamma rays) or by toxic chemicals in the environment. Mutations generally fall into two types: point mutations and chromosomal aberrations. In point mutations, one base pair is changed. The human genome, for example, contains over 3.1 billion bases of DNA, and each base must be faithfully replicated for cell division to occur. About one in every 10^{10} base pair is changed. For any particular nucleotide in a DNA sequence, the rate of mutation is extremely low, on the order of 1 in 100 million per generation. However, the low rates of mutations are the cause of genetic variations among populations.

A mutation that occurs in body cells that are not passed along to subsequent generations is a somatic mutation. A mutation that occurs in a gamete or in a cell that gives rise to gametes is special because they impact the next generation and may not affect the adult at all. Such changes are called germ-line mutations because they occur in a cell used in reproduction (germ cell), giving the change a chance to become more numerous over time. If the mutation has a deleterious effect on the phenotype of the offspring, it causes a genetic disorder. In case the mutation has a positive effect on the fitness of the offspring, it is called an adaptation. Thus, all mutations that affect the fitness of future generations are agents of evolution. Every genetic feature in every organism was, initially, the result of a mutation.

Recombination

Recombination is the process of the exchange of homologous sections of maternal and paternal chromosomes, which creates new combinations of genes. Recombination produces new genetic variation rapidly, whereas a mutation does so very slowly. Recombination creates genetic diversity at the level of genes that reflects differences in the DNA sequences of different organisms. In eukaryotic cells, which are cells with a nucleus and organelles, recombination typically occurs during meiosis. Thus, mutation is the ultimate cause of genetic variation in a population; recombination multiplies the level of genetic variation.

Hardy-Weinberg Principle

The Hardy-Weinberg theorem characterizes the distributions of frequencies of alleles and genotypes in populations that are not evolving, and is thus the fundamental null model for population genetics. The Hardy-Weinberg equilibrium model helps identify evolutionary forces that can change gene frequencies in populations. This concept was named after two scientists, G. H. Hardy and Wilhelm Weinberg, who independently described it in 1908.

The Hardy-Weinberg Theorem deals with Mendelian genetics in the context of populations of diploid, sexually reproducing individuals. The Hardy-Weinberg equilibrium is an important principle of population genetics, which states that the gene frequencies in a population will remain constant from one generation to the next in the absence of disturbing factors. When mating is random in a large population with no disruptive circumstances, the law predicts that both genotype and allele frequencies will remain constant because they are in equilibrium. Hardy and Weinberg discovery showed that changes in allele and genotype frequencies can result only from the action of individual forces on the gene pool of a population. The conditions necessary to maintain constant allele frequencies in a population, what is called Hardy-Weinberg equilibrium, are as follows:

1. Random mating (panmixia). Non-random or preferential mating, in which the probability of pairing alleles is either greater or lower than would be expected based on their frequency in the population, can change the frequency of genotypes.
2. No mutations. Mutations which add new alleles to the population or change an allele from one form to another have the potential to change allele frequencies in a population and therefore disrupt Hardy-Weinberg equilibrium.
3. Large population size. Small population size increases the probability that allele frequencies will change from one generation to the next due to chance alone. Change in allele frequencies due to chance or random events is called genetic drift. Genetic drift reduces genetic variation in populations over time by increasing the frequency of some alleles and reducing the frequency of some alleles or eliminating others.
4. No migration. Migration can introduce new alleles into a population. When gene frequencies are different in populations exchanging migrants, it will alter Hardy-Weinberg equilibrium.

5. All genotypes have equal Darwinian fitness, where fitness is the probability of surviving and reproducing. If different genotypes survive and reproduce at different rates, then gene and genotype frequencies will change in populations.

The maintenance of Hardy-Weinberg equilibrium is a conservative force and evolution is modification of this equilibrium. Genetic equilibrium is modified by mutations, natural selection, random genetic drift and migration which are major evolutionary forces.

When a population exists in Hardy-Weinberg equilibrium, the proportion of homozygotes and heterozygotes take on equilibrium values, which can be calculated from allelic frequencies in a population. For example, a particular gene has two alleles, S and A, which occur in proportions of p and q ($p + q = 1$, and therefore $q = 1 - p$). At Hardy-Weinberg equilibrium, the three genotypes that can result will occur in the following proportions.

Genotype	SS	SA	AA
Frequency	p^2	$2pq$	q^2
Allele Frequency	0.7	–	0.3
Homozygotes	$0.7^2 = 0.49$		$0.3^2 = 0.09$
Heterozygotes	–	$2 \times 0.7 \times 0.3 = 0.42$	–

The proportion of genotypes in the population total upto 1, i.e., $p^2 + 2pq + q^2 = 1$ which is a simple binomial expansion. Any deviation from these genotype frequencies indicates the presence of selection, non-random mating, or other factors that influence the genetic composition of the population. Summary of a Hardy-Weinberg equilibrium equation is shown in Fig. 8.16.

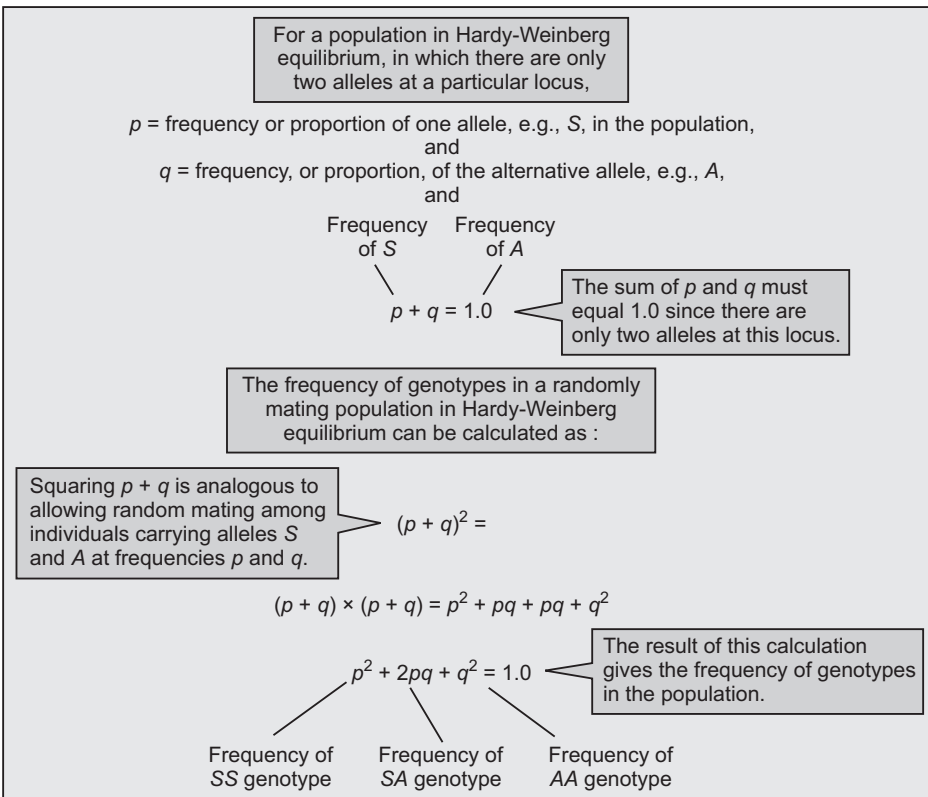


Fig. 8.16. Summary of a Hardy-Weinberg equilibrium equation. (adapted from Molles, 2008)

The Hardy-Weinberg equilibrium can be disturbed by a number of forces, including mutations, natural selection, non-random mating, genetic drift, and migration. For instance, mutations disrupt the equilibrium of allele frequencies by introducing new alleles into a population. Similarly, natural selection and non-random mating disrupt the Hardy-Weinberg equilibrium because they result in changes in gene frequencies. Because all of these elemental forces commonly occur in nature, the Hardy-Weinberg equilibrium rarely applies in reality. Therefore, the Hardy-Weinberg equilibrium describes an idealized state, and the changes in gene frequencies in nature can be measured as changes from this genetic equilibrium state.

Gene Pool

Mendelian population is essentially a group of interbreeding individuals that share in a common gene pool. Population has two important attributes: gene frequencies (allele frequencies) and a gene pool. Population geneticists study the gene frequencies and their changes within a population. The collection of all the genes and the various alternate or allelic forms of those genes within a population is called its gene pool. The gene pool is the set of genetic information, in any population, usually of a particular species. This also proves to be the basic level at which evolutionary forces operate. A large gene pool indicates extensive genetic diversity, which is associated with robust populations that can survive bouts of intense selection. Meanwhile, low genetic diversity can cause reduced biological fitness and an increased chance of extinction. However, in small populations gene frequencies may change drastically due to random genetic drift which is a non-directional force of evolution and a particular allele may be fixed or eliminated. When all individuals in a population are similar with regard to a particular phenotypic trait, the population is said to be monomorphic. When the individuals show several variants of a particular morphological trait they are said to be polymorphic.

Genetic Drift

Genetic drift is one of the major mechanisms that drive evolution. Natural selection is better known as the directional and driving force behind evolution, but genetic drift is probably an equally important mechanism particularly when population size is small. Genetic drift is the gradual change in the gene frequencies of populations over time as a result of chance (sampling error) rather than mutation, selection or migration. Drift is a random, non-adaptive change in gene frequencies in small populations. The idea of genetic drift was adumbrated by Brooks, Hagedorn and Hagedorn but the concept was especially developed by Sewall Wright, hence the random genetic drift is also known as 'Sewall-Wright effect'. The changes due to genetic drift are not driven by environmental or adaptive pressures, and may be beneficial, neutral, or detrimental to reproductive success. The statistical effect of sampling error during the reproduction of alleles is much greater in small populations than in large ones. In a relatively small population, stochastic effects can produce drastic changes in genetic composition very rapidly. In 1968, Kimura proposed the neutral theory of molecular evolution which is based on random genetic drift.

Founder Effect: A founder effect originally proposed by Mayr (1942) is the special case of genetic drift which occurs when a small group in a population splits off from the original population and forms a new colony. The new colony may have less genetic variation than the original population, and through the random sampling of alleles during reproduction of subsequent generations, continue rapidly towards fixation. This consequence of inbreeding would also make the colony more vulnerable to extinction because of reduced genetic diversity.

A population that is descended from a relatively small number of individuals who became separated from the original population shows significant differences in the degree of genetic variability from the original population. This is because its genetic inheritance depends upon a small number of Founders, who were not representative of the population from which they came, and did not possess the full range of genetic variability of the original population. As a result of the

loss of genetic variation, the new population may be distinctively different, both genetically and phenotypically, from the parental population from which it is derived. In extreme cases, the founder effect is thought to lead to the speciation and subsequent evolution of new species.

Population Bottleneck: A continued existence at low population size results in loss of genetic variation due to genetic drift. This situation is known as *population bottleneck*, an event or period in which the size of a breeding population is significantly reduced. This produces a sudden plunge in genetic variability, and greatly increases genetic drift as the population recovers. The population bottleneck condition appears to have occurred in the recent past in the populations of cheetahs in East Africa, which exist in small populations with practically no genetic variation (O'Brien et al., 1985). A population genetic survey of over 200 structural loci showed that the South African cheetah (*Acinonyx jubatus jubatus*) has an extreme paucity of genetic variability, probably as a consequence of a severe population bottleneck in its recent past (O'Brien et al., 1985). The extreme genetic monomorphism, especially at the major histocompatibility complex, and the apparent hypersensitivity of the cheetah to a viral pathogen may be related, and provide a biological basis for understanding the adaptive significance of abundant genetic variation in out-bred mammalian species (O'Brien et al., 1985).

Natural Selection

Natural selection is the gradual, non-random process by which biological traits become more or less common in a population as a function of differential reproduction of their bearers. Natural selection is the only mechanism known to cause the evolution of adaptations. The term Natural selection was introduced by Darwin in his influential book, "*On the Origin of Species*", published in 1859. Darwin was one of the first to recognize the biological significance of variation among individuals in a population. Natural selection is the means by which beneficial variations in a population tend to be preserved while unfavourable variations tend to be eliminated. Because members of a population must compete for finite resources, those better suited to their environment will out-compete those that are not as well-suited. Natural selection is the result of differences in survival and reproduction among phenotypes.

Natural selection involves five basic assumptions, which can be summarized as follows:

Struggle for existence - More individuals in a population are born each generation than will survive and reproduce.

Variation - Individuals within a population are variable. Some individuals have different characteristics than others.

Differential survival and reproduction - Individuals that have certain characteristics are better able to survive and reproduce than other individuals having different characteristics.

Inheritance - Some of the characteristics that influence an individual's survival and reproduction are heritable.

Time - Ample amounts of time are available to allow for change.

The result of natural selection is a change in gene frequencies within the population over time, that is individuals with more favourable characteristics will become more common in the population and individuals with less favourable characteristics will become less common. Natural selection is a creative force in evolution and it is the differential reproduction of genotypes in a population and brings about differential contribution of offspring to the next generation.

The Process of Natural Selection

Natural selection requires heritable variation in a given trait, and differential survival and reproduction associated with possession of that trait. Examples of natural selection are well-documented, both by observation and through the fossil record. Selection acts on the frequency of traits, and can take the form of stabilizing, directional, or diversifying selection.

Stabilizing Selection. Stabilizing selection acts against extreme phenotypes and as a consequence, favours the average phenotype. By favouring the average phenotype, stabilizing selection decreases phenotypic diversity in populations. It also leads to reduction in genetic variability. Under stabilizing selection, extreme varieties from both ends of the frequency distribution are eliminated. The frequency distribution looks exactly as it did in the generation before (Fig. 8.17). Probably, this is the most common form of natural selection, and we often mistake for no selection. A real-life example is that of birth weight of human babies. In humans, for example, the incidence of infant mortality is higher for very heavy as well as for very light babies. This type of selection is also known as centripetal or normalizing selection.

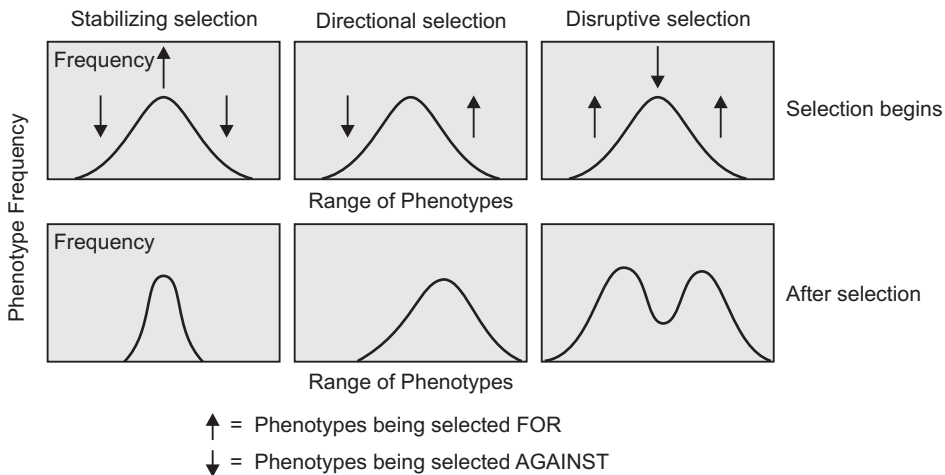


Fig. 8.17. There are three forms of natural selection. The bottom panel shows the distribution of phenotype after selection. (adapted from Kimball, 1994).

Directional Selection. Directional selection favours an extreme phenotype over other phenotypes in the population. Much of the selection practised by animal and plant breeders is of this kind and it leads to reduction in the degree of genetic variability. Under directional selection, the average of the trait under selection can change over time Fig. 8.17. The evolution of the peppered moth illustrates directional selection, and has been studied in detail. The typical form of the peppered moth (*Biston betularia*) has light-coloured wings speckled with small dark spots (Fig. 8.18). Originally, the vast majority of peppered moths had light colouration, which effectively camouflaged them against the light-coloured trees and lichens which they rested upon. But following the industrial revolution in Britain, the air in cities became polluted with dark soot released from burning coal and other fuels, which covered many surfaces, including trees. Due to deposition of soot on the trees, the bark became dark in colour in cities as compared to the countryside. When the English geneticist H. B. D. Kettlewell released moths of both types, i.e., light-coloured and dark coloured, in the woods, he observed that birds did, indeed, eat a much higher fraction of the light moths he released than of the dark-coloured moths. As a result, a dark form

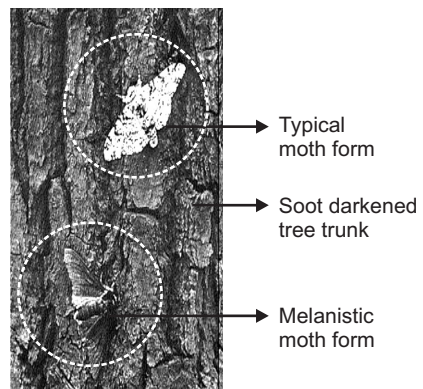


Fig. 8.18. The industrial melanism in peppered moth shows genetic change in response to selective factor in the environment. (adapted from Kimball, 1994)

of the peppered moth saw a population boom, probably because predatory birds couldn't see it clearly against sooty surfaces and preyed on the light-coloured form instead (Kettlewell, 1958). When regulations reduced air pollution in the 1970s, dark-moth populations began to fall and the lighter form resurged. Industrial melanism refers to the genetic darkening of species in response to pollutants (Majerus, 1998). Ilik Saccheri, an evolutionary biologist at the University of Liverpool, UK, and his colleagues have used molecular genetics to show that one mutation from a single ancestor causes increased dark pigment, called melanism, in the typically light-coloured moth (van't Hof et al., 2011). The selection that reduces the frequency of a deleterious mutation or eliminates it, is referred to as purifying selection which is simply directional selection in favour of advantageous genotypes.

The fossil lineage of the horse provides a remarkable demonstration of directional selection. The full lineage is quite complicated and is not just a simple line from the tiny dawn horse *Hyracotherium* of the early Eocene, to today's familiar *Equus*. Overall, though, the horse has evolved from a small-bodied ancestor built for moving through woodlands and thickets to its long-legged descendent built for speed on the open grassland. This evolution has involved well-documented changes in teeth, leg length, and toe structure.

Disruptive Selection. Disruptive selection favours two or more extreme phenotypes over the average phenotype in a population, leading to an increase in phenotypic diversity in the population. Under *diversifying (disruptive) selection*, both extremes are favoured at the expense of intermediate varieties (Fig. 8.17). This is uncommon, but of theoretical interest because it suggests a mechanism for species formation without geographic isolation (sympatric mode of speciation).

As an example, the mine tailings often contain such high concentrations of toxic metals (e.g., copper, lead) that most plants are unable to grow on them. However, some hardy species (e.g., certain grasses) are able to spread from the surrounding uncontaminated soil onto such waste heaps. These plants develop resistance to the toxic metals while their ability to grow on uncontaminated soil decreases. Because grasses are wind pollinated, breeding between the resistant and non-resistant populations goes on. But evidently, disruptive selection is at work. Higher death rates, i.e., in less resistant plants growing on contaminated soil and more resistant plants growing on uncontaminated soil. This leads to increasing divergence of the populations into two sub-populations with the extreme manifestations of this trait. Genetic diversity is increased by disruptive selection. In *Drosophila melanogaster*, divergence for a polygenic character and ethological isolation was produced simultaneously by disruptive selection for a morphological character (sternopleural bristle number) which gives evidence for sympatric mode of speciation.

CHAPTER

9

Species Interactions

A species population growing in a community cannot remain unaffected by the other co-occurring species populations. For example, plants depend on nectarivorous insects for pollination, upon seed-eating birds for dispersal, and on mycorrhizal fungi associated with their roots to absorb nutrients from the soil. The morphological and chemical diversity in plants has a broad base which provides them with defense against their natural enemies. For counter defense, herbivores and parasites have also developed morphological, physiological and behavioural adaptations as a result of co-evolutionary processes occurring over a long period of time. Predation generally develops selective pressure on prey to develop better defense and the prey defenses generate selective pressure on predators to become more effective at obtaining the prey. The result of the two types of selection pressure results in co-evolution of predator and the prey. The mutualism between plants, animals and microbes is the basis of natural ecosystem services such as pollination, seed dispersal, nitrogen fixation and nutrient recycling.

9.1 TYPES OF SPECIES INTERACTION

As shown in Table 9.1, theoretically, populations of two species may show nine types of interaction with respect to negative or positive effect on interacting species. (1) *Neutralism*, in which the associated populations are not affected by each other. (2) *Competition-direct interference type*, in which the interacting populations inhibit each other. (3) *Competition-resource use type*, in which interacting populations adversely affect each other indirectly as they compete for resources (e.g., nutrients, light) in short supply. (4) *Amensalism*, in which one of the two species populations is inhibited and the other remains unaffected. (5) *Parasitism* and (6) *Predation*, in which one species population adversely affects the other by direct attack but nevertheless depends on the other. (7) *Commensalism*, of the two species populations, one is benefited, but the other is not affected. (8) *Protocooperation*, in this both the species populations are benefited in terms of growth rate or size, but are not necessary to each other. (9) *Mutualism*, both the species populations benefit and the relations are obligatory.

Table 9.1. Types of interactions between two-species populations (from Odum, 1983)

Type of interaction	Species		General nature of interaction
	A	B	
1. Neutralism	0	0	Neither population is affected by the other
<i>Negative interaction</i>			
2. Competition: direct interference type	–	–	Each species directly inhibited by the other
3. Competition: resource use type	–	–	Indirect inhibition of both the species when common resource is in short supply.
4. Amensalism	–	0	Species A is inhibited, B not affected.
<i>Positive as well as negative interaction</i>			
5. Parasitism	+	–	Species A, the parasite, generally smaller than B, the host
6. Predation (including herbivory)	+	–	Species A, the predator, generally larger than B, the prey.
<i>Positive interaction</i>			
7. Commensalism	+	0	Species A, the commensal benefits, while B, the host, remains unaffected.
8. Protocooperation	+	+	Interaction favourable to both but not obligatory
9. Mutualism	+	+	Interaction favourable to both and obligatory

0 = no significant interaction or no effect; + benefited in terms of growth survival and other population attributes; – = population inhibited in terms of above characters.

Predation is a type of negative population interaction in which one organism consumes the other for food. In a broad sense, the predators include the carnivores and herbivores. The probability of death of the resource may depend on the duration of association between the resource and the predator (Pollard, 1992; see Ricklefs and Miller, 2000) (Fig. 9.1). Parasitism can be considered a mild form of predation as the host is not immediately killed. Parasitoids are the hyper-parasites and represent a condition between parasitism and carnivory.

Because of the widespread impact of Darwin's idea of "survival of the fittest" as an important means of bringing about natural selection, competitive aspects of nature received more attention than the cooperative or positive relationships between the species. Interactions of species which result in positive effects, such as mutualism, are as widespread as negative interactions, such as competition, predation and parasitism in nature. Both

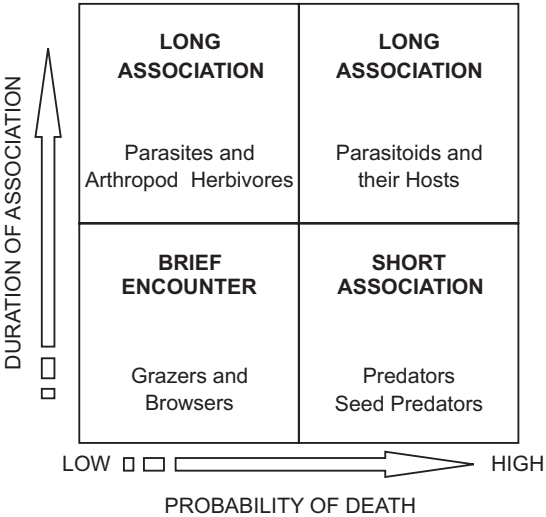


Fig. 9.1. Types of consumer resource interaction on the basis of duration of association and probability of death of resource (adapted from Pollard, 1992; Ricklefs and Miller, 2000).

negative and positive relations between populations are equally important in evolution of species and stabilization of ecosystems (Odum, 1983).

9.2 INTERSPECIFIC COMPETITION

In the broadest terms, competition refers to the interaction of two organisms competing for the same resource, e.g., nutrient, light (in case of plants), food (in case of animals) and space, particularly when the resource is in short supply. Interspecific competition involves individuals of two or more species. Opposed to this, competition among individuals of the same species is called intraspecific competition, which may be more severe due to similar demands. Competition may be for a single resource or for multiple ones. The results of interspecific competition are of great interest and have been much studied as one of the mechanisms of natural selection. To predict the outcome of competition on the basis of species characters has been one of the major goals of ecologists, right from the days of Darwin.

Competition was experimentally demonstrated long back in common garden experiment on *Galium saxatile* and *Galium sylvestre* grown in pure culture and mixed culture on acid and calcareous soils (Tansley, 1917). From these experiments, Tansley concluded that the presence and absence of species is determined by competition; the conditions of environment affect the outcome of the competition and the ecological segregation of species. A number of laboratory studies relating to protozoa, flour beetles, fruit flies and the aquatic weed (*Lemna*) provide an experimental basis to define the competitive exclusion principle. Many field experiments have illustrated the importance of interspecific competition for organisms having similar resource requirements in freshwater, terrestrial and marine environments (Smith, 1996; Ricklefs and Miller, 2000).

In general, more similar the requirements of the two species, more intense would be competition between them when grown together. Although growth of both the species could be affected by competition, usually one species is suppressed by the other species due to latter's superior competitive ability. Competitive ability is generally genetically fixed but it could vary from habitat to habitat due to the modifying effect of environment. In the case of plants particularly, there are several components of competitive ability; (i) life-form (perennials over annuals, trees over shrubs), (ii) reproductive capacity (species with greater seed output over that which produces few seeds), (iii) food reserve (large seeds over small seeds), (iv) early growth and rapid growth rate (quick sprouters after dormant season over late sprouters), (v) nutrient uptake (greater efficiency over low efficiency of uptake), (vi) vegetative reproduction (greater survival value over sexual reproduction), and (vii) genetic variability (greater number of biotypes can colonise greater number of microsites).

Lotka-Volterra Model of Competition

Lotka (1925) and Volterra (1928) developed mathematical equations relating growth of closely related competing populations. These are a pair of differential equations based on logistic equation of population growth.

For growth of species 1, the logistic equation is:

$$\frac{dN_1}{dt} = r_1 N_1 \frac{(K_1 - N_1)}{K_1}$$

For growth of species 2, the logistic equation is:

$$\frac{dN_2}{dt} = r_2 N_2 \frac{(K_2 - N_2)}{K_2}$$

In these equations, r_1 = intrinsic rate of growth of species 1, r_2 = intrinsic rate of growth of species 2, N_1 = population size of species 1, N_2 = population size of species 2, and K = carrying capacity (i.e., equilibrium population size of a species in absence of the other species). When the two species grow together, the equation assumes the forms:

For species 1:
$$\frac{dN_1}{dt} = r_1 N_1 \frac{(K_1 - N_1 - \alpha N_2)}{K_1};$$

α is the competition coefficient, i.e., inhibitory effect (competitive effect) of species 2 on species 1.

For species 2:
$$\frac{dN_2}{dt} = r_2 N_2 \frac{(K_2 - N_2 - \beta N_1)}{K_2};$$

β is the competition coefficient, i.e., inhibitory effect (competitive effect) of N_1 for every individual of N_2 .

The Lotka-Volterra equations are based on the assumptions that it works under the homogeneous environment, where migration is negligible and competition effect is instantaneous and is the only important biological interaction. The Lotka-Volterra equations have been used for modelling competition as well as other interactions like predator-prey and parasite-host interactions, mostly occurring under controlled laboratory conditions or in the environments having very negligible input and output.

When species 1 and species 2 occur on the same resource at the same time, according to the Lotka-Volterra model there are four possibilities: (i) species 2 wins and species 1 becomes extinct, (ii) species 1 wins and species 2 becomes extinct; (iii) either species 1 or species 2 wins depending upon ecological conditions at that instant of time ; (iv) neither species wins, and species are able to co-exist sharing the resources between them . The competitive exclusion and competitive coexistence are described below:

Competitive exclusion

In general, the more closely related the species, the more severe the competition. If the similarity of requirement is too great, this may result in competitive exclusion of one population by another. The principle of competitive exclusion states that no two species can co-exist on a single limiting resource for a long time. Competitive exclusion also occurs under natural conditions when accidental or intended introductions of species are made by humans.

Gause carried out studies on two closely related species of *Paramecium aurelia* and *P. caudatum*. When cultured separately, in the yeast medium, *P. aurelia* was found to have a faster rate of growth than *P. caudatum* (Fig. 9.2). When grown together in the mixed culture, *P. aurelia* out- competed *P. caudatum*. This led to the conclusion that closely related species cannot co-exist on a single limiting resource for long; the competitively superior species is able

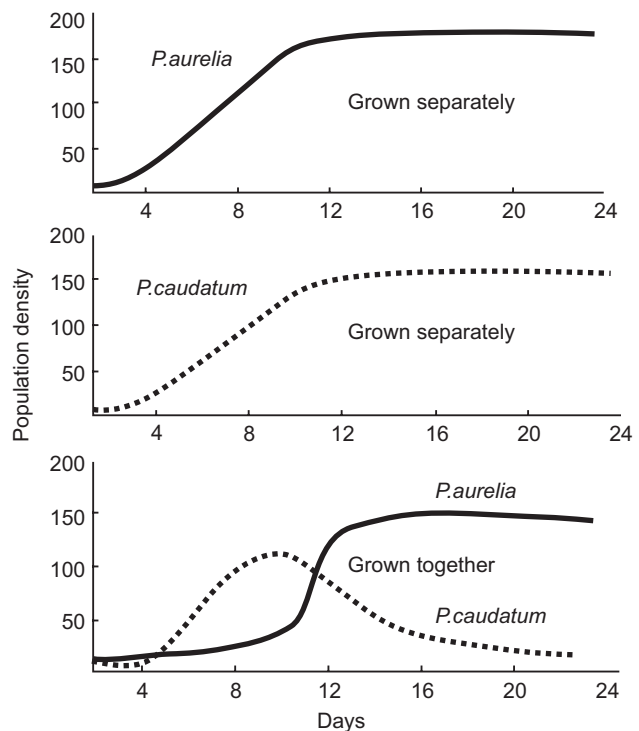


Fig. 9.2. Growth of *Paramecium aurelia* and *P. caudatum* in pure and mixed cultures. In pure culture, the two species were growing normally. In mixed culture, *P. aurelia* was competitively superior to *P. caudatum*. On the basis of this experiment, G.F. Gause defined the principle of competitive Exclusion Principle (Gause, 1934).

to survive, whereas weaker one is eliminated. Comparable results on competitive exclusion were obtained with other studies on protozoa, flour beetles and fruit flies.

Many exotic weedy plant species (e.g., *Lantana camara*, *Parthenium hysterophorus* and *Eichhornia crassipes*) have been found to displace the range of distribution of native plants. For example, the invasion of *Eichhornia* can replace the native species affecting the biodiversity in wetland and other freshwater ecosystems. *Parthenium hysterophorus* is displacing the range of the native species in constantly disturbed habitats and also becoming invasive in different cropping systems in India and other countries because of its life history characteristics such as production of large amount of seeds, varying growth strategies of plants and phenotypic plasticity to grow under a range of soil conditions (Annapurna and Singh, 2003).

Competition and Coexistence

One of the conclusions from the Lotka-Volterra equations is that when interspecific interactions are weak, then species are able to co-exist. Generally, coexistence is favoured by resource sharing and differential resource partitioning in fluctuating environments and differing life history traits. According to resource partitioning, no two species utilize the resource in exactly the same way, resources may be partitioned spatially and temporally. For example, some insectivorous birds feed in upper canopy, some in middle and some on the ground. The time during which they feed also differs.

Coexistence tends to promote resource sharing among the competing populations. Over time, the overlap in resource utilization among competing populations tends to be reduced. In other words, natural selection operates to narrow the ecological niches of both or all the competitors. The species are able to coexist by reducing the competition pressure through behavioural, physiological and genetic changes by exploiting the resources at different times or places.

Theory of Resource Capture and Resource Sharing

At present, the two theories concerning resource sharing among the species are those of Grime (1979) and Tilman (1982, 1988). Grime defines competition as the tendency for neighbouring plants to utilize the same resources and argues that success in competition is largely a reflection of the capacity for resource capture. According to this view, a good competitor is characterized by rapid growth and rapid development of absorptive surface area (also low sexual reproductive effort) which enable it to preempt both above- (e.g., light) and below-ground resources. A good competitor is a good competitor not for any particular resource, but for all resources. According to Grime, because of trade-offs a species cannot be both a stress tolerant as well as a good competitor. For example, a species cannot have traits that enable it to tolerate stress caused by deficiency in resources as well as to efficiently exploit resources and grow rapidly.

Tilman (1982, 1988) defines interspecific competition as the utilization of shared resources in short supply by two or more species. His ultimate criterion for competitive success is the ability of one species to drive the other to extinction. And the successful species would be one which can tolerate low resource levels better than others. According to his resource-ratio hypothesis, competition between species can give rise to mixed-species equilibrium if the growth of each species is limited by a different resource, and the species are replaced by each other if the limiting resources are negatively correlated. Further, a superior competitor for one resource will be an inferior competitor for the other resource. A superior competitor for a given resource is the one which has low requirement for that resource. Thus, habitats with low supply rates of resource 1 (S_1) but high supply rates of resource 2 (S_2), i.e., low $S_1 : S_2$ ratio would be dominated by species A with species B extinct at equilibrium. Conversely, habitats with high $S_1 : S_2$ ratio would be dominated by species B with species A extinct at equilibrium.

For example, on a nitrogen-poor site (low ratio of soil N : light available at the soil surface), a good competitor is the species which can extract nitrogen better when its concentration is lowest (e.g., early successional species). As succession progresses, nitrogen in soil increases and light at the soil surface becomes a limiting factor due to increased plant cover, i.e., high ratio of nitrogen : light. Therefore, in the late successional stage a species with the lowest light requirement (hence superior competitor for light) would displace the early successional species. At intermediate levels of soil N and light, both species could coexist.

Tilman's resource-ratio hypothesis is important, for it explains niche-differentiation as an important force for structuring plant communities. A few resources can generate a wide range of resource ratios.

9.3 AMENSALISM

A typical example of the situation where one species is inhibited while the other remains stable is allelopathy. Allelopathic interactions are an important factor in determining species distribution and abundance within plant communities, and are also thought to be important in the success of many invasive plants. Allelopathy is found among many terrestrial and aquatic plants. This type of interaction involves the direct effect of a toxic substance produced by one individual that causes injury to the other individual. These metabolic by-products released into the environment are, in general, selectively toxic. There are several examples of *allelopathy* among chaparral shrubs (Muller, 1966, 1969). Shrubs in Californian chaparral produce inhibitors or toxins, such as volatile terpenes in their leaves, which inhibit growth of herbs. The volatile toxins of leaves (notably cineole and camphor) accumulate in the soil during the dry season to such an extent that when rainy season commences, germination and growth of plants are not allowed to occur around shrubs. Some plants produce water soluble compounds (e.g., phenols and alkaloids), which kill other organisms and enable the producers of such compounds to dominate sites of their occurrence. Examples include *Salvia* and *Parthenium*. The leaf litter and root exudates of some *Eucalyptus* species are allelopathic for certain soil microbes and plant species. *Parthenium*, for example, releases allelochemicals from fresh plant parts and residues, such as phenolic acids (caffeic, p-coumaric, ferulic, anisic and p-hydroxy benzoic acids) and sesquiterpenes (parthenium, ambrosin, dihydro-isoparthenin and hymenin). *Lantana camara* has as many as 14 phenolic compounds with proven allelopathic potential. The tree of heaven, *Ailanthus altissima*, produces allelochemicals in its roots that inhibit the growth of many plants.

Plants release secondary compounds into the soil environment from their roots as exudates, and if the compounds released by an exotic plant are novel to a soil community, they may alter the composition and function of the soil community. Allelochemicals released from plant roots have been widely used as an explanation of the success of exotic plants in the context of plant-plant interactions (Hierro and Callaway, 2003). Diffuse knapweed (*Centaurea diffusa*) is a Eurasian species that has invaded many natural ecosystems in western North America. This species releases the chemical 8-hydroxyquinoline from its roots, which has been found to be an antimicrobial agent (Vivanco et al., 2004). Diffuse knapweed can cause shifts in the composition of the soil microbial community (Callaway et al., 2004a), possibly through the release of these allelochemicals.

Garlic mustard (*Alliaria petiolata*) belonging to family Brassicaceae, a European native, invades forests of North America, often decreasing the local plant diversity of these communities, and can also influence soil communities. The roots of this species produce glucosinolate compounds that can decrease the abundance of propagules of AMF, such as spores, and internal root structures, of native plant species in north American forests (Wolfe and Klironomos, 2005). These allelochemicals also exert long-term effects on understory and canopy plant species. (Wolfe and Klironomos, 2005).

Chemicals with allelopathic functions have other ecological roles, such as plant defence, nutrient chelation, and regulation of soil biota so as to affect decomposition and soil fertility in a number of different ways (Inderjit et al., 2011a). The ecosystem-scale roles of allelopathic chemicals can increase, decrease or modify their community-scale functions. However, the production, fate and effectiveness of allelopathic compounds in soils are influenced by environmental conditions and evolutionary history (Inderjit et al., 2011a). Study of allelopathic interactions across spatial and temporal scales could help to understand interaction among plants.

9.4 HERBIVORY

When the predator is a primary consumer or herbivore, and the prey or host is producer or plant, the interaction is called *herbivory*. Herbivory, the consumption of plants by animals (insects or vertebrates), has a long history, exceeding more than 450 million years. Two types of herbivores which have substantially different effects on plant communities are herbivore predators and grazers/browsers. The herbivore predators and seed predators consume whole plants or plant parts including seeds and seedlings. Seed predators consume from 10 to 100% of seed production. The grazers and browsers eat only a small portion of vegetation, leaving the rest to regenerate. The grazers eat grasses or other herbaceous plants, whereas browsers eat leaves and young shoots of shrubs and trees. For example, the large herbivores such as deer, wild ass and moose in terrestrial ecosystems and zooplanktons in aquatic systems are grazers (Pollard, 1992). Herbivores consume almost all plant parts. On average, at least 10% of net above-ground primary productivity of terrestrial ecosystems is consumed by grazing herbivores. Occasionally, some herbivores like insects become so numerous that they defoliate trees completely (e.g., tent caterpillars, gypsy moth) which severely reduces growth rate; repeated defoliation by insects may cause mortality. The extent of herbivory may vary from 1.5-2.5% consumption of leaf area by insects in deciduous forests to nearly 100% consumption by grazing animals in some grassland communities in arid regions at the time of food scarcity. Some nematodes and beetle larvae are specialized root feeders and exhibit a long time association with their host. The below-ground herbivory by soil nematodes may range from 6-30%. Herbivores select plant material based on nutrient content. Young leaves and flowers are often chosen because of low cellulose content and fruits and seeds because of high nutrition.

Herbivorous organisms are found in all the phyla of animals ranging from protozoa to vertebrates and these are specialized to consume various types of tissues (Crawley, 1983; Ricklefs and Miller, 2000). Many of the bright coloured fish fauna of coral reefs are herbivorous, whereas carps found in fresh water systems are herbivorous. In Galapagos Islands, the giant land tortoises and the marine iguanas are herbivorous (Ricklefs and Miller, 2000). The giant panda is a prominent herbivorous animal that feeds only on the bamboo shoots. In savannas of Africa and Asia, browsing animals like camel, giraffe and sheep are common. In grassland systems, the burrowing rodents may eat plant roots and below-ground stems. It is estimated that 50% of mammalian fauna are herbivorous; many of these specialize to eat fruits and seeds. Seed predation or granivory refers to seed consumption by rodents and seed eating birds like parrots. Many arboreal animals (e.g. tree sloth) found in rain forests are herbivorous.

Effects of herbivores on plants may be negative, neutral, or positive. Herbivores influence the distribution of plant species, and they can also stabilize plant succession, and prevent encroachment of new forms into an area. Moderate grazing may increase species diversity which is influenced by seasonality and variations in physical environment over the years. In grassland systems, heavy grazing generally by large herbivores causes a decline in productivity. Some studies also indicate that grasses tend to develop increased fitness under grazing pressure whereas defoliation by insects has adverse effects on sexual reproduction. The outbreaks of large populations of gypsy moth and budworms in temperate forests and locust in semiarid regions can have negative impact on population of trees, herbaceous species, and many facilitate spread of non-selective species.

Plant-Herbivore Interactions

This is one of the most important natural interactions in terms of species, biomass and energy transfer, as well as evolutionary impact on traits of organisms, community structure and ecosystem function (Shultz, 2002). Plants of the world are target of attack from nearly 4,00,000 species of herbivorous insects. Plants have evolved to produce chemicals to limit the insect herbivory losses, and insects in turn evolved to access new chemicals to outdo plant defences. Plants respond to attack by insects with rapid, variable, and often specific biochemical, physiological and developmental changes. A number of pathways coordinate these plant responses. For example, fatty acid signals control expressions of defence-related genes, affecting plant responses to wound. It has been suggested that phospholipid/amino acid conjugates occurring in insect guts, hence in 'spit' (regurgitant) elicit herbivore-specific volatile emissions from the host plant. These emissions, in turn, attract parasitoids, which kill the herbivorous insect. Interestingly, the chemicals that plants elicit may be similar to those that insects and other animals use, despite their wide phylogenetic differences. Theoretically, both plants and insects could manipulate each other's response mechanisms. It seems that organisms share a common code-book, that is why plants might identify insects, and they might obstruct each other's signalling. This new view of herbivory emphasizes rapid, dynamic behaviours and the factors that elicit changes and signals that coordinate them. About 90% of all herbivorous insects feed on one taxonomic family and many insect species feed on specific plants. Therefore, an understanding of co-evolutionary basis of adaptation in plant-animal interactions could be useful to manage resistance in genetically engineered crops (Rausher, 2001).

Plant defenses

During the course of evolution, plants have evolved an array of mechanical and chemical defenses, such as lignification, silicification, and development of secondary metabolites that reduce palatability. Side by side, the herbivores evolved an equally impressive array of counter measures in response to the evolution of plant defenses.

Plant defenses are divisible into three types: (i) mechanical protection, such as tough leaf texture, presence of trichomes, hairs, etc, on plant surface; (ii) complex polymers or silica which reduce plant material's digestibility to animals, referred to as *quantitative digestivity reducers* (Table 9.2); and (iii) toxins, poisonous to animals in very low concentrations, referred to as *qualitative toxins*.

The quantitative digestivity reducers, like cellulose, lignin and tannins (Table 9.2), afford protection to plants from herbivores in that their presence in high concentrations reduces palatability and digestivity and therefore lessens the attack of herbivores when other forms of herbage are available. Ruminants and other animals that can eat large quantities of tannin-rich plant tissues get little nutritional benefit. Silica (an oxide of silicon, one of the most common elements in the earth's crust) is completely indigestible to animal and also is highly abrasive to that, grinding teeth of grazers. Silica, cellulose, hemicellulose and lignin have given support and protection to plants for hundreds of million years.

Table 9.2. Some selected plant products that reduce herbivory, (from Howe and Westley, 1988)

Plant chemical* (number identified)	Description	Defensive role (i.e. the effect on consumers)
<i>Quantitative Digestivity Reducers^a</i>		
Cellulose (1 basic type), and Hemi-cellulose (1 basic type)	Sugar polymer	Animals require gut flora for digestion
Lignin (indefinite)	Phenolic polymers	Bind with proteins and carbohydrates
Tannins (indefinite)	Phenolic polymer	Bind with proteins
Silica (1 basic type)	Inorganic crystal	Indigestible
<i>Qualitative Toxins^b</i>		
Alkaloids (20,000)	Heterocyclic N-containing compounds	Many: some stop DNA and RNA production
Toxic amino acids (260)	Analogues to protein amino acids	Compete with protein amino acids
Cyanogens (23 ⁺)	Glycosides that release cyanide (HCN)	Stops mitochondrial respiration
Glucosinolates (120)	S, N and a group derived from glucose, containing organic compounds	Act as natural pesticides and as defenses against herbivores
Proteinase inhibitors (indefinite)	Proteins or polypeptides in subunits	Bind with active site of enzymes
Terpenoids (100,000 ⁺)	Polymers of C ₅ units	Many: some stop respiration

^aRequired in high quantities to reduce digestibility of plant materials to consumers; ^bPresent in minute quantity in plant tissues, poisonous to animals; *numbers in parentheses indicate the number of chemicals identified.

Apart from the above, plants particularly of the Poaceae family, have developed other defensive traits, e.g. (i) acropetal maturation order resulting into youngest tissues and intercalary meristem being at least risk of removal, (ii) elevation of apical meristems very late in the annual growth cycle, and (iii) selection of prostrate genotypes and reproduction by rhizomes and stolons rather than exposed inflorescences.

Toxins : Some plant chemicals, which may or may not have metabolic role, are toxic to animals. Toxins with a clearly protective or defensive rather than metabolic role are called *secondary compounds* (also referred to as *secondary metabolites* or *allelochemicals*; many of them have been found of medicinal use). Developing buds, young leaves, unripe fruits and other tissues that have not yet become fortified with lignin or silica and therefore are vulnerable to herbivores' consumption, generally have these toxins.

These chemicals, which occur in small concentrations (in contrast to complex polymers like tannin which affect animals due to higher concentrations), affect consumers often by blocking specific biochemical reactions. Among these toxins, *alkaloids* (heterocyclic molecules containing nitrogen) have been most studied. Approximately 20,000 different alkaloids have been isolated from plants, including such well-known compounds as nicotine in tobacco, caffeine in coffee and tea and the drugs cocaine and morphine. The proportion of species that contain alkaloids is estimated at 15-33% of the Eucotyledons (dicotyledons but for Magnoliaceae and many of them are annual plants). According to one estimate, the percentage of alkaloid bearing plants in floras declines toward higher latitudes. The flora of the Indian subcontinent is especially rich in alkaloids (about 45% species). A number of organisms including frogs, ants, butterflies, bacteria, spiders and beetle also produce alkaloids. Some butterflies use plant alkaloids (as a source of attractants), pheromones

and defense compounds. Inhibition of DNA and RNA synthesis (caffeine), inhibition of mitosis (colchicine), and stimulation of ribosome breakdown (mescaline) are some of the well known toxic effects of alkaloids. *Terpenoids* (having a repeating structure of C_5 phosphate units) are very important biologically. Some are found virtually in all higher plants because they are integral to plant metabolism (e.g., plant growth hormone gibberellic acid). Other terpenes have more restrictive roles. For example, volatile terpenes are aromatic and are often used by plants to attract pollinators. Terpenes with mainly defensive roles are restricted to some species. For example, pyrethroids, the commercial insecticides, are found in species of *Chrysanthemum*. They are also very toxic to mammals. Hydrogen cyanide (HCN) is very toxic to cellular respiration. It is released when plant tissues are crushed by herbivores. This kind of defense is referred to as *inducible defenses* (response of plant to tissue damage), in contrast to constitutive defenses, which include external structures like trichomes and hairs and chemicals that reduce digestibility (i.e. are of permanent nature). HCN can kill a cow in as low dose as 2 mg kg^{-1} body weight.

Herbivores Counter Defense Measures

Herbivores differ in size by more than 10 orders of magnitudes. Aphids (though not the smallest) has about 0.01 g weight and an elephant often exceeds 5,000,000 g in weight. Herbivores have mechanical, biochemical and behavioural countermeasures to plant defenses (Table 9.3).

Table 9.3. Common counter measures of herbivores to plant defenses.

(from Howe and Westley, 1988)

Counter measures	Modifications developed (by examples)
Mechanical breakdown of tough tissues (rich in lignin, cellulose)	Teeth (mammals); beak and/or gizzard (birds for seeds); chewing mandibles (insects and their larvae).
Breakdown of cellulose and hemicellulose	Expanded gut volume for microbial fermentation (most vertebrates and many insects) or intracellular symbionts (many insects).
Detoxification of allelochemicals	Expanded gut for microflora (above); mixed-function oxidase (MFO) systems (vertebrate liver, insect fat body and midgut); perhaps intracellular symbionts (insects).
Avoidance of noxious food plants	Well-developed taste, smell, and/or vision combined with selective foraging behaviour (all herbivores).

Mechanical breakdown: It is necessary for herbivores to break up plant tissues. Long evolutionary experience has given herbivores a variety of tools for fragmenting plant material. Grazers such as the horse and elephant which eat abrasive silica-rich grasses have high-crowned teeth that wear away slowly (Fig. 9.3). The rabbit's high-crowned teeth grow throughout the life. In comparison, omnivores (e.g., human or pig) have low-crowned teeth which wear rapidly once the cap of hard enamel is worn down to the dentine by abrasive plants (Fig. 9.3). Among the birds, some have a thick bill to crush seeds (e.g., North American cardinal), others have muscular gravel-filled gizzards to crush the kernels (e.g., Turkey's gizzards generates forces of 3.7 kg.cm^{-2} , enough to crack a hickory nut).

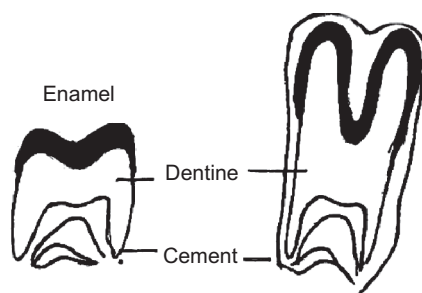


Fig. 9.3. Dentition of mammalian herbivores. Low crowned teeth is found in omnivores like man and pig and high crowned teeth is of grass-eating herbivores like cattle and horse (from Howe and Westley, 1988).

Chewing mouthparts in some insects have different plans but similar purpose. A grasshopper mouth, for instance, consists of highly modified appendages for sensing (maxilla), guiding (labrum, labium), and fragmenting (mandible) plant food (Fig. 9.4). Some insects avoid the need to break up plant tissues, such as true bugs (*Hemiptera*) and leaf hoppers and their relatives (*Homoptera*). They pierce the plant epidermis with a beak and suck plant fluids.

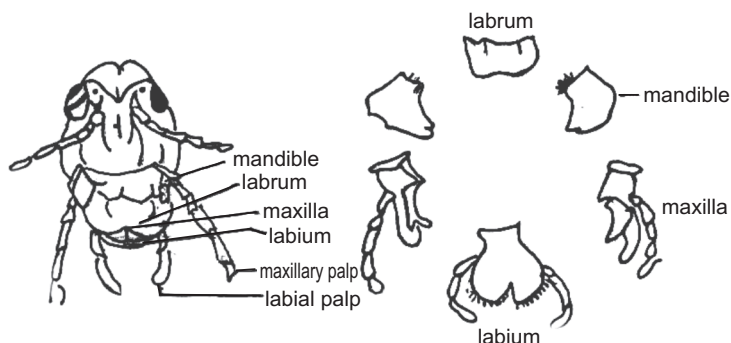


Fig. 9.4. The highly modified feeding appendages (mouth) consists of maxilla (for sensing) labrum and labium (for guiding) and mandible for fragmenting plant (from Howe and Westley, 1988).

Digestive system: It has been suggested that grasses and grazing mammals made simultaneous appearance in the Eocene, and they had an extended, intimate evolutionary history. Due to grassland formation, fruits and berries declined in abundance and the structural components increased. As a response, ungulates, which were capable of digesting such structural material, evolved. This was made possible by (i) the development of a large fermentative chamber in the form of ruminal (cattle, deer, and camel) and caecal (horse, rhinos) digestive systems, and (ii) symbiotic association with cellulolytic microorganisms in the digestive chamber.

Chemical counter-defense in herbivores: Generalized enzyme systems in some herbivores protect them against a great variety of plant toxins. Apart from the general protection, some insects have also developed specific detoxification mechanisms for toxins of their specific host plants. *Mixed-function oxidases* (MFO) which are found both in insects and vertebrates are membrane-bound enzymes that detoxify a great variety of plant and synthetic poisons. The detoxifying effect of MFO occurs very rapidly, within minutes in insects and within hours in rats and mice.

Using Plant Toxins (allelochemicals as food): Some insects have the ability to use toxins (allelochemicals) of plants as food. The *nonprotein amino acid*, L-canvanine (a toxin), occurs in many species of bean family (e.g., 12% of dry weight in seeds of the tropical bean, *Dioclea megacarpa*). By replacing the amino acid arginine in polypeptide synthesis, L-canavine produces defective polypeptide, which inhibits growth of most organisms. The toxin protects the bean against all insects, but a weevil (*Caryedes* sp.), which uses the toxin, L-canavine as a source of nitrogen. There are some other insects which accumulate plant toxins in their body, and thus become distasteful to bird predators. Thus insects use plant toxins for their own protection.

Biochemical ecology (also called chemical ecology or molecular ecology) is a newly established discipline of ecology. It deals with deciphering both the chemical structure and the information content of the mediating molecules of the chemicals the use of which is made by a great variety of organisms (ranging from microbes to insects and mammals) to deter enemies, fend off pathogens, recruit symbionts, and find mates. It is by way of molecules that most organisms communicate (send signals), the vocabulary of organisms is profoundly chemical in nature. The field focuses on the explorations at the interface of natural product chemistry and biology, and has

potential to greatly affect on view of the interactions of organisms. In brief, the signal molecules that convey information from one organism to another are synthesized under genetic control; deciphered at specific receptor sites; transduced into neuronal or endocrinal signals, and eventually translated into behavioural, physiological or morphogenetic responses (Eisner and Berenbaum, 2002).

Choice and Avoidance: Polyphagous herbivores (eating many plant species) include most vertebrate herbivores, snails, slugs and many insects. They use microbial symbionts and extensive physiological defenses to counter toxins of wide varieties of plant species they eat. Oligophagous (eating few plant species) or monophagous (eating single species, mainly insects) herbivores, on the other hand, must sense and behaviourally avoid plants that they cannot safely eat.

Plant Defense Theory

Two plant Defense theories have drawn considerable attention. One, called *apparency theory*, suggests that long-lived trees, shrubs and grasses cannot escape being discovered by herbivores. Therefore, they are well-defended by lignified, cutinized and silicified cell walls that restrict digestion, as well as by a wide variety of toxins, effective in low concentrations. Annuals and other short-lived herbs and shrubs of early successional environment rely heavily on toxins that are poisonous to all, except specialist herbivores that have counter adapted them.

The other, *resource availability* theory, emphasizes that resource availability and inherent growth rates are major determinants of amounts and kinds of defenses that plants employ. Compared to slow growing species, the fast growing species are exposed to higher rate of herbivory, they have greater ability to replace tissues, they invest less on defense, show qualitative defense (compared to quantitative defense generally employed by the slow growing species), in them turnover rate of defense molecules is higher and they show more flexible defense expression. The high turnover of small toxin molecules imposes a higher energetic cost on plants than complex digestibility-reducing polymers. This hypothesis, however, has not been fully tested.

General Effect of Herbivory

Grazing increases net primary productivity of grassland communities when consumption levels are low. Such increase in productivity (over ungrazed, control state) due to grazing is called *over compensatory* effect of herbivory. The increase in primary productivity results from increased photosynthetic rates in residual tissue due to (i) increased sink strength (enhanced assimilate demand in meristems of remaining shoots), (ii) removal of older, photosynthetically less efficient tissues, (iii) increased light intensity (due to decreased amount of dead shoots and mutual standing) on potentially more active underlying tissues, (iv) reduced rate of senescence and (v) greater flow of growth-promoting hormones to residual meristems leading to rapid leaf growth. Saliva of many herbivores contains growth promoting substances which also stimulate the growth of grazed plants.

Reports are also there to indicate increase in species diversity due to grazing. A recent study based on compilation of worldwide 236-site data suggests that changes in species composition with grazing increase with increasing aboveground net primary productivity (ANPP) and with longer, more intense evolutionary histories of grazing (Milchunas and Lauenroth, 1993). This study also indicated that it is not necessary that grazing negatively impacts root systems. However, severe grazing over a long period can reduce soil fertility by increasing soil erosion, especially as a consequence of trampling by animals. In such a situation root mass may also be reduced.

9.5 CARNIVORY

True predators are the organisms that kill the prey soon after attacking them. They range in size from microscopic protozoans to large sized animals such as lions or killer whales. Depending upon their feeding habitats, carnivore predators are sit-and-wait predators (e.g., most spiders and owls) and active predators (e.g., warblers, dragonflies). Sit-and-wait predators depend on active prey, whereas active predators actively seek out their prey, which generally are of sedentary types. A unique form of predation is cannibalism in which predator and prey are of the same species. Even plants can be predators as in the case of insectivorous plants (e.g., venus fly trap, *Utricularia*).

A tight co-evolutionary relationship develops if a prey has only one major predator. Coupled co-evolution can result in highly specialized mutual adaptations, whereas in diffuse co-evolutionary relationships, a prey has to deal with many predators. Predation along with competition is the major density-dependent factor regulating population growth. Predation also plays an important role in structuring biological community as predator-prey interactions at one trophic level affect the predator-prey interactions at the next trophic level.

Models of predator-prey dynamics

A variety of models have been used to explore predator-prey dynamics. Lotka-Volterra model is simple and valuable for understanding predator-prey relationship. This model predicts cyclic oscillations in the abundance of predator and prey populations by using a set of paired equations.

The understanding assumptions are: (i) The birth rate of the predator will increase as the population size of prey increases and (ii) the death rate of the prey will increase as the population size of predator increases.

Let N_1 and N_2 represent the number of individuals in predator and prey populations, respectively, b_1 and b_2 individual birth rates, and d_1 and d_2 individual death rates of predator and prey populations, respectively. Since individual birth rate of the predator population depends on the number of individuals in the prey population (i.e., the food available), it can be represented as $b_1 N_2$ and since individual death rate of the predator population does not depend on the number of prey individual it can be represented simply as d_1 . Therefore, the population growth of the predator will be represented by the equation:

$$dN_1/dt = (b_1 N_2 - d_1) N_1$$

In the case of prey, the individual birth rate is not dependent on the density of the predator; therefore, it can be described simply as b_2 . On other, the individual death rate of the prey population is directly proportional to the predator density, therefore, it is described as $d_2 N_1$. The prey population growth is now represented by the second of the Lotka-Volterra equations as:

$$dN_2/dt = (b_2 - d_2 N_1) N_2$$

The Lotka-Volterra model is graphically shown in Figs. 9.5 to 9.7. In Fig. 9.5, the horizontal line represents zero growth of predator population (at $dN_1/dt = 0$, $N_2 = d_1/b_1$; this equation will generate a straight line in a graph of N_1 vs N_2). To the right of this line predator population growing (more prey therefore the birth rate of the predator population exceeds its death rate) and to the left of this line it is decreasing.

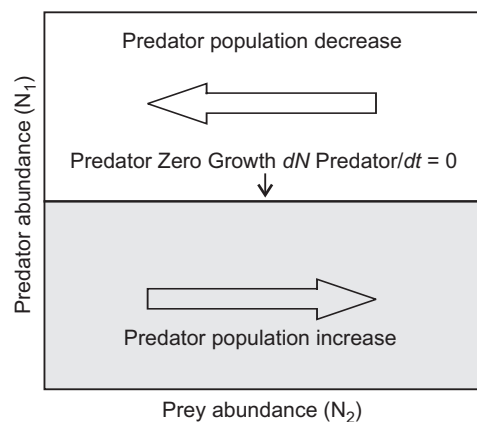


Fig. 9.5. The equilibrium isocline of predator population showing region of increase and decrease.

In Fig. 9.6, the vertical line represents zero growth of prey population (at $dN_2 / dt = 0$, $N_1 = b_2 / d_2$; this equation also generates a straight line in a graph of N_1 vs. N_2). Below this horizontal line the prey population is increasing because there are few predators (so the death rate of prey population is less than the birth rate), while above this line prey population decreases. When we combine the figures 9.5 and 9.6 into one figure (Fig. 9.7), the cyclic change in the prey and predator populations becomes apparent, and at the intersect of the vertical and horizontal lines neither population is growing and the system is at equilibrium.

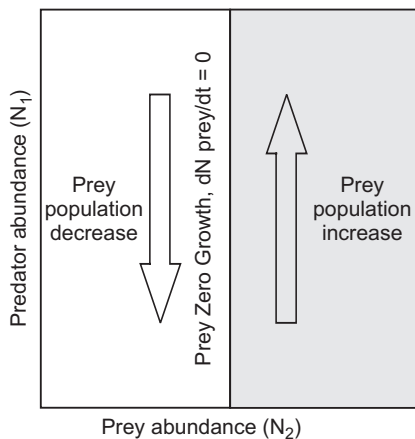


Fig. 9.6. The equilibrium isocline of prey population, showing region of increase and decrease.

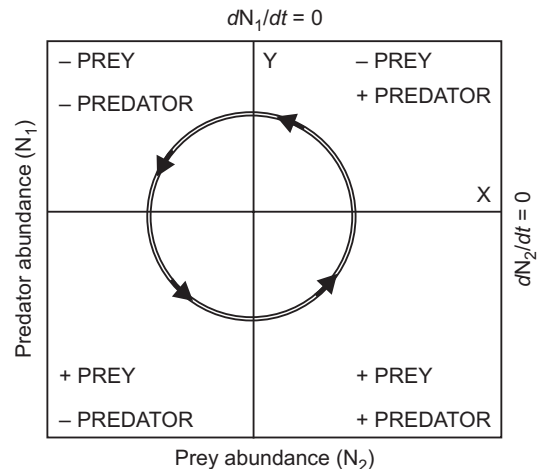


Fig. 9.7. The population trajectory combining individual changes of predator and prey populations.

When a joint instantaneous change in predator and prey population is plotted against time, it shows a regular cyclic variation in the abundance of predator and prey populations (Fig. 9.8).

The Lotka-Volterra model explains the occurrence of population cycles, but the model is so simple that it fails to represent real situations in nature. The stability in predator and prey relationships is promoted by defense strategies in prey, density limitations of predator or prey, predator seeking alternative food sources, and reduced time delays in response to changes in prey abundance. Therefore, it would be useful to know about predator behaviour in relation to functional responses, and optimum foraging strategy (profitability of preys and switching between preys) and the numerical response.

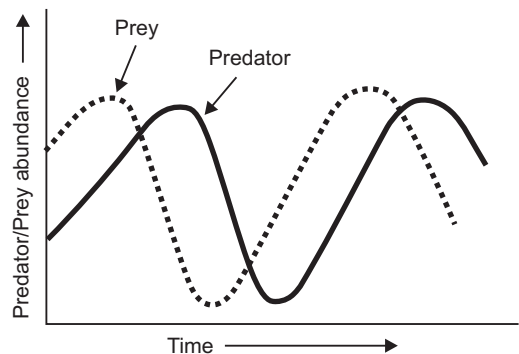


Fig. 9.8. The curves of population and predator showing changes in predator and prey population.

Predator density - functional responses

As predator density increases, the consumption rate of the predator also increases up to a maximum possible rate. This relationship is called function response (Holling, 1959, see Smith, 1996) and can be represented by three types of curves:

Type – I: The prey is consumed in direct proportion to their abundance to a maximum and when the predators processing ability is saturated the consumption rate levels off.

Type – II: The predator processing ability gradually becomes satiated as it consumes prey. As a result, number of prey eaten rises to a maximum following which the rate of increase in the consumption gradually declines with prey density.

Type – III: The number of prey eaten is low at low prey density due to low hunting ability or other factors, increases, rapidly at intermediate prey densities and levels off at high prey density.

Optimum foraging strategy

Predators tend to maximize net energy gain. Predators often concentrate their activity in the most profitable patches of prey. They generally abandon those patches where prey populations are reduced to a level which represents the average or below average profitability of the area as a whole. Thus optimum foraging strategy refers to the ability of predators to seek food efficiently and to obtain an optimal diet that includes the most efficient size of the prey for handling and energy gain.

Profitability of prey: A predator optimizes its efforts to choose the most-profitable prey given a choice between two potential prey types. It is common that predators often prefer some intermediate size between the large and small preys. For example, shore crabs (*Carcinus meanas*) prefer intermediate sized mussel (*Mytilus edulis*) which gives the greatest energy gain per unit time.

Switching between preys : Depending upon the abundance of prey species, the predators may alter or switch their preferences for a particular prey species; common prey are consumed at a rapid rate, whereas less common prey are largely ignored. Switching to alternative sources to food when prey is scarce may reduce pressure on prey. In times of scarcity of food, generally the predator switches over to alternative prey sources.

Numerical response: In addition to functional response, the predators may also exhibit numerical responses. The numerical response describes the response of a predator population to increasing prey density by population growth and immigration. Individual predators can increase their consumption of prey only to a point of satiation. Continued predator response to increasing prey density above this level can only be achieved through an increase in the number of predators, either by immigration or by population growth, which together constitute a numerical response. Immigration from surrounding areas contributes to the responses of mobile predators, which may opportunistically congregate where resources become abundant.

9.6 CARNIVOROUS PLANTS

Carnivorous plants (also called insectivorous plants) kill insects to meet their nitrogen demand. The first detailed account of insectivorous plants was given by Darwin (1875). Their presence is restricted to nutrient-poor sites, generally sunny and moist, like swamps and bogs. They are not heterotrophs as they do not capture the prey (insects) for food (as a source of carbon). This is proven by the fact that if nitrogen and phosphorus are provided externally the insectivorous plants complete life cycle without prey-capture; that the prey-capture enhances growth on nutrient-poor sites; and the prey-capture fails to cause growth in the dark. About 600 species of carnivorous plants are known belonging to six subclasses including Dionophylaceae, Nepenthaceae and Droseraceae. The carnivorous plants are of much interest as they are being used as model systems to answer various ecological questions, understanding ecological processes regulating evolutionary patterns as well as to understand extinction processes under changing environmental conditions (Ellison and Gotelli, 2001). Many carnivorous plants are threatened and need to be conserved.

To extract nitrogen, the insectivorous plants have developed (i) devices for absorbing nitrogen from dead prey, and (ii) modifications to attract, capture and digest insects. Of all insectivorous plants, pitcher plants, species of *Sarracenia*, *Darlingtonia*, *Nepenthes*, etc. are well known, especially for their pitcher-shaped leaves partly filled with fluid (Fig. 9.9). On the inner surface of the pitcher are present downward-pointing hairs or the inner surface of pitcher is extremely smooth. Both prevent insects to escape, while they wander into pitcher-shaped leaves. The liquid in pitcher is essentially an aqueous solution of proteolytic enzymes which dissolve the softer parts of the insect bodies, thus allowing leaf to absorb soluble amino acids of the body for nitrogen. Certain protozoa and insects, however, are able to survive the enzyme action and live inside the pitcher.

Sundews (*Drosera* spp.) are typical inhabitant of bog habitats. In this herb, the spatulate leaf blade is covered with prominent glandular hairs, bearing glistening droplet of a sticky fluid. Insects that alight on the leaves become stuck to the hairs; consequently the glands present on the hair tips are stimulated to secrete proteolytic enzymes. It is notable that non-proteinaceous materials, like insect wings, fail to stimulate the movement of hair or the secretion of enzymes from glands. In carnivorous plants, fast snap-trapping and suction of prey are two of the most common mechanisms to actively catch animals, mainly arthropods, for a substantial nutrient supply. Sundews (*Drosera* spp.) are known to possess leaves with glandular emergences (tentacles) which secrete glistening, adhesive glue drops for attracting and capturing prey. *D. glanduligera* is a common and widespread annual from southern Australia, which additionally features glue-free (non-sticky) snap-tentacles that can bend within a fraction of a second. Poppinga et al., (2012) have shown that *Drosera glanduligera* exhibits a sophisticated catapult mechanism to catch the prey animals, walking near the edge of the sundew triggers a touch-sensitive snap-tentacle, which swiftly catapults them onto adjacent sticky glue-tentacles. As a result, the insects are then slowly drawn within the concave trap leaf by sticky tentacles. The analysis of such catapult-flypaper traps in action highlights a unique and surprisingly complex mechanical adaptation to carnivory (Poppinga et al., 2012).

Carnivorous plants obtain most of their nutrients by capturing insects and other arthropods through their leaf-evolved biological traps. The best-known attractants in carnivorous prey traps are nectar, colour and olfactory cues. Recently, scientists from Jawaharlal Nehru Tropical Botanic Garden and Research Institute, Pacha-Palode, Thiruvananthapuram, Kerala, India, have reported a new prey capture mechanism in some species of carnivorous plants (Kurup et al., 2013). They found the existence of distinct blue fluorescence emissions at the 'capture spots' of *Nepenthes*, *Sarracenia*, and *Dionaea muscipula* at UV 366 nm (Kurup et al., 2013). *Drosera*, *Pinguicula* and *Utricularia* prey traps showed only red chlorophyll emissions at 366 nm. Masking of blue emissions on peristomes drastically reduced prey capture in *N. khasiana* pitchers (Kurup et al., 2013). These molecular emissions could be a critical factor to attract arthropods and other insects to these carnivorous traps.

The other common insectivorous plants are bladderworts (*Utricularia* spp.), which are aquatic herbs bearing 5 mm or less broad bladder-like traps interspersed among photosynthetic leaves, and the venus flytrap (*Dionaea* sp.), the leaf blade of which resembles a steel trap. The carnivorous plants differ in growth habit, type of trap and insect nitrogen requirement. The flowers

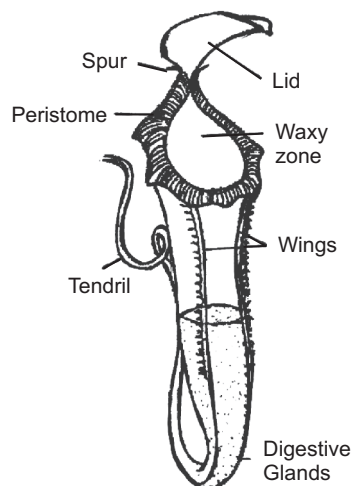


Fig. 9.9. A diagrammatic view of pitcher in *Nepenthes* (an insectivorous plant).

of most insectivorous plants are insect pollinated; the separation between flower and traps helps to distinguish between prey capture and pollination (Ellison and Gotelli, 2001). Nitrogen requirement is comparatively low in *Drosera* species and more in the case of *Darlingtonia californica* (Cobra lily) possessing more than one metre tall pitchers.

9.7 PARASITISM

Parasitism is a well known example of species interactions that results in negative effects on growth and survival of one population and a positive or beneficial effect on the other. The micro parasites including bacteria and fungi are of very small size and cause physiological and behavioural changes in plants and animals. The trypanosomes cause sleeping sickness, whereas the species of plasmodium cause malaria in human beings in different parts of the world. The macroparasites could be endoparasites (e.g., parasitic round worms and tapeworms) and ectoparasites (e.g., ticks, mosquitoes and mites). Fungi are the most common macro parasites of plants. The microorganisms like bacteria, fungi and viruses may cause disease in wild populations causing significant changes in their populations.

All parasites, whether bacteria or tapeworms and predators such as hawk or grain-eating birds, are disliked by humans as they are considered harmful, and their benefits are seldom recognized. Are predators and parasites the sole beneficiaries of the associations? Long-term studies do not suggest so. Predators and parasites help keep herbivorous insects at low density so they will not destroy their own food base (plants) and habitat. In a case of mutual “predation” in a spider host-parasite relationship, the egg parasites are eaten as they emerge from spiders’ eggs by young spiders that have escaped being parasitized (Vallerio, 1975). Removal of predator is known to have increased kaibab deer’s population dramatically (though other factors also contributed) destroying the very food base (Leopold, 1943).

Evidences are there to indicate that most rapid rise in species populations occurs when they are introduced into a new area where resources are in abundance and negative interactions are absent. The population of rabbit exploded when introduced in Australia where grazing lands were vast and rabbit’s natural predators were absent. Eventually, a parasite was used to control the irrupting rabbit population. Use of parasites to control host population was made in the case of chestnut blight which destroyed the American chestnut forests of the Appalachian region of the eastern North America.

Parasitic vascular plants

Some vascular plants grow on other vascular plants and derive their food, in part or completely, from the host plant. They are referred to as parasitic vascular plants. Parasitic plants are represented by about 4000 species accounting for 1% of angiosperm species, and can be found in most major biomes of the world (Press and Phoenix, 2005). The parasitic plants include annual and perennial herbs (*Rhinanthus* spp), vines, shrubs (e.g., Mistletoes) and trees (e.g., *Santalum album*, sandalwood tree of old world). *Cuscuta* (dodder or amarbel) is one of the well known parasites. Seeds of this twining leafless herb germinate and produce green stems over soil surface until they get their host plants about which they twine. The adventitious roots of *Cuscuta* penetrate the stem of the host, where they come in contact with xylem and phloem. The adventitious roots and other organs used by the parasite to absorb food from the host are called *haustoria*. After establishment on the host, and disconnection from the soil, *Cuscuta* grows further by extracting food and water from the host species. The stem is green, so this parasite is able to make itself some of its food.

There are others which are completely parasitic upon the roots of their host plants, e.g., broomrapes (*Orobanche*, *Conopholis*, *Epifagus* and others of Orobanchaceae), *Rafflesia* of Aristolochiaceae, a Malaysian genus. In broomrapes roots of seedlings are connected to the roots of their hosts. In some cases seeds would not germinate unless they come in contact with the root of the

suitable host. The aerial portion of these herbaceous parasites is reduced to a brownish inflorescence with no chlorophyll. Several species of Lorantheae are dwarf shrub like plants rooted on tree branches. They are called hemiparasites as they are partially autotrophic. They are injurious to trees as their haustoria absorb food from host's vascular cells. In *Rafflesia* (parasite on roots of *Vitis*) the plant is extremely degenerated, the body resembling a group of mycelium of a fungus. The entire body of the parasite remains inside the root tissues of the host. In *R. amoldi* the flower is as large as one meter in diameter, and is the only organ to be visible above ground.

9.8 COMMENSALISM

Commensalism represents the first step toward the beneficial relations between species. This relationship is quite common between sessile organisms (both plants and animals) and motile organisms, especially in habitats of ocean. A shellfish, or sponge or a worm burrow in ocean contains a commensal, an organism that requires the shelter of the host and does neither harm nor good in return ("uninvited guests"). For example, oysters sometimes have a small crab in the mantle cavity. The crabs are usually commensal, though occasionally they may take away some of host tissues. Dales (1957) has found 13 commensal species that live as guests in the burrows of large sea worms (*Erechis*) and burrowing shrimps. The commensals include fish, calms, polychaete worms, crabs and other animals which live by taking surplus or rejected food or waste materials from the host.

Vines and Lianas

Vines and lianas (woody climbers) are vascular plants which are rooted in the ground but take support of some objects (generally trees and shrubs). Thus, they are able to reach a lighted environment with minimal supporting structures of their own. Lianas are abundantly found in humid tropical forests and their diversity increases with decreasing latitude and altitude; the increase in diversity being five-fold from temperate to tropical rainforests (Richards, 1996). These are also common in early successional and disturbed forests where better light conditions prevail.

They do not take nutrients or food from the tree whose support they take. Lianas have two anatomical peculiarities: (i) the woody cylinder vertically partitioned into strands separated by parenchymatous tissues; and (ii) xylem vessels are long and wide. The first character may give suppleness to the liana, which is required to twine about the support. The second character gives superior conductance, which is required to support a foliage mass which when seen in relation to stem diameter is larger than generally found in the case of trees. Lianas have special adaptations to climb forest canopy as these possess twining stems, clasping stem tendrils, modified leaf and stem structures, thorns, spines and adhesive hair and adhesive adventitious roots (de Walt et al., 2000; Schnitzer and Bongers, 2002). The climbing mechanisms in lianas are influenced by succession stages of the forest and the disturbance regimes; the stem climbers increase in comparison to tendril climbers as the forest stands mature (de Walt et al., 2000).

Lianas often spread their canopies over those of the trees they climb, thereby affect the normal growth of the host. Lianas often negatively affect the host trees by influencing tree regeneration, and through aboveground and belowground competition for resources for light, nutrients and soil water. In fragmented landscapes and disturbed forests, lianas are likely to play greater role in forest regeneration, maintaining species diversity and regulating ecosystem level process such as whole forest transpiration and carbon budget (Schnitzer and Bongers, 2002).

Epiphytes

Plants (including flowering plants, ferns and other pteridophytes, mosses and other bryophytes, lichens, etc.) which grow perched on other plants (generally trees, but also shrubs, and larger submerged plants), and differ from parasites in not deriving food or water from the supporting

plant and from lianas in not having soil connections, are called epiphytes. For water they almost completely depend on precipitation, especially water flowing along the supporting surface (stem flow). Therefore, they must face several periods of drought. That is why they are most abundant in warm, wet climates with short periods of drought.

Epiphytes get nutrients from rainwater, which always contains some dissolved substances, in part from accumulated wind-borne particles, and in part from decaying bark surface of supporting plant (Daubenmire, 1959b). They grow on trunk, branches, and twigs and sometimes on the upper leaf surfaces (in that position called epiphylls). On a tree a wide range of epiphytes, from sciophytic (shade-loving) hygrophytes on tree trunk to bright light demanding xerophytes (including cacti) on tree tops can be found.

Many of the epiphytes also occur on ground, especially on stones. Some, as the fern *Nephrolepis*, at first are not epiphytes. They are rooted in the soil, and then by extending their rhizomes up a tree trunk and later losing connection with the ground they eventually attain epiphytic habit. This is a case of shift from climbing to epiphytic habit. Plants, such as the hemlock, *Tsuga heterophylla* and the strangling figs germinate on trees but live as epiphytes only until their roots grow down over the tree stems and develop connections with the soil.

According to Daubenmire (1959b), true epiphytes have no soil connection at any time in their life. Their adaptation features include xeromorphy (thickened cuticle, sunken stomata, succulence, etc.) and devices to hold and absorb rain water. Some epiphytes have their roots spread over the plant surface to absorb rain water film. In nest epiphytes, the roots accumulate large masses of debris that hold water like the soil. Orchids and members of arum families extend their roots outward into the air. These roots on their surface have absorptive tissue, called *velamen*. It consists of empty whitish cells that can take up water quickly even when shower is very brief and small.

In epiphytic species of *Tillandsia* (Bromeliaceae) the water absorption takes place through stem and leaves, and roots are chiefly anchorage organ (Fig. 9.10). The Spanish moss (*T. usneoides*) bears numerous scales on finely divided shoots, which collect capillary water and allow it to be absorbed by the small uncutinized spots on the epidermis. Some other species of *Tillandsia* have gutter-shaped leaves, in which water is collected in their axils. These collection points, called tanks or cisterns, contain numerous aquatic invertebrates, including mosquito larvae. *Dischidia*, a highly specialized epiphyte, has pitchers that accumulate water and leaf litter into which special absorptive roots extend. Epiphytism may evolve into parasitism, when the roots of the epiphyte penetrate bark of the host and develop haustoric, e.g. *Arceuthobium* on conifer trees.



Fig. 9.10. A trunk epiphyte (*Tillandsia*) growing on trunk of a palm.

9.9 PROTOCOOPERATION

Protocooperation is a step forward in the direction of positive interaction, in which both the organisms gain by the relationship. In sea habitat, as an example, crabs and coelenterates often associate with benefit to both. The coelenterates grow on the back of crabs or are “planted” by crabs themselves. Their presence provides protection to crabs by camouflaging and by the stinging cells present in coelenterates body. Coelenterates, in turn, are transported by the crab and get particles of food when the crab captures and eats other animals. However, in this relationship neither of the organisms is absolutely dependent on its partner. Each can survive without its associate.

9.10 MUTUALISM (OBLIGATE SYMBIOSIS)

Mutualism or obligate symbiosis represents a more advanced kind of relationship, in which the organisms have complete dependence on their associates; they cannot live without their partners. Often mutualism involves two organisms with widely different requirements, such as autotrophs and heterotrophs. Partnerships between higher plants and nitrogen fixing bacteria (*Rhizobium*) and actinomycetes (*Frankia*) are well known examples of mutualism (Table 9.4). Symbiosis is quite common between microorganisms that can digest cellulose (and other resistant plant residues) and animals that do not have mechanism (enzymes system) for this, e.g., between ungulates (cattle) and rumen bacteria; between termites and their intestinal flagellates. Many species of termites cannot digest the wood they ingest, as shown by the experiment in which flagellates were removed. The termites starved to death without flagellate symbionts, indicating extreme level of dependence.

Table 9.4. Some examples of mutualism and description of benefits to organisms that combine them

Symbiotic organisms	Benefits given to associated organism(s)
<i>Mycorrhizal symbiosis</i>	
1. Fungi	Enhances nutrient availability to plants
2. Plants	Gives a part of photosynthate to fungi
<i>Symbiotic N fixation</i>	
1. N ₂ -fixing bacteria/actinomycetes	Provides a readily assimilable supply of fixed N to plants.
2. Root nodule bearing plants	Gives energy (photosynthate) to microbes, a congenial home in the root nodule, which protects microbes from too much O ₂ which is inhibitory to N fixation
<i>Lichens</i>	
1. Fungi	Neither organism alone can tolerate desiccation
2. Algae	Give food to fungi
<i>Termite intestinal-flagellate symbiosis</i>	
1. Flagellates (Protozoans)	Digest wood cellulose ingested by termites, the “processed” food is eaten by termites
2. Termites	Ingest wood pieces used as food by flagellates in termite’s intestine
<i>Ungulates and rumen bacteria</i>	
1. Bacteria	Provides cellulose/rich food inside rumen
2. Ungulates (e.g. cattle)	Digest cellulose of food taken by animals
<i>Attine ants-fungal gardens</i>	
1. Fungi	Digest cellulose-rich food for ants
2. Attine ants	Cultivate fungal gardens on the leaves, the ants harvest and store in their nests; and fertilize fungi.

Rhizobium, the symbiotic nodule bacterium on roots of legume plants, is most well known among the N₂-fixers. The Actinomycete, *Frankia* which lives in root nodules of alders (*Alnus*) and several other woody plants is another very well known symbiotic N₂-fixer. Certain other non-leguminous woody plants fix N₂ as efficiently as do leguminous plants and alder. Most of these species occur in infertile soils.

The mycorrhizal symbiosis has been discussed earlier in Chapter 4.

Lichens are other representatives of symbiosis. They are an association of fungi and algae, which is so intimate in terms of functional interdependence and so integrated morphologically that a kind of organism different from both partners is formed. Though composed of two very different organisms, lichens are generally classified as single organism. Lichens display various steps of evolution from parasitism to mutualism. In some of the primitive lichens the fungus actually sends its hyphae into algal cells, indicating its parasitic nature (Fig. 9.11 a). In advanced forms, fungal hyphae are intermingled with algal cells or are closely appressed to them (Figs. 9.11 b and c), indicating a close harmony between alga and fungus. In this relationship, the fungus gets its food from alga. Possibly, the alga is benefited by the fungal excretions. In case the partner is a blue/green alga, it may be able to fix atmospheric nitrogen. But the truly mutualistic benefit lies in the fact that neither organism alone could tolerate desiccation which lichens endure so often on stones and stems barks.

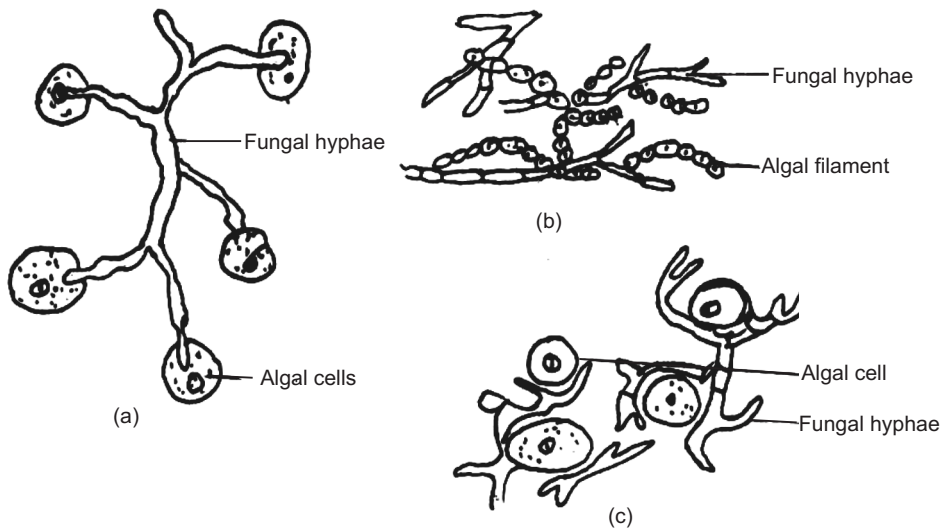


Fig. 9.11. Evolution from parasitism (a, primitive lichens) to mutualism (b and c, advanced lichens) in lichens. In (a) fungal hyphae actually penetrate into algal cells, while in (b) fungal hyphae and algal filaments are intermingled and in (c) fungal hyphae are closely appressed to algal cells but without penetrating into algal cells (adapted from Odum, 1983).

Microbial Farms: Microbial interaction with herbivores

Herbivores have developed a microflora of bacteria, flagellates and protozoans in their digestive tracts that (i) synthesize necessary vitamins, (ii) break down plant parts, and (iii) detoxify plant allelochemicals through anaerobic fermentation, making thereby poisonous food palatable. These microbes are essentially symbionts for the herbivores. Vertebrates show several structural and functional modifications in their guts for herbivory. For example, ruminants (e.g., cattle, *Bos*) swallow foliage with minimal chewing, passing the food into their four-chambered stomach. Cellulose and hemicelluloses are degraded into sugars and starches in rumen (technically, the two-part reticulo-rumen of stomach) filled with microbes. A sheep contains $16,100 \times 10^6$ bacteria and 3.3×10^5 ciliated protozoans in 1 ml of its rumen fluid. A sheep rumen is filled with 6 litres of fluid and that of large cow close to 80 litres. Non-ruminant mammalian herbivores also use microflora for fermenting and digesting plant fibres, and for housing microflora they have evolved different containers (e.g., sac-like out pocketing of the stomach analogous to rumen in monkeys, or a large *caecum* (blind sac in horses, or a large sacculated colon).

In spite of a large microbial activity, much plant material cannot be digested; while soluble carbohydrates, starch, organic acids and proteins of plants are completely or mostly digested, cellulose (43-73%) and hemicellulose (36-79%) are partly digested and lignin, silica and cutin are indigestible. Animals increase digestibility of food by (i) increasing volume of microbial fluid, (ii) retaining the partially digested food in the gut for longer period; and (iii) reducing behaviourally the proportion of indigestible plant tissues in food intake. Large mammalian herbivores retain plant digesta longer than the smaller animals using their more extensive mycoflora to extract more from food eaten. Thus they digest a higher proportion of food. As an example, white-tailed deer retains plant digesta for 45 hrs and its body weight is 48-100 kg, while American bison with body weight of 450-1350 kg and domestic cattle with similar body weight (450-900 kg) retain plant material for processing up to 80 hrs (Robbins, 1983). Compared to 56% digestion of alfalfa fibre by deer, cattle realize 70% digestion.

Small herbivores are forced to avoid fibre-rich food because of their physiological constraints, or use special adaptations to overcome limitations of a small microbial farm inside their body. Birds eat diet free of fibre, like seeds and buds. Rabbits (with body weight of 1 kg) digest only 7-9% of grass cellulose eaten for the first time, but they reingest the partially digested food pellets (reingesting feces is called *coprophagy*) which are soft and rich in protein (24-38%). They defecate fibrous pellets poor in protein.

Herbivorous insects with much smaller size cannot have a large volume of microbial symbionts for fermenting food. They have limited their diet to fewer plant species and plant parts and have evolved different and varied means of maintaining symbiotic microbes. Some insects have mutualistic microbes inside cells, called *mycetocytes* (referred to as *endosymbionts*), playing roles of (i) breaking down cellulose, fats, etc., (ii) synthesis of vitamins; and (iii) even nitrogen fixation. Some insects harbour the microbes in a special organ in digestive tract, called *mycetotomes*.

To overcome the problem of small size, some insects cultivate fungal symbionts outside their bodies - e.g., bark beetles (Coleoptera) and wood boring grubs of long-horn beetles. The bark beetles spread and cultivate fungi in burrows in wood. They eat only the wood partially digested by symbiotic fungi. The vastness of fungal gardens that the two groups of social insects, viz., some termites (Termitidae of Isoptera) and leaf-cutting ants (Attinii of Hymenoptera) maintain, rival or even surpass the volume of gut fluids of the largest ruminants. For example, leaf-cutter ants feed on the huge fungal colonies that they cultivate with leaves and flowers in vast underground nests, in addition to the plant sap. They forage for a much wider range of plant species than the insects containing small microbial farms inside their bodies. Thus the foraging behaviour of insects depends on the kind of microbial gardens they cultivate.

9.11 FACULTATIVE SYMBIOSIS

The majority of mutualisms are nonobligatory and opportunistic, where the partners may co-exist without reliance on each other. They may be diffuse, involving many species, as occurs in the case of pollination and seed dispersal by animals. The transfer of pollens from anthers to receptive stigma in flowers is called pollination. The animals that bring about cross pollination include bees, flies, beetles, ants, thrips, birds, slugs and lemurs in different types of communities. Animals pollinate flowers while foraging for reward, which includes nectar and pollens as well as oils, waxes or perfumes to be used as sexual pheromones. The floral reward has probably evolved from florivory (total consumption of flowers). Thus, the cost of animal visitation in this is only the consumption of nectar and pollens, etc. Thus plant-pollinator relationship is typically materialistic where plant receives pollination service and pollinator receives food. Some plants adopt non-rewarding approach to attract animals. Their flowers may mimic a pollinator's mate or food sources.

Pollination by animals

Cross-pollination by animals (*entomophily*) is characteristic of plants with showy and fragrant flowers. The majority of angiosperms are pollinated by animals, and nectar is the most common type of floral reward offered by plants to pollinators and a keystone resource exploited by an extraordinary variety of consumers in most terrestrial ecosystems. Insects visiting flowers get not only nectar and pollens, but also oils, waxes and volatiles all of which are a food source. Flowers can be showy due to bright colour, large size, or aggregation of the flowers into large clusters. Fragrance is due to volatile oils. Pollens of entomophilous plants are sticky and rough or spiny so that they readily adhere to insect's body. Plants tend to display different floral traits appropriate to the sensory abilities and foraging methods of visiting animals. For example, generally birds pollinated flowers are red (a colour not within the visual spectrum of many insects) and blue, bat pollinated flowers are white or green dangling in ways that are identifiable to echo-locating foragers. Moth pollinated flowers are typically white, night scented, having narrow corolla or spur, in which dilute nectar is stored. The flowers which are pollinated by bees tend to be yellow, blue or purple (but not red) are day scented having sugar rich nectar.

Flowers of many plants exhibit a great variety of specialized features that ensure services of insects and guard against self-pollination. Some of the specialized mechanisms are given below:

- Stamens remain reflexed until visiting insects cause them to spring inward and dust pollen on insect body (e.g., *Berberis*, *Kalmia*).
- In many legumes the stamens and stigma are packed in keel under tension and are freed, striking the under-part of the body, when a bee depresses the keel to get the nectar. Some of these flowers show partial self-sterility or difference in time of maturity of stamens and pistils (dichogamy) to ensure cross-pollination by insects.
- In some flowers, lobes of stigmata close together when insect contacts them, thereby guarding against self-pollination (e.g., *Tecoma*, *Utricularia*).
- Flowers of some species have perianth with deep spurs into which nectar is secreted. Normally insect must crawl into the flower and brush against the stigma, then the anthers to get the nectar (e.g., *Aquilegia*, *Delphinium*, and *Viola*).
- In *Salvia* a fine lever system occurs. Insects strike their heads, bring the anther down on the insects back as they enter the corolla tube.

Insects may become attracted to flowers also for reasons other than food. For example, in *Yucca*, the female moth, *Tegeticula*, carries balls of pollen from one flower to another, thrust them down the tubular stigma, and then use its ovipositor to place eggs among the ovules which it has just fertilized. However, a few seeds escape destruction by larvae that develop from eggs. Thus in long term the *Yucca* plant is a gainer. Flowers of certain orchids (*Ophrys*, *Cryptostylis*) are pollinated because to wandering hymenopterans insects their lip resembles their females (flowers are also scented and coloured). As a consequence, males are induced to alight and attempt to copulate. This act when done on different flowers brings about cross pollination.

Specialization relating to insect population involves parallel and complementary evolution between insects and flowers, leading to the dependence of a particular type of plant on a particular type of insect. The pollinating insect must be introduced, when a crop is introduced to an area. For example, for cultivating Smyrna figs of marketable quality in California, the pollinating wasp was also established. *Vanilla* is pollinated by hand in many areas because of lack of the pollinating insect.

Oil has also been recognized as a floral reward for bees and other insects (see Buchmann, 1987). The oil secreting organs called elaiophores (Fig. 9.12) are found abundantly in flowers of plants of neotropical savannas and forests, in at least 10 families (e.g., Iridaceae, Malpighiaceae, Orchidaceae, Scrophulariaceae, Solanaceae, Primulaceae, in about 79 genera and 2400 species). Oil flowers are pollinated by about 1.4% of 30000 to 40000 species of bee. Some bees use the lipid secretions with or in place of nectar as foodstuff for larval development. Others use floral oils also for providing water-resistant cell linings, and possibly also for adult food and nutrition.

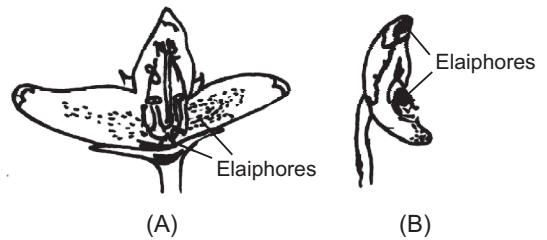


Fig. 9.12. Secretory glands (elaiophores) which secrete oils in flowering plants.

(A) Trichome elaiophores of glandular hairs on androecium and petal bases of *Lysimachia* sp. (Primulaceae),

(B) Epithelial Elaiophores, see steeply concave glands on an anther of *Mouriri* sp. (Melastomataceae).

Seed (and other disseminules) dispersal by animals (zoochory)

Animals (including birds) may disperse seeds (including fruits, inflorescences and other disseminules), both of which they eat (edible) as well as those which they do not eat (non-edible).

Seed Dispersal by Adhesion (epizoochory)

Most adhesive fruits (non-edible) adhere to animal's body or fir barbes, hooks (together called burrs) or viscid outgrowth (viscid fruits) (Fig. 9.13). Propagules are also known to disperse by clinging to mud on bird's feet. Such fruits or seeds do not have any specialized structure for adhesion, therefore are not included in epizoochory.

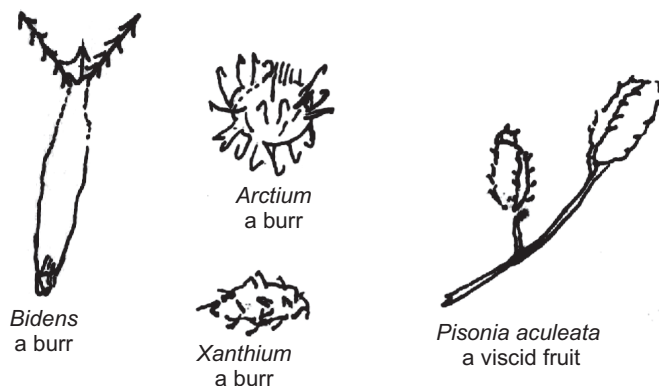


Fig. 9.13. Examples of adhesive fruits.

Back to specialized device of adhesion, structures such as hooks and barbed spines may have also some other functions than dispersal such as defence against predators. Adhesive awns and mucilage, in fact, primarily function as devices for establishment. Fruits that are not detected or cannot be groomed off can be carried by dispersing animals to long distances. Reports are there to indicate that adhesive seeds may be carried to distances of 950 km south of New Zealand or 1540 km from Java.

Adhesive fruits easily attach to passive animals, and they remain attached until animals drop off passively or animal detects and removes them. Unlike fleshy fruits which animals eat and which have conspicuous attractive device, adhesive fruits lack an attractive mechanism. It is puzzling because adhesive-dispersed spores of non-flowering plants, such as mosses and mushrooms, produce odours to attract insects to sporangia. One possible explanation for lack of attractive devices in adhesive fruits and seeds is that their dispersal is better when animals do not know that they have them on their body (Sorensen, 1986). Adhesive dispersal is uncommon, generally less than 5% of the species in different floras show adhesive seed dispersal. The modifications for adhesive dispersal are usually on the pericarp rather than on seeds themselves. Most of adhesive fruits (over 80% in a flora) have hooked or barbed spines and most of them have such structures over entire fruit surface. Viscid outgrowth is found only on a few adhesive fruits (around 10% in above sample of flora).

Wind dispersal is perhaps more ancient than adhesive dispersal. Modifications for the two types of dispersal are similar in some families. For example, in Asteraceae, some species are wind-dispersed by plumed pappus (*Senecio* spp.) and others are adhesive dispersed by a barbed pappus (*Bidens* spp.). Plant species with adhesive fruits are low growing, mostly less than 1 m. Many trees produce seeds too large to be successfully dispersed adhesively. There are some other reasons because of which adhesive fruits are rare among trees. Such adhesive fruits are absent among shrubs. In disturbed habitat species with adhesive fruits outnumber species with non-adhesive fruits. Adhesive dispersal is stronger among annuals than among perennials.

Edible seed dispersal

Dispersal of edible seeds occurs by frugivores and scatter hoarding birds and animals. The seeds of berries and other small fleshy fruits are contained in tissues (fleshy parts) used by birds and animals as food. In many of these fruits, the seeds (*Rubus*), achenes (*Fragaria*) or stones (*Prunus*) have hard coverings and so pass through the digestive tracts uninjured. Before being excreted they are carried to some distance. But the seeds whose hard cover birds are able to crack to use soft endosperm and embryo as food destroy rather than disperse them.

In some species seeds would germinate only when they have passed through the digestive tract of birds and mammals. Seed dispersal and germination are also improved by mastication and passing of seeds through digestive tracts of cattle and other such animals, if seed coat were to be broken without injury to inner organs of seed (e.g., dispersal of sweet pods of *Prosopis* by cattle). Even ants have been seen transporting seeds that are oily or have fleshy appendages.

Seed hoarding

Food storage is an important adaptation of several animal species in desert environments to the temporally variable or unpredictable food supplies that are typical of desert environments. Rodents and some birds collect and hoard seeds (cache), but then may fail to recover the hoarded seeds. If such seeds are hoarded at places favourable for their germination, the concerned plant species benefit. Failure to recover buried seeds is likely to be quite infrequent in case of rodents, and they are likely to be effective seed disperser only when seeds are abundant.

Community: Nature, Structure, Attributes and Control

Plants do not grow in isolation, in most habitats members of a species grow mingled with the members of many other species. The species mixtures or assemblages are called communities or associations. These mixtures or groups of species occur in different combinations, i.e., the communities change with the change of environment. For example, with the increase in the elevation from plains to the mountains of Nainital in the Himalaya, we come across the sal forest in the foothills, characterized by the canopy of large-leaved sal (*Shorea robusta*) trees. As we climb further, over 1000 m we notice that the sal forest is gradually replaced by the forest of a long-needle bearing pine (*Pinus roxburghii*). With further increase in the elevation increasingly greater numbers of oak trees (*Quercus* spp.) with sclerophyllous leaves are seen, and ultimately around 2000 m most of the forests are of oaks. In the above forests, *S. robusta*, *P. roxburghii*, *Quercus* spp are the respective dominants.

In each of the above forests members of the species other than those named as dominants also occur. These species are called subordinate species. For example, in the oak forests there is *Rhododendron arboreum* in the undercanopy and *Fraxinus micrantha* and *Cornus oblonga* are common deciduous associates in the canopy. Apart from tree species, the oak forest contains thickets of a small bamboo, *Arundinaria falcata*, and other undergrowths. Even the trunks and branches of the oak and other trees are covered with epiphytic ferns, mosses and orchids, and roots are associated with ectomycorrhizae. Apart from these, several birds, mammals and insects occur.

The above distinct kinds of vegetation are examples of natural communities, each structured by separate group of species fitting together in a specific manner in terms of numbers. A biotic community of an ecosystem will not only include plants and animals but also microbes.

10.1 DEFINITION

Community is a loose term and can be applied to any assemblage of organisms. As a unit of ecological study a community can be of any suitable size or durability. For example, on the one hand the epiphytes growing on tree trunks of a moist forest can be regarded as a community, and on the other, the regional vegetation type, say deciduous forest of the entire Indo-Gangetic plains, can be treated as a community.

Smith (1986) defines *community* as a naturally occurring, mutually sustaining, and interacting assemblage of plants and animals living in the same environment and fixing, utilizing and transferring energy in some manner. In this definition, a community encompasses all organisms, including plants, animals and microorganisms, of a given site. These organisms belong to at least more than one trophic level. For our purpose, here, it will be sufficient to define a plant community as an assemblage of plants commonly belonging to more than one species, occurring in a given habitat at a given time. Ecologists assume interactions and interdependence among species present

in a community (see Chapter 9). Presence of plants of different growth forms (e.g., trees, shrubs, lianas, herbs, epiphytes, etc.) having different requirements, in a community on the same unit of land, maximises the efficiency of utilization of total environmental resources, creates a variety of niches for organisms other than plants also, and imparts a layered structure, each layer with different dominants.

Stands and associations

The vegetation of a region is generally studied by examining a number of homogeneous plots of suitable sizes. The vegetation of each of these plots is a **stand**. Stand is real (or discrete); it can be observed and examined. It is defined as a particular aggregation of plants having a high degree of uniformity in composition and structure and occupying an area of essentially uniform environment. Of the several stands, some resemble one another more and, thus, form a group of stands. The stands resembling one another sufficiently, having essentially the same dominant species differing from other stands, can be grouped together into a *community type (abstract community)* or *association*. Each community type would reappear whenever equivalent environmental conditions occur. However, the term association has also been attributed by some authors not only to an abstract community but also to a concrete community or a stand (McIntosh, 1985).

A community type or abstract community is defined as a group of stands that are similar in species composition and structure and occupy similar habitats (Hanson and Churchill, 1961). Thus a plant community type of (1) a certain (relatively consistent) floristic composition, (2) uniform habitat conditions, and (3) uniform physiognomy is an *association*. This definition corresponds to the definition given by the Third International Botanical Congress. In Clements' terminology, the stable and mature (climax) communities are called associations, the developmental stages (successional stages) prior to climax stage are called *associes*. Groups of associations characterised by having dominant species of essentially the same growth form are included in a *formation*. For example, tropical deciduous forests and temperate evergreen conifer forests are formations. When animals are also considered, such large communities are called *biomes*.

In a way, the association is an abstract unit of vegetation in basic ecology as is species in taxonomy (Braun-Blanquet, 1932). Both are products of an artificial synthesis, a species of individuals and an association of stands. A number of individuals that have greater resemblance among themselves make a species. The dividing line between one set of individuals attributed to one species and the other set of individuals attributed to another species could be artificial, as is the dividing line between sets of stands allocated to different associations. However, the individual organisms of a species are genetically related populations, whereas the individual stands that are grouped into an association are not genetically related.

10.2 NATURE OF PLANT COMMUNITY

How natural the associations are as unit of vegetation or how much they owe their existence to the way the observers see them and classify? This has been one of the most controversial subjects in plant ecology until recently, and two contrasting views, as under, are often given. It may be emphasised that the two major theories described below pertain to communities that make vegetation. In this organisms other than plants are not considered.

Community unit concept

According to this view (also called 'discrete view of community'), in a given region, wherever a habitat repeats itself the same group of associated species occurs. It is to be emphasized that the consistency in the species composition is in relative terms. This view emphasizes that each community is an integrated unit with a definite boundary. Each discrete community consists of

coevolved species (with similar distributions along an environmental gradient). The pattern of species distributions along an environmental gradient as predicted by this view is depicted in Fig. 10.1. The species occupying similar range on environmental gradient (horizontal axis) with similar pattern of rise and fall of importance value (vertical axis) make a discrete community or a particular association. Boundaries of adjacent associations have little overlap forming sharp or diffused ecotones.

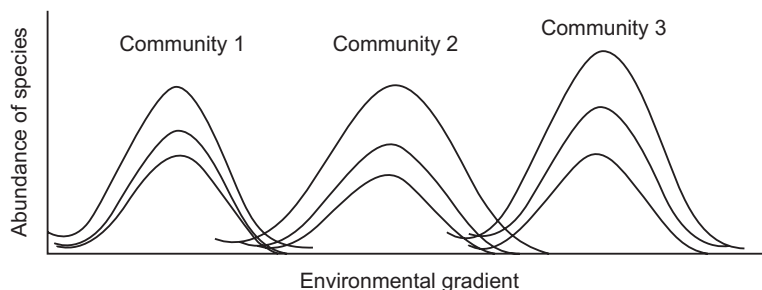


Fig. 10.1. Pattern of species distribution along environmental gradient as predicted by the discrete view of association.

F. E. Clements (1916, 1920) was the pioneer contributor to this view; he compared the community with organisms. According to the organismic concept, the community is a sort of super-organism or an organic entity; it arises, grows, matures and dies. Just as tissues have certain characteristics and functions above and beyond those of the cells that comprise them, so the community has characteristics and functions above and beyond the various populations it is constituted of. According to Clements, climax community is able to reproduce itself, repeating with essential fidelity, the stages of its development, and is the basic, integrated unit of vegetation.

Continuum concept

In contrast to the community unit theory, a different view was advanced independently in Russia in 1924 by L.G. Ramensky and in the United States in 1926 by H. A. Gleason as *individualistic hypothesis*. From this was born the *continuum view* in early 1950s largely as a result of researches undertaken by ecologists at Wisconsin University under the leadership of John T. Curtis and independently by Robert Whittaker at the University of Illinois, USA. According to Gleason (1926), each species is distributed in its own way, according to its own genetic, physiological and life cycle characteristics and its way of relating to both physical environment and interactions with other species; hence no two species are alike in distribution (Fig. 10.2). The Species do not form groupings that characterize discrete communities. In other words, distribution and abundance of a species population are independent of that of the other species populations, or dependence is loose. The vegetation of an area is regarded as a product of two factors, the fluctuating and fortuitous immigration of plants and an equally fluctuating and variable environment, with a gradual and continuous change in species composition along an environmental gradient. In other words, evolution of species in relation to one another does not produce groupings of species with similar distributions as assumed by the discrete view.

Curtis and McIntosh (1951) and Whittaker (1951, 1956) were among the first to study the actual distribution of species along environmental gradients. These researchers showed that vegetation is a continuum with a changing species composition along environmental gradients. Thus according to the continuum concept, within broadly defined habitats such as forest, grassland, and estuary, species populations replace one another continuously along gradients of physical conditions. The concept has been applied by numerous workers to almost all vegetation types of

the world. One such representation is given in Fig. 10.2. The density values of tree species were measured in a number of forest stands in the central Himalaya and then the stands were arranged along an axis of decreasing soil moisture. The density values of various species were plotted against the soil moisture values. As shown in Fig. 10.2, densities of any two species do not increase, attain peak and decline simultaneously along the axis of decreasing soil moisture. Consequently, at no point in the environmental gradient a group of species is clearly separated from other groups of species in distribution. On the other hand, segments of vegetation (the stands) continuously grade into one another in terms of species composition; the species are not really divisible into discrete units or associations. However, for the sake of convenience we can arbitrarily divide the stands on the basis of dominant species into a number of vegetation segments. In fact, Fig. 10.2 represents a simultaneous portrayal of all species response curves along an environmental gradient, in other words, a coenocline. A coenocline represents the distributions of all species as a function of environmental gradient, as well as a sequence of communities along an environmental gradient.

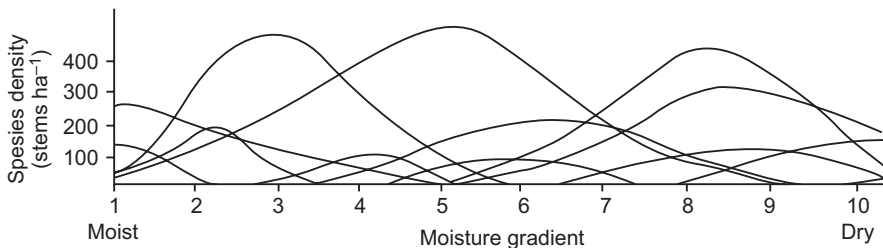


Fig. 10.2. A representation of distribution of tree species populations along moisture gradient according to continuum view. Species population curves (generally unimodal) do not have similar pattern of rise and fall, and do not form sharp boundaries along the environmental gradient.

According to the individualistic continuum theory, the species rather than the community is the basic unit of vegetation. It may be pointed out that Gleason's concept also recognises interactions among species. According to McIntosh (1985), the individualistic concept of Gleason/Ramensky, the continuum of Curtis and gradient of Whittaker are the basis of empirical and theoretical approaches to the plant community and vegetation science.

Synthetic approach

While the discrete community concept follows 'holistic approach', the individualistic view follows "reductionist approach". The discrete community concept is more applicable at a local scale and the continuum concept at a regional scale or in areas where physical environment changes gradually or sharply, such as altitudinal changes in a mountain. It is important to note that an integration of the abiotic complex with the discrete biological community gives rise to ecosystem; thus the ecosystem concept can be said to be rooted in the discrete community view.

In recent years, it is being increasingly realized that the classical association unit hypothesis and the continuum individualistic hypothesis are inadequate for explaining all vegetation patterns. Both these hypotheses assume that species have unimodal bell-shaped response curves in accordance with their realised niches. Bimodal or skewed response curves are, however, as frequent in nature as symmetrical bell-shaped response curves. Evidently, nature is highly complex, warranting more than one model to account for diverse situations.

The modern synthesis of the two classical views emphasises that species interdependence does occur, but not necessarily among all species of a community. Positive interaction between species pairs are good examples of species interdependence. They may include species with natural

root grafts, nurse or facilitative effect of one species on another, and different forms of mutualistic relationships. Keystone species (more about this in a later section) to which are associated many species, is another form of species interdependence in a community. Keystone species are not the dominant species, their abundances are lower, but their organizational control over community is disproportionately higher than their proportions in total density or cover. The positive interactions, keystone species and environmental impact on biodiversity and regeneration, all contribute to the species distribution patterns. Several hypotheses have been proposed that explain some communities, but not others (Barbour et al., 1999). They include factors as varied as competition and positive interactions or cooperation among species, episodic disturbance and chance recolonization that follows it, population dynamics and influence of selective herbivory and pathogens. Ricklefs (2008) opined that species distributions reflect the interaction of the adaptations of individual species in the ecogeographic setting of a region, including historical changes in the distribution of environments and dispersal barriers and corridors. According to Anderson et al. (2011), trait distances and mean phylogenetic distances between species interact in a multivariate context including both direct and indirect effects to detect patterns on gradients.

10.3 COMMUNITY STRUCTURE AND ATTRIBUTES

Whether associations are real or not, stands of vegetation are real. Within a stand organisms are interrelated to different degrees. For example, the partnership between a vascular plant and its mycorrhizae can be obligate. On the other extreme, we find no apparent relationship between epiphytes on the tree branches and the herbs rooted in the ground. But here too some inter-relationships may be noted. The epiphytes by utilising nutrients of stem flow may affect nutrient supply to ground flora, or leakages from epiphytes and sometimes nitrogen fixed by them may be available to the undergrowth. Structure refers to the organization of individuals that form a stand. Of the various components, growth form, stratification and coverage contribute most to the structure of a stand (Dansereau, 1957).

A community has both community-level and species-level attributes. The community structure and attributes could be divided as following:

(i) **Analytical characters** – characters which can be measured or observed directly in each stand, and (ii) **synthetic characters**, which can only be described on the basis of a number of stands and are usually derived from analytical characters. The analytical characters are further divided into two categories: (a) *qualitative characters* which are not easily quantified; and (b) *quantitative characters* which are easily quantified. The qualitative characters include: physiognomy (can be described at stand level), species composition (stand-level), stratification (stand-level), periodicity (species-level), vitality (species-level), life form and growth form (species-level but life form spectrum is shown at a stand level), sociability of individuals (species-level). Quantitative characters include; frequency, density, cover, biomass, etc., generally based on species level measurements. *Synthetic characters* include presence or constance and fidelity. In order to measure these characters we need to sample several stands of vegetation.

Barbour et al. (1999) have listed following characteristics of plant communities; physiognomy, species composition, species pattern, species diversity, change over time, productivity, nutrient cycling, and reaction of, and control over microenvironment. Of these, physiognomy can be described on the basis of vertical structure, life form pattern, cover and leaf area index (LAI) and phenology. Species composition emphasises such characteristics of species as cover, density and other forms of relative importance. Species pattern refers to spatial pattern and niche characters. Species diversity includes species richness, evenness and diversity, both within stands and between stands. Succession and responses to climatic change, and perhaps, evolution are considered in temporal changes. Productivity aspect includes biomass, annual net primary productivity, its efficiency and

allocation of net production to different plant parts. In nutrient cycling, focus is on nutrient demand, nutrient storage, and nutrient return from biomass to soil and nutrient retention and use efficiency.

Some of the parameters generally employed to describe communities are described below.

Physiognomy

Physiognomy depicts the external appearance of vegetation, and is a combination of its architectural (or biomass) structure and growth forms of its dominant species. The height of dominant species and the arrangement of the crowns and coverage of different layers differ across communities. For example, a multi-layered tropical rain forest with tall emergent trees is easily distinguishable from a simpler pine forest with one tree layer.

Since physiognomy results from the form of constituent species, one needs to characterise major kinds of form in plants, referred to as *growth form* or *life form*. Their characterisation is based on a number of features: height, woodiness, stem form, leaf form, leaf life span and deciduousness or evergreenness and others. There are a number of systems of growth forms, of which most used is that of Raunkiaer (1934), the Danish botanist, because of its simplicity. Raunkiaer considered only one main characteristic, the relation of perennating tissues to the ground surface, which reflects protection, particularly from low temperature, for dividing plants into five major life form categories (Fig. 10.3). Perennating tissues refer to buds that contain miniature twigs with leaves, and that burst and expand in spring or rainy season into new twigs, and seeds that germinate into new seedlings at the advent of favourable season.

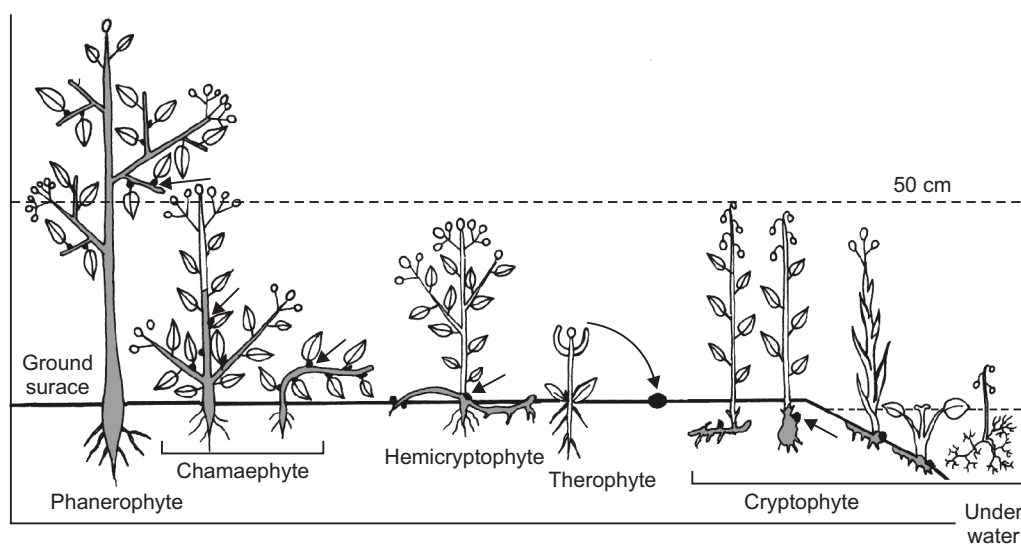


Fig. 10.3. Raunkiaer's (1934) growth form (also called life-form) categories of plants according to relative position of perennating parts in relation to ground. The lightly shaded parts persist and give rise to new growth the following year/season, but unshaded parts die back during unfavourable period of the year (cold and/or dry). Buds (darkened) are shown by arrow. In cryptophytes (also called geophytes), both terrestrial and aquatic species are shown (Derived from Raunkiaer, 1934).

Both types of perennating tissues remain inactive during unfavourable periods and thus enable the plants to survive during these periods. The location of perennating tissues on plants is the result of their evolutionary-adaptation response to climate in which they occur. In a severely cold climate or where conditions are dry during most of the year, fewer species would have buds far above the ground surface. The five major life-form categories are: phanerophytes, chamaephytes, hemikryptophytes, cryptophytes or geophytes, and therophytes. An elaborate scheme of division of these major categories is given by Muller-Dombois and Ellenberg (1974).

Some of the important features of the above life-forms are briefly summarised below.

Phanerophytes: The perennating buds are situated higher up on the aerial shoots. The plants are woody trees, tall shrubs and lianas. There may be evergreen phanerophytes (with or without bud scales) or deciduous phanerophytes (with bud scales). Phanerophytes are divided into following categories according to the height:

- (a) Nano-phanerophytes – trees less than two metres high or shrubs.
- (b) Macro-phanerophytes – trees more than 2 m high, further divided into:
 - (i) Mega-phanerophytes – over 30 m high.
 - (ii) Meso-phanerophytes – trees 8-30 m high
 - (iii) Micro-phanerophytes – trees 2-8 m high

Chamaephytes: The perennating buds are situated close to the ground. Three subtypes are recognised:

- (a) Suffruticose Chamaephytes – erect shoots die back at the onset of unfavourable condition and perennating buds occur on the lower portion of the stem (e.g., *Urtica dioica*).
- (b) Passive Chamaephytes – weakened erect shoots fall over the ground at the onset of unfavourable season and perennating buds arise along the horizontal stem at ground level (e.g., *Stellaria holostea*).
- (c) Active Chamaephytes – shoots are oriented along the ground such as creeping herbs (e.g., *Trifolium*), stoloniferous grasses (e.g., *Cynodon dactylon*) and trailing shrubs (e.g., *Cotoneaster*).
- (d) Cushion-Chamaephytes – further reduced and compact form of the active Chamaephytes (e.g., *Saxifraga*).

Carpet mosses, fruticose lichen, bog mosses and leaf succulents, (e.g., *Sedum*) are also included in Chamaephytes.

Hemicryptophytes: The perennating buds are located on the ground surface protected by the soil or dead plant parts; shoots die back at the onset of unfavourable conditions. The plants are usually biennial and perennial herbs, and also include mat forming algae, crustose lichens and thalloid bryophytes. Tussock forming plants (buds protected by old leaf sheaths, e.g., *Festuca*, *Poa*, *Dianthus*) and rosette plants (leaves arranged in a rosette protecting the buds and the aerial shoot leafless, e.g., *Viola*, *Primula*) are common examples.

Cryptophytes or geophytes: The perennating buds are located below the ground level or submerged in water. Three subtypes are recognised:

- (a) Geophytes – Terrestrial plants with perennating buds situated on rhizomes, corms bulbs or tubers. They also include fungus geophytes (e.g., tuberous fungi) and root parasites (e.g., *Orobancha*).
- (b) Helophytes – Perennating organs in soil or in mud below the water level and aerial shoots above the water level, also called amphibious plants (e.g., *Typha*).
- (c) Hydrophytes – Perennating buds under water, leaves submerged or floating. Include all other hydrophytes (except phytoplankton), such as floating plants.

Therophytes: The plants complete their life cycle (seed germination to seed maturation) within the favourable season of the year and remain dormant during the unfavourable season in the form of seeds. Apart from the seed plants, this category includes slime moulds and surface algae, annual liverworts and mosses and annual pteridophytes.

Other important life-forms are **phytoplankton**, the floating microscopic plants; **phytoedaphon**, the microscopic soil flora; **endophytes**, the microscopic parasites and those penetrating rocks e.g., *Phacidia lichenum*; and **epiphytes**, the vascular plants occurring on live plants.

The conversion of number of species in various life forms into per cent of the total species in a community or geographical region forms the **life-form spectrum or biological spectrum** (Table 10.1). The preponderance of a particular life form in the spectrum characterizes the plant climate of that region. Raunkiaer systematically selected 1000 species from the flora of the world and prepared a normal spectrum (Table 10.1). Against this normal spectrum, the biological spectra of different regions can be compared and plant climate and flora characterized. For example, if percentage of therophytes in an area is greater than that in the normal spectrum, the vegetation of that area will be called therophytic. Some tropical rain forests are almost exclusively phanerophytic. In humid climates the predominance of phanerophytes in terms of number of species continues up to warm temperate forest. However, in cold temperate forests, the percentage of phanerophytes is much lower and hemicryptophytes account for the greatest proportion. Decline in the percentage of phanerophytes is also evident with decreasing moisture level. The chamaephytes attain their maximum expression in semi-deserts, therophytes in deserts, hemicryptophytes in dry grassland, tundra and cold-temperate forest, and geophytes in mid-temperate mesophytic forest. In the entire world flora the preponderance of phanerophytes is evident; it is followed by hemicryptophytes.

Table 10.1. Biological spectrum for some regions in India. M = Megaphanerophytes, N = Nanophanerophytes, CH = Chamaephytes, H = Hemicryptophytes, G = Geophytes, HH = Hydrophytes, TH = Therophytes, L = Lianas, P = Parasites, E = Epiphytes.

Regions	% of total species										Phytoclimate
	M	N	CH	H	G	HH	TH	L	P	E	
Normal Spectrum	28.0	15.0	9.0	26.0	4.0	2.0	13.0	—	—	3.0	Therophytic
Alwar, Rajasthan (Desert)	5.1	14.7	4.3	7.6	3.4*	—	51.7	8.2	—	1.2	
Ladakh region (cold mountain desert)	2.6	4.8	26.8	9.6	21.3	0.6	33.1	—	—	—	Geo-chamae-therophytic
Kumaun Himalaya (broad elevational range)	27.3**	—	3.9	33.1	7.8	—	24.9	—	—	—	Thero-geo-hemicryptophytic
Andhra region	33.0	18.0	12.0	14.0	1.0	3.0	16.0	10.0	—	—	Phanerophytic

* Including hydrophytes. ** Including nanophanerophytes and lianas. — Absent or not considered.

Related to growth forms is the term plant *functional types*, which in recent years has been frequently used to address ecological questions at the scales of communities, ecosystems and landscapes. In this a large number of species that are similar in functional traits are placed in one group. Plant traits that have strong predictive power of ecosystem responses to environmental changes, such as climate change are determined (Cornelissen et al., 2003). However, the traits used should be easy and quick to measure for a large number of species (soft traits), and they should be correlated to traits, expected to be effective indicators of plants functions (hard traits) responsible for responses or effects at community and landscape levels, but difficult to measure (Hodgson et al., 1999; Lavorel and Garnier, 2002). For example, seed size and seed shape which are easy to quantify (soft traits) are good predictor of seed persistence in soil seed bank (hard character). The species having small and rounded seeds would have advantage over other species in the event of a major disturbance as their seeds would stay longer in soil than species having seeds different in size and shape. Some of the common soft characters which could be used as indicators to environmental changes, such as global warming, CO₂ fertilization, disturbance, soil nutrients enrichment could

be leaf span, leaf size, plant height, rooting depth etc. Plants functional traits have been shown to have several additional applications (Cornelissen et al., 2003) as following : (i) devising functional plants classification at regional and global level, (ii) providing input for dynamic global vegetation models, and large scale models for carbon, nutrients and water budgets; (iii) providing tools for understanding impacts of environmental changes; (iv) providing a basis for testing prediction about plant effects on ecosystem functions; (v) providing data for local to regional ecosystem change, and (vi) testing large scale climate-plant relationships.

Vertical structure

Since a community (even as simple as a grassland) normally comprises species of more than one life-form or growth-form, more than one vertical layer or stratum are formed. Generally, a forest has several layered or multi-storeyed physiognomy, for example, a tropical rain forest may have 5 or more layers, the topmost layer of emergent trees, two subsequent layers of smaller trees, a layer of shrubs or treelets and a layer of herbs. On the other hand, chir pine (*Pinus roxburghii*) forest is relatively simple, mostly with only one tree layer (Figs. 10.4 a and b). Each layer or stratum of the community may be composed of different species. Sometimes layers are also termed **synusia**; for example herb synusia and shrub synusia.

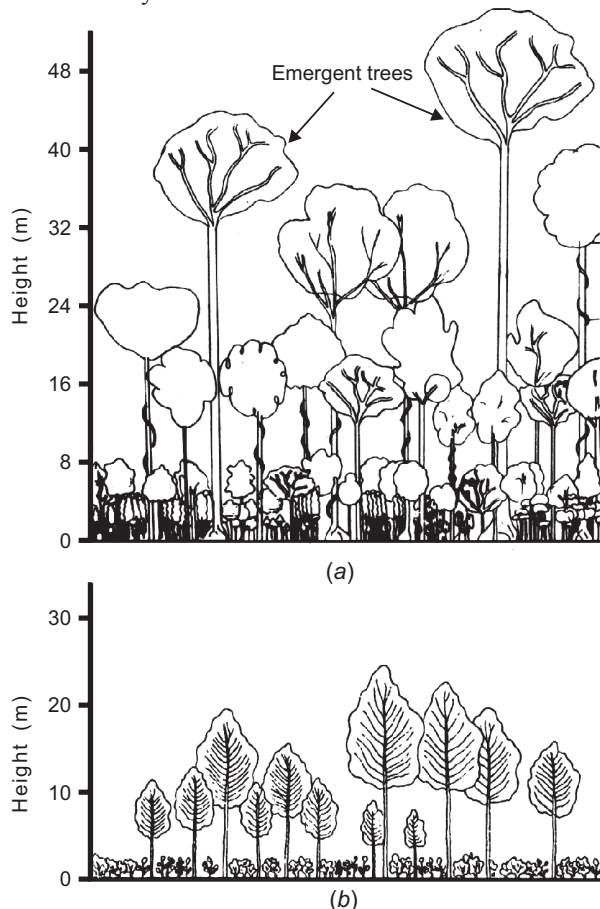


Fig. 10.4. Profile diagram of (a) a tropical rain forest of silent valley of India (Singh et al., 1984), and (b) chir pine (*Pinus roxburghii*) forest of central Himalaya. While former is multi-stratal, the latter predominantly consists of a top tree layer of *Pinus roxburghii* (Singh and Singh, 1982). In the tropical forest, emergent trees above the layer of continuous canopy give a bumpy look.

Only the canopy layer of a forest receives full sunlight with less than half of the full sunlight reaching the lower tree stratum (consisting of younger tree individuals of canopy layer and mature ones of lower tree stratum), and not more than 10 per cent of the full sunlight normally is able to penetrate these two strata of trees (Fig. 10.5). This light of low intensity (up to less than 1% of full light in a dense forest) with spectral composition, significantly different from the original sunlight is available to the herbs, growing below the shrub layer. The trees in forest are in an advantageous position as their leaves receive full sunlight. But in order to receive full sunlight they spend much of the energy of photosynthesis in growing the massive woody stem and woody branches which raise the foliage to heights where full sunlight is available. The forest herbs on the other hand are at a disadvantage as they receive low light intensity, but then they have not to spend much energy on developing supporting tissues. Along the various strata, animals also occupy different positions. For example, there would be a group of bird species feeding or nesting in tree canopy, and other groups doing the same activities on the ground. Insects are often differentiated between the crown and litter layers. Vertical stratification in response to light intensity can be observed even in phytoplankton communities of water-bodies. For example, in oceans the green algae are concentrated within shallow parts, certain brown algae occur below them and many red algae spread up to the deepest part.

Leaf area index: LAI is the ratio of total leaf area to ground area, and is a useful expression of physiognomy. It also reflects the degree to which a community is utilizing the incident light.

$$\text{LAI} = \frac{\text{Total leaf area, one surface only}}{\text{Ground area}}$$

Generally, LAI is 10-11 for tropical rain forests, which are several layered, 5-8 for deciduous forests and grasslands (having relatively fewer layers) and 1-2 for sparse vegetation of tundra and semi-arid deserts.

Horizontal Pattern: Sociability and Species Association

Sociability expresses the relation of individuals to each other. Individuals of species populations make definite patterns of distribution on the ground, clustered (or clumped or contagious), random or regular, or a combination of these. The clustered distribution shows that presence of one individual leads to the occurrence of several more individuals. In a random distribution the location of each of the individuals is determined by factors independent of those determining the location of other individuals. In regular distribution individuals are regularly spaced, perhaps showing repulsion from one another. The regular distribution is rare; random is uncommon and contagious the most common. The reasons of the contagious distribution being common in plants are: (1) clustering of individuals around their parents because of vegetative reproduction or poor seed dispersal, and

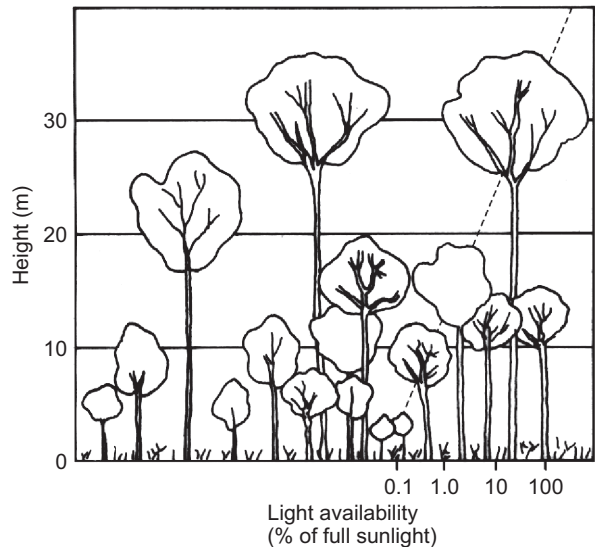


Fig. 10.5. Vertical stratification and light availability at different layers of vegetation. Canopy of trees receive 100% of full sunlight while herbs receive less than 1% of full sunlight.

(2) patchy dispersal of disseminules or patchy germination and survival resulting from variation in microenvironment. Regular distribution may occasionally occur because of (i) competitive elimination, (ii) production of toxins by older plants not allowing other individuals to establish within a sphere, and (iii) thinning of seedlings by damping off fungi (Daubenmire, 1968).

As in plants, contagious distribution is common also among animals. Social insects, such as termites, ants, some bees and wasps live in large colonies, which are so tightly integrated that they seem to function as single organism, or superorganism.

How are the distributions of different species related to one another? Whittaker (1975) points out that in natural communities associations or correlations between species for most part are not strong, and many pairs of species may show none. However, some species tend to occur together (positive association), and some tend to occur separately (negative association). Positive associations may imply that one of the two species depends on the other or both depend on each other, or they prefer similar environmental conditions. For example, epiphytic ferns that grow on the bark of a tree trunk of a particular species will show positive correlation with that tree species. An epiphytic moss and an epiphytic fern showing preference for the trunk of the same tree species and same shade intensity will show positive association among themselves. On the other hand, negative associations occur when species populations exclude each other. For example, leaves of one species may release chemicals that inhibit the growth of another species or the two species prefer contrasting environmental conditions. In grassland at Varanasi, Singh (1969) found that *Murdannia malabarica* and *Fimbristylis schoenoides* were positively associated with each other in a less disturbed site but they were distributed independently of each other on overgrazed sites. The two species seem to have similar habitat requirements but overgrazing encourages one species and discourages the other counteracting their natural tendency for positive association.

Periodicity and related aspects

Three terms, viz., periodicity, phenology and aspection, are used in reference to temporal differentiation. *Periodicity* refers to the regular seasonal occurrence of various activities, such as new leaf formation, flowering, fruiting and seed dispersal. Though determined genetically, these activities are set in motion by changes in temperature, light and water supply. *Phenology* refers to the study of biological periodicity in relation to the seasonal sequence within a year or so (Daubenmire, 1968). *Aspection* is the appearance or aspect of the community as a whole in different seasons: prevernal (early spring), vernal (spring), aestival (summer), serotinal (autumn), and hiemal or hibernal (winter).

Species of a community differ in their temporal activities within the annual cycle. In deciduous forests, a group of light-demanding herb species develop their foliage and flower early in the spring when trees are naked and canopy open. Their pollination is carried out by the insects present at that time. There are other herb species which show maximum leafing and flowering when the tree canopy is close and temperature high. A few herbs may be in bloom in autumn. Then different flower colours predominate in different seasons, and indications are there that some of the colour differences represent adaptations to attract different pollinating insects, birds and animals. In temperate forests, a sizeable portion of herb species in spring are geophytes, whereas most of summer species are hemicryptophytes. Thus a community may show different physiognomy in respect to seasonal changes.

In natural community, organisms also exhibit well-organised daily periodicities. In a forest one group of insects is active in day time, another group at night, and a third group may show maximum activity in twilight, i.e., during transitions of morning and evening, when solar light is dim. Plankton may sink or swim downward to a certain depth in day time. In the poor light in evening they swim upward and spend the night near the water surface.

Species composition

Communities, even with similar physiognomy, differ considerably in species composition. For example, Chaparral vegetation (a kind of fire resistant scrub) of California differs considerably in species composition from that of Chile, although other community characteristics are similar in them. Thus, the floristic composition of a community or of an area (listing of the species present) and importance (abundance or cover, basal area, etc.) of various species are required to compare the different communities. All species have some indicator value, though it may not be known. Species that are distinctly more important than other species in a community are called dominant. In a forest, the canopy species is considered dominant although an undercanopy species may be present in a greater number, because it is the individuals of canopy species which influence the solar radiation, precipitation, etc., available to undercanopy species. The removal of canopy species will significantly disturb the community environment.

Niche

The position of species in a community in relation to other species is termed its niche and the community may be defined as a system of interacting niches. The position is defined in relation to space and time and functional role (see Table 10.2 for the major contributions to the concept).

Table 10.2. Development of the concept of Ecological Niche.

Charles Elton	1927	The place of species in the biological environment, its relationship to food and predators.
G.F. Gause	1934	The intensity of competition between species suggests the degree to which their niches overlapped.
David Lack	1947	Niche relationships could provide a basis for evolutionary diversification of species.
G.E. Hutchinson	1959	The activity range of each species along every dimension of the environment.

This term niche tells us how a species population is specialized within a community. A similar kind of specialization we see in human societies. Individuals in society gather professional specialization in various trades, so that they are not in direct competition with others. In a township not many members of the society can survive long, if they happen to have adopted the same trade, for example, carpentry. The less efficient carpenters will not get sufficient income and will either specialize in other trade or leave the township.

Whittaker (1975) applied the Principles of Gause to communities in the form of following three statements: (i) in a stable community if two species occupy the same niche one is predicted to become extinct; (ii) no two species limited by same resources are direct competitors in a stable community, the species differ in ways that reduce competition between them; (iii) the community is an assemblage of interacting, niche differentiated species populations that complement one another, rather than directly competing. Niche complementarity has been touted as the principal cause of the positive species diversity-productivity relationships found in several experiments (e.g., Hector et al., 1999; Tilman, 1996). Here niche complementarity suggests that “a community of species whose niches complement one another is more efficient in its use of resources than an equivalent set of monocultures” (Naeem, 2002). To cite an example, when early- and late- season species and shallow and deep-rooting species are grown in mixture they would give higher productivity and would be more stable than monocultures of those species (Loreau et al., 2001; Tilman, 1999b).

Diversification of niches among the species in a community gives relative advantages to them that include reliable availability of the required resource, freeing them from competition for these resources by other species. Fig. 10.6 represents the selective processes for a pair of bird species

with the same niche characteristics. In this example, populations of two insect (prey) eating bird species (A and B) overlap broadly in their mean food size – a niche characteristic. Individuals of each species have genetically determined mean prey-size preference. When individuals of both the species are in competition, their survival would be lower for prey-sizes that are common to both the species than for the prey-sizes exclusively taken by either A or B. As a consequence, selection against genes determining common prey-size may occur resulting in a divergence of the niche with regard to the choice of prey-size.

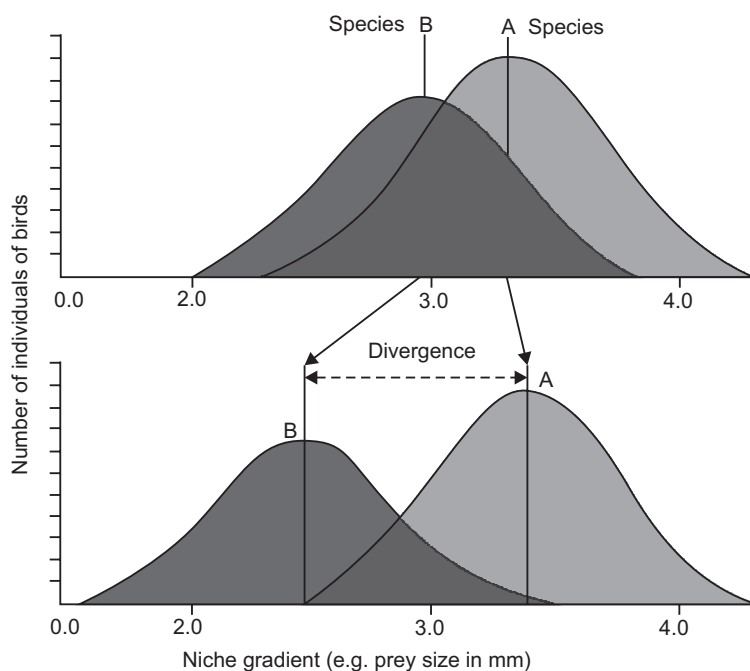


Fig. 10.6. Selective niche divergence of two bird species. Above, the two bird species show broad overlap in niche characteristic, represented by the mean prey-size choice by individuals of birds. As a consequence of competition survival of individuals choosing prey sizes around 3 mm is difficult, and hence selection is against genes determining such prey choice. Below, mean prey-size adaptations have diverged, as indicated by the arrow, and by reduced overlap.

Representation of niche differentiation of several species in a community is also possible. Whittaker and Niering (1965) have made such an attempt for Sonoran semi-desert of the lower mountain slopes in south-eastern Arizona. They identified three niche characteristics (Fig. 10.7): (i) the first axis involved relation of species to vertical height– the heights at which plant species exhibited the maximum foliage. It ranged from ground level in some herbs to 0.1-0.5 m in majority of semi-shrubs (*Encelia farinosa*, *Farnseria deltoidea*) to 0.5-2.0 m in arborescent shrubs, ocotillo (*Fouquieria splendens*), palo verde (*Cercidium microphyllum*), mesquite (*Prosopis juliflora*), to giant cactus, saguaro (*Carnegiea gigantea*) with the photosynthetic stems up to 6-9 m. (ii) The second axis represented seasonal time. Two groups of plants corresponding to two rainy seasons – one in summer and the other in winter, separated by a dry season – were differentiated. Although a majority of perennials utilized moisture of both rainy seasons, the patterns of leaf and stem photosynthesis were different for them. The plants can also be arranged along a gradient representing the longevity of leaves or any photosynthetic tissues: plants with persistent evergreen leaves, semi-deciduous plants, deciduous plants and cacti that lack leaves. Thus this gradient is of decreasing leaf persistence and increasing stem and branch photosynthesis. (iii) The third axis represented a horizontal pattern. The

extremes were represented by the species living underneath the large shrubs (primarily shaded) and those occupying open spaces between shrubs (primarily exposed).

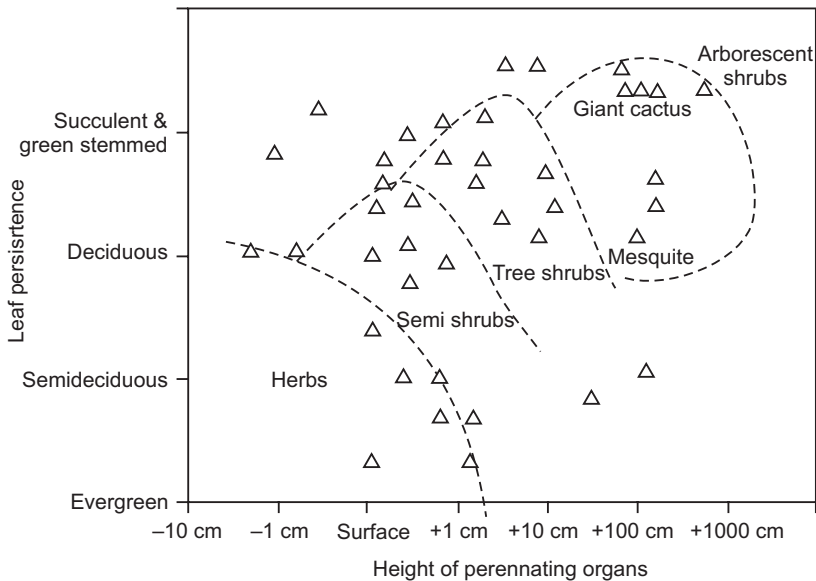


Fig. 10.7. Characterisation of the niches of several plant species in a Sonoran semidesert, Arizona. The plant species are plotted by the leaf persistence (from evergreen to leafless succulents) on the vertical axis, and plant height in relation to ground surface (on a logarithmic scale) on the horizontal axis. The plant species niches could be characterised also by their horizontal position and root depth. Some species primarily occur underneath the large shrubs, other species mostly in the open spaces between shrubs. (derived from Whittaker and Niering, 1965)

Singh et al., (1983) proposed that C_3 and C_4 grasses in the American mixed grass prairie formed two separate guilds. The former grow maximally in late spring to early summer, when temperatures are low and water abundant, the latter in early to late summer when temperatures are high and water scarce. Thus the two guilds partition the environmental resources in time.

Niche space

Obviously, species niches cannot be fully represented by considering their relations to single resource gradients. Considering that the niche can be defined by a number of variables of environment within the community, to which species must be adapted, Hutchinson (1957) proposed to measure various dimensions of niches of species. The variables can be biological (e.g., food size of birds or animals) as well as non-biological (e.g., aboveground vertical heights and seasonal times at which birds choose particular size of food). In this way the niche would be multidimensional. The variables defined by various axes on abstract multidimensional space are called niche space. In Fig. 10.8 niche response surfaces of two major macrophytic species of Lake Nainital are shown. The contour lines indicate their biomass in relation to water depth and the month of year. In this simplified representation species biomass forms a hill-shaped pattern, with values declining in all directions away from the peak. Though overlap in niche response surfaces of the two macrophytes is substantial, *Potamogeton pectinatus* realises its peak biomass at a relatively shallower site (S_4) and earlier in the season (April–May), than does *Hydrilla verticillata* (S_5 and September–October).

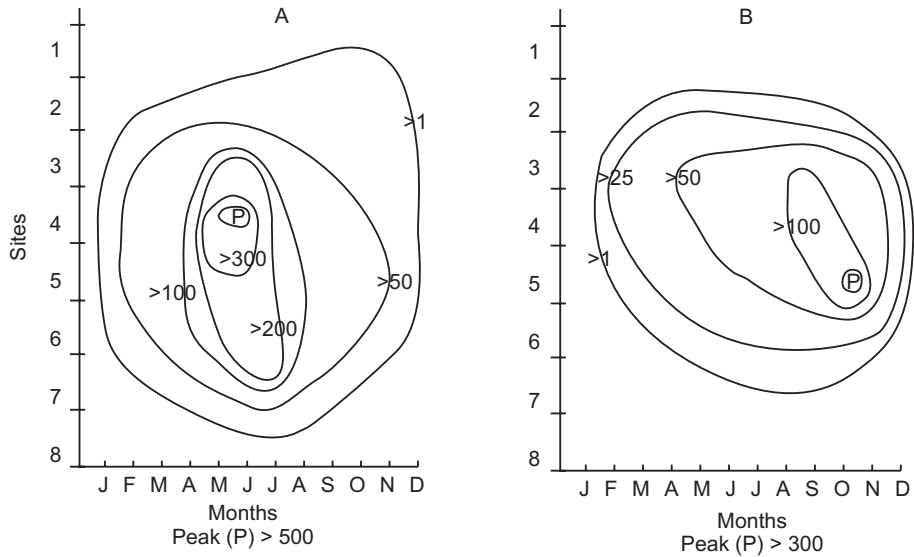


Fig. 10.8. Niche response surfaces of macrophytes of Lake Nainital (Kumaun Himalaya) in relation to two environmental gradients. The macrophytes are two submerged species - *Potamogeton pectinatus* (A) and *Hydrilla verticillata* (B). Species biomass (gm^{-2}) is outlined by contour lines in relation to time of a year and water depth, Site 1 being the shallowest and Site 8 the deepest (derived from Purohit and Singh, 1987).

Fundamental and Realised Niche

If the upper and lower limits of all the variables to which a species is adapted in the community were considered an infinite-dimensional space or hypervolume would result. For a community, the hypervolume would comprise all the hypervolumes of its constituent species (Fig. 10.9). The total range of environmental conditions that are suitable for the existence without any competition and predation from other species is called its *fundamental niche*. The part of the fundamental niche that a species actually occupies in a natural community is called its *realised niche*. Obviously, the realised niche of species is smaller than its fundamental niche because of competitive influence of the other species.

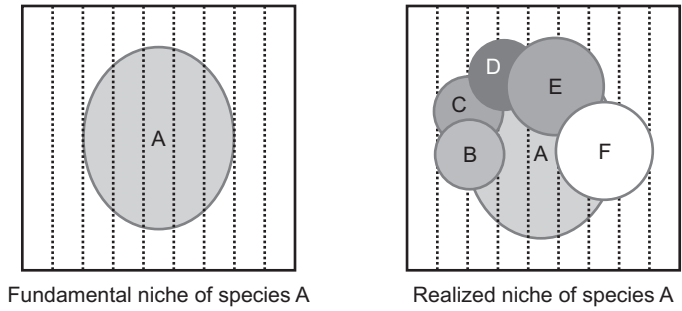


Fig. 10.9. Fundamental and realised niches of a species (A), shown as a hypervolume. The background dots represent environmental factors, such as temperature, moisture, nutrients and food, projected onto a plane. The fundamental niche of the species represents the full range of environmental conditions, both biological and physical, under which the species can successfully exist. Under the pressure of competitions, the species is displaced from part of its fundamental niche and is forced to that part of fundamental niche hypervolume to which it is most highly adapted. The portion that it occupies is its realised niche. Species B, C, D, E and F are similar species, influencing the form of realised niche of species A. Derived from the concepts given by Hutchinson (1957) who was first to formally quantify the niche concept in terms of geometric space.

However, the realised niche could turn out to be larger than the fundamental niche in presence of a species that facilitates its growth by creating beneficial conditions (see in Shouse, 2003). When competition by one or more species is removed, some portion of the niche is released.

Species Diversity

Most often species diversity is represented simply as the number of species in a community or in an area. **Species richness** can be presented as species density (number of species per unit area, calculated as $S/\ln(A)$) or number of species per individual, frequently calculated as $S/\sqrt{n}$, where A is area, n is total number of individuals and s is number of species.

Number of species is a function of area and increases with increasing area sampled; it is assumed that larger the area, greater is the resource to support a larger number of species. The rate at which the number of species increases with increasing area can be determined by species-area curve. Species-area curve can be used to compare different communities.

Arrhenius first formalised the species-area relationship as a power function (i.e., the number of species is proportional to a power of the area sampled):

$$S = cA^z$$

where S = number of species encountered; A = area; c and z are constants (coefficients) fitted to the data.

After log transformations, the species-area relationship can be written as:

$$\log S = c + z \log A$$

The species-area relationship for a dry tropical forest is depicted in Fig. 10.10 as an example. Both power-function plot (and calculated power-function equation) as well as log : log plot (when the logarithm of species number is plotted against the logarithm of the area) and the related equation are given in Fig. 10.10. The species richness of amphibians and reptiles in some tropical forest ecosystems plotted against area exemplify this relationship (MacArthur and Wilson, 1967) (Fig 10.11).

The value of the coefficient z generally ranges between 0.16 and 0.39. The species-area relationships for continental areas show an average z of 0.20 and those for islands 0.35.

The species-area relationships can also take an exponential form; $S = a + b \log A$, where a and b are coefficients.

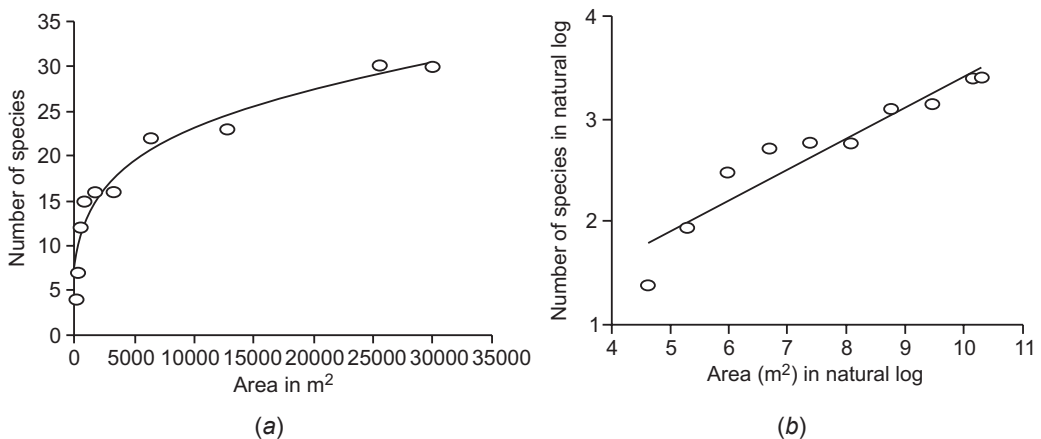


Fig. 10.10. Species-area curve for a northern dry tropical forest, India. (a) according to power function ($S = 1.50 A^{0.30}$), (b) according to log: log plot ($\ln S = 0.41 + 0.30 \ln A$), note that 'c' value is in natural log.

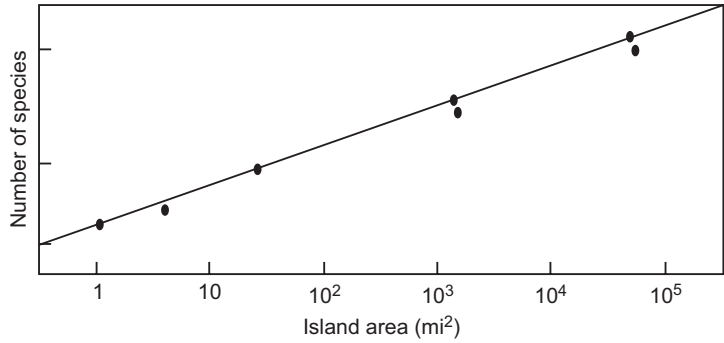


Fig. 10.11. The species area curve of amphibians and reptiles in West Indies (based on MacArthur and Wilson, 1967).

The coefficients of species-area curve, such as c , z or a and b , can also be taken as measures of species diversity, because while c and a represent the number of species in the smallest area sampled, z and b represent the slope of the curve, i.e., the rate at which number of species is increasing with the increase in area.

Number of species is also a function of the total number of individuals sampled because it is expected that sampling a greater number of individuals would yield a larger number of species. If number of species is plotted against the number of individuals a curve similar to species-area curve is obtained but it is referred to as **species accumulation curve**.

Species occurring within a sample differ in number of individuals. As shown in Fig 10.12, two hypothetical field plots, A and B, have the same floristic composition, same number of species and the same total number of individuals, but they differ in distribution of individuals among the species. In plot A, a majority of individuals are of one species, while in plot B individuals are more or less evenly distributed across the species. *Species evenness* or *species equitability* (E_c) is the distribution of individuals (or of any quantity representing importance value such as cover, basal area and biomass) among the species. This can be expressed as below:

$$E_c = s / (\log n_i - \log n_s)$$

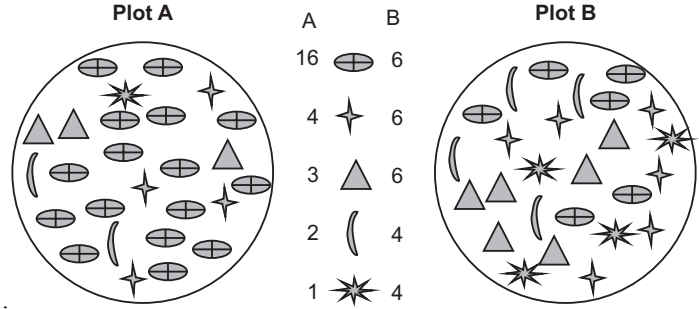


Fig. 10.12. A representation of two plots with same number of species (5 each, indicated by different symbols), but different numbers of individuals (different equitability) (numbers are indicated above against each species). Of the two plots, plot B has clearly more diversity ($H' = 1.59$ and $D = 0.80$) than plot A ($H' = 1.16$ and $D = 0.58$).

where, s is the total number of species in the sample, n_i the importance value (density, basal cover, canopy cover, biomass, etc.) of the most and n_s of the least important species.

Evenness (Ec_1) can also be expressed as the ratio of observed diversity to maximum diversity, thus :

$$Ec_1 = H' / H' \text{ max or } H' / \ln(S)$$

where, H' is the Shannon-Wiener diversity (see later) and $H' \text{ max}$ is the maximum value that H' can obtain, i.e., when there is one individual per species or $\ln(S)$, and S is the number of species.

Species diversity, a synthetic expression, is actually the product of both species richness and evenness; it is species richness weighted by species evenness. Several measures have been given to express it, of which Shannon-Wiener's index (Shannon and Weaver, 1949) and Simpson's index (Simpson, 1949) are among the most common. The Shannon-Wiener's index (derived from Information Theory and is actually a measure of uncertainty) can be written as:

$$H' = - \sum_{i=1}^s (pi) (\ln pi)$$

where H' is the index number, s represents total number of species, pi is proportion of all individuals that are of species i .

The negative sign is added to make H' come out as a positive value. H' , when natural log is used, could range between 0 and 7 but frequently it ranges between 1 and 4, species-rich complex forests have higher values than simpler vegetation.

Simpson's index of diversity is based on the concentration of dominance (λ) which is:

$$\lambda = \sum_{i=1}^s (pi)^2$$

where λ is the index number indicating concentration of dominance, and others are as in above formula. To calculate diversity (D) following equation is used:

$$D = 1 - \lambda$$

This index is an indirect measure of diversity and expresses how strongly the importance values or numbers of individuals are incorporated in the first or first two to three important species. D varies from 0 (single species) to 1. D can also be represented as $1/\lambda$, then 1 will not be the upper limit of D .

Often species richness, equitability and diversity are positively related to one another, and all the three are inversely related to dominance concentration (λ). A plant community has organisms of different sizes such as trees, shrubs, and herbs; therefore, diversity of different layers is presented separately. It is important to remember that such calculations of species diversity as above are subject to sample size, and therefore, comparisons with different studies can be made only if the sample size is the same.

Alpha, Beta, Gamma, Delta and Epsilon diversity

Alpha (α) diversity is within habitat or within community diversity. It is expressed as one or all of the following ways:

- (i) Number of species per unit area, i.e., species-area ratio, most commonly calculated as $S / \ln(A)$, where S is the number of species and A is area.
- (ii) Number of species per individual, calculated as $S / \sqrt{n}$ or $S-1 / \ln(n)$, where S is the number of species and n is total number of individuals belonging to all species.
- (iii) Shannon-Wiener or Simpson's index of diversity (explained earlier).

Beta (β) diversity (also called differentiating diversity) reflects between-habitat or between-community diversity. β diversity expresses the rate of change in number of species per unit change

in the habitat, or species turnover across an environmental gradient. Greater the heterogeneity in habitat conditions greater would be β diversity. It is commonly calculated as:

$$\beta_d = S_c / \bar{S}$$

where, S_c is the total number of species occurring in a set of samples (habitats or communities), and $\bar{S}$ is the average number of species per sample (habitat or community). For S_c each species is counted only once, whether or not it occurs more than once.

β diversity for the example in Fig. 10.13 is $11 / 4.5 = 2.44$.

Gamma (γ) *diversity* is the within-landscape diversity. It is a product of the above two kinds of diversity and can be calculated as:

$\gamma_d = \alpha \times \beta \times \text{Total number of habitats or communities}$, where α is the average value of α diversity, $\beta = \beta$ diversity of the landscape.

Several ecologists, however, represent gamma diversity by total number of species in the landscape or geographical region.

The term Delta Diversity is used for between-landscape diversity.

The term Epsilon Diversity is used for diversity within biomes (or biogeographic provinces).

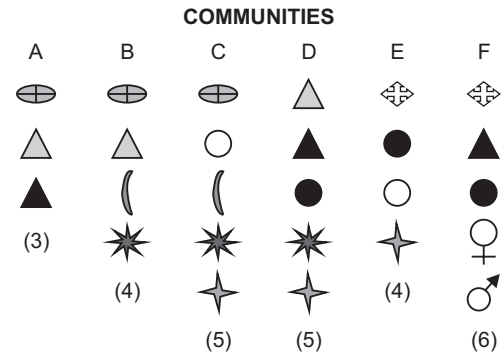


Fig. 10.13. A representation of *Beta* diversity of a landscape. A to F are six communities. Species are shown by different symbols. The mean number of species ($\bar{S}$) is 4.5, and total number of species (S_c) is 11. Thus *Beta* diversity is $(11/4.5 = 2.44)$.

Dominance-diversity curves:

We can present species abundance (or importance of species, such as relative density or relative cover) in a rank-abundance plot (Fig. 10.14). In this, abundance of each species of a given community is plotted on a logarithmic scale (vertical scale) locating species position (horizontal scale) in a decreasing order (first being the most abundant and last the least abundant species). Called *dominance-diversity* or *importance value* curve or rank-abundance model, it describes patterns of the way species of high importance, intermediate importance and rare species are assembled in a community. These curves are of three main types, and their theoretical interpretation is as follows:

- The curve given to *random niche – boundary hypothesis* (also called broken stick model) of MacArthur (1957), a mathematical ecologist, suggests that the number of rare, intermediate and common species is random. According to this model, all species and all individuals in a community are represented by a long stick broken into random pieces. In this a piece represents a species and its length number of individuals of that species. The total niche space is divided into non-overlapping, contiguous and random distributions. This broken-stick model is generally not found, only a few animals from narrowly defined, homogeneous communities, such as the nesting birds of a limited area may follow it.
- Geometric model** corresponding to the niche pre-emption hypothesis supposes that the most dominant species pre-empt most of the niche space, the next dominant species occupies the next largest niche space and so on. The resulting geometric curve is a steep straight line, as abundances of species declines in a geometric fashion. Such a distribution is approached by plant communities occupying harsh environments, such as deserts and of boreal forests with very low equitability.

- C. In the curve representing the *lognormal hypothesis* of Preston (1948), another mathematical ecologist, there would be a few species with high and low importance values and many with intermediate importance values. The curve assumes a shape intermediate between the curves given to random niche-boundary hypothesis and geometrical series. This type of curve is approached by many species-rich communities, and is said to be the best empirical description of species diversity (Barbour et al., 1999).

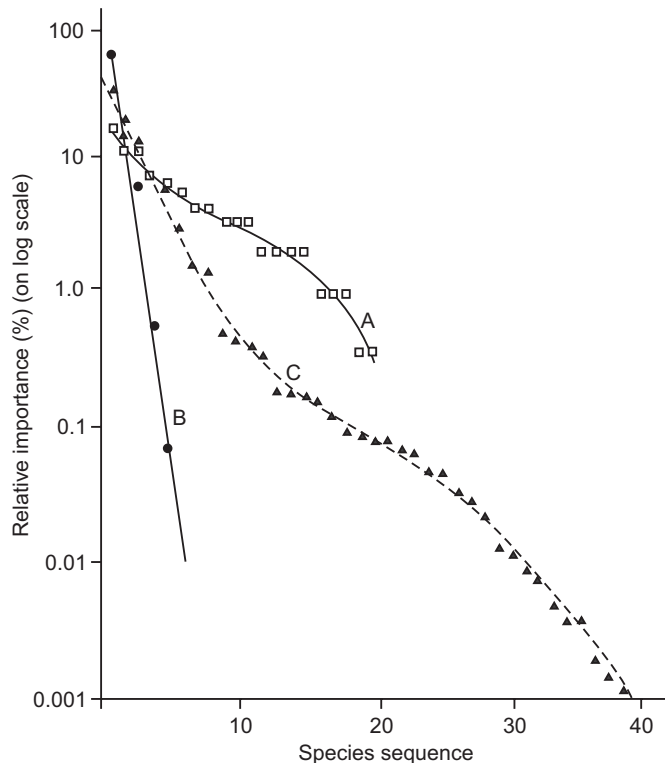


Fig. 10.14. Dominance-diversity or species importance value curves. Each species is represented by a point located by that species' relative importance (the percentage which that species represents of the total density or biomass or net production of all species in the community, on a logarithmic scale) on the vertical axis, and its position in the sequence of species from highest to lowest importance values, on the horizontal axis. A - Random niche-boundary hypothesis (densities of nesting bird pairs in a deciduous forest). B - Geometric series and niche pre-emption hypothesis (vascular plant species by net production in a sub-alpine fir forest occurring in an unfavourable environment). C - Lognormal distribution of a plant community of high species diversity occurring in a favourable environment, data are based on net productivity). (based on Whittaker, 1975).

These curves are suggested to characterize the diversity of community more effectively than synthetic or composite diversity measures, such as of Shannon-Wiener and Simpson indices.

Biomass, productivity and nutrient cycling

Communities greatly differ in biomass, allocation of biomass to different components, net primary productivity and photosynthetic efficiency. The mean standing biomass of rainforest (500 t ha⁻¹) is 5000 times that of desert communities (0.1 t ha⁻¹). A greater biomass means more leaf area to trap radiant energy to maintain growth. In Central Himalaya, biomass in an oak forest up to more than 700 t ha⁻¹ is far greater than that in a pine forest (generally up to 200 t ha⁻¹) though they

occur in similar altitudinal belts. The two forests, however, do not differ significantly in net primary productivity. The net primary productivity of cold desert communities on average is $3 \text{ kg ha}^{-1}\text{yr}^{-1}$, compared to $22000 \text{ kg ha}^{-1}\text{yr}^{-1}$ in tropical rain forests. Some communities of swamps and marshes can have net primary productivity as high as 60000 kg ha^{-1} . Across various vegetation types, the mean rooting depth varies from 40 to 150 cm, with global mean of about 102 cm. The values are lower ($<60 \text{ cm}$) for tundra, boreal forests, temperate meadows, and comparatively higher ($>140 \text{ cm}$) for Mediterranean shrublands and woodlands (Schenk and Jackson, 2002).

Of the total standing biomass, the proportion allocated to roots can vary significantly across communities, even for those occurring side-by-side. For example, in Central Himalaya, the roots account for 28% of biomass in an oak forest (*Q. leucotrichophora*), compared to 17% in a pine (*P. roxburghii*) forest (Singh and Singh, 1992). In many evergreen Himalayan forests, leaves account for more than 25% of NPP, compared to 15-25% of NPP in forests of other regions.

Communities are also characterized by nutrient cycling attributes. For example, nitrogen use efficiency (net primary production per unit of net nutrient uptake) among temperate and Central Himalayan forests varies widely from 105 in broad leaved evergreen forest of Central Himalaya to 281 in spruce forest of Rocky Mountains (see Zobel and Singh, 1997). The Central Himalayan oak forests are characterised by a low N retranslocation during leaf senescence, and massive return of N to soil through litter fall.

The mean-residence time for organic matter across various forest communities shows over 880-fold increase from tropical rain forest (0.4 yr) to boreal coniferous forest (353 yr) (Waring and Schlesinger, 1985). The percentage of soil organic carbon in top 20 cm is significantly affected by vegetation and ranges (relative to that in the first metre of soil) from 29% in cold arid shrublands to 57% in cold humid forest (Jobbergy and Jackson, 2002).

Stability

Communities are subject to disturbance or stress and vary in their ability to tolerate it. Stability is a complex term having more than one component: resistance, resilience and persistence. Persistence depends on the rate of immigration and disappearance of species. Communities in which the rate of immigration is equal to the rate of disappearance, and both the rates are very slow, are persistent. However, even if the two rates are equal but very fast, the species composition of the community will keep on changing and the community will not be persistent. Similarly, if the rate of species immigration is faster than the rate of species disappearance, the community will gain species, and if the rate of disappearance is faster than the rate of immigration, the community will lose species; under both of these conditions the community will not be persistent.

Resistance and resilience are inversely related with each other. *Resistance* shows the ability of a community to remain unchanged. Late successional species are considered more resistant than the early successional communities.

Resilience expresses the ability of communities to return to original state, following disturbance or stress. Early successional communities consisting of short-lived, r-strategy species and with low interdependence among species are considered more resilient than late successional communities having long-lived, slow-growing K-strategy species. For example, a grassland comprising r-strategy plants can easily burn (low resistance) but when burned can return to original state within a short time (high resilience) compared to a forest comprising K-strategy plants, difficult to burn (high resistance), but when burned will take a very long time to return to original state (low resilience). *Persistence* is the ability of communities to remain unchanged over time. The *Sequoia* forest along the northern coast of California is a good example, it may be less resistant and less resilient, but is persistent (Barbour et al., 1999).

Changes over time

Communities are dynamic entities, and changes in them are generally studied at three time scales: about 500 years and less referred to as succession; thousands of years referred to as climatic changes; and millions of years referred to as evolutionary changes.

With respect to succession, communities are divisible into successional or seral and climax types. Seral communities (sequence of such communities during succession is called sere) show rapid changes with time, some species declining in abundance and disappearing from the site, and others increasing in abundance. The changes may include shift from one vegetation type to another, such as from grassland to forest. A *climax* community may be in equilibrium with climate, and may show no directional change for centuries. Successional changes are primarily brought about by biotic interactions, such as competition and herbivory, and climate is assumed to be static. The earth climate changed in the past (though at a slower rate than the one expected in future due to the anthropogenic activities), and brought about striking vegetational changes. For example, temperature of full glacial eras on average was 4-6°C lower than the present temperature. At least four glaciations occurred in the Himalaya during the Quaternary alone. Between 8000 and 4500 years ago, as the climate warmed, oaks expanded and pine declined in Central Himalaya (Vishnu-Mittre, 1984). Changes in climate can be documented by undertaking dendrochronological studies (study of tree rings) of long-lived tree species. The effect of climate changes on vegetation can be studied by examining pollen accumulated at the bottom of slowly filling lakes or ponds.

Baker et al. (1996) have provided a picture of the Holocene paleoenvironment of northwest Iowa, with focus on changes in climate, communities and insect fauna (Table 10.3). Changes in rainfall seasonality seem to have played a significant role in vegetational changes. The vegetation changed from a boreal conifer forest (very cool and moist) to deciduous forest and to prairie within 9000 years with increasing summer drought.

It has been possible to study changes in vegetation over millions of years on the basis of macrofossils, such as leaf impression (pollen primarily used to study changes over thousands of years, represents microfossils) and throw light on evolutionary courses. Whittaker and Woodwell (1972) have suggested that it is unlikely that characteristics of communities, such as diversity and stability, are selected for as such. For example, during the course of evolution, species richness of a community increases by addition of species to communities, not by selection of species-rich community over species-poor community.

Table 10.3. Holocene changes in climate and biotic community in Roberts Greek, Basin north-eastern Iowa (derived from Baker et al., 1996).

Years before the present	Climate	Seasonal precipitation	Vegetation	Insect fauna
Historical period	No change	No change	Crops/pastures/ruderal species	Domination of exotic species
3500-3800	Slightly cooler	Increasing summer	Oak savanna	No change
5500- 3500	Warm, dry	Spring	Prairie	No change
9300-5500	Cool, moist	Summer, fall	Deciduous forest (<i>Quercus</i> spp., <i>Ulmus americana</i> , <i>Betula</i> , <i>Telia</i>) and some conifers.	Both forest and open ground species, warm- adapted
12500-9300	Very cool	Unknown	Conifer-hardwood forest, boreal spruce- larch forest (<i>Picea</i> spp., <i>Larix</i>)	Cool adapted

The species associations are generally loose and changeable, therefore, during the course of evolution species variously combine and recombine, and community evolution largely depends on evolution of species that make up communities.

Communities have certain emergent characters corresponding to those of species populations that make them up. That is why widely separated communities that share a similar environment, such as Chaparral in California, Chile, South America and the Mediterranean region, show convergence in their characters relating to physiognomy.

10.4 KEYSTONE SPECIES AND CONTROL OF COMMUNITY STRUCTURE

Keystone species

A keystone species is a species that has a very large effect on the community through both direct and indirect pathways (Paine, 1966, Power et al., 1996), even though its abundance or biomass in the community is low. Thus, 'A keystone species is a species that is of demonstrable importance for ecosystem function' (Cottee Jones and Whittaker, 2012). The keystone species have large ramifying effects on community structure. Single species or species guilds (functional groups of ecologically similar species) may affect the structure of communities in several ways (Ernest and Brown, 2001).

1. By altering the physical structure.
2. By affecting the prey and host populations as predators, parasites or pathogens.
3. By supplying essential services and resources as mutualists.
4. By affecting energy flow and material resources as producers and consumers.

Robert T. Paine (University of Washington) performed a predator removal study to determine the effect that the predator has on the abundance and diversity of species in the rock intertidal community in the North Pacific (Paine, 1966). The seastar (*Pisaster ochraceous*) feeds on its preferred prey, the California mussel (*Mytilus*), as well as on many other molluscs and crustaceans. The California mussel feeds on plankton in the water column, and attaches itself tightly to the rock with special threads. Many other species including four species of barnacles, a dozen species of seaweed, several limpets, snails compete with the mussels for living space on the rocks (Fig 10.15a-c). Using the removal experimental approach, Paine (1966) removed the seastar (*Pisaster*) from a large area on the shore. Where the predator was allowed to stay, it removed clumps of mussels from the rocks as it fed on them during high tide (Fig. 10.15a). Once the mussels were freed from their predator, their population increased, and they spread forming a continuous mat on rock in middle and lower intertidal zone. Wherever they spread over the rock surface, they eliminated nearly all the other species on that surface (Fig. 10.15c). Thus, preying upon mussels creates an opening for other rock dwelling species, increasing species diversity of these sessile organisms. From the results of these experiments Paine (1966) introduced the term "keystone species."

Since Paine's study, many other similar studies have been carried out to determine the effect of top predators on the species diversity of the community. Examples based on various studies, in which a keystone species is predicted to change a community composition upon removal, are given in Table 10.4. The removal of the keystone species causes at least 50% change of the assemblage considered in these studies (Mills et al., 1993). However, a 50% loss rule may not be an operational criteria for identifying keystone species in all the cases.

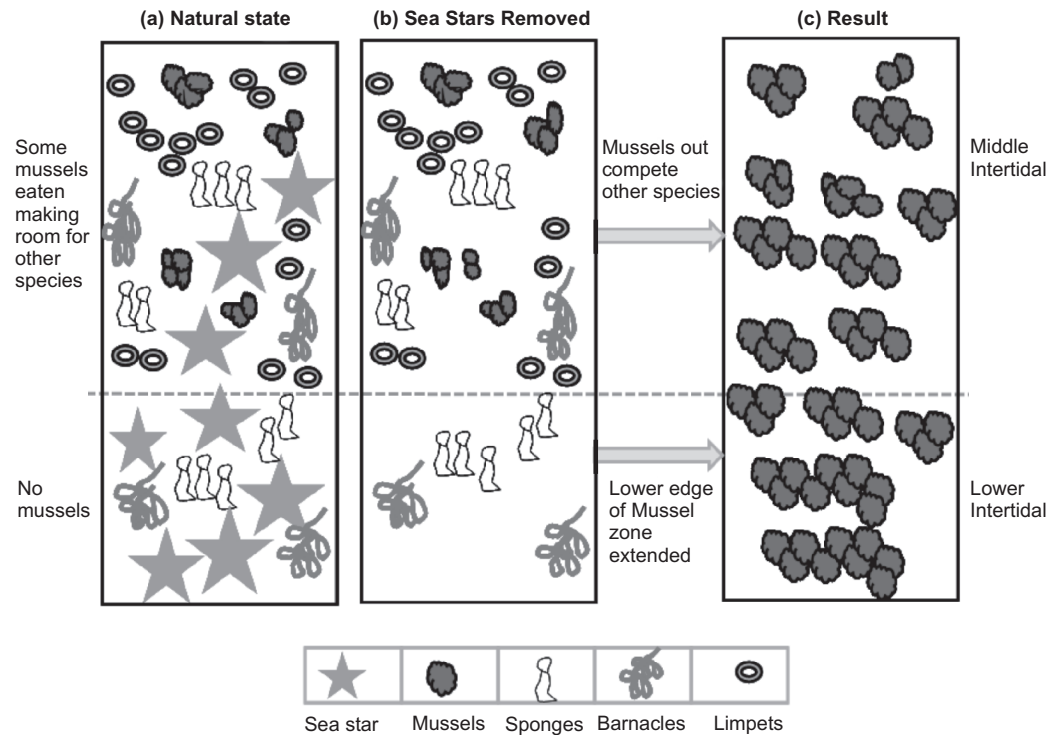


Fig. 10.15. The experimental removal of a keystone species, the predatory sea star *Pisaster ochraceus*, from a rocky intertidal zone community in western North America completely changes the structure of the community. *a.* in natural state some mussels are eaten by sea star making space for other species; *b.* the top predator, the sea star is removed; *c.* With their principal predator missing, mussels form dense beds by outcompeting and replacing other intertidal species and monopolizing all the space on the rocks. The mussel *Mytilus californianus* is an ecological dominant species that is able to compete for space successfully in the intertidal zone. (based on Paine, 1966).

Table 10.4. Predicted or observed effect of removing presumed keystones on community composition as reported in different studies (from Mills et al., 1993).

Presumed keystone (community type)	Effect of removal	Author
Starfish (rocky intertidal)	Observed reduction of system from 15 to 8 species	Paine (1966, 1969)
Sea urchins (subtidal)	Observed takeover of large fleshy algae, resulting in loss of approximately one-half of grazers	Fletcher (1980)
Palm nuts, figs, and nectar (tropics)	Predicted loss of one-half to three-quarters of total bird and mammal biomass	Terborgh (1986)
Herbivores more than 1000 kg (Pleistocene)	Hypothetical mechanism for loss of approximately half of mammalian genera during late Pleistocene	Owen-Smith (1987)

The keystone species could be divided into following categories:

Keystone carnivore predators

- The keystone predator alters the abundance of prey species which initiates a number of indirect effects. Insect-feeding birds often maintain insect diversity in forests by feeding on the most abundant species and hence preventing single-species outbreaks, which can be catastrophic.
- Sea otters in the North Pacific feed preferentially on sea urchins that feed voraciously on valuable kelp (seaweed) species that comprise the productive and diverse kelp beds of the Pacific Coast. The otters indirectly but greatly benefit the kelps and a highly diverse group of species that depend on the kelps for food and refuge (Fig 10.16). In the absence of otters, sea urchins would severely degrade the sea weed stands.
- Alligators in the Everglades feed on mid-sized preys (reptiles and mammals) as well as on small preys (mice, rats), but indirectly benefit the smaller prey by consuming the predators of these smaller preys. Alligators, by chance, also benefit a large number of water-dependent species by digging nest holes that contain water during dry seasons. This includes species that they prey on.
- Big predatory fish maintain fish diversity in rivers and lakes, by consuming abundant, easy to find species, and hence preventing dominant species from increasing to their shutout levels.
- The keystone top predators such as panthers, lions, tigers and leopards in terrestrial ecosystems have strongly influenced the communities by reducing competition among prey species.

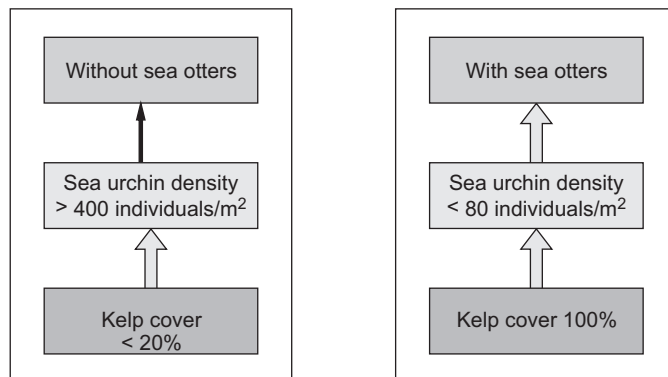


Fig. 10.16. Sea otters as keystone species in intertidal community in the North Pacific Ocean (derived from Estes and Palmisano, 1974).

Keystone seed predators

In the Chihuahuan desert system (United States) kangaroo rats are dominant seed consumers and compete with other smaller granivorous rodents. Through long-term removal experiments, the keystone guilds of the kangaroo rats have been shown to cause changes in energy flow and species composition of rodents and plants, so much so that desert shrublands were converted into grasslands. However, colonisation by a small exotic rat (*Chaetodipus baileyi*) has restored the flow of energy through the rodent community (Ernest and Brown, 2001).

Keystone mutualists

The keystone mutualist species maintain the diversity of organisms in a community by

controlling the abundance of other species through mutualistic interactions. For example, they serve as essential links in the recycling of nutrients or as major seed dispersers and pollinators. They are also known as mobile links as their foraging movements are critical to other species populations. Mobile link pollinators and seed dispersers are essential for the persistence of some plant populations. Once identified, keystone mutualists can be managed to ensure the survival of the species dependent on them. By increasing the abundance of the keystone mutualists, for instance, it would be possible to increase the populations of dependent species. Various species of bats, ants and humming birds serve as mobile links, which depend on specific hosts for their food. Flying squirrels and red-backed voles are the mammalian dispersers of the mycorrhizal fungi. Even microorganisms play a vital role. In various forest ecosystems, mycorrhizal fungi are essential for mineral mobilization and absorption. Mycorrhizal fungi carry out keystone function as a group.

The frugivore populations of primates, squirrels, rodents, and birds depend for their food on palm nuts, figs and nectar of flowers in the tropical forests. Certain species of tropical figs bear fruits during the season when no other fruit is available for frugivores. If figs are not available during periods of food scarcity many animals can die. The *Ficus* Linn. genus (Moraceae) is of central importance as a keystone resource provider in many tropical forests of south-east Asia and tropical South America. *Ficus* spp depend on obligate species-specific mutualistic wasps (Hymenoptera: Chalcidoidea: Agaonidae) for pollination. The associated round-the-year flowering makes them a keystone resource for forest fruit dispersers (see Borges, 1993). Malaysian forests are extremely rich in *Ficus* species, with about 16% of known world species. In Malaysian lowland rainforests, at least 60 species of birds and 17 species of mammals eat figs. Lambert and Marshal (1991) show that a guild of 29 *Ficus* species in a Malaysian dipterocarp rainforests act as a keystone resource provider for a wide range of frugivores. However, in forests of Western Ghats in India, *Ficus* trees are somewhat patchy in distribution, only act as keystone resource for highly mobile animals with large home range (Borges, 1993).

Keystone herbivores

Many species have been called keystone herbivores because their foraging causes drastic habitat modification. Based on the observation that large herbivores (more than 1000 kg) can readily convert closed thicket or forest into open grassy savanna, Owen-Smith (1987) proposed a keystone-herbivore hypothesis to explain the late Pleistocene extinction of approximately half of the mammalian genera with body masses of 5-1000 kg. This theory postulates that the elimination of large herbivores initiated vegetation changes that were deleterious to the fauna. Vertebrate herbivores strongly influence plant species composition. Many tropical grassland owe their origin and perpetuation to grazing animals. African elephants are known to have transformed woodlands into open grass-savanna.

Cultural Keystone Species

“Those plant and animal species whose existence and symbolic value are essential to the stability of a culture over time.” For example Cocoa in various indigenous Amazonian communities is a cultural keystone species.

10.5 WORLDWIDE TROPHIC DOWNGRADING

Recent research reveals that the loss of large apex consumers (such as lion, wolf and sea otter) is no more simply an ethical issue; it has widespread cascading effects in all kinds of terrestrial, freshwater and marine ecosystems. Related to it is the term called as “trophic cascades”, which emphasizes that top consumers have propagation impacts on their preys downward through food webs.

The empirical work of Estes et al. (2011) not only reconfirms the role of top-down forcing in ecosystems, but also shows the unanticipated and far reaching cascading effects of the loss of apex consumers on many diverse processes, namely dynamics of diseases, wide fires, carbon sequestration, spread of invasive species and biogeochemical cycles. Extinctions of ecological interactions due to the elimination of the species at the apex of trophic structure are difficult to record because their impacts are far reaching and occur at a wide range of spatial scale, from a millimetre to thousands of square kilometres, while studies on food chain and other species interactions are limited to scales from a fraction of hectare to a few hectares. Then, the impacts of their loss may become evident after several decades while studies' spans are generally much shorter.

The recent disappearance of the apex consumers from various ecosystems is largely due to the anthropogenic factors. This trophic downgrading "acts additively and synergistically" with other human induced global changes, such as climate change, habitat loss, air pollution and landuse change (Estes et al., 2011).

Despite so many problems, the effects of the loss of apex consumers have now been documented throughout the planet. The apex consumers include as varied species as killer whales (*Orcinus orca*), the great sharks, sea otters (*Enhydra lutris*), lions (*Panthera leo*), and wolves (*Canis lupus*) and cougars (*Puma concolor*). Impacts of the loss of some of these apex consumers are listed in Table 10.5.

The apex consumers primarily exert their influences on ecosystems through "top-down forcing and trophic cascades" (Estes et al., 2011). The influences are varied and many times unanticipated. In the eastern African vegetation, the disease, rind pest, reduced populations of several large herbivores including wildebeest and buffalo. Consequently plant biomass increased rapidly, resulting in wild fires during droughts. There are some evidences to suggest that the spread of diseases across the world occurs to an extent because of the elimination of apex consumers like lions and leopards in sub-Saharan Africa. The elimination of apex consumers resulted in increased populations of olive baboons (*Papio anubis*). Because of crops and food materials baboons came in contact of humans resulting in higher rates of intestinal parasites in humans and baboons. In the Himalayan region farmers are abandoning agriculture and out migrating because of the increased populations of monkeys. The damage that they cause to crops is substantial seeing that land holdings are less than 1 ha in the mountains.

The decline of vultures in India has resulted in the spread of anthrax and rabies (Markandya et al., 2008). The loss of apex consumers can substantially affect physical and chemical nature of biosphere. The predatory fishes indirectly influence phytoplankton density in marine ecosystems, and thereby their carbon uptake values. Large demersal fish substantially affects the silica supply to pelagic diatoms in the Baltic Sea. The loss of native predators leaves habitats more vulnerable to invasive alien species. In Hawaii, experimental studies showed the rise of spider population by 80% when birds were removed.

To conclude, apex consumers are not "ecological passengers riding atop the trophic pyramid", they have number of ways to have far reaching and unanticipated impacts on all kinds of ecosystems. Impacts of their loss on various ecosystems emphasize that nature functions in very complex way, clearly more than we generally perceive.

Table 10.5. Cascading effects of apex consumers: a few examples from marine and terrestrial ecosystems that have compared the condition of ecosystem in presence and absence of the apex consumers. (derived from Estes et al., 2011).

Apex consumer	Impact of the presence	Impact of the absence
Sea otter in a shallow sea floor community, Amchitka island	Primary production is high because it limits herbivorous sea urchins	Primary production is reduced because of the increased herbivory of sea urchins
Seastar (<i>Piaster ochraceous</i>) in rocky intertidal zone, California	Seastar increases species diversity by preventing competitive dominance of mussels	Mussels out compete all other herbivores, resulting in a low species diversity
Largemouth bass, Long lake, Michigan	Bass indirectly reduces phytoplankton, hence increases water clarity (bass limits small zooplankton eating fishes, thus causing zooplankton to increase)	Increased populations of zooplankton Eating fishes → Decreased Zooplanktons → Increased Phytoplankton and reduced water clarity
Arctic foxes in the Aleutian archipelago	Grasslands change into Tundra because foxes limit seabirds populations, and hence reduce nutrient inputs from sea to grasslands	Increased sea birds Populations → Increased Nutrient inputs from sea → Shift of Tundra to grasslands
Jaguar, Cougar and harpy eagle, Venezuelan forests	Maintain forests by suppressing a herbivore guild	A diverse herbivore guild established, inhibiting recruitment and survival of plants and resulting in degradation of forests
Wolf, Riparian habitat Yellow Stone National Park	Well developed willow stands as wolf keeps elk population in control	Elk population increased, which suppressed willows
Wildebeest, Serengeti Africa	Decline of woody vegetation	Woody vegetation developed when rind pest caused the decline of wildebeest, leading to wild fires due to the fuel accumulation

10.6. OTHER CONTROLS OF COMMUNITY STRUCTURE

Community structure can be described by means of food webs, which emphasizes various feeding relationships among species within a community. The number of feeding links per species and the average number of trophic levels on which a species feeds can characterise the food webs. Experimental manipulations of trophic levels in aquatic ecosystems show that consumers can depress abundance of organisms in trophic levels below them in a trophic cascade or top-down effect in the biological communities (Fig 10.17).

Bottom-up effects occur when one trophic level affects productivity of higher trophic levels. The size of the trophic level is determined by the rate of production of its food. Hansson et al. (1998) tested the bottom-up control of trophic structure in experimental microcosm communities by adding inorganic nutrients (phosphorus and nitrogen) to boost their primary production. The microcosm experiments in the aquatic systems provide evidence that increased primary production tends to augment all the overlying trophic levels. The studies have also shown that consumers could depress the size of the trophic level immediately below them and increase the number of organism two levels below.

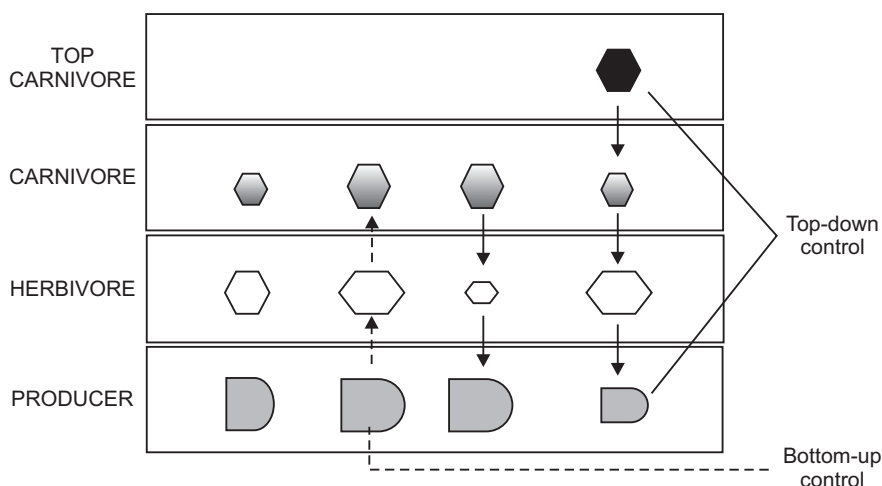


Fig. 10.17. The structure of a community may be determined by top-down or bottom-up control (modified from Ricklefs, 2001).

The top-down and bottom-up regulation of terrestrial ecosystems has been tested in predator-free forest fragments, which have been created by a hydroelectric impoundment of the Lago Guri islands in Venezuela (Terborgh et al., 2001). The vegetation of the small (0.25 to 0.9 ha) to medium islands (4 to 12 ha) is semi-deciduous tropical dry forest. The fauna of small islands is composed of the predators of invertebrates (birds, lizards, anurans and spiders), seed predators (rodents,), and herbivores (howler monkeys, iguanas and leaf cutter ants). The predators of vertebrates are absent on both small and medium islands, resulting in highly aberrant communities. These communities consist of various types of consumers without any predators. Therefore, the densities of rodents, monkeys, iguanas and leaf cutter ants are 10 to 100 times greater on the small islands than on the nearby mainland. The top-down regulation of community structure emphasizes the concept of trophic cascade, which states that perturbation at one trophic level propagates through lower levels with alternative positive and negative effects. The extreme hyper-abundance of herbivores and lower densities of seedlings and saplings of canopy trees support the top-down regulation (Terborgh et al., 2001). The major determinants of plant species diversity and their spatial distribution and relative abundance are plant-soil interactions, plant-climate interaction, plant biotic interaction and symbiosis between mycorrhizal fungi and plants. The presence of arbuscular mycorrhizal fungi affects the floristic diversity of plant communities through interplant transport of assimilates from the dominant species to the subordinate species via a common mycorrhizal network (Grime et al., 1987). Mutualist relationship between plant and arbuscular mycorrhizal system affects plant and organisms diversity (van der Heijden et al. 1998). The floristic diversity in species-rich grasslands depends upon a rich assemblage of fungal symbionts in soil that can regulate plant diversity, productivity and nutrient availability. The supply and efficient use of soil resources impart a regulatory influence on ecosystem processes, stability and resilience. Litter mediated mechanisms and soil microbial activity have strong effects on species composition in natural communities contributing to soil organic matter dynamics and nutrient mineralization.

10.7 EDGES AND ECOTONES

The transitional zone between two or more than two communities is known as ecotone. It is a narrow zone compared to the adjoining communities. Ecotones contain organisms of both the overlapping communities, as well as those restricted to ecotones only. The number of species and population density of some species are greater in the ecotone than in the adjoining community. The

tendency of the increased variety and density of organisms at the community junctions is known as edge effect. The organisms, which occur primarily or most abundantly or spend the greatest amount of time in junctions between communities, are called edge species. The differences in physical conditions of the communities result in the formation of ecotones. The inherent changes in soil and topography could form stable and permanent edges of high contrast between the communities (Fig. 10.18). Induced edges are maintained by periodic disturbance such as fire, grazing, erosion and disease. Plants of the edges are generally opportunistic and shade intolerant. The animals may need two or more communities for reproduction. Ecotones are especially prominent where sharp physical differences separate distinct communities. Some examples of ecotones are:

1. An ecotone between plant associations on serpentine and non-serpentine soils in south-western Oregon.
2. Ecotones at the transition between broad-leaved and needle-leaved forests.
3. Fire maintains sharp edge between prairies and forests in Midwestern United States.
Perennial grasses resist fire damage that kills tree seedlings, but fire cannot penetrate deeply into the moister forest.
4. Riparian zones exhibit very high edge-to-area ratios and provide a classic example of edge effect and the ecological principle of ecotones.

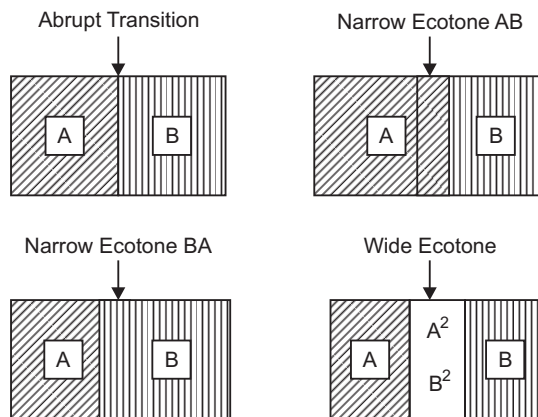


Fig. 10.18. Types of ecotones in two terrestrial communities, i.e., a forest community A and a grassland community B. (i) No ecotone development due to abrupt transition from forest to grassland; (ii) narrow ecotone, due to advancement of forest into grassland community to produce ecotone AB; (iii) narrow ecotone, due to advancement of grassland into forest community to produce ecotone BA; (iv) wide ecotone, some plants of forest as well as grassland occur together to produce ecotone A^2B^2 (modified from Smith, 1996).

Ecotones and other boundary concepts are important in biological processes (McArthur and Sanderson, 1999). According to these workers, many fundamental processes and functions occur at biological boundaries of various scales from molecular and cellular to landscapes and biomes. Boundaries offer the opportunity for interaction between and among organisms, populations of organisms, communities, groups of communities or landscapes, and biomes (McArthur and Sanderson, 1999). The interactions may be in the nature of exchange of organisms through migration or invasion or of exchange of organismic characteristics by hybridization or competition from other organisms. Hybridization between compatible taxa often occurs at ecotones. For example, members of the big sagebrush (*Artemisia tridentata*) species complex (subgenus *Tridentatae* of *Artemisia*) often meet at ecotones providing opportunities for hybridization (McArthur and Sanderson, 1999).

Methods of Sampling Vegetation and Data Analysis

An entire plant community or a segment thereof cannot be measured, even if it were small and easily accessible. Therefore, we have to depend on samples drawn from a community to approximate the structure of the entire community. This calls for a careful selection of samples, so that the data could be used to estimate the true values of the parameters as accurately as possible. In this chapter, we give an idea of some of the methods used to sample vegetation, and to analyse data obtained from such sampling.

11.1 SAMPLING APPROACHES

For describing the vegetation of an area, first it is surveyed to select representative stands or segments of vegetation cover and then within them, samples (quadrats) are placed to record data on parameters related to vegetation and species. Rapid and repeated surveys are undertaken to develop an idea about the variety of communities present, area they occupy, dominant species, important environmental variations, historical disturbance factors, such as fire and accessibility of different parts in the study area. There are several approaches to sampling of vegetation. They could be completely subjective, completely objective (random sampling) or have various combinations of regular and random distribution of samples.

In the *completely subjective* approach, which is associated with the Braun-Blanquet School of the continental Europe, a worker develops familiarity with the vegetation by extensive surveys, and then conceptualises communities. The next step consists of subjective or preferential selection of stands considered especially typical, and that of one or two representative and homogenous sample plots, called *relevés*, within them. The *relevé* is expected to represent the essence of that stand. The major criticism is that the investigator may ignore important deviations in vegetation as he will be confining his observations to the preconceived communities. This approach is biased and not suitable for statistical analysis. In the *complete random design* both the selection of stands and placement of samples (quadrats) within them are random (objective). It is unbiased and statistically suitable, but has several field problems. Some of the stands, selected randomly, may be located in inaccessible or difficult-to-access parts of the study area. Many of the samples may happen to be placed in a small portion, leaving a large area unsampled or under-sampled. Rare but important patches of vegetation are likely to be neglected. It is also being increasingly realised that plant populations are seldom randomly dispersed; they tend to be more clumped. However, objective or random location of stands is desirable where vegetation patterns are not distinct and clear in advance (Muller-Dombois and Ellenberg, 1974). On the whole, the disadvantages of random sampling outweigh its advantages. Therefore, it is infrequently used.

In a *stratified random design*, the study area is first subdivided into a number of homogenous stands or units, and then samples in such units are placed randomly (Fig. 11.1). Evidently, in this

approach all variations in vegetation are considered as samples are distributed throughout the area. This method combines advantages of both systematic and random sampling (namely accuracy and statistical validity, respectively).

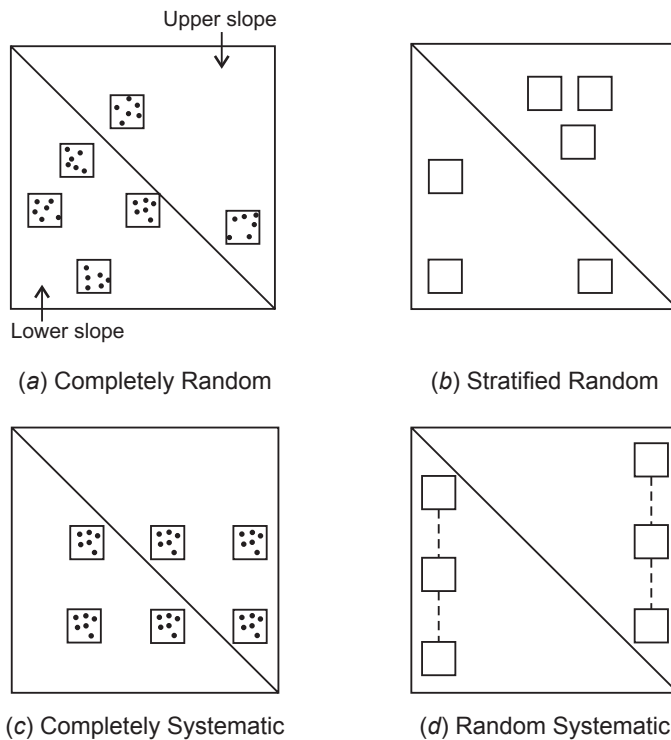


Fig. 11.1. A representation of sampling design. Squares represent the location of stands and dots within them those of sub-samples or quadrats (derived from Barbour et al., 1999).

(a) Completely random – Both samples (stands) across vegetation as well as replicate sub-samples (quadrats) within them are randomly located.

(b) Stratified random – Vegetation is divided (stratified) into homogenous areas, and then each one is sampled by equal number of randomly located stands or samples. Sub-samples within stands are located in a similar fashion (not shown).

(c) Completely systematic – Both samples across vegetation and sub-samples within them are placed at a fixed distance.

(d) Random systematic – Samples lie along convenient transects in a systematic way. A similar approach is followed for locating sub-samples within stands (not shown).

In a *complete systematic design* (Barbour et al., 1999) (completely regular) both the placement of stands and of samples within stands are systematic or regular. In this, samples are placed at fixed distances across the vegetation. As an example, stands could be located at an interval of 100 m and samples within them at 10 m interval. This method is rarely used.

In the *random-systematic design*, both random and systematic aspects of sampling are combined (Barbour et al., 1999). In this, the stands are selected by either a random or a stratified random plan. Then, within a stand the starting point (and direction) of a transect is selected by some random or stratified random plan. Along the transects samples are placed systematically. This is commonly used method in the field and is useful statistically as it incorporates some degree of randomness.

In the *gradsect method*, which is a modification of random-systematic design (developed in 1980s), a transect that represents the steepest part of environmental gradient of the study area is selectively identified. The transect is divided into homogenous parts of vegetation and each of them are sampled with quadrats or relevés in a stratified random design. The number of samples in each subdivision of the transect or vegetation type is proportional to the fraction of study area that the subdivision accounts for. For example, suppose the study area is 2400 ha, and from aerial photograph it is derived that the 1200 ha (50%) is under vegetation type A, 800 ha under B (33.3%) and 400 ha under C (16.6%). Suppose the entire area is sampled by 60 quadrats, then 30 would be placed in vegetation A, 20 in B and 10 in C. This method is less time taking and more cost effective than most others are (Barbour et al., 1999). The gradsect sampling is suggested to be an outgrowth of a purely regular sampling method initiated by (Whittaker, 1956). His subjectively chosen transects ran up in elevation along slope faces, and along the transect he selected quadrats at a regular intervals of elevation. In central Himalaya, ecologists of Kumaun University selected stands on hill base, mid-slope and ridge on various aspects. However, within stands they placed quadrats randomly (Singh and Singh, 1992).

11.2 WHAT IS MEASURED AND HOW?

Quantitative measurements generally focus on frequency, density, and cover, both of stems and canopy of species and vegetation. For these measurements ecologists have developed a variety of sampling units: two dimensional unit, called quadrats; one dimensional unit called line; dimensionless unit called points; distance or plotless methods; and variable plot method using a stick or prism. Growth form distribution, species-richness, diversity, leaf area index, biomass and productivity, niche response surface and parameters relating to nutrient cycling are some other parameters that ecologists measure. However, some of these are difficult to measure or may be very time- taking, particularly at the scale vegetation is described.

The Relevé Method

The relevé method, developed by the Swiss ecologist Josias Braun-Blanquet* (1932) in which both the selection of stands and location of samples within them are subjective.

In this method the investigators first develops familiarity with vegetation of the area to be sampled by undertaking frequent surveys. This helps in conceptualising the types of community that occur repeatedly in similar habitats. The next step is the subjective selection of stands that appear to represent a given community type followed by listing of all species encountered in each stand. Now a relevé, the sample plot, which best represents the community type is established with the following features: (i) it should be large enough to contain all species and other community characters; (ii) the habitat should be uniform, and (iii) the plant cover should be homogenous. In fact, it was assumed that a concrete stand was homogenous. That is why a common practice in the continental Europe was to sample a single relevé (McIntosh, 1985).

The relevé or the sample plot should be no smaller than the *minimal area*, i.e., the smallest area within which the species of the community are adequately represented. This is determined by developing a *species area curve*. For this generally, *nested plot technique*, in which initially a small area is lined out on the ground for example, 0.5×0.5 m (i.e., 0.25 m^2), then larger and larger plots or areas are lined out in such a manner that each larger plot encompasses all the preceding smaller plots (Fig. 11.2a, plot, as an example encompasses plots *a*, *b*, *c* and *d*) is used. All species encountered in the first plot and additional species encountered in each of the larger plot are recorded. After a certain plot size, very few additional species are encountered. When the number of species is plotted against the plot size, a point on the curve is reached where the slope most rapidly becomes horizontal (Fig. 11.2b); this defines the minimal area. Generally, a larger area than the minimal area is taken for a relevé. In many situations, it is difficult to identify the

point at which the curve levels off. In such circumstances, a point is placed at 100% values of Y and X (Fig. 11.2*b*). This point is connected with a line a_1 in Fig. 11.2*b*, passing through the origin (Y_0, X_0). Another line (a_2 in Fig. 11.2*b*) is then drawn parallel to this line such that it touches the species area curve tangentially. The tangential intersection point on the curve is then protracted to X-axis and minimal area is read off.

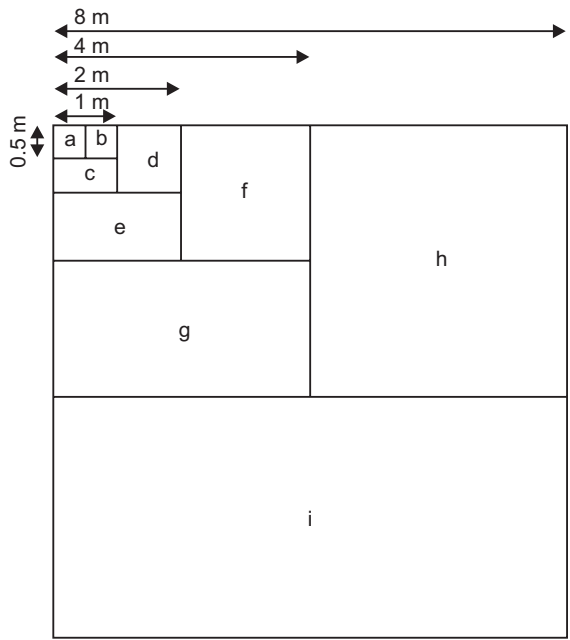


Fig. 11.2. (a) Nested plot design for determining minimal area for the species area curve. Area of a larger plot is equal to the sum of area of all smaller plots.

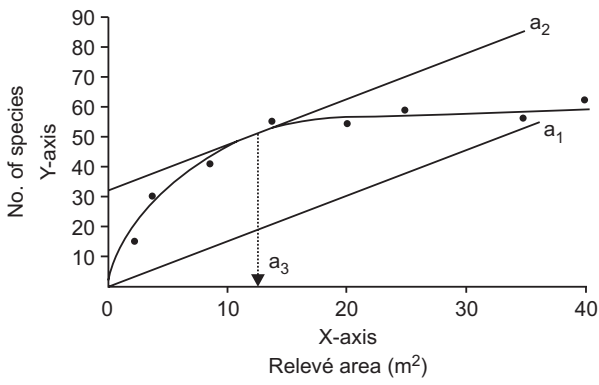


Fig. 11.2. (b) Species-area curve of a grassland, a_1 is the line joining 1 percent of values of x and y axes and their origin point; a_2 is the line parallel to a_1 that touches the species area curve tangentially; a_3 is the line protracted to the x-axis from the intersection point on the curve. The point of x-axis where a_3 meets represents the minimal area.

Some investigators define the minimal area as the area which contains a standard percentage (for example 95%) of the total species found in the stand. Minimal area is regarded also as an important community character, as it varies widely across communities (Muller-Dombois and Ellenberg, 1974).

Several pieces of information are recorded after establishing a relevé including serial number of relevé, date of sampling, location, geographic area, a tentative floristic name describing the most obvious species composition, general remarks on plant cover, particularly on number of vertical layers. A number of parameters, particularly cover, sociability, vitality, periodicity, and environmental features are determined within the relevé. The cover of species is often not measured precisely, but a number of cover categories are established and species are placed in one of them (Table 11.1).

Table 11.1. Cover classes of Braun-Blanquet and Domin-Krajina.

Braun-Blanquet			Domin-Krajina		
Class	Range of cover (%)	Mid point	Class	Range of cover (%)	Mid point
5	75-100	87.5	10	100	97.5
			9	75-99	87.0
4	50-75	62.5	8	50-75	62.5
3	25-50	37.5	7	33-50	41.5
			6	25-33	29.0
2	5-25	15.0	5	10-25	17.5
			4	5-10	7.5
1	<5	2.5	3	1-5	2.5
+	<1	0.1	2	<1	0.5
<i>r</i>	<1	*	1	<1	*
			+	<1	*

* individuals occurring seldom or only once, cover ignored and assumed to be insignificant.

The idea is that precise estimation of cover is hard to make, and values of precise estimate vary widely from investigator to investigator. *Vitality* or vigour which expresses the growth condition of species is also considered. For example in the Braun-Blanquet method following symbols are used: °° very feeble and never fruit bearing (e.g., + °°); ° feeble (e.g., 1°); * exceptionally vigorous (e.g., 3*). For normal plants no symbol is used.

In each of above, the first symbol indicates cover class, and the second vitality. For example, 3* indicates cover class 25-50% of Braun-Blanquet and plants exceptionally vigorous.

Sociability expresses dispersion of individuals of a species. Two species with same cover may be distributed in quite different manner; one may grow singly, while the other may form a large and pure crop or stand (e.g., *Typha* spp. at fringes of fresh water bodies). The five sociability or dispersion categories used by Braun-Blanquet are: 5 = growing in large, almost pure population stands; 4 = growing in small colonies or forming larger carpets; 3 = forming small patches or cushions; 2 = forming small groups and 1 = growing solitarily or singly. These sociability values are often added to cover values. For example, a species showing cover values in the range of 25-50% and growing singly will be indicated as, 3.1.

Presence and *constance* are synthetic characters, not of one stand or of one relevé, but of several of them. The percentage of stands of the given community type in which a given species is present gives the presence value of that species (suppose a species is present in 7 out of 10 stands of a community type, its presence value is 70%). Constance (sometimes called constancy) on the other hand is the percentage of relevés in which a given species is present (suppose aforesaid species is present in 6 out of 10 relevés, its constance value is 60%). Remember the size of relevé would be the same for a given community type, whereas the size of stand would vary within the community.

Fidelity or faithfulness of a species to a given association is expressed on the basis of how few stands outside the association contain them. Species are placed in one of the following five categories: 5 = species exclusive or nearly exclusive to one community; 4 = species occurring most frequently in one community and can occur rarely in a few others; 3 = species most preponderant in one association but also occur more or less abundantly in several communities; 2 = species without any strong affinity to one community; and 1 = Rare and accidental intruders.

11.3 QUADRAT METHOD

A quadrat is a sample plot of known area. In early days of plant ecology Clements was a major advocate of quadrat method, though he himself was not a user of quantitative methods.

Shape, Size and Number of Quadrats

A quadrat can be rectangular (long and narrow), circular or square. Rectangular and circular quadrats are preferable to square quadrats. Though square quadrats are often used, rectangular quadrats are better because most plant distributions are clumped, and a rectangle can best encompass patches of different species. But a rectangle should not be more than two to four times as long as it is wide; as the ratio of length to width increases, the amount of border relative to area increases, causing increased error from edge effects (generally too many individuals near border are counted). In contrast, the circular quadrats have the greatest area relative to border, but are least efficient in sampling plants distributed in clusters (Gauch, 1982). The size of the quadrat varies in accordance with the size of plants to be sampled. Greig-Smith (1964) suggests that the quadrat should be at least twice as large as the average canopy spread of the largest species of the vegetation to be sampled; and Blackman (1935) feels that normally the quadrat size should be such that it permits a species to occur in not more than 80% of the quadrats. Commonly used quadrat sizes for density measurements are as following: 10×10 m for tree layer, 4×4 m for woody undergrowth up to 3 m height, and 1×1 m for herbs (Oosting, 1956). It may be pointed out that the minimal area releve's are meant for finding out a representative species combination. Thus, their sizes are larger than the quadrats used for counting. According to the purpose of the investigation, a quadrat may be of the following kinds: *list* (for listing of all species present), *count* (for determining the individuals of species), *cover* (for determining the canopy or basal cover of species), *chart* (for mapping the plants within the quadrat), *clip or harvest* (for determining biomass or the weight of the plants) and *denuded* quadrat (for determining the sequence or development of vegetation over time, following a treatment or catastrophic event).

As a rule of thumb, the number of quadrats should be such as to sample about 20% of the vegetation of a stand. However, the required number of quadrats can be determined reasonably precisely by plotting the values of a given parameter (density, cover, etc.) against the number of quadrats (Fig. 11.3). The point at which fluctuations in the value of the parameter are minimised, would indicate the number of quadrats required. As shown in Fig. 11.3, in sampling by six quadrats the density of species A, B and C is under-represented, while that of E is over-represented. By 10 quadrats, density of most of the species levels off and a greater number of quadrats do not yield additional information.

In the *Running Mean Method* first samples of two quadrats are taken and their mean is calculated. The number of quadrats is then increased to 4, 6, 8, 10, 12 and so on, and a fresh mean is calculated each time (Table 11.2) on a graph relating mean value (no. of readings) to the sample size (quadrat number) at which the mean value ceases to fluctuate is easily determined. Values in Table 11.2 indicate that the density of *Quercus floribunda* does not fluctuate from quadrat 14 to 20. Therefore, 14 quadrats are adequate in this case.

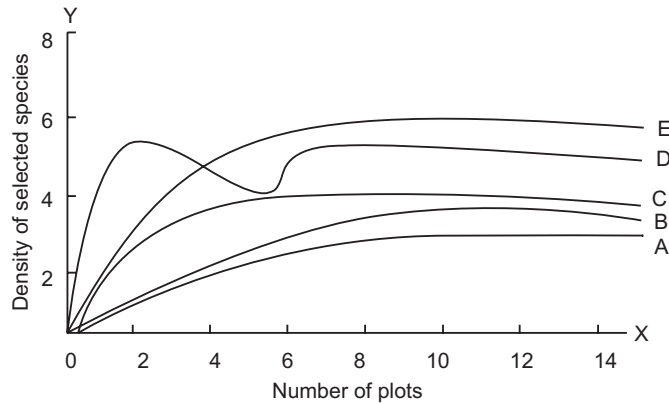


Fig. 11.3. Density of different species in relation to quadart number. Note density of all species show minimal fluctuation when 10 quadrats are taken, therefore, in this case about 10 quadrats would be inefficient for the sampling of vegetation (A,B,C, D and E are hypothetical species).

Table 11.2 Determination of minimum number of quadrats for *Q. floribunda* forest by running mean method.

No. of trees (100 m ²)	Cumulative total number	Running mean
6, 4	10	5.0
3, 3	16	4.0
5, 4	25	4.2
2, 3	30	3.7
6, 4	40	4.0
4, 5	49	4.5
4, 4	57	4.1
5, 3	65	4.1
4, 6	75	4.2
4, 4	83	4.1

Frequency, Density, Cover and Importance

Once the quadrat shape, size and number have been determined various analytical characteristics of the vegetation are quantified by laying out quadrats in stands and taking measurements.

Frequency of a species is the percentage of quadrats in which the species is present. If individuals of a species were present in 8 quadrats out of 10 quadrats studied, the frequency of that species is 80%. The presence of individuals in a quadrat can be decided in two ways: (a) the individual is rooted within the quadrat; and (b) a part of the shoot of an individual overlaps into the quadrat although the individual is rooted outside the quadrat. Frequency calculated on the basis of the former is called rooted frequency and that of b cover frequency. Frequency by quadrat method is a non- absolute measure because it varies with the quadrat size. A very large quadrat would over estimate the frequency of most species, while a very small quadrat would give underestimations. Species with similar number of individuals in an area would have different frequencies if their distribution patterns are different. A species having individuals concentrated in patches would have lower frequency than the species having individuals spread out evenly over the sampling area, even though quadrat of the same size is used. However, frequency can be made an absolute measure by reducing the quadrat to a point.

Relative frequency of a species is its frequency as a percentage of total frequency of all species. The sum of relative frequencies of all species in a stand would be 100%. In other words, in a multi-species community, even if a species has 100% frequency its relative frequency would be less than 100%.

Frequency class: Among the early plant ecologists, Raunkiaer concentrated on plant frequencies, in contrast to others who gave more importance to density. Raunkiaer's most notable contribution to community statistics, called *Raunkiaer's law* pertains to five frequency classes (20% interval): *A* = up to 20%, *B* = 21-40%, *C* = 41-60%, *D* = 61- 80%, and *E* = 81-100 %. According to his "frequency law" if the numbers of species in each of five frequency classes were plotted in sequence, the resulting distribution was a reversed J-shaped curve. Accordingly, the frequency continuously declines in first four classes, and then increases in the terminal class (81-100%). Raunkiaer studied some 8,087 frequency percentages and found that class *A* included 53% of the species, *B* 14%, *C* 9%, *D* 8% and *E* 16%. Thus the law suggested: $A > B > C = D < E$. In general, the higher the value for class *E*, the more homogeneous is the stand. Further, If the value of the ratio $(E + D) / (B + C)$ is less than one the stand is heterogeneous. Greater the value than one, greater would be the homogeneity of the stand (Fig. 11.4).

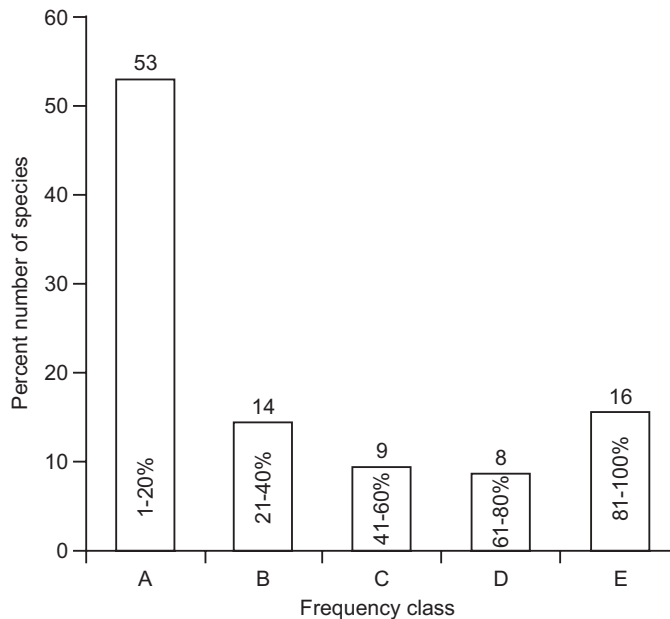


Fig. 11.4. Raunkiaer's frequency diagram, $A > B > C < D < E$, higher the value of *E* the more the

homogenous the stand. $\frac{E + D}{B + C}$, greater than 1 the more homogenous the stand.

Density of species in a stand is the number of rooted individuals per unit area. It is calculated by totalling the number of individuals recorded in all quadrats divided by the number of quadrats used. The resultant value can be extrapolated to any convenient size. For example, in a forest stand in the Kumaun Himalaya, studied by laying out 10 quadrats each of 100 m², there were on an average 4.2 trees per quadrat (100 m²) of the common oak, *Quercus leucotrichophora*. The density of this oak converts to 420 trees per ha.

The term *abundance* is variously used; one measure of this is the total number of individuals of a species recorded in all quadrats divided by the number of quadrats in which the species was

present. Remember, while calculating density the total number of individuals in all quadrats are divided by the total number of quadrats studied whether or not the species was present in all of them. Obviously, for a given species the value of abundance would be higher than that of density. Suppose in the above example, individuals of *Q. leucotrichophora* were found only in 5 of the 10 quadrats. In that case its abundance value would be $(420/5) = 8.4$ trees per 100 m². When abundance is divided by frequency the resultant ratio called, Whitfort's index, represents the distribution pattern as follows:

Range of A/F values	Distribution
< 0.025	Regular distribution
0.025 to 0.05	Random distribution
> 0.05	Contagious distribution

It may be noted that for this index the abundance must be represented as individuals per quadrat of the size used to calculate frequency. For example, if sampling was done by 50 × 50 cm quadrat, the values of abundance should be as number of individuals per 250 cm², rather than say per m².

Relative density of a species is its density as percentage of total density of all species present in the stand. The sum of relative density of all species in a stand would be 100%.

Cover or coverage of a species is the ground area covered by that species.

Canopy (or crown) cover refers to the ground area covered by the canopy when canopy is projected vertically to the ground, and *basal cover* or *basal area* is the ground area covered by the stem (Fig. 11.5). The canopy edge of an individual is rounded out (Fig. 11.6) and area under holes is not deducted from the rounded out area. Cover is given a greater ecological significance than density as it gives a better estimate of plant biomass.

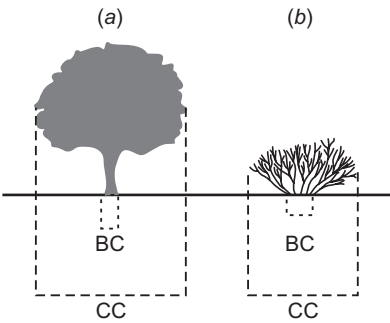


Fig. 11.5. A representation of canopy cover (CC) and basal cover (BC), (a) for a tree and (b) for a shrub.

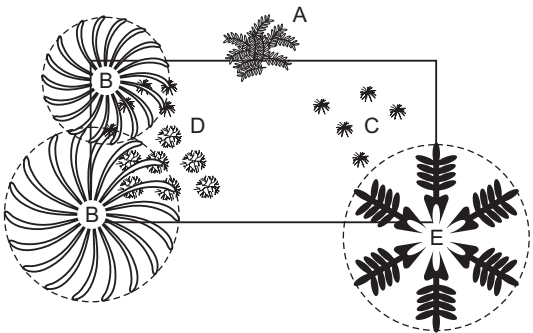


Fig. 11.6. Estimation of cover with quadrat method. In case of each individual the canopy edge is rounded out. The perimeter of circle in each plant is determined by its leaf tips. Within circles holes are ignored, assuming that their entire area is covered by leaves. Overlaps between the canopies of two individuals (such as of species B) are not counted twice. Though the canopy of E hardly penetrates into the quadrat, yet it fills in the canopy holes by connecting the radiating branches with an imaginary polygon or circle, then its cover is there.

Relative cover of a species is the total cover of that species as a percentage of total cover of all species present in the stand. Relative basal area is also called *relative dominance*. In low herbaceous vegetation such as grasslands having short-statured plants or mosses on rocks, the plant cover can be estimated by using *chart quadrats*. In this method cover is charted or mapped from small quadrats or frames, say of 1 m² subdivided in a number of squares by means of strings. On a sheet of graph paper outlines of the crown area or basal shoot systems of species are drawn to scale. It is also used to map the position of individual plants whose cover is negligible.

Pantograph system can also be used for charting the vegetation of small plants by taking small quadrats. For estimating the tree crown cover, a convenient method is photographing the tree crown cover from below. The camera contains an especially developed hemispherical, “fish eye” lens to take wide-angle canopy photography.

The crown cover of tree and other woody plants can also be measured by *crown-diameter method*. A tape is laid out from one end of the crown perimeter to the other end. This is the diameter of crown. It is necessary to measure crown diameter at least once more approximately perpendicular to the axis of first measurement. The crown cover (CC) is calculated as:

$$CC = \pi \left(\frac{D_1 + D_2}{4} \right)^2$$

where D_1 is the first reading of diameter and D_2 the second reading. The value can be expressed in square metres of crown cover per square metre of ground cover. To express the value of tree crown cover in relation to a unit of ground area, such a unit of ground area is also measured.

Measurement of *basal area* (BA), that is trunk cross-sectional area, to which canopy cover of tree is typically correlated, is more common. To obtain the basal area, trunk *diameter at breast height* (1.37 m) is measured, either by a special diameter tape that converts the circumference to diameter units or by callipers that directly measure the diameter. Diameter is measured at 1.37 m height because many trees have buttressed stems and it is convenient to take measurements at breast level. The basal area is equal to πr^2 , where r is equal to $\frac{1}{2}$ dbh. In grasses also, basal area can be measured.

Expression of dominance: Dominance is an important term but often defined ambiguously. Generally, the overstorey species with the highest basal area is regarded as the dominant. If more than one species have similar basal area (or any other such measure), such two or three species are called co-dominants. The rest of the species having distinctly lesser importance are called *subdominant* or *subordinate species*. In savanna where tree cover is too thin as compared with shrub or grass cover, the understorey species contributing most cover or density are treated as dominant. However, there are also other ways of describing dominance. In forestry, any individual tree whose crown is more than half exposed to full sun is considered dominant whether or not it is the member of species with highest basal area. This may be called physiognomic dominance of individual. Kershaw (1973) has described the dominance in terms of control of reproduction in the community, where a dominant species can be also the member of understorey. In ponderosa pine forests, as an example, the regeneration of the species, *Pinus ponderosa*, is inhibited by root competition for moisture by the understorey grass *Festuca arizonica*. Thus the grass species by virtue of controlling the regeneration of the physiognomic dominant, ponderosa pine, and hence the persistence of the community, is called *sociologic dominant*. While naming the community both physiognomic and sociologic dominants can be included.

In another view the species which at once become noticeable and visually appear to dominate the community, are regarded as dominants. For example, in some herbaceous communities, a few species become distinguishable from the rest of species because all members of them bear flowers simultaneously. Such species exhibit *aspect dominance* at a particular time. However, they may not have maximum density or cover among the species of those communities.

Importance of a species is its relative contribution to the whole community. The relative contribution can be in terms of relative density, relative cover or basal area or relative biomass, and others. Importance as used by Curtis and McIntosh (1951) is sum of relative frequency, relative density and relative basal area or cover. This sum is called *Importance Value Index* (IVI) of a species in a community. The IVI can range from 0 to 300 (the value of 300 for a species can only occur in a monospecific community). The sum of IVI of all species in a stand will be 300. The species with the greatest IVI value can also be called a *dominant species*.

11.4 LINE TRANSECT AND RELATED METHODS

A “line” theoretically has no width, it is one-dimensional. This method is usually applied to measure cover, and to study changes across an environmental gradient and ecotone. It is more useful for measuring cover of a large area, but works well only when crown cover is dense.

In a multilayered vegetation, it is recommended that cover of each layer (such as shrub layer and herb layer) should be measured separately. However, crown outlines of trees considerably larger than 15 m are difficult to measure unless one has special sighting tool, such as “moosehorn” crown closure estimator*. The sum of segments of line occupied or intercepted by a species expressed as the percentage of the total tape length used for sample, is the percent cover of that species. In this method density is not measured, although frequency can be measured by dividing the tape length into a number of segments of a suitable size and counting the percentage of segments intercepted by the species. However, alongside the tape or line, quadrats can be positioned to measure density.

A contiguous row of quadrats can be positioned to measure density. A contiguous row of quadrats that runs along the line is called the *belt transect* or *strip transect* or *line strip*. In a related method called the *bisect* method, originally applied to tropical rain forests (Davis and Richards, 1933; Richards, 1936), scale drawing of vegetation is done within a strip, say 60 m long and 8 m broad. The location of all (particularly tree) individuals, their height and canopy shape and size are depicted (Fig. 11.7). Some ecologists use symbols instead of figures. The purpose is to depict a profile of vegetation.

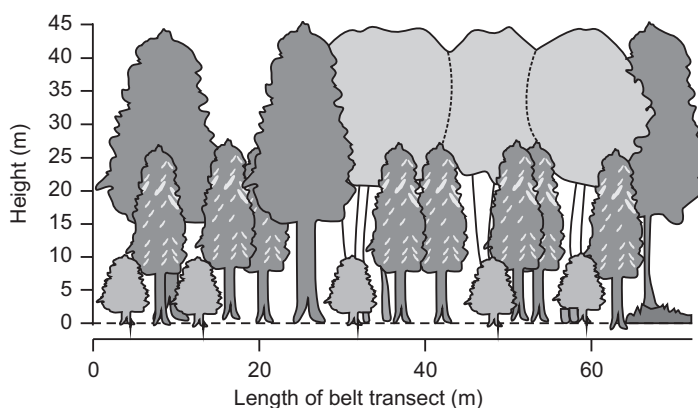


Fig. 11.7. Scale drawing of a forest vegetation profile.

11.5 THE POINT-FRAME METHOD

The point-frame measures the grass cover or the cover of other low vegetation (usually 25-50 cm tall). A point, such as, sharply pointed tip of metal pins, is a dimensionless sampling unit. Such metal pins are inserted through holes in a wooden linear frame 1 m high and 1m long. Usually 10 metal pins of each of 1m lengths are inserted at a distance of 10 cm from each other. For measuring

cover, the frame is held vertically over the strip of vegetation to be measured and pins are lowered vertically one after the other till ground is reached (Fig. 11.8). Each hit (or touch) on plant parts is recorded by species. The frame is placed randomly at several places in the stand. The frame should be held tightly so that it remains vertical. The number of hits for species individuals or a part thereof, represented as the percentage of total hits by metal pins is the percent cover of that species. While calculating crown cover or shoot cover only first touch of each needle with the shoot part is counted. Remember a needle while being lowered can hit several plant parts of the same species, or one species at higher height and another species at lower height. Basal area is calculated by counting only the hits with stem part at ground level.

This method cannot be applied to density measurements, but for measuring the cover of low vegetation, according to Goodall (1953) may be the most reliable method. Thus by this method canopy cover, leaf area index and basal cover can be determined.

11.6 DISTANCE OR PLOTLESS METHODS

In these methods, distance between individual plants or between a random point and individual plants is measured. They are also known as plot-less methods because neither quadrats nor lines or point frames are used. Original form of these methods was developed at the Wisconsin university during 1950s by Cottam and Curtis. Out of a number of distance methods (e.g., Nearest Individual Method, Point-centred Quarter Method, Random Pairs Method), we have selected *Point-Centred Quarter Method*, the most widely accepted of the distance methods.

In this method, points are located randomly, usually a rod having a compass fixed at one end is placed vertically on the ground. The needle direction becomes the first axis and the line running perpendicular to it through the point the second axis. In this way the area around the point is divided into four equal quarters. The four quarters can also be identified by placing a cross on the point. Now four distances are measured at each point. In each of the quarters distance from the random point (where rod is fixed) to the nearest tree (Fig. 11.9) is measured (the tree location is the mid point of the trunk cross section) and mean of four distances is taken for density measurement. Each tree is

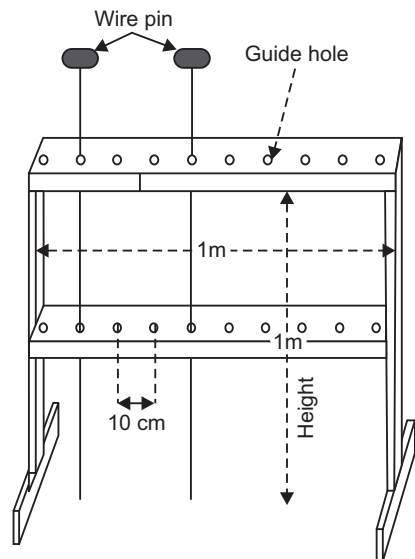


Fig. 11.8. A point-frame used to measure cover of low vegetation, like grassland.

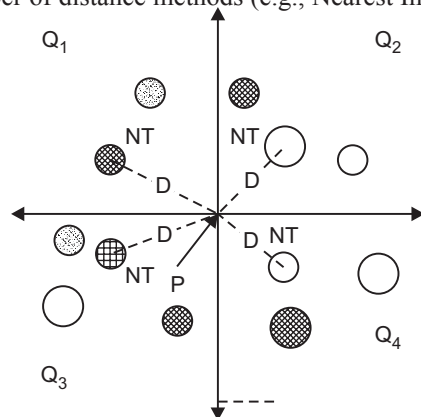


Fig. 11.9. Distance measurement in point-centred quarter distance method. P is random point from where distances (D) to the nearest of trees (NT) are measured in each of the four quarters (Q), derived either on the basis of the direction of compass needle or by placing a cross on the random point.

identified and its basal area is measured. The parameters obtained include species, density (from mean distance), and frequency (Appendix Table 11.1). Cottam and Curtis (1956) recommended that a minimum of 20 points would be required to sample a stand. Information generated by various sampling methods is briefly listed in Table 11.3.

Table 11.3. Information generated by various sampling methods.

Sampling method	Information Generated
Relevé method	Presence, constancy; species placed in cover categories (actual cover not measured), sociability and vitality and vigour of species individuals
Quadrat method	Frequency, density, canopy cover, basal area
Line transect method	Usually to measure cover, and changes along environmental gradient or ecotone; frequency measured by dividing line into segments; density usually not measured
Bisect method	For scale drawing of vegetation
Point Frame method	Mainly for cover of low vegetation (usually 25-30 cm tall); basal area and leaf area index can be measured, but not density
Distance or plot-less method	Density, basal area and frequency

In preceding section we described diverse sampling methods. However, in general, investigators have used either relevés or quadrats, particularly in case of forests (Gauch, 1982). The former was primarily used in the continental Europe while the latter is popular in the USA and several other countries.

11.7 VEGETATION ANALYSIS

The data collected from stands or other such samples described above are analysed, summarised and related to one another. Generally data are analysed by one of the following three methods: classification, direct gradient analysis and ordination (Gauch, 1982). The three methods differ in strategy of data organisation. In direct gradient analysis, first, environmental gradients are recognised and then species and community samples are arranged in relation to them. In contrast, in classification and ordination community samples with data on species importance (e.g., density, basal area and cover) are organised exclusively, leaving environmental data arrangement and interpretation to a subsequent, independent step (Gauch, 1982). The result of classification is the arrangement of community samples or species classes, which may be hierarchical or non-hierarchical. In ordination, the community samples or species are arranged in a two or three-dimensional space such that similar entities are located closely and dissimilar far apart.

Classification

Classification basically involves placing similar stands or any such samples in discrete entities, such as associations. Community ecologists vary in type of species they emphasise while classifying samples. It could be dominant species as well as minor species, individual species as well as a group of species, and characteristic species of a community as well as overall species composition. In North America, also in India and several other countries, while classifying samples into groups emphasis is on *dominant species*, generally of overstorey layer. However, in such classifications understorey dominant species is also included, and its name is separated from the overstorey dominant by a

slash, e.g., *Pinus ponderosa*/*Purshia tridentate* (tree-shrub), *P. ponderosa*/*Agropyron spicatum* (tree-grass) community and *Quercus leucotrichophora*/*Rhododendron arboreum* (tree-tree) community. This is generally non-hierarchical. In continental Europe, however, entire flora is considered to classify communities. The most widely accepted classification of this type is of Braun-Blanquet, which is based on table arrangement.

The Table Arrangement Method of Braun-Blanquet is possibly the first major classification technique in community ecology. According to van der Maarel (1975), it is also the most frequently used technique, encompassing many thousands of studies. It is a subjective method in which community types are recognised by floristic composition. Among the species occurring in a community the emphasis, however, is on the characteristic combination of *faithful* or *diagnostic* or *differential species*. It is a hierarchical classification, the basic unit being the *association*. This method involves two major procedures: (i) selection of differential species typifying a given group of samples from the species – sample matrix; (ii) selection of those samples which typify the differential species.

After collecting data for a number of stands from an area, as described in relevé method of sampling, they are presented in a raw table (Table 11.4), which contains relevé number (arranged horizontally) in the order they were sampled and names of species (vertical column in the sequence they were encountered). For each species, in all the relevés where it is found values of two parameters are entered, namely cover class and sociability class. For example, in relevé 1 *Hippophae rhamnoides* has 75-100% cover (cover class 5 according to Braun-Blanquet) and grows singly (category 1), therefore it is represented as 5.1 in Table 11.4. At the end, species number encountered in each relevé is also indicated.

Further differentiation of the table is largely facilitated by calculating the constancy of the species (i.e., number of relevés in which a species is recorded). Some species have high constancy (say above 60%), others have low (less than 10%) and still others moderate constancy (10-60%). Species both of high constancy, the ubiquitous species, and of low constancy, the accidental species are of no use in differentiating the associations. Only the species showing moderate or intermediate constancy are considered for classification. Among them a group of species occur together in certain relevés and rarely outside these relevés. For example, *Urtica dioica* and *Eurynchium praelongum* occur in relevé 1, 2, 3, 19 and 20 and rarely in other relevés. That they occur only in fewer relevés outside these five, indicate high level of *fidelity*, i.e., faithfulness to its group or association. Generally, the species occurring in 20% or less of the outside relevés, that is in relevés other than their own relevés are considered to show high fidelity to their association. These are differential or characteristic species, based on which associations can be recognised. Three such species groups (A, B and C) are recognised in differentiated Table 11.5.

It is to be noted that generally, a stand of a given association should have 50% or more of the characteristic species for the association to which it belongs. For example, for association B, there are six characteristic species, and each of the stands belonging to this association has three or more of these species. Similarly, a species being assigned to an association should occur at least in 50% of the relevés, on the basis of which a species association is recognised. It is evident that an association in this method does not consist of relevés showing total floristic similarity, but of those relevés which show similarity in differential or character species.

The relevés are arranged in another sequence according to their similarity. For example, stands 1, 2, 3, 19 and 20 are placed nearer as they make one association. Besides, the species of same association are placed one after another, therefore, species sequence in Differential Table is also different from that in Raw Table.

In a *hierarchical classification* of vegetation similar samples (see similarity index below) or species are put into groups, and then the groups are arranged into hierarchies, tree-like arrangement, called *dendrograms*. Hierarchical classification by two-way indicator species analysis (TWINSpan) is generally considered to be the most effective and robust method (Fig. 11.10).

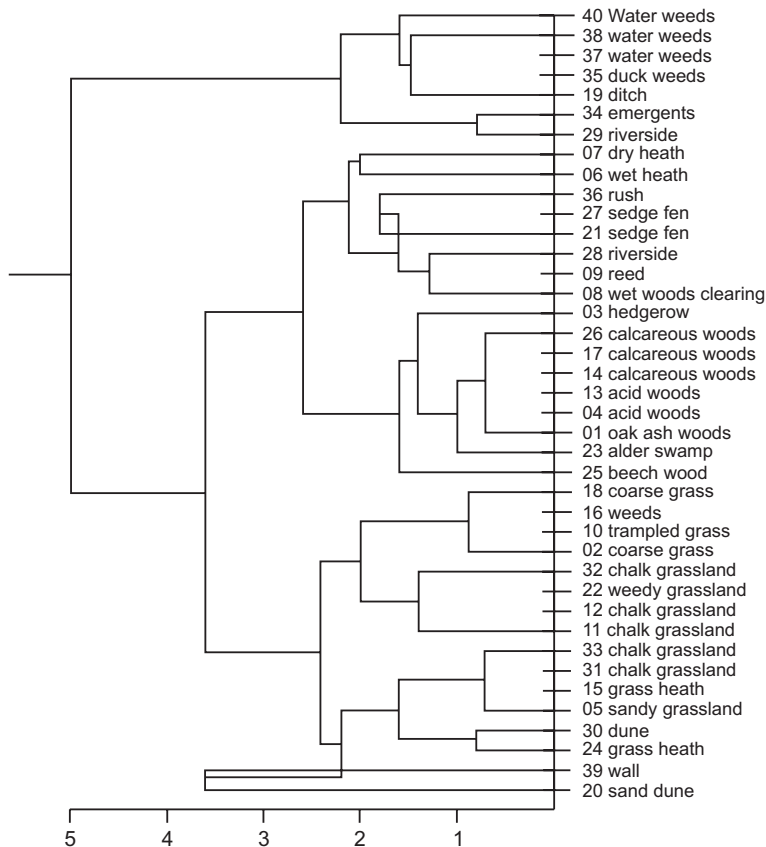


Fig. 11.10. Two-way indicator species analysis, TWINSpan, dendrogram of 40 composite samples representing the vegetation of southeast England. Sand dune and wall represent extreme of dry condition and water weeds the extreme of moist condition. Similar samples are joined at low level in the dendrogram (e.g., Chalk grassland, grass heath and sandy grassland are joined at less than 1 value) (derived from Gauch, 1982). Closeness between samples decreases from 1 to 4.

Species Association

Vegetation scientists often want to know whether species are associated in their distribution. One can use correlation coefficients for evaluating similarity in distribution of any two species among different communities or samples or quadrats. For this, species abundances (density or biomass) of each species is determined using quadrats. Correlation coefficient (r) between pairs of species is calculated. A positive and significant r indicates positive association, a negative and significant r negative association and a non-significant r indicates independent distribution of the two species. A number of indices of association (IA) have been suggested (Muller-Dombois and Ellenberg, 1974). One of the simplest to apply is Jaccards' index based on presence (I_{Ap}) data:

$$I_{Ap} = \frac{c}{a + b + c} \times 100$$

Here **a** refers to the number of samples or quadrats in which one of the two species under comparison occurs alone, **b** the number of quadrats in which the other species is found alone, and **c** is the number of quadrats in which both the species occur. Consider the following data, which is based on 20 quadrats taken from oak forest vegetation, occurring near Nainital.

Quadrat No.	<i>Q. semecarpifolia</i>	<i>Q. floribunda</i>
1	+	+
2	+	–
3	+	+
4	+	–
5	+	–
6	+	–
7	+	–
8	–	+
9	+	–
10	+	+
11	+	–
12	+	–
13	+	+
14	–	–
15	+	+
16	+	+
17	–	–
18	–	–
19	+	–
20	–	–

+ : Presence; – : Absence

If a = (number of quadrats having *Q. semecarpifolia*) **9**; b = (quadrats having *Q. floribunda*) **1**, and c = **6**, we have

$$Iap = \frac{6}{9 + 1 + 6} \times 100 = 37.5\%$$

In another formula, also based on species presence, an additional parameter, **d** which represents joint absence of the two species (number of quadrats in which both the species are absent) is also considered, as it is required for statistical evaluations (de Vries formula). For this a 2×2 contingency table is prepared which has four cells, a , b , c , and d , as follows:

Species B	Species A (<i>Q. semecarpifolia</i>)			
<i>(Q. floribunda)</i>		+	–	
	+	a	b	$a + b$
	–	c	d	$c + d$
		$a + c$	$b + d$	$n = a + b + c + d$

where a = number of quadrats having both species
 b = number of quadrats having only species B
 c = number of quadrats having only species A
 d = number of quadrats without any of the two species

A simple index of association: $I_{SM} = \frac{a + d}{n}$

Table 11.4. Contd..

Species	Relevé number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
<i>Plantago lanceolata</i>									+			+		1.1	+	1.1					
<i>Veronica chamaedrys</i>									+		+				+					+	
<i>Chamaenerion angustifolium</i>										+			1.1		+		+			2.3	+
<i>Cerastium vulgatum</i>										+			+				+	1.1	1.1		
<i>Sambucus nigra</i>																+	+				+2
<i>Cirsium vulgare</i>																+	+				
<i>Heracleum sphondylium</i>																	+				
<i>Inula conyza</i>																	+	1.1			+
<i>Cardamine hirsuta</i>																	+	+			
<i>Hypochaeris radicata</i>																					
<i>Arrhenatherum elatius</i>																				1.1	
<i>Sonchus asper</i>																					+
<i>Seedless plants</i>																					
<i>Eurynchium praelongum</i>		1.3	+3	1.3																1.3	1.3
<i>Hypnum cupressiforme</i>		+	+2	+2		+					+				+						
<i>Brachythecium rutabulum</i>			+	1.3	+					+3			+				+	+	+3		
<i>Gastrum fornicatum</i>					+	+							+				+	+			
<i>Brachythecium albicans</i>							+3			1.3		+3			+						
<i>Bryum inclitatum</i>							+3								+						
<i>Tortula ruraliformis</i>								+3							+3						
<i>Cladonia rangiformis</i>													+								
<i>Bovista nigrescens</i>																					
<i>Lophocolea heterophylla</i>															+						+
Number of species		8	9	8	8	9	11	7	10	14	12	12	10	10	22	10	11	12	12	12	13

Table 11.5. The differentiated table. Species and relevés shown in Table 11.4 have been rearranged so that similar groups (association or groups A, B, C, boxed) stand together. A large group of species that did not contribute to defining associations has been left out of the table. Also, relevé 14 has been omitted because it was a disturbed site.

Source: *The Description and Classification of Vegetation*, (Shimwell, 1972).

Revised relevé order	1	2	3	19	20	4	5	10	12	16	17	18	7	6	8	9	11	13	15
Group A																			
<i>Urtica dioica</i>	3.3	1.3	1.3	+ 0.3	2.3														
<i>Eurynchium praelongum</i>	1.3	+3	1.3	1.3	1.3														
Group B																			
<i>Montia perfoliata</i>	3.4	4.4	4.5	2.3	4.5	2.3	1.3												
<i>Stellaria media</i>	1.2	+	+	3.4	+	2.3	1.2												
<i>Geastrum fornicatum</i>	+	+		+		+	+												
<i>Cerastium vulgatum</i>				+	+	+	1.1	1.1											
Group C																			
<i>Festuca rubra</i>						+							1.1	+	1.1	+	2.3	3.3	1.3
<i>Agropyron repens</i>													+	2.3	+	3.3	+	+	+2
<i>Ammophila arenaria</i>													2.3	4.3	2.3	+	3.3	1.3	4.3
<i>Poa pratensis</i>															+	+	1.1	+	+
<i>Plantago lanceolata</i>															+	+	+	1.1	1.1

This relates the sum of quadrats with joint occurrences and joint absence of the species to the total number of quadrats sampled (Sokal and Sneath, 1963).

Taking the above example of oak forest, $a = 6$; $b = 7$; $c = 15$; $d = 4$; and $n = 32$

$$IA_{SM} = \frac{6 + 4}{32} = 0.3$$

Species B	Species A (<i>Q. semecarpifolia</i>)			
<i>(Q. floribunda)</i>		+	–	
	+	6	7	13
	–	15	4	19
		21	11	32

The 2×2 contingency table permits a comparison between the number of quadrats actually occupied by the species in question with those expected according to the hypothesis of a random distribution of individual plants. χ^2 test is used to see if the proportion of samples containing one species is significantly correlated with the occurrence of the other species. The test requires the random distribution of quadrats but not of the test species.

Calculation of the expected number of quadrats for species occurrences in the above example are made as follows:

1. Both species A and B (i.e., cell a):

$$\frac{(a + b)(a + c)}{n} = \frac{(6 + 7)(6 + 15)}{32} = 8.53$$

2. Only species B (i.e., cell b):

$$\frac{(a + b)(b + d)}{n} = \frac{(6 + 7)(7 + 4)}{32} = 4.46$$

3. Only species A (i.e., cell c):

$$\frac{(a + b)(c + d)}{n} = \frac{(6 + 15)(15 + 4)}{32} = 12.46$$

4. Neither of the two species (i.e., cell d):

$$\frac{(b + d)(c + d)}{n} = \frac{(7 + 4)(15 + 4)}{32} = 6.53$$

The observed (O) and expected (E) number of quadrats can then be compared by calculating χ^2 value which is the sum of the ratios of the squared deviations to the expected values. This is written as (Moroney, 1954):

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Using the above data the occurrence of *Q. semecarpifolia* can be compared with *Q. floribunda*

Observed values:

Species B	Species A (<i>Q. semecarpifolia</i>)			
<i>(Q. floribunda)</i>		+	–	
	+	6	7	13
	–	15	4	19
		21	11	32

Expected values:

Species B	Species A (<i>Q. semecarpifolia</i>)			
<i>(Q. floribunda)</i>		+	–	
	+	8.53	4.46	12.99
	–	12.46	6.53	18.99
		21.00	11.00	32.00

$$\chi^2 = \frac{(-2.53)^2}{8.53} + \frac{(-2.54)^2}{4.46} + \frac{(-2.54)^2}{12.46} + \frac{(-2.53)^2}{6.53}$$

$$\chi^2 = 0.75 + 1.44 + 0.51 + 0.98$$

$$\chi^2 = 3.68.$$

The χ^2 value is usually calculated from the following formula, which eliminates the need for calculating the expected and their deviations from the observed values:

$$\chi^2 = \frac{(ad - bc)^2 \times n}{(a + b)(c + d)(a + c)(b + d)}$$

Thus for the same data calculations are:

$$\chi^2 = \frac{\{(6 \times 4) - (7 \times 15)\}^2 \times 32}{13 \times 19 \times 21 \times 11}$$

$$\chi^2 = \frac{\{(6 \times 4) - (7 \times 15)\}^2 \times 32}{57057}$$

$$\chi^2 = \frac{209952}{57057}$$

$$\chi^2 = 3.68$$

The χ^2 value is looked up in a Chi-square table for one degree of freedom. The inspection of χ^2 table shows that the value is 10.8 at $P < 0.001$ and 3.84 at $P < 0.05$. Since the calculated χ^2 value, 3.68 is lower than both of these values, the two oaks are not associated with each other, rather they are independently distributed.

Similarity between communities

In many vegetation analysis schemes it is important to know how similar to each other are any two communities. Several similarity indices (also called community coefficients are available for this purpose. One of the earliest similarity index (IS_J), based on only presence and absence data, was that given by Jaccard in 1901, 1912:

$$IS_J = \frac{c}{a + b + c} \times 100,$$

where c = number of species common to the two stands/communities, a = number of species unique to the first stand/community, and b = number of species unique to the second stand/community.

Another very commonly used similarity index (IS_S), based on quantitative data, was given by Sørensen in 1948, modified form is:

$$IS_S = \frac{2W}{A + B} \times 100,$$

Where W = sum of the smaller quantitative values (density, biomass, etc.) of the species common to the two stands/communities, A = sum of quantitative values of all species in stand/community A and B = sum of quantitative values of all species in stand/community B . Each common

species will have two values, one in stand *A* and the other in stand *B* only smaller of these two values, instead of sum of both values, are summed for common species because smaller values are contained in both larger and smaller value.

Direct Gradient Analysis: In this approach the environmental gradients are recognised in advance and then communities are arranged along them. Though originated around 1930, direct gradient analysis as a method came into application around 1950 (Whittaker, 1967). To describe gradient of vegetation, generally, representative stands are sampled from a known environmental gradient, such as altitude, moisture, drainage, pH, soil depth and soil fertility. Following this approach, Singh and Singh (1992) plotted population distribution of major tree species along an altitudinal transect of Kumaun Himalaya (Fig. 11.11). Of the environmental factor complex, it is often difficult to find out which factor is most strongly correlated with community gradient, as several environmental factors vary together. Then, a factor may have more than one component. For example, soil moisture itself is complex, having components of amounts (depending on soil depth as well as on water content per unit soil), depth and duration. The gradient of communities called *coenocline*. The term *environmental complex-gradient* is used to represent gradient in environmental condition. The above two gradients together form *ecocline*, the gradient of communities and environment. Gauch and Whittaker (1972) have made 12 generalisations about species distribution and community structures in relation to environmental variations. Some of them are as follows: (i) Species abundances along the environmental gradient generally form a bell-shaped unimodal curve approximating the Gaussian curve. (ii) The modes of species differ from one another and are scattered individually along the environmental gradient. (iii) Accordingly, species dispersions vary, with some species restricted to narrow ranges along the environmental gradient, and others occurring widely. (iv) The gradient of communities (coenocline) can be shown by the amount of species turnover over the range of environmental variations (can be referred to as *beta-diversity*). (v) Species are not merely distributed in relation to physical environment, their distribution is greatly modified by competition among species.

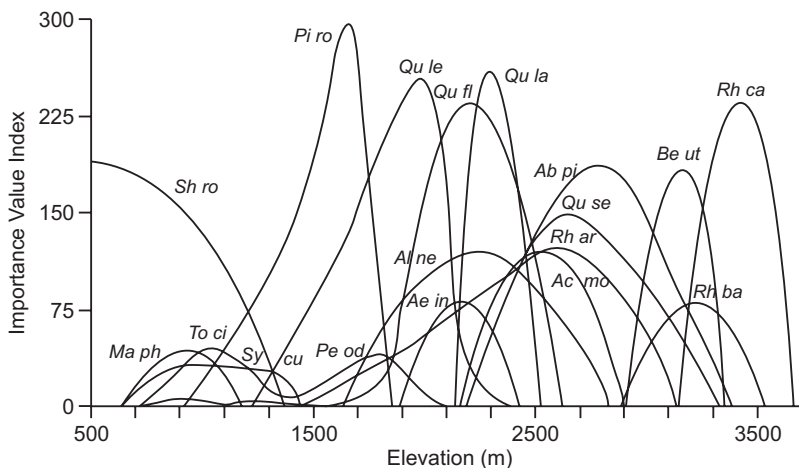


Fig. 11.11. An example of direct gradient analysis. Distribution to tree species populations along an environmental gradient (altitude) in Central Himalaya (after Singh and Singh 1992). Tree species importance values are plotted on the vertical axis, in stands (about 0.1 ha) taken along altitude (300-3300 m). *Sh ro*: *Shorea robusta*, *Pi ro*: *Pinus roxburghii*, *Qu le*: *Quercus leucotrichophora*, *Qu fl*: *Quercus floribnda*, *Qu la*: *Quercus lanuginosa*, *Ab pi*: *Abies pindrow*, *Be ut*: *Betula utilis*, *Rh ca*: *Rhododendron campanulatum*, *Qu se*: *Quercus semecarpifolia*, *Al ne*: *Alnus nepalensis*, *Rh ar*: *Rhododendron arboreum*, *Ac mo*: *Acer mono*, *Ae in*: *Aesculus indica*, *Rh ba*: *Rhododendron barbatum*, *Ma ph*: *Mallotus philippensis*, *To ci*: *Toona ciliata*, *Sy cu*: *Syzygium cumini*, *Pe od*: *Persea odoratissima*.

Ordination: While the table method aims at classifying vegetation into discrete units (association or community types), the ordination method emphasises the continuous integration of vegetation along the environmental gradients, such as, decreasing soil moisture or increasing salinity of water. The techniques employed to arrange species, communities and environments in a way that their interrelationships become apparent is called *ordination* (Goodall, 1953). According to Gauch (1982): “Ordination primarily endeavors to represent sample and species relationships as faithfully as possible in a low-dimensional space.”

The first ordination technique called *Polar Ordination (PO)* was developed by Bray and Curtis (1957), and is also known as Wisconsin Ordination. It is most appropriate for text books, as it does not involve much computation and each of the vegetational stands (or samples) is compared with all other stands in terms of total floristic composition (Appendix Table 11.2a). Beals (1984) suggests that data should be standardised before computing the distance. In the present example we have used double-centring standardisation. In this method first we take a row (species) and divide all the elements of the row with the maximum value in that row. This is repeated for all the rows (25 species in the example). Next we take the table thus obtained and sum each column (stand), then obtain the value of each element of the column as a percentage of that sum. This is repeated for all the columns (20 stands in the example), and the standardised table is thus obtained (Appendix Table 11.2b). Now, the similarity values among all the stands are calculated on the basis of quantitative values of their constituent species. One way to calculate the index of similarity (IS_s) is by Sorensen's formula described earlier.

A table is obtained that contains the values of index of similarity among all the stands. Now, if we subtract the value of each cell of this table from 100 we get the dissimilarity table (see Appendix Table 11.2c).

First, the most dissimilar stand is identified as the stand that has the maximum of sum of dissimilarities. Then, the stand showing the greatest dissimilarity with this stand is identified. These two most dissimilar stands (5 and 16 in the example, Appendix Table 11.2c) make the ends points or poles of the first axis (x -axis). Each of the remaining stands is now compared with the two endpoint stands also called reference stands, and located between the endpoints somewhere along the first axis.

A point to be noted is that the endpoint stands or reference stands are required to have less than 50% dissimilarity index with at least three stands or samples. For example, in Appendix Table 11.2c, although the most dissimilar stand is 16 (total ID = 1561.97) it does not have more than 50% similarity (or less than 50% dissimilarity) with three stands. However, stand 5 fulfils this requirement (total ID = 1338.45), and is therefore, taken as the first endpoint stand of x -axis (or zero point of the first ordination axis). The most dissimilar stand to the first endpoint stand with less than 50% dissimilarity with at least three stands is stand 10 (total ID = 1177.79), and becomes the second endpoint of x -axis.

Bray and Curtis (1957) used graphical presentation to define position of stands on x -axis. After the end point stands, all other stands are located between them by swinging arcs corresponding to their dissimilarity values with endpoint stands. Let us take the example of placing stand 1 on x -axis. As shown in Fig. 11.12a taking dissimilarity values, a compass with radius 76.69 units is swung from end-point 1 of the baseline L (70.60 units) and the other with radius of 62.35 units is swung from endpoint 2. The intersections of the two axes define a point, and a line is projected perpendicular to the x -axis. The point where that perpendicular line meets the x -axis is the position of that stand. Distance from the crossing arcs to the axis is referred to as e (Fig. 11.12a) and represents the ‘poorness of fit’ or the degree to which the ordination falls short of defining the location of a sample's point.

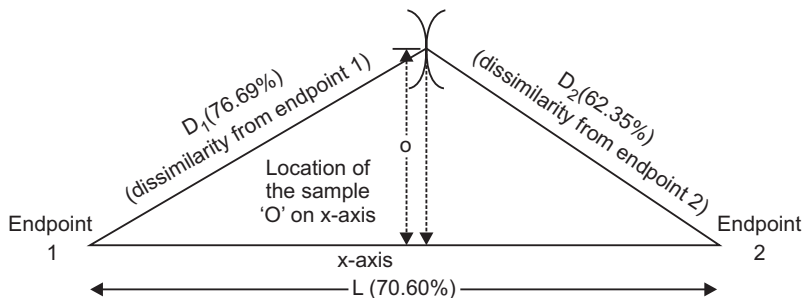


Fig. 11.12. (a) Diagram used in polar ordination values of Curtis for computing x-values. L is the distance or dissimilarity between the two endpoint stands (70.60%), D_1 is distance (or dissimilarity) of the sample O from the first endpoint (76.69%) and D_2 is its distance (62.35%) from the second endpoint. O is the sample whose x -value is being computed. e can be used to determine one of the endpoints of the next axis.

We can also use a mathematical equation for the above geometric method by which the stands are located (or ordinated) based on the Pythagorean Theorem. The equation is:

$$x = \frac{L^2 + (D_1)^2 - (D_2)^2}{2L}$$

where, L , D_1 , and D_2 are expressed as percent dissimilarities or distance values (100 – index of similarity), L is dissimilarity (70.60) in the example) or distance between the two endpoints or reference points. D_1 is the distance of the stand to be located on x -axis from the first end-point and D_2 is the same from the second end-point. These distances are a kind of *ecological ranges*, indicating the degree to which stands differ in species composition because of their separation along environment gradients or other factors. Using Pythagorean formula, x value for stand 1 would be computed as given below:

$$x = \frac{(70.60)^2 + (76.69)^2 - (62.35)^2}{2 \times 70.60} = 49.42$$

Values of x for each stand is given in Appendix Table 11.2d.

Second (Y) axis: To construct the second axis or the y -axis, e^2 ($e^2 = D_1^2 - x^2$) value for each stand is calculated (appendix table 11.2d). For example, e^2 of stand 1 = $76.69^2 - 49.42^2 = 3439.09$ (e.g., see values for stand 1 in Appendix Table 11.2d). The first-endpoint stand of y -axis is the one with the highest e^2 value and having at least three stands with over 50% similarity (or less than 50% dissimilarity). Among the stands with high e^2 values stand 12 meets this requirement (its e^2 is 2037.32), therefore, it becomes the first endpoint (y -value) of y -axis. The second end-point of y -axis should be within 10 percent of the total length of x -axis (or L , i.e., 70.60), that is, in the range 63.54–77.66. Stands 14, 18 and 19 come near this category, but none of the stands have less than 50% with at least three stands. Thus, we take another stand 2 that has 61.27 dissimilarity (nearly within the 10% range) with stand 12 (one of the end-points of y -axis), and also has less than 50% dissimilarity with at least three stands. Thus, stand 2 becomes the second endpoint of y -axis. Now, y -values of the remaining 18 stands are calculated using the same formula as used for calculating x -values (Appendix Table 11.2d).

The next step is to plot each stand's position by using x and y values. This gives a two-dimensional ordination graph Fig. 11.12b. Evidently, stands generally do not form clusters, they are separated according to their relative similarity in species composition. A lack of stand clusters supports the continuum view. Values of various environmental parameters and stand characters can be plotted to identify patterns (11.13).

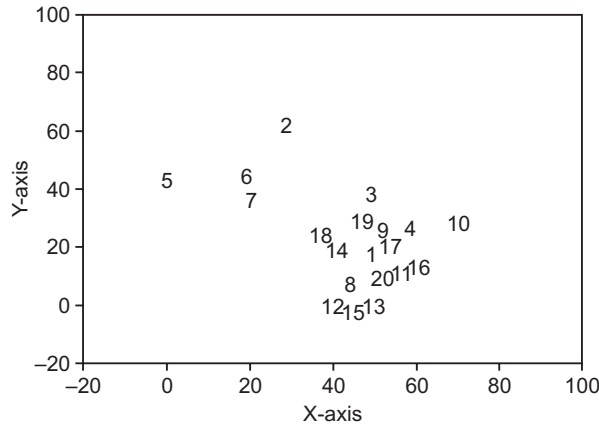


Fig. 11.12. (b) Polar ordination diagram obtained by plotting the x and y -values of the 20 stands. The triangles show location of stands. The number of stands is indicated below triangles. Stands 5 and 10 make endpoints of x -axis, and 2 and 12 endpoints of y -axis. The stands generally do not form clusters.

11.8 SUBSEQUENT ADVANCES IN ORDINATION

With the wide availability of computers in developed countries in 1970s sophisticated ordination techniques developed which truly represented multivariate analysis and were more objective. They were developed to improve ordination techniques by making them more effective in summarizing community variations and robust in handling a wide range of data sets. However, since the relationships between community and environmental variables are complex and nonlinear, they too have problems. Some of the important techniques are briefly introduced as following:

Principal Component Analysis (PCA): This is the first ordination technique in which axes are computed from data matrix, therefore there is no need to find out endpoints subjectively. The basic idea of PCA is that the data matrix of samples and species or environment variables are reduced to a fewer components or principal components by finding highly correlated combinations of species or environmental variables. For example, PCA may reduce 60 samples by 30 species data matrix in an ordination of 60 points (plots) by 2 principal components, or fundamental variables (Fig. 11.13). As in polar ordination, the distances among samples (plots, stand and quadrats) in ordination space correspond to similarity or dissimilarity in their species composition. The more the dissimilarity between samples, the more the distance in the space. However, the main features of PCA are *eigen vector* and *eigen values*. Eigen vectors are sets or scores, the weighting of each of original species or variables on each component. The eigen vector scores range from +1 to -1 (like correlation coefficients). Eigen values represent the relative contribution of each component to the explanation of the total variation in the data. Note that there is one eigen value for each component, the size of which directly indicates the importance of that component in explaining the total variation within the data set.

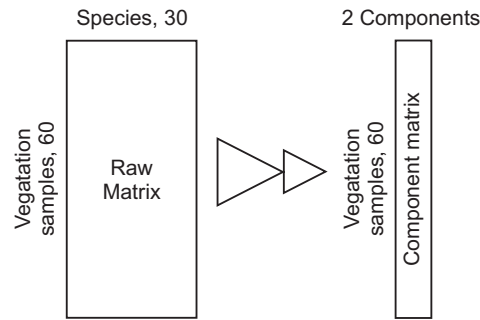


Fig. 11.13. Reduction of many species into a few components. In this examples 30 species are reduced to 2 components.

PCA, though a far more effective ordination technique than polar ordination has some serious limitations. First, PCA is primarily based on the assumption that correlations among species or environmental variable are linear, which is not the case generally. Second, the components (axes) are assumed to be completely uncorrelated with each other (orthogonal), while important environmental gradients are often themselves inter-correlated. It suffers from the arch problem. Used widely in the past by ecologists this ordination technique currently is not favoured.

Correspondence Analysis (CA) or Reciprocal Averaging (RA): The CA or RA ordination technique is conceptually related to simple weighted averaging (devised by Whittaker, 1967) but computationally involves eigenvector ordination, similar to PCA (Gauch, 1982). In this technique species ordination scores are averages of the sample ordination scores, and reciprocally, the sample ordination scores are averages of the species ordination scores. RA also ordines samples (that is stand, plot and quadrat) as well as species simultaneously. RA is found to be superior to PCA particularly when samples are heterogeneous. Like PCA it has also arch problem, but unlike PCA it does not suffer from the problems of involuted arch. RA performs a relatively objective analysis of community data, requiring no endpoint selection or weights.

Compression of the ends of the first axis relative to the middle is the second problem of this method. In CA/RA arch occurs because though the second and higher axes are uncorrelated (orthogonal), they are not independent of each other. In this, the two axes are uncorrelated because the positive correlation on one side of arch is matched by negative correlation of the other side. It is the mathematics behind the method that accounts for arch, rather than ecological content of the data.

Detrended Correspondence Analysis:

DCA (its associated FORTRAN computer programme is DECORANA) was developed to remove (detrend) above two faults of RA (Hill and Gauch, 1980). To achieve detrending the first-axis in DCA is divided into a number of segments and averaging is accomplished through a sophisticated running average procedure. The other fault of RA, compression of the first axis ends relative to the axis middle, is removed by stretching the axis ends.

Canonical Correspondence Analysis (CCA): Canonical correspondence analysis (CCA), the most recent development in ordination technique (aided by CANOCO computer programme) is different from all the preceding ordination methods. In all other methods first the ordination of the specie data alone is carried out, and then relationships associated with environment is made by superimposing environmental data on the ordination plots and looking for patterns and correlations. In contrast, CCA is the direct method, as in this, inputs include both data matrix of species $\times$ samples and environmental

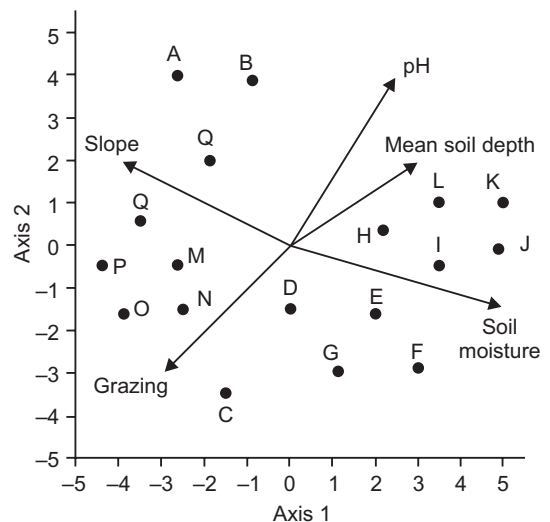


Fig. 11.14. Species/environment biplot from canonical correspondence analysis. In a hypothetical presentation, A, B, C, D are hypothetical species and points indicate their location in the graphs. Arrows represent environmental variables pointing (such as pH, grazing pressure and soil moisture) direction of maximum change. Their length is proportional to the magnitude of change.

factors $\times$ samples (Fig. 11.14). Obviously, CCA may be performed effectively if a good set of environmental data is available.

General agreement is that when environmental data are also available, CCA is the best. In the absence of suitable environmental data, DCA is recommended. Interestingly there are still some supporters of polar ordination.

Non-metric Multidimensional Scaling (NMS, NMDS): Non-metric multidimensional scaling is an ordination method that is well suited to vegetation field data, particularly if the data are non normal or are discontinuous. The objective is to iteratively search for a ranking and placement of n entities on k dimensions(axes), that is, to find a configuration of points in Euclidean space which reflects the relationship between them implied by their observed proximities, whether these are Euclidean or not. NMS, being based on ranked distances, tends to linearize the relation between environmental distance and sociological distance. It differs from PCA in that the original axes are arbitrary. It is robust to horseshoe effects.

An Example Hierarchical Classification

Vegetation classification based on species composition serves the purpose at local level. For vegetation classification at regional or global level, an hierarchical approach is required, in which both floristic and physiognomic information are used. The recently revised US National Vegetation Classification (USNVC) (Faber-Langendoer and Crawford, 2009) is a good example of hierarchical classification of vegetation. The same is summarized here. USNVC uses Vegetation criteria to define all units of classification, explicit abiotic criteria, such as “hill sal forest” in the classification of Champion and Seth (1968) are avoided. There are 8 hierarchy levels, of which the three upper levels are primarily physiognomy based; the three middle levels combine floristic and physiognomic criteria, and the two lower levels primarily consider floristic features (Table 11.6 *a* and *b*) Thus USNVC integrates both physiognomic and floristic hierarchy levels on the basis of ecological vegetation patterns. Physiognomic and structural criteria consider: (*a*) diagnostic combinations of growth forms; (*b*) ecological patterns of dominant growth forms or growth form combinations; (*c*) and vertical stratification or layering of growth forms. Floristic Criteria include (*a*) diagnostic species, or their combinations, such as differential species, character species and species with high constancy, and (*b*) vertical stratification of species which refers to layering of species found in dominant growth forms or strata, etc. Community coefficients, or similarity among samples/plots, within a type are used to groups them in an association or community type.

The upper levels of the USNVC hierarchy are based on the formation concept which has a long tradition in vegetation classification. In the USNVC, it is primarily treated as physiognomic unit. The dominant and diagnostic growth forms reflect environment at global to continent scales.

The mid-levels are based on dominant and diagnostic growth forms and compositional similarity reflecting continental to regional environmental factors and biogeography. Diagnostic and/or dominant species and compositional similarity reflecting local to regional environmental factors define lower levels of classification units.

Table 11.6. (a) Pilot example of the revised 2008 USNVC (US National Vegetation Classification) hierarchy set of types.

Upper Level (largely physiognomy based)	1.	Formation Class	Scientific Name: Mesomorphic Tree Vegetation Colloquial Name: Forest and Woodland
	2.	Formation Subclass	Scientific Name: Temperate Forest Vegetation Colloquial Name: Temperate Forest
	3.	Formation	Scientific Name: Cool Temperate Tree Vegetation Colloquial Name: Cool Temperate Forest
Middle Level (mixture)	4.	Division	Scientific Name: <i>Pseudotsuga-Tsuga-Picea-Pinus</i> Forest Division Colloquial Name: Western North America Cool Temperate Forest
	5.	Macrogroup	Scientific Name: <i>Pseudotsuga menziesii-Quercus garryana-Pinus ponderosa-Arbutus menziesii</i> Macrogroup Colloquial Name: Northern Vancouverian Montane and Foothill Forest
	6.	Group	Scientific Name: <i>Pinus ponderosa-Quercus garryana-Pseudotsuga menziesii</i> Group Colloquial Name: East Cascades Oak-Ponderosa Pine Forest and Woodland
Lower Level (largely floristic based)	7.	Alliance	Scientific Name: <i>Pinus ponderosa-Quercus garryana</i> Woodland Alliance Colloquial Name: Ponderosa Pine-Oregon White Oak Woodland Alliance
	8.	Association	Scientific Name: <i>Pinus ponderosa-Quercus garryana/Balsamorhiza sagittata</i> Woodland Colloquial Name: Ponderosa Pine-Oregon White Oak/Arrowleaf Balsamroot Woodland

Table 11.6. (b) Revised 2008 USNVC hierarchy set of types applied to dry deciduous forest vegetation of India (still tentative)

Upper Level	1.	Formation Class	USNVC: Mesomorphic Tree Vegetation C & S: Forest and Woodland
	2.	Formation Subclass	USNVC: Tropical Vegetation Forest C & S: Tropical Forest
	3.	Formation	USNVC: Tropical Deciduous Tree Vegetation C & S: Tropical Deciduous Forest
Middle Level	4.	Division	USNVC: <i>Tectona-Shorea-Anogeissus-Terminalia-Acacia-Dalbergia</i> Forest Division C & S: Tropical Dry Deciduous Forest
	5.	Macrogroup	USNVC: <i>Shorea robusta-Terminalia tomentosa-Anogeissus latifolia-Buchanania lanzan</i> C & S: Northern tropical dry deciduous forest
	6.	Group	USNVC: <i>Shorea robusta-Anogeissus latifolia</i> C & S: Dry sal bearing forests
Lower Level	7.	Alliance	USNVC: <i>Shorea robusta-Adina Cordifolia T.bellerica-Anogeissus latifolia</i> C & S: Dry Plains sal forest
	8.	Association	USNVC: <i>Shorea robusta-Mallotus philippensis-Clerodendrum viscosum</i> C & S: Moderately dry plains sal forest

USNVC = As per United States National Vegetation Classification.

C & S = Champion and Seth(1968) Classification.

Appendix Table 11.1

1. Mean distance ($\bar{d}$) = $\frac{\text{sum of all point - to - tree distances}}{\text{number of distances measured}}$
2. Mean area occupied by an individual tree (M) = $(\bar{d}^2)$
3. Density (trees per ha) (D) = $\frac{10000 \text{ m}^2}{M \text{ (in m}^2\text{)}}$
4. Relative density (RD) = $\frac{\text{No. of individuals of species A}}{\text{Total no. of individuals of all species}} \times 100$
5. Density of sp. A (per ha) = $\frac{\text{Total trees/ha} \times \text{RD of species A}}{100}$
6. Relative basal area (RBA) = $\frac{\text{BA of all trees of species A}}{\text{Total BA of all species}} \times 100$
7. Average basal area per tree = $\frac{\text{Total BA of all measured trees}}{\text{No. of trees measured}}$
8. Total BA per ha = (Density of all species) $\times$ (Average BA per tree)
9. BA per ha for a tree species = (Total BA per ha) $\times \frac{\text{RBA of the species}}{100}$
10. Frequency (F) = $\frac{\text{No. of sampling points at which sp. A occurred at least once}}{\text{Total number of sampling points}} \times 100$
11. Relative frequency (RF) = $\frac{\text{No. of points of occurrence of species A}}{\text{No. of points of occurrence of all species}} \times 100$
12. Importance Value Index (IVI) = Relative density + Relative frequency + Relative basal area

Appendix Table 11.2. (d) The obtained values for x , e^2 and y .

<i>Stands</i>	<i>X</i>	<i>e²</i>	<i>Y</i>
1	49.42	3439.05	17.85
2	28.90	1388.22	61.27
3	48.75	2138.29	38.39
4	58.76	1497.49	26.63
5	0.00	0.00	42.98
6	19.01	745.51	43.83
7	19.50	2102.12	37.26
8	44.27	2597.59	7.25
9	52.34	985.31	25.39
10	70.60	0.00	28.79
11	54.89	1622.71	11.87
12	41.22	2037.32	0.00
13	49.18	4113.32	-0.36
14	41.23	3333.50	19.21
15	43.80	5037.06	-1.49
16	56.88	6010.04	13.24
17	53.76	3592.23	20.89
18	37.78	4993.05	23.29
19	47.31	3048.42	29.68
20	52.35	6462.59	9.84

$$(i) \ x = \frac{L^2 + (D_1)^2 - (D_2)^2}{2L}$$

$$(ii) \ e^2 = D_1^2 - x^2$$

$$(iii) \ y = \frac{L^2 + (D_1)^2 - (D_2)^2}{2L}$$

Appendix Table 11.2. (a) IVI values of 25 species in 20 stands in central Himalaya.

Sl. No.	Species	WEST				EAST				NW				NORTH				SW			
		HB	LS	US	RT	HB	LS	US	RT	HB	LS	US	RT	HB	LS	US	RT	HB	LS	US	RT
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1.	<i>Shorea robusta</i>	118.6	196.9	218.5	220.1	194.9	236.1	201.9	125.7	173.0	196.8	180.8	181.3	138.7	125.1	103.6	*	150.7	136.8	107.8	*
2.	<i>Mallotus philippensis</i>	85.2	30.2	24.9	24.4	*	*	7.2	14.9	15.9	32.5	47.6	26.7	39.3	45.4	30.8	32.1	27.5	23.5	29.1	17.4
3.	<i>Syzgium cumini</i>	37.4	*	4.5	22.5	14.2	18.0	27.5	71.5	48.5	28.4	14.6	32.5	13.6	14.0	17.9	28.8	9.3	17.9	*	5.8
4.	<i>Cassia fistula</i>	14.3	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
5.	<i>Xeromphis spinosa</i>	13.5	10.0	22.1	11.3	*	*	12.6	6.9	*	*	*	5.7	*	*	*	*	*	12.3	*	*
6.	<i>Ehretia laevis</i>	11.5	5.2	4.5	10.6	*	*	3.9	41.8	37.0	19.8	35.6	*	4.0	13.3	*	*	12.8	13.7	*	*
7.	<i>Salmalia malabarica</i>	7.7	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	8.8	*	*
8.	<i>Terminalia tomentosa</i>	6.2	37.9	9.8	11.3	43.7	9.3	9.4	*	6.8	10.0	*	7.0	3.9	*	11.5	*	*	*	13.0	*
9.	<i>Lannea grandis</i>	5.6	*	5.2	*	14.3	5.4	*	7.8	6.0	*	3.5	7.4	*	*	*	*	*	*	*	*
10.	<i>Semecarpus anacardium</i>	*	9.2	*	*	8.8	5.7	4.8	*	*	*	*	*	*	*	*	*	*	*	4.9	*
11.	<i>Adina cordifolia</i>	*	*	*	*	15.7	13.1	13.7	*	*	*	*	*	*	8.5	*	*	*	7.4	2.4	*
12.	<i>Persea odoratissima</i>	*	*	*	*	8.5	*	*	31.4	5.3	4.2	3.0	*	*	4.6	*	*	*	*	*	*
13.	<i>Ougenia dalbergioides</i>	*	*	*	*	*	*	8.4	*	*	*	*	*	*	13.0	5.4	9.1	*	*	7.5	*
14.	<i>Sapium insigne</i>	*	*	*	*	*	*	4.8	*	*	*	*	*	4.5	*	*	*	*	*	*	*
15.	<i>Embllica officinalis</i>	*	*	*	*	*	*	*	*	*	4.4	*	*	*	*	*	*	*	*	4.7	*
16.	<i>Anogeissus latifolia</i>	*	10.6	10.5	*	*	12.6	*	*	7.5	3.9	*	*	*	*	*	*	*	*	*	*
17.	<i>Cedrela toona</i>	*	*	*	*	*	*	*	*	*	*	3.6	20.0	79.6	40.9	37.5	*	54.3	31.3	22.5	9.3
18.	<i>Bauhinia vahlii</i>	*	*	*	*	*	*	*	*	*	*	*	5.9	7.6	*	7.5	*	*	*	2.5	*
19.	<i>Bauhinia purpurea</i>	*	*	*	*	*	*	*	*	*	*	*	*	*	*	5.3	*	*	*	*	*
20.	<i>Bauhinia verigata</i>	*	*	*	*	*	*	*	*	*	*	*	*	*	*	5.3	20.8	*	7.7	4.9	8.0
21.	<i>Mliusa verutina</i>	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	24.2	*	*	*	*
22.	<i>Ficus retusa</i>	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	22.8	*	*
23.	<i>Bauhinia retusa</i>	*	*	*	*	*	*	*	*	*	*	*	*	*	5.6	*	*	5.2	5.0	*	*
24.	<i>Holoptelea integrifolia</i>	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	10.2	*
25.	<i>Eugenia frondosa</i>	*	*	*	*	*	*	*	*	*	*	*	*	*	*	14.1	*	9.3	5.2	*	5.2

Appendix Table 11.2. (b) Table after double-centring standardisation.

Species	Stand No.																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
<i>Shorea robusta</i>	9.42	18.56	24.10	35.90	15.57	22.96	16.09	11.46	18.73	21.97	26.63	21.22	13.08	10.77	7.85	0.00	16.63	8.69	8.46	0.00
<i>Mallotus philippensis</i>	18.76	7.89	7.61	11.03	0.00	0.00	1.59	3.77	4.77	10.05	19.43	8.66	10.27	10.83	6.46	10.35	8.41	4.14	6.33	15.07
<i>Syzygium cumini</i>	9.81	0.00	1.64	12.12	3.75	5.78	7.24	21.53	17.34	10.47	7.10	12.56	4.23	3.98	4.48	11.07	3.39	3.76	0.00	5.98
<i>Cassia fistula</i>	18.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>Xeromphis spinosa</i>	11.46	10.07	26.04	19.69	0.00	0.00	10.73	6.72	0.00	0.00	0.00	7.13	0.00	0.00	0.00	0.00	0.00	8.35	0.00	0.00
<i>Ehretia laevis</i>	5.16	2.77	2.80	9.77	0.00	0.00	1.76	21.53	22.63	12.48	29.61	0.00	2.13	6.47	0.00	0.00	7.98	4.92	0.00	0.00
<i>Salmalia malabarica</i>	16.42	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.00	0.00	0.00
<i>Terminalia tomentosa</i>	2.66	19.30	5.84	9.96	18.86	4.89	4.05	0.00	3.98	6.03	0.00	4.43	1.99	0.00	4.71	0.00	0.00	0.00	5.51	0.00
<i>Lannea grandis</i>	7.35	0.00	9.47	0.00	18.86	8.67	0.00	11.74	10.73	0.00	8.51	14.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>Semecarpus anacardium</i>	0.00	22.25	0.00	0.00	18.04	14.22	9.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.87	0.00
<i>Adina cordifolia</i>	0.00	0.00	0.00	0.00	18.86	19.15	16.42	0.00	0.00	0.00	0.00	0.00	0.00	11.01	0.00	0.00	0.00	7.07	2.83	0.00
<i>Persea odoratissima</i>	0.00	0.00	0.00	0.00	5.11	0.00	0.00	21.53	4.31	3.53	3.32	0.00	0.00	2.98	0.00	0.00	0.00	0.00	0.00	0.00
<i>Ougenia dalbergioides</i>	0.00	0.00	0.00	0.00	0.00	0.00	12.16	0.00	0.00	0.00	0.00	0.00	0.00	20.33	7.43	19.23	0.00	0.00	10.69	0.00
<i>Sapitum insigne</i>	0.00	0.00	0.00	0.00	0.00	0.00	18.82	0.00	0.00	0.00	0.00	0.00	20.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>Embllica officinalis</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.53	0.00
<i>Anogeissus latifolia</i>	0.00	18.72	21.70	0.00	0.00	22.96	0.00	0.00	15.22	8.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>Cedrela toona</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.57	6.94	22.26	10.45	8.42	0.00	17.78	5.90	5.24	8.62
<i>Bauhinia vahlii</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.45	22.26	0.00	17.65	0.00	0.00	0.00	6.10	0.00
<i>Bauhinia purpurea</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.88	0.00	0.00	0.00	0.00	0.00
<i>Bauhinia variegata</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.56	27.48	0.00	5.55	4.37	28.37
<i>Milusa verutina</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>Ficus retusa</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.00	0.00	0.00
<i>Bauhinia retusa</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.33	0.00	0.00	24.20	13.39	0.00	0.00
<i>Holoptelea integrifolia</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.53	0.00
<i>Eugenia frondosa</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.88	0.00	17.19	5.53	0.00	27.21
SUM	99.81	99.55	99.22	98.46	99.06	98.62	98.68	98.28	97.70	97.36	96.18	96.68	97.11	97.15	97.32	95.60	95.57	97.30	96.48	85.25

Appendix Table 11.2. (c) Index of dissimilarity (ID) table obtained from appendix table 11.2b.

Stands	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Sum
1.	0	67.08	56.84	50.02	76.69	74.59	66.34	57.36	60.33	62.35	51.22	54.16	71.51	70.15	78.24	79.36	72.99	54.49	82.21	77.25	1264.19
2.	67.08	0	36.04	50.27	47.16	43.09	56.24	75.01	54.08	55.92	70.15	61.27	74.49	78.21	80.68	91.92	72.03	75.67	69.21	91.46	1251.98
3.	56.84	36.04	0	37.59	67.19	39.49	63.76	63.68	42.51	51.14	54.28	47.43	73.06	76.75	78.98	90.5	70.55	73.92	79.25	89.97	1195.93
4.	50.02	50.27	37.59	0	70.36	65.88	57.94	55.44	49.67	40.47	43.97	45.12	67.58	67.22	76	77.93	62.47	69.5	79.17	81.48	1152.06
5.	76.69	47.16	67.19	70.36	0	33.27	49.82	67.51	61.03	70.6	68.09	61.13	80.82	70.94	83.4	96.15	80.52	80.13	72.71	95.93	1338.45
6.	74.59	43.09	39.49	65.88	33.27	0	47.12	73.68	46.65	58.37	61.76	58.94	80.28	73.68	82.62	94.05	79.38	80.08	73.29	93.71	1265.94
7.	66.34	56.24	63.76	57.94	49.82	47.12	0	70.79	68.78	68.65	72.76	63.05	57.64	57.85	74.09	78.39	76.49	68.15	61.62	91.76	1258.26
8.	57.36	75.01	63.68	55.44	67.51	73.68	70.79	0	29.45	57.37	42.72	52.55	77.9	71.38	83.55	84.7	72.56	71.52	87.44	89.38	1291.98
9.	60.33	54.08	42.51	49.67	61.03	46.65	68.78	29.45	0	36.31	32.88	47.77	73.1	70.26	78.39	83.61	66.09	77.95	82.27	88.24	1158.37
10.	62.35	55.92	51.14	40.47	70.6	58.37	68.65	57.37	36.31	0	43.24	53.85	67.62	64.78	75.86	78.73	62.26	77.91	59.92	82.44	1177.79
11.	51.22	70.15	54.28	43.97	68.09	61.76	72.76	42.72	32.88	43.24	0	51.2	67.62	62.13	78.95	81.8	60.38	76.15	83.01	75.06	1188.38
12.	54.16	61.27	47.43	45.12	61.13	58.94	63.05	52.55	47.77	53.85	51.2	0	41.84	68.68	50.72	79.48	63.2	69.47	68.36	76.27	1126.5
13.	71.51	74.49	73.06	67.58	80.82	80.28	57.64	77.9	73.1	67.62	67.62	41.84	0	61.29	52.06	84.95	53.51	74.68	70.95	74.64	1318.52
14.	70.15	78.21	76.75	67.22	70.94	73.68	57.85	71.38	70.26	64.78	62.13	68.68	61.29	0	64.88	65.17	37.92	50.77	65.34	74.31	1265.72
15.	78.24	80.68	78.98	76	83.4	82.62	74.09	83.55	78.39	75.86	78.95	50.72	52.06	64.88	0	76.23	55.09	67.4	56.64	54.2	1362.98
16.	79.36	91.92	90.5	77.93	96.15	94.05	78.39	84.7	83.61	78.73	81.8	79.48	84.95	65.17	76.23	0	87.66	86.06	77.73	51.55	1561.97
17.	72.99	72.03	70.55	62.47	80.52	79.38	76.49	72.56	66.09	62.26	60.38	63.2	53.51	37.92	55.09	87.66	0	52.34	79.14	58.4	1279.97
18.	54.49	75.67	73.92	69.5	80.13	80.08	68.15	71.52	77.95	77.91	76.15	69.47	74.68	50.77	67.4	86.06	52.34	0	74.16	72.74	1371.09
19.	82.21	69.21	79.25	79.17	72.71	73.29	61.62	87.44	82.27	59.92	83.01	68.36	70.95	65.34	56.64	77.73	79.14	74.16	0	82.46	1423.89
20.	77.25	91.46	89.97	81.48	95.93	93.71	91.76	89.38	88.24	82.44	75.06	76.27	74.64	74.31	54.2	51.55	58.4	72.74	82.46	0	1521.26

Note:Stands 5 and 10 are endpoints for x-axis; and Stands 2 and 12 are endpoints for y-axis.

CHAPTER

12

Community Dynamics: Succession

The gradual change in species composition and processes of communities over time is known as ecological succession or community development. Understanding the process, rates and pattern of ecological succession is important for the management of ecosystems and for understanding vegetation potential and dynamic changes in the landscapes. Community development is a normal natural process, which occurs due to continuing struggle between species with different adaptations for physical factors (light and space) and resources for survival (food and nutrients). Ecological succession can result in a change in the relative abundance of dominant species or a complete replacement of species from immature, unstable communities to more mature, self sustaining communities over a period of time. From the stand point of time, ecosystem and community changes can be divided into two types: (i) changes occurring over geological time scale (million years), and (ii) those occurring over medium time scale, say in 1-1000 years.

Community changes occurring over geologic time period are called Palaeo-ecological changes. These changes are synthesised on the basis of fossil records, e.g., of leaves, twigs, cones, pollens and seeds. For example, fossil evidences indicate that in Rajasthan Desert of India the vegetation during the Tertiary period consisted largely of tree species ascribed to humid environment. Later on, in response to drier climate the desert plants prevailed in this area.

The time required for succession to proceed varies with the nature of climate and initial quality of soil. On the basis of radiocarbon dating method, it may take for the Beech-maple forest on the Michigan sand dunes a time period of about 1000yrs (Olson, 1958). The succession is very slow on a bare rock. Climax stages of western grasslands are reached in 20-40 years of secondary succession. In humid tropics, forest communities may regain their climax vegetation within 100years after clear cutting.

The term succession is generally used for directional and continuous change in species composition and abundance following disturbance of a site (McCook, 1994) or creation of a bare area. The process usually leads to a sequence of communities before a stable and steady- state community develops. The directional changes refer to an overall increase in biomass and change in the life-form and life-history characters of species.

Even a severe site like an exposed rock, devoid of soil and water, does not remain without plants for long. As time proceeds patches of lichens appear, then follow successive plant communities as those of mosses and Jungermanniales, annual herbs, perennial herbs, shrubs and finally trees. During this process of changing vegetation the environment also changes. For example, patches of lichens disintegrate the rock surface, mechanically and chemically, and collect loose debris forming a thin layer of soil. This makes the growth of mosses possible. The mosses modify the physical environment further by adding more organic matter, making the soil layer thicker and improving moisture conditions so that annual herbs are now able to grow. The environment continues to change in a direction that makes the entry of perennial tall herbs and shrubs possible.

12.1 TYPES OF SUCCESSION

Succession occurring on sites previously unoccupied by vegetation such as on bare land created by volcanic activity, glaciers, landslides, coastal dune formation, and drainage of lakes, is termed *primary succession*. Succession that occurs on sites previously occupied by plants but now devegetated is called *secondary succession*. Storms, fire, tree cutting and cultivation are common causes of destruction of vegetation. These disturbances remove or kill most live aboveground biomass but leave some soil organic matter and plants or plant propagules in place. The severity of disturbance is probably the major factor determining the rate and trajectory of vegetation development after disturbance.

In both, succession begins with a group of pioneer plants, which appear first on the site and are adapted to grow in harsh environments. Another group of species better adapted to the changed environment subsequently replaces this group. The process of replacement of groups of plants or communities is continued until the area comes under the control of a community, which is able to perpetuate itself indefinitely. The first community to appear is the *pioneer community* and the last is called the climax. The whole sequence of the temporary communities that occurs prior to the climax community is termed *sere*. Floristically or structurally distinct segments of a *sere* are termed *seral stages*, such as grass stage, forb stage, etc.; or herb stage, shrub stage, tree stage, etc. Succession initiated with the establishment of pioneers on surfaces, which have scarcity of water, such as rocks, is termed *xerarch succession*, and that initiated with the establishment of pioneers on wet surfaces, such as, ponds is termed *hydrarch succession*. The successional stages of the former are called *xeroseres* and those of the latter *hydroseres*. Succession may result from two main types of processes, i.e., autogenic changes, which are self-regulatory, brought about by the plants themselves at the site, and the allogenic changes which occur due to external disturbance factors (e.g., fire and human activity).

Successional changes brought about by the organisms themselves constitute autogenic succession and those brought about by environmental factors and not affected by organisms constitute allogenic succession. Nitrogen enrichment in soil by nitrogen fixing plants, shade cast by tree canopy and change in pH by detritus are examples of autogenic inputs, while filling up of a pond due to human activity is an example of allogenic input. However, no site would be free of any external disturbance. Therefore, no succession would be purely autogenic. The normal succession starting with the green plants or lichens or blue green algae is called autotrophic succession. The succession occurring in a hypertrophic lake or a stream with heavy input of detritus, where heterotrophs dominate over autotrophs, is called heterotrophic succession.

On the basis of ratio of rate of gross photosynthesis (GP) to rate of community respiration (CR), succession can thus be divided into *autotrophic succession* and *heterotrophic succession*. At the beginning of succession GP exceeds CR in the former, while in the latter CR exceeds GP. The heterotrophic succession occurs in unusual situations, such as hypertrophic lakes where heterotrophs prevail over autotrophs and majority of energy input is through detritus. In any case, the GP/CR ratio narrow down and tends to approach 1 as succession proceeds.

12.2 MECHANISM OF SUCCESSION

Classical Approach: Clementsian Hypothesis

Among the first and definitely most influential ecologists to suggest the mechanism of succession was F.E. Clements. He recognized three major phases and six basic processes of succession:

Phases	Processes	Changes
1. Initiation	Nudation	The creation of a bare substrate
	Migration	Immigration of species to the new area
	Ecesis	The establishment of plants through germination, growth and reproduction
	Reaction	The modification of habitat by organisms
2. Continuation	Competition	Intra- and inter-specific competition between organisms
3. Termination	Stabilization	The community in equilibrium with the habitat

Nudation refers to the formation of a bare area, such as due to the appearance of a raw geological formation (*primary bare area*) or due to destruction of existing vegetation (*secondary bare area*). The primary bare areas are mostly produced by physiographic processes, such as erosion by water, wind, gravity or glaciers; deposition by above agents and volcanism; emergence of new land surface; submergence of land surface; and production of unstable strata. The secondary bare areas are mostly produced by climatic phenomena (drought, wind, and lightning) or by biotic agents such as deforestation, cultivation, grazing, insect outbreaks, etc. (Daubenmire, 1968b). Migration refers to the arrival in the bare area of *disseminules* (live detached reproductive units, such as seeds) or *propagules* (reproductive units attached with parent plants such as stolon) of plants growing in the adjacent area. In the secondary bare area, the disseminules or propagules may already be present as buried seeds or rootstocks – these are termed *residuals*. *Ecesis* refers to germination, establishment (development of immigrants to maturity) and multiplication (addition of new plants from the progeny of immigrants). The increased size and number of plants now lead to *competition* for available resources, such as light, water and nutrients. All the activities of plants affect the environment. For example, plants dampen the amplitude of moisture and temperature variations, they increase the organic matter in the soil, which in turn, modifies the porosity of soil and increases its capacity to hold water. Such autogenic effects by the plants on the habitat are termed reaction. The changed habitat, as a consequence of reaction, now favours a different set of migrants and may become unsuitable for the early immigrants. With the arrival of new migrants all of the above steps are repeated. This mechanistic repetition of the above processes will continue until the species admixture is able to maintain itself (*stabilization*) as a climax (dynamic *steady state*) with no further detectable change in species composition with time.

The Clementsian hypothesis (Clements, 1916) considered climax or terminal community to be a super-organism and succession the embryonic development of that organism (see chapter 10). “As an organism, climax arises, grows, matures and dies. The climax is capable of reproducing itself, repeating with essential fidelity stages of its development. That is, every time a climax is reproduced, the essential steps are similar”. Thus, succession is linear, orderly, predictable and developmental. This is called the Holistic view.

In the above view, the situation is similar to a relay race. The pioneer community (like a runner at start) hands over the site after a period of time to the next seral community, as the runner at the start of a relay race hands over the baton to the next runner after covering a certain distance (Fig. 12.1a). This process is repeated till the site is handed over to the climax community (as is stick handed over to the last runner). Therefore, this view is also termed *relay floristics* (Egler, 1954). According to this model, there is a complete replacement of one group of species by the other over a period of time.

Initial Floristic Composition

Egler (1954) proposed a contrasting model, called *initial floristic composition* (IFC) model (although essentially for secondary succession). In this, the site may contain seeds or vegetative material of the pioneer as well as that of several of seral stages. Species develop largely from these materials with or without the invasion of additional species. Although most of the species are present right from the beginning of succession (initial floristic pool), groups of species rise and fall in importance synchronously in sequence during the course of succession because of their variable growth rates (Fig.12.1*b*). Even the group of species that constitute the terminal stage of succession may have been present right from the beginning; they grow slowly and dominate the site later, replacing the smaller earlier quick growing species.

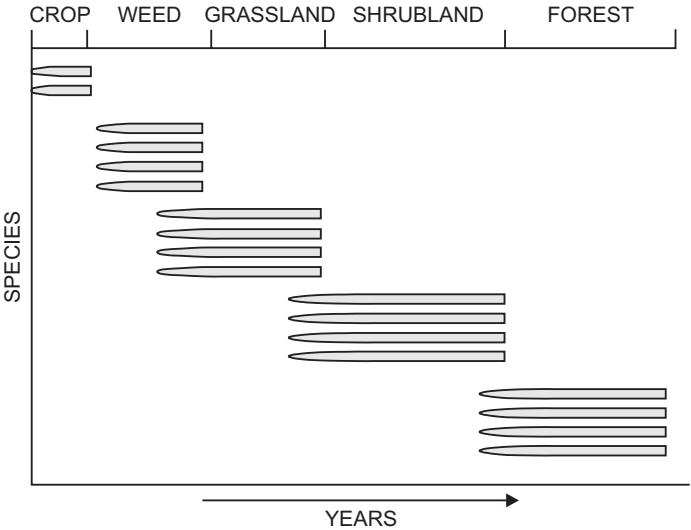


Fig. 12.1. (a) Relay floristics model; the site is handed over by one seral community to the other until a climax community developed.

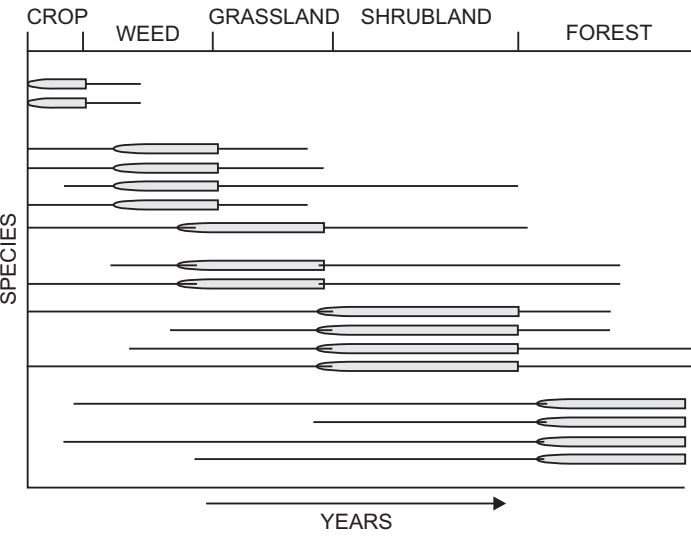


Fig. 12.1. (b) Initial floristics composition view predicts that species of late successional stages can be present on the site from the beginning but attain dominance only later because of slow growth. (Egler, 1952-1954).

Disequilibrium Theories of Succession

The Gleason's (1917, 1926) individualistic concept (also called reductionist approach) challenged the super-organism concept of succession proposed by Clements. According to Gleason, succession "is an extraordinarily mobile phenomenon whose processes are not to be stated as fixed laws, but only as general principles of exceedingly broad nature and whose results need not and frequently do not occur in any definite predictable way". Succession occurs due to individual response of different species. But not until 1970s his ideas were appreciated and received acceptance. The disequilibrium theories of succession emphasise that:

- There are many possible stable vegetation community end points
- All communities are unique, and will contain internal species variations due to competition as well as patch dynamics on a local scale.
- Ecosystems are not "super-organisms".
- There are no long-lasting climatic climaxes - changes always occur.
- Communities are ever changing, temporary alliances, just happen to grow together.
- Succession is multidirectional, as there are different pathways and outcomes of succession even on sand dunes and glacial forelands communities.
- At any stage of seral development, there are differences between sites because of stochastic processes (chance) i.e., due to disturbance, competition or new arrivals, etc.

Evidence for the disequilibrium theories of succession come from modern studies of recolonisation following 'sterilization' of landscapes, e.g., Krakatua in Indonesia (Whittaker et al., 1989) and palaeoecological studies of infilled lakes (Walkér, 1989). The disturbance is a feature of all vegetation communities. For example, strong winds may topple trees or fires may clear large areas of forest creating gaps through which any species may enter purely by chance. Once a species enters there is a race to maintain the lead. The disequilibrium view of succession states that there is no lasting climax. Individual species make temporary alliances so that succession may run in any or several directions. The Krakatau islands in Indonesia seem to support this theory (Whittaker et al., 1989). The eruption of the volcano sterilized the three surrounding islands in 1883 and a new island appeared in 1930. About 150 years have passed since the beginning of the observation of vegetation succession on the islands. The vegetation change on all the three islands is patchy because of tree fall, fires and diseases that affect the vegetation. The opportunities for dispersal of seedlings, etc. are also different on the islands. The islands changed pace at different times with multidirectional change rather than Clement's unidirectional model (Whittaker et al., 1989).

As shown in Fig. 12.2. the succession of *Nelumbo* community may proceed in several directions depending upon unpredictable circumstances. van der Valk (1982) while studying succession in wetlands found that the succession from one to another stage may not be as straightforward and predictable as described above. He considered following three life history features of greater significance related to establishment of species in changed environment condition as succession proceeds:

(i) Life span: (i) Annuals, referred to as **A** species; (ii) perennial, referred to as **P** species; and (iii) vegetative growth, referred to as **V** species.

(ii) Propagule longevity: (i) Species with long lived propagules (usually seeds) already present in substrates, that enable them to establish whenever favourable environment is available, referred to as seed bank species (**S**); and (ii) species with short lived propagules usually with less than one year longevity that depend upon dispersal mechanism of propagule to reach a site at favourable time, referred to as dispersal dependent species (**D**).

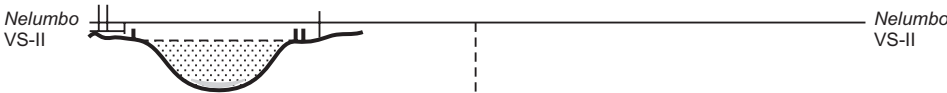
(iii) **Propagule establishment requirements:** (i) species requiring exposed substratum (i.e., during drawdown), referred to as type *I* species; and (ii) species requiring flooded or submerged condition, referred to as type *II* species.

All possible combinations of these features will produce 12 life history types. Of which, the four annual types are shown as AS-I, AS-II, AD-I, AD-II; the four perennial types are shown as PS-I, PS-II, PD-I, PD-II; and the four vegetatively reproducing types are shown as VS-I, VS-II, VD-I, VD-II. AS-I, as an example, is an annual species with long lived propagules present in seed bank of the site that can establish the species whenever exposed substratum is available.

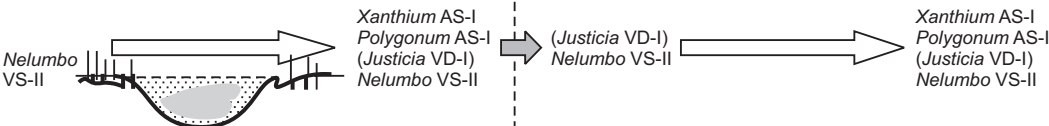
This model is illustrated by taking an example of *Nelumbo* community in a shallow lake (see Fig. 12.2.) For the sake of simplicity, in addition to *Nelumbo pentapetala* (VS-II), i.e., a vegetatively propagating seed bank species requiring flooded condition for establishment, only three representative species viz. *Justicia americana*, *Xanthium americana* and *Polygonum pennsylvanicum* are considered.

The response of *Nelumbo* community to possible different water level regimes even for as short a period as two years can be several: (i) If water level remains unaltered, *Nelumbo* will remain the sole member of the community. (ii) A summer decline in water level, exposing the substratum will secure the establishment of *Xanthium* and *Polygonum*, which are annual, seed bank species requiring exposed substratum; their propagule were able to reach the site at opportune time. (iii) A water level fluctuation, involving exposition of substratum and subsequent flooding will eliminate *Xanthium* as it fails to withstand reflooding.

(A) Constant Water Level



(B) Summer Decline In water Level



(C) Fluctuating Summer Water Level

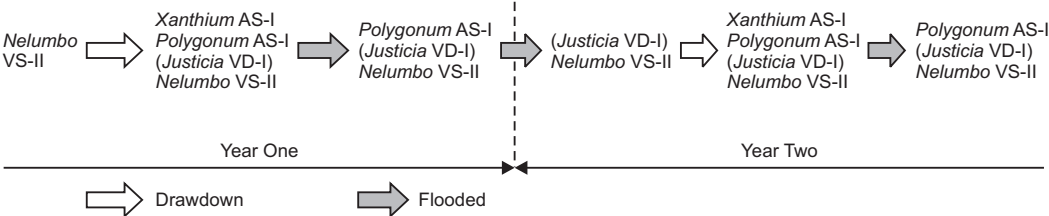


Fig. 12.2. Predicted successional sequence under three water level regimes in a *Nelumbo* community in a shallow pond.

Life History types: A species = Annuals, and V species = Perennials with vegetative growth. 'S' stands for seed bank species, and 'D' stands for dispersal dependent species; Annual types: AS-I; Vegetative reproducing perennial type: VS-I, VS-II, VD-I.

Connell and Slatyer's Models

Connell and Slatyer (1977) proposed three alternative models of succession, viz., Facilitation or 'relay floristics', Tolerance and Inhibition models.

In brief, these models suggest that early colonizers of a disturbed site either make the condition more favourable for later ones (facilitation), have no effect (tolerance model) or inhibit them (inhibition model). Here, the late and early species refer to the type of species, not to the timing of arrival on the disturbed site. The early species have certain characters that enable them to colonise a bare site. In the facilitation pathway, only a few early successional species can establish themselves on a bare area, while in the other two pathways any species (able to survive as adult) can establish.

	(a) Facilitation model	(b) Tolerance model	(c) Inhibition model
	Which kinds of the species that arrive in the bare area establish themselves?	Any species that survive as adults.	As in tolerance model
What is the effect of the modification of environment by early occupants on subsequent recruitments?	The modified environment is less suitable for subsequent recruitment of ESS and more suitable for the recruitment of Late Successional Species (LSS); thus ESS facilitate the recruitment of LSS.	The modified environment is less suitable for subsequent recruitment of ESS, but has little or no effect on the subsequent recruitment of LSS.	The modified environment is less suitable for the subsequent recruitment of both ESS and LSS.
What is the effect of the presence of the individuals of ESS on subsequent colonizers?	Since the modified environment is less suitable for ESS, and also due to competition from LSS, the ESS are eliminated.	Juveniles of LSS (already present or that invade) attain maturity and multiply despite the presence of healthy individuals of ESS, thus LSS tolerate the presence of ESS and eventually the latter are eliminated.	Early colonists exclude or suppress subsequent colonists of all kind of species as long as their individuals persist undamaged and/or continue to regenerate vegetatively.
What is the continuation of the above sequence?	The sequence of species replacement continues until resident species no longer facilitate the invasion of other species.	The sequence continues until no species is there that can invade and grow in the presence of the residents.	As above.

Fig. 12.3. Three models of the mechanisms producing the sequence of species in succession (Connell and Slatyer, 1977). In each, succession starts as disturbance opens a relatively large space (bare area), releasing resources.

Facilitation: The facilitation model suggests that the entry and growth of the later species is dependent upon the earlier species Fig 12.3a. The earlier species "prepare the ground" for arrival of the later species, in other words, they prepare conditions or the availability of resources (nutrients, water, and soil composition) that favour the arrival of species that replace them. Evidences for

facilitation model are based on the studies from the primary succession of the glacial foreland at Glacier Bay, Alaska, and sand dune succession along lake Michigan. This model follows Clement's hypothesis described earlier.

Tolerance:- The tolerance model suggests that a predictable sequence is produced by the existence of species having different strategies for exploiting resources (Fig 12.3b). The late successional species can invade and grow to maturity in the presence of those that preceded them. With passage of time, the earlier species will be excluded. Some old field secondary successional patterns support the tolerance model of succession.

Inhibition:-The inhibition pathway suggests that no species necessarily has competitive superiority over another and whichever colonises the bare area first can hold it against all comers (Fig 12.3c). Later colonists can invade and reach maturity only after the first occupants die, or are damaged. Studies indicate that all of the three models may be correct but applicable in different situations.

Unified Model of Succession

MacMahon (1980) has proposed a unified model to account for successional sequences in all situations (Fig. 12.4). In this model, boxes represent the states S_0 ----- S_3 of the plot over time. Cycling can occur back through process (e.g., deforestation). The S_0 represents a point where a plot has substrate available for colonization (i.e., bare area). Subsequent succession may be primary (if bare area was a primary bare area), or secondary (on secondary bare area). The only difference is that the secondary bare area will already have 'residuals' (seeds, spores, bulbs, etc.) remaining from the predisturbed state. As the residuals survive and endure the environment (environmental driver E in Fig.12.4) and have their effect on the environment (Reaction (R) in Fig.12.4), they are joined by a series of migrants from the biotic pool available from the adjacent area. Subsequent succession depends upon the species, which gets to the plot at this time. Species with highly vagile (efficient dispersal) seeds (e.g., annuals or herbaceous perennials) or species growing near the plot will reach it earlier than the distant species, or those with heavy seeds. If the size of the plot is small, late successional species may occupy the plot quickly and the succession is shortened. Timing of the creation of the bare area is also important. For example, if the bare area is created in a year of a good seed crop of a species, that species will have a greater chance of colonizing the bare area and of pre-empting the plot through shading out, or allelochemicals, than if the bare area was created in a low seed crop year. The processes of reaction and biotic interaction (not only competition but also processes such as mutualism, symbiosis, predator-prey relations, allelopathy) occur simultaneously. The net result is that a particular species will occur on the plot until such time as the nature of the plot (which also keeps on changing) violates one or more of its tolerance axes. There is also a constant barrage of disseminules (or propagules) reaching the plot. Each migrant undergoes through the same test of persistence. The process of reaction in which the occupying species change the nature of the plot-classically to a state where they themselves cannot survive – may or may not be true depending upon the cause of the creation of bare area and the species and environment involved. And thus the state of the plot will move from S_1 to S_2 to S_3 , etc. This approach combines classical Clementsian concept with that of Gleason's individualistic concept. As MacMahon (1980) puts in "Thus, processes are undoubtedly, with some alteration as Clements outlined them, but they are really working on the biota in a highly individualistic way".

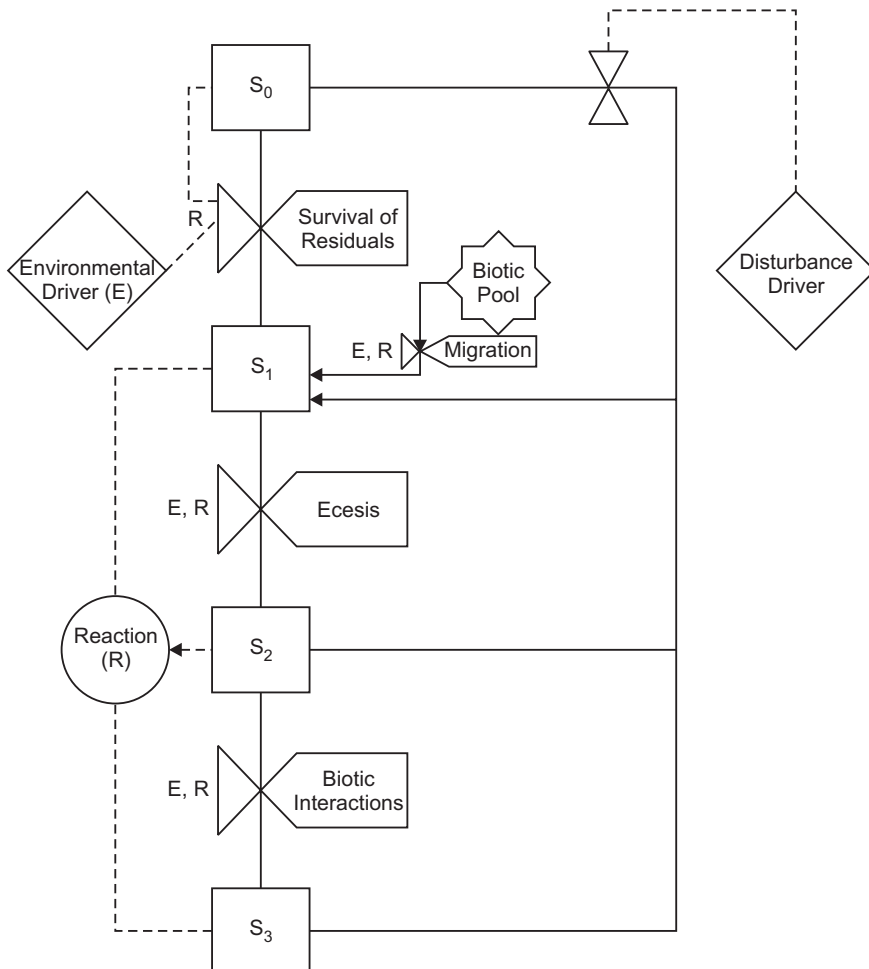


Fig. 12.4. A model of the change in the status of the components of a plot of ground overtime. Boxes are status of the plot at any instant. Diamonds are system drivers. The circle is an intermediate variable. Dashed arrows show information flows. Letters next to control gates replace dotted lines from that point to the control for the sake of graphic simplicity (MacMahon, 1980).

Three-Plant Strategy model and Competition

Grime's (1977) plant life-history strategies are described in chapter 8. Ruderals are members of early successional stage and competitive species increase in dominance as succession advances. In Grime's view a superior competitor would be a superior competitor for all resources, which is for light (aboveground) as well as nutrients (belowground). Here competition is defined as the tendency for neighbouring plants to utilize the same resources, and success in competition is largely a reflection of the capacity for resource capture. Competitive traits are part of life history characteristics of species, and competition is what competitors do best, i.e., exploitation of resources. Coupled with low sexual reproductive effort (lower proportional allocation of energy to seeds), the rapid growth of good competitors translates into a rapid development of absorptive surface area which enables them to pre-empt both above- and belowground resources. Species tolerant to shortages

of resources (stress) are unlikely to be good at resource capture. For they require contrasting life history characters.

Grime (1977) expanded his theory of the three life history traits to explain mechanism that drives secondary succession. According to him, the path of succession varies in relation to potential of productivity. In all paths, ruderals are established on a disturbed site, but subsequent species vary. In a favourable environment, with potential for high productivity, competitive species play a more important role and succession involves a higher species change. When conditions are stressful (infertile soil or/and any climate), competitive species play less important role and species changes are less. Stress tolerant species replace competitive species as stress increases.

Resource-Ratio Hypothesis

Tilman's (1980) resource ratio hypothesis (described in chapter 9) predicts that interspecific competition for resources and long-term pattern of supply of limiting resources are two key elements in succession, which result from a gradient through time in the relative availabilities of limiting resources. Plants that are superior competitors for one resource must be the inferior competitors for another resource that becomes limiting. When succession begins on a nitrogen-poor site, the species that has the lowest nitrogen requirement may outcompete other species with high nitrogen demands. As succession proceeds the soil may become rich in nitrogen, but light at the soil surface becomes limiting with increasing plant biomass. According to the resource ratio hypothesis, the early successional species which are superior competitors for nitrogen should be inferior competitor for light (that is, they require high light), while the late successional species would be superior competitor for light (low light requirement) and inferior competitor for nitrogen.

Because of trade-off, a good competitor for nutrients cannot be a good competitor for light, as they require contrasting traits – more biomass allocation to roots for more nutrient capture, and more allocation to shoots for light capture. This is the directional change in the supply of limiting resources that results in directional replacement of one species by the other.

12.3 CHANGES INVOLVED IN ECOLOGICAL SUCCESSION

As succession proceeds, changes occur not only in the biotic community but also in physical environment and overall characteristics of ecosystems (Margalef, 1968; Odum, 1969) in a holistic manner. That is why Odum considered succession as ecosystem development. On the other hand, succession is considered an ecocline in time (Whittaker, 1975) and the pioneer and climax communities make the end points of the ecocline. Thus the continuum concept of communities is the basis of Whittaker's view of succession.

Odum's (1969) list of characters that change as succession proceeds are given in Table 12.1 with some modifications.

Table 12.1. Differences among early and late stages of succession. Based on Odum (1969) and Barbour et al. (1980).

Earlier stages		Later stages
1. Gross production (GP)/community respiration (R) ratio	Greater or less than 1	Near 1
2. Net community production	High	Low
3. Biomass	Low	High
4. Size of organisms	Small	Large
5. Strategies	<i>r</i> -strategies with the preponderance of ruderals	<i>K</i> -strategies, with the preponderance of competitors
6. Food chain	Linear	Web like
7. Role of detritus	Low	High
8. Mineral cycles	Open (more leakages) with rapid exchange between organisms and environment	Closed (less leak-ages) with slow exchanges between organisms and environment
9. Intrabiotic control of nutrients	Low	High
10. Species diversity:	Low	High
11. Internal symbiosis	Low	High
12. Stability:		
Resistance	Low	High
Resilience	High	Low
13. Niche specialization	Broad	Narrow

The pioneer species are usually strong light demanders, produce an abundance of small seeds, having capability to disperse widely and to withstand unpredictable (e.g., diurnal and seasonal variations in temperature and moisture) environment on barren and open sites by developing prolonged dormancy and ability to germinate in wider range of environments. In the late successional stage, the environment is more predictable, competition is for density-related factors and individuals allocate more energy to non-reproductive activities. The former species are called *r*-strategists and the latter *K*-strategists. A comprehensive account of *r*- and *K*-strategy is given in chapter 8.

As the stress conditions ameliorate with the growth of plants and the environment becomes relatively more stable and predictable, perennial species that can store biomass year-after-year, and those that are specialist to a narrow range of environmental conditions attain preponderance. Much of their production is diverted to storage and maintenance. Such species have high root-to-shoot ratio, for example, a well-developed large root system is required to extract nutrients from the pool for which several species are competing.

During succession, since species that replace those of the earlier stages are long-lived, their dimensions, massiveness and biomass would also increase. Therefore, community biomass increases as succession progresses. For example, in the Brookhaven Oak-Pine forest, New York, the biomass is around 1 kg m⁻² (dry weight) in the annual and meadow stages and 30-40 kg m⁻² in fully mature forest (cited in Whittaker, 1975). With advancing succession, the biomass: production (B/P) ratio tends to increase from 1.0 in the annual stage, and 2-4 in the meadow stage, to 4-7 in shrub stage, 10 in 55-years old forest and probably 25-35 in the mature forest. Accordingly, the amount of standing biomass supported by the available energy flow should increase as succession proceeds, and turnover of organic matter would be relatively slower in later stages of succession.

Compared to pioneer stages, the stock of nutrients in biomass is greater at late successional stages mainly because of greater standing crop. In other words, as succession progresses, the intrabiotic/extrabiotic ratio increases with regard to nutrients.

Whether or not diversity continues to increase through the entire course of succession remains controversial. According to Margalef's (1968) postulations, diversity will tend to peak during the early or middle stages of succession and then would decrease in climax. Such a pattern was found in a forest succession at Brookhaven, New York. Auclair and Goff (1971) showed that trends of diversity are not the same on different sites. At intermediate disturbance level, species diversity is suggested to peak. Huston and Smith (1987) emphasize the life-history based explanation of succession and focus on individual plant interactions. On xeric sites diversity increases through succession, on mesic sites the maximum diversity is attained much earlier than mature stages, and on intermediate sites it approaches an asymptote late in succession.

In a well-lighted early successional environment multilayered trees do better, whereas in the low light condition of late successional stage monolayer species are favoured (Fig. 12.5). In the multi-layered condition, leaves of upper branches shade leaves of lower branches, which is not a useful character in the low light environment. A common observation is that within an individual tree, branches growing in shade or near the ground often assume a monolayer habit (e.g., in *Fagus*, *Abies* and *Taxus*), whereas branches from the crown show a multilayered form.

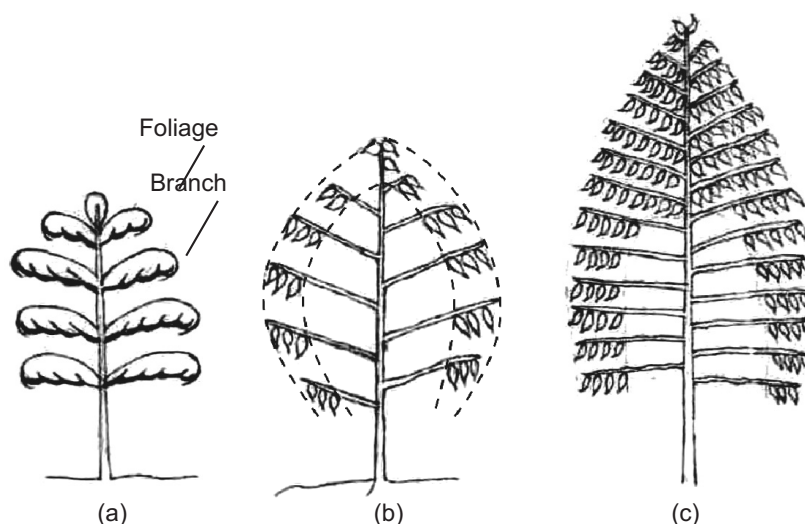


Fig. 12.5. (a) Multi-layered, (b) Mono-layered, (c) Multilayered at the top where sun light is in abundance and with lower branches which has been in shade showing mono-layered condition.

Early successional shrubs show rapid lateral spread of a crown close to the ground. Since little supporting tissue (strong stem) is required, a dense horizontal cover develops at relatively low costs. However, this growth form bears the increased risk of damage from ground-browsers. That is why, a number of early successional shrubs (e.g., *Rubus*, *Prunus*, *Rosa*, *Crataegus*) have thorns, which are absent in late successional plants with stout and tall stems, which keep branches at heights unapproachable to browsing mammals of the ground (Fig. 12.7). The early successional shrubs are mainly “explorative”, producing runners and suckers, which enable them to expand rapidly on the ground. In contrast, the late successional species are “exploitative” of resources, many of which become increasingly limiting as succession proceeds. (Kuppers, 1990). The cost of gaining

space by growth of canopies, expressed as volume/dry matter or volume/N mass in tissues strongly decreases from early to late succession (Fig.12.6). As light availability decreases with succession, the photosynthetic efficiency, i.e., the maximum assimilation rate of sun leaves declines, and so does their carbon budget.

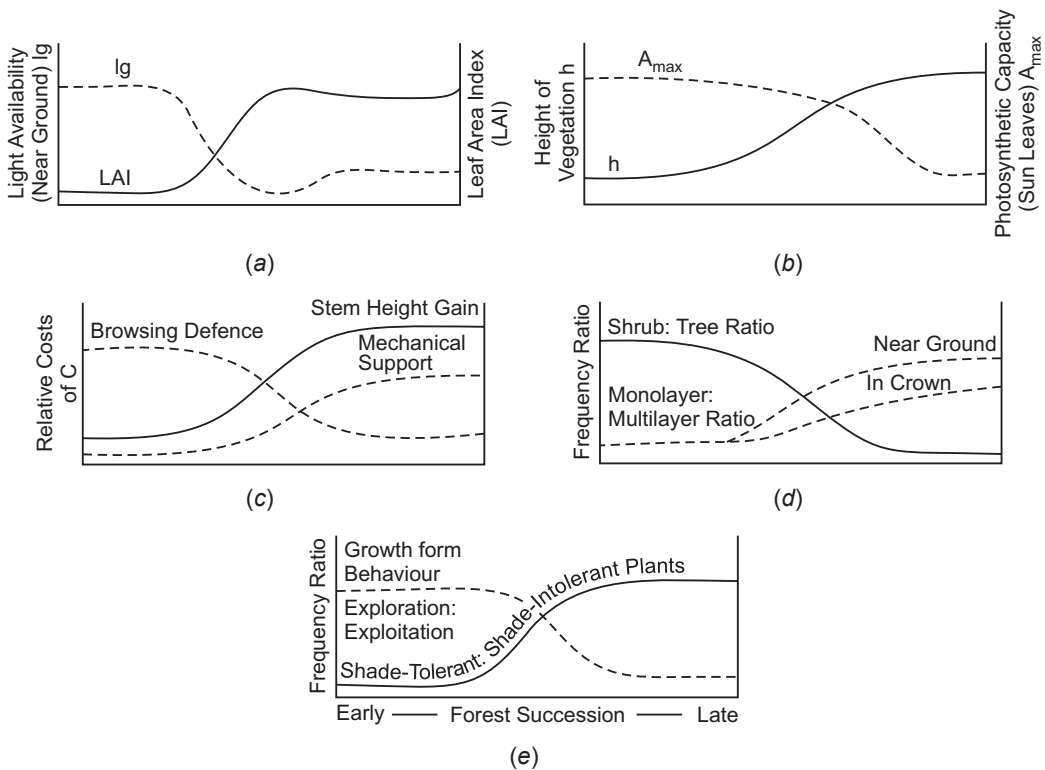


Fig. 12.6. Some biotic and abiotic features that change with progression of secondary succession where one growth form is replaced by another (e.g., forest succession in central Europe starting after abandonment of agriculture). Relative cost in (c) is in terms of cost of carbon per unit gain in features described. Browsing defence in (c) considers ground browsing animals only. In (e) the ratio of growth form with exploration habit (runners, suckers) to those with exploitative habit decrease as succession progresses, whereas of shade-tolerant to shade intolerant increases (from Kuppers, 1990).

In pioneer stages, organisms are linked together through food in relatively simple and linear manner. In the late stages, food chains become complex webs, several organisms occupy more than one trophic level, inter-related through several food links. Along this course, there is an increase in the role of detritus pathway of energy flow, as there occurs the development of indigestible supporting tissues, such as large trunk and branches (with tissues rich in lignin and cellulose). The feedback control between plants and herbivores acts against the grazing pathway of energy flow.

Differences in nutrient cycles between young and mature ecosystems are also discernible. Because short-lived individuals of young stage return most nutrients contained in them to decomposers and soil within a year or so, the rate of nutrient exchanges between biotic and abiotic components is rapid. Nutrient cycling changes through early primary succession as nitrogen fixers establish and add nitrogen to the ecosystem. Other elements cycle in proportion to the cycling of nitrogen. In contrast, in mature systems, the nutrients are locked up in persistent parts such as trunk and branch and only a small fraction returns to soil annually through leaf fall; and other shedding parts (flowers, fruits, barks, small twigs, etc.). Recycling within the ecosystem is potentially strongest

in mid-succession, when rates of nutrient mineralization limit the rates of uptake by vegetation.

As succession progresses, interactions among the components of ecosystem grow, and forms of symbiosis develop in such unrelated organisms as coelenterates and algae (in coral reefs), and between mycorrhizae and trees in forests. These forms of symbiosis tighten the nutrient cycle and hence stability of ecosystems.

The time taken in the recovery to original state after disturbance is shorter (i.e., greater resilience) for the early successional community compared to the late successional community (lower resilience). For example, in a forest succession, the early successional stage may be recovered in few years after clear cutting of vegetation, as it may consist primarily of annuals. But a late successional community, where tree species dominate may take 100 to 1000 years to attain the original species composition.

During primary succession, remarkable changes occur in the soil : (i) soil is accumulated, (ii) organic matter is added by the vegetation, (iii) water-holding capacity increases, (iv) soil porosity and aeration increases, (v) bulk density decreases, (vi) water run off decreases and infiltration increases, (vii) more of the plant available nutrients are released, and (viii) soil profile develops.

12.4 CONCEPT OF CLIMAX

The climax community is in equilibrium with its environment and represents a steady state of species composition, community structure and energy flow. However, the steady state indicates the dynamic nature of climaxes with changes occurring in the community continuously. These changes tend to perpetuate the community rather than alter it. The climax community has the following characteristics:

- The climax community is able to tolerate its own reactions.
- The climax tends to be mesic for the climate in which it occurs.
- There is a high degree of biological organization due to complex food webs, more mutualism and stratification.
- Mature communities generally have high species diversity.
- The resistance of climax communities is high but resilience is low, therefore, they can be considered fragile.

Three major theories to define, identify and interpret the climax communities are as follows:

Monoclimax Theory

Clements' assumption was that all non-climax vegetation would eventually develop into climax vegetation since it is not in equilibrium with the climate. In other words, in a given area all successions beginning on such diverse habitats as ponds, bogs, old fields, rock surface, river valley silt, etc., will eventually converge towards a single climax community, which is decided by the climate of the area and is termed *climatic climax*. This is the monoclimax theory of Clements.

Clements suggested that the characteristic appearance of each climax is because of the dominants that control it. The dominants belong to the same life form, which represents the highest possible level that the climate can support. Each climax is direct expression of its climate: the climate is the cause; the climax is the effect, which in turn reacts upon the climate (Weaver and Clements, 1938). According to this interpretation, all communities, other than climatic climax are seral stages. This concept reduces, for example, all wetland vegetation to the status of seral communities, i.e., relatively short-lived stages in a hydrarch successional sequence terminating in a forest climax. But it has been noted that several wetlands are not short-lived. Properties of water bodies that matter most to plants viz. aeration, turbulence, temperature, turbidity are relatively little affected by

general climatic changes which affect much more the terrestrial plants. In fact, palynological studies have shown that wetland vegetation is often less changed over geologic time than the terrestrial vegetation in an area. In Clement's concept such wetlands whose development are arrested and held indefinitely (before climax) are sere climax.

In order to account for a variety of more or less stable communities but different from regional climatic climax, Clements proposed four other terms: (1) *subclimax* - succession gets arrested at a stage which persists for a long time in response to physiographic or edaphic factors, before being replaced by the climatic climax; (2) *Disclimax* - the vegetation replacing the true climax as a result of some persistent biotic disturbance in the environment, e.g., the grasslands in the Gangetic Plains which occur due to grazing in a deciduous forest climax; (3) *Preclimax* - slightly drier localities in a given climatic climax area may have a self-perpetuating community different from the climatic climax, e.g., some of the pine forests in the Himalaya; (4) *Post-climax* - slightly "better moisture" areas in the climatic climax region may support a different yet self-perpetuating community, e.g., *Terminalia arjuna* community growing near river banks in a dry deciduous forest climax.

Polyclimax Theory

Some ecologists realized that even when undisturbed by man, several types of stable communities could develop in an area. For example, a south facing-slope of a mountain in Northern Hemisphere may have one type of stable community because it would be warmer and less mesic, and a north facing other type in the same locality. Such variations may result from differences in rock types (e.g., granite vs. limestone) also. Different kinds of climaxes may develop even on the same parent material in the same locality. For example, in northern California on the same parent material (beach deposit and sandstone) stunted pygmy forest and giant red wood forest grow side by side. The stunted nature of pygmy forest is due to an iron-cemented hardpan located about 50 cm below the surface. The soil above the hardpan is excessively acidic (pH 2.8-3.9) and low in Ca, Mg, K and P. The impervious hardpan also restricts the deeper root penetration. On the other hand, soil below red wood forest is sandy. Thus due to the edaphic differences we find two climax communities differing not only in species composition but also in stature of vegetation. The influence of physical factor is so strong that successions occurring on two aforesaid sites cannot converge toward one climax type, even in indefinite period. Similarly, a wetland as mentioned earlier may have its own stable community, not the forest as Clements thought (van der Valk, 1982).

Thus, the convergence of successions in an area is only partial at the best; differences continue to exist among the climaxes that develop in different habitats. Tansley (1920) recognised the existence of a number of climax communities, forming a mosaic corresponding to the mosaic of habitats and suggested a polyclimax theory. However, this view too identified a climatic climax community that is the stable and most extensive in different habitats for an area. Tansley recognised following climax types:

- 1. Climatic climax:** Climax under normal conditions of climate, soil and topography and no disturbance.
- 2. Edaphic climax:** Substrate peculiarities are well pronounced to produce self-perpetuating vegetation, which is different from the climatic climax of the area.
- 3. Topographic climax:** Changes in topography enough to cause variant microclimates, each giving rise to self-perpetuating vegetation.
- 4. Fire climax:** Recurrent burning of vegetation eliminates fire-sensitive species and self-perpetuating vegetation develops.
- 5. Zootic climax:** Self-perpetuating community in response to zoological factors e.g. grazing gives rise to zootic climax of grassland.

Climatic, edaphic and topographic climaxes are primary climaxes, while fire and zootic climaxes are secondary disclimax.

The Climax Pattern Theory

The third concept is due to Whittaker (1953, 1967). According to the population pattern concept, the climax community represents a pattern of populations which corresponds to and changes with the pattern of environmental gradients to form ecoclines (gradients of communities and environments). However, in this concept also a climatic climax was suggested, this is the central or most extensive community, which comprises the largest share of habitats that are not special or extreme for the area.

12.5 TRANSIENT AND CYCLIC CHANGES

Non-cyclic changes

The transient successional sequences develop in any regional climax. Year to year changes involve random fluctuation about a form or average over large or small areas. Fluctuations involve changes in kind of species and importance values of constituent species. Any habitat, how much stable it may be, is never static. It is subject to fluctuations in environment in response to which changes in individuals and populations occur.

Cyclic climax

Within a community the loss of individuals due to other than catastrophic causes (such as fire and landslide) and growth in their place of individuals of species of usually the same community is called replacement change. Mosaic pattern of vegetation characterises any climax community where death of individuals alters the environment. When an individual of a given species within a community dies, it makes the way for the appearance of another individual, either of the same species or of another species belonging to the same community. After the death of an old tree several seedlings appear but normally only one reaches to maturity. Thus, normally the change is on individual basis, and continuation of such a change does not bring about large changes in community characters.

The cyclic replacement involves a series of vegetation and habitat changes through which community maintains and regenerates itself. It can be explained best by the example of *Calluna vulgaris*. When *Calluna* dies in a patch; *Cladonia sylvatica* grows and becomes dominant on the dead stems of *Calluna* stems, if protection is provided. The disintegration of *Cladonia* and creation of bare patches of exposed soil follow this and is the termination of the downgrade series. The upgrade series is

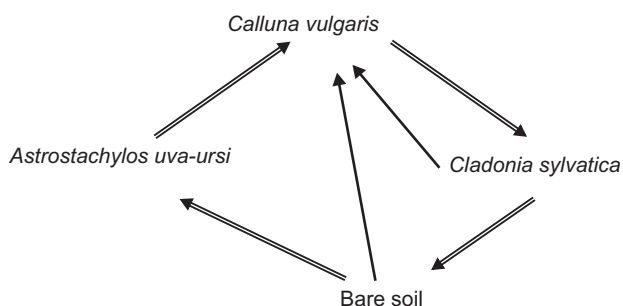


Fig. 12.7. A cyclic succession in the *Calluna* community in Scotland (Watt, 1947).

initiated on the bare patches with the appearance and subsequent occupation by *Arctostaphylos uva-ursi*. This stage is subsequently invaded by *Calluna*, which becomes dominant again completing the upgrade series. As shown in Fig. 12.7, other cycles may also occur. For example, *Cladonia* may be directly replaced with *Calluna* in an upgrade series.

Mosaic patterns are common in many communities where death of individuals alters the community. Treefalls open forest canopy and help to create patches of the habitat that are dry, hot and receive more sunlight. The forest opening is invaded by the early successional species. Thus tree fall due to the action of wind, frost or disease creates a shifting mosaic of seral stages within an otherwise climax forest community, as has been reported for the high altitude balsam fir forests (Sprugel and Bormann, 1981). In time, these patches of early successional are reclaimed by the late successional.

12.6 EXAMPLES OF SUCCESSION

Primary Succession

Rock outcrop succession (xerarch succession): A recently exposed barren rock exhibits a very severe environment. Being devoid of soil, the rock surfaces can undergo violent fluctuations of moisture and temperature within a day's time. Succession beginning on such a rock is primary and of xerach type.

Normally two types of micro-sites can be distinguished on a barren rock, the plane surface devoid of soil and the depressions with a trace of mineral soil (Fig.12.8), resulting from exfoliation. While the pioneer plants of the former are usually crustose lichens and some mosses, on the latter they are mainly mosses, such as *Bryum* and *Grimmia* and higher plants such as *Selaginella* and *Sedum*. Obviously, lichens can endure prolonged desiccation to the point of becoming brittle. In contrast, the leaves of higher plants die when their water content drops below about 50%. Having osmotic pressures unto 1000 atm, these lichens are able to imbibe moisture even from humid air and reactivate photosynthesis. But even humid air conditions like this do not last longer. Consequently active durations of photosynthesis are too short and far in-between the inactive photosynthesis durations. Because of this, lichens are among the slowest-growing organisms.

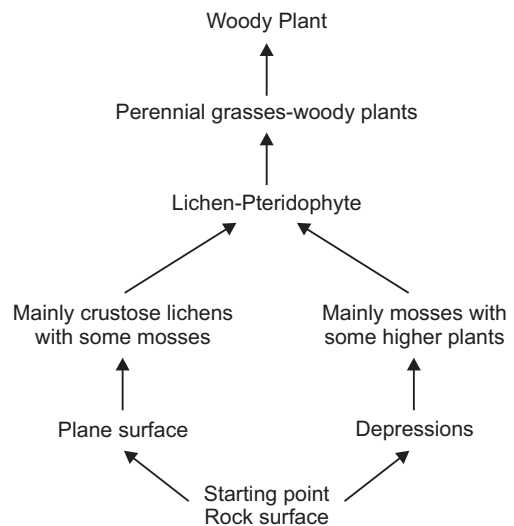


Fig. 12.8. Major successional stages starting in granite rock in eastern North Carolina (based on Oosting, 1956). The pioneers of plane surface are different from those on depressions.

Soil and a loose film of debris start accumulating on the rock surfaces, as a consequence of which moisture and nutrients in the habitat increase. These changes promote the establishment of several herbs, for example *Agrostis* and *Hypericum*, resulting in a lichen-annual herb stage. Further increase in moisture, nutrients and soil invites perennial grasses (e.g., *Andropogon* spp), and in crevices with deeper soil, woody plants may appear. In the condition of frequent and unpredictable changes in moisture and temperature mortality is very high and it takes several generations and considerably long time before the woody species (e.g., *Juniperus virginicus*, *Rhus* sp., *Vaccinium* sp.) can establish.

Succession from aquatic bodies (hydrarch): Misra (1946) has studied the hydrarch succession starting from a pond in the upper Gangetic plains (Fig. 12.9). In the pioneer phytoplanktonic

community, *Euglena*, *Chlamydomonas*, *Microcystis* and *Ulothrix* are common. The sedimentation and decay of plankton start the build up of an organic layer on bottom and subsequent regeneration of nutrients (*autogenic influences*). A deep water submerged associates (water is still clear) primarily consisting of *Chara*, *Nitella*, *Potamogeton* and *Myriophyllum* succeed the phytoplankton stage. The pond continues to become gradually shallower due to inputs of eroded soil, organic matter and other wastes generated in the catchment (*allogenic influences*). Now shallow water submerged associates consisting of *Hydrilla*, *Ceratophyllum*, etc. emerges.

Further inputs of organic matter and sediments make the lake still shallower, nutrient-rich and turbid. This encourages the colonization of more nutrient demanding plants, which have roots in substratum and floating leaves on water surface (e.g., *Nelumbo*, *Nymphaea*) or those macrophytes which float on water surface (e.g., *Trapa*). This floating leaf associates eliminates the submergent macrophytes by curtailing the light with their cover of floating leaves. This stage also diversifies the animal life by providing greater and varied habitats (for example, hosts of insects occupying undersurface of floating leaves).

As time proceeds, the pond becomes so shallow that seasonal fluctuation in water level intermittently exposes and floods the plants. Unable to withstand this condition, floating leaf plants give way to more tough textured plants which depend less on protection of water, namely the amphibious or emergent macrophytes. The common plants of this marsh are *Polygonum*, *Typha*, *Scripus*, etc. These plants utilize best of the aquatic and terrestrial worlds, i.e., water of the former and air and light of the latter. This is one of the reasons for their being highly productive. The amphibious condition supports animals which require submergent condition for one activity of life or at one stage of life (for example, larval stage) and aerial condition for another (for example, adult stage).

Further build up of substrate is due to the death and decay of dense root system and that of overall biomass. To this are added sediments reaching from outside pond. Gradually the pond becomes too shallow and is colonized by moist meadow associates consisting of species such as *Cynodon*, *Panicum* and *Cyperus*.

As the bottom of the pond further rises (being filled up by allogenic and autogenic influences) and assumes a level equivalent to that of the surrounding land, meadow grasses such as *Dichanthium* and *Bothriochloa* colonize it. Because of persistent grazing the succession is often arrested at this stage in the form of an almost stable meadow association (disclimax). However, if these biotic factors are withdrawn, the meadow association is replaced by a shrub stage, which eventually leads to the climax deciduous forest of the area.

Secondary Succession: Because of the rapidity of secondary succession and because of the disturbances, both natural (due to the landslides, by natural fire, storms etc.) and man-made

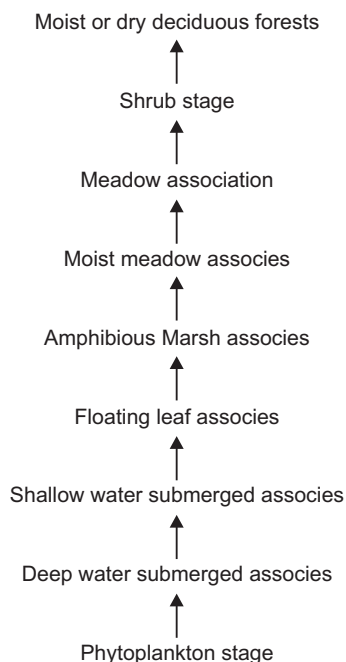


Fig. 12.9. Hydrarch succession in Gangetic Plain (after Misra, 1946). Succession at Meadow associations may be arrested due to grazing resulting into disclimax of grassland.

(road construction, man-managed fire, clear felling, mining etc.), that occur frequently, secondary succession has been studied by several workers. Apart from the disturbed sites, an abandoned crop field is a suitable starting point for secondary succession. To illustrate the pattern of secondary succession we cite some examples of forest succession after disturbance.

Forest secondary succession: In Oak forest (*Quercus leucotrichophora*) of Central Himalaya, subsequent to a landslide as secondary succession proceeds, fine soil contents, organic matter and nutrients, like N, P and K increase rapidly. Soil carbon and N reach the original forest levels by 45 years and various nutrients at 28 to 60 years (Fig.12.10.) Annual and biennial herbs colonize and dominate in the early stages, 1-6 years of age, and perennial herbs gain dominance over annuals at 15 years (Fig.12.11.) Species richness increases during secondary succession, and herb species content and cover reach the level of original oak forest at 15 and 35 years, respectively.

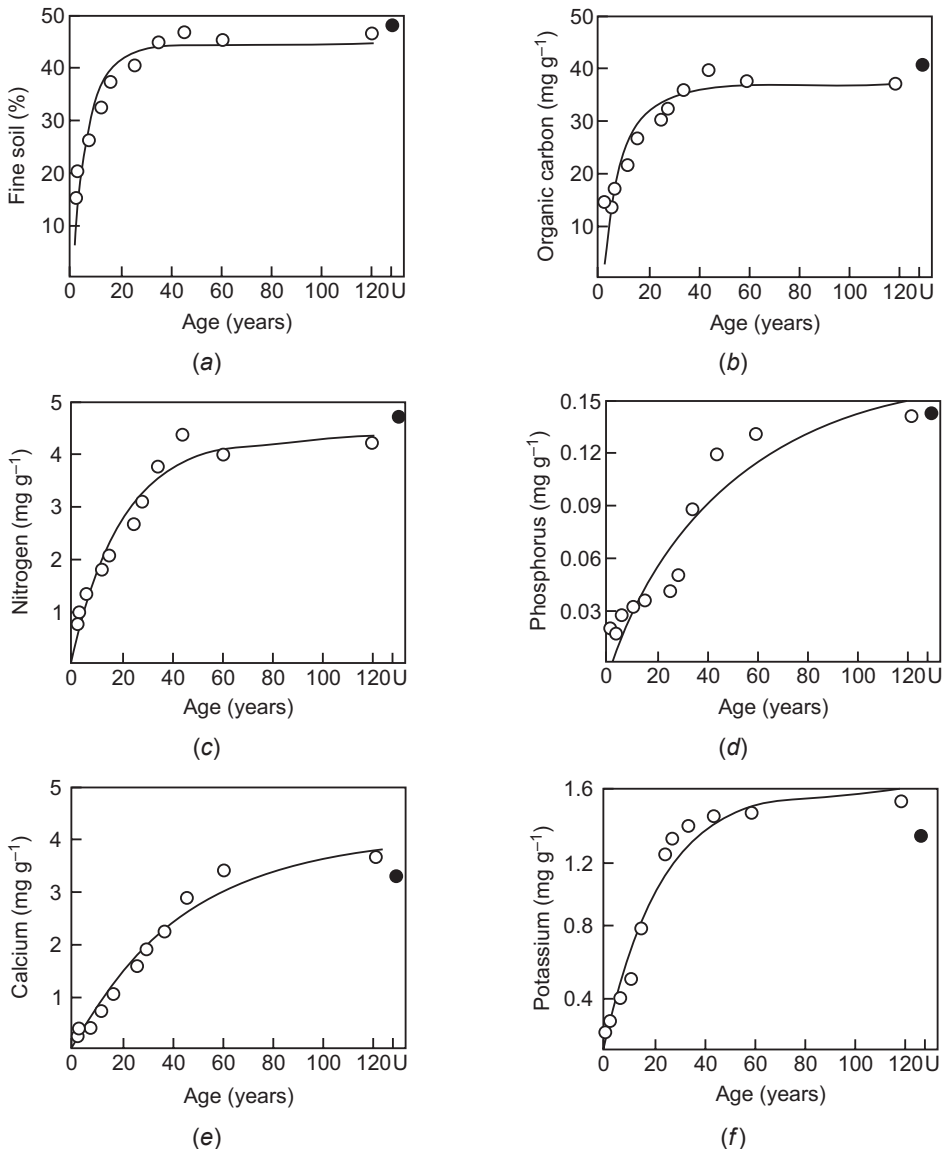


Fig. 12.10. Changes in fine soil, organic matter, and inorganic nutrients during secondary succession series in oak forest zone of central Himalaya. Age (in yrs) is in relation to a landslide event. Solid circles indicate the average values for undamaged sites(from Reddy and Singh,1993).

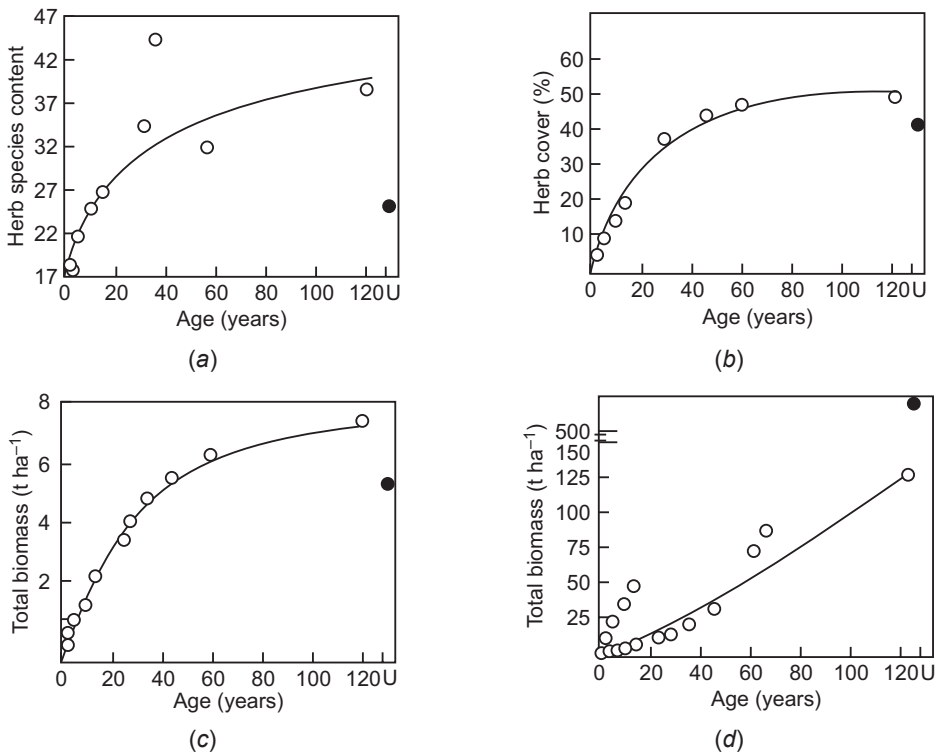


Fig. 12.11. Changes in (a) herbs species content, (b) herb layer, (c) total herb biomass, (d) total stand biomass during succession in oak forest zone. Solid circles indicate the average values for undamaged sites (from Reddy and Singh, 1993).

A number of leguminous shrubs and root nodule bearing tree, *Alnus nepalensis* (root nodules contain nitrogen fixer Actinomycete, *Frankia*) occur at different stages of succession, and enhance the soil nitrogen content. Shrubs, which first, appear at 11 years of age increase in species content and cover up to 35 years. Tree seedlings of most species are seen as early as 10-15 years after landslide.

A phase of rapid increase in herb biomass corresponds to a phase of rapid increase in soil nitrogen. The tree biomass and total biomass of vegetation increase with age throughout the succession.

Secondary succession in an alpine Meadow. In the Central Himalayan alpine meadows, vegetation consists of short-stemmed perennial herbs. Sites laid bare as a result of night camping of sheep provide good opportunity to study secondary succession starting on a nutrient-rich site (Rikhari et al., 1993). Changes in species composition along the age gradient of 1 to 35 years (Fig. 12.12) indicated that the mechanism of succession could be explained by the tolerance model of Connell and Slatyer (1977). The early occupants of the bare site, littered with dung, *Rumex nepalensis* and *Poa partensis* modified the environment, so that it became less suitable for the subsequent recruitment of their individuals. Individuals of the late successional, *Trachydium roylei* had ability to grow to maturity in the continued presence of early successional species. The relative cover of *T. roylei* was directly correlated with the age of site, and those of early successional species, *R. nepalensis* and *Poa partensis* were negatively correlated. With age, the relative per cent cover by different growth forms also changed, cushion and sprawling forbs (growing close to ground) dominating the late successional stage (Fig. 12.13). The species richness increased with age, but species diversity (Shannon Wiener Index) peaked at the middle of succession (Fig. 12.13)

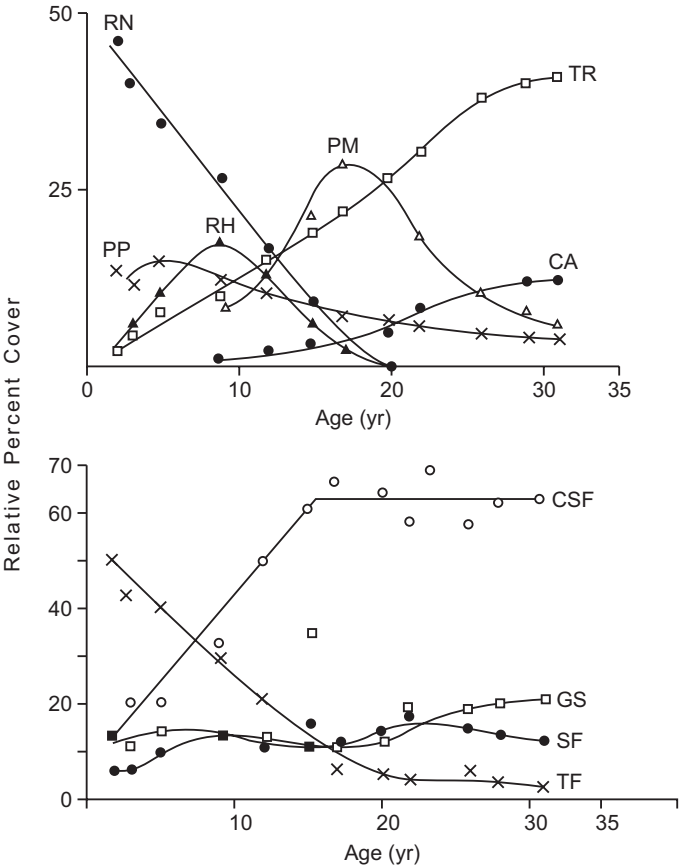


Fig. 12.12. Species and growth form distribution in an alpine meadow of the Central Himalaya. The relative per cent cover of major species (a) and the relative per cent cover of growth forms (b). RN = *Rumex nepalensis*; PP= *Poa pratensis*, RH = *Ranunculus hirtellus*, PM = *Plantago major*, CA = *Carex alpina*, TR = *Trachydium roylei*, TF = tall forb, SF = short forb, CSF = cushion and sprawling forb, GS = grass and sedge. (developed from Rikhari et al., 1993).

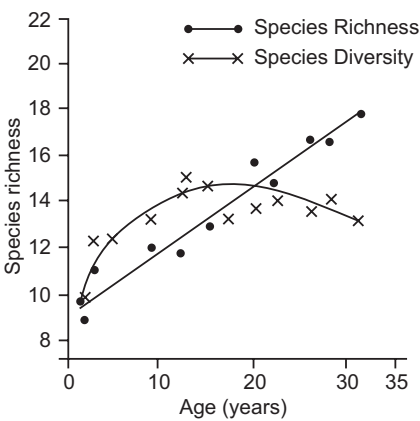


Fig. 12.13. Relationship between age and species richness, and age and species diversity in secondary succession in an alpine meadow of Central Himalaya (developed from Rikhari et al., 1993).

12.7 METHODS OF STUDYING SUCCESSION

Piecing Together Evidences of Nature and Relics

As the earliest map of the vegetation of Minnesota (USA) shows, there were isolated bits of forest dotting the grassland area, each tending to be of crescent form and fitted to the easterly margin of a lake, pond or marsh. Each stand had a complete complement of forest undergrowth, but the stands were too widely separated from one another for such a distribution pattern to be explained on the basis of chance bringing the seeds of so many species to each of the restricted plots. The remarkably reduced size of forest stands was due to extensive fire, which spread under influence of westerly winds. The remarkably reduced size of forest stands was due to extensive fire, which spread under influence of westerly winds. Fires induced the spread of grasslands, which in turn magnified the intensity of fire, as a result very small forest stands (dots of forest stands) could survive only where natural barriers, i.e., water bodies did not allow fires to encroach upon.

The form of very old trees of *Quercus macrocarpa*, a fire resistant species, left adjacent to ecotone between forest stands and grasslands, was characterized by broad low-branched crown resembling trees in an old orchard. The new forest developed from sprouts of species which had been killed back to the ground several times by fire, and as they attained sizeable stature and density they forced the old *Q. macrocarpa* trees to produce shoots rising vertically. To summarise, the vegetation changed from forest to savanna and then back to forest.

Time Interval Observations on Permanent Plot

Repeated observations on same plots at certain time-intervals are a safe method for documenting the changes in plant communities. But succession is such a slow process that in order to encompass the entire range of successional steps on a permanent plot, studies by a sequence of workers of several generations would be required. Detailed vegetation maps, particularly in early stages when plants are small in stature would help in documenting the changes in importance values of species.

Piecing Together Communities of Different Ages

Sites in the same general area but disturbed at different times can be used to interpret the successional changes in the vegetation, which followed the disturbances. For example, a site disturbed 5 yr. ago will have a relatively early successional community, the one 20 yr old will have a mid-successional community and so on.

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Part – III

ECOSYSTEM ECOLOGY

- ♦ ECOSYSTEM: STRUCTURE, FUNCTION AND PROCESSES
- ♦ PRIMARY PRODUCTIVITY AND DETRITUS POOL
- ♦ ENERGY TRANSFERS BEYOND PRIMARY PRODUCERS
- ♦ GLOBAL BIOGEOCHEMICAL CYCLES AND HUMAN IMPACT
- ♦ ECOSYSTEMS NUTRIENT CYCLING
- ♦ TERRESTRIAL BIOMES AND AQUATIC AREAS

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CHAPTER

13

Ecosystem: Structure, Function and Processes

13.1 ECOSYSTEM CONCEPT

In nature an organism does not exist in isolation; it is associated with organisms of its own and of different kinds (community) as well as with the physico-chemical environment (habitat). If the organism is a plant it depends for its growth on the aerial environment for light, temperature and CO₂, on the soil environment for inorganic nutrients and water. If it is an animal it depends on the plants for food and often shelter. If it is a microorganism it depends on the dead organic matter from plants and animals for its food, and releases the nutrients tied in the organic matter for the reuse of plants. There are thus continuous and causal interactions among the organisms and between the organisms and the habitat resulting into an integrated, functional unit. This unit was termed ecosystem by Tansley (1935), who defined it as, “a system resulting from the integration of all living and non-living factors of the environment”. Odum, E.P., to whom goes the credit of firmly establishing the concept of ecosystem in ecological literature, defined it in 1953 as a “basic functional unit of nature which includes organisms and their non-living environment, each interacting with the other and influencing each other’s properties, and both necessary for maintenance and development of system” (Odum, 1953). Later, Odum (1971) gave a more elaborate definition: “Any unit that includes all of the organisms (i.e., “the community”) in a given area interacting with the physical environment so that a flow of energy leads to clearly defined trophic structure, biotic diversity and material cycles (i.e., exchange of materials between living and non-living parts) within the system is an ecological system or ecosystem”.

More simply, ecosystems can be defined as assemblages of organisms and their associated environment, functioning collectively as a system to exchange material and energy (Reichle, 1975). Thus, habitat which helped to define a community becomes an interacting part of an ecosystem. Some European scientists use the term *biogeocoenosis* for ecosystem.

Implicit in all definitions is the concept of “interacting functional unit” in which living and non-living components are variously coupled through processes leading to exchanges of materials and energy such that a self-correcting homeostasis is generated. This “wholeness” is developed to such an extent that separating the ecosystem into parts would seem artificial. The ecosystem thus is the level of biological organisation unique to ecology as sustained life on earth is a characteristic of ecosystems, not of individual organisms or population. The ecosystem concept has provided a conceptual framework for the study of nature. The laws that govern the flow of energy and matter through organisms and environment have been studied within the framework of ecosystems. A lake, an island or a watershed are good examples of ecosystems in the context of systems theory of ecosystem analysis. The results of analysis of whole ecosystems have been found to have direct and relevant management applications (Likens, 1992, 1998). Ecosystem concept has proved to be

of practical value to understand the complexity of natural systems and generating a holistic view of ecosystem properties, besides being an effective communication link to the public for nature conservation (O'Neill, 2001).

13.2 ECOSYSTEM STRUCTURE

The ecosystem can be visualized as having two major kinds of components: (1) *Abiotic* (non-living) and (2) *Biotic* (living) components (Fig. 13.1). Ecosystems have structure due to the interaction of abiotic and biotic components, which vary over space and time. These two components sometimes are referred to as *ecotope* and *biocoenosis*, respectively. The climate (*climatope*) and soil (*edaphotope*) factors are dealt within the abiotic and the plant (*phytocoenosis*), animal (*Zoocoenosis*) and microbial (*microbocoenosis*) life are dealt within the biotic components.

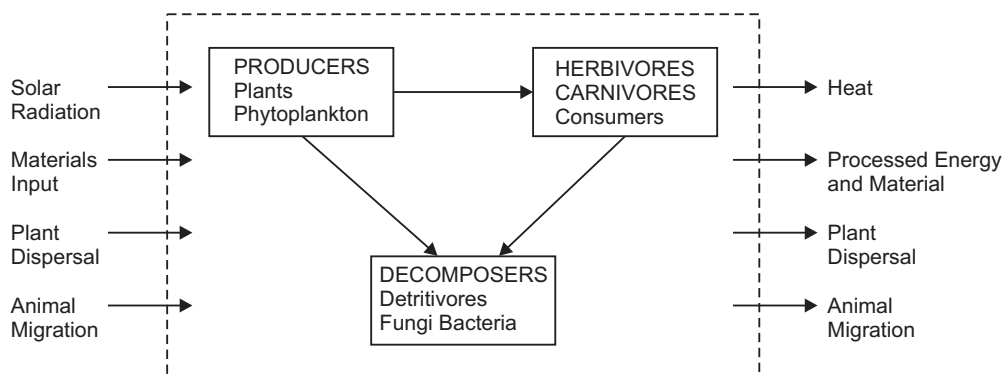


Fig. 13.1. Schematic representation of an ecosystem showing abiotic and biotic components. The dashed lines represent the boundary of an ecosystem.

Abiotic components

The abiotic structure is characterized by (1) the quantity and distribution of non-living materials (inorganic substances, e.g., C, N, CO₂, H₂O, etc.; and organic compounds, e.g., proteins, carbohydrates, humic substances) and (2) the climate regime, such as temperature, light, etc. Confusion arises when we try to classify inorganic and organic nutrients. Most compounds like proteins and carbohydrates are not only bound in organisms, but also in soil and atmosphere, they continuously move to and fro between biotic and abiotic components. There are, however, some substances that are unique to one or the other state. For example, adenosine triphosphate (ATP) occurs only in living component, its existence in non-living component is negligible, on the other hand humic substances, the resistant end-product of decomposition never occur in cells. Among others, although chlorophyll and the genetic material, DNA, occur both inside and outside the cells, they are non-functional outside the cell.

Biotic components

The organisms that make up the living part of the ecosystem (biotic community) are divisible into two major categories, viz., *autotrophs* (producers) and *heterotrophs* (consumers). This division is based on the function of the organisms and not on taxonomy. Autotrophs, the green plants and algae (and autotrophic bacteria) by capturing the solar energy and utilizing the simple inorganic substances, manufacture the food, that is utilized by the heterotrophs, which cannot manufacture their own food. The heterotrophs are divisible again into *phagotrophs* (phago = to eat) or *macroconsumers*, and *microconsumers* or *saprotrophs* (sapro = to decompose). The former include chiefly

the animals that ingest other organisms or particulate organic matter (e.g., snail that ingests organic particles). The saprotrophs chiefly include bacteria and fungi which break down complex dead protoplasm, absorb some of the decomposition products, and release inorganic nutrients that are reused by the producers; they are also called *decomposers*. The phagotrophs may be herbivores (ingesting plants, e.g., goat, deer), or *carnivores* (ingesting other animals - e.g., tiger, lion) or *omnivores* (ingesting both plants and animals, e.g., bear, man).

The microconsumers (bacteria and fungi are relatively immobile, with high rates of metabolism and turnover (each population is short lived, and one is replaced by another continuously and rapidly hence biomass is never large). They also display biochemical specialization hence their significance cannot be assessed by counting and weighing them. Macroconsumers are active food gatherers and therefore are usually much more mobile, with complex sensory neuromotor, respiratory and circulatory systems.

The biotic structure of the ecosystem is characterized by the composition of the biological community including species numbers, biomass (i.e., weight per unit area), life-form, life-history and spatial distribution of species. Species diversity at the levels of autotrophs and macroconsumers, as well as the ecosystem level diversity (complexity; reflecting the inter-relationships among the different functional groups) is an important structural characteristic of the ecosystems.

There are fundamental structural similarities among the ecosystems. For instance, consider an aquatic (e.g., pond or lake) and a forest (e.g., oak forest) ecosystem. Both have the three necessary biological components : autotrophs, macroconsumers and microconsumers. Both need the same vital materials such as N, P, trace elements, and are regulated or limited by the same conditions of existence, e.g., light, temperature, etc. More often than not the function of the autotrophs and heterotrophs are separated in space imparting a stratified structure to the ecosystem. In lake Nainital, for example, almost all photosynthetic activity by phytoplankton (microscopic algae) is confined to the top 1 m belt. As these producers die, they settle on the bottom where they are acted upon by the decomposers, which leads to release of nutrients to be utilized by plants again. Thus the 'green belt' where almost whole photosynthesis occurs is separated in space from the 'brown belt' lying on the bottom, where most of the decomposition occurs. Similarly, in the forest ecosystem the crowns form the green belt and the soil and litter the brown belt. The leaves after a period of active photosynthesis senesce and fall on the ground, where dead matter is subjected to decomposition.

However, there are structural differences also between the two. In the case of the forest the autotrophs are trees, shrubs and herbs, while in the pond they are phytoplankton and hydrophytic angiosperms. Similarly, the consumers in the forest may be such animals as deer and lion compared to zooplankton and fishes in the pond. The roles of producers, consumers and even decomposers are carried out by taxonomically different groups of organisms and their sizes and longevity differ in the two ecosystems. Also the predominant medium in the pond is water and that in the forest is air. Consequently the pond is more oxygen-limited than the forest.

An ecosystem may be characterized on several grounds. A habitat approach may help us to characterize an ecosystem as a desert ecosystem or a tundra ecosystem or a fresh-water ecosystem. Physiognomy of the vegetation and kinds of plant species present may help us to recognise a forest ecosystem or a grassland ecosystem.

13.3 ECOSYSTEM BOUNDARY

We cannot see an ecosystem as we see for example an elephant. The skin clearly marks the boundary of the elephant or of other animals; the ecosystem may not have such a "definite" boundary. In some cases, the boundaries of the ecosystem are reasonably "apparent", as is in the case of a pond or a lake. At other times, for example between short- and mixed-grass, and between mixed and tall grass prairies, the boundaries are not that sharp. The pond or a lake also is always

associated with a catchment or watershed, from which it receives the inflow of water as well as of a variety of inorganic and organic substances. The watershed may be grassland or a forest and, in its own right, it deserves to be treated as an ecosystem. However, since the existence of the lake depends on its watershed, the entire drainage basin (lake + watershed) may be considered as a single ecosystem. In the same way, the whole biosphere (that part of the earth and its atmosphere which is inhabited by living things) may be treated as one huge ecosystem. Traditionally, earth's ecosystems have been divided into categories for comparing their structure and function.

The ecosystem boundaries are determined by the environment, i.e., by what forms of life can be sustained by the given environmental conditions (Smith, 1996). For example, the lake environment will permit the occurrence of phytoplankton and fish but not of trees and tiger which can occur in the environment of the watershed. Similarly, a close examination will reveal differences in physical environment (e.g., soil, rainfall and temperature) and species composition among short-, mixed- and tall grass prairies. Although there exist strong linkages and continuity among the adjacent ecosystems, it is possible to delimit them for the purpose of study and description. Flows occur across ecosystem boundaries (Fig. 13.1) by import and export of matter and energy. It is an open system having a definite input and output environment which must be considered an integral part of the ecosystem concept (Fig. 13.1). The input environment not only consists of solar energy input but also includes the materials and organisms that are imported. Similarly, organisms migrate out of the ecosystem, and energy and materials are exported to other ecosystems. For example, a lake receives leaf litter and associated microbes if its catchment is forested. If the lake has an exit, nutrients and plankton are exported to adjacent systems. Migration of large animals, like tiger from forest to grassland and vice-versa is well known.

13.4 ECOSYSTEM FUNCTION

The two fundamental functions of an ecosystem are energy flow and biogeochemical cycling (Fig. 13.2).

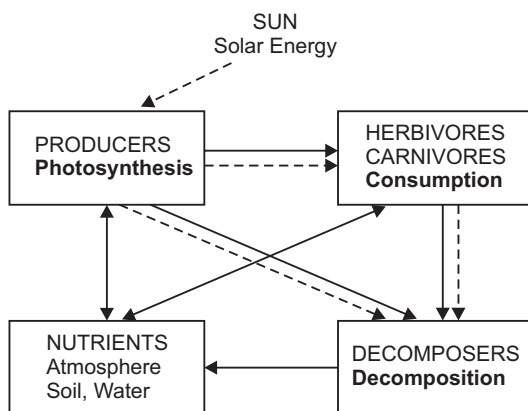


Fig. 13.2. The interaction of abiotic and biotic components in an ecosystem showing the flow of energy and cycling of nutrients.

Energy flow

The green plants capture the solar energy and convert it through the process of photosynthesis into chemical energy of food and store it into their body. This process is called *primary productivity*. Heterotrophic organisms depend upon this energy. The herbivorous animals derive the energy by ingesting plants or plant parts. These animals are eaten by carnivorous animals (first order carnivore) which in turn are eaten by other carnivorous animals (second order carnivore). In this chain of

eating and being eaten away, the green plants form the *first trophic level* (or food level because they constitute food for herbivores), the herbivores the *second trophic level* (they are food for carnivores) and the second order carnivores the third trophic level, and so on. Such a linear arrangement of trophic levels is called *food-chain* along which the energy flows. Such a food chain constitutes the *grazing pathway* or the *grazing food chain*:

Sun → green plants → herbivores → first order carnivores → second order carnivores

Examples of this would be:

Sun → shrub or grass → deer → lion

or

Sun → phytoplankton → zooplankton → fish → man

At each trophic level some energy is lost as heat and respiration and consequently available energy decreases as one moves away from the first trophic level. Therefore, the number of trophic levels in a food chain is limited. Also organisms at each trophic level die and generate waste organic material. This constitutes *detritus* (particulate non-living organic material) and provides energy to a different pathway called *detritus pathway* or *decomposer food-chain*. Examples of such food chains are:

dead grass → fungi → collembola → predatory mite

or

dead grass → fungi → bacteria → virus

Both the grazing and detritus types of food chains are present in all ecosystems; the magnitude of energy transferred along them may however, differ. For example, in grassland ecosystems grazing pathway predominates and in forests the detritus pathway is dominant. These food-chains operate such that each unit of energy captured by the plants is ultimately dissipated to the environment as heat or respiration. This dissipated energy cannot be reused by the organisms and therefore energy flow is a one-way process (Fig. 13.2). An ecosystem contains several food-chains of each of the above types; often these food chains are inter-linked forming a *wood web*. For example:

Sun → grass → grasshopper → insectivorous bird → hawk

↓ ↓

Sun → grass → Frog → snake

In the above example two food chains, viz., grass → grasshopper → insectivorous bird → hawk, and grass → grasshopper → frog → snake → hawk are interlinked.

Biogeochemical cycling

Of the 30 to 40 elements necessary to life, six rank as the most important: carbon, hydrogen, oxygen, nitrogen, sulphur and phosphorus. These materials move from non-living to the living and back to the non-living again in a cyclic manner (Fig. 13.2). This movement is described as *bio* - (for living) *geo* - (for earth and water) *chemical cycling* or nutrient cycling. These materials (with the exception of H₂O where all organisms take it up directly also) are first taken up by the plants, bound in the organic matter and move along the food chain to different trophic levels and ultimately from all trophic levels, with the detritus, to the decomposer food-chain. The decomposers break down the complex organic compounds and release the nutrients to the soil from where they are again taken up by the plants.

13.5 ECOSYSTEM METABOLISM

Ecosystems can be categorized and compared through certain metabolic parameters. Some ecosystem-level parameters are (Woodwell and Botkin, 1970): (1) Gross primary production (GPP), i.e., total photosynthetic fixation of carbon; (2) Net primary production (NPP) = GPP - R_A;

where R_A is autotrophic (plant) respiration and equals $GPP - NPP$; (3) Heterotrophic respiration (R_H) = consumer + decomposer respiration; (4) Ecosystem respiration (R_E) = $R_A + R_H$; and (5) Net ecosystem production (NEP) = $GPP - R_E$. Reichle (1975) compared these parameters for four divergent ecosystems in terms of functional metabolic ratios and found more or less a consistent pattern across the ecosystems (Table 13.1), particularly production efficiency (R_A/GPP) was very similar across widely different biomes (Coleman, 2010).

Table 13.1. Comparative metabolic parameters of four contrasting ecosystems. All values in $gm^{-2} yr^{-2}$ (After Reichle, 1975) (symbols are explained in the text).

Property	Mesic forest	Xeric forest	Prairie	Tundra
GPP	1620	1320	635	240
R_A	940	680	215	120
NPP	680	600	420	120
R_H	520	370	271	108
NEP	160	270	149	12
RE	1460	1050	486	228
Production efficiency (R_A/GPP)	0.59	0.52	0.34	0.50
Effective production (NPP/GPP)	0.42	0.5	0.66	0.50
Maintenance efficiency (R_A/NPP)	1.38	1.13	0.51	1.00
Respiration allocation (R_H/R_A)	0.55	0.54	1.26	0.90
Ecosystem productivity (NEP/GPP)	0.10	0.20	0.23	0.05

13.6 ECOSYSTEM PROCESSES

In any ecosystem, inputs of energy and nutrients are handled by four processes (i) photosynthesis, (ii) herbivory, (iii) carnivory, and (iv) decomposition.

Photosynthesis

In photosynthesis, part of the electromagnetic energy of the solar radiation of wave lengths approximately 400-700 nm is absorbed and transformed into the energy of chemical bond. For each gram atomic weight of carbon taken up, 112 kcal potential energy is obtained.

This simple representation of the process in words is:

Carbon dioxide + Water + Light energy in presence of chlorophyll associated with enzyme system results in glucose + Oxygen.

The whole process is divisible into three groups of processes: (i) photochemical processes occurring in the presence of light, (ii) purely enzymatic processes not requiring light (i.e., the dark reactions), and (iii) the process of diffusion causing the exchange of CO_2 and O_2 between the chloroplasts and the external air. All the green plants, algae and some bacteria accomplish photosynthesis. The light reactions involve two photosystems which are composed of photosynthetic pigments and electron carriers. Chlorophyll a, which occurs in several forms each with a different photo activity and absorption maxima in the red region of the spectrum, is the most common pigment and occurs in both photosystems I and II. The chl a form which absorbs longer wavelengths predominates in photosystem I, while the short wavelength-absorbing forms together with most of the chl b, c or d and other accessory pigments (carotenoids and phycobilins) are components of photosystem II. The energy trap for photosystem I is a porphyrin pigment known as P700 and that for photosystem II is the hypothetical P690 (or P680 or chl- a_2). The trapping molecule is

excited by the absorbed energy which results in the transfer of an electron against a gradient of redox potential. The change of energy status results in the production of highly active oxidants and reductants. When photosystem II is sensitized by light, photoreaction 2 forms a primary oxidant X^+ and a primary reductant Q^- . The primary oxidant causes splitting of water and the evolution of oxygen with the consequent restoration of P690 in the unexcited form. The electron from Q^- is transferred through a chain of electron carriers, comprising plastoquinone, cytochromes and plastocyanin, that joins the two light reactions. During these steps part of the energy from the electron is utilized in the formation of ATP.

In photosystem I, P700 accepts light energy and donates an electron to an acceptor Z, a ferredoxin-reducing substance. The oxidised P700 accepts the low energy electron from photosystem II and returns to its ground state. The reduced acceptor Z^- transfers the electron through ferredoxin and ferredoxin-NADP reductase to $NADP^+$ where $NADPH_2$ is formed. The ATP and $NADPH_2$ are subsequently used in reduction of CO_2 to carbohydrates through light-independent reactions. The uphill transfer of hydrogen atoms from the redox potential level of +0.8 (splitting of water) to a level of -0.4 volt (reduction of CO_2 to CH_2O) is the essence of the energy-storing photochemical stage in photosynthesis.

Photosynthesis by green plants is the only process by which solar energy is bounded in the energy of food on which depend all other organisms. The limits of photosynthesis can be roughly calculated as follows: Ten quanta of light energy are required to reduce one CO_2 molecule. Ten quanta supply 520 kcal per molecule of CO_2 and 1 molecule of CO_2 stores about 105 kcal; then the quantum efficiency of photosynthesis is $(105/520 = 0.2)$ 20%. Leaves get light-saturated between 10 and 20% of full sunlight in terms of photosynthesis but can absorb up to 80% of full sunlight. So if leaves absorb 80% of the light, and use 20% with an efficiency of 20%, then the overall efficiency is $(0.20 \times 0.20 \times 0.80 = 0.032)$ 3.2%. This efficiency is also influenced by several factors, e.g., soil moisture, temperature, length of growing season, type of photosynthetic pathway and so on. Usually 1% or less of the energy in the solar radiation is considered to be fixed in photosynthesis. Since vegetation usually does not use 99% of solar radiation, it would appear to be highly insufficient in transferring energy into the ecosystem. Billings (1964) has cited an interesting calculation: To produce 100 bushels of corn seed per acre (= 45 quintals per hectare), about 20,000 pounds (= 9099 kg) of CO_2 are needed. The corn plants on one acre must process about 21000 tons of air to get this amount of CO_2 . This calculation relates only to corn seeds, for the production of whole crop the value will be much greater. Such calculations show the miraculous efficiency of plants.

While in all green plants it is the water which is oxidized during the oxidation reaction with the release of gaseous oxygen, in green and purple sulphur bacteria (*Chlorobacteriaceae* and *Thiorhodaceae*) it is inorganic compounds such as hydrogen sulphide (H_2S); and in purple and brown non-sulphur bacteria (*Athiorhodaceae*) it is the organic compounds. In the case of bacteria oxygen is not released during photosynthesis. Compared to plants, the bacterial photosynthesis which mostly occurs in aquatic bodies is of minor importance in terms of production of organic matter; it usually contributes 3-5% of the total aquatic production in Japanese lakes (Takahashi and Ichimura, 1968). However, it is of significance because these bacteria can work in environments where green plants cannot reach. Then their photosynthesis is responsible for the return of certain minerals from the deep aquatic sediments. The most well known example is of the role of sulphur bacteria in sulphur cycle. Below the layer of green algae on the tidal mudflats, one can see the prominent pink or purple coloured layers formed by sulphur bacteria. Since they function in absence of oxygen (obligate anaerobes), they are found usually in reductive zone (upper edge) of an aquatic body, where only faint light can reach.

The non-sulphur photosynthetic bacteria can function both in presence and absence of oxygen (facultative anaerobes) and when light is not available they function as heterotrophs. It is expected that both types of photosynthetic bacteria play important role in highly polluted and eutrophic water bodies when oxygen is either absent or scarce.

Chemosynthetic bacteria are also regarded as producers. They obtain their energy for carbon dioxide assimilation not by chemical oxidation of simple inorganic compounds, for example, ammonia to nitrite, nitrite to nitrate, sulphide to sulphur, and ferrous to ferric ions. They can be called *chemotrophs* instead of autotrophs and their role is intermediate between autotrophs and heterotrophs. The sulphur bacteria *Beggiatoa*, which are abundantly present in sulphur springs and several nitrogen bacteria that carry out certain steps of nitrogen cycle and can grow in dark but with exception require oxygen. The chemosynthetic bacteria help in recycling of minerals; they also save the energy being lost from the ecosystem. The hydrothermal vents in some parts of ocean floor produce large amounts of superheated ocean water and hydrogen sulphide gas. In the dark and hot environment, specialised producer bacteria carry out chemosynthesis to convert inorganic hydrogen sulphide to nutrients, they require for their growth. Recent studies have shown that aerobic anoxygenic bacteria play an important role in photosynthetic activity and exhibit large microbial diversity in the marine ecosystems (Karl, 2002; Beja et al., 2002), Box 13.1.

Box 13.1. Photosynthetic Pathways in Marine Ecosystems

Aerobic bacteria containing chlorophyll-a are widely distributed in marine plankton. These bacteria require oxygen, but do not generate it as a by product. They may be accounting for 11% of total microbial community and 5% of the photosynthesis (Kolber et al., 2000) and exhibit a large diversity as analyzed using methods of molecular biology (Beja et al., 2002). In the open sea, there is a complex relationship between sunlight, biological energy production and dissolved organic matter. All the three modes of photosynthesis, i.e., oxygenic photosynthesis, (OP), anaerobic anoxygenic photosynthesis (AnAnP) and aerobic anoxygenic photosynthesis (AAnP) are found. Besides these, the bacteria also have rhodopsin-based and phytochrome-based light driven processes, all these processes control the flow of energy in global oceans. Oxygenic photosynthesis is found in low nutrient oceans, whereas AnAnP, AAnP and DOM (dissolved organic matter) metabolic processes occur in deep oceans. The discovery of wide variety of aerobic anoxygenic bacteria is important for understanding carbon cycling and food webs in the complex marine ecosystems (Karl, 2002).

Respiration

In a broad sense respiration is an energy yielding biotic process; it can be of three types: *Aerobic respiration*; in which molecular oxygen is the hydrogen acceptor, i.e., the oxidant; *Anaerobic respiration*; in which an inorganic compound is the electron acceptor; and *Fermentation*; in which an organic compound is the electron acceptor.

The aerobic respiration is displayed by all higher organisms, and in a sense is the reversal of photosynthesis. A portion of energy thus released is used for the maintenance of life processes (*maintenance respiration*) and another part for supporting new growth (*growth respiration*).

Anaerobic respiration and fermentation are usually displayed by saprophages such as bacteria, fungi and protozoans. These organisms can be *obligate anaerobes*, i.e., respiring always without oxygen, or *facultative anaerobes*, i.e., respiring with and without oxygen. The methane bacteria that decompose materials through reducing both organic and inorganic carbon compounds are good example of obligate anaerobes. As a result of respiration of these bacteria methane gas (CH_4), called “swamp gas” is released. They are active in such diversified habitats as swamp and in rumen of the cattle.

The anaerobic *Desulfovibrio* bacteria are also of ecological importance because they reduce SO_4 in deep sediments to H_2S gas, which when rises to upper sediments is acted on by photosynthetic sulphur bacteria. Yeast is a well known fermentor. It ferments the organic plant residues of soil.

In the case of facultative anaerobes, the end product would not be same when they respire with and without oxygen. For example, when *Aerobacter* was grown on glucose as carbon source, all of the glucose was converted into its protoplasm and CO_2 , in the presence of O_2 , and a series of incompletely decomposed organic compounds (butanediol, ethanol, formic acid, etc.) in the absence of O_2 . In the presence of O_2 , 40-50% of original carbon of glucose was converted into bacterial protoplasm, whereas in the absence of O_2 , the same was only about 15% (Pirt, 1957). Further, rate of respiration in terms of energy yield per unit weight of substrate is generally considerably lower in absence of O_2 than in its presence. However, the anaerobes are not less important than the aerobes, though they are comparatively less efficient in acting on dead matter. They play ecologically significant role, for, they alone function in deep sediments of ecosystem, where O_2 and light are absent. Although they are inefficient decomposers, they increase the overall decomposing efficiency in the ecosystem. Without them a sizeable portion of energy and nutrients would remain unutilized. It should be recognised that ultimately all energy released in respiration escapes to the environment.

Herbivory and carnivory

Herbivory refers to ingestion of living plants or plant parts by heterotrophs, while carnivory refers to the ingestion of herbivores and other heterotrophs by animals. These also include such feeding habits as sap sucking (e.g., aphids on plants), blood sucking (e.g., leach from animals), and plant and animal parasitism. These processes result in *secondary production* (after assimilation) as photosynthesis results in *primary production*.

In a typical terrestrial ecosystem, herbivory may remove about 10% of net primary productivity (percentage varies: 2-3% for desert scrub, 4-7% for forest, 10-15% for temperate grasslands, 30-60% for tropical grasslands and grasslands managed for cattle raising). However, episodic outbreaks of herbivores (e.g., tent caterpillars, locusts, rabbits, lemmings and pathogens) may consume much higher percentages of net primary productivity, even causing a complete decimation of vegetation.

The controlling effect of these processes on the ecosystem is greater than the amount of consumption would indicate (O'Neill, 1976). For example, the loss of wood production is far more than the amount of leaf consumption by caterpillars in an oak forest. In forest communities, insects increase the production by opening the canopy by removing weaker plants which in turn permits vigorous growth of younger plants (Mattson and Addy, 1975). Consumers also influence the ecosystems by affecting nutrient cycling. For example, their faeces and urine increase nutrient turnover rates. Partial breakdown of plant parts by consumers stimulates microbial activity. Role of herbivory on plant distribution (through seed dispersal) and on plant diversity is well known. Studies have indicated that consumers play an active role in the maintenance and regulation of energy flow through the ecosystem and hence contribute to its persistence (O'Neill and Reichle, 1980).

Decomposition

Decomposition is the breakdown of dead organic matter into simpler inorganic compounds with concomitant release of energy. A majority of the net primary production in the ecosystem (70-85%) is metabolized by the decomposers. The major groups of litter-decomposing organisms are bacteria, actinomycetes, fungi, protozoa, nematodes, microarthropods, macroarthropods, enchytraeid worms, and lumbricid worms. The microorganisms secrete enzymes into dead plant and animal materials to carry out specific chemical reactions. They absorb some of the products resulting from the chemical action of enzyme on detritus, and some portion is left to be acted upon by other sets of organisms. Thus a succession of organisms occurs on decomposing material. A common sequence may be:

mould fungi and non-spore forming bacteria, spore forming bacteria, *cellulose* myxobacteria and actinomycetes. A single group of organisms cannot complete the process of decomposition.

Detritivores (feeding on dead organic matter) by their various activities contribute to the process of decomposition. Several detritus or litter feeding invertebrates such as earthworms, millipedes, isopods and the larvae of beetles and flies in terrestrial ecosystems, mollusks, crabs and larvae in aquatic systems comminute the litter and thus increase the surface area of the decomposing materials. With the increased surface area, microbial population and their activities are also enhanced. In this way, these litter-feeders increase the rate of decomposition. They can utilize only the easily digestible protein and carbohydrates, which make up on average less than 10% of the plant material ingested. The litter feeders such as earthworms may also enrich the decomposing material with B_{12} , and such an enriched matter is more vigorously acted upon by microbes. In addition to this, there are other types of organisms, such as nematodes, collembolans, larval forms of beetles (Coleoptera), flies (Diptera) and mites (Acarina) which feed on microbes (bacteria, fungi) and algae and can be called microbial consumers. Such organisms are organised in food webs (such as, bacteria → protozoa → nematodes; fungi → collembola → mite). By feeding on microbes, these consumers regulate their population as well as increase the total decomposition activity. The microbial population during the process of organic matter decomposition immobilize nutrients in its biomass. Thus nutrient mobilization (release) and hence nutrient cycling is regulated. This contributes to the homeostasis of the ecosystem. Experiments have shown that the exclusion of litter-feeders and microbial consumers from litter mass slows down the rate of decomposition significantly.

An accepted pathway of decomposition is: (1) development of phylloplane microflora, (2) colonization by saprophytic microorganisms, (3) comminution and ingestion by invertebrates, (4) microbial colonization and utilization of faeces and comminuted litter resulting in chemical degradation, and production of complex phenolic polymers (humic acids), and (5) the formation of stable organomineral complexes (humus). Superimposed on this are the decomposer food chains and several of the above steps (e.g., 2, 3 and 4) are concomitant. Most important factors affecting the rate of decomposition are moisture, temperature, and chemical composition of the substrate (Singh and Gupta, 1977; Swift et al., 1979). Moisture and temperature increase the rate of decomposition in an interactive manner. Nitrogen content of the substrate positively affects the rate while the effect of lignin (which is hard to decompose) is negative. "Decomposition is a process of equivalent importance as photosynthesis and needs to be understood in its full detail" (Heal et al., 1997). During the last two decades, decomposition studies in terrestrial ecosystems have focused on ecosystem processes, soil biology, nutrient and organic matter management in agriculture, and forestry (Heal et al., 1997). Biodegradation of detritus in marine systems is attracting the attention of marine biologist to understand their role in oceanic carbon cycle, biological diversity and the importance of detritus food webs in nutrient regeneration in the oceans.

During the process of decomposition organic substances of hormonal properties, that are referred to as "ectocrine" (Lucas, 1947) (or "exocrine" or "environmental hormone") are released. Some of the substances are known to be inhibitory for some organisms. Popular antibiotic, penicillin, produced by the fungus *Penicillium* is one good example of this type. Others which include various vitamins and growth substances (thiamine, vitamin B_{12}) are known to stimulate the growth of other organisms. The ectocrines also constitute a control mechanism for ecosystem functions.

Time separation among the processes

The fate of energy fixed in photosynthesis can be as follows: (i) some of it is respired instantaneously; (ii) some of it is consumed by herbivores; (iii) some of it decomposes long after it is produced (the tissue of trunk of large trees may start decomposing only after hundreds or thousands of years); (iv) the organic matter ingested by herbivores may pass on to carnivores and eventually may decompose, (v) the dead plant matter may become part of detritivores.

Each unit of energy fixed in photosynthesis and not spent in respiration, is ultimately dissipated through decomposition. The temporal separation of the process of photosynthesis (primary production) and decomposition is a basic property of ecosystem. This holds the energy in usable form for varying periods of time allowing the existence on it of varied life.

13.7 ECOSYSTEM STABILITY

In any area all the plants, animals and microorganisms, soil and climate influence one another in varying degrees, and certain properties emerge from the whole unit, which cannot be known by summing up the properties of the components. The components are not in random aggregations, they are organized in certain ways, which are the result of long evolutionary time. Over a very long evolutionary time scale (about 3 billion years) the shift from a non-oxygenated reducing atmosphere to an oxygenated one has completely changed the ecosystem processes. Evolution affects ecosystem processes through the diversity and adaptation of organisms, particularly in situations where only one species is capable of performing an ecosystem process (Jones and Lawton, 1995).

Ecosystem can be visualized as a set of interdependent subsystems and sub-sub systems at many spatial and temporal scales conforming to the hierarchical organization. It is an open system with built-in homeostatic mechanisms. Because of the requirement of an outside input in the form of solar radiation and an output to the environment (e.g., heat of respiration), for continued operation, the ecosystem is an *open system* (Fig. 13.3). Each subsystem or object receives stimulus or input from the environment and produces a response or output. For example, one can consider solar energy (stimulus) as input to the plant subsystem and the chemical energy of food produced by the plants as the output (response). The output from one subsystem may become the input for another subsystem (Fig. 13.3). For example, the food output from plants becomes input (food intake) for a herbivore. The object or the subsystem (*holon*) together with its input environment (*creaon*) and output environment (*genon*) has been defined as *enviro* and portrayed as an “elementary particle” for ecology by Patten (1982). In this example, if the input (solar radiation) ceases the subsystem (plants) ceases, to function. And if the plants cease to function the ecosystem will cease to function.

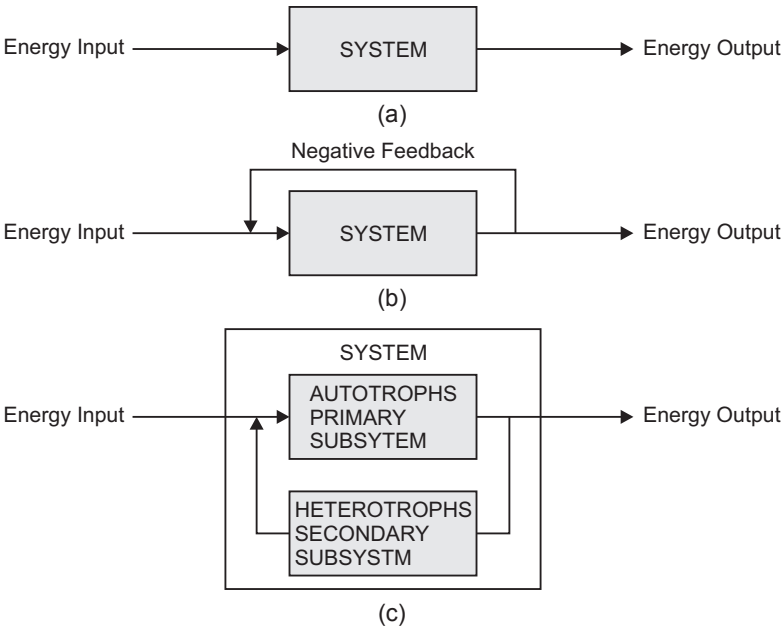


Fig. 13.3. An ecosystem is: (a) an open system receiving input of energy and having the output; (b) and (c) a cybernetic system having negative feedback control.

Natural ecosystems are complex networks of interacting feedbacks (DeAngelis and Post, 1991). In open systems some of the output or the output information may be fed back as input to control the functioning - when this happens the system is called a *cybernetic system* (Smith, 1996). A system responds to inputs and has outputs. A specialised kind of system response is called feedback (Fig. 13.3.) This feedback is of two types: *positive feedback* and *negative feedback*. In the positive feedback, increased output results in increased input and therefore in further increased output and so on. A serious situation arises when an increase of the environmental degradation leads to positive feedback. For example, deforestation, pollution, erosion and fire act as positive feedbacks and therefore have destabilizing effects.

In the negative feedback the input is reduced which results in subsequent decrease in output. Let us take the example of a forest tree population (Fig. 13.4.) It produces millions of seeds each year. Suppose all seeds were to germinate into seedlings (output) and suppose all seedlings were to grow into mature trees. These mature trees will form the input and the population of that tree would expand (positive feedback) to such an extent that no place will be left for any other species. However, hardly a fraction of seedlings produced are actually able to grow into mature trees. Because of *environmental resistance* in the form of limiting resources and crowding effects a majority of seedlings would die and hence future input to the population will be reduced (negative feedback) (Fig. 13.4.) The positive and negative feedback mechanisms act in such a fashion that the populations or subsystems of an ecosystem oscillate around a *homeo-* (= same) static- (= standing) *plateau*. This is dynamic equilibrium or dynamic stability. The feedbacks do not occur only at the population but also at biochemical levels. Similar controls are exercised through energy flow and nutrient cycling functions.

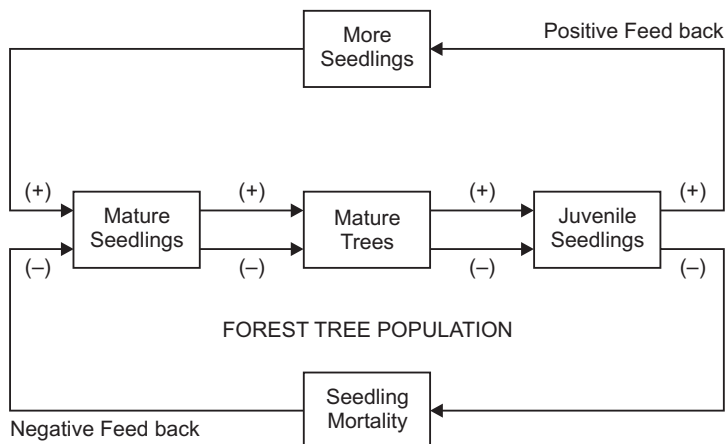


Fig. 13.4. The role of negative and positive feedback mechanism to regulate tree population in a forest. The positive (+) sign shows increase in quantity due to greater output (juvenile seedlings) resulting in higher population density. The negative sign (–) indicates lower output (juvenile seedlings), resulting in regulation of tree population in the forest.

Negative feedbacks occur when two components of a system have opposite effects on one another. Consumption of prey by a predator, for example, has a positive effect on the consumer but a negative effect on the prey. The negative effect of predators on prey prevents an uncontrolled growth of a predator's population, thereby stabilizing the population sizes of both predator and prey (Fig. 13.5). For example, the population of deer in our food-chain example could be controlled on one hand by the amount of grass or browse available and on the other by the population of its predator, the lion. The population of lion in turn, will depend on the available food (deer population) directly and on the grass or browse indirectly. There are also positive feedbacks in ecosystems in which both

components of a system have a positive effect on the other. For example, plants provide their mycorrhizal fungi with carbohydrates in return for nutrients promoting the growth of both components of the symbiosis until they become constrained by other factors (Fig. 13.5). If the negative feedbacks are weak or absent (a low predation rate due to predator control, for example), population cycles can amplify and lead to extinction of one or both of the interacting species.

Diversity within and among the trophic levels is yet another mechanism contributing to the stability of the ecosystem. For example, when several autotrophic species are present in the ecosystem, elimination of one would free other species from competition and the total photosynthesis of the system may remain unchanged. Heterogeneity (species diversity, spatial heterogeneity, genetic heterogeneity) has been viewed as essential to preventing catastrophic consequences due to perturbations (O'Neill and Reichle, 1980). The storage and release of nutrients provides a control through nutrient availability for growth. Odum (1971) states, "The interplay of material cycles and energy flows in large ecosystems generates a self correcting homeostasis with no outside control or set point required".

The ecosystem stability has two components: *resistance* and *resilience* (Holling, 1973). The resistance implies the ability of the system to resist the forces that tend to cause it to leave its equilibrium. For example, a forest ecosystem with large biotic structure can resist a fire better than a grassland ecosystem with smaller biotic structure. The resilience on the other hand is the rapidity and ease with which the system returns to its original equilibrium state following perturbation. Resistance is associated with the dominance of "*K* - selected" and resilience with "*r*-selected" species (O'Neill and Reichle, 1980). For example, a burned grassland can quickly recover to its original state but a burned forest may take hundreds of years to recover or may not recover at all to its original condition. The tropical rain forest ecosystems are said to be highly *fragile* because they have poor resilience; once greatly

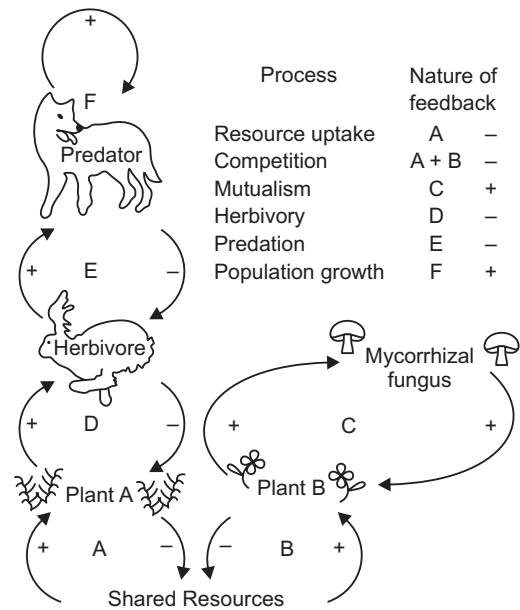


Fig. 13.5. Examples of linked positive and negative feedbacks in ecosystems. The effect of each organism (or resource) on other organisms can be positive (+) or negative (-). Negative feedbacks resist the tendencies for ecosystems to change, whereas positive feedbacks tend to push ecosystems toward a new state. (adapted from Chapin et al., 2002.)

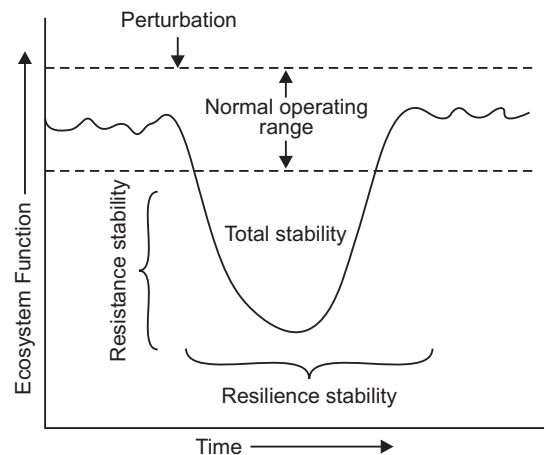


Fig. 13.6. The resistance and resilience stability of an ecosystem. The degree of deviation from normal operating range of the ecosystem function is resistance stability; the time required for recovery of normal processes is resilience stability (modified from Odum, 1983).

perturbed they may not ever return to the original state (Gomez-Pompa et al., 1972). The concept of resilience and resistance stability as applicable to an ecosystem is shown in Fig. 13.6. Mycorrhizal symbionts act as positive feedbacks by amplifying processes that increase ecosystem stability and self sustaining ecosystem processes (van der Heijden et al., 1998a, b)

13.8 CONTROLS ON ECOSYSTEM PROCESSES

Interactions among terrestrial ecosystems, the atmosphere and the oceans

The development in the field of ecosystem ecology has come a long way from early descriptive approach of ecosystem structure to analysis of ecosystem functions. A number of studies have focussed on the interactions among structure and functions, and among interlinked ecosystems and temporal changes (Likens, 1998). With the availability of modern tools and approaches such as radioactive and stable isotopes, remote sensing, experimental manipulation of entire ecosystems, comparative analysis and simulation modelling, the role of ecosystems in the global processes is becoming increasingly evident.

Human activities have caused a number of changes in a variety of ecosystems through agriculture, forestry, urbanization and industry. The most obvious impact of humans is the loss of biological diversity and the extinction of biological species. The frequency of species extinction is directly related to resource use, land use changes and environmental degradation at various levels. The impacts of human activities on earth system as a whole are being analyzed to understand how terrestrial ecosystem processes affect the atmosphere and oceans. The ecosystem impacts occurring at local level may have global impacts. For example, the conversion of a forest to a grazing land is a local activity. It also causes loss of carbon to the atmosphere, having large impacts on local and regional rainfall. But the loss of organic matter has effects on the global carbon cycle. The information on global patterns of CO₂ and atmospheric pollutants provide telltale evidence of major locations and causes of the global problems. Therefore, the changes in earth system do not operate in simple and cause-effect relationships. A single type of change can cause large number of responses in the earth system through teleconnections in the climate system (IGBP, 2001). These changes also interact with the natural variability to become feedback for further changes.

State factors, interactive controls and ecosystem processes

The systems approach for studying ecosystems has emphasized general properties of ecosystems, with little emphasis on mechanisms and interactions regulating ecosystem processes. Disturbance caused by fire, wind, insect outbreaks and other factors and human activities act as important controls on ecosystem stability. The role of state factors as defined by Hans Jenny (1980) and that of interactive controls (Chapin et al., 1996) in the ecosystem framework recognises that:

- (1) Ecosystems are a part within landscapes.
- (2) Ecosystems are open to flow of energy, materials, and biota and are not tightly bound in space and time.
- (3) Ecosystem structure is governed by five state factors (Jenny, 1980) - climate, topography, parent material (the rocks that give rise to soils), potential biota (the organisms present in the region and could potentially occupy the site) and time (Fig. 13.7). Global variation in climate explains the distribution of biomes (see Chapter 18). Within each biome, parent material influences soil development and ecosystem processes at the regional scale (see Chapter 4). The potential biota has a major control on biodiversity of organisms at a site. Time regulates the development of soil, evolution of organisms over longer time scale, and responses of ecosystems to disturbance and environmental change over varying time scales.

- (4) There are five dynamic interactive controls regulating ecosystem processes (Chapin et al., 1996). The five interactive controls, e.g., local climate, supply of resources, functional groups of organisms, disturbance regimes and human activities, act as control and are controlled by ecosystem properties (Fig. 13.7). The physical and chemical components of climate affect the activity of organisms and ecosystem processes. Resources are the energy and materials used by the organisms; the type of organisms influence ecosystem processes. Functional groups are species having similar role in ecosystem processes. A gain or loss of functional types can change ecosystem properties and processes through changes in species traits, resource supply and disturbance regime variables (Hooper and Vitousek, 1997). Disturbance is critical for sustaining ecosystem structure and the process rates. The intensity and frequency of disturbance can cause long-term ecosystem change. Human activities are the primary ecological drivers of degradation and rehabilitation of ecosystems.
- (5) There are linkages between primary ecological variables and the state factors. The interaction between state factors, interactive controls and ecosystem processes determines the structure and functioning of an ecosystem at various spatial and temporal scales (Fig. 13.7). Conservation and management of ecosystems depends on a clear understanding of human impacts on the interactive controls as they are the primary ecological drivers of ecosystem degradation.

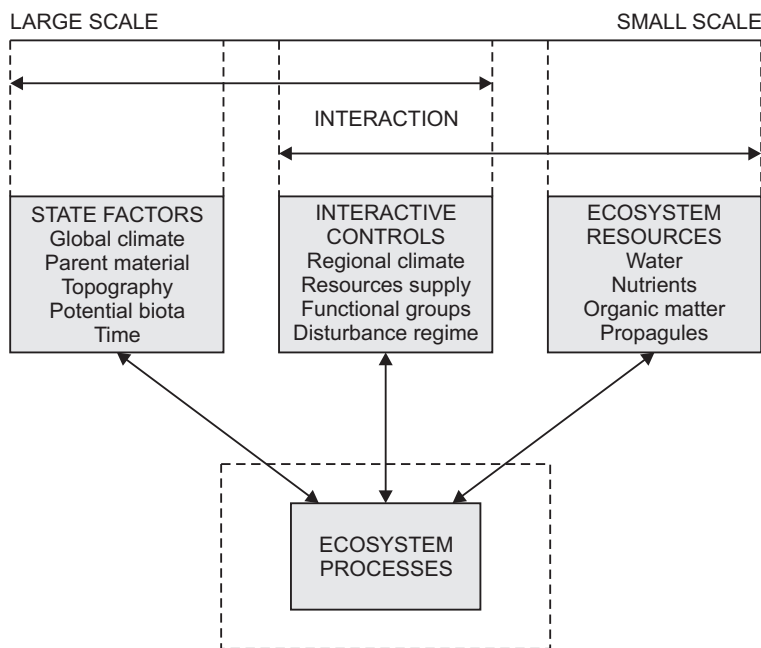


Fig. 13.7. The relationship between state factors, interactive controls and ecosystem processes. (modified from Chapin et al., 1996).

Jenny's state factor approach was a major conceptual contribution to ecosystem ecology by emphasizing the controls over processes and suggesting an experimental approach to test the importance and mode of action of each control (Chapin et al., 2002). A logical way to study the role of each state factor is to compare sites that are as similar as possible with respect to all but one factor. For example, a *chronosequence* is a series of sites of different ages with similar climate, parent material, topography, and potential to be colonized by the same organisms. In a *toposequence*, ecosystems differ mainly in their topographic position.

Sites that differ primarily with respect to climate or parent material allow us to study the impact of these state factors on ecosystem processes. A comparison of ecosystems that differ primarily in potential biota, such as the Mediterranean shrub-lands that have developed on west coasts of California, Chile, Portugal, South Africa, and Australia, illustrates the importance of evolutionary history in shaping ecosystem processes (see Chapin et al., 2002).

Integration of humans into ecosystem framework

Integration of humans and their social, economic and ecological needs into ecosystem framework is essential for ecological sustainability. As shown in Fig. 13.8, the ecosystem resources and services are dependent on energy, biogeochemical and hydrological cycles and the various types of human uses, which have an effect on these flows. Human activities have an increasing impact on all the processes that govern ecosystem properties. The human use of ecosystem services and support, both temporal and spatial scales, depends on the proper functioning of local ecosystems linked to other multifunctional ecosystems (Folke, 1998).

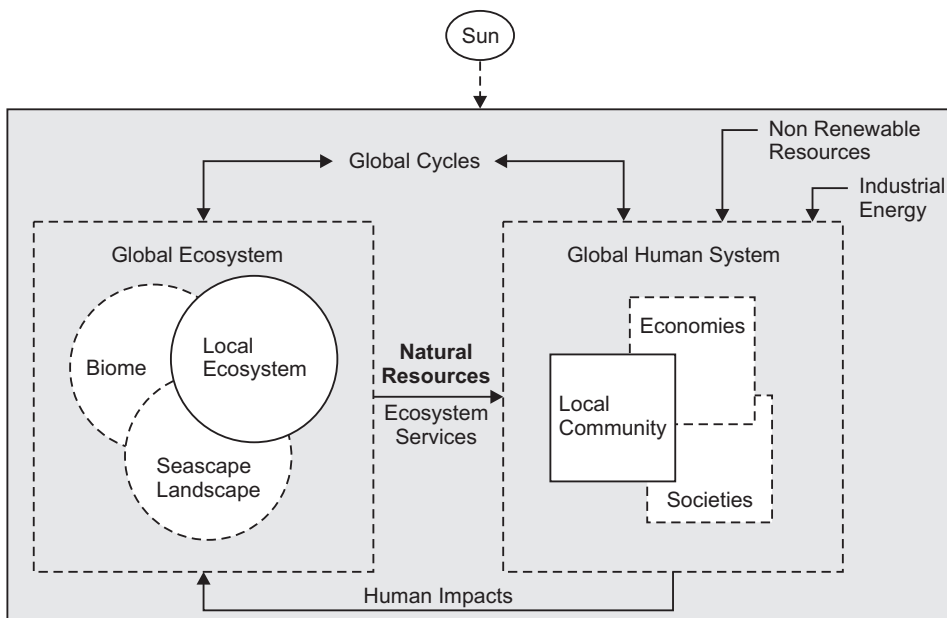


Fig. 13.8. The relationship natural ecosystems and the human system. The global cycle refer to the climate and ocean circulation. The global cycles are the hydrological and biogeochemical cycles. There are cross scale interactions between various flows, natural resources and ecosystem services. (modified from Folke, 1998).

13.9 NEW PARADIGM IN ECOSYSTEM CONCEPT

Norberg et al. (2001) have modelled ecosystems as complex adaptive systems. In their non-equilibrium model, evolutionary dynamics has been found to play a key role to regulate productivity in ecosystems (Norberg et al., 2001). There is a clear link between environmental fluctuations, persistence of trait diversity and ecosystem functioning (Norberg et al., 2001). The results of recent studies on biological diversity on ecosystem functioning, perhaps are a step forward defining a unified ecological theory based on evolutionary, population and ecosystem perspectives (Tilman, 2001).

According to O'Neill (2001), some of the major assumptions of the current ecosystem concept, which lead to weaknesses, are (1) specific spatial unit (but the spatial distributions of the constituent populations may be much larger than the perceived ecosystem boundaries), (2) Spatial homogeneity (but it is the spatial heterogeneity which maintains the full range of populations required for stability), (3) species substitutability (but it leads to anomalous conclusions when applied to phenomena other than stability), (4) minimized role of natural selection (but the natural selection is the most powerful predictive theory available to ecology), (5) humans are external to ecosystem (but humans have altered the physical environment of the ecosystem as no other organism has done). O'Neill (2001) feels that ecosystems are collections of interacting populations, which are shaped by natural selection; and the natural selection is one of the processes that determines ecosystem stability. O'Neill (2001) has proposed a set of principles for the new paradigm of the ecosystem concept, with a major focus on the question of scales, variability, short-term and long-term sustainability and population processes. These principles have been summarised in Box 13.2.

Box 13.2. Principles of the new paradigm of Ecosystem concept (based on O'Neill, 2001)

- An ecological system is composed of a range of spatial scales ranging from the local system to the potential range of dispersal of all the constituent species.
- The biotic and abiotic factors, dispersal barriers and dispersal mechanisms determine the potential dispersal range.
- The potential dispersal range can change with climate, soils, and topography and dispersal barriers.
- Within the potential dispersal range is an effective dispersal range that is time-scaled to the problem under study.
- Within a local ecosystem, such as a watershed, forest stand or a lake, populations interact to form complex networks and feedback loops, and interact to maximize biotic potential.
- As the local ecological system approaches the biotic potential of the site, it increases local stability but decreases global sustainability.
- The balance between rates of change in environmental conditions and rates of change in the biota govern the stability of the local ecological systems.
- The balance between the size distribution of disturbances and the effective dispersal range of the biota govern the stability of the local ecosystems.
- *Homo sapiens* is a keystone species that changes system stability by altering environmental conditions, rate processes and biotic structures.

Primary Productivity and Detritus Pool

The flow of energy through ecosystems parallels with the carbon balance as light energy is used to reduce carbon dioxide to carbon containing compounds during photosynthesis. Virtually all life forms on the earth depend on a very small fraction of solar radiation for the synthesis of food by green plants. The solar radiation is fixed through the process of photosynthesis in the form of organic compounds; a portion is utilized by the plants for their own maintenance and the rest supports the diverse array of heterotrophic organisms. From plants, those heterotrophs, which ingest living matter get the energy, and the remaining is converted into detritus pool to support those heterotrophs which derive their energy from non-living organic matter.

14.1 SOLAR RADIATION AND ENERGY UNITS

Solar radiation, a portion of which is transformed by plants into chemical form in photosynthesis, reaches the earth's outer atmosphere at a relatively constant rate, called *solar flux* or solar constant. This is the amount of energy of all wavelengths per unit time from the sun incident upon a surface placed outside the earth's atmosphere perpendicular to the sun's rays at the mean distance of the earth from the sun. This is estimated to be 2 cal/cm²/min (or 1.39 kW/m² or 1.39 kJ/m²/second). About 42 per cent of incoming radiation is reflected (33 per cent from clouds and 9 per cent from dust), an additional 10 per cent of the solar flux is absorbed by ozone, oxygen, water vapour and carbonic acid or is diffused and scattered by air molecules and smaller particles. Only about 48 per cent of this flux reaches the earth after passing through the troposphere. This energy averages about 3000 to 4000 kcal/m²/day (= 1.1 – 1.5 million kcal/m²/yr). However, this value is not the same at all places; it declines towards poles as the latitude increases, the value being less than half in areas beyond 60-70° latitude than in areas between 0-30° latitude. Since most of the ultraviolet radiation, below 380 nm is absorbed by the high level ozone layer, solar radiation reaching the earth surface consists of only 10 per cent ultraviolet rays. The remaining solar radiation is equally divided into visible (380-780 nm) and infrared (780-3000 nm) radiations.

The fate of the average annual input of solar radiation on the earth can be summarised as, 30 per cent reflected, 46 per cent directly converted to heat, 23 per cent dissipated in evaporation and precipitation and 0.2 per cent in wind waves and current. The rest of it which normally is merely 0.8 per cent is fixed in photosynthesis by autotrophs (Hulbert, 1971). The energy content of various types of organisms and their average energy requirements differ widely (Box 14.1).

14.2 CONCEPT OF PRIMARY PRODUCTION

The quantity of plants present in a community at a given point of time in a given area is called plant *biomass*. This quantity is generally expressed in terms of dry weight per unit area (e.g., g/m², kg/m² or t/ha). While the term biomass is usually reserved for live plants, the term *standing crop*

can be used both for live and dead plants. The term *yield* is used to express the weight of the harvest at any given time, mostly at the end of the growing season. *Primary production* of a community or any part thereof, is defined as the total radiant energy or $\text{CO}_2\text{-C}$ fixed by photosynthetic or chemosynthetic activity of producer organisms (chiefly the green plants; according to Chapin III, et al., 2006, from an energetic perspective, chemoautotrophy is better classified as a component of secondary production, rather than GPP) in the form of organic substances in an unit area in a given period of time. It is usually expressed as dry weight per unit area per unit time (e.g., $\text{g/m}^2/\text{yr}$ or t/ha/yr), or as energy (e.g. $\text{cal/m}^2/\text{yr}$) by determining the caloric values of plant tissues with a Bomb calorimeter. Land plants average about 4.25 kcal/g dry wt, aquatic macrophytes 4.50 and phytoplankton 4.9 kcal/g dry wt. The concept of metabolism and productivity has been illustrated in Fig. 14.1. The total energy or $\text{CO}_2\text{-C}$ fixed by photosynthesis in a given area (or volume) per unit time, including the photosynthate (organic matter) used up in respiration, is the *gross primary production* (GPP). A significant part of this energy is released by respiration of plants themselves (R_{SA}) for use in their metabolic activities. GPP minus R_{SA} is the *net primary production* (NPP). In Fig. 14.1, $\text{NPP} = \text{S1} + \text{S2} + \text{S3}$. This is defined as the rate of storage of energy (or organic matter) in plant tissues in excess of the respiratory utilization by the plants during the period of measurement.

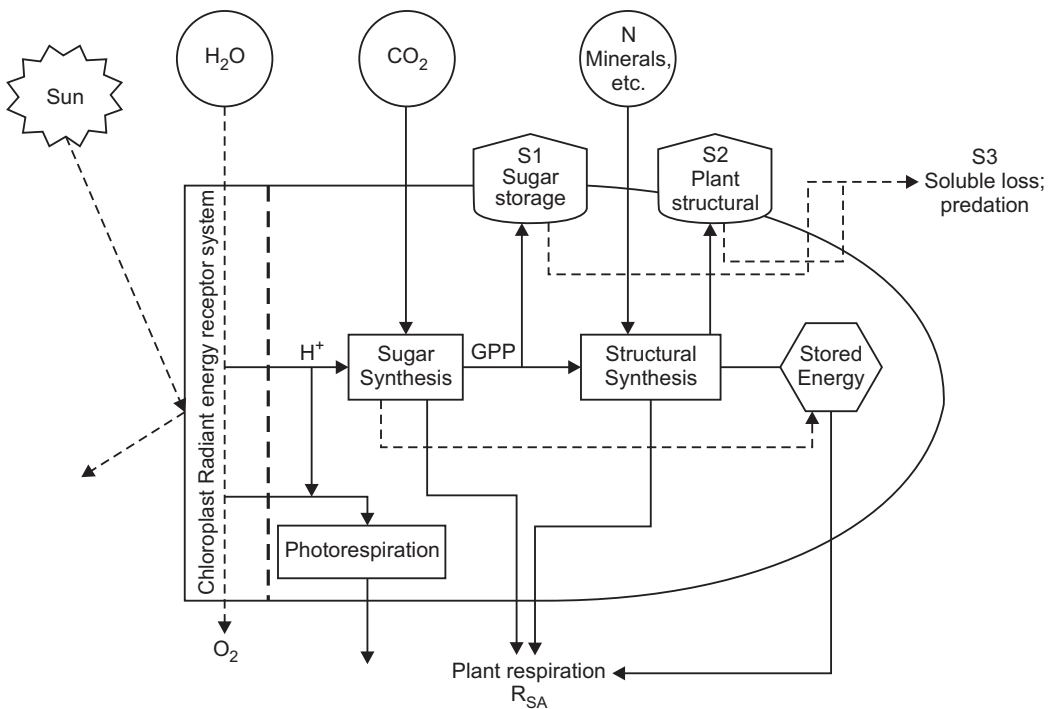


Fig. 14.1. Schematic representation of metabolism and productivity at the level of plants in an ecosystem; GPP = Gross primary production, $\text{S1} + \text{S2}$ = Net community production and $\text{S1} + \text{S2} + \text{S3}$ = Net primary production (redrawn from Hall and Moll, 1975).

Thus while $\text{GPP} = \text{NPP} + R_{\text{SA}}$, $\text{NPP} = \text{GPP} - R_{\text{SA}}$. Some of the organic matter or energy stored in the community in net primary production is consumed by the heterotrophs (Hc) during the measurement period. Thus, the *net community production* (NCP) is the amount of stored organic matter or energy per unit area per unit time not used by heterotrophs, i.e., $\text{NCP} = \text{NPP} - \text{Hc}$. Further, during the measurement period, a portion of the material produced may die and is either deposited at the soil surface or is decomposed.

Box 14.1. Some important energy units**Work**

1 cal = 4.18 J, 1 kcal = 1000 cal = 4180 J,

1 W.s = 0.239 cal = 107 erg; 1 W.h = 3.6 kW. s = 3.6 kJ = 0.86 kcal

Radiation

1 W/m² = 1 J/m²/s = 1.4×10^{-3} cal/cm² min

1 erg/cm²/s = 1.43×10^{-6} cal/cm²/min = 10^{-3} W/m²

1 klx = 4 – 10 W/m² (depending on light source)

Average Energy content of organisms (in kcal/g dry weight)

Terrestrial plants 4.5, seeds 5.2, Algae 4.9, insects 5.4, other invertebrates 3.0, vertebrates 5.6

Average Food energy requirements

Humans: 40 kcal/kg live weight/day or 2800 kcal/day for a 70 kg adult

Small birds and mammals: 1.01 kcal/g live weight/day

Insects: 0.5 kcal/g live weight/day

Average energy content of fossil fuels

1 g coal = 7.0 kcal, 1 g gasoline = 11.5 kcal,

1 cubic foot natural gas = 250 kcal,

1 litre kerosene = 11628 kcal, 1 litre diesel 10814 kcal,

1 litre petroleum = 9419 kcal

Average energy content of food

grains = 3865 kcal/kg, pulses = 4070 kcal/kg, fresh leaf vegetables = 676 kcal/kg, roots and tubers = 942 kcal/kg, other fresh vegetables = 574 kcal/kg, fresh fruits = 848 kcal/kg, fresh fish = 1414 kcal/kg, goat meat = 1548 kcal/kg, lamb meat = 2295 kcal/kg, cow milk 749 kcal/kg, buffaloe milk = 1278 kcal/kg.

The net community production can, thus, be represented as:

$$NCP = \Delta B + L + Hc,$$

where ΔB refers to increment in biomass, L refers to mortality and Hc refers to biomass consumption by herbivores, within the measurement period. Most of the values referred to as net primary production in literature are usually net community production.

Another term, *net ecosystem production* (NEP) has been used to represent annual increment of biomass to the ecosystem and is defined as $NEP = GPP - (R_{SA} + R_{SH})$ or $NEP = NPP - R_{SH}$, where R_{SA} is plant respiration and R_{SH} is heterotrophic respiration. The term NEP was first introduced by Woodwell and Whittaker (1968) to represent the difference between ecosystem level photosynthetic gain of CO_2-C (i.e., GPP) and ecosystem (plant, animal and microbial) respiratory loss of CO_2-C (i.e., ecosystem respiration, ER). NEP was also thought to represent the rate of C accumulation in ecosystems, that is the imbalance of photosynthesis and ER. However, as pointed out by Chapin III et al. (2006), in several ecosystems dissolved, volatile, and depositional organic and inorganic C fluxes are substantial and therefore, imbalance between GPP and ER may not equal the rate of C accumulation in ecosystems. Chapin III, et al. (2006) propose a new term, Net Ecosystem Carbon Balance (NECB) to be applied to the net rate of C accumulation in (or loss, when NECB is negative, from) ecosystems. Here NECB represents the overall ecosystem C balance from all sources and sinks—physical, biological, and anthropogenic (e.g., non-respiratory loss of C by fire). Extrapolation of NECB to larger spatial scales is termed Net Biome Production (NBP).

To summarise, at the primary trophic level a part of the solar energy that reaches the plants is absorbed while another part is not absorbed. Of the absorbed energy, only a small fraction, usually less than 1 per cent is assimilated. Of the assimilated energy (GPP when rate is indicated) a fraction is respired (R_{SA}) by the plants and the remainder is converted into plant tissues (i.e., NPP when rate is indicated). The percentage of respiration, for example may be only around 25 per cent of GPP in a cornfield and as high as 50-70 per cent in some forests.

14.3 FOUNDATIONS OF PRIMARY PRODUCTIVITY

Photosynthesis, Photosynthetic pathways and their adaptive significance

The basis for all energy transfers in the ecosystem is the process of *photosynthesis* in which green plants utilize incident solar energy and CO_2 to produce O_2 and complex organic materials through a combination of biophysical and biochemical processes driven by radiation of wavelengths approximately 400-700 nm which are absorbed by the chloroplasts. The series of steps comprise interdependent light and dark reactions as briefly outlined in Chapter 13.

The actual fixation of CO_2 may be by one of the two major pathways; the (i) Calvin-Benson, and (ii) Hatch and Slack. Plants with Calvin-Benson pathway are referred to as C_3 plants because the first stable product formed is a 3-carbon compound, phosphoglyceric acid (PGA). CO_2 is initially bound to ribulose-1, 5-diphosphate (RuDP) through the action of the enzyme RuDP carboxylase (Rubisco).

In C_4 photosynthesis (Hatch and Slack), CO_2 is bound to phosphoenolpyruvate (PEP) by the enzyme PEP carboxylase to form oxaloacetate. The latter is immediately converted to either malate or aspartate, the 4-carbon acids. These acids are transported to thick-walled bundle sheath cells where CO_2 is released. The thick walls of the bundle sheath cells act as physical barrier to CO_2 diffusion, hence CO_2 is concentrated in these cells and it subsequently enters the Calvin-Benson cycle. Thus the C_4 cycle enhances the C_3 pathway (Fig. 14.2). Additionally, the photosynthetic activity in the bundle sheath is carried out close to the supply of water from the vascular bundle. Effects of water stress, to which the photosynthetic apparatus is highly sensitive, are thus minimized. The C_4 plants have large dense chloroplasts in the bundle sheath cells around vascular bundles in leaves, a distinctive morphological feature, called Kranz anatomy, noted a century earlier.

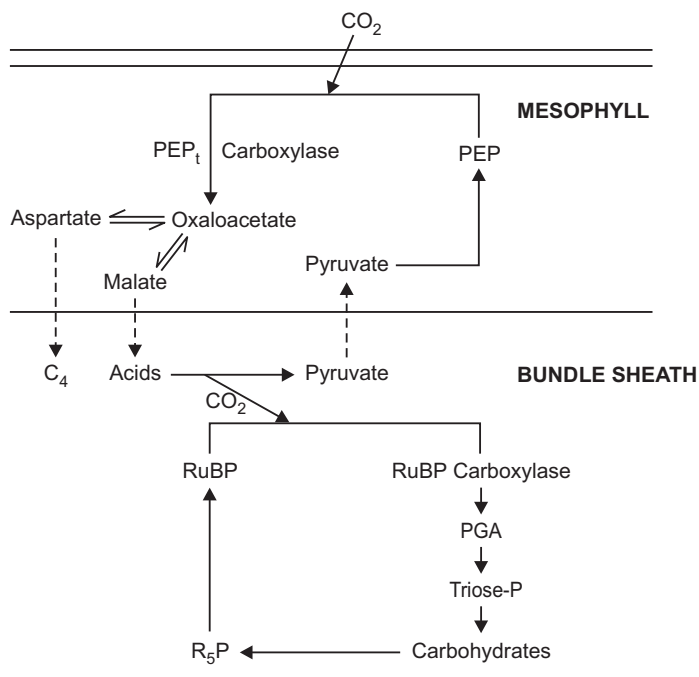


Fig. 14.2. Spatial separation of C_4 and C_3 cycles in C_4 plants. The C_4 cycle operates in the mesophyll and the C_3 cycle in the bundle sheath.

Of great ecological significance is the fact that the C_3 and C_4 plants respond differently to light (Fig. 14.3a), temperature (Fig. 14.3b) and water. When photosynthetic rate per unit of leaf surface is compared, we find that C_3 plants tend to attain the peak at moderate light intensities and temperatures. Both high light intensities and high temperatures tend to inhibit photosynthesis of these plants. In contrast, C_4 plants are adapted to high light intensities and temperatures, greatly exceeding the photosynthetic rates of C_3 plants under these conditions. The maximum growth rate of the former is four times greater than that of the latter. The C_3 plants also use significantly lesser amount of water for each unit of dry matter produced (less than 400 g water compared to 400-1000 g water per gram of dry matter produced in C_3 plants).

The greater net CO_2 fixation rates of C_4 plants are probably important in terms of survival and adaptation, but, more importantly, the performance of these plants under extreme conditions of stress is definitely of selective advantage. The C_4 species survive and are better adapted than many C_3 species under

conditions of high water stress, high temperature, high irradiance and low CO_2 concentrations. In hot dry climates, such as in tropics, stomata may be only partially open during a major period in day time in order to check transpiration leading to reduced CO_2 diffusion to chloroplast. This results in lower CO_2 , which may be in very low concentration, concentrations at the photosynthetic site. The RuDP carboxylase needs a high concentration of CO_2 . Therefore, in C_4 plants the enzyme PEP carboxylase traps CO_2 , which may be in very low concentration, first into oxaloacetate and then the CO_2 is released in the bundle sheath area at high concentration for RuDP carboxylase to act upon. The C_3 cycle requires three ATP molecules and two NADPH molecules to reduce one molecule of CO_2 , while the C_4 cycle requires five ATPs and three NADPHs. However, the capability of C_4 plants to utilize higher light intensities, to function efficiently at lower CO_2 concentrations, to use lesser water per gram dry weight, and to exhibit no or little photorespiration, compared to C_3 plants, compensates for additional energy requirement for C_4 cycle.

C_4 plants are mostly of tropical origin. In terms of net primary productivity, C_4 grass canopy is more efficient at lower latitudes (towards tropics) while C_3 grass canopy performs best at higher latitudes (temperate and alpine) (Fig. 14.4). When C_4 plants occur in the temperate region, they grow best during the warm season. Because of the evolutionary trends, the C_4 grasses are referred to as warm-season grasses, whereas the C_3 grasses are called cool-season grasses (Sims et al., 1978; Singh et al., 1983). In the northern plains of India, farmers grow many C_4 crops, viz. maize, sorghum, sugarcane, etc. during the warmer part of the year and C_3 crops viz., wheat, mustard, gram, pea, etc. during the cooler part. It may, however, be noted that all tree species are C_3 plants.

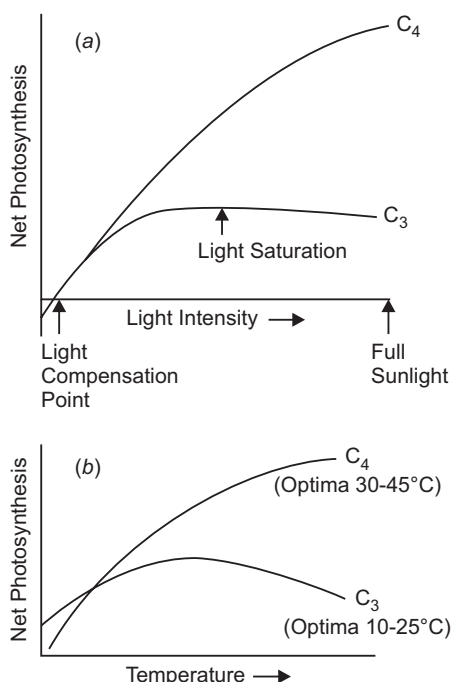


Fig. 14.3. Response of net photosynthesis in C_3 and C_4 plants to variation in light intensity and temperature. (a) C_3 plants become saturated at levels considerably lower than full sunlight but C_4 plants are often not saturated at full sunlight. (b) Optimal temperatures for C_3 plants are lower (10-25°C) and for C_4 plants are higher (30-45°C).

Conditions in very dry areas with wide extremes of temperature, going up to 40°C in the day (with leaf temperatures up to 50°C), and down to <15°C at night, perhaps necessitated a further modification in the photosynthetic pathway. In such conditions, if the stomata were to open appreciably during the day, water loss would occur at a detrimental rate. The problem of providing CO₂ for photosynthesis in this situation has been solved by another carbon fixation scheme, referred to as crassulacean acid metabolism (CAM). A number of plants having succulent photosynthetic tissues, including members of Crassulaceae, Cactaceae and Euphorbiaceae occurring in deserts, show this mechanism. About 10% of the earth's

flora possess CAM photosynthesis. These plants commonly grow in xeric habitats or in high light environments such as epiphytes or plants in the canopies of tropical forests. They keep their stomates closed during the hot day time and open them in relatively cool night when the rate of water loss will be much lower. Carbon dioxide, absorbed in the night, is fixed by PEP carboxylase and stored as organic acids, mainly in the vacuoles, by a pathway similar to that in C₄ plants. In the day time, when light is available, the organic acids release CO₂ which, then, is fixed through Calvin-Benson cycle. The dark CO₂ fixation has a low temperature optimum (10-15°C). The closed stomates and delayed photosynthesis restrict the water loss and enable the plants to maintain their water status. This cycle is important to plants in dry habitats due to temporal separation between nocturnal CO₂ fixation and CO₂ metabolism during the day. CAM photosynthesis allows some plants to fix carbon under dry conditions and allows ecosystems carbon gain even at very low soil moisture. These plants resemble C₄ plants in having no differentiation of mesophyll cells into palisade and spongy parenchyma, higher optimum temperatures for photosynthesis (30-35°C), low photorespiration, occurrence in open, warm and often saline sites; they resemble C₃ plants in having lower light saturation point for photosynthesis and no requirement of Na; and apart from having unique scheme of carbon dioxide fixation, they differ from C₃ and C₄ plants in having comparatively very low maximum growth rate (about 0.02 g dm² day⁻¹) and very high water use efficiency (about 80 g H₂O g⁻¹ CO₂ fixed). While in C₄ plants, the C₄ and C₃ cycles are spatially separated (C₃ occurring in bundle sheath cells), in CAM plants these are temporally separated (C₄ occurring at night and C₃ in day). Important characteristics of C₃, C₄ and CAM plants are given in Table 14.1.

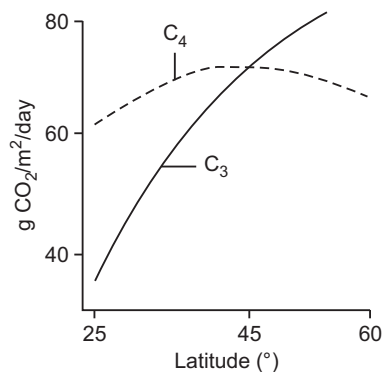


Fig. 14.4. Daily carbon gain by C₃ and C₄ grass canopies at different latitudes in northern hemisphere during July (based on Larcher, 1982).

Photosynthetic carbon fixation in phytoplankton

Energy is the prime limiting factor in marine environments. Therefore, phytoplankters do not rely on active uptake of inorganic carbon from the abundant HCO₃⁻ pool, which is about 90% in the sea water. During the process of photosynthesis, mostly passive uptake of CO₂ occurs through the cell membranes of phytoplankters at sea water CO₂ concentration of 10-15 μmol/liter. To ensure efficient operation of the enzyme ribulose biphosphate carboxylase/oxygenase (Rubisco), an active uptake of CO₂ or HCO₃⁻ also occurs into the algae cells and/or in the chloroplasts by using the enzyme carbonic anhydrase, which plays a key role as a carbon concentrating mechanism. These organisms also absorb carbon from the small and variable pool of CO₂ dissolved in water. The various groups of phytoplankton show differences with respect to the CO₂ specificity of the carboxylating enzyme, Rubisco. Highest Rubisco specificities are found in the most recently evolved groups of phytoplankton, the diatoms; progressively lower specificities occur in coccolithophorids, green algae, dinoflagellates and most ancient phytoplankters, the cyanobacteria (Riebesell, 2000).

Table 14.1. Some traits related to photosynthetic pathways. (Compiled from Black, 1973 and other sources).

No.	Trait	C ₃	C ₄	CAM
1.	Taxonomic range	very wide, algae to higher plants	No algae, lower vascular flowing plants	Some species in about 26 families <i>Welwitschia</i>
2.	Habitat preference	None in particular	Open, warm	Open and warm, also saline
3.	Leaf anatomy	Festucoid, Palisade layer present, palisade cells not differentiated	Kranz, palisade layer absent, bundle sheath cells present with chloroplasts	Mesophyll not differentiated, big cells with large vacuoles
4.	Photosynthetic light saturation	3000-6000 ft-C	Near or above 8000-ft-C	Between C ₃ and C ₄ plants
5.	Optimum temperature for photosynthesis	10-25°C	30-45°C	30-45°C
6.	CO ₂ compensation point	30-70 ppm	Less than 10 ppm	0-200 ppm
7.	Stomatal behaviour	open in daytime, closed at night	open in daytime, closed at night	Closed in daytime, open in Night time
8.	Carboxylating enzyme	RuBP carboxylase	PEP carboxylase in mesophyll and RuBP carboxylase in bundle sheath cells	PEP carboxylase in night and RuBP carboxylase in day time
9.	CO ₂ acceptor	RuBP	PEP carboxylase in mesophyll and RuBP in bundle sheath cells	PEP in night and RuBP in day time
10.	First stable product of photosynthesis	PGA (a C ₃ acid)	Malate, aspartate (C ₄ acids)	Malate
11.	¹² C/ ¹³ C ratio	High	Low	Low
12.	Pathway separation	None	Spatial (mesophyll/bundle sheath)	Temporal (night/day)
13.	Photorespiration	High	None or very low	Low
14.	Maximum photosynthetic rate (mg dm ⁻² hr ⁻¹)	15-35	40-80	1-4
15.	Maximum growth rate (g dm ⁻² day ⁻¹)	0.5-2.0	3-5	0.01-0.02
16.	Water efficiency (g H ₂ O g ⁻¹ CO ₂ fixed)	450-950	250-350	50-55

Reinfelder et al. (2000) have shown for the first time that a marine microalga, the coastal diatom, *Thalassiosira weissflogii*, uses the C_4 pathway for photosynthesis. In the single-celled diatom, the two processes could occur simultaneously, one in the cytoplasm and the other in chloroplast (Reinfelder et al., 2000). The existence of C_4 photosynthesis in diatoms suggests that they could be flexible to respond to environmental variability, as well as important to studying foodweb structure and the marine carbon cycle (Riebesell, 2000; Reinfelder et al., 2000).

Photosynthetic Capacity

Plants differ markedly in their photosynthetic capacity. Photosynthetic capacity is the maximum rate of net photosynthesis under natural CO_2 concentration (0.03 per cent by volume) and optimal conditions with respect to all other external factors (saturating light intensity, optimal temperature and adequate water supply). Larcher (1982) has compiled a list of plant groups showing their photosynthetic capacities (Table 14.2).

Table 14.2. A cross section of values for photosynthetic capacity of plant groups (after Larcher 1982).

Plant group	CO_2 uptake (mg/dm ² illuminated leaf surface/hour)
Herbaceous C_4 plants	30-80
C_3 crop plants	20-45
C_3 heliophytes	20-40
C_3 sciophytes	4-20
CAM plants (in light)	3-20
CAM plants (in dark)	10-15
Tropical and subtropical forest canopy trees	12-24
Sun leaves of deciduous trees	15-25
Shade leaves of deciduous trees	5-10
Evergreen conifers	5-18
Subtropical evergreen broad-leaf trees (sun leaves)	10-18
Desert shrubs	6-20
Mosses	upto 3
Emergent hydrophytes	28-40
Submerged hydrophytes	2-6

Some of the trends that have emerged from comparison of photosynthetic capacities of plant groups are (i) C_3 plants photosynthesize at markedly lower rates than C_4 plants, (ii) among C_3 plants, crop plants photosynthesize at higher rates, perhaps due to selection breeding, (iii) photosynthetic capacity of thallophytes and submerged aquatics is very low, (iv) herbs out perform woody species, (v) shade plants manage only half to one-third of the carbon assimilation of sun plants, and (vi) conifers and succulents photosynthesize only at moderate rates. Similarly, plants differ in specific respiratory rates (rate of dark respiration measured at a standard temperature). Herbaceous species respire twice as rapidly as the foliage of deciduous trees, the latter respire approximately at five times the rate of foliage of evergreen trees, and heliophytes respire distinctly more rapidly than sciophytes (Larcher, 1982). Because of the variable photosynthetic capacity, the efficiency of foliage to produce organic matter differs from species to species. This is illustrated by data on the fresh weight and surface area of foliage needed to produce 1 m³ of solid wood in certain tree species of Switzerland (Table 14.3). Evidently, foliage of Scots pine is more efficient compared to that of spruce or fir.

Table 14.3. Efficiency of foliage of certain tree species in merchantable wood production in Switzerland (based on Trendelenburg and Mayer-Wagelin, 1955).

Species	Fresh weight of foliage required to produce 1 m ³ wood (kg)	Area of foliage needed to produce 1 m ³ of solid wood (m ²)
Scots pine	1200	6600
Spruce, even aged	3000	16500
Spruce, uneven aged	4000	20800
Fir	2400	13440
Oak	1100	13640
Beech	800	14080

Relative growth rate (RGR) and net assimilation rate (NAR) are two important measurements in context of growth and production of plants (Box 14.2).

Box 14.2. Relative growth rate

The daily rate of gain in dry weight, the relative growth rate (RGR), is governed by two sets of factors. First, it is a function of the net assimilation rate (NAR) which can be defined as the net diurnal efficiency of assimilation per unit of photosynthetic surface. Secondly, there is the leaf area ratio (LAR) which is the ratio of the total assimilation surface to the total plant weight. Estimates of RGR and NAR can be made by the following formulae, provided that the leaf area is not more than doubled in the time t :

$$RGR = \frac{\ln W_t - \ln W_0}{t}$$

$$NAR = \frac{W_t - W_0}{t} \times \frac{\ln A_t - \ln A_0}{A_t - A_0}$$

where the measurements are over a period of time t (in days), the initial values of the plant weight and leaf area are W_0 and A_0 , respectively, and the final values W_t and A_t , respectively.

Allocation of Net Primary Production and Accumulation of Biomass

The photosynthate is allocated to different components of the plant body according to survival and growth strategies. The annual plants initially allocate a greater proportion of assimilates to leaves for quick development of photosynthetic canopy, and later to reproductive parts to ensure successful seed production. The perennial herbs, after their photosynthetic canopy is formed, tend to allocate more to the non-photosynthetic tissues (for support structure) and/or to storage organs from which the reserves can be remobilized to initiate growth in the beginning of the next growing season. A marked amount of assimilates is also used to develop the perennial root system. In trees, a significant proportion of the photosynthate is used to develop the massive support organs such as trunk and branches. Initially, there is a large allocation to foliage which declines later to a lower asymptote (Fig. 14.5). Peak allocation to branches and trunk usually lags

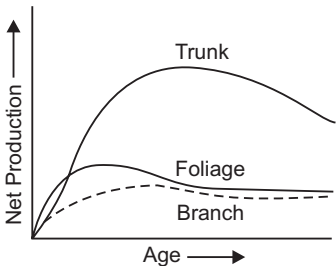


Fig. 14.5. Changes in allocation of net primary production (NPP) to foliage, branch and trunk in scots pine (based on Albrectson, 1980).

behind that to leaves. In deciduous trees, initial growth of foliage and new shoots perhaps occurs at the expense of labile material stored from the preceding season; this reserve is again rebuilt in the later part of the growing season. Because of the need to build massive support structures, prolific seed formation in trees may occur only at an interval of few years (the good seed years), when due to specially favourable conditions a large amount of assimilates is formed which can be allocated to reproduction.

Table 14.4 shows the share of foliage, branches and trunk in the aboveground biomass and net above ground primary production (ANP) of the tree layer of four forests. Leaves, which represent the photosynthetic machinery, account for 37 to 41 per cent of the tree layer ANP although they comprise only 3.4 to 4.6 per cent of the above ground tree biomass. Thus, most of the ANP used in the development of foliage would find its way to detritus each year through litter fall. The tree trunks account for 25 to 46 per cent of the ANP but from 43-77 per cent of the biomass. The dry deciduous forest and the evergreen oak forest allocate 38 to 33 per cent of the ANP to branches while the pine and the old-growth sal forests allocate only 14 to 26 per cent ANP for these organs. In contrast to the forests, grasslands allocate maximum ANP to the leaves. It is evident that the short-lived components account for a greater proportion of ecosystem function and the long-lived components a greater proportion of ecosystem structure (Singh and Singh, 1993).

Table 14.4. Allocation of biomass (B) and net aboveground primary production (ANP) to different tree components of four forest types.

	Dry deciduous ¹		Old growth Sal ²		Pine forest ³		Oak forest ⁴	
	B	ANP	B	ANP	B	ANP	B	ANP
Percent allocation								
Foliage (%)	4.3	37.2	3.4	40.7	4.3	39.4	4.6	39.7
Branches (%)	52.6	38.0	20.0	26.2	19.5	14.2	44.5	32.8
Trunk (%)	43.1	24.8	76.6	33.1	76.2	46.4	50.9	27.8
Total aboveground								
tree biomass (t/ha)	62.94	—	561.8	—	171.7	—	301.5	—
Total tree ANP (t/ha/yr)	—	8.15	—	12.9	—	11.9	—	12.8

¹Singh and Singh (1991,1993); ²Singh and Singh (1989); ³Chaturvedi and Singh (1987); ⁴Rawat and Singh (1988).

Allocation of net primary production to roots is important. In temperate grasslands more than 70 per cent of net production is translocated to belowground plant parts (Sims and Singh 1978). In the Pacific silver fir (*Abies amabilis*) ecosystem, roots and mycorrhizal fungi account for 70-80 per cent of net primary production (Vogt et al., 1982). A significant proportion of the net photosynthate allocated to roots is either diverted to mycorrhizal symbionts or exuded into the soil. Photosynthates exuded from roots and mycorrhizal hyphae support a diverse community of soil organisms, and the exudates are important for maintaining the physical and biological structure of soils.

Different layers of the vegetation (e.g., herbaceous and ligneous or woody) contribute to the net primary production according to the availability of light. Table 14.5 compares the relative contributions of herbaceous and ligneous (trees and shrubs) layers to the net primary productivity of a savanna, a dry deciduous forest and an evergreen Oak forest. The sparse tree population in the savanna permits abundant light to the ground surface, as a result the herbaceous layer is well developed and contributes as much as 71 per cent ANP against 29 per cent of the tree layer.

Table 14.5. Net aboveground primary production of ligneous and herbaceous layers in three vegetation types.

	Savanna ¹	Dry deciduous ²	Oak forest ³
Ligneous layer (Tree + shrub) (%)	28.6	90.7	92.4
Herbaceous layer (%)	71.4	9.3	7.6
Total ANP (Trees + herbs) (t/ha/yr)	10.4	11.0	15.3

¹Pandey and Singh (1992); ²Singh and Singh (1991); ³Rawat and Singh (1988).

BOX 14.3. Allometric Theory

In a series of papers, Enquist and Niklas (2001, 2002) have discussed the importance of allometric theory to interrelate attributes of organisms, community and ecosystem across temporal and spatial scales. They have applied allometric relationships on the worldwide data sets, the interspecific data sets of the angiosperms and gymnosperms and the intraspecific data sets, and found highly significant relationships between biomass of roots, stems and the leaves. The biophysical models are the following set of equations:

- (1) M_L proportional to $M_S^{3/4}$ proportional to $M_R^{3/4}$; where M_L = biomass of leaf, M_S = biomass of stem, M_R = biomass of root.
 - (2) M_A proportional to $M_R + M_R^{3/4}$; where M_A = aboveground biomass, M_R = belowground biomass.
- The equations described as single power functions seem to apply widely across major phylogenetic divisions (conifers *versus* angiosperms) and across diverse ecological conditions (Enquist and Niklas, 2002). However, the increase in biomass may not be proportional among the three components due to greater allocation of biomass to the stems in large plants (Zens and Webb, 2002; Enquist and Niklas, 2002). In small plants, the aboveground biomass is comparatively high because of greater contribution from the leaves. On the other hand, in larger plants, the contribution of leaves is smaller.

According to these workers, allometric theory can provide a valuable quantitative tool for studying a great variety of form and growth rate phenomenon of individual plants within and across communities. It can give a better insight into the mechanisms regulating the structure of ecological communities across varying environmental conditions. Allometric theory also has wider implications to “predict past and present-day structure and function relation at the level of individual, community and the whole ecosystem” and to understand the mechanistic connections between numerous features of organismic biology, ecology, ecosystem studies and evolutionary biology (Enquist and Niklas,2001; Enquist and Niklas, 2002).

As the tree layer becomes dense, less light reaches the ground and the importance of herb layer as a contributor to net production declines. Thus, in the evergreen Oak forest less than 8 per cent ANP is due to the herb layer, ligneous vegetation contributing as much as 92 per cent ANP. Closure of canopy results in a substantial increase in the leaf area index (LAI).

The net primary production is usually maximum at LAI = 4. However, in mature forests, LAI may be greater than 8 when GPP assumes maximum rate and NPP declines (due to increasing cost of respiration as the

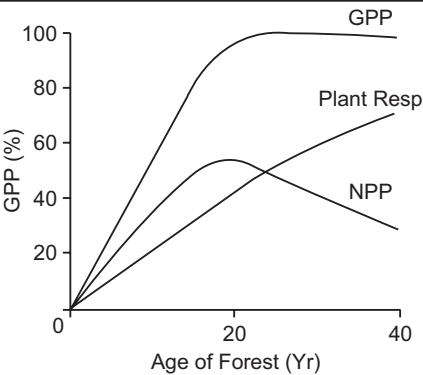


Fig. 14.6. Changes in the ecosystem metabolism during development of a forest from establishment to maturity (based on Waring and Schlesinger, 1985).

photosynthetic capacity of shaded leaves declines). However, net production in communities having plants with steeply inclined leaves (80 degrees) may continue to increase up to LAI of 7 or more. The influence of LAI is also partly reflected in age-NPP relationship (Fig. 14.6). GPP reaches an asymptote with canopy closure, respiration continues to increase with age, while NPP declines after attaining a peak at or just before canopy closure. The relationship between net production and leaf area index is also influenced by the light intensity. For example, a stand of *Trifolium subterraneum* shows optimal LAI (i.e., LAI at which net production was maximum) of less than 4 when light intensity is 100 cal/cm²/day, and of 7 when the solar radiation is 700 cal/cm²/day.

14.4 MEASUREMENT OF PRIMARY PRODUCTIVITY

Gross primary production is never directly measured, its estimates are usually derived from direct measurements of net primary production plus estimates of plant respiration. Some common methods to measure primary production are discussed here for both terrestrial and aquatic ecosystems.

Harvest Method

For a crop field, where a single annual species grows, a single harvest when crop matures may be sufficient to measure the approximate ANP. Here the initial dry matter may be regarded as zero or it may be regarded as equal to the weight of seeds sown. In the latter case, the seed weight is subtracted from the oven-dried biomass of mature crop to determine the ANP. Even in such simple communities, ANP may be underestimated because the leaves that are produced and shed before the crop matures are not accounted for.

In natural communities, things are more complicated as all species occurring may not reach their peak growth at the same time. For multispecies herbaceous communities, such as grasslands, plants are harvested from plots randomly located in a grid. The harvested material is separated into species and into live and dead shoots. Samples are oven dried to constant weight. Sampling is repeated several times within an annual cycle, using different plots each time. Changes in standing crop in time are calculated to estimate the ANP according to one of the following methods:

1. ANP = Peak standing crop (live + dead shoots) – initial standing crop (live + dead).
2. ANP = Sum of peak biomasses (peak dry weights of live shoots) of all species sampled – initial total biomass.
3.
$$NPP = \sum_{t=1}^n LS + DS$$
, that is, sum of increments in biomass of live shoots plus increments

in the standing crop of dead shoots within the same sampling intervals, in an annual cycle.

The first method is usually sufficient for simple communities where most of the species realize peak growth at the same time. Methods 2 and 3 are suitable for multiple species communities. Different methods and their advantages and disadvantages are discussed in Singh et al. (1975); the method 3 is more commonly used than the other two methods.

The above methods provide estimates of net primary production for the aboveground vegetation only. Estimates for root production are obtained by excavating monoliths of pre-fixed size. The monoliths are washed to remove soil particles, and the belowground plant parts are oven-dried. The sampling is repeated several times as in the case of aboveground vegetation. Increments in the biomass of all belowground plant parts (roots, tubers, rhizomes, etc.) within the annual cycle are then summed to represent annual below ground net production (BNP). The sum of net primary production of the aboveground vegetation (ANP) and that of belowground plant parts (BNP) provides the estimate for annual total net primary production (TNP, often simply referred to as NPP).

Dimension analysis

Estimation of net primary productivity of natural forests and of shrub communities having individuals of different ages and large weights requires an elaborate approach. This method involves the following steps:

(1) Sample trees of different size classes are felled and measured for diameter at breast height (dbh), and height. Dry weights of the tree parts, viz., stem, branch, twigs, leaves, stump roots, lateral roots, flowers and fruits are determined for each sample tree.

(2) The harvest data obtained as above are subjected to regression analysis to relate the dry weight of each component with the dbh of the tree. This approach is termed dimension analysis (Whittaker and Woodwell, 1967) or allometry (Kira and Shidei, 1967). The regression equation most frequently used is:

$\log Y = a + b \log X$, where Y = dry weight of the component (kg tree^{-1}), X = dbh ($\text{cm}^2 \text{ tree}^{-1}$), and a = intercept and b = slope or regression coefficient. These equations can be developed for each component of the tree or for the tree as a whole in which case Y will be the dry weight of the entire tree.

Dbh squared x height (i.e., d^2h), instead of only dbh, has been favoured as an independent variable by Kira and Shidei (1967). The error is often minimized when the independent variable includes tree height (Whittaker and Marks, 1975). The above two steps result into biomass equations for different species in the forest being studied. Once the regression equations for different species have been developed, no further felling of trees is required.

(3) Mean dbh value for each species for a girth class is used in the regression equation to get an estimate of the mean tree biomass for that girth class. This value is multiplied by the density of trees in that class. Total biomass of a species is calculated by summing the biomass values across diameter classes of that species. Sum of biomass values across species yields stand biomass.

(4) Marked trees in sample quadrats are measured twice, first at the start of the study and the next time usually after one year. Usually trees are divided into number of girth or diameter classes and mean diameter increment for each class is then calculated. If d^2h is used in the biomass equations, tree heights are also measured.

(5) Using the allometric equations as explained in step 3 the biomass for different components of trees is calculated from the dbh or d^2h measurements. For example, B_1 is biomass in 1980 and B_2 in 1981. The net change in biomass ($B = B_2 - B_1$) yields annual biomass accumulation. The sum of B values for different components yields net biomass accretion in the trees.

(6) In evergreen tree species annual leaf fall is added to the foliage biomass for estimating annual leaf production. For deciduous species foliage production can be estimated directly through leaf fall or through foliage biomass if biomass equations were developed at the full-foliage period. Wood and miscellaneous litter fall is also added to the biomass of twigs/branches.

(7) The above estimates do not include fine root production. The production of fine roots is estimated using the monoliths as explained in the harvest method. Fine root production is then added to the values obtained above.

A small fraction, usually 1.5 per cent of leaves is eaten by insects, and some small amount of shrubs and herbs are browsed by herbivores; similarly some roots are eaten by burrowing animals and some wood by boring insects. But these amounts are generally negligible. Nevertheless if estimates for the above losses are available they should be added to the net production estimates.

Production of larger trees can also be approximated through estimated volume increment (EVI) which is one-half of the increase in wood area at breast height times tree height.

These equations having only CBH or DBH and height as the estimator, developed on the basis of trees harvested from a single site may not be suitable for regional estimates. These equations pose a significant problem for regional-scale comparisons of tree biomass estimates, because variation caused by environmental, structural, and compositional gradients may be confounded with variation resulting from the use of different regression equations. Therefore, for the study of regional variations of tree biomass estimates, those aspects of forest structure that vary significantly at regional scales should be used in the estimator. Wood specific gravity (ρ) is one such characteristic that varies widely between tropical forest tree species, and is closely related to differences in life history strategies and differences in diameter growth rates (Chaturvedi et al., 2010). It is also recognized as an important determinant of differences in tree biomass over successional gradients. Baker et al. (2004) incorporated ρ as a simple multiplication factor in the biomass equations based on tree diameter and was able to demonstrate regional differences in the estimates of tropical forest above ground biomass. Nelson et al. (1999) was able to reduce the error of biomass estimate by incorporating ρ in the estimator. Chaturvedi et al. (2010) observed that ρ -based equation (Stem biomass (kg tree^{-1}) = $0.5(\pi/4) \rho D^2 H$, where, 0.5 is the assumed form factor, defined as the ratio of stem volume to the volume of a cylinder with the H (m) and DBH (cm) of the tree) gave results closer to actual estimates as compared to the CBH-based equations. These authors concluded that the non-destructive equation with ρ , height and DBH can be used instead of allometric equations having CBH as estimator developed on the basis of harvested trees. For juvenile trees, Chaturvedi et al. (2012) use the following equation: $\ln Y = a + b (\ln \rho D^2 H)$, where Y is stem biomass (g), D is diameter (cm) and H is height (cm).

Carbon dioxide method

Changes in CO_2 concentration may be used to determine the CO_2 uptake in photosynthesis or its loss in respiration in terrestrial conditions. A transparent plastic box or a transparent polyethylene tent is placed over the community, air is drawn through the enclosure and CO_2 concentration in incoming and outgoing air is measured through infra-red gas analyser. Net production (Net CO_2 exchange) is measured during the day and respiration during the night. Odum and Pigeon (1970) encircled an area in forest by a large tent open at top and bottom, and equipped with fan at the bottom to bring about one-way air movement. CO_2 concentrations at the bottom and at the top were used to calculate net production.

The vegetation in the enclosure can also be labelled with ^{14}C , by releasing $^{14}\text{CO}_2$ within the enclosure and allowing the plants to absorb $^{14}\text{CO}_2$ for a fixed period of time. ^{14}C activity is then counted in the plants and the amount assimilated is calculated.

The build-up of temperature and an increase in speed of air movement within the enclosure during the measurement, however, affect the estimates of NPP. The aerodynamic method in which periodic measurements of CO_2 are made is an advance over the enclosure method. Using CO_2 sensors placed at intervals along a vertical pole, differences in the CO_2 concentration above and within the autotrophic layer (green layer) are measured. From this a measure of net primary production is obtained. The method has been used satisfactorily for crops, grasslands and forests. This method has an advantage over the enclosure method because natural conditions are maintained during the course of CO_2 measurements.

Light-and-dark bottle method for Aquatic Systems : Oxygen Measurement

In this method, oxygen production is measured by using a pair of bottles (typically of 125-300 ml capacity), one of which is covered with black tape or aluminium foil to exclude light (dark-bottle), thus permitting only respiration to occur. In the other, the light bottle, both photosynthesis

and respiration occur. Water samples of lake, pond, ocean, etc. from given water depth are taken and are distributed into pairs of light and dark bottles. The water contains usually phytoplankton and some consumers, viz., zooplankton and bacteria. The bottles are suspended, at the depth from which samples were taken and allowed to incubate from 4 to 24 hours. Oxygen is measured by Winkler method, involving fixation with MnSO_4 , H_2SO_4 and alkaline iodine, which releases elemental iodine in proportion to oxygen. The iodine is titrated with sodium thiosulphate solution at a concentration calibrated to estimate mg oxygen per litre, which is g/m^3 water or ppm (parts per million). The initial concentration of dissolved oxygen (O_2IB) falls to a lower value (O_2DB) in the dark bottle due to respiratory consumption, and increases to another value (O_2LB) in the light bottle depending upon the difference between photosynthesis and respiration. The difference $\text{O}_2\text{IB}-\text{O}_2\text{DB}$ represents respiration over the time interval involved, the difference $\text{O}_2\text{LB}-\text{O}_2\text{IB}$ represents net photosynthesis, and the difference $\text{O}_2\text{LB}-\text{O}_2\text{DB}$ gross photosynthesis. These equations can be written as:

$$\text{Net photosynthesis} = \text{O}_2\text{LB}-\text{O}_2\text{IB}$$

$$\text{Respiration} = \text{O}_2\text{IB}-\text{O}_2\text{DB}$$

$$\text{Gross photosynthesis} = \text{O}_2\text{LB}-\text{O}_2\text{DB}$$

Nowadays, oxygen electrodes are commonly employed to measure changes in oxygen concentration of water. By calculating the oxygen changes in the light bottle after incubation of 4 or 8 or 12 or 24 hrs, net primary productivity is determined. Community respiration is determined from changes in oxygen concentration of dark bottle. The combined values of the two approximate the gross production. Since water samples also contain heterotrophs, the net primary productivity can be considered as NEP.

^{14}C Radioisotope Technique

The ^{14}C technique is particularly useful for measuring the productivity of plankton communities in waters of low productivity such as oligotrophic lakes and open ocean (actually such estimates of productivity can also be argued as that of NEP). The ^{14}C can also be used in light and dark-bottles. In this method, a known quantity of ^{14}C , usually in the form of bicarbonate, such as $\text{NaH}^{14}\text{CO}_3$, is introduced into bottles containing water samples. They are incubated at the water depth of measurement usually for 6 hrs. The total amount of CO_2 in water sample is measured before incubation so that the ratio of normal $^{12}\text{CO}_2$ to $^{14}\text{CO}_2$ can be determined. The samples are filtered and counted in a scintillation counter. Difference between the counts (radioactivity) in light and dark bottles represents ^{14}C assimilated by phytoplankton. The total carbon uptake is calculated as ^{14}C assimilated/ ^{14}C available $\times$ ^{12}C available $\times$ appropriate constants for volume and time. Estimates of NPP in aquatic ecosystems based on ^{14}C are intermediate to the theoretical rates of NPP and GPP because phytoplankton respire some but not all of the newly fixed, ^{14}C -labeled organic C.

Chlorophyll measurement

The net primary productivity of both the terrestrial and aquatic ecosystems can also be estimated by determining the chlorophyll content provided the assimilation ratio (rate of production per gram of chlorophyll) and the available light intensity and duration are known.

The concentration of chlorophyll determines the rate of photosynthesis and has been frequently used as an indicator of oceanic plant biomass and productivity. Marine algae assimilate 3.7g carbon /g chlorophyll / hr (Ryther and Yentesch, 1957). The concentration of chlorophyll has been found to be low in ocean gyres ($< 0.05 \mu\text{g/litre}$) and high in upwelling areas, along continental shelf regions and coastal areas and estuaries (1 to $10 \mu\text{g/litre}$) (Jeffrey et al., 1997). In aquatic ecosystems, chlorophyll concentration has been estimated using spectrophotometric, fluorometric and high performance liquid chromatographic methods. *In situ* monitoring of fluorescence, turbidity

and water colour (spectral emission) of the phytoplankton have been used to measure chlorophyll content over large areas. Chlorophyll measurements have been used to determine diurnal, seasonal and long-term changes in productivity of different oceanic regions. Satellite remote sensing has been used to map surface chlorophyll and productivity in the oceans (Jeffrey et al., 1997).

14.5 PLANT BIOMASS AND TURNOVER

Among the various ecosystems, forests accumulate greatest plant biomass and the marine ecosystem the least (Fig. 14.7). In general, forest biomass increases towards equator. However, exceptions do occur. For example, old-growth *Sequoia sempervirens* forest of the Pacific North West may accumulate 3461 t/ha stem biomass (Fujimori et al., 1976) while dry tropical forests may accumulate only 78-320 t/ha total biomass (Murphy and Lugo 1986). In general, biomass of mature forests is in the range of 20-60 kg/m², in woodlands 4-20 kg/m², in shrublands 2-10 kg/m² and in deserts 0.5-3 kg/m². The perennial aerial structures in the forests account for nearly three-fourth of the total biomass, while the roots account for only about one-fourth (Table 14.6). On the other hand, grasslands and shrubby vegetation of tundra store a substantial proportion of plant biomass in roots.

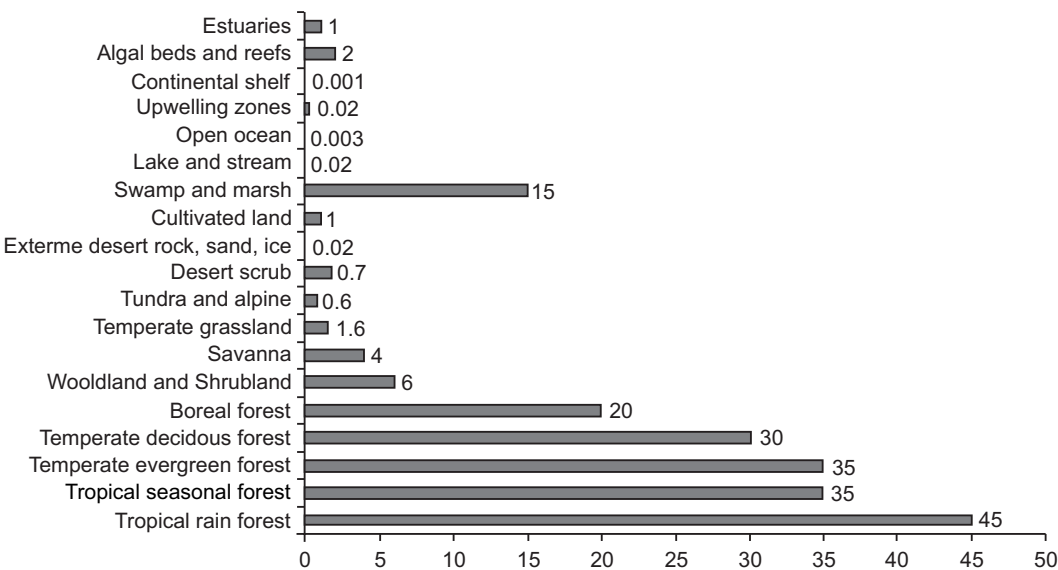


Fig. 14.7. Trend in the variations in biomass of different terrestrial and marine ecosystems (based on data from Whittaker and Likens, 1975).

Table 14.6. Per cent distribution of plant biomass in different components (compiled from various sources).

	Green parts	Perennial aerial parts	Roots
Shrubby Tundra	13	12	75
Boreal forest	7	71	22
Temperate deciduous forest	1	74	25
Temperate tall grass prairie	17	0	83
Tropical sub humid grassland	45	0	55
Tropical dry deciduous forest	3	73	24
Wet tropical forest	8	74	18

It is useful to assess the rate at which biomass of a given component is replaced. The rate of replacement of a component of a system or of a substance in a component is referred to as *turnover rate*. According to Odum (1971), turnover may be defined as the ratio of throughput to content and can be expressed as rate fraction or as a *turnover time*. Turnover time is the reciprocal of *turnover rate*. Turnover time broadly indicates the *residence time* or *transit time* which refers to the time a given amount of substance remains in a designated compartment of a system. The turnover rate is, thus, the ratio of production to biomass (P/B). For example, the foliage biomass in a dry deciduous forest is 2.7 t/ha and the net foliage production is 3.05 t/ha/yr (Table 14.4). The P/B ratio, thus, is 1.13, and the turnover time (reciprocal of P/B ratio) is 0.88 yr. This means that in less than a year (above 10.5 months) the foliage is completely replaced. In the same example, biomass of the branches is 33.07 t/ha and net branch production is 3.12 t/ha/yr. The turnover rate of the branch biomass is then 0.094 (i.e., 9.4 per cent per year) and the turnover time is 10.6 yr. Thus, the more dynamic components of the ecosystem have a faster turnover rate. Similarly, the total aboveground tree biomass of the dry deciduous forest is 62.9 t/ha and net aboveground primary productivity 8.2 t/ha/yr. These values indicate that the aboveground tree biomass in this forest is replaced at the rate of 13 per cent per year ($P/B = 0.130$) and the turnover time is 7.7 yrs. In contrast, the turnover rate of the aboveground tree biomass of the Oak forest is only 4.2 per cent per year and the turnover time is 23.6 yr. The P/B ratio of phytoplankton ranges between 20-40 and turnover time (B/P) from 0.05-0.025 yr (or 9-18 days). Phytoplankton biomass turns over at a faster rate compared to plant biomass of a grassland which in turn is replaced at a faster rate than that of a forest (Table 14.7). Low P/B ratio indicates extensive development of supporting tissues in the forest. High P/B ratios of phytoplankton and grasslands indicate greater energy availability to herbivores. Greater longevity and larger accumulation of biomass, usually result in slow turnover. Plant biomass in ecosystems with warmer conditions turns over at a faster rate compared to that in cooler ecosystems (Table 14.7).

Table 14.7. Mean biomass, net primary production (NPP) and turnover of biomass in selected terrestrial ecosystems (Data from various sources).

Ecosystem	NPP (t/ha/yr)	Biomass (t/ha)	Turnover rate	Turnover time (yr)
Boreal forest	7.5	200	0.037	26.7
Temperate deciduous forest	11.5	350	0.033	30.4
Temperate tall grass prairie	11.0	24	0.46	2.2
Tropical sub humid grassland	21.0	22	0.95	1.05
Tropical dry forest	15	200	0.07	14.3
Tropical wet forest	30	500	0.06	16.7

14.6 NET PRIMARY PRODUCTION ON LAND

About one-third of the land, comprising tundra and alpine ecosystem, desert, semi-desert scrub and extreme desert (rock, sand and ice) has very low primary production ranging from 0-250 g/m²/yr (Table 14.8). On the other hand, several forests of warm and humid region, often called tropical rain forests, may realize as high as 3,000-3,500 g/m²/yr net production that is the maximum production rate on the land on a large area basis (Whittaker, 1975). Similar high values for production occur in swamp and marshes which account for a very small fraction of the area, and in crops like sugarcane it can be still higher (up to 4000-6000 g/m²/yr). The environment of these communities is characterized by high moisture availability and relatively warm temperatures, and often regular inputs of nutrients. Between these extreme values are the net productivity values (250-2500 g/m²/yr) of temperate ecosystems.

Table 14.8. Net primary productivity and related characteristics of terrestrial biomass (from Whittaker and Likens, 1975).

Ecosystem type	Area (10 ⁶ km ²)	Primary productivity (dry matter)			Biomass (dry matter)		
		Normal range (g m ⁻² yr ⁻¹)	Mean (g m ⁻² yr ⁻¹)	Total (10 ⁹ t yr ⁻¹)	Normal range (kg m ⁻²)	Mean (kg m ⁻²)	Total (10 ⁹ t)
Tropical rain forest	17.0	1000-3500	2200	37.4	6-80	45	765
Tropical seasonal forest	7.5	1000-2500	1600	12.0	6-60	35	260
Temperate forest:							
evergreen	5.0	600-2500	1300	6.5	6-200	35	175
deciduous	7.0	600-2500	1200	8.4	6-60	30	210
Boreal forest	12.0	400-2000	800	9.6	6-40	20	240
Woodland and shrubland	8.5	250-1200	700	6.0	2-20	6	50
Savanna	15.0	200-2000	900	13.5	0.2-15	4	60
Temperate grassland	9.0	200-1500	600	5.4	0.2-5	1.6	14
Tundra and alpine	8.0	10-400	140	1.1	0.1-3	0.6	5
Desert and semi-desert scrub	18.0	10-250	90	1.6	0.1-4	0.7	13
Extreme desert:							
Rock, sand, ice	24.0	0-10	3	0.07	0-0.2	0.02	0.5
Cultivated land	14.0	100-4000	650	9.1	0.4-12	1	14
Swamp and marsh	2.0	800-6000	3000	6.0	3-50	15	30
Lake and stream	2.0	1000-15000	400	0.8	0-0.1	0.02	0.05
Total continental	149		782	117.5		12.2	1835
Open ocean	332.0	2-400	125	41.5	0.0-0.005	0.003	1.0
Upwelling zones	0.4	400-1000	500	0.2	0.005-0.1	0.02	0.008
Continental shelf	26.6	200-600	360	9.6	0.001-0.04	0.001	0.27
Algal beds and reefs	0.6	500-4000	2500	1.6	0.04-4.0	2.0	1.2
Estuaries	1.4	200-4000	1500	2.1	0.01-4.0	1.0	1.4
Total marine	361		155	55.0		0.01	3.9
Grand total	510		336	172.5		3.6	1841

Temperature is one of the major determinants of productivity pattern in the world. From Tropics to Arctic regions productivity values decline as mean annual temperature decreases from 25-30°C to less than zero. The positive effect of the increase in temperature is greater on GPP, for it also increases respiration, say from 50-60 per cent in most temperate forests to 70-80 per cent in tropical forests. In fact, because of higher respiration in tropical communities there is a considerable overlap in the range of NPP between the two regions. Much of the adverse effect of low temperature on productivity may be due to the shorter growth period in cold climates. It is partly because of shorter effective growth period, that deciduous forests, at least in temperate zone, boreal forests, woodlands, shrublands, savanna, temperate grasslands, lakes and streams and most of the croplands realize low productivity. Of these ecosystems, many of the forests, some grasslands and some productive crops which realize productivity in the range of 1000-2500 g/m²/yr, occupy relatively favourable environments, and most of the woodlands, shrublands and temperate grasslands unfavourable environments. Broadly speaking, primary productivity increases from polar regions, to the equator except for the intervening strongly water-limited deserts. The greater

productivity of tropical regions (Fig. 14.8) to a large extent, is due to the favourable combination of high incident solar radiation, warm temperatures, and abundant rainfall. These factors result into longer, almost year-round growing season.

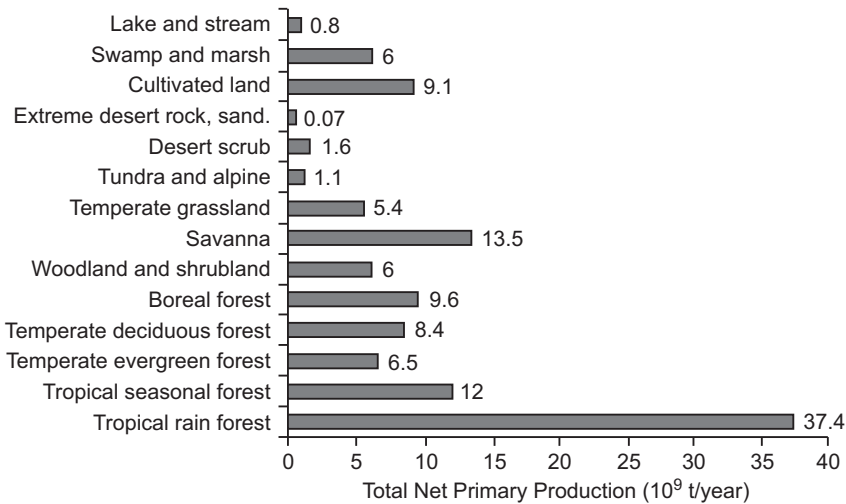


Fig. 14.8. Total net primary production in different types of terrestrial and freshwater systems.

Because the stomata must be open if CO_2 is to be taken in photosynthesis, and through open stomata enormous transpiration loss of water occurs, plants on land need enormous amounts of water, up to 700-1000 g or more per g of net production. Therefore, water is a major determinant of production on land. In drier climates (with less than 1000 mm rainfall), the production almost increases linearly with increasing annual precipitation. In a dry climate evapotranspiration may equal precipitation, while in wet climates it is less than precipitation but would increase as temperature increases. Therefore, dry deciduous species realize lower production than their evergreen counterparts, in spite of the fact that production efficiencies of leaves of deciduous species are greater than those of the evergreen leaves. Evergreen tissues are reported to have 30-60 per cent lower maximum photosynthetic rates per unit surface area than deciduous leaves (Mooney, 1972). Deciduous plants have low leaf area index (LAI) during periods of leaf development in the spring and may miss a substantial period of favourable temperature. Similarly, the period of favourable temperature in later part of summer may be lost as a consequence of photoperiodically induced leaf drop. The needle-leaf bearing evergreens are also known to have twice as much LAI as deciduous species.

Precipitation appears to be the most important factor for biomass accumulation in seasonally dry tropical forests. Studies have indicated that biomass in mature tropical forest generally increases along precipitation gradients. Recently, a review of 229 estimates of aboveground biomass from 44 studies in seasonally dry tropical forest confirmed the paramount importance of precipitation (Becknell et al., 2012). In this analysis, above ground biomass (AGB) of mature forests ranged from 39 to 334 Mg ha^{-1} and exhibited a positive relationship with mean annual precipitation (MAP) which explained >50% of the variation in AGB (Fig. 14.9 a), according to, $\text{AGB} (\text{Mg ha}^{-1}) = 0.1872 \text{ MAP (mm)} - 52.5608$ ($R^2 = 0.55$). AGB was also negatively related to the length of growing season (Fig. 14.9 b) and was positively related to log-transformed stand age. Consequently, sites with lower MAP had a lower maximum attainable biomass and biomass accumulation tended to level off around 40 yrs of age. Because of differences in water availability (depending upon soil texture, other soil properties, and rainfall), species composition and intensity of biotic disturbance, the biomass in dry tropical forest is not uniformly distributed across the forest but exhibits patchy distribution

(Chaturvedi et al., 2011). Chaturvedi et al., (2013) found that stomatal conductance is the most important variable which accounted for the greatest amount of variability (62%) in RGR of tropical tree seedlings, emphasizing the role of stomatal conductance in shaping growth patterns across spatial and temporal gradients of soil water availability. Stomatal conductance and soil moisture content together explained 64% variability in RGR.

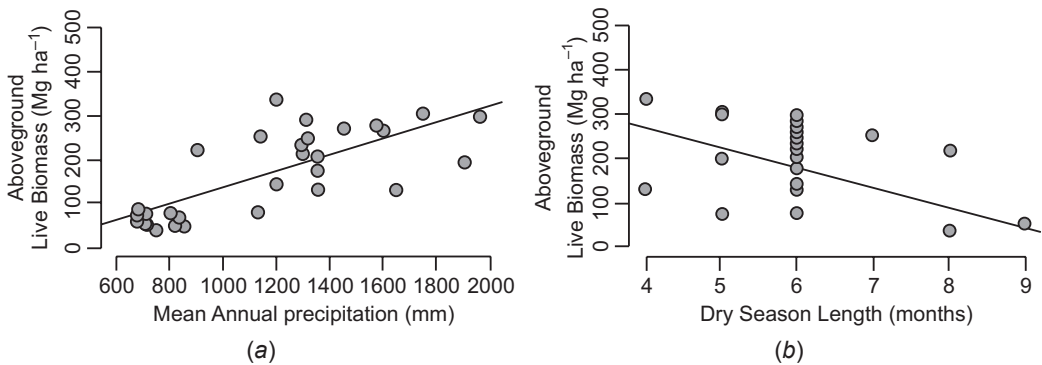


Fig. 14.9. Above ground biomass as a function of (a) mean annual precipitation, and (b) length of dry season (after Becknell et al., 2012)

Webb et al., (1983) have analysed relationships between the variations of aboveground net primary production (ANP) and abiotic as well as biotic variables, using data from 15 forest sites, 11 grassland sites and 4 desert sites in the United States. The sampled sites covered an extremely wide range of precipitation from 116 mm at a desert site to 2592 mm at a forest site, and mean annual temperature, from 5.5 to 21.4°C. PET (potential evapotranspiration) ranged from the average of 663 mm/yr for forests to 1452 mm/yr for the deserts and short grass steppes. The annual water deficits ranged between 153-659 mm in forests and between 924-1821 in grasslands and deserts. Aboveground standing crop (ASC) ranged between 0.70 t/ha to 882 t/ha, showing a difference greater than three orders of magnitude between desert and forest. Similarly, NPP (net primary productivity) varied from 0.16 t/ha/yr at a desert site to 32.8 t/ha/yr at a forest site. A similar trend was found for peak foliage standing crop (FSC).

The study indicated that differences in ANP are closely related to differences in abiotic variables but that community composition also influences ANP. This observation is based on the following: (i) The ANP increases with the increasing peak foliar standing crop (FSC), forests with large FSC realizing far greater ANP (716-3279 g/m²/yr) than grasslands (82-452 g/m²/yr) and deserts (16-292 g/m²/yr) (Fig. 14.10). The aboveground standing crop (ASC) decreases with increasing water deficit, expressed as PET (annual potential evapotranspiration) - PPT (annual precipitation) (Fig. 14.11). (ii) However, the ANP per unit of ASC increases as water availability decreases

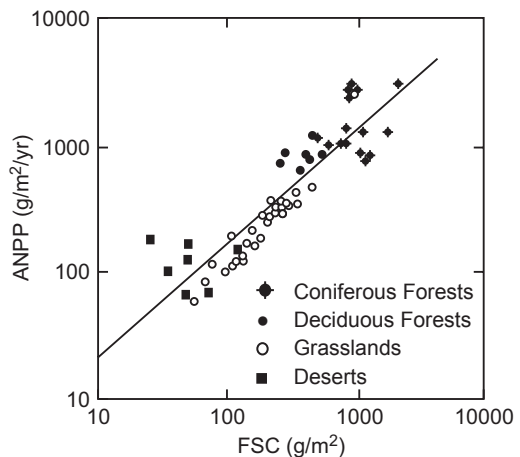


Fig. 14.10. Relationship between aboveground net primary production (ANPP) and peak foliar standing crop (FSC) of forests, grasslands and deserts in the United States (redrawn from Webb et. al., 1983).

$$\ln \text{ANPP} = 0.76 + 0.9 \ln \text{FSC}$$

$$R^2 = 0.794$$

(Fig. 14.11). Webb et al., (1983) suggest that, possibly severely water-stressed systems require more ANP per unit of ASC for internal ecosystem processes. (iii) Similarly, production per unit foliage standing crop (ANP/FSC) tends to increase with decreasing water availability from forest to desert (Fig.14.12). Comparison between the forests indicates that deciduous forests produce more per unit of FSC than do the coniferous forests. (iv) When production is seen as percentage of solar radiation (a form of photosynthetic efficiency) forest sites with better water availability are clearly more productive than water-limited sites like, grasslands and deserts. (v) However, when photosynthetic efficiency is normalised per unit of FSC, the difference almost disappears, and deserts with sparse foliage turn out to be as efficient in converting radiant energy to organic matter as do highly productive forests.

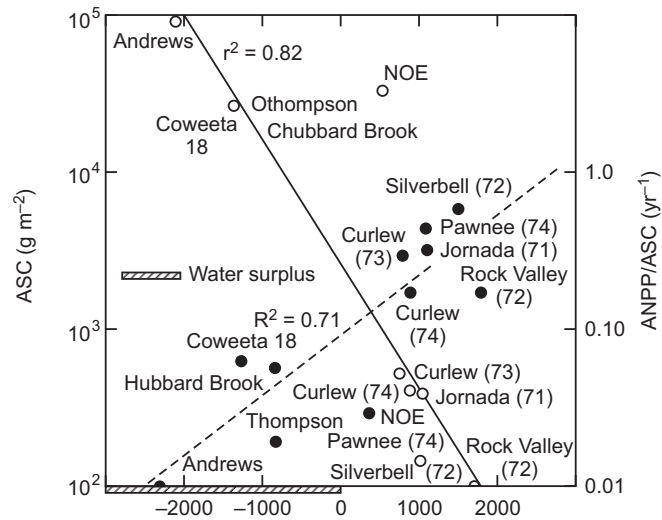


Fig. 14.11. Relationship between annual water availability (Potential evapotranspiration, PET minus Precipitation, PPT) and ratio of aboveground net production to aboveground standing crop (ASC) for forests, grasslands and deserts in the United States (redrawn from Webb et al., 1983. —○— ASC, —●— ANP/ASC, // Water surplus

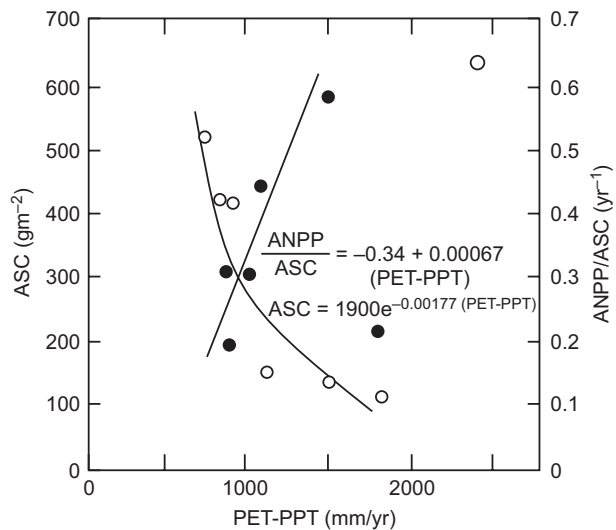


Fig. 14.12. Relationship between annual water availability (PET-PPT) and ratio of aboveground net production in aboveground standing crop (ANPP/ASC) for water stressed systems in the United States (redrawn from Webb et al., 1983).

Productivity models

Some regression models have been developed to illustrate the relationships of net primary production with temperature and precipitation (see Lieth, 1975 for details). The Miami model comprises a pair of correlation models of temperature and precipitation *versus* net productivity, and was first presented at the Miami Symposium. It is assumed that average net productivity does not exceed 3000 g/m²/yr.

The relationship between mean annual temperature and productivity (Fig. 14.13) is expressed as:

$$Y = \frac{3000}{1 + e^{1.315 - 0.119x}}$$

where, Y is net productivity (g/m²), x is mean annual temperature (°C), and e is the natural log base.

The optimum temperature for productivity, 15°–25°C (Fig. 14.13), agrees with the optimum temperature range for photosynthesis, and the productivity almost doubles every 10°C between –10° and 20°C. The curve in Figure 14.13 has a sigmoid shape.

The relationship between precipitation and productivity (Fig. 14.14) is expressed as:

$$Y = 3000 (1 - e^{-0.000664x})$$

where Y is the net productivity (g/m²), x is the annual precipitation (mm), and e is the natural log base. This curve is also sigmoid with an initial sharp increase in productivity with increase in precipitation. Both these equations are used for predicting the net production from the environmental variables of a given site, however, the lower of the two calculated values is used. The relationship between net primary productivity and actual evapotranspiration (AET) and the length of the photoperiod are shown in Figs. 14.15 to 14.16. The NPP clearly increases with AET (a measure of temperature and water availability) and the length of the photoperiod.

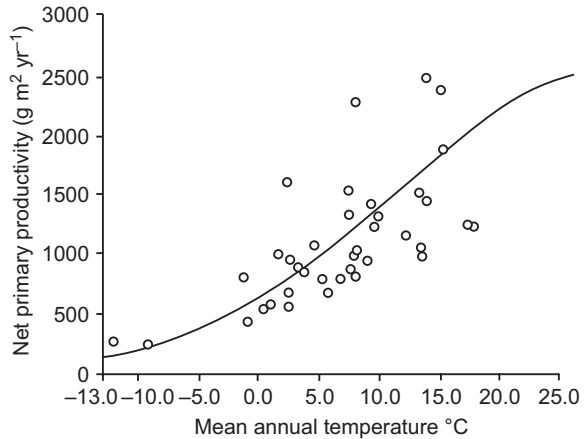


Fig. 14.13. Net primary productivity on land in relation to mean annual temperature (adapted from Lieth, 1975)

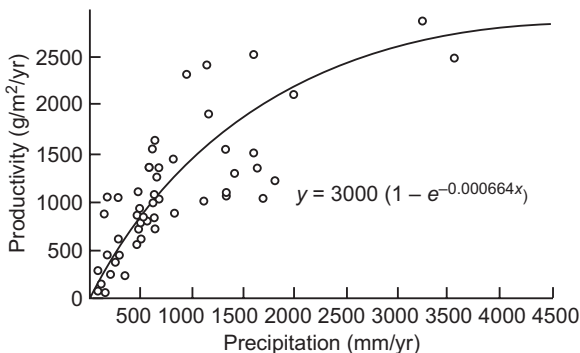


Fig. 14.14. Net primary productivity on land in relation to mean annual precipitation. (adapted from Lieth, 1975)

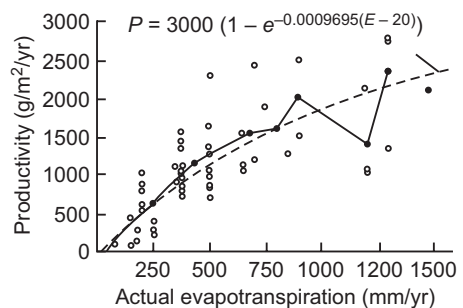


Fig. 14.15. Net primary productivity in relation to actual evapotranspiration (AET) (from Lieth, 1975).

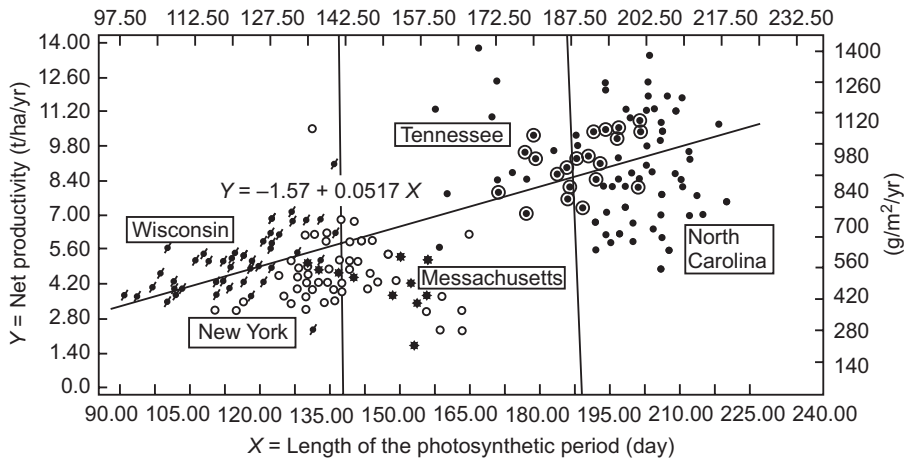


Fig. 14.16. Net primary productivity in Eastern United States in relation to the length of photosynthetic period (PP) on land in relation to actual evapotranspiration (AET). (adapted from Lieth, 1975)

Other Popular Models

BIOME-BGC model

BIOME-BGC is a biogeochemical model that simulates the storage and fluxes of water, carbon and nitrogen in terrestrial ecosystems using a daily time step and at different ecological scales. Initially, the model developed as FOREST BGC model (Running and Gower, 1991) for dynamic C allocation and nitrogen budgets in forest ecosystem. Later it was generalized as an ecosystem process model for other biomes, BIOME-BGC; 34 parameters are used to separate the different biomes (Running and Hunt, Jr, 1993). The model requires daily weather data including radiation, maximum and minimum temperatures, precipitation and daytime vapour pressure deficit (VPD). It also requires information describing the soil properties and eco-physiological traits of vegetation. The net ecosystem CO_2 exchanges, representing the net accumulation or loss of carbon (CO_2 flux) by the entire soil-stand system, are determined by:

$$F_c = A - R_a - R_h$$

where F_c ($\text{mmol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$) represents the CO_2 flux between the reference height and canopy air space. A ($\text{mmol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$) is the photosynthesis or assimilation rate, R_a ($\text{mmol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$) the autotrophic respiration and R_h ($\text{mmol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$) the heterotrophic respiration. The model has been successfully used by several authors worldwide. Kumar and Raghubanshi (2011) used it for dry tropical forest and found that among the eco-physiological parameters, canopy light extinction coefficient, canopy average specific leaf area (projected area basis), new stem C : new leaf C, maximum stomatal conductance (projected area basis), C:N of fine roots, all-sided to projected leaf area ratio and canopy water interception coefficient showed the strongest effect on simulated NPP.

Production efficiency models

Production efficiency models (PEMs) are based on the theory of light use efficiency (LUE) which states that a relatively constant relationship exists between photosynthetic carbon uptake and radiation receipt at the canopy level (McCallum et al., 2009). In general, all PEMs employ a similar basic methodology to calculate NPP. Typically this involves two steps, first calculating GPP and then subtracting Autotrophic Respiration (Ra) to derive NPP:

$$GPP = PAR * FAPAR * LUE * Scalars$$

$$NPP = GPP - Ra$$

where: GPP Gross Primary Productivity (gC m^{-2}), PAR Photosynthetically Active Radiation (MJ m^{-2}), FAPAR Fraction of Absorbed PAR (dimensionless %), LUE Light Use Efficiency (gC MJ^{-1}), Scalars Temperature, (VPD) Vapour Pressure Deficit, etc (0-1), NPP Net Primary Production (gC m^{-2}) and Ra Autotrophic respiration (gC m^{-2}).

Kale and Roy (2012) used such a model which uses LUE and intercepted photosynthetically active radiation (IPAR) and is said to be one of the efficient ways to estimate the NPP at different scales as (i) the modelling follows the basic principles of the process of photosynthesis and (ii) is satellite driven. It is the relationship of ground based IPAR with satellite derived indices mainly normalised difference vegetation index (NDVI), which makes PEM satellite driven. The PEM based NPP is estimated through, $NPP = \varepsilon * (\Sigma IPAR)$ where, ε is the weighted (relative density and relative basal area of tree species can be used as weight) LUE and IPAR is the Intercepted Photosynthetically Active Radiation ($\mu\text{mol/m}^2/\text{s}$). The LUE (or 'energy conversion efficiency') is the efficiency of green plants to convert the radiation incident on them to energy and $IPAR = I_o - T_c$, where, I_o is the total incident PAR and T_c is the PAR transmitted through the canopy. The IPAR is then related to NDVI. See Kale and Roy (2012) for the use of PEM model to estimate net primary productivity of dry tropical forests.

14.7 PRIMARY PRODUCTION IN OCEANS AND OTHER AQUATIC SYSTEMS

Because oceans occupy two-third of the global area, their productivities are of special interest. In most oceanic areas, the suspended cells of algae, the phytoplankton, are the only producers. The depth at which 1 per cent of full sunlight reaches can be approximately regarded as the lower limit of productive, lighted euphotic zone of water. For, approximately at this amount of light, respiration equals photosynthesis (i.e., *compensation point*). On this basis the depth of productive belt of the oceans can generally be considered 30-120 m. The marine phytoplankton community represents a deeper photosynthetic system than any land community, including forests. Depth of light penetration in oceans is largely determined by its absorption by the plankton and dead organic particles, together called *sestron*, as well as by water itself. This absorption of light is lower in tropical oceans, hence, generally, light penetration is more in them than in the temperate oceans. However, the maximum photosynthesis in oceans does not occur at the highest light intensity, but at intermediate light intensities below the water surface where phytoplankton cells attain the highest photosynthetic rate (Fig. 14.17). Chlorophyll a optimum also occurs below the surface. For Arabian sea, Gomes and Pant (1989) have reported the following values: At 100 per cent surface light intensity chl a averaged 0.39 mg/m^3 and primary productivity $0.37 \text{ mg C/m}^2/\text{yr}$; at 30 per cent surface light intensity chl a was 0.7 mg/m^3 and primary productivity $0.97 \text{ mg C/m}^3/\text{hr}$, and at 1 per cent surface light intensity chl a was 0.43 mg/m^3 and productivity $0.21 \text{ mg C/m}^3/\text{hr}$. Different groups of phytoplankton show photosynthetic peaks at different light intensities (Fig. 14.18).

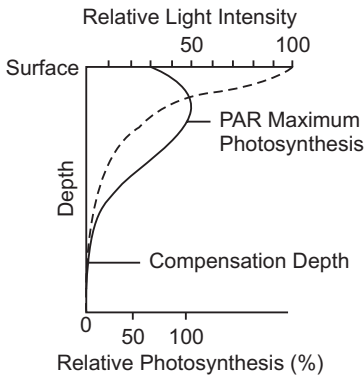


Fig. 14.17. Photosynthetic activity of phytoplankton in relation to light intensity and depth of water (based on Talling, 1970).

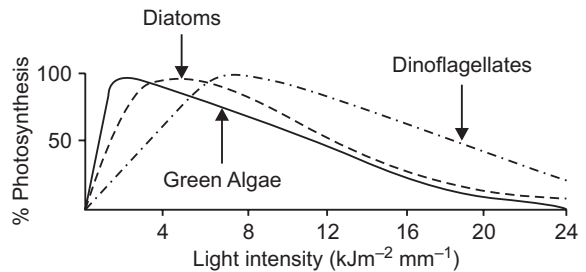


Fig. 14.18. Photosynthetic activity of marine algal groups in relation to light intensity (based on Ryther, 1956).

Unlike the terrestrial productivity which is primarily determined by water and temperature, the marine productivity is primarily limited by nutrients. In summer, when most of production occurs in all oceans, the light intensities are of similar magnitudes for tropical and temperate regions, and the plankton species of the two regions have similar overall light use efficiencies (Whittaker, 1975). Because open oceans account for over 92 per cent of the marine area, some details of the factors which influence their productivity are required. The euphotic zone of the open oceans is subjected to loss of nutrients by two major phenomena. One is the sinking of plankton organisms and their dead remains to deeper waters below the lighted zone, and the other is the thermal stratification of water. Loss of plankton organisms to deeper water means also the loss of nutrients contained in them. The thermocline acts as a barrier to the mixing of deeper water with the upper waters. Because of this separation, nutrients deposited in the deeper water as a consequence of sinking plankton organisms become unavailable to the lighted belt, where phytoplankton organisms require them. When thermocline is established, only small amounts of nutrients can return to lighted belt as a result of turbulent movement of water at and below the lower limit of the lighted zone. Permanent thermocline does not exist in temperate oceans, as a consequence during the spring nutrient overturn occurs and phosphorus and nitrogen are recirculated to the euphotic zone from the deeper layers.

Phosphorus is the most critical nutrient element in many open ocean waters, however, nitrogen may be in others. In the open ocean, productivity ranges between 2-400 g/m²/yr and the mean is placed at 125 g/m²/yr (Whittaker and Likens, 1975). The difference between tropical and temperate oceans is not marked. While the increased stratification and the increased sinking of plankton (both related to high temperature) limit the productivity in tropics, low winter light intensities in winter season keep the productivity at low levels in the temperate regions.

In certain oceanic regions, surface currents of coastal water turn away from shore and move out to open sea. This sudden seaward movement of water is caused by the *Coriolis force* supplemented by wind currents. The sudden seaward movement of surface water causes a vertical movement of some of the deeper nutrient-rich water toward the surface. This vertical movement of water is called upwelling. Where upwelling occurs productivity is higher, up to 400-600 g/m²/yr and supports important commercial fisheries in the world.

Among the remaining types of marine ecosystems which together constitute almost 8 per cent of the marine area, productivities in continental shelves (7.3 per cent of total marine area) are not different from those of upwelling areas. In the continental shelves where upwelling also occurs, productivity may range up to 1000 g/m²/yr or more. Productivity in estuaries, algal beds and reefs

ranges up to 500-4000 g/m²/yr. The giant kelp which forms a thick bed along the Californian coast, and some tropical coral reefs are among the most productive ecosystems of the world. In all these cases, productivity is supported by nutrients carried past the organisms by moving water. From the above, it is apparent that marine productivity covers almost the same wide range of values as terrestrial productivity. However, much of the marine area (92 per cent under open ocean) has a very low productivity, consequently the total marine productivity is lower than total terrestrial productivity.

Primary productivity of fresh water habitats is usually similar to the comparable marine habitats. For fresh water Himalayan lakes, Kaul and Handoo (1989) have reported a range of primary productivity from 1.3-8.2 t C/ha/yr for the microphytes and from 10-41 t dry matter/ha/yr for the macrophytes.

14.8 EFFICIENCY OF ENERGY CAPTURE

The efficiency of the conversion of radiant energy to chemical energy through photosynthesis is called *efficiency of energy capture* (EC) or *efficiency of photosynthetic energy utilization* or *efficiency of energy conversion* or *photosynthetic efficiency*. This is given by the formula:

$$\text{EC (\%)} = \frac{\text{Energy stored in primary production}}{\text{Incident photosynthetically active radiation}} \times 100$$

The photosynthetically active radiation (PAR) (390-710 nm) averages 47 per cent (45-50 per cent) of the total incident short-wavelength radiation (300-3000 nm) in the northern hemisphere. Large algal cultures and individual leaves may attain as high as 15 per cent EC (in some C₄ grasses up to 24 per cent) under non-limiting favourable conditions. It is usual and more meaningful to calculate EC on the basis of net primary production than gross primary production for stands or communities of plants, and for the growing season. Under natural field conditions there are several factors such as sub-optimal levels of soil water, nutrients, light (leaf shading) and temperature which impose limitations on photosynthetic efficiency. Most of the terrestrial plant communities show efficiency of energy capture between 1 to 3 per cent of the PAR during growing season. For temperate grasslands, growing season EC ranges from 0.65 to 1.4 per cent (for desert grassland 0.17 per cent) and for tropical grasslands from 0.23 to 2.1 per cent on annual basis. Cooper (1970, 1975) reported 2.2-7.8 per cent EC for canopies of temperate grasses (C₃) during maximum growth periods and up to 9.3 per cent for tropical grasses (C₄). For the entire growing season EC ranges from 1 to 3 per cent. For forests the general range in EC seems to be from 0.9 to 2.9 per cent. The phytoplankton communities show a general range of EC from 0.5 to 1.1 per cent. In the emergent macrophytes, EC may range from 0.25-3.32 per cent.

14.9 HUMAN USE OF PRODUCTIVITY

Total production of the earth is estimated at 172.5×10^{15} g of which 2/3 is produced by the terrestrial habitats. Although tropical forests cover less than 7 per cent of the earth's surface, they account for about 25 per cent of the total production. Humans use photosynthetic products from plants, directly as food and fibre and indirectly as food for domesticated animals. According to the estimates of Vitousek et al. (1986), only 4% of the terrestrial net primary production is used directly by humans and livestock as food, fibre and fuel wood. About 44% of productivity is co-opted by humans as non-edible production or destroyed by human activities such as permanent settlements, burning of biomass, clearing of forests and desertification. Thus, only 50% of terrestrial and most of the aquatic production is available to all other life forms. Globally, about 5 to 8% of the aquatic productivity is harvested for human use, particularly from regions having productive estuaries, lakes and streams. Recently, Rojstaczer et al. (2001) have given an estimate of 10 to 50% utilization of terrestrial photosynthetic products based on large scale measurements as well as taking into account

uncertainties associated with area of agricultural lands, forests permanently cleared for human habitation and the exact extent of shifting cultivation in secondary tropical forests. Therefore, it is important to understand the energy needs of natural food webs in different types of natural and agricultural systems for maintaining the natural balance.

14.10 LITTER FALL IN TERRESTRIAL ECOSYSTEMS

A large fraction of the terrestrial aboveground net primary production finds its way to the soil surface through the process of litter fall. Senesced leaves, dead portions of twigs and branches bark, dead boles, flowers and fruits periodically or continually fall to the ground and constitute aboveground litter or detritus. Similarly, fine and coarse roots form the belowground litter or detritus. Detritus is the starting point for the saprophagic or detritus food chain.

Determining litter fall

Litter can be collected with the help of litter traps. Litter traps can be of different shapes and sizes. These usually consist of wooden or metal boxes open at the top and at the bottom. The bottom is fitted with a nylon or wire mesh screen to provide for free drainage of water. Each box is fitted with legs so that when placed on the forest floor, the bottom is about 10-15 cm above the ground level. The traps are placed randomly on the forest floor. Litter which falls in the traps is collected at periodic intervals, separated into fractions (such as leaves, bark, twigs, fruits, etc.) and by species. These components are then dried and weighed. Total litter fall is then measured for annual cycle by summing the values of periodical collections.

Aboveground litter fall

Species, and consequently communities, vary in the pattern of litter fall (Figure 14.9). In evergreen forests, both equatorial as well as temperate coniferous, litter fall is more or less a continuous process, although there may be distinct peak(s). In deciduous forests most of the litter fall is confined to a few months of the year.

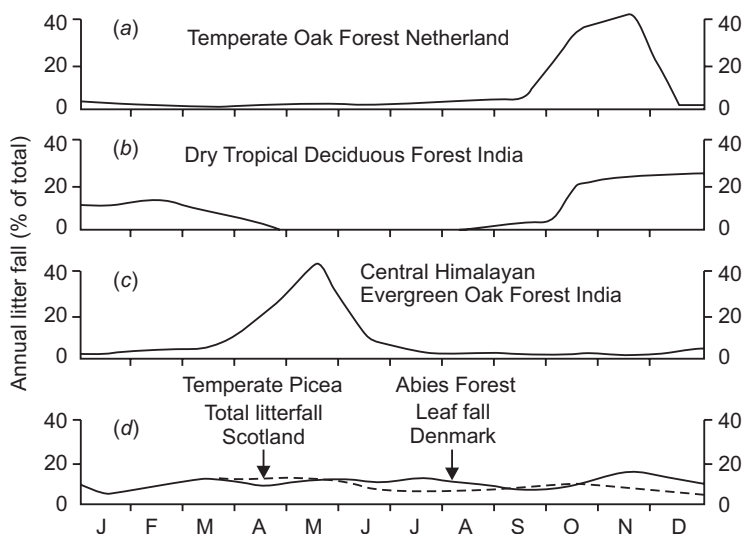


Fig. 14.19. Pattern of leaf fall within a calendar year in contrasting forest types (a) Temperate deciduous oak forest, Netherlands, (b) Dry tropical deciduous forest, India, (c) Central Himalayan evergreen oak forest, India, (d) temperate *Picea*-*Abies* evergreen forest, Denmark (leaf fall) and Scotland (total litter fall).

The amount of litter fall closely parallels the biomass of foliage. In general 70 per cent material produced as litter world wide is leaf matter. Bray and Gorham (1964) found a negative linear relationship between total (leaf + non – leaf) litter fall and latitude for a range of latitudes from equator to about 70° (Fig. 14.20.) The mean annual litter fall for various climatic regions is given below:

	Leaf fall (t/ha)	Total litter fall (t/ha)
Arctic-alpine	0.7	1.0
Cool temperate	2.5	3.5
Warm temperate	3.6	6.6
Equatorial	6.8	10.9

Litter fall also indexes the net primary productivity. For example, Bray and Gorham (1964) found that annual leaf fall approximates total net primary production if multiplied by 4.9 for cool temperate evergreen forest, by 3.7 for cool temperate deciduous forest and by 3.3 for equatorial forest.

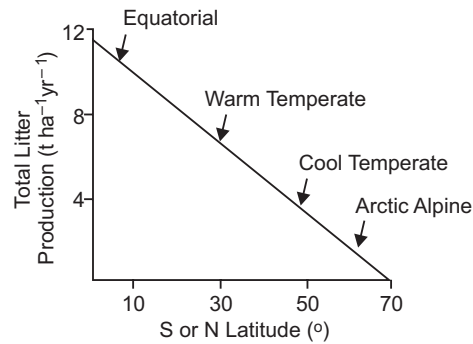


Fig. 14.20. Relationship between annual total litter fall and latitude (based on Bray and Gorham, 1964).

Actual evapotranspiration (AET) which represents all possible combinations of energy and moisture availability is also related to litter fall (Fig. 14.21.) as shown by Meentemeyer et al., (1982). The statistical models relating litter fall to AET are:

$$Y_1 = 0.5805 X - 78.55$$

$$Y_2 = 1.0066 X - 183.92$$

where Y_1 and Y_2 are respectively leaf fall and total litter fall ($\text{g/m}^2/\text{yr}$), and X is AET (cm/yr).

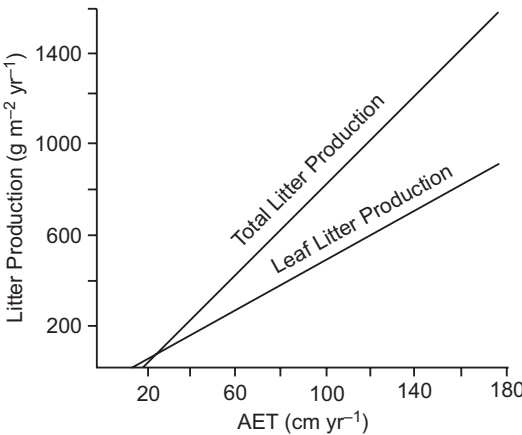


Fig. 14.21. Relationship between annual litter fall and actual evapotranspiration (AET) for forests of the world (based on Meentemeyer et al., 1982).

In a recent careful study involving data from some 319 forest sites, Lonsdale (1988) developed the following statistical model:

$$\text{Log } Y = -0.012 L - 0.000072 A + 1.04,$$

where Y is the total litter fall (t/ha/yr), L is latitude (degrees), and A is altitude (m). The plot of this equation (setting altitude as zero) is shown in Fig. 14.22. For the leaf litter fall, the model was:

$$\text{Log } Y = -0.009 L + 0.80$$

where Y is leaf litter fall and L is the latitude. While the total litter fall decreases with increase in altitude, leaf fall apparently remains unaffected. Latitude determines the mean annual temperature and rainfall; these quantities decrease with increasing latitude. This indicates that the litter fall decreases from equator towards pole.

Below-ground litter

Importance and magnitude of the belowground contributions to total detritus pool is only now beginning to be recognised for forest ecosystems, although the importance of roots as detritus contributors in grasslands (because of greater proportion of biomass) has long been known. According to Vogt et al. (1986), fine root inputs to total detritus may range from 20-77 per cent. Belowground inputs to total detritus transfer may be 2.3 to 5.2 times the aboveground transfers in yellow poplar stand, douglas fir stand and silver fir stands.

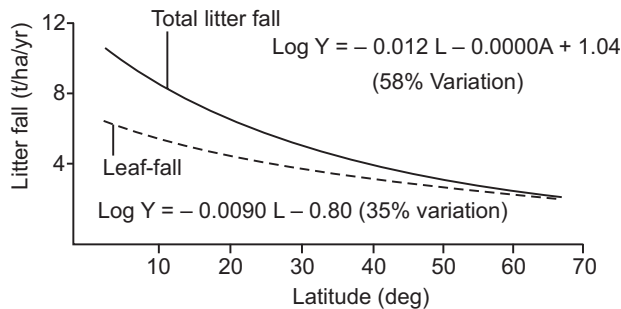


Fig. 14.22. Relationship between annual litter fall and latitude. Solid curve = total litter fall regression line assuming altitude zero, broken curve = leaf fall. (based on Lonsdale, 1988).

Energy Transfers Beyond Primary Producers

Primary producers convert the energy of solar radiation into chemical energy of food and create two major kinds of *Energy (food) pools*: the living organic matter (plant biomass) and the non-living organic matter (plant detritus). These pools make the starting points for two major pathways of energy transfers (Fig. 15.1). The *biophagic* or *grazing pathway* begins with the ingestion of living organic matter by herbivores and the *detritus* or *saprophagic pathway* with the consumption of the detritus by the detritivores or saprovores. Organisms positioned along the grazing pathway comprise *biophagic subsystem* and those participating in the saprophagic pathway constitute *decomposition subsystem*. All heterotrophic organisms subsisting on the food produced by the autotrophic organisms (primary producers), are called *secondary producers* and the organic matter synthesized by them represents *secondary production*.

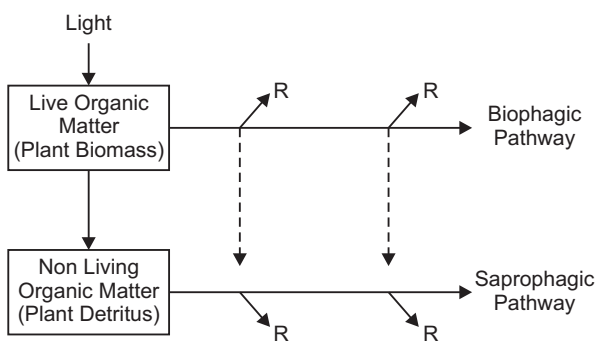


Fig. 15.1. The two major food energy pools and related pathways of energy transfer.

15.1 FOOD CHAIN AND FOOD WEBS

Food chains and Trophic Levels

We are all familiar with the idea that one organism eats another and in turn is eaten by still another organism. For example, a frog eats an insect which ate plants, and the frog itself is eaten by a snake. In this example, plant is the food for insect, the insect is food for frog and the frog is the food for snake. Plant, insect, frog and snake, thus form a food *chain*. In each step of feeding and being fed upon, energy is transferred from one organism to the other, or in other words, the transfer of energy through a series of organisms occurs along a food chain. Thus:

Plant → insect → frog → snake

Plant → insect → insectivorous bird → hawk

Plant → deer → lion

The organism which is eaten is called *prey* and the organism which eats it is called *predator*. Thus insect is the prey for bird and the bird is the predator for insect. A predator for one organism can be a prey for another organism. For example, the insectivorous bird is the predator for insect but prey for the hawk. The animal which feeds on plants is called herbivore and the animal which feeds on the *herbivore* is called carnivore. Herbivore is thus a *primary consumer* (first order consumer),

and the *carnivore*, a secondary consumer (second order consumer or first order carnivore). A carnivore eating another carnivore is a *tertiary consumer* (second order carnivore). Different parts of the same plant may be consumed by different herbivores, for example

Flower → nectar eating insect ; Leaf → leaf-eating insect

Vascular tissue → sap sucking aphids

Root → root-eating nematodes.

In turn, these herbivores support different food chains via different carnivores. The food chain can now be generalized as:

Plants or primary producers (Trophic level 1)	→	Herbivores or primary consumers (Trophic level 2)	→	First order carnivores or secondary consumers (Trophic level 3)	→	Second order carnivores or tertiary consumers (Trophic level 4)
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This generalized food chain represents *grazing food chain* because at each step a living organism or its part is consumed.

Positions of organisms along a food chains are called *trophic levels*, and various species which obtain food energy by the same number of steps from the sun are called members of the same trophic level. Thus, all green plants (producers) belong to first trophic level, plant eaters (primary consumers) to second trophic level, secondary consumers to the third trophic level and so on. The highest carnivore in the food chain is known as the top predator. Connections involving feeding are known as the trophic connections. Trophic levels provide the framework for studying ecological energetics and efficiencies (Lindemann, 1942) as well as for studying the cascading trophic interactions.

The trophic classification is based on functions of species, not on their taxonomic selection. However, a species can occupy more than one trophic level. For example, we human beings are herbivores or primary consumers, or belong to trophic level 2 when our diet consists entirely of vegetable and grains, we are secondary consumers i.e. trophic level 3 when our diet is meat of chicken or goat, and we are tertiary consumers (trophic level 4) when we eat carnivorous fish. Thus, man is a kind of *omnivore*, and so is crow, the familiar occupant of our court-yards, whose food can be cereals, insects or animal flesh. The red fox is a well investigated omnivore, which feeds mostly on insects and fruits during summer, and on rodents (rats, rabbit, etc.) in spring and winter (Scott, 1955).

Energy available for each trophic level is partitioned in a number of ways. A part of the energy is ingested and a part is left unutilized. A portion of the ingested energy is assimilated and the remaining is lost as egested (faecal) matter. The assimilated energy is partitioned into production and respiration. Further, a portion of the production can be stored (e.g., stored fat in animals) for re-use at a later time (e.g., reserve energy used by predator in the search for prey), and another portion can be exuded or excreted. The remaining portion of the energy is utilized in the growth of the biomass and is available for use by the next trophic level. Thus, because of the losses, the amount of energy available to succeeding trophic levels decreases. This is in accord with the second law of thermodynamics which states, in effect, that each energy transfer is accompanied by dispersion of energy into unavailable form (i.e., respiratory heat loss). However, if all losses are summed the energy input balances energy output as required by the first law of thermodynamics.

Food Webs

In nature, food-chains are interconnected into a net-work called *food-web* through euryphagous organisms and omnivores. The term food web has its origin from Elton's "food cycles", which describe the patterns which species eats what other species in a community. The analysis of

food webs is important to understand ecosystem dynamics. A combination of field observations, experimental field studies, mesocosm and small scale laboratory experiments have been used to understand ecological relationships and dynamics of food webs (Ricklefs and Miller, 2000).

Eating and being eaten within a food web has an important connotation. Hairston et al., (1960) argued that since plants are able to accumulate, by and large, a lot of biomass (world is green), the grazers are being controlled by carnivores (thus plants are resource-limited and herbivores are predator-limited). Smith (1969) and Fretwell (1977) argued that in even-length food chains (e.g., A in Fig. 15.2), plants will be controlled by grazing and in odd-length food-chains (e.g., B in Fig. 15.2) herbivores will be limited by carnivores and plants will be limited by resource. If a fourth trophic level (tertiary consumers) is added to the food-chain, plants will again be grazer-limited while secondary consumers (1st order carnivores) will be limited by the 2nd order carnivores. Thus, resource and predation limitations provide for a *balance in nature*. Further, it should be easy to visualise how a unit of energy assimilated at the base of the food web, is retained in biologically useful and available form through the processes of transfer and reassimilation by organisms in different food chains and at different trophic levels for as long as possible (the 2nd law of thermodynamics would not allow this to happen indefinitely), and in doing so the entire biological diversity is sustained.

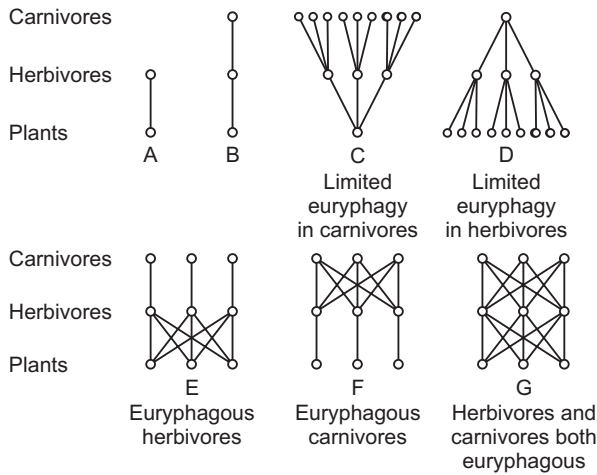


Fig. 15.2. Range of trophic structure in food webs with varying levels of competition and interconnections.

A variety of arrangements of food chains into a food web is possible; six such possibilities are shown in Fig. 15.2.

Food-web of an exposed intertidal rocky shore is shown in Fig. 15.3. Note that organisms such as *Mytilus edulis* derive energy both from plankton and detritus, and organisms like *Pisaster* and *Leptasterias* are euryphagous and prey upon all 7 herbivores in the system. The carnivore *Thais* is also eaten by *Pisaster*, the latter thus deriving food energy from both herbivore as well as carnivore levels. This food-web has 4 trophic levels and 25 food-chains (e.g., plankton - *Mytilus* - *Thais* - *Pisaster*, plankton - acorn barnacles - *Leptasterias*, etc.). Individual food chains in this web vary in length from 3 to 4 trophic levels.

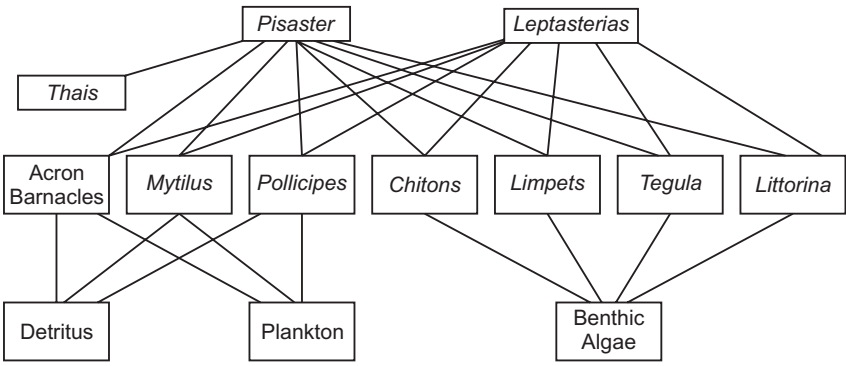


Fig. 15.3. Food web of an exposed intertidal rocky shore (based on Briand, 1983).

Box 15.1. Tracing food chains and webs

Two most common methods used are (i) analysis of gut contents, and (ii) application of radioisotopes. In the first method, representatives of all species occurring in a given habitat are collected and their gut contents are examined. Whole organisms or their parts present in the guts are identified and feeding relations established.

In the second method, a suitable radioisotope, such as P^{32} , is used to label a number of individuals of a single plant species at a time. Following the labelling, samples of animals occurring in the vicinity of the plants are periodically collected and the amount of P^{32} in them is assayed. Herbivores first show the peak radioactivity followed by the predators. Next time individuals of another plant species are labelled and the entire procedure is repeated. Thus in due course of time the entire food web is elucidated.

Types and Patterns of Food webs

Theoretical approaches have been used to analyze the interactions of populations in food webs (May, 1975). Experimental studies by Robert T. Paine established the role of individual species in the maintenance of community structure (Paine, 1966, 1980). The interactions in food webs have also been analyzed using the pulse (adding the individuals) and press experiments (removing the individuals) by Bender et al. (1984). Most of the food webs have been analyzed in aquatic habitats; only a few food webs have been worked out in prairies, forest and desert ecosystems. Analysis of food web patterns have been done by a number of workers (Briand, 1983; Cohen 1984; Pimm, 1982; Cohen et al., 1990; Williams and Martinez, 2000; Kitching, 2001). The main food web characteristics relate to the length of food chains, the amount of omnivory, the extent of compartmentalization and the ratio of predator to prey (Pimm, 1982; Cohen and Briand, 1984). Famous experiments by Kitching and Pimm in aquatic communities embedded in plants (phytotelmata, such as tree holes and pitcher plants) have been the basis of food web theory.

According to Paine (1980), there are three types of food webs:

Connectedness webs : These are based on the concept who eats whom, emphasizing the feeding relationships. These are also known as topological food webs.

Energy flow food webs: In this type of food webs, energy flow through the food web is estimated. The connections between populations are quantified by the flux of energy between a resource and its consumer.

Functional food webs: Paine (1980) developed the idea of functional food webs on the basis of the impact of species on the structure of community. The interaction food webs show the feeding relationships within topological food webs.

The food webs have also been described as source web, sink web and community web.

Source webs: The source web also refers to one or more kinds of organisms and organisms that eat them, their predators and so on.

Sink webs: These refer to one or more kinds of organisms and organisms that eat them, their other prey, and so on, (i.e., descending order of trophic levels).

Community webs: A group of species within a defined area of habitat.

15.2 FOOD WEB COMPLEXITY AND ECOSYSTEM STABILITY

Does increasing complexity of food web (C-G in Fig. 15.2) provide for greater stability of the system? This question does not have a unique answer as yet. Intuitively, it has been argued that the more complex a food web the greater will be stability. Complexity can be assessed in terms of connectance of the web. The topological food webs are composed of nodes (trophospecies), trophic levels (basal species, top predators and intermediate species) and cross links (established

through competition, predation, parasitism and mutualism). The average links per species is called the linkage density. The total number of links in the food web divided by the number of possible links is called connectance, which is the ratio of actual number of interspecific interactions/ potential number of interspecific interactions.

Connectance is thus, represented by the fraction of possible pairs of species which interact directly through predator-prey links or competition (all pairs of predators which share a prey species are assumed to be in competition). (The *connectance* in food web depicted in Fig.15.3 is 0.46). Several recent modelling exercises and field studies indicated that connectance decreases with increase in species richness, and that there is an overall tendency for inherent stability to increase as complexity decreases. Thus, more complex communities are less likely to return to their original state following a perturbation. Among situations E-G depicted in Fig. 15.2, G indicates maximum connectance hence minimum stability. This is contrary to the conventional wisdom that greater complexity leads to greater stability. A situation where a large number of species compete at the trophic level to which energy flows (C in Fig.15.2) would be more stable according to Southwood and Way (1970), than the situation where larger number of food species are available to an organism (D in Fig. 15.2). However, species deletions at the base (plants) in D and at the top in C will have less effect on the system than vice-versa. Briand (1983) analysed 40 food webs and found that connectance was generally lower in fluctuating environments than in constant environments; in other words constant environments supported more complex (hence less stable or more fragile) communities compared to fluctuating environments which harboured more stable and less fragile communities. Nevertheless, if the food web is organised into a number of sub-units (compartments) with high connectance within the subunits and loose connection between subunits, the system may be quite stable. Compartmentalization in the food webs emphasizes strong interactions within several groups of species within a subunit. For the food webs of invertebrates living in *Nepenthes* pitcher plants in different regions, the complexity of food webs is correlated with increasing species diversity (Beaver, 1985).

Limitation of Food Chain Length

The total number of links from top predators to basal species is called chain length. Food-chain length is a central characteristic of ecological communities because it regulates community structure, ecosystem processes and contaminant concentrations (Post, 2002). Pimm (1988) has analysed this question in detail by considering the length of food chains from such diverse systems as phytotelmata to open oceans. Phytotelmata are habitats made up of plant-held waters such as in axils of leaves, petals and bracts, in the internode spaces of bamboos, in tanks within bromeliads, in pitcher plants, and in tree holes. Conventional wisdom tells us that at each step of energy transfer in the food chain, 80-90 per cent of the potential energy is lost as heat, and therefore less and less energy is available to support an additional trophic level as we move along the food chain from primary producers to predators. Since each link in the food chain decreases the available energy by about one order of magnitude, the number of such links in the food chain is limited. In fact, analyses of a large number of food webs from rain forests to phytotelmata indicate that food chain lengths are mostly limited to 3-4 trophic levels. However, if the hypothesis of energy constraint as outlined above were correct, the length of food chains should vary markedly, highly productive systems supporting longer food chains and low productive systems shorter food chains. This does not happen in nature. From a modelling exercise, Pimm and Lawton (1977) suggested that webs with longer food chains may undergo severe population fluctuations facilitating extinction of top predators, and as the number of trophic levels increases so the average recovery time also increases. Also, higher the flux of energy more quickly the effects of the perturbation are swept from the system and so higher would be the resilience (i.e., short recovery time). Longer food chains may decrease this flux and therefore increase the recovery time. Since less resilient systems are unlikely to persist in non-

constant environment, only systems with few trophic levels are found in nature. According to Pimm (1988), the length of food chains is thus determined by a balance between factors which shorten the chains (disturbances) and the species ability to recover from these disturbances.

Other hypotheses advanced in order to explain short food chains are (i) design constraint - e.g., it may be impossible to design a predator that is regularly fast enough to catch a tiger and fierce enough to kill it, (ii) home range constraint - top predators need to forage over a large home range because they are large and not 100 per cent efficient in finding and converting prey into predator biomass, (iii) predators are likely to attack a prey that is of right size and in a right place, regardless of its trophic level (Begon et al., 1986) so there is no need to have long food chains. Recent analysis of food webs suggests that resource availability limits food-chain length only in systems having very low resource availability. Predator to prey mass ratio (PPMR) determines the length of food chains, particularly in oceans, with smaller PPMR leading to longer food chain length in a community of given size composition (Barnes et al., 2010).

The Niche Model of Food Webs

The niche model describes the biological mechanisms responsible for the structure of food webs and predicts quantitative aspects of most complex food webs that underlie a wide range of ecological systems from deserts to islands, lakes and estuaries in the primary literature (Williams and Martinez, 2000). The “niche model,” is based on the premise that each trophic species (a group of animal types that share the same predators and prey, such as some lizards) fits a specific niche based on what it eats and, in turn, what eats it.

To test their theory, Williams and Martinez (2000) selected seven of the most complex food webs available in the literature, including the Little Rock Lake in Wisconsin, Chesapeake Bay, Skipwith Pond and the Coachella Desert. They have analyzed the trophic complexity using only two parameters, i.e., the number of functionally distinct species (diversity) and the density of trophic interactions (connectance). By plugging these two parameters into the model, they accurately predict a set of 12 characteristics, such as the length of food chains, the number of omnivores and cannibals, the distribution of specialists and generalists, and the vulnerability of species within each food web (Williams and Martinez, 2000). The most widely accepted model to date -the cascade model- successfully predicts only a few food web properties: top species, such as predators; intermediate species, such as herbivores; and basal species, such as plants (Cohen et al., 1990). The niche model performed similarly across the different food webs and consistently predicted their individual properties more accurately than the cascade model. (Williams and Martinez, 2000).

15.3 ENERGY FLOW MODELS, ECOLOGICAL EFFICIENCIES AND ECOLOGICAL PYRAMIDS

Energy flow models

Now if we link the trophic levels with each other showing the inputs and losses of energy at each trophic level, an *energy flow diagram* (or energy flow model) results as given in Fig. 15.4. Lindeman (1942) was the first to propose such a model (Fig. 15.5) assuming that (i) plants and animals can be arranged into trophic levels, and (ii) the laws of thermodynamics hold for plants and animals. In this model, the standing crop of energy of any trophic level is designated by the Greek capital letter lamda (Λ) using a subscript to identify the trophic level (e.g., Λ_1 for producers, Λ_2 for herbivores and so on). If Λ_n is taken as energy content of a given trophic level, Λ_{n-1} represents that of the preceding trophic level and Λ_{n+1} the succeeding trophic level. The contribution of energy from one trophic level per unit time to the other trophic level is represented by little lamda (λ_n). Thus, λ_n represents the energy input to Λ_n from Λ_{n-1} . Energy lost to a trophic level as respiration (R_n) plus the energy (λ_{n+1}) transferred by this trophic level to next higher trophic level (Λ_{n+1}) comprise total energy lost (λ_n) from the trophic level Λ_n . Thus, in Figure 15.5, λ_1 is total loss of

energy from Λ_1 (i.e. $\lambda_2 + R_1$). The energy flow from one trophic level to the other or the rate of change of the energy content (= net production) at any trophic level (Λ_n) can be represented as: $\lambda_n + \lambda_{n'}$. For example, in Fig. 15.5 the rate of change of the energy content will be: $\lambda_2 + \lambda_{2'}$.

$$\delta . \Lambda_2 / \delta . t = \lambda_2 + \lambda_2$$

The above formulation helps us to calculate certain important *ecological efficiencies* or *transfer efficiencies*.

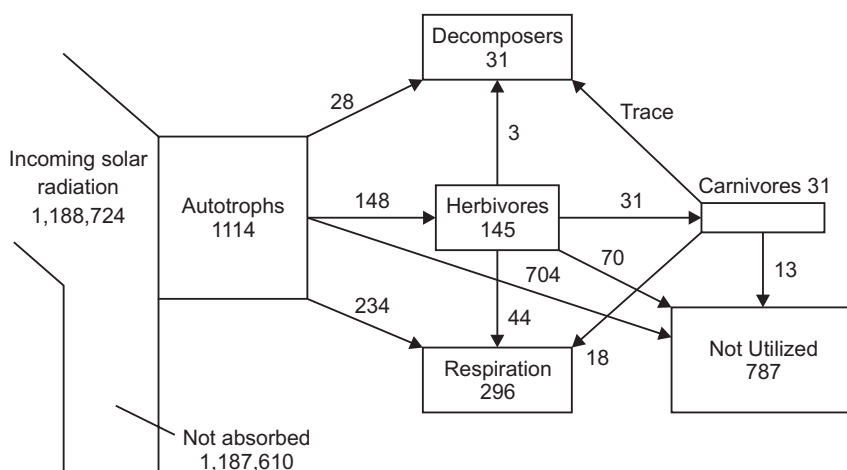


Fig. 15.4. Energy flow diagram for Cedar Bog Lake, Minnesota, in $\text{kcal m}^{-2} \text{yr}^{-1}$ (based on data of Lindeman, 1942).

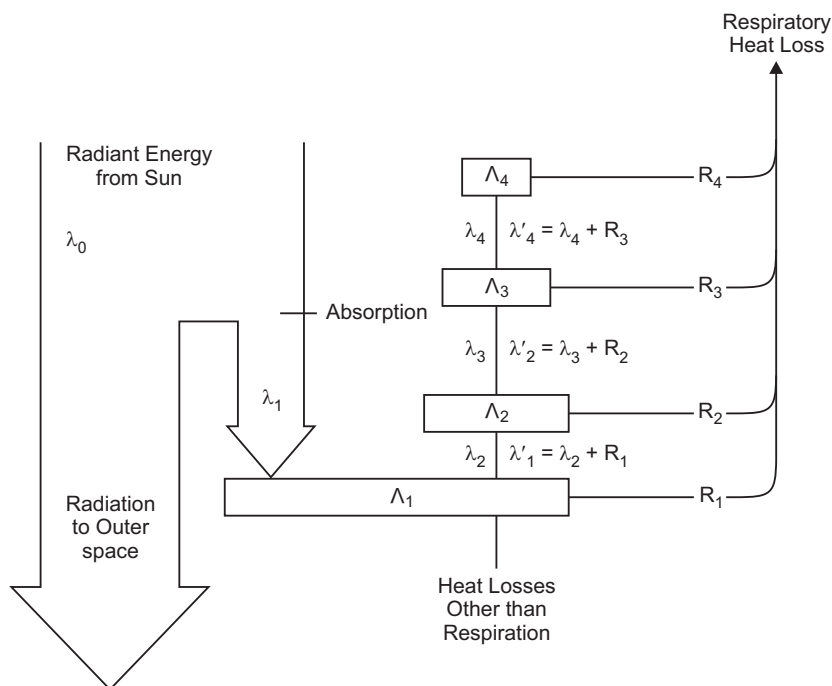


Fig. 15.5. Diagrammatic representation of trophic-dynamic concept of Lindeman (1942).

Combining Biophagic and Saprophagic Pathways of energy flow

We have seen from our examples of food web that several organisms link the detritus and grazing food chains. An insectivorous bird not only preys upon phytophagous insects but also on saprophagous insect or even earthworm. Wiegert and Owen (1971) have proposed an energy flow model which takes care of this (Fig. 15.6). In this model (i) the first order biophages utilize living autotrophs, (ii) the first order saprophages utilize non-living organic matter from autotrophs as well as organic material egested by the first order biophages, (iii) first order biophages can be utilized by predators and parasites (second order biophages) or they can die and form the energy source of second order saprophages as can also the first order saprophages, (iv) biophages and saprophages of the same order constitute one trophic level, thus each trophic level comprises both biophages and saprophages.

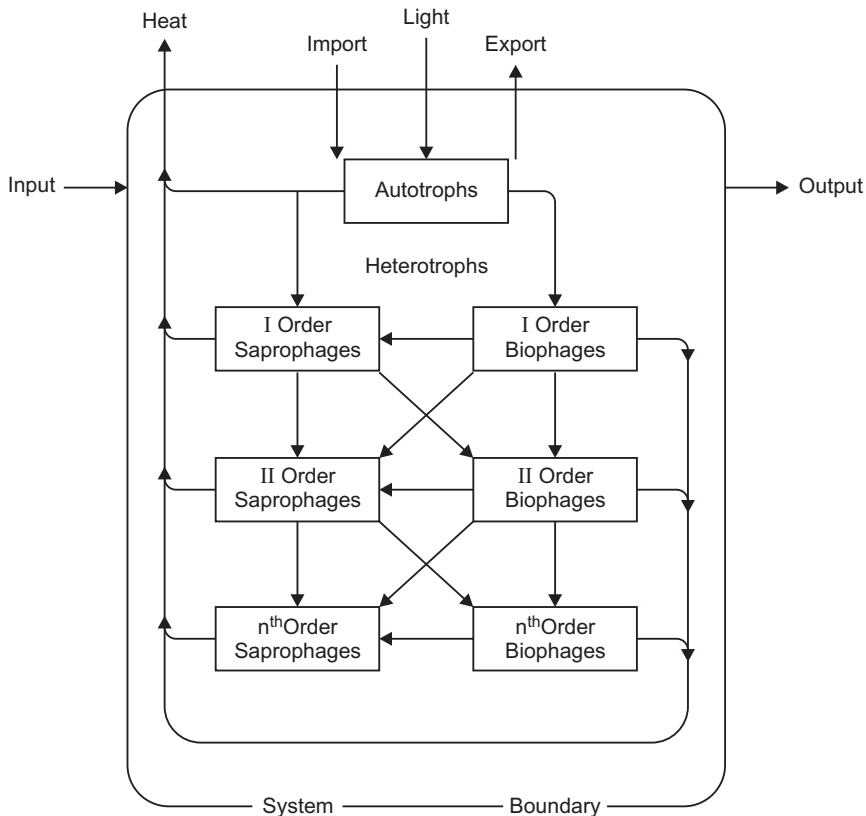


Fig. 15.6. A representation of energy flow through combined biophagic and saprophagic food chains in ecosystems (based on Wiegert and Owen, 1971).

In Y-shaped model also, grazing and detritus food chains are shown together (Fig. 15.7). Odum (1983) regards this model as more realistic than the single channel flow model (Fig. 15.4) because - (i) it conforms to the basic stratified structure of ecosystems, (ii) direct consumption of living plants and utilization of dead organic matter are usually separated in both time and space, and (iii) the macroconsumers (phagotrophs) and microconsumers (saprotrophs) differ greatly in size-metabolism relations.

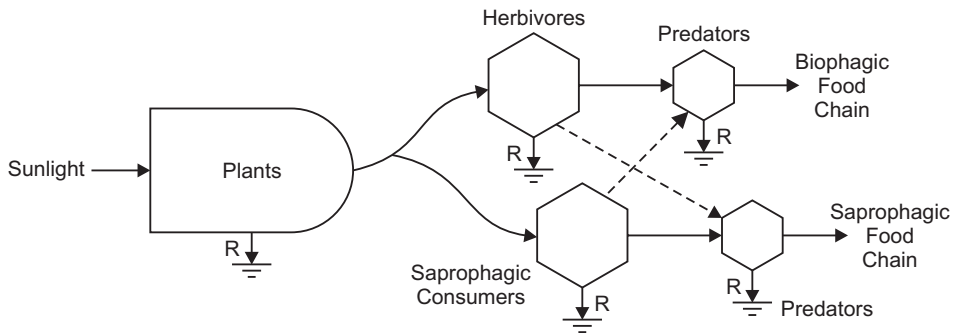


Fig. 15.7. The Y-shaped energy flow diagram showing separation and linkages in grazing and detritus food chains (based on Odum, 1983).

As an example, in a lake the lighted zone, representing the “green belt”, is the space where grazing chain is predominant (phytoplankton are grazed by zooplankton and other organisms). On the other hand, the lake bottom represents the “brown belt” where dead organisms are deposited and they are acted upon by bacteria, fungi and other organisms, depending on the type of energy use. Most of natural ecosystems, such as forests and marshes, operate as detritus systems, for, over 90 per cent of primary production is not consumed by heterotrophs until plant parts die and reach water, sediments and soils. This delay in consumption of primary production increases structural complexity of the ecosystem. In contrast, in heavily grazed grasslands or pastures and in some shallow waters, 50 per cent or more of the net primary production may pass down the grazing path.

Energy Partitioning in the Individual / Species populations

A model for energy partitioning in the individual or the species population shows the living structure or biomass (Fig. 15.8). I represent the energy input which is light in the case of autotrophs and food in the case of heterotrophs. The usable part of the input is assimilated (A) and the unusable part is ejected or not utilized (NU). A large portion of assimilated energy must always be respired to provide maintenance or existence energy to keep the body functioning and repaired. A part of energy can be used for growth and reproduction. S is the stored energy, which is used to accept further inputs of energy.

The partitioning of energy between P and R is of vital importance to the individual and species. Large organisms require more maintenance energy than small ones as they have more biomass to maintain. The warm-blooded animals (birds and mammals) require more energy than the cold-blooded animals. Predators, generally, must expend a large percentage of assimilated energy in respiration to find and overcome the prey. Through natural selection, organisms maintain a favourable benefit-cost ratio of energy. Therefore, it is of interest to study optimum foraging strategy in different kinds of organisms and under different conditions. Species adapted to unstable, recently derived or under-populated areas

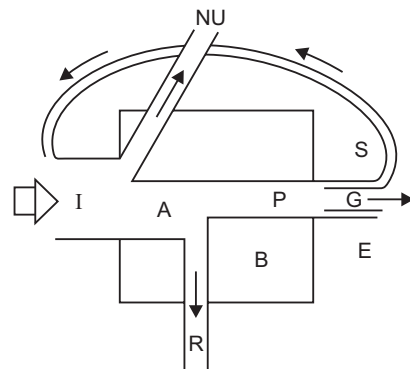


Fig. 15.8. Components for a ‘universal’ model of energy flow.

I = input or ingested energy;
 NU = not utilized energy;
 A = assimilated energy;
 P = production; R = respiration; B = biomass;
 G = growth; S = Stored energy
 E = excreted energy (based on Odum, 1963)

generally allocate a large portion of their energy to reproduction. The species adapted to stable and more favourable habitats, allocate little energy to reproduction.

Energy Flow in Stream and River

In streams, photosynthetic input of energy from stream algae is often small. There is a large input of dead organic matter into the stream from the surrounding areas. Detritivores (organisms that feed on dead organic matter) are common in streams and feed mainly on this dead organic matter. Some of the animals are shredders while others are filter feeders. Many stream predators are larvae of land-dwelling insects, such as dragonflies. In the flood plain of the Amazon River basin, a unique type of food chain has been observed based on external food inputs. The fish feed on fruits and nuts carried into the streams during the rainy season.

Energy Flow in Oceanic Systems

In open oceans, the first trophic level is composed of many species of floating algae, which are known as phytoplankton. These are ingested by zooplankton, which, in turn, are fed upon by larger organisms, such as fish or other filter feeders (Fig. 15.9). This is the grazing food chain. A substantial portion of the carbon fixed by photosynthesis (> 50%) is released into the surface water column as dissolved organic matter (DOM). The dissolved organic matter is utilized by the bacterioplankton (heterotrophic bacteria), which are an important component of the phytoplankton community. In the aquatic food webs, this pathway of energy transfer is referred to as microbial loop. In this loop, bacterioplankton assimilate energy to produce new biomass, known as secondary production. The secondary production is a major pathway for the transfer of energy to higher trophic levels in the aquatic environment. Another type of food chain is based on dead organic material that is produced at the surface and sinks toward the ocean floors. These form the energy base for both carnivores and detritivores. This forms the energy base for both carnivores and detritivores. Thus, microbes play a central role in regulating the fluxes of matter and energy in marine ecosystems (DeLong and Karl, 2005).

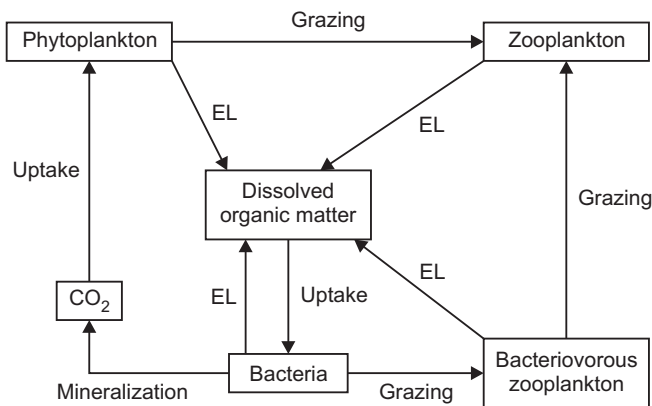


Fig. 15.9. A simplified planktonic food web in marine ecosystems showing the microbial loop. Excretion and lysis (EL) of the phytoplankton, zooplankton and bacteriivorous zooplankton is the major pathway for the transfer of dissolved organic compounds to the dissolved organic matter compartment. The microbial loop is the pathway in which the dissolved organic matter is utilized. The bacterioplankton mineralize nutrients contained within organic compounds and convert a portion of the dissolved carbon into biomass. Grazing by bacteriivorous protozoans establish a link to higher trophic levels (modified from Fuhrman, 1992).

Highly productive areas in the oceans occur in coastal regions and in areas of the upwelling water, accounting for about 10% of the ocean area and 25% of the ocean's total primary productivity of practical importance. These coastal and upwelling regions supply more than 95% of the estimated marine fishery yield. In coastal marine ecosystems, trophic relations of the fauna have undergone drastic changes due to over-fishing over the historical times, and recent human disturbances (see Box 15.2).

Box 15.2. Changes in Trophic Relations in the Coastal Marine Ecosystems

In the coastal marine ecosystems like seagrass beds, kelp forests and coral reefs, major structural and functional changes have occurred due to over-fishing of the vertebrate mega fauna, including sharks, sea turtles, sea cows, seals and whales (Jackson and Sala, 2001; Jackson et al., 2001). Kelp forests in the oceans on northern latitudes have experienced widespread reductions in numbers of trophic levels due to hunting of the sea otters and subsequent population explosion of the sea urchins. Loss of sea otters, cod, and other predatory fishes has also resulted in increase in populations of free living invertebrates and pathogenic bacteria (Jackson and Sala, 2001). The effects of fishing on the coral reefs have been most intense in the Caribbean. The corals of the Great Barrier Reef in Australia have been adversely affected due to the outbreak of the crown-of-thorn starfish (*Acanthaster planci*), which feeds on the corals. The seagrass beds of tropical and subtropical oceans have also shown major decline in mega fauna. The complex and diverse, three dimensional biological habitats in coastal areas are being reduced to simple, two dimensional sediments and rocky surfaces; vertebrate fauna are being replaced by smaller and smaller invertebrates and abundant microbes (Jackson and Sala, 2001). The food webs have shown changes in the proportion of vertebrates and invertebrates. This may have far reaching implications for the sustainability of coastal ecosystems in view of natural and human disturbances such as nutrient loading and eutrophication, hypoxia, disease, storms and climate change (Jackson et al., 2001).

Chemosynthetic Energy Flow in the Ocean

In the depths of oceans, the food chain is largely based on chemosynthesis, in which the source of energy is the hot, inorganic sulphur compounds emitted from the vents of ocean floor. The energy base for the deep-sea hydrothermal vent ecosystem is chemolithoautotrophy by sulphur oxidizing bacteria. The sulphur-laden water is emitted from hot water vents at depths of 2500 to 2700 m associated with areas where flowing lava causes sea floor spreading. A rich biological community exists in and around the vents, including large calms, brown mussels and white crabs. The clams and mussels filter chemosynthetic bacteria and particles of dead organic matter from the water, crabs graze on bacterial mats, others, including at least one of the fishes, are predators. The tubeworms form symbiotic association with sulphur - oxidising bacteria. The entire food web in a hydrothermal vent community is based on chemoautotrophy. This ecosystem is remarkable, as the primary production is not based on sunlight but on geothermal energy. The organisms of the vent ecosystems have a great capacity to tolerate heat.

Does energy cycle?

We are used to saying that energy flows while nutrients cycle. It is true in as much as the energy once dissipated as respiratory heat cannot be reused. However, the previous discussion has shown that some portion of energy, before it is finally dissipated, is definitely cycled in the food web. Examples are (i) storage and reuse of energy at almost all trophic levels, (ii) cycling at predator level - one is familiar with the example of a wounded wolf being devoured by other individuals of the wolf pack, and the devourer may become the prey next day, (iii) cycling in the saprophagic

food chain - dead bodies of organisms are recycled by the organisms of the same or other trophic levels, and (iv) inputs from saprophagic pathway into biophagic pathway. Rigler (1975) said, "Traditionally we say that energy flows unidirectionally, whereas materials cycle. This is only a crude approximation of the real difference, because energy and materials both cycle and both flow out of the ecosystems irreversibly". According to Wiegert (1988), while inorganic nutrients cycle very efficiently, energy cycles relatively inefficiently.

Ecological Efficiencies

The amount of energy reaching each trophic level depends on the net production of the primary producers at the base of the food chain and the extent of energy transfer at each trophic level. The production efficiency of each trophic level varies from 5% to 20% of that of the level below it. Some of the important efficiencies are given below:

1. Assimilation efficiency

$$\text{For plants, } \frac{\text{energy fixed by plants}}{\text{light absorbed}} \times 100$$

(This is efficiency of energy capture discussed in the previous chapter).

$$\text{For consumers, } \frac{\text{Food energy absorbed (assimilated)}}{\text{food energy ingested}} \times 100$$

This is a measure of the efficiency with which a consumer population extracts energy from the food ingested. This efficiency varies widely; 37 % in ground squirrel, 91% in old field mouse, 94 to 95% in spiders and weasel. In general, the efficiency is low for herbivores (20 to 50%) and high for carnivores (about 80%).

2. Utilization or consumption efficiency

$$= \frac{\text{ingestion at trophic level } n}{\text{net production at trophic level } n-1} \times 100$$

This represents the proportion of total productivity available at a trophic level that is actually consumed by the organisms of a succeeding trophic level. The utilization efficiency of net primary production by the herbivores (i.e., proportion of net primary production transferred to herbivores) varies from 1.5 to 2.5% in mature temperate deciduous forests, 28 to 60% in grazed grasslands, 30 to 45% in managed rangelands and 60 to 99% in plankton communities of ocean waters.

3. Growth or production efficiency

$$= \frac{\text{Production at trophic level } n}{\text{assimilation at trophic level } n} \times 100$$

This production/assimilation ratio is a measure of the efficiency with which the assimilated energy is incorporated into the protoplasm. This efficiency ranges from typically 1 per cent in endothermic birds and mammals, 10 per cent in long-lived ectotherms like fish, to as much as 60 per cent for carnivorous invertebrates. The production efficiencies for the herbivores, carnivores and detritivores vary from 21% to 36%, the value being greatest in the case of detritivores.

4. Ecological growth efficiency

$$= \frac{\text{Production at trophic level } n}{\text{ingestion at trophic level } n} \times 100$$

The ecological growth efficiency also varies widely from 1.9 per cent in ground squirrel to 14.17 per cent in grasshoppers and spiders.

5. Transfer efficiency

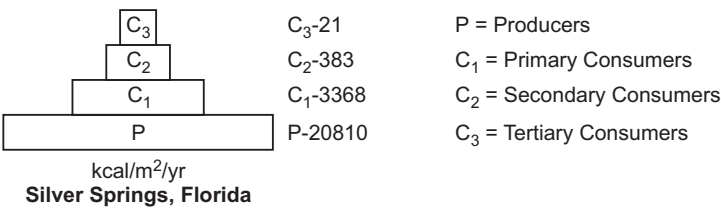
Another efficiency term frequently used is ‘transfer efficiency (TE)’. It is calculated as: production at trophic level n divided by the production at trophic level n–1.

Ecological pyramids

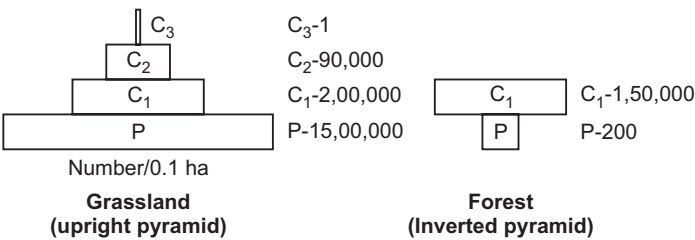
Elton (1927), who introduced ecological pyramids as a way to represent the distribution of abundance and biomass among body sizes, noted that larger organisms higher in food chains were less abundant than smaller ones lower down (Trebilco et al., 2013). Now the layers are defined by trophic level rather than by body-size class. A graphic representation of *trophic structure*, in which the first trophic level or producer level forms the base and the higher trophic levels form successive tiers (Fig. 15.10) is called ecological pyramid. Three types of ecological pyramid are generally described:

- (1) the pyramid of numbers, in which individuals at each of the successive trophic levels are counted per unit area and their numbers are plotted ;
- (2) the pyramid of biomass in which the total biomass existing at each of the successive trophic levels is measured in terms of dry weight or caloric value, per unit area and plotted;
- (3) the pyramid of productivity, in which productivity or energy flow per unit time at each of the successive trophic levels is measured and plotted.

a. Energy Flow Pyramid



b. Pyramid of Number



c. Pyramid of Biomass

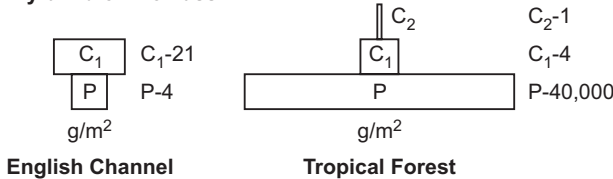


Fig. 15.10. Examples of ecological pyramids (adapted from Odum, 1983 and references therein).

The *pyramid of productivity* or *energy flow* is of the fundamental significance, for, it depicts the actual functional relationships between the trophic levels. Since the productivity or energy flow is expressed on per unit time basis, the pyramid is always and for all ecosystems is in an upright

position. Governed by the second law of thermodynamics, energy flow declines from the producer level to successive higher trophic levels. Consequently, the energy flow pyramid necessarily assumes a typical pyramidal shape, with a broad base sloping toward the top, reflecting a steep, stepwise decrease in production towards increasing trophic level (Fig. 15.10a). This implies that there is always greater production of primary producers compared with herbivores, and greater production of herbivores compared with primary carnivores, and so on.

The two other types of pyramid viz. *pyramids of number* and *biomass* are subject to reversals, the top becoming heavier than the base i.e. “wrong side up”. The numbers of individual organisms as indicated in Fig. 15.10b for a grassland generally decreases up the sequence of trophic levels, number of plants is greater than that of herbivores and that of herbivores is greater than that of carnivores. But in a forest where thousands of insects may feed on a single tree, or where hundreds of parasitic worms depend on one host, the pyramid of number would be reversed (Fig. 15.10c). Compared to this, the pyramids of biomass (Fig. 15.10c) are less frequently reversed, but animals at times exceed plants in biomass, such as in plankton communities of temperate region in winter. Very often in the sea, secondary consumers, such as fish and tertiary consumers, such as shell fish are large and out-weigh the phytoplanktonic producers. According to Trebilco et al. (2013), inverted biomass pyramids may occur under extreme ecological conditions, such as low trophic-level subsets of communities, such as in planktonic size classes with the biomass of larger heterotrophic zooplankton outweighing that of smaller autotrophic phytoplankton or in detritivorous communities. Misra et al. (1968) argued that greater energy conservation in aquatic animals and greater energy dissipation in land animals may result in inverted biomass pyramid in open waters, while on land, biomass pyramid may remain upright.

The shape of the pyramid of productivity is not affected by size or rate of metabolism of organisms, while the other two pyramids are affected by them. Animals may have larger biomass per unit area than plants, but their production per unit time per unit area would be smaller than that of plants.

15.4 SECONDARY PRODUCTIVITY

The rate of energy storage at consumer (animals and saprotrophs) level is called *secondary productivity*. It may be represented as formation of dry weight (or equivalent energy) of new organic matter by heterotrophs per unit area per unit time. Both primary and secondary productivity involve the synthesis, by a particular kind of organic matter, of more of that particular kind (i.e. biological autocatalysis). But, while primary productivity involves the direct assimilation of energy from sunlight, secondary productivity requires the conversion of one kind of organic substance, that is that produced by a particular species, into another kind. Only small fractions of leaf weight (equivalent to 1-5 per cent NPP of forests) or other plant materials eaten by herbivores become their tissues (Fig. 15.11). Of the food ingested (eaten) by animals, large fractions are egested (faeces and urine) and make no contribution to animal production. Some of the digested food is spent on respiration to support animal's metabolism and activities. Therefore, this also does not appear as new organic matter. Only the remainder fraction becomes new organic matter and adds to animal's weight and is referred to as secondary production. This can be compartmentalized as new growth of existing individual and reproduction to form new individuals.

Though the terms “gross” and “net” productivities are not used (consumers are not actual producers but depend on plant's production), assimilation and secondary production respectively are analogous to them.

The net primary production is manufactured and stays *in situ* and can be dealt as vegetation, whereas secondary production seldom stays *in situ* and is relatively rapidly spread out in scattered, discrete units; we have no word comparable to vegetation to apply to animals. Thus, the determination

of secondary productivity is far more difficult than it is of primary productivity. In most cases, each major animal species must be measured in detail for population density and age composition, food consumption and utilization, and growth and reproduction. Often the field data are supplemented with laboratory data particularly on growth and metabolism which are very difficult to measure in the field. It is relatively easier if an animal population consists of one or more easily identifiable age classes, for, their growth can be followed with more ease. For example, if a chicken population of a given age is measured for its mean weight once at the beginning and then after one year, the mean weight increase times chicken density would be secondary productivity per yr per unit area.

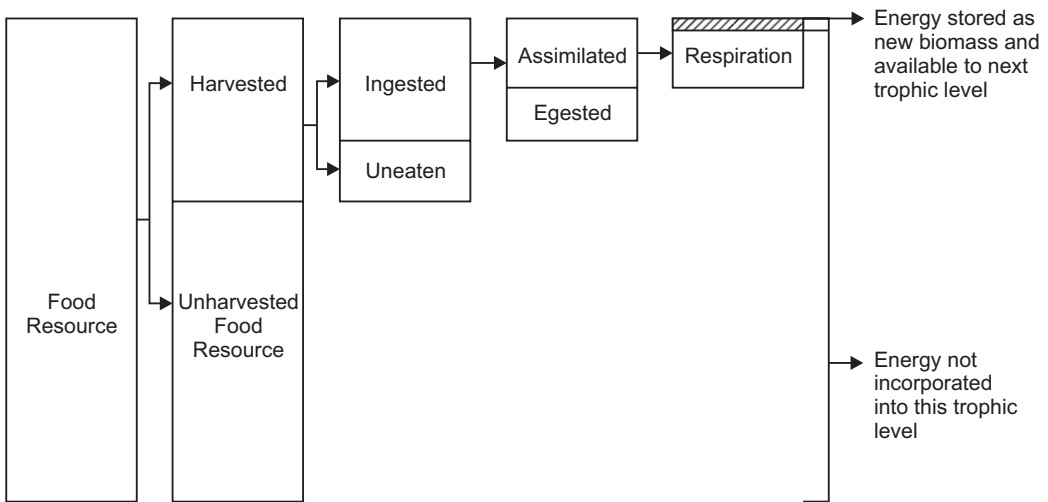


Fig. 15.11. A schematic representation of energy transfer and losses within trophic levels. At the secondary trophic levels, only a small fraction of the energy is incorporated into biomass, which is available for the next trophic level (based on Smith, 1996).

Secondary productivity would be the greatest when animal's reproduction is maximum and young individuals are growing rapidly. When animals, like deer and birds attain mature sizes they cease to grow and the entire assimilation is spent in respiration in support of life activities, including the maintenance of body temperature in warm-blooded animals. Here the secondary productivity is zero. However, at population level the P would be usually more than zero, for it normally consists also of some young and growing individuals. In spider, a carnivore, and old-field mouse, a herbivore, most of ingested energy (I) is assimilated (A) - A/I ratio being above 0.9 (Table 15.1). In contrast, in salt marsh grasshopper and African elephant, the herbivores, more than half of assimilated energy is egested, and A/I ratios for them are reported as 0.37 and 0.44, respectively. It seems that carnivores have usually high A/I ratio while herbivores can have low as well as high A/I ratio. In a majority of the populations, almost entire assimilated energy is consumed in respiration (R), the R/A ratio ranging between 0.97-1.00. Only in salt marsh grasshopper (0.63), plant hopper (0.74) and small spider (0.84), some substantial portion of A results in secondary production (Table 15.1).

Table 15.1. Secondary production of some consumer populations (kcal/m²/yr).

Species	Ingestion (I) (kcal m ⁻² yr ⁻¹)	Assimilation (A) percentage of I	*Respiration as percentage of A	Production (kcal m ⁻² yr ⁻¹)	Reference
Herbivore					
Harvest ant	34.5	90	99	0.1	Odum et al. (1962)
Plant hopper	41.3	67	75	7.0	Wiegert (1964)
Salt marsh grasshopper	3.7	37	63	0.51	Teal (1962), Smally (1960)
Old field mouse	7.4	91	98	0.1	Odum et al. (1962)
Ground squirrel	5.6	68	97	0.11	Wiegert and Evans (1967)
Meadow mouse	21.3	82	97	—	Golley (1960)
African elephant	71.6	44	100	—	Petrides and Swank (1966)
Carnivore					
Spider, small < 10 mg	12.6	94	84	1.9	Moulder and Reichle (1974)
Spider, large > 10 mg	7.4	95	104	-3.0	
Weasel	5.8	95	—	—	
omnivore					
Savanna Sparrow	4.0	90	100	—	Odum et al. (1962)

There are thus four major factors which determine secondary productivity: (i) quantity and quality (including nutrient status and digestibility) of net primary production, (ii) degree of utilization of net primary production by consumers, (iii) the assimilation/ingestion efficiency of consumers, and (iv) the production/assimilation efficiency of the consumers. The relationship between primary and secondary productivity is illustrated in Figure 15.12.

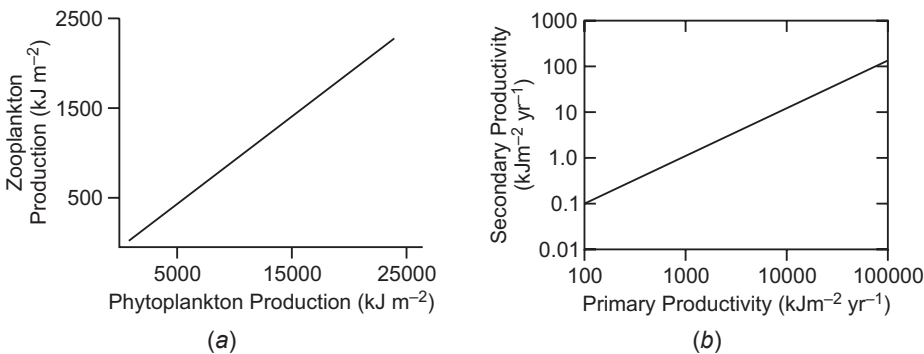


Fig. 15.12. Relationship between secondary production and net primary production for (a) Lakes (based on Brylinsky and Mann, 1973) and (b) African grassland (based on McNaughton et al., 1989).

Whittaker and Likens (1973) and Woodwell and Pecan (1973) have tried to give values of secondary productivity and animal biomass in various natural ecosystems of the earth (domestic animals not included). Though the values are rough approximations, some observations can be made:

- (1) Mean consumptions of net primary production by animals are 1 per cent in cultivated fields (man tries to remove all herbs), 2-3%t in deserts and tundras, 4-7 % in forests, and 10-15% in grasslands (in intensively grazed grasslands 15-60% . Compared to these consumptions, percentages are significantly higher in oceans, 40% in open areas (up to 90% in marine area), 30-35% in upwelling areas and continental shelf water and 15% in algal beds and estuaries. The overall mean values come to 7 and 36% for land and oceans, respectively.
- (2) Secondary productivity as the percentage of assimilation or ingestion may be zero for mature animals having ceased to grow, 5-15% for many herbivores and still higher for young carnivores.
- (3) The animal's secondary production is likely to be only 1% of net primary productivity on the land and 5-6% in the sea. This means almost three quarters of animal secondary production (excluding domestic animals) may be in the ocean, where zooplankton organisms form most of that. Since man cannot harvest these organisms, he has to depend on carnivores' production which may be as much as 10% of the zooplankton production.
- (4) From the above it is apparent that of the total net primary production, 93% on the land and 63% in the oceans go to reducers (decomposers) or to detritus food chains. On the basis of growth efficiencies (new protoplasmic carbon/substrate or detritus carbon), it is suggested that the reducer production is 24 times greater than animal production on land and half or about as much as animal production in oceans (growth efficiency is taken as: 30-40%t for fungi; 5-10% for aerobic bacteria and 2-5% for anaerobic bacteria on land, and 5-10% for marine reducers). However, values of secondary production of bacteria and fungi may be even more tenuous than that of animal production.
- (5) In most land communities, most of the animal biomass is due to the short-lived arthropods and soil animals like mites, springtails and annelids, large animals, easily seen by us, constitute little. Earthworms account for most of animal biomass in forests, however, vertebrate biomass may exceed invertebrate biomass in some grasslands. In the seas also microscopic zooplankton concentrate greater biomass, larger than animals like fishes.

Man's enormous impact on the organic layer is well known, the point we want to make here is that its biomass has now attained considerable amount, perhaps has exceeded earthworm's biomass. Humans harvest animals to the tune of 90×10^6 t/yr (including milk and eggs) on land and about $20 \times 90 \times 10^6$ t/yr in the oceans. In addition to these, humans harvest biomass to meet the needs of fibre and to create space for settlements and the industries.

15.5 DETRITUS PATHWAY OF ENERGY FLOW AND DECOMPOSITION PROCESSES

Decomposer organisms and their trophic interactions

In all ecosystems a substantial portion of the net production moves to the *detritus* pool consequent to mortality of organisms and parts thereof (e.g., leaf litter), and transfer of unused and unassimilated material by the biophages to soil or sediment. Detritus is the primary energy source for a group of organisms called *decomposers* which also organise themselves into food chains along which transfers of energy occur. The main groups of organisms involved in decomposition are given below. Their trophic classification cannot be rigid as several of them may function at more than one trophic level.

Saprotrophic micro-organisms

These include bacteria, fungi and actinomycetes. Some of the micro-organisms are unicellular and form colonies (bacteria and yeast) and others are mycelial. These organisms secrete extracellular enzymes which break down complex organic matter into smaller molecules which are then absorbed and utilized intracellularly. Many fungi and few bacteria possess cellulases and lignin-degrading enzymes. The arbuscular mycorrhizal fungi have been found to enhance litter decomposition by producing cellulases as well as increasing nitrogen capture from complex organic materials into the soil by breaking down proteins into amino acids (Hodge et al., 2001).

Saprotrophic invertebrates

This group includes earthworms and enchytraeids, slugs and snails, arthropods such as Diptera, Collembola and Acari, Orthoptera and Isoptera, etc. The food (dead plant parts mainly) is first comminuted and then swallowed for digestion in the gut system. Some of these detritivores such as cockroach and termites digest the structural plant polysaccharides with the aid of symbiotic bacteria and protozoa which possess the cellulolytic enzymes. These detritivores often ingest the dead matter together with the micro-organisms growing there on. The exclusion of soil invertebrates from the litter bags reduces decomposition rates (Gupta and Singh, 1977).

Saprotrophic vertebrates

Vertebrates such as game herds of savanna or cattle occasionally or for a period of time may feed on dead vegetation. A number of vertebrates feed on dung and carrion. The hyenas and most of the canidae (foxes, jackals, etc.) are predators as well as carrion feeders.

Microbivores

These are specialized microfauna principally the protozoans (specially amoebae) and nematodes. Some of these graze particularly the bacteria, others fungal hyphae. Cryptostigmata, Collembola and Enchytraeids also consume micro-organisms.

Invertebrate carnivores

Opiliones (harvest spiders), Chilopoda (centipedes), Araneida (spiders) are examples of carnivores which derive their food from saprotrophic invertebrates. Other predators include nematode-eating soil fungi, pseudo scorpions, staphylinid and carabid beetles and the prostigmatid and mesostigmatid mites.

Wiegert et al. (1970) recognised three functional groups in the decomposer subsystem: (i) Fungi and bacteria saprophages, (ii) Animal saprophages, and (iii) Animal predators (Fig. 15.13) with close interaction and repeated use of material between the first two groups. Heal and MacLean (1975) used a more elaborate model (Fig. 15.14) comprising (i) microbial and invertebrate saprophages, (ii) 1st order carnivores (including microbivores), and (iii) 2nd order carnivores, with energy and matter being recycled between detritus pool and its consumers. Distinctions are made between vertebrates and invertebrates, between microbial saprophages and detritivores, and between microbivores and other 1st order carnivores.

Two important differences between biophagic and saprophagic pathways are to be noted: (i) The trophic level interactions in the grazing food-chain depend on the densities of both the predator (recipient) and the prey (donor), while those in the saprophagic chain are largely donor (detritus) controlled. (ii) Unassimilated matter and dead bodies are lost to the grazer subsystem but those from the decomposer subsystem are sent back to the base of the food chain (detritus pool in Fig. 15.14). Thus a considerable amount of organic matter (or energy) is *cycled* until completely metabolized (e.g., dead bodies of microbivores or other carnivores are acted upon again by microbial saprophages).

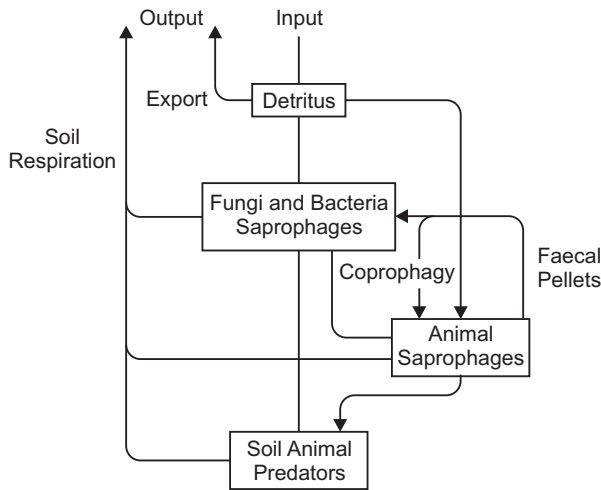


Fig. 15.13. A representation of the functioning of organisms in the decomposition subsystem (based on Wiegert et al., 1970).

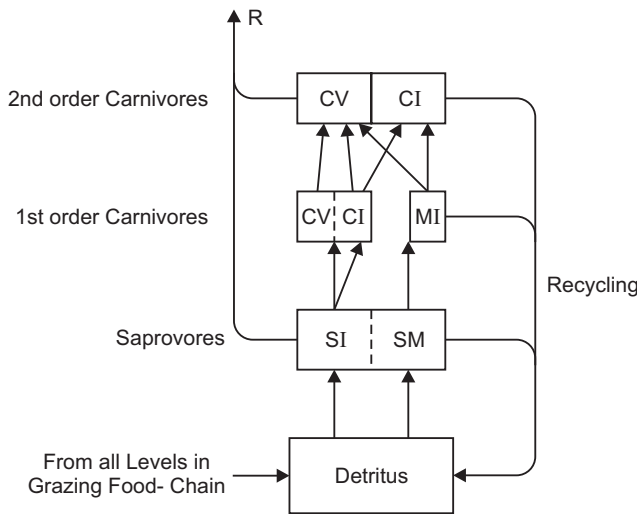


Fig. 15.14. A representation of energy flow through decomposition subsystem. SI = saprovores invertebrates, SM = saprovores microorganisms, CV = vertebrate carnivores, CI = invertebrate carnivores, MI = Microbivores (based on Heal and MacLean, 1975).

Detritus food web

The decomposer community forming the soil food webs is functionally complex as compared to the plant-based food chains. Therefore, the controls on the process of decomposition are multiple and complex. Decomposition results in the release of substrate and nutrients for growth of other organisms. Trophic interactions among various groups of the soil fauna are known as detritus food webs. The soil food webs have a high degree of omnivory, high connectivity and greater species richness as compared to the aboveground food webs (Cohen et al., 1990). The detritus food web of short grass prairie ecosystem (Fig. 15.15) shows up to nine trophic levels including plant, microflora, and micro- and meso-fauna (Hunt et al., 1987). The microfauna groups, i.e., protozoa and nematodes mainly feed on plant roots, microflora and microfauna groups. The interactions between plant roots and microfauna aid in the release of plant available nutrients (Fig. 15.15).

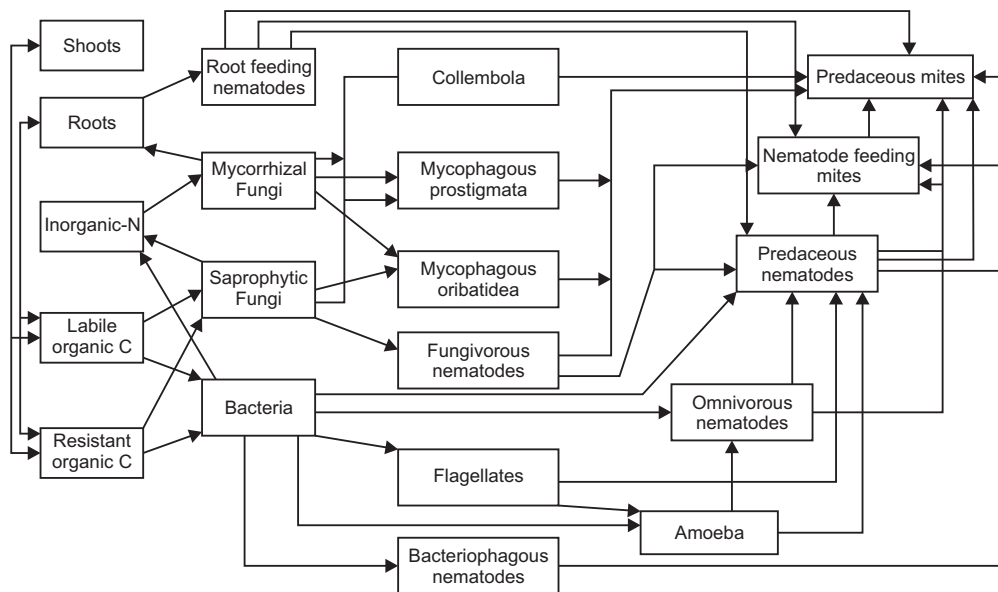


Fig. 15.15. A simplified food web for the soil organism community in a short grass prairie ecosystem (redrawn from Hunt et al., 1987).

The detritus based food webs show the presence of distinct within habitat compartments, maintained by morphological differences between bacteria and fungi and differences in habitat niches to soil moisture and management practices (Hendrix et al., 1986). The high degree of omnivory leads to greater connectivity in soil food webs. In a microcosm experiment, Wardle (1995) showed that connectivity remained unaffected even through species richness increased. The bacterial feeding and fungal feeding in the detritus food webs are generalist feeders. Independent of species numbers, connectivity in fungi-based compartments is less than in bacteria-based system (Wardle, 1995). The linkages between plant litter quality, soil biota and decomposition processes have been analyzed by Lavelle (1997). In soil, bacteria and fungi are the primary decomposers and the mass of these components (microbial biomass) relates to the resource quality. Plant litter quality regulates the diversity of soil biota and the nature of soil biotic interactions, by operating through “micro foodwebs”, litter transformation system and ecosystem engineers for nutrient immobilization- mineralization. Micro foodwebs mainly comprise micro-fauna (nematodes and protozoa) which are predators. The litter transformers show a mutualistic relationship with the microflora based on internal rumen. Ecosystem engineers, such as earthworms and termites, have mutualistic relationship with the microflora in their gut.

Decomposition processes

It is difficult to arrange the steps of decomposition process in a chronological sequence because of (i) overlapping feeding habits of the decomposer organisms, (ii) cycling of material within and between trophic levels, (iii) differences in the substrate quality of the detritus and the change in the state of the decomposing material with time, and (iv) marked effect of abiotic environment on the rate of the process. In general, the rate and pathways of decomposition are determined by the qualitative and quantitative composition of the decomposer community, their physical environment and the substrate quality. A possible pathway of decomposition of plant debris is as follows:

1. Colonization by saprotrophic micro-organisms.

2. Comminution and ingestion by saprotrophic invertebrates resulting into an increase in the surface area and enrichment of the surface by proteins and growth substances.
3. Eating and re-eating (coprophagy) of the comminuted litter with adhering micro-organisms by the saprotrophs. Each time the saprotroph digests the high quality microbial biomass and egests the remaining detrital particles which are recolonised by microbes. In insect-micro-organism symbiotic systems (e.g., termites and betsy beetles) considerable change in chemical composition of the substrate occurs.
4. Utilization of comminuted litter and faeces by micro-organisms resulting into chemical degradation of tissues and production of complex phenolic polymers.
5. Formation of stable organo-mineral complex (humus and humic acids).

Box 8.3. Measurement of litter decomposition

The decomposition rate can be expressed by the rate parameter k which in part measures the effectiveness of decomposer organisms in breaking down organic materials in a given set of environmental conditions. In a steady state condition, the amount of litter fall would be equal to the amount of litter decayed in an annual cycle. The rate parameter k (i.e. instantaneous fractional loss rate) is given by:

$$k = L/X$$

where L is litter fall and X is standing crop of litter on the forest floor at steady state. In this formulation it is assumed that litter fall (L) occurs at a steady rate.

In cases where most of the litter falls at the end of a growing season (such as a deciduous forest), k' is calculated as:

$$k' = \frac{A}{A + F}$$

where A is the litter fall and F is the minimum standing crop of litter on the forest floor within the annual cycle. The instantaneous decay rate can be calculated as:

$$k = -\ln(1 - k')$$

In the experimental approach, samples of litter are air dried and placed into plastic or fibre grass mesh bags. The mesh size should be such as to allow free entry of soil microflora and fauna. The bags are then placed on the forest floor. A few bags are recovered after different time intervals and the mass of litter remaining is dried and weighed. The k can be calculated as:

$$k = -\ln(X/X_0)$$

Where X_0 is the initial weight of litter enclosed and X is the weight remaining at the end. In this case it is assumed that there occurs a constant fractional weight loss and when values of mass remaining are plotted against time on a semilog graph a constant negative slope, $-k$, is obtained.

The time by which 50 per cent litter will decompose is estimated by $0.693/k$, 95 per cent by $3/k$ and 99 per cent by $5/k$.

Decomposition rate

Decomposition is a major driver of ecosystem organic matter dynamics. The rate of decomposition depends on the origin and quality of detritus, and environmental conditions. From a meta-analysis of global data on relative mass fractions and annual turnovers of leaf, fine stems and fine roots of plants, Freschet et al., (2013) estimated that a grassland species would on average shed only 26% of its total annual litter production as leaves, against 41% as fine stems (culms, flowering stems) and 33% as fine roots in contrast to a forest species which would shed 41% of its total annual litter production as leaves, compared to 11% as fine stems and 48% as fine roots. The authors estimated that on average below-ground litter represents approximately 33 and 48%

of annual litter inputs in grasslands and forests, respectively. The analysis also revealed that fine roots and stems had lower decomposition rates relative to leaves (approximately 1.5 and 2.8 times slower, respectively, than leaf litter). Further, a resource acquisitive plant community (nutrient rich plants on nutrient rich sites) produces litter of consistently higher decomposability than a resource conservative plant community (nutrient-poor plants on nutrient poor sites). Overall, the analysis suggested a major role for below-ground litter as a driver of ecosystem organic matter dynamics and provided evidence that within ecosystems, the relative inputs of above- versus belowground litter strongly control the overall quality of the litter entering the decomposition system.

Different constituents of the plant detritus decompose at different rates according to relative resistance to attack by saprotrophs. For example, Fig. 15.16 shows that phenols, waxes and lignin are more resistant than cellulose, hemicelluloses and sugars. In general, the resistance varies in the order: sugars < starch < hemicelluloses, proteins and pectins < cellulose < lignins < suberins < cutins. Because different detritus contain these components in different proportions they decompose at different rates (Fig. 15.17). There is generally an inverse relationship between decomposition rate and lignin content of the litter (Upadhyay et al., 1989). Also, the heterogeneity of material results into a succession of micro-organisms in time. Many species of sugar fungi (e.g. *Penicillium*, *Mucor*, *Rhizopus*) and some bacteria grow fast on the substrate during early phases of decomposition. After the easily metabolizable sugars are exhausted, the residual material becomes more recalcitrant, and the sugar fungi are succeeded by cellulolytic and ligninolytic forms. Thus Ascomycetes, Basidiomycetes and Actinomycetes possessing specialized metabolism follow the sugar fungi. However, because of the interactions between micro-organisms and detritivores, and changes in the substrate quality during decomposition, an universal successional pattern cannot be given. One thing is certain that as the resistant materials increase in proportion the rate of decomposition slows down.

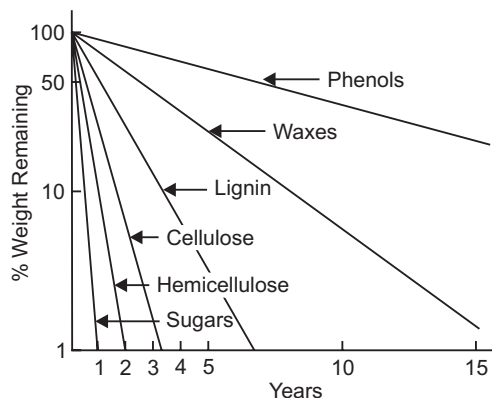


Fig. 15.16. Decomposition of selected litter constituents (based on Minderman, 1968).

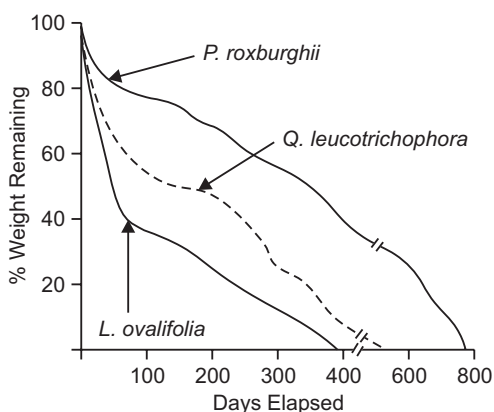


Fig. 15.17. Decomposition of three leaf litter species in Central Himalayan forests with varying ligning content. Lignin contents are 23% for *Pinus roxburghii*, 17% for *Quercus leucotrichophora* and 15% for *Lyonia ovalifolia* (based on Upadhyay and Singh, 1989).

The abundance of different kinds of decomposers varies with the environment and ecosystem types (Table 15.2). Bacterial cell counts and hyphal lengths increase with decreasing latitude. Tropical soils with low organic matter have low nematode and protozoan populations and low numbers of mesofauna. However, termites increase in the tropical habitats and determine major energy and nutrient flux pathways because of their high cellulolytic efficiencies brought about by symbiotic associations with protozoa or microflora. Macroarthropods are generally more abundant in

tropical ecosystems. From the body size/food consumption relationships, biomass and accentuating influence of temperature on invertebrate metabolic rates, Swift et al. (1979) predict much higher litter decomposition rates in the tropics compared to temperate and polar latitudes. Meentemeyer (1978a) has indeed reported a positive relationship between actual evapotranspiration (which is greater in lower latitudes), reflecting both moisture and energy availability, and decomposition rate (Fig. 15.18). As with latitude, the decomposition rate is inversely related also to altitude (Upadhyay et al., 1989). Meentemeyer (1978b) further suggested that the greater the abundance of energy and moisture (AET), the faster the decay rate for a given lignin content but the higher the lignin content, the more energy and moisture are required to accomplish litter break down in a unit time. Thus, the substrate quality and AET have a combined effect on the rate of decomposition (Fig. 15.19). Needless to say the effect of AET and substrate quality on decomposition rate is realized through their effects on the quality, quantity and activity of the decomposer organisms.

Table 15.2. Soil organism populations in major ecosystem-types (abridged from Swift et al., 1979).

Organisms	Boreal forest	Warm temperate forest	Tropical forest
Microflora Bacteria (no/g soil)	5×10^6	6×10^6	55×10^6
Fungi (m hypha/g soil)	4000	3000	6000
*Microfauna (<100 μm)			
**Nematoda + Protozoa (no/m ²)	270×10^6	200×10^6	65000
Mesofauna (100 μ m -2 mm) (no/m ²)	1,151,000	71,500	22,000
Macrofauna (2 mm-20 mm)	820	1400	1650

* Mesofauna include Collembola, Acari, Enchytraeidae, Isoptera, Diplura, Symphyla, Protura, Diptera.

** Macrofauna include Earthworms, Diploda, Isopoda, Coleoptera, Mollusca, Orthoptera, Dermaptera, etc.

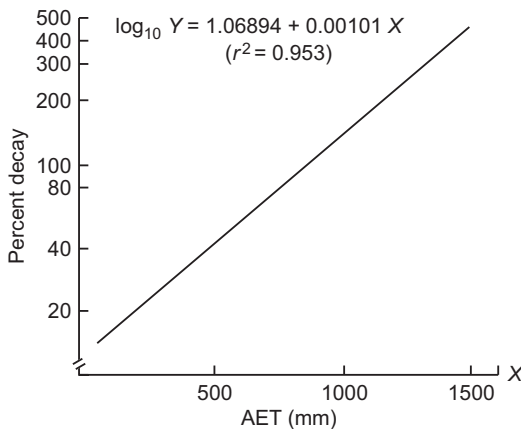


Fig. 15.18. Relationship between actual evapotranspiration (AET) and annual weight loss of leaf litter. Decay rate of 100% means that all leaf litter will decompose in one year, 200% complete decay in 6 months, 300% in 4 months, and so on (based on Meentemeyer, 1978a).

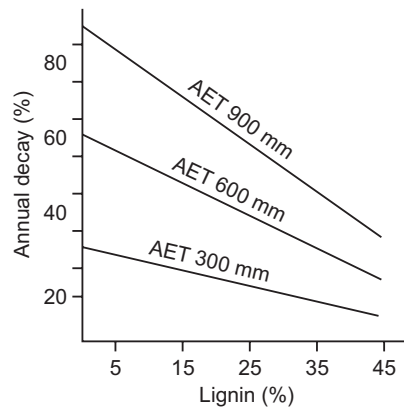


Fig. 15.19. Change in the relationship between lignin content and decomposition with actual evapotranspiration (AET) (based on Meentemeyer, 1978b).

Global Biogeochemical Cycles and Human Impact

Of the 90-odd chemical elements occurring in nature, 30 to 40 are known to be required by living organisms. The elements circulate in the biosphere in characteristic circular paths from environment to organisms and back to the environment. We refer to these paths as biogeochemical cycles ('bio' refers to organisms and "geo" to the earth including rocks, air and water). Cycling of mineral elements is also called nutrient cycling. The six most important elements necessary for life are: carbon, hydrogen, oxygen, nitrogen, sulphur, and phosphorus. The first three are combined in energy-rich materials such as carbohydrates and fats. These together with N and S are essential ingredients in all proteins. P is needed for the transfer of chemical energy within protoplasm where this energy is used for activity or growth.

The biogeochemical cycles are driven by energy flow and are crucial for the maintenance of life on earth in its present form, and biological processes largely determine the main features of these cycles. In each biogeochemical cycle, two compartments or pools are recognized: (i) the large, slow moving reservoir pool, and (ii) the small, actively moving exchange pool which exchanges rapidly between organisms and their immediate environment. The reservoir pool is sometimes called *unavailable pool* and the active moving pool, the *available pool*.

The biogeochemical cycles fall into two basic groups, viz., gaseous types, in which the reservoir is in the atmosphere or hydrosphere (ocean), and sedimentary types, in which the reservoir is in the lithosphere. The gaseous cycles such as nitrogen and carbon have capacity to self-adjust rather easily because of the large atmospheric or oceanic reservoirs or both. They can be considered well-buffered globally because of their better capacity to adjust to change. Sedimentary cycles, which involve mineral elements like phosphorus and sulphur, tend to be easily disturbed because a large portion of the nutrient is in inactive or immobile reservoir in the earth's crust.

16.1 THE CARBON CYCLE

Accounting for about 49% of organism's dry weight, carbon is only next to water in significance to the living world. The distribution of carbon in major components of the biosphere, as estimated for early 1980s is shown in Table 16.1. The carbon cycle is characterized by small but active atmospheric pool (about 700 GT; 1 GT, gigaton = 1000 million tons = 1 petagram, Pg or 1×10^{15} g; more recent estimates put the value around 800 Pg). The cycle involves a gaseous phase, the atmospheric CO_2 . The basic movement of CO_2 is from the atmospheric reservoir to plants (by photosynthetic or chemosynthetic fixation) and animals (mostly by herbivory and some directly from atmosphere by biochemical fixation), and from them to decomposers and then back to atmosphere (Fig. 16.1). Land plants fix about 120 GT and marine biota about 92 GT of carbon per year. Only about 0.2 % of the total carbon contained in the atmosphere, held in the biomass of plants and animals and dissolved in water turns over rapidly, i.e., once in a year. There is a reservoir

of carbon in the ocean, in addition to atmosphere. The oceans contain more than 50 times as much carbon (35×10^3 GT) as the air (Table 16.1) and act as the regulator of carbon in the atmosphere.

Table 16.1. Major carbon reservoirs and fluxes (after McCarthy et al., 1986/87).

Reservoirs	GT carbon
Atmosphere	700
Oceans	
Total inorganic	35,000
Particulate organic (biota)	3
Land biota	600
Soil humus	3,000
Marine sediments	
Organic	10,000,000
Calcium carbonate	50,000,000
Fossil fuels	5,000
Fluxes	GT carbon yr ⁻¹
Atmosphere-marine biota	45
Atmosphere-land biota	70
Deposition in oceans	1-10
Fossil fuel combustion	5

1 GT = 1 billion metric ton (or 1×10^{15} g or 1 petagram)

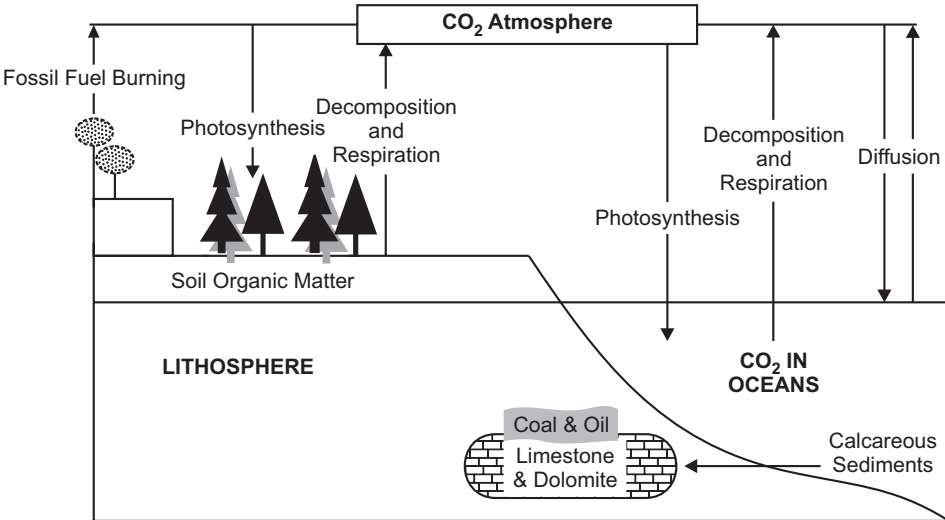


Fig. 16.1. A simplified representation of the carbon cycle. Biota include typical plant-consumer relations. Lithospheric C is not shown because of its very low fluxes.

Plants are the major force behind the global carbon cycle as they fix carbon through the process of photosynthesis. The carbon fixed by the plants is used for their own growth and then enters the food chains as animals eat the plants. About twice as much carbon dioxide is taken up by photosynthesis on land as in the oceans and rivers. Land plants are estimated to contain about 250

times as much carbon as aquatic plants, most of it in the form of forest growth. In the oceans, only a small proportion of the carbon is held in living organisms. Most of it is in the detritus left behind by these organisms, as well as some dissolved carbon dioxide mainly in the form of bicarbonates.

Respiratory activity of plants and animals, and, more importantly, of the decomposers in processing of dead remains of other organisms and the waste materials, results in the return of biologically fixed carbon as CO_2 to the atmosphere. At the global level, the total respiration and decomposition are thought to be equal to total photosynthesis. The fire, by burning the organic materials, provides (nonbiological combustion), another route of carbon return from the biota to the tune of about 2-4 GT per year. The characteristic turnover times vary from a few weeks for marine biota to about 1000 yrs for the CO_2 in the ocean.

The geological components involving deposition of organisms as coal, oil and animal remains (such as mollusc shells and protozoan tests as carbonate rocks) make up a significant constituent of the system. The geochemical part of the carbon cycle stores carbon for far longer period of time and comprises two main pools: the deep ocean water, with an estimated 36,000 billion tonnes, and **rocks**, especially limestone, which hold an estimated 90 million billion tonnes. Other rocks also lock away carbon, notably those that produce fossil fuels. This carbon too originates in living things - plants provide the carbon for coal, whereas oil and gas also come from marine animals.

The lithosphere including continental and oceanic crusts (carbonate and noncarbonate sediments, and igneous and basaltic rocks) account for about 65×10^6 GT carbon. On weathering and dissolution of carbonate rocks, the combustion of peat, coal and oil (fossil fuels), and volcanic activity involving deposits of carbonate rocks and fossil fuels, the bound carbon is returned to atmospheric-aquatic reservoir of carbon cycle. The characteristic turn over times for C held in the lithosphere are hundreds of millions of years.

The interchange between oceanic and atmospheric carbon dioxide plays a significant role in carbon cycling and, therefore, needs a brief consideration. This interchange occurs mainly through diffusion, and hence, its direction depends upon the concentration of CO_2 . However, input of CO_2 in aquatic phase also occurs through precipitation. A litre of rain water contains about 0.3 ml gaseous CO_2 . The dissolved CO_2 combines with water to form carbonic acid (H_2CO_3), which dissociates into hydrogen (H^+) and bicarbonate ions (HCO_3^-). Since these reactions are reversible and depend upon the concentrations of the components, a local depletion of atmospheric CO_2 results in a net movement of CO_2 from the aquatic body into the atmosphere. Similarly, the intake of HCO_3^- in photosynthesis by aquatic plants would tend to shift the equilibrium the other way.

Moore and Bolin (1986/87) have summarised the processes in oceans which play an important role in the global carbon cycling. These are (i) consumption of CO_2 in primary production in surface waters, (ii) enrichment of deep waters in CO_2 due to decomposition and dissolution of detrital matter from surface waters, (iii) sinking of water in polar regions, (iv) upwelling of this water in the equatorial regions with corresponding outgassing of CO_2 . The processes (i) and (ii) constitute the biological pump, and (iii) and (iv) the physical (solubility) pump (Fig. 16.2). The biological pump lowers the partial pressure of CO_2 in surface waters and increases it in deep waters. In low production areas, however, partial pressure is greater than that in the atmosphere, so CO_2 may be released from the sea surface. Carbon is fixed by phytoplankton through photosynthesis in the upper oceanic layers. Some of this carbon is cycled through the food-web within this layer, but a majority is exported to the deeper layers and sediments as dead organic matter, to be slowly remineralized back to CO_2 . The net result is an enrichment of the deep waters (main CO_2 storage area) with CO_2 . Since the rate of uptake of CO_2 by phytoplankton is determined by the quantities of available N, P, Si and Fe, concentrations of these nutrients govern the efficiency of the phytoplankton-powered biological pump.

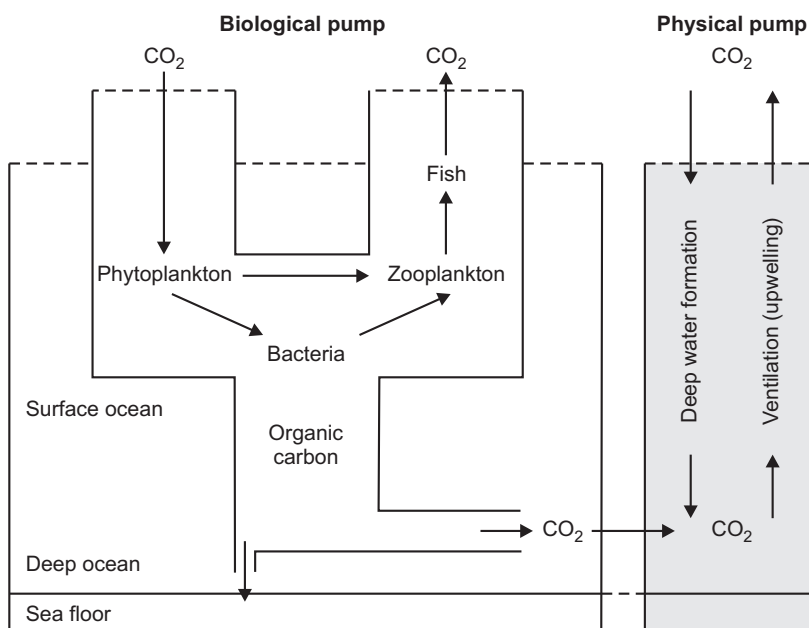


Fig. 16.2. The regulation of ocean-atmosphere CO_2 exchange through biological and physical pumps

In the polar regions, ice formation leaves much of the salt behind increasing the salinity and density of cold water. These cold and dense water masses absorb atmospheric CO_2 (solubility of CO_2 increases with a decrease in temperature) and sink, transferring the CO_2 to abyssal waters. The CO_2 -enriched water moves slowly towards equatorial regions, where upwelling (vertical movement) occurs. The upwelled water warms, CO_2 becomes less soluble and some of it is released back to the atmosphere (the complete cycle taking about 1000 yr). Thus, the two pumps, biological and physical, tend to concentrate CO_2 in the deeper layers of the oceans.

Human Impact on Carbon Cycle

Current estimates of annual fluxes between global carbon pools are given in Fig. 16.3. During the pre-industrial age, the various pathways collectively formed self-regulating feedback mechanisms resulting in a homeostatic system. Additions and deletions would be readily compensated, resulting in a steady state of CO_2 concentration in the atmosphere. However, this balance has been disturbed and CO_2 in the atmosphere has increased from about 280 ppm (parts per million) in 1750 to about 400 ppm in May, 2013. All investigators agree that the major cause of the increase in atmospheric CO_2 has been the combustion of fossil fuels (plus kilning of limestone for cement production). The uncertainty regarding the contribution of deforestation is quite large. For example, some ecologists observe that destruction of forests are releasing nearly as much CO_2 into the atmosphere as are industrial processes (Bolin, 1977; Woodwell et al., 1978). Human activity produces approximately 8 billion tonnes of C each year (6.4 from fossil-fuel burning and 1.6 from deforestation particularly in the tropics), of this, only 3.2 billion tonnes remain in the atmosphere to warm the planet?. The question is where is the missing carbon? It is estimated that 2.2 billion tonnes are stored in the oceans, about 0.7 billion tonnes are taken up by the northern mid- to high-latitude forests, and the balance by the tropical forests due to CO_2 fertilization effect. Stephen et al. (2007) showed that tropical forests are the net source of a mere 100 million tonnes of carbon annually and carbon sequestration in the tropics is substantial enough to almost counterbalance the effects of deforestation.

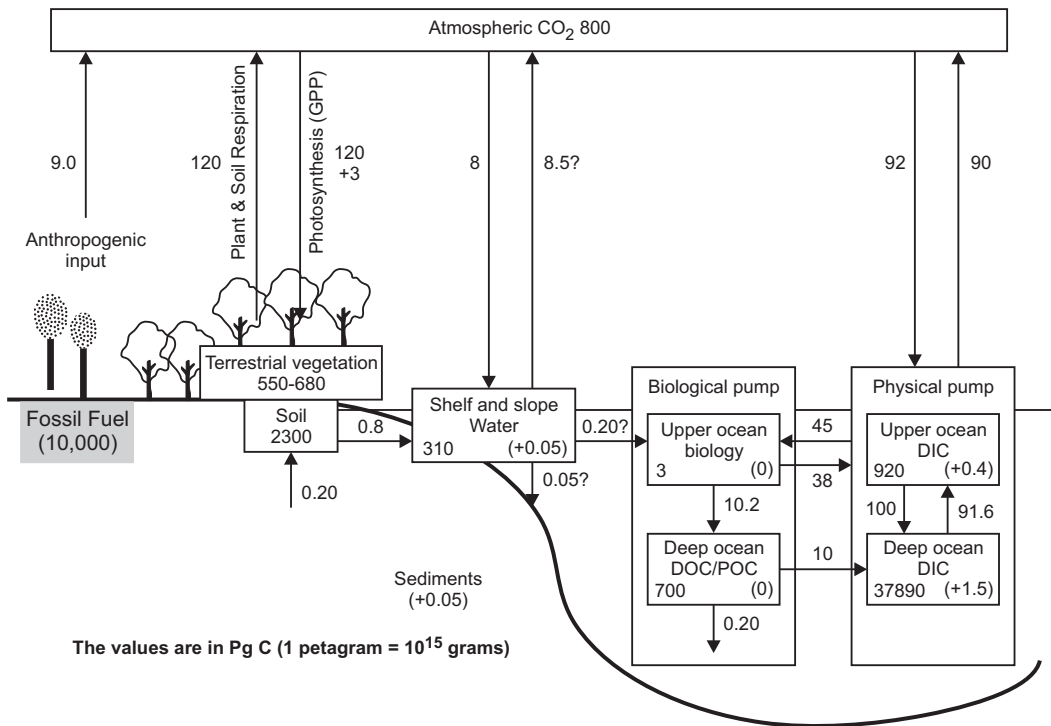


Fig. 16.3. Carbon cycle with current estimates of pools and fluxes. The flux values are in petagrams of carbon per year, values in boxes represent global inventory and those in parentheses average annual increases in the inventory due to anthropogenic input. (based on Balino et al., 2000; US DOE, 2008; Le Qu'ere' et al., 2013)

Le Qu'ere' et al. (2013) have quantified all major components of the global carbon budget, including their uncertainties for the period 2002-2011. Global CO₂ emissions from fossil fuel combustion and cement production has reached 8.3 ± 0.4 PgC for the period 2002-2011 (Box 16.1). The global CO₂ emissions were dominated by emissions from China (28% in 2011), the South Africa (16%), the EU (27 member states; 11%), and India (7%). Global CO₂ emissions from Land-Use Change activities were 1.0 ± 0.5 PgC. Atmospheric CO₂ growth rate was 4.3 ± 0.1 PgC. The ocean CO₂ sink was 2.5 ± 0.5 PgC yr⁻¹. The terrestrial CO₂ sink calculated as the residual from the carbon budget was 2.6 ± 0.8 PgC in 2011. The evolution of the CO₂ budget over time can serve as a key indicator of the anthropogenic perturbation of the climate system as well as to monitor the evolution of human induced climate change (Le Qu'ere' et al., 2013).

The problem of net CO₂ release as a consequence of forest clearing and expanding agriculture perhaps now largely lies in the tropics. It may be pointed out that forests are important carbon "sinks". On the other hand, in agriculture a greater release of CO₂ into atmosphere than the removal in photosynthetic fixation especially results from frequent ploughing and from the fact that plants are active only for a short period within an annual cycle. The release of CO₂ as a consequence of rapid oxidation of humus has also other effects. Because of rapid water run off in landscapes dominated by agricultural fields (instead of water filtering down through humus layers in soil, which normally is the case in forests), the rate of dissolution of many trace elements has declined. Consequently, the need for adding trace elements in agroecosystems in order to maintain the yield is being increasingly felt.

Box 16.1

The five components of the anthropogenic CO₂ budget for the period 2002–2011. All values are in PgC yr⁻¹. All uncertainties are reported as ± 1 sigma (adapted from LeQu'ere' et al. 2013).

Emissions	Mean (PgC yr ⁻¹) 2002-2011
Fossil fuel combustion and cement production	8.3 ± 0.4
Land-Use Change emissions	1.0 ± 0.5
Partitioning	
Atmospheric growth rate	4.3 ± 0.1
Ocean sink	2.5 ± 0.5
Residual terrestrial sink	2.6 ± 0.8

16.2 NITROGEN CYCLE

The significance of nitrogen lies in its being the major builder of proteins and nucleic acids, the essential constituents of organisms and regulators of biological functions. Fig. 16.4 is a simple representation of the nitrogen cycle.

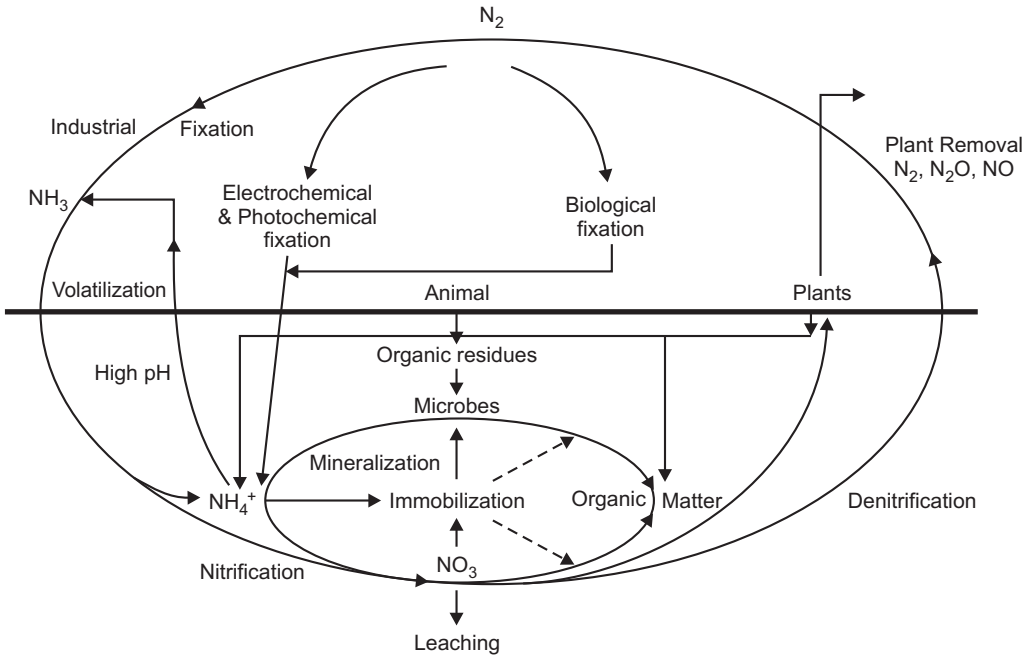


Fig. 16.4. A simplified representation of the nitrogen cycle.

Although we live in an atmosphere which is largely N₂ (about 79%, total of 39 × 10⁵ Pg), the gaseous N₂, unlike CO₂, cannot be directly used by most organisms. Atmospheric nitrogen enters the protoplasm of plants only when it is converted into usable forms, viz., nitrate (NO₃) and ammonia (NH₃). The fixation of gaseous nitrogen in forms usable for plants involves breaking of the triple bond of molecular N₂ (N ≡ N), and is particularly energy consuming.

Biological N fixation

In a major pathway, called biofixation, N_2 is fixed by some microscopic prokaryotic algae and bacteria. In this process, N_2 and H_2O combine to form NH_3 . This strongly endothermic reaction is catalyzed by the nitrogenase enzyme system, and may require from 50 to 200 g glucose per g of nitrogen fixed. The N_2 -fixer organisms may be free living (non-symbiotic), symbiotic or associative. Free living fixers are less efficient than symbiotic fixers, and may fix from 0.1-2 kg N/ha/yr in subarctic and arctic regions to 50-70 kg/ha/yr in warm and damp environments (blue-green algae in rice fields). The symbiotic bacteria (e.g., *Rhizobium*, the nodule bacteria of legumes) are more efficient and may fix up to 200 or more kg N/ha/yr, using about 4 g of C in the form of carbohydrates from the host plant per g of nitrogen fixed in the form of amino acids and amides, at the time of greatest activity. The nodule bacteria fix more N_2 than needed for the legume plant and the bacteria. The surplus is then secreted and fertilizes the soil. Free living bacteria colonising root network and mycorrhizae and maintaining associative symbiosis with host plants may fix considerable amounts of N (2 – 25 kg N ha⁻¹yr⁻¹ *Azotobacter paspali* in associative symbiosis with *Paspalum notatum* can fix 90 kg N₂/ha/yr. Some economically important microbes involved in biological N fixation are given in Table 16.2.

Table 16.2. Some economically important microbes involved in biological N fixation.

Organisms	General Properties	Use in Agriculture
<i>Azotobacter</i>	Aerobic, Nitrogen fixers	Hormonal effect on root and plant growth
<i>Azospirillum</i>	Aerobic, N fixer in association with roots of grasses	Increases crop yield, Inoculation benefits crops
<i>Rhizobium</i>	Fixes N in legume- <i>Rhizobium</i> symbiosis	Legume crops benefited
Actinomycete <i>Frankia</i>	Fixes N in symbiosis with woody trees - Alder, <i>Myrica</i> , <i>Casuarina</i>	Potentially important in reforestation wood production
Blue-green alga <i>Anabaena</i>	Fixes N in aquatic and terrestrial ecosystems	Enhances crop yield of rice

Nitrogen fixing organisms

(1) **Free-living bacteria** – This group includes both aerobic (e.g., *Azotobacter*) and anaerobic bacteria (e.g., *Clostridium*), purple bacterium (*Rhodospirillum*) and some other photosynthetic bacteria; and soil bacteria like *Pseudomonas*. The rates of nitrogen fixation in free-living bacteria vary from 1 to 2 kg N ha⁻¹yr⁻¹.

(2) **Symbiotic nodule bacteria on legume plants** - for example, *Rhizobium*, which is largely tropical in origin. The leguminosae is a large family of 750 genera and 20,000 species. Legumes have ecological significance as they provide nitrogen inflow to the natural habitats such as tropical rain forests. There is specificity between *Rhizobium* species and the host plants, i.e., a given *Rhizobium* will inoculate some legumes but not others. The rate of fixation of nitrogen depends on the soil and climatic conditions. The nitrogen fixed by *Rhizobium* in the root nodules can be used directly by the host plant. N passes from root nodules and plant roots into the soil or incorporated into the bodies of decomposer organisms. Some strains of *Rhizobium* are capable of forming nodules on stems as in the case of *Sesbania rostrata* and *Aeschynomene afraspera*. A tropical tree, *Parasponia* (Ulmaceae) is also nodulated by *Rhizobium*.

(3) **Symbiotic nodule actinomycetes on non-legume plants** - The actinomycetes-induced nodules were first discovered in alders (*Alnus*), subsequently these have been recorded in several dicot woody species, including *Myrica*, *Casuarina*, *Hippophae*, *Shepherdia*, *Ceanothus*, *Comptonia*, *Elaeagnus*, *Coriaria*, and gymnosperms, *Araucaria* and *Ginkgo*. About 160 species of various families including Betulaceae, Elaeagnaceae, Coriariaceae, Zygophyllaceae and Rubiaceae are known to form nodules. Unlike the legume nodule bacteria (*Rhizobium*), this is temperate in origin. Though said to be less evolved than the *Rhizobium*-bearing nodule, the actinomycetes (mostly genus *Frankia*)-bearing nodule is equally efficient in nitrogen fixation.

(4) **Blue-green algae (also known as Cyanobacteria)** - They include the members of Nostocales, viz., *Anabaena*, *Nostoc* and others. They can be free-living fixers, as well as symbiotic fixers, or a partner of fungus in the lichens, or on fronds of water fern, *Azolla*. Because of the nitrogen fixation, the *Azolla-Anabaena* combination has enabled the farmers of south-east Asia to grow paddy crops without chemical fertilizers for centuries; the nitrogen fixation rates could be as high as 100-200 kg N ha⁻¹yr⁻¹.

(5) **Bacteria living on leaves of several plants** - They can also fix nitrogen, e.g., *Gunnera* species, found commonly in parts of South and Central America, is an unusual angiosperm that contains nitrogen fixing Cyanobacteria (*Nostoc punctiformae*) in the pockets at the base of petioles of the leaves.

(6) **Unicellular, filamentous Cyanobacteria in aquatic environments**. Some filamentous Cyanobacteria have special oxygen excluding cells, called heterocysts, which function as sites of nitrogen fixation. Planktonic Cyanobacteria (*Synechococcus*, *Trichodesmium*, and *Oscillatoria*) are common in lakes as well as in open oceans and in the Baltic Sea. In fresh water systems, Cyanobacteria consist of heterocystous species (e.g., *Anabaena*), whereas in brackish water of Baltic Sea, blooms of *Nodularia* and *Aphanizomenon* are common. Heterocystous species are also found as symbionts in some marine diatoms. Interestingly, the major nitrogen fixing Cyanobacteria in oceans is an aerobic, non-heterocystous species, *Trichodesmium*; the annual rates of nitrogen fixation by *Trichodesmium* in oceans could be equivalent to those of the legume crops. Tropical oceans are low in nitrate-nitrogen and NH₄-nitrogen, therefore activity of Cyanobacteria is important. They are also important nitrogen fixers in polar and sub-polar regions. Thus, Cyanobacteria make a major contribution to the global nitrogen cycle.

N addition from the atmosphere

Though in lesser quantity, N₂ is also fixed in the atmosphere with the aid of lightning and sunlight energy, referred to as electrochemical and photochemical fixation, respectively. In these processes, nitrogen combines with oxygen to form NO₂. N compounds in the atmosphere are returned to the soil in rainfall as NH₃, NO₃, NO₂, nitrous oxide and organic N. Soil has a pronounced capacity for adsorbing NH₃ gas from the atmosphere. The NO₃ is taken up by plant roots when it is washed to the ground with rain and snow. N₂ fixation through these processes is in the order of 35 mg/m²/yr.

From the standpoint of commercial agriculture, the industrial fixation of N is the most important source of N as a plant nutrient. The production of N by industrial fixation is based on the Haber-Bosch process, in which H₂ and N₂ gases react to form NH₃:



This process requires high amounts of fossil fuel energy. In this, water initially combines with N₂ to form NH₃, which is later converted into ammonium salts, used as fertilizers.

Fate of fixed N

The biological N_2 fixation on global basis is about 44-200 Tg ($1 \text{ Tg} = 1 \times 10^{12} \text{ g}$) on land and 1-130 Tg in ocean, atmospheric fixation 0.5-30 Tg and industrial fixation 60 Tg/yr. A group of bacteria called nitrite- and nitrate-bacteria convert NH_3 into NO_3 , in a process referred to as Nitrification. The NO_3 -N and NH_3 -N are absorbed by the roots of higher plants and are subsequently bound in the protoplasm of organisms as protein and other compounds. From plants, the nitrogen in the bound form moves to animals, and both from plants and animals to detritus.

Several bacteria and fungi participate in the break-down of protoplasm. As decomposition proceeds, amino acids and organic residues are formed, and from these NH_3 is released in the process of ammonification performed by a group of bacteria called ammonifying bacteria. This NH_3 is either directly taken up by higher plants, or is converted into NO_3 by nitrification and then taken up by plants or is denitrified.

One can guess easily that if the cycle of nitrogen were to be complete then the NO_3 must return to the atmosphere. The denitrifying bacteria living in soil and water, break-down NO_3 , and use the energy released in the process, to support themselves. The process is called denitrification. Denitrifying bacteria are anaerobic because they grow in environments where there is little or no oxygen, such as sediments and deep layers in the soil near the water table, flooded soils, etc. The gaseous N_2 released as a by-product then escapes and enters the atmosphere where from it came. Thus, the cycle is completed.

The various reactions of nitrogen cycle are summarized in Table 16.3. The global nitrogen reservoirs and fluxes are given in Box 16.2, and the estimated quantities for various components are shown in Fig. 16.5.

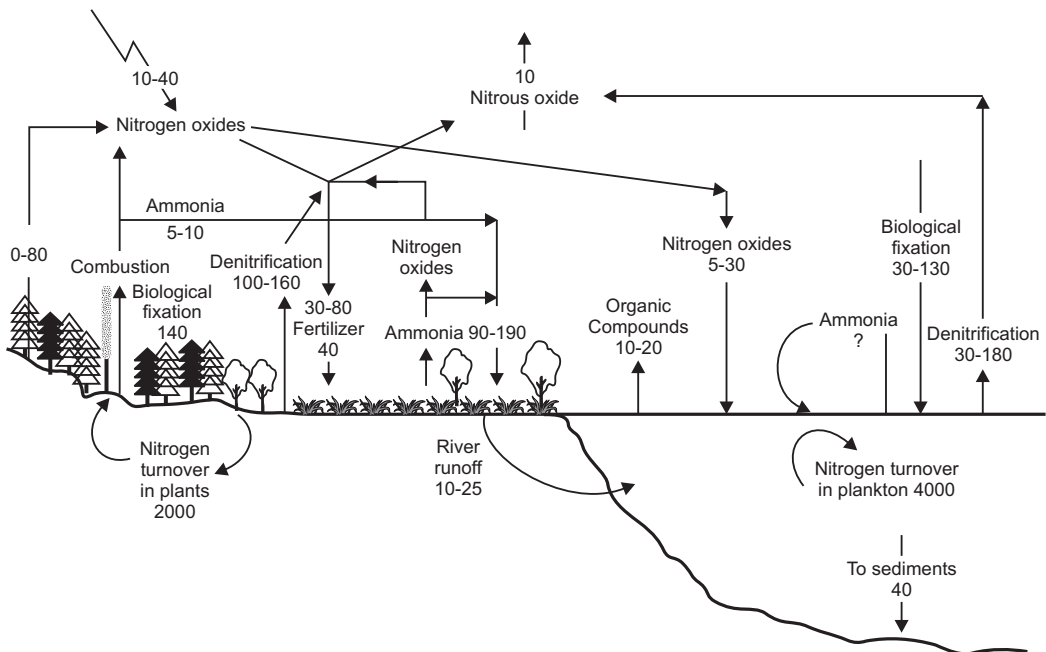


Fig. 16.5. The global nitrogen cycle (based on Rosswall, 1983). The values are in Tg N/year. $1 \text{ Tg} = 10^{12} \text{ g}$.

Box 16.2					
Global nitrogen reservoirs and annual fluxes. Values for reservoirs are in petagrams (1 Pg = 1×10^{15} g) and those of fluxes are in teragrams nitrogen per year (1 Tg = 1×10^{12}) (based on Rosswall, 1983).					
Reservoirs					
Atmosphere		Ocean		Land	
N ₂	39×10^5	Plants	0.30	Plant	11-14
N ₂ O	1.4	Animals	0.17	Animals	0.2
NH ₃ + NH ₄	174×10^{-5}	Microbes	0.02	Microbes	0.5
NO + NO _x	6×10^{-4}	Dissolved detritus	530.0	Detritus	1.9-3.3
NO ₃	1×10^{-4}			Soil organic matter	300
Org N	1×10^{-3}	Particulate detritus	3-240	Soil inorganic	160
		N ₂ (dissolved)	22000		
		N ₂ O	0.2		
		NO ₃	570		
		Nitrite	0.5		
		NH ₄	7		
Fluxes					
Biological N fixation		44-200 (land)			
		1-130 (ocean)			
Atmospheric fixation (lightning)		0.5-30			
Industrial fixation		60			
Industrial combustion and fuel burning (NO _x , N ₂ O)		10-20			
Fire		10-200			
Biogenic NO _x production		0-90			
Denitrification		43-390 (land)			
		0-330 (ocean)			
Denitrification (N ₂ O)		16-69 (land)			
		5-80 (ocean)			
Nitrification N ₂ O production		4-10 (ocean)			
Ammonia volatilization		36-250			
Dry and wet NH ₃ /NH ₄ deposition		110-240			
Dry and wet NO _x deposition		40-116			
Dry and wet deposition of organic N		10-100			
River runoff		13-40			

Table 16.3. Some biochemical processes and organisms involved in the nitrogen cycle.

Process	Examples of organisms
Nitrogen fixation	
(1) $\text{N}_2 + 8 \text{H}^+ + 6\text{e}^- \rightarrow 2\text{NH}_3 + \text{H}_2$	Some blue green algae, <i>Azotobacter</i>
Nitrification	
(2) $\text{NH}_4 + \frac{1}{2} 2 \text{O}_2 \rightarrow \text{NO}_2^- + 2\text{H}^+ + \text{H}_2\text{O}$	<i>Nitrosomonas</i>
(3) $\text{NO}_2^- + \frac{1}{2} \text{O}_2 \rightarrow \text{NO}_3^-$	<i>Nitrobacter</i>
Denitrification	
(4) $4\text{NO}_3 + 2\text{H}_2\text{O} \rightarrow 2\text{N}_2 + 5\text{O}_2 + 4 \text{OH}$	<i>Pseudomonas denitrificans</i>
(5) $2 \text{NO}_3 + \text{H}_2\text{O} \rightarrow \text{N}_2\text{O} + 2\text{CO}_2 + 2 \text{OH}$	Anaerobic sulphur bacteria
Ammonification	
(6) $\text{C}_2\text{H}_5\text{NO}_2 + \frac{1}{2} \text{O}_2 \rightarrow 2\text{CO}_2 + \text{H}_2\text{O} + \text{NH}_3$ (glycine and amino acid)	Many bacteria, most plants and animals

Human Impact on Nitrogen Cycle

The current nitrogen biogeochemical cycle is strongly affected by the agricultural system (Sutton et al., 2013a). The estimates around the years 2000-2010, globally humans introduced 120 Tg N per year of new N as synthetic fertiliser to sustain crop and grass production and as feedstock for many industrial processes (Galloway et al., 2008), in addition to 50-70 Tg N which is fixed biologically by the agricultural system. Only 20-30% of the N originally introduced in agricultural soils ends up in food for human consumption as cereals, vegetables and fruits, while the rest is used to sustain the livestock production (Billen et al., 2013). Losses are represented mainly by ammonia (NH_3) emissions from agricultural and livestock production systems (37 Tg N per year, Sutton et al., 2013a), soil denitrification (25 Tg N, Billen et al., 2013), and N leaching and runoff (95 Tg N, Billen et al., 2013). Fresh waters receive around 39-95 Tg N per year from agricultural soils (Bouwman et al., 2011; Billen et al., 2013). Part of this remains in superficial aquifers, part is lost to the atmosphere by denitrification, contributing to the emission of the strong greenhouse gas nitrous oxide (N_2O), and part is discharged to coastal waters (40-66 Tg N per year, Voss et al., 2011).

Combustion is responsible of the emission of 30-40 Tg N per year as NO_x (Fowler et al., 2013), which is about five times the NO_x naturally produced in the atmosphere by lightening (Fowler et al., 2013). Burning fossil fuels produces a significant additional Nr resource (~20% of human Nr production) that could be captured and used, but which is currently wasted as emissions of nitrogen oxide (NO_x) to air, contributing to particulate matter and ground-level (tropospheric) ozone that adversely affect human health, ecosystems and food production systems. Ammonia emissions from agricultural systems are 37 Tg N per year with a further 15 Tg N from biomass burning, industrial and various waste sources, as compared to the 13 Tg N per year from natural systems and oceans, with a total annual emission of 65 Tg N (Sutton et al., 2013b). Finally, N wet and dry deposition is also influenced by N emissions and is estimated to be around 70 Tg N on terrestrial ecosystems and 30 Tg N on oceans annually (Fowler et al., 2013). Global N fluxes are summarized in Box 16.3.

Box 16.3. Global nitrogen fluxes around year 2000-2010 (Source : Sutton et al., 2013a)

Global Nitrogen Fluxes	Tg N/yr	References
Fertiliser consumption	120	Fowler et al. (2013)
N ₂ crop fixation	50-70	
Crops & grass production	122	Billen et al. (2013)
Crops & grass for livestock production	100	Billen et al. (2013)
N back to agricultural soils	57	Billen et al. (2013)
NH ₃ emissions–agricultural system (total)	37	Sutton et al. (2013)
Riverine input to oceans	40-66	Voss et al. (2013)
Surplus in agricultural soils	120	Billen et al. (2013)
Input from agricultural soils to aquifers and rivers	95	Billen et al. (2013)
Soil + aquatic systems denitrification	25+52	Billen et al. (2013)
NO & N ₂ O emissions from soils	23	Fowler et al. (2013)
N ₂ natural fixation in terrestrial ecosystems	58	Fowler et al. (2013)
NH ₃ emissions–	20.8	Sutton et al. (2013)
Combustion	70	Fowler et al. (2013)
Wet and dry deposition on oceans	30	Fowler et al. (2013)
Denitrification in oceans	100-250	Voss et al. (2013)
N ₂ Fixation by oceans	140	Voss et al. (2013)
Burial in oceans	22	Voss et al. (2013)
Flux from coastal ocean to open ocean	390	Voss et al. (2013)
Flux from open ocean to coastal ocean	450-600	Voss et al. (2013)
Wet and dry deposition of NH _x and NO _x on agricultural and soils	69	Duce et al. (2008)

Many temperate and boreal ecosystems are N limited. In these regions, atmospheric N deposition has caused the acidification of soils and waters, loss of soil cations, a switch from N to P limitation, a decline in the diversity of plant communities adapted to low N availability, and increases in carbon uptake and storage (Vitousek et al., 1997c). Atmospheric N deposition in terrestrial landscapes can be transformed through soil microbial processes to biologically available nitrogen, which can result in surface-water acidification, and loss of biodiversity. NO₃⁻ and NH₄⁺ inputs from terrestrial systems and the atmosphere can acidify freshwater systems because of reduced buffering capacity due to soil acidification (Galloway et al., 2004)

Nitrogen deposition over tropical land area increased during the past decades and is projected to increase even further (Galloway et al., 2004). Natural N availability is much greater in many tropical forests than in most temperate forests due to high rates of N fixation by heterotrophic soil microbes and rhizobia associated with legumes, which are abundant in many tropical forests. Hietz et al. (2011) have shown that leaf N and the ratio of N isotopes ($\delta^{15}\text{N}$) increased substantially in a moist forest in Panama between ~1968 and 2007, and the tree-ring $\delta^{15}\text{N}$ in a dry forest in Thailand over the past century. A decade of fertilization of a Panamanian forest with N also caused increases in leaf N and $\delta^{15}\text{N}$, indicating regional increases in N availability due to anthropogenic N deposition (Hietz et al., 2011).

Aquatic ecosystems exhibit varied responses to nitrogen enrichment. NO_3^- loading from N saturated, terrestrial ecosystems can lead to acidification of downstream freshwater systems and eutrophication of downstream marine systems. Freshwater acidification can cause aluminum toxicity and mortality of pH-sensitive fish species. Because marine systems are generally nitrogen-limited, excessive N inputs can result in water quality degradation due to toxic algal blooms, oxygen deficiency, habitat loss, decreases in biodiversity, and fishery losses (Rabalais, 2002).

16.3 PHOSPHORUS CYCLE

Being the constituent of nucleic acids, cell membranes, phospholipids, phosphorylated compounds, bone and teeth, phosphorus is a nutrient of major importance to plants and animals. It often becomes a limiting or a regulating element for plant productivity. This is because the ratio of phosphorus to other elements in organisms is markedly higher than the ratio of phosphorus in available and primary sources.

There are three main groups of organic phosphorus compounds, which are found in plant and are also present in soils: (1) Inositol phosphates; (2) Nucleic acids; (3) Phospholipids. Inositol phosphates are phosphate esters of sugar-like compounds, which are most abundant and form 10 to 50% of the total organic P in soil. These are thought to be of microbial origin and tend to be stable in acidic and alkaline conditions. These can also interact with higher molecular weight humic compounds. Nucleic acids constitute 1 to 5 % of total organic phosphorus in soil. Phospholipids are fat like compounds of microbial origin that account for 0.2 to 2.5% of organic P. Some micro organisms are capable of solubilizing immobilized phosphorous. Phosphate is assimilated by these microbes, and in the process, organic acids are produced. Some chemolithotrophic bacteria such as *Nitrosomonas* and *Thiobacillus* mobilize organic phosphates by producing nitric and sulphuric acids.

Fluxes of phosphorus involve plant-available forms such as dissolved phosphate ion (PO_4^-), dissolved organic P, and particulate inorganic P (rather bicarbonate-extractable P). In fact most of P in the biosphere exists as phosphate. Due to the abundance of calcium, aluminium and iron, all of which form phosphate salts with very low solubility in water, most of the earth's P is immobilized in rock, soil or sediment. Natural rates of mobilization of P from the insoluble salts are very low. In terrestrial ecosystems, biota (plants and animals), available P of soil, and humus and microbial P of soil, and in oceans dissolved P, biota, and detritus form tightly coupled compartments for P cycling (Fig. 16.6).

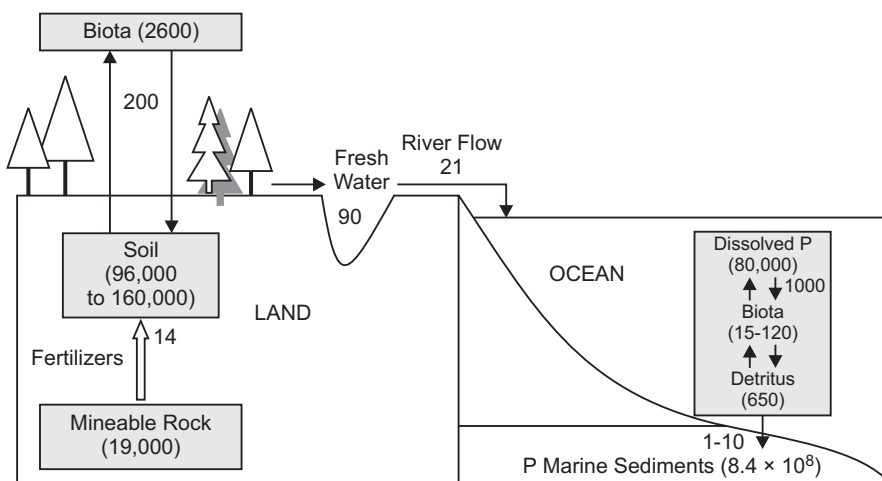


Fig. 16.6. The global phosphorus cycle (based on Richey, 1983). The values are in Tg P for reservoirs and Tg P/year for fluxes.

The rocks and their deposits formed in the geological past constitute the major reservoir of phosphorus. The land biota contain only a minute amount of P (about 2600 Tg P compared to soil (more than 96,000 Tg) or mineable rocks (19,000 Tg) (see Box 16.4). Similarly although total dissolved P in the oceans is 80, 000 Tg, marine biota contain only 50-120 Tg P. Further, although land plants have 20-50 times as much P as marine biomass, they cycle only 200 Tg P/yr compared to 600-1000 Tg/yr cycled by marine biota. As a result of erosion of soils and rocks, or leaching of dissolved forms, phosphates are released into rivers and lakes, etc. Some of them are precipitated in lake sediments while the majority escapes into the sea, where part of it is deposited in the shallow sediments and part of it is lost to the deep sediments. As organisms die and sink to lower levels, surface waters become depleted of phosphate supplies, but upwelling of deep waters returns some phosphorus to the surface. Through the uplifting of sediments (on the human scale this is virtually zero) and the harvest of fish from sea the phosphorus returns from sea to land. According to one estimate, through the fish we eat, 60,000 tons of elementary phosphorus returns annually. Sea birds by depositing their faecal material (guano deposits) on land also contribute significantly to the return of phosphorus to the cycle. But this is insufficient to compensate for the loss from the land to the sea.

Box 16.4. Global phosphorus reservoirs (Tg P) and fluxes (Tg P/yr) (based on Richey, 1983). Reservoirs						
Atmosphere			Land		Ocean	
Particulates over land	0.025	Biota	2.6×10^3		Biota	50-120
		Soil	$96 \times 10^3 - 160 \times 10^3$	Dissolved (inorganic)	80×10^3	
Particulates over oceans	0.003	Mineable rock	19×10^3		Detritus (particulate)	650
		Fresh water	90	Sediment	840×10^6	
Fluxes						
Atmosphere (land) overland	→	Atmosphere (ocean) over ocean	1.0	Marine (dissolved)	→	Biota 600-1000
Atmosphere over ocean	←	Atmosphere over land	0.3	Marine detritus	→	Sediment 2-13
Atmosphere	→	Land	3.2	Terrestrial biota	→	Soil 200
Atmosphere	→	Ocean	1.4	Mineable rock	→	Soil 14
Land	→	Atmosphere	4.3	Soil	→	Freshwater 4-7
Ocean	→	Atmosphere	0.3	Fresh water (dissolved)	→	Ocean 1.5-4
				Fresh water (particulate)	→	Ocean 17

In brief, the phosphorus cycle can be summarised as: (1) Major reservoir of P is found in the lithosphere. (2) From rock, phosphorus is released by the process of weathering. (3) From weathering, the released phosphorus is transported to soil by wind or water as inorganic phosphate. (4) Inorganic phosphate is absorbed and assimilated by plants. In most soils, the amount of available phosphorus

is about 0.01% of the total phosphorus in soil. (5) From plant, phosphorus moves through the food chain in the organic form. (6) The debris of plant, animals and microorganisms return organic phosphates. (7) The dead organic matter is acted upon by the phosphatising bacteria to release inorganic phosphorus from bound organic form. (8) Loss of Phosphorus occurs in run-off water in deep-ocean sediments. (9) Phosphorus is returned from shallow marine deposits in fish harvest and guano deposits in fish eating birds and geological uplift. (10) Turnover of organic phosphorus occurs due to phosphatase activity associated with root activity and microbial populations. (11) The precipitation of phosphorus in marine habitats limits primary productivity. (12) Phosphorus limited lakes are oligotrophic.

The global phosphorus fluxes are described for the year around 2000-2010 (see Sutton et al., 2013 and references therein). Based on this report, the world's annual rock phosphate mining in 2011 amounted to 191 Mt (Scholz and Wellmer, 2012), which corresponds to about 25 Tg P per year. Most of the P mined is added to agricultural soils as synthetic fertilisers (18 Tg P for 2011, Heffer and Prud'Homme, 2012; and 14 Tg P in 2000, Bouwman et al., 2011), whereas a small quantity is used by industry especially for detergents and food additives.. Accumulation of P in agricultural soils is estimated to be around 12 Tg P per year .About 4-7 Tg P per year are embedded in food for human consumption, part of which is lost with food waste. Human excretion accounts for 3-5 Tg P annually, and the quantity of P reaching the rivers by sewage systems is 1-3 Tg P, including detergents. Weathering releases 15-20 Tg P per year, and annual soil erosion is estimated to be 15-26 Tg P. A share of the P made available by weathering and erosion reaches fresh waters by leaching and runoff amounts to 2-7 Tg P per year. Once in surface continental waters, part of P is trapped in sediments in lakes and reservoirs and about 9 Tg P per year are delivered to coastal waters where it can contribute to local eutrophication. The available studies suggest that annual P burial in sediments could be between 3 and 38 Tg P (Sutton et al., 2013a).

Human impact on P cycle

Humans have markedly increased the rate of phosphate mobilization by extracting phosphate fertilizers from mineable rocks. However, a major part of phosphorus added as fertilizer is immobilized in the soil because of abundant supplies of calcium, aluminium and iron which bind with it in the presence of oxygen. A portion of P is harvested in crops and is later released to water bodies through sewage and waste. In due course of time, P moves to sediment. Without massive use of phosphate fertilizers in the modern agriculture, yields can hardly be increased or sustained. So the use of fertilizers has continued to increase. The known phosphate reserves are expected to last for only 60-160 years at projected levels of use in future.

16.4 SULPHUR CYCLE

Though the sulphur cycle has a gaseous phase also, it is mainly recognized as a sedimentary cycle. In contrast to carbon and nitrogen cycles, the sulphur cycle has the large reservoir in soil and sediments and the smaller one in the atmosphere, and it does not cycle smoothly. Reservoirs of sulphur are fossil fuels and elemental sulphur deposits in rocks and minerals. It is among the ten most abundant elements in the earth crust. Inert sulphur reservoirs are metal sulphides in rocks, elemental sulphur deposits and fossil fuels. The concentration of sulphur is low in the atmosphere, where it exists in gaseous form as H_2S and SO_2 and in particulate form as sulphates and sulphides. The residence time of sulphur in the atmosphere is very small. Organic sulphur is found in soil and the body of living organisms and as inorganic sulphate in the ocean. The combustion of fossil fuels releases SO_2 and other sulphur gases and solids into the atmosphere. Lithosphere contains about 24×10^9 Tg S, ocean water 1.3×10^9 Tg, soil (including land plants) 2.7×10^5 Tg and atmosphere only 4.8 Tg. The global reservoirs and fluxes of sulphur are shown in Box 16.5, and the global

sulphur cycle in Fig. 16.7. It is clear that the main source of mobile sulphur is the lithosphere and its ultimate sink is the ocean, with limited recycling within the continental atmosphere-soil system. The sulphur flux from the continental atmosphere to the oceanic atmosphere is 81 Tg S/yr against the opposite flux of 20 Tg S. This imbalance is on account of the anthropogenic sulphur emission to the continental atmosphere.

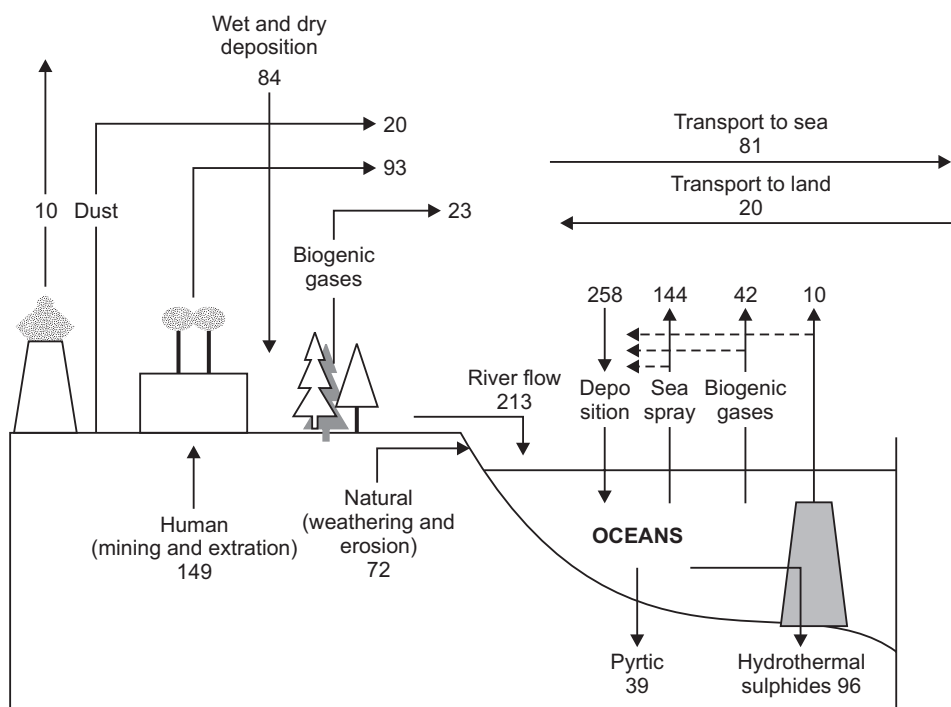


Fig. 16.7. Major components and flux pathway of sulphur. The values are in Tg S/yr (based on Brimblecombe et al., 1989).

Sulphur has unique role in plant and animal metabolism because it is essential for the synthesis of amino acids such as cysteine, methionine, cystine and co-enzymes-thiamine and biotin and certain vitamins. Sulphur groups of amino acids determine the structure of proteins. The roots of green plants absorb sulphur from the soil mainly in the sulphate (SO_4) form, like nitrate and phosphate. However, sulphur does not limit plant growth as much as does nitrogen and phosphorus, nor it is required as much as nitrogen and phosphorus by the ecosystem.

Sulphur is incorporated in plants as -SH (sulfhydryl) group in proteins that passes to the heterotrophs through the food chain and is released into the soil through the dead tissues. The biologically incorporated sulphur is mostly mineralized into inorganic SO_4 by ordinary decomposing process. However, some is directly reduced to sulphides, H_2S by bacteria (e.g., some species of *Escherichia* and *Proteus*) (Fig. 16.8). Organo-sulphur decomposition in soils and sediments yields mercaptans and H_2S , a process known as desulphuration. Within the soil, sulphur oxidation produces large amounts of sulphuric acid, which solubilizes and mobilizes phosphates and other minerals beneficial to the growth of microbes. Some micronutrients associated with insoluble iron sulphide can become available to plants. Sulphur oxidizing bacteria are active in soils and aquatic habitats; sulphates produced from their activity are assimilated by other organisms. Under aerobic conditions, decomposition by bacteria and fungi (e.g., species of *Aspergillus* and *Neurospora*) produces inorganic sulphur.

Table 16.4. Some chemical reactions of sulphur cycle.

1. Reduction into hydrogen sulphide in an aerobic condition	
$\text{SO}_4^{2-} + 2\text{H}^+ \rightarrow \text{H}_2\text{S} + 2\text{O}_2$	Organism <i>Escherichia, Proteus</i> bacteria
2. Oxidation to elemental sulphur without oxygen	
$6\text{CO}_2 + 12\text{H}_2\text{S} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + \text{S}$	Chemosynthetic autotrophic bacteria, e.g., <i>Thiobacillus</i>
Energy liberated in oxidation of H_2S is used for CO_2 assimilation	
3. Oxidation to sulphate	
$6\text{CO}_2 + 12\text{H}_2\text{O} + 3\text{H}_2\text{S} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 3\text{SO}_4^{2-} + 6\text{H}^+$	Photosynthetic purple bacteria

In marine environments, major decomposition product of organic sulphate is dimethyl sulphide (CH_3S , DMS). DMS is released during zooplankton grazing and also during the decay processes. 90% of total sulphur flux from marine environments to the atmosphere occurs in the form of DMS. DMS, H_2S and mercaptans are subject to photo-oxidative reactions that lead to the formation of sulphate, SO_2 and Methane sulphonie acid (MSA). Water vapours can condense around MSA and SO_4^{2-} . In open oceans clouds are formed in this way.

Micro organisms are able to use elemental sulphur, and convert sulphur from one form to another. Sulphur oxidizing bacteria are active in soils and aquatic habitats and the sulphates produced from their activity can be assimilated by other microbes within the soil. The role of microorganisms in oxidative and reductive sulphur transformation, as shown in Fig. 16.8, is discussed below:

Oxidative Sulphur Transformation

1. Anaerobic S Oxidation : Sulphates are produced during the anaerobic oxidation of sulphides and thiosulphates by purple and green photosynthetic bacteria. These photosynthetic bacteria use H_2S and S as reducing agents for CO_2 fixation. Green bacteria (*Chlorobium*) oxidise H_2S to sulphur and purpler sulphur bacteria (*Chromatium*) oxidise sulphur to sulphate. During this oxidation process, elemental sulphur is deposited either intracellularly or extracellularly (*Chlorobium*).

2. Oxidative Sulphur Transformation : In the presence of oxygen, reduced sulphur compounds are capable of supporting chemolithotrophic microbial activity. Elemental sulphur and sulphides are used as energy source by chemolithotrophs. *Beggiatoa* and *Thiothrix* are found in sulphide rich environments. These bacteria use hydrogen sulphide as a source of electrons in their metabolism. Hydrogen sulphide is oxidised to elemental S, which either accumulates in the cell or oxidised to sulphate.

Thiobacillus thiooxidans is an acidophilus bacterium, which grows on sulphur containing coal deposits. There are obligate chemolithotrophs that obtain their energy from oxidation of sulphur. There are obligate aerobes, which require molecular oxygen for the oxidation of inorganic sulphur. These aerobically oxidize elemental sulphur to sulphides and sulphates as well as oxidize H_2S and other reduced sulphur components. Due to this reaction, H_2SO_4 is leached from coal deposits into mine water. Acid formation by microbes seriously damages the environment.

3. Reductive Sulphur Transformations : Of particular interest is the reduction of SO_4 to elemental sulphur or to sulphides, such as H_2S by anaerobically respiring heterotrophic bacteria, *Desulfovibrio* and *Desulfomaculum*. This is of ecological importance because *Desulfovibrio* bacteria work in deep sediments and in deep anoxic waters. The H_2S gas that rises to shallow sediments or

to surface water is thus made available to organisms like, photosynthetic bacteria. *Desulfovibrio* is an obligate anaerobe common in aquatic environments that uses sulphate as a terminal electron acceptor. The reduction of sulphate results in the production of H_2S and SO_4 . H_2S is highly toxic to the cytochrome systems and plant roots and can kill the plants. Sulphate reduction is involved in some elemental sulphur deposits. Sulphate reduction contributes to the atmospheric cycling of sulphur. H_2S , dimethyl sulphide and other reduced volatile sulphur compounds, which reach the atmosphere, are subject to oxidation and photo-oxidation reactions, converting them to sulphate. Sulphur reducing bacteria use sulphur, sulphate and other oxidized compounds as electron acceptors in anaerobic respiration.

There is a considerable ecological interest in some steps occurring in the sediments of ecosystems which regulate the cycling of other nutrients. Sulphur precipitates in the presence of such cations as iron (Fe) and calcium (Ca) as highly insoluble ferrous sulphide (FeS) and ferric sulphide (Fe_2S_3), called pyrite, or not so highly insoluble calcium sulphate (CaSO_4). To the formation of ferrous sulphide under anaerobic conditions, is also related the unavailability of the biologically important nutrients, such as copper, cadmium, zinc and cobalt. In contrast, the binding of the ferrous and ferric sulphides converts phosphorus from unavailable to available forms.

Some organic sulphur enters the atmosphere as sulphur dioxide (SO_2) through incomplete burning of fossil fuels, particularly coal. Normally, SO_2 is a transitory phase in sulphur cycle and is present in flow concentrations well within the limits, but increased burning of fossil fuel has resulted in a serious atmospheric pollution in many localities. A large portion of the sulphur in the biosphere is found in fossil fuels. When these are burned, the sulphur dioxide, elemental sulphur and hydrogen sulphide in atmosphere are oxidized to sulphur trioxide (SO_3). Combining with water, this forms sulphuric acid (H_2SO_4) causing acid rain.

16.5 GLOBAL WATER CYCLE

Water constitutes 70 per cent weight of organisms; it is a medium of biological activity; it is a geological agent eroding soil from one place and depositing it in another; it carries nutrients in dissolved state and acts as an agent of their distribution; it is an agent of energy transfer and use; it is an important factor in controlling variations in temperature because it has the highest specific heat (1 cal/g/°C) and latent heat (80 cal/g heat is required to convert 1 g of water to ice, and 536 cal/g is required for converting 1 g of water at 100°C to water vapour with no change in temperature).

The water cycles rapidly but a majority of it (about 90 per cent) is locked up in the lithosphere. The 0.127 geogram of water (1 geogram = 1×10^{20} g) which is the source of all precipitation, constitutes an infinitesimal proportion of the global water ($266,069.88 \times 10^{20}$). The atmospheric vapour is equivalent to 2.5 cm of rain, i.e., world's 9 days supply and thus turns over every 9 days. The water in the rivers turns over every 12 days. It is estimated that by the year 2100 man will have used or polluted all fresh water reserves, and by the year 2230 he may have to depend only on precipitation. (For certain facts about water in India, see Box 16.6).

Many studies on the global water cycle deal with only specific aspects (Trenberth et al., 2007). Reliable data on the surface water budget are often available only over certain regions. Relatively few studies (e.g., Trenberth et al., 2007, 2011; Oki and Kanae, 2006) have attempted to provide a synthesized, quantitative view of the global water cycle. The quantitative knowledge of the various components and their variability of the global water cycle are still fairly limited because of a lack of reliable data for surface evapotranspiration, oceanic precipitation, terrestrial runoff, and several other fields. The major pathway of hydrological cycle involving an interchange between earth surface and atmosphere via precipitation and evaporation is in steady-state (Fig. 16.9). However, when considered separately, more water evaporates than returns through precipitation in oceans (precipitation-evaporation ratio, P/E ratio about 0.9), while in the case of land the reverse is true (P/E = 1.6). Obviously, some of the rainfall that the land receives is due to the evaporation of water

from oceans.

Box 16.6. Facts about Hydrological Cycle of India (Source: National Commission on Agriculture, 1976, Part V. Govt. of India).
(All values are in million ha m/yr)

1. Average precipitation	400
2. Annual evaporation	70
3. Surface runoff	115
4. Soil water + ground water recharge	215
5. Regenerates as surface flow	45
6. Brought by springs and rivers from other countries	20
7. Total surface flows (3 + 5 + 6)	180
8. Surface water stored in reservoirs, etc.	15
9. Evaporation from reservoirs, etc.	5
10. Water flow in rivers	165
11. River flow utilized through diversion works	15
12. River flow that goes to sea and neighbouring countries (10-11)	13
13. Of the 215 m ha m percolating into soil, about 50 m ha m (12.5 per cent of total precipitation) percolates as ground water and the rest is retained in soil as soil moisture. Ground water receives another 5 m ha m from flood flows and 12 m ha m from the irrigation system (thus 67 m ha m ground water is replenished annually). Of the total ground water 45 m ha m regenerates into rivers, 13 m ha m is extracted for various uses and 9 m ha m increases water table or is lost as evapotranspiration.	
14. Transpiration (55 crops + 55 forests) 110 m ha m.	

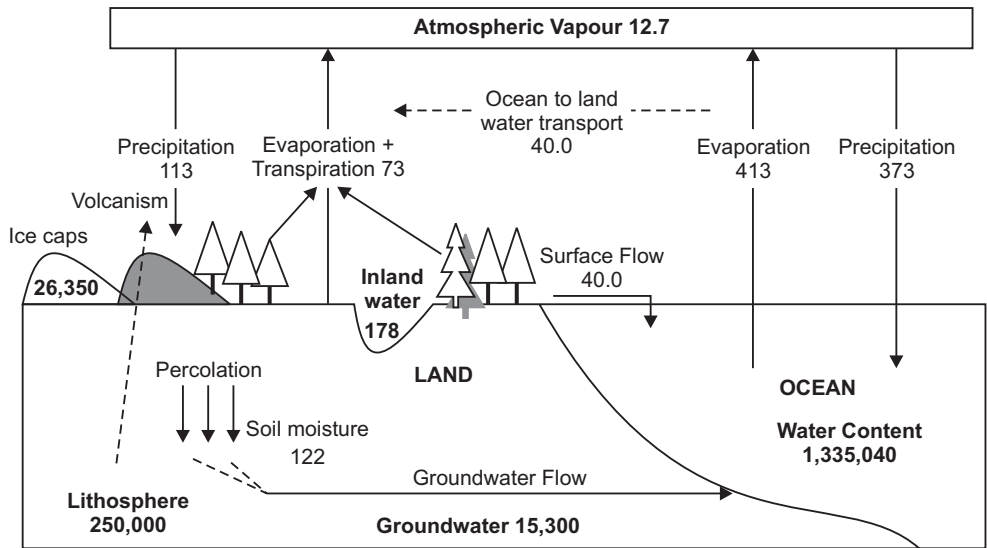


Fig. 16.9. The hydrological cycle. Estimates of the main water reservoirs, given in cubic km (10^3 km^3), and the flow of moisture through the system, given in cubic km/yr ($10^3 \text{ km}^3 \text{ yr}^{-1}$), equivalent to Eg (10^{18} g) yr^{-1} . The values of pools and flows adapted from Trenberth et al. (2007)

It is of interest to know how energy drives the hydrologic cycle. For the uphill movement of the water via evaporation enormous amounts of solar energy are needed. To evaporate 1 g of water 0.536 kcal energy is required, total annual evaporation is 378×10^{18} g water, so 2×10^{20} kcal (i.e., $0.536 \times 378 \times 10^{18}$) of energy is spent annually for evaporation. This is one-fifth of the total energy income at the surface of the earth.

Human Impact on Global Water Cycle

As the climate changes partly from human activities, the water cycle is also changing. Moreover demand for water continues to increase owing to growing population, enhanced agricultural and industrial development, and other human activities such as transformation of landscape and construction of dams and reservoirs. This affects the disposition of water when it hits the ground: its way to rivers or infiltration into the soil and percolation to depths to replenish the underground water reservoirs. The main impacts of a warmer climate on global water cycle (Trenberth and Asrar, 2012) include the following:

- 1. With warming, higher atmospheric temperatures increase the water holding capacity of the atmosphere by about 7% per degree Celsius.
- 2. Over the ocean an increase has occurred in total column water vapour of about 4% since the 1970s (Trenberth et al., 2007).
- 3. With more heat in the earth system, the evaporation is enhanced resulting in more precipitation.
- 4. Potential evapotranspiration has increased, and in dry areas this means drying and more intense and longer lasting droughts.
- 5. Greater warming over land versus the ocean is changing monsoons.
- 6. More intense rains and snow potentially results into more intense storms; More precipitation occurs as rain rather than snow.
- 7. There is a risk of more extremes, such as floods and droughts.

Interactions of Biogeochemical Cycles

The cycles of the essential elements are coupled because they intersect where biological processes occur. The C, N, and P seem to be incorporated into the biomass of primary producers in relatively constant ratios. The concept of *stoichiometric proportion* (also called *Redfield ratio*) was developed by Redfield (1958) who found mean elemental composition of marine plankton communities to be C : N : P = 106 : 16 : 1, roughly proportional to the atomic weights of the carbon, nitrogen and phosphorus atoms. The preferential loss of ammonium and phosphate following death and initial mineralization changes this initial ratio. Suess and Muller (1980) reported the C : N : P ratio for the particulate organic matter in the north-east pacific to increase from 106 : 16 : 1-106 : 7.9 : 0.25 within euphotic zone to 106 : 3.5 : 0.11 deeper in the water column. Most estuaries and open water marine ecosystems have very low N : P ratios and therefore are nitrogen limited. The C : N : P ratio of the organic matter actually being decomposed in shallow water sediments is 106 : 19 : 3.3 and 106 : 4.6 : 0.37 in offshore sediments (Martens et al., 1978).

Some C : N : S : P ratios are given below (compiled from Bolin et al., 1983; and Melillo and Gosz, 1983).

	C	N	S	P
Land plants	790	7.6	3.1	1
Forest vegetation (tissue)	1500	10	1	1
Land animals	78	11	1.1	1

Forest litter	500	10	2.9	1
Ocean animals	93	12	0.7	1
Soil organic matter	54	3	1.2	1
Soil	120	9.4	1	1

Organic matter entering a soil system is different from that entering a marine sediment because land plants are markedly enriched in carbon (preponderance of structural compounds). The ratio of total P to C and N in soils is quite different from ratios of these elements in marine sediments, because most P in soil is unavailable to plants. Also because of the abundance of organic C in detritus entering the soil, C cycles more rapidly than N and P.

Interactions of C, N, S and P are quite apparent during soil and ecosystem development. N-fixation peaks relatively early in the soil development, it is reduced as the availability of N in soil is increased or as P and S become limiting to N-fixers. In later stages, relative availability (and hence cycling) of N is more closely coupled to that of P and S.

Cultivation of soil for 50-70 years in the temperate zone has resulted in a loss of 50 per cent C and N, 30-40 per cent S and 10-30 per cent P (Stewart et al., 1983). These losses in tropics are faster. Reduced C inputs decrease the capacity of microbial biomass to immobilize increased levels of N and S, which are released due to faster decomposition in the cultivated soils. This excess amount is easily leached to streams or ground water.

The changes in the ratios of nutrients due to anthropogenic factors (e.g. concentrations of dissolved inorganic N and P in the world rivers have increased by a factor of 2 in the past 30 yrs) have lead to major biotic changes. The entry of large amounts of P relative to N and Si has caused a replacement of silicon containing diatoms by blue-green and brown algae (Wollast, 1983). Enhanced nutrient loading, through manipulation of biogeochemical cycles, has caused eutrophication of water bodies which has become a major problem world-wide. Since more of N or/and P is available more carbon is fixed. Melillo and Gosz (1983) have estimated that fertilization of forests (N-eutrophication) with N from fossil fuel sources may result in the storage of 300 Tg C in these systems annually.

Atmospheric interactions of C, N and S compounds have attracted considerable attention (Crutzen, 1983). The composition of the atmosphere is strongly influenced by the photochemistry of ozone in the troposphere (10-17 km of the atmosphere). Its photolysis by ultraviolet light produces highly reactive hydroxyl (OH) radical which is the only tropospheric compound capable of attacking such gases as CO, NO_x (NO + NO₂), H₂S and hydrocarbons. Above a critical NO concentration O₃ is created in the troposphere by the photochemical oxidation of hydrocarbons to CO and then to CO₂. Some compounds that are inert in troposphere, such as N₂O, CFCl₃, CF₂Cl₂ and CCl₄, are broken down photochemically in the stratosphere (10-50 km of the atmosphere) leading to formation of reactive radicals NO, NO₂, Cl and ClO that attack ozone through catalytic chains of reactions. As N₂O is the main source of NO_x in the stratosphere, increase in N₂O through anthropogenic effects on nitrogen cycle is disastrous.

It is clear that ecosystems across the biosphere are subject to rapid changes in elemental balance and climatic regimes. A major force structuring ecological responses to these perturbations lies in the stoichiometric flexibility of systems (Sistla and Schimel, 2012). Stoichiometric flexibility of systems expresses the ability to adjust their elemental balance whilst maintaining function (Sistla and Schimel, 2012). According to these authors, this flexibility can result from two mechanisms (i) by varying stoichiometry within individuals, since organisms can alter tissue chemistry to

adjust their overall stoichiometry, and (ii) by shifting dominance among organisms that differ in their average stoichiometry. Although stoichiometry may vary significantly within or between organisms and across trophic groups, the chemically coupled feedbacks between primary producers, heterotrophs and their abiotic environment suggest that biota in terrestrial ecosystems will tend towards relatively convergent C : N : P ratios when moving from local to global scales. Consider the example of soil microbial community. Changing soil conditions may drive the microbial community towards a more fungal-like (higher microbial biomass C : N ratio) or more bacterial-like (lower C : N ratio) composition. The shifting between bacterial and fungal dominance has implications for both ecosystem nutrient cycling and subsequent responses to perturbation. Although soil microbial biomass C, N and P concentrations vary globally over several orders of magnitude, microbial biomass C : N : P ratios are conserved at a global scale (C : N average = 8.6, N : P average = 6.9). Similarly, despite biome-level differences trending towards greater foliage N : P with decreasing latitude, at global level forest leaf C : N : P ratios are broadly constrained worldwide (C : N average = 43.6; N : P = 27.8).

Ecosystem Nutrient Cycling

Plants take up nutrient elements from the atmosphere, water, soil and sediments. The macronutrients including N, P, K, Ca, Mg and S are important biochemically as these are required for the synthesis of various types of structural and non-structural components of plants. N is a major constituent of proteins, nucleic acids and chlorophyll; P is required for the formation of ATP, nucleic acids and cell membranes; and S is a constituent of many plant proteins. Among others, K has specific role in stomatal function, Ca in cell walls and Mg in chlorophyll. These nutrients are also known for their ability to stimulate the rate of enzymatic reactions. Then there are several micronutrients, Fe, Cu, Zn, Mn and others which are involved in enzymes and coenzymes, particularly in the reactions of photosynthesis. There are some micronutrients which play a role in the metabolism of other nutrients. For example, molybdenum (Mo) is essential for N metabolism in plant tissues, and symbiotic nitrogen fixation. Cobalt, Sodium and Silicon are also beneficial elements. For example, cobalt is essential for microbes involved in N fixation, including those species which have symbiotic relationship with plants. These elements are acquired by plants from various sources, and transferred through ecosystems in patterns which are broadly similar to those of energy transfers.

Nutrient fluxes occur in biological structures (self organizing systems) and operate at different scales, ranging from microbial aggregates through ecosystems to landscapes and the entire biosphere (Lavelle and Spain, 2001; Lavelle et al., 2005). Regulation of nutrient cycling operates at different scales of time and space, allowing the flow of nutrients released by microbial activities to adjust to plant demand, thus limiting losses to other parts of the ecosystems or to different ecosystems (Lavelle et al., 2005). In natural ecosystems, this “synchrony” between release of nutrients and their use by microorganisms and plants is regulated by complex interactions among physical, chemical, and biological processes.

17.1 THE NUTRIENT CYCLE MODEL

The ecosystems occupy arbitrarily defined volumes of the biosphere at the earth-atmosphere interface (Bormann and Likens, 1967). The boundaries of an ecosystem are frequently defined by the boundaries of a biological community, e.g., forest ecosystem, grassland ecosystem, and sometimes by the boundaries of the physical environment, e.g., lake ecosystem. The ecosystems are open systems and they participate in the various larger biogeochemical cycles through a system of inputs and outputs. The pathways of loss and replacement of nutrients connect one ecosystem with another (Fig. 17.1).

The inputs into an ecosystem are derived from meteorologic (e.g., nutrients in gases, precipitation, dust, etc.), geologic (dissolved or particulate matter carried in through water and soil movements e.g., renewal of floodplain nutrients), and biologic (e.g., deposition of material

gathered by animals elsewhere) sources. Similarly, nutrients leave the ecosystem (outputs) through meteorologic (e.g., transport of gases and particulate matter), geologic (e.g., dissolved and particulate matter in moving water and soil), and biologic (removal by animals and humans) pathways. These inputs and outputs constitute the extrasystem nutrient transfers or extrasystem nutrient cycling which effectively connect different ecosystems.

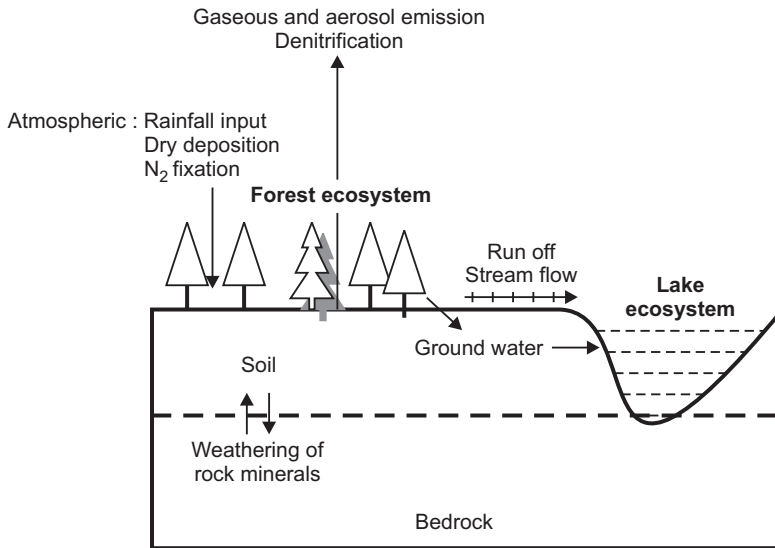


Fig. 17.1. The pathways of nutrient movement connecting ecosystems.

Within an ecosystem, e.g., a terrestrial ecosystem, we can visualize nutrients residing in five compartments, viz. plant biomass, animal biomass, detritus, soil and rock minerals, and available nutrients (Fig. 17.2). Nutrients move from the soil compartment to the plants through root uptake from available nutrients pool, and then move to the animal and detritus compartments. Some nutrients are recycled within the plants (internal recycling) by resorption during senescence. Detritus from the plant and animal compartments is decomposed in the soil, releasing nutrients for further uptake by plants. The nutrient cycling within an ecosystem is called intrasystem nutrient cycling. In some ecosystems, nutrient transfers through the animal compartment can be particularly important. For example, in productive grazed grasslands, a large proportion of plant production (including associated nutrients) is consumed by herbivores (some times more than 90% of the entire aboveground production) and nutrients are subsequently deposited in faeces and urine. Also, herbivory accounts for a three-fold greater proportion of nutrient transfer in pelagic than in terrestrial ecosystems (Cyr and Pace, 1993).

The biogeochemical cycling of nutrients thus involves an external (i.e., extrasystem) and an internal (i.e., intrasystem) cycle. The external cycle comprises those processes that add or remove nutrients from the ecosystems, for example, in the case of N these processes are N-fixation, dry and wet deposition, N fertilization, N leaching, runoff erosion, denitrification and NH₃ volatilization. The internal cycle comprises those processes that convert the nutrient from one chemical form into another or transfer it between ecosystem pools. Reverting to the example of N cycling, these processes include plant assimilation of N, return of N to soil in litterfall and root turnover, N mineralization (conversion of organic N to inorganic N, microbial immobilization (uptake of inorganic N by microorganism) and nitrification (production of nitrite and nitrate from ammonium or inorganic N) (Hart et al., 1994).

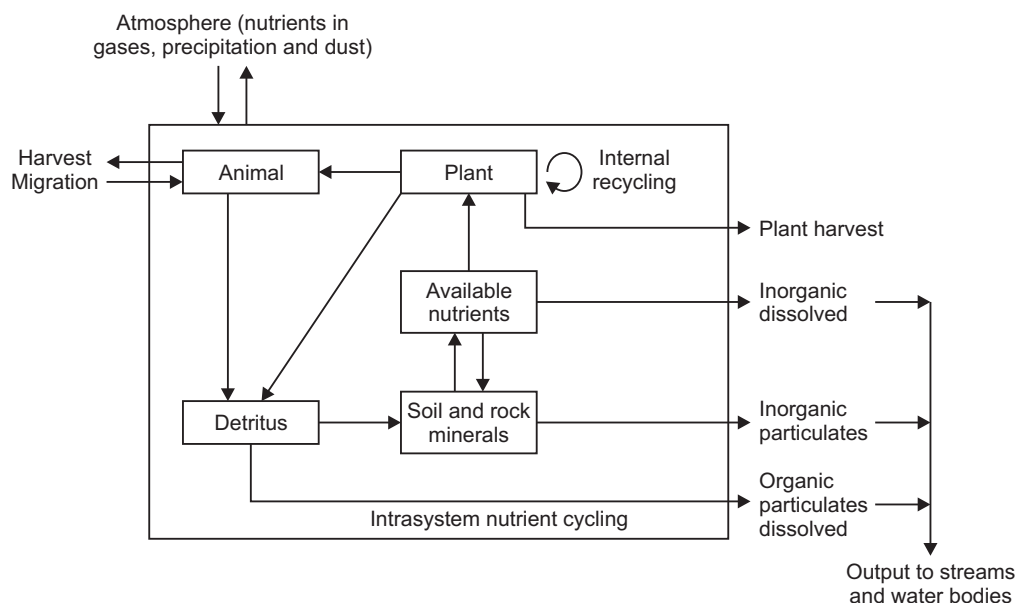


Fig. 17.2. A compartment model of intrasystem cycling of nutrients and extrasystem nutrient transfers.

The extrasystem transfers of nutrients account for only a small fraction of quantities that are cycled through the intrasystem cycling, and hence the latter tends to be a closed system particularly in mature ecosystems, where the difference between the input and the output tends to become negligible. Mature natural forests and woodlands have nearly balanced budget and can attain nutrient recycling rates up to 90%, as in the case of rainforests of Panama (Golley et al., 1975). The ecosystem N budget studies also show that N_2 -fixation is only a minor contributor to the total N supply for plant growth in ecosystems with successional mature vegetation (Aerts and Chapin, 2000). Increased anthropogenic inputs of N in recent years (e.g., 10-20 g/m²/yr in northeastern United States and 50-100 g/m²/yr in central Europe) are, however, tending to make the intrasystem N cycle, a relatively more open system. The intrasystem cycling is also relatively more open in grasslands, aquatic ecosystems and disturbed forest ecosystems, where ecosystem inputs and outputs constitute a good proportion of nutrients being cycled.

17.2 ECOSYSTEM INPUTS OF NUTRIENTS

Atmosphere

Nearly all of N, and a majority of S and Cl in terrestrial ecosystems are primarily derived from the atmosphere.

Precipitation contains dissolved and particulate nutrients, including substantial amounts of N. Leaves and branches of trees act as filters for atmospheric particles and gases which contain nutrients. Trace gases, particularly oxidized forms of N and S, are also a minor source of plant nutrients for land plants. These sources include direct additions of oxidized N (due to lightning) and S (from volcanic eruptions). In recent years, industries have added large amounts of NO₂ and SO₂, and other gases to the atmosphere, which in turn lead to increased deposition of N and S in the ecosystems.

Both oxidized and reduced forms of these gases are readily dissolved in rainwater and are transported to the ground as dissolved ion which is available for plant uptake. Rainfall also contains nutrients derived from the oceans. The aerosols originating from oceans contain important seawater constituents, especially Na, Mg, Cl, and SO_4 . Smaller aerosols (<1 mm) may be transported over great distances, from oceans to forest lands.

The relative efficiency of “scavenging” of nutrients by rainwater is calculated as washout ratio.

$$\text{Washout} = \frac{\text{Ion concentration in rain mg l}^{-1}}{\text{Ion concentration in air mg m}^{-3}}$$

In Central England, as an example, washout ratios during 1971 were 338 for Na ($2.3 \text{ mg l}^{-1} / 0.0068 \text{ mg m}^{-3}$ in air), 250 for Ca and 80 for Mn (Waring and Schlesinger, 1985). The deposition of nutrients through rainfall is called wet deposition or wetfall.

Dust in windstorms in drier regions also contains atmospheric aerosols rich in Ca, K and SO_4 . Volcanic eruptions and fires also inject gases and particulate matters, which can be transported to great distances. Dry deposition or dry fall is the result of gravitational sedimentation of particles during rainless periods, and the rate of deposition depends on particle size.

The deposition velocity can be calculated as given below:

$$\text{Deposition velocity} = \frac{\text{Rate of dry fall mg cm}^{-2} \text{ sec}^{-1}}{\text{Concentration in air mg cm}^{-3}}$$

Though wet fall is the predominant process, dry fall can also be important. For example, even in humid forests of the southeastern United States, 19-64% of the total annual atmospheric depositions of ions such as Ca, Na, K and Mg and up to 89% of P deposition are reported to be due to dry fall (Swank and Henderson, 1976). Dry fall contains more ions derived from continental sources than those derived from the ocean.

Apart from wet and dry fall, the vegetation surface of forests and other natural vegetation also captures nutrients of fog droplets (interception) and aerosols (impaction) from horizontal air streams. The interception may be sizeable in coastal and montane forest ecosystems. Plant leaves and soil can also directly absorb reactive gases, such as SO_2 and NH_3 . These two processes (impaction and gaseous absorption) are reported to account for one-third of the atmospheric deposition of S in the northern hardwood forest at Hubbard Brook, New Hampshire (Eaton et al., 1978).

Weathering of Rock Minerals

Rock weathering, a term used for breaking down of the parent rock, is a major source of several nutrients. Rock is considered external to the biotic system (beyond the rooting zone), therefore some workers consider nutrient supply from rock weathering as a part of extra-system cycle. Rock weathering involves (i) mechanical weathering, that is fragmentation of material with no chemical change, and (ii) chemical weathering, which occurs when the parent material and its fragments react with water and other chemicals dissolved in water and mineral constituents are released as dissolved ions. Chemical weathering also includes formation of new, secondary minerals, which are not readily available to plants. Unavailability of phosphate due to its becoming complexed with iron is an example of this phenomenon.

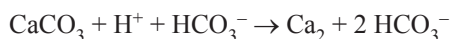
In forest ecosystems, weathering generally accounts for 80-100% of the input of Ca, Mg, K, P and Fe, which are rock-forming elements. In weathering reactions, soluble ions are released, so they are available for uptake and cycling within the forest. Chemical weathering reactions are far more rapid in warm and moist conditions of tropical forests than in temperate or boreal forests, and among various ecosystems, they are more rapid in forests compared to dry grasslands and deserts.

Rock type also affects the rate of chemical weathering and nutrient release. Metamorphic rocks, such as gneiss, schist, quartzite, and many igneous rocks, such as granite, were formed deep in the earth, and are composed of primary silicate minerals. Of them, quartz is the simplest (consisting of only silicon and oxygen in a tetrahedral crystal), but very resistant to weathering. Sedimentary rocks (underlying 75% of the land area), such as shales, limestones and sandstones, are formed close to the ground, generally as a result of sediment accumulation under water. Such N is primarily derived from the burial of organic matter in marine and freshwater sediments, where it is incorporated into rock as organic N or as ammonium in silicate minerals. It may be pointed out that globally, sedimentary rocks contain 10^{21} g of fixed N, considerably more than the 10^{19} g of fixed N in the total biosphere (Galloway, 2004). Contrary to the general belief that new N enters terrestrial ecosystems solely from the atmosphere, Morford et al. (2011) have shown that N-rich bedrock comprises a hitherto overlooked source of ecologically available N to forests, the N inputs from rock weathering can be substantial and may range from 3-10.9 kg ha⁻¹ yr⁻¹. More often than not, sedimentary rocks are easily eroded. Thus, weathering of N-rich sedimentary rocks may even provide substantial amount of N in some ecosystems.

Carbonation, which is the reaction driven by the formation of carbonic acid (H₂CO₃) in the soil solution, is by far the most important chemical weathering process in a majority of terrestrial ecosystems.



Carbonation reactions are specially important in the weathering of limestone, leading to the release of Ca in soluble form.



Weathering of silicate rocks yields nutrient cations, K, Ca, Mg, and Fe. For example, during the carbonation of potassium feldspar, the primary mineral (K-Al-silicate) is converted into a secondary mineral (Al-silicate) releasing K and silica in soluble forms. Weathering of carbonates, which is more rapid, yields abundant Ca and Mg in soils. Other weathering reactions are simple solubilization of minerals, oxidation, and hydrolysis.

The rate of weathering is very difficult to measure, that is why only a few studies have been made (Likens et al., 1977). In the Hubbard Brook Experimental Forest in New Hampshire, USA, the amounts of different nutrients released in weathering (kg ha⁻¹ yr⁻¹) amounted to Ca (21.1), Si (18.1), K (7.1), Na (5.8), Mg (3.5) and Al (1.9) (Likens et al., 1977). The values at other sites may vary depending upon the composition of parent material and the rate of weathering. The nutrients that are released by weathering, if present in the root zone, are absorbed by the plants. In highly weathered tropical soils, a substantial amount of nutrients may be lost in drainage water.

Hydrologic inputs

Stream and river waters can bring significant amounts of nutrients into some terrestrial and swamp forest ecosystems. In cypress (*Taxodium distichum*) swamp - forests of the southeastern United States, as an example, net primary productivity is greatly dependent on P input through stream flow from upland ecosystem. The net primary productivity is 5 t/ha/yr in closed bogs without stream water (that is without P input) and 20 t/ha/yr in the swamp forest receiving P at the rate of 100 g/m²/yr (Brown, 1981). The terrestrial ecosystems of the floodplains receive substantial amounts of nutrients along with alluvium deposition. The legendary productivity of the flood plains of the Gangetic Plains is largely due to nutrient renewal each year through flood waters transporting nutrients from the Himalaya and other adjoining uplands. Hydrologic inputs are most important for the streams and other aquatic ecosystems.

Biological inputs

Many bacteria and blue-green algae (prokaryotes) have the enzyme nitrogenase, which fixes atmospheric N_2 (i.e., converts N_2 into NH_4^+ , which is readily available to plants). Some of them are symbiotic (nodulated plants), and others are free-living (see Chapter 16). Associative N_2 -fixation is the third form. In this, plants are associated with bacteria, such as *Azospirillum*, *Klebsiella*, *Bacillus*, *Acetobacter*, and *Pseudomonas* - like bacteria in their rhizosphere (the zone around the roots) but no differentiated structures, such as nodules, are formed on plant roots (Elmerich et al., 1992). Associative N_2 fixation has been studied in certain pines, other conifers and grasslands. Bormann et al. (1993) estimated about $50 \text{ kg ha}^{-1} \text{ yr}^{-1}$ associative N_2 fixation by *Pinus rigida* and *P. resinosa*, while the symbiotic N_2 fixation by nodule-forming *Robinia* and *Alnus* on the same site was two and five times greater. The symbiotic N_2 fixation is more efficient because of rapid transfer of photosynthate (i.e., greater availability of labile organic carbon) to the symbiotic organisms.

Nitrogen input to ecosystems by nodulated trees and shrubs is especially high on infertile, early successional sites (disturbed sites, such as forests regenerating after clear-felling or burning), savanna, etc. The N_2 fixation is an energy-intensive process and therefore N_2 -fixing trees grow well and contribute substantially to N input only in disturbed, relatively open ecosystems (high light environments). Leguminous trees even if present in a closed forest, e.g. in some tropical forests, seldom nodulate or fix nitrogen. Actinomycete association with plant roots is called Actinorrhizae or actinorrhizal relationships. These are formed by association of *Frankia* strains with about 160 species from 24 genera and 8 plant families. The plants can fix nitrogen, and the relationship is particularly important in the life history of woody and shrubby plants on infertile sites. *Frankia* symbioses are frequent in temperate and circum-Polar Regions and heath environments. The nodules of some actinorrhizae are small (*Comptonia*), whereas plants like *Alnus*, *Coenothus* and *Casuarina* have large nodules. *Casuarina* and *Coenothus* are important non-legume nitrogen fixers occurring in tropical regions. In temperate regions, *Alnus* and *Myrica gale* are important nitrogen fixers. In soils planted with *Alnus* and *Casuarina*, the amount of nitrogen fixation has been estimated at about $50 \text{ kg N ha}^{-1} \text{ yr}^{-1}$. This association provides a unique opportunity for microbial management of soil fertility to improve plant growth processes in forestry. Nitrogen fixation by free-living microbes generally ranges between 0-3 kg/ha/yr in most temperate soils (Roskoski, 1980).

17.3 BIOTIC ACCUMULATION AND STORAGE OF NUTRIENTS IN PLANTS

Accumulation of nutrients in plant biomass is rapid during forest development subsequent to disturbance, such as clear-cutting and burning (Fig. 17.3a, b). There are also parallel increases in floor litter mass. As long as net ecosystem production is positive, plant nutrients will be stored in organic matter. Greater the rate of nutrient accumulation in ecosystem components, the lesser is the nutrient loss from the ecosystem. During the accumulation phase, nutrient loss through stream flow will be lower than the extra system nutrient inputs. In old growth forests, where little net nutrient accumulation in organic matter occurs, total ecosystem losses (outputs) should, therefore, balance inputs of nutrients. Comparison of different forests indicates that organic matter and nutrient storage in forest floor increase from tropical to boreal forests, due to slower decomposition in cold climate, while the storage in aboveground vegetation decreases from tropical to temperate to boreal forest. For example, while wet tropical forest stores in the aboveground vegetation about $1404 \text{ kg N ha}^{-1}$ (Edwards and Grubb, 1982), the boreal forest stores only 116 kg N ha^{-1} (Cole and Rapp, 1981). Conversely, only 214 kg N ha^{-1} occurs in the forest floor of the tropical forest compared to 617 kg N ha^{-1} in the boreal forest.

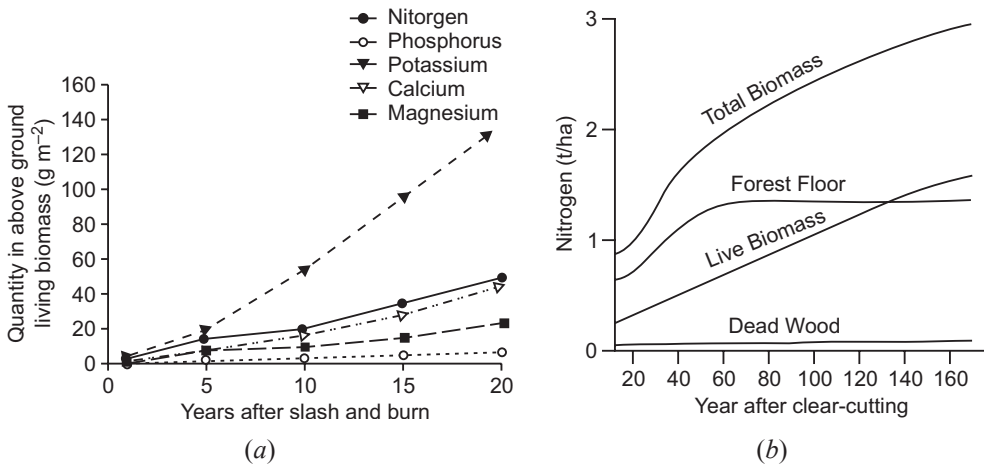


Fig. 17.3. (a) Quantities of nutrients in the redeveloping aboveground vegetation following slash and burn (Data from Toky and Ramakrishnan, 1983). (b) Estimated accumulation of nitrogen in different components of the Northern Hardwood Forest Ecosystem after clear-cutting (based on Bormann and Likens, 1979).

17.4 NUTRIENT OUTPUTS (ECOSYSTEM LOSSES)

The pathways of nutrient output from an ecosystem include: (i) transport of dissolved and particulate substances in streamwater, and in some areas also to ground water; (ii) gaseous release and particulate losses to the atmosphere; (iii) losses due to fire; and (iv) harvest of biomass.

Stream Water Losses

Of the various means of loss of nutrients from ecosystems, the loss through streams is the most obvious. Nutrient concentrations in stream water are directly related to the weathering of soil minerals in a forested watershed. However, concentrations of Na, Cl and SO₄ in stream water may indicate the effect of atmospheric deposition as well. Nutrients carried in streams are divisible into two fractions: (i) dissolved ions, and (ii) particulates. Particulate matter includes, inorganic as well as organic particulates, ranging in size from colloidal clays to large boulders. Pieces of leaves and decaying litter, are typical examples of organic particulates. In undisturbed forests the loss of organic particulates is generally equal or less than 1% of the net primary productivity.

In an undisturbed forest, nutrient losses are predominantly in dissolved form (Table 17.1). Because of carbonation reactions in weathering, generally bicarbonate is the dominant anion in the stream waters of an undisturbed forest. The total removal of nutrient (loss or output) is largely determined by the amount of stream water flow. The rate of increase in the loss of cations with the increase in streamflow amount is greatest for Ca, followed by Na, Mg and K (Fig. 17.4), although bedrock type can cause changes in this pattern. The annual loss of Ca can be especially high on limestones.

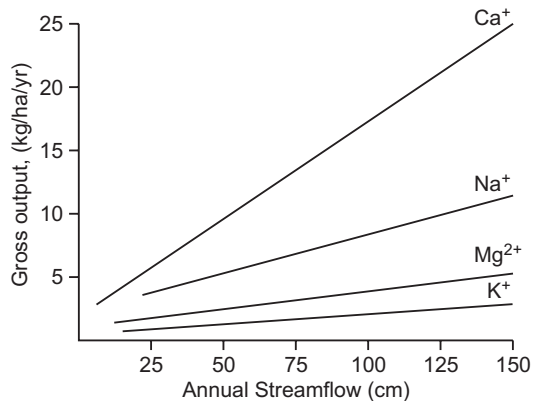


Fig. 17.4. Increasing annual streamwater loss of major cations with increasing annual streamflow in the Hubbard Brook Forest, New Hampshire (based on Likens et al., 1977).

Table 17.1. Average annual gross output of some elements as dissolved and particulate substances from a watershed of the Hubbard Brook Experimental Forest, New Hampshire (data from Likens et al., 1977). Except for P and Fe, all elements are predominantly lost as dissolved substances.

Element	Total (particulate + dissolved) (kg/ha)	% Particulate	% Dissolved
Mg	3.34	5.7	94.3
Ca	13.93	1.7	98.3
N	4.01	2.7	93.3
K	2.40	21.7	78.3
S	7.48	3.3	96.7
P	0.019	63.2	36.8
Fe	0.64	100.0	0

The nutrients which are in short supply are retained more strongly by the accumulating biomass than those which are easily available. That is why stream water losses of N and P in most ecosystems comprise a small percentage of the total inputs. Impact of nutrient uptake on streamflow losses of nutrients is also indicated by the fact that the losses are lower when plant growth is active than when plants are dormant. In many regions, there are nutrient losses to ground water which are difficult to measure. Usually, the volume of groundwater flow can be estimated as: Groundwater flow = Precipitation – (streamflow + evapotranspiration). The rock-forming elements, Ca, K, P, Mg and Fe may circulate through an ecosystem for many years, before being eventually lost in stream water. When deposited in ocean sediments, they once again become part of the rocks. These rocks are uplifted in a geological time-scale. This leads to the beginning of a new cycle.

Disturbances in natural terrestrial ecosystems accelerate the loss of nutrients through runoff soil and water. For example, the losses of N, Ca, K and P from an undisturbed oak forest through overland flow and sediment output were, respectively 0.16, 0.49, 0.27 and 0.04 kg/ha per monsoon season. However, from landslide-damaged oak forest, the losses were 25.65, 143.14, 86.43 and 5.21 kg/ha, respectively, for N, Ca, K and P (Singh and Singh, 1992). In the monsoon climate with a highly seasonal precipitation, most of the nutrient loss may occur within a short period of two to three months (mid-June to mid-September). In contrast, in temperate forests nutrient losses are greater during the spring snowmelt and during winters when evapotranspirational losses are minimal, resulting in most precipitation to flow through streams.

Losses to the atmosphere

Nutrient losses from ecosystems to atmosphere occur both in gaseous and particulate forms. Due to the anaerobic bacterial activity in the wetland and flooded terrestrial soils, reduced gases, such as methane and hydrogen sulphide are frequently released. Similar bacterial reactions have recently been recognized to occur temporarily in thick layers of soil organic matter of undisturbed upland forests during periods of heavy rainfall. Soil pits produced by tree fall are also the microsites of such a bacterial activity and consequently release reduced gases (Sextone et al., 1985). In these locations NO_3^- of the soil can be reduced to N_2 and N_2O as a result of denitrification reaction (Fig. 17.5) brought about by denitrifying bacteria such as *Pseudomonas*. In many ecosystems, N_2 and N_2O (produced by denitrification) may be regarded as the forms by which available N in the soil is lost from the ecosystem. The rate of denitrification largely depends on the rate at which NO_3^- , the substrate for this reaction, is available in soil. Low oxygen and a high supply of organic carbon promote the N loss through denitrification. A study on a large-scale regional N budget indicated that denitrification caused a greater loss of N than the hydrologic losses in Western Africa (Robertson

and Rosswall, 1986). Some amount of NO and N₂O also originate during the aerobic process of nitrification and are lost to the atmosphere. Certain ecosystems, such as grazed grasslands on N-rich soils and fertilized agricultural fields, emit a substantial amount of NH₃, through volatilization.

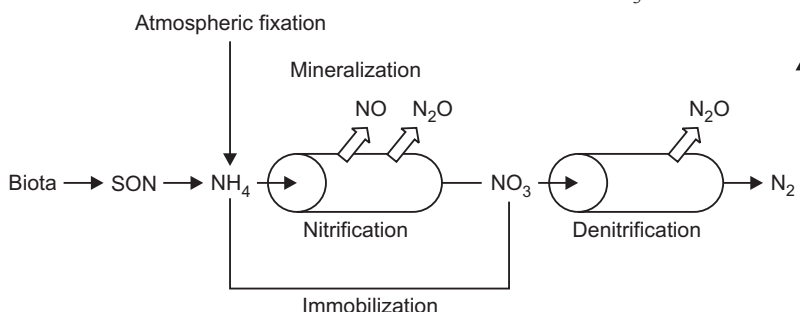


Fig. 17.5. Nitrogen transformations producing gases during nitrification and denitrification. SON = soluble organic nitrogen. All the steps are mediated by soil microbial communities.

Gaseous and particulate losses may also occur directly from plants to the atmosphere. These include volatile emissions of hydrocarbons, such as terpenes from forests (it causes summer time haze). Some volatile compounds may contain N and S. Leaves are also known to emit NH₃ during senescence.

Nutrient losses due to fire

When terrestrial communities are burnt, nutrients are lost in gases (e.g., N and S). Other nutrients left in the ash are lost through the post-burn increased runoff and erosion from the bare soils. The nutrient losses (percentage of the amount present in vegetation and litter before fire) generally follow the order: N > K > Mg > Ca > P.

Nutrient losses in forest harvest

Harvest of ecosystem products, such as timber harvesting, other tree components and herbage, also involves nutrient loss. Blowdown of trees in forests can be considered “natural harvest”. Wood harvest alone may account for only a small fraction of loss of the total nutrients in forest ecosystems of temperate zone. For example, saw-log harvest in a Tennessee mixed-oak forest removed only 0.1 to 7% of nutrient pool of N, P, K and Ca (Johnson et al., 1982). Tree components left, such as leaves and twigs have much higher nutrient concentrations than the wood. They also decompose quickly, resulting in a massive release of nutrients subsequent to harvests. The nutrient loss that occurs due to this type of harvest is replenished, before the harvest cycle is completed.

However, when aboveground biomass is removed, many changes may occur that may affect nutrient transfers. For example, reduced transpiration may dramatically increase nutrient losses through increased stream water runoff and soil erosion.

17.5 INTRASYSTEM CYCLE

Nutrient uptake from the soil solution by plants, nutrient allocation to different plant components (e.g., foliage, stem, branch and root) and nutrient return to soil are the major processes of intrasystem nutrient cycles. These processes involve rather large movements of nutrients compared to the amount of inputs and outputs from intersystem transfers. For example, Sollins et al. (1980) have shown that in an old-growth Douglas fir forest the circulation of N and P within the forest stand was 6.5 and 18.2 times greater than these nutrients received from the atmosphere and rock weathering. Similarly, nutrient transfers through the biotic component are larger than the losses to the stream water in the forest ecosystems.

Availability of nutrients in soil solution

Availability of nutrients to plants in soil solution is determined by several factors.

1. The exchange reactions are the dominant chemical processes that occur in soils. The silicate clay minerals have surface negative charges that attract and hold soluble nutrient cations from the soil solution. The negative charges possessed by silicate clay minerals contribute to soil cation-exchange capacity (CEC), expressed as milliequivalents per 100 g of soil. Iron and aluminium hydroxides, on the other hand, possess surface positive charge at most of the normal levels of soil pH. Nutrient anions, particularly PO_4^{-3} and SO_4^{-2} , are held on these particles. In addition to clay particles, organic matter also increases CEC of soil. The CEC and base saturation, that is the percentage of exchange sites actually occupied by nutrient cations, increases during the initial soil development on newly exposed parent materials, and declines in an intensely weathered soil such as soils of tropical forests. When CEC is high, as in fertile soils, losses of nutrients to runoff water are low. Tropical forest soils dominated by iron and aluminium oxides have low CEC.
2. More than 95% of nitrogen occurs in soil in organic forms. The insoluble organic N is converted into soluble organic N which is mineralized to $\text{NH}_4\text{-N}$ and then under certain conditions to $\text{NO}_3\text{-N}$ (Fig. 17.5). Both $\text{NH}_4\text{-N}$ and $\text{NO}_3\text{-N}$ are freely available for uptake by the plants. $\text{NO}_3\text{-N}$ is highly mobile in the soil, whereas $\text{NH}_4\text{-N}$ is relatively immobile due to its presence on cation exchange sites.
3. Soil pH is known to affect the availability of several nutrient elements. For example, at low pH, PO_4^{-3} is strongly bound by Fe and Al in soils reducing its availability to plants. Roots of some plants secrete chelates (e.g., citrate) into the soil, which solubilize phosphate by binding the Fe from insoluble iron phosphates (Lambers et al., 1998). At low pH, clay surfaces also strongly adsorb PO_4 . At high pH, PO_4 combines with Ca and forms insoluble phosphates. Maximum availability of P is, thus, around pH 6.5. In forest soils, more than 50% of total P in surface soils is available in organic form. Therefore, mineralization of organic P (i.e., cleaving of the ester bond, C-O-P, by the phosphatases produced by plant roots and mycorrhizae) regulates phosphorus availability as determined by the competition between biological and geological sinks for the phosphate anions (Attiwill and Adams, 1993). Solubility of other ions in soil solution is also affected by pH. For example, availability of N and K declines when pH is in acidic range (below 5.5).
4. Because more soil nutrients occur as positive ions than negative ions, uptake of ions by plant roots would develop a charge imbalance. Plants that use NH_4^+ as N source are reported to acidify the immediate zone around their roots (Nye, 1981). The uptake of NO_3 causes an opposite effect as a result of plant releases of HCO_3^- and organic acids to balance the negative charge (Waring and Schlesinger, 1985).

Nutrient supply and uptake

In aquatic ecosystems, plants such as *Elodea*, *Lemna* and *Vallisneria*, absorb most nutrients dissolved in water through their submerged leaves. In terrestrial plants also, leaves are able to absorb nutrients dissolved in the rain water or those present in the gaseous form. However, the major source of all nutrients for plants in terrestrial ecosystems is soil. Nutrients are taken up by plant roots from the soil solution; nitrogen is taken up as NH_4^+ and NO_3^- , phosphorus as H_2PO_4^- and HPO_4^{2-} and sulphur as SO_4^{-2} and metals as cations. Ca, Mg, K and Na, are the major exchangeable cations in soils. An exchangeable cation is held adsorbed on a negatively charged surface of both organic and inorganic colloids. In the exchangeable form, cations are not soluble but can be brought into solution for plant uptake by exchange of protons from plant roots.

Nutrient supply rates often determine the nutrient uptake rates. Nutrients are supplied to plant roots by the following mechanisms: (i) Interception of exchangeable nutrients by roots growing into the soil; this is particularly important for Ca and $\text{NH}_4\text{-N}$. (ii) Mass flow of ions with the movement of soil water in response to transpiration; this is important for Mg, SO_4^{2-} , Fe and Ca. (iii) Diffusion of ions toward root surface; this occurs when the uptake reduces the nutrient concentration at the root surface and mineralization increases the concentration away from root surface in the soil solution. Nutrients such as N, P, K move along this concentration gradient to the root surface. In most soils, the rate of diffusion of phosphate ion is markedly lower than that of K^+ and NO_3^- .

How do nutrients enter the plant? Some nutrients enter the plant passively following the flow of water, but the majority of nutrient uptake is largely an energy-dependent active process. Uptake of nutrients is controlled at root cell membranes, not all elements get entry, toxic nonessential, and excess elements are often excluded or allowed to have limited entry. It is evidenced by the presence of a large quantity of Na^+ in root exudates of several hardwood trees (Smith, 1976). It is suggested that nutrient uptake involves enzyme-mediated reactions and presence of specific ion carriers in root cell membranes (Ingstead, 1982). The ion carriers recognize nutrient ions in the soil solution and transport them to the root's vascular tissues. As concentrations of ions increase in soil solution, active uptake increases, the maximum rate reaching with the saturation of all carriers. Competition among groups of ions with the same charge, for the ion carrier may greatly affect the uptake of a particular ion. For example, the presence of large amounts of K^+ in soil solution may reduce the uptake of NH_4^+ .

The N uptake deserves special attention because its demand often exceeds its availability. Species differ in their preferences for uptake of the different available forms of N, viz., ammonium (NH_4^+), nitrate (NO_3^-) and dissolved organic N. The cycling of nitrogen is rather conservative as nitrogen may be immobilized in litter of older forests which prevents the loss of nitrogen into stream water. Species of the sites where microbial nitrification is slow or inhibited often show superior growth with NH_4^+ . A rapid nitrification seems to favour NO_3^- uptake. For example, of the two major forest-forming species of the central Himalaya, banj oak (*Quercus leucotrichophora*) shows preference for NO_3^- , while chir pine (*Pinus roxburghii*) equally utilizes NO_3^- and NH_4^+ . Nitrification is more rapid in banj oak site than in chir pine site. It is puzzling why several plants do not clearly prefer NH_4^+ which is more easily assimilated than NO_3^- . Conversion of NO_3^- to amino groups ($-\text{NH}_2$) requires more plant energy than conversion of NH_4^+ to amino groups inside plants. Some of the explanations for the lack of preferential NH_4^+ uptake are as following: (i) The rate of delivery to the root by diffusion or mass flow is higher for more mobile NO_3^- than for NH_4^+ . (ii) The plant uptake of NO_3^- has no competition that occurs between NH_4^+ and other positively charged nutrient ions. (iii) NH_4^+ is potentially toxic to plants. However, Aerts and Chapin (2000) have reviewed a large number of studies to show that (i) species which occur in habitats rich in $\text{NO}_3\text{-N}$ preferentially absorb nitrate, (ii) those which occur in habitats rich in $\text{NH}_4\text{-N}$ preferentially absorb ammonium, and (iii) both mycorrhizal and non-mycorrhizal plants are able to absorb soluble organic N in several terrestrial ecosystems.

Role of mycorrhizae

Symbiotic association with mycorrhizal fungi is widespread among trees and other higher plants, and is important to the nutrient transfer, especially for P. Absorption of N and other nutrients with low diffusion rates, is also facilitated by mycorrhizae. Presence of mycorrhizae allows the root to explore a greater volume of soil for nutrients, as the absorption surface increases several fold due to fungal hyphae. Many species owe their successful growth on nutrient-poor sites, such as mine spoils and rocky surface, with little soil, to mycorrhizal association. Because of their smaller diameter and larger surface: volume ratio, mycorrhizae acquire nutrients at a lower carbon cost than roots (Aerts and Chapin, 2000). Interestingly, in the presence of N-fixing bacteria, P uptake is enhanced, illustrating thus an interaction between N, P and C in the nutrition of higher plants.

What is the cost of mycorrhizal symbiosis to the higher plants? In a study on the Pacific silver fir (*Abies amabilis*) stands in Washington, the annual growth of mycorrhizae accounted for nearly 15% of the annual net primary production (Vogt et al., 1982). The tree may decrease allocation to roots in the presence of mycorrhizae however, resulting in a better overall growth. Why is mycorrhizal infection more in a nutrient-poor site? According to one explanation, when plants are subject to nutrient-deficiency, their growth usually slows but photosynthesis continues to be relatively high (Chapin, 1980). It is known that plants from nutrient-poor habitats allocate more C to the roots (Hobbie, 1992). The resultant increase in the carbohydrate concentration in roots may stimulate mycorrhizal infection in nutrient-poor sites.

Nutrient concentrations and storage in vegetation

Nutrient concentrations differ widely among plants and in different components within plants. In a majority of trees, the concentrations in the aboveground component follow the order: leaf > twig > branch > bole, and in the belowground components: fine root > lateral root > stump root. Among different growth forms, the leaves of evergreens have lower concentrations than forbs and graminoids, and deciduous species have the highest N concentrations (Aerts and Chapin, 2000). The foliar N : P mass ratio (Redfield ratio) of 10 : 1 appears optimal for plant growth. Plant growth is N-limited at N : P ratios less than 14 : 1 and P-limited at ratios greater than 16, and colimited by N and P at intermediate ratios (Aerts and Chapin, 2000). Ecosystems vary enormously in the quantities of different nutrients held in the plant biomass and in litter compartments (Table 17.2). In most ecosystems, the quantities follow the order: $N \geq Ca > K > P$. The variation among ecosystems is accounted for by the dry matter accumulation and the nutrient concentration of the accumulated biomass.

Table 17.2. Nutrient content (g/m²) in vegetation and litter in certain forest ecosystems.

Forest	Component	N	P	K	Ca	Reference
<i>Pinus banksiana</i> Canada	Vegetation	16.5	1.4	8.2	11.2	Foster and Morrison (1976)
	Litter	32.8	4.3	38.8	50.0	
<i>Pinus roxburghii</i> , India	Vegetation	114.5	14.8	37.8	54.0	Chaturvedi and Singh (1987b)
	Litter	13.1	1.1	2.4	8.3	
Post oak-black jack oak, USA	Vegetation	107.1	9.5	119.6	433.3	Johnson and Risser (1974)
	Litter	8.6	0.6	6.2	21.6	
Oak forest, India	Vegetation	336.6	14.0	99.7	452.6	Rawat and Singh (1988b)
	Litter	9.4	0.4	1.2	10.2	
Tropical dry deciduous, India	Vegetation	53.4	3.5	24.2	26.8	Singh and Singh (1991b)
	Litter	3.3	0.2	1.3	1.1	

Seasonal changes in foliage nutrients and nutrient resorption

In woody plants, concentration of N, P and K declines continuously from bud-break to leaf senescence. Calcium, magnesium and iron do not follow the above pattern, their concentrations, with few exceptions, generally increase continuously until leaf abscission. The decline in N, P

and K concentrations occurs as a result of increasing accumulation of photosynthetic products and thickening of cells as leaves expand and mature (dilution effect), and then due to retranslocation of nutrients to other parts of trees during leaf senescence. The mass of N, P and K in leaves (nutrient concentration multiplied by leaf dry mass) increases during the growth period, as leaves expand and gain in mass, and then declines as leaves begin to senesce (Fig. 17.6). The fraction of nutrient mass (N, P, K) of leaves that is withdrawn generally ranges between 20% and 80% across woody species of different habitats. Species show variation in the level of resorption for different nutrients. For example, *Quercus prinus* (chestnut oak) retranslocates at leaf fall 78% of N mass, 59% P, 53% K and 8% Ca present in mid-August (Ostman and Weaver, 1979). In some species the mass of Ca increases instead of showing withdrawal during leaf senescence. The resorption of nutrients from leaves represents an intraplant or internal nutrient recycling. The internal nutrient reserve resulting from nutrient resorption is readily available for new growth, and reduces the dependence on nutrient uptake from soil. Lal et al. (2001a) found that the N-mass resorbed in the previous year was sufficient to support the full expansion of the leaf area and more than 50% of the maximum leaf weight in the succeeding growing season in several dry tropical species. Resorption is not associated only with senescing tissues. In trees with leaves of several years' life-span, new leaf formation occurs by utilizing nutrients resorbed from leaves formed in the preceding year, in some species from leaves of less than six months age (Nambiar and Fife, 1987).

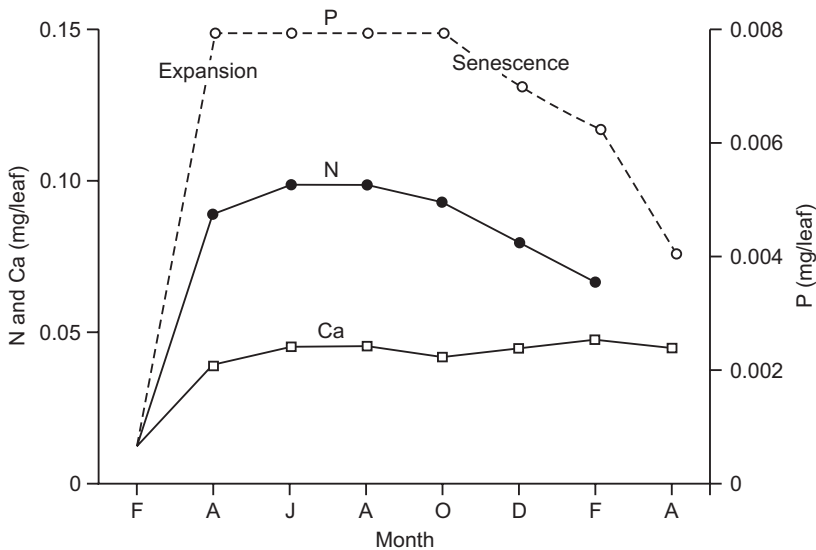


Fig. 17.6. Nutrient mass (expressed as mg/leaf dry weight) changes in *Pinus roxburghii* (Central Himalaya) leaves within an annual cycle (data from Ralhan and Singh, 1987).

Current generalizations suggest that (1) evergreen species are less efficient in N resorption than deciduous species (Eckstein et al., 1999), and (2) species from nutrient-poor environments have a higher nutrient use efficiency (Hobbie, 1992). Data collected from a large number of tree species occurring in a dry tropical environment suggested that greater pools of N and P were resorbed in deciduous species than evergreen species at a nutrient-rich site, but the deciduous species of the nutrient-rich site exhibited a significantly greater resorption efficiency for N and P than the deciduous species of the nutrient-poor site (Table 17.3). Evidently, species follow an individualistic adaptational strategy with respect to nutrient cycling, and resorption efficiencies have no consistent pattern across gradients of nutrient availability. Nutrient resorption is, nevertheless, a conservation measure because nutrients deposited in litter could be exposed to losses in runoff and other processes.

Table 17.3. Mean resorbed nutrient pools and resorption efficiencies of dry tropical tree species growing on nutrient-rich Inceptisol and nutrient-poor Ultisol (based on Lal et al., 2001b).

Resorbed nutrient pool (mg/cm ² leaf area)	Ultisol, deciduous	Inceptisol deciduous	Inceptisol, evergreen
N	101.8	120.0	91.8
P	7.6	11.2	7.8
Resorption efficiency (%)			
N	51.8	65.8	54.7
P	48.1	56.3	42.4

Nutrient return from vegetation to soil

Nutrient return to soil from the aboveground vegetation takes place through (i) litterfall (fall of leaves, twigs reproductive structure, etc.), and (ii) leaching from canopy by rainwater through stemflow and throughfall. Throughfall contains nutrients washed out and leached from leaf surfaces, and stemflow waters, those leached mainly from bark surfaces. Though nutrient concentrations in stemflow waters are much higher than in throughfall waters, much water reaches the forest floor as throughfall. Thus, the throughfall water comprises about 90% of nutrients that reach soil as a result of leaching from vegetation. Total contribution of leaching (throughfall + stemflow) to nutrient return is substantial for most nutrients and is the greatest for K (Table 17.4). Leaching of nutrients generally occurs in the order: K > Ca > Mg > P > N. Species differ in leaching induced nutrient movement from canopy to soil as leaching seems to depend on leaf nutrient concentration, surface area-to-volume ratio, surface texture, leaf age and amount of rainfall. Leaching is particularly pronounced at the time of senescence.

Table 17.4. Nutrient return from vegetation to soil in some forests to compare the contributions of litter fall and throughfall + stem flow.

Forest	Process	Nutrient return (kg/ha/yr)					Source
		N	P	K	Ca	Mg	
Mixed deciduous forest, Great Britain	Litter fall	63.5	2.6	19.0	83.3	9.7	Cole and Rapp (1981)
	Throughfall + stemflow	9.2	0.73	35.1	28.0	15.3	
Sal forest, India	Litter fall	69.8	5.4	31.4	80.7	NA	Mehra et al. (1985)
	Throughfall + stemflow	3.8	0.7	6.4	6.0	4.4	
Pine mixed broadleaf forest, India	Litter fall	47.7	4.1	22.8	50.8	NA	Mehra et al. (1985)
	Throughfall + stemflow	4.3	0.6	10.0	5.8	5.1	
Pine forest India	Litter fall	56.9	6.4	37.1	74.2	NA	Mehra et al. (1985)
	Throughfall + stemflow	5.2	0.7	12.9	10.5	6.2	
Tilonj dominated mixed oak forest, India	Litter fall	70.0	5.8	31.9	91.6	NA	Mehra et al. (1985)
	Throughfall + stemflow	6.3	0.6	17.8	10.4	6.4	

NA: Data not available.

Since leaf fall accounts for most of the nutrient return through litter fall, resorption of nutrients before leaf abscission greatly influences the amounts of nutrients that return from canopy to soil. Calcium is highly mobile within the plant, where it is incorporated within the cell walls. Therefore 80% of the total annual aboveground cycle of Ca is mediated through litterfall. Potassium is highly mobile and therefore leaching from crowns accounts for about 30% of the total annual aboveground cycle of K.

Since growth and death of fine roots are difficult to measure, information on nutrient return through root litter fall is scanty. Nevertheless, the fact that the fine root production can exceed the leaf production in some forest ecosystems, indicates the great importance of this pathway of nutrient return. For example, in yellow poplar (*Liriodendron tulipifera*) forest in Tennessee, the belowground processes, particularly root death contributed twice as much N and seven times as much K as did the aboveground processes (Cox et al., 1978). Similarly, in a tropical forest in the Amazon, the quantities of N and P added to the soil by turnover of fine roots were 343 kg and 11 kg/ha/yr, respectively. These compare with 61 and 0.8 kg ha⁻¹ yr⁻¹, respectively, added by fine aboveground litter.

Decomposition and Nutrient Release

The breakdown of litter and soil organic matter releases nutrients to plants and microbes, and thereby completes the intrasystem nutrient cycle. Litter decomposition in tropics is fast mainly because of warm and moist conditions, but highly variable, the rates overlapping at their lower end with decomposition rates in temperate forests. Litter turnover (*k*), calculated as annual litter fall divided by litter standing crop on soil surface has been reported in the range of 0.34-1.1 for various tropical forests (see Vitousek and Sanford, 1986). On the other extreme, in boreal peatlands (very cold), *k* could be as small as 0.001 (Olson, 1963). Consistent with the rate of decomposition, mineral release decreases from tropics to temperate regions and to boreal forest areas. Nutrient turnover time which is reciprocal of *k*, i.e. (1/*k*), for the major forests of the world and three forests of India is given in Table 17.5. Aerts and Chapin (2000) found that in both temperate and tropical regions, the leaf litter decomposition rates of evergreen species were lower than those of deciduous species.

Table 17.5. Mean turnover time (yr) for nutrient content of the forest floor litter mass in different forests of world (data from Waring and Schlesinger, 1985; Chaturvedi and Singh, 1987; Rawat and Singh, 1988b; Singh and Singh, 1991).

Forest	Mean turnover time (i.e., 1/ <i>k</i> , in yr)			
	N	P	K	Ca
Boreal coniferous	230	324	94	149
Boreal deciduous	27.1	15.2	10.0	13.8
Temperate coniferous	17.9	15.3	2.2	5.9
Temperate deciduous	5.5	5.8	1.3	3.0
Tropical rainforest	2.0	1.6	0.7	1.5
Pine forest, India	2.44	2.50	1.43	2.17
Oak forest, India	1.31	1.30	1.22	1.56
Dry deciduous forest, India	1.33	1.27	1.39	1.20

Most of the N and P available to plants pass through the soil microbial biomass, which acts both as a source and a sink of plant-available nutrients. The soil microbes are known to immobilize N when C : N ratio is greater than 25 : 1, and excrete N below this ratio. Thus, the microbial community in decomposing litter of high C : N ratio immobilizes available N from the soil solution. Thus, the pattern of nutrient release from the decomposing litter involves an early immobilization of

nutrients in the microbial biomass, followed by a net nutrient release consequent to death, decay and cell lysis of microbial organisms. Soil microbial component of infertile soils, particularly receiving material with high C : N ratios, may immobilize N and P causing severe scarcity of plant-available nutrients, at least temporarily.

The lysis of microbial biomass may result in a pulsed release of nutrients in some ecosystems which may be synchronized with active plant growth (Singh et al., 1989). However, if this pulsed release is not synchronized with plant uptake, nutrients may be lost by the ecosystem (Lodge et al. 1994).

Nutrient use efficiency

We define nutrient use efficiency (NUE) as net primary production per unit nutrient uptake (net uptake).

$$\text{Thus,} \quad \text{NUE} = \frac{\text{Net primary production}}{\text{Net nutrient uptake (i.e., adjusted for resorption)}}$$

Recognizing that the annual losses of nutrients must be replaced by uptake, we can also calculate NUE by dividing net primary production by nutrient losses through litter fall and leaching.

Because of convenience, often the expression given below is used:

$$\text{NUE} = \frac{\text{Dry mass of litter fall}}{\text{Nutrient content of litter fall}}$$

It is easy to collect data on the parameters used in this expression.

Over a broad range of species, photosynthesis is positively correlated with leaf N concentration. Adaptations to minimise nutrient losses through leaching and to increase internal nutrient use (retranslocation from senescing leaves and other such organs) may be effective in increasing NUE. Communities on sites of low fertility are often said to use nutrients more efficiently (high biomass production per unit of nutrient taken up from the soil) than those of high fertility. Coniferous forests which occupy relatively nutrient-poor sites show higher NUE than deciduous forests, often associated with nutrient-rich sites (Table 17.6). However, this may not be always true. For, trees on nutrient-rich sites generally are taller and have more wood than those on nutrient-poor sites, and wood has much lower nutrient concentrations than leaves. Thus, as a result of a relatively more wood production per unit nutrient used, the nutrient use efficiency on fertile sites may not be lower than that for the communities on nutrient-poor sites (Grubb, 1989). NUE increases with age of trees because of the proportional increase in wood production and withdrawal of certain nutrients, particularly P and K, from older wood, at least in some species.

Table 17.6. Comparison of nutrient use efficiency (NUE) of coniferous and deciduous forests. NUE is = Net primary production (kg/ha/yr) divided by net uptake of nutrient (kg/ha/yr) (data from Cole and Rapp, 1981).

Forest type	NUE			
	N	P	K	Ca
Deciduous	143	1859	216	130
Coniferous	194	1519	354	217

Nutrient budget

A mass-balance approach is used to represent the budget of nutrients in an ecosystem. Nutrient uptake, as an example, is measured indirectly by adding annual nutrient accumulation and losses. This is referred to as gross nutrient uptake. When the value of this uptake is adjusted for resorption of nutrients from senescing leaves and other litter components, the value is referred to as net uptake

(net uptake = gross uptake – reabsorption). Nutrient budgeting is shown for four forests in Table 17.7 as examples. Such a budgeting makes comparisons among ecosystems easy. Table 17.7 shows that nutrient storage in the vegetation compartment followed the order : sal > oak > pine > dry deciduous forest. In the sal forest, the vegetation pool accounted for nearly half of N and two-third of P stored in the ecosystem. On the other hand, gross uptake of N was highest in oak and decreased through sal, pine and dry deciduous forest, and for P the order of uptake was: pine > sal > dry deciduous and oak forest. Substantial amounts of N and P were resorbed from senescing tissues. The resorption was maximum in pine and minimum in the oak forest. In all these ecosystems there was a substantial retention of nutrients in the vegetation over the annual cycle, showing the aggrading nature of these ecosystems.

Table 17.7. Nitrogen and phosphorus budgets for the tree layer of certain Indian forests (data source: Dry deciduous, Singh and Misra, 1979 and Singh and Pandey 1981; Sal, Singh and Singh 1989; Pine and Oak, Singh et al., 1984).

	Forest types							
	Dry deciduous		Sal		Pine		Oak	
	N	P	N	P	N	P	N	P
Compartment pools (g m ⁻²)								
Vegetation	65.6	5.4	445.0	40.3	110.6	14.5	329.9	13.5
Litter	3.6	0.3	5.0	0.3	13.1	1.0	9.4	0.4
Soil	297.0	11.9	401.0	15.1	396.4	21.8	580.0	18.6
Total	366.2	17.6	851.0	55.8	520.3	37.4	921.2	32.5
Accumulation in net production (g m ⁻² yr ⁻¹)	14.1	0.9	15.4	1.26	14.3	2.0	18.5	1.0
Reabsorption before senescence (g m ⁻² yr ⁻¹)	3.3	0.1	3.4	0.24	5.3	0.5	1.8	0.1
Net uptake from soil (g m ⁻² yr ⁻¹)	10.8	0.8	12.0	1.01	9.1	1.0	16.7	0.9
Return (g m ⁻² yr ⁻¹)								
Litter fall (leaf + wood)	6.3	0.4	7.38	0.67	5.4	0.5	9.4	0.6
Removal from canopy in stemflow	–	–	–	–	0.006	0.003	0.01	0.001
Removal from canopy in throughfall	–	–	–	–	0.4	0.06	0.5	0.04
Root mortality	2.1	0.2	0.5	0.04	0.09	0.02	0.76	0.05
Total	8.4	0.6	7.9	0.7	6.0	0.6	10.7	0.7
Retention in vegetation components (g m ⁻² yr ⁻¹)	5.7	0.3	7.5	0.5	3.1	0.4	6.0	0.2

Ecosystem nutrient budgets can also be presented in the form of compartment models as shown in Figs. 17.7-17.9. In these models, the standing state of nutrients are given in compartments and flux rates on the arrows connecting one compartment with the other. This makes the comparison among ecosystems still easier. For example, a comparison of these figures would show that much more N and P are cycled in the grassland occurring on a fertile, alluvial soil (Fig. 17.8) compared to the quantities cycled annually through the dry deciduous forest occurring on a nutrient-poor soil (Fig. 17.7), although the storage of nutrients in the vegetation compartments is much more in the forest than in the grassland. In general, plants from a fertile habitat take up and lose nutrients rapidly, produce fast-decomposing litter, and sustain a greater magnitude of herbivory, resulting into rapid rates of nutrient cycling (Hobbie, 1992). While comparing Figs. 17.8 and 17.9, one can easily differentiate between the nutrient cycling in tall tropical and short temperate grasslands. The

proportion of N cycled through the root compartment in the shortgrass prairie is substantially greater than the proportion cycled through the shoot compartment (Fig. 17.9), while the reverse is true for the tropical tall grassland (Fig. 17.8).

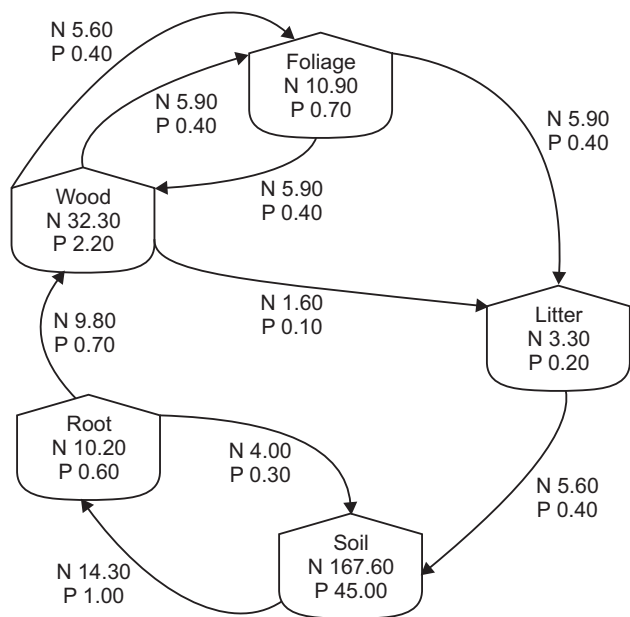


Fig. 17.7. A compartment model of intrasystem cycling of N and P in a tropical dry deciduous forest, India. Values in compartments (tanks) are mean standing crop of nutrient in g/m², and those on arrows are net annual flux rates in g/m²/yr (based on Singh and Singh, 1991b).

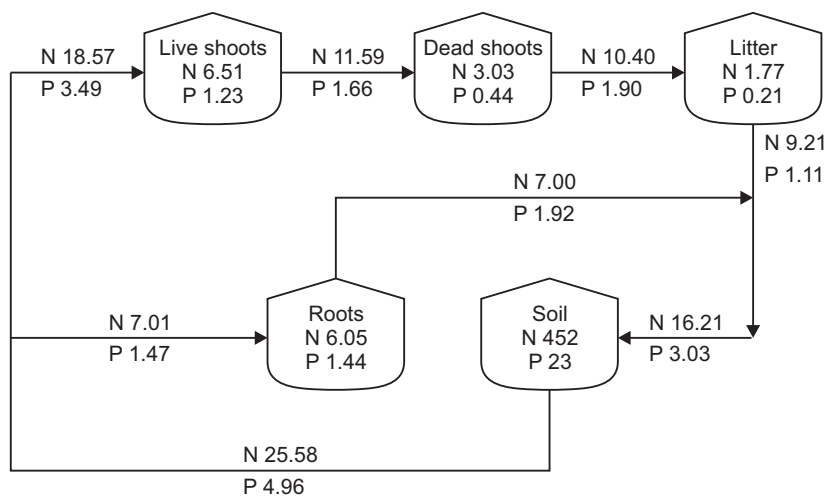


Fig. 17.8. A compartment model of N and P cycling in a dry subhumid alluvial grassland, India. Values in compartments (tanks) are mean standing crop of nutrient in g/m², and those on arrows are net annual flux rates in g/m²/yr (based on Singh et al., 1979).

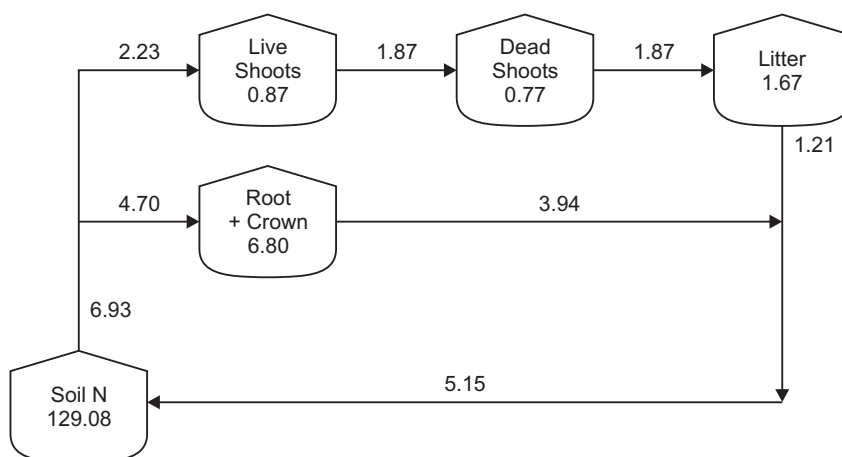


Fig. 17.9. A compartment model of N cycling in a shortgrass prairie, U.S. Values in compartments (tanks) are mean standing crop of nutrient in g/m^2 , and those on arrows are net annual flux rates in $\text{g/m}^2/\text{yr}$ (based on Bokhari and Singh, 1975).

17.6 NUTRIENT CONSERVING ADAPTATIONS IN OLIGOTROPHIC SOILS

The temperate and tropical forest ecosystems differ in nutrient cycling patterns because of the differences in weathering, soil properties and the decomposition of detritus. The soils in tropical forests are deeply weathered and are low in clay content with low cation exchange capacity (ability to retain nutrients). Therefore, in many tropical forests, most of the nutrients occur in living biomass. The high productivity of tropical forests is supported by rapid regeneration of nutrients from the detritus. The various nutrient conserving mechanisms on oligotrophic soils are discussed as follows:

1. **Large root mass:** Large root mass and high root : shoot ratio are characteristic of some species adapted to infertile sites. Large root masses occupy more fully the volume of soil where nutrients are held after release from decomposition and thus increase surface area on which nutrients can be strongly adsorbed. Lower allocation to root mass may be compensated for by developing high specific root length (i.e., root length per unit root mass).
2. **Root concentration near the soil surface:** In nutrient-poor soils, roots may be strongly concentrated in surface layer to tap the nutrients before they are leached down the soil profile. Formation of a thick root mat on top surface is often characteristic of infertile tropical humid forests. Concentration of roots near the soil surface where nutrients are released from decomposing litter gives further advantage to the species for obtaining nutrients. Equally important is the advantage such species get in competition with decomposers themselves for the available nutrients. Thus, in the “root mat” nutrients may move directly from decomposing matter to roots, without intermediate storage in mineral soil.
3. **Nutrient absorption by aerial roots:** Adventitious tree roots are found in many tropical moist forest trees. Abundant root hairs, unsubsized root tips and presence of endomycorrhizae suggest that these roots play a role in transferring nutrients from the dead part of mats formed on stem and branches of trees by bryophytes, lichens, ferns, bromeliads, orchids, etc.

4. ***Mycorrhizal association:*** Forests and other terrestrial communities on nutrient-poor sites generally have intimate association with mycorrhizae. A majority of C allocation below-ground in low-nutrient environments supports mycorrhizae (Hobbie, 1992). Arbuscular endomycorrhizae (AM) which actually penetrate the cortex of feeder roots are more common in tropical forests. Mycorrhizae may provide a “direct” pathway for nutrients from decomposing litter to roots, bypassing the involvement of the known decomposers. Attachment of mycorrhizae with decomposing litter can be seen in some tropical forests. Mycorrhizae seem to completely close nutrient cycling in tropical rain forests also by deriving nutrients from dying roots, and establishing possibly an interspecific transfer of nutrients. The distribution of species with symbiotic association for nutrient supply seems to be affected by water factor. In savannas, trees with sheathing mycorrhizae decline in importance with decreasing rainfall, while the importance of nodule-bearing species (N-fixers) increases with decrease in water (see Grubb, 1989).
5. ***Evergreen, sclerophyllous leaves:*** Species of nutrient-poor soils generally have evergreen, scleromorphic (tough and generally narrow) leaves. Their leaf P concentration, as an example, is below 0.3%. The tough, evergreen leaves with low nutrient concentrations are also resistant to herbivory. Leaves of long life-span may be advantageous to plants of infertile soils, because frequent replacement of leaves entails high costs. In broadleaved species, sclerophilly is more common in tropical evergreen forests than in temperate forests. The long-lived leaves of high specific leaf mass (leaf dry weight per unit area) have possibly high cost of production, low but long period of photosynthesis, and are therefore referred to as having a high cost, slow profit strategy, in contrast to low cost, quick profit strategy of short-lived deciduous leaves.
6. ***Nutrient scavenging epiphylls:*** Epiphylls are plants such as mosses, lichens and algae which are present on tree leaves especially in humid tropics. They can derive nutrients (scavenge) from rainwater moving across leaves, whereas some of the lichens and algae may also fix N.
7. ***Secondary compounds as chemical defense:*** The role of secondary plant compounds such as phenols as defenses against herbivory by mammals, insects and pathogens is well known in trees and other plants of the tropics. It is argued that the cost of producing new leaves is greater than that of producing secondary compounds. Secondary compounds may also affect nutrient cycling by inhibiting activity of decomposing bacteria and those involved in denitrification.
9. ***High silica levels:*** Phosphate availability is often low in tropical soils, while many species of trees and grasses contain high levels of silica. It can have two roles: (i) Silicon partially substitutes phosphorus when it (P) is low; (ii) Silicates replace phosphates bound with iron and aluminium in the soil, thereby releasing the phosphates in soluble form.
10. ***High nutrient storage in biomass:*** Large accumulation of nutrients in wood and total biomass, relative to soil is suggested to be the characteristic feature of some humid tropical forests. It is suggested that high biomass accumulation of nutrients would minimize nutrient leakages from the ecosystem. However, evidences contrary to the proportionately greater nutrient storage in biomass in tropical forests than in temperate forests are also seen.
11. ***Nutrient retranslocation during senescence:*** Nutrient resorption before the abscission of leaf minimises nutrient leakages from the ecosystem, but whether the proportion of nutrient resorption is more on infertile sites than on the fertile sites has not yet been established. In fact, studies indicate that resorption may be equally high in plants occurring on fertile soils.

17.7 EFFECTS OF N- AND P-ENRICHMENT ON BIODIVERSITY

There has been the unprecedented increment in the anthropogenic N and P inputs in the biosphere since the beginning of the Industrial Revolution (from 15.3 to 259 Tg N year⁻¹ and < 0.3 to 16 Tg P year⁻¹; Peñuelas et al., 2011). This large-scale nutrient enrichment is now considered to be one of the most important threats to biodiversity conservation (Tilman et al., 2001). The mechanisms underpinning species loss include toxic effects, increased susceptibility to pests and environmental stresses, altered mutualistic relationships, soil acidification and eutrophication (Bobbink et al., 2010). Particular attention has been paid to eutrophication, i.e., increased primary productivity through an enhanced supply of production-limiting nutrients, and the subsequent competitive exclusion of species with low productivity (Hautier et al., 2009). Considering the prediction of a further 2.4- to 2.7-fold increase in agriculturally driven eutrophication with N and P by the year 2050 (Tilman et al., 2001), there is an urgent need to understand the relative contribution of increased N and P to biodiversity loss.

On the basis of data on soil, productivity and plant species from 132 seminatural grasslands located along a gradient of nutrient availability and atmospheric N deposition in north-east England and southern Scotland ($n = 40$), central France ($n = 40$) and Belgium ($n = 53$), Ceulemans et al. (2013) have shown higher species richness in P-limited grasslands. Their study highlights negative effect of P enrichment on biodiversity as compared to N enrichment in at least some terrestrial ecosystems. They have used linear mixed models to investigate the relation between soil nutrients, acidity, limitation and productivity, and indices for plant species richness. Soil P was significantly negatively related to total species number, forbs and endangered species. Soil N was only significantly negatively related to number of forbs and endangered species. Furthermore, these workers have provided novel evidence of higher species richness in P-limited grasslands than in N-limited grasslands Ceulemans et al. (2013).

CHAPTER

18

Terrestrial Biomes and Aquatic Areas

The biosphere is the thin film of the living organisms that occur on earth. For the sake of convenience, it can be divided into: (i) *Geobiosphere*, comprising terrestrial ecosystems, and (ii) *Hydrobiosphere*, comprising aquatic ecosystems (Table 18.1). The biome includes both, the characteristic regional vegetation as well as the characteristic assemblage of animals, for vegetation alone the term formation is used. Being the primary independent factor of the environment, climate can be used as a basis for further dividing the geobiosphere. The seven climate belts are: (i) The equatorial rain zone, (ii) the summer rain zone, (iii) the subtropical dry zone, (iv) the subtropical winter-rain zone, (v) the temperate zone with year-round precipitation, (vi) the subpolar zone, and (vii) the polar zone. The temperate zone could be further divided, whereas sub polar and polar zones could be combined to form a single Arctic zone. The nine climate zones, termed zonobiomes (ZB) recognized by Walter (1979), are given in Table 18.2. Each zonobiome has a clearly defined climate, and to a large extent the zonobiomes largely correspond to soil types and zonal vegetation. A zonobiome is a large and relatively uniform environment within the geobiosphere. Zonobiomes of Asia are shown in Fig. 18.1.

Table 18.1. Divisions of Geobiosphere and hydrobiosphere (based on Walter, 1979; Whittaker, 1975)

Divisions	Main Characteristics
Geobiosphere (Terrestrial Ecosystems)	
Zonobiomes (biomes)	Large ecological systems with distinct vegetation
Zonoecotones	Zones of overlap between zonobiomes
Orobiomes	Mountains
Pedobiomes	Extreme soil environments
Hydrobiosphere (Aquatic Ecosystems)	
Freshwater biomes	
Lentic	Standing water systems-lakes and ponds
Lotic	Flowing water systems- Rivers and streams
Freshwater wetlands	Shallow standing water areas for a part of the year
Marine biomes	
Marine sandy beaches	Littoral zone
Marine mud flats	Zone of high and low tides in bays and estuaries
Coral reefs	In warm water of tropical and subtropical oceans
Marine surface pelagic communities	Open Ocean water
Deep open benthos communities	Extends from boundary of continental shelf to deepest oceans

Table 18.2. Climate and vegetation characteristics of major terrestrial biomes (based on Walter, 1979; Whittaker, 1975).

Zonobiomes	Climatic zones	Biomes	Characteristics of plants	Important features
ZB I	Equatorial- high temperature and high rainfall	Tropical Rainforest	Broad leaved evergreen trees, epiphytes, lianas, orchids	Greatest diversity of species Nutrient poor soils
ZB II	Tropical- high temperature and seasonal drought	Tropical seasonal forest/ savanna	Broad leaved deciduous or evergreen trees; grassland with scattered trees/shrubs; monsoon forests	Species adapted to fire and drought Periodic burning important
ZB III	Subtropical- High seasonal arid climate	Subtropical desert	Succulents, annual herbs, sparse vegetation	Plants adapted to low moisture and high temperature
ZB IV	Mediterranean- winter rainfall and summer drought	Woodland/ shrubland	Evergreen or summer deciduous, drought resistant trees/ shrubs	Chaparral of California; marquis around Mediterranean sea
ZB V	Warm temperate- cool winter and warm summers	Temperate rain forest	Coniferous and broad leaved evergreen trees Maximum tree height (60to 120m) High tree longevity (500 yr)	Species diversity low; evolutionary adaptation to survive fire/ reproduce after fire; winter rainfall and no summer – drought season
ZB VI	Nemoral- Moderate climate with winter freezing	Temperate deciduous forest	Mesophytic and deciduous forest, many perennial herbs	Species diversity low (western Europe) to high (Eastern Asia)
ZB VII	Continental – (Arid temperate with cold winter)	Temperate grassland/ desert	Perennial bunch forming and sod forming grasses	Very high impact of agriculture
ZB VIII	Boreal – (Cold temperate, cool summer and cool winter)	Boreal coniferous forest (Taiga)	Largely evergreen coniferous trees; mosses and lichens form ground cover	Soils highly acidic and low in nutrients; permafrost (permanently frozen soil)may be present
ZB IX	Polar- very cold winter	Arctic Tundra Alpine Tundra	Low shrubs, grasses, sedges, lichens Rosette shrubs, grasses, sedges, heath plants	Treeless vegetation on permafrost Plants above treeline on mountains, permafrost absent



Fig. 18.1. Asia with zonobiomes I-IX. I. Equatorial with diurnal climates, II. Tropical with summer rains, III. Subtropical-arid (desert climate), IV. winter rain and summer drought, V. Warm temperate (maritime), VI. Tropical temperate with a short period of frost, VII. Arid temperate with a cold winter (continental), VIII. Cold temperate (boreal), IX. Arctic (including Antarctic) (based on Walter, 1979).

Whittaker (1975) combined the vegetation structural characteristics and climate parameters (temperature and precipitation) into a single scheme of biome distribution (Fig. 18.2). Then he depicted the approximate boundaries of the biomes with respect to climatic factors. In the temperature range of 20° to 30°C, vegetation types range from rainforests to deserts depending upon the amount of rainfall (Fig. 18.2). Seasonal forests, short dry forests and scrublands have been distinguished in the range of 250 to 150 cm annual rainfall. In the temperate latitude, the vegetation has been distinguished into four types (temperate rainforests, temperate seasonal forests, woodland and shrubland, and cold deserts) along the gradient of decreasing rainfall. In the cold climate, i.e., taiga region, the vegetation types are not distributed on the basis of climate. At temperatures below -5°C, all the vegetation types are classified as tundra (Fig. 18.2).

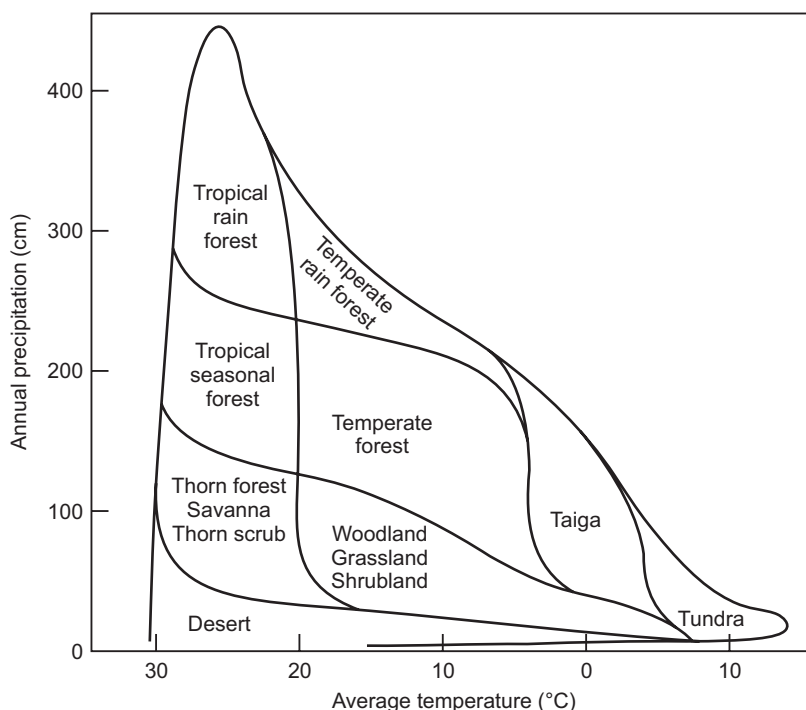


Fig. 18.2. The biomes as delineated on the basis of average temperature and precipitation (based on Whittaker, 1975).

18.1 MAJOR TERRESTRIAL BIOMES

A biome is a fundamental unit of which larger ecological systems are made up. It is a major kind of community on a given continent with characteristic physiognomy (Whittaker, 1975). As an environment, a biome is a uniform area belonging to a zonobiome, orobiome, or pedobiome. The dry deciduous forest of India, for example, is a biome of ZB-II (tropical with summer rains) type, the evergreen forests of Western Ghats represent a biome of orobiome I type, and the Himalayan oak forests represent a biome of orobiome II type, and so on. Each biome has its own physiognomy, or vegetation structure, that has developed in response to its major feature of environment. There are six principal physiognomic types on land: forest, grassland, woodland (small trees, scattered or in small groups), shrubland (where shrubs dominate, and total plant cover is over 50%) and semi-desert scrubs (semi-arid communities with shrub and other plants forming sparse vegetation cover, often below 10%). In each physiognomic type, more than one biome-type could occur in response to major climatic differences. For example, alpine meadow, temperate grassland and tropical savanna are different biomes though all are grasslands with grass as dominant plants. Similarly, tropical rainforest, dry deciduous forest and temperate conifer forest are different biomes occurring in different climates. On the other hand, tundra communities are dominated by more than one growth-form; and their dominance varies with environment in a given area. Thus the tundra biome type is defined by environment as well as a range of physiognomic types.

The major terrestrial biomes along with climate and vegetation characteristics are summarised in Table 18.2. The characteristics of major biome types are given largely on the basis of Whittaker (1975). An attempt has also been made to give an idea of the range of variations within each biome. It is important to recognize that a variety of local ecosystems may occur within each biome.

1. Tropical Rainforests

These forests are found in humid tropics, where dry season (months with less than 100 mm of rain) is very short or absent (Fig. 18.3), and temperature shows a limited range within an annual cycle (around mean monthly temperatures 24-25°C). Throughout the year, days are of about 12 hours duration, unlike temperate regions where days are much longer in summer and much shorter in winter.

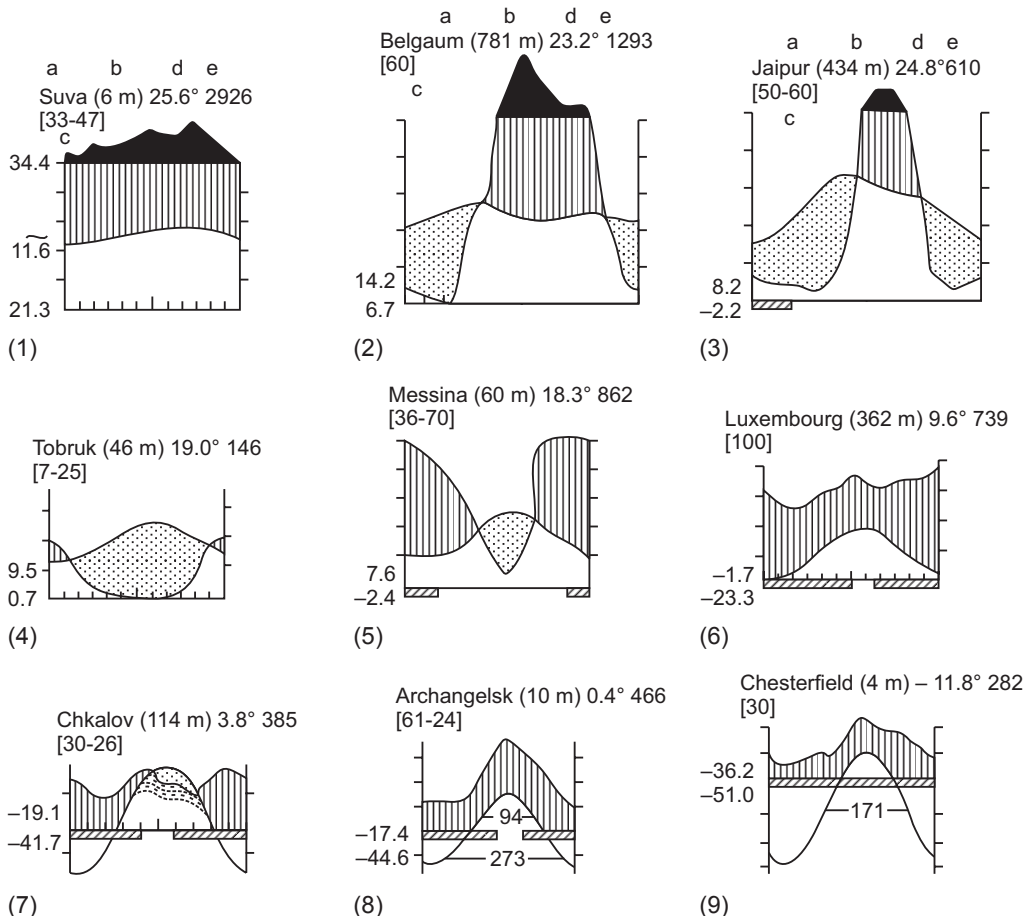


Fig. 18.3. The climatic diagrams for stations located in areas of 1. Tropical rainforest; 2. Tropical seasonal forest; 3. Thornwood; 4. Subtropical desert; 5. Mediterranean climate; 6. Temperate evergreen forest; 7. Temperate grassland; 8. Taiga forest; 9. Tundra. In climatic diagrams, the letters and numbers (shown for station (1 to 3), apply to all the stations) indicate the following : (a) station; (b) height above sea level; (c) number of observations of temperature and the second precipitation; (d) mean annual temperature in degrees centigrade; (e) mean annual precipitation (mm). (from Walter, 1979.)

South and Central America, Africa, Southeast Asia and North-Australia are the regions where these forests occur. The largest continuous areas of tropical rain forests are found in the Amazon basin of South America. In India, they are found mainly in the Western Ghats region. Ignoring recent volcanic soils and alluvial deposits, the soils are usually very old (up to Tertiary period); laterization is common and pH is in acidic range. Trees are tall (generally up to 50-55 m), of numerous species (40

to 100 tree species per ha is common), many with buttressed roots which may reach as far as 9 m up the trunk. Trees are smooth-barked with no bud protection, which is not required in a round-the-year favourable climate. Many species resemble one another, so much so that it is difficult to differentiate; they generally have very similar oval, medium sized leaves, with drip tip (Fig. 18.4) and entire margin. Three or more tree strata are distinguishable, with large woody climbers or lianas extending from the ground to the canopy; orchids, other epiphytes, and stranglers are abundant. Strangler figs are “hemiepiphytes”, which germinate on branches of trees, then rapidly grow down the trunk of supporting trees up to the ground and enmesh it. As the strangler fig roots thicken, they weaken the secondary growth of supporting host-tree, which is eventually strangled and killed. The canopy is concentrated with species, unlike the temperate forests, where more species occur in the undergrowth. Invertebrate fauna is very rich, striking in colour, sizes, and adaptations for defence by mimicry and other means. Mammals and reptiles are predominantly arboreal (living in canopy) and nocturnal. Birds are present in great diversity in the upper strata of the rain forest.

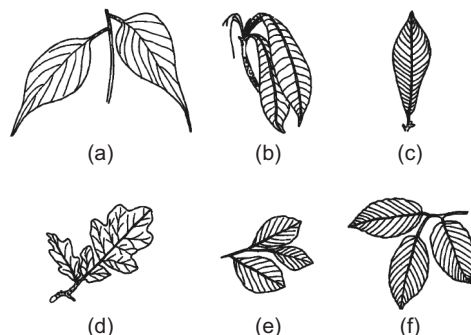


Fig. 18.4. The shapes of leaves in tropical (a – c) and temperate (d – f) forest ecosystems.

Where soil is nutrient-poor, oxisols, a large fraction of the ecosystem nutrients is stored in the plant biomass. The net primary production (NPP) often reaches $20\text{--}30\text{ t ha}^{-1}\text{ yr}^{-1}$. Decomposition is rapid, and nutrients, rapidly mineralised, may be immediately taken up by the surface root mat, having well-developed mycorrhizal association. Frequent tree fall, gap formation, and colonization of gaps by light demanding species make the system very dynamic. The tropical rainforests are being exploited for subsistence agriculture, timber harvest and extraction of non-timber forest produce. Timber harvest is a source of valuable timber like mahogany as well as fire wood.

Elfinwoods is a tropical biome occupying the sub alpine belt (above 2500 m elevation) of mountains of Africa, South America, and New Guinea. Because of high elevation, the climate is cold but because of the low latitudes, seasons are not sharply distinguishable; rain fog and clouds are constant features. Because of the latter, they are also called “cloud forests”. In elfinwoods, evergreen broadleaf shrubs or small trees show a characteristic thicket-like growth, with contorted branches and low canopies. Branches are festooned with mosses, lichens and ferns, and when dense, the hanging mosses make the stands visually and physically impenetrable. Thus, the overall impression is of a miniature rainforest, where luxuriance (impenetrable thicket-like growth) combines with stunted small trees and shrubs.

2. Tropical seasonal forests

These include monsoon forests, and several other forest communities, semi-evergreen or deciduous in leaf habit. They occur in Southeast Asia, India, south of Amazon basin in South America, Africa, on both sides of equator, Northern Australia and the West Indies. The annual rainfall is generally high, but majority of it (up to three-fourth or more) occurs during a short period of the late summer, generally from mid-June to mid-September in the monsoon climate (e.g. India), thus the dry season is quite long (Fig. 18.3). Depending upon the moisture availability, communities may be semi evergreen or totally leafless during the dry season. But species which shed leaves with the onset of dry season begin to leaf out and flower well before the end of the summer drought. Trees are usually 30–35 m tall, but become shorter in dry climate, where forests are also reduced to open woodlands. The NPP often reaches $15\text{--}20\text{ t ha}^{-1}\text{ yr}^{-1}$, and standing crop 200–500 t ha^{-1} . In India, these forests occupy extensive areas in the Central Highlands and Deccan Peninsula.

The tropical dry forests have infertile soil (Alfisols and Ultisols). The major land uses of this biome are timber harvest, grazing and agriculture. Due to overexploitation for a long time and population pressures, only a fraction of the tropical dry forests have remained in their original state. Less than 2% of the dry tropical forests remain intact in Central America and are one of the most threatened ecosystems of the biosphere (Janzen, 1988a).

3. Tropical Broadleaf Woodlands and Thornlands

These biomes are composed of small trees, which replace tropical seasonal forests in areas toward drier climates (and toward less favourable soil conditions). Cerrado that occurs extensively in Brazil includes a range of communities from woodlands with canopy typically at 4 to 7 m height to a thicket or scrub less than 3 m tall on arid soils of low fertility. Canopy may range from a closed one to open scrub, with or without scattered trees. Leaves of trees and shrubs are large, rigid, semi evergreen; branches are markedly twisted, and have thick fire-resistant bark. Spinose plants and succulents are generally absent, but palms and leguminous trees with small leaflets are usually present. In broad terms, Cerrado occupies intermediate position between seasonal forests (in more favourable conditions) and thorn woods (less favourable conditions).

Miombo, are evergreen and semi-evergreen woodlands and thickets, which occur extensively in the interior southern Africa, northern parts of South America, the West Indies and in southeast Asia.

Thornwoods occur in tropics toward more arid climates than those of seasonal forests and broadleaf woodlands. Widespread in South America as the Brazilian '*Caatinga*', thorn woods also occur in areas of Burma, India and Thailand, Africa and Madagascar, and on limestone in the West Indies and Middle America. Spiny species of *Acacia* and other genera which become leafless during the dry season are dominant and occur extensively. Succulent plants are also present, and increase in proportion towards the dry limit of the biome. The physiognomy ranges from thorn woodlands approaching forest in density and structure, through more typical woodlands of smaller trees with more open canopy, to dense scrub of large shrubs or shrub thickets.

4. Tropical Savannas and Grasslands

Savannas are tropical grasslands with scattered trees (called tree-savanna) or shrubs (called shrub-savanna), or without woody plants. Grasses of genera like *Panicum*, *Pennisetum*, *Andropogon* and *Imperata* form the dominant cover. Trees (mostly *Acacia* spp., and other legumes) that occur in savannas are totally different from those of the tropical rainforests. Savannas have attained their maximum expression in Africa, where world's richest fauna of grazers and browsers (e.g., zebra and giraffe) are found. Because of these hooved grazers, there also occur predators such as lions. Savannas also occur in Australia, South America and southern Asia, but communities are less rich and less extensive. In India, they interface with the dry deciduous forests and often represent degraded stages of these forests. Deep, loamy sand and summer rain less than 50 cm, down to as little as 20-10 cm favour the appearance of savanna. The savanna climate is too dry for forests. Savannas occur also in less arid climates, but only when fire is frequent or soil conditions unsuitable for forests. Roots of woody species penetrate deeper into soil than the grasses. Since the species must adapt to both drought and fire, species richness in savannas is far lower than in the tropical rainforests.

Annual fire in the vegetated areas and grazing by herbivores affect the balance between grasses and trees in the savannas. The forage quality of the grasses is low. The vegetation is grazed by both wildebeests as well as the livestock. There are annual migrations of enormous herds of wildebeests and other ungulates on the Serengeti Plain in Africa for the vitality and sustainability of the grassland systems. The large mounds build by some termite species are a prominent feature of most savannas. Termites play an important role in litter decomposition in the savannas.

5. Deserts

The *true deserts* are primarily subtropical, occurring generally in climates drier than those of semidesert scrub. In extremely arid climate, with precipitation below 12 cm per year, vegetation cover is very sparse and the bare ground surface (sand, stone and salt crust) vast. Plant forms range from typical low shrubs, to some very distinctive plants and lichens in areas with fog. True deserts are always poor; both in flora and fauna. The vast land area of Sahara in northern Africa, the Arabian Peninsula, the west coast of South America, and parts of Australia, India and southwestern United States are the typical areas of true deserts.

Warm semideserts occupy large areas also in South America (from Argentina through Chile to Peru), in surrounding areas of the Sahara in northern Africa, and in areas of Arabian Peninsula, Iran and the Thar desert of India. In this biome, as the data for North American semideserts indicate, annual precipitation is between 5-30 cm, and the summer mean maximum and minimum temperature are between 34-40°C and 19-26°C, respectively. *Warm semidesert* scrubs occupy dry warm-temperate and subtropical climates. An open scrub of creosote bush (*Larrea divaricata*) is most widespread in North America, occurring in both major parts of warm deserts, the Sonoran and the Chihuahuan from eastern Arizona to Texas and south through Chihuahuan, and in others.

6. Temperate Rainforest

These are giant forests occurring in cool maritime climate, with winter rainfall and occur at higher latitudes than tropical rainforest. A belt of mixed coniferous rainforests along the Pacific coast of North America from Oregon to Alaska is a typical example. They are called “rainforests” not because of the year-round rainfall, but because of the fogs which provide water to leaves in the dry summer. The redwood (*Sequoia sempervirens*) trees in temperate rain forests of the Pacific Coast (north-western United States) easily attain height of 60-90 m, with some trees up to 100 m. The 10 genera of largest trees are found in the Pacific North Coast conifer forests: *Abies*, *Chamaecyparis*, *Larix*, *Calocedrus*, *Picea*, *Pinus*, *Pseudotsuga*, *Sequoia*, *Thuja* and *Tsuga*. The Sitka spruce (*Picea sitchensis*) forests of Alaska are dwarf. Unlike the tropical rainforests, species richness is low. The aboveground standing crop averages 2,000 t ha⁻¹ (compared to 200-500 t ha⁻¹, generally found in other forests). Longevity is easily more than 500 years, and may exceed 1000 years. Other temperate rain forests such as *Nothofagus* (broad-leaf southern beech) of South America (pacific coast of southern Chile), and southern-hemisphere conifers in northern New Zealand (*Podocarpus*) are not that tall. Australian temperate rain forests of *Eucalyptus regnans*, are perhaps the tallest in the world.

Temperate evergreen forests show varied structure in response to varied climates they occupy. Species with evergreen sclerophyll leaves (tough, relatively small and broad leaves) dominate in less humid, maritime (summer-dry) climates in California, the Mediterranean region, and southern Australia. On the other hand, needle-leaved evergreen forests (conifers) occupy large areas in continental climates (summer-rainfall, with strong seasonal variations, that is characteristic to interior parts of continents of temperate region) of the western United States. Leaf area index (LAI) is as high as 16-40 in some forests. Douglas fir and white fir forests prevail in relatively much humid habitats and various pine forests are common in drier habitats. The mixed forest of Sierra redwood (*Sequoia sempervirens*) with pine, fir, and incense cedar (*Libocedrus decurrens*) in the Sierra Nevada is another representative of evergreen forest in the United States. In the eastern Asia, evergreen forests occur where soil conditions or fire frequency or both favour pines. In Chile, Southern hemisphere conifers dominate temperate evergreen forests, while in Australia *Eucalyptus* prevails. In India, temperate evergreen forests are found at high altitudes in the Himalaya and are dominated by conifers. The oak forests of the Himalaya are also evergreen.

7. Temperate Deciduous Forests

These forests are associated with a moderately humid (600-2000 mm precipitation), continental temperate climate with a marked (but not too prolonged) cold season. They are mostly

confined to the Northern Hemisphere. Minimum winter temperature may drop to -30°C or so, and the maximum may reach 38°C . Growth period is 4-6 months long, and clear-cut wet and dry seasons do not exist. Here leaflessness of broadleaf species (oak, maple, beech, ash, etc.) is adapted to cold, instead of drought. Species richness shows a wide range. While the mesophytic forests of southern Appalachians in North America and deciduous forests of eastern Asia (eastern China) are relatively rich in species, those of the Western Europe are species-poor. Tree height is typically 35 m, although at places trees can be as tall as those of the tropical rainforests (up to 50-65 m). These forests are relatively open permitting more light to reach the forest floor than temperate conifer forests. Life starts with snow-melt in March-April and ends in autumn, with rapid leaf fall. Geophytes are first to appear, subsequent to snow melt when trees are still leafless, and this may account for 18% of all species, which is three times of the Raunkiaer's world average. A high percentage of tree species have leaves with non-entire margins (i.e., serrated or lobed) (Fig. 18.4). Litter decay is relatively rapid, litter half-life being in the order of one year, resulting in more rapid mineral cycling. Net production of $13\text{--}15\text{ t ha}^{-1}\text{ yr}^{-1}$ is typical.

Several species bear pulpy fruits and nuts, such as acorns and beech nuts. The Virginia deer, bear, gray squirrels, gray fox, and wild turkey are the animals of the original forests of North America, Europe, and some of the other regions.

8. The Taiga Boreal Coniferous Forest

The most extensive conifer forests are found in this biome. They occupy the cold edge of the climatic range that can support forest on the earth. A synonym is boreal coniferous forest. The forest type commences at the point where climate becomes too unfavourable for broadleaf deciduous species or evergreen temperate forests described above, that is to say when summers become too short and winters too long, with number of days with daily average temperature above 10°C dropping below 120 days. The taiga extends around the world in the northern part of North America and Eurasia and extends to the south at higher elevations of mountains as subalpine forest. Soils largely belong to the spodosol soil order.

The dominant trees are evergreen spruces (e.g., *Picea glauca*), firs (e.g., *Abies balsamea*), larches (*Larix* spp.) or pines in most areas, however, in some areas deciduous trees of *Populus* and *Betula* are dominant. Species richness is low everywhere. The forest is close-canopied, with LAI of 9-11; tree size of about 20 m; and longevity 200-300 years. Shrubs and under-canopy trees are uncommon, but dwarf shrubs, club mosses (*Lycopodium*), true mosses and lichens may form 100% cover on the ground. Thus, the forest is essentially two-layered. The vegetation is reduced to open pine woodlands in arid mountains, bogs or muskegs in humid subarctic areas, woodland of scattered trees with tundra-like undergrowth in the far North, and wind-sheared shrub community (Krummholz) toward the timberline.

In general, trees are shallow rooted with predominant ectomycorrhizal associations, required for efficient nutrient uptake. Decomposition is very slow, and as much as 60% of all ecosystem carbon remains locked up in humus. Characteristic animals are moose, wolverine, lynx, and several kinds of snowshoe hares, finches and wood warblers.

9. Temperate Woodlands and Shrublands

These include communities of smaller trees than those of forests. However, communities range from nearly full canopy cover to open woodlands with scattered trees, which may eventually merge into treeless grasslands in drier conditions. Climate of the temperate woodlands is too dry for a true forest, and less dry for the occurrence of grasslands, shrubland and semidesert. Growth-form of trees also varies considerably, as needle-leaved sclerophyllous, and deciduous broad-leaved trees, and their combinations can be found. Pygmy conifer woodlands, dominated by pine and juniper, in western United States, oak woodlands in the Great Valley of California, evergreen oak and oak-

pine woodlands in the south western states of North America and Mexico are some of the major communities of this biome. Communities, similar in structure occur around the Mediterranean and in the southern continents.

Temperate shrublands include the shrublands with sclerophyllous leaves, which develop in climates of the Mediterranean type (moderately dry, warm temperate, maritime climates with little or no rainfall in summer season). The sclerophyll shrublands can have both arborescent shrubs (2-5 m tall) with a closed canopy, as well as shrubs less than 1 m tall and quite open, and several intermediate forms. Most familiar among these communities are the Mediterranean Sea *maquis* and Californian *chaparral*. The cape scrub or fynbos in South Africa, and those in Chile and west and South Australia are equivalent biomes. Frequent fires are a characteristic feature of these shrublands. Most of the aboveground plant parts are burnt off, and roots that survive the fire produce new shoots subsequently and canopy is formed again.

Chaparral like deciduous shrublands of some inland mountains (often treated as separate formation) is another important shrubland community of the temperate zone. Other similar temperate shrublands are aromatic shrublands of the drier Mediterranean climates, heaths mostly of cooler maritime climates, and distinctive shrublands in the Southern Hemisphere.

Scrub communities can be quite species-rich and diverse, as is the case with those occurring on mountain slopes and the upper margins of desert plains in North America. They may include prickly pear, *Acacias*, *Agave*, etc., or a tree woodland. Lizards and snakes are common fauna in chaparral, though mammals may also be many such as gray fox, wood rats, kangaroo rats, and small rodents.

10. Temperate Grasslands and Cool Deserts

Typically include the prairies and steppes, covering large areas in North America and Eurasia, respectively. Broadly speaking, they occupy areas where rainfall (25-75 cm per year) is intermediate between those of deserts and forests. The P/E ratio in the grassland drops below 1, and soils become deep and finer, and favour fibrous roots of grass. Climate is typical continental, with long, cold winters and hot summers. For example, in tall prairie of Illinois mean monthly temperature is -3.2°C in January and 24.2°C in July, but in short grass prairie the minimum can drop to -14°C or lower. Effect of soil moisture is well documented on the communities within the prairies. Along the declining moisture gradient from east to west in North America there occurs a gradient of communities; tall grass (1.5-2.75 m) prairie, mixed grass prairie, short grass (15-50 cm) prairie and bunch grass prairie. Productivity declines along this gradient. Roots of most species in all prairies go deep (up to 2 m), and root weight in climax perennials is several times more than that of shoots. Some of them, such as big bluestem (*Andropogon gerardii*), wheat grass (*Agropyron smithii*) and buffalo grass (*Buchloe dactyloides*) have underground rhizomes and are thus sod-formers. Fire has helped grasses to out compete woody plants that invade the grassland. Forbs (composites, legumes, etc.) are constantly present, but are always less important constituents of prairies. The warm season grasses have many C₄ species, while in cold season C₃ species prevail. Soil fauna, particularly earthworms are prominent, and play significant role in mixing the soil. Temperate grasslands also occur in Africa (*velds*) and Argentina (*pampas*). In comparable climate of Australia, however, there occur grassy woodlands with *Eucalyptus*.

Despite the limitation of vegetation to single principal layer, plant diversity is higher than in most temperate forests. However, presence of only one layer is the cause of lower bird diversity than in forests. Mammals are divisible in two principal types, the larger, running type, and smaller, burrowing type, such as ground squirrel and prairie dogs. Among large mammals, ecological equivalents in different continents include the bison and pronghorn in North America, the wild horse and ass and taiga antelope of Eurasia, the larger kangaroos of Australia, and the zebras, antelopes, and others of the warm-temperate grasslands of Africa.

Cool semideserts occur extensively in North America (Great Basin, between Rocky Mountains and the Cascades in the western United States) as cool semidesert scrub, and parallel communities are found in Central Asia and Iran, South America (Patagonia and Andes) and Australia. In North America, sagebrush (*Artemisia* spp.) is dominant and perennial grasses are also common. The biome may also include dry grasslands, but as a natural climax, the semidesert scrub occurs in climates drier than those of grasslands. Animals are black-tailed jack-rabbit, pronghorn, sage grouse, sage sparrow, and sage thrasher; and kangaroo rats are common jumping animals.

Arctic-alpine semi deserts are the biome of arid climates above timberline. Spiny cushion plants, succulents (in open growth), low, spiny cushion plants formed by *Astragalus* and other genera, and small sage brushes make communities of this biome. The semiarid areas of the Arctic are generally treated as part of Tundra, but they are more desert-like in physiognomy, plant adaptations and in sparseness of their animal life.

11. Tundras

The arctic tundra occurs in the regions where cold conditions throughout the year do not support the growth of trees. This condition is found in the Arctic (including Eurasia, North America, Greenland and Iceland) and Antarctica regions on northern and southern hemispheres. These are the treeless arctic plains (between forests to the south and polar ice cap to the north), with vegetation often characterized by varied and complex patterns of dominance by lichens, mosses, grasses and sedges, and dwarf shrubs. Usually, plants do not exceed 10 cm height because of low temperatures (mean minimum in winters often below -50°C) and short growing season (50-90 days). There are about 6 months in the year with a mean temperature above 0°C , and sometimes this period is as short as 55 days (Fig. 18.4). Strong winds ($15\text{--}30\text{ m sec}^{-1}$) and low evaporation, with P/E ratio much above 1 are other characteristic climatic features. Histosols (bog soils, and organic soils with more than 20% organic matter) and entisols, with little profile development are major soil orders. The top (15-60 cm) soil thaws in summer and freezes in winter, and beneath this soil is the permanently frozen parent material, called permafrost.

As expected, in this severely unfavourable climate, few plant species are able to survive. At wet sites where vegetation is most luxuriant the plant cover may be 100% but even there LAI is limited to 1-2. Belowground biomass is much more than the aboveground biomass, the ratio being 4.1. In many of the perennial forbs, aboveground parts are condensed into a strikingly compact canopy of xerophytic foliage, referred to as cushion. Some species are able to resume growth while still under snow cover. Flowers are variously coloured and attractive, insect pollination occurs in such species. Seeds are minute, less than 1 mg in three-fourth of species.

Muskox, caribou and reindeer (in Eurasia), polar bears, the arctic hare, and lemming are predominant mammals; longspurs, ptarmigans, snow bunting, snowy owls, and horned larks are characteristic birds. The majority of large vertebrates leave Tundra in winters; the birds migrate to the south, and only lemmings and arctic ground squirrels stay. Existing mainly upon the young buds of the *Cyperaceae*, lemmings remain active throughout the winters beneath the hard snow cover.

12. Alpine Tundras (Shrublands and Grasslands)

These systems occur in mountain areas above tree line, the upper limit of tree growth. Such areas called "alpine" typically support grasslands, though low shrublands also occur. In humid mountains of South America the alpine belt, called paramo has alpine shrubland with dominance of rosette shrubs in large areas, but there are also alpine grasslands, particularly in drier areas. In some cases, the rosette shrubs are arborescent, growing 3-5 m tall. Alpine tundras experience extreme temperatures, short growing season, thin soils, high wind and slow growth of plants. These regions contain many endemic species as the plants have evolved in isolation. They usually occupy humid areas of paramo, and decline toward higher elevations, but occasional presence up to

timberline can be seen. Shrublands, very similar in physiognomy occur in the African mountains. *Senecio*, a composite is a common shrub of the alpine shrublands. In the Himalaya, heaths such as *Rhododendron* are common and these are also found in parts of African mountains. Further, alpine and subantarctic shrubland, in New Zealand resemble large heaths, rather than the rosette shrub. In the case of *Rhododendron*, prostrate forms prevail, and in between these shrubs, plants of *Cassiope* and similar genera occur.

Alpine grasslands, as mentioned above, are the principal communities of mountains above timberline. In north temperate mountains, sedges are dominant in alpine meadows. With decreasing plant cover toward higher elevations, cushion forming species, often increase in dominance, exceeding sedges and grasses. In the Central Himalaya, tussock forming grasses (*Danthonia* spp.) usually prevail. Tussock forming grasses which occur in the zone of temperate climate extend up to alpine zone in New Zealand.

Generally, miniature shrub communities, of dwarf willows, rhododendrons (heaths) or other woody plants also occur along with meadows in alpine zones of different regions. Pikas, hoary marmot, mountain goat and sheep, and grizzly bear are common animals of the alpine zone. In winters, many of larger animals migrate to lower elevations, while smaller animals either hibernate or remain in shelter in meadows.

Major forest types of India are given in Box 18. 1, and major grassland types in Box 18.2.

Box 18.1. Major forest types of India (based on Champion and Seth, 1968).

I. Moist Tropical Forests

Group 1: Tropical wet evergreen forests; tall (≥ 45 m) dense, entirely evergreen, rainfall > 3000 mm, dominated by species of *Dipterocarpus* among others.

Group 2: Tropical semievergreen forests; dominants are deciduous, but evergreens predominate in lower or middle canopy, rainfall 2000-2500 mm, species of *Magnolia*, *Cinnamomum*, *Terminalia*, *Xylia*, etc.

Group 3: Tropical moist deciduous forests; dominants deciduous, lower storey trees usually evergreen, may reach 40 m height, rainfall 1200-2500 mm. May or may not bear teak (*Tectona grandis*), or sal (*Shorea robusta*).

Group 4: Littoral and swamp forests; mainly evergreen, include mangrove and freshwater swamp forests.

II. Dry Tropical Forests

Group 5: Tropical dry deciduous forest; entirely deciduous or nearly so, rainfall 800-1200 mm, some forests bear teak or sal, others are devoid of these species and have species of *Anogeissus* and *Terminalia*, etc.

Group 6: Tropical thorn forests; deciduous with low thorny trees, rainfall 200-800 mm, *Acacia*, *Balanites*, *Prosopis*, *Salvadora*, etc. predominate.

Group 7: Tropical dry evergreen forests; hard-leaved evergreen trees, rainfall 870-1200 mm, dominated by *Manilkara*, and *Memecylon*, etc.

III. Montane Subtropical Forests

Group 8: Subtropical broad-leaved hill forests; largely evergreen, rainfall 1000-3000 mm (Cherrapunji, 11000 mm), in southern India species of *Calophyllum*, *Rhododendron nilagiricum*, etc., and in eastern Himalaya, species of *Quercus* predominate.

Group 9: Subtropical pine forest; pine associations predominate in low altitudes of Himalaya, rainfall 1000-3000 mm, *Pinus roxburghii* (western Himalaya), *P. kesiya* (eastern Himalaya).

Group 10: Subtropical dry evergreen forests; low xerophytic forest and scrub in the foothills of western Himalaya, *Olea* and *Acacia* spp.

IV. Montane Temperate Forests

Group 11: Montane wet temperate forests; closed evergreen high forests, rainfall 1500->5000 mm, in southern India shola forests (*Ternstroemia*, *Ilex*, *Rhododendron*), in eastern Himalaya *Quercus* forests with *Machilus*, *Acer*, *Rhododendron*, etc.

Group 12: Himalayan moist temperate forests; all along Himalaya (1500-3300 m altitude), rainfall 1000-2500 mm, several conifers (e.g. *Abies*, *Cedrus*, *Picea*, *Tsuga*) and broadleaved species (*Quercus*, *Rhododendron*, *Acer*, *Ulmus*, etc.), composition differs with altitude and location (western vs. eastern Himalaya).

Group 13: Himalayan dry temperate forests; essentially open formations, along inner valleys of Himalaya, rainfall <1000 mm, conifers like *Pinus* (e.g. *P. gerardiana*), *Picea* and *Juniperus*, predominate.

V. Subalpine Forests

Group 14: Subalpine forests; stunted deciduous or evergreen forests with conifers (2900-3500 m altitude), mean annual temperature 2°C, rainfall 83-600 mm, species of *Abies*, *Betula* and *Rhododendron* predominate.

VI. Alpine Scrub

Group 15: Moist-Alpine Scrub, throughout Himalaya, above timber line to 5500 m altitude, species of *Juniperus*, *Rhododendron* predominate.

Group 16: Dry-Alpine Scrub, in low rainfall areas, upto 5500 m altitude, xerophytic, species of *Juniperus*, *Eurotia*, *Salix* predominate.

The variability in environment, and species composition and leaf characteristics led Singh and Singh (1987) to recognise 11 forest formation types in the Himalaya.

Box 18. 2. Major grassland types of India (Based on Dabadghao and Shankarnarayan, 1973)

- I. *Sehima-Dichanthium* type; covers Peninsular India, including the central Indian Plateau, the Chhota Nagpur Plateau and the Aravalli ranges; topography undulating, rainfall 300-635 mm; dominant component comprises 24 perennial grasses along with 129 herbaceous species including 56 legumes; on gravely soils and under protection or light grazing *Sehima* dominates and on level soils *Dichanthium* is the dominant grass.
- II. *Dichanthium-Cenchrus-Lasiurus* type; covers northern portion of Gujarat, whole of Rajasthan, western Uttar Pradesh and Punjab; topography broken here and there by spurs of the southern hill ranges and by sand dunes in western Rajasthan, monsoon rainfall 100-700 mm; eleven important perennial grasses with 45 herbaceous species including 19 legumes; *Cenchrus setigerus* and *C. ciliaris* form communities where soil is not too dry, and *Lasiurus indicus* on loose sandy soil in areas receiving less than 300 mm rainfall.
- III. *Phragmites-Saccharum-Imperata* type; covers parts of the Gangetic Plains, the Brahmaputra valley and extends to the plains of Punjab; topography level characterised by low-lying, ill-drained soils; 19 principal grass species and 56 herbs including 16 legumes, predominate.
- IV. *Themeda-Arundinella* type; appears around the altitude 350 m and extends up to 2100 m, rainfall 1000-2000 mm, snow fall in winter towards upper ranges; 16 perennial grasses (including species of *Bothriochloa*, *Chrysopogon* and *Cymbopogon*) and 34 herbaceous species including 9 legumes, are important constituents.
- V. Temperate-Alpine type; occurs above 2100 m in the northern montane belt and above 1500 m in the eastern belt, snow fall is common, 35 important perennial grasses (including species of *Agrostis*, *Danthonia*, *Phleum*, and *Poa*), 68 herbaceous species including 6 legumes are important vegetation components.

There is a variety of communities, with different dominants, in response to variability in environmental conditions including grazing, in each of the above type (Yadava and Singh, 1977; Singh and Gupta, 1992).

18.2 ZONOECONOMES, OROBIOMES AND PEDOBIOMES

Zonoeconome

The zonobiomes are not discrete units, one zonobiome gradually grades into another showing broad overlaps along the gradient of climate zones. Thus a gradient of zonobiomes can be visualised along the gradient of climate zones, each representing a continuum. The overlap between the zonobiomes, transitional zones between them, is known as *zonoeconomes* (ZE) (Fig. 18.5). To elaborate, an ecotone is an area of ecological tension, over which one type of vegetation is gradually replaced by another, e.g., deciduous forests by steppes in temperate regions; dry deciduous forests by thornwoods or savannas in tropical regions. In zonoeconomes, species from both types of vegetation occur side by side under the same general climatic conditions, and are in a state of keen and continuous competition. Depending upon the conditions of microclimate, relief, soil texture and others, vegetation types are replaced by one another, resulting in a diffuse mixture or a mosaic of the two types of vegetation. Across a zonoeconome, at first one kind of vegetation is better represented, then both types are roughly equally represented, and finally, the second type becomes prominent, with the first becoming increasingly sparse. Zonoeconomes can have, however, their own species, not found in either of the two flanking zonobiomes. Zonoeconomes are referred to by the zonobiomes they link. For example, ZE II/III is the zonoeconome linking the zonobiomes, tropical with summer rains and sub-tropical arid (desert climate). In Indian subcontinent a part of Rajasthan falls within this zonoeconome.

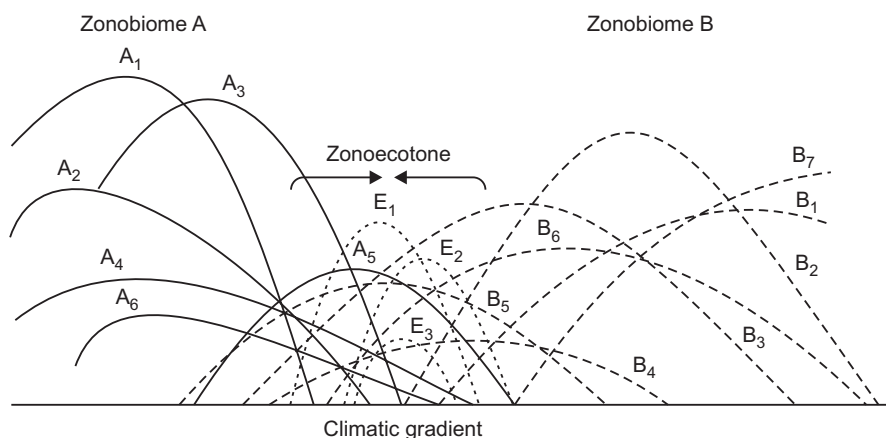


Fig. 18.5. A representation of zonoeconome, the transitional zone of two zonobiomes (indicated as A and B). In the zonoeconome species of both zonobiomes are represented (A1—A6 of zonobiome A and B1—B7 of zonobiome B). In addition, the zonoeconome may have its own species (E1-E3) not found in the either zonobiome. (based on Walter, 1979).

Orobiomes

We have until now, seen horizontal division of biosphere, generally along the gradient of latitude, but geobiosphere can also be divided vertically along the altitude of mountains. Mountains have different climates than the zones from which they arise, therefore, should be considered separately for dividing into biomes. The mountain biomes are called *orobiomes* (OB), and are identified by dividing mountains into altitudinal belts. Temperature declines altitudinally, as it does latitudinally. But the rate of decline is far more rapid along the altitude (Fig. 18.6). In Euro-North Asiatic region, the temperature decline with 100 m altitudinal rise is approximately equal to that occurs in 100 km shift from south to north. Thus, differentiation of vegetation zones is about

100 times more rapid in mountains than in plains. But the vegetation pattern resulting from the increase in altitude is not the same as that from the increase in latitude. It is because the gradients of environment in the two are different (Fig. 18.6). For example, day length does not change with the rise in altitude, but it does from lower to higher latitudes.

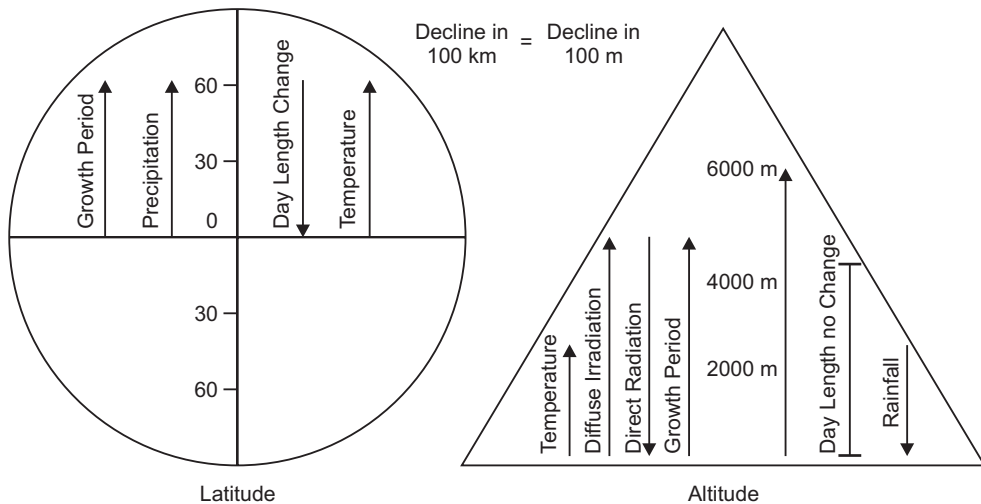


Fig. 18.6. Changes in environmental conditions along the gradient of latitude (shown for southern hemisphere) and altitude (mountains). Arrows indicate the direction of decline in various parameters.

In the Himalaya, there occur zonobiomes such as Arctic ($2B_1X$) which is very different from the zonobiome, tropical with summer rains prevalent in the plains of the subcontinent. Every mountain within a zonobiome is an ecological entity with a typical sequence of altitudinal belts. From lower to higher altitude, they are typically, colline (lowermost), montane, alpine and nival (highest, permanent snow). But the pattern differs from one mountain to another. For example, the pattern of altitudinal belts in zonobiome ZB-I (Equatorial with diurnal climate) is entirely different from that in the ZB-VI (typical temperate with short period of frost). For this reason, orobiomes are distinguished according to the zonobiomes they belong to, e.g., orobiome I, orobiome II, and so on.

Further, an orobiome may extend over more than one zonobiome, or may rise from the boundary between two zonobiomes. In other words, orobiomes may be uni- multi- and interzonal. The Himalaya is an example of interzonal orobiome (between the arid temperate with cold winter in the north and tropical with summer rain in the south). The Himalayan mountain ranges, in fact, form a barrier, creating a cold rain shadow areas in the north, which is much different from the areas of summer rain climate in the south.

Pedobiomes

The pedobiomes are associated with certain extreme types of soil. In this, soil is so much extreme or special that its effect overrides the climatic effect on vegetation. For example, the soil may contain salts (NaCl , Na_2SO_4) up to a harmful level to most species, or may have so little nutrients that vegetation differs from that typical to the zonobiome. It is possible that the influence of soil on the development of vegetation is so strong that two or three zonobiomes, differing sharply in climatic conditions may have similar vegetation in areas having an extreme soil type. This vegetation is also called *azonal vegetation*. Naming of the Pedobiomes is done according to soil type: *lithobiomes*, developing on stony soil, *halobiomes* on salty soils, *psammobiomes* on sandy soils, and *peinobiomes* on nutrient-poor soils (peina, Greek = hunger, lack).

18.3 FRESHWATER AQUATIC ECOSYSTEMS

Freshwater bodies may be conveniently considered in three categories as follows: (i) *Lentic* (Lentus = calm, sluggish) or standing water, which includes lake and ponds. (ii) *Lotic* (Lautus = to wash) or running water, which includes springs, streams and rivers. (iii) *Wetlands* which include marshes and swamps, where water levels frequently rise and fall, seasonally as well as annually.

1. Freshwater Lentic (Lakes and Ponds) Water bodies

Communities occur in a vast range of water bodies and nutrient conditions in diverse climates. Most of them are young in geological sense. Ponds may be seasonal with life span of few weeks and months, or perennial with age of several hundred years. Lakes date back as far as the ice ages. Lake Baikal of Russia and some others are very ancient.

Typically, three zones are distinct in ponds and lakes: (i) *littoral zone*, a belt close to shoreline, light reaching the bottom, supporting rooted plants; (ii) *limnetic zone* of open water, where light intensity is adequate and plankton (free floating microscopic algae and animals) are the dominant biota; (iii) *profundal zone* of deep water with light inadequate or absent, and where only heterotrophs occur (Fig. 18.7). The bottom of the water body is referred to as benthic zone. Lakes of temperate regions generally become thermally stratified due to differential heating and cooling. In the summer thermal stratification, the upper layers of water become warm and do not mix with cooler water of deeper layers, because of difference in the density of water. The cool water has higher density than the warm water. The warmer part of the lake with small temperature gradient, is called *epilimnion* (limnion = lake) and the cooler part of the lake is called *hypolimnion*. These two layers are separated by metalimnion, characterized by a steep decline in temperature and includes thermocline, that is the zone of sharp change in temperature, which checks the mixing of waters of the two zones.

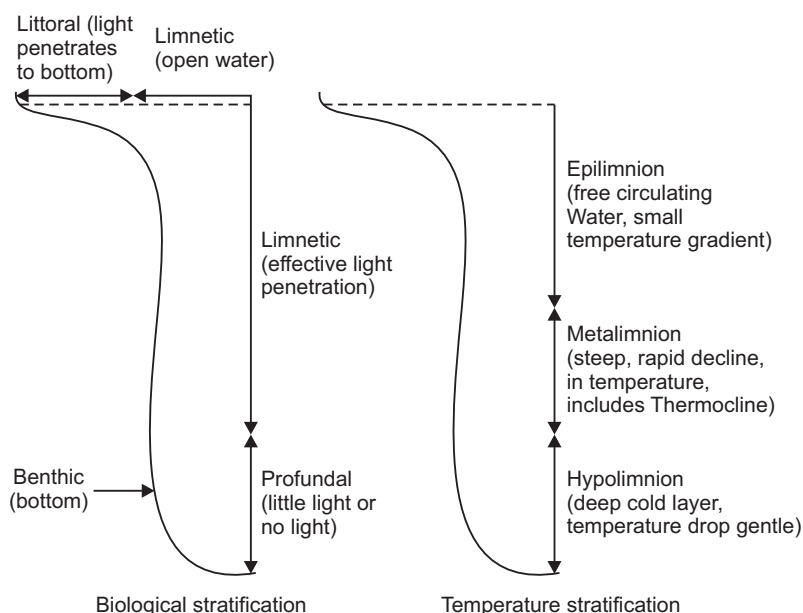


Fig. 18.7. Zonation and pattern of temperature and oxygen content of water along the depth of a lake.

Tropical shallow waters may exhibit poikilothermy (middle layer cooler) or dictiothermy (middle layer warmer). When the surface waters, rich in dissolved oxygen, attain a temperature of 4°C , they sink to the bottom and the bottom waters, rich in nutrients and dissolved CO_2 , move upward resulting into mixing of two waters. This is the phenomenon of **overturn**, and at this time thermocline disappears (Fig. 18.8). With respect to overturns, lentic water bodies are divisible into: (i) monomictic (one overturn per year, in polar regions during summer, and in subtropical regions during winter), (ii) dimictic (two overturns, one in spring and the other in fall, in temperate regions), (iii) polymictic (more or less continuous mixing, in high altitude lakes in equatorial regions), (iv) oligomictic (rarely mixed, in tropical regions), and (v) meromictic (permanently stratified due to salts liberated from sediments creating a permanent density difference between epilimnion and hypolimnion waters).

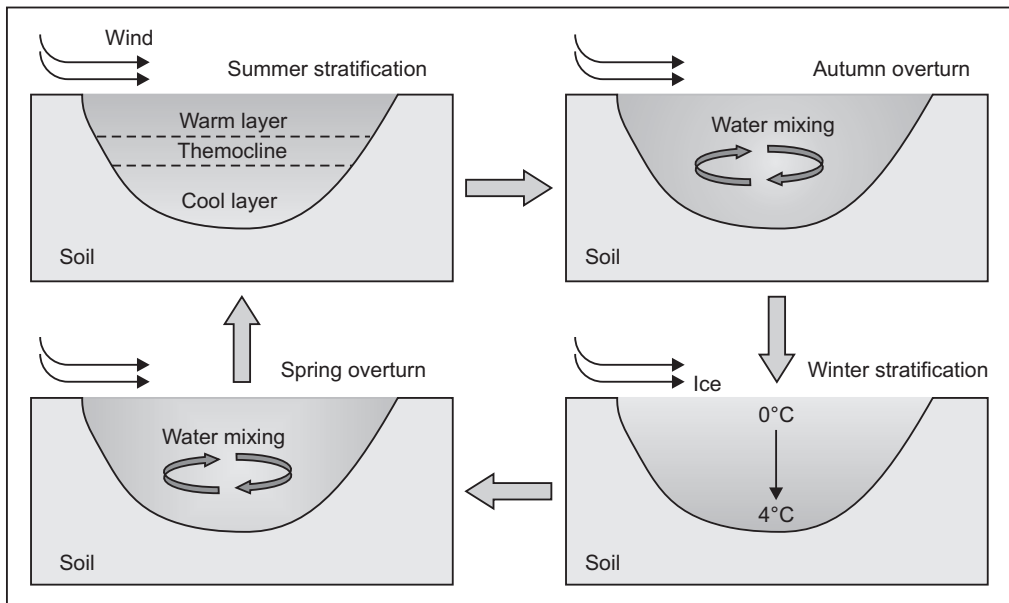


Fig. 18.8. Thermal stratification occurs in lakes. The seasonal mixing patterns of a temperate lake are determined by its temperature profile. In summer and winter, water is thermally stratified, while in autumn and spring water of all depths intermixes. This leads to autumn and spring overturns.

It is difficult to group communities of lakes and ponds into biomes because of many reasons including that climate is not the major factor of their development. Species diversity is generally low. Only in very ancient lakes (Lakes Baikal, Victoria, Nyssa, and others) flora is rich, which has developed over a long period of time.

2. Freshwater Lotic Water bodies

The lotic water bodies have markedly different biota from lakes and ponds. Lotic bodies, such as streams are exceptionally open systems, exchanging nutrients and energy with much larger areas, than the lentic water bodies. A stream of the forested area, for example receives large amounts of leaf litter, fallen wood and dissolved organic matter, from the adjacent forests (Fig. 18.9). In running water, currents are major controlling factor, and lack of thermal stratification is an additional factor which differentiates it from stagnant water. Generally, two zones are distinguishable within a stretch of stream: (i) a rapid zone, where current is fast enough to keep the bottom clear off silt and other loose material, consequently the bottom is firm; (ii) a pool zone, where current is slow, water is deeper, and sand and silt are deposited on the bottom. In the zone of fast currents, only specialized

species that become firmly attached or cling to hard bottom (e.g., caddis fly and black fly larvae) are able to survive. Among fishes, those which can swim against the strong current exist. In the zone of soft bottom, burrowing animals, rooted plants and swimming animals flourish. However, plankton occur only where water is deep and the current is gentle.

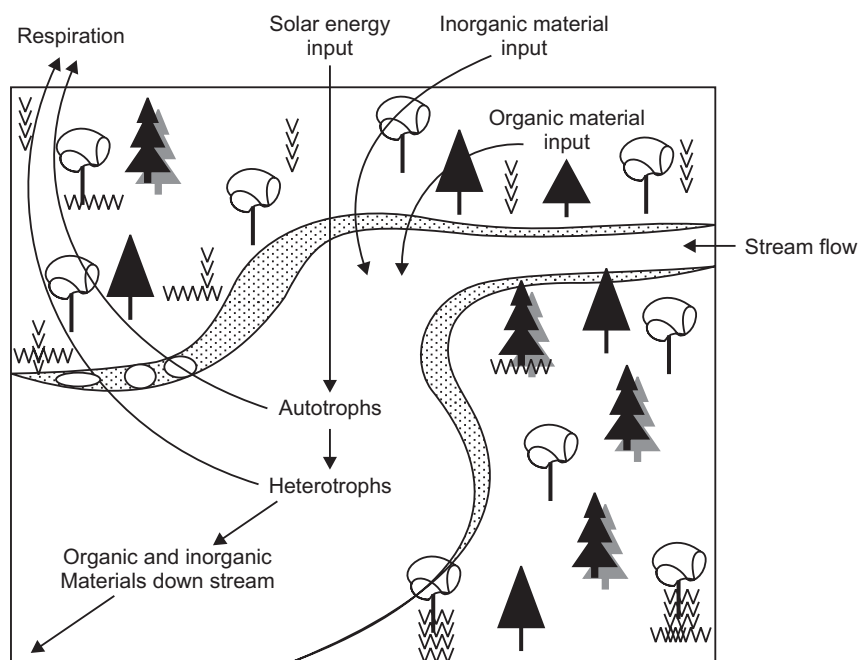


Fig. 18.9. Stream is an open ecosystem, receiving energy and nutrients from outside in various ways.

The flowing water systems are the most threatened ecosystems as most of the human civilizations have flourished along the major rivers in the world. Due to ever-increasing demands of the population for food and development, the degradation of the flowing water systems is increasing. The major threats to freshwater flowing water systems are habitat alteration (water diversion, impoundment and channelization), introduction of exotic species and pollution from both point and non-point sources.

3. Freshwater Wetlands (Marshes and Swamps)

The freshwater wetland communities occur in shallow areas of freshwater bodies or such areas which are inundated by water for at least part of the year. Soil is saturated with water continually. They lie at the interface between the land and open waters and include a wide spectrum of habitats, subjected to water level changes of such large magnitude and frequency that are not seen in terrestrial or true aquatic ecosystems. Generally, wetlands are also very open systems; and can be connected with rivers (riverine wetlands, located on low lying depressions and floodplains and lakes (lacustrine; locus = lake). Further, there are palustrine (paludis = marsh) wetlands, variously called marshes, bogs, fens, and temporary ponds that are located in depressions not directly connected with lakes or rivers. Following Shaw and Fredine (1956), Smith (1996) has divided wetlands, both freshwater and marine into 20 categories listed in Table 18.3.

Table 18.3. Freshwater and marine wetland types.

Types of wetlands	
(A) Inland fresh water areas	
1. Seasonally flooded basins or flats	5. Open fresh water
2. Fresh meadows	6. Shrub swamps
3. Shallow fresh marshes	7. Wooded swamps
4. Deep fresh marshes	
(B) Coastal fresh water	
9. Shallow fresh marshes	11. Open fresh water
10. Deep fresh marshes	
(C) Inland saline areas	
12. Saline flats	14. Open saline water
13. Saline marshes	
(D) Coastal saline areas	
15. Salt flats	18. Regularly flooded salt marshes
16. Salt meadows	19. Sounds and bays
17. Irregularly flooded salts marshes	20. Mangrove swamps

Gopal (1987) has grouped Indian fresh water wetlands into two major types depending upon the duration of water logging. (i) The perennial wetlands, with water logging throughout the year include habitats, such as river banks and margins of large lakes or reservoirs, and (ii) seasonal wetlands which dry up completely for varying periods of time depending upon the vagaries of monsoon. Examples are numerous, village ponds, fish ponds, paddy fields, river flood plains, etc. Biswas (1984) has reported 1193 wetlands in India covering a total area of 3.9 million hectares in 274 districts of India. Mangroves along the Indian coast cover a total area of about 356500 ha, of which more than half are in West Bengal and continuous with those in Bangladesh (Blasco, 1977). Wetlands support a large variety of flora and fauna. Besides planktonic and periphytic algae which are common in aquatic habitats, more than 300 species of macrophytes are found in Indian wetlands.

Wetland communities are highly productive, as they enjoy favourable conditions of both aquatic (abundant water) and terrestrial (light and nutrients) habitats. Though occupying only about 2 per cent of the area of the world, wetlands contain 10 to 14 per cent of the carbon storage (Armentano, 1980) and they are valuable habitats for a variety of breeding, wintering and migratory species of invertebrates, fish and wildlife (Mitsch and Gosselink, 2000).

18.4 MARINE ECOSYSTEMS

Following features of oceans are of major ecological interests: (i) oceans occupy 70 per cent of earth's surface; (ii) they are deep and life extends to all depths, often beyond 5,000 m depths (possibly there is no zone without life), though it is much more concentrated around margins of the lands; (iii) the oceans are continuous, they are connected with one another (not separated as are land and freshwater bodies), (iv) they have special chemistry with salinity (or salt content), 35 parts per 1000 parts of water (by weight) or 3.5 per cent are salt (Table 18.4), compared to 0.05 per cent in freshwater. The water is alkaline (pH 8.2) and is strongly buffered, that is resistant to change in pH; (v) the oceans are continuously circulating under the influence of strong wind, such as trade wind, and that caused by difference in water density (related to variation in temperature and salinity), (vi) tides produced by the pull of moon and sun, which are especially important factors in the shoreward zones, is the major feature. Zonation in the oceans is shown in a simplified manner in Figure 18.10.

Polar regions of the oceans are highly productive as upwelling brings nutrients to the surface. The nutrient rich waters support the growth of plankton which is eaten by small crustaceans known as krill. Krill is a principal food of the baleen whales. Thus, the plankton nourished by nutrient upwelling form the basis of an important marine food web. Tiny phytoplankton provide 50% of the oxygen on earth and form the basis of the ocean food chain up to fish and marine mammals, and ultimately human consumption.

Table 18.4. The chemistry of the sea water (adapted from Odum, 1983).

Parts per thousand (or grams per kg)	
Cations	
Sodium	10.7
Magnesium	1.3
Calcium	0.4
Potassium	0.4
Anions	
Chloride	19.3
Sulphate	2.7
Bicarbonate	0.1
Carbonate	0.007

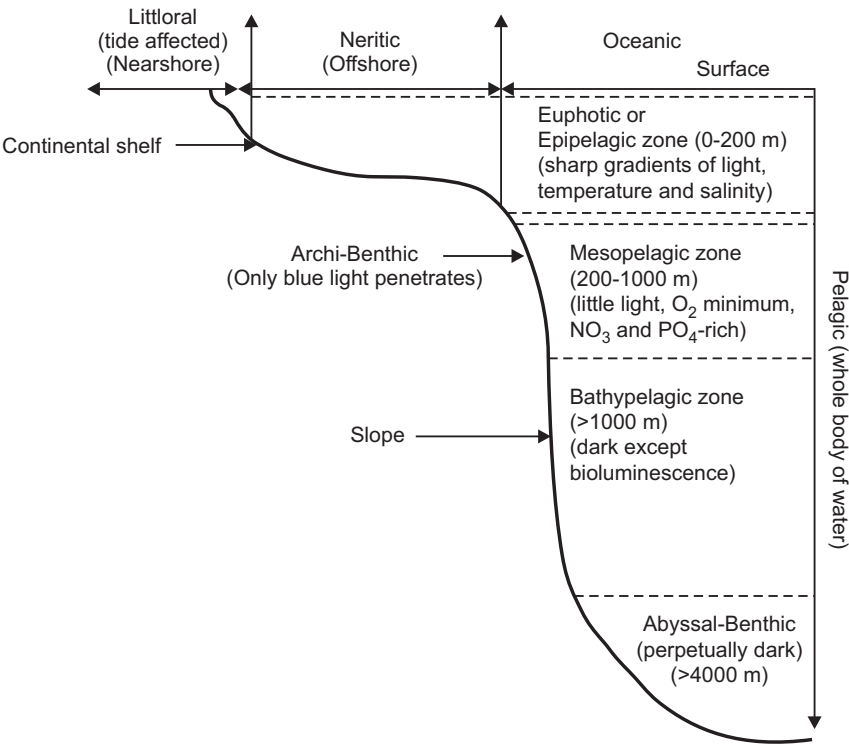


Fig. 18.10. A simplified representation of horizontal and vertical zonation in marine ecosystem.

The marine ecosystems are faced with principal threats due to pollution, over fishing, habitat destruction, invasion of exotic species and global climate change.

Marine communities at the land and sea interface

These include the littoral, marine sandy beach and marine mud flat communities. Estuarine, mangrove and coral reef communities are quite important ecologically as well as economically. Coastal systems such as mangroves, salt marshes and seagrass meadows have the ability to absorb, or sequester, carbon at rates up to 50 times those of the same area of tropical forest.

The littoral communities occupy coastal belts washed by waves and alternatively exposed and submerged by tides. The biota is characterised by barnacles, oyster, limpet, kelp (e.g., *Fucus*, *Laminaria*) or red algae (e.g., *Gigartina*). In India, seagrass beds are found along the coasts of Tamil Nadu, Lakshadweep islands, Andaman and Nicobar Islands and the Sundarbans. Eight hundred and forty four species of seaweeds are found in shallow waters all along the Indian coast particularly in Tamil Nadu, Gujarat, Goa, Maharashtra and Lakshadweep (Venkataraman et al. 2012).

Marine sandy beaches occur around the world in the littoral belt. The tides keep on submerging and exposing sandy beaches. The surface sand keeps on moving under the influence of tides. Thus, the environment is highly unstable and difficult. The organisms are alternately exposed (low tide) and submerged (high tide). Beach animals are largely dependent on plankton and detritus or dead particles brought in by river waters, though non-diatoms and other algae grow on the beach itself. Crustaceans and worms burrow in the sand, and microscopic organisms like copepods, rotifers and protozoan live in the water that fills spaces between sand particles. Fish and shrimps keep on moving up and down with water flow, and clams and snails occur at low tide levels.

Marine mud flat communities occur in bays and estuaries, where water is quieter, unlike on beaches where water keeps on moving up and down. However, like the beaches, animals and microbes of mud flats receive their food from outside as dead remains of salt marsh plants, and as plankton and dead particles from tides and river drainages that meet. However, primary production also occurs on mud flats, with microscopic algae in the upper surface and photosynthetic bacteria below that. Among the animals, snails occur on the surface of mud, and numerous forms of organisms, such as clams, distinctive worms, specialized crustaceans, and microscopic forms occur within the mud.

Estuaries, where fresh water of rivers meets the salty water of the ocean (resulting in brackish water, intermediate in salt content and water density), are complex ecosystems of high productivity, often combining salt marsh and mud flats with deeper water, and linking freshwater communities with those of oceans. Estuaries function on both planktonic-based and detrital-based food webs (see Smith, 1996). About 80 per cent of commercial marine fisheries depend in one way or other on estuaries (as a source of food, as a spawning ground, etc.).

Mangrove refers to an ecological group of halophytic plant species as well as to a variety of complex plant communities dominated by these species. The mangrove ecosystems are found in the inter-tidal zones of sheltered shores, estuaries, creeks, backwaters, lagoons, marshes and mud-flats, along the tropical coastlines of Asia, Australia and Americas.. along the tropical coastlines of Asia, Australia and Americas. The dominant salt-tolerant, sclerophyllous broadleaved trees form a unique ecosystem with associated plants, including epiphytic and terrestrial ferns, orchids, lichens, non-mangrove halophytes, sea grasses and seaweeds, and fauna such as fish, shrimp, shellfish, crabs, lobsters, reptiles and birds. These provide habitats, spawning grounds, nurseries and nutrients to a number of animals and harbour several endangered species ranging from reptiles (e.g., crocodiles, iguanas and snakes) and amphibians, to mammals (tigers, deer, otters and dolphins) and birds (herons, egrets, pelicans and eagles). The mangrove trees show a variety of adaptations such as

knee roots, prop roots, pneumatophores, vivipary, etc. and their trunks, leaves and roots provide habitats for diverse fauna. Many organisms colonise mangrove roots such as sponges, clams, algae and oysters. The coastal zone of the mainland of India and that of Andaman and Nicobar Islands is endowed with the presence of extensive and diverse mangroves. These ecosystems are found along both the east and west coasts of India. Important mangrove areas are in the Sundarbans, Bhitarkanika, Krishna and Godavari delta of Andhra Pradesh, Andaman and Nicobar Islands, Gulf of Kachchh, and the Pichavaram-Vedaranyam area of Tamil Nadu coast.

Coral reefs are built around coral, a coelenterate animal, related to the jellyfish and sea anemones in the warm waters of tropical and subtropical oceans. The animal forms a colonial structure embedded in a skeleton of calcium carbonate or its mass. Within the cells of the coral there occur numerous cells of symbiotic endozoic algae, dinoflagellates. Thus, the coral represents a partnership, with the algae providing food, and the coral giving structural support and nutrients to algae. The coral also feeds on plankton of water, and by doing so procures nutrients to be used by its partner, the dinoflagellates. By maintaining a highly efficient nutrient cycling and high level of nutrient conservation, the coral reefs attain high productivity. Like tropical rainforests, their species diversity and richness in form and colour are extraordinary. The coral reef communities include calcareous bryozoans, polychaete worms, molluscs (all occurring in coral mass), sea anemones, sponges, crabs, snails and echinoderms (occurring in abundance in association with the ecosystem). The variety is further enhanced by the varied colours of coral itself, and diverse array of fishes, which are brilliant and multicoloured, and blessed with colour warning signals and mimicry. The coral reef fishes are both herbivorous and predatory.

Coral reefs are usually categorised into: (i) barrier reefs (which parallel shorelines along continents; e.g. the Great Barrier Reef of Australia), (ii) fringing reefs (around islands), and (iii) atolls (horseshoe-shaped reefs surrounding a lagoon). In India, reef building corals occur at Palk Bay, Gulf of Mannar near Sri Lanka, Gulf of Kutch, central west coast, Andaman and Nicobar Islands, Lakshadweep islands (Laccadives), and Gaveshani bank some 100 kilometres off the coast from Mangalore.

Coral reefs are the most productive and diverse ecosystems of the world. About 58% of the world's coral reefs have been degraded due to human activities such as coastal development, overexploitation and pollution from marine and land based resources (Bryant et al., 1998). Areas of greatest species richness have experienced greater impacts than the less species-rich regions. The species rich marine areas on the basis of diversity of reef fish, corals, snails, and lobsters have been designated as biodiversity hotspots (Roberts et al., 2002). These hot spots are: Philippines; Gulf of Guinea (off the west coast of Africa); Sunda Islands, Indonesia; southern Mascarene Islands (Indian Ocean); eastern coastal waters of South Africa; northern Indian Ocean; waters of southern Japan, Taiwan, and southern China; Cape Verde Islands (off West Africa); western Caribbean; and the Red Sea and Gulf of Aden. There is an urgent need to protect these most threatened ecosystems by creating marine reserves under the existing provisions on International law (Roberts et al., 2002).

Continental shelf (Neritic) Benthos Communities

The neritic benthos communities occur below the littoral zone on the bottom. These communities also change, with changing water depth. In shallower waters, light reaches up to the bottom, which may be sandy with silt and clay depositions brought by rivers. Here temperature changes with seasons. Availability of light leads to the development of extensive algal communities, ranging from giant kelp “forests” through smaller communities of green, brown or red algae, to a film of unicellular algae (*Phytoplanktons*). Rich fauna of fish, crustaceans and others live and feed there. The depletion of light leads to development of different flora and fauna complements on the deeper bottoms.

Marine Surface Pelagic Communities

These include plankton and the larger swimming animals that occur in open ocean waters. They may be separated into surface pelagic and deeper-water pelagic communities. In the surface pelagic zone, light supports photosynthesis by the phytoplanktons such as diatoms and dinoflagellates. Zooplanktons are mostly copepods that feed on the phytoplanktons. Apart from these, marine zooplankton includes a diverse mixture of other animals, such as shrimp like euphausiids (krill), arrow worms, comb jellies, tunicates etc.

The marine deep pelagic communities occur below the lighted surface waters. Obviously, they are heterotrophic, depending largely for food on the settling of plankton and dead particulate organic matter from the surface waters. Many of the fish and crustaceans which swim between the surface and deeper waters ingest and excrete organic matter to be used by communities of deeper waters. In this dark pelagic zone, carnivorous copepods and other crustaceans occur abundantly, along with colonial bacteria. In the deepest part of the oceans, animals may be without functional eyes and may have light-producing organs or luminescence.

The Deep Open Benthos Communities

These extend from the boundary of continental shelf to the deepest oceans. In the absence of light, plants are absent and the communities are entirely heterotrophic, consisting of animals living on and in the mud surface, and bacteria. The mud consists mainly of the skeleton of organisms or red or brownish clay of volcanic origin. Some of the animals are able to rise above the mud, while others are embedded in the mud, and there are still others which are able to swim above the mud. The deep ocean benthos are rich in biodiversity though strange, occupying permanent cold and dark water surviving under intense pressure and scarcity of food that moves down from the distant surface waters.

The oceanic systems deeper than 1000 m, cover about 62% of the planet. Unregulated fishing in the deep oceans is a cause of great concern as it is posing a major threat to the deep-ocean life. The World Wide Fund for Nature and the World Conservation Union in the recently published report have stressed the need for protecting large areas of sea (WWF/IUCN, 2001; Roberts, 2002). It is possible to protect large areas of deep sea by creating marine reserves within the Framework of International Law of the Sea (WWF/IUCN, 2001; Roberts, 2002).

The Deep Sea and Hydrothermal Vents

The abyssal plain of the oceans has undulating topography and exhibits high species diversity. Hadal trenches, the deepest places on Earth, plunge to as deep as 11,000 metres below the surface of the sea. First seen in 1977, hydrothermal vents occur in volcanically active areas of the seafloor like mid-ocean ridges, where tectonic plates are pushing and pulling above magma hotspots in Earth's crust. In these areas super-heated gases and chemically rich water erupt from the ground at temperatures of up to 400°C. Microbial organisms are able to withstand these extreme temperatures to create energy from the chemical compounds being forced up through the vents, particularly hydrogen sulfide, via a process called chemosynthesis. Some of these microbes live symbiotically inside tubeworms, while others form large mats, which attract progressively larger organisms that graze on them. So far, over 500 species that live only at hydrothermal vents have been discovered; it is possible that these communities are the oldest ecosystems on Earth and the place where life began (SCBD, 2012).

Part – IV

ENVIRONMENTAL POLLUTION

- ♦ AIR AND NOISE POLLUTION
- ♦ WATER POLLUTION
- ♦ SOIL POLLUTION – METALS, PESTICIDES, ISOTOPES AND WASTES
- ♦ ECOTOXICOLOGY

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CHAPTER

19

Air and Noise Pollution

The atmosphere is a mixture of gases that forms about 400 km thick layer around the earth. The lower portion of the atmosphere (mostly troposphere), extending from 8-16 km height from the earth surface, contains more than 90% of the gases and is particularly affected by air pollution. In the troposphere, the air mass is constituted of approximately 78% nitrogen and 21% oxygen and the remaining 1% is carbon dioxide and several other trace gases (argon, neon, helium, methane, krypton, hydrogen, nitrous oxide, carbon monoxide, xenon, ozone, ammonia, nitrogen oxide, sulphur dioxide and radon).

Broadly speaking, air pollution is the alteration of atmospheric chemistry by air pollutants from natural and anthropogenic sources. The emission of SO_2 from volcanoes, O_3 from lightning, and particulates and CO_2 from naturally occurring fires, etc. are the natural sources of air pollution. Anthropogenic sources of air pollutants are from both mobile (the transport vehicles) and stationary sources (e.g., power plants, metal smelters, and industry, etc.). Air pollution is a complex problem because numerous chemical substances are involved, which occur in all three states, gaseous (about 90% of total weight of air pollutants), liquid and solid.

Air pollution thus is the degradation of air quality and natural atmospheric conditions due to addition of gaseous and particulate air pollutants from both natural and anthropogenic sources. Air pollution cannot be regarded as a local phenomenon, the geographic extent of air pollution has been found to involve vast area due to transboundary movement of air pollutants. The interaction of many pollutants can occur simultaneously. Air pollution effects interact with other environmental stresses like drought, nutrient deficiency, herbivory leading to the problems of multiple stresses.

The acute exposure of pollution was the focus in the past, but chronic form of air pollution is now a bigger threat to human health. The potential for cumulative effects of low but chronic exposures (low and continuous) to pollutants is high. The effects of certain pollutants on soil chemistry and ecosystem nutrient levels are known to be significant. The scales of pollution range from individual leaves to whole plant canopies and the ecosystems (Taylor et al., 1994). Air pollution has direct or indirect consequences on human health, physiology and growth of plants and animals, and structure and processes of ecosystems (see Winner, 1994).

19.1 AIR POLLUTION

The burning of fossil fuels (oil and coal), and destruction of forests are the main causes of air pollution. Air pollution sources are divisible into two categories: (i) mobile sources, and (ii) stationary sources. The mobile sources of air pollution are plentiful, and widely dispersed. Sources of major air pollutants are listed in Table 19.1. Transportation, electric power generation and industry, incineration of garbage, and material processing and fire are the greatest air polluters. Barry Commoner (1971) in his book *The Closing Circle* points out that modern technology pollutes more than the technologies it replaces.

Table 19.1. Sources of major air pollutants.

Pollutants	Sources
Sulfur oxides SO _x	Combustion of coal, oil and other sulphur containing fuels, petroleum refining, metal smelting, paper production, Industrial processes.
Nitrogen dioxide NO ₂	Incomplete Combination of atmospheric nitrogen and oxygen, at high combustion temperature (automobile engines), by-products in fertilizer manufacturing, combustion of natural gas and coal.
Hydrogen sulphide H ₂ S	Metal smelting, oil-wells and refineries
Carbon monoxide CO	Combustion of fuels such as gasoline (mostly automobiles)
Particulate matter	Fuel combustion, industrial processes, construction, forest fires, refuse incineration, automobile traffic, etc.
Volatile organic compounds	Evaporation of organic solvents, industrial processes, transportation
Hydrogen Fluoride	Aluminium production, ceramics, coal gasification, burning of coal in power plants
Lead	Combustion of leaded fuels, paints with lead, lead-smelting
Mercury	Manufacturing of paper, chemicals, and paints; pesticides, fungicides

Transportation, the mobile source of air pollution

The combustion of gasoline and other hydrocarbon fuels in automobiles, trucks, and jet airplanes produces several primary pollutants: nitrogen oxides, gaseous hydrocarbons, and carbon monoxide, as well as large quantities of particulates, chiefly lead. In terms of weight, automobiles generate two third of human caused air pollution. Transportation as a whole, accounts for 70% of CO, 44% of oxides of nitrogen, 33% volatile organic compounds, and 21% particulate matter annually generated into air. In urban areas of India, vehicular pollution is of particular concern as these are ground level sources of pollution affecting the health of people. The number of motor vehicles in the world has increased from 0.3 million in 1951 to 37.2 million in 1997 and more than one billion in 2010. The number of vehicles in China increased 20 times from 2000 to 2010. The second largest growth rate was in India which had 20.8 million cars in 2010, compared to 78 million in China and about 240 million in the USA. Carbon monoxide (CO) and hydrocarbons (HC) account for 64% and 23% of the total emission, respectively. In Delhi alone, the daily pollution load has increased from 1450 to 3000 tonnes during 1991-1997 (MoEF, 1997; UNEP, 2001a).

The Stationary Sources

The principal stationary sources of air pollution include thermal power stations, various industries, office buildings and homes. The massive combustion of coal in thermal power stations and smelters generates tons of ash, soot and other particulates of carbon, silica, aluminium and iron oxides besides oxides of sulphur and nitrogen, hydrogen fluoride and carbon monoxide. Assorted hydrocarbons and metals such as lead, mercury, cadmium, selenium, vanadium, and zinc are also produced from factories. Air pollution from cities also affects the rural areas. Petroleum refineries are responsible for hydrocarbon and particulate pollution. In India, industries such as iron and steel, thermal power plants, copper/zinc/aluminium smelters, fertilizers, pesticides, pulp and paper mills, cement and oil refineries are major stationary sources of pollution (CPCB, 2000a). From these industries, the emission of SPM (suspended particulate matter) has increased considerably. The petroleum refineries, textiles, pulp and paper and industrial chemicals account for about 87% of SO₂ and 70% of NO₂ emissions.

Other sources

Solid waste disposal has increased considerably with our dependence on packaging of products, ranging from food to toys. Both packing and disposal of packages generate pollution. Package is often disposed of by incineration, releasing combustion products of packages into the atmosphere. Incineration of solid wastes emits CO, aldehyde, hydrocarbons, particulates, etc. Incineration and open burning of plastic wastes contributes to particulate, volatile organics and CO emission.

Un-insulated high-voltage power lines ionise the adjacent air and produce ozone and other hazardous pollutants. Insecticides, herbicides, radioactive fallout and dust from fertilizers, mining operations and livestock feedlots are other sources of air pollution.

19.2 TYPES OF AIR POLLUTANTS

The air pollutants can be conveniently divided into two types, i.e., primary and secondary. A *primary air pollutant* is a harmful chemical that is directly added to the air, e.g. carbon monoxide (CO) and sulphur dioxide (SO₂). When carbon containing substances such as coal, oil and natural gas are burnt partially, CO enters the air ($2C \rightarrow O_2 = 2CO$). Burning of oil and coal containing sulphur impurities releases SO₂ ($S + O_2 \rightarrow SO_2$). The examples of trace gases and their concentration in polluted and non polluted air samples are given in Table 19.2.

Table 19.2. Examples of the trace gases, and their concentration in polluted and unpolluted air samples.

Gas	Concentration (μ mol/mol)		Pollution sources
	Unpolluted air	Polluted air	
CO ₂	350	700	Fossil fuel* combustion, deforestation, etc.
CO	0.02	35	Fossil fuel combustion
O ₃	0.01	0.12	Phytochemical reaction with fossil fuel combustion and other gases
NO ₂	0.001	0.05	Fossil fuel combustion
SO ₂	0.0002	0.14	Fossil fuel combustion

* Fossil fuels include oil and coal.

Secondary air pollutants are harmful chemicals formed due to reaction of primary air pollutants with other air components. For example, SO₂ reacts with O₂ to produce sulphur trioxide ($2 SO_2 + O_2 \rightarrow 2SO_3$). The SO₃ can then react with water vapour of air to produce droplets of sulphuric acid, another secondary air pollutant ($SO_3 + H_2O \rightarrow H_2SO_4$).

‘Criteria air pollutants’ are those air pollutants that have been regulated and are used as indicators of air quality. The regulations or standards are based on criteria that relate to health and/or environmental effects. Criteria pollutants are the only air pollutants with national air quality standards that define allowable concentrations of these substances in ambient air. The six pollutants recognized by the United States Environmental Protection Agency (EPA) as criteria pollutants are ozone, carbon monoxide, nitrogen dioxide, particulate matter, sulphur dioxide and lead. Air pollution standards set by the EPA for the four major criteria or primary pollutants (CO, O₃, SO₂ and NO₂) are given in Table 19.3.

Table 19.3. Air pollution standards set for criteria pollutants by the US Environmental Protection Agency (EPA). P = Primary standards are set to protect human health, and S = secondary standards, are set to protect human well being, including vegetation.

Pollutants	Averaging period (hour)	Concentration (μ mol/mol)*
SO ₂		
Primary standard	24	0.14
Secondary standard	3	0.50
CO		
Primary standard	1	35.0
Secondary standard	8	9.0
O ₃		
Primary and Secondary	8	0.03
NO ₂		
P and S	Annual mean	0.053

* Annual maximum not to be exceeded more than once a year.

Three criteria pollutants, SPM [suspended particulate matter and RPM (respirable particulate matter)], SO₂ and NO₂ have been used for monitoring air quality in various cities in India (CPCB, 2000a). The range of concentration of these pollutants in major cities of India is shown in Figs 19.1 to 19.3. The suspended particulate matter (SPM) is one of the most prevalent air pollutants in most cities and often exceeds the permissible limits.

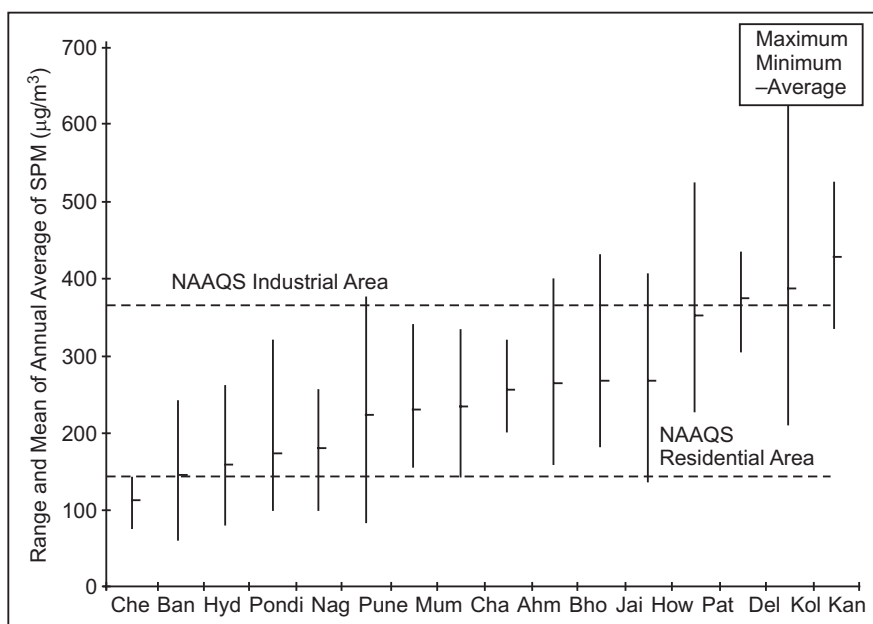


Fig. 19.1. Range and mean of annual averages (1990-1998) of SPM in various cities of India. Che: Chennai; Ben: Bangalore; Hyd: Hyderabad; Pondi: Pondicherry; Nag: Nagpur; Pune: Pune; Mum: Mumbai; Cha: Chandigarh; Ahm: Ahmedabad; Bho: Bhopal; Jai: Jaipur; How: Howrah; Pat: Patna; Del: Delhi; Kol: Kolkatta; Kan: Kanpur. (source: CPCB, 2000a).

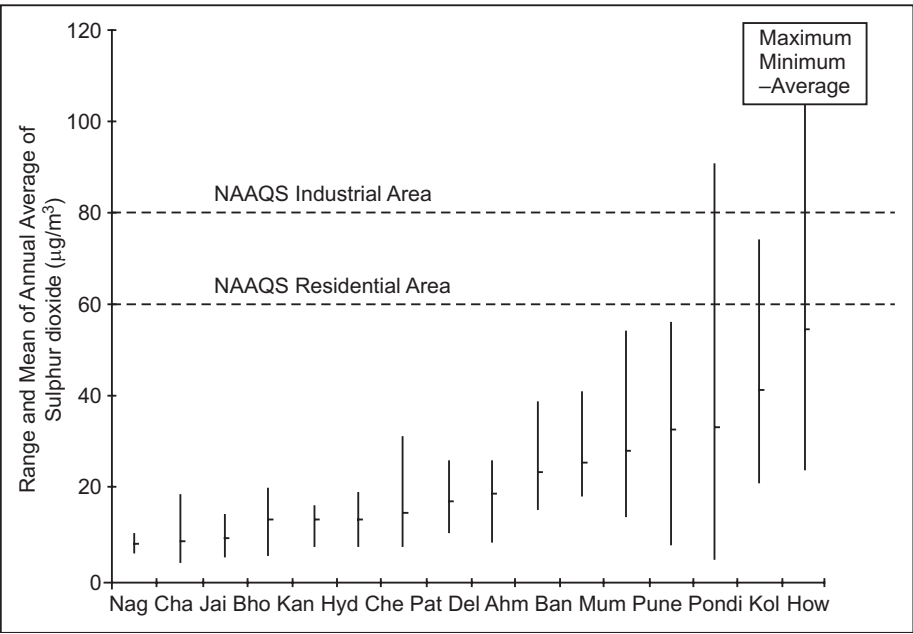


Fig. 19.2. Range and mean of annual averages (1990-1998) of SO_2 in various cities of India. Che: Chennai; Ban: Bangalore; Hyd: Hyderabad; Pondi: Pondicherry; Nag: Nagpur; Pune: Pune; Mum: Mumbai; Cha: Chandigarh; Ahm: Ahmedabad; Bho: Bhopal; Jai: Jaipur; How: Howrah; Pat: Patna; Del: Delhi; Kol: Kolkatta; Kan: Kanpur. (source: CPCB, 2000a).

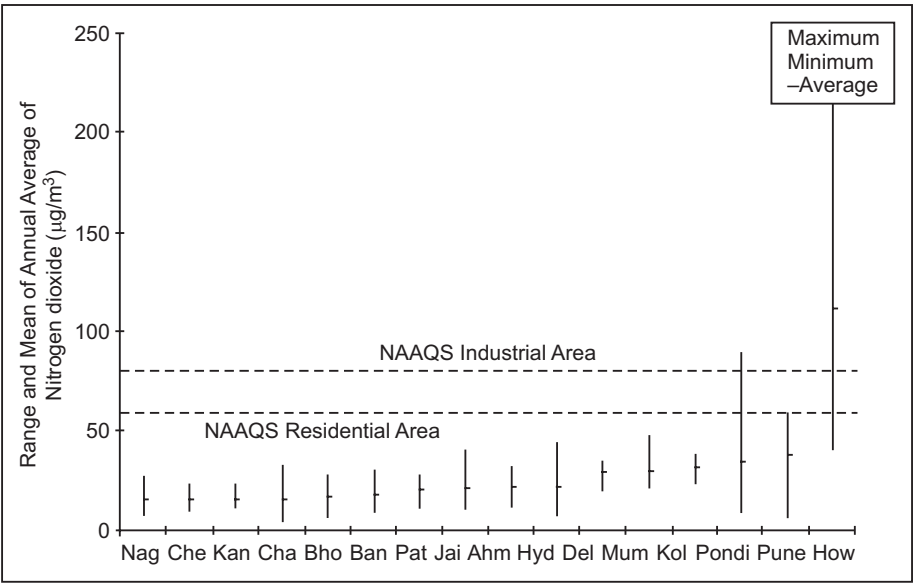


Fig. 19.3. Range and mean of annual averages (1990-1998) of NO_2 in various cities of India. (from CPCB, 2000a).

The standard for particulate matter less than 10 μm in aerodynamic diameter (PM_{10}) (i.e., respirable particulate matter) as specified by EPA (on 14 December 2012) is 24-h average concentration of 150 $\mu\text{g}/\text{m}^3$. The EPA also concluded that particles with aerodynamic diameter less than 2.5 μm ($\text{PM}_{2.5}$) had a greater association with mortality and morbidity rates than PM_{10} .

and established an annual $PM_{2.5}$ standard level of $15 \mu\text{g}/\text{m}^3$, and a 24-h $PM_{2.5}$ standard level of $35 \mu\text{g}/\text{m}^3$. Particles between 2.5 and $10 \mu\text{m}$ get deposited, in the upper respiratory tract. Particles with aerodynamic diameter smaller than $2.5 \mu\text{m}$ ($PM_{2.5}$), also called fine particles, correspond to the respirable particle fraction that is capable of penetrating the alveolar region of the lung.

Besides the “criteria pollutants”, other primary air pollutants in the atmosphere are CO_2 , NH_3 , CH_4 , non-methane hydrocarbons, H_2S and nitrogen oxides, hydroxyl radicals (OH), some heavy metals (e.g., Hg), volatile organic hydrocarbons (VOCS).

Aerosols, Particulate Matter and Black Carbon

Aerosol refers to the particulate/air mixture, as opposed to the particulate matter alone, however, the term aerosol has also been used for the particulate component alone. Aerosol particles could be solid, liquid or mixed, and smaller than $20 \mu\text{m}$ in diameter. Dust which has larger particles ($20\text{--}30 \mu\text{m}$) is not included in it. The term is now customarily applied to deposits of particulate matter, as collected on filters or particle size separating impactor plates. Soil erosion is the main source of solid aerosol particles, and sea spray from the ocean surface is the main source of liquid aerosol. In this, sea salt crystals are formed in an aqueous solution upon evaporation. All aerosol particles have some moisture and they can grow into cloud or fog droplets when vapour pressure of water exceeds saturation point. Such aerosol particles are called cloud condensation nuclei, but fogs and clouds are generally not included in aerosols.

Aerosols which are directly injected from earth and water surface in the atmosphere are called primary aerosol, and those which are formed after chemical conversion in the atmosphere from pre existing aerosols are called secondary aerosols. Solid or liquid particles suspended in either gas or liquid medium are called particulate matter. They are transported to long distances. Usually the term applies to solid, however, liquid particles suspended in gases are also particulate matter.

Particulate pollution includes many kinds of things – sulphate salts, sulphuric acid droplets, metals like lead or oxides of iron, dust from finely divided particles of carbon or silica, liquid sprays and mists. Particulates occur as sprays, mists, and dusts from spraying and grinding activities, land clearing, and building of highways. Soot and fly ash are emitted from power plants and factories. Forest and agriculture fires are also major sources of particulates. Gases in the air react to produce secondary particulate particles, both solid and droplets. Then substances act as nuclei onto which other materials condense to produce new chemical entities. Hydrocarbons, for example, react with oxidants in the atmosphere to produce peroxide radicals, that eventually form large molecules, which in turn join together to form small droplets or solids.

Particulate matters constitute about 5% of the weight of all air pollutants. In size, particulate matters range from 0.005 to $100 \mu\text{m}$ in diameter (μm or micrometer is one millionth of one meter). Dust, which constitutes particles of large sizes accounts for half the total mass of particulate matter. Being coarse (larger than $2.5 \mu\text{m}$) and heavy, the dust particles settle out of the atmosphere quickly, and otherwise do not get to delicate lung tissue. As stated earlier fine particulate matter with less than $2.5 \mu\text{m}$ diameter (as defined by the EPA) is a considerably greater health problem. In fact, particles smaller than $2 \mu\text{m}$ diameter are not effectively removed even by rain, and can be transported over long distances. Continuous exposure to particles smaller than $2.5 \mu\text{m}$, called $PM_{2.5}$ is known to cause diabetes and cardiovascular diseases. Particulate matter from diesel exhaust can cause biological responses related asthma. Smaller particles (diameter less than $1.0 \mu\text{m}$) remain suspended in air for weeks and are hazardous to health because they can easily reach non-ciliated portion of lung, and even alveoli.

Anthropogenic aerosols such as dust, smoke, fly ash, and soot which result from incomplete combustion are recognized as air pollutants with severe consequences on environment and human health. The anthropogenic emissions of these pollutants, particularly of black carbon (BC), have

increased across the Asian countries by 18.8% to 127.7% from 2000 to 2006. They are major health hazards, causing severe respiratory diseases. At global scale each year some 16 million women die prematurely because of kitchen pollution. Since aerosols can be transported to great distances and heights, their intercontinental mixing is possible. Aerosols and other particles are suspected to be the major cause of drop in monsoon precipitation in India. To this, the contribution of aerosol pollution of Europe is also suspected to be a contributor.

Aerosols affect climate by changing the radiative properties of the atmosphere. Much of the positive radiative forcing from aerosols is because of black carbon (BC) in soot. The direct radiative forcing of black carbon is greater than that of CH_4 , and its global warming effect could be highest after CO_2 in terms of direct forcing (Jacobson, 2000). Globally, the main sources of BC are open biomass burning (3.5 Tg/yr), residential burning for cooking and heating (2 Tg/yr), and fossil fuel combustion in vehicles (1.5 Tg/yr). Burning in agriculture areas during dry seasons in developing countries results in a release of considerable amount of BC each year. Though anthropogenic sources of BC occur in all parts of world, they are largely concentrated in tropics (84% contribution) where solar radiation is intense (Ramnathan and Carmichael, 2008).

The aerosol mix having BC can form 3-5 km thick brown clouds across several continents. The brown clouds curtail solar radiation and have the so called “dimming effect” which can change the regional precipitation pattern. Some aerosols thus have cooling effect, but because of the positive radiative forcing of BC, the net effect is warming of the atmosphere. Then BC when deposited on glaciers, reduce albedo, and speed up snowmelt. In cities of India the BC concentration peaks during winters and spring.

Oxides of sulphur (SO_x)

Oxides of sulphur are produced due to combustion of fossil fuels; particularly coal in power plants, which generally has 1-3% S (can be as high as 8-10%). Oxides of sulphur are also produced during petroleum refining, metal smelters. Sulphur dioxide (SO_2) is itself a poison, but it can also form sulphuric acid (H_2SO_4), a secondary pollutant produced as a result of reaction of SO_2 with hydrogen peroxide, water vapour and other substances in the atmosphere.

Oxides of Nitrogen (NO_x)

In nature, soil bacteria produce oxides of nitrogen but concentrations remain low and harmless. However, in industrial cities their concentrations may easily reach harmful levels, up to 100 times greater than in rural areas. Oxides of N are produced during combustion of coal, oil, natural gas and motor vehicle fuel. The O_2 and N_2 of air combine to form nitric oxide ($\text{N}_2 + \text{O}_2 \rightarrow \text{NO}$), by and large a harmless gas. However, under the influence of sunlight NO combines with O_2 to produce a toxic gas, nitrogen dioxide ($2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$). Like SO_x, NO_x is a major contributor to acid rain problem. Its toxicological effects are also similar to those of SO_x, if not identical. Concentration of NO_x in relation to their effects as described by the US Environmental Protection Agency (EPA) is as given below:

Atmospheric background	4 ppb
City air background	up to 80 ppb
EPA's air quality standard (annual mean)	0.05 ppm
Increased respiratory rate in rats	0.8 ppm (a few hrs)
Pungent odour noticeable	1-3 ppm
Occupational limit (8 hr/day)	5 ppm
Pulmonary edema, fatal	100-1500 ppm (1 hr)

Carbon monoxide

Carbon monoxide (CO) is the product of incomplete combustion of fuel, largely in automobiles. Contribution of humans is only 10% of the CO load dumped into atmosphere (in nature CO is released mainly with methane and other substances in marsh gases and other gases, as a result of decay; forest and grass fires and volcanoes, and reactions in the upper atmosphere are other sources of CO). In the air of polluted cities, nearly 95-98% of CO is from man-made activities. In terms of amount, it is the most important air pollutant and source of major effects on human health.

Hydrocarbons (HCs) or volatile organic compounds (VOCs)

Any compound composed of carbon (C) and hydrogen (H) is called hydrocarbon. Hydrocarbons are of great variety. Some, such as certain polycyclic hydrocarbons with multiple-ring structures, have chemical carcinogenicity (causing cancer). Of the carcinogenic hydrocarbons, benzopyrene is a major threat as tons of this is dumped into air each year as a by-product of burning of coal and petroleum. Some atmospheric hydrocarbons of the more general category of VOCs have also indirect polluting effects. They include unburned gasoline and evaporated solvents especially from petroleum refineries.

Lead

Atmospheric lead contamination comes largely from the automobiles and the high-compression engine. Lead smelters and paints are other sources of lead. In the 1920s tetraethyl lead was introduced to motor fuel as means to reduce engine knock and engine wear. Nearly all of that lead used in motor fuel was released into atmosphere. With the use of unleaded fuel, concentration of atmospheric lead has declined in many developed countries. However, where leaded petrol is still in use, urban air can have thousand times more lead than in clean air over mid-ocean.

Other air pollutants

Hydrogen sulphide (H_2S) is a colourless gas with the characteristic smell of rotten-egg, and is emitted as a result of incomplete combustion of fossil fuels as well as from petroleum refining industries, natural gas plants and petroleum units. Its average lifetime ranges for 24 hrs in summer to 42 days in winter.

Radioisotopes, asbestos, fluoride components, air toxins released from accidents (such as from a Union Carbide plant in Bhopal, India in 1984) are other pollutants which affect air quality. Asbestos has become an important pollutant because of its use in buildings for fire-proofing and for improving acoustics. Fluoride compounds are released into the atmosphere from smelting of iron and aluminium and processing of phosphate fertilizers. They are harmful to both humans and their animals. Fluorine is a very reactive oxidant, and because of this property affects calcification of bones and teeth.

19.3 FACTORS AFFECTING DISTRIBUTION OF AIR POLLUTION

Intensity of pollution is affected by several factors such as weather, meteorological conditions and topography. Light, humidity, temperature changes, topographical features all can influence distribution of pollutants, hence its level (Table 19.4).

Temperature affects air pollution in several ways. With increase in temperature rate of chemical reactions increases. At higher temperature air holds more water, which in turn, dissolves more pollutants. However, temperature also affects condensation of water vapour.

Normally, air temperature usually drops as altitude increases. Under atmospheric thermal inversion condition, a layer of warm air is formed over a cold layer of air (Fig. 19.4), and consequently

acts like a lid, preventing pollution to disperse. The air is effectively trapped together with pollutants as long as the inversion persists. Inversion inhibits movement of air and can contribute significantly to air pollution by limiting the natural upward dispersal of urban-industrial pollutants. Thermal inversion causes pollution to stay longer.

Table 19.4. Factors affecting distribution of air pollutants.

Factors	Effects on pollution	Typical effects
Precipitation	–	Cleanses the air
Humidity	+	Dissolve many pollutants, and makes pollution more visible
Fog	+	Remains the same
Sunshine	+	Initiates oxidation
Wind velocity	–	Less pollution near the source but faster and wider distribution
Increasing temperature with increasing height	+	Less dispersion
Barometric pressure	+	Lighter wind less dispersion
Height of emitting source	–	Wider dispersion and dilution of pollutants
Valleys	+	Trap pollutants
Mountains, hills	+	Break force of winds but promote light winds
Plains	–	Wider dispersion

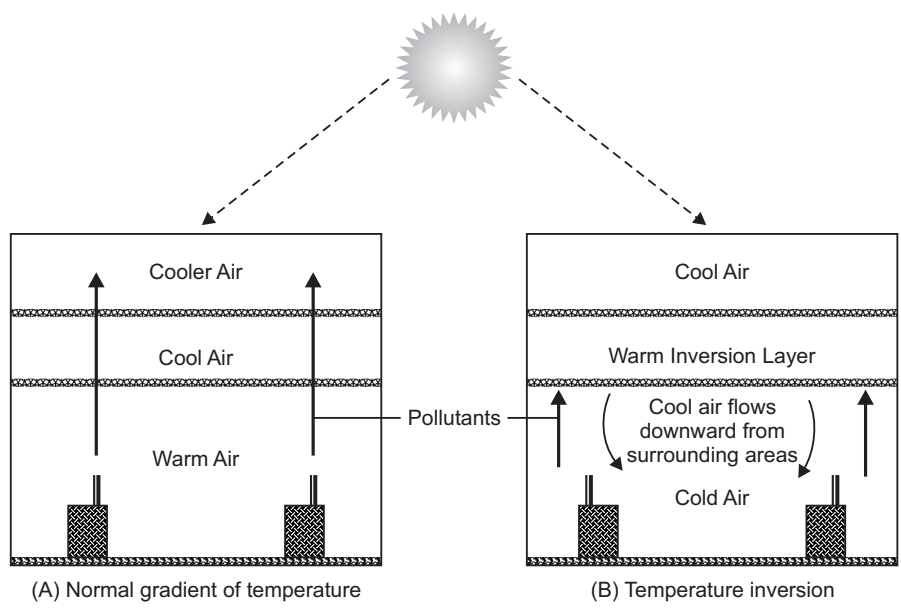


Fig. 19.4. Formation of Thermal Inversion, (A) Normally warm layer rises diluting the pollutants; (B) in temperature inversion, a warm layer of air gets trapped between cool layers of air resulting in the trapping of air pollutants.

Reactions in the atmosphere also affect pollution level. Both ultraviolet radiation and sunlight directly activate chemical reactions, thus causing pollutants to react with one another. Photochemical oxidation is the most well known radiation induced chemical reaction.

Cities at the bottoms of topographic depressions, such as valley bottom, are very prone to dangerous pollution levels, particularly when temperature inversion occurs. In fact, mountains themselves contribute to thermal inversions.

19.4 ATMOSPHERIC DEPOSITION OF POLLUTANTS

The atmospheric deposition of air pollutants occurs through three separate processes: (i) wet deposition – the deposition of material contained in precipitation; (ii) dry deposition – the direct deposition of particles and gases of atmosphere on vegetation, soil and surface water; and (iii) cloud deposition – the deposition of non-precipitating droplets of clouds and fog.

Wet deposition is simply the result of gravity acting on droplets that have grown large enough to fall as rain or snow. The atmospheric particles and gases are incorporated into cloud droplets, or raindrops and snowflakes. Cloud deposition results when cloud droplets (generally 5-100 μm , mostly 10-20 μm) containing dissolved ions come in contact with terrestrial surfaces. This is seen in mountain and coastal areas. Particles in dry deposition are on a larger side of the size spectrum ($>5 \mu\text{m}$), and are deposited by gravitational sedimentation and inertial impaction.

In general, the relative deposition rates of gases (which in part depend on their water solubility) are in the order: $\text{HNO}_3 > \text{NH}_3 > \text{SO}_2 > \text{O}_3 > \text{CO}_2 > \text{NO}_2 > \text{NO}$. Particulates are mainly removed by physical processes, through wet and dry deposition (Table 19.5). Rain and snow sweep particulates out of atmosphere. Gases either are removed by adsorption into solids or by undergoing chemical reactions in the air.

Table 19.5. Sinks for some atmospheric pollutants (processes involved are spontaneous chemical transformation, microbial degradation and physical processes).

Pollutants	Sink
SO_2	Precipitation, scavenging, followed by oxidation to sulphate particles
H_2S	Oxidation to SO_2
NO	Atmospheric oxidation to NO_2
NO_2	Precipitation, scavenging and oxidation to nitrate
Hydrocarbons	Oxidised to CO_2 , absorption in soil followed by microbial degradation and photochemical degradation
Ozone	Photochemical reaction (reversion to O_2)
CO	Uptake by soil and conversion to CO_2 by microbes

The natural sinks for air pollutants (and other types) could occur due to spontaneous chemical transformation – e.g., oxidation; microbial degradation of pollutants; physical processes – e.g. sedimentation and solubilization.

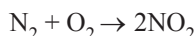
19.5 SECONDARY AIR POLLUTION

Photochemical smog

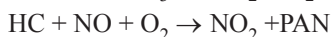
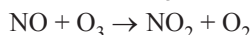
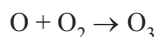
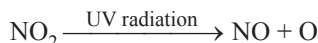
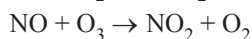
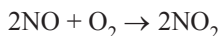
In the presence of sunlight, nitrogen oxides combine with hydrocarbons to form photochemical oxidants consisting of ozone and the eye-irritating peroxyacetyl nitrate (PAN) and NO_x (Fig. 19.5). In the light, NO_2 splits into NO and atomic oxygen (O) due to the influence of UV radiation. The active atomic oxygen reacts with molecular oxygen (O_2) to form ozone (O_3). As they are affected by sunlight, the photooxidants (NO , NO_2 and O_3 all formed in the presence of sun) exhibit marked diurnal and seasonal changes in their concentrations. The ozone, in turn, oxidizes the photolytically

produced NO to NO₂. Nitrogen oxides also react with oxygen in the air to form nitrogen dioxide, a foul-smelling brown gas. Photochemical smog is formed in big cities under warm conditions and intense solar radiation. Automobiles play an important role in O₃ and PAN formation in the urban areas. The reactions leading to the formation of ozone are as follows:

Reactions occurring inside the engine



Reactions occurring in the atmosphere:



Being a relatively small molecule, PAN gets distributed over great distances.

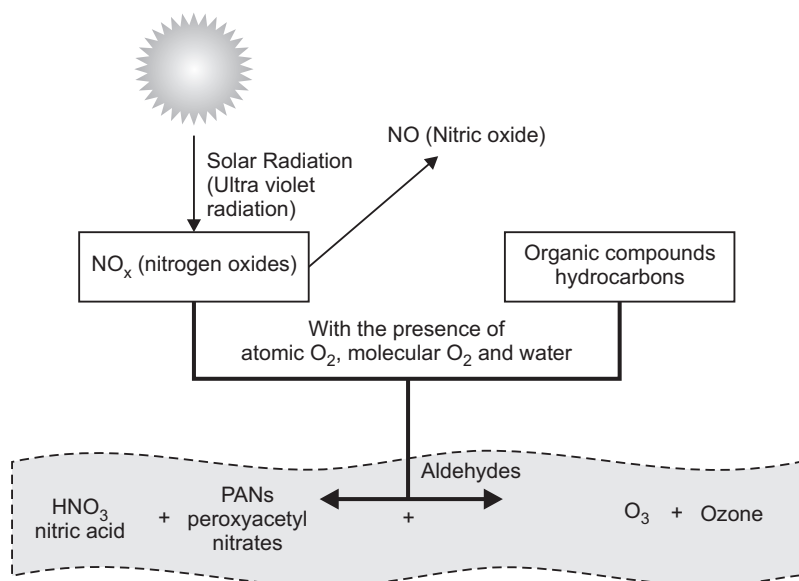


Fig. 19.5. Simplified scheme of formation of photochemical smog. (based on Botkin and Keller, 1995).

Acid Deposition

Acid deposition is not a new phenomenon. Robert Angus Smith, a British chemist, coined the term acid rain in 19th century (Smith, 1872). Acid deposition emerged as an ecological issue in the late 1960s and early 1970s, with the reports of acid precipitation in Sweden and around Scandinavia. Acid deposition is a serious environmental problem in large parts of North America, Canada, Central Europe and Russia. Even in a less industrialized country like India, a few instances of acid rains have been recorded in industrial parts of cities like Mumbai (Bombay) and Delhi. In 1974, when first instance was recorded in Trombay and Chembur area the rain had pH 4.5-4.8. Acid deposition in the later decades of the 20th occurred in many other temperate and tropical countries that became rapidly industrialized (Rhode et al., 1988). About 7 to 17 % of worlds terrestrial natural/

seminatural ecosystems are negatively affected by acid deposition (UNEP/RIVM, 1999). Though acid rain does not make headlines any more, it is still a problem as the pollutants that cause acid rain are also linked to other issues such as coastal eutrophication, change in nutrient balance of entire ecosystem, climate change and troposphere ozone (see Driscoll et al., 2001; Greaver et al., 2012). Sulphur deposition stimulates microbes to methylate Hg, which results in the introduction of Hg into food chain and in its bio accumulation.

Acid deposition, which has two forms, wet and dry, is a broad term used to describe several ways by which acid-forming substances fall out of the atmosphere to the surface of the earth. Wet deposition refers to acidic rain, (also fog, and snow) due to the presence of sulphuric acid and nitric acid. Dry deposition refers to acidic gases and particulates that are washed from trees and other surfaces by rainstorms.

Measurement of acid rain: Acid rain is measured using a pH scale, which has values from 0 to 14. A pH of 7 is neutral, whereas a pH less than 7 indicates an acidic solution (Fig 19.6). The lower a substance's pH, the more acidic it is. Pure water has a pH of 7.0, normal rain is slightly acidic because carbon dioxide dissolves into it, so it has a pH of about 5.5. The pH scale is logarithmic. Therefore, a solution with a pH of 6 is ten times more acidic than a solution with a pH of 7. A solution of pH 5 is 100 times more acidic than a solution with a pH of 7.

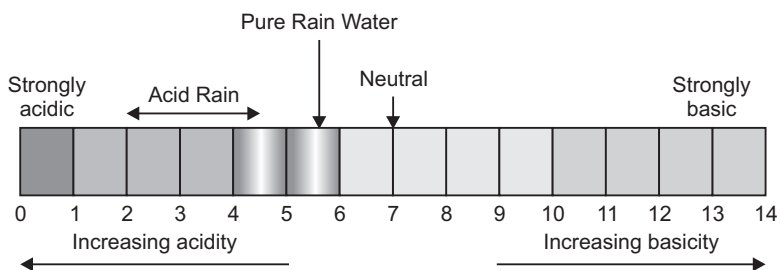


Fig. 19.6. The pH scale includes values between 0 and 14, expresses the concentrations of acidic and basic solutions. Pure water is neutral, has a pH of 7.

Formation of acid rain : The sulphur dioxide (SO_2) and nitrogen oxides (NO_x) are the primary causes of acid rain. Coal burning power plants, large smelters and industrial boilers are the main sources of sulphur dioxide emissions (Fig. 19.7). Motor vehicles are a major source of nitrogen oxides. Atmospheric winds can transport oxides of sulphur and nitrogen to thousands of kilometres from tall smokestacks of big industrial set ups. This causes the problem of acid deposition away from its source of generation. Acid rain occurs when these gases react in the atmosphere with water, oxygen, and other chemicals to form various acidic compounds.

The reactions that convert the gases, SO_2 and oxides of nitrogen, into acid are triggered by sunlight and depend on the atmosphere's abundant supply of oxygen and water. The troposphere, the lowest 10 or 12 kilometres of the atmosphere, is the place where the reaction cycle occurs (Fig. 19.7). The reaction starts when light strikes ozone (O_3), which may have moved downward from the ozone layer in the stratosphere or may have been formed in the troposphere by the action of nitrogen- and carbon-containing pollutants. The result is a molecule of oxygen (O_2) and a lone oxygen atom (O), which is highly reactive. It combines with a water molecule (H_2O) to form two hydroxyl radicals (HO). This transforms nitrogen dioxide (NO_2) into nitric acid (HNO_3) and initiates the reactions that transform SO_2 into sulphuric acid (H_2SO_4). Each hydroxyl radical can oxidise thousands of sulphur-containing compounds. Sulphuric acid formed through gas-phase reaction condenses to form microscopic droplets (0.1-2 micrometer in diameter, that is millionth of a meter), which is one of the components that cause summer time haze in some industrial areas. Some of the

sulphate particles settle to the ground, in a process called dry deposition (dry deposition is used also in reference to the capture of SO_2 gas by vegetation). Most of sulphate particles, however, are incorporated into clouds.

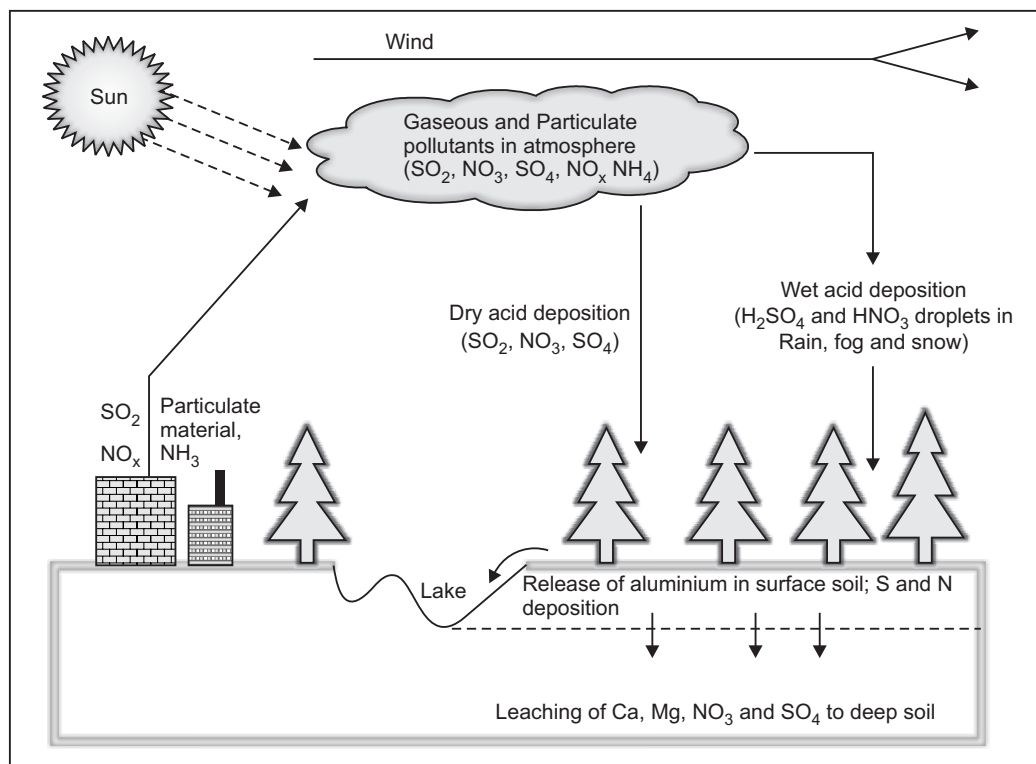


Fig. 19.7. Dry or wet acid deposition or acid rain.

Acid deposition returns these substances to the ground, causing the pH of surface waters and soil to decrease. Sulphur deposited as SO_2 or SO_4 containing aerosols contributes directly to acidification. Nitrogen deposition as ammonium or nitrate as well as nitrification may also contribute to soil acidification (Van Breemen et al., 1982). Acidification by nitrate and ammonium deposition depends on the transformation of N compounds in the soil. For example, plant uptake of ammonium produces acidity, while uptake of nitrate (NO_3) reduces acidity.

The acid rain may fall hundreds of miles from the pollution source. For example, 92% of sulphur deposition in Norway comes mainly from England and Central Europe. Wherever acid rain falls, it undergoes a new round of changes, which can reduce the acidity of water that reaches lakes and streams. Alkaline soils rich in limestone can neutralize the acid directly. Vegetation and soil can immobilize the acid, as they retain sulphate of sulphuric acid and nitrate of nitric acid. Acidity can also be reduced by the cation (positive ion) exchange. In this, the ions of calcium, magnesium and other metals of soil replace acid's hydrogen ions.

19.6 EFFECTS OF AIR POLLUTION

The air pollutants differ in their chemical reactivities (Table 19.6). Air pollutants have become part of nutrient cycling, both at global scale and at the scale of local forest and other natural ecosystems.

Table 19.6. A summary of chemical reactivities of air pollutants.

SO ₂	A water soluble, rapidly diffusing, acid forming oxidising agent. With water can form sulphurous and sulphuric acids which react with organic matter, metals, etc.
Particulate matter	Vary in property depending upon their chemical nature. Solid particles can absorb various chemicals onto their surfaces.
NO ₂	An oxidising agent, forms nitric acid with water, less soluble in water, that is why travels further into respiratory system. Nitric acid is a very powerful oxidising agent, reacting with almost all metals and organic compounds particularly lipids. NO ₂ is involved in ozone production in the atmosphere.
Hydrocarbons	Combines with many kinds of compounds with varied reactive properties, carcinogenic hydrocarbon compounds can combine with DNA, causing cancer and mutation. Volatile hydrocarbons are also involved in atmospheric reactions generating ozone.
Ozone and other photochemical oxidants	Unstable; very reactive oxidizing agent, combining with many organic compounds in tissues, also with rubber and other materials.
H ₂ S	Behaves like a weak acid, can also behave like a reducing agent, mildly reactive with organic compounds and metals.
Fluorides	Very reactive reducing agents, can combine with many compounds of very low reactivity. HF is toxic at ppb level and fluoride ions can act as free radicals.
CO	Only slightly soluble in water, but very dangerous because of its very high efficiency for haemoglobin, even more than that of oxygen.
Nitric oxide	Little direct effect, being insoluble in tissue water, but can be converted into nitrogen dioxide.

Air Pollution Effects on Plants

Plant responses to air pollutant like SO₂, O₃ and HNO₃ generally include: (i) development of chlorotic and necrotic spots on leaves; (ii) increase in leaf senescence and drop (e.g. due to O₃ exposure), (iii) reduced photosynthesis (SO₂ particularly inhibits the oxidising side of photosystem II); (iv) reduced stomatal conductance; (v) change in root-to-shoot ratio (e.g., SO₂ and O₃ exposure); and (vi) reduced growth. These important changes for plants may significantly modify productivity and functioning of both managed/unmanaged ecosystems.

Evidence has accumulated to show that exposure of plants to air pollution (and also to other environmental stresses) results in the formation of free radicals which can oxidize the fatty acids in membrane lipids leading to membrane damage and eventual cell death. These free radicals include superoxide, super hydroxide, hydrogen peroxide, ozone and oxygen.

Nitrogen dioxide is a reddish-brown gas and like SO₂ has a considerable environmental impact. NO₂ enters leaf through stomata just as readily as does SO₂, but diffuses through cuticle much faster. It is converted into inorganic nitrates and peroxyacetyl nitrate (PAN). Necrosis occurs when the concentration of NO₂ exceeds 1900 µg m⁻³. Nitrogen dioxide reacts with water and forms nitric acid (HNO₃), which is equal to H₂SO₄ in its capacity to do harm. Being a strong oxidizing agent, nitric acid reacts with nearly all metal, most organic compounds, especially with lipids. In plants, nitrite ions are assimilated into amino acids (catalyzed by nitrite reductase), thus its effect is detoxified. The atmospheric nitric oxides are in fact additional source of N for plants but they can become harmful to leaves by acidifying them. Sulphur dioxide increases the toxic effect of NO₂ by inhibiting the catalyst nitrite reductase. This is a good example of synergistic effects of pollutants.

PAN enters the leaves through stomata, affects chloroplasts by oxidizing sulphhydryl groups of proteins, and induces free radical formation to oxidize fatty acids in the membranes. Chlorosis is observed when concentration of PAN increases beyond $989 \mu\text{g m}^{-3}$.

Compensatory stress responses to air pollutants: Some plant responses to air pollution have also been characterized as forms of stress “compensation”. Compensatory stress responses are physiological adjustments that result in increased productivity in response to a stress. Reduction in root-to-shoot ratio of plants exposed to pollutants, SO_2 , O_3 , and others which limit photosynthesis is a common example of compensatory response.

Populus hybrids grown in air containing $0.1 \text{ m mol/mol O}_3$ dropped 75% of their leaves, compared to 10% drop in those grown in normal air (Matyssek et al., 1993). Increased rates of leaf ageing due to the O_3 exposure are compensatory, if metabolites, including nitrogen, are metabolized and transported from older leaves. The O_3 caused increases in leaf fall in *Populus* were accompanied by increases in N, P, and S content of the remaining leaves. Thus the senescence of older leaves may not be premature for the environmental conditions, but rather may be a compensatory mechanism by which *Populus* plants respond to air pollution stress. For example, in loblolly pine O_3 fumigation reduced biomass but increased the level of N, S and P in leaves and photosynthesis rate above control plants (Kelly et al., 1993).

Plant-pathogen interaction: Air pollution can affect the susceptibility of a species to pests or pathogens by causing lower carbon allocation for producing defense compounds in plants. For example, exposure of *Pinus ponderosa* to increased O_3 since early 1950s in Los Angeles basin resulted in: (i) foliar injury; (ii) premature leaf fall; (iii) changes in photosynthesis and stomatal conductance; and (iv) reduced nutrient retention in leaves. As a consequence of premature leaf fall, whole plant photosynthetic capacity and subsequently diameter increment of trees declined. The weakened trees were more susceptible to root rot (by *Fomes aunosus*) and pine beetles (*Dendroctonus brevicomis*). With the death of many pine trees, the understorey composition changed to include O_3 -tolerant shrub and oak species. Characters of the entire pine ecosystem changed, including fluxes of water, carbon and nutrients in response to elevated O_3 .

Plants resistance to SO_2 and other air pollutants: The plants resist SO_2 pollution with the help of various passive and active processes: (1) by lowering the toxic concentration of SO_2 either (i) by reducing SO_2 uptake or (ii) by reducing acidification due to the high buffering capacity. (2) Restricting toxin stress (i) by detoxifying cation process (e.g., release of H_2S) or (ii) by maintaining enzymes stability, and (iii) by incorporating S in metabolic products, such as cysteine and methionine. Some characters, though not causally related to the pollutant (i.e., are non specific), have protective roles against SO_2 stress : (i) in some thallophytes structural, chemical and ecological characters check the SO_2 entry; (ii) vascular plants have mechanism for stomatal closure; (iii) herbs may survive SO_2 pollution because of short leaf lifespan and regular leaf production; (iv) winter deciduous species escape severe pollution load of winters; (v) plants which have increased uptake of alkali- and alkali-earth cations, tolerate toxic stress due to the higher buffering effect; (vi) binding of SO_2 to secondary products is another means of reducing stress; (vii) the C_4 plants are more resistant to SO_2 stress, reason being that PEP-carboxylase (C_4 plants) is less sensitive to SO_2 than RuBP-carboxylase (C_3 plants). This is the reason for the occurrence of C_4 grasses in the vicinity of volcanic S spray.

Sulphur dioxide enters the leaf through open stomata almost as readily as CO_2 , and when stomata are closed it can enter leaf even via cuticle. At low SO_2 concentration in air (about $45 \mu\text{g m}^{-3}$) transpiration is increased, but at higher concentration (about $1300 \mu\text{g m}^{-3}$) it is reduced.

The diffusion pathway of SO_2 is the same as of CO_2 . SO_2 and H_2S and some other compounds are not foreign substances for plants. In the cell walls, SO_2 is dissolved in water (SO_2 , H_2O), forming hydrogen sulphite (HSO_3) and sulphite (SO_3^{-2}). These are distributed inside the cell

between chloroplasts, cytosol and vacuole. Incorporation of S is mainly affected by photosynthetic S metabolism, the end product of which are the S-containing amino acids, cysteine and methionine, and some others. Surplus SH groups and sulphite may be converted into sulphide and lost as H_2S gas exchange. This is the way detoxification of plants takes place, but the reaction can work only within a limit.

The degree of resistance to SO_2 and other pollutants differs with species varieties and ecotypes and life stages. C_3 members of Polygonaceae are particularly less sensitive to SO_2 stress. We need to know SO_2 resistant plants for planting them in areas exposed to SO_2 pollution (Table 19.7). SO_2 can be fatal at 600 ppm concentration. Among plants, radishes and green leaves are known to be more vulnerable to the toxic effects of H_2S than SO_2 . It causes rapid discolouration of lead based paints.

Table 19.7. Tree and shrubs suitable for planting in SO_2 polluted localities (from Larcher, 1995).

Highly useful	Moderately useful
Broad-leaved trees (Deciduous)	
<i>Acer plantanoides</i>	<i>Castanea sativa</i>
<i>Buxus sempervirens</i>	<i>Ginkgo biloba</i>
<i>Celtis australis</i>	<i>Magnolia hypoleuca</i>
<i>Platanus hybrida</i>	<i>Populus candicans</i>
<i>Quercus</i> sp.	<i>Robinia pseudoacacia</i>
Conifers	
<i>Juniperus</i> sp.	<i>Chamaecyparis</i> and <i>Thuja</i> sp.
<i>Picea omorika</i>	<i>Pinus mugo</i> , <i>P. nigra</i>
<i>Taxus baccata</i>	<i>Picea pungens</i> f. <i>glauca</i>
Shrubs	
<i>Calluna vulgaris</i>	<i>Berberis</i> sp.
<i>Erica carnea</i>	<i>Prunus laurocerasus</i>
<i>Gaultheria shallon</i>	<i>Rosa rugosa</i>

Plant species which are relatively resistant to air pollutants rapidly form abscisic acid (ABA), which causes stomates to close. Lower concentrations and shorter exposures to toxic substances may reversibly increase the respiration rate. This increase in metabolism reflects mobilization of plant defense mechanisms such as repair of cellular damage by synthesizing enzymes and antioxidants. These enzymes include superoxide dismutase, catalase, ascorbic acid oxidase, polyphenol oxidase, etc.

Sensitive plants: Certain mosses and lichens and many fungi pathogenic to plants (e.g., *Rhytisma acerinum*, *Diplocarpon rosae*, *Gymnosporangium* species, *Puccinia graminis*) are very sensitive to SO_2 (and HF and HCl). Among the lichens, gelatinous lichens are more sensitive than crustose ones. Certain lichens, however, are even tolerant to SO_2 (e.g., *Stereocaulon* species occurring in volcanic craters). But in general, even one-hundredth of the SO_2 concentration injurious to higher plants is enough to suppress lichens.

As a powerful oxidizing agent, O_3 can damage crops and natural vegetation. Necrosis and stunting occurs once O_3 concentrating exceeds $157 \mu\text{g m}^{-3}$. Ozone dissociates rapidly in plant tissues to form atmospheric O_2 and peroxides. These peroxides harm plasmalemma and then all other biomembranes, resulting eventually in necrosis of tissues. Exposure to ozone is known to cause early leaf senescence and leaf drop. Changes in root to shoot ratio is another important effect of

O₃ exposure. These effects on plants may modify community structure and ecosystem functioning. Plants sensitive to certain air pollutants are given in Table 19.8.

Table 19.8. Terrestrial vascular plants as selective pollution-sensitive indicators (from Larcher, 1995; collected from different sources).

Pollutant	Plant species
SO ₂	<i>Populus tremula</i> , <i>Medicago sativa</i>
H ₂ S	<i>Pseudotsuga menziesii</i> , <i>Spinacia oleracea</i>
HF, F ₂	<i>Prunus armeniaca</i> , <i>Gladiolus communis</i>
HCl	<i>Syringa vulgaris</i> , <i>Fragaria vesca</i>
NH ₃	<i>Taxus baccata</i> , <i>Brassica oleracea</i>
NO _x	<i>Apium graveolens</i> , <i>Petunia x hybrida</i>
O ₃	<i>Nicotiana tabacum</i> , <i>Phaseolus vulgaris</i>
PAN	<i>Petunia x hybrida</i> , <i>Phaseolus vulgaris</i> , <i>Poa annua</i>

Methylation of mercury induced by sulphur deposition

Methyl mercury (Me Hg) is severely neurotoxic. Organisms take up Hg and accumulate in methylated form, which is facilitated by the presence of SO₄⁻² and HG. The process of methylation is largely accomplished by sulphate-reducing bacteria. In a methylated form Hg is readily taken up by microbes, zooplankton, and macro invertebrates. Then, Hg is bioconcentrated up the food chain.

Air Pollution Effects on Ecosystems

Air pollution can influence communities and ecosystems by direct toxicity or by causing physiological changes in species. However, because of global nature of air pollution all ecosystems have been affected, no ecosystem is pristine or virgin. Evidences are there to indicate that air pollution affects litter decomposition, biochemical cycles, biodiversity, primary productivity and species composition.

As for aquatic ecosystems, their acidification is primary caused by S, though N also contributes. By 2000 nearly one-third of the American lakes and streams had become acidic. Though SO₄⁻² deposition has been decreasing since 1989, water bodies remain acidic (Greaver et al., 2012).

Nitrogen deposition is the main new nitrogen source to many high altitude lakes. Nitrogen leaching indicates that ecosystems are unable to retain it. The impact of N deposition can vary from one ecosystem to another. It decreases CH₄ uptake in coniferous forests, increases CH₄ production in wetlands, promotes biogenic N₂O emission in a wide range of ecosystems including forests, grasslands and wetlands (Liu and Greaver, 2009).

Community structure: There are a number of well documented evidences of direct phytotoxicity of point source pollutants (SO₂ and heavy metals from industry) affecting community structure. Of greater worry is a change in productivity or community structure resulting from chronic air pollution stress distributed over a large geographical area and affecting a number of terrestrial landscapes. For example, in a grass community sensitivity of clover to SO₂ at chronic level reduced its ability to obtain nutrient, water, and radiation resources, consequently, ryegrass and other species already present grew better in response to decrease in clover (see Taylor Jr. et al., 1994). Thus, the structure of the community changed.

Decline of *Picea rubens* (red spruce) in the eastern United States is suggested to be due to linkage between loss of winter hardiness of tree species and presence of SO₄ in acid mist or O₃ at low concentrations (chronic level).

Increased input of nitrogen due to pollution is reported to lead to decrease in species diversity, root-to-shoot ratio and mineralization. Because of higher above ground to below ground ratio, plants become susceptible to droughts, and fires, and to wind damage as structural anchorage weakens (Greaver et al., 2012). Some other effects of N deposition are : decreased mycorrhizal diversity, shift toward graminoids in alpine communities, decrease in fine root biomass due to N leaching, expansion of exotic species, change in diatom assemblage in alpine lakes and dominance of N-tolerant lichens

Nutrient cycling: There are many mechanisms by which deposition of N, S, and H can affect and interact with nutrient cycling in an ecosystem. Foliar leaching, decomposition of litter, uptake of nutrients and soil leaching are quite common. As for the effect of N deposition on ecosystems, it can vary considerably: it can stimulate plant growth and increase carbon storage, however, it can cause nutrient imbalance that decreases productivity, bring about change in species composition, and deplete biodiversity (Greaver et al., 2012). Simultaneously, N may act as a nutrient as well as strong acid anions in some ecosystems, the full effect of them, however, is still unknown.

Foliar leaching: Since air pollutants are first deposited upon the canopy of forests and other communities, one of the first processes to be affected is cation leaching from leaves by H^+ and NH_4^+ .

Decomposition: Decomposition and nutrient mineralization can be affected by pollutants (e.g., H^+ , S or N) either directly by their deposition or indirectly via their adverse effect on litter quality.

Uptake: Increase in concentrations of nutrient cations in soil solution could increase cation uptake. Particularly, increases in concentrations of strong – acid anions (SO_4 , NO_3 and Cl) in soil solution must result in corresponding increases in cation concentrations. Because most unmanaged ecosystems have been N-deficient, it is quite likely that increased N deposition has increased N uptake and biomass in both grasslands and forests.

Effects of Acid Rain

The strength of the effects depends on acidity of the water, the chemistry and buffering capacity of the soils involved, and the types of fish, trees, and other living organisms that rely on water. Acid deposition has a variety of effects, including damage to forests and soils, fish and other living things, materials, and human health. Acid rain also reduces visibility through the air. Acid rain causes acidification of lakes and streams and contributes to damage of trees at high elevations (for example, red spruce trees above 2,000 feet) and the sensitive forest soils.

Effects on terrestrial ecosystems: Trees like, spruce, pine, ashes and birch appear especially vulnerable to acid deposition. Studies have shown extensive dieback, i.e., reduced growth leading to tree mortality in high elevation red spruce (Siccama et al., 1982; see Driscoll et al., 2001) and sugar maples (Houston et al., 1999; see Driscoll et al., 2001) Acid rain does not usually kill trees directly. Instead, weakens the trees by damaging their leaves, limiting the nutrients available to them, or exposing them to toxic substances slowly released from the soil. Acid rain causes slower growth, injury, or death of forests. Quite often, injury or death of trees is a result of these effects of acid rain in combination with one or more factors (Fig 19.8). Reduction in the rate of photosynthesis and bacterial nitrogen fixation, increased sensitivity to drought and diseases and death of roots due to the increased uptake of toxic aluminium released from soil under the leaching influence of acid are some of the direct effects of acid deposition on plants of terrestrial ecosystems. Evidences indicate that acid rain's contribution to forest decline is a stress-causing factor due to interference with calcium nutrition and calcium dependent cellular processes.

Acid rain may erode cuticle of leaves and leach away nutrients from leaf. The waxy coating of leaf, which is waterproof, is easily removed by acid rain. Forests in high mountain regions often are exposed to greater amounts of acid than other forests because they tend to be surrounded by acidic clouds and fog that are more acidic than rainfall. When leaves are frequently bathed in this acid

fog, essential nutrients in their leaves and needles are stripped away. This loss of nutrients in their foliage makes trees more susceptible to damage by other environmental stresses like acid deposition, troposphere ozone, ultraviolet radiation, insect attack or cold winter weather (Mohnen, 1988).

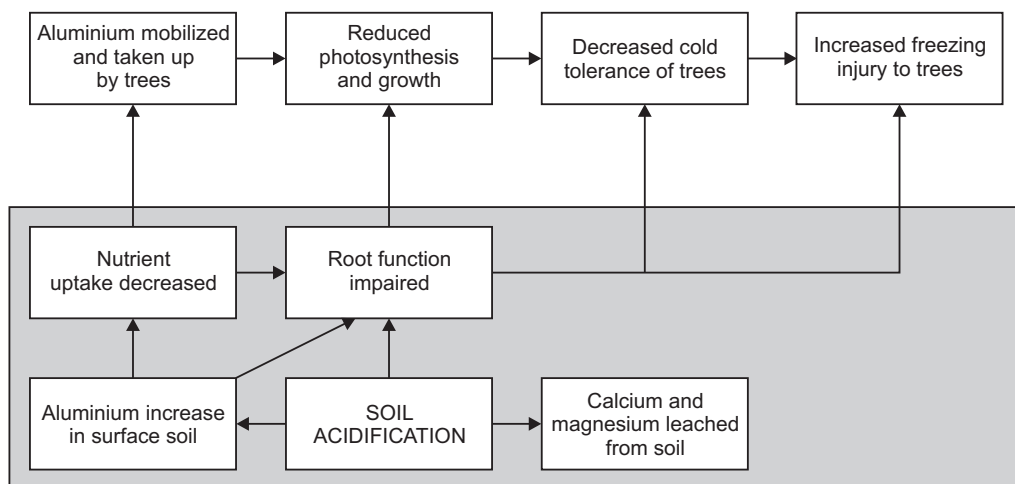


Fig. 19.8. Acid deposition effect on trees and soil chemistry.

Acidification affecting the seminatural ecosystems has been found most prominent in the industrialized countries of Western Europe, Eastern Europe, and the eastern parts of North America and Canada (UNEP/ RIVM, 1999).

Soil acidification: Soil acidification is the most thoroughly investigated effect of atmospheric deposits on biogeochemical cycling. It may be mentioned that processes such as natural production and leaching of carbonic, organic and nitric acids; humus accumulations in soils; weathering of primary minerals; and uptake of cations and anions need to be considered while estimating potential for atmospheric deposition to acidify soils.

Soil buffering capacity: The soil may neutralize some or all of the acidity of the acid rainwater. This ability is called buffering capacity, and without it, soils become more acidic. The buffering capacity of soils depends on the base saturation (amount of positively charged cations) in the soils. The higher the base saturation, the more buffering capacity of soils have to absorb the H^+ ions. The differences in soil buffering capacity are an important reason why some areas that receive acid rain show a lot of damage, while other areas that receive about the same amount of acid rain do not appear to be harmed at all. The ability of forest soils to resist, or buffer, acidity depends on the thickness and composition of the soil, as well as the type of bedrock beneath the forest floor. Acidic water dissolves the nutrients and helpful minerals in the soil and then washes them away before trees and other plants can use them.

Leaching of soil nutrients: Acid rain causes the release of substances that are toxic to trees and plants, such as aluminium, into the surface layer of soil. A combination of loss of soil nutrients and increase of toxic aluminium may be one way that acid rain harms trees. Such substances also wash away in the runoff water and are carried into streams, rivers, and lakes. Acid deposition alters the chemistry of soils (Fig. 19.9). The surface of the soil particles is negatively charged. In normal soil, cations such as calcium and magnesium are attracted to the negatively charged soil particles. In acidified soil, hydrogen ions from acid deposition displace the positively charged cations. In addition, aluminium ion released when soil becomes acidic also adheres to the soil particles. Essential elements like calcium and magnesium are leached easily from the acidic soil. Death of microbes also results in nutrient loss from soil. There is a direct link between forest damage and change in soil chemistry due to acid deposition.

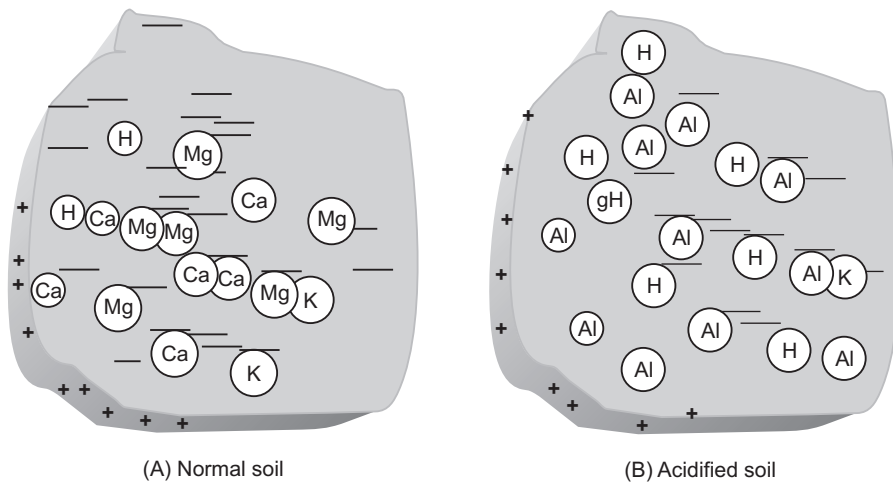


Fig. 19.9. Effect of soil acidification on cations (A) in normal soil negatively charged clay particles have mineral ions such as calcium and magnesium; (B) in acidic soil, negatively charged clay particles have aluminium.

Effect of acid rain on aquatic ecosystems

The ecological effects of acid rain are most clearly seen in the aquatic environments, such as streams, lakes, and marshes. Acid rain flows to streams, lakes, and marshes after falling on forests, fields, buildings, and roads. Acid rain also falls directly on aquatic habitats.

Surface water chemistry. Most lakes and streams have a pH between 6 and 8, although some lakes are naturally acidic even without the effects of acid rain. What happens when acidified water through runoff or ground water reaches a lake or stream? Water bodies may have bicarbonate and other basic ions, which can neutralize acids, thus not allowing pH to fall below a value of about 5. The content of such neutralizing ions in water is known as its acid-neutralizing capacity (ANC). The ANC value gives an idea of how susceptible or less susceptible the water body is to acid rain. A lake with zero ANC is acidified directly by any input of acid rain.

Several regions in the U.S. containing many of the surface waters are sensitive to acidification. The lakes and streams in the Adirondacks and Catskill Mountains in New York State exhibit both chronic and episodic acidification (see Driscoll et al., 2001). In areas like the northeastern United States, where soil buffering capacity is poor, some lakes now have a pH value of less than 5 due to atmospheric deposition. The acidification problem increases in magnitude due to “episodic acidification”. It refers to brief periods during which pH levels decrease due to runoff from melting snow or heavy downpours. Emissions from U.S. sources also contribute to acidic deposition in eastern Canada, affecting thousands of lakes.

In Sweden, one of the most severely affected countries, even wells have acid water. In combination with N fertilizers, acid rain is affecting soil biology and nutrient cycling of crop fields in Sweden. Nearly 25% of the country’s land and water is either acidified or threatened with acidification. In Sweden and many industrialized countries, acid rain problems increased dramatically from 1950’s, attaining peak in 1970’s.

Effects on aquatic organisms. The harmful effects of acid water in aquatic bodies may include reduction in species diversity and reproduction of many organisms. Formation of deformed embryos and young ones, destruction of food base (Crustaceans food, insect larvae, etc.) of fishes, leaching of heavy metals from soil into water bodies; inhibition of bacterial activity in decay resulting in

immobilization of nutrients in organic matter can also result due to acid rain. Mollusks which require CaCO_3 , are first to disappear with the increasing acidity as they are vulnerable to pH even as moderately low as 5.

Table 19.9. Summary of effects of major pollutants on materials.

Chemical	Material attacked	Damages
CO_2	Building material – limestone, marble, slate, mortar	Deterioration, leaching and weakening of materials
Sulphur oxides	Ferrous metals	Corrosion
	Copper	Corrosion to CuSO_4 (green)
	Aluminium	Corrosion to AlSO_4 (white)
	Leather	Embrittlement
	Paper	Disintegration embrittlement
	Textile	Reduced tensile strength deterioration
H_2S	Silver	Tarnish
	Copper	Tarnish
	Paint-leaded	Blackened due to formation of lead sulfide
Ozone	Rubber	Cracking, weakening
	Textiles	Weakening
	Paints	Fading
Nitrogen oxides	Paints	Fading
Hydrogen fluorides	Glass	Etches, opaque
Solid particulates (soot tars)	Building materials	Soiling
	Painted surfaces	Soiling
	Textiles	Soiling

Effects on Food chain. Acid rain causes a cascade of effects that harm or kill individual fish, reduce fish population numbers, completely eliminate fish species from a water body and decrease biodiversity. As acid rain flows through soils in a watershed, aluminium is released from soils into the lakes and streams located in that watershed. So, as the pH in a lake or stream decreases, aluminium levels increase. Both low pH and increased aluminium levels are directly toxic to fish and zooplankton. In addition, low pH and increased aluminium levels cause chronic stress that decreases the survival, size and diversity of fish. At pH 5, most fish eggs cannot hatch.

The plants and animals living within an ecosystem are highly interdependent. For example, frogs may tolerate relatively high levels of acidity, but if they eat insects like the mayfly, they may be affected as a part of their food supply may disappear. Due to connections between the many fish, plants, and other organisms living in an aquatic ecosystem, changes in pH or aluminium levels also affect biodiversity. In Precambrian shield lakes in Canada, zooplankton food webs showed decreases in species diversity, when the pH dropped below 5.0 in the 1970s (Locke and Sprules, 1994). An increase in pH by two units helped to improve species diversity in terms of species richness, and number of predatory and competitive links in 1990s as compared to the 1970s values. Thus, as lakes

and streams become more acidic, the numbers and types of fish and other aquatic plants and animals that live in these waters decrease.

Effects on monuments and buildings

One of the costly impacts of the acid rain is accelerated weathering of many precious statues and buildings, especially limestone marble. The damaged monuments need special care to save them from acid rains. Metals also corrode when exposed to acidic water. The damage to historical monuments due to acid rains is a matter of great concern. One of the often quoted examples is the badly damaged sandstone statue of the Augsburg Cathedral, Germany. The statue is covered with black coating that contains large amount of gypsum (a produce of reaction between limestone and sulphuric acid). Sulphur dioxide from power generating station and heating systems is known to cause the problem. The oxides of sulphur react with aluminium to form aluminium sulphate which corrodes wall surface. Pollutants damage leather, paper, and silver waste by reacting with them (Table 19. 9).

Air pollution Effect on Human Health

Oxides of Sulphur: The effects of SO_2 on human health are related to its concentrations as described below:

Atmospheric background	0.2-0.4 ppb
Nonindustrial city air	0.01 ppm
EPA's 24-hr primary standard	0.14 ppm
People with asthma begin to experience distress	0.5 ppm (1 minute)
Level at which even normal people experience bronchial spasm impaired lung function, occupational limit (8 hours exposure)	1.0 ppm (1 hr)
Water logged lungs (pulmonary oedema) and permanent damage	20 ppm

Sulphur dioxide is quickly absorbed and acts in the upper respiratory tract, reacts with cellular constituent chemicals such as enzymes. Sulphuric acid which SO_2 forms, lowers pH and thus impairs enzyme functions and destroys various functional molecules.

Physiological effects of acute exposure to SO_2 on man include the following:

- (i) Gives rise to irritation reactions which cause blood capillaries to dilate and exude fluids. This leads to tissue fluid accumulation (called edema) and swelling, bronchial spasms, shortness of breath.
- (ii) General physiological reaction of SO_2 is similar to allergic asthma, i.e., impaired pulmonary function via increased airway resistance, impaired lung clearance, and increased susceptibility to infection. An exposure of one hour to 1-5 ppm of SO_2 is enough to cause irritation to humans.

Chronic pollution aggravates lung diseases like chronic bronchitis, pulmonary fibrosis are some of the more common symptoms of SO_2 pollution.

H_2S is primarily absorbed through lungs, and is extremely toxic to humans. Emphysema, chronic bronchitis, lung cancer, dust-induced human problems, and other diseases are linked to air pollutants. Air pollution can cause various diseases in human beings (see Kupchella and Hyland, 1989) : (i) pulmonary irritation and acute impairment of lungs; (ii) cancer; (iii) structural changes; (iv) systematic toxicity (lead poisoning, a common example) etc. Asphyxiation is a disease associated with CO pollution. The affinity of CO for haemoglobin is more than 200 times than that of oxygen,

therefore it limits the availability of oxygen by competing with it for haemoglobin. Even if blood carries as little as 5% of the haemoglobin, it can cause physiological impairment. About 10 ppm CO is likely to reduce drastically the ability to carry oxygen. At 100 ppm CO, humans experience headache, dizziness and impaired perception, and at 300-400 ppm even for a few minutes, vision is impaired, and one may feel nausea and abdominal pain. Hydrogen sulphide also causes asphyxiation, but indirectly, by paralyzing the breathing control centre.

Effect of Lead: Brain and bone marrow cells are most severely affected by lead. Mental retardation, lowered IQ, and behavioural abnormalities are some of the known effects of lead. In red blood cells precursors, lead inhibits a key steps required for synthesis of haemoglobin. The effect of lead is ultimately manifested in impaired ability of blood to carry oxygen to the tissues. Thus a kind of anaemia results. Lead can also affect kidneys, and is known to cause tumours in rats. Its health effect threshold level in blood has been brought down from 50 mg/dl (decilitre) some years ago to 15 µg/dl in children. By eliminating lead from petrol, millions of children can be saved from brain damage, and lowered IQ.

Effect of suspended particulates: Suspended particles are serious health hazards as well as affect ecosystems and the biosphere. Particulates block sunlight and may cause changes in climate that could have long-term effects on the biosphere. The cancer causing chemical benzopyren is associated with particles of soot. Pneumoconiosis, used for all dust diseases, result from certain air pollutants which cause structural changes in lungs. Silicon dioxide in air is known to cause silicosis and coal dust causes black lung disease.

Suppression of body resistance to diseases and changes in human behaviour

People exposed to air pollution such as (nitrogen dioxide) become more vulnerable to diseases than those living in clean air. Epidemiological studies (pertain to relative frequency of disease within a population or its subgroup) suggest that pollution in general, weakens human immune system and causes undesirable changes in human behaviour. Reports are there to indicate higher frequency of diseases in more polluted areas than in less polluted areas for respiratory ailments, cases of asthma, and lung cancer. Some studies show that smog increases forgetfulness and irritability.

Synergistic effect of air pollution

The synergism pertains to different impact of union of two factors than a sum of their separate impacts. A good example is the combined impact of asbestos pollution and cigarette smoking. Medical researchers found that asbestos workers who smoked had more lung cancer than those who did not smoke, or those smokers who were not asbestos workers.

Effect of air pollutants on cell constituents and enzyme activity

For most biological reactions to occur enzymes are necessary, and pollutants can damage them. In many enzymes, metals are integral part of their structure. Zinc as a pollutant can interfere with enzymes that normally have magnesium; if zinc is substituted for magnesium other metals and pollutants can also combine to enzymes, leading to structural deformities (see Kupchella and Hyland, 1989). Pollutants also react with cell membrane and with other cell parts. Cell membrane thus damaged may allow essential cellular constituents to escape or permit things normally kept out to get in. Pollutants that react with DNA, may kill cells or cause mutation and cancer.

Effects of air pollution on animals

Most studies on the effect of air pollution pertain to animals of economic interest. Fluorides are reported to cause worldwide damage to farm animals, particularly those pastured around aluminium smelters. Lead poisoning and harmful effect of molybdenum mercury and selenium on farm animals have also been reported.

19.7 CONTROL OF AIR POLLUTION

There are three major aspects of pollution control: Technological, economic, and legal, which are described as follows;

Technological approaches for air pollution control

The approach of cleaning air is based on how to use energy. These may vary from simply using a technology to check the emission of fumes from a stack (such as installation of electrostatic precipitator control system) to a broad approach based on energy conservation in life style. Some of the measures suggested for keeping the air clean are listed below:

1. A number of alternatives have been suggested, including adopting a life style that uses less energy and pollutes less, or using energy sources that produce lesser amount of pollutants as byproduct. The economic lifestyle also needs to be modified to meet the challenges of air pollution. For example, the cost of pollution control needs to be passed on to consumers.
2. Reduce the energy use without changing the way of life. For example, we can use smaller cars or more energy efficient cars and Replace existing bulbs with CFL bulbs. A few states and cities have started the replacement with CFL.
3. Remove pollutant generating stuff from energy sources, e.g., remove sulphur from coal before burning it.
4. Check the generation of pollutants where burned, e.g., by putting catalytic converters in automobiles and installing scrubbers.
5. Protect the objects from pollutants, e.g., coat statues with protective materials, breed varieties of plants resistant to air pollutants.
6. Planting rows of trees along the road sides and around pollutant-emitting industries. The width of these shelter belts and height of trees are decided by the source and level of pollution and wind conditions.
7. Constructing walking and bicycle lanes, and encouraging mass transportation.

By improving mass transit, the number of vehicles can be reduced. The technological measures include reduction in vehicle size, horse power and compression ratio of internal engine of automobiles. Replacement of gasoline engines with electric or steam engines; and technological improvements for controlling emissions from automobiles power plants and other source of pollution. Sulphur dioxide pollution by power plants can be reduced by (i) using low sulphur fuels, (ii) keeping smoke stacks high (but it only diverts pollutants of one place to another place), (iii) chemically scrubbing sulphur dioxide and other sulphur compounds from the effluents going up stack, and (iv) removing sulphur from fuels.

BOX 19.1. Air Pollution Control Devices

Bag House: It consist of large filter bags made up of fibres which trap particles by forcing air containing dust and other particles.

Cyclone: It removes heavy particles by causing the particle-containing gas to swirl and removes the particle through centrifugal force.

Electrostatic precipitator: It removes particles by putting electric charge on them and then attracting them to an electrode of opposite charge.

Scrubbers: It uses a liquid spray to remove aerosol and greenhouse substances from air stream.

Catalytic converters: Used in cars, they remove organic compounds from streams of exhaust by completing their oxidation into CO_2 and H_2O in the presence of a catalyst.

Afterburner: It keeps undesirable compounds from escaping by burning them.

Certain technologies for pollution control are already available but their use needs to be properly implemented. Several kinds of particulate control device are available (see Box. 19.1.). The scrubbers can effectively remove gaseous pollutants from air. The automobile emission can be controlled by the catalytic conversion of pollutants to relatively harmless products.

The filter bags remove the dust particles by trapping these on a porous mesh of cotton, glass fibre or asbestos cellulose (Figs. 19.10 and 19.11). The effluent gases enter the bottom of bags and clean air escapes through the sides. The filter bags remove 99% of dust particles from the stack gases. Cyclone filters remove the heavy dust particles under the action of gravity and downward spiralling winds (Fig. 19.12). Electrostatic precipitators are commonly used to remove dust, fly ash, asbestos fibres and lead salts from the gaseous effluents. Pollutants pass between the pairs of positively and negatively charged electrodes. The dust particles pick up the negative charge and collect on the accumulating plate with a positive charge (Fig. 19.13). The electrostatic precipitators are very efficient to remove dust up to 99%.

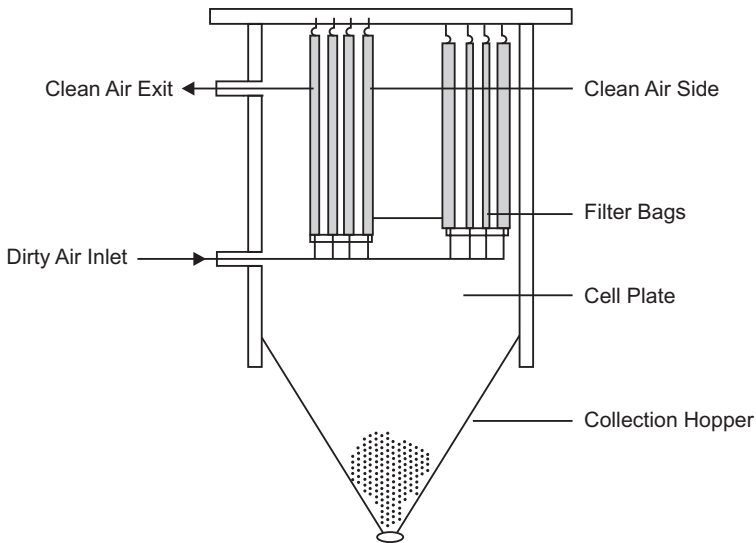


Fig. 19.10. A Filter Bags system for removal of particulate matter (based on Manahan, 2000).

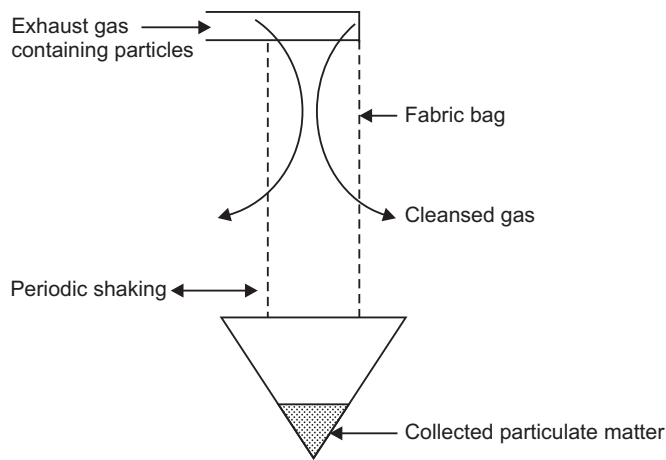


Fig. 19.11. Bag house collection of particulate emissions (based on Manahan, 2000).

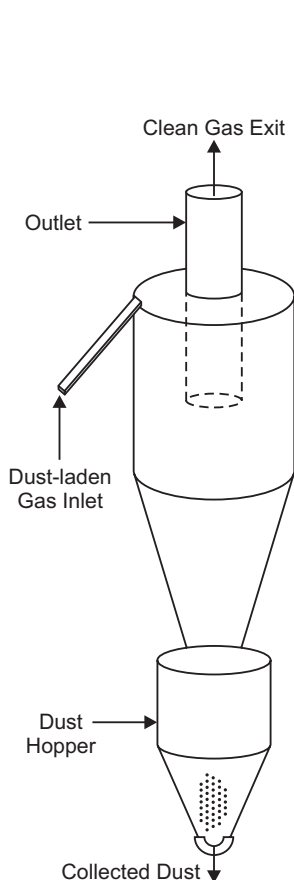


Fig. 19.12. A cyclone collector for the dust collection (based on Manahan, 2000).

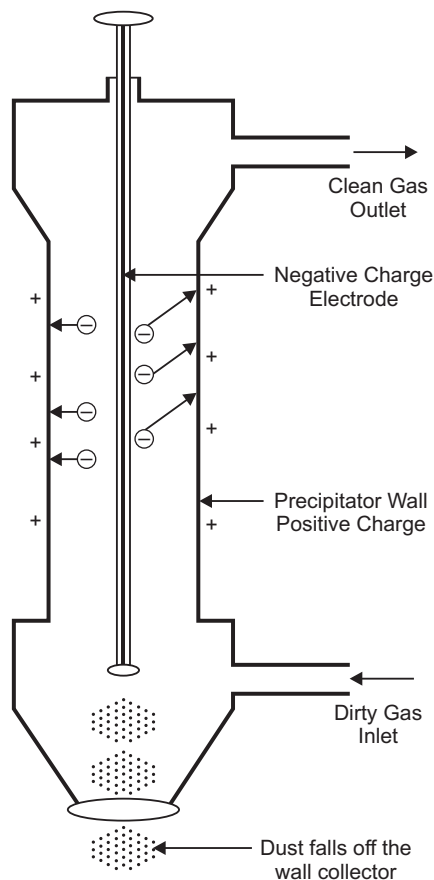


Fig. 19.13. Electrostatic precipitator. (based on Manahan, 2000).

Flue gas desulphurization or scrubbing is widely used for removing sulphur dioxide from smokestack gases of power plants and other industries (Fig 19.14). In this method, the gases are treated with a slurry of lime (calcium oxide; CaO) and limestone (CaCO_3 , calcium carbonate). The sulphur oxides react with the calcium to form insoluble calcium sulphides or sulphates, which are collected and disposed in a landfill site. This method can remove 95% of SO_2 and nearly 100% SPM from the gases. Nitrogen oxides produced during high temperature combustion processes in industry can be removed from stack gases by burning the fuels at slightly low temperature. Modification of furnaces and engines to bring about complete combustion helps to control carbon monoxide and hydrocarbons. In India, scrubbers have been installed in many industries to reduce emission of suffer dioxide.

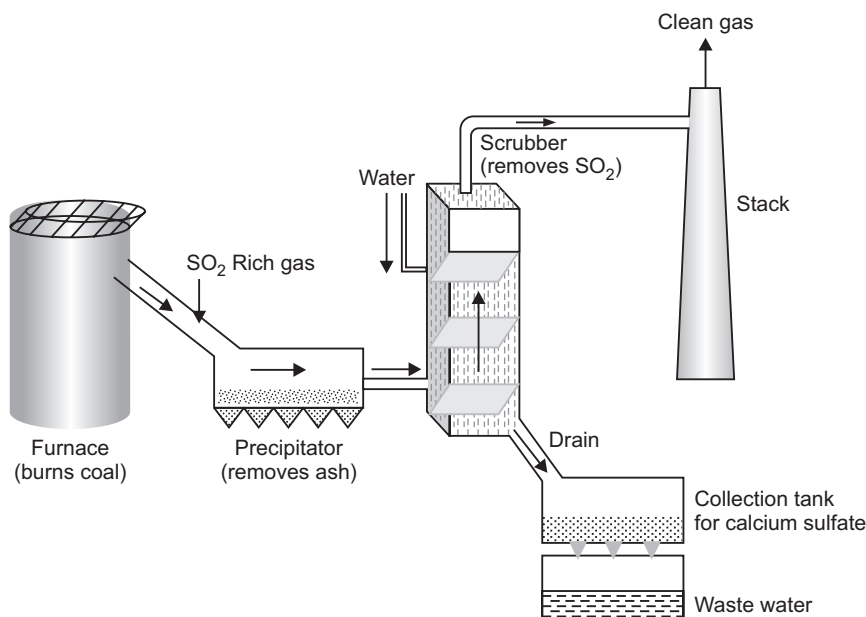


Fig. 19.14. Scrubber used to remove sulfur oxides from the effluent gases (based on Botkin and Keller, 1995).

Control of vehicular pollution

To deal with the problem of vehicular pollution both fuel and engine quality need to be improved. For example, to get the best results we need unleaded petrol in combination with catalytic converters, and diesel with 30 ppm sulphur in combination with de-NO_x converter or a particulate trap. Emission of large amount of particulate matter is a great problem with the use of diesel. Recently, an exhaust filter has been developed that traps and burns particulate matter emitted from diesel engine. With the use of this trap the emission of particulate matter is reduced from 1000 units to 1 unit. Unlike India and the USA, in Europe diesel cars are in more use and are found more fuel efficient than petrol cars. An alternate fuel is *Compressed natural gas* (CNG). This is a fossil fuel substitute for petrol, diesel, and propane/LPG and is a more environmentally “clean” alternative to conventional fuels. It is being increasingly used in India to combat vehicular pollution.

Catalytic Converters: The catalytic converter is one of the most effective emission controls which processes the exhaust to remove pollutants. Vehicles with catalytic converters require unleaded fuel, since lead forms deposits that block the catalytic converter. The catalytic converter comprises a ceramic support, a wash coat (usually aluminium oxide) to provide a metal (platinum, rhodium, and palladium are most commonly used) to perform the catalyst function (Figs. 19.15 to 19.16). Catalysts containing palladium are more sensitive to the sulphur content of gasoline than platinum/rhodium catalysts. Two types of catalytic converters are commonly used in automotive engines: oxidation (two-way) catalysts control hydrocarbon and carbon monoxide emissions and oxidation-reduction (three-way) catalysts control hydrocarbons, carbon monoxide, and nitrogen oxides. The catalytic converter is attached to the exhaust system of a car. It is coated with catalysts of platinum and rhodium. As the auto exhaust gases pass over the catalysts, carbon monoxide is oxidized to carbon dioxide, oxides of nitrogen are reduced to nitrogen, and unburnt hydrocarbons are oxidized to water and carbon dioxide. As a result, the gases released to the air are mostly free from the pollutants.

Two-way catalysts use platinum, palladium, or both to increase the rate of reaction between oxygen, unburned hydrocarbons, and carbon monoxide in the exhaust (Fig. 19.15). This reaction would normally proceed slowly. Catalyst effectiveness depends on its temperature, the air-fuel ratio of the mixture, and the mix of hydrocarbons present. Three-way catalysts generally use a combination of platinum, palladium, and rhodium (Fig. 19.16). In addition to promoting the oxidation of hydrocarbons and carbon monoxide, palladium and rhodium promote the reduction of nitric oxide (NO) to nitrogen and oxygen.

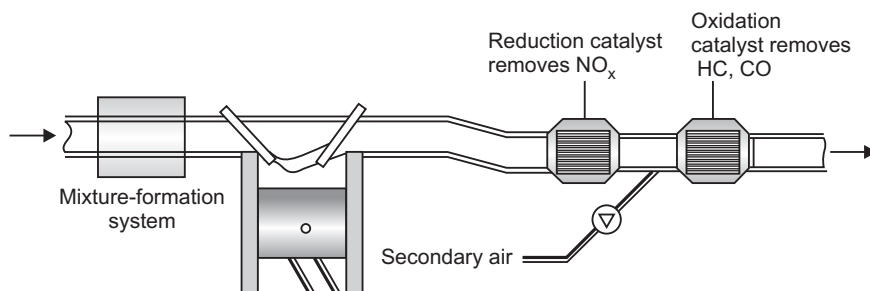


Fig. 19.15. Dual bed oxidation catalytic converter (adapted from Faiz et al., 1996).

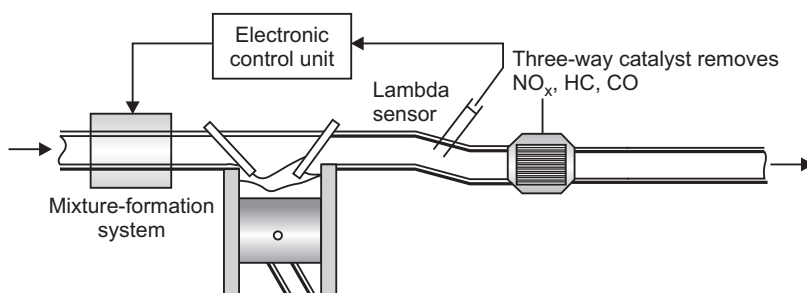


Fig.19.16. Single Bed three-way catalytic converter (adapted from Faiz et al., 1996).

Alleviating the Problem of Acid Deposition

The technological solutions to the problem of acidic rain can be both on a short and/or a long-term basis.

Shot-term methods: Liming is one of the ways to reduce the effects of acid rain. Lime is added into the lakes to neutralize the acid so that plants and animals can return to their habitats again. Liming raises the pH of medium affected by acid rain. However, this is a short term solution as liming needs to be done regularly.

Long-term methods: The only way to ameliorate the effect of acid rain on a long term basis is to reduce the amount of sulphur and nitric oxide from the waste gases of power plants and automobile exhaust. This approach has already brought about drastic reduction in the sulphur emission in the US, Germany and Japan. The various long-term approaches include using coal that is naturally low in sulphur, desulphurization to remove sulphur dioxide and other contaminants with flue-gas, and using catalytic converters in vehicles. Finding alternatives to fossil fuels for energy and reducing total demand on fossil fuel energy seem to be a long term solution to reduce gaseous pollutants causing acid rain problems.

19.8 AIR POLLUTION LEGISLATION

The National Ambient Air Quality Standards (NAAQs) prescribe specific standards for residential, rural, industrial and other sensitive areas. Industry specific standards apply to iron and steel plants, cement plants, fertilizer plants, oil refineries and the aluminium industry. The Central Pollution Control Board (CPCB) has laid down strategies for maximum permissible limits of different pollutants for different categories of industries and these standards have been notified by the Ministry of Environment and Forests (MoEF) under the Environmental Protection Act 1986. To empower the central and state pollution control boards, Air (Prevention and Control of Pollution) amendment Act, 1987 was enacted.

In 1988, the Motor Vehicles Act, was enacted to regulate vehicular traffic which also dealt with various aspects of vehicular pollution and was notified under the Environmental Protection Agency (EPA) of 1986. The emission standards for new vehicles were first notified in the year 1991 in India. Stringent emission norms were laid down by the Bureau of Standards in 1996 for gasoline and diesel. Euro I norms are applicable from 1 April 2000 and Euro II norms will be applicable all over India from April, 2005 (CPCB, 1999). Stringent Euro I and Euro II emission norms were notified by the Supreme Court in April 29, 1999 for the city of Delhi. The notification made it mandatory for car manufactures to obey the norms of Euro I and Euro II by May 1999 and April 2000, respectively, for non-commercial vehicles sold in Delhi.

Apart from safety standards several more steps are required for maintaining air clean. These include (1) measuring the amount of pollutants in air; (2) devising plan for bringing existing air quality levels into compliance with the standard, and (3) developing ways of enforcing compliance. Local agency charged with monitoring air pollution must place measuring instruments at appropriate locations. Emission standards which are the maximum amount of pollutants that may be discharged from specific sources are determined and enforced.

Air Pollution Standards

The first major step taken was to set Air Pollution Standards. They were a kind of ambient air quality standards with a margin of safety such that most sensitive people would suffer no adverse health effect. They set limits beyond which air quality would not be allowed to deteriorate. In 1971 Environmental Protection Agency (EPA) in USA identified 6 pollutants requiring some national ambient air quality standards, based on the existing knowledge about air in USA. They were ozone, carbon monoxide, nitrogen dioxide, particulate matter, sulphur dioxide and lead. Obviously, they were most known for their harmful effect on human health vegetation, material, visibility and other factors related to public welfare. Standard for each of the pollutants set in the USA were of two categories: (1) primary standards, which intended to provide a safety limit with respect to human health; and (2) secondary standards, which intended to take effects on all things related to public welfare including vegetation, buildings and other materials. Standards for some pollutants are shown in Table 19.10. The standards change as and when more knowledge is made available about the harmful effects of pollutants. In India, we have a Central Pollution Control Board for dealing with the air pollution problems. The National Ambient Air Quality Standards created by this board are similar to those set by the US. India air quality standards however, are different for industrial, residential, sensitive and rural areas on 24 hourly and annual basis (Box 19.2).

Table 19.10. National Ambient Air Quality Standards (NAAQS) set by EPA.

Pollutant [Final rule cite]		Primary/ Secondary	Averaging Time	Level	Remarks
Carbon Monoxide [76 FR 54294, Aug 31, 2011]		primary	8-hour	9 ppm	Not to be exceeded more than once per year
			1-hour	35 ppm	
Lead [73 FR 66964, Nov 12, 2008]		primary and secondary	Rolling 3 month average	0.15 µg/m ³	Not to be exceeded
Nitrogen Dioxide [75 FR 6474, Feb 9, 2010] [61 FR 52852, Oct 8, 1996]		primary	1-hour	100 ppb	98th percentile, averaged over 3 years
		primary and secondary	Annual	53 ppb	Annual Mean
Ozone [73 FR 16436, Mar 27, 2008]		primary and secondary	8-hour	0.075 ppm	Annual fourth-highest daily maximum 8-hr concentration, averaged over 3 years
Particle Pollution Dec 14, 2012	PM _{2.5}	primary	Annual	12 µg/m ³	annual mean, averaged over 3 years
		secondary	Annual	15 µg/m ³	annual mean, averaged over 3 years
		primary and secondary	24-hour	35 µg/m ³	98th percentile, averaged over 3 years
	PM ₁₀	primary and secondary	24-hour	150 µg/m ³	Not to be exceeded more than once per year on average over 3 years
Sulphur Dioxide [75 FR 35520, Jun 22, 2010] [38 FR 25678, Sept 14, 1973]		primary	1-hour	75 ppb	99th percentile of 1-hour daily maximum concentrations, averaged over 3 years
		secondary	3-hour	0.5 ppm	Not to be exceeded more than once per year

Box 19.2 Ambient Air Quality Standards (INDIA)

Pollutants	Time-weighted average	Concentration in ambient air		
		Sensitive Area	Industrial Area	Residential, Rural & Other areas
Sulphur Dioxide (SO ₂)	Annual average *	15 µg/m ³	80 µg/m ³	60 µg/m ³
	24 hours **	30 µg/m ³	120 µg/m ³	80 µg/m ³
Oxides of nitrogen as NO ₂	Annual *	15 µg/m ³	80 µg/m ³	60 µg/m ³
	24 hours **	30 µg/m ³	120 µg/m ³	80 µg/m ³
Suspended Particulate Matter (SPM)	Annual *	70 µg/m ³	360 µg/m ³	140 µg/m ³
	24 hours **	100 µg/m ³	500 µg/m ³	200 µg/m ³

Respirable Particulate matter (RPM), (size less than 10 μm)	Annual*	50 $\mu\text{g}/\text{m}^3$	120 $\mu\text{g}/\text{m}^3$	60 $\mu\text{g}/\text{m}^3$
	24 hours**	75 $\mu\text{g}/\text{m}^3$	150 $\mu\text{g}/\text{m}^3$	100 $\mu\text{g}/\text{m}^3$
Lead (Pb)	Annual*	0.50 $\mu\text{g}/\text{m}^3$	1.0 $\mu\text{g}/\text{m}^3$	0.75 $\mu\text{g}/\text{m}^3$
	24 hours**	0.75 $\mu\text{g}/\text{m}^3$	1.5 $\mu\text{g}/\text{m}^3$	1.00 $\mu\text{g}/\text{m}^3$
Carbon Monoxide (CO)	8 hours**	1.0 mg/m^3	5.0 mg/m^3	2.0 mg/m^3
	1 hour**	2.0 mg/m^3	10.0 mg/m^3	4.0 mg/m^3

*Annual Arithmetic mean of minimum 104 measurements in a year taken twice a week 24 hourly at uniform interval.

**24 hourly/8 hourly values should be met 98% of the time in a year. However, 2% of the time, it may exceed but not on two consecutive days.

A national ambient air quality monitoring has been undertaken by the National Environmental Engineering Research Institute (NEERI), Nagpur in accordance with the World Health Organisation's selected parameters in major cities of India since 1978. The objective of the air quality programme has been to reduce and control air pollution, and to formulate guidelines for Central and State Pollution Control Boards to minimize the adverse impacts of air pollution on environment and human beings.

19.9 ECONOMIC CONSIDERATIONS FOR POLLUTION CONTROL

Cost consideration of air pollution involves two factors: (i) cost of controlling air pollution and (ii) cost of not controlling air pollution. Determination of cost of the first is straight forward but that of second is difficult and can be subjective to some extent. There are evidences, however, that suggest, that cost of not controlling air pollution exceed the cost of controlling air pollution. A report on pollution control prepared in the USA (see in Kupchella and Hyland, 1989) looked at the impact of air pollution in terms of cost associated with health and damage to vegetation and crops, materials and residential properties. Certain assumption are applied in regard to cost of being sick, the cost of missing work, the value of human life, and the relationship between pollution and sickness and pollution and material damages. Losses of crop can also be enormous for example, ozone is reported to cost the US farmers between one and five billion dollars a year in lost crop production (values are of 1987, see Kupchella and Hyland, 1989). Freeman report estimated that benefits of controlling air pollution ten years later in the USA amounted to 21.4 billion dollars. The estimate made hitherto hardly consider the economic impact of air pollution on ecosystems, and even the effects on animals are inadequately considered.

Air Pollution in India

Though per capita CO_2 emission of India is only about 1.46 t/yr (compared to 4.76 t of China and 19.28 t of the USA), India ranks lowest in air quality among the 132 countries whose environments are annually surveyed with satellite data by Yale and Columbia Universities with the support of dozens of outside scientists. Sky in the northern half of India is covered with thick aerosol layer each winter. India stands 125th among the 132 countries surveyed when over all environment is considered, even behind Pakistan and Bangladesh. According to a World Bank report of 2013. India's pollution costs 5.7 percent of its GDP, i.e, 375 L cr (375 lakh crore) a year, of which air pollution costs 52%, cropland degradation 19%, water supply, sanitation 14%, pasture degradation

11% and forest degradation 4%. Globally, about 2 million people die prematurely every year because of air pollution, and many more suffer from breathing ailments, heart diseases, lung infections and cancer. Many of them are expected from India. Coal and wood fires and unfiltered diesel engines are major sources of fine dust particles which cause lethal pollution.

Overall, the major sources of air pollution are transport, cooking, household heating and power stations. Indoor air problem is quite severe in India. CNG did improve the look of air in Delhi. However, according to a central pollution control board report 2011 even CNG is not without problems. Retrofitted CNG car engines emit hazardous carboxyl and green house gas methane (most of the CNG cars in India have retrofitted engines).

PM₁₀ (particulate matter < 10 microns called respirable particulates); affects human health, as it can be inhaled and accumulated in respiration system. As the data of in 2008 indicate, it is a serious problem in 52% of 63 cities sampled in the country. In general, concentrations were more than 1.5 times of the standard permissible level. Records of PM_{2.5} (particulates <2.5 µm in diameter; they pass into blood stream and cause additional problems) are not available, but are likely to be high in the air of our cities (China has just begun to monitor PM_{2.5} in 2013).

In spite of improvement in air quality since passing the Air (Prevention and Control) Pollution Act in 1981, Indian cities are still highly polluted. India heavily depends on coal (more than 80% of energy produced from coal), which emits large amount of pollutants. Then, there are about one lac of brick kilns in the country, which produce soots in all parts of India (except in mountains). Fuel wood and open biomass burning, fuel adulteration, inefficient vehicles and traffic congestion are considered to be main reasons of high air pollution. There is a persistent haze and smoke cover in the country

Fuel wood and biomass burning

There are some 100 million households where firewood and various forms of biomass are used in cook stoves (Chullas'). Of this, more than 60% consist of twigs, dung cake and agricultural waste (thus firewood being only 40%). The country level magnitude of this biomass is equal to 148.7 million tonne of coal replacement per year. They burn at a low efficiency, releasing high levels of smoke, PM₁₀ particulate matter, NO_x, SO_x, polyaromatics, polyanomonia, formaldehyde, CO, etc (164 million tonnes of combustion products annually), causing 300,000-400,000 premature deaths annually. The smoke released by the traditional cooking fuel is expected to contribute at least 4.2 million tonnes of fine particles annually as per conversion rates of Environmental Protection Agency (EPA) of the USA. India's air pollution is largely because of poverty and inefficiency. Access to clean cooking energy can address much of this problem. Urbanization brings about this change to an extent, as of the total cooking fuel of an average household, the traditional fuel (firewood, dung cake, crop residue, etc) accounts for 24% in urban areas and 90% in rural areas. Animal dung could be used to produce biogas, a clean cooking energy, and the slurry of biogas plants could be used to make fertilizers for crops.

Apart for cooking, biomass is burned for heating and disposal of waste material, including crop residues. Small forest fires are common in all areas where people live in and around forests. In mountain fuel wood consumption increases with altitude from less than 1 kg/person in foothills

to 2.7 kg/person at about 2500 m altitude in Himachal state of India (Rawat et al., 2009). The total biomass, combustion in India annually emits 0.25 Tg black carbon, 0.94 Tg organic matter, 2.04 Tg $PM_{2.5}$. The black carbon release from biomass burning is about three times as much as from other sources, namely vehicles and industrial sources.

Transport

India's per vehicle emission is among the highest in the world. Many vehicles particularly two-wheel and three-wheel vehicles lack catalytic converters. Since 2005, India has adopted emission standard of Bharat Stage IV vehicles, which is equivalent to Euro IV standard. Adulteration of gasoline and diesel with cheap fuel like Kerosene is common in India and much of South Asia. Such an adulterated fuel raises the emission of hydrocarbons, CO NO_x, particulate matter and sulphur. The high sulphur level can deactivate catalyst of vehicles, and once it is deactivated the emission of all pollutants is raised. Because of inadequate roads (account for 20-22% of a city area is in developed countries, compared to less than 10% in India), traffic congestion is a serious problem. Compared to the car emission at the speed of 55-75 km/hour, it is twice at the speed of 20-40 km/hr, and 4-8 times at the speed of 20 km per hr.

India's green house gas emissions

Even with a very low per capita CO₂ emission, India is the third largest emitter (1.65 Gt per year) after China (6.9 Gt) and the USA (5.2 Gt per year). India's thermal power plants are highly inefficient. Through CO₂ emission per unit GDP purchasing power parity basis has improved by over 10% from 1990 to 2009, it is still high, compared to those of developed countries.

Present air quality status:

Indian Pollution Control Board now has a fairly elaborate pollution monitoring network. It consists of 308 monitoring stations distributed across 115 cities/towns in 25 states and 4 union territories, which routinely measure SO₂ NO_x, suspended particulate matter (SPM) and respirable particulate matter (PM₁₀). As per 2010 data analysis the key points are as following:

- Concentrations of PM₁₀ and other particulate matter are higher in Indian cities than air quality norms of India and world.
- Sulphur dioxide concentration is on decline in several cities largely because of the introduction of clean fuel standards and, access to LPG as a cooking energy, and use of CNG in vehicles.
- Monsoon acts as a 'scavenger' of air particles including black carbon that is why air is the cleanest in September-October, and dirties during winters when open burning is common.
- Mining operations can cause severe local environmental stress. For example, stone mining has several occupational stresses causing pneumoconiosis, silicosis, tuberculosis, asbestosis, and asthma. These problems are very common in Rajasthan which produces 50 million tones of dimensional stones per year. Aluminum refinery in Orissa is a source of heavy pollution in surrounding villages.

19.10 NOISE POLLUTION

Noise pollution can be defined as the disturbing and harmful loud sound into the ambient atmosphere that interferes with hearing, causes physiological stress and hampers the work efficiency. Noise is the most common health occupational hazard in the cities and big industry. Sound travels in

pressure waves and affects our ear drums. The intensity of sound (loudness) is measured in decibel (dB) or modified decibel scale called the decibel – A (dB_A). The dB_A scale takes into account high pitched sounds to which human ear is more sensitive. The threshold level for hearing in human beings is 0dB_A, a level at which the ambient sound intensity is equal to the reference intensity of the noise. Noise can range from 0 to more than 120 dB_A, at which point physical discomfort begins. The dB_A scale is logarithmic, i.e. 10, 20 and 100 decibels represent 10 times, 100 times and 10^{10} times the threshold intensity, respectively. Some examples of sound levels are shown in Fig. 19.17 and their effects are included in Tables 19.11 and 19.12.

Table 19.11. Some examples of noise pollution in decibel A scale.

Example	dbA	General effects/perception
Hearing threshold	0	Hearing threshold
Normal Breathing	10	Very quiet
Average home	40	Quiet
Television/vacuum cleaner	70	Moderately loud
Washing machine/Typical factory	80	Very loud
Jet aircraft at 300 m	100	Very loud - uncomfortably loud
Chain saw	120	Uncomfortably loud painfully loud
Jet at take off	150	Painfully loud

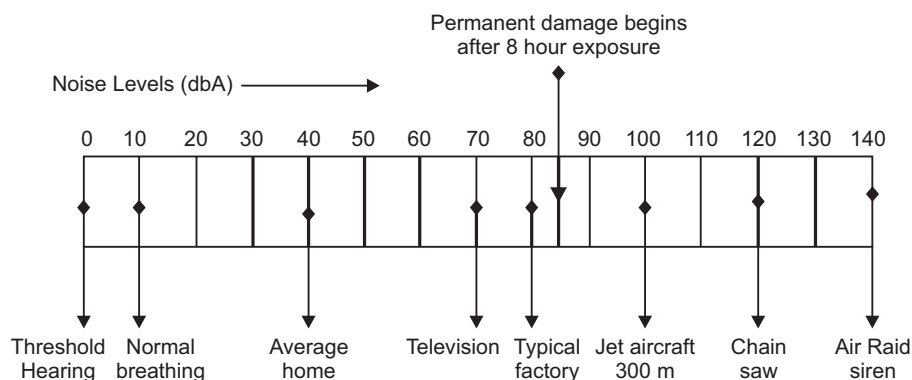


Fig. 19.17. Noise level of some common sounds (based on Miller Jr,1988).

Sources and effects

Most of the noise that travels in the atmosphere is of human origin from vehicles, sound amplifiers, from industrial machines, and mine site detonation, etc. The landing and takeoff of various types of aircrafts create a lot of noise pollution in residential areas located near the busy airports.

Human ears can take sound up to 60 dB_A without damage or hearing loss. At a noise level of 85 dB_A, damage to hearing can occur after an exposure of eight hours. Very loud noises (more than 140 dB_A) cause pain and permanent hearing loss. Prolonged exposure to excessive noise seriously

affects heart beat, peripheral circulation, hypertension, muscle tension, headache and causes other physiological effects.

Control of Noise pollution

In India, the Central Pollution Control Board has recommended the zone-wise ambient noise levels (Table 19.12). A ‘silent zone’ around 100 metres of hospitals and schools can give comfort to ailing patients and allows smooth studies of students. Putting up some type of shields between the noise producer and the hearer can help control noise pollution. Forests and dense growth of hedge of bushes can effectively act as noise barriers. Industrial workers and runway traffic control personnel may use earplugs to protect themselves from unwanted noise exposure. In heavy industries, soundproof insulating jackets or filters are used by workers to reduce noise from machines. Efforts are needed to create an awareness among the general public about the harmful effect of noise pollution.

Table 19.12. Zone-wise ambient noise levels according to Central Pollution Control Board in India.

Zone	Day (dbA) 6 hr - 21 hr	Night (dbA) 21 hr – 6 hr
Industry	75	70
Commercial	65	55
Residential	55	45
Silent zone	50	40

Anthropogenic noise and wildlife conservation

Recent discoveries on the impact of anthropogenic noise on various organisms have led to the emergence of a new discipline of applied ecology. Species and individuals within a population differ in their responses to noise stimulus (Francis and Barber, 2013). Anthropogenic noise is divisible into two types: (I) chronic and frequent noise which interferences with animals’ abilities to detect important sounds, and (II) intermittent and unpredictable noise which animals perceive as a threat. Both can lead to fitness costs. A noise may not affect an organism’s every activity. For example, an animal might show no response to noise with regard to habitat occupancy or foraging, but may fail to pair because of physiological stress, and hence have fewer offspring. Most noise-related impacts involve behavioural responses across following four categories: (i) Changes in temporal patterns—e.g., foxes avoid crossing roads when sound is high; European robins start singing in night to avoid day time acoustic interferences. (ii) Changes in spatial distribution —e.g., site abandonment and decreases in spatial abundance both when low frequency noise interferences with communication, and when high variable sound is perceived as a threat. (iii) Decrease in foraging efficiencies coupled with increased vigilance. (iv) Change in mate attraction (Francis and Barber, 2013) —e.g., in noisy areas female great tits (*Parus major*) can hear male songs only at very high frequency, resulting in smaller clutches and fewer fledglings.

In nature, even very low anthropogenic sound may severely affect animals, though their environment might be filled with high sounds. In oceans anthropogenic sounds can reduce abundance of fishes, affect migration routes, and cause injuries to animals. There are evidences to indicate that even deep marine animals are getting affected by noise (due to shipping and offshore human activities). Even at low frequency sound some whales find difficult to communicate. Cephalopods like *Octopus cyanea* lose sense of balance and position. Sound causes lesions due to the damage to hair cells (Andre et al., 2011).

Water Pollution

Water is the most important natural resource and essential for life as it provides habitat for diverse types of aquatic life in rivers, lakes and oceans and it makes 65% of human body. The great historical cities grew around rivers and lakes because of human dependence on water. The expansion of agriculture and industrial development has not only increased water consumption considerably but it has affected water quality also. Water is easily polluted because of its great ability to dissolve substances. Even before raindrops strike the earth, they start picking up pollutants. Once on the ground, water picks up things rapidly, and becomes contaminated. The various human activities and industries not only require water in large amounts, they also pollute it while using it. Apart from industries, water is polluted by agricultural and domestic or municipal sources. The water pollutants vary in nature; they include biological agents, chemicals that make water rich in nutrients, chemicals that poison water, sediments, heat and radioactive waste. Nearly all water bodies are affected by pollution, including ground water. In many developed countries, water pollution is a major problem and many river basins have been found to show high organic matter concentration (UNEP/ GEMS, 1995). In rapidly industrializing countries like India, China and Brazil, untreated sewage and industrial wastes create substantial pressure on water quality (Shiklomanov, 1997; UNEP/GEMS, 1995). Polluted water loses its economic and aesthetic value. Therefore, monitoring the quality of water is an important task. Studies have been carried out by the UNEP GEMS/ water programme on a global scale (UNEP/GEMS, 1995) and several other programmes in different regions of the world. Over the years, water pollution has emerged as an important issue in India as most of the rivers are polluted, which are having substantial negative impact on human health and aquatic life. The Central Pollution Control Board (CPCB, 1999; 2000b,c) and the Ministry of Environment and Forests in India have undertaken studies on the state of water quality of major river basins in India. In developing countries 80-90% of waste water discharged in rivers daily is untreated, contributing to the death of over two million people annually from diarrhoeal diseases. Polluted water kills people more than violence of all kinds combined. Controlling water pollution is urgent for ecological sustainability of water resources as well as for underlying economic reasons and human health. Therefore, it is important to learn about various types of water pollution, their sources, effects on human health, water quality, and the technological and legislative measures for controlling water pollution in aquatic systems.

20.1 TYPES OF WATER POLLUTANTS

Defining water pollution

Water pollution can be defined as any unfavourable change in physical, chemical and biological properties of water that makes it harmful to humans and other life forms. Human actions reduce the usefulness of water as a resource. However, the decrease in usefulness is in a relative term. A

water source may be fit for navigation and fishing, but not for irrigation or drinking which requires pure water. Tolerance limits for selected substances/characteristics in water to be used for drinking purposes are given in Box 20.1, and selected water quality criteria of fresh waters for various uses are given in Box 20.2. Examples of polluted water include, red, lifeless, acid mine drainage water, raw sewage-laden waters that enter the Ganga in Varanasi, or any other city, turbid domestic waters full of detergents containing phosphate, the rust-coloured, opaque water of a lake, full of toxic metals, *Microcystis* and *Escherchia coli*, or a highly nutrient-rich village pond water, coated with *Lemna* or *Azolla*, the heated water from the cooling towers of an electric-generating plant, and of the lake superior contaminated with asbestos fibres. Thus water pollution is common in all the regions of world, the magnitude and the type of pollutants may vary.

Box 20.1. Tolerance Limits for Drinking Water as per IS-10500-1991.

Substance or Characteristic	Requirement (Desirable Limit)	Undesirable Effect Outside the Desirable Limit	Permissible Limit in the Absence of Alternate Source
pH Value	6.5 to 8.5	Beyond this range the water will affect the mucous membrane and / or water supply system	No relaxation
Iron (mg/l, Max.)	0.3	Beyond this limit taste / appearance are affected, has adverse effect on domestic uses and water supply structures, and promotes iron bacteria.	1.0
Chloride (as Cl mg/l, Max.)	250	Beyond this limit, taste corrosion and palatability are affected	1000
Dissolved Solids (mg/l, Max)	500	Beyond this palatability decreases and may cause gastro intestinal irritation	2000
Calcium (as Ca) (mg/l, max)	75	Encrustation in water supply structure and adverse effects on domestic use	200
Sulphate (as SO ₄) (mg/l, Max)	200	Beyond this causes gastro intestinal irritation when magnesium or sodium are present	400
Nitrate (as NO ₃) (mg/l Max.)	45	Beyond this methaemoglobinemia	100
Nitrite (mg/l)	0.3	—	—

Types of water pollutants

Water pollutants can be divided in various ways, one simple way is given below:

1. Organic matter – In water, organic matter consumes oxygen in the process of decomposition.
2. Biological agents – e.g., *Vibrio cholerae* that causes cholera.
3. Nutrient enrichment e.g., high contents of phosphates and nitrates in water causes eutrophication.
4. Toxic chemicals or chemical poisons – e.g., arsenic, mercury and polychlorinated biphenyls from insecticides; oil and petroleum products; plastics and industrial chemicals.
5. Physical agents – e.g., heat that raises temperature of water used for cooling purposes in industries; suspended particles with inorganic and organic matter.

6. Radioactive wastes – e.g., radioactive materials that make their way into water supplies from the processing of uranium ore or from nuclear power plants.
7. Salts – e.g., agents of hardness such as calcium salts.

Box 20.2 Primary water quality criteria for various uses of fresh waters, as laid down by the central board for the prevention and control of water pollution (1979)

Sr. No.	Characteristics	A*	B*	C*	D*	E*
1.	Dissolved Oxygen (DO) (mg/l, min.)	6	5	4	4	–
2.	Biochemical Oxygen demand (BOD) (mg/l, max.)	2	3	3	–	–
3.	Total coliform organisms** (MPN/100 ml. Max.)	50	500	5000	–	–
4.	pH value	6.5-8.5	6.5-8.5	6.0/8.5	6.5/8.5	6.0-8.5
5.	Free ammonia (as N)(mg/l. Max.)	–	–	–	1.2	–
6.	Electrical conductivity (micro mhos/cm max.)	–	–	–	–	22.50
7.	Sodium absorption ratio, SAR max.	–	–	–	–	26.00
8.	Boron (mg/l, Max.)	–	–	–	–	2.00

* Use classes

** There should be no visible discharge of domestic and industrial wastes into Class 'A' waters. In case of class 'B' and 'C' the discharge shall be so regulated/treated as to ensure maintenance of the stream standards.

Fresh water classes according to some criteria and permissible uses.

A. Source of drinking water, without conventional treatment, but after disinfection.

B. Out door bathing.

C. Source of drinking water with conventional treatment & disinfection.

D. Fisheries, propagation of wild life.

E. Irrigation, industrial cooling, controlled disposal of waste.

Biological oxygen demanding wastes

Oxygen is added to water by diffusion from atmosphere and photosynthetic activity of green plants. The amount of dissolved oxygen in water is an indicator of its quality and the kind of life it can support. Water with dissolved oxygen content of 8 mg/l and more can support many fish and other desirable forms of aquatic life (Fig. 20.1). Water with dissolved oxygen content of 2 mg/l can only support worms, bacteria and other decomposers. The primary source of organic matter in lakes and rivers are waste water from industrial effluents and domestic sewage. Pollution load due to sewage or other organic wastes is measured in terms of their biological oxygen demand (BOD) and it is an expression of oxygen depleting effect of a given amount of organic matter via the action of decomposers. The decomposers use organic matter as source of energy and nutrients and in the process of oxidation use up the available oxygen. BOD is usually expressed in mg of dissolved oxygen content per litre of water for a specific number of days at a given temperature. Large amount of sewage in water generates high BOD and lower level of dissolved oxygen (Fig. 20.1). BOD of clean fresh water is 2 mg/l, whereas values exceeding 5 mg/l indicate pollution. Several factors affect BOD. Cool water dissolves more oxygen and, in it the rate of decomposition of organic matter is slower than in warm water. The dissolved oxygen should not drop below 3 ppm at any time; in fact should remain above 5 ppm for the normal health of an aquatic system. The large plant biomass

accumulation and decomposition cause organic matter pollution and anoxic conditions. Among the common organisms of water affected by organic pollution is *Tubifex* worm. In polluted waters of Indian rivers, the BOD content has been found to be more than 3 mg/ litre in 57% samples, 3-6 mg/ litre in 28% samples and above 63 mg/ litre in 15 % samples (Fig. 20.2).

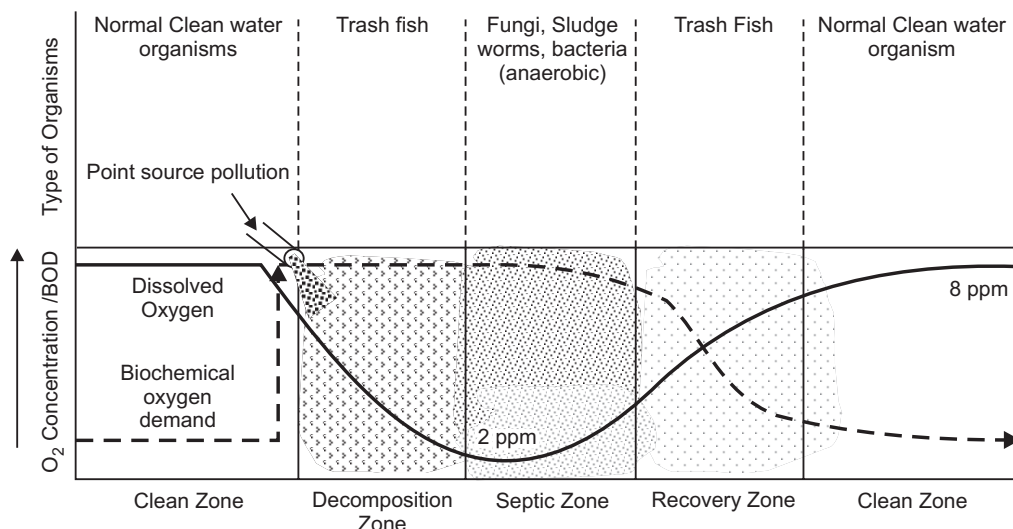


Fig. 20.1. The Oxygen sag curve (solid line) and biological oxygen demand curve (dotted line) for a stream receiving pollutants from a point source.

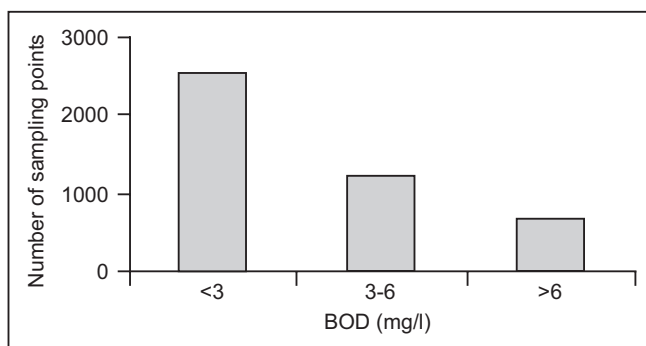


Fig. 20.2. The level of dissolved oxygen content in water at different locations in India (based on data from CPCB, 1999; UNEP, 2001a).

The organic matter load of water is also indicated by the chemical oxygen demand (COD). COD is the oxygen equivalent of the organic matter in the water that is susceptible to oxidation by a strong chemical oxidant. COD is always higher than BOD.

Eutrophication

Natural eutrophication is a process that occurs to all lakes over time as the weathering of rocks and soils from the surrounding catchment area leads to an accumulation of nutrients and associated sediments in the water. Young lakes (and man made reservoirs) usually have low levels of nutrients and correspondingly low levels of biological activity. Such lakes are oligotrophic (Greek word *oligos* meaning little or few), meaning little-nourished. Old lakes usually have high levels of nutrients and correspondingly high levels of biological activity. Such lakes are referred to as being eutrophic (Greek word *eu* meaning well), literally *eutrophic* means well-nourished.

The natural time scale for the aging of a lake from being oligotrophic to eutrophic is of the order of thousands of years. However, a high rate of input of nutrients (from human activities) can increase the rate of aging significantly resulting into eutrophic conditions. This cultural or anthropogenic eutrophication has already happened in many parts of the world in both temperate and tropical regions. Both phosphorus and nitrogen contribute to eutrophication. The renewal of all the water in a lake may take up to a hundred years compared to a few days for the renewal of the water in a river. Consequently, lakes are particularly susceptible to pollution such as artificial eutrophication.

Among the nutrients, phosphorus and nitrogen are most associated with eutrophication and water pollution, however in many lakes excessive P inputs are the primary cause. Human activities add excessive amounts of plant nutrients (primarily phosphorus, nitrogen and carbon) to streams and lakes in various ways (Fig. 20.3). The run-off from agricultural fields, feed lots, urban lawns and golf courses is an important source of the nutrients, particularly of $\text{NO}_3\text{-N}$, to the aquatic systems. Untreated or partially treated domestic sewage is another major source and usually has a high $\text{PO}_4\text{-P}$ content. Sewage is a major source of the phosphorus to the lakes when detergents contain large amounts of phosphates. Some detergents contain large amounts of tripolyphosphates. Up to 25% of phosphate and nitrate fertilizers applied to crop fields make their way into water, and cause eutrophication. These nutrients, particularly phosphates are generally found in inadequate quantities in nature, therefore their abundance triggers rapid growth of algae and macrophytic plants (Fig. 20.3). Cultural eutrophication also stimulates blooms of algae (e.g., cyanobacteria in freshwater systems; dinoflagellates and other phytoplankton in coastal waters). Algal blooms cause turbidity of water and block the penetration of sunlight, which causes the death of the underwater plants that provide shelter to the aquatic animals. The dissolved oxygen content of water is reduced and decomposition of excessive blooms releases methane, and ammonia gases. The cyanobacteria also produce toxins which may even be lethal to animals and humans. The major differences in oligotrophic and entrophic waters are listed in Box 20.3.

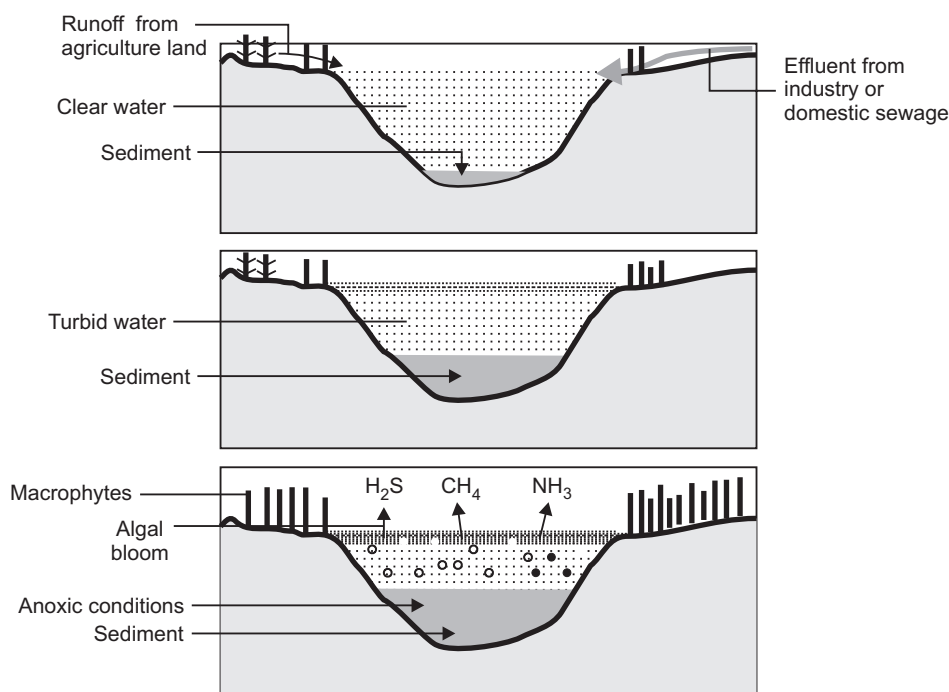


Fig. 20.3. Eutrophication of lake due to run-off from agriculture field, and addition of effluents from industry and domestic sewage.

Box 20.3 Differences between oligotrophic and eutrophic waters

Parameter	Oligotrophic	Eutrophic
Phytoplankton		
Quantity	Poor	Rich
Variety	Many species	Few species
Distribution	To great depths	Trophogenic layer
Migration	Extensive	Limited
Water blooms	Very rare	Frequent
Characteristic algal groups	Chlorophyceae, Diatomaceae, Chrysophyceae	Cyanophyceae, Diatomaceae
Physico-chemical features		
Surface to volume ratio	Low	High
Mean depth	Shallow	Deep
Transparency (average secchi disc depth)	High	Low
Total dissolved solids	Low (around 100 ppm or less)	High (> 100 ppm)
Dissolved oxygen	High, all depths all year	Depletion in hypolimnion
Specific conductance (mhos at 18°C)	< 200	> 200
Biological oxygen demand	Low	High
Chemical oxygen demand	Low	High
NO ₃ -N	Low	≥ 300 µg/l
PO ₄ -P	Low	≥ 10 µg/l

The phosphates act as water softeners to improve the cleaning action, but they are also powerful stimulant of algal growth as they are washed into lakes. The excessive growth of algal blooms is promoted by the nutrient addition which change water quality, cause oxygen depletion and result in fish kills. Many native species can disappear or replaced by species more adapted to the changing conditions. Rotting and stinking algae foul beaches and shorelines. Increased growth of algae and vegetation may impede water flow and the movement of boats; the water may become unsuitable for drinking even after treatment; decrease in the amenity value of the water (e.g., it may become unsuitable for water sports such as sailing); disappearance of commercially important species (such as trout). There is a decrease in health and diversity of indigenous fish, plant and animal population. The spawning and nursery habitat is destroyed and waterfowl has less availability of food.

On the basis of nutrient concentration, the lakes are classified as: oligotrophic, mesotrophic, eutrophic and hypertrophic. An eutrophic lake with heavy influx of sewage and runoff from rural and urban areas may become exceptionally rich in nutrients and converted to a hypertrophic water body. When large amounts of extrasystem organic matter (particularly humic materials) enter into a water body staining the water brown - the system is termed dystrophic. The hypertrophic and dystrophic water bodies become extinct move quickly than the water body undergoing natural eutrophication.

A reduction in the extent of artificial eutrophication and restoration of dying water body can in principle be achieved by:

- Management of catchment area to reduce soil erosion.
- Reducing the use of phosphates in detergents.
- Reducing the use of nitrate containing fertilizers.
- Using sewage treatment methods to remove phosphate and nitrate before discharge of the effluent into the water body.
- Directing wastewater away from lakes to rivers and the sea.
- Aerating the hypolimnion to prevent oxygen depletion without disturbing thermal stratification.
- Removing phosphate-rich sediments by suction dredging. The sediments can be stored in nearby basins, diluted and treated with aluminium sulphate to precipitate phosphorus and then returned to the lake.
- Treating sediment *in situ*. An oxidizing solution of calcium nitrate is injected into the top sediment layer. The denitrifying bacteria mediate the reduction of NO_3 whereby organic matter is oxidised to CO_2 and H_2O , releasing gaseous N_2 in the atmosphere. During this process iron compounds are oxidized and phosphorus is precipitated.
- Harvesting macrophytes.
- Removal of root-felt from the upper layers of sediment before raising the water level.

Disease - causing organisms

These are infectious organisms come into the water body from the wastes of humans and animals. Municipal wastewater contains many bacteria, viruses, protozoa, parasitic worms and other infection agents that cause human and animal diseases. Some human diseases transmitted by polluted water are given in Table 20.1.

Table 20.1. Types of biological agents responsible for water-borne diseases (Lund, 1978).

Types of organisms	Species	Diseases
Bacteria	<i>Vibrio cholerae</i>	Cholera-Paracholera
	<i>Salmonella typhi</i>	Typhoid fever
	<i>Escherchia coli</i>	Enteritis
	<i>Shigella dysenteriae</i>	Dysentery
	<i>Clostridium perfringens</i>	Enteritis
Protozoans	<i>Entamoeba histolytica</i>	Amoebic dysentery
Helminths (multicellular)	<i>Schistosoma</i> of different types (flukes)	Schistosomiasis
	<i>Taenia saginata</i> (beef tapeworm)	Taeniasis
	<i>Ascaris</i> (round worm)	Ascariasis
	<i>Anchylostoma</i> (hookworm)	Anchylostomiasis
Viruses	Polio- viruses, Coxsackie- viruses, Echo-viruses	Fever, headaches, nausea, diarrhoea, muscular pains, meningitis, paralytic polio, hepatitis
	Hepatitis-A- virus	Infectious hepatitis

The common intestinal bacterium, *Escherchia coli* is typically used as an indication of the indirect measure of sewage present in water and as a direct measure of the presence of disease causing agents. The water sample is first passed through a filtering apparatus. The filter disk is

then placed on a medium that supports coliform bacteria for a period of 24 hr. After inoculation, the number of greenish bacterial colonies is counted. Each colony of *E. coli* arises from a single coliform bacterium in the original water sample. In the drinking water the amount of coliform bacteria should not be more than one/100 ml (Fig. 20.4). In polluted water, the number of coliform bacteria is high. In polluted water of Indian rivers coliform bacteria have been found to vary from 500 to more than 5000/100 ml water (Fig. 20.5).

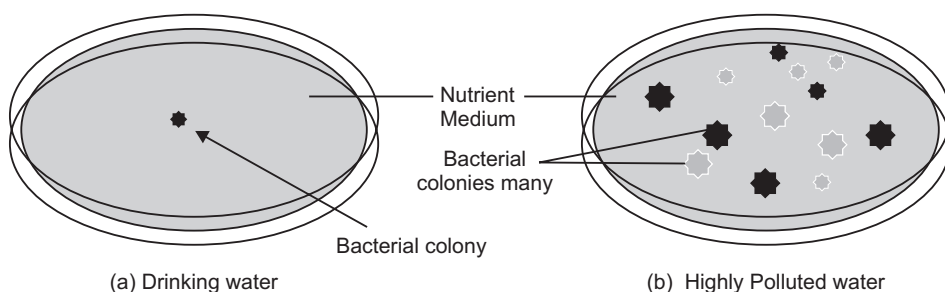


Fig. 20.4. Coliform bacteria growing on nutrient medium in a pettier dish. (a) drinking water, one colony per 100ml water (b) highly Polluted water, many colonies per 100m water.

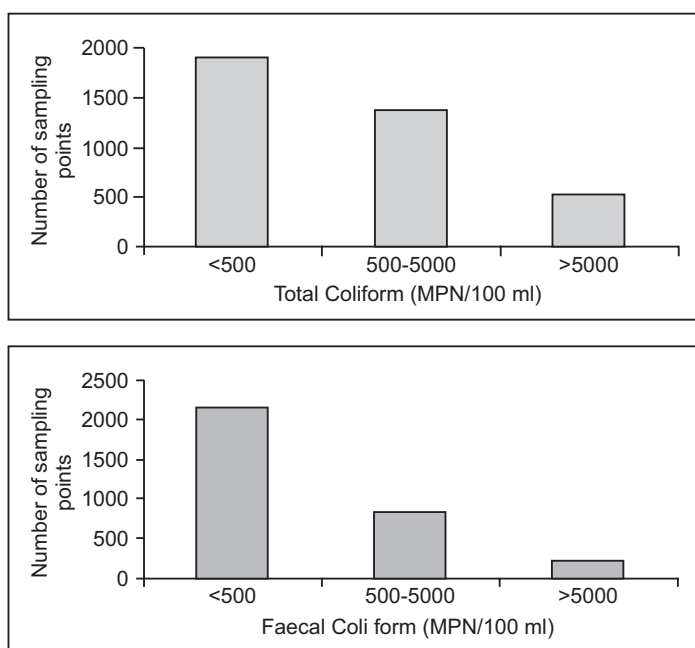


Fig. 20.5. The range of the most count number total coliform and fecal coliform bacteria in water at different locations in India. (based on data from CPCB, 1999; UNEP, 2000a).

Toxic chemicals

The toxic chemicals get into water bodies from various sources, such as arsenic from insecticides, cadmium from electroplating operations, and chromate from different industrial processes. Lead, mercury, selenium, cyanide, chromium, zinc, copper and nickel have varied sources. They are toxic to fish, other aquatic life forms and humans as they interfere with the action of enzymes and other biochemicals. WHO guide values for selected chemicals in drinking

water are given in Box 20.4. Biological magnification (increase in concentration with increasing trophic level, the maximum being in the top carnivores) along the food chain further complicates the problems of metal toxicity (See also Chapter 21). The mercury introduced to food chain killed nearly 50 fish eating people in the sea coast village of Minimata Bay, Japan, and many were affected by neurological disorder. This disease was called Minimata disease. The methylation of mercury (incorporation of inorganic mercury into an organic methyl (CH_3) compound by microbes) in the lake sediment or sediment of other aquatic bodies is the key to the transport of mercury into food chains. The organic toxins include pesticides like DDT and chlordane, hydrocarbons, such as benzene which can cause cancer, and other chemicals causing genetic defects. Polychlorinated biphenyl (PCBs) compounds are similar to DDT in their extreme stability. These extremely toxic organic compounds originate in a great variety of manufacturing processes, such as manufacture of brake lining, grinding wheels, glass, ceramics, and various types of coating, flame-proofing paint, varnishes, sealants, electric equipments, tyres, plastic coating, soap, water treatment chemicals and others. Because of the extreme toxicity, the accepted level of PCBs in fish to be eaten is only 2 ppm. In higher concentration it causes death, its long-term effect in lower concentrations is not yet known. The biological magnification of PCB in the Great Lake region of United States is shown in Fig. 20.6. Similarly, fish growing in sea water with just 0.1 ppb DDT will have 57 mg DDT per kg body weight and the seagulls which feed on these fish will have 800 mg DDT per kg body weight. Due to high DDT content the shell of the seagull eggs becomes thin and very fragile.

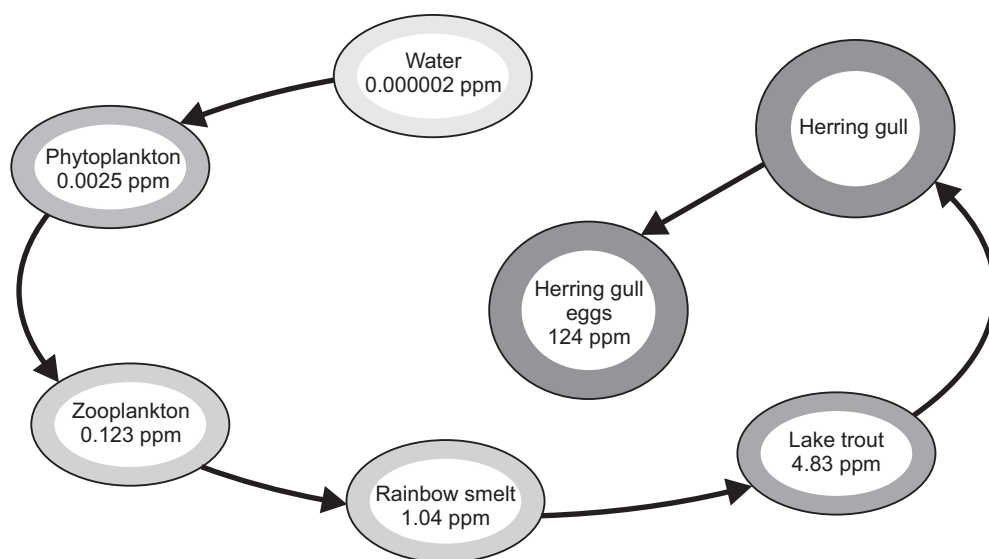


Fig. 20.6. Biological magnification of polychlorinated biphenyls (PCB) in lake superior in the great lakes region in USA.

Thermal Pollution

Since water is able to absorb large amounts of heat with small increases in its own temperature, it is much in use as a cooling medium. This explains the location of many industrial plants along rivers. The heat in water body is regarded as a pollutant, as high temperatures may disrupt aquatic life and ecosystem processes. Since species vary in their temperature tolerance ranges and levels at which they perform the best, rise in water temperature leads to change in species composition, often undesirable ones. In life cycle of an organism, certain stages are particularly sensitive to change in temperature. For example, trout eggs take 165 days to hatch at 3°C , but only a month at 1°C . A temperature rise may not affect growth of plant, but may suppress growth of pollen tube.

In a lake, the rise in temperature may lead to disappearance of diatoms and dominance of blue-green algae. Increase in BOD and consequent oxygen depletion is one of the common and principal consequences of heat. Less oxygen dissolves in warm water than in cold water; with increase in temperature from 0° C to 40°C, the dissolved oxygen capacity of water decreases (Fig. 20.7). In a well mixed water, oxygen content is about 14.6 ppm at 0°C, but it is only 6 ppm at 40°C.

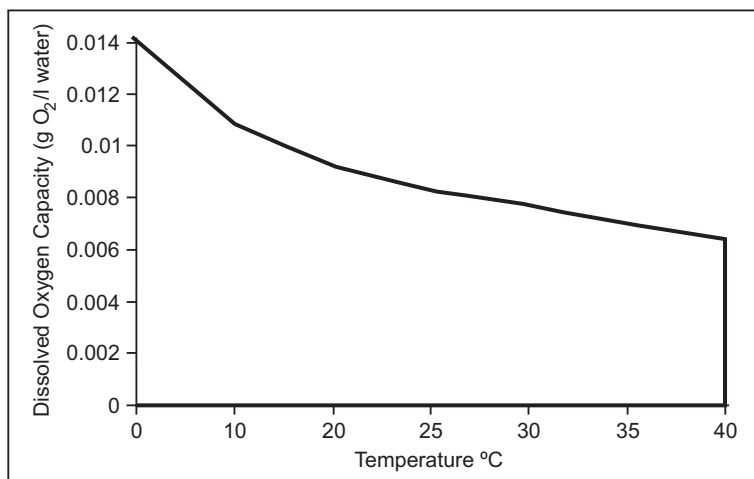


Fig. 20.7. The dissolved oxygen content decreases with increase in temperature.

Radioactive substances

Problems of radioactive wastes are discussed in detail in Chapter 21. Here we simply list the ways they enter aquatic bodies.

1. Water from nuclear power plants (they are also source of heat waste).
2. Fallout from nuclear weapons testing.
3. Uranium ore processing.
4. Wastes from research laboratories, hospitals that use isotopes in diagnosis and therapeutic procedures.

Low-level wastes from hospitals may be dumped illegally. Laundering of contaminated clothing from laboratories is another source of radioactive pollution.

Other pollutants

The other forms of water pollutants include those which change colour of water, (e.g., effluents from dye manufacturing and textile industries, printing presses, etc.), floating liquids and solids, suspended or sedimentary solids, foam formation, and inorganic salts, acid and alkali. People object more to colouring agents than to more harmful pollutants. Acid mine discharges derived from iron oxide and sulphate make the water red and yellow. Oil, grease and a number of other materials that float on water surface are pollutants in the aesthetic sense. But some of them can also be toxic, and floating objects can reduce light penetration and diffusion of gases. Sediment pollution consists of excessive amounts of suspended soil particles that eventually settle down and accumulate at the bottom of a body of water. Sediment pollution comes from erosion of agricultural lands, forest soils exposed by deforestation, degraded stream banks, overgrazed rangelands and construction, etc. Undissolved solids or solids which take long time to dissolve, lower the water quality physically. Silt and other particles close waterways, fill up dams, and make water muddy. These solids can also choke fish gills and filter feeders, like clams, and make water unsuitable for industrial use. Thus, sediment pollution may adversely affect the shipping industry.

Addition of some inorganic salts, such as that of calcium to water bodies contributes to the hardness of water (hardness is the degree to which certain dissolved salts make it difficult to produce a soapy lather in water). These salts can become deposited within pipes and equipments; hardness interferes with dyes in the textile industry and causes problems in the beer-brewing industry. Magnesium sulphate has a cathartic effect, causing a chronic chemical type of diarrhoea.

Acids produced as by product of many industries can disrupt the functioning of aquatic ecosystems. Alkalies, such as sodium hydroxide, are produced from soap manufacturing and the tanning of leather.

20.2 SOURCES OF WATER POLLUTION

Sources of pollution are divisible into two categories: (i) *point sources* – pollution sources which have clearly identified location, such as a channel through which wastes are discharged (e.g., pipes and sewers), (ii) *non-point sources* – when location of pollution sources is diffuse and difficult to mark, such as runoff from urban areas and agricultural fields. The former are easy to control and regulate, but the latter pose major difficulties.

The common sources of water pollution are agricultural areas, municipal sewage, industries, electrical utilities, mines, and oil production, its use and transport. The status of water pollution and the sources of water pollution in major rivers of India are summarised in Table 20.2. The exiting water quality of D and E classes is very poor as compared the desired classes of A, B and C. The Yamuna river is most polluted in the country having very high BOD and coliform in the stretches between Delhi and Etawah.

Table 20.2. Some examples of highly polluted stretches of various rivers in India (CPCB, 1999). (see Box 20.2 for definition of classes C, D and E)

River	Polluted stretch	Desired class	Existing class	Critical parameters	Possible source of pollution
Godvari	Downstream of Nasik and Nanded	C	D/E	BOD	Wastes from sugar industries, distilleries and food processing industries
Krishna	Karad to Sangli	C	D/E	BOD	Wastes from sugar industries and distilleries
Yamuna	Delhi to confluence with Chambal	C	D/E	DO/BOD	Industrial wastes from hosieries, tanneries, electro-planting and engineering industries and domestic waste from Ludhiana and Jalandhar
Chambal	Downstream of Nagda and downstream of Kota	C	D/E	BOD, DO	Domestic and industrial waste from Nagda and Kota
Damodar	Downstream of Dhanbad	C	D/E	BOD, Toxicity	Industrial wastes from Dhanbad, Durgapur, Asansol, Haldia and Burnpur
Gomti	Lucknow to confluence with Ganges	C	D/E	DO, BOD, Coliform	Industrial wastes from distilleries and domestic wastes from Lucknow

Agricultural pollution sources

Intensification (high inputs that particularly include chemical fertilizers) and expansion have made agricultural areas a major non point source of water pollution in rivers of India. Runoff of chemical fertilizers, manure, pesticides, sediment and salt from crop fields and body wastes of farm animals pollute the water bodies. Much of eutrophication of water bodies is due to the runoff from agricultural areas that includes large amounts of nitrates, phosphates and other fertilizers. Much of these losses occur because N and P inputs are in mobile form and are not synchronized with demands of crops. Furthermore, the disrupted biological community of crop field soil has low ability to retain nutrients. Increased concentration of NO_3 in drinking water, causing methaemoglobinemia in humans is quite common in many agricultural areas. Up to 30% NO_3 input to crops eventually enters water bodies. A significant fraction of pesticides such as chlorinated hydrocarbons (dichlorophenyl trichloroethane i.e., DDT being the most known) applied to agricultural systems move into adjacent ecosystems via leaching or aerial drift.

Municipal sewage

Municipal sewage is the major source of water pollution in India. Life in many lakes and in parts of rivers adjacent to cities has been damaged because of the high BOD generated by the municipal sewage. In developing countries much of the sewage remains untreated, and even in the most developed countries the treatment level is inadequate.

Industry

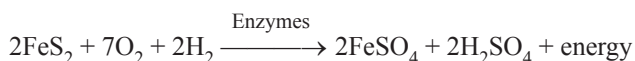
Industries are great users of water. It is used for cooling, processing and boiler feed and sanitary services. Pulp and paper industry producing organic wastes is one of the examples of the principal water polluting industries. Water is used in removing fibres from wood that go into paper making. In these processes sulphide liquors with high oxygen demands are produced. Even recycling of paper is not free from pollution. In this de-inking and bleaching processes yield pollution.

Large and medium size industrial units, including paper, sugar mills and chemical, leather, distillery and pharmaceutical industries, have contributed to much of water pollution in Ganga and Yamuna rivers in India.

Electrical utilities generate a huge amount of the waste heat that is discharged into water supplies. Here steam is used to make turbines to spin. The steam must be cooled and condensed before it can be used in the boiler for another cycle. Water is used to carry away the heat of steam. A 100-megawatt plant requires 8-11 million litres of cooling water every minute. The wastewater from electronic industries contains high levels of heavy metals like copper, lead and manganese. Industrial pollution affects dissolved oxygen content, temperature and pH, etc. And some industrial effluents cause toxicity to humans and various forms of aquatic organisms.

Water pollution from coal mines and coal processing

Streams in coal mine areas become polluted from acid drainage, which may completely destroy the aquatic life. As water passes through coal mines various iron compounds are oxidized. Among them iron sulphide is most important. Chemosynthetic bacteria accelerate the acidification process by oxidizing inorganic chemicals (iron sulphides in this case). These microbes oxidize iron sulphides in presence of water which results in the formation of sulphuric acid, as given below.



Thiobacillus thiooxidans is the most common chemosynthetic bacterium of coal mine.

It is difficult to clean up stream water polluted by acid drainage. However, control of acid mine

drainage is possible. Nemerow (1978) has given a number of methods for preventing formation of acid mine water and for neutralizing the acid in these waters. Some of them are as follows: sealing coal mines to check water flow in or out; keep oxygen out by flooding mines with water and then holding them there to check oxidation of iron sulphide; using limestone or lime to neutralize the acid, and covering strip mines with earth. The mined coal is not used directly; it is subjected to washing out impurities, which again causes pollution.

Pollution from oil production, transport, and use

Well over half of the oil produced annually is moved over the ocean from well heads to the consumers. Although only about 1% of this oil ends up as pollutant, this amounts to considerable contamination, for annual worldwide oil production is about 12 trillion litres. Nearly 6.1 million tons of oil enters the oceans annually from various sources as given in Table 20.3. Accidents of oil tankers though account for only 3.3% of oil pollution, they release huge amounts of oil suddenly, often more than 100,000 tons, causing acute disturbance in marine ecosystems. For example, the 250,000 t oil released in the tanker accident of 1978 in the Brittany coast of France, contaminated 300 km of sea coast. Some fractions of the oil spills are very volatile and are lost into the atmosphere immediately, but the others can stay in marine life for years. The processes that decrease the oil pollution are emulsification (braking into fine droplets), dispersion, oxidization by sunlight, degradation by microorganisms and deposition in ocean sediment. Accounting of the oil after the first few months of the 1978 accident showed that 13% became incorporated in water column from surface to bottom, 4.5% degraded quickly in the water, 8% ended up in tidal sediments, 28% washed into the intertidal zone; 30% evaporated, 20% remained unaccounted (Gundlach et al., 1983). After three years no oil was left but level of hydrocarbons was high in estuaries and marshy areas of the ocean. However, more of the oil pollution is due to the routine operations than from major accidents involving big tankers. The coastal areas are faced with most severe oil pollution problems.

Table 20.3. Sources of the oil pollutants that end up in the oceans worldwide (based on study of the U.S. National Academy of Sciences, 1975 and Kupchella and Hyland, 1989).

Source	Amount (million tons per year)	Per cent of total
River and urban runoff	1.90	31.1
Tanker operation	1.33	21.8
Coastal facilities (sewage plants, refineries, etc.)	0.80	13.1
Natural seepage	0.60	9.8
Atmospheric fallout	0.60	9.8
Other transportation activities	0.60	9.8
Tanker accidents	0.20	3.3
Offshore petroleum production	0.80	1.3
Total	6.11	100.0

The oil that gets dumped into oceans such as through leakage of offshore wells, floats and coats various objects and marine organisms, the benthic life of tidal zone being particularly vulnerable. Birds with oil coated feathers are unable to fly and retain less body heat. Thus they die of exposure. An accident in 1976 killed some 20,000 birds. In this about 4 million litres of heating oil was dumped into Chesapeake Bay.

Long term effect of oil is not known, however, some of the chemicals of crude oil are known to be carcinogenic.

How to deal with oil spills?

Many strategies have been applied to deal with oil spills but with a limited success: (i) using detergents to disperse oil, but it can create more problems; (ii) using straw as absorbent to hold oil, but it creates problem of straw disposal, (iii) burning oil, but it leaves the more fire-resistant chemicals behind, (iv) deploying barriers that give time to siphon up, skim off, or recover oil in some other way; and (v) removing hydrocarbons in petroleum by using microbes and providing them conditions that promote their growth. The microbes that can handle oil spills and petrochemical effluents secrete powerful biosurfactants which emulsify oils, and then use them as carbon source.

Lake Erie, an example of water pollution and pollution control

We need to know water pollution problem also in relation to a given water resource to widen our understanding. Lakes, rivers and oceans are the main kinds of our water resource. The Great Lakes of North America are the largest fresh water reservoirs on the face of earth. They are connected to one another and eventually to the ocean. Of these lakes, Lake Erie serves the greatest number of people and industrial centres. It is used for drinking water, navigation, recreation, fishing, generating hydroelectric power, etc. Draining some 30,000 square miles of North America (both USA and Canada), it is used for dumping treated as well as untreated sewage of about 13 million people with a highly consumptive life style. The sediment input from the Detroit River alone is close to 5000 m³ on normal days, mostly from agricultural fields. On stormy days the sediment load becomes 10-fold larger. The lake biology has been drastically changed. The BOD demand of detergents and other eutrophicants has caused stress on many organisms by creating severe oxygen depletion over long periods. In brief, the lake faced the problem of accelerated ageing under the cultural influences. This was brought about by over enrichment from phosphates, nitrates, organic matter, and other by products of civilization and through sediments eroded from catchment's surfaces.

A number of curative measures were taken subsequently: by early 1980s three-fourths of the more than 100 industrial polluters in American side of the lake were meeting at least a minimum treatment requirements; major cities such as Detroit and Windsor, Ontario banned phosphate-containing detergents (the major eutrophicant) in the early 1970s; billions of dollars were spent cleaning up the effluents; and restocking and fish management programmes were started to bring fish back.

Though lake recovered considerably from its eutrophication problems, concentrations of toxic, persistent chemicals reached problem levels. Sources of the toxic compounds like DDT, dioxin, toxaphene, PCBs were still there in farms and the leaky disposal sites located in the drainage area continued to pollute the lake. By 1985 much success was attained in controlling the pollution from the point sources. However, pollutants from non point sources, such as leaking land fills and airborne contaminants remained high and showed increasing trends in some cases. Chemicals like DDT were reported to be transported through atmosphere to the lake Erie from as far away as Mexico and Central America. Evidently, in spite of expenditure of billions of dollars yearly, the water pollution problems persist, so large and varied have become the human enterprises.

Groundwater Pollution

The available groundwater is about 13-20 times as much as water available on surface. The groundwater consists of water contained in the soil or aquifers, i.e., areas underlain by impermeable rock layer, called unconfined aquifer or areas between two impermeable rock layers, called confined aquifers or artisan (Fig. 20.8). Through the existing technology about one-fourth of the ground water is extractable globally. We greatly depend on groundwater and because of its excessive use,

water table levels are reported to drop in many areas of the world. For example, in the USA the groundwater supply is 22% of all freshwater use, and 50% of the total population and 95% of rural residents depend on domestic water supply on groundwater. Water table drops when more groundwater is extracted than the amount recharged. In many areas groundwater that is drawn upon is non-renewable because of being fossil water. The fossil water accounts for nearly three-quarters of the current water supply in Saudi Arabia. Groundwater is primarily used as drinking water; the pollution from different sources can cause serious human health problems (Shiklomanov, 1997).

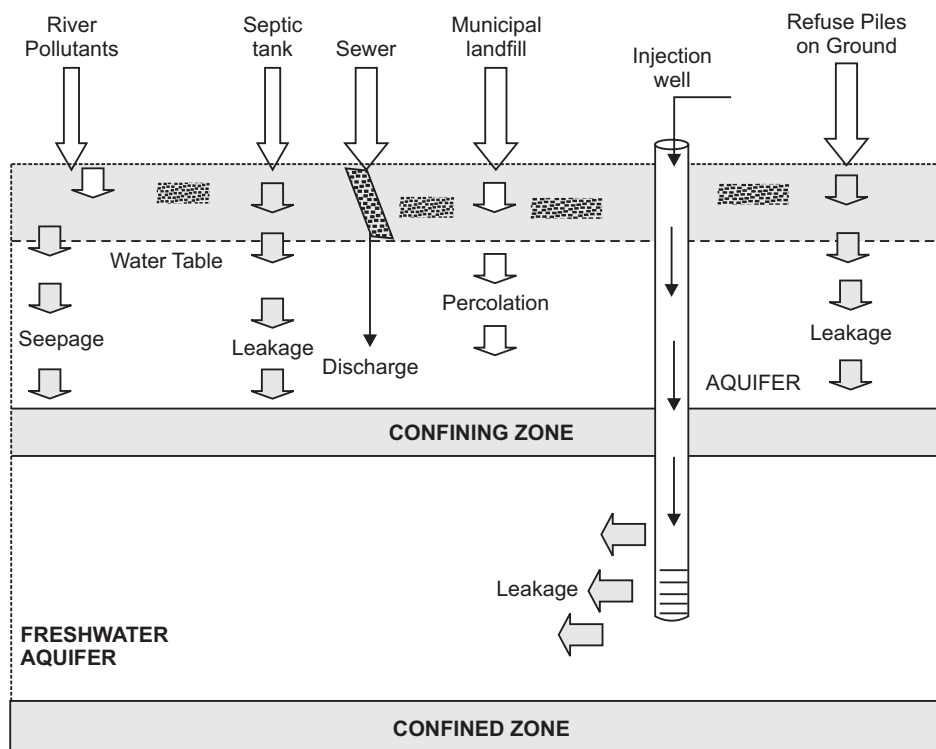


Fig. 20.8. Ground water receives seepage from land fills or open refuse pile, septic tank, sewer and pollutants from rivers.

Once contaminated, groundwater may remain contaminated for centuries. The main pathways of groundwater pollution are as follows:

- (1) Water seeping through the hazardous waste dumped in leaky land fills or open ordinary garbage dumps. These garbage piles amount to billions of tons and contain thousands of tons of toxic wastes.
- (2) Agricultural waste, domestic waste and deliberate deep-well disposal.
- (3) Septic tanks and sewer leakage.
- (4) Salt water intrusion
- (5) Natural hydrological pollution

Groundwater contamination is reported from all developed countries and from many large urban areas of India and other developing countries. For example, 39 public wells in California, USA were closed by public health officials because they were contaminated by trichloroethylene, known to cause cancer. The four most common pollutants of groundwater are reported to be nitrate, chloride, heavy metals and hydrocarbons (Pye and Patrick, 1983). According to a recent report,

200,000 people were suffering from arsenic poisoning from tube wells along Indian-Bangladesh border and more than 10 million Indians could be at the risk of drinking arsenic contaminated water (Bagla and Kaiser, 1996). High levels of arsenic leached from underground sources into village wells, the concentration of arsenic in water being up to 3.7 mg/l, against the World Health Organization (WHO) permissible level of 0.01 mg/l and EPA's permissible level of 0.05 mg/l. The arsenic poisoning caused lesions, pigmentation, and keratoses on palms and soles of the people.

Pollution of groundwater supplies from synthetic fertilizer application is also a problem in parts of India. Groundwater samples in the states of Uttar Pradesh, Haryana, and Punjab were found to have between 5 and 16 times the prescribed safe amount of nitrate, with one site in Haryana almost 30 times the prescribed limit. Groundwater in these areas is also being progressively depleted because of over extraction for irrigation (TERI, 1998).

Remedial Measures for Contaminated Groundwater

The remedial measures are divisible into three types: (i) surface controls, (ii) groundwater controls, and (iii) leachate collection and treatment. The surface controls include (i) sealing of surface, which decreases surface water infiltration; (ii) revegetation, as root uptake and evapotranspiration reduce volume of leachates; and (iii) grading, which promotes surface runoff, resulting in to reduced infiltration and leachate generation. The groundwater controls are based on (i) checking the movement of groundwater flow towards wastes; (ii) adsorption or neutralization of certain groundwater contaminants; (iii) groundwater pumping to lower the water table to prevent the groundwater contact with buried or impounded wastes, and microbial removal of contaminants, particularly petrochemicals. In the leachate collection and treatment, leachates are not allowed to migrate to groundwater or they are collected and transported for treatment or for recirculation.

Pollution in the Oceans

The oceans make up 70% of the earth's surface and appear so large that no waste matter disposal can pollute them. However, ocean pollution is on the rise due to deep sea dumping of industrial wastes, pesticides and sewage and run off of eroded soil and fertilizers from land sources (Fig. 20.9). About 85 % of all marine pollution is washed from the land rather than dumped or accidentally spilled in the mid ocean. In the open oceans, litter amounting to millions of tons from ships including those of military, loss of plastic nets and fishing apparatus, etc. cause pollution, affecting life of turtles, fish and other animals. Animals die from oil pollution and oceans can retain toxic and carcinogenic elements of oil for years. Heavy metals and some chemical compounds like DDT and PCBs are persistent. As they pass through the food chain, their effects are magnified which are a big threat to sea birds and other marine life.

Many coastal areas of the oceans are faced with severe pollution problems due to oil spills, sewage and other wastes. This localised marine pollution is because about 60% of the human population is concentrated within 100 km from the sea margin. Estuaries are the most productive

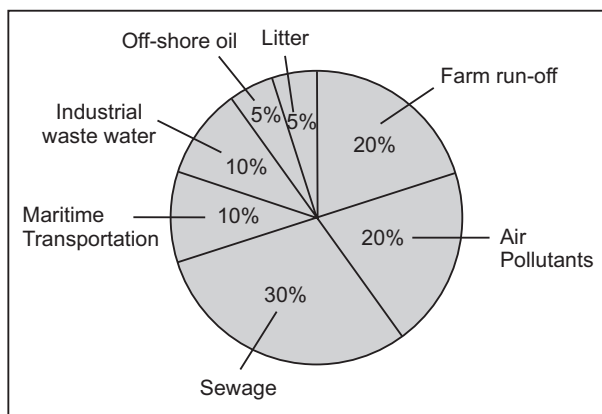


Fig. 20.9. Type of pollutants entering the oceans (National Geography, 2002).

ecosystems and they serve as the breeding areas for many species of key positions in the oceanic food chains. They are severely affected by pollution problems. Because of this, commercial and recreational fish industry is threatened in some regions of the world. High levels of organic chemicals, disease-causing organisms, high nutrient concentrations, and low dissolved oxygen have become characteristic of coastal waters in general. Coastal areas of oceans are also affected because rivers carry into the oceans much of the industrial and municipal wastes dumped into them.

Nearly 50% of mangrove ecosystem worldwide have been destroyed or drastically modified by human activities (see Vitousek et al., 1997). In recent years, the frequency, extent, and duration of harmful algal blooms in coastal areas have increased considerably. Nutrient enrichment and temperature increases are the main contributors to the extensive algal blooms. By causing oxygen depletion and releasing toxins, algae blooms kill deep-dwelling fish. For example, the collapse of the Baltic Sea cod fishery in the early 1990s is attributed to nitrogen pollution and resultant eutrophication (see Moffat, 1998). Excessive multiplication of dinoflagellates due to organic pollution gives the water a red tinge. The “red tide” causes large scale mortality of fish and severe poisoning in fish-eating humans.

Indian government has recently approved an integrated Coastal Zone Management (ICZM) project. Coastal line of India is poorly known from environment stand point. Occasional incidents of oil spill, dumping of waste oil, and other toxics make news headlines. However, environmental status of our marine ecosystems and adjoining shoreline are poorly investigated.

20.3 CONTROL OF WATER POLLUTION

Earlier, water pollution control was considered only the matter of engineering, now importance of using ecologically sound practices is being also recognized. For example, apart from treating sewage water, we need to reuse and recycle water. Control of water pollution costs a lot, but costs of not doing so may be higher. It depends on how you do your cost calculations. Some common terms used in relation to water treatment are given in Box 20.5.

Box 20.4 : Some terms related to water treatment

Adsorption – used to remove organic matter from waste water with activated carbon.

Aerial tank – used for injecting air/oxygen into water.

Coagulation – the clumping together of solids to get them settled down rapidly with the help of lime, alum, and iron salts.

Comminutor – used for catching and shredding of heavy solid matter in primary wastewater treatment.

Electrodialysis – used to separate soluble minerals from the water with the help of current and membranes.

Flocculation – process by which dissolved chemicals are precipitated by physical, chemical or biological action.

Sand filters – used to remove some suspended solids from sewage; sludge accumulating on sand bed is removed periodically.

Sedimentation tanks – to remove solids from sewage by pumping it to tanks. Solids settle on bottom or float on top as scum. Solids are finally disposed off by using techniques such as incineration, digestion and filtration.

Septic tanks – underground tanks into which domestic wastes are piped. Bacteria decompose the organic wastes. The sludge is pumped out of tank and effluents blow out of tank onto the ground. This is used when a sewer line is not available.

Trickling filter – a support medium for bacterial growth, usually a bed of stones and rocks.

Primary and secondary sewage treatment

In sewage treatment plants, water pollutants are removed by physical (primarily sedimentation), biological (based on oxidation) and chemical (stripping) methods. The three stages of sewage treatment and methods involved are given in Table 20.4. Sewage water after the primary treatment (in which primarily solid materials are removed) is still not clear. The sludge and solids captured in the process are disposed of by some means, including spreading it in land fills or incinerating it.

Table 20.4. Summary of sewage treatment systems.

Sewage treatment	Target to achieve	Methods	Final product
Primary (physical) treatment	Physical removal of solids from polluted water	Involves screening to catch floating materials, filtration, grinding, sedimentation in tanks, and chlorination	Still unclean water, containing lot of dissolved organic matter, BOD reduced by about 20% and nutrients by 10-20%.
Secondary (biological) treatment	Removal of dissolved organic matter (that constitutes BOD in sewage), employing aeration and microbial decomposition	Activated sludge* Or Trickle-filter** Followed by anaerobic digestion at high temperature	Much cleaner water with only 10% of original BOD, suspended solids removed by 90%, total N by 50% and total P by 30%
Tertiary (chemical) treatment	Removal of chemical remains	Precipitation of P compounds (by adding iron or aluminium salts also removal of N compounds including ammonia	Clean water

*A process in which aerobic microbes decompose organic matter dissolved in waste water.

**In this wastewater is trickled downward over the oxygen-rich stone surface, so that bacteria can break down the organic wastes.

The secondary treatment of water is mainly for removing dissolved organic matter that is responsible for the BOD. The organic matter is decomposed in containers with arrangement for air supply (Fig. 20.10). In the activated sludge process, the aerobic microbes which decompose organic matter increase in number while so doing (Table 20.4). In a trickle filter technique, organic matter is decomposed in an oxygen rich environment into CO₂, H₂O and mineral salts (Fig. 20.11). Effluents from secondary treatment are usually disinfected with chlorine or ozone to kill harmful bacteria before releasing the water in rivers or streams.

The solid collected by sedimentation in the primary treatment is digested anaerobically, and the treated sludge is dried and disposed of by incineration or land disposal. Removal of chemicals in water by chemical methods, characterizes tertiary treatment. For example, phosphorus compounds are precipitated through the addition of iron or aluminium salts.

In India, the gap between the generation of waste (29,000litre/day) and treatment of waste water (6000 million litre/day) alarmingly wide: The waste water treatment plants are few, and whatever is available, only a small fraction of them are functional. A 1992 World Health Organization report shows that out of India's 3,119 towns and cities, only 209 have partial sewage treatment plants and only 8 have full treatment facilities.

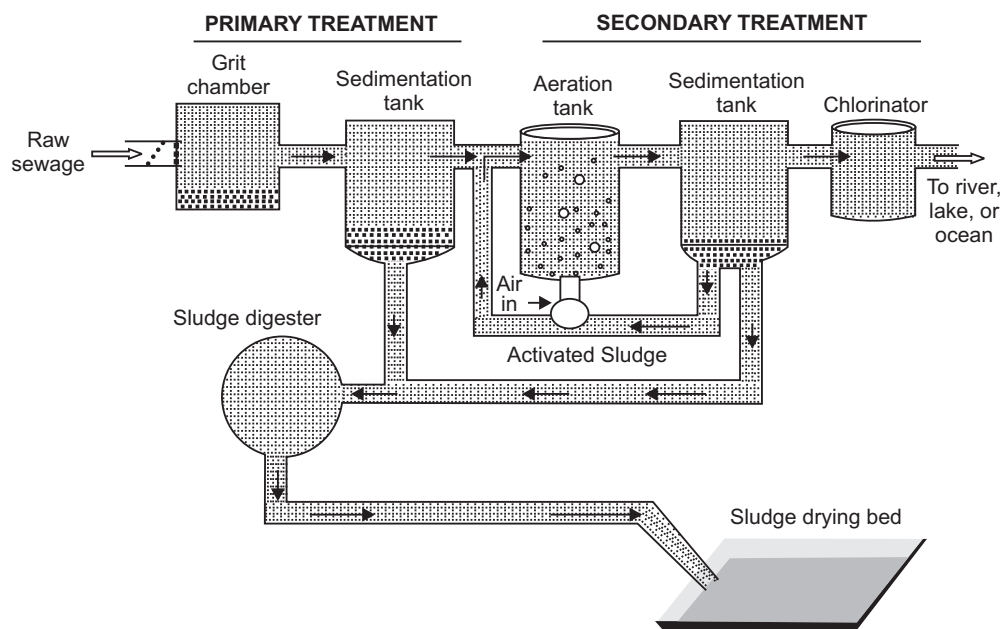


Fig. 20.10. Primary and secondary treatment of waste water.

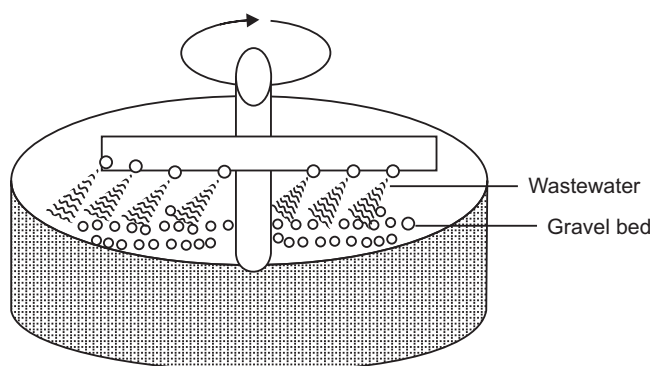


Fig. 20.11. Trickling filter for secondary waste treatment.

Physico-chemical treatment systems

In recent years, physical-chemical treatment systems (tertiary treatments) have been developed which basically depend on clarification, filtration, absorption, and disinfection (Fig 20.12). The processes help to remove dissolved metals, phosphorus, nitrates, pathogens and pesticide residues from the treated water (Fig. 20.12). In clarification, chemicals such as alum or lime are used to precipitate wastes into particulate masses that settle out. Filtration accounts for removal of the remaining solids, and absorption for removal of dissolved organic matter by chemical charge attraction using activated carbon and other such materials. Some systems depend on electrodialysis, in which charged poles attract negatively and positively charged ions from wastewater, causing them to pass through tiny pores in membranes and then allowing the water to move on relatively free of dissolved ions. The ion exchange process is used for removal of inorganic compounds by passing the water successively over a solid cation exchanger and a solid anion exchanger

(Fig. 20.13). In the process, the cations and anions are replaced by hydrogen ions and hydroxyl ions, respectively. The cation exchanger is regenerated with strong acid and the anion exchanger with strong base. The ion exchanger produces water of very high quality, but the operational cost is high and also has some problem of waste generation. The physical-chemical systems are not affected by toxins and can work even in cold weather. In these aspects they are better than the treatment system described earlier.

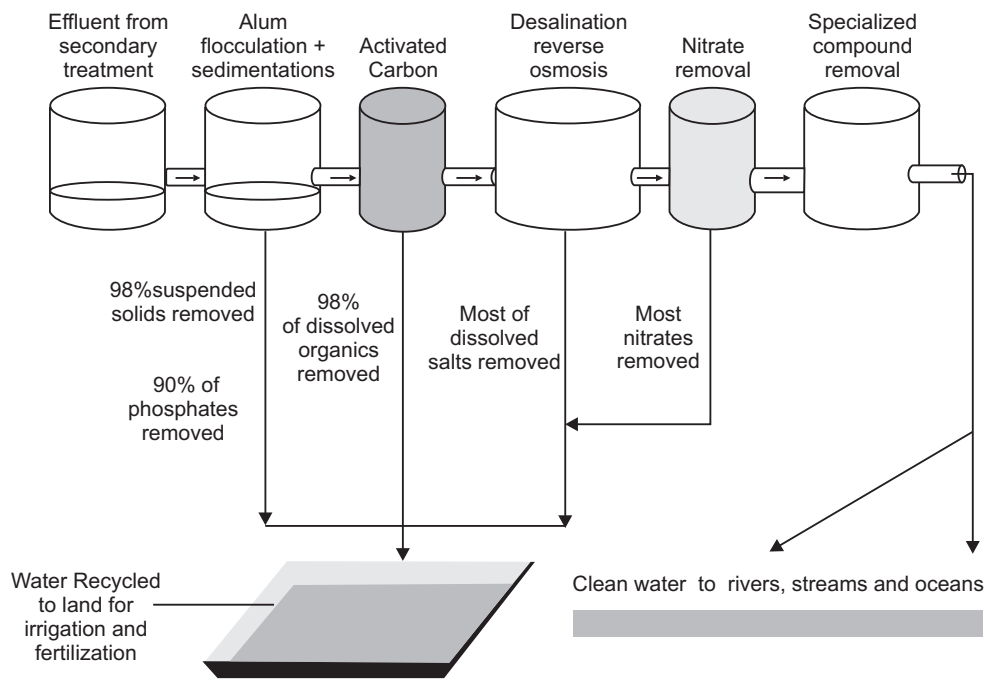


Fig. 20.12. Advanced sewage treatment for removing suspended solids, dissolved organics, phosphates, nitrates and other specialized compounds like DDT.

Gaseous chlorine is added to water as a powerful and a short-lived germicide. Ozone is sometimes used as a disinfectant in place of chlorine, in view of the negative effects of chlorine ions on ozone layer as well as human health. In the ozone water treatment system, air is filtered, cooled, dried and pressurized which is subjected to an electric current of about 20,000 volts. The ozone thus produced is pumped into the contact chamber for 10-15 minutes (Fig. 20.14). Because of the low decomposition of ozone in water, chlorine is also added in the contact chamber to maintain disinfection throughout the water distribution system.

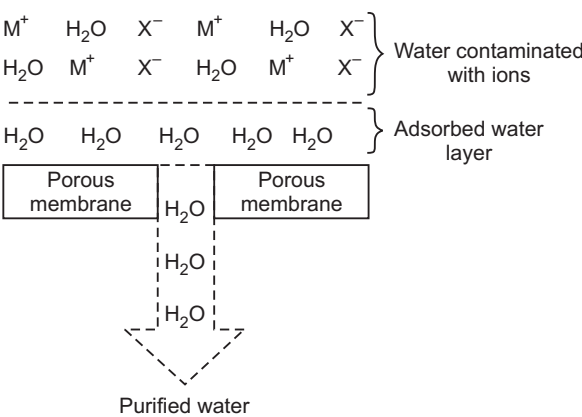


Fig. 20.13. Solute removal from water by reverse osmosis.

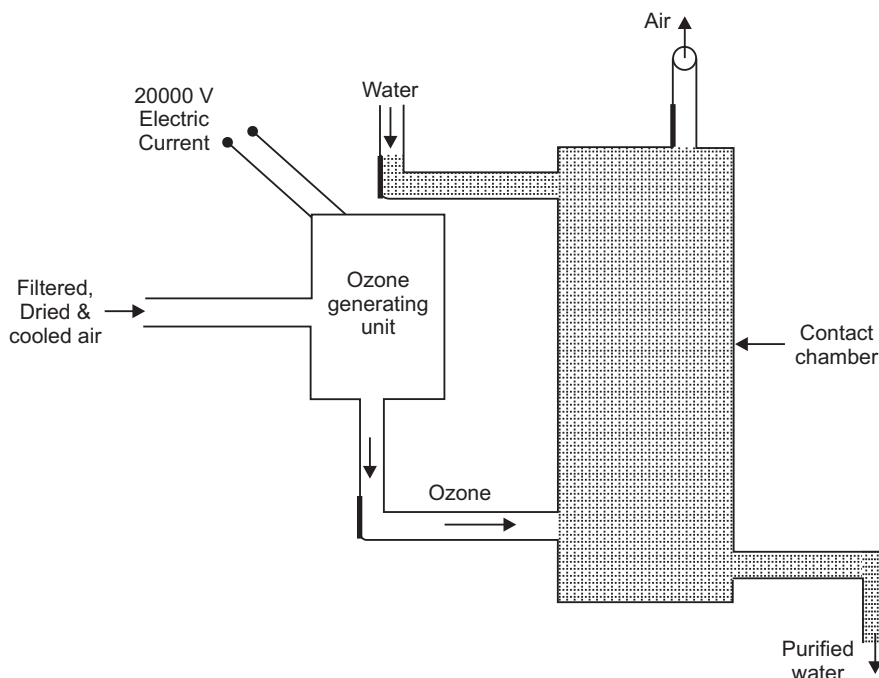


Fig. 20.14. A schematic diagram of ozone water treatment water system.

Biological Treatments

They are very effective for removal of biological oxygen demanding wastes and pathogens from primary treated wastewater. Two types of systems are described below:

Solar Sewage Plant: It is an ecological waste engineering method in a green house that can be used to purify waste water. In the green house a lagoon is created. The sewage flows into rows of large aquarium tanks covered with plants such as water hyacinth (*Eichhornia crassipes*), *Typha* and *Phragmites*. (*Lemna*, *Wolffia*, *Spirodela*), submerged plants like *Hydrilla* and *Ceratophyllum* are also used for wastewater treatment. The water hyacinth can readily absorb and concentrate heavy metals such as lead, cadmium, mercury and nickel (Wolverton and McDonald, 1978). Other free floating and submerged plants have the capability, to decontaminate and detoxify the wastewater. The algae and microbes decompose waste into nutrients that are absorbed by the plant. Toxic metals are absorbed in plant biomass. Water filters through the sand, gravel and plants which also filter out algae and organic residues. The tanks also contain zooplanktons which are consumed by cray fish and other types of fishes. After a fortnight, clean water filters out of the systems.

Constructed Wetlands: Constructed wetlands that mimic the waste-absorption function of natural wetlands are low cost systems and provide an economically viable solution for wastewater management. The constructed wetlands consist of emergent macrophytes like *Typha latifolia* and *Phragmites karka*. Roots of these plants help in cleaning of organic pollutants; the plants can reduce BOD of water by 67-90%, N by 65% and also decrease the P concentration in water. Plants which are selected for constructed wetlands have high photosynthetic rates, high O_2 transport ability, and tolerance to pollution as well as pollution assimilative capacity around root zone. Plants should be resistant to pests and diseases and easy to manage at high plant density. In the system developed by the Max Planck Institute of Germany, the untreated wastewater is first passed through a bed of gravel topped with sand in which common reed (*Phragmites* sp) is planted. Solid particles are trapped on

the sand as the water percolates downward. Dissolved inorganic and organic material is absorbed by the reeds or decomposed by the microbial community associated with the roots and in the gravel. The air passages in the body of the reed plant transport oxygen down through the plant and outward from the roots to the soil to supply the massive needs of micro-organisms. The water then flows to a gently sloping bed or gravel planted with *Scirpus* sp. This acts as a living filter absorbing more organic and inorganic materials. Bacteria and other pathogenic organisms in the sewage are greatly reduced. Floating wetlands/engineered wetlands are used for controlling pollution from non-point sources, such as the pollution of coastal zones or river bank, flood plains, etc.

Industrial water pollution control

Industrial water pollution may contain toxic metals and toxic organic chemicals which can be removed from effluents chemically. A number of strategies other than directly removing waste from the effluent have been given (Nemerow, 1978); (i) volume reduction, (ii) strength reduction, (iii) neutralization, and (iv) equalization of discharge. In India, the Central Pollution Control Board, (CPCB) has fixed general standards for discharge of pollutants in industrial effluents (Tables 20.5 and 20.6).

Table 20.5. General standards (upper permissible concentrations) for discharge of environmental pollutants: Effluents according to the Environment (Protection) Rules, India, 1986 (additions inserted in 1988, 1993). (from Anonymous, 1997).

Parameters	Standards			
	Inland surface water	Public sewers	Land for irrigation	Marine coastal areas
Suspended solids (mg/l)	100	600	200	(a) For process wastewater –100 (b) For cooling water effluent 10 per cent above total suspended matter of effluent
Particular size of suspended solids	Shall pass 850 micron IS Sieve	–	–	(a) Floatable solids, max. 3 mm (b) Settleable solids, max. 850 microns
pH	5.5 to 9.0	5.5 to 9.0	5.5 to 9.0	5.5 to 9.0
Temperature	Shall not exceed 5°C above the receiving water temperature	–	–	Shall not exceed 5°C above the receiving water temperature
Oil and grease mg/l max.	10	20	10	20
Total residual chlorine, mg/l max.	1.0	–	–	1.0
Ammonical nitrogen (as N), mg/l max.	50	50	–	50
Total kjeldahl nitrogen (as NH ₃): mg/l, max.	100	–	–	100
Free ammonia (as NH ₃) mg/l, max	5.0	–	–	5.0

Biochemical oxygen demand (5 days at 20°C), mg/l max.	30	350	100	100
Chemical oxygen demand, mg/l max.	250	–	–	250
Arsenic (as As), mg/l max.	0.01	0.01	–	0.01
Mercury (As Hg), mg/l max.	0.01	0.01	–	0.01
Lead (as Pb), mg/l max.	2.0	1.0	–	2.0
Cadmium (as Cd), mg/l, max.	2.0	1.0	–	2.0
Hexavalent chromium (as Cr ⁶⁺), mg/l, max.	2.0	2.0	–	2.0
Copper (as Cu) mg/l, max.	3.0	3.0	–	3.0
Zinc (as Zn) mg/l max.	5.0	15	–	15
Selenium (as Se), mg/l, max.	0.05	0.05	–	0.05
Nickel (as Ni) mg/l, max.	3.0	3.0	–	5.0

Table 20.6. Upper permissible concentration (standards) for compliance by small scale industries to check water pollution (Central Pollution Control Board, 1988). (from Anonymous, 1997).

Parameters	Concentration not to exceed
Total concentration of mercury (Hg) in the final (combined) effluent	0.01 mg/l
pH	5.5-9.0
Suspended solids	250 mg/l
Biochemical oxygen demand	150 mg/l
Temperature	Shall not exceed 5°C above the ambient temp. of the receiving body
Free available chlorine	0.5 mg/l
Oil & grease	10 mg/l
Cu (total)	3.0 mg/l
Iron (total)	3.0 mg/l
Zinc	5.0 mg/l
Cr (total)	2.0 mg/l
Phosphate (as P)	5.0 mg/l
Bioassay test	90% of test animals after 96 hours with 1:8 dilution
Sulphide (as S)	2 mg/l
Phenolic compounds (as C ₆ H ₅ OH)	5 mg/l

Hexavalent chromium (as Cr)	0.1 mg/l
Nickel (as Ni)	3.0 mg/l
Cadmium (Cd)	2.0 mg/l
Chloride (as Cl)	9000 mg/l
Sulphate (as SO ₄)	1000 mg/l
Cynides (as CN)	0.2 mg/l
Ammoniacal nitrogen (as N)	50 mg/l
Lead (as Pb)	0.01 mg/l
Total metal	10.0 mg/l

The above standards may be relaxed in case where the water from small-scale industrial units are collected and treated in a terminal treatment plant.

*Above the ambient temperature of the receiving body.

** After 96 hours with 1:8 dilution

For dissipating waste heat, a system consisting of wet and dry towers is used. In this the heat is dissipated into atmosphere not in water.

Deep well injection

Hazardous waste requires more safe sites. One way of providing safe dumping is to inject such wastes into limestone or sandstone strata, sandwiched between relatively impermeable layers, well below the water table (Fig. 20.15). There are several hundreds of deep well sites in the United States.

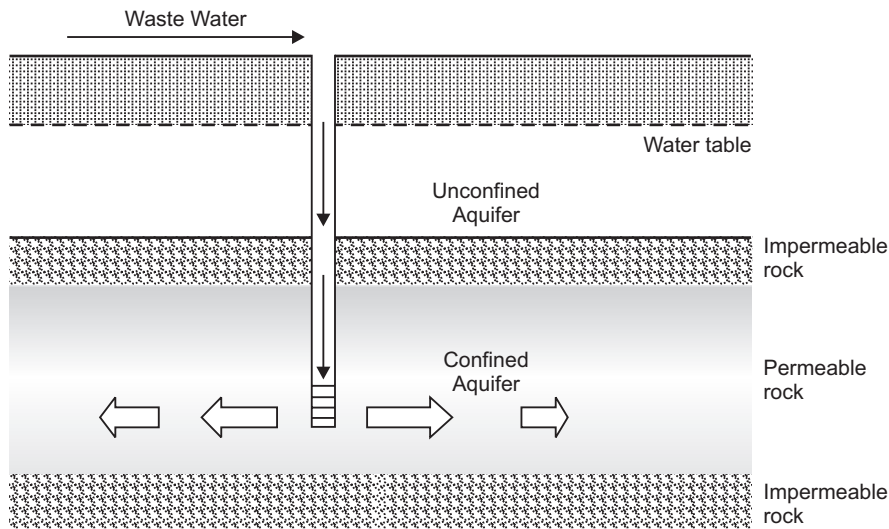


Fig. 20.15. Deep well injection of waste water causes pollution of ground water aquifer.

Recycling water and pollutants

Recycling of matter is an ecological approach as it helps conservation. By reusing water over and over, its consumption can be reduced substantially. For example, by recycling, the water consumption for steel manufacture is reduced 65 times. The pollutants of water are resources out of place; they can be useful when removed from water. For example, zinc generated in rayon production can be recovered from wastewater and used again.

Another way of conserving water is using water for more than one thing in the same use cycle. Using treated sewage water for irrigation is one such example. The risk of contamination and infection can be a serious problem, however. Sewage water or partially treated water has been found particularly useful in revegetating strip-mined areas. Apart from quickly leaching the acid out of the soil, the water provides a wet environment and adds organic matter.

Bioremediation

Bioremediation is the use of entire organisms (mostly microbes) or enzymes to transform a toxic substance into benign or less toxic substance. A number of micro-organisms are now known which have the specific ability to detoxify toxic chemicals. The majority of strains like *Pseudomonad* handle detoxification of organic combines like PCBs by a number of enzymatic steps encoded by genes contained in megaplasmids. Phytoremediation uses plants to clean up pollution in the environment. It is an environmentally safe and cost effective method of cleaning up the contaminants. Plants can help to clean up many types of pollution including metals, pesticides, explosive and oil from the soil as well as water system. For example, plants can also help to prevent the spread of the pollution from one site to other areas. Phytoremediation is an *in situ* method to control pollution. Plants remove harmful chemicals from the groundwater where their roots take in water and nutrients from soil, streams and groundwater. Plants can clean up chemicals as deep as their roots penetrate and these chemicals are stored in roots, stems and leaves. The chemical pollutant can be changed into less harmful chemicals within the plants or some of the pollutants can be changed into gas and released into the environment. Dense growth of hybrid poplar – helps to control groundwater plumes. White polar trees are being used to absorb arsenic from mine wastes in white wood creek in South Dakota, USA. Riparian buffer zone (an area of vegetation running parallel to a water body) helps to prevent pollution with the presence of deep roots. The applications of phytoremediation for controlling pollution are given below:

- (i) It is an effective technique for cleaning polluted soil, waste water stream and groundwater.
- (ii) It can be used both for organic and inorganic containments.
- (iii) It can be used for checking industrial and agricultural run off.
- (iv) It also prevents leachate formation from land fills and other containment sites.

Treatment of drinking water

Pure water is required for drinking purpose; therefore it requires more intense and complete treatment. For example, even harmless gases, if they give water a bad taste, are to be removed. To clean the water for drinking purpose, a combination of following steps is taken:

- (i) Spraying water into the air to release gases such as H_2S that give bad taste and smell.
- (ii) Removal of finely suspended particles, proteins, and other materials by using coagulants, such as aluminium sulphate (alum). The coagulants clump together the materials having electrical charges. Subsequently, coagulated and suspended solids are allowed to settle in sedimentation basin.
- (iii) Taking out unsedimented materials as well as bacteria (even viruses) from water by passing it through filters (e.g., sand filters) and adsorbants.
- (iv) Killing of the remaining microorganisms using chlorination or UV.

The condition of drinking water supply in our country is poor. Even the level of research on water quality is low. There are indications for the presence of as many as 200 harmful chemicals, mostly organic, and some carcinogenic (causing cancer). Nearly one-third of the city populations live in slums, with limited or no safe water availability.

20.4 LEGISLATION FOR WATER POLLUTION CONTROL

The Government of India has enacted The Water (Prevention and Control of Pollution) Act, 1974 for preventing and abating water pollution. Under the purview of water act, the central and state pollution control boards were constituted to maintain location and some specific standards for the discharge of effluents in water ways. The Water (Prevention and Control of Pollution) Cess was introduced in 1997 for the levy and collection of cess or a fee on water consuming industries and local authorities as specified by the state pollution control boards. The Water (Prevention and Control of Pollution) Cess Rules of 1978 contain the standard definitions and indicate the kind of and location of meters that every consumer of water is required to affix. In 1991 Coastal Regulation Zone Notification was issued by the Government of India which specified regulations on various activities, including construction. It gives some protection to the backwaters and estuaries.

20.5 MONITORING AND INITIATIVES FOR CONTROLLING POLLUTION IN INDIA

India's Central Pollution Control Board has a National Water Quality Monitoring Network comprising 1429 stations across 27 states and 6 union territories. The network collects samples from 293 rivers, 94 lakes, 9 tanks, 41 ponds, 8 creeks, 23 canals, 18 drains and 411 wells. Water samples are analyzed for dissolved oxygen, bacterial count, and others, the total number of parameters being 28. Apart from these, 9 trace metals and 28 pesticide residues are also measured. Monitoring from 1995 to 2008 indicates a high organic and bacterial contamination. Almost all rivers have high Biochemical Oxygen Demand (BOD). In 2008, values of BOD were: 590 mg/l in river Markanda, D/s Kala Amb, 364 mg/l in River Kali (Western) D/S Muzaffarnagar, 353 mg/l in Amlakhadi at Ankleshwar, 247 mg/l in Western Yamuna Canal D/s Yamuna Nagar, 70 mg/l in Yamuna canal at Okhla after meeting Shahdara drain, 58 mg mg/l in Betwa at Nayapur D/s (a water sample with 5 day BOD between 1-2 mg/l is considered polluted) (Central Pollution Control Board, Annual Report, 2008-2009). Obviously, all of the above rivers are heavily polluted.

The total coliform and fecal coliform densities, which indicate water contamination with fecal material of humans, livestock, pets etc, were between 500 and 100,000 MPN/100 ml (ideally it should be absent, but can be tolerated up to 105/MPN/100 ml). The total annual average coliform density in India now looks like declining as values above 500 MPN/ 100 were found in 47% samples in 2006, but only 33% in 2008. Since 2005, the market of water treatment plant has been increasing at the rate of 10-12 percent per year largely supplied from the US.

Initiatives for controlling pollution in the rivers Yamuna and Ganga: Yamuna in Delhi has been reduced to "a putrid ribbon of black sludge", with nearly 57% of the city's waste entering Yamuna. Only about 55% of about 15 million population are connected to the sewage system. About 75% of Yamuna pollution is due to the raw sewage. The population along the river Ganga is about 400 million, a majority of them do not have toilet, and nearly 2 million people bathe daily in the river.

Under the National Water authority and the Ministry of Environment and Forests, two big projects have been undertaken to control pollution in rivers Ganga and Yamuna in India. These are known by the name Ganga Action Plan (GAP) and Yamuna Action Plan (YAP) and funded by World Bank and Japanese government currently. The Ganga Action Plan was launched in 1985 to clean up the River Ganga. The Yamuna Action Plan was launched on the world environment day in June 1993 for the pollution abatement in towns of U.P. and Haryana. The Ganga Action Plan and the Yamuna Action Plan are briefly described below:

Ganga Action Plan (GAP)

The river Ganga is highly polluted at number of places like Haridwar, Kanpur, and Varanasi, etc. For example at Haridwar, small and big pharmaceutical companies discharge various types of toxic chemicals into river. At Kanpur over 300 tanneries discharge large quantity of sludge and

liquid wastes into the river containing high concentration of Pb, Cd and Cr. At Varanasi, domestic organic wastes and industrial wastes with high number of coliform bacteria are added to the river water. The Phase I of GAP was completed in 2001 and the objectives of GAP are as follows:

- (i) Construction of interceptor drains and sewers to divert flow of sewage and other liquid waste away from the river.
- (ii) Low cost sanitation schemes in adjoining areas of the river.
- (iii) Biological conservation measures.
- (iv) Prevention and control of Industrial pollution.
- (v) Regular and continuous monitoring of water quality at several places.

Although the Ministry of environment and Forests (MoEF, 1999a) claims a general increase in dissolved oxygen (DO) and a decrease in BOD in the downstream of Ganga water in major cities (e.g. Kanpur DO 6.7 in 1986 to 7.5 in 1999, BOD 8.6 in 1986 to 6.5 in 1999; Varanasi DO 5.9 in 1986 to 8.4 in 1999, BOD 10.6 in 1986 to 3.7 in 1999, all values are average summer values in mg/ l), the phase I of GAP is considered not very successful. The industries at places like Haridwar, Kanpur and Varanasi have been discharging the effluents into the water. Phase – II of GAP was launched in 1993 with an objective to clean the tributaries of Ganga that include Yamuna, Gomti and Damodar. In Phase – II, some Japanese agencies are taking part to monitor the level of pollution in the river. GAP II was expanded in 1996 into the National River Conservation Plan (NRCP), which presently covers polluted stretches of 36 rivers in 20 States in the country. Recognising the need to revamp the river conservation strategy, the Central government has given Ganga the status of a ‘National River’ and has set up the National Ganga River Basin Authority (NGRBA) on 20 February 2009.

Yamuna Action Plan

The Yamuna Action Plan was launched for the Pollution abatement in towns of Haryana and U.P. Entire length of Yamuna river (Approximately 1400 km), out of which Delhi – Okhla stretch of 22 km is worst in terms of water quality having high levels of BOD (30 mg/l) and millions of coliform bacterial are present per 100 ml. Wastewater contributes to 80% of pollution which is generated in Delhi, less than 50% of total effluents are treated. Moreover, inadequate water flow in the river leads to poor dilution of pollutants. Phase – II of YAP is going on in 8 towns of Uttar Pradesh and in 6 towns of Haryana to control pollution in Yamuna. Following steps have been taken to control pollution in river Yamuna;

- (i) Construction of sewage treatment plants.
- (ii) Installing of sewage pumping stations for pumping sewage into treatment plants.
- (iii) To control industrial pollution and to prevent effluent discharge into the river.

The Yamuna Action Plan Project Phase II is regarded as the core project under the National River Conservation Plan of Government of India. The project addressed the abatement of severe pollution of the River Yamuna by raising sewage treatment capacity, caused by rapid population growth, industrialization and urbanization in the towns of the river basin, which includes Delhi, the capital of India. Building new and expanding capacity of old sewage treatment plants and laying and rehabilitating sewers will be done to enhance the treatment capacity particularly in Delhi and Agra.

The JICA assisted Yamuna Action Plan Phase - III project at Delhi under National River Conservation Plan

Government of India has been implementing Yamuna Action Plan (YAP) with assistance from Japan International Cooperation Agency, Government of Japan since 1993 to supplement the

efforts of State Governments in addressing the problem of pollution of river Yamuna. A total of 276 schemes including 38 sewage treatment plants have been completed in 21 towns of Uttar Pradesh, Haryana and Delhi and 753.25 million litres per day of sewage treatment capacity has been created so far. On 01-December, 2011, The Cabinet Committee on Economic Affairs approved the proposal of “Japan International Cooperation Agency (JICA) assisted Yamuna Action Plan (YAP) Phase - III project at Delhi”, under the Centrally Sponsored Scheme of National River Conservation Plan at an estimated cost of Rs 1656 crore; being implemented over a period of seven years.

Under the YAP-III, it is proposed to rehabilitate the damaged trunk sewers to maximize the utilization of available treatment capacity and modernize the Sewage Treatment Plants (STPs) in three catchment areas of Delhi namely Okhla, Kondli and Rithala, Construction of a new state of art STP in place of old STP of 136 million litres per day capacity at Okhla and to equip them with tertiary level treatment facilities to achieve treated effluent quality of 10 mg/l for Biochemical Oxygen Demand (BOD) and Total Suspended Solids (TSS) so as to improve the water quality of river Yamuna. These works will be integral to the Sewerage Master Plan being prepared by Delhi Jal Board (DJB). The project will be implemented by DJB under the supervision of the Department of Urban Development, Government of NCT of Delhi. The implementation of the project shall improve the water quality of the river along with improved aesthetics of the area through tackling of sewage generated from identified catchments of Delhi.

National River Action Plan

The National River Action Plan has been set up to monitor water quality in rivers of India. Under the National River Action Plan (NRAP), the CPCB has identified the stretches of major rivers in India with high or intermediate levels of pollution. At present 156 towns in 14 states along 22 interstate rivers are considered under the NRAP, out of which 74 towns are located on the river Ganga and 21 on the river Yamuna (MoEF, 1999).

Soil Pollution-Metals, Pesticides, Isotopes and Wastes

Soil pollution refers to any physical and chemical change in the soil that adversely affects the growth of plants, and other organisms living in the soil. Soil pollution originates from waste dumping, use of agrochemicals including fertilizers and pesticides, mining operations, processing of petroleum products and release of heavy metals, etc. Soil pollution affects all the components of ecosphere as the pollutants tend to move in lakes, ponds and rivers; percolate down into groundwater and enter the atmosphere. Heavy metals and certain organic compounds are phytotoxic even at low levels and can adversely affect soil productivity over a long period of time. The decontamination of soils having high concentration of heavy metals is one of the most challenging environmental problems. Pesticides disturb balance in nature because they affect prey-predator relationships, non-target organisms and show biological magnification through the food chain. Various technological options are available and need to be applied to reduce the risks of heavy metals, and pesticides. Bioremediation, integrated pest management and biological control of disease causing organisms are being increasingly popularised to tackle the problems of metal and pesticide pollution.

Unscientific disposal of solid wastes in open areas creates unhygienic conditions and spreads disease causing organisms in the environment. The industrial and hazardous wastes have a negative impact on human health and ecosystem integrity. Main aspects of waste management involve source reduction, treatment and recycling. Some reduction measures include process modification, changes in management practices and increase in the efficiency of the equipment processing the wastes. The technologies to manage wastes can be chemical, biological, physical, thermal or recycling of disposable wastes. Recycling is the use and reuse of chemical waste as an effective substitute for a commercial product. The solid wastes management is an issue of great concern particularly in a country like India. The challenge has to be met judiciously by adopting a combination of approaches of reuse, recycling and safe disposal. Disposal comprises the discharge, deposit, injection or placing of a waste into a landfill facility and used for generating energy in an incinerator.

21.1 TYPE OF METALS AND NATURE OF POLLUTION

Heavy metals create long-term pollution problems, as they circulate in food chains, and persist in organisms and sediments. Main sources of heavy metals are their ores, waste products, vehicular traffic, metal processing industries, refuse dumps and sludge. Metals are mostly obtained from the earth's crust; already 310 million tonnes of copper (Cu), 250 million tonnes of zinc (Zn), 240 million tonnes of lead (Pb), and 20 million tonnes of nickel (Ni) have been mined and wasted in environment. Some metals are essential for the well-being of plants, animals and human beings, others are not required. For example, for plants and animals zinc, copper, manganese (Mn), and cobalt (Co) are essential, chromium (Cr) and Ni are essential for animals, and lead, silver and cadmium are not required by any of them. The actual toxic action varies from one metal to another, but generally it involves metals binding to some metabolically-active groups. Toxic and accessible

metals affect human environment most. Metals occurring at less than 1000 ppm levels in the earth's crust are called *trace elements*. The metals with densities more than 5 g cm^{-3} are called *heavy metals*, and those with lower densities called light metals.

Metals like mercury and lead (see also chapters 19 and 20) can pollute water, air and soil and get into human tissues through food chains (Fig. 21.1) Mercury is a global threat to human and environmental health. Current anthropogenic sources are responsible for about 30% of annual emissions of mercury to air. Another 10% comes from natural geological sources, and the rest (60%) is from 're-emissions' of previously released mercury that has built up over decades and centuries in surface soils and oceans (UNEP, 2013). On the basis of this report, anthropogenic emissions of mercury in to the environment are summarized in Box 21.1.

Box 21.1. Anthropogenic Emissions of Mercury in to the Environment

Total anthropogenic emissions of mercury to the atmosphere in 2010 are estimated at 1960 tonnes because of human activities, which is about 30% of the total mercury that was emitted and re-emitted from anthropogenic and natural sources in that year. About 62% of the annual total anthropogenic emissions to air are associated with artisanal and small-scale gold mining (ASGM) and coal burning, other major sectors include ferrous and non-ferrous metal production and cement production. East and Southeast Asia are responsible for about 40% of global anthropogenic emissions. About 75% of the mercury from this region comes from China, which is about one-third of the global total.

Concentrations of mercury in Arctic marine animals today are about 10-12 times higher than in preindustrial times (i.e., prior to about 1800). This means that on average about 92% of the mercury in marine predators such as seabirds, seals, and whales is anthropogenic in origin. The upper 100 meters of the oceans have twice the mercury that they did a century ago. Intermediate and deeper waters have 10-25% more mercury on average. Seawater concentrations will thus be especially slow in responding to changes in mercury inputs from atmospheric deposition and river flow. As a result, mercury concentrations in marine biota are likely to increase slowly for decades to centuries, even without an increase in atmospheric emissions. Further increases in atmospheric emissions of mercury will have long-term consequences for commercial fisheries and all consumers of marine and freshwater foods.

Source : UNEP (2013) : Global Mercury Assessment 2013

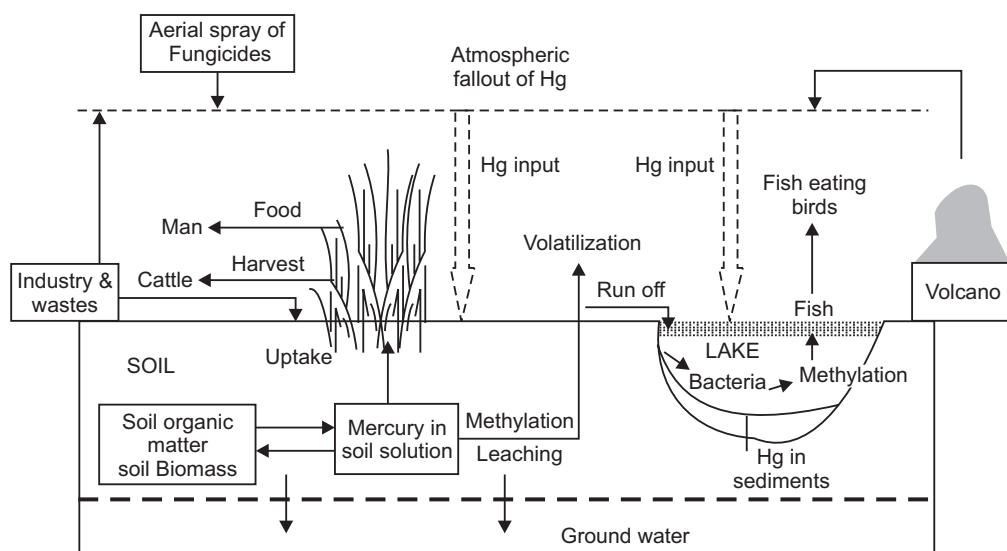


Fig. 21.1. Mercury into land, soil and water and food chain.

Concentrations of some elements are known to increase from lower to higher trophic levels, a phenomenon called *biological magnification*. The incidence of neurological disorder affecting many fish eating people in the industrial city of Minamata in Japan is a good example of biological magnification of mercury. Apart from the toxicity and increase in concentration from lower to higher trophic level, the *persistence* of the metals in environment makes metal pollution difficult to control.

Sources of pollution, harmful effects and harmful levels in environment of some metals are summarized in Table 21.1. The toxic effects of metals to human beings are generally related to their interference with enzyme actions and metabolism in general (Table 21.1). The major ailments include disturbance of central nervous system, and ailments of lungs, kidney and liver. Many metals are carcinogenic.

Plant resistance to heavy metals

Plants are generally sensitive to heavy metals in high concentrations in soil and water. However, certain plants are able to grow in contaminated habitats by employing avoidance mechanisms (Larcher, 1995), as follows:

- (i) Low uptake of heavy metals.
- (ii) Immobilization of metals in cell walls, preventing the contact of heavy metals with protoplasm.
- (iii) Low permeation across the boundary layer of protoplasm.
 - (a) Compartmentalization and formation of complexes with organic and inorganic acids, phenol derivatives and glycosides in the vacuole.
 - (b) Chelation in the cytoplasm to sulphur containing polypeptides and to SH-containing proteins.
 - (c) Retranslocation.

Phytochelatin (PCs), the trace-metal-binding peptides play an important role in metal tolerance. PCs protect plant enzymes from trace-metal poisoning; in metal tolerant strains PCs are accumulated at higher levels than in metal-sensitive strains.

Bioindicators of Pollution Impact

Bioindicators are organisms or communities which show their sensitivity to pollution stress by alterations in their vital processes (response indicators) or by considerably accumulating pollutants (accumulation indicators). Examples of hyperaccumulators are given in Table 21.2 In response to changes in their surroundings as a result of pollution, *indicator organisms* may decline, disappear or, conversely, may show more growth and reproduction. Being sedentary, plants are continuously exposed to pollution impact, therefore, serve as better bioindicators. Changes in floristic composition and disappearance of sensitive species are good indicators of pollution stress. Lichens are among the well known examples of bioindicators of air pollution and deposition of heavy metals.

Organisms that show particularly high sensitivity to certain pollutants are called *test organisms* (Table 21.3). They are used to indicate the presence of phytotoxic substances (and even the amount present). They are grown under standardized condition, and changes in biochemical, physiological and morphological characters are recorded. Microorganisms, animals, algae, submerged aquatic plants, tree seedlings, tissue and cell cultures are used as test organisms (see Table 21.3 for examples of macrophytes).

Monitoring organisms are employed for qualitative and quantitative detection of pollution stress because of their specific responses to pollution. Organisms which accumulate pollutants in large amounts without being damaged are useful monitoring organisms. Accumulation indicators include bacteria, fungi and various plant groups, higher land plants, aquatic plants, mosses lichens

Table 21.1. Summary of pollution related characters of some toxic metals.

Metals	Pollution sources	Harmful effects		Harmful levels in environment
		Human health	Plants and other organisms	
Aluminium (Al) (third most abundant element, non-ferrous)	Al, second most used in industry; waste from mining and manufacturing industries.	Bioavailability of ATP and casein in animals reduced; enzymes inhibited; suspected to cause neurological diseases	Injurious to plants when pH less than 5; roots become thick and lose fine branches	Al concentration should not exceed 1.5 mg/l in water
Arsenic (As) (a metalloid)	Arsenical pesticides, smelting ores containing As, combustion of coal, exposure from drinking water.	Tends to accumulate in lipid-rich tissues, hair, nails and skin, lung tissues; arsenites bind strongly with sulph-hydryl (-SH) groups of proteins resulting in enzyme inhibition.	Leguminous plants, crops like onion, rice and paddy are vulnerable to As toxicity	Cases of arsenical dermatosis are reported in India, but harmful level in environment is not known precisely
Cadmium* (Cd) (a rare, toxic heavy metal)	Electroplating, paints, Ni-Cd batteries, PVC, burning of Cd-containing materials, such as automobile tyres, motor oils, plastics and coal.	Highly toxic to humans; enters through food, but the workers in mining and smelting industries are constantly exposed.	Toxicity to plants similar to that of Pb, Ni, and Cr; Crustaceans in aquatic bodies get affected.	About 1-2% of ingested, 11% of inhaled Cd is retained in body. At 10 ppm in water Cd kills fishes in one day and at 10 ppm in 10 days. Less accumulated in fish muscles, so less harmful to fish eaters.
Chromium (Cr) essential to humans and animals in trivalent form	Leather tanning, explosives, ceramics, paint pigments, photography and wood preservation. Wastes of these industries pollute water by dissolving Cr as chromate salts.	The hexavalent forms highly toxic to humans, causing dermatitis, allergic skin reactions, etc.	Cr is less toxic to plants than Cd, Cu, Pb, Hg, Ni and Zn	Cr at 10 ppm concentration in water is lethal to some algae. In a survey of 60 towns in India Cr in drinking water was found from 0.002-0.005 ppm, but up to 0.1 ppm in potable waters. The upper limit for safe water is 0.05 ppm.
Copper (Cu), a heavy metal necessary for organisms as a trace element, but toxic at higher concentrations	The third most important in the industries. Air from industrial units producing non-ferrous metals, wood combustion, iron-steel production, spray of Bordeaux (a fungicide having 1-2% CuSO ₄).	Ingestion of 15-75 mg Cu causes gastro-intestinal problems.	Inhibits plant growth even at 0.1 mg/l; Cyanobacteria particularly vulnerable, as Cu suppresses N ₂ -fixing ability	Concentration higher than 1 mg/l in water harmful

Lead (Pb) trace element	Manufacture of pipes, pottery, alloys, in making battery plates, paints, varnishes, pesticides (lead arsenate), plastic industries, etc.	Air borne Pb is readily absorbed through lungs; about half of the inhaled particulate lead (0.9 µm) is absorbed, and smaller particles (0.1 mm) can be absorbed by 90%. A chronic exposure may cause weight loss, loss of teeth, blue line in gums, etc.; Pb poisoning cause anaemia, injurious to nervous system and dysfunctioning of kidney.	Inhibits microbes resulting in slower decomposition of organic matter, leafy vegetables pick up readily. Toxic effects more at low pH	Pb concentration has increased from about 2 mg in pre-industrial man to 12 mg in the present day adults; the blood Pb concentrations in large cities of India ranges from about 11 to 36 mg/dl. It is related to traffic situation and ranges in air from 0.016 to 0.656 mg/m ³ .
Mercury (Hg) rare element	Manufacture of plastics, electronic industries. But air pollution is mainly due to Hg released in fossil fuels combustion.	In inorganic forms, Hg attacks mainly liver and kidney but organo-mercurial compounds concentrate especially in brains. Cereals, pulses and vegetables, fish, milk and eggs can cause intake of Hg by human beings.	Hg concentrates up through the food chain, Inorganic Hg is converted into more toxic methyl mercury compounds by anaerobic bacteria in sediments and mud that affects aquatic organisms.	A person should not consume more than 0.5 kg fish a week if it has 0.5 ppm Hg concentration**
Nickel (Ni) essential micronutrient for animals and microbes	Forms about 0.008% of the earth's crust; used in electroplating and in making alloys and steel (which contains 8% Ni) etc.	Dietary intake of Ni averages 300-600 µg/day which is generally recognized as safe (cooking in steel utensils tea, etc., are main dietary sources)	Toxicity includes nickel-itch and respiratory disorders in animals;	
Selenium (Se), essential trace element, being a natural component of enzyme glutathione peroxidase	Fossil fuel burning accounts for up to 90% of Se in air; in water pollution is due to its presence in rock phosphates, flyash, solid wastes disposed by the coal-firing power plants	Daily intake of 50-200 mg is essential for humans, but higher levels affect central nervous system, but heart and lungs may also be affected.	Neutral to alkaline soils have problems of Se toxicity. It is mutagenic. Animals exposed to Se develop problem of vision impairment.	
Zinc (Zn) essential trace element	Smelting of ores causes air pollution, refuse from municipality and motor automobiles are other sources.	Zn at 15 mg/m ³ in air in the work environment may produce metal fume fever.		

*Permissible intake of Cd in food is about 57-71 µg/day. In Japan people living along Juntsu River used to take 100-1000 µg Cd/day/person because rice was irrigated with water having 0.35-3.5 ppm Cd. This resulted in about 100 deaths by the end of 1965. The disorder caused by Cd was called *Itai-Itai* (resulting from a kidney malfunctioning).

**Minamata Episode – the toxic effect of methyl mercuric compound came to the notice when many people were reported to be affected by the then mysterious neurological disorder in the industrial city, Minamata. The methyl mercuric compound-rich effluents generated from the industrial unit of the city which produced vinyl chloride and acetaldehyde by catalytic conversion of acetylene. Some of the HgCl₂ catalyst was converted into methyl mercuric compound, which became bioconcentrated by fish and shellfish in Minamata Bay. Those who consumed fish showed neurological disorder. It was confirmed by 1958.

and others. For example, the lichen, *Hypogymnia physodes* accumulates many mineral pollutants (Pb, Zn, Fe, Mn), and is highly sensitive to SO₂ and other gaseous pollutants. Because of their heavy metal accumulating ability, some fungi and bacteria are used to retrieve them from sewage sludge and wastewater. Many fungi (e.g., *Amanita muscogora*, *Lactarius* spp.), lichens (e.g., *Cetraria* spp., *Cladonia* spp.), mosses (e.g., *Bryum*, *Thuidium*) and vascular plants were found to be radioactive due to accumulation of ¹³⁷Cs, ¹³⁴Cs, ¹⁰⁵Ru, ⁹⁰Sr, following the accident of Chernobyl nuclear reactor plant.

Table 21.2. Examples of plants known to be hyperaccumulator of metals, also called metallophytes and toxicophytes (particularly which accumulate As, Se).(based on Larcher, 1995 and other sources as indicated).

Species and their occurrence	Metal	Concentration (mg metal kg ⁻¹ dry matter)
<i>Eichhornia crassipes</i> (tropical waters)	Fe	14400
<i>Thlapsi caerulea</i> (British Isles)	Zn (roots)	25000
<i>Minuartia verna</i> (Europe)	Cu (roots)	1850
	Pb (roots)	26300
	Cd (leaves)	348
<i>Jasione montana</i> (British isles)	As (leaves)	3100
<i>Mechovia grandiflora</i> (Congo basin)	Mn (leaves)	7000
<i>Acrocephalus robertii</i> (Congo basin)	Co (leaves)	1490
<i>Psychotria douarrei</i> (New Caledonia)	Ni (roots)	92000
<i>Pearsonia metallifera</i> (E. Africa)	Cr (roots)	1620
<i>Astragalus preussi</i> (N. America)	U (leaves, roots)	70
<i>Astragalus acemosus</i> (N. America)	Se (leaves)	15000
<i>Alyssum bertholonii</i> (Brooks and Redford 1978)	Ni (shoots)	13400
<i>Miconia lutescens</i> (Bech et al., 1997)	Al (shoots)	6800
<i>Pteris vittata</i> (Chinese brake fern) (Gumaelius et al., 2004)	As (fronds)	27000

Table 21.3. Submerged macrophytes as test plants for metal pollution (based on Nobel et al., 1983).

Pollutant	Macrophytes		
	<i>Potamogeton lucens</i>	<i>Potamogeton crispus</i>	<i>Elodea canadensis</i>
Lead (Pb)			
Threshold concentration for injury (ppm)	10.4	2.1	10.4
50% lethality (ppm)	—	—	—
Cadmium (Cd)			
Threshold concentration for injury (ppm)	0.01	0.01	0.01
50% lethality (ppm)	0.6	0.6	0.6
Copper			
Threshold concentration for injury (ppm)	0.06	0.03	0.06
50% lethality (ppm)	0.3	0.06	0.03
Zinc (Zn)			
Threshold concentration for injury (ppm)	0.7	4.9	0.7
50% lethality (ppm)	3.3	6.5	3.3

Decontamination of Polluted Soils

Physico-chemical and thermal methods: Current techniques of decontamination of soils polluted with heavy metals and some organic pollutants are based on physico-chemical extraction, such as acid leaching and electro-osmosis, or on *in situ* immobilization, for example, by vitrification, a thermal process in which metals are fused with the silica at high temperature (2000°C) to form inert glass.

Bioremediation: Bioremediation refers to the method of using naturally occurring biodegradative processes to treat the contaminated soils. *In situ* bioremediation is the treatment of soils in place. *Ex situ* remediation involves implementation of physical removal of soil from a contaminated site and then treating it. For bioremediation, oxygen and nutrients are added at the site of contamination. Bioventing is the technique of adding oxygen in the unsaturated zone. Many of the remediation processes used for inorganic contaminants could also be appropriate for organic chemicals. Organic compounds are of several types and therefore require more than one method of treatment. Actual practice of bioremediation is limited but is rapidly gaining popularity.

Phytoremediation: '*In situ*' methods like phytoremediation involving phytostabilization and phytoextraction make use of plants for remediation of a contaminated site. Some plants are metal-tolerant and they markedly concentrate metals in their tissues. These plants, called *hyperaccumulators* of metals, (also *metallophytes* or *toxicophytes*), can contain in excess of 100 $\mu\text{g g}^{-1}$ dry weight of Cd, 1000 $\mu\text{g g}^{-1}$ Ni, Cu, Co, and 10,000 $\mu\text{g g}^{-1}$ Zn or Mn (Baker and Brooks, 1989). The hyperaccumulator plants can be used in phytoextraction of metals from metal-polluted soils and water (e.g., Brown et al., 1994) and for phytomining (Nicks and Chambers, 1995). Biomass of such plants is harvested, disposed of as hazardous waste or is incinerated at low temperature. Metals concentrated in ash are recovered. Phytoextraction is an experimental approach to remove contaminants from the soil by plant uptake. Various types of soil amendments have been used to improve the establishment and maintenance of vegetation on contaminated sites.

There are evidences to suggest that plants which markedly accumulate metals are protected from insect herbivory and from fungal and bacterial infections (Boyd and Martens, 1998). Now nearly 400 hyperaccumulating plant species have been identified, majority of which accumulate Ni (for examples, see Table 21.2). Many hyperaccumulators of metals are members of family Brassicaceae. *Thlaspi caerulescens* is reported to take up 30 kg Zn ha⁻¹ and up to 2 kg ha⁻¹ Cd (see Dobson et al. 1997). As for tissue concentration, *Thlaspi caerulescens* can accumulate Zn to more than 5%, Cd to 0.2%, and Pb to 1% concentration on dry weight basis. Indian mustard (*Brassica juncea*) and *Alyssum* have also been found to do well in extracting heavy metals. Phytoextraction is being recognised a useful technique also for removing radionuclides such as ¹³⁷Cs and ⁹⁰Sr from contaminated soils, and for the removal of excess Na, Se and B from saline soils.

In situ phytoremediation of organic contaminants such as petroleum hydrocarbons involves promoting plant growth on contaminated soils. The plants reduce the concentration of contaminants through rhizospheric effect by enhancing soil microbial activity. Plants like tall fescue (*Festuca arundinacea*), white clover (*Trifolium repens*) or *Cynodon dactylon* have been found to reduce total petroleum hydrocarbons in contaminated soils by 40 to 50% (Pierzynski et al., 2000). The other promising techniques that involve plants and other organisms are *degradation* and *stabilization* of pollutants in the rhizosphere. *Fagopyrum esculentum* secretes oxalic acid into the soil, oxalic acid complexes with Al and forms Al-oxalate which is not phytotoxic. In phytostabilization, use of plants is made to eliminate bioavailability of toxic metals in soils, e.g., cultivars of *Agrostis tenuis*, *Festuca rubra*, etc.

Phytovolatilization in which inorganic contaminants are extracted and then dispersed into atmosphere by volatilization from aerial plant parts is another phytoremediation technique found useful for removing Se and also metals like Hg from the contaminated soils. The plants that

metabolize some organic contaminants or volatile organic carbon compounds release these into the air through transpiration.

Many types of microorganisms can be used to bioremediate the organic contaminants. Under aerobic conditions, fungi could play an important role in the degradation of organic substances. Under anaerobic conditions, bacteria are important decomposers of the contaminants. Experimentally, genetically engineered microorganisms (GEMS) have been developed for specific purpose of bioremediation. Bioreactors are generally used to ensure optimal growing conditions for effective bioremediation.

21.2 PESTICIDE POLLUTION

The insects, microbes, rodents like rats, and weeds must be killed to save the crops we grow. Insects like locusts are known to cause famine by destroying every blade of grass growing. They are called pests; a term used for organisms that are unwanted by man for one reason or another. Chemicals like DDT and other chlorinated hydrocarbons and organophosphates used to kill such pests are called *pesticides*. Depending on the type of target organisms killed, the pesticides can be divided as *herbicides* (plant killers), *insecticides* (insect killers) and *fungicides* (which kill fungi). All of these are called *biocides*.

Nature of Pesticide Chemicals

They are synthetic chemicals (made in laboratories) that pollute all components of ecosphere, water, soil, and air, sometimes simultaneously. Today the total worldwide use of the various pesticides amounts in millions of tons.

Among the pesticides, insecticides are the best known. They are divisible into three subgroups: (i) **chlorinated hydrocarbons** of H, C and Cl - e.g., methoxychlor, aldrin, dieldrin, lindane, toxaphene, heptachlor, DDT, DDE, DDD, (ii) **organophosphates**, containing C, H, and P - e.g., diazinon, malathion, parathion, methyl-parathion, (iii) **carbamates** containing arrangements of C, H, and N - e.g., Aldicarb, carbaryl. Chlorinated hydrocarbons are of persistent nature, they do not easily break down, they are fat soluble (not water soluble) and accumulate in fatty tissues of animals. Organophosphates easily break down, are water soluble, and more toxic to animals and humans. More than 60% of the pesticides used relate to agriculture, the other uses include public health, industry, and home application. Agriculturists are using pesticides not only to produce more, but also to produce cosmetically flawless fruits and vegetables (spotless, nice in look). Consumers demand perfectly spotless apple or tomato. But they forget that invisible chemicals used to make them good looking are bad for the health. Pesticides are also used in manufacturing (e.g., as wood preservative), some of them have toxins.

Pesticide Ecology

Pesticides have become major pollutants of our environment. The special features of pollution caused by these chemicals (and also of other such chemicals) are given below.

Slow break down and persistence: Organisms do not have enzymes to degrade synthetic chemicals, so these chemicals persist in nature. For example, DDT may persist for 10 years or more in soil.

Biomagnification: Some pesticides (also some other chemicals, metals and radioactive materials) move into storage components of human and animal bodies, such as fat and bone, and are passed up the food chain in higher and higher concentrations. The increase in concentration of these materials from lower to higher trophic level along the food chain is called *biomagnification* (see also chapter 20). As shown in Fig. 21.2, DDD (a metabolite of DDT) is in harmless concentration

in water, its concentration increases by 250 times in plankton, and from plankton to the fish-eating bird by more than 300 times. The fish eating grebes may not be able to tolerate such a high level of DDD concentration. It must be understood that all top carnivores, from fish to humans, are at risk from biological magnification. Therefore, it is important to test the pesticides for their persistence and accumulation in the environment.

Varied pathways of spread in the environment: Once in the environment, pesticides and the other such hazardous chemicals spread into soil, water and air, and cycle between nonliving and living components through various routes (Fig. 21.3). The pesticides enter the atmosphere as they are sprayed on the agricultural fields and then evaporate. When sprayed, a substantial portion may affect the non-target areas through precipitation, erosion and leaching, surface run-off, seepage from ground water and crop harvest by humans. Pesticides worked into soil may also get into the air by volatilization. Pesticides are directly applied to water bodies to control insects and weeds. However, runoff from soil can also carry pesticides along with soil into water bodies or they may drift over and get deposited on the water surface. Chlorinated hydrocarbons are known to adhere to soil particles, in lake bottom on sediment residues and return to water as a result of seasonal turnover of water. In many industrial countries most surface waters contain pesticides.

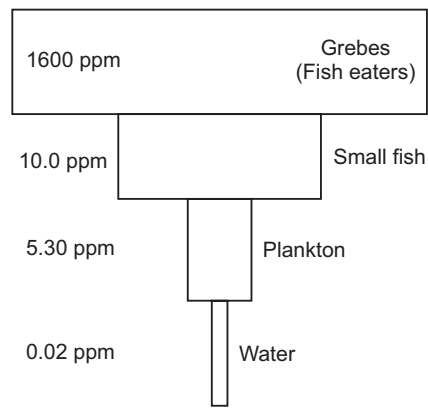


Fig. 21.2. Biomagnifications of the pesticide, DDD that is the rise in concentration of the pesticide from lower to higher trophic level along the food chain in Clear Lake, California (based on Connel and Miller, 1984).

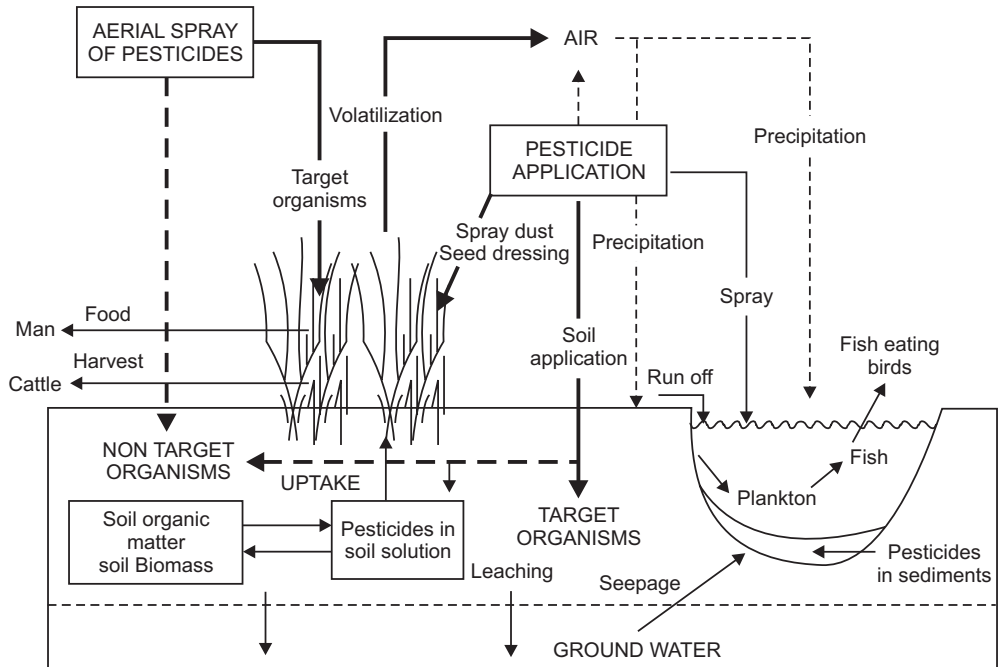


Fig. 21.3. The multiple pathways of pesticides in the environment through land, surface water and food chain.

Harmful effects on desirable species: Application of pesticides has drastically reduced the incidence of many diseases, including malaria (mosquito), sleeping sickness (tsetse fly), and bubonic plague (rat flea). It helped to raise food grain production dramatically and saved many forest species by killing their pests. But being non-specific insecticides these often also kill non-target, desirable species. For example, in California, DDT not only killed the insect pest of orange, the cottony cushion scale (*Icerva purchasi*), it also killed the predator which used to control the insect population and save orange trees.

Indiscriminate use of pesticides has been reported to increase the mortality of honey bee, and to increase many insect diseases, because of elimination of their predators. A million of bee hives are destroyed each year, largely because of indiscriminate use of pesticide.

Resistance to pesticides: When pesticides are sprayed, the pests and parasites show a decrease. After sometime as the populations become resistant they increase in size above the economic threshold (Fig. 21.4). Indiscriminate use of pesticides causes natural selection of resistant strains of pests, and thus disrupts the natural predator-prey relationships. The result is that more pesticides are used. During the last 90 years, more than 550 species of insects, mites and ticks worldwide have evolved resistant (to pesticides) strains (Dreistadt and Dahlsten, 1987) and more than 400 species of microbes and weeds have developed resistance to pesticides (Fig. 21.5). Resurgence of malaria in many countries is attributed to mosquito resistance to insecticides.

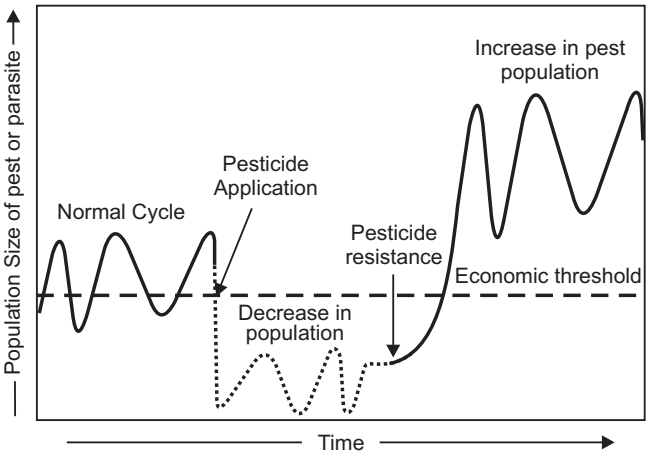


Fig. 21.4. The effect of pesticide application on pest or parasite population.

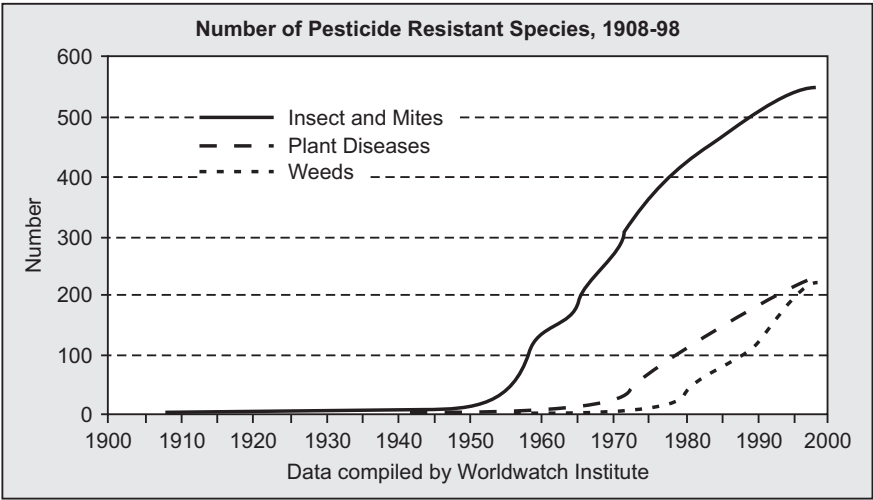


Fig. 21.5. Insect, plant disease causing organisms and weeds becoming resistance to pesticides from 1908-1998 (reproduced from www.worldwatch.org).

Pesticides and Health Problems

Humans: In developed countries, about 50% of the people's food intake may contain measurable residue of chlorinated hydrocarbons (Pimentel et al., 1977). Generally, chlorinated hydrocarbons interfere with the transmission of nervous impulses, and by doing so affect the functioning of nervous system of humans. Symptoms of DDT poisoning are many and varied: facial numbness, vomiting, dizziness, confusion, tremor, etc. Chlorinated hydrocarbons, such as DDT, chlordane, aldrin, dieldrin and heptachlor are suspected carcinogens. The immediate effects of exposure to organophosphates include dizziness, vomiting, heavy sweating, memory loss, sleepwalking, speech difficulty and depression.

Body burdens: Pesticide residues are present in the body of most humans. It is largely because our food sources are affected by pesticides. The *tolerance level*, which is the concentration in food, considered safe for human consumption varies from one pesticide to another (Table 21.4). The *acceptable daily intake* (ADI) of pesticides which is the amount considered on daily basis, also varies (Table 21.4) (Both are set by the WHO and FAO).

Table 21.4. Typical tolerance levels and acceptable daily intake of selected pesticides (as recommended by WHO/FAO).

Pesticide	Typical tolerance (ppm)	Acceptable daily intake (mg/kg body weight)
Organochlorines		
Dieldrin	0.1	0-0.0001
Lindane	1	0-0.01
Toxaphene	7	
Heptachlor	0	0-0.0005
DDT	1	0.0025
Organophosphates		
Malathion	8	0-0.02
Parathion	1	0-0.002
Carbamates		
Carbaryl	10	0-0.01
Carbofuran	0.1	0-0.01

Wildlife: Pesticides find entry into our bodies and those of animals by ingestion, inhalation and absorption through exposed surface. Lethal effects of chlorinated hydrocarbons on animals, such as fish, birds and mammals usually involve the disturbance of central nervous system. Before death, respiratory problems, sluggishness, and neurological complications occur. These pesticides also damage kidney and liver. The sub lethal effects of these chemicals include decrease in reproductive capacity, resulting from the mortality of fry in fish and failure of eggs to survive to maturity in birds. DDT has been reported to cause thinner egg shell in ducks and falcon. DDT is known to bring about change in marine ecosystem structure by diminishing phytoplankton photosynthesis and affecting food chain. Though more toxic, organophosphates are less likely to cause chronic problem because they easily break down. Because species are exposed simultaneously to many pesticides, they can have synergistic effects, which are different from the effects of any one pesticide independently.

How to Reduce Dependence on Pesticides ?

Worldwide 2.5 billion kg pesticides are applied annually in agriculture to control pests. However, of the 70,000 pest species which attack agriculture crops throughout the world, 99% are controlled by their natural enemies. Each insect pest has 10-15 natural enemies. Technology should be developed to control pests keeping in view all natural processes and ecological principles. In recent years an approach called integrated pest management (IPM) is being promoted to reduce the use of pesticides. In this approach a combination of several measures is advocated: (i) host plant resistance, (ii) biological control, and (iii) cultural controls along with pesticides.

Millions of dollars have been saved annually by transferring genes resistant to pathogens to productive crop varieties from their wild relatives and even from unrelated species. Biological control is primarily based on use of parasites' natural enemies, such as predators of pests, pathogens and herbivores to suppress pest populations. The effect of introduction of a biological control on the prey populations shows decrease in the number below the economic threshold level, indicating an effective biological control (Fig. 21.6). For example, for the biological control of insects, vodalia beetles (*Rodalia cardinalis*) have been used to control cottony cushion scale that infected citrus plantations in California in nineteenth century. The other biological measures include application of sterile male technique, use of sex attractant pheromones to draw insects into traps. Tritrophic control (host plant emits pheromone like chemical when attacked by a pest, which attracts predators of the pest) is also being experimented upon. Several cultural methods are applied to make the agroecosystem unfavourable for the pest species. These include: crop rotation to prevent build up of pests year after year; tilling a field so that eggs are exposed to drying; strip cropping to break up a large stretch of monoculture. For example, some pests attack both cotton and alfalfa but more the latter, by planting alfalfa in strips between rows of cotton, pest may be kept away from this economically important crop of cotton. It is possible to manage natural enemies of insect pests to share agricultural landscapes. More efforts are required to develop crop varieties resistant to diseases or having structures such as hair on leaves which limit herbivory.

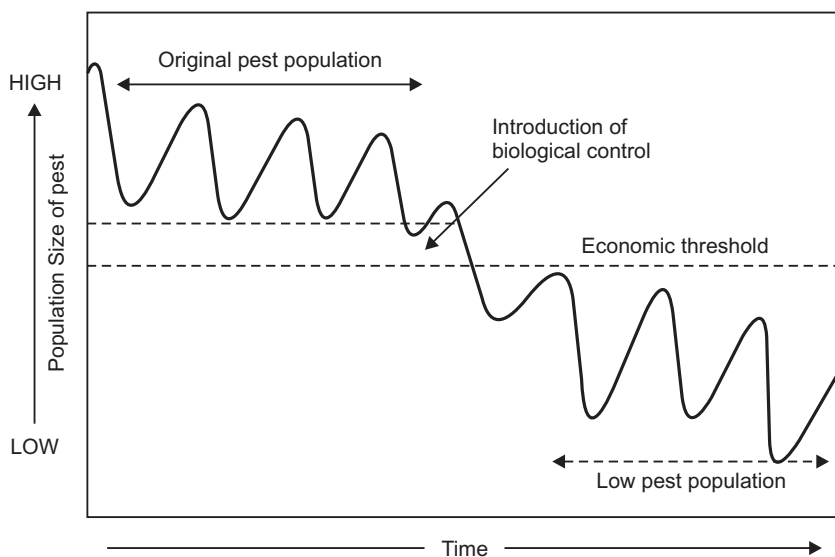


Fig. 21.6. Control of pests using biological control that tends to keep the population below the economic threshold.

Bans and Restriction on Pesticides

As of 1987, EPA had banned four pesticides in the USA, viz., EDB, dinoseb, 2, 4, 5-T, and silvex. The most infamous of these is 2, 4, 5-T, or agent orange, used as defoliant in Vietnam. It contains dioxins, which are carcinogenic and cause birth defects. EDB (ethylene dibromide) is in the same family as DDT, chlordane, heptachlor, etc. It is used to fumigate soil to drop larvae of harmful insects attacking stored grains, logs, and fruits. Since then many pesticides have been banned in USA and other developed countries. Some pesticides are banned in India, also.

In India, the pesticides are regulated by the Insecticides Act 1968 and Insecticides Rules of 1971. Over the past more than 30 years, about 23 pesticides have been phased out in India, but more than 32 pesticides banned in many countries are still in use; these include chlorpyrifas, lindane, endosulfan, carbaryl and paraquat.

21.3 RADIOISOTOPE CONTAMINATION

Radioactive wastes include spent fuel elements and high level waste, transuranic wastes, low level waste and uranium mill tailing. The disposal of radioactive wastes is one of the most sensitive environmental issues. Many radioactive wastes are extremely toxic and the nuclear wastes can contaminate the environment. Radioactive wastes can be introduced into the water as liquid effluents discharged from commercial nuclear fuel processing plants.

Isotopes are forms of an element that have identical atomic structures, but different numbers of neutrons in nucleus. Isotopes are stable (deuterium, the heavy hydrogen), as well as unstable. The unstable isotopes undergo spontaneous disintegration and are radioactive, therefore are called radioisotopes. Of the three radiations given off by disintegrating radioactive elements – alpha, beta and gamma rays, the last is most harmful to human body. A radioisotope has a definite half-life that is the time required for 50% disintegration of radioactive element. For example, half lives of iodine-131, cesium-137, carbon-14 and plutonium-238 are 8 days, 27 years, 5,600 years and 24,000 years, respectively. Induced radiation is caused by technological development, such as atomic bomb, and nuclear power plants account for nearly 15% of natural radiation since 1940s. Radiations of natural sources called background radiations are milder and non-hazardous. For example, natural radiation exposure level in millirem/yr is 35 from cosmic rays, 5 from air, 34 from building materials, 25 from food, 11 from ground, 5 from coast to coast jet flight, 1 from colour TV, 50 from chest X-ray, 0.001 from living within a 50 miles radius of nuclear power plant. Before 1963, when aboveground tests of atomic weapons were not banned, considerable amounts of radioactive materials were released into the atmosphere. Circulating around the earth, the radioactive materials produced substantial increases in radioactivity in rain and snow. Nuclear plants installed for generating energy are also the source of potential hazards of radioactive materials (see Box 21.2). Radioactive materials in environment are toxic pollutants, and protoplasm in cells; especially the nucleus of organisms when exposed to them can be damaged, or adversely affected by ionizing radiations. Effects of ionizing radiations occur on three scales: (a) intense and short-term exposure producing severe effects (causes sickness, hair loss, tissue damage); (b) lower-level and often chronic exposures producing delayed consequences including cancer, rapid ageing, intestinal bleeding (all were recorded in Hiroshima and Nagasaki years after bombing in the II World War); and (c) effects which can occur from any level of exposure on the genetic materials. The last one often results in mutations, which are mostly hazardous.

Organisms are affected by the radioisotopes released into air, soil and water. The radioisotopes are both of the elements which are parts of the chemistry of organism (such as carbon-14, phosphorus-32, calcium-45, potassium-40, hydrogen-3 or tritium, iodine-131), as well as of those not connected with their chemistry (such as, strontium-89 and 90, cesium-134 and 137, thorium-232, plutonium-239). Some, such as radioiodine (iodine-131) cesium-134 and strontium-90 are of special

concern because like some of the pesticides they accumulate in tissues, and increase in concentrations along the food chain (biological magnification). Iodine-131 has been found in concentrations several times more in thyroid gland of man than in environment. Most unexpectedly, cesium became a problem to the people of so distant land as arctic tundra who had nothing to do with nuclear test and modern technology. From soil and lichens it reached the reindeer feeding on lichens, and to Eskimos eating reindeer. Strontium-90 is of special interest because it behaves like calcium. Since children add calcium in their growing bones, they are especially vulnerable to strontium-90. Milk and other food materials are sources of strontium-90 to humans.

Problems associated with use of nuclear energy are as follows:

- (1) Even with a tight control on the release of radioactive materials, the accumulated effect of all the nuclear plants (now in hundreds) can be significant at global level.
- (2) As the accidents of Chernobyl and others indicate, foolproof control is very difficult to achieve. Then, terrorists can easily make a bomb from nuclear material.
- (3) We have no idea of safe sites for the burial of highly radioactive wastes, including junks of nuclear plants. Radioactive wastes must be disposed of in a manner that they are not released into biosphere for they may take hundreds or thousands of years until they decay to harmless levels.
- (4) Damage to man and his ecosystems can be enormous.
- (5) Nuclear power plants generate 40 per cent more waste heat per kilowatt than the fossil fuel-fired plants.
- (6) Water demand of nuclear plants may cause shortage of water for other purposes. Large scale use of water for cooling would cause heat-pollution of water including that of oceans. Severe heat pollutions may change the biota destroying the organisms of economic values.

Box 21.2. Nuclear Reactors and Nuclear Energy

The nuclear reactor is designed to produce heat that can then be used to generate electric power. Reactors based on nuclear fission (splitting of atomic nucleus, generating lot of electric energy) are of two types: burner reactor and breeder reactor. The problem with the burner reactor is that it consumes the fissionable Uranium-235 isotope while converting only small amount of the plentiful Uranium-238 into fissionable plutonium. Since the Uranium which it consumes quickly, is in limited quantity, burner reactors cannot serve us for long. The breeder reactor generates more fissionable material than it burns, while at the same time generating electricity. It is more preferred because its fuel consumption is less, but from it nuclear weapons can be made easily. Fusion is opposite of fission. When developed it is believed to be inexhaustible source of clean energy. It is not possible to make nuclear weapons from it, technological problems of producing fusion energy are still difficult to deal with.

21.4 WASTE DISPOSAL IN THE ENVIRONMENT

Solid Wastes

Waste generation is associated with human civilization; the higher the human consumption the greater and more varied the waste materials. The average refuse per person per day in the USA has increased from 0.91 kg in 1920 to over 2 kg at the present time, and Americans generate about 4 billion tons of waste per year. In UK the present rate is 1 kg per head per day and in developing countries less than 0.5 kg per head. According to the World Bank data, urban Asia produced 2.7 million m³ municipal solid waste per day in the year 1998-99, and in 2025 this will increase to 5.2 million m³. In many industrialised countries much of the waste material is generated from mining

(75% of total in the USA), next in importance are agriculture, industries and municipalities. Paper, food, metals and glass are major components in developed societies. *Solid waste* includes not only a solid, but also a liquid or even a liquefied gas. Besides polluting air, soil and water, the improper handling of solid wastes and their disposal promotes the growth of micro organisms and vectors that spread diseases or produce noxious odours. In urban areas, solid wastes clog the drains creating unhygienic conditions and over flow in rainy season.

Currently, 3.4-4 billion tonnes of municipal and industrial waste are produced every year, of which non-hazardous industrial waste accounts for 1.2 billion tonnes (Chalmin and Gaillochet, 2009). A major share of the waste generated is MSW originating from urban settlements (1.7-1.9 billion tonnes, or 46 per cent of the total waste generated) with 0.77 billion tonnes of this being produced by 25 Organisation for Economic Cooperation and Development (OECD) countries alone (UNEP, 2010). As a country develops and becomes wealthier, the composition of its waste stream typically becomes more varied and complex. Fig. 21.7 illustrates the high proportion of organic-rich MSW in middle and lower income countries with a gross national income per capita of less than US\$ 12,196, while the high-income countries' MSW streams contain a large proportion of paper and plastics.

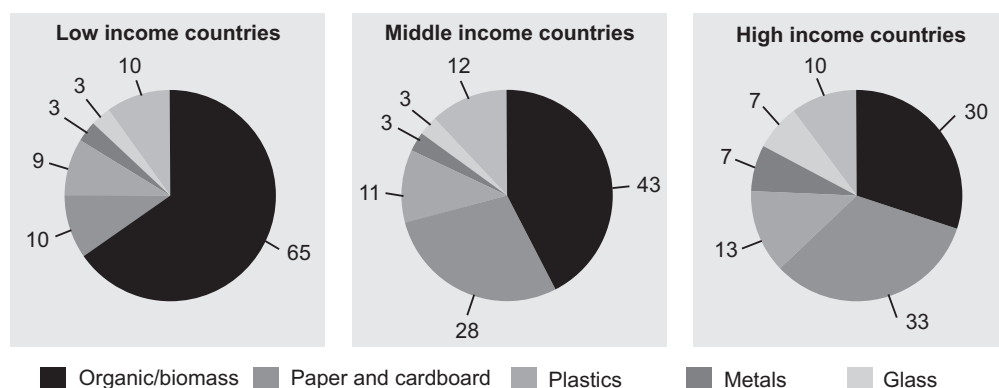


Fig. 21.7. Composition of municipal solid waste by national income. (based on Chalmin and Gaillochet, 2009).

Electronic waste (e-waste) continues to increase dramatically amid growing global demand for electronic and electrical goods. It is estimated that in 2004 alone, 315 million personal computers (PC) became obsolete globally and 130 million mobile phones were estimated to have reached their end of life in 2005 (UNEP, 2005). The USA produces most electronic scrap, reportedly 3.16 million tonnes in 2008 (EPA, 2009). The total e-waste generated worldwide rose from 6 million tonnes in 1998 to 20-50 million tonnes in 2005 (UNEP, 2005).

There has been a significant increase in the generation of municipal solid wastes in India, over the last few decades. India generates about 36.5 million tonnes of municipal solid waste per year. In Indian cities, urban population produces more wastes per capita than the rural population. The daily per capita solid waste generation is about 100 g in small towns and 500 g in large towns; of the total waste generated, the recyclable wastes account for 13 to 20% of the total (see UNEP, 2001a). With the improvement in living standards, the nature of wastes is changing in composition; more non biodegradable wastes such as metal, plastic and glass are being produced as compared to the amount of organic wastes (Fig 21.8). Increasing urbanization and rising income of people will be producing more wastes per capita in future. Figure 21.9 illustrates the generalized composition of solid wastes in India. It is evident that the compostable wastes are the most abundant of these solid wastes.

Due to lack of any legislation, the biomedical wastes generated from clinics, hospitals, pathological laboratories and veterinary centres are also disposed of along with municipal solid wastes.

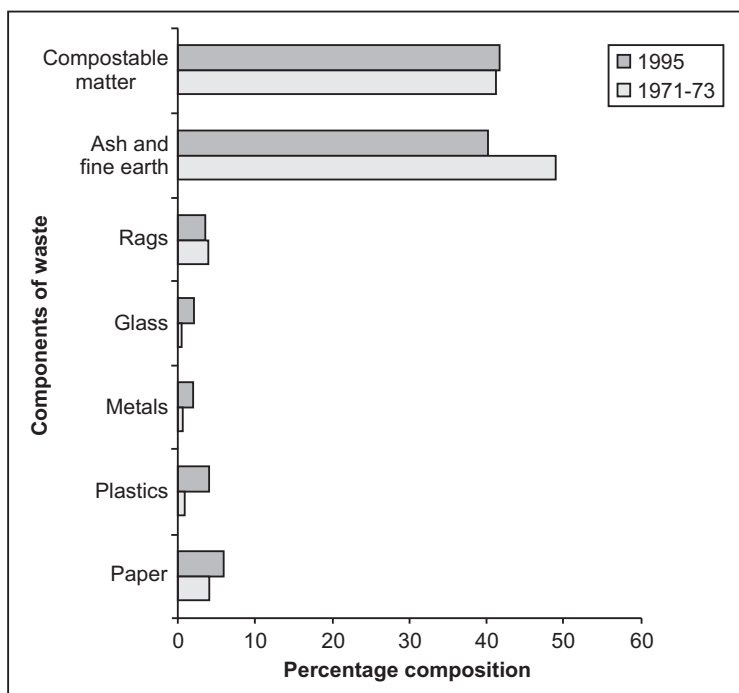


Fig. 21.8. The change in composition of wastes produced in India over the years (based on data from UNEP, 2001b).

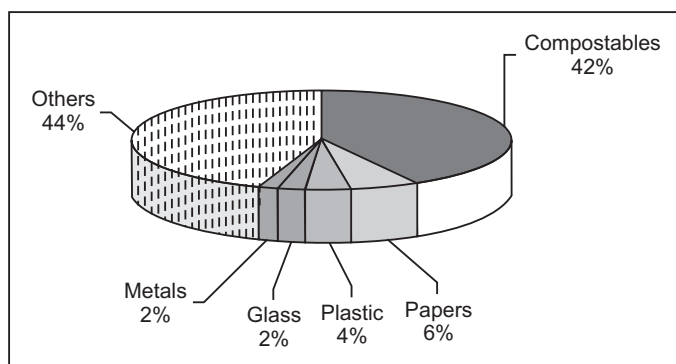


Fig. 21.9. Composition of urban solid waste in India (world bank urban development sector Unit Solid Waste Management in Asia, 1999).

Hazardous or toxic wastes

Hazardous or toxic wastes could be liquid or solid, those are dangerously reactive, corrosive and explosive. **Hazardous waste** includes unwanted and useless material that may pose a threat to health of humans, other organisms, and to ecosystem as a whole. These wastes contain substances that are fatal to humans or animals in low doses, and are carcinogenic, mutagenic or teratogenic to human or other life forms.

The *composition of waste material* has changed over time depending on food habits, cultural traditions, life style and climate etc. For example, in a typical American town the domestic refuse primarily consisted of fine dust in 1935 (57%), whereas paper and rag became much more important in recent years (53% in 1975, 46% in 1980). In Indian cities, there is substantial change in the type of wastes produced over the years (Fig. 21.8) Packing wastes have increased considerably in industrialised countries. Sewage sludge, the solid residue from wastewater settling tanks was earlier considered harmless waste material, but now it is often composed of toxic heavy metals, synthetic organic compounds, pathogens, and other hazardous materials. In India, around 7.2 million tonnes of hazardous wastes are generated which comprise recyclable wastes, incinerable and disposable wastes (MoEF, 2000; CPCB, 2000d) (Fig 21.10). According to the Ministry of Environment and Forests, there are 323 hazardous waste recycling units in the country. Some industrial units in the country also recycle the imported wastes such as battery scrap and galvanised zinc. The studies from Gujarat, Maharashtra, Delhi and Rajasthan have shown that industrial areas generate large amounts of hazardous wastes which have both direct and indirect effects on soil and water of rivers, groundwater and coastal areas. In addition to hazardous wastes, industries also generate around 147 million tonnes of high volume non-hazardous wastes (see UNEP, 2001a). The non-hazardous wastes mostly include coal ash (thermal power plants), plant furnace stage (steel mills), red mud and tailings (aluminium, zinc and copper industries) and gypsum (fertilizer plants). The quantities of industrial wastes produced per annum from these industries are shown in Figs 21.11 and 21.12.

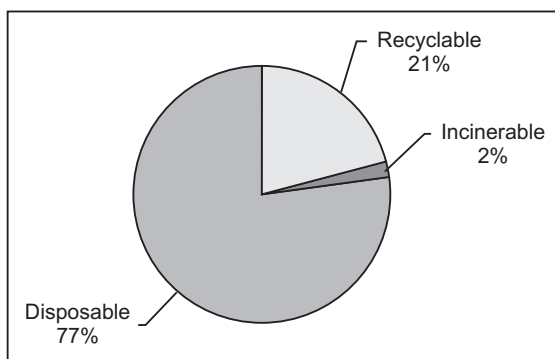


Fig. 21.10. The percentage of total Recyclable, Disposable and Incinerable Hazardous waste produced in different states of India (on the basis of data from UNEP, 2001b).

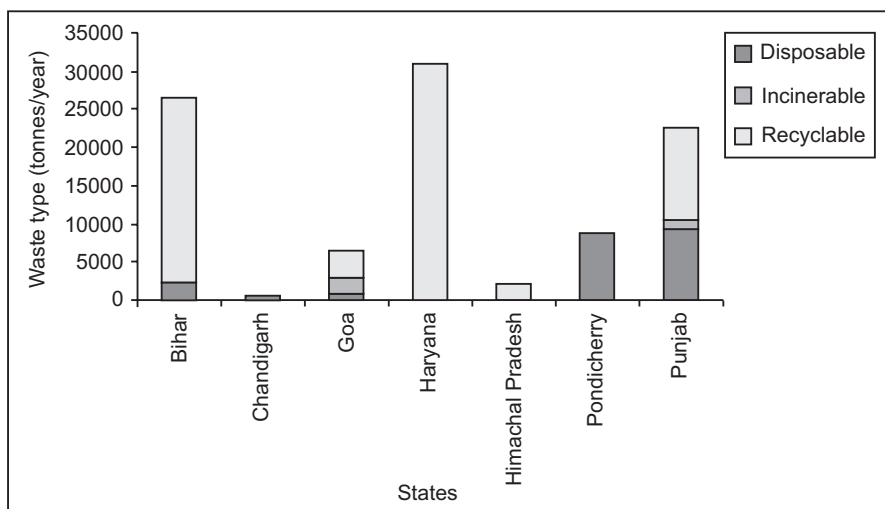


Fig. 21.11. The amount of hazardous waste generation in different state of India (on the basis of data from UNEP, 2001b).

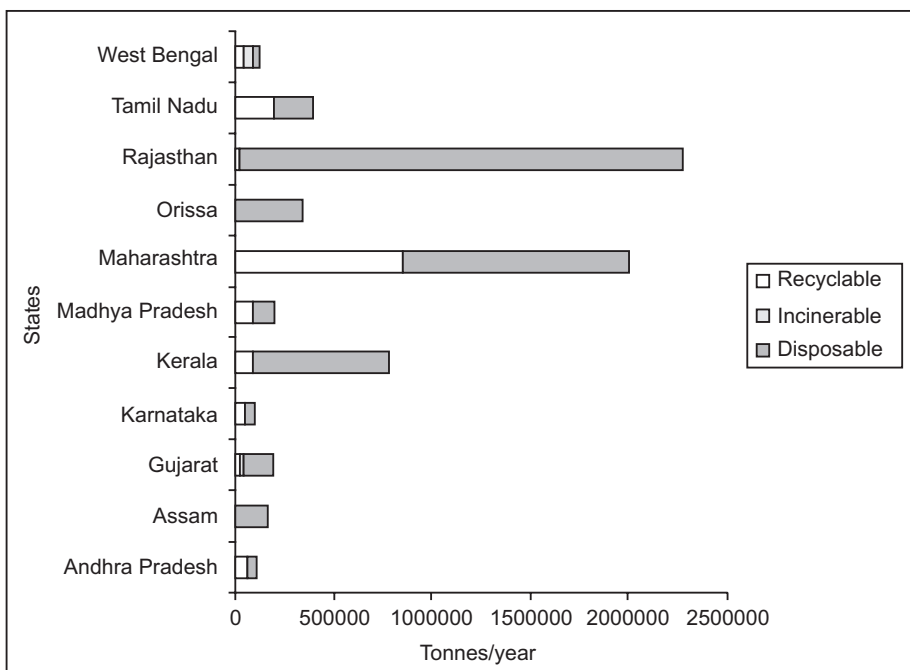


Fig. 21.12. The amount of hazardous waste generation in different states of India (on the basis of data from UNEP, 2001b).

Plastics, the Controversial Solid Waste

During the last fifty years, plastics have replaced more traditional manufacturing materials. Many different types of plastic are in use which is a big threat to environment. The worldwide rise in the use of plastics during the last 2-3 decades has been extraordinary because of their toughness, versatility and low cost. The non-biodegradability makes plastics tough, but also highly objectionable as waste. In landfills the rate of decomposition in the first 10 years is only 0.2%, however, photodegradation may cause 10% decomposition. The total amount of household waste has approximately doubled from 1960 to 1993, but plastic waste has grown nearly 48 times. Another major problem is that the recovery through recycling of plastics is much lower than other wastes: 3.5% while for most others the percentage is more than 20. There are two major categories of plastics, viz., thermosets, that solidify and harden when heated, and thermoplastics, that melt when heated. Plastics of thermosets type are difficult to recycle as they cannot be melted after they have been heated to hardening. Plastics left as litter or disposed off in landfill sites do not decompose easily. During incineration, the plastic wastes produce pollutants and some highly toxic emissions. Waste products from plastic manufacturing industries are highly toxic in nature. It is difficult to recycle the mixed plastic wastes.

Waste Disposal and Management

Waste management is based on the following three approaches:

- Integrated waste management.
- Disposal of residue in a manner least harmful to environment.
- Recovery of energy from the combustible fraction of waste.

Integrated waste management

Integrated waste management is defined as a set of management alternatives including source reduction, reuse, recycling, composting, land fill and incineration (Fig. 21.13). The aim is to reduce the amount of urban and other wastes. Physical, chemical and biological processes are used for various recovery methods. The main objectives of integrated waste management in developing countries are waste avoidance, maintaining the hazardous substances at very low level and devising environmentally sound treatment and disposal methods (UNEP, 2001b). Greater attention is at segregation of different wastes at collection point which can reduce the cost of disposal. Though the cost is high, integrated waste management is a practical solution for management of solid wastes.

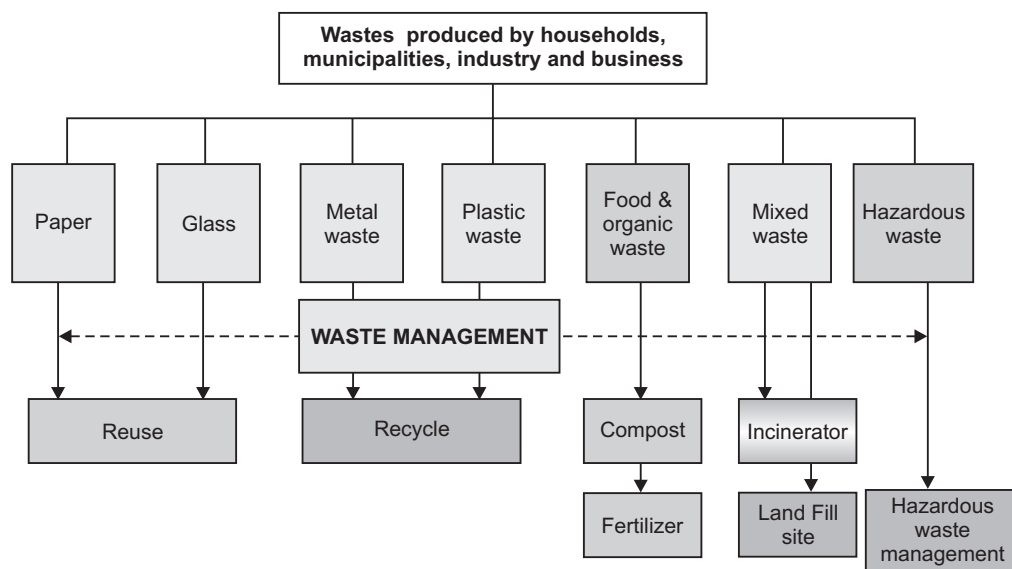


Fig. 21.13. Integrated waste management through reuse, recycle, composting and incineration and disposal in land fill.

Waste Recycling

Recycling is the process by which used or waste materials are put back into productive use. Waste material can be recycled directly as well as indirectly. A good example of direct cycling is the reuse of glassware in the factory where it is manufactured (in-house cycling). The glassware is returned to the furnace for remelting and re-use or in plate glass manufacture broken glass is recycled. In indirect cycling some properties of the material are recycled. Some familiar examples of indirect cycling are recovery of heat generated in wood waste or tyre incineration and its use in space or process heating, use of wood working wastes into chipboard manufacture, and conversion of waste plastic containers into coarse products, such as fence posts.

Sorting of waste and use of recycling bins for collecting different types of wastes such as plastics, aluminium and glass, facilitates the recycling process (Fig. 21.13). If operated efficiently, recycling can reduce pollution problems from waste disposal, provide alternative energy and help to conserve natural resources. Recycled glass, making of paper from waste pulp and remelted aluminium can save substantial amount of energy. Recycling of metals as a whole has grown world wide to save energy and resources. Iron, steel, tin, copper and aluminium are the most easily recycled metals. Local authorities can provide more environmentally conscious waste disposal and recycling schemes. The biodegradable wastes can be treated locally through composting to produce organic fertilizers.

Re-use of articles and volume reduction

Where costs of transporting and handling are low, articles can be recycled for repeated uses – e.g., returnable bottles, glass fibres, steel shuttering used on building sites. Recycling and reuse of materials is crucial for conserving resources and reducing environmental degradation. Reclaiming waste materials for reuse gives an economic advantage. Reuse of items like paper and plastic bags, wood and metal items is possible (Fig. 21.13). Some industries and non-governmental organizations popularize the reuse of various items.

Where a large amount of refuse is generated in the form of empty containers, cartons, packing materials, etc. its volume is reduced before transporting. The volume reduction can be achieved by making bundles, compaction (now machines are available for this), crushing and comminution, pulverisation (converting into dust), briquetting of wood wastes and composting (used for organic refuse).

Disposal of Residue in a Manner Least Harmful To Environment

Composting: Composting is a feasible option when biodegradable and non-degradable wastes are handled separately. Biodegradable solid wastes from food processing plants, kitchen and lawn waste, and municipal sewage sludge are mixed with soil and decomposed by aerobic soil organisms. Composting can be done on a local scale or in large plants. In India, composting of agricultural wastes is a common practice in rural areas. Compost, also known as farmyard manure is used as an organic fertilizer in the agricultural systems.

Sanitary landfill: A sanitary landfill is a waste disposal method for urban wastes. The wastes are confined to an appropriate small area of land; the wastes are spread in thin layers, compacted and covered with a fresh layer of clay (Fig. 21.14). Covering of the wastes restricts the entry of insects, rodents and other animals to the land fill site. To avoid the problems of leachate pollution to surface and ground water, the landfills are lined with clay and plastics before being filled with garbage and biodegradable organic wastes. The leachates can be pumped from the bottom of land fill sites, and sent to leachate treatment facility (Fig. 21.14). When the landfill is full, it is covered with soil and clay. Methane gas produced by anaerobic decomposition in the sealed landfill is collected and burned to produce steam or electricity. In arid regions, land fill sites are relatively safe because of less amount of leachate being produced. In humid regions, wastes are buried above the water table in a relatively impermeable environment. A safe sanitary land fill has a multiple barrier design including monitoring, plastics and clay liners, and biogas and leachate collection system (Fig 21.14).

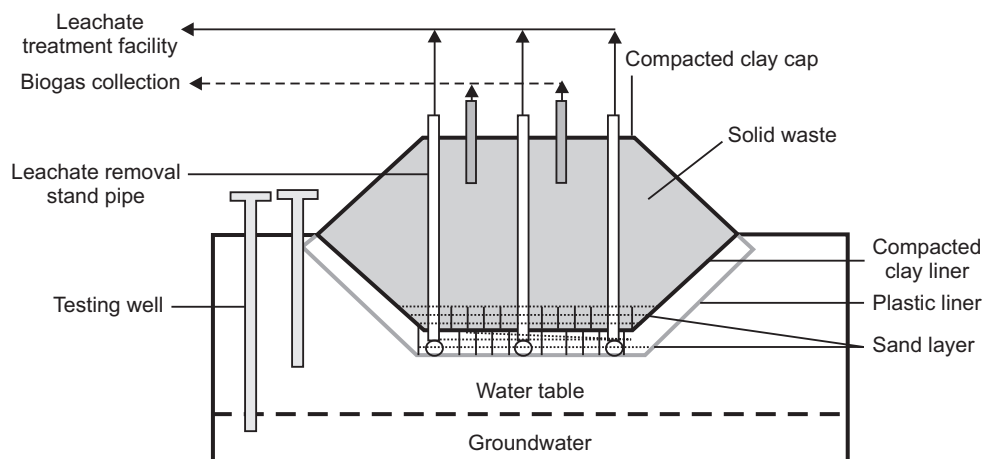


Fig. 21.14. A schematic diagram of a sanitary landfill showing multiple design, monitoring system, and leachate collection system.

A secure landfill is an extension of sanitary land fill which is used for hazardous wastes to confine the wastes to a particular region, for collection and treatment of leachate, and to detect possible leaks.

However, landfills have some drawbacks of producing toxic gases and leachates. Emissions from landfills are reported to contain more than 113 toxic chemicals that easily accumulate in human body. Therefore, the landfills are provided with vent pipes to collect these gases and leachate. The decomposition of organic wastes in landfills produces greenhouse gases. The leachates can pollute soil and water.

Solidification and stabilization: These processes also help to minimize ground water pollution from land disposal of hazardous wastes and mobilizing most of heavy metals present in the sludge, contaminated sites and other wastes. The ash from incineration process can be treated by the processes of solidification and stabilization. Encapsulation is a specific form of waste stabilization, used for radioactive materials, and for high level radioactive wastes; in the process highly toxic wastes could be made less harmful to the environment.

Fly ash use for making concrete: Re-utilisation of fly ash earns carbon credits in India . In India, about 26,000 hectares of land is covered by ash ponds. This land contains nearly 90 million tonnes of fly ash that is generated annually in the country. It is estimated that every tonne of fly ash reused to make concrete reduces 1 tonne of CO₂ equivalent GHG emissions. Lafarge India Pvt. Ltd. has implemented a CDM project activity through fly ash reuse to replace clinker in Arasmat Cement Plant in Chattisgarh, India. By increasing the fraction of flyash (to replace clinker) added to blended cement procured from a thermal power station, the project activity has been successful in reducing approximately 69,359 tonnes of CO₂e per year, with a potential to earn CERs worth US\$ 0.9 million (UNFCCC, 2006).

Rural energy supply from waste: Agri-business ventures promoting conversion of organically-rich biomass waste into biogas have great potential to supply power to rural regions. The Asian Development Bank (ADB) has supported the installation of over 7,500 biogas digesters in more than 140 rural villages in China and has suggested potential models for agri-business ventures such as community-based, small-scale industries, small- and medium- scale industries and large-scale industries for the Greater Mekong sub-region (GMS) (Owens, 2009).

About 500 rural households in the Indian state of Bihar have been benefiting from off-grid power generated from rice husk since 2008. Three quintals (300 kg) of rice husk are used per day in a power plant to generate 32 kilowatts of power. The rice husk costs Rs 60 (US\$ 1.3) per quintal. The production cost per plant per month is about Rs 20,000 (US\$ 426). There is sufficient electricity for a household to light up two rooms and charge a mobile phone for about US\$ 2 per month.

Energy recovery methods

Major methods to recover energy include incineration, pyrolysis and fluidised bed combustion.

Incineration: In incineration, the waste matter is combusted to ash at high temperature (>1000°C), resulting in 90% reduction by volume and 60% by weight. Therefore, transportation and dumping are convenient. The energy generated in the combustion is usable by adding waste heat boiler or warm air furnace to the incinerator (Fig. 21.15). It is particularly useful for some manufacturing industries, such as furniture and other wood based articles where wood waste output (e.g., chips, off-cuts, wood dust and sand dust) is large. However, while operating, air pollution should be avoided by using electrostatic precipitators and scrubbers to remove the particulates and sulphur containing effluent gases .The liquid wastes are further treated and appropriately disposed of.

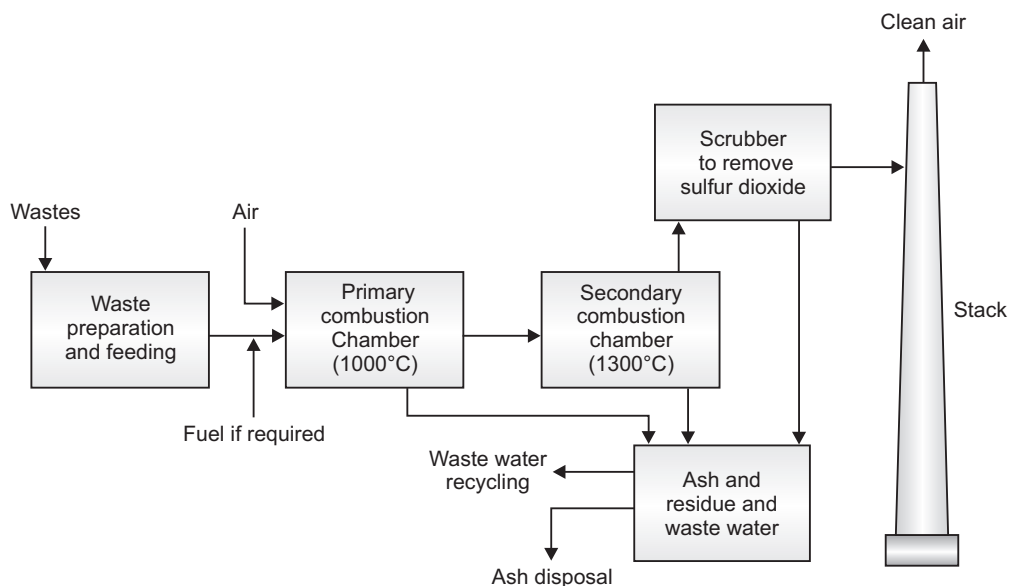


Fig. 21.15. Major components of a high temperature hazardous waste incineration system.

Incineration of wastes is useful for municipalities, hospitals and manufacturing industries faced with problems of combustible wastes, such as volatile organic compounds, saw dusts, tyre and solvents. Incinerators can be designed to handle wastes in any physical state and have proven effective in treating solids, liquids, sludge, slurries, and gases. Incinerators are very effective for destroying organic compounds in wastes. Incineration of solid and liquid chemical wastes, such as polychlorobiphenyl materials, is a common and highly efficient method of destruction at temperature of about 800°C (1470°F). Suitable for the purpose of incineration of different type of wastes are stationary incinerators, cement kilns, incinerator ships, and smaller mobile thermal destruction units. Incineration is probably the best method to manage infections medical wastes.

The energy potential in solid waste is considerable. For example, one ton of solid waste from municipalities is approximately equal to 250 litres of kerosene oil. Frankfurt, Germany, produces 7% of its electricity by incineration. However, environmental impact of incinerating waste needs to be fully determined. Both air pollutants and ash generated can be toxic.

Pyrolysis: It is a recycling process which relies on heating the waste in a retort at a high temperature in an atmosphere deficient in oxygen. It is a process of destructive distillation in the absence of oxygen with the simultaneous removal of organic compounds. The end-products like charcoal, organic liquids, ash and metal residues can be used as solid, liquid or gaseous materials. Non-combustible items, such as glass and metals are removed from the equipment for subsequent re-use or disposal. Pyrolysis is useful for tyres and plastics, and suitable for municipal, hospitals, and organic wastes.

Fluidised bed combustion: This is a system of combustion in which waste (a kind of fuel) is burnt continuously in an atmospheric fluidised bed which offers much flexibility in that it can accept a low grade fuel. The fluidised bed operates at low temperatures. Emissions of polluting sulphur dioxide can be checked by addition of limestone. It is suitable for destructing hospital wastes, and for dealing with organic wastes.

Dumping of wastes in oceans: In the past substantial quantities of wastes such as dredged spoils, industrial wastes, mine tailings and sewage sludge have been dumped at sea. Dumping of wastes, particularly in oceans needs to be carefully investigated as it is an area of concern keeping

in view the well-being of the marine environment. Now, there are strict regulations and licensing of waste disposal at sea. The International Convention for the Prevention of Marine Pollution by dumping of wastes and other matter, also known as London Dumping Convention (LDC), has banned sea disposal of certain hazardous wastes and has imposed prohibition of incineration at sea and a complete ban on dumping of industrial wastes (UNEP, 1993).

In fact, any solutions to the waste problem should be solidly based on ecological principles and long-term monitoring of waste sites should be part of any strategies to deal with wastes. We have not even started assessing long-range impact of new chemicals which we allow to enter the biosphere.

21.5 WASTE AS A RESOURCE

The highly consumptive *Homo sapiens* generates tons of wastes in many forms. Disposal of human excreta and waste matter generated from a household remains a serious challenge in urban management throughout the developing world. Most of the people in poor countries still burn waste matter in open or bury it, the same way as our ancestors did several millennia ago. In a world where an efficient use of resources is getting a wider acceptance, trash is being increasingly seen as a resource – e.g., as a feedstock for energy and several products. It could be used to reduce the pressure of human consumption on our ecosystems. For example, crop residues and yard clippings can be used to make fuels, medicines and cosmetics (Wigginton et al., 2012). There is a lot of scope to recycle plastics and metals several times. Carbon dioxide, the principal green house gas too could be “captured and used creatively” (Wigginton et al., 2012). Technologically, waste water can be used to produce energy and clean water. However, working with waste involves several challenges not only concerning technology but also policy, politics and psychology and economics. For example, people find it difficult to drink clean water obtained by treating toilet water.

This section briefly describes the technological innovations that are already in use or are likely to be invented and used to manage waste by converting into resources.

Forms of waste

Waste that humans produce is varied and huge in amount. For example, the United States produces about 12 billion tons or about 400 tons of waste per person per year, of which only 350 million tons or about 1.2 tons, per person per year are Municipal Solid Waste (MSW), and the rest comes from mining, road building, and other construction activities and farming. In addition to these, there are human excreta, waste matter dumped into water ways, and pollutants that humans spew into the air. In a broad term, CO₂ that the various anthropogenic activities emit is a waste. Electronic waste is now being recognized as another serious problem. It accounts for 1-3% of total municipal waste generation in the USA. In Europe it is 5-7 million tonnes per year, i.e., about 14-15 kg per person per year, and is rising at a rate of 3-5% per year. In India electronic products amount to 1,46,180 tons per year. Mumbai, Delhi, Bangalore, and Chennai are the cities from which most of such waste as ,computers, TV, telephones, etc., is generated. With rapidly changing technology in TV and computers, the reuse of their parts is declining.

The amount of waste increases with the income of countries, and so is the percentage of waste matter recovered (Fig. 21.16). The composition of the waste also changes with income (Table 21.5). For example, the percentage of organic and biomass component in municipal waste declines from 65% in low income countries to 30% in high income countries, while the contribution of paper increases from 10% in low income countries to 33% in high income countries. Worldwide, about 11.2 billion tonnes of solid waste is collected annually.

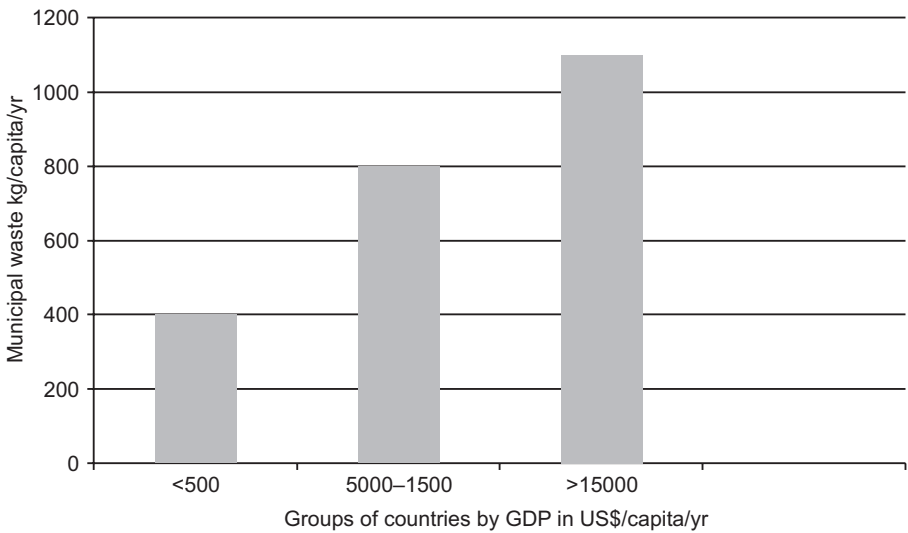


Fig. 21.16. Waste generation as a function of GDP.

Table. 21.5. Waste composition changes with income in US\$ (UNEP, 2012).

	Low (<5000) % of total waste	Middle(5000-15000) % of total waste	High (>15000) % of total waste
Organic/biomass	65	43	30
Paper & Cardboard	10	28	33
Plastics	9	11	13
Metals	3	3	7
Glass	10	12	10
Others	97	97	93

People have begun to use the term, waste industry, as with the varying forms of waste matter, its use and disposal are becoming diverse. There are now plants which process 75 tons of recyclable material an hour. General Motors in the USA now has the concept of zero waste to land fills. Though very difficult to achieve, General Motors has already cut the amount of waste per vehicle manufactured by 28% to 304 kg in 2010.

In certain cases cost effective collection of waste is the major challenge. Then recycling has its cost, and it may not be always the best solution.

Whether to burn waste matter to generate electricity?

Where land is in plenty land fill is the most common means of solid waste disposal, an alternative to it is waste-to-energy plant (incinerators). However, the two can be combined for waste disposal. In Montgomery (USA), nearly 1800 tons of waste is burned a day and 52 megawatts of electricity is generated. This helps offsetting operating cost of plants. By so doing, the waste transported to land fills is reduced by 90%. However, the incinerator is said to produce more pollutants per unit of power generated than coal-fired power plants. The supporters of waste-generated electricity argue that electricity is not their primary purpose, it is in addition to the disposal of waste. There is always a net gain with regard to greenhouse gas emission with waste-to-energy approach, as by diverting wastes from land fills, methane emission from land fill is also reduced.

Waste water reuse

Waste water use is quite common; In Israel, as an example, 73% of municipal sewage is treated and reused for irrigating agricultural crops, which amounts to about 5% of the Israel's total water use and 13% of municipal supply (Schnoor, 2008). Effluent from waste stabilization ponds can be used for irrigation purpose, or to recharge ground water aquifers. And, then, ponds which are used for gravitational sedimentation and flocculation of pollutants provide wet land like services. Waste water reuse, however can have some problems: accumulation of contaminants and salts in soil in the case of waste water irrigation, and exposure of people to disease causing microbes.

Human excreta and water

The flush toilet has helped us to maintain our hygiene for more than hundred years. But it is highly wasteful as it uses a lot of drinkable water to carry away human urine and faeces. The problem is more acute in dry regions, and developing countries where building sewage systems is generally very expensive. Even in developed countries, discharging untreated waste into downstream ecosystems is known. It may be pointed out that while irrigated water is consumed by crops, water used to flush toilets is available for further use. Some possibilities are as below:

1. **Composting toilet using aerobic bacteria, which generates enough heat to destroy pathogens.** But it has not caught on in private households, partly because transporting the finished compost away calls for some management.
2. **A toilet design that keeps urine and faeces separate.** Since urine contains fewer pathogens so needs less efforts to disinfect than faeces. Furthermore, urine is a rich source of nitrogen and phosphorus, and it is less difficult to obtain them from separated urine and use than when urine is mixed with faeces. At the same time faeces not mixed with urine and water (dry faeces) is easier to process both chemically and biologically. Drying alone disinfects faeces.

Overall products include water, fertilizer and methane that can be used for cooking or to generate electricity (see in Vogel, 2012). Then, it is possible to make clean water from sewage. In California, engineers have gone a step forward. They use a battery of filters and purifiers to further clean sewage waste water which is clean enough to drink as per government standards. But there is psychological problem. People are not willing to accept water out of the sewer, therefore the ultra pure water is first pumped into the ground, and then piped to people's kitchen sinks. While Chinese have been using human faeces for fertilizing their crops for centuries, Indians are not as fecophilic as Chinese, they are not inclined to use water or fertilizers coming from sewers.

Microbial electrochemical technologies for conversion of waste into bioelectricity and useful chemicals

Organic waste materials have a considerable amount of energy which remains untapped. In fact, currently the treatment of organically rich wastes itself consumes energy – e.g., wastewater treatment consumes about 15 GW, or 3% of all electrical power produced in the United States (see in Logan and Rabacy, 2012). Wastewater (domestic, industrial, and animal wastewater together) in the US contains about 1.5×10^{11} kilowatt-hour (KWH) of potential energy, i.e., about 17GW of power. Then, agriculture generates a huge amount of biomass waste (1.34 billion tons of biomass in the US, enough to generate 600 GW of continuous power), which can be used to generate electricity without any adverse effect on the amount of food production. Furthermore, these biomass sources could be industrially used to produce chemicals.

In recently developed microbial electrochemical technologies, microbes catalyze electrochemical reactions, e.g., microbial fuel cells that generate electrical power. In electrochemical processes a fuel to provide electrons and an electron acceptor or oxidizer are required. In

electrochemical technologies, organic matter is the fuel and oxygen works as the primary oxidizer (however, in others it could be nitrate, sulphate and CO_2) for bacterial aerobic respiration. Bacteria (e.g., *Geobacter* and *Shewanella* spp.) work as exoeletrogens, that is they transfer electrons outside the cell to insoluble acceptors, such as iron and other metal oxides or to electrodes in bioelectrochemical systems (Logan and Rabacy, 2012).

The exoelectrogenic microorganisms in microbial fuel cells use any source of biodegradable organic matter (organic sources include acetate, ethane, glucose, hydrogen gas, and polymers, such as polysaccharides, proteins and cellulose) or inorganic matter in water that does not directly require oxygen as a part of degradation process to generate electric power. In microbial fuel cells, bacteria give rise to electrons to the anode, and protons into solution. This causes a negative anode potential. Generally oxygen in air works as an oxidizer at the cathode. For oxygen reduction in microbial fuel cells activated carbon is quite useful because it is inexpensive and renewable (produced from any biomass source).

Improving ways to denitrify waste water

About four decades ago, it was noticed by engineers that when ammonia of waste water was converted into nitrates, sometimes the amount of nitrates was lower than expected. The cause was discovered in 1990s by microbiologist, Gijs Kuenen of Delft University and his group of scientists. The waste water had a bacterium, called anammox (named for anaerobic ammonium oxidation), which converted ammonia into nitrogen gas in the absence of oxygen. This reaction was thought to be impossible before that. Unlike several other bacteria which divide every half an hour, anammox divides after every two weeks. Thus, to have its culture enough to clean waste water at an industrial scale, one might have to wait for years. Then, the problem was that the bacteria was not found in the wild. Later on anammox and its relations were found in waters of Black sea. Now it is regarded as the integral part of global nitrogen cycle, accounting for 50% of the annual nitrogen turnover. Now many believe that the use of anammox could bring about marked improvement in the removal of waste water nitrogen. It is realized that anammox should be grown in conjunction with another bacterium that changes ammonia to nitrite. Then anammox converts nitrite to nitrogen gas and water.

Exclusion of bacteria that make nitrate at normal temperature was another challenge to be encountered. This seems to be possible by maintaining larger granules. With the use of molecular techniques, the commercial use of anammox to remove waste water nitrogen is likely to be developed soon.

Taking a cue from tropical corals to lock CO_2 into concrete and cement

Carbon dioxide is the ultimate industrial waste product (Service, 2012), and world continues to struggle to mitigate it. Brent Constantz, a Californian entrepreneur and biomineralization expert thinks that it is possible to combine sea water with CO_2 to manufacture cement and concrete (manufacture of cement is a major source of CO_2 emission). The idea draws on the chemistry that corals use to build their hard shells (Service, 2012). Constantz had already tried to commercialize it, but results were mixed. Currently, at global level 2.5 billion tons of Portland cement is produced per year, and 12.5 billion tons of aggregate. Making a ton of concrete entails the generation of a ton of CO_2 . In all, the concrete production accounts for about 5% of all carbon dioxide emission at the global level which is huge, as 35 billion tons of CO_2 comes from burning fossil fuels alone.

In 2007 Constantz got the opportunity to turn his idea into a company and developed a pilot cement plant. The CO_2 rich flue gas was carried into the base of 33.5m high tower from the top of which seawater was poured. As the gas streamed up the tower it met sea water droplets, resulting in calcium and magnesium carbonates. The white powder thus formed was used to make cement. This

sequestered one-half tons of CO_2 per ton of concrete formed. This appeared to be an extraordinary success, but the reality was different. Ken Caldeira, a climate scientist at Stanford University argued that getting calcium and magnesium ions to bind with CO_2 to precipitate out water is not easy, as carbonate ions are stable in water only at pH higher than 9.5 (clearly higher than seawater pH of about 8.1). It was discovered that engineers operating the plant were adding sodium hydroxide to raise pH to 12 or 13. The higher the pH the more CO_2 dissolve into water and the more speeded up the precipitation of calcium and magnesium carbonates. Maintaining a high pH level has a heavy cost. However, Constantz did not give up. Some other companies began to use the Constantz idea to make cement and address the problem of CO_2 rise in atmosphere. Shell-forming organisms generally build up a microenvironment of high pH around them.

Constantz tried again taking a cue from role of the enzyme Carbonic Anhydrase (CA) in whisking CO_2 out of human blood. It is similar to corals which pull out CO_2 out of the seawater to build their hard shells. The CA raises the rate of the process 1 million-fold (i.e., without CA the process is very slow). It also seemed that by speeding up the rate the reaction could be managed at seawater pH, or with inexpensive bases such as coal ash, a waste product from coal-fired electric plants. A few other companies are also exploring CA to make seawater cement, and Constantz is not alone in his dream of converting billions of tons of CO_2 waste into useful products like cement. Several scientists think that people would be premature to dismiss Constantz.

A number of toxic chemicals are produced naturally by plants, animals and fungi, which could be poisonous. A toxicant is a substance having detrimental effects on living organisms and their biological processes. The science of toxicology relates to the study of nature of poisons and the effects of harmful chemicals on the living organisms. The Swiss scientist, Paracelsus (1499-1541) recognized that “everything is poison, yet nothing is poisonous”, depending upon the concentration of a substance. Paracelsus observed that a large concentration of a toxic substance was toxic, whereas in small amounts it was harmless. The Spanish physician, Mattieu Orfila (1787-1853) studied the toxicity of natural substances and recognized a close relationship between pathology and chemical content of the tissues (see Wright and Welbourne, 2002; Jolly et al., 2003). The experimental work of the French physiologist Claude Bernard on animals provided an analytical approach for the development of science of toxicology. With the advent of chemical revolution during nineteenth century, testing of new pharmaceutical products and identification of potentially harmful effects of the chemical was the main focus of toxicology. Rachel Carson’s book “*Silent Spring*” highlighted the dangers of indiscriminate use of chemicals upsetting the balance of nature (Carson, 1962).

In the late 1960s, the term ecotoxicology was used to the study of environmental toxicity caused by both natural and artificial substances on living organisms occurring in nature (Truhaut, 1975; see Jolly et al., 2003). Newman (1998) has defined ecotoxicology as “the science of contaminants and their effects on the constituents of biosphere including humans”. Ecotoxicology is the study of toxic effects of chemicals such as hazardous wastes, industrial chemicals, heavy metals, pesticides and fertilizers on the organism, population, community or ecosystems including human beings. It is a multi- step process to describe the sources, pathways and effects of chemicals that are harmful in the environment. The effects are seen at every level from the molecule to populations and to the biosphere. Environmental monitoring has enabled to quantify the harmful effects of toxic substances in the terrestrial and aquatic environments from anthropogenic sources. Risk assessment to environment from man-made toxins is important for undertaking suitable measures to minimize the harmful effects of contaminants.

22.1 TYPES OF ECOTOXICANTS

With the progress in synthetic chemistry, a variety of new chemicals are being produced for human use. Thousands of chemicals are manufactured worldwide on a large scale. A wide and increasing range of chemicals from industry, agriculture, medicine and other sources are contributing to the chemical load on natural ecosystems. These chemical substances are detrimental to the living organisms. There are more than 200, 000 man-made organic compounds and toxicity of majority of these is not known (Wright and Welbourne, 2002). The developments in the field of environmental toxicology and the analytical techniques has enabled to study the speciation of contaminants in different constituents of the environment (Jolly et al., 2003). For example, spectroscopic methods

have enabled to determine trace metals, whereas high performance liquid chromatography (HPLC) has been most widely used separation technique.

Broadly speaking, the chemical agents are of two types, i.e., hazardous and toxic. Hazardous chemicals generally include various flammable substances, explosives, sensitizers, irritants, acids and caustics. Toxic compounds affect individual health by affecting various physiological processes and pose a threat to ecosystem integrity. The commonly encountered environmental toxics which are of concern include pesticides (insecticides, herbicides, and fungicides etc.), heavy metals (mercury, lead and cadmium etc.), oil and petroleum products, domestic and industrial wastes and radioactive substances. Toxic compounds may be of general nature killing many cells or they may be extremely specific in their target and mode of action.

Xenobiotic substances are those that are foreign to a living system. The various kinds of inorganic and organic toxic substances, along with their harmful effects on human health, are listed in Tables 22.1 and 22.2. Some environmentally significant toxic substances are briefly described as follows:

Table 22.1. Toxicology of environmentally significant inorganic compounds (compiled from Manahan, 2000).

Inorganic elements and compounds	Toxicological effects
Heavy metals and toxic elements	Irritates eyes, upper respiratory system and lungs
Ozone	
Elemental white phosphorus	Systemic poison
Elemental fluorine (F ₂)	Toxic irritant and attacks skin
Bromine (Br ₂)	Irritating to respiratory tract eyes
Beryllium	Causes lung fibrosis and pneumonitis
Cadmium	Cause painful bone disease a kidney damage
Lead	Inhibition of synthesis of hemoglobin affects central and peripheral nervous system
Arsenic	Coagulates proteins and forms complexes with co-enzymes, asresenicosis manifest in skin discolouration and other symptoms
Elemental mercury	Cause insomnia, depression and irritability
Toxic inorganic compounds	Rapidly acting poisons
Cyanide	
Carbon monoxide	Causes disorders of respiration system and heart
Nitrogen oxides	Pulmonary edema, central nervous system depressant
Hydrogen fluoride	Fluorosis (bone abnormalities and mottled soft teeth)
Inorganic silica compounds	Silicosis results from exposure to silica dust
Asbestos	Asbestosis (a pneumonia condition)
Phosphine (PH ₃)	Depression of central nervous systems
Hydrogen sulfide	Headache, dizziness, affects central nervous system.
Sulfuric acid	Corrosive poison and dehydrogenate in concentrated liquid forms
Inorganic sulfur compounds	Irritants, corrosive toxic irritants, affects central nervous system
Monochloride, Tetrachloride	
Carbon disulfide	

Toxic elements and elemental forms

These include ozone, elemental white phosphorus, elemental halogens (Fluorine, F_2 and Bromine, Br_2), heavy metals (cadmium, lead, arsenic and elemental mercury) and beryllium. Lead is one the oldest and most prevalent poisons found in the earth's atmosphere. Lead inhalation produces serious health problems, ranging from lowered IQs to heart disease in human beings. Lead inhalation and poisoning can come from automotive exhausts and house paint. Other sources include: lead pipes, seams of food cans, ceramic glazes, batteries, and cosmetics, as well as lead mining, smelting, and coal combustion.

Toxic inorganic compounds

The hydrogen cyanide (HCN) and cyanide salts, which contain N^- ions, are rapidly acting poisons. Carbon monoxide (CO) is a common accidental poison. Oxides of nitrogen (NO and NO_2) are toxic. Hydrogen halides (HF, HCl, HBr and HI) are toxic gases.

Silica (SiO_2 , quartz) occurs in a variety of rocks. Silicosis results from human exposure to silica dust and is a common occupational disease. Asbestos is a group of fibrous silicate materials found in soil throughout the world. Asbestos is a natural fibrous mineral. Asbestos is used widely as a fire-retardant, and has good insulating qualities. It is used in many features of buildings, including flooring, walls, insulation, and more. Friable, or easily crumbled asbestos, can send nearly microscopic particles into the air. The inhalation of asbestos causes asbestosis (a degenerative lung disease). Lung cancer has been linked to the inhalation of microscopic asbestos fiber.

Phosphine, Tetra phosphorus decoxide is an inorganic phosphorus compound toxic to living organisms.

Hydrogen sulfide is a colourless and highly toxic foul smelling gas. Sulfuric acid is a severely corrosive poison and dehydrating agent in the concentrated liquid form. A number of inorganic sulfur compounds like tetra fluoride, Hexafluoride, monochloride, tetrachloride, carbon disulfide etc. are highly toxic.

Toxic Organic compounds

Some environmentally significant organic chemicals and their ecotoxicological effects are summarized in Table 22.2.

Radionuclides

Radionuclides are unstable isotopes of elements that undergo spontaneous decay and in the process emit one or other types of radiation. Radionuclides continue to disintegrate in sequence until a stable atom is formed. The radionuclides may be naturally occurring radio isotopes or produced with nuclear power generation, development and use of nuclear weapons or during their use for research and medical purpose. Uranium (U) and Zn are normally present in plants and soils at very low concentrations. The radioactive isotopes of C, H, P and S are present are also naturally present in plants and aquatic animals. Many other examples of radionuclides are isotopes of americium (Am), Cesium (Cs), krypton (Kr), plutonium (Pu), and thorium (Th). The isotopes that do not undergo radioactive decay are called stable isotopes. Some environmentally important isotopes, their source and principal mode of decay are listed in Table 22.3.

Table 22.2. Toxicology of environmentally significant organic compounds (compiled from Manahan, 2000).

Organic compounds	Toxicological effects
Hydrocarbons Alkane Hydrocarbons (gaseous methane, ethane, propane, n-Butane and isobutene)	Inflamed, dry, scaly skin
Alkene and alkyne Hydrocarbons, (ethylene, propylene and acetylene)	Phototoxic to plants, irritant to eyes and respiratory system.
Benzene and Aromatic Hydrocarbons	Skin irritant, fatigue, headache, Appetite loss
Toluene	Headache, Nausea, impaired coordination
Naphthalene	Exposure to naphthalene causes anemia and reduction in red blood cells.
Polycyclic Aromatic Hydrocarbon	Benzo [a] pyrene is the cause of cancer
Phenols Phenols, (O Cresol, m-cresol, p-cresol 2-Nitrophenol Pentachlorophenol)	Protoplasmic poison, cause malfunctioning of respiration and circulatory system
Aniline organonitrogen compound	Methemoglobinemia
Organonitrogen pesticides insecticides carbaryl, carbofuran, paraquate herbicide	Systemic Insecticide systematic poison
Organohalide Pesticides DDT Aldrin, Dieldrin, endosulfan etc.	Affect nervous system, loses of memory Act on brain, can cause cancer.
Chlorophenoxy herbicides (2,4-D, 2,4,5-T Agent orange) and Silvex	Nerve damage, brain damage
Organosulfur compounds (methane thiols, methyl sulfuric acid and dimethyl sulfate)	Headache, Nausea, carcinogenic
Organophosphorus compounds Parathion Malathion	Skin and respiratory disease
Organophosphorus military poison 'nerve gas' include Sarin and VX	A systemic poison to the central nervous systems

Table 22.3. Some environmentally important radionuclides.

Isotope	Half-life	Principal radiation	Main source
^{14}C	5760×10^3 years	Beta	Natural and Nuclear reactors
^{40}K	1.3×10^9 years	Beta	Natural
^{222}Rn	3.8 days	Alpha	Nuclear reactors
^{89}Sr	52 days	Beta	Nuclear reactors
^{90}Sr	28 years	Beta	Nuclear reactors
^{131}I	8 days	Beta, Gamma	Nuclear reactors
^{134}Cs	2 years	Beta, Gamma	Nuclear reactors
^{137}Cs	30 years	Beta, Gamma	Nuclear reactors
^{235}U	7.1×10^8 years	Alpha	Nuclear reactors
^{238}U	4.5×10^9 years	Alpha	Nuclear reactors

Sources of Radionuclides Some radionuclides like Uranium–238 series and Thorium–232 series occur naturally. Uranium–238 has 14 radioactive daughter products with half-life of microseconds to thousands of years. The inert gas Radon (^{222}Rn) is one of the daughter products produced during the decay of Uranium daughter products. Uranium mining and refining operations are responsible for concentrating naturally occurring radioactive materials. Carbon-14 and Potassium 40 are other naturally occurring radionuclides with very long half life. Uranium–235 is the fissionable nuclide and forms the basis for most reactor fuel (a purified Uranium oxide). Uranium dioxide pellets are fabricated into fuel elements or rods. The production and testing of nuclear weapons also enriches significant quantities of radioactive materials. It is known that testing of nuclear weapons is a source of low doses of radioactivity worldwide. The use of radio nuclides in research and medical treatment also produces large quantities of radioactive materials.

There are three types of radioactive wastes: low level, intermediate level and high level. Low level wastes consist of clothing and equipment used in hospitals, not generally regarded as a serious public health hazard. These radioactive wastes can be disposed of in shallow burial trenches or by ocean dumping. Intermediate – level wastes are a product of the nuclear energy industries and includes materials contaminated by plutonium. High level nuclear wastes are extremely hot and dangerously radioactive. They must be completely isolated from the environment for at least a thousand years. It accounts for 90% of the total radioactivity discharged by human technologies.

Radioactive decay Radioactivity of materials is generally measured by the rate of disintegration or decay. The rate of decay is described in terms of its half-life, which may be defined as the time required for one-half of a given amount of the isotope to decay to another form. Each radioisotope is characterized by its own half-life, the value range from milliseconds to million of years. For example, Uranium-235 has a half life of 700 million years. The rate of radioactive decay of ^{14}C has a half life of 5760 years, whereas radon-222 has a half life of 3.8 day. There are three major kinds of nuclear radiation: alpha particles (α), beta particles (β) and gamma (γ) rays. Alpha and beta particles have low penetration in air and human body. Gamma rays are of short wavelength and can penetrate deep.

The rate of radioactive decay is represented as Becquerel (Bq) which is defined as decay per second. This unit is named after a French scientist, Henri Becquerel, who discovered radioactive decay in 1896. The rate of radioactive decay can also be described in terms of the unit curie (Ci); being the rate of decay of one gram pure radium (^{226}Ra) or 3.7×10^{10} disintegrations per second. There is equivalence between curie and becquerel, and can be shown with the keep of equation

$$1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq}$$

22.2 DISTRIBUTION AND FATE OF TOXIC SUBSTANCES

There are many routes by which chemicals enter the bodies of living organisms. These include water, food, air, occupational exposure, food, medical drugs, and incidental and accidental exposure. Solubility primarily regulates the movement of toxic chemicals through the body of living organisms and the various components of the environment. Water soluble compounds move rapidly in both in aquatic and terrestrial systems and within the bodies of living organisms. Oil soluble toxics tend to penetrate into tissues and cell as these are not used during the metabolic processes. These partly soluble compounds are likely to accumulate and stored in lipid deposits.

Persistence

Many chemical substances are persistent for long times in the environment. Many commonly used chemicals, such as chlorofluorocarbon. PVC plastics, chlorinated hydrocarbon, pesticide, asbestos are resistant to degradation.

Chemical interactions

Many materials have additive effects when they occur together in exposure. Perhaps the greatest concern is the synergistic effects because in this interaction one substance multiplies the effects of another.

Acute and chronic toxicity

Acute effects are caused by a single exposure to the toxin and result in an immediate health crisis of some sort. Chronic effect can result from a single dose of a very toxic substance, or it can be the result of continuous or repeated sub-lethal exposure. Acute local exposure occurs at a specific location over a time period of few seconds or a few hours. Acute systemic toxicity occurs during a brief exposure to a single dose. The chronic effects of toxicity many span over a period of several years.

22.3 DOSE – RESPONSE RELATIONSHIP

The dose (the amount of chemical) determines the extent of an adverse effect of a toxic substance on an organism. The dose – response relationship is one of the key concepts of toxicology to examine the time dependent effects of exogenous and endogenous chemicals and physical agents. Dose is the amount of a toxicant usually per unit of body weight to which an organism is exposed. For example, a dose of lead to a bird might be 5 mg Pb/kg body weight. Sometimes dose is expressed as a dose per unit time (e.g., 5 mg lead/kg body weight/day). The dose response is a quantitative relationship which recognizes that the effect of a certain chemical depends upon the concentration of the toxic factor. Response is the effect upon an organism due to exposure to a toxicant. At lower dose, a few organisms show the response, whereas at high doses all the organisms exhibit response.

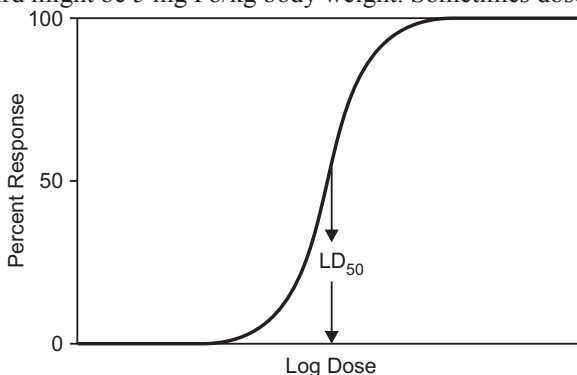


Fig. 22.1. Illustration of a dose-response curve.

The dose dependency can be shown with the help of dose-response curve (e.g., by giving different doses of a toxicant to a uniform population of test-animals). Then the cumulative percentage of deaths is plotted as a function of the log of the dose (Fig. 22.1). In general, the dose-response curves are S-shaped. The dose corresponding to the midpoint is the statistical estimate of the dose that would kill 50% of population and written as LD_{50} , which is a crude approximation of a chemical toxicity, LD_{50} is generally determined for new synthetic chemical as a way of estimating their toxic potential. The dose-response curve may differ among different kinds and strains of organisms, types of tissues and populations of cells. Acute toxicity oral Ld_{50} rates (mg/kg) of some chemical substances are given in Table 22.4.

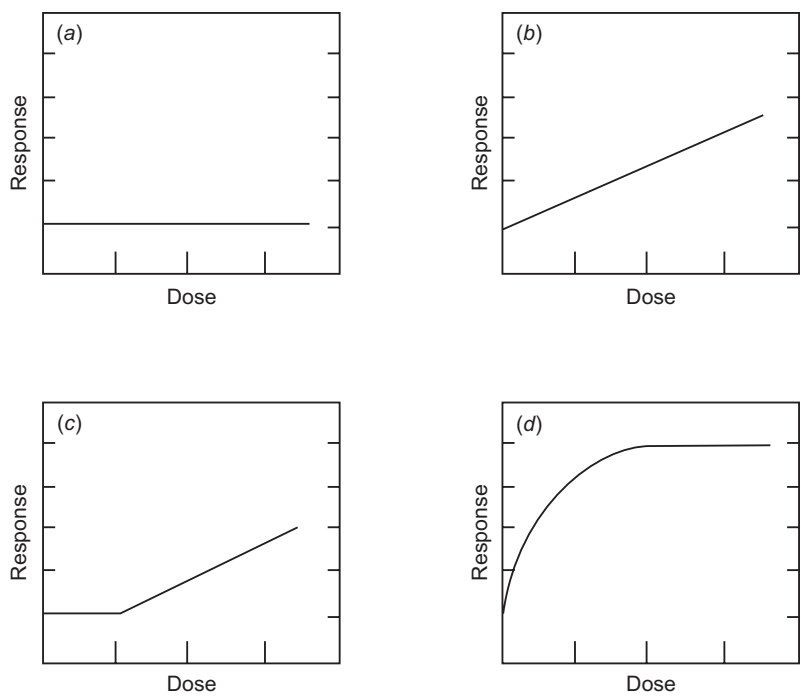


Fig. 22.2. Four general shapes for dose-response curves. Curve (a) reflects no response, curve (b) is a linear response, curve (c) is a threshold response, and curve (d) is an asymptotic response. (redrawn from Pierzynski et al., 2000).

Table 22.4. Acute toxicity oral Ld_{50} rates (mg/kg) of some chemical substances (based on Kornhauser et al., 1997).

Substance	Acute toxicity oral Ld_{50} rates (mg/kg)	Long-term effects
Benzene	930	Tumorigen Mutagen Reproductive effectors Primary irritant
DDT	80	Tumorigen Mutagen Reproductive effector
Phenol	310	Tumorigen Mutagen Reproductive effector Primary irritant
Parathion	2	Tumorigen Mutagen Reproductive effector

The dose response curve shows that some subjects are very sensitive to a particular poison when these are killed at a dose of LD_5 because of a hypersensitive response. In certain cases, organisms can survive even at a dose of LD_{95} and the response would be hyposensitive. The mid-range of dose-response curve is termed normal.

The ED-50 (effective dose 50%) is the dose that affects 50% of the population or observed subjects. The ED-50 is more refined concept than the LD_{50} and can be used to indicate responses such as reduced enzyme activity, decreased reproductive success or onset of r disease symptoms.

For non-carcinogenic chemicals, the assumption is that some threshold exists below which there is no toxic response or in other words at very low dose, there is no adverse effect (Fig. 22.2). On the other hand, carcinogens are non-threshold substances as exposure to any amount of carcinogen will cause some problem of cancer.

The absorbed dose of radiation by living tissues is represented in terms of gray (Gy) or radiation absorbed dose (rad). These units characterize the amount of energy that is deposited in living tissue as radiation passes through ($\text{Gy} = 100 \text{ rads}$). One rad is equivalent to 100 ergs of energy per gram. The level of biological damage differs for different kinds of radiation. To account for the different biological effectiveness, the concept of dose equivalent was introduced with the unit "rem" (roentgen equivalent man) or sievert ($\text{ISv} = 100 \text{ rem}$). For small doses, the mSv ($1000 \text{ mSv} = 1 \text{ Sv}$) are used. The whole body exposure to 250 mSv for a short period of time can have measurable effects on white blood cell counts, while 4000 mSv will result in death of half of the exposed population. A dental or chest X-ray film supplies 10 mSv per film.

Response or effects can very widely ranging from non-observable effects to temporary and reversible to chronic effects. The goal of a dose-response assessment is to obtain a mathematical relationship between the amount of a toxicant to which an organism is exposed and the risk of an adverse outcome from that dose. Toxic effects due to a combination of chemical could be :

Independent: Chemical produces different effects or has a different mode of action.

Additive: The combined effect of two or more chemicals is equal to the sum of the individual effects of the chemicals.

Synergistic: The effect is more than additive as two or more chemicals can react to create a greater effect than any one of them alone.

Antagonism: The effect is less than additive.

22.4 TOXIC EFFECTS

The effects of various types of toxic chemicals can be described under the following headings.

Teratogens Teratogens are chemicals that cause birth defects during embryonic growth and development. One of the well known examples of teratogens is thalidomide (a sedative compound). When this drug is used by pregnant women, it results in abnormal foetal development. It leads to a condition known as phocomelia in which a baby has seal like limbs, without any legs or arms. Alcohol is the most prevalent teratogen as it can lead to foetal alcohol syndrome during pregnancy in women. The biochemical mechanisms of teratogens include enzymatic inhibition, non availability of vitamins to foetus and alterations in the permeability of placental membranes.

Carcinogens: A carcinogen is a particular kind of toxin that causes cancer. Carcinogenesis is a multi step, multifactor process involved in the development of malignant tumours. Such substances affect the metabolic process in DNA. Many elements in the environment act as carcinogens. These include synthetic chemicals, sunlight, air pollutants, heavy metals, chemical pesticides, and cigarette smoke. Heavily industrialized areas often exhibit a higher cancer rate due to exposure to various types of chemicals.

Unburnt tobacco contains substances which are carcinogenic. Tobacco and smokeless tobacco products (e.g., chewing tobacco and oral snuff) have the risk of oral cancer among users (IARC, 1987). Tobacco smoke is an important carcinogen and has large impacts on human mortality worldwide by causing lung cancer. Carcinogens may be physical, chemical or biological agents. Biological agents are retroviruses, human-immunodeficiency virus-1 and virus-2 etc. A large number of identified carcinogenic agents are industrial chemicals. Exposures to various types of industrial chemicals that increase the risk of cancer are related with occupational health hazards. Some carcinogenic chemical agents include nitrosamines and polycyclic hydrocarbons. The role of xenobiotic chemicals in causing cancer is called chemical carcinogenic. Some physical phenomenon that are carcinogenic are ultraviolet and ionizing radiations.

Mutagens: Mutagens are agents that cause sudden change in the genetic structure of living organisms. A number of chemicals, pesticides and mutagens are suspected of promoting mutations in many groups of living organisms. Mutagenic damage is due to the absorption of high energy radiations, reaction with chemicals or reaction with an invading virus or other parasite (McGregor and Partensky, 1998). Chemicals reacting with DNA are often mutagens and frequently also carcinogens. A number of industrial chemicals and pesticides are suspected of promoting mutations in humans and need to be screened for mutagenic effects.

Estrogens: A number of xenobiotic substances can have adverse effects on animals or human reproductive systems by mimicking or interfering with the action of estrogens. A number of compounds including phthalates, alkyl phenols, organ chlorine compounds and polycyclic hydrocarbons are exoestrogens. Recent reports have indicated about the possibility of fall in sperm counts of man and of the rise in hormone-related cancers, such as breast cancer in women due to environmental hormones—estrogen (Arnold et al., 1996). Observations of the effects of environmental estrogens are paving the way to understand a larger phenomenon of environmental signaling. It is believed that environmental estrogens are the paradigm for a greater understanding of the health effects of external signals in the environment (McLachlan and Arnold, 1996).

The effects of chemical substances can occur at the organisms, community or ecosystem levels due to contaminant transfer between living organisms along with their toxicological effects at various levels (Fig. 22.3).

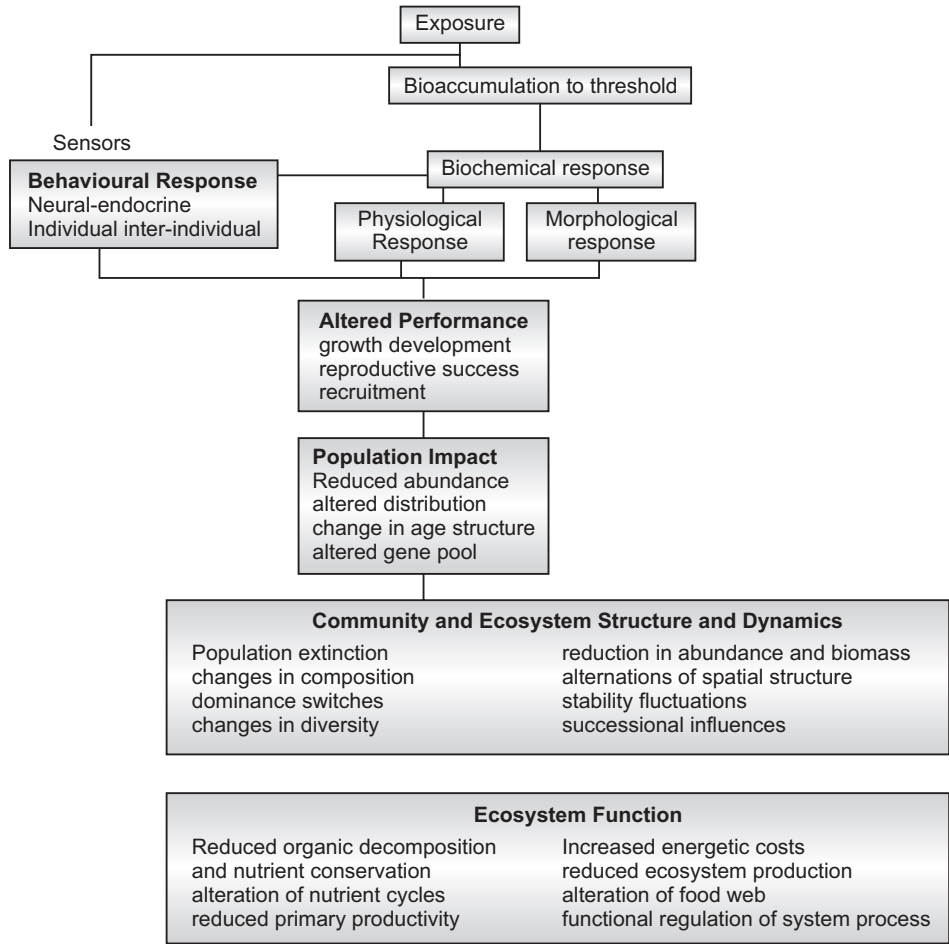


Fig. 22.3. The induced effects following exposure to toxic pollutants, having impacts from individual to ecosystem (based on Sheehan, 1984; see Boudo and Ribeyre, 1997).

Organism level – The responses are based on direct biological effects including mortality, behavioral changes, physiological impairment.

Population level – Change in birth rate and mortality rates, increased dispersal and local extinction. Competition between and within species for resources modifies the effects of chemicals on populations (De Laender and Janssen, 2013). It has been reported that the presence of a competitor at limiting food availability increases pesticide effects on the abundance of sensitive zooplankton species and reduces its recovery rate when pesticide exposure stops. In fact, competition reduces food availability to the least successful competitor, and effects of chemicals are typically more severe when food is scarce.

Community level – Changes in community structure and functional components and habitat destruction.

Ecosystem level – Changing species diversity and number of trophic level as well as production of biomass. The ecosystem functions are affected at concentrations that are higher than those where structural changes are observed (De Laender et al., 2011).

The chemicals, present in soil, air and water, come in contact with living organisms; and may cause mutagenic effects (affect at DNA Level) or may adversely affect the organisms through exposure.

The effects of contaminants on ecological systems can be influenced by a variety of environmental factors such as temperature, pH, salinity, water hardness and soil characteristics. Temperature affects the chemical activity of contaminant and physiological processes of organisms, which in turn affects toxicity. The pH within the soil, surface water or groundwater can affect the form, solubility and toxicity of some contaminants. Hardness of water refers to the amount of calcium, magnesium and ferric carbonate in water and it affects the toxicity of heavy metals. In terrestrial ecosystems, clay content and amount of organic matter commonly affect toxicity and bioavailability of elements.

Effects of toxicants on terrestrial environment

From an ecological point of view, toxic substances affect the structure and functioning of the ecosystem which are the basic protection goal in ecological risk assessment. The various types of toxic compounds such as heavy metals and pesticides affect plant growth in organisms like invertebrates, micro-organisms. Effects on plant biomass production, related to contamination of soil or air including deposition on plant surfaces. Plants are the source of food for the whole system (including humans) and have additional roles in terms of land protection, nutrient cycles, equilibrium of gases in the atmosphere, etc. The toxic compounds also affect above-ground and foliar invertebrates, which represent food for other organisms, and perform essential roles as pollinators, detritivores, saprophages, pest controller etc.

The effects on terrestrial vertebrates exposed to contaminated food, soil, air, water or surfaces, has obvious economic and/or social consequences. Poisoned birds and mammals probably constitute the highest social concern, while reproductive effects, although less evident, represent a higher ecological hazard. Accumulation of toxic compounds in food items and through the food chain represents an additional concern related to the consumption of this food by humans and domestic animals.

These concerns combine human and ecological interests. Direct human interests include managed species (cultivated plants and trees, bees, domestic animals) but also wild species essential as source for supplies (e.g., forest, pasture), landscape conservation (e.g., vegetation cover), or even for leisure (from gaming to bird-watching).

Effects on aquatic systems

The aquatic ecosystems including seas, estuaries, rivers and lakes receive pollutants and toxicants as a result of human activity from land based sources. In aquatic organisms, there is a greater affinity for gill, liver and muscles to accumulate lead, cadmium and mercury, respectively. It is well established that vertebrates and invertebrates are capable of accumulating heavy metals (Cd, Cu, Pb and Zn) from the aquatic-environment (Hellou et al., 1992). Mercury tends to accumulate in greatest concentration in calms. Pesticides affect crustaceans, fishes and mollusks. Echinoderms are very sensitive to oil pollution (Newton and McKenzie, 1995). It is known that heavy metals and pesticides can accumulate in aquatic organisms through food chain concentration.

Effect of Toxic chemicals on human health

Toxicants can harm human beings, when they are absorbed and reach the organs that are the target of their toxicity. This can happen through ingestion, inhalation, and absorption through the skin. In humans and other animals, toxicants usually affect one or more target organs such as the lungs, skin, or gastrointestinal tract. For example, if a person inhales asbestos fibers, the fibers get stuck in the airways of the lungs and irritate the lung lining, causing lung impairment over time. Dermatitis can result if the asbestos fibers irritate skin cells. Sometimes the toxicant crosses from the external environment of the lung, skin, or gastrointestinal tract into the bloodstream. Many parts of the human body are designed to absorb chemicals quickly and effectively. The stomach, intestines, and colon absorb nutrients from our diet. These organs easily absorb nutrients and other chemicals because of their large surface area, thin diffusion distance, and high blood flow. The lungs also are designed for rapid absorption. Chemicals that are inhaled are quickly absorbed into the bloodstream through the thin walls of the air sacs in the lungs. The skin protects the body from harmful agents in the environment. However, the skin is in direct contact with the environment. While the dense outer layer of skin cells is a good barrier to chemical absorption, it is not perfect, even when intact. When the skin is cut or abraded, it absorbs chemicals very rapidly.

Toxins once absorbed in the body are metabolized, transported or excreted. During the metabolic process, they have adverse biochemical effects and cause manifestation of poisoning. These processes are characterized by kinetic phase and a dynamic phase. In the kinetic phase, a toxic substance may undergo absorption, metabolism, temporary storage, distribution and excretion (Fig. 22.4). When a toxicant remains unchanged during the metabolic process, it is available for further biochemical interactions as an active parent compound. Some toxicant can be detoxified and excreted or converted to a nontoxic active metabolite.

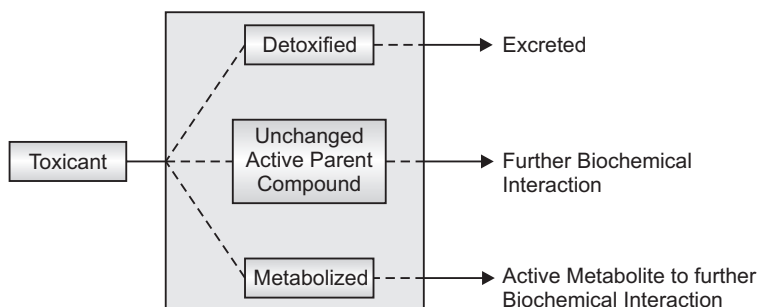


Fig. 22.4. Processes involving toxicants in the kinetic phase (redrawn from Manahan, 2000).

In the dynamic phase, a toxicant or toxic metabolic interacts in its cells, tissues, or organs in the body leading to various types of toxic effects. The dynamic phase is comprised of primary reaction with the receptor or target organ, biochemical effects (e.g., impairment of enzymatic reactions or alternations of cell membranes and interference with respiration etc.) and observable effects in terms of behavioural and physiological responses (Fig. 22.5). The immediate and readily observable manifestation of poisoning is changes in temperature, pulse rate, respiration rate and blood pressure. Central nervous systems poisoning may be manifested by convulsions, paralysis, hallucinations and ataxia (lack of coordination of voluntary movement of the body). The chronic response to toxicant exposure is mutations, cancer, birth defects and effects on immune system.

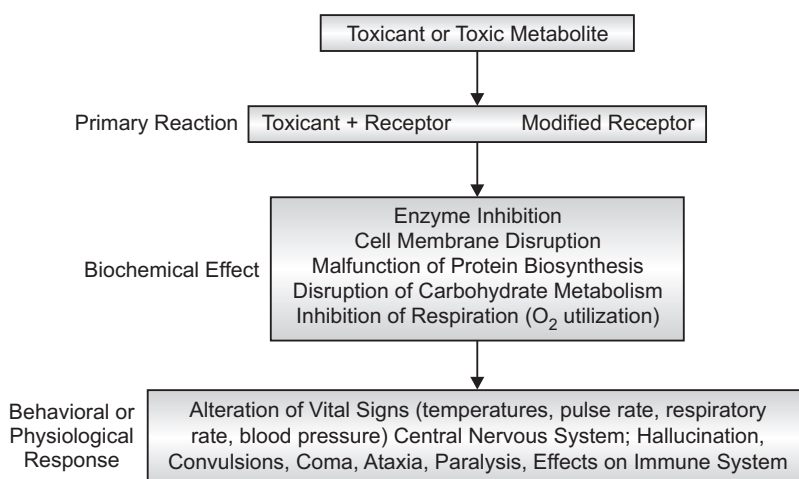


Fig. 22.5. The dynamic phase of toxicant action (adapted from Manahan, 2000).

Neurotoxins

The neurotoxin chemicals specifically affect nerve cells. For example, lead and mercury kill nerve cells and sometimes result is permanent neurological damage.

Effect of Lead

Exposure to lead can be extremely harmful because of damaging effect on brain, kidneys, reproductive system, and cardiovascular system. Lead poisoning, even in small doses, can also significantly reduce the IQs of young children. Still more adverse effects are: aggressive behavior, delinquency, and attention disorders in children, and increased blood pressure and hypertension in adult conditions that increase the risk of heart disease.

Effects of radionuclides

Environmental problems from radionuclide are due to mining and processing of uranium ores, production of weapons and nuclear power plants, nuclear weapon testing, nuclear power plant accidents and occurrence of radon gas in homes. The mining of uranium for nuclear fuel poses a great threat to environment as well as occupational health problems to workers. Radioactive dust and waste water from nuclear power plant can affect the ecosystems as a whole. Radioactive wastes are often disposed in open pits which contaminate soil, sediment and ground water.

There are two types of radiations – ionizing and non-ionizing. Non-ionizing radiations from the sun are vital to life on the planet. Exposure to ionizing radiations produces more complex biological effects. Alpha radiation has a low penetration power; beta particles can penetrate up to a depth of centimeter, whereas gamma radiations pass right through living organisms and can only be blocked by a thick concrete and lead.

Exposure to radioactive radiations causes leukemia, eye cataracts, various forms of cancer due to high degree exposure. The hazard is greatest when radionuclide enter the food chains. Radionuclides of cesium and strontium have a long half life. In the food chain, these are hazardous substances are taken up by grazing animals and humans. Some radionuclides are selectively concentrated within the body of living organisms. For example, iodine is selectively observed in the thyroid gland, whereas Cs, strontium (Sr) and P in bones. The internal exposure through food chain transfer or inhalation pathways is generally of great concern.

Effect of radioactive radon

In recent years, the occurrence of radioactive Radon (Rn) gas in homes has received considerable attention. Most of it comes from natural sources and is a daughter product of decay of ^{238}U . Radon is a gas that is produced by the breakdown, or radioactive decay, of uranium naturally found in most earth materials. It is a colourless, odorless, and tasteless gas that has been proven to be a large indoor air health threat, which can greatly increase the risk of lung cancer. When radon is inhaled, particles trapped in the lung continue to break down and release radiation that can damage lung tissue. Prolonged exposure to this gas significantly increases the chances of developing lung cancer. This danger increases as the duration of exposure increases and can be especially serious in the case of smokers.

Bioaccumulation and Biomagnifications

Bioaccumulation is the selective absorption and storage of great variety of molecule inside living cells. Biomagnification is the tendency of some chemicals to become increasingly concentrated at successive higher trophic levels of a food chain or food webs. Due to this process, toxic compounds which are low in concentrations in the environment can reach dangerous levels inside cells and tissues of living organisms. The effects of DDT, PCB, heavy metals and radioactive materials get magnified through the food webs. DDT or PCB's can accumulate through the food chain as these are fat soluble. Due to biomagnification, DDT also interferes with the reproduction of peregrine falcons, bald eagles, brown pelicans and other predatory birds by causing thinning of eggshells.

The radioactive dust that falls on the earth after atomic explosions is called radioactive fallout. Fallout radionuclides (especially ^{90}Sr and ^{137}Cs) have been and are being passed on to man via food chain. For example in Arctic and sub-Arctic of Alaska and Northern Finland, where reindeer and caribou meat is consumed, ^{137}Cs in man was higher during 1965. From the air, Cs-137 gets deposited in the reindeer moss, lichen that is a primary producer. This lichen is a primary winter food for the caribou; it also showed high concentration of ^{137}Cs in winter. ^{137}Cs also showed food chain concentration which doubled with each trophic levels in the food chain (Fig. 22.6).

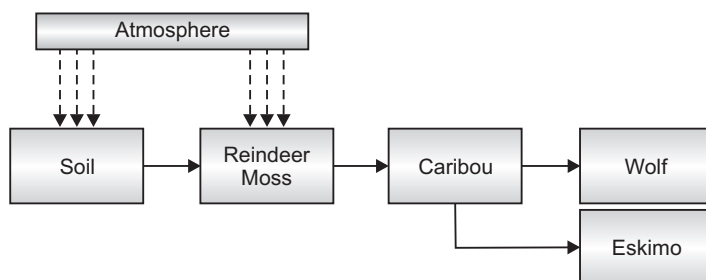


Fig. 22.6. $^{137}\text{Cesium}$ released into atmosphere by atomic bomb test gets deposited on the soil and reindeer moss, from where it gets magnified through the food chain.

The most-severe nuclear reactor accident occurred on April 16, 1986 at the Chernobyl Nuclear Power plant in the Ukraine. The accident of Chernobyl produced a nuclear cloud and deposited much of contamination over arctic Eurasia (Davidson et al., 1987). The isotopes of ^{131}I , ^{134}Cs , ^{137}Cs , and ^{90}Sr were emitted and people living by the nuclear plant were most adversely affected. Much of the nuclear fall out from the Chernobyl site occurred in Scandinavian countries, Poland and some Northern European countries. ^{137}Cs caused much exposure to the general public. There were significant effects of the fallout on the forage and agricultural crops. Radioactive contaminated lichens passed strontium and cesium to reindeer. In vertebrate food chains, strontium and cesium usually accumulate, particularly at higher levels. In arthropod food chains, radioactive elements like potassium, sodium and phosphorus accumulate; strontium and cesium do not accumulate. Because half-life of ^{137}Cs is about 30 years, its effects will persist for many years on the site which have been exposed to nuclear fall out.

22.5 ECOTOXICOLOGICAL TESTS

The environmental risks of toxic substances can be assessed and managed by measuring the concentration of these compounds in the environment or by testing the effects of chemicals on biota using toxicity tests. Ecotoxicological tests are widely practiced standards in the process of regulation of new chemicals used in industry, agriculture or plant protection. Therefore, reproducible test protocols are used which are simple and carried out on artificial systems under laboratory conditions. The effects are studied on mortality, reproduction, physiological performance or sometimes behaviour of organisms. Efforts are made to represent in these tests both terrestrial and aquatic systems, including different trophic levels, organisms and processes so as to analyze the implications for the functioning of natural ecosystems.

There are international efforts to adopt strategies globally at United Nations (UN) level, in the organizations like Economic Co-operation and Development (OECD). OECD provides the test guidelines, strategies and principles for testing the various properties of chemical (OECD, 1993). The method for environmental analysis is harmonized internationally in the International Standard Organisation (ISO) for water, sediment, soil and wastes. They include toxicity and biodegradation testing, microbiological quality analysis and ecological quality monitoring. European Union (EU) member countries use their own ISO standards and European standards (EN) for environmental analysis and chemical safety.

Some examples of tests according to international standardized methods for assessment of the acute and chronic toxicities are summarized in Table 22.5 on the basis of the report of the Technical Group of Basel convention. To find out whether the exposure is chronic or acute, effects of contaminants are segregated in to lethal and sub lethal efforts. Lethal effects refer to direct mortality of exposed individuals. Sub lethal refers to the adverse effects other than death for exposed individuals. Sub lethal effects include reduced rates of growth, behavioural changes and reproductive impairment.

Table 22.5. Types of Ecotoxicological tests (based on Technical Working Group of the Basel Convention, 2001).

Organism	Parameter for Test
Aquatic systems	
<i>Daphnia magna</i>	48 hr acute lethality
<i>Daphnia magna</i>	21 days lethality and reproduction
Algal systems	72 hr Growth inhibition
Terrestrial systems	
Higher plants	14 day, germination
Earthworm	14 day lethality
Collembola	Lethality and reproduction
Microbial processes	Short term processes of soil microflora, N-cycle

A number of microbiological toxicity tests have been designed using bacterial strains, mixed populations of environmental bacteria and some unicellular algae for both terrestrial and aquatic systems. The internationally harmonized tests by OECD for testing the effect of chemical, especially pesticides on microbes are given in Table 22.6. These tests are based on mineralization and inhibition of respiratory activity.

Table 22.6. Internationally harmonized tests used to testing the effects of chemicals on microbes. (reproduced from Jukka, 2002).

Test	Organism (s)	Principle	Standardization	Applications	Reference
Carbon mineralization test in soil	Indigenous microbes in soil	The degradation of added organic material in soil	OECD TG217	Pesticide and chemical testing	OECD, 2000
Nitrogen mineralization test in soil	Indigenous microbes in soil	The degradation and nitrogen mineralization from added organic material in soil	OECD TG216 ISO 14238	Pesticide testing chemical testing	OECD, 2000 ISO, 1997
Activated sludge respiration inhibition	Activated sludge	Oxygen consumption inhibition in activated sludge	OECD TG209	Chemicals	OECD, 1984

Worms of the family Enchytraedidae (Oligochaeta, Annelida) are small, inconspicuous animals, which can play an important role in terrestrial ecotoxicology (Rombke, 2004). These animals are highly sensitive to anthropogenic stress factors under field conditions. The Enchytraedidae reproduction Test (ERT), a chronic laboratory test using artificial soil, has been developed for testing of chemicals, as well as for soil quality assessment as per guidelines of ISO and OECD (Rombke, 2004).

22.6 ECOLOGICAL RISK ASSESSMENT

Risk assessment may be defined as the process of determining the probability and magnitude of adverse effects to organisms following exposure to toxic materials. Ecological evaluation of risks from contaminants can be beneficial in identifying exposure pathways for human population, such assessments are important for setting permissible limits of chemical toxins in the environment. For toxic chemicals, the risk assessment consists of four basis steps that are briefly explained as under:

Hazard identification

This process consists of defining the nature of likely cause of environmental health problems. For example, to identify a contaminant to understand the extent of toxic radiation produced from the nuclear power plant or from uranium mines. It also involves documenting the toxic effects on humans.

Exposure assessment

Ecological exposure occurs when a chemical reduces an organism and is in a term that is bioavailable. The lower the bioavailability, the lower the risk that will be posed by the contaminant. The bioavailability differs among species, and can change under different environmental conditions. This step mainly deals with determining the concentration of a contaminating agent in the environment as well as estimating the intensity, duration and frequency of human exposure to a particular toxin. Exposure to a particular toxin can occur via inhalation, ingestion of food or through the skin. Contaminant sources, release mechanisms, transport and transformation processes are all important aspects of exposure assessment.

Dose – response assessment

This step involves quantifying the adverse effects arising from exposure to a hazardous reagent or toxin based on the degree of exposure. Sometimes, it is difficult to show a relationship between dose of a toxic substance and a response being produced by it.

Risk characterization

During this step, potential impacts of a chemical are estimated based on severity of its effects and degree of exposure to a particular toxin. Once risks are characterized, regulatory options are evaluated in a process called risk management which attempts to integrate risk with technical, legal, political, social and economic issues.

New approaches for predictive risk assessment

Adverse outcome pathways: There is an increasing demand for ecological risk assessment to assess more chemicals, with greater speed and accuracy, and using fewer resources and experimental animals; there is need to move away from in vivo testing and make greater use of computational, molecular, and in vitro tools (Ankley et al., 2010; Villeneuve and Garcia-Reyero, 2013). Ankley et al. (2010) have proposed a conceptual construct, Adverse Outcome Pathway (AOP), that portrays the existing knowledge concerning the linkage between a direct molecular initiating event (e.g., a molecular interaction between a xenobiotic and a specific biomolecule) and an adverse outcome at a biological level of organization. AOPs thus represent a sequence of events that begins with a molecular initiating event, spans multiple levels of biological organization, and ends with an adverse outcome (such as effect on survival, development, and reproduction in individual organisms

and impacts on populations). Sequence of events may include: Toxicant (chemical properties) to macromolecular interactions (such as DNA binding, protein oxidation) to cellular responses (e.g., gene activation, protein production or depletion) to organ responses (e.g., altered physiology, tissue development or function) to organism responses (such as lethality, impaired reproduction) to population responses (such as recruitment, extinction). According to Villeneuve and Garcia-Reyero (2013), “if we identify a molecular initiating event in an AOP, we have defined a critical interaction that can then be modeled to develop quantitative structure–activity relationships (QSARs) that predict the likelihood of a chemical interacting with a specific target.” These authors suggest “transitioning from a chemical-by-chemical approach to regulatory testing and risk assessment, toward a more integrated system focused on cross-chemical understanding of pathway perturbation motifs i.e. recurring patterns of biological response associated with specific types of perturbation” for future.

Ecotoxicoproteomics: It is believed that proteins are the first functional stage directly or indirectly affected by toxicants; all living organisms respond to even the most subtle environmental changes through alterations in the expression of multiple genes and proteins. Proteomic profiles (proteome refers to the proteins encoded by a given genome) embody the link between effects at both the molecular level and the whole organism level. Lemos et al. (2010) have argued that ecotoxicoproteomics (environmental proteomics) is a powerful tool to spot early molecular events involved in toxicant responses, which are responsible for the adverse effects observed at higher levels of biological organization, thus contributing to elucidate the mode of action of stressors and to identify specific biomarkers. Stress-induced changes at the gene/protein expression level occur earlier in time and at lower doses than those causing alterations at the organismal level. For example, in humans the defence responses at sub-pathological doses have been successfully monitored via proteomics. Biomarkers are measures at the cellular, biochemical or molecular level that show whether a key organism has been exposed to a stressor and also the magnitude of the organism response to the contaminant. Single proteins that are consistently up- or down-regulated after toxicant exposure could be identified and correlated to a specific toxicant or “biomarker pattern” (“protein expression signatures”) is determined without the need of protein identification. Lemos et al. (2010) state “The “biomarker approach” assumes that effects seen at higher levels of biological organization (populations and communities) are the consequence of the sum of effects on individuals, which resulted from impacts at the cellular and molecular levels. This makes these lower biological level and early responding endpoints potential tools for rapid diagnoses of toxicity”.

Trait-based assessment: Population vulnerability to toxicants are often extrapolated from data available from a given laboratory model species assuming that the model species is representative of a larger group, because it is not feasible to test all species and chemical combinations. Rubach et al. (2011) have proposed a framework for ecotoxicological assessment using species traits rather than taxonomic species. These authors feel that information about how a given species’ traits might contribute to its susceptibility may eventually permit better prediction of the potential for adverse effects to a broader range of species. Among several traits, they found, through literature review, strong trait-linkage to population vulnerability and ecotoxicological processes for the following traits: Food choice, Active avoidance, Size of organism, Assimilation efficiency, Toxicant elimination ability, Generation time, Life span, Clutch size, Survival to reproduction.

Part – V

GLOBAL CHANGE

- ♦ GLOBAL ENVIRONMENTAL CHANGE
- ♦ BIOLOGICAL INVASION: PROCESS, IMPACTS AND PREVENTION

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Global Environmental Change

Earth's environment has been undergoing significant changes due to increasing human population and its activities, especially during the last century. Some of the important changes that are occurring in the global environment relate to increased resource consumption; depletion of fossil fuel reserves that were generated over several hundred million of years ago in the geological past; and the large scale changes in land cover and land use. Excessive use of terrestrial productivity (>40%) and accessible fresh water (54%), to meet the demands of food, fodder and fibre of the ever increasing human population is having a large impact on the natural systems. So much so that another Geological Era, called the Anthropocene, is suggested to have begun since 1850 with the following features: (i) fossil fuel reserves created over hundreds of millions of years are being exhausted within a few human generations, (ii) about 50% of the land surface has been transformed, (iii) more nitrogen is fixed synthetically and applied as fertilizers than is fixed naturally, (iv) more than half of all accessible freshwater is directly or indirectly used by humans, and the underground water resources are being fast depleted, (v) concentrations of greenhouse gases have substantially increased, and (vi) extinction rates are increasing alarmingly (IGBP, 2001). In this Era, the human-caused transformations are pushing the Earth System outside of its normal operating domain, the biophysical and human components of the Earth System are indeed tightly coupled. The Earth is considered a system comprising interacting physical, chemical, and biological global-scale cycles and energy fluxes that result in conditions necessary for life; it is a system that life itself helps to control.

Among the most significant changes brought about by human activities are the increase in concentration of carbon dioxide and other greenhouse gases in the troposphere, and depletion of stratospheric ozone. The naturally occurring greenhouse gases in the lower atmosphere regulate the global temperature, whereas ozone in the stratosphere filters out the harmful ultraviolet radiations reaching the Earth's surface. Thus, any change of ozone in the stratosphere or gain in greenhouse gases in the lower atmosphere may lead to global climate change, having its large scale impact on climate and weather conditions, biodiversity and food production as well as human health. However, it must be understood that global change is more than climate change. For example, direct effects of increasing CO₂ concentration, depletion of stratospheric ozone due to release of chlorofluorocarbons, impact of nitrogen deposition on biodiversity, spread of invasive plants, may not have a direct influence on the climate system.

The UN Intergovernmental Panel on Climate Change (IPCC), in its Fourth Assessment Report (IPCC, 2007), has provided a strong scientific evidence of global warming and the necessity for urgent and comprehensive reductions in greenhouse gas emissions. Rising temperature and frequent floods, droughts and storms are impacting the lives of millions of peoples. The recent research on global climate change has shown continuing ice-loss from ice sheets and glaciers, yielded new

predictions of accelerating sea level rise, underlined the implications of declining rainforests due to climate change in the tropics, as well as the future trends for major tropical storms and monsoon seasons.

In this chapter, we aim to discuss about the sources and the increasing concentration of greenhouse gases, and the effects of climate change on terrestrial and marine ecosystems, and the concerns about the stratospheric ozone depletion and its effects on organisms.

23.1 GREENHOUSE EFFECT

Short wave radiation from the sun (0.2 to $4\ \mu\text{m}$) reaches the earth's surface unhindered but the outgoing long wave radiation radiated from earth (4.0 to $100\ \mu\text{m}$) is partially retained by the atmosphere. The energy that is radiated from earth either passes through the atmosphere to outer space through the region of atmospheric window (approximately 10% of the energy). The atmosphere absorbs a substantial portion of the long wave radiation emitted by the earth (approximately 90% of energy radiated from earth), and radiates energy back to it. This downward flux of the long wave radiation, called greenhouse flux, keeps the earth warm. Naturally occurring greenhouse gases in the atmosphere absorb these longer wave radiations, forming a blanket over the earth and controlling the escape of heat from the earth's surface to outer space and keep our planet warm and hospitable (Fig. 23.1). Although this analogy to a greenhouse is loose, it points out that the situation described above is comparable to the glass-enclosed greenhouse that is transparent to incoming radiation, but retains some of the outgoing longer wavelength radiation; allowing suitable temperatures for the growth of plants inside the greenhouse.

Thus the phenomenon of trapping and reradiating of heat by greenhouse gases in the atmosphere is referred to as *greenhouse effect*. The 'natural greenhouse effect' keeps Earth's surface warm and at roughly constant mean global temperature of $+15^\circ\text{C}$. The gases that are most active in absorbing the radiant heat energy are known as greenhouse gases or radiatively active gases. Nitrogen and oxygen are most abundant gases in the atmosphere accounting for about 78% and 20.9% of the total gaseous volume but they are not greenhouse gases. The remaining 1% is composed of argon, water vapour, carbon dioxide, ozone and other trace gases. Water vapour, CO_2 , N_2O , CH_4 , CFCs and ozone occur in minute quantities in the atmosphere but play a critical role in maintaining even temperatures on the earth. In the absence of greenhouse gases in the atmosphere, temperatures on this planet would be much lower than they are today, about -20°C .

The capacity of the atmosphere to keep the earth warm depends upon the concentration of greenhouse gases. The excessive increase in concentrations of greenhouse gases in the atmosphere will retain more infrared radiation, resulting in '*enhanced greenhouse effect*'. The consequent increase in the global mean temperature is referred to as *global warming*.

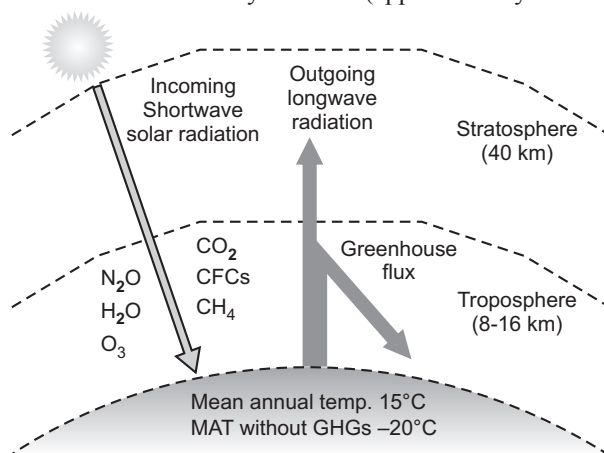


Fig. 23.1. The greenhouse effect. The atmosphere is transparent to the incoming short wave radiations (0.2 to $4\ \mu\text{m}$). It is translucent to the long wave infrared radiation (8 to $50\ \mu\text{m}$). Some Long wave radiations are absorbed by the naturally occurring greenhouse gases in the atmosphere maintaining earth's temperature at 15°C .

23.2 GREENHOUSE GASES

Water vapour and carbon dioxide (CO_2) are the most important naturally occurring greenhouse gases present in the atmosphere. Methane (CH_4), nitrous oxide (N_2O), chlorofluorocarbons (CFCs), and tropospheric ozone (O_3) are the other greenhouse gases, of these CFCs are solely man-made. Greenhouse gases reduce the emission of thermal radiation to space, thereby affecting the energy budget everywhere around the globe, i.e., warming the surface. Among the different agents of climate change, water vapour, anthropogenic greenhouse gases and aerosols have larger roles (IPCC 2001, 2007). The abundance of greenhouse gases in the atmosphere is increasing due to industrial development, exponential increase in energy production and use, destruction of forests, land use changes globally and intensification of agriculture. The change in land use is a complex phenomenon having its regional impact on climate and weather conditions. For example, the conversion of a forest to grazing land or a cropland is a local activity, causing loss of carbon to the atmosphere. However, the loss of soil organic matter has global effects on the carbon cycle and the concentration of greenhouse

gases. Similarly, the biomass burning associated with agricultural practices in South-East Asia is an important regional activity, but has global environmental impacts due the production of carbon monoxide, which is linked to increasing methane concentration in the troposphere. The temporal pattern of atmospheric concentration of two greenhouse gases and inferred temperature over the last four glacial-interglacial cycles is shown in Fig. 23.2. The atmospheric concentration of carbon dioxide, methane and nitrous oxide over the past 1000 years indicates an exponential growth in the abundance of these gases since around the year 1850 (Fig. 23.3).

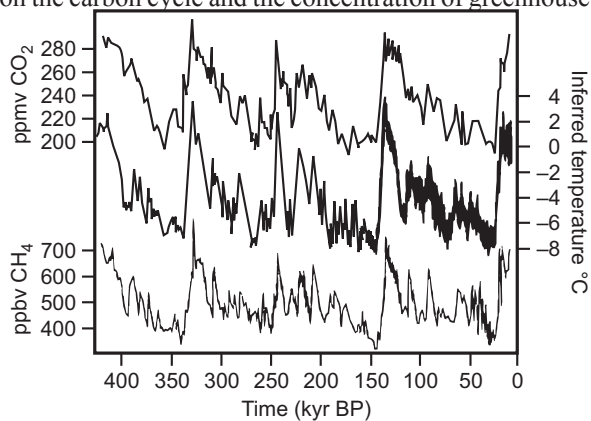


Fig. 23.2. Atmospheric concentrations of carbon dioxide, methane and inferred temperature over the last four glacial-interglacial cycles. (from Vostoc Ice Cores, after Steffen et al., 2004)

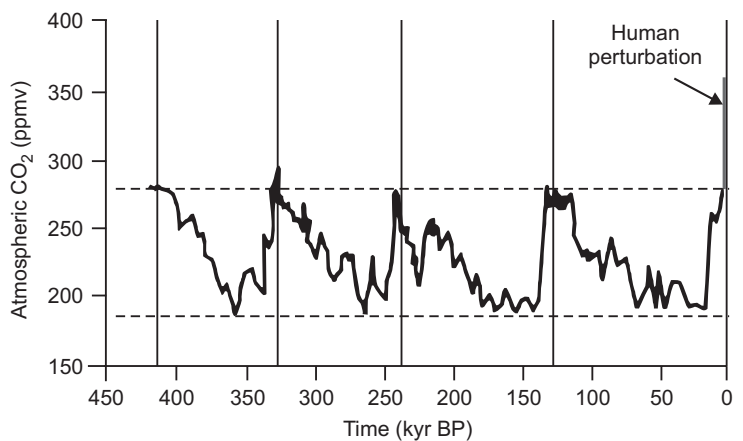


Fig. 23.3. Atmospheric carbon dioxide concentration over the past 420,000 years, with the recent human perturbation superimposed. (updated from Petit et al., 1999).

The sources and trends of increase in the concentration of greenhouse gases in the atmosphere are summarized Table 23.1. The trends in the change of greenhouse gases are described below.

Table 23.1. The increase in the concentrations of greenhouse gases in the atmosphere as affected by human activities. (based on T.J. Blasing, *cdiac.ornl.gov/pns/current_ghg.html*).

Greenhouse gas	Pre-industrial ~ 1750 AD Tropospheric Concentration	Recent Tropospheric Concentration ¹	Atmospheric life time ² (years)
Carbon dioxide (CO ₂)	280 ppm	392.6* ppm	~100
Methane (CH ₄)	700 ppb	1874 ³ /1758 ³ ppb	12
Nitrous oxide (N ₂ O)	270 ppb	324 ³ /323 ³ ppb	114
CFC-11 (trichlorofluoromethane) (CCl ₃ F)	zero	238 ³ /236 ³ ppt	45
CFC-12 (CCl ₂ F ₂)	zero	531 ³ /529 ³ ppt	100
HCFC-22 (CHClF ₂)	zero	226 ³ /203 ³ ppt	12

ppm – parts per million; ppb – parts per billion; ppt – parts per trillion

*CO₂ averages for year 2012 (www.esrl.noaa.gov/gmd/ccgg/trend/)

1. Concentration for other greenhouse gases for 2011 (<http://cdiac.ornl.gov/ndps/alegage.html>)
2. The atmospheric lifetime (IPCC, 2007)
3. The first value in a cell represents Mace Head, Ireland, a mid-latitude Northern-Hemisphere site, and the second value represents Cape Grim, Tasmania, a mid-latitude Southern-Hemisphere site.

Carbon dioxide

The increase in CO₂ concentration in the atmosphere has been largely the result of fossil fuel burning, deforestation and change in land use. The principal cause of increase in atmospheric carbon dioxide in recent years has been combustion of fossil fuels (coal, oil, etc.) which releases about 21 billion tons of CO₂ in the atmosphere annually. The other known major source of carbon dioxide is deforestation which is estimated to account for 15-30% of annual carbon dioxide emissions. Every year about 0.8 per cent of the total global forest area is cleared for other use. Forest clearing releases 0.3-2.6 Gt (1 Gt = 10⁹ t) of carbon annually as CO₂.

There is a clear evidence that the composition of greenhouse gases in the lower atmosphere has changed during the last century (IPCC, 2001) and over the long time scales of glacial and interglacial periods (Barnola et al., 1987; Petit et al., 1999). Large ice cores have been extracted from Vostok and Siple sites (Antarctica) and Summit site in Greenland. The air trapped from the past in the glacial ice has been analyzed for CO₂. Glacial ice is a time capsule as it traps the air bubbles of the air at the time of its formation. The analysis of the air trapped in the ice showed that CO₂ concentration has varied between 180 to 280 ppm (280 CO₂ molecules per million molecules of air) in the past 160,000 to 420,000 years during the glacial and interglacial periods (Barnola et al., 1987; Petit et al., 1999). The CO₂ concentration in the atmosphere varied in parallel with temperatures; the concentration being higher when temperatures were high (Fig. 23.2). The concentrations of CO₂ were about 25% lower during the glacial periods than during the preindustrial interglacial periods.

Since 1958, CO₂ is being measured at the Mauna Loa Observatory in Hawaii. Mauna Loa is an extinct volcano, situated at high altitude, is an ideal place for measuring exact concentration of CO₂ in the atmosphere. It has the longest continuous record of the atmospheric CO₂ concentration and its

seasonal and annual trends (Keeling and Whorf, 2002) (Fig. 23.4). There is a seasonal fluctuation in CO_2 of about 5 ppm during the year. In the Northern Hemisphere, CO_2 concentration peaks in summer just before plant growth begins and falls off when the growing season is at its peak (Fig. 23.4, inset). This results from the net uptake of CO_2 by vegetation during the growing season and its net release during the less favourable season, representing the seasonal variations in vegetation metabolism. In recent times, atmospheric concentration of carbon dioxide has increased from pre industrial levels of 280 ppm to about 392.2 in 2011 (Table 23.1, and Fig. 23.4). For the first time in human history, the concentration of climate-warming carbon dioxide in the atmosphere has passed the milestone level of 400 parts per million (ppm) on 9th May in 2013. Thus, the CO_2 levels in the air currently are higher than at any time in the past history of Earth, indicating a pronounced human impact (Fig. 23.5).

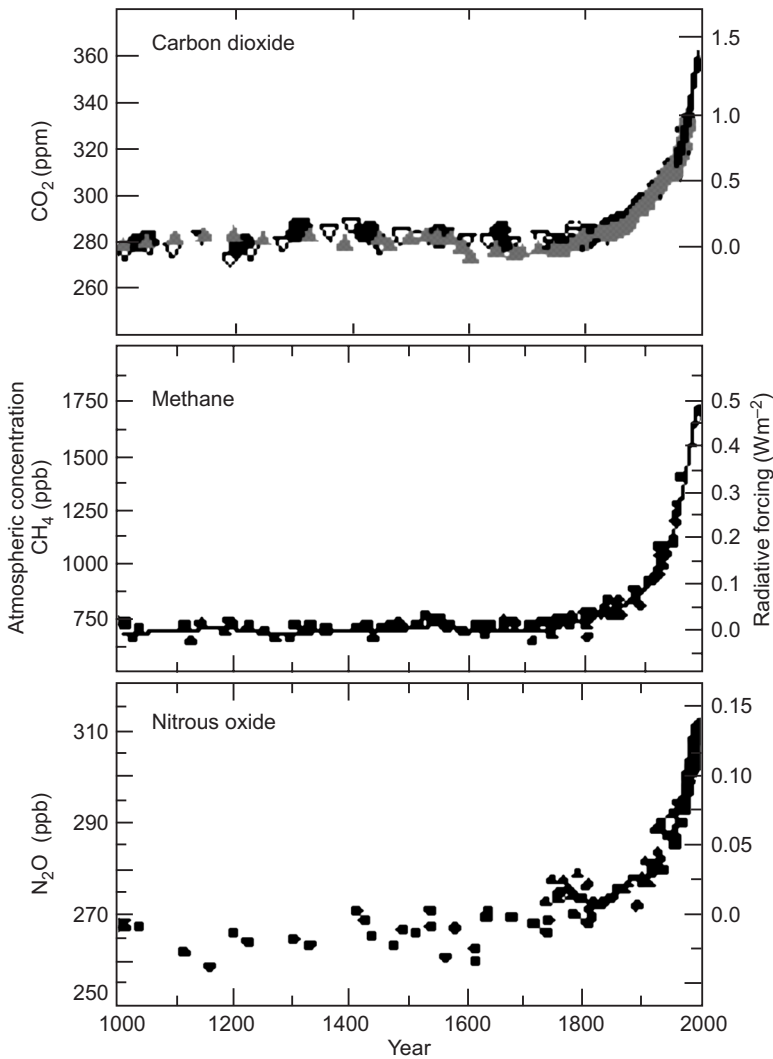


Fig. 23.4. Changes in atmospheric concentrations of carbon dioxide, methane, and nitrous oxide over the past 1000 yrs (from IPCC, 2001).

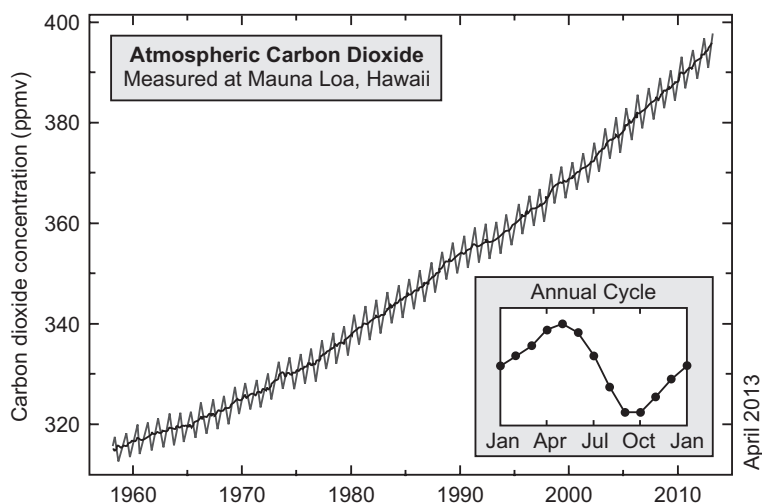


Fig. 23.5. Atmospheric carbon dioxide measured at Mauna Loa Observatory Hawaii from 1958 to 2012.

Methane

The Methane concentration has more than doubled (1874/1758 ppb, global average 1813 ppb) since its concentration of 740 ppb during the preindustrial times (Table 23.1, and Fig. 23.6). CH_4 now is the most abundant non- CO_2 GHG in the atmosphere today; its present concentration being 2.5 times higher than that observed in ice cores dated to AD 1000–1750 and is higher than that observed throughout the existing ice-core record, which spans the past 800,000 yr, the increase in concentration from pre-industrial times implies anthropogenic emissions of $340 \text{ TgCH}_4 \text{ yr}^{-1}$ or $8.5 \text{ GtCO}_2\text{-eq yr}^{-1}$ (Montzka et al., 2011). Methane is very effective in causing warming as it absorbs infrared radiation of different wavelength than CO_2 . On a molecule for molecule basis, methane is 21 times more powerful than CO_2 at trapping heat in the atmosphere. During the past 100 years, the concentration of CH_4 has increased 100% and over the geological time scale its concentration has shown variations corresponding to that of atmospheric CO_2 .

Emission sources of CH_4 are mostly biological and associated with human activities like landfills, livestock management (enteric fermentation in ruminants), and anaerobic respiration in soils of wetlands ($150\text{--}180 \text{ TgCH}_4 \text{ yr}^{-1}$), and rice agriculture, and biomass (wood fuel) burning. About 80% of atmospheric methane has biological sources. Agriculture and fossil fuel exploitation together account for about $230 \text{ TgCH}_4 \text{ yr}^{-1}$ ($5.8 \text{ GtCO}_2\text{-eq yr}^{-1}$), or two-thirds of all human-derived CH_4 emissions (Montzka et al., 2011). Fresh water wetlands produce CH_4 due to incomplete decomposition of organic matter in oxygen poor environments, whereas termites produce CH_4 during the digestion of cellulose. Methane is also released due to anaerobic activity of methanogens in flooded rice fields, bogs, tundra, wetlands and lakes. Increase in methane (about 20%) is also related to production and use of fossil fuels. It leaks from oil and gas exploration, recovery and distribution of methane and also released during coal mining. The temporal trend of CH_4 is depicted in Fig. 23.6.

Chlorofluorocarbons

CFCs (CFC-11, CFC-12 and CFC-113) are highly stable and man-made gaseous compounds of carbon and halogens, which are considered to be most dangerous to planet's stability. The tropospheric concentrations of chlorofluorocarbons have increased dramatically over the past 50 years (Walker et al., 2000; Tarasova et al., 2012) (Table 23.1). While CFCs and most halons are

decreasing, hydrochlorofluorocarbons (HCFCs) and hydrofluorocarbons (HFCs), which are also potent greenhouse gases, are increasing at rapid rates, although they are still low in abundance; SF₆ (Sulphur hexafluoride) mixing ratio has increased to double that observed in the mid-1990s (Fig. 23.6). Major sources of CFCs are leaking air conditioners, refrigeration units and cleaning of electronic components), production of plastic foams and propellants in aerosol spray cans. The CFCs in the atmosphere has a long residence time of 12 to 100 years in the atmosphere. On a molecule for molecule basis, CFCs are 12000-15000 times more effective than CO₂ at trapping infrared radiation. It absorbs radiation in the region of atmospheric window, where the naturally occurring greenhouse gases do not absorb the infrared radiation.

Nitrous Oxide (N₂O)

Nitrous oxide is increasing in the atmosphere and causes as much as 5 to 6% of the anthropogenic greenhouse effect. The annual rate of increase of N₂O is 0.2 to 0.3% per year and has a long residence time of 114 yrs in the atmosphere. The main sources of N₂O are agricultural activity biomass burning and industrial processes. Anthropogenic N₂O emissions at present account for 6.7 TgNyr⁻¹ (3.1 GtCO₂-eq yr⁻¹), or about 40% of all N₂O emissions, and arise primarily from agricultural activities (1.9 GtCO₂-eq yr⁻¹) and industrial processes including fossil fuel combustion (0.8 GtCO₂-eq yr⁻¹). Natural N₂O sources are predominantly terrestrial (75%), and these emissions are highest in the tropics (Montzka et al., 2011). N₂O is produced during nylon production, burning of nitrogen rich fuels, denitrification of nitrogen rich fertilizers in soil and nitrate-contaminated ground water. Deforestation and biomass burning are also significant sources of N₂O to the atmosphere. N₂O is 310 times more powerful than CO₂ at trapping heat in the atmosphere. Nitrous oxide is now the third most important contributor to radiative forcing of long-lived greenhouse gases, recently surpassing CFC-12, and its impact on climate integrated over 100 years is 298 times greater than equal emissions of carbon dioxide (CO₂) (Tarasova et al., 2012). The trend of N₂O increase is shown in Fig. 23.6.

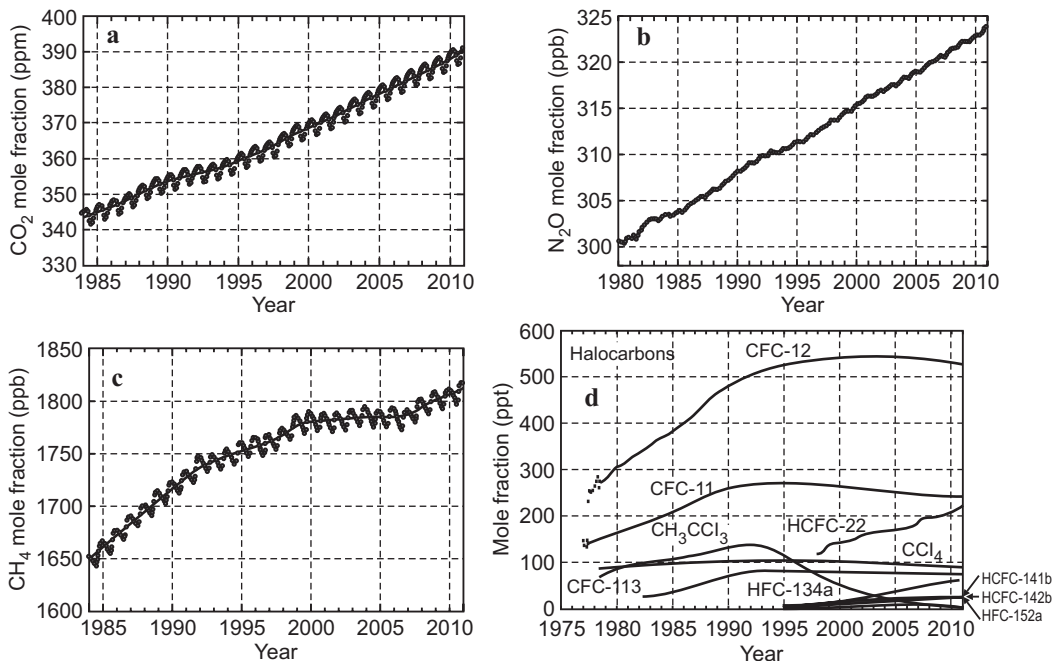


Fig. 23.6. The globally averaged concentrations of (a) carbon dioxide, (b) nitrous oxide, (c) methane, and (d) halocarbons, in atmosphere. Source: WMO, 2012. (www.wmo.int/gaw).

Black carbon and ozone in the lower atmosphere

Black carbon and ozone in the lower atmosphere are harmful air pollutants that have substantial regional and global climate impacts. They disturb tropical rainfall and regional circulation patterns such as the Asian monsoon, affecting the livelihoods of millions of people. 3. Black carbon’s darkening of snow and ice surfaces increases their absorption of sunlight, which, along with atmospheric heating, exacerbates melting of snow and ice around the world, including in the Arctic, the Himalaya and other glaciated and snow-covered regions. This affects the water cycle and increases risks of flooding. Tropospheric ozone and black carbon (BC) contribute to both degraded air quality and global warming (Shindell et al., 2012).

Radiative Forcing and Warming Potential of Greenhouse Gases

Radiative forcing is the change in average net radiation at the top of the troposphere by changes the balance between incoming and outgoing solar radiations. A positive radiative forcing leads to warming of the Earth’s surface and the lower atmosphere. An increase of CO₂, CH₄, N₂O, and CFCs in the atmosphere leads to retaining of more infrared radiations and acts like a positive radiative forcing. The NOAA Annual Greenhouse Gas Index shows that from 1990 to 2010, radiative forcing by long-lived greenhouse gases increased by 29%, with CO₂ accounting for nearly 80% of this increase. Radiative forcing of N₂O exceeded that of CFC-12, making N₂O the third most important long-lived greenhouse gas (Tarasova et al., 2012).

Carbon dioxide contributes about 47% of total warming; the contribution of CH₄ and CFCs being 14% and 20%, respectively (Fig. 23.7). The relative contribution of different sources to greenhouse gases (GHG) is shown in Fig. 23.8. The contribution of developing nations to GHG emissions is less than half of that emitted by the developed countries, the latter have been responsible for more than 60% of the stock of GHGs (Fig. 23.9). The fastest growth in CO₂ emission until 2025 is projected to occur in developing countries, whose collective emissions are projected to rise 84 per cent (compared to 35 per cent growth for industrialized countries) (Fig. 23.10). By 2025, the developing country share of global emissions is projected to be approximately 55% (compared to 48 per cent in 2000). Among developing countries, the largest relative growth until 2025 is forecast for Mexico (124 per cent) and for China (118 per cent). China is projected to surpass the United States as the world’s largest emitter.

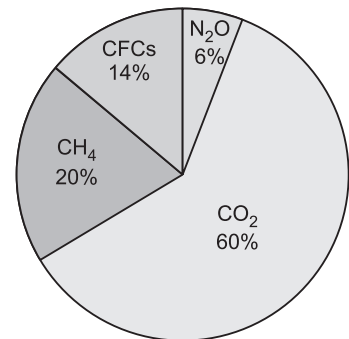


Fig. 23.7. The relative contribution of different greenhouse gases to global warming.

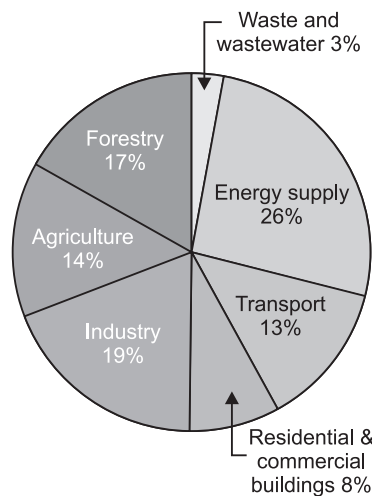


Fig. 23.8. The relative contribution of various human activities to the increase of greenhouse gases in the atmosphere from different sources. I PCC (2007). *Climate Change 2007: Mitigation of Climate Change*.

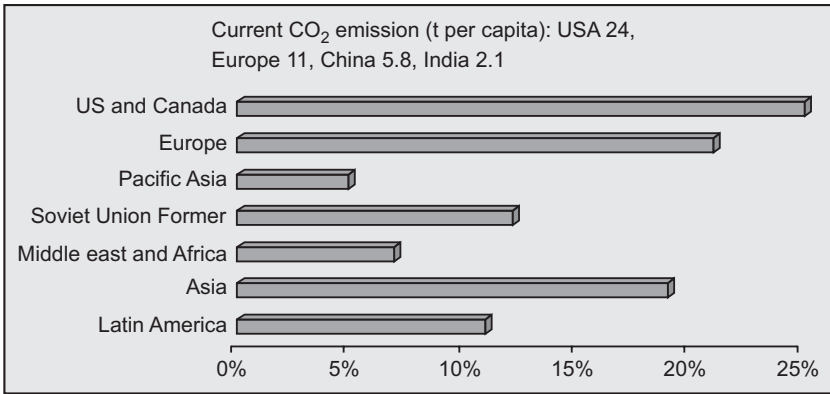


Fig. 23.9. Regional contribution to cumulative CO₂ emission for industrial sources and landuse change (1900-1999) (based on Marland et al., 2000).

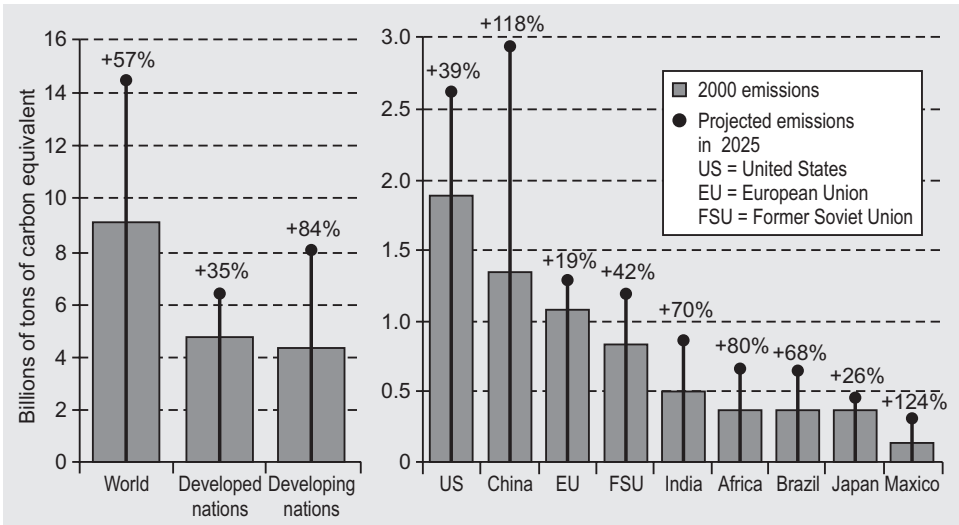


Fig. 23.10. Projected greenhouse-gas emissions in 2025: World Resources Institute, Navigating the numbers (http://pdf.wri.org/navigating_numbers.pdf).

Role of Water Vapour and Aerosols

Water vapour in the atmosphere is the most important greenhouse gas as it results in two-third warming of the earth's surface (IPCC, 2001; Kump, 2002). The water vapour absorbs radiation of about the same wavelength as CO₂. The amount of water vapour in the atmosphere, cloud cover and albedo (reflectivity) of clouds, and the rainfall intensity and duration have significant effects on the climatic processes.

Aerosols have both direct (reflect and absorb solar radiation) and indirect effects (modify cloud properties) on the climate system and the hydrological cycle. Anthropogenic aerosols include urban haze, smoke from vegetation fires, dust and oceanic aerosols. Most aerosols have a regional distribution, exhibit a high degree of spatial and temporal variation and have a short residence time of about a week in the atmosphere. Urban haze has its origin from urban and industrial pollution, industry, and cooking and fire places, generally from highly populated areas. Smoke from vegetation fire is dominated by fine organic particles with varying amount of light absorbing black carbon. Forest fires produce large amount of smoke, whereas burning of grasses emits large quantity of black carbon.

Dust is emitted from dry regions of deserts and from the disturbed soils in different regions; the dust emission being highest in regions of Sahara, East Asia and The Saudi Arabian deserts. Oceanic aerosols are composed of coarse salt particles and fine sulfate particles produced from oceanic emissions. The net result of aerosols is the cooling of the earth's surface (IPCC, 2001) However, more information is needed for understanding the effects of aerosols on clouds and precipitation both on regional and global scale and its impact on the climate system (Kaufman et al., 2002).

The carbon footprint

The carbon footprint is a measure of the impact of human activities on the environment in terms of the amount of greenhouse gases produced, and measured in units of carbon dioxide equivalent. According to The Carbon Trust, “A carbon footprint measures the total greenhouse gas emissions caused directly and indirectly by a person, organisation, event or product” (<http://www.carbontrust.com/>). A carbon footprint is measured in tonnes of carbon dioxide equivalent (tCO₂-e). The carbon dioxide equivalent (CO₂-e) allows the different greenhouse gases to be compared on a like-for-like basis relative to one unit of CO₂. CO₂-e is calculated by multiplying the emissions of each of the Kyoto Protocol six greenhouse gases, i.e., Carbon dioxide (CO₂), Methane (CH₄), Nitrous oxide (N₂O), Hydrofluorocarbons (HFCs), Perfluorocarbons (PFCs) and Sulphur hexafluoride (SF₆) by its global warming potential (GWP). Throughout a product's lifetime, or lifecycle, different greenhouse gases may be emitted, such as methane and nitrous oxide, each with a greater or lesser ability to trap heat in the atmosphere. These differences are accounted for by calculating the global warming potential of each gas in units of carbon dioxide equivalents (CO₂-e), giving carbon footprints a single unit for easy comparison.

Countries vary widely in their contribution to the emissions that are driving up atmospheric stocks of greenhouse gases. With 15% of world population, rich countries account for almost half of emissions of CO₂. The carbon footprint of the United States is five times that of China and over 15 times that of India (UNDP, 2008). In Ethiopia, the average per capita carbon footprint is 0.1 tonnes of CO₂, compared with 20 tonnes in Canada. Differences in the depth of carbon footprints are linked to the history of industrial development in different countries. But, they also reflect the large ‘carbon debt’ accumulated by rich countries—a debt rooted in the over-exploitation of the Earth's atmosphere. The distribution of current emissions points to an inverse relationship between climate change risk and responsibility. The carbon footprint of the poorest one billion people on the planet is estimated at around 3 per cent of the world's total footprint, but these the poorest billion people are highly exposed to climate change threats for which they carry negligible responsibility.

23.3 IMPACT OF GLOBAL WARMING AND CLIMATE CHANGE

Box 23.1. Potential Effects of Global Warming

- Elevated temperatures of the biosphere
 - melting of polar ice
 - increase in sea level (flooding of major cities)
 - increase of methane from permafrost
- Weather extremes
 - more rainfall during shorter periods
 - more evaporation and soil moisture deficiencies
- Ecosystem disruption
 - stress and death of vegetation
 - migration of species
- Human Health
 - heat stress
 - migration of disease vectors

It is generally accepted that the increase in concentration of greenhouse gases in the atmosphere would have profound impact on the climate of earth, including global warming, disturbed precipitation pattern and increased storm intensities (Box 23.1). The predicted global warming would cause rise in sea level and shift in the range of distribution and phenology of organisms.

The changes in the distribution of species are likely to have marked effect on species diversity and ecosystem functioning of coral reefs, mountain ecosystems, coastal wetlands, tundra, boreal and temperate forests. There is possibility of increasing invasion by exotic weed species as well as decoupling of species interactions in ecosystems, which may affect community structure and ecosystem functioning. These changes would have serious impacts on the sustainability of several ecosystems and the services they provide to human society.

Weather and Climate

The increase in concentration of carbon dioxide and other greenhouse gases causes the earth's average air temperature to increase. Averaged over all land and ocean surfaces, temperatures have warmed roughly 1.33°F (0.74°C) over the last century (IPCC, 2007). There is variability of global temperature over various time intervals (such as year to year or between decades) as well as the long-term increase since 1880 (Fig. 23.11). More than half of this warming—about 0.72°F (0.4°C)—has occurred since 1979. Because oceans tend to warm and cool more slowly than land areas, continents have warmed the most (about 1.26°F or 0.7°C since 1979), especially over the Northern Hemisphere. There are slight differences in global records between groups at NCDC, NASA, and the University of East Anglia. Each group calculates global temperature year by year, using slightly different techniques. However, analyses from all three groups point to the decade between 2000 and 2009 as the hottest since modern records began more than a century ago. Temperatures in the 2010s have been running slightly warmer still. For the next two decades, a warming of about 0.2°C per decade is projected for a range of SRES emission scenarios (IPCC, 2007). Even if the concentrations of all greenhouse gases and aerosols had been kept constant at year 2000 levels, a further warming of about 0.1°C per decade would be expected.

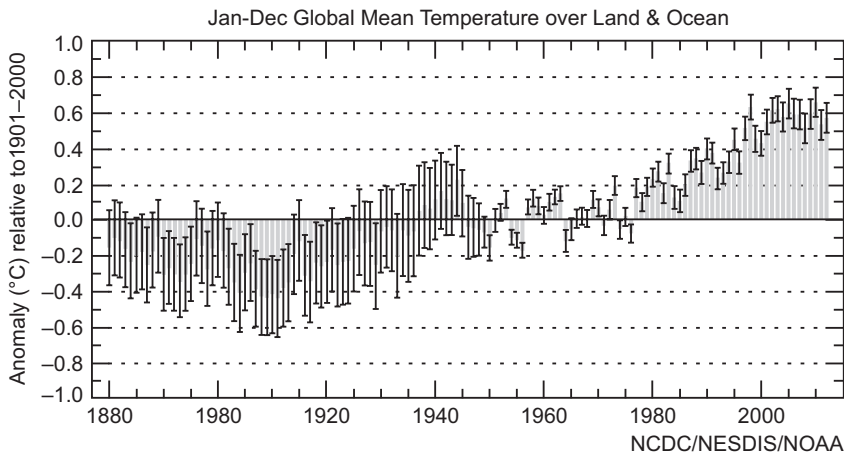


Fig. 23.11. Global average temperature since 1880. This graph from NOAA shows the annual trend in average global air temperature in degrees Celsius, through December 2012. For each year, the range of uncertainty is indicated by the vertical bars. (adapted from NOAA's *National Climatic Data Center*).

This increase in the average temperature may not be uniform and vary over different places on earth. For example, the tundra biomes may experience greatest increase in temperature and in comparison rise in tropical rain forest zone is likely to be much small.

Climate changes from late 1960s have caused an increase in precipitation (0.5 to 1%) during autumn and winter mostly in the northern hemisphere. In the subtropics, precipitation has decreased by 0.3% per decade. The IPCC report predicts that globally average precipitation, particularly winter precipitation will increase in future in the northern hemisphere. Further the frequency of extreme events (e.g., drought, floods, etc.) will substantially increase. Rapid glacial melt can cause serious flood damage in heavily populated regions of Himalaya, whereas increase in floods from Asian monsoons has implications for the low lying areas.

The most worrisome feature is the increase in the frequency of extreme events, e.g. devastating floods (high rainfall events) or droughts, heatwaves, or heavy snowfall, etc.

Melting of Polar Ice Caps and Glaciers

Perennial surface ice on land is comprised of ice sheet, ice shelf, glacier, and ice cap (Box 23.2). The earth's ice cover acts as a protective mirror, reflects a large portion of the sun's energy that goes back into space and keeps the planet cool. Glaciers in mountain areas and Polar ice caps are particularly sensitive to temperature shifts. Accelerated shrinking of mountain glaciers on every continent, rapid reduction of Arctic sea-ice, disintegration of floating ice shelves, and increased melt rates of Earth's three Ice Sheets—Greenland, West Antarctic, and East Antarctic—provide compelling evidence of our changing climate.

Box 23.2. Perennial surface ice on land

Ice sheet: A mass of land ice of continental size, and thick enough to cover the underlying bedrock topography. Its shape is mainly determined by the dynamics of its outward flow. There are only two continental ice sheets in the modern world, on Greenland and Antarctica; during glacial periods there were others too.

Ice shelf: A thick, floating slab of freshwater ice extending from the coast, nourished by land ice. Nearly all ice shelves are located in Antarctica.

Glacier: A mass of surface-ice on land which flows downhill under gravity. In general, a glacier is formed and maintained by accumulation of snow at high altitudes, balanced by melting at low altitudes or discharge into lakes or the sea. The term 'glacier' is used for different types of surface land ice masses including outlet glaciers, valley glaciers, mountain glaciers and glacierets.

Ice cap: Dome-shaped ice mass with radial flow, usually covering the underlying topography.

Source: WGMS 2008, IPCC 2007

It is estimated that the arctic ice sheet has shrunk by about 6% between 1978 and 1996. The arctic Greenland ice sheet (containing about 8% of world's ice) has thinned more than a meter per year on the average since 1993 along its southern and eastern regions. During 2013, surface melting of the snow and ice of the Greenland Ice Sheet had a slightly late start, but quickly spread over a significant area, extending over more than 20% of the ice sheet in early June, 2013 and reaching above 2,000 meters (6,500 feet) elevation in some areas (*National Snow and Ice Data Center National*, <http://nsidc.org/greenland-today/Snow>). Surface melting on the ice sheet reduces the reflectivity of the snow, and this change is measured by albedo, the ratio of the amount of light reflected from a surface to the amount of incoming light. Satellite data have been used to track this change (Stroeve et al., 2013, in revision), showing that albedo has dropped by as much as 0.3 (30% reflectivity) in some areas. Arctic sea ice reached its maximum extent for the year on March 15, 2013 at 15.13 million square kilometres.

Box 23.3. The fate of glaciers in Himalayas and surrounding regions

The information reported by the Intergovernmental Panel on Climate Change (IPCC) that all glaciers in the Himalayas could disappear by 2035 (the IPCC retracted its statement subsequently), created a lot of controversy and lowered its reputation as the statement was not based on a proper research publication. Himalayan glaciers are not going to disappear by 2035, but by and large they are shrinking.

The Hindukush Himalayas (Himalayas plus Hindukush region located in and around Afghanistan) has more glaciers (>54000 glaciers with total area >61,000 km², and total snow area 7,60,000 km²) than any other region on the planet outside the two polar caps. Referred to as the water tower and the third pole, the Himalayan region is warming more (0.01-0.03°C per year or 1-3°C in century) than global average rate, and the rate of temperature rise within the region increases with altitude, the highest being between 4800-6200 m altitudes. With further increase in altitude, warming rate stabilizes or declines slightly.

An analysis of 75 glaciers of the HKH region shows that (see in Singh et al., 2011) (i) 63 glaciers are shrinking, and 12 which are from Karakoram region (located at the western side of the Himalayan arc) are slightly growing. As for methods, there were 11 mass balance-based studies, and 14 area -based studies, and all the 25 showed glacier shrinkage. All Karakoram glacier studies were based on glacier tip measurements (the degree of confidence is the highest in mass balance measurement and the least in of tip measurement).

Recently Yao et al. (2012) have investigated status of glaciers over the past 30 years: glacier retreat of 82 glaciers, area reduction of 7,090 glaciers and mass-balance change of 15 glaciers. They found most intensive shrinkage in Himalayas (excluding Karakoram). The shrinkage in glaciers decreased from Himalayas to Eastern Pamir, where the glacier retreats and area reduction were least, and mass balance was positive. They explain that the increase in precipitation due to westerlies was the main cause of positive mass balance in the glaciers of Pamir regions, while the rapid shrinkage in Himalayas was because of decrease in precipitation. Of course, warming was a common factor to both. The point of contention is why estimates differ across the glaciers and measurements of different investigators. There are several methodological complexities and environmental variables 1. As referred to earlier, mass balance data are scarce. 2. Only a very small number of 54000 glaciers that occur in Himalayas have been investigated. 3. It takes 4-5 days to reach a glacier site from road head, therefore, generally field observations are on smaller glaciers located at lower altitudes where temperature are warmer, hence the higher rates of glacier melt. 4. Measurements using satellites cover much larger numbers of glaciers (measurements on 7100 glaciers were taken for 7 years by Gravity Recovery and Climate Experiment, GRACE satellite mission) indicate that majority of glaciers have been shrinking in the past 30 years, but at the rates about one-tenth as fast as previously estimated (i.e., 50 gigatonnes of ice a year). 5. But scientists using remote sensing techniques may differ in interpretation of data. In another study based on the remote sensed data, the of ice loss rates were found to be 12 gigatonnes ice a year from 2003 to 2008. The lower rates of glacier thinning as estimated by the GRAE satellite could be because they cannot tell the difference between ice and liquid water. Consequently, a glacier lake could be mistaken a glacier mass. In Tibet region, as an example the area of glacial lakes has increased by about 26% since the 1970s. 6. The climate and characters of glaciers vary considerably along the Himalayan arc. The Eastern and Central Himalayan Glaciers generally gain in mass from summer-monsoon snow fall (summer accumulation type), while in the west, such as Karakoram region glaciers gain in mass during winters. During 1970-2010 precipitation in much of the Himalayas which is under the influence of Indian monsoon has decreased, while in Tibet Plateau and Pamir region, which is under the influence of westerlies has somewhat increased 7. In some glaciers their tips are covered with

Contd...

debris because terrain is steep and avalanche activity is strong. 8. The extent of warming varies with altitude, therefore, glaciers located in altitudinal belt where warming is high, melt more.

To conclude, it may be pointed out that all models project substantial mass losses in coming decades for most glaciers in the Himalayan region, but complete region-wide disappearance of glaciers is unlikely to occur even by 2100.

How important is glacier melt water with regard to river discharge? The current thinking is that the contribution of glacier melt is small, less than 10% of total discharge in river Ganga, as its basin's precipitation is dominated by monsoon rain. The proportional contribution of glacier melt is relatively higher in the Indus which is away from the monsoon influence. But remember these interpretations are based inadequate on sampling.

* based mainly on Singh et al. (2011), Bolch et al. (2012), Kaab et al. (2012) and Yao et al. (2012)

In Antarctica, there is massive ice cover (2.3 kilometres thick), which represents 91% of earth's ice; the land-based ice sheets may not show much changes. Although average sea ice extents around Antarctica have been increasing over the last several years, land-based ice sheets are melting at fast rates.

In some parts of the continent, the Antarctic ice is melting just as rapidly as the ice in the Arctic Circle. In fact, the Antarctic Peninsula was one of the earliest climate change "hot spots", where rapid warming caused enormous segments of glaciers to break off into the ocean. In a recent study in the journal *Nature Geoscience*, new ice core data from James Ross Island, located at the northern tip of the peninsula, shows that warming on the peninsula has reached its highest rate in 1,000 years (Abram et al., 2013). The ice melt in peninsula has increased 10-fold over the last 600 years, with the largest increases occurring in the last half century. Summer melting is now at a level that is unprecedented over the past 1,000 years (Abram et al., 2013).

Outside the poles, most ice melt has occurred in mountain and sub polar glaciers. Large scale ice melt would raise sea level and flood the coastal areas. As mountain glaciers recede, large regions that rely on the glacial run-off for water supply could experience severe water shortage. For example, in Northern India, a large population depends on the tributaries of the glacier-fed Indus and Ganga Rivers for irrigation and drinking water. But as the glaciers melt in the Himalayan region (Box 23.3), these rivers are expected to swell initially which would be followed by very low levels of water in the subsequent years. The volume of summit glacier on Kilimanjaro Mountain in Africa has decreased by 81% from the year 1912 to 2000. The permafrost (permanently frozen soil) in the tundra and boreal forests has also started melting; having effects on vegetation and infrastructure.

Many glaciers and ice caps in polar, temperate, and high altitude tropical regions are experiencing retreat and volume loss. Diminishing glacier and ice cap volumes not only dominate current sea-level rise but also threaten the well-being of approximately one-sixth of the world's population who depend on glacier ice and seasonal snow for their water resources during dry seasons (WGMS, 2008). Most icecaps and glaciers in the mountains of tropical Africa are expected to disappear by 2030. Loss of permanent ice from Africa's mountains will have profound effects on surrounding ecosystems, as well as the hydrology and temperature regime of Africa's unique highland cold-water lakes. In the Himalaya particularly, glacier loss increases the threat of glacier lake outburst floods.

Sea Level Changes

Sea levels are rising as a result of global warming. This rise is predominantly a consequence of three factors—thermal expansion of warming ocean water, addition of new water from the ice sheets of Greenland and Antarctica and from glaciers and ice caps, and the addition of water from land surface runoff. All three potential sources are being affected by human activities. Since at least

the 19th century, sea-level changes have been measured directly by tide gauge records and, since the 1990s, by satellite altimetry. Sea-level changes over longer periods of time, thousands to millions of years, are inferred from geologic evidence (Rohling et al., 2009). The average rate of global mean sea-level rise over the 20th century was approximately 1.7 millimetres (mm) per year as recorded by tide gauges (Church and White, 2011). Measurements from satellite altimetry data, beginning in the early 1990s, suggest that this rate has increased to approximately 3.2 mm/yr (Ablain et al., 2009, Church and White, 2011). Satellite measurements reveal important variations in global mean sea level between and within ocean basins. As the present warming trend is expected to continue, global mean sea level will continue to rise.

Four scenarios of global mean SLR ranging from 0.2 metres, to 2.0 metres by 2100 have been identified (Parris et al., 2012) (Fig. 23.12). These scenarios provide a set of plausible trajectories of global mean SLR for use in assessing vulnerability, impacts, and adaptation strategies. None of these scenarios should be used in isolation, and without consultation with experts and coastal managers.

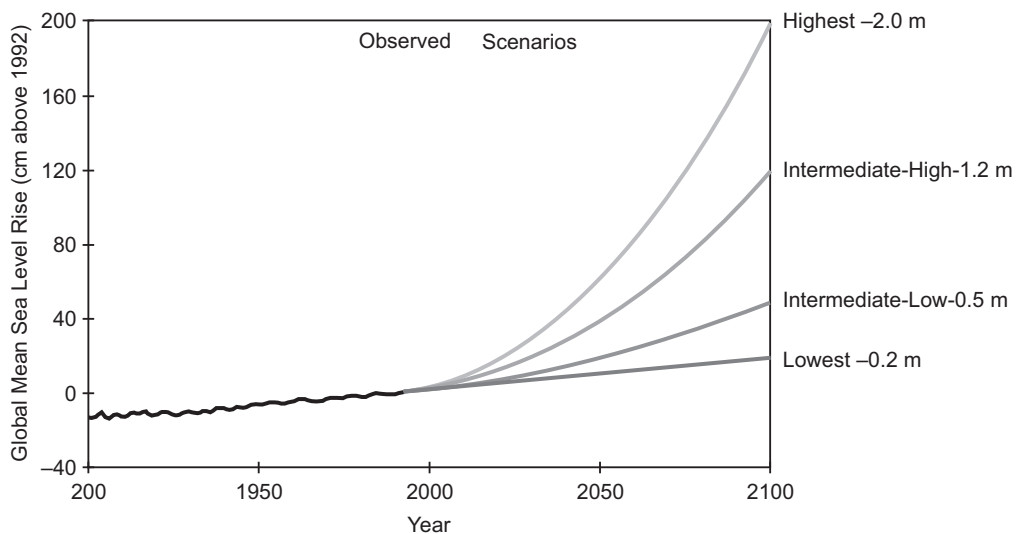


Fig. 23.12. *Global mean sea level rise scenarios.* Present Mean Sea Level (MSL) for the US coasts is determined from the National Tidal Datum Epoch (NTDE) provided by NOAA. The NTDE is calculated using tide gauge observations from 1983 – 2001. Therefore, we use 1992, the mid-point of the NTDE, as a starting point for the projected curves. The Intermediate- High Scenario is an average of the high end of ranges of global mean SLR reported by several studies using semi-empirical approaches. The Intermediate Low Scenario is the global mean SLR projection from the IPCC AR4 at the 95% confidence interval.

Source: *Global Sea Level Rise Scenarios for the United States National Climate Assessment*, Parris et al. (2012).

The impacts of sea-level rise include increased frequency and severity of flooding in low-lying areas, erosion of beaches, and damage to infrastructure and the environment, including wetlands and inter-tidal zones, and mangroves, with significant impacts on biodiversity and ecosystem. Mangroves are taxonomically diverse group of salt tolerant, broad-leaved evergreen trees and shrubs that grow primarily in tropical and subtropical coastal regions of Asia, Australia and America. Mangroves are made up of salt tolerant evergreen trees and act as natural buffers against tropical cyclones. Mangroves have special aerial roots, prop roots and buttresses to live in muddy, shifting and saline conditions. These mangrove systems are the nurseries of many invertebrate species and fish. Possible climate change impacts on mangroves are summarized in Table 23.2.

The projected sea level rise is expected to have negative impacts on human settlements, tourism, freshwater supplies, fisheries, exposed infrastructure, agricultural and dry lands, and wetlands; having social and economic losses to millions of people living in coastal areas. For example, Maldives is chain of more than thousand islands in the Indian Ocean and is at high risk due to sea level rise. Bangladesh is particularly vulnerable because of its dense population living in low lying river deltas. A major portion of the most fertile agricultural land of Bangladesh may get submerged and affected by salinity, causing massive out migration. Projections by the IPCC indicate that a sea-level rise of 1 metre — expected sometime within a century — would inundate up to 20% of the country’s land area and displace 14% of the total population.

The Sundarban mangrove forest, covering an area of 16,000 km² of land and water, is the longest mangrove forests in the world. Sundarban mangrove lies within the delta of the Ganges, Brahmaputra and Meghna rivers in the Bay of Bengal; 60% of the area lies in Bangladesh and 40% area in India. It is a biosphere reserve and the core area (Sundarban National Reserve) is world Heritage site. Under the project Tiger, the Government of India has established a Tiger Reserve in Sundarban. The impact of climate change on Sundarbans mangrove is highlighted in Box 23.4.

BOX 23.4. Impact of climate change on Sundarbans mangrove

The Sunderbans mangrove harbours a rich biodiversity of plants and animals, and is the only habitat for the endangered Royal Bengal Tiger. There is great abundance of invertebrates, birds, insects, monkeys and reptiles (Gopal and Chauhan, 2006). The mangrove ecosystem provides income to the local people from the collection of mollusks, crustaceans, fish and honey. A large number of species of prawns and shrimps are exploited commercially. According to IPCC report 2007, sea level rise is the greatest threat to the Sundarban mangrove system because of flooding of low-lying delta, retreat of shoreline, salinization and acidification of soil, changes in water table and adverse effect on livelihood of local people (IPCC, 2007). Mangrove will be affected greatest by sea-level rise. A 45 cm rise in global sea level will cause a destruction of the 75% of the Sundarbans mangrove. Some climate change models also predict increased frequency of tropical cyclones in coastal regions that will impact the biodiversity and resilience of the mangroves. Three small islands in the Sunderban area have already been submerged.

Table 23.2. Possible climate change impacts on mangroves. (source: IPCC, 2007; Gopal and Chauhan, 2007)

Parameter	Possible climate change impacts
Temperature increase	Seasonal patterns of reproduction of plants and animals
Elevated CO ₂	Species specific variations
Sea-level changes	Decrease in geographic distribution Change in species composition Species diversity of mangroves on small islands
Tropical storms	Increased frequency and intensity of storms will affect salinity levels, flooding of low lying areas and wetland sediment budget
Precipitation change	Effect on plant growth and aerial extent

Range of Distribution and Phenology of Organisms

Animals and plants have a specific range of distribution. The likely shift in temperature ranges would affect altitudinal and latitudinal distribution pattern of organisms. With increasing global warming many species are expected to shift slowly from lower altitudes to higher altitudes and from lower latitudes to higher latitudes. In the northern hemisphere, an increase of 500 m in the

altitude means a decrease of about 3°C in mean temperature. So if temperature were to increase by 3°C , species distribution may shift by about 500 m up a mountain side (Fig. 23.13). Similarly, the northern limits of species ranges are expected to move pole-ward by about 500 km. There are greater chances of extinction of local populations along range boundaries at lower latitudes or higher elevations. The range of distribution of endotherms and the global distribution of diversity of large herbivores will also change. The modeling approach has proved that little brown bats will be showing a northward expansion in response to changing climatic conditions in the next century. These changes in the distribution of species are likely to have marked effect on species diversity and ecosystem functioning of coral reefs, mountain ecosystems, coastal wetlands, tundra, boreal and temperate forests. There is possibility of increasing invasion by exotic weed species as well as decoupling of species interactions in ecosystems, which may affect community structure and ecosystem functioning. Outbreaks of insect pest and diseases are likely to occur frequently.

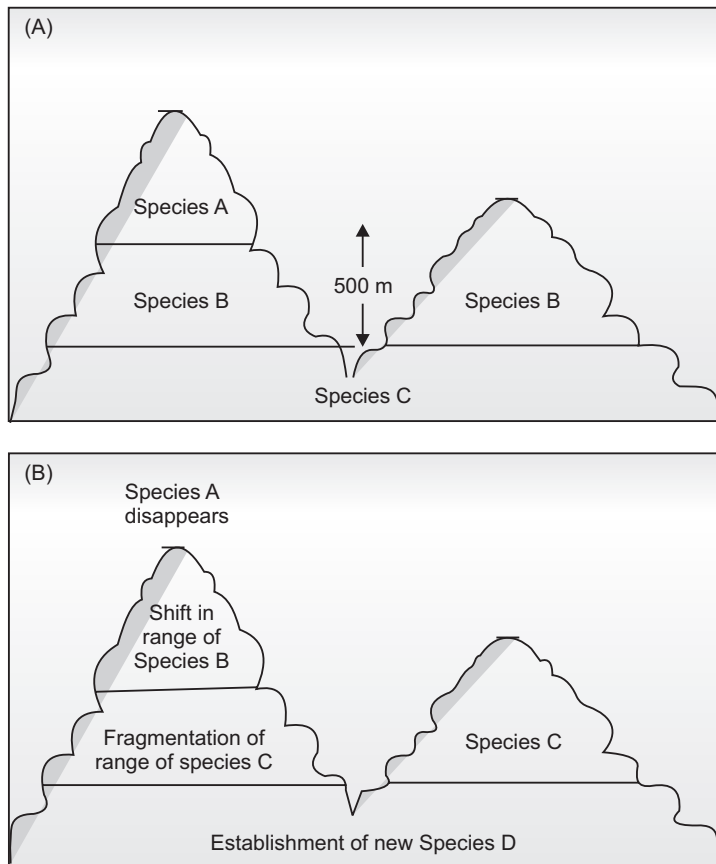


Fig. 23.13. The possible effect of climate change on the elevational distribution of species. (A) Ranges of species before climatic warming and (B) after predicted warming. (redrawn from Ricklefs, 2001).

The effect of recent climatic changes on the distribution and phenology and the physiology of the organisms have been analyzed (IPCC, 2007; Penuelas and Filella, 2001; Walther et al., 2002). The warming trends during the last 30 years or so have shown that the spring activities of organisms, like breeding and singing of birds, arrival of migrant birds, migration of butterflies and flowering in plants, have advanced a few days, mostly in temperate countries (Fig. 23.14). Similarly, changes in autumn activity such as the late onset of leaf colour change and the extension of growing season of

plants has also been recorded for some temperate trees. The changes in spring phenology in response to climate change trends have been found to be more conspicuous for non-tree plants as compared to trees in higher latitudes (Root et al., 2003). The advancement in phenology was found to be 4.2 days per decade (for plants in latitudinal belt from 32 to 49.9°N) and 5.5 mean days per decade (for plants in latitudinal belt from 50 to 70°N) (Root et al., 2003). There are clear trends of northward and upward elevation shifts in range boundaries of birds, butterflies, and alpine herbs in response to climate change trends (Parnesan and Yohe, 2003). In middle and high latitudes, the spring phenology (spring bud bursting, flowering dates and advanced greenness) of deciduous trees in temperate and boreal ecosystems has been found to respond to climate change trends (Badeck et al., 2004).

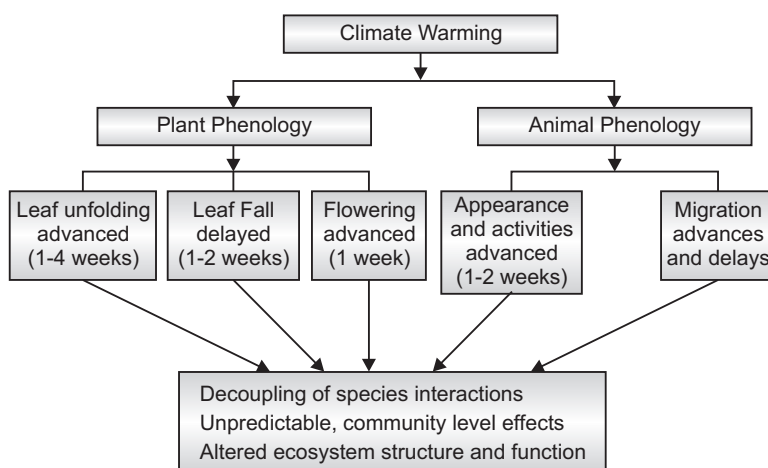


Fig. 23.14. Effect of climate warming on plant and animal phenology in recent times (redrawn from Penuelas and Filella, 2001).

Several meta analyses of data and observations, strongly support the idea of phenological variations as influenced by climate change (Penuelas et al., 2002; Badeck et al., 2004; Parmesan, 2007). Recent warming trends have been associated with earlier onset of vegetation activity in spring (e.g., leaf-out or flowering date) and an overall extension in the length of the active growing season (Richardson et al., 2013 and references therein). These trends have been observed across a diverse range of plant taxa, including herbs and grasses as well as trees and shrubs, and analogous patterns have been reported for amphibians, birds, fish, and mammals (Richardson et al., 2013).

The longest known phenological records assembled to date track the flowering of Japanese cherry trees back to the 9th century (Primack et al., 2009). Analysis of these data provides insight into phenological variability on annual, decadal, and century time scales, and also provides a pre-industrial context for recent phenological shifts. Spring blooming is now occurring earlier than at any other point in the past 1200 years (Primack et al., 2009). In semi-arid and arid desert ecosystems, phenological shifts resulting from climate change may occur both as a result of changes in the timing and amount of precipitation, and increases in temperature.

Richardson et al. (2013) have emphasized the importance of phenology as a factor that mediates vegetation feedbacks to the atmosphere and climate system through a diverse set of processes and mechanisms; these include albedo, surface roughness length, canopy conductance, water and energy fluxes, photosynthesis and CO₂ fluxes, and fluxes of biogenic volatile organic compounds (BVOCs). However, there is need to better understand of the drivers of phenology, particularly in under-studied biomes (e.g., tropical forests) as well as a mechanistic understanding of the role of photoperiod, even in well-studied biomes. Furthermore, climate change impacts on both community structure and ecosystem function will be partially controlled by the response of the phenology of individual organisms to climate change.

Impact on Soils and Nutrient Cycling

The soil is the major pool of organic carbon. The world's soils contain twice as much carbon as the atmosphere and therefore, small increases in organic carbon loss from soils could greatly enhance carbon dioxide concentrations in the atmosphere, potentially creating a positive feedback on climate (Smith and Fang, 2010). Since temperature is an important factor regulating CO₂ release from the soil due to the activities of soil organisms and root respiration, global warming may accelerate the decomposition of soil organic matter, the CO₂ thus formed would lead to further warming and changes in soil organic pools (Jenkinson et al., 1991). At the global level, a meta-analysis of 20-yr data by Bond-Lamberty and Thomsons (2010) shows that over the past 20 yrs soil respiration has increased with climate warming and thus the study supports the hypothesis that soil carbon fluxes will increase in a warming climate. Jenkinson et al. (1991) have analyzed the effect of global warming on the release of CO₂ from soils. They have predicted that a global temperature rise of 0.3°C per decade will release an extra 61×10^{15} gCO₂-C from the soil into the atmosphere by the year 2050. This will account for 19% of the total CO₂ release from fossil fuel combustion. Thus, a change in the soil organic carbon content will affect the CO₂ concentration in the atmosphere. The impact of climate change is predicted to be greatest in the boreal and tundra zones, which are characterized by large pools of organic matter in soils. The increased emission of CO₂ and CH₄ from boreal and tundra soils will have positive feedback effects on climate change. The carbon pools in vegetation and soil will show major changes in response to elevated CO₂ in the atmosphere and N-deposition. Decomposition of organic matter, soil microbial activity and nutrient cycling patterns may also shift modifying soil fertility and the plant species diversity.

On the other hand, an increase in retention of carbon in the older soil carbon pool might be expected under elevated CO₂. The soil carbon stability may increase due to higher rates of carbon input to soil and at the same time lower rates of carbon withdrawal. On the basis of several agricultural FACE studies, Kimball et al (2002) have indicated an increase of 17% (Nitrogen non-limiting condition) and 3% (N limiting condition) of soil carbon due to increase of 300 ppm of CO₂ in the air. Evidently, assessing the balance between increased soil carbon inputs through greater plant growth due to climate warming, and increased carbon losses through higher decomposition rates, should be a research priority (Smith and Fang, 2010).

Effect on Coral Reefs

The coral reefs are found in the nutrient poor tropical oceans. These are complex systems based on anthozoan corals and their symbiotic endozoan dinoflagellate algae and the coralline algae. Though coral reefs generally occur in nutrient poor tropical oceans, they are among the most productive ecosystems and their varied habitats support a high diversity of colourful invertebrates and vertebrate life. They occupy 15 to 30% of the continental shelf area between 40°N and 40°S. Coral reefs are threatened systems because of fishing practices, pollution, increased disease outbreaks and invasive species, and coral bleaching.

Coral reef bleaching is a reduction in the density of dinoflagellate algae (marine microscopic plants) that are found in reef ecosystems. Warmer water temperatures can result in coral bleaching. When water is too warm, corals will expel the algae (zooxanthellae) living in their tissues causing the coral to turn completely white. Bleaching also represents a reduction in algal pigments that renders reef corals bleached or white in appearance. Thus, coral reef bleaching represents an uncoupling of the mutually life-sustaining association between algae and corals. Observations indicate that mass coral reef bleaching has coincided in space and time with the warmest season and with warmer than usual sea surface temperatures (generally in excess of an approximate temperature threshold of 30°C). In 2005, the U.S. lost half of its coral reefs in the Caribbean in one year due to a massive bleaching event when the warm waters centered around the Northern Antilles near the

Virgin Islands and Puerto Rico expanded southward. Over the last 30-40 years 80% of coral in the Caribbean have been destroyed and 50% in Indonesia and the Pacific. Bleaching associated with the 1982-83 El-Nino killed over 95% of coral in the Galapagos Islands, and the 1997-98 El-Nino alone wiped out 16% of all coral on the planet. The number of regions reporting mass coral bleaching has increased substantially in recent years (Fig. 23.15) the mass bleaching received global attention in 1997-98, when increased sea surface temperatures associated with El Niño resulted in extensive bleaching of the world's reefs (NOAA and IUCN, 2006). As tropical seas continue to warm, coral reef ecosystems are likely to undergo significant changes. These changes include losses to biological diversity and coral cover, as well as economic losses to the fisheries and tourism sectors. Thus, there is need to integrate mass bleaching phenomena into management efforts aimed at sustaining the value of coral reef ecosystems (NOAA and IUCN, 2006). More than 30% of coral reefs are considered to be effectively lost worldwide through a combination of anthropogenic effects and extensive coral bleaching episodes (Wilkinson, 2000).

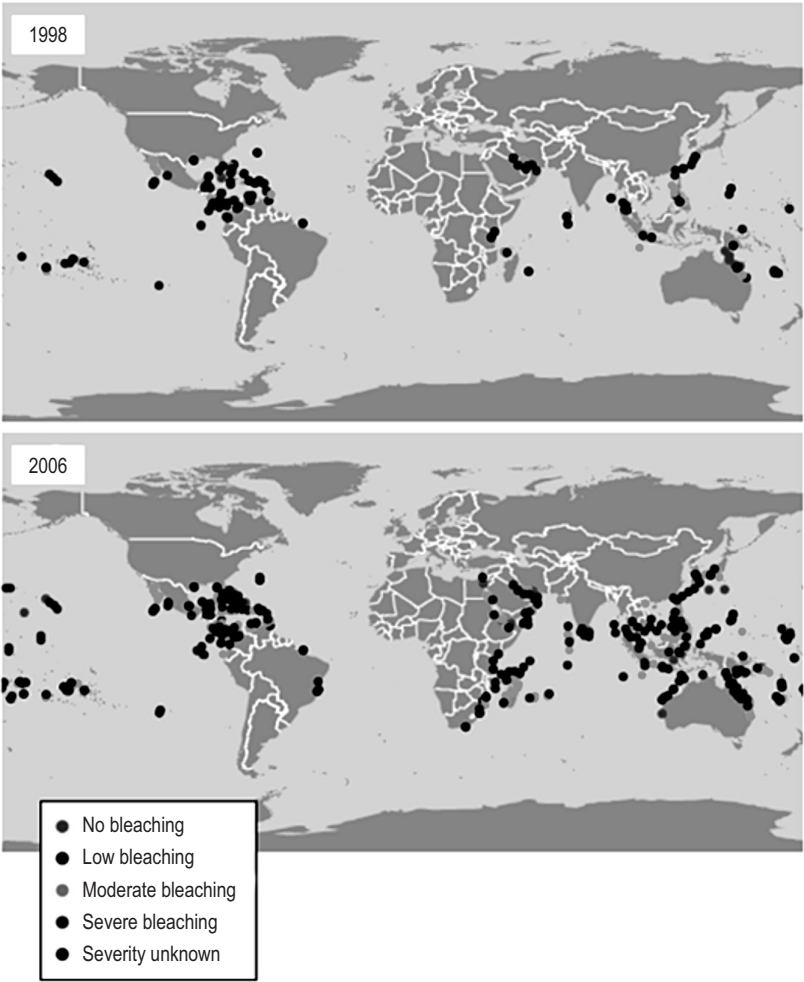


Fig. 23.15. Global trends in the extent and severity of mass bleaching of corals The extent and severity of mass coral bleaching events have increased worldwide over the last decade. (adapted from NOAA and IUCN. 2006. *A Reef Manager's Guide to Coral Bleaching*. US - The World Conservation Union. Great Barrier Reef Marine Park Authority, Townsville, Australia.

Temperature increase, sea level rise and changes in sea water chemistry will be having drastic effects on coral reefs. The CO₂ related changes on coral reefs are summarized in Table 23.3. The formation of calcareous skeletons is a widespread phenomenon in reef building corals, coralline algae, and planktonic foraminifera. The corals are most conspicuous calcifying organism in the oceans and account for about 10% of the global CaCO₃ production. In seawater, calcium reacts with carbon dioxide to form calcium carbonate. During the formation of coral, carbon dioxide held in solvents as CaCO₃ is removed by algae cells of the corals. Coral reefs and their associated carbonate environments have been damaged by recent atmospheric CO₂ rise. The calcium carbonate from the foraminifera also forms white limestone. Ocean acidification will affect coral reefs and the ecosystems that depend on them in the more temperate latitudes because the processes involved are more robust in colder water. Coral reefs in warmer waters are subjected to the threat of coral bleaching. Combined with higher water temperatures, ocean acidification is thought to be a major cause of coral bleaching, destroying coral reef ecosystems around the world. Some studies projecting a rapid contraction of tropical coral reefs by 2050 (Logan, 2010).

Table 23.3. The CO₂ related changes on coral reefs (based on Kennedy et al., 2002).

CO ₂ Related Change	Mechanism	Effect of Coral Reefs
Sea Water Chemistry	Dissolution of CO ₂ results in lowering of pH lowering and carbonate ion concentrations	Reduction in calcification rates in reef building corals and algae.
Sea Water temperature	Higher sea-surface temperatures	Coral bleaching due to increase in annual maximum sea-surface temperature of 1-2°C
Sea Level	Melting of ice caps and glaciers, and thermal expansion of oceans	Some reefs may benefit; others may suffer
Storm frequency or intensity	Some climate models indicate an increase in storm frequency or Intensity	Some species may decline while others may benefit.
Dust deposition	Increased delivery of dust borne iron may enhance phytoplankton growth	Increased phytoplankton production, macro algae may compete with corals and calcifying algae for space.

Surface Ocean Chemistry and Biogenic Calcification

In oceans, production of calcifying coccolithophorids (*Emiliana huxaleyi* and *Gephyrocapsa oceanica*) and non-calcifying groups of phytoplanktons result in the export of organic matter from surface layer to deeper layers, constituting the biological pump. Coccolithophorids that form dense blooms covering large areas of the oceans are the most productive calcifying organisms in the oceans. Their activity is also the driving force for calcium carbonate production in surface layer and its transport to depth, also known as calcium carbonate pump. The biological (organic carbon) and calcium carbonate pumps control the pattern of CO₂ exchange between the oceans and the atmosphere (Fig. 23.16).

The increase in CO₂ concentration in the atmosphere may also impact the biota by affecting the marine biogenic calcification. A doubling of CO₂ is expected to cause a 20% to 40% reduction in biogenic calcification of the predominant calcifying organisms including corals, foraminifera, and coccolithophorids. Biogenic calcification was found to be reduced in *Emiliana huxaleyi* and *Gephyrocapsa oceanica* under elevated CO₂ of the water (Riebesell et al., 2000).

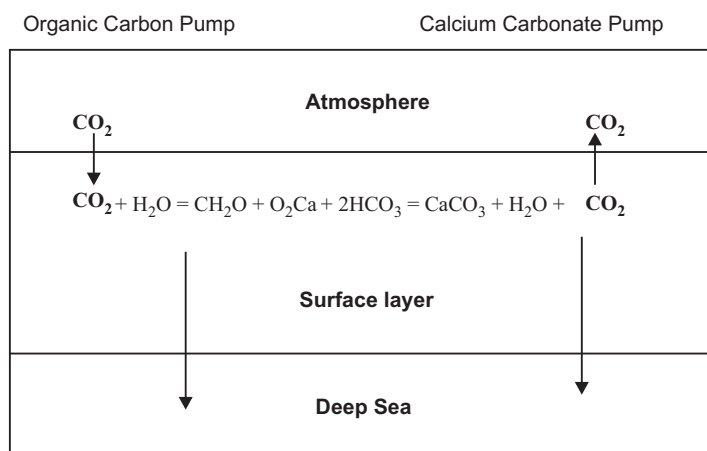


Fig. 23.16. The biological carbon pump generates a CO₂ sink in the ocean. The calcium carbonate pump transports calcium carbonate to ocean depth and releases CO₂ in the surface layer. These two processes regulate the biologically mediated ocean–atmosphere CO₂ exchange.

Variations in atmospheric CO₂ content results in immediate changes in CO₂ concentration of the oceanic surface layer and carbonate precipitation. Changes in carbonate chemistry are likely to have impacts on individual phytoplankton groups (Riebesell et al., 2001). The CO₂ sensitive taxa such as calcifying coccolithophorids would benefit more from the present increase in atmospheric CO₂ concentration compared to non-calcifying diatoms and Phaeocystis, (Riebesell et al., 2001). The shift in dominance of coccolithophorids would enhance the strength of the carbonate pump and in turn would lower biologically mediated CO₂ uptake from the atmosphere. At a sea water pH of 8.2, the contribution of HCO₃⁻, CO₃²⁻, CO₂ and H₂CO₃ to total dissolved inorganic carbon (DIC) are about 90%, 9%, 1% and 0.001%, respectively. The increase in CO₂ concentration will cause reduction in carbonate concentration and a decrease in sea-water pH. During the glacial times, lower CO₂ concentration has been associated with the changes in oceanic carbon sinks through the biological pump (Rahustorf, 2002).

Sea-air is a coupled system that affects and is affected by climate and environmental change. Climate change will alter the air-sea exchange of trace gases. The flux of carbondioxide from the oceans is one of the most important natural factors determining the rate of future climate change. Recent studies show that ocean warming reduces CO₂ solubility and leads to increased ocean stratification, thus reducing the ocean's ability to absorb carbon. The ocean will continue to be a carbon sink as long as atmospheric CO₂ concentrations continue to rise. But as the ocean's capacity to act as a carbon sink diminishes, atmospheric CO₂ may over centuries stabilize at higher levels than would otherwise have been the case.

The oceans also have a key role in the fluxes of nitrous oxide and methane, which are highly effective greenhouse gases.

Ocean Acidification

The oceans annually absorb a substantial proportion of anthropogenic carbon dioxide (CO₂), which reacts with water to form carbonic acid, some of the carbonic acid molecules react with water molecules to yield bicarbonate ions and hydronium ions, thus increasing the ocean's H⁺ ion concentration. The mean surface ocean pH has already decreased from a preindustrial average of about 8.2 to a present value of 8.1. Since the pH scale is logarithmic, this change represents approximately a 30 per cent increase in acidity. Feely et al. (2010) project a pH decrease to a mean

of about 7.8 by 2100. By the end of this century the surface waters of the ocean could be nearly 150 per cent more acidic, resulting in a pH that the oceans haven't experienced for more than 20 million years; thus ocean acidification may be approaching the planetary boundary (Rockstrom et al., 2009). Increased ocean acidity affects marine animals with carbonate shells and skeletons, calcareous algae and other organisms. Affected organisms include reefbuilding corals as well as animals critical to ocean food webs, including several important human food sources such as crabs and molluscs.

23.4 CO₂ FERTILIZATION EFFECT ON PLANTS

CO₂ is a raw material for photosynthesis, the process by which plants transform the sun's energy into a chemical form and create carbohydrate molecules. If the concentration of atmospheric CO₂ available to plants increases, plants may be able to increase their rate of photosynthesis and thus grow more if no other resource, like water or soil nutrient, is limiting. This response of plants to elevated concentrations of CO₂ is known as the "*carbon dioxide fertilization effect*."

During the last three decades, there has been considerable scientific effort to determine plant responses to rising atmospheric CO₂. Numerous CO₂ enrichment studies have been carried out in greenhouses, growth chambers, and open-top chambers. The open top chambers are structures within which plants are grown in CO₂ enrichment experiments, generally possessing transparent side-walls and open tops, through which either ambient or CO₂ enriched air is continually pumped. Free Air CO₂ Enrichment (FACE) is a technique that has been used to deliver CO₂ enriched air to vegetation located out-of-doors in a natural field setting. The CO₂ delivery system is usually constructed out of pipes and tubes that are arranged in a circle. Because these units do not employ walls or enclosures, FACE experiments are considered to be on the cutting edge of CO₂ enrichment technology. The free-air CO₂ enrichment technology helps maintain elevated atmospheric CO₂ levels over fairly large areas using computer controlled, vertical vent pipes arranged in circular ways. To study, the responses of tall trees to elevated CO₂, a new canopy free-air enrichment system, known as WEB-FACE has been used. For example, pure CO₂ was released from small laser-punched holes in a network of small tubes woven into the forest canopy of a *Fagus sylvatica* dominated mixed deciduous forest (Pepin and Korner, 2002). Information from FACE research on crop, forest, and other ecosystems has enabled to study the impacts (both positive and negative) of future global change on primary productivity, leaf CO₂ assimilation, leaf nitrogen content and belowground processes more precisely (Nowak et al., 2004; Pendall et al., 2004).

Various reviews (Idso and Idso, 2001; Jablonski et al., 2002; Loladze, 2002; Woodward, 2002), have shown that in a CO₂ enriched world of the future, the elevated CO₂ will be affecting the growth, reproductive effort, transpiration and stomatal conductance and chemical composition of wild and agricultural plants. The various effects are briefly described below.

The CO₂ enrichment studies in greenhouses, growth chambers, and open-top chambers have suggested that growth of many plants could increase by about 30% on average with a doubling of the atmospheric CO₂ concentration, in the short-term, under favourable conditions of water, nutrients, light and temperature. In C₃ plants, net photosynthesis increases due to reduced photorespiration, increased efficiency of Ribulose biphosphate carboxylase, besides increased water use efficiency. The CO₂ enrichment also reduces the damage caused by photo inhibition in plants with C₃ photosynthetic pathway as well as increases the water use efficiency. Several studies have shown that there may not be much effect on the C₄ plants as they have physiological mechanism of concentrating low levels of ambient CO₂, special adaptations to conserve water and prevent photorespiration.

The studies have shown that C₃ plants (rice, wheat, soybean and oats) and C₄ plants (e.g., corn, sugarcane and maize) respond differently to increased CO₂ concentration. After analyzing approximately 40 and 80 individual responses of C₄ and C₃ grasses to elevated CO₂, respectively,

Wand et al. (1999) found that both C_4 and C_3 grasses respond favourably to atmospheric CO_2 enrichment. Photosynthetic rates, for example, increased by an average by 25% and 33% for C_4 and C_3 grasses, respectively, in response to a doubling of the atmospheric CO_2 concentration. In addition, atmospheric CO_2 enrichment increased total biomass of C_4 and C_3 grasses by 33 and 44%, respectively. In a face experiment, the net primary production of maturing loblolly pine (*Pinus taeda*) increased by 25% on a site of moderate soil fertility (DeLucia et al., 1999).

Biomass Allocation and Reproductive Effort: Crop plants grown under elevated CO_2 conditions showed a 28% to 35% increase in crop seed production as there is linkage between vegetative physiology and reproductive effort in crop plants at high CO_2 . The meta-analysis by Jablonski et al., (2002) has revealed that increased levels of the carbon dioxide increases yields of crop plants, being greater in C_3 non-legumes and non-legumes followed by C_4 plants.

Stomatal Conductance and Water Use Efficiency: Stomata are small pores in plant leaves through which water vapour and carbon dioxide diffuse during transpiration and photosynthesis (carbon fixation), respectively. Stomatal conductance is a property related to the ease with which water vapour escapes from plant leaves through the stomata. Stomatal density is a measure of the number of stomatal pores per unit area of leaf. Carbon dioxide concentration is a major determinant of stomatal conductance to maintain balance for CO_2 uptake as well as minimizing the loss of water. Some long-term field-based studies on tree species showed an overall 21% reduction in stomatal conductance as observed more clearly in the long-term studies than in the short-term studies (Woodward, 2002). An analysis of long-term measurements of whole-ecosystem carbon and water exchange indicated a substantial increase in water-use efficiency (the ratio of water loss to carbon gain) in temperate and boreal forests of the Northern Hemisphere over the past two decades and the observed increase was most consistent with a strong CO_2 fertilization effect (Keenan et al., 2013). The increase is associated with trends of increasing ecosystem-level photosynthesis and net carbon uptake, and decreasing evapotranspiration.

Nutrient Quality of Plants: The meta-analysis of various studies showed that there was no reduction in grain N concentration in legumes and rice on the average under CO_2 enrichment in spite of a mean increase of 38% in the mass of rice grains produced in the high CO_2 air and 58 % increase in plant biomass of legumes (Jablonski et al., 2002).

Community Level Effects: CO_2 enrichment can also affect plant communities directly. For many plant species, increased CO_2 concentrations lead to increased rates of net photosynthesis and improved water-use efficiency, resulting in larger plants. This effect is greatest in C_3 plants and is small in C_4 plants. Where plant communities consist of both C_3 and C_4 species, the different responses of these two groups can lead to changes in plant community composition overtime. Recent studies by Morgan et al. (2011) on CO_2 and warming effects on a mixed prairie indicated that elevated CO_2 favoured C_3 grasses and enhanced stand productivity, whereas warming favoured C_4 grasses. However, combined warming and CO_2 enrichment stimulated above-ground growth of C_4 grasses in 2 of the 3 years of experiment when soil moisture most limited plant productivity. The results indicate that in a warmer, CO_2 enriched world, both soil water content and productivity in semi-arid grasslands may be higher than previously expected. Biogeochemical theory predicts that soil nitrogen (N) scarcity may limit natural ecosystem response to elevated CO_2 concentration, diminishing the CO_2 fertilization effect on plant productivity in terrestrial ecosystems. Langley and Megonigal (2010) found that N addition enhanced the CO_2 stimulation of plant productivity of herbaceous brackish wetland in the first year of a multi-year experiment, indicating N-limitation of the CO_2 response. They further found that N addition strongly promotes the encroachment of C_4 plant species that respond less strongly to elevated CO_2 concentrations. Overall, Langley and Megonigal (2010) found that the observed shift in the plant community composition ultimately suppresses the

CO₂ stimulation of plant productivity by the third and fourth years. These authors suggested that trade-offs between the abilities of plant taxa to respond positively to different perturbations may constrain natural ecosystem response to global change.

At elevated CO₂, more photosynthates are allocated to roots which are found to promote mycorrhizal associations and nitrogen fixation by the plants. In CO₂ rich environment, the plants will be having higher C/N ratio, thereby affecting decomposition and herbivory through both direct and indirect effects.

Changes in the number of flowers, fruits and seeds and their nutritional quality could affect reproductive success and seedling survival of plants. Such changes could also have long-term effects on ecosystem functioning. Seed nitrogen could affect establishment dynamics by affecting germination, growth rate and competitive interactions among co-occurring species of plants. These effects could be magnified trophically through their impacts on pollinators, frugivores and granivores. Thus community level interactions could alter biodiversity, plant community dynamics and ecosystem functioning. The differential effect on reproductive effort and seed quality across species and functional groups has implications for both natural and agricultural systems.

Photosynthetic acclimation under elevated CO₂ : Agricultural species grown in elevated CO₂ environments often, but not always (Gavito et al., 2000), exhibit some degree of photosynthetic acclimation or down regulation of net photosynthesis.

23.5 TOOLS TO STUDY GLOBAL CLIMATE CHANGE

Palaeoclimate Records

The Palaeoclimate records give important clues about the climate changes in the past due to natural phenomenon. A wide variety of archives such as marine and lake sediments, ice caps, tree rings, long-lived corals and archeological remains give useful information about the past climate changes. One of the most interesting cases is the examination of glacial ice. Snow changes to glacial ice after recrystallization and increase in the density. The process also traps air bubbles that are analyzed to give information about the concentration of CO₂ at the time of ice formation. The glacial ice is a time capsule that provides information about the atmospheric conditions in the past. To study the ice, long cores of ice extracted and trapped air bubbles are carefully sampled. This method has been used to analyze CO₂ and CH₄ contents of the atmosphere up to 460,000 years (Petit et al., 1999). The 420,000 year Vostok ice core record of atmospheric CO₂, methane concentration and global temperature through four glacial –interglacial cycles has clearly demonstrated a regular pattern and a tight coupling of their temporal dynamics throughout the record (Petit et al., 1999). The main maximal and minimal of temperature and atmospheric trace gases concentration follow a regular pattern through time, each cycle spanning approximately 100,000 years. The upper and lower bounds of all the three variables are tightly coupled, which are typical of a self regulating system (Petit et al., 1999).

Trees record climate change. Each year a tree forms a new layer of tissue. The width of each year's ring reflects growing conditions such as moisture or sunlight. A wide ring indicates faster growth (good conditions), whereas a narrow ring indicates slower growth (poor conditions). When the tree is cut or samples taken by using a corer, these rings can be "read" like a diary of the climate.

General Circulation Models

The climate models describe the climate situations in numerical terms. Most climate models examine what the climate would be likely in a few decades or a century from now. The general circulation models have been used to predict the future consequences of climate warming on biosphere and its life support system. The GCM's predict atmospheric changes by using the

variables, such as temperature, relative humidity and wind conditions on a global scale (IPCC, 2001). To organize the data for calculations in GCMs, the surface of the earth is divided into large cells measuring several degrees of latitude and longitude. Simulation models are used for making predictions of future atmospheric circulation and climate. In atmosphere-ocean general circulation models, the ocean and atmosphere components are coupled recognizing that oceans play a very important role relating global climate (IPCC, 2001). Now, however, emphasis is on developing regional circulation models so that the predictions can be more region-specific.

Remote Sensing and Geographical Information Systems

NASA launched the EOS (Earth Observing System) Terra mission in December 1999, a new satellite series, to describe, understand and simulate the functioning of the Earth System (Ranson and Wickland, 2001). The Terra fly in the near-polar, sun synchronous orbit and its crossing time of the equator is about 10.42 am. The Terra has different scientific instruments for quantitatively monitoring the Earth's lands, oceans and atmosphere. This mission holds a great promise to understand and simulate the functioning of the Earth System.

Earth CARE is a planned space mission by the European and Japanese space agencies. The project is part of ESA's Living Planet Programme. Earth CARE – the largest and most complex Earth Explorer mission to date – is being developed as a joint venture between ESA and the Japan Aerospace Exploration Agency, JAXA. The satellite will make measurements useful for better understanding the Earth's thermal and solar radiation balance. In particular a combination of active (lidar and radar) and passive radiometers and imagers) instruments will enable Earth CARE to simultaneously measure the vertical and horizontal distribution of clouds and atmospheric aerosols along with Top-Of-Atmosphere (TOA) Long- and Short-wave fluxes. The satellite is to fly in a 400 km high polar orbit at an inclination of 97° degrees, weighing 1700 kg.

Maps have traditionally been used to explore the Earth and to exploit its resources. Observation of land-cover change from space is important for analyzing global change phenomenon. Geographical Information Systems (GIS) technology is becoming an essential tool in the effort to understand the process of global change. Various map and satellite information sources can be combined in models that simulate the interactions of complex natural systems. The condition of the Earth's surface, atmosphere, and subsurface can be examined by feeding satellite data into a GIS.

23.6 APPROACHES TO DEAL WITH GLOBAL WARMING

The economic impact of global warming is quite high. Studies estimate that if we do not act, we may be losing at least 5% of global GDP each year, now and forever (Stern, 2007). If wider range of risks and impacts is taken in to account, the estimates of damage could even rise to 20% of the global GDP or more. The action, in terms of reducing the emission of greenhouse gases to avoid the worst impacts of climate change will cost only about 1% of global GDP each year (Stern, 2007).

In order to stabilize atmospheric concentrations of greenhouse gases, immediate reduction in man-made greenhouse gas emissions is necessary. The various mitigation strategies that could help to reduce the risk of global warming include reduction of energy use by developing more efficient systems, reducing use of fossil fuels and replacing by alternative energy sources, increasing vegetation particularly forest cover, minimizing the use of nitrogen fertilizers for reducing N₂O and improving rice cropping practices for reducing CH₄ emission. Biotechnology can help improve crop production with more efficient resource use, breeding programmes can use genetic diversity such that plants can respond to climate change, low or no tillage can increase carbon sequestration in the soil. There are various mitigation strategies that could help to reduce the risk of global warming (see also chapter 30).

Some practical tools for climate change mitigation include monitoring climate change, quantifying greenhouse gas emissions, communicating on environmental impacts, promoting good practice in environmental management and design, protecting natural ecosystems, increasing investment in renewable energy, biological carbon sequestration, opening markets for energy efficient technologies, and improving the livelihood opportunities to rural poor.

Renewable and Alternative Sources of Energy

Energy planning based on reduction of energy consumption and exploiting alternative sources of energy (solar power, wind power and geothermal power) and switching to hydrogen fuels will reduce emission of greenhouse gases. Eliminating the use of CFCs will reduce global warming as well as protect the ozone layer. Tapping and burning of methane from landfills will provide energy as well as it can reduce the emission of CH_4 from these sources.

Geoengineering Approaches for Carbon Sequestration

Large-scale physical interventions, or technological fixes on a planetary scale, are proposed actions addressing climate change grouped under the term geoengineering. The various geoengineering approaches for carbon sequestration are shown in Fig. 23.17. have discussed. The various strategies for mitigating climate change on the basis of the Climate Change Science Compendium (SCOPE and WMO, 2009) and Lenton and Vaughan (2009) are summarized below:

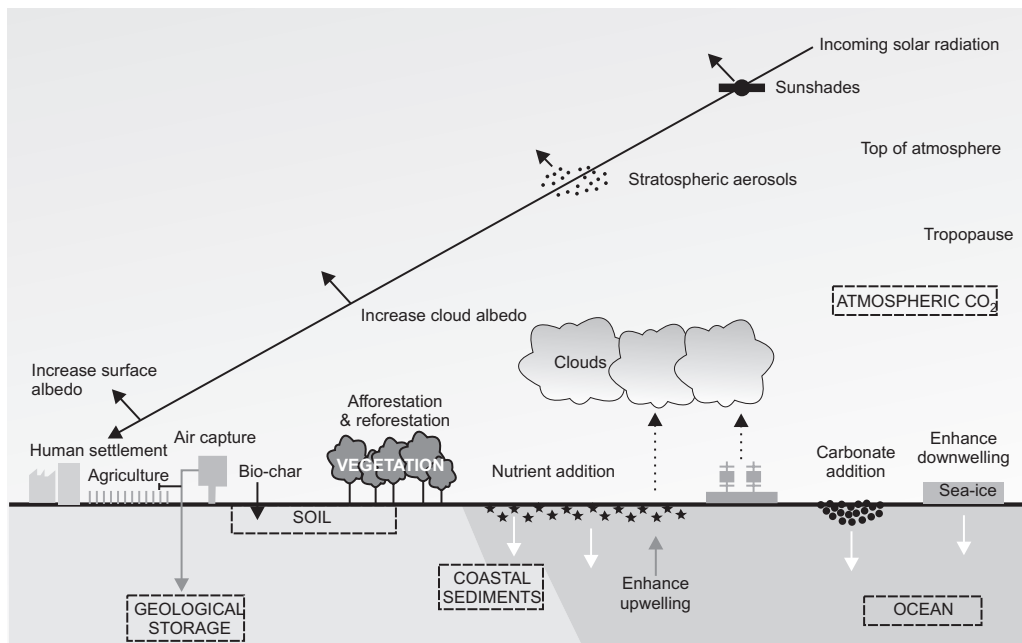


Fig. 23.17. Schematic overview of the climate geoengineering proposals considered. Black arrowheads indicate shortwave radiation, white arrowheads indicate enhancement of natural flows of carbon, grey downward arrow indicates engineered flow of carbon, grey upward arrow indicates engineered flow of water, dotted vertical arrows illustrate source of cloud condensed nuclei, and dashed boxes indicate carbon storage. (adapted from Lenton and Vaughan, 2009).

Carbon Management in Underground Geological Formations: Carbon Capture and Storage (CCS) is a method for the geological sequestration of carbon dioxide (CO_2). CCS systems are designed to capture CO_2 emissions where they are most concentrated at industrial point sources such

as coal power generation plants and to transport it to storage reservoirs. CCS will require significant expenditures on equipment and infrastructure to capture, transport, and store the compressed CO₂.

This is one of the ways to separate and capture the carbon dioxide produced during the combustion of fossil fuels and then sequester it away from the atmosphere. Carbon sequestration (C uptake in bicarbonate system) in underground geological formations seems to be viable option. Technology is available to sequester CO₂ in geological formations at depths of several kilometres, though it may cost 20 to 30 % more than that of the conventional coal plant.

Carbon Dioxide Removal (CDR) techniques are designed to extract CO₂ from the atmosphere (Royal Society, 2009). CDR concepts are based on the carbon sequestration accomplished by nature through photosynthesizing plants and other organisms.

‘Iron Fertilization’: This exploits the CO₂ sequestration potential in parts of the ocean that are nutrient rich but do not support plankton growth due to a lack of iron. Supplying large amounts of iron to these areas of the ocean could stimulate plankton blooms that theoretically will bind carbon molecules and eventually sequester them on the deep sea-floor.

Aerosol Injection Schemes: These are designed to artificially increase aerosol levels in the stratosphere, causing an overall increase in planetary reflectivity. The method using sulphate aerosols simulates the effect of large volcanic eruptions on global climate in reducing incoming solar radiation and has been the subject of climate geoengineering proposals for sometime.

Sunshade Geoengineering: The proposed installation of space-based sunshields or reflective mirrors will be able to deflect a proportion of incoming solar radiation before it reaches the atmosphere. Solar Radiation Management (SRM) techniques reflect a small percentage of the Sun’s light back into space to offset the effects of increased GHG concentrations (Royal Society, 2009).

Carbon dioxide removal methods may be preferable to solar radiation management methods because they involve fewer risks and uncertainties (SCOPE and WMO, 2009). The necessary management practices to respond to climate change include a switch to environmentally sound energy sources; a halt to deforestation in the tropics; sustainable management of fisheries, forests, agriculture, and other ecosystem services; and the development of innovative approaches to sequester carbon from the atmosphere over decades to millennia.

Carbon Sequestration in Vegetation

The potential for carbon sequestration in vegetation is quite high for mitigating the effects of global warming. It is known that the amount of carbon in the soils of terrestrial ecosystems is approximately three times that of the atmosphere. The amount of carbon is also 700 times more than the estimated annual increase of CO₂ in the atmosphere. The terrestrial ecosystems are a huge natural biological scrubber for the CO₂ from the various fossil fuel emission sources such as automobiles, power plants, and industrial facilities. Carbon sequestration in terrestrial ecosystems can occur in living aboveground biomass, living biomass in soil (roots and microbes) and recalcitrant organic and inorganic carbon in surface soil. For effective carbon sequestration, the increased photosynthetic carbon fixation must occur in the long-live pools. Trees take up CO₂ during photosynthesis and can store carbon in the roots and stems for longer periods of time. In Central Amazonian, forests represent about 45% of total carbon storage and 30% of aboveground net production, with a mean residence time of 80 years (Chambers et al., 2001). Thus the rate of carbon sequestration in intact rain forest could be 0.2 to 0.3-petagram carbon per year as a long-term storage capacity.

It is estimated that trees could remove 10 to 15% of the excess carbon dioxide in the atmosphere by creating large-scale tree plantations. The carbon may remain sequestered away from the atmosphere for at least 50 to 100 years. The potential for carbon sequestration could be enhanced by using improved agricultural practices, reducing oxidation of soil organic matter, enhancing soil texture to trap more carbon as well as protecting the wetlands.

In agroecosystems, conservation agriculture, improved varieties of crop plants, increase in plant cover on agricultural soils and improving soil fertility can contribute appreciable carbon sequestration in agricultural soils.

Carbon Trade

Carbon trade which relates to making a provision for the payment to the industries, community and others for undertaking activities that sequester carbon through plantation and conserving forest is another approach to mitigating the problem of rise in atmospheric CO₂.

Imposing Carbon Tax

Several nations have imposed taxes on greenhouse gas emissions on input basis. According to European Commission, 50% tax is imposed on energy production and 50% tax on carbon emission. Some European countries (Sweden, Finland and Netherlands) have carbon based taxes on all types of fossil fuels like oil, gas and coal. The taxes can also motivate industrial emitters to improve efficiency and to develop carbon dioxide capturing technologies.

23.7 STRATOSPHERIC OZONE

Ozone is a molecule made up three atoms of oxygen and is naturally found in minute quantities in the earth's atmosphere. Most of the ozone in the atmosphere (about 90%) is found in the stratosphere, which extends from about 12 to 50 km above the earth's surface. The concentration is greatest in the lower stratosphere (15 to 30 km). The large amount of ozone in the stratosphere is often referred to as the "ozone layer." The ozone layer extends over the entire globe with some variation in altitude and thickness. The ozone layer in the stratosphere is known as "good ozone" due to its protective role for the living organisms by filtering out the ultraviolet radiation (UV-B) emanating from the sun. Only 10% of ozone is found in the troposphere, the region closest to Earth (from the surface to about 10–16 kilometres). In the troposphere it is a harmful pollutant (hence called bad ozone) and a constituent of the urban smog, which affects plants, building materials and human health unfavorably. Chlorofluorocarbons and biogenic emissions of CH₄ and N₂O cause stratospheric ozone loss.

There appears to be a direct relation between stratospheric ozone and global warming. The amount of stratospheric ozone can be affected by the increases in the concentration of greenhouse gases, which lead to decreased temperatures in the stratosphere and accelerated circulation patterns, which tend to decrease total ozone in the tropics and increase total ozone at mid and high latitudes.

Formation and Measurement of ozone

In the stratosphere, the formation and destruction of ozone is a continuous phenomenon. The rate of formation is equal to the rate of destruction by the natural processes. In recent times, ozone in the stratosphere is being depleted by various types of man made chemicals. In stratosphere photo dissociation of ozone into O₂ and O occurs by UV radiation (Fig. 23.18). O₂ and O recombine to form ozone. Thus ozone dynamics dissipates the energy of UV as heat. Equilibrium is established between the generation and

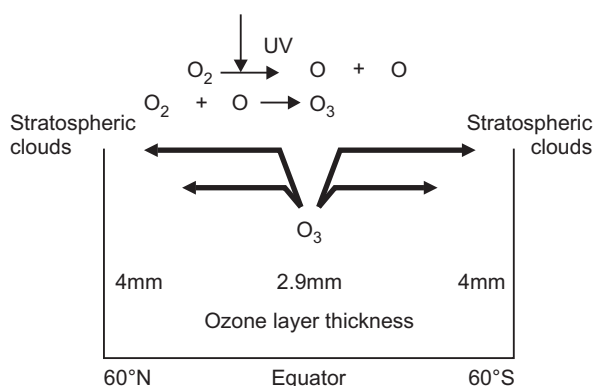


Fig. 23.18. The process of ozone formation.

destruction of ozone, leading to a steady state concentration of ozone layer between 20 and 26 km above sea level. This absorbs solar UV radiation with increasing effectiveness at shorter wavelengths so that essentially no radiation below 0.295 μm penetrates the atmosphere to the ground.

Most of the O_3 is produced above the equator where solar irradiance is maximal; ozone thus formed at these latitudes subsequently diffuses in the stratosphere towards the poles. Thus, while O_3 concentrations above the equator may average 0.29 cm (a thickness of vertical column of stratospheric O_3 condensed to standard temperature and pressure), O_3 above poles may exceed 0.40 cm at the end of the winter season. Concentration of stratospheric O_3 tends to be highest in late winter and early spring and lowest in late summer and early fall. Absorption of UV-radiation by ozone increases exponentially with O_3 thickness. As a result, the UV-B radiation is greater in tropics (i.e., near equator) and decreases towards poles.

Measurement of Ozone in the atmosphere

Ground based instruments (Dobson Spectrophotometer) have been measuring ozone since 1957-59 at about 100 ground stations in different regions of the globe and the satellite based instruments since 1970s. Dobson unit is commonly used to measure the concentration of ozone in a column of air, one Dobson unit being equivalent to a concentration of one part per billion (1 ppb O_3). A typical reading for a healthy polar ozone layer varies from 300 to 450 ppb. Instruments on balloons and aircraft-flying at 15 km height make it possible to measure ozone *in situ* (in place, where it occurs). Satellite measurements give simultaneous information about the ozone levels over the entire globe, for a long period of time. The TOMS (Total Ozone Mapping Spectrophotometer) is a satellite borne instrument. It is used to gain a global picture of ozone levels on a daily basis.

A global network of total-ozone observing stations was established in 1957 as part of the International Geophysical Year. Today, there are about 100 sites located around the world ranging from South Pole, Antarctica (90°S), to Ellesmere Island, Canada (83°N), that routinely measure total ozone. The accuracy of these observations is maintained by regular instrument calibrations and inter comparisons. Data from the network have been essential for understanding the effects of chlorofluorocarbons (CFCs) and other ozone-depleting substances on the global ozone layer, starting before the launch of space-based ozone-measuring instruments and continuing to the present day. Ground-based instruments with excellent long-term stability and accuracy are now routinely used to help calibrate space-based observations of total ozone.

Ozone-Depleting Substances (ODSs)

ODSs are manufactured for specific industrial uses or consumer products, most of which result in the eventual emission of ozone depleting gases to the atmosphere. A large fraction of ODS emissions reach the stratosphere and lead to ozone depletion because of the action of chlorine and bromine atoms.

The ozone depleting substances are chlorofluorocarbons, halons, carbon tetrachloride and methyl chloroform. The compounds that contain chlorine, fluorine and carbon are called chlorofluorocarbons (CFCs). Halons contain carbon, bromine and fluorine and are mainly used as fire extinguishers. The CFCs are non-toxic and non flammable and are chemically inert. The CFCs have a number of applications including airconditioning, refrigeration, foam blowing, cleaning of electronic components. ODS emissions have increased substantially from the middle to the late 20th century, reached a peak in the late 1980s, and are now on decline. These chemicals are highly stable and can persist for decades or even centuries in the atmosphere (Table 23.4). Once released into the lower atmosphere through the use of aerosol spray cans, cleaning solvent or leakage of refrigeration units, these chemicals diffuse into the stratosphere. CFCs release chlorine atoms and halons release bromine atoms in the presence of ultraviolet radiation in the stratosphere. It is the atomic form of

chlorine and bromine that destroys ozone, not the intact molecules. The effectiveness of the atomic form is high; one molecule of CFC on the average destroys 100,000 molecules of ozone. Nitric oxide (NO) and hydroxyl ions (OH⁻) also cause O₃ depletion.

Table 23.4. The types, uses and residence time of commonly used ozone depleting substances in 1989 (source: World Watch Institute, Washington).

Type	Primary use	Residence time (yr)
CFC-12	Refrigeration and airconditioning	139
CFC-11	Foam blowing refrigeration, aerosol propellant	76
CFC-113	Solvents	92
Carbon tetrachloride	Solvents	67
Methyl chloroform	Solvents	8
Halon 1301	Fire extinguishers	101
Halon 1211	Refrigeration, foams	22

In 1974, Sherwood Rowland and Mario Molina from University of California Irvine discovered that the presence of chlorofluorocarbons in the earth's atmosphere can deplete ozone molecules in the stratosphere. They along with Paul Crutzen were honoured with Nobel Prize in chemistry in 1995 for their work on the formation and decomposition of ozone, the CFCs threat to the ozone layer and for protecting human welfare. Paul Crutzen described that methane and nitrous oxide also destroyed ozone. Had bromofluorocarbons been used instead of chlorofluorocarbons, the result would have been catastrophic, as an atom for atom basis, bromine is 100 times more effective than chlorine in destroying ozone. Thus, according to Crutzen, mankind has indeed been lucky.

The ODSs that contain bromine are the halons and methyl bromide (CH₃Br). Halons are halocarbon gases originally developed to extinguish fires. Halons were widely used to protect large computer installations, military hardware, and commercial aircraft engines. As a consequence, halons are often released directly into the atmosphere upon use. Halon-1211 and halon-1301 are the most abundant halons emitted by human activities (Table 23.4).

Methyl bromide (CH₃Br), which is the source of bromine, used as a soil fumigant in agriculture, and is emitted from marine phytoplankton and during biomass burning. Much of the methyl bromide is destroyed in the troposphere. Only a very small percentage reaches the stratosphere. The free radicals of chlorine and bromine are not consumed during ozone destruction.

The supersonic jets release nitric oxide, a pollutant that reacts with ozone to form nitrogen dioxide and oxygen. The jets that travel through the ozone layer have the greatest impact. However, the threat to ozone layer from high flying jets is much lower than from the CFCs.

Halogen source gases are compared in their effectiveness to destroy stratospheric ozone using the Ozone Depletion Potential (ODP). A gas with a larger ODP destroys more ozone over its atmospheric lifetime. The ODP is calculated relative to CFC-11, which has an ODP defined to be 1. Halon-1211 and halon-1301 have ODPs significantly larger than CFC-11 and most other emitted gases because bromine is much more effective (about 60 times) on a per-atom basis than chlorine in chemical reactions that destroy ozone. The gases with small ODP values generally have short atmospheric lifetimes or contain fewer chlorine and bromine atoms.

Ozone Depletion and Ozone Hole

Ozone loss was first detected in the Stratosphere over the Antarctica on the basis of ground based measurements at the Halley Bay research station. Since then the ground based observations and satellite borne instruments have confirmed the loss of ozone in the Antarctica region during each spring (October). The British Antarctica survey team, led by Joe Farman, in 1985 reported

that ozone layer over Antarctica had shrunk each year since 1970s. The spring time ozone layer thickness above Antarctica has declined rapidly from 280-325 Dobson unit (DU), between 1956-70 to about 94 DU in 1995 (Fig. 23.19).

Ozone hole refers to the thinning of ozone layer, i.e., depletion in ozone concentration, which now regularly occurs during Antarctica spring. In September 1998, the hole was biggest in terms of area (25 million km²) with high levels of ozone destruction at an elevation of 15-21 km; ozone throughout the stratosphere dropped to a level as low as 92 ppb. The ozone hole of the 1999 was the second largest and the strongest ozone hole phenomenon. Currently, the ozone thins from 40% to 55% during the Antarctica spring. During the 1999-2000, Arctic ozone depletion varied from 20 to 30% poleward from 65°N. Ozone depletion has also been recorded in the mid latitudes (23°-60°) by about 4% per decade during 1979 to 1991 in both the northern and southern hemispheres.

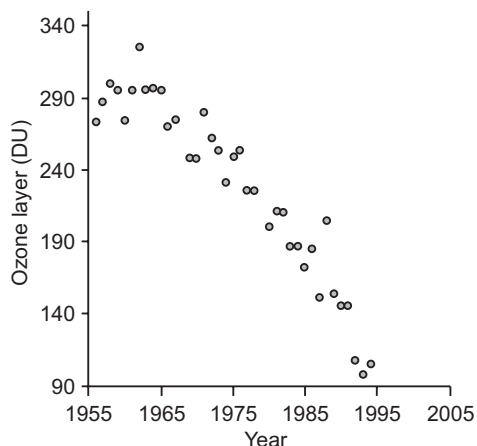


Fig. 23.19. Total ozone observed over Halley Bay in Antarctica in October 1956-94.

Most of stratospheric ozone is created in the tropics, which is then transported by stratospheric air currents to the Arctic and Antarctica. Winds play a key role in ozone destruction in the polar Regions. During the polar winter, the cold air over the Antarctica creates a huge whirlpool of fast moving air encircling Antarctica creating the polar vortex. These winds circulate about the pole creating the 'Polar Vortex'. This vortex effectively insulates Antarctica from rest of the atmosphere. Such polar vortex is also formed over the arctic. Because of its extreme isolation, temperature inside the polar vortex drop to -80 to -90°C. The strong and stable vortex prevents migration of ozone in to stratosphere over the Antarctica. The conditions of Antarctica winter and spring that favour ozone destruction are:

1. Freezing of NO₂ and CH₄ so that they cannot be up with chlorine.
2. The polar stratospheric clouds at low temperature provide heterogeneous surfaces for the reaction of free chlorine to occur.
3. The ozone depleting reactions are very fast. Chlorine attacks ozone molecules under simultaneous co-occurrence of sunlight and freezing conditions in the early spring of Antarctica.

The decline in ozone concentration of the total ozone column varies from 1.7 to 3% over the northern hemisphere since 1969. Greatest decline of 5 to 10% has been observed over Antarctica and Argentina in the southern hemisphere. Global average decline of ozone in the stratosphere was about 3% for the period 1997-2001 compared to the 1980 average value.

Arctic Ozone Hole

Ozone hole was also confirmed above Arctic in 1990. However, the losses of ozone over the Arctic are lower (5 to 10%) compared to that over Antarctica (60 to 70%) during the spring time. Moreover, the polar vortex over Arctic is not as tight as that over the Antarctica due to the presence of land masses with high mountain ranges in the northern hemisphere. The Arctic stratosphere warms faster in the spring, leaving less time for the critical overlap between cold and sunlight, necessary for ozone depletion.

An unprecedented chemical destruction of stratospheric ozone inside the Arctic polar vortex has been observed in March 2011, identifiable as an Arctic "ozone hole" (Manney et al., 2011).

For the first time, destruction of ozone over the Arctic in the springtime was comparable to that over Antarctica. Unusually long-lasting cold conditions in the lower stratosphere led to destruction of more than 80% of the ozone between 18 and 20 kilometres above sea level (Manney et al., 2011). In March 2011, over 40% of the vortex area had ozone levels less than that threshold. Colder temperatures lead to increased presence of polar stratospheric clouds; ice crystals in these clouds speeds up chemical reactions that break down ozone. It is difficult to predict such severe Arctic ozone depletion in the future. However, if it becomes more severe, effects of increased UV on organisms may be increased by reduced protection due to decreases in snow or ice-cover resulting from global warming (Manney et al., 2011).

The Chlorine and Bromine Reactions that Destroy Stratospheric Ozone

Under intense UV light, the CFC molecule breaks down and releases chlorine atom. The process of ozone depletion is documented in Fig. 23.20. Stratospheric ozone is destroyed by reactions involving *reactive halogen gases*, which are produced in the chemical conversion of *halogen source gases* e.g. CFCs. The most reactive of these halogen gases are chlorine monoxide (ClO), bromine monoxide (BrO), and chlorine and bromine atoms (Cl and Br). These gases participate in three principal reaction cycles that destroy ozone.

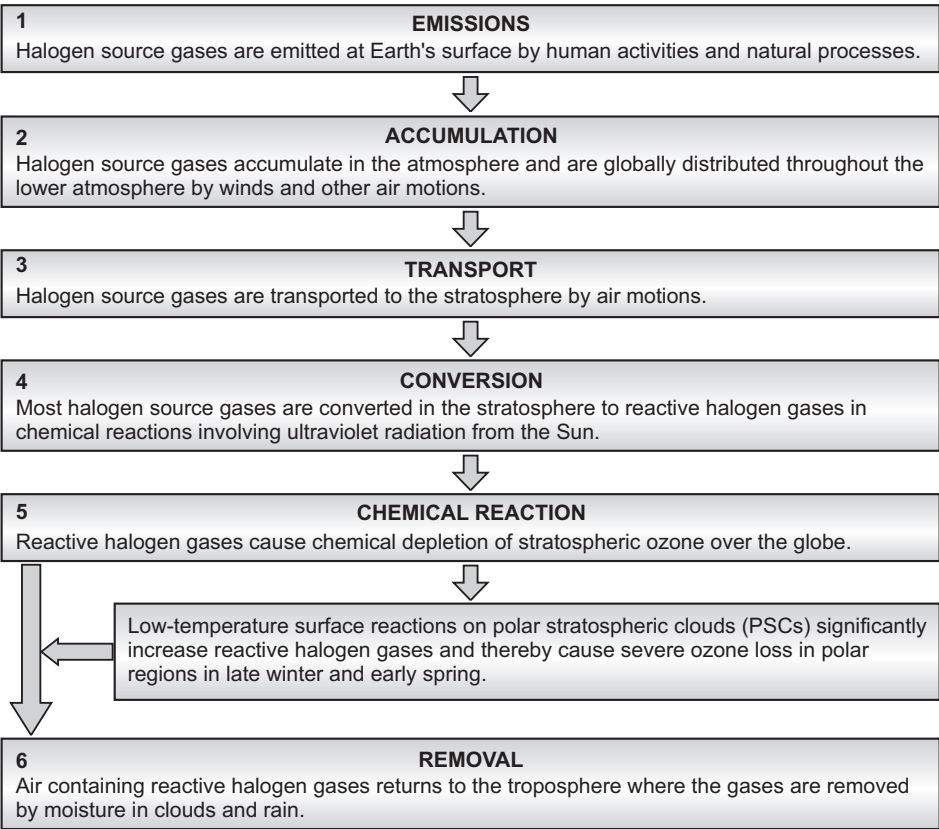


Fig. 23.20. Principal steps in stratospheric ozone depletion. (1) The stratospheric ozone depletion process begins with the emission of halogen source gases by human activities (ODSs) and natural processes. The various steps are (2) accumulation, (3) transport, (4) conversion, (5) chemical reaction, and (6) removal. Ozone depletion ends when reactive halogen gases are removed by rain and snow in the troposphere and deposited on Earth's surface (adapted from Fahey and Hegglin, 2010).

Cycle 1. The cycle is made up of two basic reactions: $\text{Cl} + \text{O}_3$ and $\text{ClO} + \text{O}$. The net result of Cycle 1 is to convert one ozone molecule and one oxygen atom into two oxygen molecules (Figs. 23.21, 23.22). In the stratosphere, UV light releases a chlorine atom from a CFC molecule. The chlorine atom reacts with an ozone molecule, breaking it into molecular oxygen and atomic oxygen. Atomic oxygen combines with chlorine to form chlorine monoxide (ClO). The ClO is highly reactive and unstable. It takes up a free oxygen atom forming oxygen molecule and releasing a free chlorine atom (Fig. 23.21 and Fig. 23.22). Therefore, most important reaction mechanism involves dimerization of ClO radicals at very low temperatures to form Cl_2 and O atoms which combine with oxygen atom to form chlorine monoxide and reduce the chance of O_2 formation. A single chlorine atom can destroy thousands of ozone molecules before it is neutralized. Methyl chloroform and carbon tetrachloride, which are common industrial solvents, act in a similar fashion, although quantities of chlorine are some what lower.

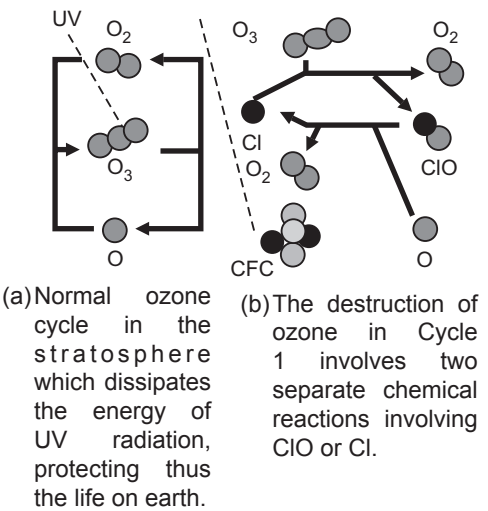


Fig. 23.21.

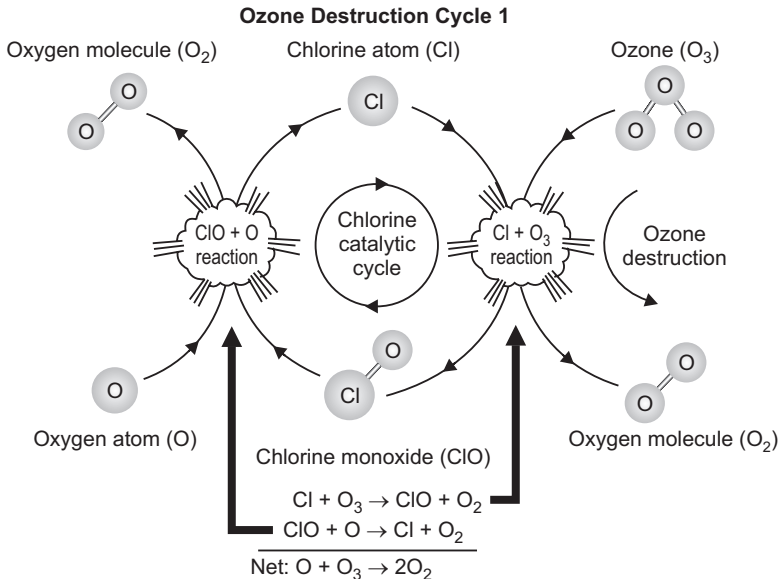


Fig. 23.22. The destruction of ozone in Cycle 1 involves two separate chemical reactions involving ClO or Cl (adapted from Fahey and Hegglin 2010).

Ozone Destruction Cycles in Polar Regions	
Cycle 2	Cycle 3
$\text{ClO} + \text{ClO} \rightarrow (\text{ClO})_2$	$\text{ClO} + \text{BrO} \rightarrow \text{Cl} + \text{Br} + \text{O}_2$
$(\text{ClO})_2 + \text{sunlight} \rightarrow \text{ClOO} + \text{Cl}$	$\text{ClO} + \text{BrO} \rightarrow \text{BrCl} + \text{O}_2$
$\text{ClOO} \rightarrow \text{Cl} + \text{O}_2$	$(\text{BrCl} + \text{sunlight} \rightarrow \text{Cl} + \text{Br})$
$2(\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2)$	$\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2$
Net: $2\text{O}_3 \rightarrow 3\text{O}_2$	$\text{Br} + \text{O}_3 \rightarrow \text{BrO} + \text{O}_2$
	Net: $2\text{O}_3 \rightarrow 3\text{O}_2$

Fig. 23.23. Polar ozone destruction cycles 2 and 3 (adapted from Fahey and Hegglin, 2010).

Cycles 2 and 3. The abundance of ClO is greatly increased in polar regions during winter as a result of reactions on the surfaces of polar stratospheric clouds (PSCs). Cycles 2 and 3 (Fig. 23.23) become the dominant reaction mechanisms for polar ozone loss because of the high abundances of ClO and the relatively low abundance of atomic oxygen (which limits the rate of ozone loss by cycle 1). Cycle 2 begins with the self-reaction of ClO.

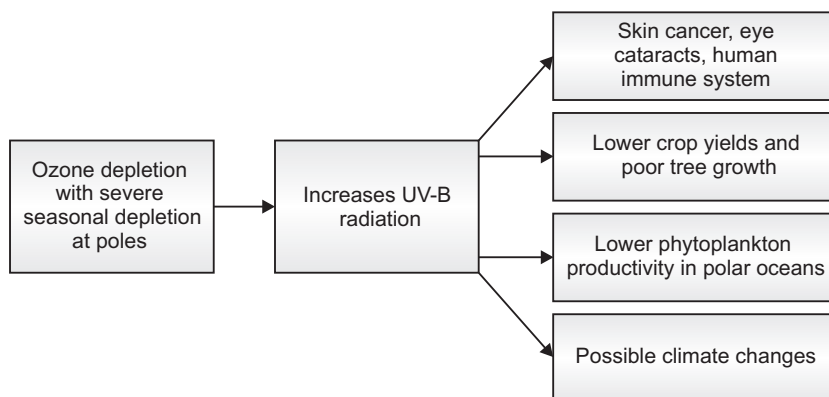


Fig. 23.24. The influence of increased UV-B radiation absorption on human health, plants and aquatic ecosystems. (based on UNEP, 1998)

Cycle 3, which begins with the reaction of ClO with BrO, has two reaction pathways to produce either Cl and Br or BrCl (Figure. 23.23). The net result of both cycles is to destroy two ozone molecules and create three oxygen molecules. Cycles 2 and 3 account for most of the ozone loss observed in the Arctic and Antarctic stratospheres in the late winter/early spring season. At high ClO abundances, the rate of polar ozone destruction can reach 2 to 3% per day in late winter/early spring.

The quantity of bromine is very low as compared to chlorine in the stratosphere. However, Br atoms are more effective than Cl in attacking O₃ in the chlorine rich stratosphere. About 5 to 10% of total stratospheric ozone depletion could be attributed to methyl bromide.

Effects of Ozone Depletion

On the basis of wave length, there are three types of ultraviolet radiations; including UV-A (0.320 to 0.400 μm), UV-B (0.280 to 0.320 μm) and UV-C (0.100 to 0.280 μm). As the Sun's radiation passes vertically through the atmosphere, all the UV-C and approximately 90% of the UV-B is absorbed by ozone molecules in the stratosphere. UV-A radiation (which is mostly harmless to living organisms) is less affected by the atmosphere. Therefore, the UV radiation reaching Earth's surface is composed mainly of UV-A with a small UV-B component. UV-C radiation is lethal to the living organisms and UV-B radiations are damaging to living organisms. The thinning of ozone layer causes increase in the UV-B radiation reaching the earth surface; UV-B level has increased at the ground level by 2 to 11% in the southern latitudes and by 0.1% to 7.5% in the northern latitudes. The increase in UV-B radiation has been greatest at 65° N and S latitudes.

A 5% loss of ozone results in a 10% increase in UV-B radiation reaching the earth surface. In humans, the UV-B radiation may cause skin cancer and eye cataracts, and diminished functioning of the human immune system (UNEP, 1998). Elevated levels of UV-B radiation may also decrease the yield of many crop plants and affect aquatic food chain in the oceans, (Fig. 23.24). These effects are described as following.

Skin Cancer and Eye Cataracts

In humans and animals, exposure to UV-B radiation occurs principally through the skin and eyes due to absorption of solar radiation by the chromophores present in the cells. Exposure to UV-B radiation due to stratospheric ozone depletion may cause skin cancers, photoaging, and sunburn, especially in light skinned population. Basal cell carcinoma and squamous cell carcinoma result from the malignant transformation of keratonocytes, the major structural cell of skin (Longstreth et al., 1998). Cutaneous melanoma results from the malignant transformation of melanocytes, which are the skin pigment producing cells. Photoaging is also likely to increase with stratospheric ozone depletion and increase in UV –B radiation. Sunburn is probably the most widely experienced form of acute solar damage to skin due to UV-B radiation. The effects on the skin occur largely through the outermost layer of the skin because UV-B radiation is absorbed by melanin (a principal pigment of skin) and transuronic acid, a molecule found in large amounts in the outermost layer of skin. Two amino acids, tyrosine and tryptophan are largely responsible for UV absorbance of proteins.

The cells and tissues that are exposed to UV-B radiation are associated with the cornea, the iris and the lens. The lens of eye is a strong absorber of UV-B radiation. The link between UV-B exposure and development of eye cataract is well established and may result in increased incidence of eye cataracts due to increased exposure to UV-B radiation.

The Solar UV Index

The solar UV Index (UVI) describes the level of solar UV radiation relevant to human sunburn (erythema). The UVI was originally used in Canada. The maximum value in the South of the country is 10 at midday in the summer, and about 1 at midday in the winter. In other locations and under different conditions the UVI can be higher or lower. In the tropics at sea level the UVI can exceed 16 and peak terrestrial values of 25 can occur at high altitudes. The higher the UVI, the greater the potential for damage, and the less exposure time it takes for harm to occur. For fair-skinned individuals a UVI of more than 10 can cause skin damage (erythema) from an exposure of about 15 minutes. Outside the protective layer of Earth’s atmosphere (altitude > 50 km), the UVI can exceed 300. The combination of total ozone, aerosols, clouds, air-pollution, altitude, surface reflectivity and solar zenith angle (that is determined by the geographical position, season and time of the day) are the main factors resulting in variation in the UV Index. UV levels increase with altitude, and therefore the UVI is higher at mountainous locations (as seen, for example, in the Andes and Tibet). The presence of “patchy clouds” or snow-covered ground can result in larger UV Index values., where UV index of 18 is common in the high altitude desert *Puna de Atacama* in Argentina during January and December, with a maximum of 20 and even more on occasional days. The UV radiation exposure categories (www.unep.org/PDF/Solar_Index_Guide.pdf) are as follows:

Exposure Category	UVI Range
Low	< 2
Moderate	3 to 5
High	6 to 7
Very High	8 to 10
Extreme	11+

Effect on the Human Immune System

The increasing amount of UV-B exposure can damage DNA and also affect the proteins like tyrosine and tryptophan. Excess exposure of UV-B radiation may cause diminished functioning of immune system in human beings. The immune system is affected through langerhans or antigen presenting cells that reside or migrate through the epidermis (Longstreth et al., 1998).

Effect on the Terrestrial Ecosystems

Changes in UV radiation caused by global environmental change can have very important consequences for terrestrial ecosystems. These variations in UV radiation (both UV-B and UV-A) will affect a large range of both natural and cropland ecosystems (Fahey and Hegglin, 2010). The effects are briefly described as follows:

1. A wide range of field studies suggests that increased UV-B radiation reduces terrestrial plant productivity by about 6% (Fahey and Hegglin, 2010). This reduction results from direct damage and increased diversion of plant resources towards protection and acclimation.
2. Region-specific changes in cloud cover and vegetative cover (in response to increased aridity or deforestation) predicted for the coming decades are likely to have large impacts on the levels of UV radiation received by terrestrial organisms.
3. The breakdown of dead plant material and consequently carbon loss to the atmosphere are affected by UV-B radiation.
4. Variations in UV-B radiation caused by climate change and ozone depletion can have large effects on plant interactions with pests, with important implications for food security and food quality.
5. UV-B radiation may improve the quality of food, for example, through increased antioxidant activity, flavour and fibre content. Knowledge gained in this area could be used in the design of agricultural systems that take advantage of these natural plant products to increase nutritional value.
6. Changes could occur in microbial biodiversity with consequences for soil fertility and plant disease, which in turn can alter rates of decay (an important ecosystem process that contributes to soil fertility). Changes in species composition of microbes by UV-B radiation can affect susceptibility of living plants to fungal infections.
7. Increase in levels of ultraviolet radiation may cause dramatic decline in commercial crops such as corn, rice, wheat and soybean. Legumes are particularly sensitive to UV-B radiation.

Effect on the Aquatic Ecosystems

Detrimental effects of solar UV-B radiation have been shown for many aquatic organisms (see Fahey and Hegglin, 2010); the effects are briefly described as follows:

1. UV-B radiation has a greater impact on aquatic organisms in sites polluted by crude oil and heavy metals such as cadmium, selenium or copper.
2. Climate change alters the exposure of aquatic organisms to solar UV radiation by influencing their depth distribution as well as the transparency of the water.
3. Enhanced solar UV-B radiation in conjunction with rising global temperatures may negatively affect seaweeds that have ecologic and economic importance. The vertical distribution of seaweeds in their ecosystem is strongly determined by solar UV radiation.
4. Climate-driven changes in environmental conditions may influence the capacity of protective strategies of aquatic organisms to adapt to solar UV radiation.
5. The rise in atmospheric CO₂ concentrations increases the acidity of the water, making calcified organisms more vulnerable to solar UV-B radiation.
6. UV-B radiation inhibits photosynthesis in most phytoplankton as they may penetrate through 10 m clear open ocean waters. This in turn can affect the whole food chain of organisms that depend on phytoplankton—such as zooplanktons, krill, squids, fish and whales, etc. Studies suggest that productivity of phytoplankton has decreased by 6 to 12% since 1987 in monitored areas of Antarctica Ocean.

7. Enhanced exposure of phytoplankton to UV-B radiation may lead to changes in species composition of phytoplankton community, decrease in the availability of nitrogen and reduced uptake of carbon dioxide in the Antarctica and Arctic oceans (Hader et al., 1998). As phytoplanktons absorb CO₂ from the atmosphere, a decrease in phytoplankton production could possibly increase global warming.
8. Ultraviolet radiation may also inhibit primary production by bacterioplankton as ultraviolet radiation (290–400 nm) from the sun penetrates to significant depths in many marine environments. Studies have shown that both UV-B and the longer-wavelength UV-A affect marine carbon cycling, primarily through effects on phytoplankton and bacterioplankton, which comprise the base of oceanic food webs. While phytoplankton are responsible for carbon fixation, bacterioplankton play a vital role in the mineralization of nutrients and provide a trophic link to higher organisms.

Effect on Materials

Ultraviolet light is harmful to many products, paints, and plastics which deteriorate when exposed to ultraviolet light. Polymers commonly used in solar cells and optoelectronic devices are degraded by solar UV radiation, affecting their service lifetimes and cost. Semi-conducting polymers such as poly (3-hexylthiophene) (P3HT) and poly (3-octothiophene) (P3OT), which are used in active layers of solar cells as well as the plastics used in enclosures of solar panels, degrade on exposure to solar UV radiation, thus limiting their service life.

Upward Trend in Total Ozone over Antarctica

The success of the Montreal Protocol has been demonstrated in controlling the production and consumption of ozone-depleting substances (ODSs). This has protected the ozone layer. The resulting reductions in atmospheric abundances of ODSs also reduced the human influence on climate because all ODSs are greenhouse gases. By protecting both ozone and climate, the Montreal Protocol has provided a *dual benefit* to society and Earth's ecosystems (Fahey and Hegglin, 2010).

A recent analysis of total ozone data from multiple satellite instruments (1979–2009), reported that an upward trend in total ozone over Antarctica since the late 1990s is detectable and is presumably due to reduction in CFCs (Salby et al., 2011; Andrady et al., 2012). This positive trend was identified after removing the effect of circulation changes on ozone. However, significant uncertainties exist in the stability of the satellite instruments over the long periods considered, the combination of measurements from different satellites, and the isolation of effects from circulation. The latter is particularly difficult as changes in circulation may affect the temperature in the stratosphere and the formation of polar stratospheric clouds that are involved in the depletion of ozone. Further confirmation of these findings and/or longer data sets would be valuable (Andrady et al., 2012).

23.8 STRATEGIES FOR PROTECTING STRATOSPHERIC OZONE

A large number of substances are now being used as substitutes for ozone depleting substances that are being phased out under the Montreal Protocol and its amendments. The ozone-friendly products are now manufactured widely in all the industrialized countries as well as many developing countries such as India, China, and Brazil. Some of the current alternatives to CFCs in refrigeration, foam sectors, industry and fire fighting sectors are given in Table 23.5.

Table 23.5. Substitutes of chlorofluorocarbons (based on UNEP, 1998; and Miller, 1997).

Types	Benefits	Consequences
Hydrochlorofluorocarbons (HCFCs – 21,22,124,141b,152a)	Atmosphere residence time (2-20 years) Pose less danger to ozone layer; can be used in aerosol sprays refrigeration, air conditioning, foam and cleaning agents.	Are greenhouse gases, If used in large quantities will deplete ozone; Low to acute inhalation toxicity
Hydrofluorocarbons HFCs -23, 32, 135, 134a, 227ea	Atmosphere residence time (2-20 years) Contains fluorine in HFCs is less effective in ozone depletion Can be used in aerosol sprays refrigeration, air conditioning, foam and cleaning agents.	Are greenhouse gases Some toxicity upon repeated inhalation exposure
Hydrocarbons (Propane and butane)	Cheap and readily available can be used in aerosol sprays refrigeration, air conditioning, foam and cleaning agents.	Can be flammable and poisonous increase ground level air pollution
Ammonia	Simple alternative for refrigerators	Toxic if inhaled, needs careful handling

Refrigeration Sector

In the new refrigeration units, the alternatives that are being used are HFC134a, Hydrocarbons, HFC blends, and Ammonia. Air-cooled air conditioning units have HFC 134a, HFC blends. Automobile air conditioners have HFC blends, HFC-134a. In the new motor vehicles, HFC 134a is an ozone safe refrigerant used in air conditioning.

Foam Sector

For foams, zero ODP alternatives include carbon dioxide, hydrocarbons, HFC-152a or HFC-134a. In some markets, HCFCs are used for rigid thermal insulating foams. HCFCs have significantly lower ozone depletion potential than the CFCs, it is strong greenhouse gas. Therefore, they serve only interim replacements and must be phased out. HFCs are other potential foam blowing substitutes and are being investigated for their greenhouse gas potential or any toxic effects.

Aerosols and Solvents

The alternatives include hydrocarbons such as propane, butane and iso-butane, HCFCs, HFCs. Other ozone friendly, CFC-free products are manual pump sprayers and dry powder inhalers. Alternatives for CFC-113 and methyl chloroform include aqueous and semi-aqueous cleaning hydrocarbons, non-ozone depleting chlorinated solvents, perfluorocarbons and non-solvent cleaning processes.

Fire-fighting Sector

Halons are considered necessary only in a few critical cases such as fire-fighting systems for control rooms, aircraft, etc. Alternative fire-fighting equipment contains carbon dioxide or dry powder. Apart from gaseous alternatives to halons new systems such as water mist technologies and combustion generated aerosols are developed as alternatives to halons fire-fighting equipment.

The currently used ozone friendly products including FCs, HCFCs, HFAs and HFCs are being tested for the toxic effects and global warming potential. It is reported that all the three HFAs show a low level of inhalation toxicity. Upon acute and repeated inhalation exposure in standard toxicological testing protocols, all the three HFCs, -32, -125, and -132a show low toxicity. HCFC-141 is generally characterized by low toxicity, whereas, all the three HCFCs (21, 124, 141b) show low to acute inhalation toxicity.

The atmospheric abundances of the CFCs are showing a decline because of the Montreal Protocol's provisions; their greenhouse-warming contributions will also decline. On the other hand, use of hydrofluorocarbons (HFCs) and HCFCs as substitutes for CFCs would cause the greenhouse-warming contributions of these new compounds to increase. Indeed, global observations of many HFCs and HCFCs, as well as of hydrogen fluoride (HF), confirm that their contributions as a greenhouse gas are currently increasing. Thus the issues of ozone depletion and climate change are interconnected.

23.9 INTERNATIONAL INITIATIVE FOR MITIGATING GLOBAL CHANGE

The longer term challenge of stabilizing the atmospheric concentrations of greenhouse gases requires that global emissions be significantly lowered than today. Risks from potential global warming are significant and international action is needed. International climatic policies are fairly a recent phenomenon and briefly discussed below.

Intergovernmental Panel on Climate Change (IPCC)

In June 1989, the world conference on changing atmosphere in Toronto led to the formation of Intergovernmental Panel on Climate Change (IPCC), which was jointly established by United Nations Environmental Programme (UNEP) and the world Meteorological Organization. Intergovernmental Panel on Climate Change (IPCC) is a consortium of over 2000 scientists and policy analysts from over 100 nations and has produced three major reports the various aspects of the climate change (IPCC, 1990, 1995 and 2001).

The aim of the IPCC was to provide the international Community with technological guidance to deal with the problems of climate change. The initial IPCC reports provided the basis for the discussion at The Second World Climate Conference in November 1991. The work of the IPCC continued through the 1990s with the supplementary assessment in 1992 and a major reassessment in 1995. The Third and Fourth Assessment Reports of the IPCC (2001, 2007) give most comprehensive information on the atmospheric abundance of greenhouse gases and its possible impacts on climate, ecological systems, global carbon cycle and other related issues.

Fourth Assessment Report of IPCC has had major impact in creating public awareness on various aspects of climate change. The Nobel Peace Prize for the year 2007 was awarded to the IPCC jointly with Mr Al Gore, Dr R.K. Pachauri (From Tata Energy and Resources Institute, TERI, New Delhi), and presently Chairman of the IPCC received the prize on behalf of IPCC. While delivering the acceptance speech, Professor Pachauri applauded the pioneering work of thousands of experts and scientists contributing to the climate change science.

The Working Group I contribution to the IPCC's Fifth Assessment Report (AR5) considers new evidence of climate change based on many independent scientific analyses from observations of the climate system, paleoclimate archives, theoretical studies of climate processes and simulations using climate models. Some of the main findings are summarized in Box 23.5.

Box 23.5. Extract reproduced from Working Group I Contribution to the IPCC Fifth Assessment Report

Warming of the climate system is unequivocal, and since the 1950s, many of the observed changes are unprecedented over decades to millennia. The atmosphere and ocean have warmed, the amounts of snow and ice have diminished, sea-level has risen, and the concentrations of greenhouse gases have increased. The atmospheric concentrations of carbon dioxide (CO₂), methane, and nitrous oxide have increased to levels unprecedented in at least the last 800,000 years. Human influence on the climate system is clear. This is evident from the increasing greenhouse gas concentrations in the atmosphere, positive radiative forcing, observed warming, and understanding of the climate system.

Continued emissions of greenhouse gases will cause further warming and changes in all components of the climate system. Global surface temperature change for the end of the 21st century is *likely* to exceed 1.5°C relative to 1850 to 1900. It is *virtually certain* that there will be more frequent hot and fewer cold temperature extremes over most land areas on daily and seasonal time scales as global mean temperatures increase. It is *very likely* that heat waves will occur with a higher frequency and duration. Occasional cold winter extremes will continue to occur. The high latitudes and the equatorial Pacific Ocean are *likely* to experience an increase in annual mean precipitation by the end of this century under the RCP 8.5 scenario. In many mid-latitude and subtropical dry regions, mean precipitation will *likely* decrease, while in many mid-latitude wet regions, mean precipitation will *likely* increase by the end of this century. Extreme precipitation events over most of the mid-latitude land masses and over wet tropical regions will *very likely* become more intense and more frequent by the end of this century, as global mean surface temperature increases. While monsoon winds are *likely* to weaken, monsoon precipitation is *likely* to intensify due to the increase in atmospheric moisture. Monsoon onset dates are *likely* to become earlier or not to change much. Monsoon retreat dates will *likely* be delayed, resulting in lengthening of the monsoon season in many regions. Global mean sea-level rise for 2081–2100 relative to 1986–2005 will *likely* be in the ranges of 0.26 to 0.55 m, 0.32 to 0.63 m, 0.33 to 0.63 m or 0.45 to 0.82 m for different scenarios. The rise by the year 2100 is 0.52 to 0.98 m, with a rate during 2081–2100 of 8 to 16 mm yr⁻¹. Thermal expansion accounts for 30 to 55% of 21st century global mean sea-level rise, and glaciers for 15 to 35%. Climate change will affect carbon cycle processes in a way that will exacerbate the increase of CO₂ in the atmosphere. Further uptake of carbon by the ocean will increase ocean acidification. A large fraction of anthropogenic climate change resulting from CO₂ emissions is irreversible on a multi-century to millennial time scale, except in the case of a large net removal of CO₂ from the atmosphere over a sustained period.

Source: *Climate Change 2013: The Physical Science Basis Summary for Policymakers*

(http://www.climatechange2013.org/images/uploads/WGIAR5-PM_Approved27Sep2013.pdf)

United Nations Framework Convention on Climate Change

The United Nations Conference on Environment and Development (UNCED, Earth Summit), held at Rio de Janeiro, Brazil in 1992, established the principles for reducing greenhouse gas emission and adopted the framework convention for climate change. The United Nations framework convention on climate change (UNFCCC) came in March 1994 for full discussion. The objective of the convention is to achieve stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system. The second conference of parties to the UN climate change convention has led to the establishment of Global Environmental Fund. It was aimed to support projects in developing countries to prepare inventory of sources and sinks of greenhouse gases.

There are now 195 parties to the convention taking part in climate change negotiations. All parties to the UNFCCC are represented at the Conference of Parties (COP) at which they review the implementation of the convention and any other legal instruments that the COP adopts and take decisions to promote the effective implementation of the convention.

At the UN Climate Change Conference in Doha, Qatar (COP18/CMP8) during December, 2012, governments have taken the next essential step in the global response to climate change. Countries launched a new commitment period under the Kyoto Protocol, agreed to a firm timetable to adopt a universal climate agreement by 2015 and agreed to raise necessary ambition to respond to climate change. They also endorsed the completion of new institutions and agreed to ways and means to deliver scaled-up climate finance and technology to developing countries.

Kyoto Protocol

The third session of the conference of parties of the United Nations framework convention on climate change took place in Kyoto in December 1997 and approved the *Kyoto Protocol*, specifying the commitments of different countries to mitigate climate change. This protocol requires countries to take appropriate measures to reduce their overall greenhouse gas emissions to a level at least 5% below the 1990 level by the commitment period 2008-12 (Bolin, 1998). Methane and nitrous oxide reductions are to be from 1990 levels. For the three long-lived industrial gases—the hydrofluorocarbons, perfluorocarbons and sulfur hexafluoride, the baseline can be 1990-95.

The Bonn conference held during July, 2001 has made the Kyoto Protocol rectifiable. So far 37 countries have ratified the Kyoto Protocol. The seventh session of the Conference of Parties of Climate Change Convention (COP-7) has been hosted by the Government of Morocco in Marrakech during November 2001. The eighth conference of parties (COP-8) meeting on climate change was held in New Delhi in October 2002. During these two meetings, some progress was towards implementing the Framework Convention on Climate Change. As of June 2013, there are 192 parties to the *Kyoto Protocol* to the United Nations Framework Convention on Climate Change. This total includes 191 states (189 United Nations member states as well as the Cook Islands and Niue) and one supranational union (the European Union) aimed at combating global warming.

The Kyoto Protocol, as the only existing and binding agreement under which developed countries commit to cutting greenhouse gases, has been amended so that it will continue as of 1 January 2013.

Montreal Protocol on Substances that Deplete the Ozone Layer

The Vienna Convention for the Protection of the Ozone Layer and its Montreal Protocol on Substances that Deplete the Ozone Layer are dedicated to the protection of the earth's ozone layer. With 197 parties, they are the most widely ratified treaties in United Nations history, and have, to date, enabled reductions of over 97% of all global consumption of controlled ozone depleting substances (measured in ODP tonnes).

Montreal protocol is a series of international agreements on the reduction and elimination of CFC's and other ozone depleting substances. In 1987, 27 industrialized countries signed the Montreal Protocol, a landmark international agreement to protect the stratospheric ozone by agreeing to limit the production and use of ozone-depleting substances, phasing out of ozone-depleting substances and helping the developing countries to implement use of alternatives to CFCs. In the Beijing meeting agreement during 1999, it has been agreed to a phase out of all types of ozone depleting substances including bromochloromethane, methyl bromide and HCFCs in the near future within 20 to 40 years times. However, due to long residence of various ozone depleting substances in the atmosphere, ozone layer will take a long time (around 50 years or so) to recover.

23.10 GLOBAL ENVIRONMENTAL CHANGE RESEARCH PROGRAMMES

The International Council for Science (ICSU) is the common scientific sponsor of the four international global change programme. These are the International Geosphere-Biosphere Programme (IGBP) International Human Dimensions Programme on Global Environmental Change (IHDP) World Climate Change Research Programme (WCRP) and DIVERSITAS.

International Geosphere-Biosphere Programme (IGBP), began in 1987, is one of the world's leading scientific research programmes studying changes to the global environment (IGBP, 2001). The overall goal of IGBP is to describe and understand the interactive physical, chemical and biological processes that regulate the total Earth system, the unique environment that it provides for life, the changes that are occurring in this system, and the manner in which they are influenced by human activities. IGBP's vision is to provide essential scientific leadership and knowledge of the Earth system to help guide society onto a sustainable pathway during rapid global change.

International Human Dimensions Programme on Global Environmental Change (IHDP) is an international, interdisciplinary non-governmental research programme situated at the interface between the social and natural sciences. The aim is to understand the factors, which lead to modifications of the earth's physical and biological systems and their effects on human quality of life and sustainable development in different parts of the world. Sustainability science has a major focus on management strategies for the global sustainability, appropriate methodologies and institutional needs; political agenda of sustainable development and the nature society interactions (Kate et al., 2001).

The World Climate Change Research Programme (WCRP) encompasses studies of the global atmosphere, world oceans, cryosphere (including continental ice sheets, ice caps, glaciers, sea ice and snow cover), and the land surface and groundwater systems, which together constitute the Earth's physical climate system. The goal of DIVERSITAS is to unify the various approaches to the study of biodiversity. The aim is to understand the role of biodiversity in ecosystem functioning and the socioeconomic aspects of the conservation of biodiversity.

In the next ten years (2010-20), IGBP, working with partners, will provide the knowledge, expertise and coordination to identify and assess risks posed to society and ecosystems by major changes in the Earth's biological, chemical and physical cycles and processes, and communicate this to society.

23.11 INDIA'S INITIATIVE FOR MITIGATING CLIMATE CHANGE

India has signed and ratified the United Nations Framework Convention on Climate Change (UNFCCC). The Ministry of Environment and Forests is the nodal agency for climate change issues in India. The ministry has working groups on the UNFCCC and the Kyoto Protocol. Current initiatives in India for reducing the greenhouse gas (GHG) emissions involve the estimation of the sectoral GHG emissions, National inventory of GHG sources and sinks, transfer of environmentally sound technology, conservation of forests and biodiversity and coastal zone management besides the involvement several governmental and non-governmental organizations in climate change research.

India has taken numerous response measures to balance development plans with economic and environmental concerns. Various measures are being taken in coal, oil and gas sectors, renewable energy, transport and residential, power and industry sectors to lower the emission of greenhouse gases. The Ministry of Energy has a very active programme to promote the use of renewable energy from solar power, wind energy, biogas generation, and increasing the use of small hydropower. In the transport sector, a major initiative has been the upgradation of vehicular emission norms by introducing the Bharat Stage II motor cars and passenger vehicles and the CNG run vehicles in the National Capital region of Delhi.

The major energy consuming industries in India are steel, cement, caustic soda, aluminium and electric power generation. In these industries as well as in the power sector, promotion of energy efficient practices and equipment are being introduced through technology upgradation and energy conservation measures. The National Forest Policy Action Programme has been formulated for sustainable forest development. Large scale afforestation programmes are being undertaken under various schemes for regenerating degrading lands, and ecologically fragile areas.

Carbon management in multifunctional forests keeping in view the socioeconomic needs of the people and a proper valuation of ecosystem services could be effective climate change mitigation strategy particularly in the Indian context (Pandey, 2002). The major focus has to be on restoration of degraded lands, expansion of agroforestry, conservation agriculture and restoration of wastelands for the sequestration of carbon dioxide. With reforestation and other suitable measures of ecological rehabilitation, the organic carbon pools of soils and vegetation could be increased. Soils and vegetation represent the potential sinks for the CO₂ and reforestation can be used as a possible means of mitigating global warming. Afforestation of wastelands can sequester large amounts of carbon in vegetation and soils. Besides carbon storage, afforestation will be improving vegetation cover and provide a useful resource of environmentally responsive wood products. The various initiatives for managing climate change in India are summarized as follows:

1. To reduce the emissions intensity of greenhouse gases by 20-25% by the year 2020 on a 2005 reference level.
2. Diversification in energy sector by developing clean energy technologies.
3. Green India Mission to increase the quality and quantity of forest cover in 10 million hectares of degraded lands .
4. Comprehensive Climate Change Assessment.
5. Regional partnerships with neighbours and other countries to deal with climate change.

Sensitizing Youths

For reconstructing economies and revive growth, it is the need of the time to create more jobs for the youth. All future jobs will be green because of increasing problems due to environmental degradation. Young people had a presence at the Rio summit in 1992. Even since then youths are finding way to attend United Nations Climate Conferences to make their voices heard.

Earth hour was observed in 160 countries in 2009. Many young people come forward to speak about climate change. Earth hour presents many opportunities—to be heard, to take action and to be connected to millions of people all over world.

Many youths attended the crucial climate change negotiation in Copenhagen, Denmark in December 2009. The energy of youth can accelerate the environmental movement faster than any other force. India is a land of the youth with more than sixty per cent of the population of 1.4 billion falling under the age of thirty-five. The opportunities for education in science and technology are growing. Thus, it is important to take initiatives that promote low carbon economies and lifestyles by increasing use of renewable energy, afforestation and new tree plantations on wastelands, restoring degraded ecosystems, managing the wastes and conserving our biodiversity.

Biological Invasion: Process, Impacts and Prevention

The alien invasive species contribute enormously to global change because they harm human health; pose a threat to biological diversity, and cause enormous economic losses (Vitousek et al., 1997a; McNeely, 2001). Biological invasions now operate on a global scale and will undergo rapid increase in this century due to interactions with other global changes such as globalization of markets, explosive rises in global trade, travel, tourism and exchange of goods (Hobbs, 2000). The human caused breakdown of geographical barriers to species dispersal is the primary cause of species invasion in the new environments. Then they become pests in their naturalized homes. The spread of invasive species has been made worse by climate change, and is now one of the greatest threats to the ecological and economic well-being of our planet. These invasions now pose an enormous challenge to the conservation of biodiversity and natural resources (MEA, 2005b; TEEB, 2010).

Invasive species may lead to loss of species diversity, increase erosion and affect the physical structure of the habitat. At the continental and global scale, the invasive species diminish the regional distinctiveness of flora and fauna (Vitousek et al., 1997) through homogenization (redistribution of species between and among continents) and depauperization (loss of native taxa due to predation, competition and disease) (McNeely, 2001). The Convention on Biological Diversity (CBD) recognizes invasive species as a global problem. After habitat destruction, the exotic species are now considered the second greatest threat to biological diversity (Simberloff, 1995). Invasive species can threaten biological diversity by reducing genetic variation and eroding gene pools, through the extinction of endemic species, and by altering habitat and ecosystem functioning (Hulme et al., 2008). The cost of damage caused by invasive species is estimated at 1.5 trillion dollars per annum in environment and economic damage, which is close to 5% of global GDP (GISP; Global Strategy, 2008 – 2010).

24.1 NATURE AND STATUS OF INVASIVE SPECIES

Invasive species are those that occur outside their natural range, spread rapidly and cause harm to other species, communities, or entire ecosystems and to human well-being. They are known by several names such as non-natives, introduced, non-indigenous, exotic and foreign species. However, all non-natives are not harmful or invasive. “Invasive alien species” (IAS) are those subset of alien species of plants, animals, fungi, viruses, and bacteria whose establishment and spread threatens ecosystems, habitats, or species with economic or environmental significance (McNeely et al., 2001). According to Richardson et al. (2011), invasive species are alien species that sustain self-replacing populations over several life cycles, produce reproductive offspring, often in very large numbers at considerable distances from the parent and/or site of introduction, and have the potential to spread over long distances, possibly leading to adverse effects on invaded habitat (‘impacts’).

Invasive species can be plant, animals, and microorganisms and they are found almost in all types of ecosystems. At least 10% of world's vascular plants (300,000) have the potential to invade other ecosystems and affect native biota in a direct or indirect way. The 100 worst invasive species in the global database are given in Table 24.1. The yellow crazy ant, the small Indian Mongoose, the Nile Perch, Strawberry guava and the South American water hyacinth are among the worst invasive species (Lowe et al., 2000).

Table 24.1. Some examples of the world's worst invasive alien species (based on Lowe et al., 2000).

MICRO-ORGANISM		LAND INVERTEBRATE (CONTINUED)	
Avian malaria	(<i>Plasmodium relictum</i>)	Argentine ant	<i>Linepithema humile</i>
Banana bunchy top virus	(<i>Banana bunchy top virus</i>)	Asian long horned beetle	<i>Anoplophora glabripennis</i>
Chestnut blight	(<i>Cryphonectria parasitica</i>)	Asian tiger mosquito	<i>Aedes albopictus</i>
Crayfish plague	(<i>Aphanomyces astaci</i>)	Big-headed ant	<i>Pheidole megacephala</i>
Dutch elm disease	(<i>Ophiostoma ulmi</i>)	Common malaria mosquito	<i>Anopheles quadrimaculatus</i>
Frog chytrid fungus	(<i>Batrachochytrium dendrobatidis</i>)	Common wasp	(<i>Vespula vulgaris</i>)
Phytophthora root rot	(<i>Phytophthora cinnamomi</i>)	Crazy ant	(<i>Anoplolepis gracilipes</i>)
Rinderpest virus	(<i>Rinderpest virus</i>)	Cypress aphid	(<i>Cinara cupressi</i>)
AQUATIC PLANT		Flatworm	(<i>Platydemus manokwari</i>)
Caulerpa seaweed	(<i>Caulerpa taxifolia</i>)	Formosan subterranean termite	(<i>Coptotermes formosanus shiraki</i>)
Common cord-grass	(<i>Spartina anglica</i>)	Giant African snail	(<i>Achatina fulica</i>)
Wakame seaweed	(<i>Undaria pinnatifida</i>)		
Water hyacinth	(<i>Eichhornia crassipes</i>)	Golden apple snail	(<i>Pomacea canaliculata</i>)
LAND PLANT		Gypsy moth	(<i>Lymantria dispar</i>)
African tulip tree	(<i>Spathodea campanulata</i>)	Khapra beetle	(<i>Trogoderma granarium</i>)
Black wattle	(<i>Acacia mearnsii</i>)	Little fire ant	(<i>Wasmannia auropunctata</i>)
Brazilian pepper tree	(<i>Schinus terebinthifolius</i>)	Red imported fire ant	(<i>Solenopsis invicta</i>)
Cogon grass	(<i>Imperata cylindrica</i>)	Sweet potato whitefly	(<i>Bemisia tabaci</i>)
Cluster pine	(<i>Pinus pinaster</i>)	AMPHIBIAN	
Erect pricklypear	(<i>Opuntia stricta</i>)	Bullfrog	(<i>Rana catesbeiana</i>)
Fire tree	(<i>Myrica faya</i>)	Cane toad	(<i>Bufo marinus</i>)
Giant reed	(<i>Arundo donax</i>)	Caribbean tree frog	(<i>Eleutherodactylus coqui</i>)
Gorse	(<i>Ulex europaeus</i>)	FISH	
Hiptage	(<i>Hiptage benghalensis</i>)	Brown trout	(<i>Salmo trutta</i>)
Japanese knotweed	(<i>Polygonum cuspidatum</i>)	Carp	(<i>Cyprinus carpio</i>)

Kahili ginger	(<i>Hedychium ardnrianum</i>)	Large-mouthbass	(<i>Micropterus salmoides</i>)
Koster's curse	(<i>Clidemia hirta</i>)	Mozambique tilapia	(<i>Oreochromis mossambicus</i>)
Kudzu	(<i>Pueraria Montana</i>)	Nile perch	(<i>Lates niloticus</i>)
Lantana	(<i>Lantana camara</i>)	Rainbow trout	(<i>Oncorhynchus mykiss</i>)
Leafy spurge	(<i>Euphorbia esula</i>)	Walking catfish	(<i>Clarias batrachus</i>)
Leucaena	(<i>Leucaena leucocephala</i>)	Western mosquito fish	(<i>Gambusia affinis</i>)
Melaleuca	(<i>Melaleuca quinquenervia</i>)	BIRD	
Mesquite	(<i>Prosopis glandulosa</i>)	Indian myna bird	(<i>Acridotheres tristis</i>)
Miconia	(<i>Miconia calvescens</i>)	Red-vented bulbul	(<i>Pycnonotus cafer</i>)
Mile-a-minute weed	(<i>Mikania micrantha</i>)	Starling	(<i>Sturnus vulgaris</i>)
Mimosa	(<i>Mimosa pigra</i>)	REPTILE	
Privet	(<i>Ligustrum robustum</i>)	Brown tree snake	(<i>Boiga irregularis</i>)
Pumpwood	(<i>Cecropia peltata</i>)	Red-eared slider	(<i>Trachemys scripta</i>)
Purple lossestrife	(<i>Lythrum salicaria</i>)	MAMMAL	
Quinine tree	(<i>Cinchona pubescens</i>)	Brush-tail possum	(<i>Trichosurus vulpecula</i>)
Shoebuttan ardisia	(<i>Ardisia elliptica</i>)	Domestic cat	(<i>Felis catus</i>)
Siam weed	(<i>Chromolaena odorata</i>)	Goat	(<i>Capra hircus</i>)
Strawberry guava	(<i>Psidium cattleianum</i>)	Grey squirrel	(<i>Sciurus caralinesis</i>)
Tamarisk	(<i>Tamarix ramosissima</i>)	Macaque monkey	(<i>Macaca fascicularis</i>)
Wedelia	(<i>Wedelia trilobata</i>)	Mouse	(<i>Mus musculus</i>)
Yellow Himalayan raspberry	(<i>Rubus ellipticus</i>)	Nutria	(<i>Myocastor coypus</i>)
AQUATIC INVERTEBRATE		Pig	(<i>Sus scrofa</i>)
Chinese mitten crab	(<i>Eriocheir sinensis</i>)	Rabbit	(<i>Oryctolagus cuniculus</i>)
Comb jelly	(<i>Mnemiopsis leidyi</i>)	Red deer	(<i>Cervus elaphus</i>)
Green crab	(<i>Carcinus maenas</i>)	Red fox	(<i>Vulpes vulpes</i>)
Marine clam	(<i>Potamocorbula amurensis</i>)	Ship rat	(<i>Rattus rattus</i>)
Mediterranean mussel	(<i>Mytilus galloprovincialis</i>)	Small Indian mongoose	(<i>Herpestes javanicus</i>)
Northern Pacific sea star	(<i>Asterias amurensis</i>)	Stoat	(<i>Mustela erminea</i>)
Spiny water flea	(<i>Cercopagis pengoi</i>)		
Zebra mussel	(<i>Dreissena polymorpha</i>)		

Plants, mammals and insects constitute the most common type of invasive alien species in terrestrial environment, where grasses and trees are a particularly widespread problem of bioinvasions. Invertebrates (mollusks and crustaceans) and algae are predominant alien species in marine and coastal areas. Fish, aquatic weeds and other invertebrates are the frequent invasive alien species of fresh waters.

Although large number of exotics have become naturalized in India and have affected the distribution of native flora to some extent, only a few have conspicuously altered the vegetation patterns of the country. *Chromolaena odorata* (*Eupatorium odoratum*), *E. adenophorum*, *Lantana camara*, *Mikania micrantha*, *Mimosa invisa* and *Parthenium hysterophorus* among terrestrial exotics, and *Eichhornia crassipes* and *Pistia stratiotes* among aquatics, have posed serious threat to the native flora (Singh et al., 2001). Khurro et al. (2011) have presented an inventory of the alien flora of India, to show that there are 1,599 invasive species, belonging to 842 genera in 161 families, constituting 8.5% of the total Indian vascular flora. According to these workers, the three most species-rich families to which invasive species belong are Asteraceae (134 spp.), Papilionaceae (114 spp.) and Poaceae (106 spp.), and the three largest genera are *Eucalyptus* (25 spp.), *Ipomoea* (22 spp.), and *Senna* (21 spp.). Biogeographically, more than one-third (35%) of the alien flora in India has its native ranges in South America, followed by Asia (21%), Africa (20%), Europe (11%), Australia (8%), North America (4%); and cryptogenic (1%). When 869 species (cultivated 812 spp., casual 57 spp.) were excluded from the analysis, 31% (225 spp.) of the total naturalized species (730 spp.) belonged to the invasive category (Khurroo et al., 2011).

24.2 INVASION PROCESS AND HYPOTHESES RELATED TO SUCCESS OF INVASION

The invasion is not necessarily a smooth process; major episodes of population expansion may be punctuated by uneventful periods. The statistical rule holds that 1 in 10 of imported species becomes introduced (Williamson, 1993), 1 in 10 of those introduced becomes established, and 1 in 10 of those established becomes a pest. Of the three transitional stages namely “escaping”, “establishing” and “becoming a pest”, each has a probability of $\approx 10\%$ and is known as 10’s rule (Williamson and Fitter, 1996).

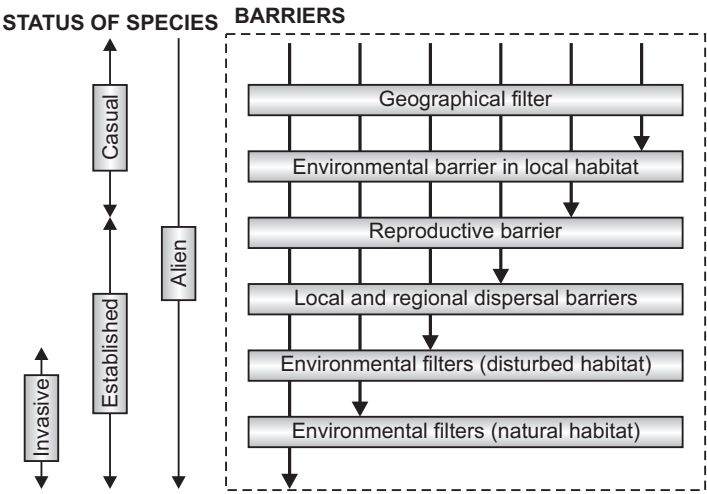


Fig. 24.1. Barriers or filters to be overcome by invasive species (based on Richardson et al., 2000; see McNeely et al., 2001). ‘Established’ is also taken as ‘Naturalized’.

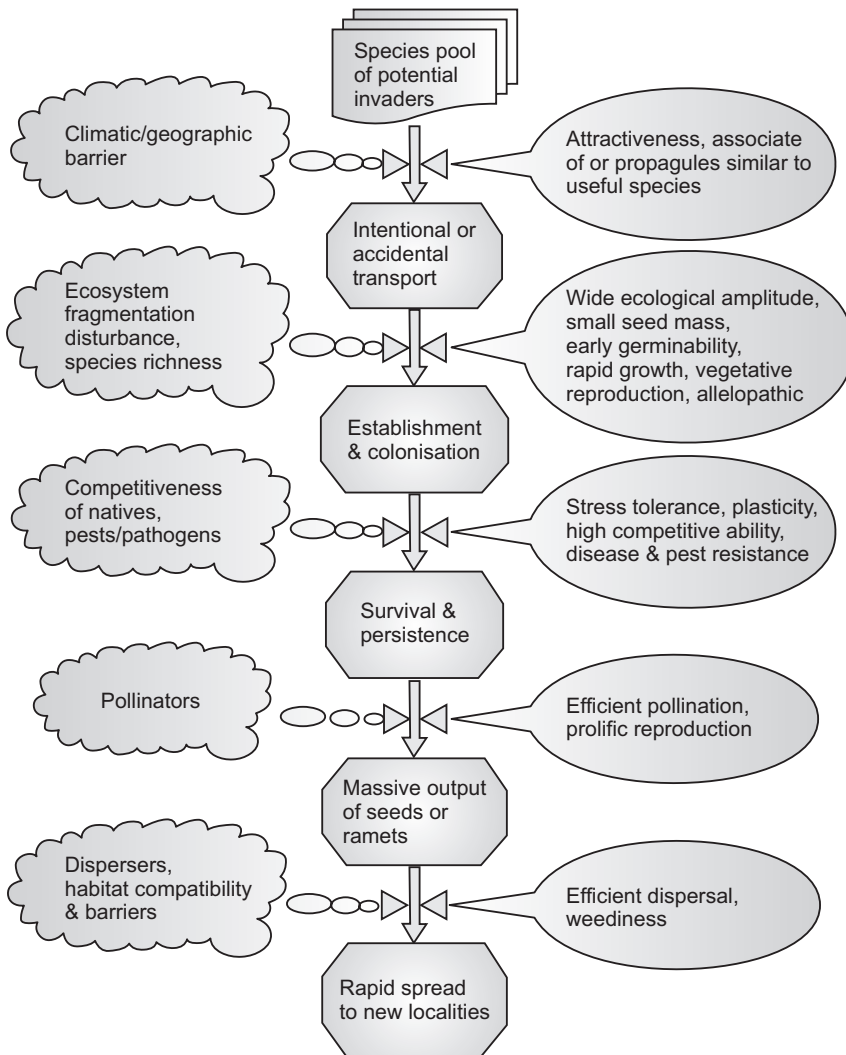


Fig. 24.2. The process of biological invasion. Stages are shown in the centre, environmental determinants on the left, and traits which help the species to cross various barriers on the right (based on Mantri et al., 2002).

Fig. 24.1 shows various barriers (filters) that a plant must overcome to become alien, casual, naturalized (established) or invasive. The process of invasion as shown in Fig. 24.2 can be summarized as:

- (i) For the invasion to take place, a taxon overcomes various geographic and climate barriers, through intentional or accidental transport.
- (ii) Establishment as a result of successful reproduction overcoming all the barriers.
- (iii) Survival and persistence overcoming the biotic and local environmental barriers.
- (iv) Expanding the size of population so as to spread to new regions overcoming the dispersal and habitat barriers.
- (v) Adapting to abiotic environment and biota in the new area.

Catford et al., (2009) recognize the following stages in the process of invasion:

1. Transport: Movement of plants or plant propagules to new location
2. Introduction: Arrival of plant or plant propagules into new location
3. Colonization: Survival of introduced plants
4. Naturalization: Survival and reproduction enabling pioneer population to be self-sustaining
5. Spread: Dispersal of propagules and spread of populations outside of area where first introduced.

The alien species so established can become invasive immediately or after a lag period of several years.

Several hypotheses have been proposed to explain the success of species as invaders but most of these can explain the success of only some invasives to some degree in some circumstances (Catford et al., 2009). Multiple factors often contribute to invasions, and therefore there can be interactions and feedbacks among the various hypotheses. Further, different hypotheses may operate at different stages of the invasion process, and therefore, the invasiveness of invaders can only be explained when a set of hypotheses is applied in complementarity. In fact various stages in the invasion process are parts of an introduction-naturalization-invasion continuum. Some of the hypotheses are briefly described below.

Propagule Pressure (PP) and Global Competition (GC) Hypotheses: Large supply and high frequency of introduction of plant propagules increase chance of successful invasion due to high genetic diversity, continual supply of propagules, and higher probability of reaching a favourable environment. Of special importance are the quantity, quality, composition, and rate of the supply of propagules. With an increasing number of species introduced, the higher the likelihood that a competitive species will be in the invading species pool (GC).

Empty Niche Hypothesis (ENH): Due to a limited indigenous species pool, the recipient, community and ecosystem are unsaturated so invaders can use the spare resources and occupy the unused niches (i.e. there is room for the invaders).

Limiting Similarity Hypothesis (LSH): Successful invaders are functionally distinct (have novel traits) from species in the recipient community, so encounter minimal competition and can fill an empty niche. Ability to fix nitrogen (e.g., soil–biota mutualisms) is an example of a novel trait. Functionally distinct invaders have high per capita impact on ecosystems being invaded.

Enemy Release Hypothesis (ERH) and Evolution of Increased Competitive Ability (EICA): Upon entry into a new range, invader loses its natural enemies (herbivores, pathogens) that limit its population size in its home range. This release from enemies results in an increase of population size in the introduced range. When species lose enemies that they have defended against in their native range, resources previously used for defense are reallocated to growth and reproduction, thereby facilitating invasion due to the evolution of increased competitive ability (EICA). Even partial release (reduction of number of enemies) from enemies can help the invasion process.

Novel Weapons Hypothesis (NWH): Novel-weapons hypothesis poses that some invaders possess allelopathic, anti-herbivore, or antimicrobial chemicals to which native organisms have not adapted, and that these novel compounds provide invaders with direct or indirect competitive advantages and thus these novel chemical weapons, enhance the invader's competitive ability and success in the new range. The NWH was first proposed in the context of allelopathy as a potential mechanism for the success of *Centaurea diffusa* as an invader in North America (see Callaway et al., 2004 a; Inderjit et al., 2011 a). Some invasive plant species appear to strongly suppress neighbours in their nonnative ranges but much less so in their native range, using volatile organic compounds. In controlled experiments, Inderjit et al., (2011 b) found that volatiles from *Ageratina*

adenophora (syn. *Eupatorium adenophorum*), an aggressive Neotropical invader, and litter caused higher mortality of species native to India and China, but not of species native to Mexico.

Disturbance (DS) and Increased Resource Availability (IRA)/Fluctuating Resource Hypothesis (FRH): Species require resources for colonization and establishment so an increase in resource levels provides opportunity for invasion. These hypotheses assume that resources are fully utilized under 'normal' conditions by the native plants. Resource levels increase by either an increase in supply (e.g., abiotic disturbance, eutrophication) or a decrease in resource use (e.g., die back of resident plants). Subsequent to disturbance the plant community becomes more susceptible, as there is an increase in the amount of unused resources. Disturbance events increase resource availability and reset succession, giving invading species an equal chance of success at colonization and establishment. Disturbance is crucial for establishment and is thus influential in mediating naturalization, whereas plentiful resources are needed for a species to become abundant, corresponding with invasion (Richardson and Pyšek, 2012).

Adaptation (ADP) and Phenotypic Plasticity Hypotheses (PPH): Pre-adaptation of invader to ecosystem conditions of the introduced range, or post-introduction adaptation, enable it to be successful in the new range because of its specialization and associated competitive ability. For example, the tetraploid populations of *Centaurea maculosa* in the native European range of the species are pre-adapted in the sense that they tolerate a wide range of climates and therefore have succeeded as invasives when introduced to North America. New habitat challenges may also cause the expression of potentially beneficial traits through phenotypic plasticity. Thus, being plastic may allow a population of invaders to quickly adapt to the new environment and to maintain higher fitness relative to native taxa.

Local Pathogen Hypothesis (LPH): Invader accumulates generalist pathogens while remaining less affected than the native plant species, thus enhancing plant invasiveness. For example, accumulation of local pathogens contributes to the invasive success of *Ammophila arenaria* in North America (Inderjit and van der Putten, 2010).

Soil Communities Hypothesis (SCH): Inderjit and van der Putten (2010) have argued that soil communities contribute to the invasion success of an exotic plant by (i) accumulating native soil pathogens in the presence of an exotic plant, (ii) disrupting mutualistic associations between native mutualistic symbionts and seedlings of native plants, (iii) degrading innocuous novel chemicals released by exotic plants into toxic by-products, and (iv) releasing nutrients from exotic litter.

Genomic Attributes Hypothesis (GAH): Having multiple sets of chromosomes, i.e. polyploids (such as *Parthenium hysterophorus*), is often associated with a plant's vigour and the ability to adapt to different environments, enabling the plant to be more competitive and potentially invasive in a new environment (Pandit et al., 2011). Many widespread, successful plant invaders are recently formed allopolyploid hybrids (Ellstrand and Schierenbeck, 2000).

Ideal Weed Hypothesis (IWH): Life history, weedy characteristics and traits of the invading species facilitate invasion by enabling them to outcompete indigenous species. Some traits that have been correlated with invasiveness include ruderal life history, small seed size, high genotypic and phenotypic plasticity, rapid growth, high and early fecundity and fertility.

Faunal Interaction Hypothesis (FIH): Many introduced plants rely on mutualism with local biota in the new range e.g. animal-mediated pollination and seed dispersal.

Successful invasion in the new environment is facilitated by the absence of natural controls such as diseases, parasites, interaction with specific soil microbes and predators. The native species do not have adaptations to resist competition from introduced species; invasion in combination with other types of disturbances can weaken their competitive ability. Observations and experimental studies indicate that invasive species are less parasitized than native populations in the introduced range (Mitchell and Power, 2003; Torchin et al., 2003). As many as 473 plant species naturalized

to the United States from Europe, were tested for diseases; on the average 84% fewer fungi and 24% fewer virus pathogens infected each plant species in its naturalized range as compared to its native range (Mitchell and Power, 2003). The number of parasites found in native populations is twice greater than that found in exotic populations of molluscs, crustaceans, fishes, birds, mammals, amphibians and reptiles (Torchin et al., 2003).

24.3 CHARACTERISTICS OF INVASIVE SPECIES

Generally, the end result of invasion depends on (i) attributes of invasive species, (ii) vulnerability of the invaded systems, and (iii) complexity of natural systems. Similar ecological or climatic conditions in native and new locations result in pre-adaptation of invasive species; species with specialized pollinators are not generally invasive; species can establish and colonise the new habitat if it has appropriate traits to tolerate the range of abiotic and biotic conditions of the new habitat, and it must have efficient pollination mechanisms, prolific reproduction and efficient dispersal mechanisms for spread in newer territories.

The plant characteristics which are indicative of potential invasiveness of species are : (i) high input of viable propagules, (ii) short (<2 years) development time to reproductive maturity, (iii) seeds or other reproductive units with prolonged (>5 years) period of dormancy/residency in seed bank, (iv) high rate of aerial or subterranean biomass production, (v) dense and spreading foliage/canopy, (vi) efficient long distance (>1 km) dispersal capabilities, (vii) presence of inter specific allelopathic properties and/or absence of intraspecific allelopathic properties, (viii) successful colonizer of disturbed or bare ground, (ix) reproductive strategy that facilitates survival in fire prone environments, and (x) broad distribution over a wide range due to distinct climatic types or ecotypes. It has been argued that species which are not pre-adapted to the new environment by evolution in similar climates in their source regions are less likely to invade the new environment also those species which do not develop adaptive characters under unfavourable conditions get excluded by the native species (see Mantri et al., 2002).

Some of the biological attributes associated with invasive plants are summarized below (see also Sharma et al., 2005):

- (i) **Fitness homeostasis:** The ability of an individual or population to maintain relatively constant fitness over a range of environment is fitness homeostasis, and most likely contributes to the invasive potential of an alien species.
- (ii) **Seed size and seed weight:** Small seed is also associated with large seed production as in *Parthenium hysterophorus*. Dispersal ability also contributes to invasiveness according to Rejmanek and Richardson (1996), small seed mass (<50mg) short juvenile period (<10 yrs) and short interval between large seed crops (1-4 yrs), are associated with invasiveness of woody species in disturbed landscapes.
- (iii) **Animal dispersal:** Vertebrate dispersal is responsible for the success of many woody invaders in disturbed as well as undisturbed habitats. For example, seeds of *Lantana* are widely dispersed, predominantly by fruit-eating birds, sheep, goats, cattle, foxes, jackals, and monkey, leading to its spread.
- (iv) **Geographical range:** One of the likely predictors of species invasiveness is the size of its native geographical range. The propagules of the species, having wide spread distribution have a high probability of transport to other countries or continents. A wider geographical range also means wider ecological amplitude and the possibility of population differentiation resulting into ecotypes adapted to different habitats within the geographical range.
- (v) **Alternative mode of reproduction:** Vegetative reproduction is responsible for increased

habitat compatibility and therefore, for successful invasion. Vegetative reproduction is particularly important for dispersal in aquatic habitats. The invasiveness of *Eichhornia crassipes* is mainly attributed to its free-floating life form and asexual reproduction by stolons. A combination of sexual and vegetative reproduction contributes to the invasive potential, e.g., *Mikania micrantha*, *Lantana camara*.

- (vi) **Competition with native species:** Alien species belonging to exotic genera (and therefore possessing traits different from those of resident species) are more likely to be invasive than the alien species with native congeners. An exotic species can also outcompete its native congener. For example, the alien *Agropyron desertosum* outcompetes the natives *A. spicatu* due to more efficient use of soil water in the Great Basin in California.
- (vii) **Allelopathy:** Allelopathy is one of the several attributes of a plant permitting it to invade and establish in a new ecosystem. For example, allelopathic properties of *Parthenium hysterophorus* and *Lantana camara* inhibit the germination and growth of other plant species. The potentiality of *Eupatorium riparium* and *E. adenophorum* to dominate other plant species in Meghalaya has been attributed to their allelopathic properties (Tripathi et al., 1981). Volatile organic compounds from leaf litter of *Ageratina adenophora* are associated with invasiveness of a Neotropical weed in Asia (Inderjit et al., 2011).
- (viii) **Phenotypic plasticity:** It is the ability of a genotype to modify its growth and development in response to changes in environment. *Parthenium hysterophorus* showed plastic response to soil quality leading to two different contrasting strategies which contribute to its success as an invader (Annapurna and Singh, 2003). The first strategy resulted in tall, fast growing competitors with small seed mass, appropriate for rapid population expansion. The second resulted in short plants with high seed mass for persistence in less favorable habitat leading to slow build-up of population, with a gradual increase in size of the seed bank (Annapurna and Singh, 2003).
- (ix) **Introduction of Novel Traits:** Invasive organisms occupying any trophic position can transform ecosystems when they introduce novel traits to the ecosystem (Wardle et al., 2011). Examples are: (a) Invasion of Brazilian cerrado by *Pinus elliotii* and elimination of native species, (b) Invasion of *Pinus contorta* in grassland in Patagonia (Chile), which produces acidic litter that alters soil communities and impairs decomposer processes, (c) Invasive fallow deer (*Dama dama*) in northern New Zealand; removes palatable shrubs with high litter quality, this leads to domination of the ground layer by low litter quality grasses and alterations of soil food webs, (d) Dense understorey in *Acer saccharum* forest in Wisconsin is greatly reduced after invasion by the burrowing earthworm *Lumbricus terrestris*.
- (x) **Genetic Traits and Ploidy Level:** Recent work suggests that genetic attributes may be strongly associated with plant species status. Pandit et al. (2011) collated data on chromosome numbers for 640 endangered species (worldwide) and their 9005 congeners, and for 81 invasive species and their 2356 congeners. They related ploidy (diploid versus polyploid) to endangerment and invasiveness, and found that endangered plants are disproportionately likely to be diploid and to have lower ploidy ratios, whilst invasive plant species are generally found to have high chromosome counts and to be polyploid. They found that invasive plants are 20% more likely to be polyploids than diploids; and a species is 12% more likely to be invasive as chromosome number doubles. On the contrary, endangered plant species are 14% less likely to be polyploids than diploids.

As the capacity to invade varies among alien species, depending upon the above characteristics, so does the susceptibility to invasion from habitat to habitat and from community to community. Thus in order to understand the dynamics of plant invasions insights on traits of the plant (elements

of species invasiveness) and features of the environment (components of community invasibility) are required, and none of the components can be fully assessed without reference to the other. So to understand plant invasion in totality one must consider the complementarity of the species invasibility and community invasibility. A habitat characterized by disturbance is more prone to invasion than an undisturbed habitat. The pace of invasion has accelerated during the 20th century because of rapid modification of natural habitat. The natural disturbance (storm, earth quake, volcanic eruptions, fire and climate) and management regimes have been found to make the ecosystem more susceptible to invasion.

24.4 ENTRY PATHWAYS AND VECTORS

The rate at which communities are reshuffled as species move in and out of them through geographic range shifts has been greatly accelerated by human activities. Disease causing organisms (e.g., rabies virus and chestnut blight) have spread far and wide by human activities. The routes by which invasive alien species enter new habitats are known as pathways and the means by which they travel to new destinations are known as vectors. The information concerning the specific mode of transport for an invader is required for effective control as well as help to identify and regulate human activities that promote the spread of harmful invaders. Humans are chiefly responsible for the intercontinental movement of non-indigenous species and continue to play important roles in regional spread of these species after they become established. For example, shipping constitutes the pathway and the ship ballast water as vector for the introduction of many coastal and marine invasive alien species. Pathways and vectors are numerous and comprise both intentional and unintentional human activities.

The source of population is found where species occur naturally in native ecosystems, whereas invaded ecosystems form the secondary sources. The destinations are locales, where the new species arrive intentionally or inadvertently. Natural pathways for the dispersal of non-native species include wind and water currents. The man-made pathways are intentional which include deliberate transport of living seeds, whole plants and pets. The unintentional addition has occurred in blast water discharge of ships. For example, the red tide organisms have been transported in this way from one part of the ocean to another part of ocean. The comb Jelly was introduced to the Black Sea via shipping industry in 1980s. It now comprises 95% of the biomass of the black Sea. It feeds on zooplanktons, eggs and fish larvae and has reduced Anchovy fisheries substantially. Soils associated with the trade of nursery stock have resulted in the introduction of fire ants from Argentina to Europe. Importation of fruits and vegetables has resulted in the introduction of plant pests. Intentional movement of people causes the spread of pathogens, and humans are vectors of new types of disease causing pathogens. International trade has introduced the Asian tiger mosquito which carries dengue fever to the Americas and Africa and has spread life threatening bacteria *Escherichia coli* in meat exports.

Many exotic species have been introduced to control other exotic species that were introduced earlier. For example, ferrets and weasels were introduced in New Zealand and mongooses in Hawaii for controlling rats and rabbits. These predators caused still worse-situations by having negative impact on ground nesting birds, while contributing little to the control of rats and rabbits. In Australia, red foxes were introduced to control introduced rabbit; the rabbits thrived, but the foxes caused decline of native marsupials.

24.5 IMPACT OF INVASIVE SPECIES

The introductions of non-native species can have a major impact on the structure and functioning of terrestrial, freshwater and marine systems as well as islands and urban areas. They are a serious impediment to conservation and sustainable use of global, regional and local biodiversity. Many exotic grasses, shrubs, trees and even exotic bird species are living in environments manipulated

by human activities. In most cases, their impacts on native biota are not very severe. However, many invasive species cause economic and ecological harm to the native species and ecosystems. Many reports and documents deal with the status, impacts and trends of invasive alien species (see McNeely et al., 2001; UNEP/CBD/SBSTTA, 2001). The status of invasive alien species in CBD (Convention on Biological Diversity) thematic areas and other areas is discussed below:

Forests

Temperate forests have been damaged extensively by the introduced disease causing organisms such as Dutch elm disease, chestnut blight and gypsy moth. Within tropics and subtropics, tree plantations and agroforestry are important sources of biological invasions. *Eucalyptus* from Australia is widespread in Southeast Asia, India, California, and various parts of Africa.

Grasslands and savannas

Many non-native grasses including *Bromus* spp., *Pennisetum* spp., *Eragrostis* spp. and *Imperata cylindrica* have widely invaded grassland ecosystems in different regions. *Lantana camara* and *Pinus* spp. are common woody invasive species.

Fresh water ecosystems

Lakes, springs and head waters of streams are particularly vulnerable to invasive alien species. Invasive species are invading large areas of freshwater systems in both temperate and tropical regions affecting the diversity of native species and water quality. Alien water weeds like water hyacinth and water lettuce (*Eichhornia crassipes* and *Pistia* species) are a global problem.

Coastal and marine environments

In coastal and marine environments, invasive species are common and highly significant agents of change. Many groups of organisms including protozoan, sponges, tunicate fish and sea weeds have caused major invasions in recent years. Increase in shipping activity worldwide has made it the most important pathway for the spread of invasive alien species in marine environments.

On the oceanic islands, invasive alien species are a cause of the irreversible species extinctions. In the Galapagos Islands, the number of introduced species is approaching the number of native species. The impact of introduced plants and animals on Pacific Islands has been very severe. The long time isolated islands of Hawaii, Guam, Mauritius and New Zealand have been severely affected by invasive species. In New Zealand, roughly two third of the land surface is covered by exotic plants; whereas half of the Hawaii's wild species are non-natives.

Ecosystem Level Effects

The ecological consequences of some invasive species as predators and herbivores, pests and disease causing organisms, competitors and having ecosystem level effects are briefly described below:

Predators and Herbivores: The Brown tree snake, a nocturnal snake, is native to Eastern and Northern Australia, Indonesia and New Guinea. The brown tree snake (*Boiga irregularis*) was introduced from Indonesia to Guam, an island in the west Pacific Ocean. Now, there is a large population of brown tree snake. Many native species of birds have become extinct in the rainforests of Guam as the snake consumes them in large numbers (Savidge, 1987). The Guam rail, a flightless bird became extinct in nature during 1986 due to the introduction of brown tree snake. The population of small reptiles and mammals has been drastically reduced.

Sea lamprey (a primitive jawless fish) is a native to Atlantic Ocean. It gained access to great upper lakes in 1900s. Sea lamprey preys upon native fish species and has caused extinction of several species. The population of lake trout and several other commercially important fishes has

been reduced substantially due to sea lamprey.

Lake Victoria (bounded by Uganda, Tanzania and Kenya) is the world's largest tropical lake. Before the 1920s, lake Victoria contained more than 350 species of fish from cichlid family, of which 90% were endemic. The lake has one of the most diverse and unique assemblage of fish in the world. Due to the introduction of exotic species, the Nile Perch and Nile Tilapia, more than half of the native species of cichlids are either extinct or found in very small numbers. These introduced species feed on, and out compete the cichlids for food.

Some generalist introduced species (for example goats, pigs and rabbits) have been the most destructive herbivores on the islands.

Pests and Diseases: Agricultural weeds can contaminate seed crops and reduce their value. Invasive insects can harm crop plants and reduce their productivity and act as vectors of disease among livestock and in human population. From an economic perspective, introduced insects (e.g. boll weevil and corn borer) are the most destructive pests of crop plants. Apple snails are threatening the rice plants and wetlands in Asia.

The chestnut blight reduced the chestnut trees, once major components of temperate deciduous forests in America and Europe. Avian malaria and avian pox are the introduced diseases affecting the birds, threatening their biodiversity particularly on the islands. The Asian Tiger mosquitoes, which are vectors of yellow fever and encephalitis, have spread to new continents.

Large developmental projects such as dams, irrigation schemes, land reclamation, and population settlement programmes, have contributed to the invasion of diseases such as malaria, dengue, schistosomiasis and trypanosomiasis. For example, the prevalence of lymphatic filariasis in southern Nile Delta increased by 20 fold since the building of the Aswan dam in the 1960s because of increase in breeding sites for the mosquito vector of the disease following extension of irrigation.

Competitors: The effects of exotics as competitors are most conspicuous with plants and sedentary animals. For example, some invasive species such as Kudzu vine, Zebra mussel, purple loose fire, and water hyacinth compete with other native species for space, water, nutrients and light.

In the Thar Desert of India, *Prosopis juliflora* has been rapidly colonising grassland habitats because of its high competitive ability under water stress conditions. *Prosopis* spp have invaded large areas of *karoo* and arid lands in South Africa (McNeely, 2000). The introduction of salt cedar (*Tamarix* spp.) in south western United States is having greatest impact on riparian habitats with in desert ecosystems. *Lantana camara* has established in the tropics and subtropics as a weed of pastures and disturbed natural ecosystems in about 50 countries. *Lantana* leads to decreased productivity in pastures and forests and poisons cattle. It invades disturbed natural ecosystems and excludes understorey species.

Eichhornia crassipes and *Salvinia* spp. are the worst invasive plants in aquatic ecosystems. Water hyacinth was introduced in India from South America as an ornamental plant; now, it has become a nuisance in many aquatic systems and is also having negative effects on native species. Another well known example of alien species in freshwater system is the introduction of zebra mussel in great lakes of United States in late 1980s. It is a native to Caspian Sea and is a highly invasive species because it clogs and colonizes pipes and has displaced several native species of mollusks. The zebra mussel is causing wide-spread die-off of native fresh water mussels in North America.

Ecosystem structure and functioning: At ecosystem level, alien species affect structural diversity, cause local extinction of native species, change fire and nutrient regimes and nutrient cycling patterns as well as produce allelochemical substances (Table 24.2). The invading species alter the composition of the native biological community and a variety of ecosystem properties such as productivity, nutrient cycling, natural disturbance regimes and vegetation structure (Vitousek, 1986). The effect of alien species on community structure, nutrient cycling, hydrology and fire regime in some ecosystems are summarized in Table 24.2.

Table 24.2. Impact of invasion on community structure and ecosystem processes (based on Sharma et al., 2005).**(a) Plant Community structure**

- Reduced species diversity in area invaded by *Heracleum mantegazzianum* compared with un-invaded region in Czech Republic.
- Lower richness of seedling and sapling in area invaded by Norway maples
- *Orbea variegata* reduced the diversity of annual plant and performance of chenopods in Australian shrubland. (*Orbea* impact was mediated via reduced water availability)
- Impact of *Tatarian* honeysuckle on the diversity and cover of native understory in New England forest due to light competition
- A succulent perennial *Carpobrotus edulis* reduces soil water availability to native shrub which reduces the growth and reproduction in coastal chaparral
- *Bromus tectorum* invasion reduces the amount of soil water available for other plants in Nevada grasslands
- Reduction in competitive performance of *Fynbos* shrub due to increased nitrogen availability caused by *Acacia* invasion

(b) Higher trophic level

- Increased nest predation of *Turdus migratorius* in dry deciduous forest of Illinois by introduction of *Rhamnus cathartica* and *Lonicera maakii* due to low nest height and absence of sharp thorn on the exotic species
- Himalayan *Impatiens glandulifera* produces more nectar than native *Stachys palustris* and receives more visitations by Bumble-bee in Europe
- Increase in earthworm densities under the introduced *Berberis thunbergii* and *Microstegium vimineum* than native *Vaccinium* species in New Jersey
- Increase in earthworm densities under *Myrica faya* in Hawaii
- Enhanced activity of feral pigs and high rate of nitrogen mineralization under *Myrica faya* in Hawaii
- Lower richness, fewer Fungi and invertebrates and higher abundance of active bacteria in site invaded by *Bromus tectorum* in Utah

(c) Nutrient cycling

- *Myrica* and *Acacia* tend to increase available nitrogen in system they invade in Hawaii
- *Bromus tectorum* reduces nitrogen mineralization rates by having greater carbon-nitrogen & lignin-nitrogen ratios than native species, while similar litter quality effects did not explain nitrogen mineralization under invasive *Hieracium* in New Zealand grassland
- *Hyparrhenia rufa*, lowers rate of nitrogen cycling in Costa Rica

(d) Hydrology

- *Tamarix* in south western North America increases evapotranspiration by 300-450 mm/yr due to high leaf area
- Increase in water use upto 105-120 mm/yr by *Centaurea solstitialis* in annual grasslands of western North America due to its active growth period in summer
- Decrease in community water use due to water loss by invasion of *Bromus tectorum* in New Zealand.
- Exotic annual grasses in California displace competitively deep rooting native perennials
- Excessive water use by *Acacia* and *Hakea* sp. in South Africa has lead to major water loss (estimated at 3×10^9 m³/y) and for the reason many rivers do not flow at all or flow only infrequently

(e) Fire regimes

- Introduced grasses have increased fire frequency more than three folds in seasonally dry shrubland and woodland, effecting biodiversity in Hawaii
- Invasion of annual grasses from Mediterranean grasslands in desert shrubland, converting shrubland to grasslands there by effecting biodiversity

Differences between native and exotic plant species in their requirements and modes of resource acquisition and consumption may cause a change in soil structure, its profile, nutritional content and moisture availability. Nitrogen fixing exotic plants can alter the soil chemistry of the environments they invade. For example, the introduction of the nitrogen fixing tree *Myrica faya* to nitrogen limited ecosystem of Hawaii resulted in marked increased in nitrogen inputs. This in turn has changed the structure and function of native forests (Vitousek et al., 1987) inviting still more invasive species. Fire prone exotic plants can allow fires to burn more extensively as in the case of the invasion of cheatgrass (*Bromus tectorum*) into Western North America. The introduction of deep rooted salt cedar (*Tamarix* spp) to the Mojave and Sonoran deserts of North America has lowered water table because of high transpiration rates.

Ecosystem consequences

Ehrenfeld (2010), Pyšek and Richardson (2010), Simberloff (2011) and Strayer (2012) have evaluated and reviewed the information on the consequences and impact of biological invasions on ecosystems. Their findings are briefly recapitulated below. (i) Some communities and/or ecosystems are more invasible than others. Their inherent invisibility depends on the level of resources available at the time of invasion, which is closely linked to the disturbance level, but also on the presence of herbivores, pathogens, and predators that can act as a constraint to the establishment of new species. (ii) In many but not all cases, invasive species have higher values of resource-acquisition traits, larger size, and/or higher growth rates, which imply higher pool sizes of C, N, etc. and affect primary and secondary productivity, production and destruction of engineered structures such as reefs and burrows; frequency and severity of fires and availability of light and temperature. (iii) Plant invasions result in much higher pools of aboveground (133% higher) and belowground (5%) carbon (C), aboveground (85%) and belowground (112%) nitrogen (N), litter C (49%), soil C (7%), and soil N (19%) (iv) Litter decomposition rates are on average 117% higher for exotic than co-occurring native species, and the higher litter masses imply higher element input rates to the soil (iv) N fluxes (N mineralization and nitrification) within terrestrial ecosystems affected by plant invasions tend to be increased by invasive plant species (v) Species that act as engineers or keystones in their native range (e.g., beavers, earthworms, colonial bivalves, densely growing grasses) are very likely to carry out the same activities in an invaded range. Such invaders often lead to structural habitat change with ecosystem-wide impacts. (vi) Although the expected direction of invaders' effect on ecosystem is through population and community but many invading species, called ecosystem engineers, affect ecosystem characteristics directly without being mediated by populations or communities. Such engineering effects can then have strong effects on populations and communities. Examples of non-native engineers include eucalyptus (*Eucalyptus* spp.), Australian pine (*Casuarina equisetifolia*), cordgrass (*Spartina* spp.) in coastal wetlands, beavers (*Castor canadensis*) in Patagonia and elsewhere, various species of freshwater and marine mussels, wild boars (*Sus scrofa*), common carp (*Cyprinus carpio*), etc. (vii) Four pathways for change in trophic web structure are (a) trophic cascades in which invasive predators reduce populations of consumers that transport nutrients between habitats, reducing inputs and altering within-system cycling; (b) invasive predators that alter the species composition and abundance of detritivores, altering litter decomposition rates; (c) invasive plants, whose litter is more or less palatable to detritivores than native species' litter ; and (d) exotic animal modification of detritivore habitats within soil or water, as seen for invasive ants and worms and zebra mussels. (viii) The joint impacts of two or more species are more severe than would have arisen if each species had invaded alone. For example,

ponds created by the beavers have increased amounts of suitable habitat for three salmonid species introduced to Tierra del Fuego, brown trout (*Salmo trutta*), rainbow trout (*Oncorhynchus mykiss*), and brook trout (*Salvelinus fontinalis*). The phenomenon is called ‘**invasional meltdown**’.

Undisturbed Hawaiian rainforests are dominated by the ohia tree (*Metrosideros polymorpha*), slow-growing native trees. These forests are invaded by the tropical ash (*Fraxinus uhdei*) and the Canary Island fire tree (*Morella faya*), fast growing tree (Asner et al., 2008). The invasion by *Morella faya* altered soil composition, changed the amount of carbon stored in ecosystems, and affected the ecosystem services. The invader shaded out native understory plants in rainforest zones. There was decrease in maximum aboveground biomass (AGB) in the woodland–savanna ecosystems and aboveground carbon stocks.

According to Fridley (2012), the non-native shrubs, lianas and some herbs in deciduous forest in eastern North America have been found to cause extended leaf phenology and the autumn niche. Non-native species are extending the autumn growing season by an average of 4 weeks compared with that of natives. The higher autumnal activity of invaders may lead to significant shifts in nutrient cycling, affecting leaf nitrogen resorption of non-natives, and causing significant changes in forest-floor litter quality. Invasions by non-native shrubs, lianas and some herbs may be a major driver of anthropogenic ecosystem change in forest ecosystems (Fridley, 2012).

In 1980s, *Caulerpa taxifolia* was accidentally released into Mediterranean Sea. This alga has spread like a dense carpet over the flow of the Mediterranean Sea. On the ocean flow, it causes damage to the biologically diverse sea floor communities of native grass species, sponges, corals, sea fans, anemone, sea stars and lobsters. *Caulerpa* is also toxic to many species of Mediterranean species. Nile perch and comb jelly are other examples of invasive species causing ecosystem-level effects. Such invaders make the environmental conditions unfavourable for native flora and fauna by.

The ‘fynbos’ area in South Africa is a global hot spot of biodiversity, where 68% of plant species are endemic. In this biologically diverse ecosystem, the invasion of woody species such as *Hakea sericea*, *Acacia* spp. and *Pinus* spp. have caused extinction of many indigenous plant species and pose a great threat to many more species (Huntley, 1996). Alien species are a significant direct threat affecting 30% of all threatened birds, 15 % of all threatened plants and 10% of all threatened mammals.

Economic losses: The economic losses due to invasive species can amount to billions of dollars every year as their management costs include not only costs to preservation, control and mitigation, but also indirect costs due to significant impacts on the goods and services provided by ecosystems (Table 24.3). Besides the direct cost, the economic cost also includes their indirect environmental consequences and various non-market values. It is estimated that about 137 billion US dollars are lost per year from an array of invasive species (Pimentel et al., 2000). Recent estimates indicate that damages from 79 alien species cost US\$120 billion for the United States; weeds alone cost AUS\$3.9 billion per year in lower farm income and costs to central and local government on monitoring, control, management, and research on weeds was at least AUS\$116.4 million each year; for Europe the total costs of invasive alien species is at least €12.5 billion per year and probably over €20 billion per year if extrapolated (see Pysek and Richardson, 2010 for references).

Table 24.3. Indicative costs (US \$) of some invasive alien species (based on Pimental et al., 2000).

Species	Economic variable	Economic Impact
Introduced disease causing organisms	Annual cost to human, plant, animal health in USA	\$41 billion per year
A sample of alien species of plants and animals	Economic costs of damage in USA	\$137 billion per year
Salt Cedar (<i>Tamarix</i>)	Value of ecosystem services cost in western USA	\$7-16 billion over 55 years
Knapweed (<i>Centaurea</i> spp and Leafy spurge (<i>Euphorbia esula</i>))	Impact on economy in three US states	\$40.5 million per year direct costs \$89 million indirect cost
Zebra mussel (<i>Dreissana polymorpha</i>)	Damages to US and European industrial plants	Cumulative costs 1988-2000 = \$750 million to 1 billion
Six weed species	Costs of control in Australia agroecosystems	\$105 million / year
<i>Pinus</i> , <i>Hakea</i> , and lowland acacias	Costs of restoring South African fynbos to pristine condition	\$2 billion
Water hyacinth	Costs of clearing in 7 African countries	\$20-50 million / year
Rabbits (<i>Oryctolagus</i>)	Costs in Australia	\$373 million / year (agricultural losses)
Varroa mite	Economic cost to beekeeping in New Zealand	\$267-602 million

The impacts of alien species on ecosystem services

Invasive species may cause changes in ecological services by impacting hydrological cycle, recycling of nutrients, conservation and regeneration of soils, pollination of crops and seed dispersal (McNeely et al., 2001). Examples of impact types of invasive species in Europe, are discussed by Binimelis et al., (2007) and Vilà et al., (2010). The various mechanisms by which invasive species impact ecosystem services include competition, predation, herbivory, parasitism, changing soil chemistry, hydrology, fire frequency or nutrient cycling. Invasive species affect ecosystem services at four levels of complexity, i.e., species, communities, ecosystems and atmosphere.

Delivering Alien Invasive Species Inventories for Europe (DAISIE) Project has facilitated the development of the first pan-European assessment of the impacts of alien species on ecosystem services (Vilà et al., 2010). These workers have reported 1094 invasive species with documented ecological impacts and 1347 with economic impacts. The two taxonomic groups with the most species causing impacts are terrestrial invertebrates and terrestrial plants. Terrestrial plants create greater ecological impacts than economic impacts (Table 24.4). Terrestrial invertebrates create greater economic impacts than ecological impacts. Alien species from all taxonomic groups affect “supporting”, provisioning”, “regulating”, and “cultural” services and interfere with human well-being. Biological invasions by alien species often result in a significant loss in the economic value, biological diversity and function of invaded ecosystems (Vilà et al., 2010). The impacts of invasive species on ecosystem services are summarized in Fig 24.3. However, The full economic impact of invasive species on ecosystem services is often incompletely assessed, and not accounted for in decision making.

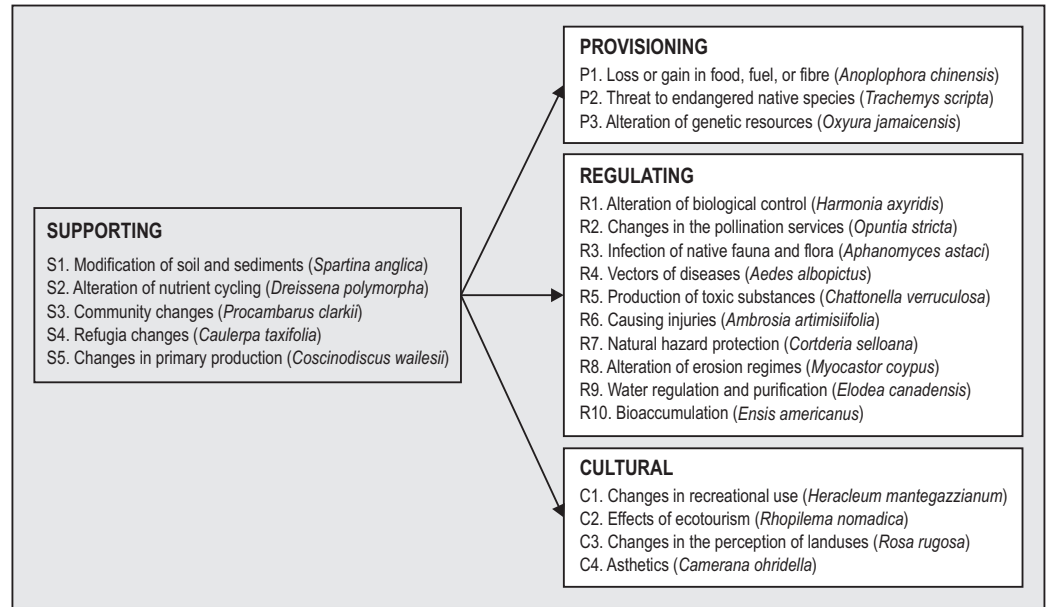


Fig. 24.3. Examples of impacts of invasive species on ecosystem service (adapted from Vilà et al., 2010).

Table 24.4 Total number and percentage of alien species known to have an ecological or economic impact for different taxonomic groups in Europe. Source: DAISIE database search at 12 February, 2008.

Taxonomic group	Total	Ecological impact (%)	Economic impact (%)
Terrestrial plants	5789	326 (5.6)	315 (5.4)
Terrestrial invertebrates	2481	342 (13.8)	601 (24.2)
Terrestrial vertebrates	358	109 (30.4)	138 (38.5)
Freshwater flora and fauna	481	145 (30.1)	117 (24.3)
Marine flora and fauna	1071	172 (16.1)	176 (16.4)

Strayer (2012) states “Thus, the effects of species invasions on ecosystem functioning can be as large as those of any human actions (e.g., addition of nutrients and toxins, changes to disturbance or hydrologic regimes, harvest of organisms), and can affect most ecosystem functions.”

Soil Communities and Plant Invasion

The effects of one exotic plant species on soil biota may be manifested by one of these mechanisms or by several mechanisms acting simultaneously (Fig. 24.4). In numerous ecosystems, the invasion of exotic plant species has caused major shifts in the composition and function of soil communities. Soil organisms, such as pathogenic or mutualistic fungi, have direct effects on the establishment, growth, and biotic interactions of exotic plants (Callaway et al., 2004c, Wolfe and Klironomos, 2005). An integrated understanding of how aboveground and belowground biota interact with exotic plants is necessary to manage and restore communities invaded by exotic plant species (Wolfe and Klironomos, 2005).

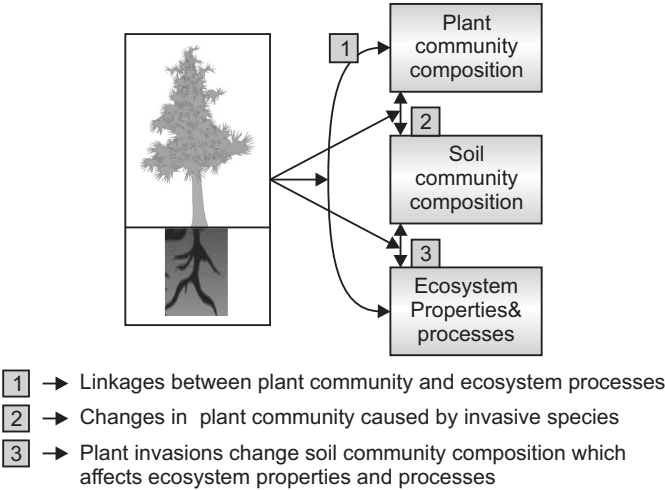


Fig. 24.4. Plant invasions, Soil Biota and Ecosystem Processes. The composition and functioning of soil biota are closely linked with aboveground composition and functioning and exotic plants can directly or indirectly disrupt these links. (after Callaway et al., 2004c; Wolfe and Klironomos, 2005).

Plants release secondary compounds into the soil environment from their roots as exudates, and if the compounds released by an exotic plant are novel to a soil community, they may alter the composition and function of the soil community. Allelochemicals released from plant roots have been widely used as an explanation of the success of exotic plants in the context of plant–plant interactions (Hierro and Callaway, 2003), Diffuse knapweed (*Centaurea diffusa*) is a Eurasian knapweed species that has invaded many natural ecosystems in western North America. This species releases the chemical 8-hydroxyquinoline from its roots, which has been demonstrated to be an antimicrobial agent (Vivanco et al., 2004). Diffuse knapweed can cause shifts in the composition of the soil microbial community (Callaway et al., 2004b), possibly through the release of these allelochemicals (Fig. 24.5).

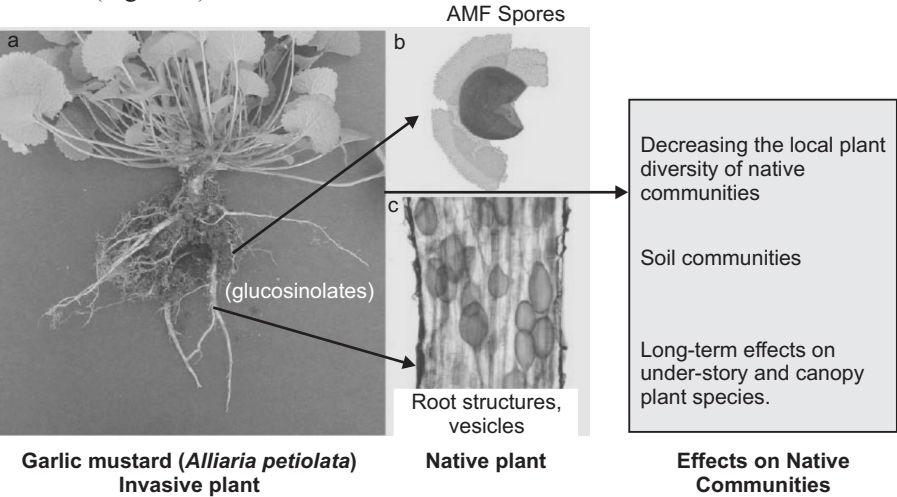


Fig. 24.5. (a) Garlic mustard (*Alliaria petiolata*) and (b, c) arbuscular mycorrhizal fungi (AMF) of native plant. Garlic mustard, a European native, invades forests of North American. Native mycorrhizal fungi and understory plants are affected by the chemical 8-hydroxyquinoline released from the roots of garlic mustard (derived from Wolfe and Klironomos, 2005).

Soil-borne communities and *Ammophila arenaria* invasion

Soil borne communities (nematodes and fungi) restrict the growth of *Ammophila arenaria* in Europe, its native range. *Ammophila arenaria* is invasive in North America, South Africa, Southern Australia and New Zealand. Specialist pathogenic nematodes are absent in the introduced range (Van der Putten et al., 2005). However, the plant's escape from enemies appears to be only partial considering the negative soil feedback it experiences in California (Beckstead and Parker, 2003). In South Africa, *Sporobulus virginicus* harbors pathogenic fungi in its rhizosphere, and these offer biotic resistance to *A. arenaria* (Beckstead and Parker, 2003). Thus, *A. arenaria* invasion combines aspects of the enemy release, biotic resistance and accumulation of local pathogens hypotheses as shown in Fig. 24.6 on the basis of Inderjit and Van der Putten (2010).

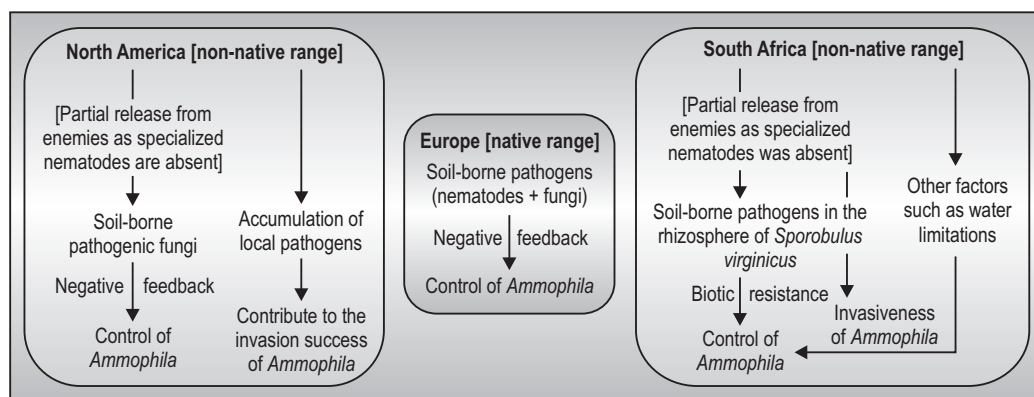


Fig. 24.6. Interactions between *Ammophila arenaria* and soil organisms for its control and abundance in the native and non-native ranges. (after Inderjit and van der Putten, 2010).

24.6 PREVENTION AND MITIGATION OF ALIEN SPECIES INVASION

When a potential or actual alien invasive species has been detected, in other words, when prevention has not been successful, steps to mitigate adverse impacts include eradication, containment and control. Eradication aims to completely remove the alien invasive species. A strategically important focus for eradication is to identify points of vulnerability in the major invasive pathways, such as international ports and airports, for monitoring and eradication activities. Control aims for the long term reduction in abundance or density of the alien invasive species. A special case of control is containment, where the aim is to limit the spread of the alien invasive species and to contain its presence within defined geographical boundaries.

There are four major steps to deal with problems of alien species, i.e., prevention, early detection, eradication and control (Fig. 24.7). Prevention of introduction is the first and most cost-effective option; the various possibilities are:

1. Interception based on regulation enforced with inspection and fees.
2. Treatment of material suspected to be contaminated with non indigenous species.
3. Prohibition of the particular commodities in accordance with international regulations.

For example, The Tansy ragwort (*Senecio jacobaeae*) was introduced from Eurasia and is an invasive species in rangelands of North and South America. This plant has displaced species of forage plants and poisoned livestock with its pyrrolizidine alkaloids. The introduction of natural enemies i.e., cinnabar moth (*Tyria jacobaeae*) from France and the ragwort flea beetle from Italy has successfully controlled the ragwort in rangelands.

There is a considerable potential for the biological control of *Eichhornia crassipes*, *Salvinia molesta* and water lettuce (*Pistia stratiotes*) using their natural predators from South America. *Salvinia* is an indigenous species in south-east Brazil and its associated weevil (*Cryptobagous salviniae*) was introduced into Australia in 1980. The weevil has provided an effective control of the *Salvinia* fern. It has now been introduced into Australia, Kenya, India and Malaysia for the control of *Salvinia*. Since 1971, two South American weevils, *Neochetina eichhorniae* and *N. bruchi* have been widely introduced in Australia, Asia and Africa for the control of water hyacinth; providing effective control in some areas. Similarly, *Neohydronomous affinis* from South America has provided effective control in Australia, South Africa and United States. Thus, there is strong possibility of controlling the invasive species using the natural predators and parasites as indicated in the recent studies (Mitchell and Power, 2003; Torchin et al., 2003).

Habitat management to control invasives involves measures such as prescribed burning, grazing and other activities. Integrated pest management involves mechanical, chemical, and biological control methods based on ecological research, regular monitoring, and careful coordination (Wittenberg and Cock, 2001). Education and public participation have to be a mainstay of any approach dealing with prevention, eradication and control of invasives.

Management Strategy against Invasive Species: New Directions

Much of invasion biology focuses on ecological effects, predicting spread and developing control methods rather than documenting economic and social damage to society from impacts on ecosystem services. Because ecosystems provide life-support services to all of human society, using the ecosystem-service framework for prevention and control of IAS has potential for reaching a diverse audience and giving them a stake in the outcome of IAS introductions. The next generation of IAS science and policy should reflect the fact that invasive alien species, similar to habitat loss and climate change, are emerging as a major driver of global environmental change, with grave consequences for biodiversity and human well-being (Pejchar and Mooney, 2009).

Lantana has adapted very well to the ecosystems it has invaded, often transforming their natural state. According to Bhagwat et al. (2012), for controlling *Lantana camara*, emphasis in Australia has been on biocontrol, whereas in South Africa and India mechanical removal has been a preferred option. Its bioclimatic niche and therefore potential for its expansion might include much more land area in Australia, India and South Africa than it has currently invaded. The future management of invasive species requires an adaptive management approach to their invasion (Bhagwat et al., 2012).

Simberloff et al. (2013) suggest that optimal management strategy against invasive species should evolve with time since introduction, as the management efficiency decreases and management cost increases with time, in the direction of Prevention (Information, regulation and legislation, and quarantine measures) to Early detection (Interception, monitoring and surveillance, and removal) to Management (Eradication, containment, and control). Guiding principles on invasive species adopted by the Convention on Biological Diversity (2002) reflect these findings: prevention is the priority response; early detection, rapid response, and possible eradication should follow when prevention fails. Long-term management is the last option. Stringent biosecurity based on risk assessment, as applied in New Zealand and Australia, has significantly reduced the number

of invasions. Europe and New Zealand had similar invasion rates of non-native mammals through the 19th century, but no invasions occurred in New Zealand after public perceptions shifted and biosecurity policies were adopted.

24.7 INTERNATIONAL INITIATIVES FOR CONTROL OF INVASIVE SPECIES

Convention on Biological Diversity: The convention on Biological Diversity recognises the importance of invasive species as a global problem. The Convention on Biological Diversity in article 8th calls on governments “to prevent the introduction of, control or eradicate those alien species which threaten ecosystems, habitats or species.”

Global Invasive Species Programme: The global Invasive Species Programme (GISP) has been formed in 1997 for controlling alien species that threaten ecosystems, habitats and species. The GISP is the component of *Diversitas* and coordinated by SCOPE (Scientific Committee on the Problem of Environment), IUCN, UNEP and CABI (Centre for Agriculture and Biosciences International). GISP is preparing new tools and approaches for the management, control and mitigating the impacts of invasive species on local, national and global scale.

Invasive Species Specialist Group of IUCN: The Invasive Species Specialist Group (ISSG) is a part of the Species Survival Commission (SSC) of IUCN. The ISSG is a global group of scientific and policy experts on invasive species from 46 countries. The group’s activities primarily focus on invasive species that cause biodiversity loss and threaten oceanic islands. The aim is to increase awareness about biological invasions and suggest ways to prevent control or eradicate invasive species.

Part – VI

NATURAL RESOURCES AND BIODIVERSITY

- ♦ NATURAL RESOURCES : HUMAN IMPACTS AND CONSERVATION
- ♦ BIODIVERSITY : MAGNITUDE, FUNCTIONS AND SERVICES
THREATENED SPECIES AND CONSERVATION
- ♦ REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS

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Natural Resources: Human Impacts and Conservation

A resource is anything useful or can be made useful to humans to meet their needs and wants. The resource which is directly available for use from nature is called *natural resource*, and includes fresh air, freshwater in lakes and rivers, soil, land, forest, grasslands, biodiversity, fisheries, ecosystems, minerals, salts, fossil fuels and others. In other words natural resources are the goods and services supplied by our environment. Some resources are not directly available, but they become resource because of our economic systems and cultural beliefs.

Earlier when people lived at subsistence level the exploitation of resources was limited to local scale, now the spatial scales have become much larger, much beyond the local scale. Tree bark (e.g., *Taxus*), forest fruits, resin, plant dyes, pulp for hand-made paper, berries used to give flavour to wine, roots of many inconspicuous species of a region, lichens, all are being harvested by entrepreneurs, and are marketed to distant places. Humans have begun to degrade all kinds of natural resources because of increasing demands of the growing population, use of advanced technology for resource use as well as increase in per capita resource consumption.

According to the State of the World 2004, the global consumer class (users of televisions, telephones, and the Internet, along with the culture and ideals these products transmit) constitute more than a quarter of the world (some 1.7 billion people). Almost 50% of this class now lives in developing countries, which also have the greatest potential to expand the ranks of consumers. Forests, wetlands, and other natural places are shrinking to make way for people and their homes, farms, and factories. More than 90 percent of paper still comes from trees, which accounts for consumption of about one fifth of the total wood harvest worldwide. Cars and other forms of transportation account for nearly 30% of world energy use and 95% of global oil consumption.

Therefore, it becomes important to prevent degradation of natural resources and carefully manage them to ensure their sustainable utilization. Resource conservation involves using, managing and protecting resources so that they are not wasted, depleted or degraded and are available on sustainable basis for the present and future generations.

25.1 RESOURCES: TYPES, DEGRADATION AND CONSERVATION

Natural capital consists of natural resources, natural ecosystems and ecological processes, which provide many services to people. Natural resources are divisible into two principal categories: (i) the *renewable* --unconditionally renewable, conditionally renewable, and (ii) *non-renewable* (Fig. 25.1).

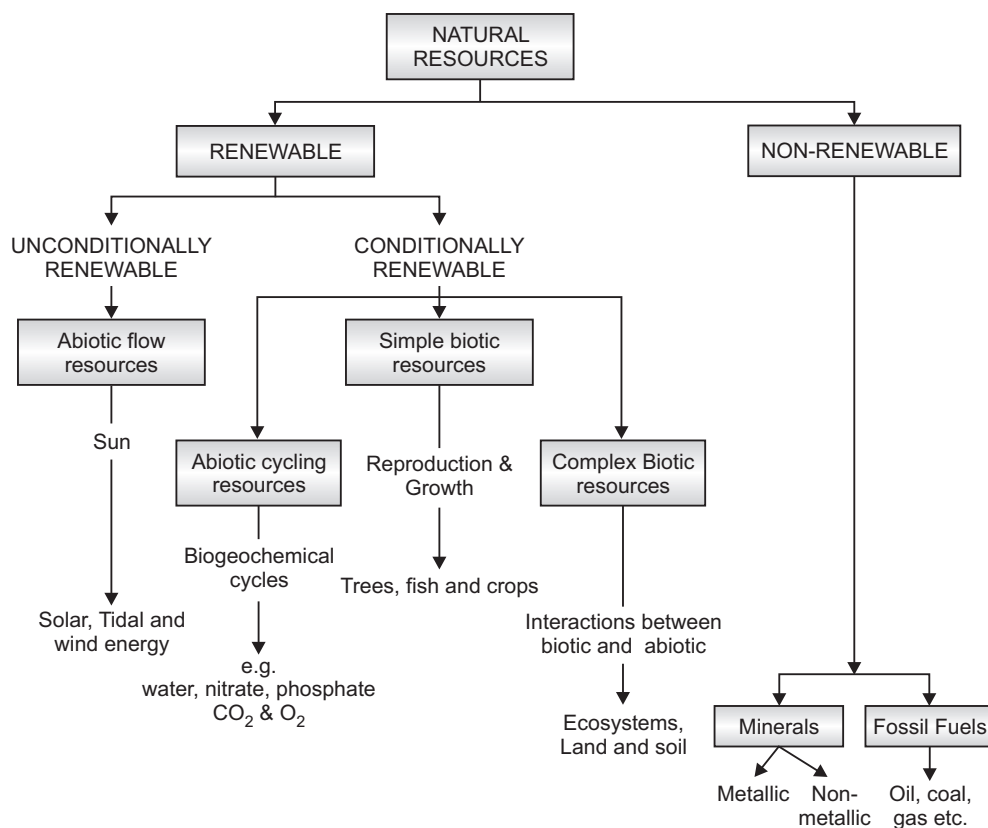


Fig. 25.1. The nature of natural resources (adapted from Gilbert et al., 1990; see Young, 1995).

Renewable resources

Perpetual resources or unconditionally renewable resources including direct solar energy, winds, and tides are virtually inexhaustible resources on human time scale (Fig. 25.1). In general terms, *conditionally renewable* resources consist of those that can be renewed or reproduced, and increased in amount through natural processes, such as, freshwater, fresh air, fertile soil, trees in a forest, and grasses in grasslands, fruits and fibres. Most resources derived from living things (simple biotic resources and complex resources) are conditionally renewable resources (Fig. 25.1). However, if a renewable resource is used faster than its replenishment, it can be depleted and converted to non-renewable resource.

Non-renewable resources

They consist of minerals and fossil fuels, which cannot be renewed in time span meaningful for humans (Fig. 25.1). These are exhaustible resources that exist in fixed amount (stock) and are replaced slowly than they are used (e.g., coal and oil). Though coal, oil and natural gas are biotic resources as they were produced by photosynthetic activity of plants millions of years ago, they cannot be renewed in practical terms. The stock of petrol or iron ore is not reproduced and does not increase in our times. Once consumed, they are practically gone forever.

Resource degradation

The highest rate at which a renewable resource can be used without decreasing its potential for natural replacement is its natural replacement rate or sustained yield. If the sustained yield of a

biological resource is exceeded, the base supply of resource begins to decline- a process known as resource degradation. A non-renewable resource is considered economically depleted when 80% of its total estimated supply has been removed and used. However, some resources can be recycled or reused to extend supplies (e.g., copper, aluminium, iron). The depletion of natural capital could have significant irreversible and adverse consequences for present and future generations.

The exploitation of one type of resource can affect the other type. For example, if a forest is cut or burned, not only the living resources, such as wood and medicinal plants are affected, but soil and nutrients are also lost, polluting adjoining streams and river water.

Resource conservation

For maintaining the supply of renewable resource, it is important to use the resource at or below its sustained yield. The resource conservation aims to (i) maintain ecological processes and life-support systems, (ii) ensure continuous yield of plants, animals, and materials, (iii) preserve the quality of environment, (iv) conserve biological diversity, (v) achieve sustainable use of species and ecosystems. The collective resource management programmes with the involvement of local social groups have been found to be effective for the management of watershed, forest, fishery, agriculture, etc. as well as for improving community well being (Pretty, 2003).

25.2 HUMAN IMPACT ON NATURAL RESOURCES

With increasing human domination on earth's ecosystems, natural resources are declining both in quantity and quality on a global scale. Nearly half of the earth surface has been transformed by human enterprises, such as agriculture, industry and commerce. The human appropriation of terrestrial net primary productivity is 39-50%; of the total accessible fresh water more than half is affected by human use; due to the human enterprises more nitrogen is fixed than by all natural terrestrial sources combined, the effect of which is being felt on varied natural resources; already about one-quarter of the bird species are gone and air chemistry is clearly different now than before industries began to expand dramatically. Vitousek et al. (1997b) have documented human impact on various natural resources at the global scale (Fig. 25.2).

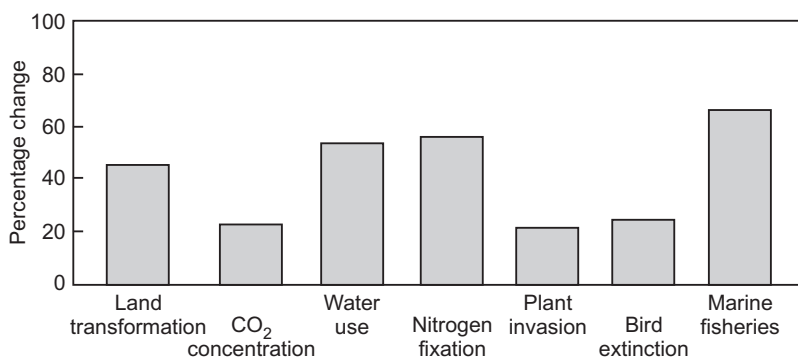


Fig. 25.2. The extent of human impact on the Earth's natural resources (land, water, fisheries) and various components (atmospheric CO₂ concentration, nitrogen fixation, biota). The impact is shown as per cent of total land surface affected by human activities; CO₂ concentration is the change due to man-made deforestation and burning of fossil fuels; water use is the percentage of accessible fresh water that humans use; nitrogen fixation is the percentage of terrestrial nitrogen fixation that humans realize in industrial fixation and by growing nitrogen fixing crops; plant invasion represents percentage of plants in Canada introduced by humans; bird extinction represents humans caused extinction during last two centuries; and marine fisheries represent percentage of major marine fisheries fully exploited, overexploited or depleted (from Vitousek et al., 1997b).

The aquatic environment and its productivity are also on the decline. About 58 percent of the world's coral reefs and 34 percent of all fish species are at risk from human activities. An estimated 75 percent of global fish stocks are now fished at or beyond their sustainable limit and are experiencing declining yields. During the last 15 years “the global ocean has lost more than 90% of the large predatory fishes” due to advancement in fishing technologies and inadequate governance (Myers and Worm, 2003).

25.3 LAND RESOURCES

Land distribution and Land use

The global land area is 13.2 billion ha. Of this, 12 percent (1.6 billion ha) is currently in use for cultivation of agricultural crops, 28 percent (3.7 billion ha) is under forest, and 35 percent (4.6 billion ha) comprises grasslands and woodland ecosystems. Low-income countries account for about 22 percent of the land area (FAO, 2011) (Table 25.1). Cultivated land is a leading land use (a fifth or more of the land area) in South and Southeast Asia, Western and Central Europe, and Central America and the Caribbean, but is less important in sub-Saharan and Northern Africa, where cultivation covers less than a tenth of the area.

Table 25.1. Regional Distribution of Main Land Use Categories in 2000 (adapted from Fischer et al., 2010 and FAO, 2011)

		Cultivated land		Forest land		Grassland and woodland ecosystems		Sparsely vegetated and barren land	
	Global share of land %	Mha	%	Mha	%	Mha	%	Mha	%
Low-income	22	441	15	564	20	1 020	36	744	26
Middle-income	53	735	11	2 285	33	2 266	33	1 422	21
High-income	25	380	12	880	27	1 299	39	592	18

Land is a precious resource, which humans have used to produce various goods and services. The use of land results in changes in structure and functioning of ecosystems, and their ability to interact with atmosphere and water bodies (Vitousek et al., 1997b). Human exploitation of land is quite diverse. The occupation of the earth by urban-industrial areas and agricultural crops (10-15% of the Earth) and by managed grazing lands (25 % of the Earth's land area) are examples of drastic alteration of the land. But such omnipresent is the impact of humans that even the seemingly pristine and remote forest areas have been subjected to hunting in the past, and now have their air chemistry changed. Land use varies with climatic and soil conditions and human influences. The fourth *Global Environment Outlook (GEO-4)* (UNEP, 2007) reports that increased demand for water, waste disposal and food has resulted in unsustainable patterns of land use. Forest cover and composition, cropland expansion, intensification of agriculture, desertification and urban development are the key components in land-use change (UNEP, 2007). Presently, the share of world land sources that are degraded is 25%; that are moderately degraded 8% , and that are improving 10%. Land degradation is the temporary or permanent lowering of productive capacity of land. The sum of all degraded lands is close to 5 billion ha or about 43% of Earth's vegetated surface.

The degraded lands have reduced capacity to supply products of direct uses (food, seed, biomass energy, fibre, timber, biodiversity, etc.). Land degradation has affected about 2000 million hectares of soil world-wide (see UNEP, 2002 a). The loss of potential productivity due to soil erosion worldwide is estimated to be equivalent to some 20 million tons of grain per year (UNEP, 1999). The main types of soil degradation are water erosion (56%), wind erosion (28%), chemical degradation (12%) and physical degradation (4%). The direct causes of land degradation and subsequent soil erosion in relative terms at the global level are overgrazing (35%), deforestation (30%), other agricultural activities (28%), over harvest for firewood (7%), and industrial activity (1%).

Land degradation is needed to be checked and degraded land rehabilitated for the following reasons: (i) to produce more food biomass and other similar goods; (ii) for balanced biogeochemical cycles that regulate greenhouse gas fluxes and determine Earth's total energy balance; (iii) for preserving biodiversity, which in part, depends on high net production; and (iv) for maintaining economic growth and social structure.

The current state of agricultural land, forests, grasslands, drylands, and recent changes to these land covers and uses are discussed below.

Agriculture

Food security is one of the major concerns because of the increasing world population, diminishing land and water resources, and climate change effects on agriculture. Today, approximately a billion people are chronically malnourished (Foley et al., 2011). Now, it is a big challenge to produce sufficient food for the increasing population and minimize the negative environmental effects of crop cultivation. In 2009, there were approximately 3.3 billion hectares of pasture and 1.5 billion hectares of cropland globally, with the extent and proportion of total land area varying greatly across regions (FAO, 2011a). The average availability of cultivated land per capita in low-income countries is less than half that of high-income countries and the suitability of cultivated land for cropping is generally lower. High-income countries, as a group, cultivate more than twice the land area per capita (0.37 ha) than either middle income (0.23 ha) or low income (0.017 ha) countries (Fischer et al., 2010; FAO, 2011a). Some facts about agriculture land-use are (FAO 2011a) :

- (i) Total world land area suitable for cropping: 4.4 billion hectares
- (ii) Percent of the total world cultivated area that is rainfed: 80% (1.2 billion ha)
- (iii) Total land area currently being cultivated: 1.6 billion hectares of which 20% (0.3 billion ha) is on marginally suitable lands
- (iv) In several regions, soil quality constraints affect more than half the cultivated land base, notably in sub-Saharan Africa, Southern America, Southeast Asia and Northern Europe.

Foley et al. (2011) reviewed studies to show that worldwide agriculture has already cleared or converted 70% of the grassland, 50% of the savanna, 45% of the temperate deciduous forest, and 27% of the tropical forest biome. Today, agriculture is mainly expanding in the tropics, where it is estimated that about 80% of new croplands are replacing forests. The current yields of wheat, maize and rice have been estimated at 64, 50 and 64% of their potential yield, respectively. Thus there is a tremendous yield gap on underperforming lands. The average global crop yields increased by only 20% between 1985 and 2005, although in the past 50 years, the world's irrigated cropland area roughly doubled, while global fertilizer use increased by 500% (over 800% for nitrogen alone). Intensification has also caused water degradation, increased energy use, and widespread pollution. Foley et al. (2011) suggest strategies for increasing food production while reducing negative environmental impact of agriculture: (i) halting agricultural expansion, (ii) closing 'yield gaps' on underperforming lands, (iii) increasing cropping efficiency, (iv) shifting diets (grain vs. meat) and (v) reducing postharvest waste.

Agricultural productivity is limited by biophysical and other factors. Extending conventional agriculture into uncultivated lands requires mechanization to modify the surface, and supplements in the form of fertilizers, herbicides, pesticides and irrigation water. Excessive use of machinery and chemical supplements, however, breaks up soil structure, increases erosion, chemically pollutes soil, contaminates groundwater and surface water, changes greenhouse gas fluxes, destroys habitat and builds genetic resistance to chemical supplements.

Conservation agriculture includes no-tillage, zero-tillage, direct seeding, and minimum tillage, and aims to improve livelihoods of farmers through the application of minimal soil disturbance. Conservation tillage practices leave crop residues on the soil surface to reduce soil erosion. The area under conservation tillage is estimated to be over 111 million-hectares worldwide (Derpsch et al., 2011), this is about 8% of global cropland including diverse farming systems under temperate, subtropical and tropical conditions. The extent of zero-tillage adoption worldwide for the year 2008-2009 is shown in Fig. 25.3. Several studies have shown an increase in soil organic carbon sequestration under conservation agriculture in different climatic regions of the world (Govaerts et al., 2009; Lal, 2011; Corsi et al., 2012). The impact of conservation agriculture on soil organic matter is summarised in Fig. 25.4.

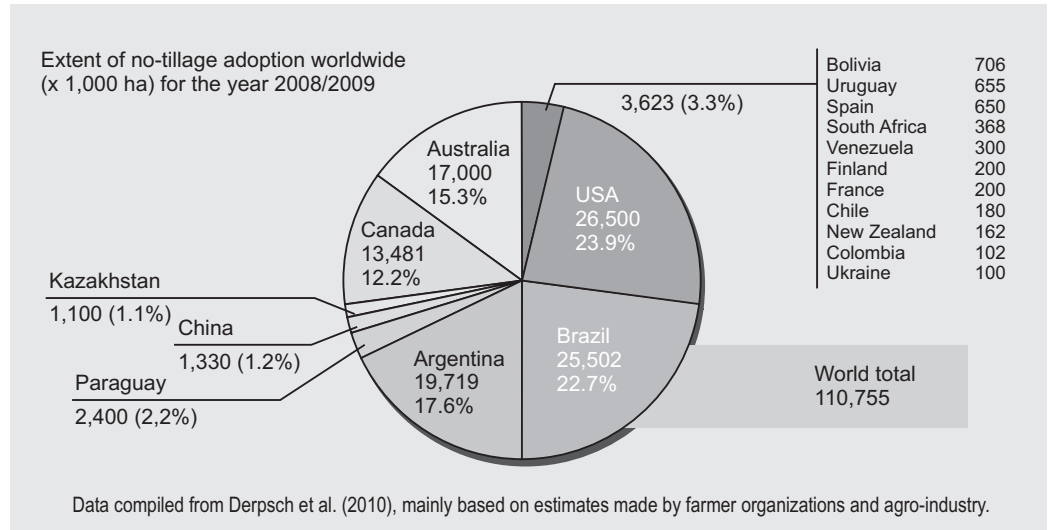


Fig. 25.3. Extent of no-tillage adoption worldwide (x 1,000 ha) for the year 2008/2009. (reproduced from A Gattinger et al., 2011 www.misereor.org).

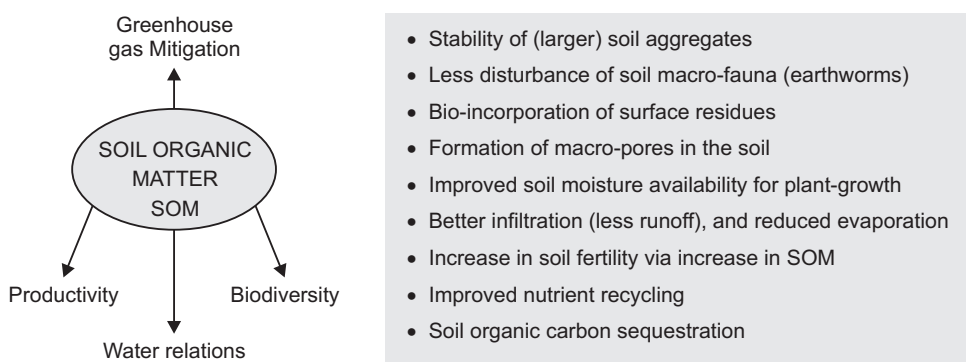


Fig. 25.4. Conservation agriculture impact on the soil.

In recent years, zero-tillage with wheat crop succeeding rice crop is now the most widely adopted resource conserving technology in the Indo-Gangetic plains of India, including regions of Haryana and Punjab (RWC, 2005). However, the soils of the Indo-Gangetic plains are poor in the organic carbon and hold a great potential for soil carbon sequestration in agriculture systems under zero tillage. Direct-seeded rice has been shown to be a viable technology to save water, to mitigate green-house gas emission and adapt to climatic change (Singh and Ladha, 2011).

Forest resources

FAO's Global Forest Resources Assessment 2000 has given a common definition of the forested lands as those areas having an extent of at least 0.5 ha with tree crown cover of more than 10% (FAO, 2001a). In the past, forests spread over half the world's land surface. Today forests and woodlands cover just over 4 billion hectares, 31 % of the Earth's land area (FAO, 2011b). The majority of these are boreal forests extending across northern and central Russia and much of Canada and Alaska. Temperate forests remain in a patchy distribution across the United States, Europe and the Asian mid-latitudes. Large expanses of tropical forest are found in the Amazon, Africa's Congo Basin and parts of South East Asia.

The distribution of forests in different regions of the world is given in Table 25.2. The four most forest-rich regions of the world (the Russian Federation, Europe, North and central America, South America) account for 83% of the total forest area. (Fig. 25.5). The five most forest-rich countries (the Russian Federation, Brazil, Canada, the United States of America and China) account for more than half of the total forest area (53%); whereas about 64 countries with a combined population of 2 billion people have forest on no more than 10% of their land area (FAO, 2010a). Some facts about forests are given below (The State of World Forests Report; FAO, 2012) :

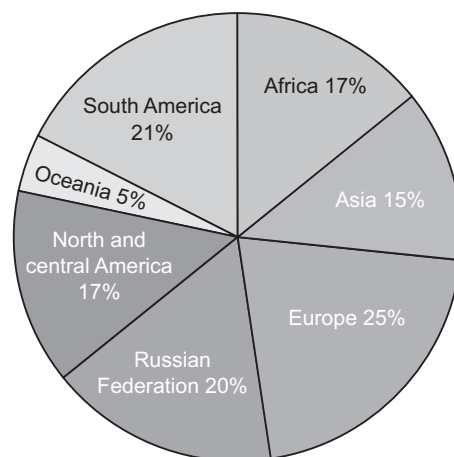


Fig. 25.5. Global Forest Area (million ha) in different regions of world (based on FAO, 1998).

- At present 12% of the world's forests are designated to conserve biodiversity.
- Globally, around 13 million hectares of forests were converted to other uses or lost through natural causes each year in the decade of 2000-2010.
- At the regional level, the largest area of forests designated for conservation of biological diversity is found in South America (116 million ha), followed by North America and Africa.
- On average, one percent of all forest land was reported to be significantly affected each year by forest fires.
- Outbreaks of insect pests damage nearly 35 million ha of forest annually, primarily in the temperate and boreal zones.
- Woody invasive species (trees, shrubs and vines) affected more than 34 million ha of forest in the United States and 1.6 million ha in Sudan.

Table 25.2. Distribution of forests in major regions of world, 2010 (based on FAO, 2010a).

Region	Forest Area		Area Of Planted Forest	
	1000 ha	% of total forest area	1000 ha	% of total forest area
Africa	674, 419	17	15, 409	2.3
Asia	592, 572	15	122, 775	20.8
Europe	1, 005, 001	25	69, 318	6.9
Russian Federation	809, 090	20	–	–
North and central America	705, 393	17	38, 661	5.5
Oceania	191, 384	5	4, 101	2.1
South America	864, 351	21	13, 821	1.7
Total	4, 033, 0603869	100	264, 084	6.6

Forest cover in India

At the beginning of the 20th century about 30% of land in India was covered with forests. But by the year 2011 the forest and tree cover has been reduced to 23.81%; 21.05% forest cover plus 2.76% tree cover (Table 25.3; Fig. 25.6). This is considerably less than the national goal of having a minimum of one-third total geographical area (and at least 67% in the hills) of the country under forest or tree cover as per the National Forest Policy (1988). Of the existing forests, about 2.54% are very dense forests, 9.76% moderately dense forests, 8.75% open forests, 1.28% scrub forests in 2011 (FSI, 2011). The National Forest Policy emphasizes the maintenance of environmental stability through preservation and where necessary, restoration of the ecological balance that has been adversely disturbed by serious depletion of the forests, and conserving the natural heritage of the country by preserving the remaining natural forests with the vast variety of flora and fauna, which represent the remarkable biological diversity and genetic resources of the country. National Forestry Action Programme (NFAP) of the Government of India focuses on five interrelated issues: (i) Protect existing forest resources, (ii) Improve forest productivity, (iii) Reduce total demand, (iv) Strengthen policy and institutional framework and (v) Expand forest area.

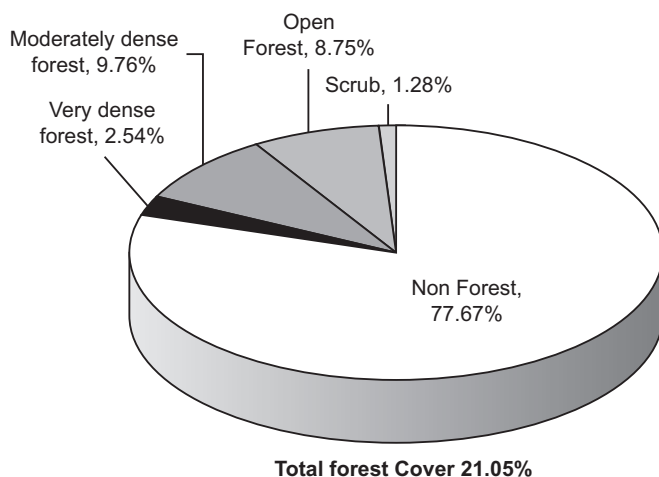
**Fig. 25.6.** Forest Cover in India in 2011 (based on FSI, 2011).

Table 25.3. Forest and tree Cover in India in 2011 (based on Forest Survey of India, 2011).

Class	Area (km ²)	% Geographic area
Very Dense forest	83, 471	2.54
Moderately dense forest	320, 736	9.76
Open forest	2, 87, 820	8.75
Total Forest Cover	6, 92, 027	21.05
Tree cover	90, 844	2.76
Scrub	42, 177	1.28
Non-forest (other land use)	2, 553, 059	77.67
Total	3, 287, 263	100

Key functions of forests

More than 80% of biodiversity and nearly 50% of organic carbon in terrestrial vegetation (1200 Pg carbon; Pg = petagram is 10^{15} g) are stored in forests of the world. About 170,000 plant species or two-thirds of all plant species of the earth occur in tropical forests alone (Noble and Dirzo, 1997). Forests are precious resources having both consumptive and non-consumptive uses and provide important environmental services to mankind. These services originate from the following key functions of the forest:

(i) Productive functions include production of wood, fruits and a wide variety of compounds, such as resins, alkaloids, essential oils, latex, pharmaceuticals, etc. Apart from timber and firewood, forests are source of numerous non-timber forest products (NTFPs) including resin, dye, oil seeds, nuts, medicinal plants, silk, honey, wax, natural rubber, tannins, essential oils, fodder and fibres. The leaf litter is collected from the forest floor and composted for manuring crops in many areas, which serve as a source of nutrients.

(ii) Protective functions include conservation of soil and water; preventing drought, sheltering against wind, cold, radiation, noise, sights and smells, etc. Costanza et al. (1997), and Noble and Dirzo (1997) estimated that forests of the world on average provide services worth \$ 1000 ha⁻¹ yr⁻¹.

(iii) Regulative functions involve absorption, storage and release of gases (CO₂, O₂), water, mineral elements and radiant energy. Such regulative functions improve atmospheric and temperature conditions, and enhances the economic and environmental value of the landscape. Forests effectively regulate floods and drought, and the global biogeochemical cycles, particularly of carbon.

Deforestation

On the basis of UNEP (2012) and FAO (2011b), the rate of forest loss from both deforestation and natural causes is slowing, but remains alarmingly high. At the global level, annual forest loss decreased from 16 million hectares in the 1990s to approximately 13 million hectares between 2000 and 2010 (FAO, 2011b). The highest rates of tropical forest loss over this period occurred in South America and Africa (Fig. 25.7). Although much of the developed world has experienced net reforestation since the late 1800s as a result of rural-urban migration and farm abandonment, natural factors such as drought, forest fire and insect attacks have increased forest loss in recent decades.

The key drivers of forest loss are population growth, poverty, economic growth, land pricing, international demand for timber and other forest products, insecurity of the rights of local people, and incomplete valuation of forest ecosystems. (UNEP, 2012). The main causes of deforestation

are expansion of agriculture, urbanization, industrialization, excessive commercial use of timber, fuel wood, non-timber forest products and cattle grazing. Export of forest products is responsible for deforestation in Africa and South America (10%) and South Asia (30 to 40%). If this rate of deforestation continues it is feared that remaining tropical forests may disappear within a century.

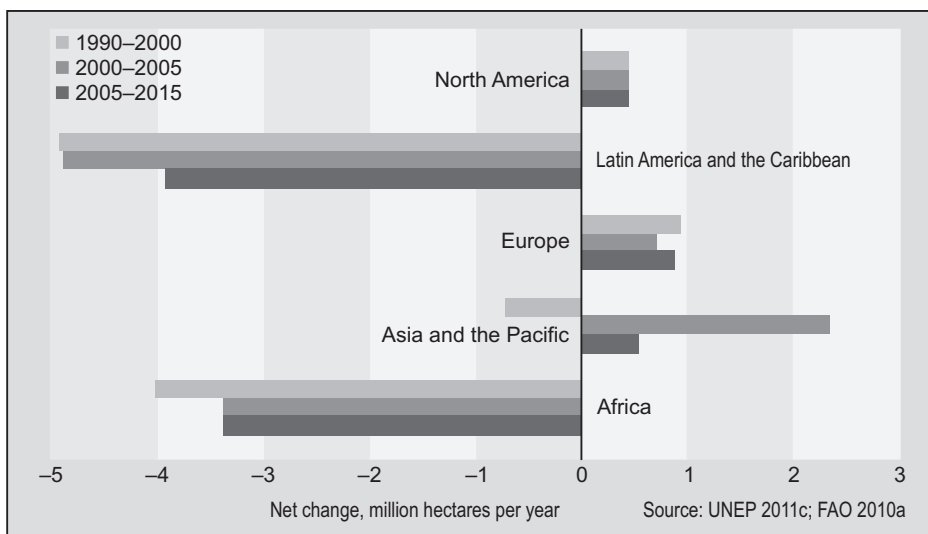


Fig. 25.7. Change in forest area by regions, 1999-2010. (adapted from FAO, 2010b; UNEP, 2011).

Impact of deforestation

- Forests, particularly on mountains, provide considerable protection from floods by trapping and absorbing precipitation, and slowly releasing it later. When the forest is removed, the amount of runoff water flowing into rivers and streams increases several folds, which may cause problems of flash floods and deposition of sediments in river beds.
- Deforestation results in increased soil erosion and loss of soil fertility. In drier areas, deforestation can lead to desertification.
- Deforestation causes extinction of plant, animal and microbial species. It also threatens indigenous people whose culture and physical survival depends upon the forests.
- Deforestation also induces regional and global climate change. Generally, rainfall declines in deforested areas and droughts become common. Deforestation contributes to global warming by releasing stored carbon into the atmosphere as carbon dioxide, which is a greenhouse gas.

Forests greatly affect budgets of greenhouse gases, and at least temperate zone forests are net sinks of C ($0.7 \pm 0.2 \text{ Pg C yr}^{-1}$). Tropical forests are currently thought to be a net source of C (2.2 Pg C yr^{-1} in the 1990s) because of forest clearing and conversion to other land uses (Houghton, 2003). However, tropical deforestation is currently estimated to release only around 1.1×10^{15} grams of carbon per year, or about 12% of total anthropogenic CO_2 emissions (Friedlingstein et al., 2010).

Forest conservation and management

Forest conservation and management programmes should ensure: (i) sustainable supply of tree products, non-timber forest products and ecological services, and (ii) maintenance of long term ecological balance through protection, restoration and conservation of forest cover. As per agricultural statistics, about 173.6 million ha land in India is degraded and these lands may be

utilized for some kind of tree plantations and agroforestry systems to meet the requirements of forage, fuel-wood and other forest products. Some methods of forest conservation and management are described below.

Conservation forestry

It involves protection of degraded forests to allow recoupment of their flora and fauna. Well-stocked forests are managed scientifically for producing timber and other forest products without causing any negative environmental impact. Forest areas designated as national parks and sanctuaries are protected from human interference.

Forest plantations

Extensive planting of trees through afforestation programmes is needed to save the diminishing forest cover. To achieve these goals the production or commercial forestry has been widely adopted in different regions of the world. Production or commercial forestry aims to fulfil the commercial demand, without causing denudation of natural forests, through intensive plantation in available land. Forest plantations, generally cultivated for industrial purposes, constituted 6.6% of the world's total forest area or around 264 million ha in 2010 (FAO, 2010b, 2011b). (Table 25.2). Compared to multispecies undamaged forest, monoculture plantations tend to provide fewer ecosystem services that include timber, carbon and water storage and soil stabilization.

In terms of composition, *Pinus* (20%) and *Eucalyptus* (10%) remain the dominant genera worldwide, although the diversity of species planted has been increasing (Table 25.4). Some valuable hardwood plantations (long rotation and slow growing) in tropical countries are that of teak (*Tectona grandis*), mahogany (*Swietenia* spp.) and rosewood (*Dalbergia* spp.), while in temperate countries oak (*Quercus* spp.), ash (*Fraxinus* spp.), walnut (*Juglans* spp.), and hard maple (*Acer* spp.) predominate (FAO, 2001a).

Table 25.4. Annual plantation rates and plantation areas by region and species group (based on FAO, 2001 a).

Region	Plantation areas by species groups (000 ha)							
	<i>Acacia</i>	<i>Eucalyptus</i>	<i>Hevea</i>	<i>Tectona</i>	Other broadleaf	<i>Pinus</i>	Other conifer	Unspecified
Africa	345	1,799	573	207	902	1,648	578	1,985
Asia	7,964	10,994	9,058	5,409	31,556	15,532	19,968	15,365
Europe	–	–	–	–	15	–	–	32,000
North and Central America	–	198	52	76	383	15,440	88	1,297
Oceania	8	33	20	7	101	73	10	2,948
WORLD TOTAL	8,317	17,860	9,885	5,716	33,556	37,391	20,743	53,618

Social forestry and Agroforestry

In Social forestry, the trees and shrubs are planted on unused and fallow land to provide fuelwood, fodder, etc. thereby reducing pressure on existing forests. For example, unused farmland, community land, road and rail sides, etc. are planted with suitable indigenous and/or exotic tree species.

Agroforestry combines the use of trees and other woody perennials in association with agricultural crops, pastures or livestock on the same unit of land either at the same time or in time

sequence. It is thus a combination of agriculture and forestry technologies to create integrated, diverse and productive land use systems. Farming systems that incorporate trees have a potential of being more productive and sustainable. Shifting cultivation is the oldest system of agroforestry, which is a rotation system of agriculture and fallow of natural vegetation (forest or savannah) for maintaining soil fertility.

On the basis of the components present, agroforestry systems have been classified as agrisilvicultural (woody plants with crops), silvipastoral (trees with pastures and or animals) and agrisilvipastoral having crops, pastures and or animals and woody plants (Nair, 1985). The scattered trees on cropland in the arid and semiarid regions have been found to increase soil fertility as well as increase crop yield beneath the tree canopies. For example, *Prosopis cineraria* in the semiarid regions of India has been found to increase yields of millet crops, *Paulownia elongata* is grown with wheat in temperate zones of China, and *Alnus nepalensis* is grown with cardamom (*Amomum subulatum*) in India (see Young, 1997).

Agroforestry has both productive and service functions. Agroforestry systems serve to diversify the outputs including fuel wood, fodder, timber and medicinal products, other non-timber forest products. The service functions of agroforestry systems are in control of soil erosion, maintenance and improvement of soil fertility and carbon sequestration in soil. The third assessment report of the intergovernmental panel on climate change stresses the importance of agroforestry (particularly on unproductive grasslands and croplands) for carbon sequestration and mitigating climate change by storing carbon through enhanced build up of soil organic matter and carbon storage in roots and deep soil layers (IPCC, 2001).

For soil and water conservation, the different types of agroforestry systems are windbreaks and shelterbelts, contour hedgerows and reclamation agroforestry. Contour hedge rows reduce soil and water erosion; some commonly used species are *Leucaena leucocephala*, *Inga edulis* and *Calliandra calothyrsus*. *Tithonia diversifolia* is a common hedge species found at mid elevation throughout East Africa and South-East Asia. Some trees and other woody perennials (e.g., *Moringa oleifera* and *Balanites roxburghii*, and *Zizyphus* species) are used to protect agricultural lands from erosion in dry areas prone to wind erosion. The shelter belts are large wind breaks of several rows of trees (e.g., *Eucalyptus*, *Cassia* and *Prosopis*).

After pre-treatment of saline soils (by lowering water table and providing adequate drainage) and sodic soils (treatment with gypsum and protection of natural vegetation), the salt tolerant species like *Prosopis juliflora*, *Acacia nilotica* and *Casuarina equisetifolia* have been used in reclamation agroforestry. The integration of trees with monocropping or native vegetation has been found to be ecologically sustainable land-use system for improving soil fertility as well as medium to long-term carbon sequestration in soil organic matter and the perennial components of the trees and grasses (Kaur et al., 2002b).

Joint Forest Management

Local community participation in the management of forests has been gaining pace since the late 1970s. In Nepal, about 36 per cent of forestry investment in forestry sector is earmarked for community forestry. The forest users' groups protect, manage, and use the forest area, share all benefits among users and possess exclusive rights to forest income.

In India, Joint Forest Management (JFM) has emerged as a major forest management strategy, which was introduced in 1990. The community provides any labour required to improve degraded areas and protects the forest while it regenerates. A portion of the income from selling timber is given to the community in addition to the right to gather non-wood forest products. In India, there were officially 84,632 JMF groups protecting and managing over 17 million hectares of state forest lands as of September 2003; there are also Self-Initiated Forest Protection Groups managing state-owned forests in India on the principle of participatory forestry.

The positive impacts of the JFM programme have included (i) an improvement in the relationship between the forest department, staff and local communities, (ii) increased income for participating communities and (iii) an improvement in the condition of forests. However, JFM has had several shortcomings, including a lack of firm legal basis, domination of JFM groups by the village elite, inequitable sharing of benefits within communities and in some cases the inter community conflicts. Despite these problems, JFM still represents a significant improvement in forest management in India (FAO, 2001b).

Non-wood Forest Products (NWFPs)

NWFP are products of biological origin other than wood derived from forests, other wooded land and trees outside forests. These provide a wide range of goods for domestic use and for the market, which includes fruit, nuts, medicinal herbs, forage and thatch and are available in open-access or semi-open access circumstances, particularly for the resource poor people. Some 80 percent of the population of the developing world uses NWFPs for health and nutritional needs. Several million households worldwide depend heavily on these products for subsistence consumption and/or income. At the local level, NWFPs also provide raw materials for large-scale industrial processing, including processing of internationally traded commodities such as foods and beverages, confectionery, flavourings, perfumes, medicines, paints and polishes. At present, at least 150 NWFPs are significant in terms of international trade; they include honey, gum arabic, rattan and bamboo, cork, forest nuts and mushrooms, essential oils, and plant and animal parts for pharmaceutical products. From a positive perspective, NWFPs can also be viewed as a safety net because these serve as a source of emergency sustenance in times of hardship when crops fail, when economic crises hit, in times of conflict or war, or when floods wash away homes (FAO, 2001a). The value of NTFPs exceeds that of timber and economic systems and needs to be considered in full valuation of forest products (Jansen et al., 1991).

Forest certification

Forest certification is a voluntary, market-based instrument that enables consumers to identify forest products with high environmental standards (FAO, 2001b). The aim is to focus on quality of forest management rather than the quality of forest products. It has contributed to the growing trend to define production and process standards for social and environmental performance in resource management. By the beginning of 2004, about 5 per cent of the global forest had been certified for sustainable forest management. About 92 per cent of these forests were located in Europe (43%) the United States (49%). At the same time, only four countries with tropical moist forests (Bolivia, Brazil, Guatemala and Mexico) have large areas under certified forests. More than 120 countries have some form of forest certification, many with their own national systems. Forest certification and related eco-labelling are innovative policy instruments for assuring the sustainability and multifunctional role of forest assets for human well-being.

Three main certification approaches (FAO, 2001b) are in operation:

1. Forest Stewardship Council (FSC) accreditation — an international system which requires producers to meet a global set of Principles and Standards for good forest stewardship and provides a trademark for product labelling;
2. Certification of the Environmental Management System (EMS) under the International Organization for Standardization (ISO) 14000 series; and
3. National certification schemes, some of which also incorporate elements of the FSC and ISO approaches.

Carbon sequestration in forestry systems

The world's forests store more than 650 billion tonnes of carbon, 44 percent in the biomass, 11% in dead wood and litter, and 45% in the soil. More than 75% of the total terrestrial biomass carbon stock and more than 40% of the soil organic carbon stock are found in forest ecosystems (Jandl et al., 2007). Boreal forests store more carbon in their soils than tropical forests, while tropical forests store much more carbon in their plant biomass. While sustainable management, planting and rehabilitation of forests can conserve or increase forest carbon stocks, deforestation, degradation and poor forest management reduce them. Pan et al. (2011) estimate that global forest systems constituted a total carbon sink of 2.4 to 0.4 billion tonnes of carbon per year from 1990 to 2007.

Grassland resources

Grasslands (also called rangeland) provide forage and habitat to domestic animals and wildlife. Grass cover is extremely effective in binding soil particles with the help of highly branched fibrous root system, thereby significantly reducing soil erosion. In rural areas of India, dried hay removed from grasslands, particularly from tall grasses, is also used as fuel or thatching material.

According to White et al. (2000), grasslands cover some 40% of the earth's surface (excluding Greenland and Antarctica) and are found in every region of the world but most commonly in semi-arid zones (28% of the world's grasslands), followed by humid (23%), cold (20%), and arid zones (19%); Sub-Saharan Africa and Asia have the largest total area in grassland; The five countries with the largest grassland area are Australia, the Russian Federation, China, the United States, and Canada and the five countries with the highest percentage of grassland area, all in Sub-Saharan Africa, are Benin, Central African Republic, Botswana, Togo, and Somalia. Human populations are highest in the dry grasslands (arid, semi-arid, and dry sub-humid) of Sub-Saharan Africa followed by Asia. Human populations are lowest in the dry grasslands of Oceania. Managed grazing systems are defined as any geographically extensive operation designed for the production of animals for consumption, including for meat, milk, and any major animal products. Managed grazing land covers more than 33 million square kilometres (25%) of the global land surface and has larger geographic extent than any other form of land use (Asner et al., 2004).

In India the area under various kinds of grass cover, including fallow and wastelands, is estimated to be about 18% of the total land area. If we include the forested area (about 19% of total land), most of which also supports grazing, about 37% of land can be said to be available for grazing. The average annual production of dry grass or hay in India is about 250 million tons.

Degradation of grasslands

Degradation or destruction of grassland is mainly related to overpopulation. To enhance food production, grasslands possessing fertile soils have been ploughed and converted to agricultural lands (e.g., North American prairies); temperate grasslands have experienced heavy conversion to agriculture. Three regional syndromes i.e., desertification, woody encroachment, and deforestation are inherent to global grazing (Fig. 25.8). These syndromes have widespread and differential effects on the structure, biogeochemistry, hydrology, and biosphere-atmosphere exchange of grazed ecosystems as well as represent a major component of global environmental change (Asner et al., 2004).

In southern Asia and Africa, grassland areas are frequently overgrazed. For example, the number of animals grazing in the arid and semi arid regions of India has been found to be 2-10 times greater than the capacity of the grassland to feed the animals. The lack of plant cover due to overgrazing causes soil erosion by water and wind. When overgrazing occurs in combination with extended periods of drought, once fertile grassland can be converted to a desert. The conversion of grassland (or forest) to desert is called desertification.

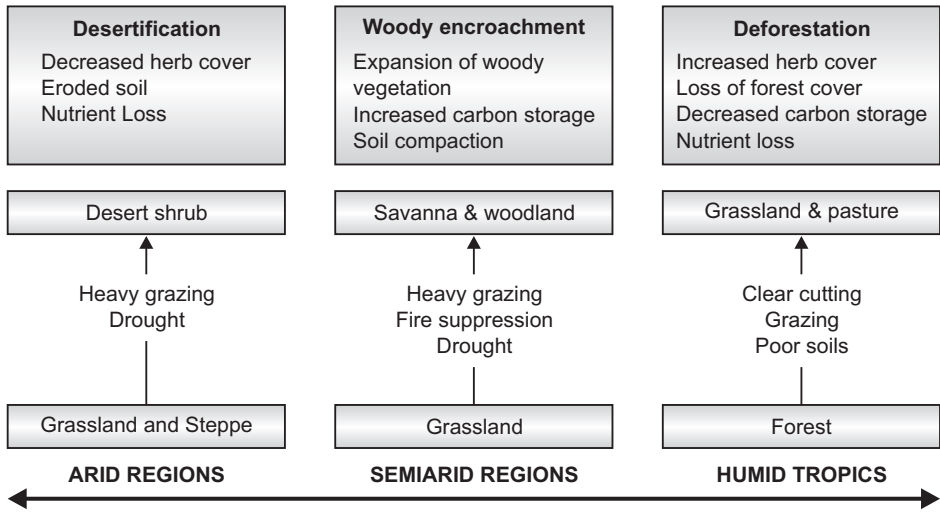


Fig. 25.8. The managed grazing practices in arid, semiarid and humid tropical regions results in desertification, woody encroachment and deforestation (the three regional syndromes) on a global-scale (adapted from Asner et al., 2004).

Many grazing systems experiencing desertification in the southwestern United States, Australia, and Africa are now dominated by one or a few woody shrub species, with nearly none of their herbaceous canopy remaining on the landscape. Once established, a combination of biogeochemical and hydrological feedbacks sustains these shrub systems in a new stable state, very different from the prior grassland (Schlesinger et al., 1990; Fig. 25.9). Wind erosion removes soil nutrients and carbon from shrub interspaces (Okin et al., 2001).

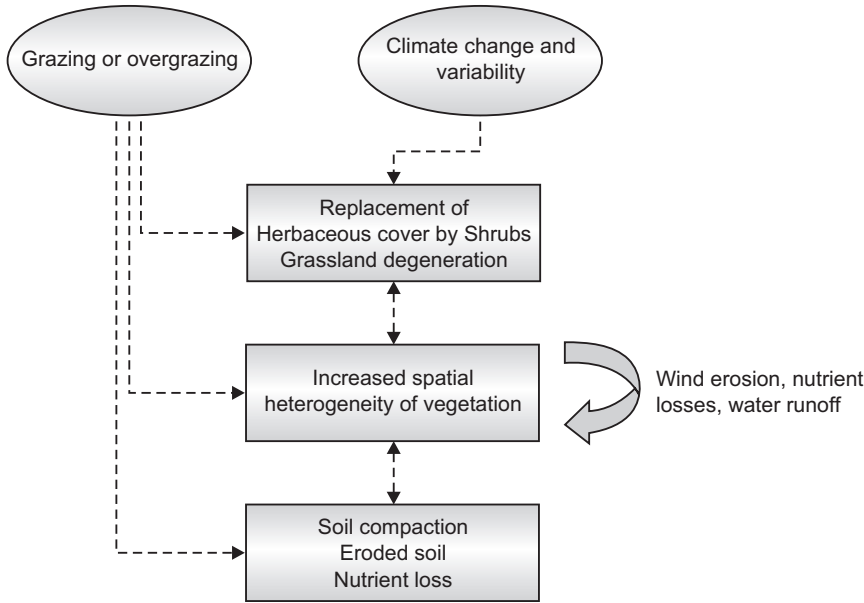


Fig. 25.9. Processes leading to desertification in grazing lands of arid regions (redrawn from Schlesinger et al., 1990; see Asner et al., 2004).

Nutrient loss from grazing areas is also associated with burning, soil erosion, volatilization and export of livestock products depending upon grazing history and the inherent levels of fertility of soils. The nutrient depletion in some herder-based pastoral systems (e.g., in Australia and New Zealand) can be an insidious factor of grassland degradation.

Grassland management

The aspects considered in grassland management include the landscape (configuration, distance to water, size and terrain), abiotic conditions (climate, soils, and topography), socio-cultural conditions (culture values, technology development), forage quality and quantity (species composition and distribution), animal nutrient requirements (species, breed type, physiological status), and grazing management decisions (stocking rate, seasonal or all year grazing, continuous, rotational or deferred grazing, water points and forage supplementation).

There are several methods to improve the productive potential of the grassland: (i) grazing at stocking rates limited to the carrying capacity (i.e., the number of animal units that can be maintained per unit area without causing deterioration of the grassland), (ii) control of shrubs, and seeding of forage species, (iii) conservation of soil and water by reducing loss of soil and water from the grassland, (iv) removal of woody bushes or shrubs and weeds, which usually adversely affect the productivity of grasses, (v) use of controlled burning to promote recycling of nutrients stored in dried mulch and to reduce woody species invasion, and (vi) reservation of certain areas for producing hay. The grazing systems are: (i) continuous grazing, (ii) deferred grazing, (iii) rotational grazing, and (iv) deferred-rotational grazing. In the continuous grazing system the grassland is not divided into compartments and the animals move and graze in the whole area round the year. In the deferred system the grazing area is divided into compartments and at least one of these is rested until seed setting. In the rotational grazing no compartment is rested and all of them are grazed in rotation of specified duration. The deferred-rotation system is a mix of the above two types and is considered as the best system of grazing.

Grassland management in India

Since the major cause of deterioration of grasslands is unplanned and intensive grazing, the first step for improvement is to control and regulate grazing. Continuous grazing at controlled rates based on carrying capacity, or an early deferment for four months (August to November) has been found beneficial for semiarid and arid grasslands. For *Sehima* grasslands, deferred-rotational grazing has been found to be better for maintaining range productivity than continuous grazing (see Singh and Gupta, 1992). Results from a simulation-optimization model showed that a combination of moderate grazing with an occasional light-grazing year (to allow the legumes to recuperate) is the best grazing strategy over a 12-yr grazing plan, for a subhumid grassland (Swartzman and Singh, 1974). Various procedures which are helpful in reducing the grazing pressure on existing grasslands include enhancement of the fodder supply by seeding and fertilizing pastures; provision of silvipastoral top feed; production of fodders (maize, cow peas and berseem) on marginal agricultural land; and rotation of short-term leys of grass/legume pasture with cash crops on arable land.

Reseeding the deteriorated rangeland with higher-producing grasses and legumes (both native and introduced) has also been tried with success in India. Broadcast seeding without tillage gave better results with *Lasiurus hirsutus* than sowing in rows, whereas rows resulted in better establishment of *Cenchrus ciliaris*. The monsoon season is the best season for reseedling. Before species are introduced they should be screened carefully for yield and resistance to drought and grazing. Planting of grasses with trees having edible leaves such as *Acacia nilotica* var. *indica*, and *Prosopis cineraria* along the boundaries of arid rangelands can provide nutritionally rich fodder and shade for animals as well as serve as windbreaks (see Singh and Gupta, 1992).

25.4 DRY LANDS AND DESERTIFICATION

Drylands ecosystems cover over one third of the Earth's land surface and are critical for global food security. They support over half the world's livestock and cover about 44% of all cultivated areas. According to MEA (2005c), 2.1 billion people live in the world's deserts and drylands, 90% of the population in developing countries is in drylands, 50% of world's livestock is supported by dry rangelands, 46% of global carbon is stored in drylands, 44% of all cultivated land is in drylands and 30% of all cultivated plants came from drylands.

Drylands contain many unique habitats and many endemic species, with a species richness of flowering plants that is amongst the highest in the world. About 51% of the world's conservation hotspots include drylands. Some 10 to 20% of drylands are already degraded, and ongoing desertification threatens many of the world's ecosystems, including those inhabited by some of the poorest human populations. Therefore, desertification is one of the greatest environmental challenges today and a major barrier to meeting ecological and human needs in drylands. Severe land degradation is now affecting 168 countries across the world, according to new research released by the UN Desertification Convention (UNCCD) <http://www.unccd.int/en/programmes/Event-and-campaigns/WDCD/>.

Drylands are often categorised into four sub-types – dry sub-humid, semi-arid, arid and hyper-arid (deserts) – all of which are included in IUCN's programme of work (Table 25.5). These are diverse landscapes that cover vast areas and often encompass other ecosystems, such as wetlands or high-altitude forests. They also include significant urban areas and host mega-cities such as Mexico City, Los Angeles, Cairo and New Delhi. Drylands are characterised by highly uncertain precipitation, which means that water availability can vary greatly not just between seasons, but also between years and between areas. Other natural phenomena can characterise drylands and both herbivores and fire can play an important ecological role.

Table 25.5. Global figures for the four types of drylands (adapted from Safriel et al., 2005).

Dryland sub-habitat	Aridity index*	Share of global area (%)	Share of global population (%)	% rangeland	% cultivated	% other (including urban)
Hyper-arid	<0.05	6.6	1.7	97	0.6	3
Arid	0.05–0.20	10.6	4.1	87	7	6
Semi-arid	0.20–0.50	15.2	14.4	54	35	10
Sub-humid	0.50–0.65	8.7	15.3	34	47	20
Total	–	41.3	35.5	65	25	10

* The ratio of precipitation to potential evapotranspiration.

Desertification

Desertification is defined as “land degradation in arid, semi-arid and dry sub-humid areas resulting from various factors, including climatic variations and human activities” (UNEP, 1994). This phenomenon contributes to loss of biodiversity and to poverty in dry areas. Sustainably managing drylands can contribute greatly to conservation of biodiversity and to mitigating climate change as well as to strengthening human resilience. Such a degraded land looks like a desert, and its productivity is drastically lower than the potential value.

Desertification is occurring in many areas of the world (Fig. 25.10). About 6.46 billion ha of the terrestrial land surface is made up of dry lands (UNEP, 1997) distributed over more than 110 countries, which are potentially threatened by desertification. Africa, Asia and Latin America are the

most threatened by desertification. 2/3 of the continent is desert or drylands in Africa and is strongly linked to poverty, migration, and food security. Degraded areas include expanding deserts in China, India, Iran, Mongolia and Pakistan, the sand dunes of Syria, the steeply eroded mountain slopes of Nepal, and the deforested and overgrazed highlands of the Lao People's Democratic Republic. 30% of the land in the United States is affected by desertification.

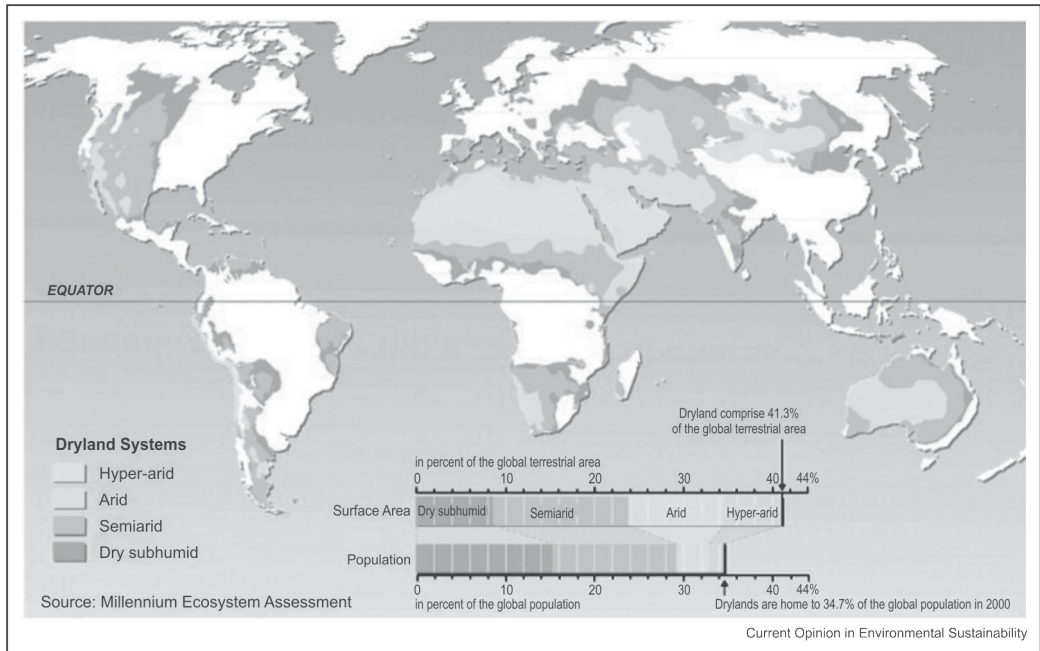


Fig. 25.10. Present day drylands and their categories. Ecosystems and Human Well-being: *Desertification Synthesis*. © 2005 Millennium Ecosystem Assessment.

Across the world, desertification affects the livelihoods of millions of people who rely on the benefits from dryland ecosystems. In drylands, water scarcity limits the production of crops, forage, wood, and other services ecosystems. Some 10 to 20% of drylands are already degraded, and ongoing desertification threatens the world's poorest populations and the prospects of poverty reduction. Therefore, desertification is one of the greatest environmental challenges today and a major barrier to meeting basic human needs in drylands.

There are many estimates for desertification. They vary, but are reliable enough to indicate gravity of the problem: (i) Deserts are increasing at a rate of 6 million ha annually in the world. The rates of desertification seem to accelerate particularly in developing countries, and each year 9 to 11 million ha drylands are being abandoned due to degradation (see in Daily, 1995). (ii) The desert of India (in Rajasthan) has advanced 1500 km eastwards during last 800 years, and the Sahara Desert in West Africa expands southward at the rate of 10 km a year. (iii) About 900 million ha land has become desertified during the past half century. (iv) At least 50 million people, half of them in Africa, are finding it difficult to feed themselves because of desertification. In India, the total area under desertification is 81.45 mha; water erosion (26.21 mha), wind erosion (17.77 mha), vegetal degradation (17.63 mha) and frost shattering (9.47 mha) are the major processes of desertification (Ajai et al., 2009). Area-wise Rajasthan, J&K, Gujarat and Maharashtra have high proportions of land undergoing degradation.

Impact of desertification

- In a heavily disturbed environment even where climate is not dry, the proportion of annual plants in vegetation increases, reducing the ability of land to restore itself.
- Because of vegetation loss, desertification makes areas more prone to floods.
- Desertification has huge economic consequences – the World Bank estimates that at the global level, the annual income foregone in the areas affected by desertification amounts to US\$ 42 billion each year, while the annual cost of fighting land degradation would cost only US\$ 2.4 billion a year.
- Land degradation brings hunger and poverty. People living in areas threatened by desertification are forced to move elsewhere to find other means of livelihood. Usually they migrate towards urban areas or go abroad. Mass migration is a major consequence of desertification.
- About one third of the earth's land surface (4 billion hectares) is threatened by desertification and over 250 million people are directly affected by desertification
- Cherrapunji has become world's wettest desert because subsequent to deforestation, soils were carried away by destabilizing force of unusually heavy rainstorms.

Development paths and desertification

The Millennium Ecosystem Assessment (2005c) developed four plausible scenarios to explore the future of desertification and human well-being until 2050 and beyond. In all four scenarios, the desertified area is expected to increase, though not at the same pace. Poverty and unsustainable land use practices will continue to be the main factors driving desertification in the near future, and climate change will also play a role. Effectively fighting desertification will help reduce global poverty and will contribute to meeting the Millennium Development Goals.

Desertification can be slowed down or even halted, and desertified land can be reclaimed by employing improved management practices. The strategy to combat desertification involves several steps: (i) preparation of desertification hazard map, (ii) reducing the pressure of grazing and harvesting, and bringing down within a sustainable range, (iii) stabilizing sand dunes, (iv) construction of wind breaks or shelter-belts in the windward side of the cropped land and settlements, and raising other plantations; and (v) treatment of saline soil and water-logged land.

United Nations Convention to Combat Desertification (UNCCD)

The United Nations Conference on Environment and Development, Rio de Janeiro and Agenda 21 called on the UN General Assembly to set up an inter-governmental committee to prepare for a legally binding instrument that addresses the problem of desertification. The ratification of the United Nations Convention to Combat Desertification (UNCCD) by June 1994 was a major post-Rio accomplishment, providing a mechanism to assess and respond to the problems of dryland degradation around the world. The UNCCD was adopted during 1994 and June 17 becomes the World Day to Combat Desertification. On a global scale efforts are under way to reclaim desertified areas, prevent further degradation, and preserve traditional knowledge about how to care for the land. Many of these efforts are encouraged by the U.N. Convention to Combat Desertification. The Convention's 195 parties work together to improve the living conditions for people in drylands, to maintain and restore land and soil productivity, and to mitigate the effects of drought. As the dynamics of land, climate and biodiversity are intimately connected, the UNCCD collaborates closely with the Convention on Biological Diversity (CBD) and the United Nations Framework Convention on Climate Change (UNFCCC), to meet these complex challenges with an integrated approach and the best possible use of natural resources.

The Convention to Combat Desertification has emphasized the full participation of people most affected by desertification and specifically women. Unlike the traditional system, in which women had no role, the current effort aims to be participatory and equitable, making sure that women are included in decision-making. Among other things the Convention calls on nations to protect “traditional and local knowledge, know-how, and practices”. In the 10-Year Strategy of the UNCCD (2008-2018) that was adopted in 2007, Parties to the Convention further specified their goals: “to forge a global partnership to reverse and prevent desertification/land degradation and to mitigate the effects of drought in affected areas in order to support poverty reduction and environmental sustainability”.

Soil erosion

Soil formation is a natural process, so soil is a renewable resource. But formation of an inch of soil layer generally takes 200-1000 yrs, and soil erosion may occur faster than rate of soil formation, so it can become a non-renewable resource. Soil erosion is one of the greatest environmental problems. It is associated with agriculture and is the principal cause of land degradation and has severe effects on soil functions. Increasingly, extensive cultivation, overgrazing, and deforestation, aggravated by drought, are destroying vegetation and leaving precious topsoil exposed to wind and water erosion. The global dust storms are the result of desertification, which now affect 70 percent of all drylands on earth.

Worldwide 75 billion tons of soil is removed annually by water and wind erosion, mostly from agricultural lands. This results in a world-wide destruction and abandonment of 12×10^6 ha of arable land annually (at present 1.5×10^9 ha of land is being cultivated annually in the world). The major causes of accelerated soil erosion in agricultural area are: (i) agriculture on steep slopes, (ii) reduced amount of living and dead biomass left in crop fields, and (iii) diminishing organic matter. In a watershed of Nepal Himalaya, Carson (1985) estimated rate of soil erosion by water, $440 \text{ t ha}^{-1} \text{ yr}^{-1}$, compared to the continental averages of $40 \text{ t ha}^{-1} \text{ yr}^{-1}$ and below $1 \text{ t ha}^{-1} \text{ yr}^{-1}$ from land covered with natural vegetation. Some soil erosion values are given in Table 25.6. Soil loss from barren lands can be 100 times or more of that covered with sod or other vegetal cover. Nearly 90% of crop residue in Bangladesh is removed, and situation is not different in India.

Table 25.6. Average soil erosion rates (t/ha/yr) in different landuses and continents. Soil formation rate (i.e. rate of conversion of parent material into A, E and B layers of soil), 1 t/ha/yr. (derived from data given in Pimental et al., 1995).

Landuse	Rate of erosion (t/ha/yr)
Undisturbed forests	0.004-0.05
Agricultural fields	
Asia, Africa, South America	30-40
Europe, North America	17

Erosion adversely affects soil productivity by reducing soil water storage, water holding capacity of soil and water infiltration rates, organic matter soil nutrient (Pimental et al., 1995). The soils eroded by water sediment rivers, canals and dams, increasing thus severity of floods, pollute water bodies with fertilizers and pesticides that cling with soil particles. In India, life of many dams is getting shortened because of much higher sedimentation rates than the assumed rates (Table 25.7). The wind erosion may cause the problem of air pollution by enhancing the amount of suspended particulates in the air.

Table 25.7. Sedimentation rate of reservoirs far exceeds the assumed rate (based on Valdiya, 1987).

Reservoir	River on which dam constructed	Rate of sedimentation (ha m/100 km ² /yr)	
		Assumed	Observed
Pong	Beas	4.29	15.10
Bhakra	Sutluj	4.29	6.14
Kalagarh	Ramganga	4.29	17.30
Mayurakshi	Mayurakshi	3.61	20.09
Matatila	Betwa	1.43	3.5
Gandhi Sagar	Chambel	3.61	5.29
Nizam Sagar	Marjna	0.29	6.57
Panchet	Damodar	2.47	9.92
Maithon	Barkar	1.62	13.02
Hirakund	Mahanadi	2.52	3.55

The worldwide cost of soil erosion both on-site and off-site is 400 billion dollar each year (Pimentel et al., 1995). This cost is estimated in terms of: cost of energy pumping additional water, cost of repairing damage of water bodies, etc.

Soil conservation

Terracing: The ancient Chinese, the Mayans of Central America, and the Incas of the Andes independently devised methods of terracing -- planting crops on a staircase of flat steps to slow the flow of water down a hillside to prevent from erosion from agricultural lands. In hilly terrain, erosion can be controlled by using terracing. In these regions, for example, farmers place lines of rock and debris on the contour, making sure the barriers are porous enough to let water seep through slowly.

Live fences and sand fences: In areas such as Thailand and the Philippines, farmers plant hedges of *Vetiveria zizanioides*, a tall, deep-rooted perennial grass, to form easy-to-maintain living fences along sloping lands. In arid regions of Africa blowing sand is a constant problem, sand fences often made of crop residue such as millet and sorghum stalks bound together along with large boulders to keep the sand in place. Straw grids and drought-tolerant shrubs and trees also help decrease the action of wind on loose soil.

Windbreaks: Windbreaks made out of tall trees reduce the drying effects of the wind to increase soil moisture for crops. Strip cropping, in which strips of grass or small grains are planted between other crops, can greatly reduce water erosion on cultivated fields.

Conservation tillage: Erosion control technologies, such as no-tillage or low-tillage, rotation of crops, and use of mulch cover can greatly reduce the erosion rate. Frequent ploughing accelerates soil erosion, and low tillage checks erosion and help water storage (Table 25.8).

Table 25.8. Effect of conservational technologies on soil loss (t soil loss/ha/year) from crop fields (based on Pimental et al., 1995). Data are from different states of the United States. Crops include wheat, cotton and soybean.

Practice for crop cultivation	Soil loss
Conventional	36-91
Conventional, continuous cropping	20-47
Conventional, plow-disc	27-47
Conservational practices	
Conservational, rotation*	7-9
Conservation, contour	6
Conservation, no-till	0.02-1.3
Conservation, mulch	1.7
Natural vegetation	
Undisturbed grass	0.07
Undisturbed forest	0.02

*Rotation ensures that some part of crop field is continuously covered by plants.

25.5 WATER RESOURCES

Freshwater resources are finite upon which we depend for our survival. Water sustains aquatic biodiversity, economic development and the ecological integrity of environment. We use renewable freshwater for drinking, irrigating the crops, and industrial uses, transportation, recreation and waste disposal. Water is a powerful indicator of ecological sustainability and economic prosperity.

Major pools of water

About three-fourth of earth's surface is occupied by oceans containing 97.5 percent of earth's water which is salty and not usable. Freshwater in rivers, groundwater, snow and ice, is about 2.5 percent or about 35 million (35,029,000) km³ only (Shiklomanov, 1998). Most of the freshwater is either in the form of permanent ice or snow in Antarctic/Arctic regions (about 69.55%) and is not readily accessible for use (Fig. 25.11). At any time most of freshwater is stored underground in the form of deep underground basins/aquifers, soil moistures etc. (30%) whereas less than 1% occurs in lakes, marshes and wetlands and rivers and only 0.1% occurs in the atmosphere as water vapour (Fig. 25.11).

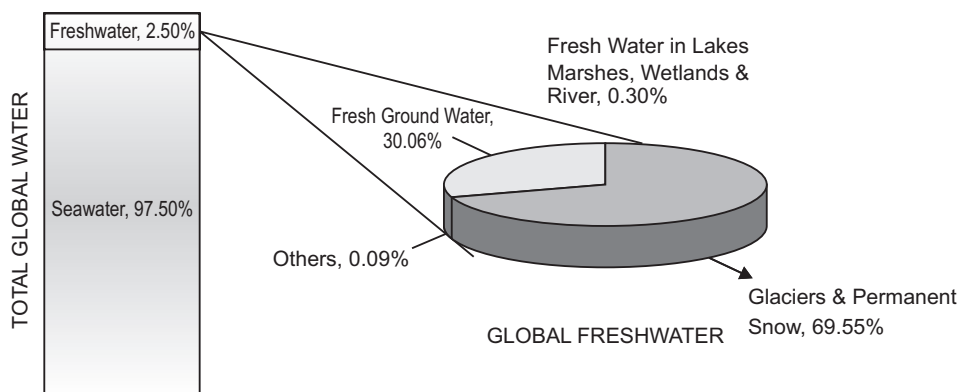


Fig. 25.11. Global freshwater distribution (based on Shiklomanov, 2000).

The renewable and usable water

Total usable freshwater supply to ecosystem and humans from river system, lakes, wetlands, soil moisture and shallow groundwater is less than 1% of all freshwater and only 0.01% of the global water resource on earth. As per WHO estimates, only 0.007% of all water on earth is readily available for human consumption all over the world. This indicates that freshwater on earth is finite.

The renewable and usable water consists of only the water flowing through the solar-powered hydrological cycle (fossil ground water is non-renewable, its extraction means depletion of the reserves). The amount of precipitation on land (P_{land}) is the amount of terrestrial renewable fresh water ($RFWS_{land}$, see in Fig. 25.12), which subdivides into evapotranspiration from land (ET_{land}) and runoff to sea (R). Thus, $RFWS_{land} = P_{land} + ET_{land} + R$ (see in Postel et al., 1996). The accessible run off totals at 12, 500 km^3/yr , representing 0.001% of the estimated global water. The difference between precipitation and evaporation from the land surface ($119,000 - 74,200 = 44,800 \text{ km}^3/yr$) represents the total runoff of the Earth's rivers ($42,700 \text{ km}^3/yr$) and direct groundwater runoff to the oceans ($2,100 \text{ km}^3/yr$), however with uncertainties in estimates on account of different reasons (Shiklomanov, 2000; see Sophocleous, 2004).

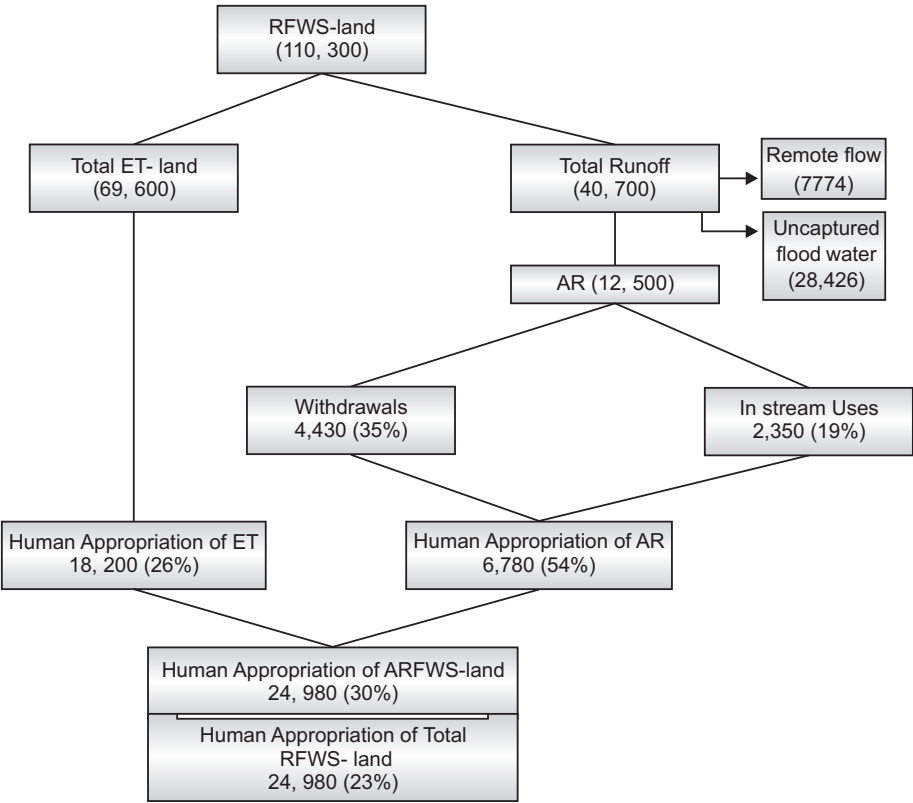


Fig. 25.12. Flow diagram of an analysis of human appropriation (i.e. act of appropriating water to specific use or purpose) of renewable fresh water supply on land. RFWS-land = terrestrial renewable fresh water supply, ET = evapotranspiration, AR = accessible runoff. All values are in $km^3 \text{ yr}^{-1}$. (based on Postel et al., 1996).

Human use of freshwater

Globally, annual withdrawal of water for human use including ground water amounts to at least 3,600 km³ (FAO, 2002). Presently, humans are using about 54 percent of worlds accessible freshwater contained in rivers, lakes and aquifers. Globally, consumption of water has increased 4-8 percent per year since 1950 as a result of population growth and change in life style of people. About 69% of total water is used in agriculture, 10 percent used in domestic and municipal supplies and 21% consumed by various industries (FAO, 2003). If per capita consumption of water resource continues at the present rate, humankind could be using about 90% of all available water within 25 years (Hinrichsen and Robey, 2000; see Sophocleous, 2004). In many regions of the world; the availability and quality of water to meet human needs are very limited. All over the world, more than billion people lack access to clean drinking water; three billion people lack basic sanitation services.

It is runoff water (e.g., water in river) which humans generally divert or withdraw for irrigating crops, in industrial production, and household uses (see Fig. 25.13). We depend on this water also for maintenance of aquatic life (fisheries), navigation, dilution of pollutants and electricity generation. Plants' evapotranspiration (ET) is about 530 kg water per kg of biomass production. To irrigate crop field for production of food grain we co-opt about 18,200 km³ water or 27% of total terrestrial ET. The distribution of global runoff in relation to human population is highly uneven (Table 25.9). Asia with 60% of world population shares only 36% of global runoff, while South America with about 5% of global population accounts for 25% of runoff.

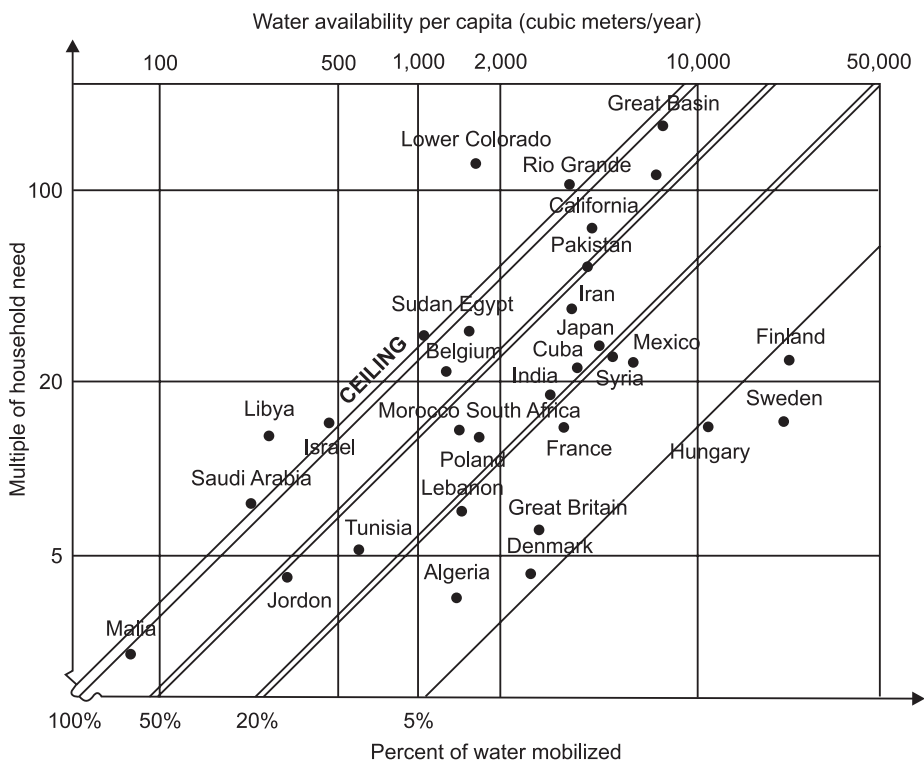


Fig. 25.13. Water demand relative to water availability (renewed through rainfall or inflow from upstream countries). Countries above ceiling are using more water than available. The excess water in them may come either from non-renewable aquifers (fossil water) or is transported from other nations. Household need (H) is taken as 100 l/d/person. Countries like Finland and Sweden have much more water than household demand. India stands mid-way (from Falkenmark and Widstrand, 1992).

Table 25.9. Continent-wise distribution of global runoff in relation to population share (based on Postel et al., 1996). Total global runoff is 40,700 km³/yr.

Continent/region	Total River Runoff* (km ³ year)	Share of global river runoff (%)	Share of global population** (%)
Europe	3,340	8	13
Asia	14,550	35.8	60.5
Africa	4,320	10.6	12.5
North and Central America	6,200	15.2	8.0
South America	10,420	25.6	5.5
Australia and Oceania	1,970	4.8	0.5
Total	40,700	100.0	100.0

* Runoff Estimates from L'Vovich et al., (1990); ** Population Estimates from 1994 world populations data sheet (Population reference Bureau, Washington, DC, 1994)

Nations widely differ in their household water demand relative to water availability (Fig. 25.13). For example, Libya, Lower Colorado, Rio Grande, Israel and Saudi Arabia are using more water than is renewed each year through rainfall or inflow from upstream countries. This excess water comes from ancient, non-renewable aquifers or transported from other areas.

Much of the runoff is remote and inaccessible for use, particularly in tropics and high northern latitudes. For example, the Amazon River contains 15% of global runoff, but 95% of its runoff is inaccessible, and only 0.4% of world population has access to its water. Most of the river flows in Tundra Taiga are remote from human populations. The control over runoff water is difficult to achieve, therefore a substantial amount of runoff is lost as flood water even in accessible areas. Approximately 27% of the total runoff is renewable ground water and base river flow, much of the remaining 29,600 km³ is flood water difficult to capture by constructing dams. At present large dams actively regulate 3500 km³ of river runoff.

Table 25.10. Estimated global water use and consumption by sector, values are of 1990 (based on Postel et al., 1996).

Sector	Use (km ³ /yr)	Consumption as approximate % of use
Agriculture*	2880	65
Industry*	975	9
Municipalities (household uses)*	300	17
Reservoir losses**	275	100
Subtotal	4430	51.5
Steam flow needs	2350	0
Total	6780	34
Total as a percent of AR (12,500 km ³)	54%	18

*Assumes averaged applied water use of 12,000 m³/ha and consumption equal to ~ 65% of with draws.

**Assumes evaporation loss equal to 5% of gross reservoir storage capacity.

* Estimates are from Shiklomanov (1993).

AR = Accessible runoff.

Of the accessible runoff (AR) the greatest human use is in agriculture (Table 25.10). Industrial water use has levelled off or declined in developed countries, but is increasing rapidly in most of developed countries ($\sim 975 \text{ km}^3$ use globally). Where runoff water is accessible, human intervention has greatly affected river systems through diversion and impoundments. In the USA, as an example, only 2% of the rivers run unimpeded. Major rivers, such as the Ganga, the Nile and the Colorado are used so much that little water reaches the sea. This change severely affects the life in sea. For example, reduction in volume of the Aral Sea resulted in loss of many fish and other species.

Global withdrawals have tripled over the last 50 years to meet the demands of a growing population with increasing wealth and consumption levels. Projected water demands may reach planetary boundaries in the coming decades (Rockstrom et al., 2009). Some facts about water-use are summarised (FAO, 2011): 70% of all water is withdrawn from aquifers, streams and lakes by agriculture: 60% of total world crop is produced by rainfed agriculture systems: 40% of world population is now living in water-scarce regions: 11 countries currently use more than 40% of their water resources for irrigation each year, a threshold that is considered critical: 8 countries are withdrawing 20% of their water resources annually, indicating substantial pressure and impending water scarcity; 100 percent or more of renewable water resources are currently used in Libya, Saudi Arabia, Yemen, and Egypt, while only 1% of total renewable water resources is currently used in South America.

Agricultural, industrial and domestic water withdrawals have steadily increased. Agriculture is by far the largest global water user with withdrawals for this purpose being unsustainable in many places due to unbalanced long-term irrigation water budgets (M EA 2005). Between 1960 and 2000, global ground water withdrawal increased from 312 km^3 to 734 km^3 per year, resulting in groundwater depletion increasing from 126 km^3 to 283 km^3 per year (Wada et al., 2010). Many globally important agricultural centres are particularly dependent on groundwater, including northwest India, northeastern China, northeast Pakistan, California's Central Valley, and the western United States (Fig. 25.14) (Wada et al., 2010).

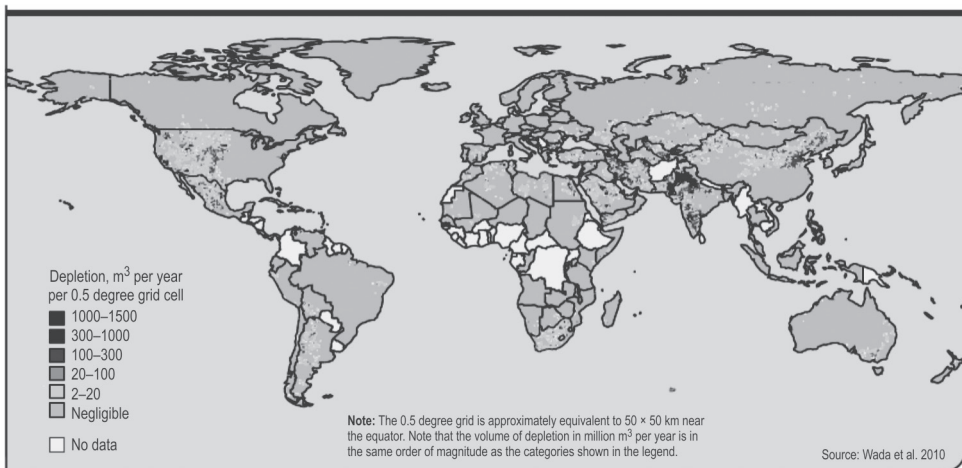


Fig. 25.14. Global annual ground water depletion, 2000 (all in $\text{mm} \cdot \text{a}^{-1}$). (adapted from Wada et al., 2010).

Water Foot Print

In the parlance of water resource management, water is termed 'blue water' and 'green water' depending on the source. Blue water is located in visible sources of water like rivers, lakes, ground water, etc. and green water resides in the pore spaces in the soil. When rain drops on earth it is

divided into blue water and green water; the global ratio of blue vs. green water is 35:65 (this ratio is also true for India). The blue water is actually the runoff water but a portion of it is converted into green water when we irrigate the cropland. The water that we contaminate is called 'grey water'. On the other hand, 'virtual water' is the amount required to grow a product either in agriculture or in industry. The Water Footprint of a product is the volume of freshwater appropriated to produce the product, taking into account the volumes of water consumed and polluted in the different steps of the supply chain. Most cultivated plants for example need 900-1200 (some up to 4000) moles of H_2O to fix 1 mole of CO_2 . This virtual content of water of rice, wheat and potato are 2850, 1400 and 1000 litres per kg, respectively. Virtual water content of 1 kg boneless pork comes to at least 15000 litres, one cup of coffee requires 140 litres of virtual water. Water foot print which is the consumption based assessment of water utilized by individual, community, product, industry, state or a country, is calculated on the basis of virtual water. The global water footprint in the period 1996-2005 was 9087 Gm^3/yr (74% green, 11% blue, 15% grey), agricultural production contributed 92% to this total footprint.

The water footprint of the global average consumer in the period 1996-2005 was 1385 m^3/yr . The average consumer in the US has a water footprint of 2842 m^3/yr , while the average citizens in China and India have water footprints of 1071 m^3/yr and 1089 m^3/yr , respectively. China, India and the US are the countries with the largest total water footprints within their territory, with total water footprints of 1207, 1182 and 1053 Mm^3/yr , respectively. India is the country with the largest blue water footprint within its territory: 243 Mm^3/yr , which is 24% of the global blue water footprint. Irrigation of wheat is the process that takes the largest share (33%) in India's blue water footprint, followed by irrigation of rice (24%) and irrigation of sugarcane (16%). China is the country with the largest grey water footprint within its borders: 360 Mm^3/yr , which is 26% of the global grey water footprint.

External water foot print is : virtual water import minus virtual water export; so the total water foot print of a country is internal water foot print + external water foot print. The internal footprint of a country is sum total of agriculture water use, domestic water use, and the industrial water use minus the virtual water export in the form of products produced in the country. Virtual water export from US is 314 Gm^3/yr , of China 143 and of India 125 Gm^3/yr . A country can externalise its water footprint by importing products (e.g., grains) from other countries. Major virtual water importer is US (234 Gm^3/yr) followed by Japan (127 Gm^3/yr), Germany (125 Gm^3/yr) and China (121 Gm^3/yr). More information can be found in Mekonnen and Hoekstra (2011).

Human Impact on Water Resources

1. Approximately 99 percent of all freshwater is in underground aquifers and about 25% of the world population draws its water from the ground water supplies for drinking and irrigation. The ground water resource is being depleted globally; the reserves are being overexploited (ten times faster).
2. Poor management has resulted in the salinization of about 20% of the worlds irrigated lands in the arid and semiarid regions.
3. The pollution of freshwater systems limits the range of possible use of freshwater even in the most prosperous countries. Industrial pollution has been found to affect the entire ecosystem and the aquatic food chains.
4. About 60% of the world's biggest rivers (numbering 227) have been fragmented because of construction of dams, diversion or canalization, which have negative impacts on freshwater systems.

5. With global warming, greater volumes of water will evaporate from plants, soil and water bodies. Some decrease in snow and ice cover has already been observed in mountain regions of the world.
6. About 150 of the world's 200 major river systems are shared by at least two countries. Conflicts between countries occur both because of water extraction and pollution.
7. Dam building and river control significantly benefit humans, providing flood protection, reliable water supplies and hydroelectric power. But dams can also have detrimental impacts on ecosystems, including channel fragmentation and flow modification, altering ecosystem processes and affecting aquatic organisms, particularly migratory species.
8. Dam density is highest in industrialized countries, although construction in developed regions has slowed because of availability of suitable locations. However, dam building is being actively pursued in many developing countries to secure water and electricity supplies.
9. Salinization of overexploited aquifers, especially in coastal areas, is another serious concern, particularly for communities dependent on groundwater for drinking.
10. Groundwater nitrate concentrations are increasing, especially in areas of rapid urbanization, inadequate sanitation and/or heavy agricultural fertilizer use. Nitrate in groundwater contributes to eutrophication and has direct human health impacts. Both naturally occurring arsenic and that mobilized by human activities threaten drinking water quality in many countries. Groundwater contaminated with arsenic from natural geologic sources affects 35–75 million people.

Groundwater contamination

Indicators

Arsenic, nitrate and salinization

Global trends

Very little progress in some areas; deterioration in others

Most vulnerable communities

Populations in rapidly urbanizing areas with inadequate sanitation

Regions of greatest concern

Arsenic is of particular concern in Bangladesh, India, highly populated river deltas in South East Asia, North America and Eastern Europe

How to increase water availability

To capture and store more flood runoff and to desalinate sea water are the principal options. Towering icebergs seem to be impractical for at least next 30 years (Postel et. al., 1996). The construction of new large dam reservoirs is going to decline in future because of rising costs, and social and environmental problems. During 1950 and the 1970s on an average 1,000 large dams per yr were constructed, at present number has dropped to 260 per year (WCD, 2000). Pollution prevention from both point and non-point sources may play significant role in increasing availability of AR and in safeguarding ecological functions. Increase in water use efficiency in all the sectors, changes in crops, and ecological restoration of marginal lands in place of cultivating irrigated agricultural crops are some other measures which could help slow the growth of human use of AR.

There is need to evolve a combination of measures to ensure adequate supplies of water for different sectors with due consideration to socio- economic characteristics of water, increasing the

efficiency and productivity of water use, adoption of adaptive management in water resources based on monitoring and re-evaluation and protecting the integrity of aquatic ecosystems (UNEP, 2002b; Sophocleous, 2004).

25.6 FISHERIES AND OTHER MARINE RESOURCES

Fisheries are important natural resources, accounting for 19% of the total human consumption of protein, and providing direct employment to 200 million people worldwide (Botsford et al., 1997). Over one billion people worldwide depend on fish as their main source of animal protein. Fishing is the catching of aquatic wildlife and harvesting of natural production, therefore catches must not exceed the natural population capacity for regeneration. The number of over fished population accounts for 22% of recognized fisheries (Vitousek et al., 1997b) and about 47% of fisheries are at their sustainable limits of exploitation (FAO, 2002). Total fish production in 2001 was 130.2 million tonnes, out of which 37.9 million tonnes was from aquaculture (FAO, 2002). The global fish catch has remained static at 100 million tonnes per year from 1995, except the variations due to catches of Peruvian anchoveta.

Threats to fisheries

Fishing methods such as trawling and drift nets capture everything in their way indiscriminately; sometimes 70% of the catch is thrown away. This commercial fish catch disturbs 27 million metric tonnes non-target marine animals annually (accounts for one-fourth of world's marine fish catch). Dredges and trawls also adversely affect the marine habitats.

The increase in fishing-fleet capacity is one of the main threats to the survival of marine capture-fishery resource. The fish populations in the world ocean suffer from overexploitation, habitat destruction, incidental mortality of non-target species, evolutionary shifts in population demographics, pollution, global climate change and change in ecosystem structure and function (FAO, 2002; Pikitch et al., 2004).

Globally, humans use 8% of the marine primary productivity, but removal is nearly 25% in upwelling areas and 35% in temperate continental shelf systems. In Japan, as an example, humans predominantly depend on marine food. The continental shelves constitute 7 % of the total ocean area of earth but generate most of the biological production supporting over 90 % of the global fish catches except the tuna and other oceanic organisms that gather their food from the less productive open oceans (Pauly et al., 2002). Clearly, maintaining the biodiversity of healthy reefs is the key maintaining sustainable reef fisheries.

Fisheries management and conservation

Humans have failed to conserve marine fisheries as fisheries management to date has focussed to maximise the catch of a single target species, often ignoring habitat, predator and prey of the target species as well as their interaction with other ecosystem components. Studies have suggested putting an upper limit on fishing mortality to prevent overexploitation of fishing grounds. Following methods can be effective for sustainable fisheries:

Marine protected areas and Reserves: The marine protected areas (MPAs) can play an important role to help rebuild the depleted fish stocks (Roberts et al., 2001; Pauly et al., 2002). In most cases, the successful MPAs have been used to protect the sedentary species, which eventually could sustain the fisheries outside the reserves by exporting juveniles or adults (Roberts et al., 2001; Pauly et al., 2002). The MPAs can also help to rebuild the complexity of the habitat, decreasing mortality of juveniles, and prevent extinction of species just like forest or other natural terrestrial habitats (Pauly et al., 2002). Thus zoning of the ocean into marine reserves and areas with limited

fishing, establishing reserves of appropriate size, location of marine reserves and their networking as well as managing functionally diverse ecosystems (e.g. coral reefs and mangroves) could be practicable strategy for sustainable fisheries (Pauly et al., 2002).

Ecosystem-based management: Ecosystem-based fishery management emphasizes ecosystem management rather than the target species for sustaining healthy marine ecosystems and the fisheries they support (Pikitch et al., 2004). The strategies for ecosystem-based management should focus on : (i) avoiding degradation of ecosystems, (ii) minimizing irreversible changes in assemblages of species and ecosystem processes (iii) to maintain record of long-term socioeconomic benefits, (iv) to understand ecosystem processes in relation to human impacts, (v) to reduce excessive killing of nontarget species and juvenile life, (vi) using more selective and less damaging fishing technologies and managing target species keeping in view the state of the system, protected species and non-target species. (Pikitch et al., 2004).

Aquaculture: Aquaculture offers a potential solution to the dilemma of depleted ocean fisheries as well as meeting the growing demand for seafood. Aquaculture now accounts for one-third of the world's total supply of food fish (White et al., 2004). Aquaculture is the farming of aquatic organisms including fish, molluscs, crustaceans and aquatic plants; and is the fastest growing sector of the world food economy, and increasing by 10% per year (FAO, 2002; White et al., 2004). The vast majority of aquaculture takes place in Asia, over 70% of worldwide aquaculture production was in China alone (FAO, 2002). However, the current 'blue revolution' of aquaculture has adopted the industrial mode of production with a focus on farming high-value carnivorous fish (e.g., salmon) often requiring intensive use of resources, having adverse environmental impacts as well as social conflicts. The environmental impacts of aquaculture are summarised in Table 25.11. Ecological aquaculture need to be promoted with a focus on developing the farming systems that preserve the environments in which they are situated, while enhancing the quality of the environments as well as maintaining a productive culture system. Thus ecological aquaculture needs to be incorporated into sustainable fisheries management and coastal zone management (White et al., 2004).

Table 25.11. Aquaculture Systems and their environmental impacts (reproduced from White et al., 2004).

System Type	Common Species	Environmental issues
Mollusk culture	Oysters, clams, mussels	Introduction of exotic species (target and non-target) and diseases. Can be minimized with planning and Management.
Inland pond	Catfish, carp, yellow perch, tilapia,	Moderate environmental impact due to some waste discharge and use of exotic species.
Coastal pond	Shrimp	High environmental impact. Discharge of wastes. Destruction of mangroves.
Net pen and/or cage	Salmon, sea bass, sea bream, grouper, tuna, snapper	High environmental impact. Discharge of biological and chemical wastes, exotic species, and transfer of diseases.

25.7 ENERGY RESOURCES

Energy use

Energy has been described as the ability to do work, transfer heat, or to set things in motion. Energy as food is fundamentally important for all organisms. But humans require huge amount of additional energy to sustain their civilization: to run their cars and industries, to process materials and to keep their houses warm or cool. To maintain our current civilization we have largely depended on fossil fuels. Electricity is a favourite mode of delivery of energy for human use. It can be delivered without much technical problem directly to power appliances and machines. Electricity use has shown 100-fold increase during last 2-3 decades in many countries. Over the years the efficiency of generation of electricity from fossil fuels has increased from 20 to 35%, and electricity is likely to be with us forever as a delivery form of energy. Apart from burning fossil fuels, electricity can be generated by burning wood, garbage, trash, as well as from flowing water, wind, the earth's heat, sunlight, and nuclear fission.

All energy used by man originates in one of the following sources: (a) *radiant energy* emitted by the sun (solar energy); (b) *geothermal energy* from the interior of earth; (c) *tidal energy* originating in the gravitational pull from the moon; (d) *nuclear energy*.

The dominant is solar energy, thousands of times larger than all the others and inexhaustible as long as the sun shines (approximately 4.5 billion years). The energy available on earth with present energy consumption and fossil fuel reserves is compared in Table 25.12.

Table 25.12. Energy available on Earth (source: Hulbert 1971).

Available Energy	x 10 ¹² watts
Solar	174 000
Geothermal	32
Tidal	3
Fossil fuel reserves (mainly coal)	~ 2000
Winds, waves convection and currents	370
Present world energy consumption*	12.7
Photosynthesis	40

* World's average primary energy consumption (2001) is 2.1 kW per capita).

Overall, energy resources fall within the following broad categories:

Non-renewable energy Fossil fuel includes all forms of stored solar energy including crude oil (petroleum), natural gas and coal. These are considered primary sources of energy. The distribution of proved total energy reserves and oil reserves is greatest in Middle East, Europe and Eurasia (Figs. 25.15 a, b and 25.16).

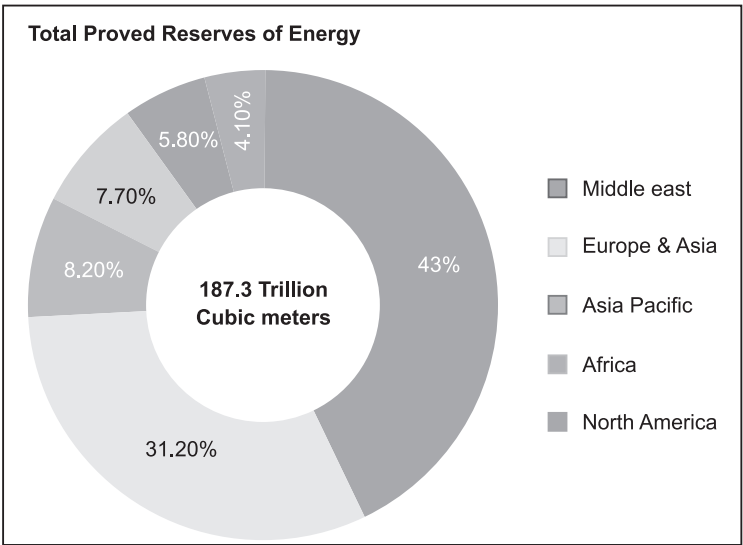


Fig. 25.15. a. Proved total Energy reserves at the end 2012 (Trillion cubic meters) worldwide. (based from British Petroleum Company, 2012 BP Statistical Review of World Energy).

Nuclear energy is energy produced from nuclear fission (splitting) or fusion (combination). Nuclear power is a relatively clean form of energy in the sense that it does not cause air pollution, and requires very little fuel. Earlier it was held out that it is going to solve our energy crisis. In some countries it has already become important energy source. However, its contribution to total energy use in the world is still small and its future uncertain because of several factors: (i) high cost of building nuclear plant, (ii) security problem (how to keep uranium and plutonium out of hands of the wrong people), (iii) waste disposal problem, and (iv) nuclear accidents, particularly of the Chernobyl-size. We have yet to come up with safe way of storing nuclear wastes including old reactors. From bottom of ocean to space, all kinds of site have been proposed for storing nuclear wastes, but with no progress as far as giving practical shape is concerned.

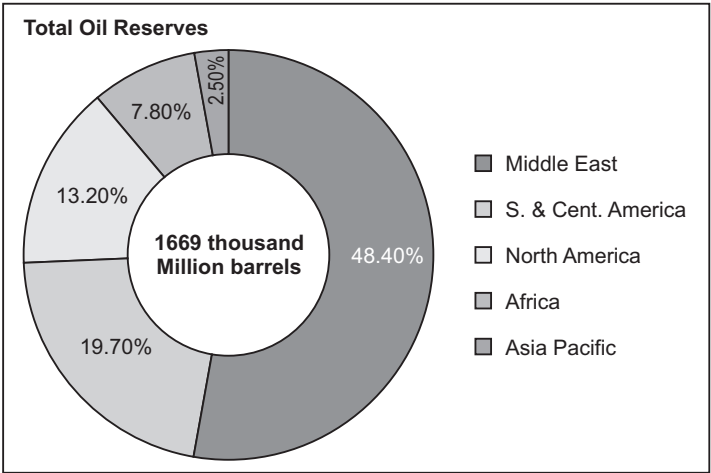


Fig. 25.15. b. Proved total oil reserves at the end 2012 (Trillion cubic meters) worldwide. (based from British Petroleum Company, 2012 BP Statistical Review of World Energy).

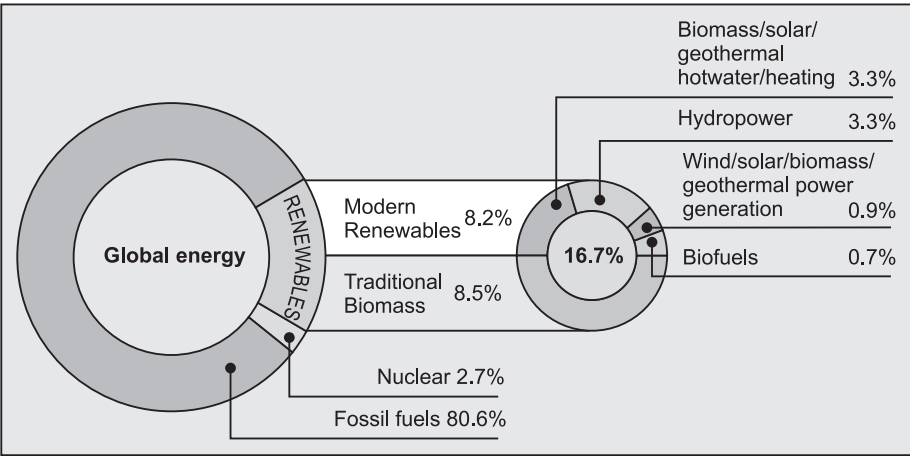


Fig. 25.16. Renewable energy share of global final energy consumption, 2010.

Renewable Energy

The renewable energy sources and technologies include solar energy, wind energy, hydropower, biomass energy including biofuels, and geothermal energy. **Biofuel** is any fuel that is derived from recently living organisms or their metabolic byproducts. It includes traditional and modern biomass.

In 2010 total renewable energy accounted for 16.7% of total energy supply. Biomass heat accounted for 11.4%, and hydropower 3.3% (Fig. 25.17) of total energy consumption. Modern renewable energy can substitute for fossil fuels in four distinct markets: power generation, heating and cooling, transport fuels, and rural/off-grid energy services.

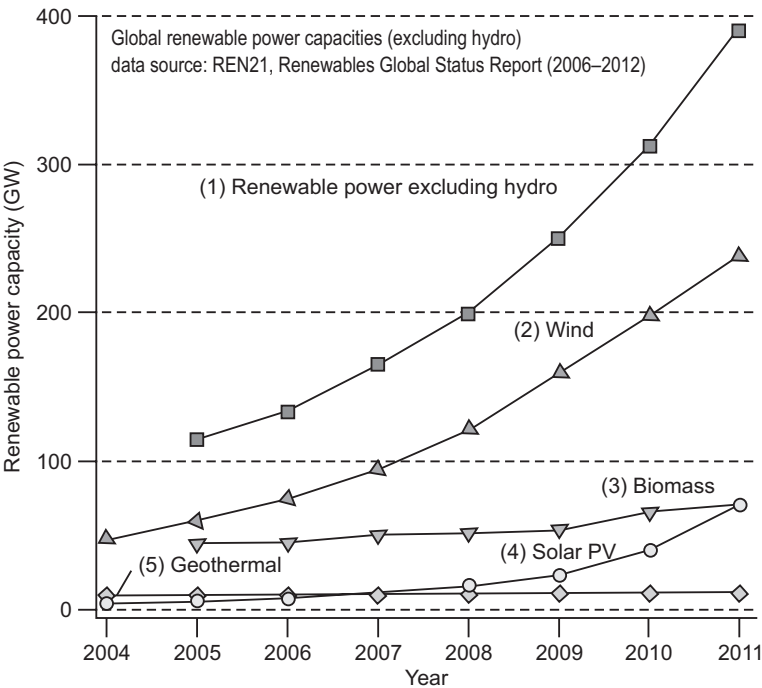


Fig. 25.17. Global renewable power capacities (excluding hydropower) (adapted from REN21, 2012).

During the period from end-2006 through 2011, total global installed capacity of many renewable energy technologies grew at very rapid rates (e.g., Solar photovoltaics (PV) grew the fastest of all renewable technologies during this period, with operating capacity increasing an average of 58% annually (REN 21, 2012). It was followed by concentrating solar thermal power (CSP), which increased almost 37%, growing from a small base, and wind power, which increased 26% (based on diverse sources, see REN 21, 2012). For the first time ever, in 2011 solar PV accounted for more new electric generating capacity in the European Union than did any other technology(European Wind Energy Association (EWEA), “Wind in Power: 2011 European Statistics” (Brussels: February 2012; see REN 21, 2012).

Hydropower

Hydropower is indirectly linked to sunlight which evaporates water of the oceans which precipitates in the form of rain in the continents forming rivers. Dams are built on rivers forming reservoirs, which guarantee a steady supply of water for electricity generation. A large dam is a dam with a height of 15 meters or more from the foundation. Large hydropower plants flooding large areas might displace people and have undesirable ecological or social impacts.

Hydropower is produced in 150 countries, the Asia-Pacific region generated 32 percent of global hydropower in 2010. China is the largest hydroelectricity producer, with 721 terawatt-hours of production in 2010, representing around 17 percent of domestic electricity use. Brazil, Canada, New Zealand, Norway, Paraguay, Austria, Switzerland, and Venezuela have a majority of the internal electric energy production from hydroelectric power. Paraguay produces 100% of its electricity from hydroelectric dams, and exports 90% of its production to Brazil and to Argentina. Norway produces 98–99% of its electricity from hydroelectric sources

Hydroelectric power is electricity generated via turbines as water passes through holes in dams. Somewhat less than 20% of the world’s hydroelectric capacity is currently being trapped, so there is a large scope for trapping much more. But it does not mean, all is good with hydroelectric power, this, in fact, has many negative points.

Controversies associated with dams are many: (i) the lakes (or water reservoirs) formed behind the dams cover a large area of land, including good agricultural bottom land, (ii) the ecosystems behind dams including the aquatic ones are disturbed and modified, (iii) precious biodiversity may be destroyed because of habitat loss, (iv) deep lakes with low water flow support aquatic life different than do the fast flowing rivers, (v) development costs of hydroelectric projects are very high, partly because dams get filled up with silt much before the life time projected, (vi) displacement of people causes great social problems. More about problems associated with dams is given in the box.

Solar energy

Solar energy is delivered to the earth every day whether we use it or not. World over, the amount of solar energy of a day is equal to that the world uses in 30 years. But many technological and economic factors stand in the way of achieving even a small fraction of this potential. Solar energy can be used for space and water heating, or it can be converted into electricity via steam, and fed into the existing electrical, delivery/distribution systems or power grids. Direct solar energy technologies harness the energy of solar irradiance to produce electricity using photovoltaics (PV) and concentrating solar power (CSP). Solar panel electricity systems, also known as solar photovoltaics (PV), capture the sun’s energy using photovoltaic cells. These cells don’t need direct

sunlight to work - they can still generate some electricity on a cloudy day. The cells convert the sunlight into electricity, which can be used to run household appliances and lighting. Based on the 'photovoltaic effect' of semiconductors, PV cells generate power by converting solar irradiation to electrical direct current.

CSP systems are used to convert solar irradiation to heat energy, which is then converted to electrical energy. The sun's rays are concentrated onto a small area through reflection and refraction using mirrors and lenses, and the heat produced is used to power turbines (generally steam) which generate electricity. In CSP systems, a field of heliostats is employed. Each heliostat consists of a mirror like reflector and a control system (Fig. 25.16). Solar energy is still very expensive; its storage is still limited, cloud covers can limit its usefulness on many days. Because of these reasons solar energy contributes negligible to our energy use below 1%. However, because of its being environment friendly its use need to be encouraged. To deal with cloudy days, satellites orbiting at thousands of miles above cloud cover can be used to convert sunlight into microwaves. The microwaves beamed to earth would be converted into electricity (Fig. 25.19).

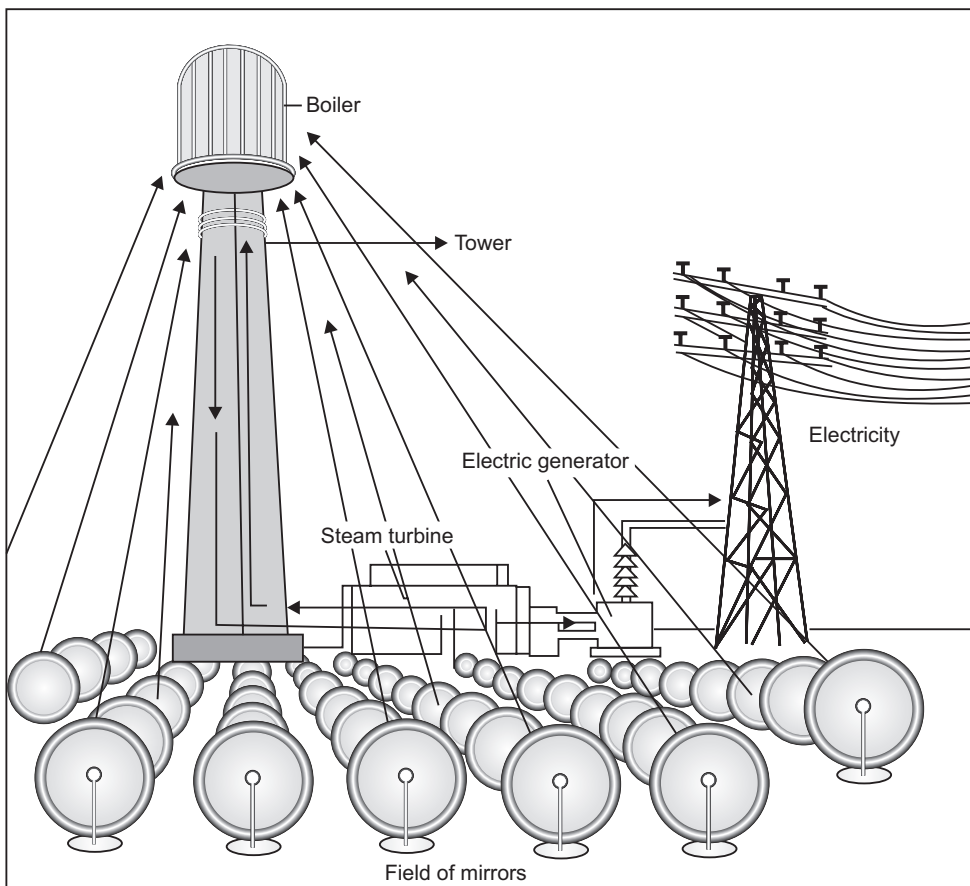


Fig. 25.18. A solar power plant based on a field of heliostats, each consisting of a mirror like reflector and control system. The control system is used to continuously focus reflected solar rays into a central boiler mounted on a tower. To produce electricity a turbine is used. The fluid heated in the tower expands through the turbine, resulting in electricity generation (from Kupchella and Hyland, 1989).

Wind energy

Harnessing wind energy by constructing large fan like structures has been known for long. Such structures were common in use to lift water from wells for irrigating crops. Now, fans are used to drive turbines and generate electricity. Wind is free and does not pollute, but it is intermittent and not available everywhere. Visual pollution, hazards to air planes and migrating birds, noise and interference with television and radio are the other problems. Its use is still negligible.

Tidal power

Like wind energy tidal power is free and clean. The tidal energy can be used to generate electricity in some coastal areas and is a renewable source of energy. But it is expensive to build and maintain structures required to harness the tidal energy. Building dams across harbours or inlets can cause navigational problems. Though tidal power came into use as early as 11th century, it is very unlikely that it would ever be an important energy source.

Ocean thermal energy

Oceans of the world are enormous collectors of solar energy, absorbing energy equivalent of 250 billion barrels of oil a day. This energy can be trapped to generate electricity. This is based on either using the warm sea water to evaporate a fluid with low boiling point, or to evaporate sea water itself by subjecting it to a vacuum. In both the conversion of liquid to a gas creates pressure that is used to drive turbines for generating electricity. This renewable and clean energy can be promoted. Technological improvements, particularly for dealing with the corrosiveness of sea water are required for its wider use.

Geothermal energy

The energy derived from heat of the earth is called geothermal energy. As we go deeper from the earth surface temperature on average increases approximately at the rate of 3.3°C for every 100 m from the surface. In some places this rise in temperature is much more, and molten lava and heat from the earth's core come close to the surface. This heat can be used in generating electrical energy. There are 600 active volcanoes on the earth. However, with a geothermal plant there may be associated environmental problems, such as release of the huge amounts of heat, water vapour and sulphur. Hydrothermal power plants and thermal applications of geothermal energy are mature technologies.

Energy from biomass

Traditional biomass refers to unprocessed biomass-based fuels such as crop residues, firewood and animal dung used frequently with very low efficiencies for cooking and heating in

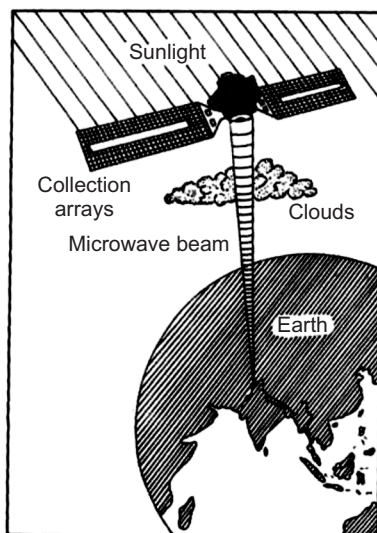


Fig. 25.19. Satellites fixed over the same point on the earth's surface could be used to convert sunlight into microwaves, which in turn, could be beamed to earth and converted into electricity. Thus solar energy could be used on cloudy days and in nights (from Kupchella and Hyland, 1989).

many developing countries. In developing countries fuelwood (a biomass energy source) is still major cooking energy. Biomass energy accounted for an estimated 14% to 11% of the world's final energy consumption during 2000 to 2001, which is comparable to energy consumption through electricity (15%) and natural gas (16%) (IEA, 1998, 2003). At a regional level, the developing regions (Africa, Asia and Latin America) record high levels of biomass energy consumption (18-49%) as compared to developed regions (3 to 5%) (IEI, 2002). Biomass is the energy source for the poor. It is estimated that approximately 50% of the population in developing countries relies on biomass energy, with some regions recording higher proportions of population (e.g., 73% in Africa).

Modern biomass refers to biomass produced in a sustainable way and used for electricity generation, heat production and transport. This includes wood and forest residues from reforestation and/or sustainable forest management, rural (urban and agricultural) and urban residues (including solid waste and liquid effluents). The organic matter wastes, such as municipal refuse and agricultural and forestry residues can also be used as biomass energy. This is the type of biomass most commonly used in OECD countries.

Bioenergy can be produced from a variety of biomass feedstocks, including forest, agricultural and livestock residues; short-rotation forest plantations; energy crops; the organic component of municipal solid waste; and other organic waste streams. Through a variety of processes, these feedstocks can be directly used to produce electricity or heat, or can be used to create gaseous, liquid, or solid fuels. Some examples of commercially available bioconversion technologies include small- and large-scale boilers, domestic pellet-based heating systems, and ethanol production from sugar and starch. Advanced biomass integrated gasification combined-cycle power plants and lignocellulose-based transport fuels are examples of technologies that are at a pre-commercial stage, while liquid biofuel production from algae and some other biological conversion approaches are at the research and development (R&D) phase (IPCC, 2012b). From biomass several useful energy materials are derived, viz., biogas, alcohol and petrol. Animal wastes, such as cow dung has been found useful for generating methane (CH_4), which is used as cooking gas. Considerable efforts have been made in India to make people accept use of biogas plants, but with a limited success.

Biofuel is any fuel that is derived from recently living organisms or their metabolic by-products. Examples of biofuels include ethanol, which is produced from sugar cane and methane that is released by decaying garbage. Methanol (CH_3OH) or wood alcohol and ethanol ($\text{C}_2\text{H}_5\text{OH}$) or grain alcohol represent common alcohols from biomass. As for methanol, it can be made from almost any kind of existing fuel, ranging from wood to agricultural residue to coal. A mixture of gasoline and ethanol called gasohol is also used. Ethanol has made a valuable contribution to the development of Brazil's agro-industry. Moreover, the increased use of alcohol as a transport fuel appears to have contributed to the reduction of air pollution in mega-cities such as São Paulo.

Global energy consumption

World primary energy consumption grew by 1.8% in 2012, well below the 10-year average of 2.6%. Consumption in OECD countries fell by 1.2%, led by a decline of 2.8% in the US (the world's largest decline in volumetric terms). Non-OECD consumption grew by 4.2%, below the 10-year average of 5.3%. Global consumption growth was below average for each fossil fuel and for nuclear power; regionally growth was below average everywhere except Africa. Oil remains the world's leading fuel, at 33.1% of global energy consumption.

The five principal energy use sectors are: domestic, transportation, industrial, commercial and agricultural. By 2100, six to eight billion additional people, or more than double today's population, will need access to affordable, reliable, flexible and secure energy services. According

to the World Energy Outlook (WEO, 2004) and International Energy Outlook (2013), demand for oil will continue to expand, whilst gas use is projected to double by 2030, largely because it will be the fuel of choice for electric power generation. Coal will continue to supply a fifth of world energy needs, mostly in power g There has occurred more than 20-fold increase in total energy use since the industrial revolution , with the growth caused slightly more by population increase than by expansion of per capita consumption (Fig. 25.20) (see Ehrlich et al., 2012).

According to the International Energy Outlook (IEO, 2013), world use of petroleum and other liquid fuels grows from 87 million barrels per day in 2010 to 97 million barrels per day in 2020 and 115 million barrels per day in 2040 (Fig. 25.21). With world GDP rising by 3.6% per year, world energy use will grow by 56% between 2010 and 2040. Half of the increase is attributed to China and India. Renewable energy and nuclear power are the world’s fastest-growing energy sources, each increasing by 2.5% per year; however, fossil fuels continue to supply almost 80 percent of world energy use through 2040 (IEO, 2013).

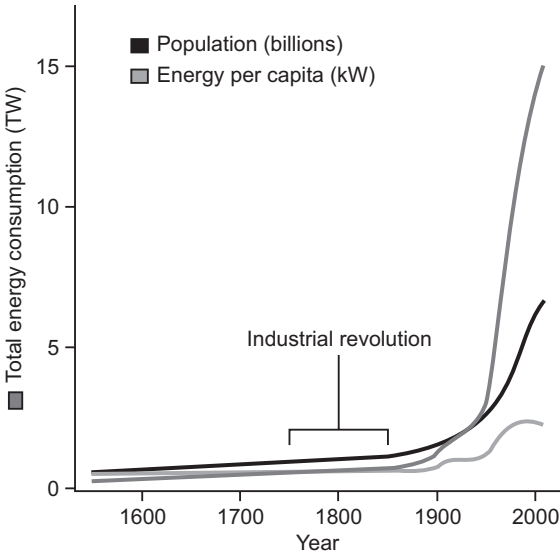


Fig. 25.20. History of growth in world population and environmental impact of Homo sapiens, indicated by its surrogates, per capita and total human energy use. (adapted from Ehrlich et al., 2012; Nature 486 : 68-73).

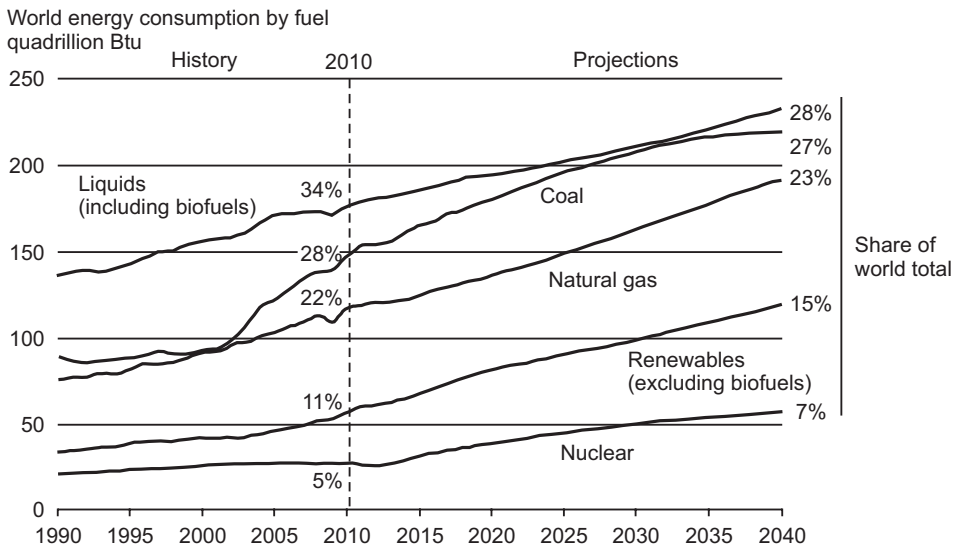


Fig. 25.21. World energy consumption by fuel category (quadrillion Btu) (adapted from EIA, International Energy Outlook, 2013).

All the growth in the use of liquid fuels is in the transportation and industrial sectors. In the transportation sector, in particular, liquid fuels continue to provide most of the energy consumed.

The transportation sector accounts for 63 % of the total increase in liquid fuel use from 2010 to 2040, and the remainder is attributed to the industrial sector, where the chemicals industry continues to consume large quantities of petroleum throughout the projection. The use of liquids declines in the other end-use sectors and for electric power generation. To satisfy the increase in world liquids demand, liquids production increases by 28.3 million barrels per day from 2010 to 2040, including the production of both petroleum

Coal, oil, natural gas and hydropower constitute the main sources of commercial energy consumption in India (Fig. 25.22). Traditional energy resources in the form of fuelwood, agricultural wastes and animal residues account for 20% of total energy consumption in India. The industrial sector is the largest consumer of commercial energy, whereas transport sector is the main consumer of petroleum products. Total energy consumption by sources in India is shown in Fig. 25.23. The primary energy consumption in India is dominated by coal and hydrocarbons, with 5% of energy accounted by other sources like hydro, renewables and nuclear and 20% energy from solid biomass and wastes (EIA, 2011). In 2011, oil and gas accounted for around ~31% of India's total primary energy consumption, next only to coal, which accounts for ~41% (EIA, 2011).

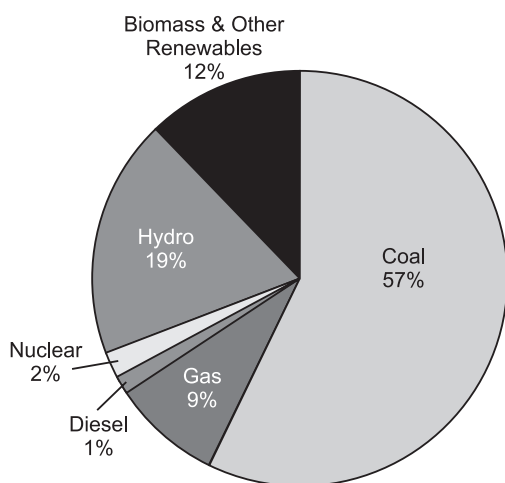


Fig. 25.22. India installed power capacity, 2011. Adapted from U.S. energy Information Administration, International Energy Statistics, India Central Electricity Authority.

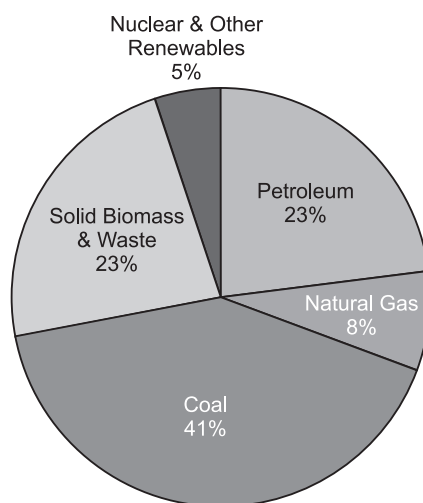


Fig. 25.23. Total energy consumption in India. Adapted from U.S. energy Information Administration, International Energy Statistics, India Central Electricity Authority.

Among the industries that consume about 80% energy are the fertilizer, aluminium, textile, cement, iron and steel, pulp and paper. With one-sixth of the world population, India's per capita energy consumption was 585 kilograms of oil equivalent (kgoe) in 2009, far lower than the global average. With expected high growth rate in the future, India's per capita energy consumption is expected to be more than double by 2031-32, around 1,124 kgoe, which will still be lower than the 2009 world average of 1,797 kgoe.

Energy conservation

Our immediate task is to conserve (save energy) by using meaningful devices and by perfecting new, environmentally sound energy-use technologies:

1. Energy storage and transmission: Increase storage and transmission efficiencies

2. Lighting: Reduce the illumination level in urban areas, where it is higher than required and replace incandescent light by fluorescent light, for a 40-watt fluorescent bulb gives off more light than a 100 watt incandescent bulb.
3. Heating and cooling: Adapt to higher temperatures in summer and lower temperatures in winters than the temperatures presently maintained, for keeping home warm, insulate the rooms, use heat pumps for heating and cooling rooms. Heat pumps are used to collect heat from outside and pump it in during winters, and to do the reverse during summer.
4. Housing designs: To save energy used to keep temperatures of houses comfortable following measures are used; in cold areas smaller houses in high densities, thicker walls and lighter and better sealed joints for more heat insulation, windows as passive solar collectors to keep rooms warm, better control of moisture to protect insulation and cooling in hot areas, insulated foundation walls in cold climates, use solar space heater, appropriate face of house with properly designed overhang- a south-facing house in the northern hemisphere would receive more sun in winters.
5. Transportation: Mass-transportation is required to be promoted, e.g., cars be replaced by buses, and buses by rail trains, improve fuel efficiency technologically, use alternate fuels such as hydrogen, solar batteries, etc.

Indian Programmes of Renewable energy

The Indian Renewable Energy Programme of the Ministry of Non-Conventional Energy Sources (MNES) in collaboration with the Indian Renewable Energy Development Agency (IREDA) has a focus on promoting the renewable energy development through research and development, demonstration projects, government subsidy programmes, etc. Solar thermal and solar photovoltaic technologies are being implemented by the MNES under the Solar Energy Programme, which is regarded as one of the largest in the world. India currently has a total grid installed PV capacity of 1,050 MW, most of which has been installed in Gujarat and Rajasthan. The Solar Photovoltaic programme aimed particularly at rural and remote areas to develop street lighting, solar lanterns, home lighting systems, water pumps, power plants, etc.

Solar water heating has been applied in individual residences, hotels and some industries. Solar air heating has been utilised in various parts of the country for drying agricultural produce and in timber kilns.

Announced in November 2009, the Government of India launched its Jawaharlal Nehru National Solar Mission under the National Action Plan on Climate Change with plans to generate 1,000 MW of power by 2013 and up to 20,000 MW grid-based solar power, 2,000 MW of off-grid solar power and cover 20 million sq metres with collectors by the end of the final phase of the mission in 2020.

India has 150 GW of renewable energy potential, about half in the form of small hydropower, biomass, and wind and half in solar, cogeneration, and waste-to-energy. Developing renewable energy can help India increase its energy security, reduce the adverse impacts on the local environment, lower its carbon intensity, contribute to more balanced regional development, and realize its aspirations for leadership in high-technology industries.

The principal objective of the Solar Thermal Programme is the market development and commercialization of solar water heaters, solar cookers, etc. The MNES has been promoting the sale of box solar cookers since the early 1980's. This type of cooker is designed to prepare food for up to 4-5 people and can be supplied with or without electrical back-up. However, the Dish Solar Cooker designed for 10-15 people and the Community Solar Cooker for 35-40 people have also been developed. In March 1999 the world's largest Solar Steam Cooking System was installed at

Mount Abu, Rajasthan; it is a hybrid system with back-up of oil-fired boilers and is designed to prepare food for 10,000 people.

According to the Ministry of New and Renewable Energy (MNRE), there exists a potential of about 1700 MW from urban waste (1500 from MSW and 225 MW from sewage) and about 1300 MW from industrial waste. The ministry is also actively promoting the generation of energy from waste, by providing subsidies and incentives for the projects. Indian Renewable Energy Development Agency (IREDA) estimates indicate that India has so far realized only about 2% of its waste-to-energy potential. A market analysis from Frost and Sullivan predicts that the Indian municipal solid waste to energy market could be growing at a compound annual growth rate of 9.7% by 2013.

25.8 MINERAL RESOURCES

Mineral resources are non-renewable, as the rate of their regeneration is slower than the rate at which they are used. Minerals include chemical elements, inorganic compounds, complex aggregation of elements and compounds (e.g., iron is element asbestos is a mixture of compounds). Deposits of minerals that have been identified and are recoverable are called mineral reserves. It is possible to extend life of mineral reserves by searching new deposits, mining sea, using recycled materials, finding substitutes, improving extraction of minerals. However, the lasting solution is to consume less.

New deposits are often uneconomic to exploit. Mining the deep sea floor is suggested to be feasible for cobalt, nickel, manganese, copper, etc., and near shore mining has been ongoing for sand and gravel, calcium carbonate, phosphorites, sulphur and tin. The mineral utilization has two major problems: (i) how to reduce the amount of energy required to extract minerals? (ii) several environmental problems are related to mineral utilization. Pollution of air and water, and damage of land are major environmental problems associated with processes involved in mining – viz., extraction, processing, transportation, fabrication and discarding.

Using recycled materials has many advantages including: (i) energy saving, (ii) less pollution, (iii) less mining wastes and (iv) decreased water use. Recycling is easy when products are basically made of one mineral, such as aluminium or steel. But most products have more than one mineral and separating one from another is difficult. Technological innovations are required to find substitutes for minerals. Already success has been considerable. For example, in automobiles chromium has been replaced by aluminium and plastics (however, plastics have created new environmental problems) (Kupchella and Hyland, 1989).

In developed countries like USA per capita consumption is 20 to 50 times greater than developing countries, and USA, meets more than 60% of its consumption by importing more than 20 minerals and metals. Mineral deposits are highly uneven in distribution. For example, Soviet Union and South Africa produced 55% of the world's manganese, 62% of chromite, and 93% of platinum group metals, and Zaire and Zambia produce about two-thirds of the world's cobalt.

Biodiversity : Magnitude, Functions and Services, Threatened Species and Conservation

Biodiversity is essential for human survival and economic well being, and plays a prominent role in modulating ecosystem function and stability. Biodiversity is not uniformly distributed on the earth, and could comprise 5 to more than 50 million species. Biodiversity refers to the totality of genes, species, and ecosystems of a region. The term “biological diversity” was first defined as including two related concepts, genetic diversity (the amount of genetic variability within species) and ecological diversity (the number of species in a community of organism) by Norse and McManus (1980). A tropical and conservation biologist, Thomas E. Lovejoy, used the term biological diversity during 1980 to express primarily the number of species. The term biodiversity was coined by Walter Rosen for the 1986 National Forum organized by the National Research council under the title ‘Biodiversity’ (Wilson, 1988). Since 1986, the terms and concepts related to biodiversity are of widespread use among biologists, environmentalist, political leaders and concerned citizens world-wide. The UN Convention on Biological Diversity defines biodiversity as “the variability among living organisms from all sources, including, *inter alia*, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems” (UNEP, 1992). Partel et al. (2011) have argued that absence of a species on a site is as important as presence. Therefore, biodiversity should include all species in the region that can potentially inhabit the particular ecological conditions of a site; this set of absent species is called ‘dark diversity’. There has been a great deal of worry regarding the loss of biodiversity because it represents the potential source of wealth in the form of (a) new crops, (b) pharmaceuticals, (c) petroleum substitutes, (d) biocides and other products. Organisms are required for the proper functioning of the earth’s ecosystems. Loss of biodiversity would check the evolutionary capability of biota to cope up with environmental changes. Biologically-rich and unique habitats are being destroyed, fragmented, and degraded due to problems caused by increasing human population, resource consumption and pollution. Biodiversity loss is now one of the world’s most pressing crises, and has emerged as a major driver of ecosystem change. used a suite of meta-analyses of published data to show that For example, the effects of species loss on productivity and decomposition—two processes important in all ecosystems—are of comparable magnitude to the effects of many other global environmental changes (Hopper et al., 2012). Intermediate levels of species loss (21–40%) reduced plant production by 5–10%, comparable to previously documented effects of ultraviolet radiation and climate warming. Higher levels of extinction (41–60%) had effects rivalling those of global warming, ozone and acidification. At intermediate levels, consumer species loss generally had equal or greater effects on decomposition than did elevated CO₂ and nitrogen addition.

26.1 MAGNITUDE OF BIODIVERSITY

Systematic work on identifying and naming species has been in progress for the last 250 years, but still we have collected, described and named far less number of species than the actual number

present. Table 26.1 gives a summary of the number of known species on the earth distributed among major taxonomic groups. The known and described number of species of all organisms on the earth is 1.78 million, whereas the predicted number of total species varies from 5 to 30 million (Gaston, 1998) and averages at 14 million. The species are unevenly distributed across taxonomic groups. About 61% of the known species are insects. Out of 1,025,000 species of insects, approximately 62% species are comprised of beetles. Only 4650 species of mammals are known to science. A large number of plant species and vertebrates are known (Gibbs, 2001). But even the number of presently unknown plant species is thought to be 10 to 20 per cent of the number of known plant species (Joppa et al., 2011). There are many more species that have not yet been described, especially in the tropics. Information about bacteria, viruses, protists and Archaea is just fragmentary. Thus, our knowledge of diversity is remarkably incomplete. A more recent summary for eukaryotes gives lower and upper estimates of 3.5 and 108 million species, respectively (Gaston, 2010; Singh, 2012). New species are being described at a rate of about 13,000 per annum, or about 36 species on the average day but there is little immediate prospect of greatly reducing the numbers that remain unknown to science. New species are being discovered faster than ever before due to the efforts of projects under the United Nations Environmental Programme. The World Conservation Union and the World Conservation Monitoring Centre. The Species 2000 is an Internet-based global research programme that aims to create an index of the world's known species. The Global Biodiversity Information Facility aims to ensure interoperability among the various databases of biodiversity studies (Sugden and Pennisi, 2000).

Table 26.1. Approximate numbers of species which have been described and identified from all over the world (based on Gibbs, 2001).

Group	Number of species
Higher plants	270,000
Algae	40,000
Fungi	72,000
Bacteria (including cyanobacteria)	4,000
Viruses	1,550
Mammals	4650
Birds	9700
Reptiles	7150
Fish	26,959
Amphibians	4780
Insects	1,025,000
Crustaceans	43,000
Mollusks	70,000
Nematodes and worms	25,000
Protozoa	40,000
Others	110,000

26.2 GLOBAL ACCUMULATION OF BIODIVERSITY

The accumulation of biodiversity on earth reflects the difference between the rates of speciation and extinction (Fig. 26.1). The overall natural extinction rate estimated from fossil data was 10^{-7} species per species year (Ehrlich and Wilson, 1991). The background rate of extinction as calculated by Pimm et al. (1995) was 1 to 0.1 species per million species per year. Accumulation of biodiversity occurred because speciation rate must have been several fold higher than the background extinction rate.

The global accumulation of biodiversity has occurred due to a range of evolutionary processes through time. Analysis of the fossil record of microbes, algae, fungi, protists, plants and animals shows that diversity of both marine and continental life increased exponentially since the end of the Precambrian reaching to 2400 families in Pleistocene and Holocene (Benton, 1995, Fig. 26.2). According to Knoll (1984), there were about 25,000 flowering plant species some 125 million years ago, this number swelled to 100,000 by the late Cretaceous and today there are more than 270,000 flowering plant species. The proliferation of biodiversity reflects gradual accumulation of genetic differences between lineages in response to selection for adaptation to different environments. Accumulation of genetic variation leading to reproductive isolation results in speciation.

The formation of new species occurs mainly due to cladogenesis (branching of lineage with time) or anagenesis (change of entire lineage). Modes of speciation include allopatric, parapatric and sympatric (as discussed in chapter 7). Polyploidization, mutation, genetic drift, recombination, gene flow, and chromosome duplication in hybrids could also lead to speciation. As many as 45% of higher plants (43% of 12,000 dicotyledonous species and 58% of 5000 monocotyledon species) appear to have evolved through polyploidy (Grant, 1963).

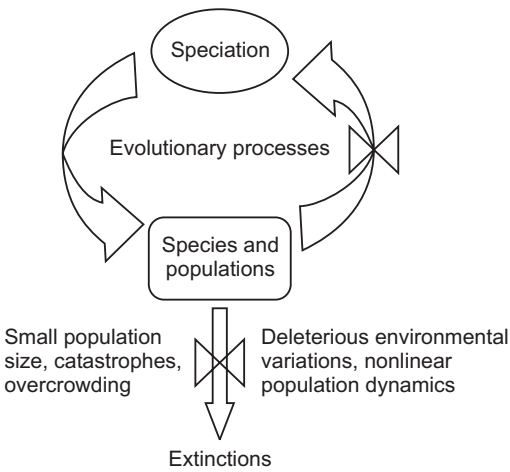


Fig. 26.1. Accumulation of global biodiversity as a balance between rates of speciation and extinction. Evolutionary processes act in a positive feed back manner. Both evolutionary and extinction processes are driven by several factors.

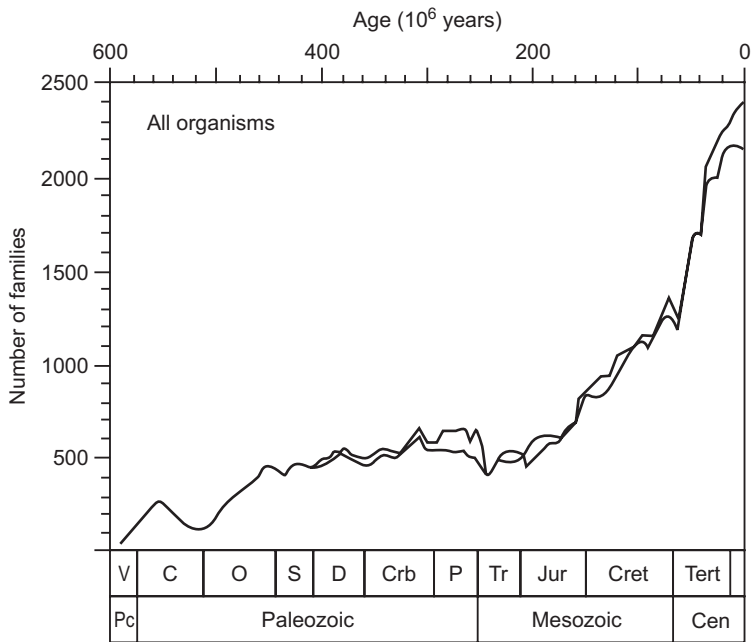


Fig. 26.2. Diversification through time: All organisms (after Benton, 1995).

The origin or evolutionary success of taxa is often attributed to key innovations -aspects of organism phenotype that promote diversification by evolutionary change in individual traits. These innovations may enhance competitive ability, relax adaptive trade-offs or permit exploitation of a new productive resource base. There could also be a positive effect of area on speciation rate. From an analysis of evolutionary-species-area-relationship for lizards on Caribbean islands, Losos and Schluter (2000) found that speciation rate on islands smaller than a threshold island size above which the rate of species proliferation increased with the area of the island.

The taxonomic richness of vascular plants through time indicates the following trends (Fig. 26.3): a gradual increase between mid-Carboniferous and, end-Permian, a minor reduction at the end of Permian, a rapid rebound to pre-Mesozoic level, a slow increase up to early-Cretaceous, previously dominant groups were replaced by Coniferophytes, Cycads, Cycadeoids which dominated the plant biota until the mid-Cretaceous, the Angiosperms appeared in mid-Cretaceous and rapidly diversified in the Cenozoic (Signor, 1990).

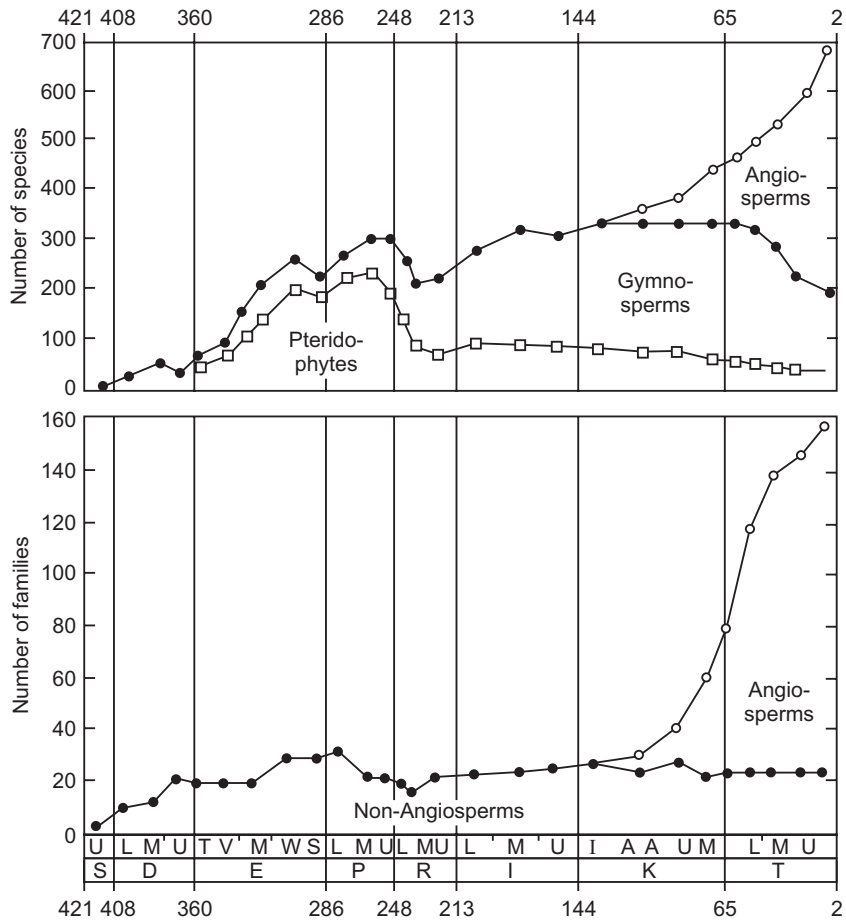


Fig. 26.3. Taxonomic richness of vascular plants through time (after Signor, 1990).

26.3 ELEMENTS OF BIODIVERSITY

Three groups of basic building blocks of biodiversity, viz., genetic diversity, organismal diversity, and ecological diversity have been recognized (Gaston, 2010; Singh, 2012) (Table 26.2). Genetic diversity encompasses the components of the genetic coding that structures organisms

(nucleotides, genes, chromosomes) and variation in the genetic make-up between individuals within a population and between populations, organismal diversity encompasses the full taxonomic hierarchy and its components, from individuals upwards to populations, subspecies and species, genera, families, phyla, and beyond to kingdoms and domains, and ecological diversity encompasses the scales of ecological differences from populations, through habitats, to ecosystems, ecoregions, provinces, and on up to biomes and biogeographic realms (Gaston, 2010; Singh, 2012).

Table 26.2. Elements of biodiversity, focusing on those levels that are most commonly used (Adapted from Gaston, 2010; Singh, 2012).

Ecological diversity		Organismal diversity
Biogeographic realms		Domains or kingdoms
Biomes		Phyla
Provinces		Families
Ecoregions		Genera
Ecosystems		Species
Habitats	Genetic diversity	Subspecies
Populations	Populations	Populations
	Individual	Individual
	Chromosomes	
	Genes	
	Nucleotides	

Biodiversity can be measured also at higher level of taxa. This takes into consideration the genetic diversity above the species level, e.g., genus, family, order, and phylum. The phylogenetic diversity considers the relatedness of different species on large-scale evolutionary history. An area poorer in species-level diversity may be richer in phyla-level diversity. For example, about 80% of known species are terrestrial; the marine diversity at phyla-level is expected to be twice as much as the terrestrial diversity. There live more than 70 phyla of all life forms from bacteria to vertebrates. Of these, 20 are exclusively marine and 18 exclusively terrestrial. As many as 23 other phyla contain marine species, while only 10 more contain terrestrial species.

26.4 LEVELS OF BIODIVERSITY

Biological diversity includes three hierarchical levels: (i) Genetic diversity, (ii) Species diversity, and (iii) Community and Ecosystem diversity. These levels of biodiversity are interrelated, yet distinct enough to be studied separately to understand the interconnections that support life on earth.

Genetic Diversity

Genetic diversity refers to the variation of genes within species; the differences could be in alleles (different variants of same genes), in entire genes (the traits determining particular characteristics) or in chromosomal structures. The genetic diversity enables a population to adapt to its environment and to respond to natural selection. If a species has more genetic diversity, it can adapt better to the changed environmental conditions. Lower genetic diversity in a species leads to uniformity, as is the case with large monocultures of genetically similar crop plants. This has advantage when increased crop production is a consideration, but can be a problem when an insect or a fungal disease attacks the field and poses a threat to the whole crop.

The amount of genetic variation is the basis of speciation (evolution of new species). It has a key role in the maintenance of diversity at species and community levels. The total genetic diversity of a community will be greater if there are many species as compared to a situation where there are only a few species. Genetic diversity within a species often increases with environmental variability. We know that each species, varying from bacteria to higher plants and animals, stores an immense amount of genetic information. For example, the number of genes is about 450-700 in *Mycoplasma*, 4000 in *Escherichia coli*, 13000 in *Drosophila melanogaster*, 32000-50000 in *Oryza sativa*, and 35000 to 45000 in *Homo sapiens sapiens*.

Each species consists of many organisms, and generally no two organisms of a species are genetically identical due to the high levels of genetic polymorphism across many of the gene loci. Then there is another level of variation. Wide-ranging species are made up of multiple breeding populations that have complex patterns of geographic variation in genetic polymorphism. Therefore, merely saving a few individuals of an endangered species would not be adequate, for losing numerous of its individuals amounts to loss of much of its internal diversity. When populations expand from a few individuals due to conservation, they would be much more genetically uniform than ancestral populations.

Species diversity

Species diversity refers to the variety of species within a community or a region. Simplest measure of species diversity is a count of the number of species present in an area i.e. species richness. Number of species often increases with the area of the site leading to different species-area relationships. Relative commonness and rarity of the individual species, i.e. equitability (species evenness), is another important aspect of species diversity.

In nature, the number and kind of species as well as the number of individuals per species vary, leading to greater diversity (see Chapter. 10 for a detailed discussion). Further, not all species of a community are equally different. It is possible to classify species on the basis of their functions. *Functional types* are those species which perform different ecological functions; and *functional analogues* represent distinct taxa performing the same or very similar ecological functions.

Community and Ecosystem Diversity

Different communities may share some species; however the relative abundance of species may vary depending on the nature and intensity of interspecific interactions among the species. Diversity at the level of community and ecosystem has three perspectives: alpha, beta and gamma diversity (see Chapter 10). Ecosystem diversity describes the number of niches, trophic levels and various ecological processes that sustain energy flow, food webs and the recycling of nutrients. It has a focus on various biotic interactions and the role and function of keystone species. As discussed in chapters 7 and 17, the number of habitats or ecosystems can vary within a geographical area. We also know that savannas, rain forests, deserts, lakes and wetlands and oceans are major ecosystems, where species live and evolve. The number of habitats/ecosystems present in a region is also a measure of biodiversity.

26.5 USES OF BIODIVERSITY

Humans derive many direct and indirect benefits from the living world. Biodiversity is the source of food, medicines, pharmaceutical drugs, fibres, rubber, timber and natural products that sustain global trade. The biological resources contain potentially useful resources as well. The diversity of organisms also provides many ecological services (see below) free of charge that are responsible for maintaining ecosystem health and human welfare. The diversity of phytoplankton is essential for controlling CO₂ in the atmosphere as discussed earlier in chapter 23. The uses of biodiversity are briefly described below.

Source of Food and Improved Varieties

Biodiversity provides food from crops, livestock, forestry and fish. Biodiversity is of use to modern agriculture in three ways: (i) as a source of new crops, (ii) as a source material for breeding improved varieties; and (iii) as a source of new biodegradable pesticides.

Of the several thousand species of edible plants, less than 20 plant species are cultivated to produce about 85% of the world's food. Wheat, corn and rice, the three major carbohydrate crops, yield nearly two-third of the food sustaining the human population. Fats, oils, fibres, etc. are other uses for which more and more new species need to be investigated. Marine organisms are a source of food such as fish, invertebrates, algae etc. Currently about 100 million metric tons of aquatic organisms, including fishes, molluscs, and crustaceans, are taken from the wild every year and represent a vital contribution to world food security. Some facts about food from plants, animals, fish and aquatic ecosystems are summarized in Box 26.1.

Wild relatives of commercial species must continuously be crossbred with these species to improve crops for cultivating in different areas for different uses. Genes of wild species are used to confer new properties such as disease resistance or improved yield upon domesticated species. For example, rice grown in Asia is protected from the four main diseases by genes yielded by a single wild rice species (*Oryza nivara*) from India. Natural chemical defenses against herbivory is quite common in wild plants, especially of tropical rainforests. These plants can be used as poisons and medicines. For example, methyl carbamate insecticides owe their development to calabar bean (*Physostigma venenosum*) of West Africa.

Box 26.1. Food from plants, animals, fish and aquatic ecosystems

Plants account for over 80% of the human diet

- 30 000 terrestrial plants are known to be edible.
- 7 000 are cultivated or collected by humans for food
- 30 crops feed the world.
- 5 cereal crops provide 60% energy intake of the world population.
- Crop improvement accounts for 50% of the world's food.

Food from Animals, fish and aquatic ecosystems

- Only 14 of the more than 30 domesticated mammalian species provide human food supply from livestock. There are about 8 300 breeds, of which 8% are extinct and 22% at risk of extinction
- Fish provides about 3 billion people with most of their intake of animal protein
- The world's aquatic ecosystems contain over 175 000 species of fish, molluscs, crustaceans and aquatic plants
- 10 species account for about 30% of the world marine capture fisheries production and 10 species account for about 50% of the aquaculture production

Source : www.fao.org/nr/cgrfa/en/ April 2013 Accessed 15 July, 2013

Drugs and Medicines

Biodiversity is a rich source of substances with therapeutic properties. Several important pharmaceuticals have originated as plant-based substances, which are of incalculable value to human health. Examples of plant-derived substances developed into valuable drugs are: Morphine (*Papaver somniferum*), used as an analgesic; Quinine (*Chinchona ledgeriana*) used for the treatment of malaria; and Taxol, an anticancer drug obtained from the bark of the yew tree (*Taxus brevifolia*, *T. baccata*) (Table 26.3). Currently, 25% of the drugs in the Pharmacy are derived from a mere 120

species of plants. But, throughout the world, traditional medicines make use of 10,000 to 20,000 plant species. Plants can also be used for the manufacture of innumerable synthetic products, called botanochemicals.

Table 26.3. Certain plant-derived substances that have been developed into valuable drugs (compiled from various sources, see Singh and Khurana, 2002).

Substance	Use	Original source
Codeine, Morphine	Analgesic, Antitussin	<i>Papaver somniferum</i>
Digitalin	Cardiotoxin, Antimalarial	<i>Digitalis purpurea</i>
Vincristine, Vinblastine	Alkaloids used for the treatment of Hodgkinson's disease and lymphocytic leukaemia	<i>Catharanthus roseus</i>
Quinine	Antimalarial	<i>Chinchona ledgeriana</i>
Taxol	Ovarian cancer	<i>Taxus brevifolia</i>
Castanospermine	Anti HIV activity	<i>Castanospermum australis</i>
Artemisinin	Antimalarial agent	<i>Artemisia annua</i>
Theokal	Arrhythmic	Hawthorn
Thisilyn	Liver damage	Milk thistles

Industry

The industrial products like timber, oils, lubricants, food flavours, industrial enzymes, cosmetics, perfumes, fragrances, dyes, paper, waxes, rubber, latexes, resins, poisons and cork can all be derived from various plant species. Supplies from animal origin are wool, silk, fur, leather, lubricants, waxes. Animals are also used as a mode of transportation. The marine algae are a source of polysaccharides, feed for livestock and building materials. Biodiversity may be a source of energy (such as biomass and biofuel).

Tourism and recreation

Biodiversity is a source of economical wealth for many areas, such as many parks and forests, where wild nature and animals are a source of beauty and joy, attract many visitors. Ecotourism in particular, is a growing outdoor recreational activity.

Aesthetic and Cultural Benefits

Biodiversity has also great aesthetic value. Examples of aesthetic rewards include ecotourism, bird watching, wildlife, pet keeping, gardening, etc. Throughout human history, people have related biodiversity to the very existence of human race through cultural and religious beliefs. In a majority of Indian villages and towns, plants like *Ocimum sanctum* (Tulsi), *Ficus religiosa* (Pipal), and *Prosopis cineraria* (Khejri) and various other trees are planted, which are considered sacred and worshipped by the people. Several birds, and even snake, have been considered sacred. Today, we continue to recognize plants and animals as symbols of national pride and cultural heritage.

Scientific role of biodiversity

Biodiversity is important because each species can give scientists some clue as to how life evolved and will continue to evolve on Earth. In addition, biodiversity helps scientists understand how life functions and the role of each species in sustaining ecosystems.

26.6 BIODIVERSITY, ECOSYSTEM SERVICES AND FUNCTIONS

In recent years the links between biodiversity and ecosystem services and the benefits people derive from these have received increasing attention (CBD, 2010; TEEB, 2010; MEA, 2005a; MEA, 2005b). The ecosystem/ecological services include maintenance of gaseous composition of the atmosphere, climate control by forests and oceanic systems, natural pest control, pollination of plants by insects and birds, formation and protection of soil, conservation and purification of water, and nutrient cycling, etc. These ecosystem services for 16 biomes have been valued in the range of 16 to 54 trillion (10¹²) US dollars per year (Constanza et al., 1997) which is 1.8 times of global GNP. Economies of the Earth would grind to a halt without the services of ecological life-support system – so the total value to the economy is infinite.

After the Earth Summit in Rio de Janerio in 1992, interest grew rapidly in understanding how biodiversity loss might affect ecosystem functioning and services. Hundreds of experiments were carried out all over the world on relationships between biodiversity and ecosystem functions (B-EF), and they contributed to the development of two new types of research approaches: Meta–Analysis, and Coordinated Distributed Experiments (CDE) or meta-experiments. Meta–analysis is a quantitative approach that integrates reviews and summarizes large numbers of independent studies. For data analysis it applies quantitative statistical methodologies. Since methods employed in independently carried out studies vary, combining data in a useful way is difficult. In Coordinated distributed environmental experiments (CDEs) standardized controlled experiments are run in parallel by several research groups in multiple locations. B-EF subsequently led to the agenda of *Biodiversity and Ecosystem Services* (B-ES) – biodiversity is beneficial to humans. While B-EF was concerned with experiments based on combinations of species growing at places, B-ES focused on large-scale patterns across landscapes more relevant to evaluation (Bradley et al., 2012).

Biodiversity and Ecosystem services (BES)

In 2005 Millennium Ecosystem Assessment (MEA) for the first time highlighted the condition of ecosystem and ecosystem services. The MEA defined four categories of ecosystem services, i.e., provisioning services, regulating services, cultural services and supporting services that contribute to human well-being (MEA, 2005b) (Fig. 26.4). The ecosystem services approach can save many ecosystems with high biodiversity and willingness of society to protect their biodiversity.

PROVISIONING SERVICES	SUPPORTING SERVICES
The supply of goods of direct benefit to people: ● Food ● Wood and fibre ● Medicinal plants ● Fuel 	Essential to the functioning of ecosystems: ● Nutrient cycling ● Primary production ● Soil Formation
REGULATING SERVICES	CULTURAL SERVICES
Rarely given a monetary value in conventional markets: ● Climate regulation ● Flood control ● Water purification ● Disease prevention ● Pollination 	Contribute to wider needs and desires of society: ● Aesthetic ● Spiritual ● Educational ● Recreational

Fig. 26.4. Types of Ecosystem Services (MEA 2005b).

Several provisioning ecosystem services increase with increase in diversity (Bradley et al., 2012). For example, intraspecies or genetic diversity increases the yield of crops; a higher tree species diversity results in more wood production in plantations; in grasslands species diversity contributes to fodder production. The more known regulating services that biodiversity improves include fish yield stabilities, resistance of ecosystems to exotic invasive species, above ground C sequestration, and soil organic matter and nutrient mineralization. Most of these are linked to ecosystem functions to which biodiversity contributes.

Mixed observations have been recorded with regard to the impact of biodiversity on long term carbon storage, pest abundance, and impact of animal diversity on animal diseases. Little data are available about the effect of fish species diversity on fish yield in many studies. No impact of biodiversity has been recorded on water quality. Oligotrophic water bodies which have fewer species attract tourists because of their clean water. Moreover, diverse natural enemy communities may inhibit biocontrol.

The problem is that B-EF, are based on small experimental studies thoroughly done, while B-ESs operate at larger scales, difficult to study. It may be pointed out that ecosystems give multiple services, and there may be tradeoffs among them. For example, by planting fast growing tree species carbon sequestration can be increased, but that may decrease water discharge and supply.

Biodiversity and Ecosystem Functioning

Relating biodiversity to ecosystem function is among the most intractable problems in ecology and a subject of hot debate among ecologists (Singh and Khurana, 2002). According to one school of thought, there are clear causative relationships between diversity and ecosystem functioning, while according to the other, ecosystem properties are not necessarily driven by species diversity per se, but rather the main drivers of ecosystem properties are the key functional attributes or traits of the dominant species present and the composition of the functional types. Four hypotheses have been proposed to describe the relationship of biodiversity to ecosystem functioning (Singh, 2002; Singh and Khurana, 2002).

- (i) **Rivet-Popper hypothesis:** According to the rivet- popper hypothesis, species in an ecosystem are something like the rivets on an airplane, with each species playing a small but significant role in the working of the whole. The loss of each rivet weakens the plane by a small but noticeable amount - until it loses airworthiness and crashes. In this view, progressive loss of species steadily damages ecosystem function.
- (ii) **Redundancy hypothesis:** According to the redundancy hypothesis, most species are superfluous - all that matters is that the biomass of primary producers, consumers, decomposers, etc. is maintained and if it is, only a few species are needed to keep the system in motion.
- (iv) **Diversity-stability hypothesis:** It predicts a linear relationship in which rate of ecosystem processes is maximum with greatest number of species.
- (v) **Idiosyncratic hypothesis:** This hypothesis acknowledges that there may be no pattern or an intermediate relationship between species diversity and ecosystem function. According to this hypothesis, the contribution of species to function is unpredictable, since both the identity and order of deletion will affect function differentially.

Naeem et al. (1994) performed an elegant experiment by using 14 model, terrestrial microcosms maintained in separate chambers of an Ecotron. In different ecotrons, plant and animal diversity was manipulated to create low, intermediate, and high diversity microcosms, with 9, 15 and 31 species, structured in such a manner that the set of microcosms with lower-diversity communities functionally resembled depauperate descendants of higher-diversity communities that

had lost species uniformly across all trophic categories. In this study, higher-diversity communities consumed more CO₂ than lower-diversity communities and productivity increased two- to three-fold as biodiversity was increased. The high-diversity ecosystems intercepted more light because they had better three-dimensional space-filling canopies which probably accounted for the differences in productivity. This study supports the ecological tenet that diversity begets productivity.

Biodiversity is also implicated with ecosystem stability. Ecosystem stability consists of two components: resistance to perturbation and resilience, i.e., the speed of recovery following a perturbation. Tilman and Downing (1994) examined this relationship for a grassland. They found that the most species-rich plots produced about half of their pre-drought biomass during the drought, whereas the most species-poor plots produced only about one-eighth. Thus more diverse plots were more resistant to drought. These authors also found that the species-poor plots took longer to return to pre-drought conditions and thus had lower resilience compared to species-rich plots. Thus, a small number of abundant species may account for a large fraction of ecosystem function under normal conditions, but in situations of extreme or unusual events, a variety of species may be needed to maintain the ecosystem function.

Combination of a variety of plant species in a forest resulted into considerable diversity in the timing and duration of phenophases and maximized the period of food availability to pollinators, frugivores and seed predators in return for their pollen transfer and seed dispersal services which are crucial for perpetuation of the system (Singh and Singh, 1992).

Widely accepted findings about the contribution of biodiversity to ecosystem functioning

There is a general agreement among ecologists on certain contributions of biodiversity to ecosystem functioning. They are based on many experiments carried out in different countries, including several parallel experiments using same methods. In treatments of these experiments all other environmental factors (e.g., soil, nutrients and light) were the same, only species number varied.

1. Biodiversity loss in ecosystems reduces the efficiency by which essential resources (nutrients, water, light, prey) are captured, biomass produced, litter decomposed and nutrients recycled (Balavanera et al., 2006; Cardinale et al., 2011). The effect of species diversity can be seen across various kinds of organisms and ecosystems and among different trophic levels. Diversity increases complementary use of resources, such as light and nutrients. For example, differences in plant morphology between species make a diverse community structure complex, and thus increases utilization of light. In diverse communities species vary in time and portion of soil pool from where they uptake nutrients. More diverse ecosystems are more efficient in nitrogen uptake where there is more nitrogen fixation or soil N is not limited (Spehen et al., 2005). Non-nitrogen fixing plant species take up nitrogen fixed by nitrogen fixing species.
2. With the increase in biodiversity, the *ecosystem stability* through time increases. The mechanisms include over-yielding and compensatory – asynchrony in species responses to environmental fluctuations. If everything else is equal, the loss of productivity due to a drought would be lower in a species-rich community than a species-poor community. In a drought year the increase in biomass of deep rooted species is likely to compensate for decrease in the biomass of shallow rooted species. In a diverse community, presence of species with varied root depths is more likely to occur than in a species-poor community. Differences in species responses to environmental changes thus have compensatory effect on ecosystem functions.

3. Biodiversity impact is non-linear and saturating (Cardinale et al., 2006; Cardinale et al., 2011), the effect of initial biodiversity loss is small on ecosystem functioning, but the rate of change in ecosystem functions accelerates as more species are lost (Fig 26.5). To know the full impact of species loss on ecosystem function, it is essential to carry out long-term studies.

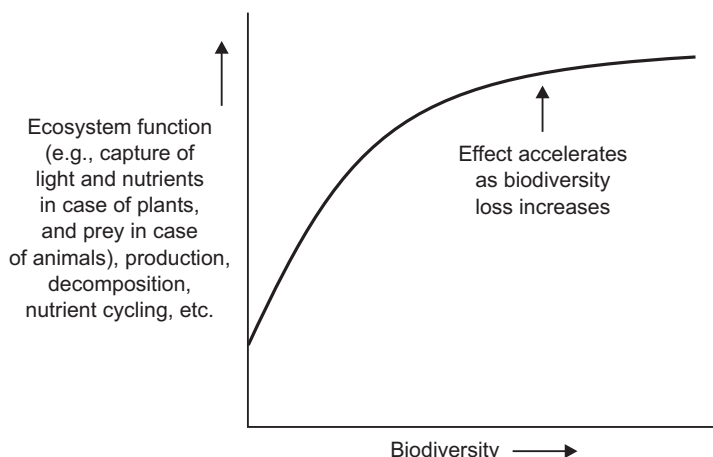


Fig. 26.5. Conceptual relationship between biodiversity and ecosystem functions (derived from Bradley et al., 2012).

4. Diverse communities are more productive because they have species which have large effects on productivity. Differences in functional traits among species increases the overall resource capture by communities (Huston, 1997; Loreau and Hector, 2001). Complementarity among species may represent niche partitioning, or positive interactions among them.
5. Loss of diversity across trophic levels is more detrimental to ecosystem functions than diversity loss within trophic levels. Loss of higher consumers can dramatically reduce plant biomass as its impact cascades through a food web.
6. Functional traits of species strongly influence ecosystem processes. How to estimate the extent of difference in functional traits of species and linking it to ecosystem processes is a challenging research area.

Emerging Trends from Biodiversity and Ecosystem functioning research

Trend 1

The impact of species loss on ecological processes might be large, large enough to rival impacts of many other global environmental drivers, such as drought, spread of invasive alien species, global warming, ozone hole, acidification of oceans, elevated CO₂, fire and nutrient pollution (Hooper et al., 2012; Tilman et al., 2012).

Trend 2

Diversity effects grow stronger with time and spatial scale (Tilman et al., 2001 b; Stachowicz, et al., 2008). Impact of diversity on ecosystem functions may be much larger in natural conditions than what we observe by conducting small spatial-scale experiments for a short period (Fig. 26.6). In large areas, environmental heterogeneity is more and frequency of environmental fluctuations

increases in a longer run of time. All these variations increase niches which species can exploit, hence more the species more effective would be ecosystem functioning. The more heterogeneous the environment, more the number of species required for efficient functioning of ecosystems.

Trend 3

Maintaining multiple ecosystem processes in multiple places and time periods requires higher levels of biodiversity than does single process at a single place and time (Hector and Bagchi, 2007). Therefore, for an overall efficient ecosystem functioning, more species are required than the numbers estimated on the basis of a single ecosystem process (Fig. 26.7). Researches generally deal with one diversity–function relationship at a time, whereas multiple ecosystem processes occur simultaneously, and they require many more species than estimated on the basis of single character research. Then, species performances also differ from year to year and from location to location. Isbell et al. (2011) show that 84% of the 147 grassland plant species studied in 17 biodiversity experiments promoted ecosystem functioning at least once and different species promoted ecosystem functioning during different years, at different places, for different functions and under different environmental change scenarios. Therefore, although species may appear functionally redundant when one function is considered under one set of environmental conditions, many species are needed to maintain multiple functions at multiple times and places in a changing world.

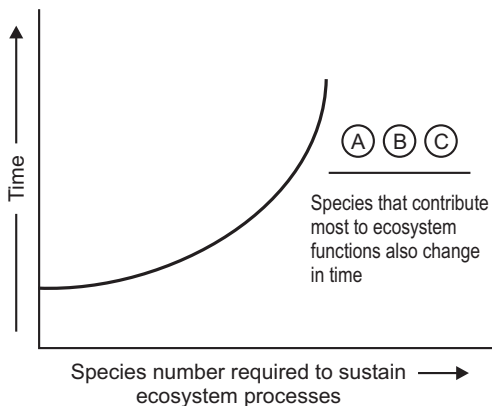


Fig. 26.6. Relationship between time and species number required to sustain ecosystem processes.

Trend 4

Evolution that results in variation in genetic traits among organisms contributes to ecosystem processes. Phylogenetic distances among species present in an ecosystem may account for biomass production more effectively than taxonomic differences.

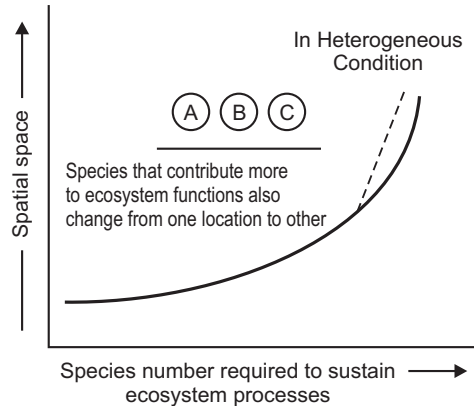


Fig. 26.7. Number of species required to sustain an ecosystem process increases rapidly with time and spatial space. A species (A) may perform well in a given year but not in the following year, (B or C may replace A). Similarly, species which account for much of biomass at one place may be replaced by another species at the other place (A replaced by B or C).

26.7 DISTRIBUTION OF BIODIVERSITY

Biodiversity is not uniformly distributed across the geographical regions. Certain regions of the world are megadiversity zones. The IUCN Plant Conservation Office has identified 234 areas of especially high plant diversity around the world. These are located in north-western South America, Central America, tropical Africa, eastern Mediterranean region and southeast Asia/Malaysia.

In recent years, tropical forests have drawn worldwide attention because of their high biodiversity and rapid destruction of habitats. Covering only 7% of the Earth's land surface, these

forests contain over 70% of the world’s species of all organisms. In a plot of just 100 m² of a lowland tropical forest in Costa Rica, Whitmore et al. (1985) counted 233 vascular plant species, which is equivalent to about one-sixth of the total flora of the British Isles. And this did not include full sampling of species-rich crowns containing epiphytes. Then there were bryophytes, macro- and microlichens which could not be sampled. Wilson (1987) collected 43 species of ants belonging to 26 genera, about equal to the entire ant fauna of the British Isles, from a single leguminous tree in the Tambopata Reserve of Peru. There are several such data that indicate the astonishing diversity of these forests: 700 species of trees in 10 selected 1-hectare plots in Borneo (the same as in all of the North America), 3000 species of beetles from only five plots of the Amazon basin, etc. Many of the tropical rainforest areas also show a high level of endemism. The flora of the Afrotropical realm probably includes some 40,000 vascular plant species. The areas that contain them have high level of endemism, up to 75% species.

Though tropical rainforests are unusually rich in species, some sites in Mediterranean climate region also show high richness on area basis. The richest Mediterranean-climate sites are those with some degree of current disturbance. Even in other areas the Mediterranean-climate sites have more species number per unit area than the temperate zone sites (Table 26.4).

Table 26.4. Mean vascular species number per 0.1 ha sample area in some terrestrial ecosystems. Non-Mediterranean sites include only woody plants over 2.5 cm diameter at breast height. Values in parentheses are for site number (from various sources given in Mooney, 1988).

Forest/zone	Mean species no./0.1 ha	Site of maximum species
Dry tropical forest (4)	41-67	Venezuelan coast
Moist tropical forest (3)	88-125	Panama canal zone
Wet tropical forest (3)	118-236	Costa Rica
Temperate zone (6)	21-48	Australia, forests and woodlands
Mediterranean zone (8)	35-139	Israel, open shrubland

The floristic diversity (higher plants) and level of endemism in some areas of South Africa (the fynbos biome) and Mediterranean zone is especially impressive. The wet tropical and moist tropical forests in South America may contain 8,550 vascular plants in an area of 75,000 km². Three-fourth of the species are endemic. Similarly, the South African Cape contains 6000 higher plant species within 46,000 km² area. This subregion has been considered one of the world’s six distinctive floristic regions.

26.8 GRADIENTS OF BIODIVERSITY

Biodiversity varies with change in latitude or altitude. As we move from high to low latitudes (i.e. from the poles to the equator), the biological diversity increases. This pattern can be seen in terrestrial, marine and freshwater habitats (Gaston, 2000; Macdonald, 2003; Willig et al., 2003). Latitudinal gradients of diversity in mollusks at species, genera and families are well established. As shown in Figs. 26.8 and 26.9, such a correlation between diversity and latitude is also found for a wide variety of taxonomic groups, such as trees,

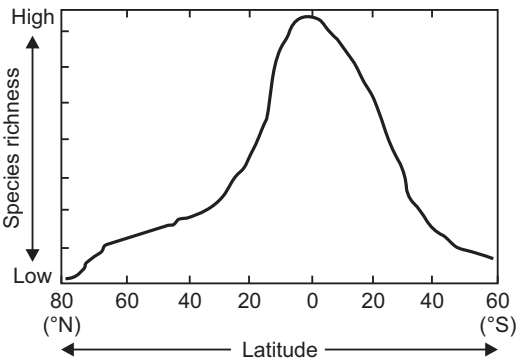


Fig. 26.8. A generalised latitudinal gradient in species richness of birds, mammals, lizards, butterflies and trees.

terrestrial vertebrates, ants, birds, butter-flies, and moths, and marine bivalves etc. However, there are differences in the patterns of species richness in Northern and Southern hemispheres; the gradients are generally steep and significant in northern hemisphere, whereas less strong in the Southern hemispheres. The gamma diversity of mammals in the new world reveals the classical pattern of latitudinal gradient (Fig. 26.9 *b*). Marine and freshwater fishes also show tropical increases in species richness (see Willig et al., 2003).

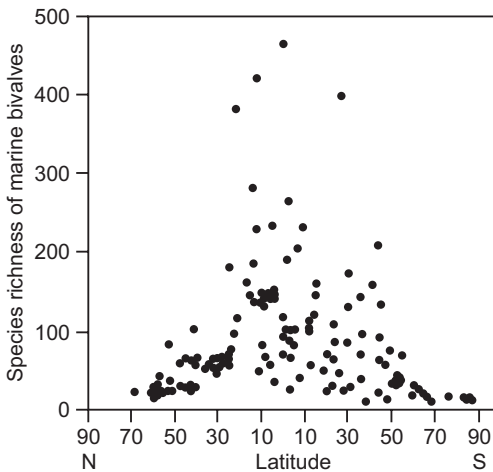


Fig. 26.9. (a) Latitudinal patterns in species richness of marine bivalves. (adapted from Flessa and Jablonski, 1995 and Townsend et al., 2003).

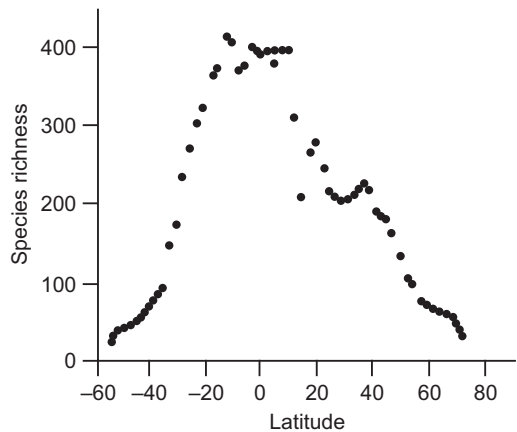


Fig. 26.9. (b) Spatial gradient of mammalian species richness in the continental new world defined by 2.5° latitudinal bands (adapted from Willig et al., 2003).

Elevation gradients basically reflect precipitation and temperature gradients and thus, elevational trends in species richness are generally thought to mimic those that occur along latitudinal gradients. We generally notice a decrease in species diversity from lower to higher altitudes on a mountain in terrestrial environments. A 1000 m increase in altitude results in a temperature drop of about 6.5°C. This drop in temperature and greater seasonal variability at higher altitudes are a major factor that reduces diversity. For example, the birds in tropics (Terborgh, 1977), the birds and mammals in Himalayan mountain of Nepal (Hunter and Yonzon, 1992) and plants of Sierra Manantlan, Mexico (Vazquez and Givnish, 1998; see Townsend et al., 2003) show that species diversity decreases monotonically with increase in altitude.

There is evidence to indicate that although species diversity tends to decrease with increasing elevation, it spikes at some intermediate level of elevation, giving rise to a humped shaped relationship. On the basis of fifty-six data sets, the elevation gradient of nonvolant small mammals revealed mid-elevation peak in species richness (McCain, 2005). A maiden study on the Himalayan fishes revealed that total and nonendemic fish species richness monotonously decrease with increasing elevation, while endemics peaked around mid elevations (700–1500 m) (Bhatt and Pandit, 2012).

The latitudinal and altitudinal gradients of species diversity are two master gradients, although regional and taxa-related exceptions occur. For example, according to one estimate, for every 100 cm increase in rainfall 50 tree species are added. Bird diversity increases with increase in vertical layering of the vegetation that is from grassland, which has only one principal layer (herbs) to forests which have herbs, shrubs and one or two tree layers. Also it is expected that more complex and heterogeneous the physical environment, more complex and diverse will be the flora and fauna.

In aquatic environments, the species richness changes with depth. In large lakes, the abyssal depths contain only a few species as compared to the illuminated surface water. In benthic communities of oceans, diversity increases from shallow parts to intermediate depth and then declines in very deep parts. Because of rapid and frequent variations in water levels, conditions are unstable in shallow parts. In the deepest part conditions are stable but light, nutrients and food are limiting.

The highest bio-diversity is found in Tropical Indo-Western pacific (waters off the coast of Asia, East Africa and Northern Australia and pacific islands). In these regions, highest levels of marine species are found off the coast of Philippines, Indonesia and New Guinea. The reef fish diversity is high in ocean water surrounding Polynesia, portion of Indian Ocean and Caribbean islands.

Hypotheses for the observed global patterns of biodiversity distribution

Several hypotheses have been proposed to account for the observed patterns of biodiversity distribution. Some of these are enumerated in Table 26.5. The evolutionary time theory (i.e. older communities hold greater diversity than young communities) and the climatic stability theory (i.e., more stable the environment more diverse will be the biological life) can explain the high biodiversity of tropical rain forests. The steep decline in species diversity from equator to poles has been explained also on the basis of decline in solar energy input, as measured by warmth or sunshine. Such a correlation has been found for birds, butter-flies, moths, etc. The older, stable climate is expected to support high speciation rates due to more sedentary populations and hence geographical isolation, larger number of generations per year and more opportunities for selection. On the other hand, greater spatial heterogeneity would result in low extinction rates due to greater specialization of taxa, more resources, less competition and smaller size of populations.

Table 26.5. Hypotheses to account for global patterns of biodiversity distribution (see Singh, 2002).

Hypothesis	Explanation	Reference
Evolutionary time Hypothesis	Older communities have high diversity compared to newly evolved ones, because more time has been available in the former for development of biota in an uninterrupted fashion.	Fischer (1960)
Climate stability Hypothesis	Uniformity of climate as well as relative constancy of resources in a stable community results in the formation of numerous niches allowing more species to occupy the unit habitat space, by evolution of finer specializations and adaptations.	Connell and Orias (1964)
Spatial heterogeneity Hypothesis	Availability of a wide range of ecological niches in more spatially complex environment favours coexistence of more species, due to greater availability of resources and less competition between the species.	Simpson (1964)
Productivity Hypothesis	Diversity of a community is determined by the amount of energy flowing through the food web.	Connell and Orias (1964)

26.9 HOT SPOTS OF BIODIVERSITY

The biodiversity hot spots also illustrate the uneven distribution of biodiversity on the earth. Myers introduced the term hotspots in 1988 for the geographical regions particularly rich in endemic, rare and threatened species found in relatively small areas but facing significant threats of habitat loss. According to Myers et al. (2000), to qualify as a hotspot, an area must contain 0.5% or 1500 of the world's 300 000 plant species and should have lost 70% or more of its primary

vegetation. Myers et al. (2000) identified 25 terrestrial biodiversity hot spots (Table 26.6). Myers et al. (2000) also recognized 9 leading hot spots (Tropical Andes, Sundaland, Madagascar, Brazil's Atlantic Forest, Caribbean, Mediterranean Basin, Indo-Burma, and Philippines) which contain 30% of all plants, 25% of all species in four vertebrate groups, and cover only 0.7% of earth's land surface. The leading hotspots are richer in endemics than other hot spots. The eight hottest hot spots are Madagascar and Indian Ocean Islands, Philippines, Sundaland, Brazil's Atlantic Forest, Caribbean, Indo-Burma, Western Ghats and Sri Lanka, and Eastern Arc Mountains and coastal forests of Tanzania and Kenya.

Table 26.7. Global hot spots and the number of endemic species of higher plants. The hot spots' are areas showing exceptional concentrations of species with exceptional levels of endemism and that face exceptional threat of depletion. Source: Myers et al. (2000).

Hotspots	Endemic Plants	Endemic Vertebrates
Tropical Andes	20000	1567
Mesoamerican forests	5000	1159
Caribbean	7000	779
Brazil's Atlantic Forest	8000	567
Choc/Darien of Panama Western Ecuador	2250	418
Brazil's Cerrado	4400	117
Central Chile	1605	61
California Floristic Province	2125	71
Madagascar and The Indian Ocean islands	11600	771
Eastern Arc and Coastal Forest of Tanzania / Kenya	1500	121
Western African Forests	2250	270
Cape Floristic Province	5682	53
Succulent Karoo	1940	45
Mediterranean Basin	13000	235
Caucasus	1600	50
Sundaland	15000	701
Wallacea	1500	529
Philippines	5832	518
Indo-Burma Eastern Himalayas	7000	528
South-Central China	3500	178
Western-Ghats Sri Lanka	2180	355
South-western Australia	4331	100
New Caledonia	2551	84
New Zealand	1865	136
Polynesia / Micronesia	3334	223
Total	133,149	9645
EASTERN AFROMONTANE	2,356	113

The Conservation International in the year 2005, has added nine more biodiversity hotspots to the 25 originally reconised hotspots by Myers et al. (2000). These newly recognized hot spots are East Melanesian Islands, Eastern Afromontane, Horn Of Africa, Irano-Antololian, Japan, Madrean Pine-oak Woodlands, Maputaland-Pondoland-Albany, and Mountains of Central Asia. The 34 hotspots cover only 2.3% of earth’s land surface, have lost 70% or more of its primary vegetation, contain 75% of most threatened mammals, birds & amphibians and 50% plants; and over 50% of the world’s plant species and 42% of vertebrate species are endemic to these hotspots (Conservation International, 2005; Ebert, 2005). The approximate locations of 34 hotspots are shown in Fig. 26.10.

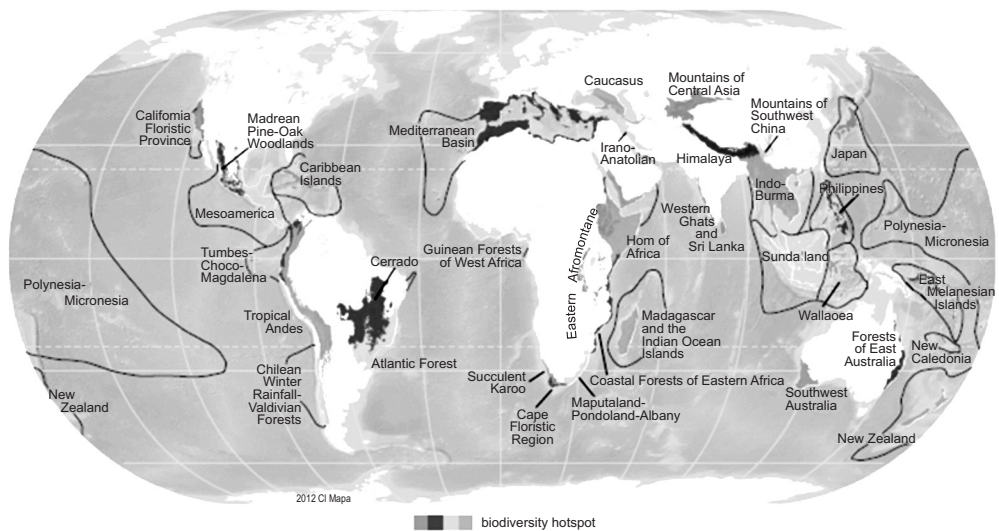


Fig 26.10. The terrestrial biodiversity hotspots (based on Conservation international).

Marine Hotspots of Biodiversity

In marine environments, Roberts et al. (2002) have identified 18 richest multitaxon centers of endemism which cover only 0.028% of the world ocean (Table 26.8). They have also identified 10 marine biodiversity hot spots which account for 15.8% of the world’s coral reefs but they contain 44.8 to 54.2% of the restricted range species. Interestingly, eight of the 10 marine hot spots and 14 of the 18 centres of endemism are adjacent to terrestrial biodiversity hot spots.

Table 26.8. Marine hotspot. The marine biodiversity hotspots are centres of endemism with average threat score above 1.67 (based on Roberts et al., 2002).

Marine Hotspot Location	Number of ≤ 10 *cell range species present	Area of coral reef in hotspot (km ²)	Adjacent terrestrial Hotspot
South Japan	75	3136	Philippines
Western Australia	56	1713	Southwest Australia
Gulf of Guinea	45	N.A.	West African Forests
Great Barrier Reef	43	23972	None
Hawaiian Island	35	442	Polynesia
North Indian Ocean	31	4790	Western Ghats and Sri Lanka
New Caledonia	31	5303	New Caledonia
Eastern South Africa	30	178	Cape floristic province

Philippines	17	21983	Philippines
South Mascarene island	16	205	Madagascar
Sunda Island	13	12639	Sunda land
Easter Island	11	N.A.	Polynesia

* cell is a geographical unit (each cell = 50,000 km²).

26.10 REGIONAL PATTERN OF BIODIVERSITY

The uneven distribution of biodiversity is also illustrated at regional and local scales. The local/regional species richness represents a balance between the rates of species immigration/invasion and species extinction/disappearance. Although all immigrants may not be invaders, their success at colonization imposes competitive effects on extant organisms resulting in disappearance of some of them from the site, and if the disappearing species was an endemic, this results in global extinction. Invasion frequently has a deleterious impact on the local environment, causing a cascade of extinction events.

Several hypotheses have been proposed to account for locally high species richness; these are summarized in Table 26.6 (see Singh, 2002; Singh and Khurana, 2002 for more details). However, the ability of a large number of species which apparently compete for the same few resources to coexist in relatively homogenous sites, is still poorly understood.

Table 26.6. Hypotheses to account for especially high biodiversity at local/ecosystem scales (compiled from various sources, see Singh, 2002).

Hypothesis	Tenets
Intermediate disturbance (Horn, 1975)	Highest diversity is maintained at intermediate levels of disturbance due to a mosaic of habitats, permitting both pioneers and late successionalists to coexist.
Herbivore-driven forest diversity (Janzen, 1970)	Establishment of young trees in the vicinity of their parents is constrained by the activity of tree-specific herbivores; however, as herbivores are specific to a tree species, other tree species may become established in the vicinity of herbivore's target, resulting in extremely high diversity.
Equilibrium model of resource competition (Tilman, 1988)	With an increase in the availability of resources, more species are supported which tap the resources efficiently, but as resource availability increases further, competition for space and light allows only the best competitors to dominate, while others are excluded.
Neighbourhood recruitment limitation (Hubbell et al., 1999)	Due to poor dispersal ability, low local abundance or chance events, many species may be absent from a neighbourhood and such absentees forfeit any chance of competitive victory at the site, permitting inferior competitors to win by default. This can lead to essentially unlimited diversity.

26.11 THREATS TO BIODIVERSITY

Human activities have greatly accelerated the global rate of species extinctions (Pimm et al., 2001). Anthropogenic threats leading to extinction of species and consequent loss of biodiversity are: habitat loss and fragmentation, introduction of non-native species, overexploitation, soil, water and atmospheric pollution, and intensive agriculture and forestry, and climate change. N-deposition and CO₂ loading of atmosphere are among the factors expected to cause biodiversity loss in future. Some alarming figures about biodiversity from the United Nations Environmental Programme (UNEP,

2012) Global Environment Outlook-GEO-5 are: With more than 30 per cent of the Earth's land surface used for agricultural production, some natural habitats have been shrinking by more than 20 per cent since the 1980s; The world lost over 100 million hectares of forest from 2000 to 2005, and has lost 20 per cent of its sea grass and mangrove habitats since 1970 and 1980, respectively; In some regions, 95 per cent of wetlands have been lost. Two-thirds of the world's largest rivers are now moderately to severely fragmented by dams and reservoirs; Vertebrate populations have declined on average by 30 per cent since 1970; around 20 per cent of vertebrate species are now under threat; The extinction risk is increasing faster for corals than for any other group of living organisms, with the condition of coral reefs declining by 38 per cent since 1980. Rapid contraction is projected by 2050.

Habitat Loss and Fragmentation

The destruction of habitats is the primary reason for the loss of biodiversity in terrestrial and coastal ecosystems (Sala et al., 2000). Habitat loss could be attributed to conversion, habitat degradation and fragmentation. . When people cut down trees, fill a wetland, plough grassland or burn a forest, the natural habitat of a species is changed or destroyed. These changes can kill or force out many plants, animals, and microorganisms, as well as disrupt complex interactions among the species. Habitat Loss and degradation is the most pervasive threat to birds, mammals and plants, affecting 86% of all threatened birds, 86% of the threatened mammals assessed and 88% of the threatened amphibians (Baillie et al., 2004).

The alteration of land use pattern has resulted in fragmentation of habitats, ecosystems and landscapes in most parts of the world. Habitat fragmentation is a leading cause of biodiversity loss. Forest fragmentation replaces large areas of native forest by other ecosystems leaving isolated forest patches, with deleterious consequences for most of the native forest biota (Murcia 1995). Fragmentation increases the local rate of extinction by (i) reducing population sizes or colonization from similar habitats, (ii) eliminating keystone predators or mutualists, (iii) enhancing stochastic phenomenon and genetic bottlenecks, (iv) promoting edge effects, and (v) interrupting landscape-scale processes such as wildfire. A forest patch surrounded by croplands, orchards, plantations, or urban areas is an example of fragmented habitats. With the fragmentation of a large forest tract, species occupying deeper parts of forests are the first to disappear. With an increasing number of habitats becoming fragmented, we are faced with the task of conserving a patchy world. There are two seemingly contrasting options: 'preserve optimal habitat quality wherever it can be found' as opposed to 'preserve a large number of well-connected patches' (Roslin, 2002).

Introduction of Alien Invasive Species

Introduction of invasive species may cause disappearance of native species through changed biotic interactions. Invasive species are considered second only to habitat destruction as a major cause of extinction of species. Alien Invasive Species are a significant threat to globally threatened species affecting 326 bird species (30% of all), 212 amphibians species (11% of all), 760 mammals (8% of all) and 361 plant species (15% of all threatened plants) (IUCN, 2003; Baillie et al., 2004). Exotic species are having large impact especially in island ecosystems, which harbour much of the world's threatened biodiversity.

Disturbance and Pollution

Communities are affected by natural disturbances, such as fire, tree fall, and defoliation by insects. Man-made disturbances differ from natural disturbances in intensity, rate and spatial extent. For example, man by using fire more frequently may change species richness of a community. Then, some human impacts are new, never before faced by biota, e.g., the vast number of synthetic

compounds, pesticides, massive releases of radiations or spillover of oil in sea. These impacts lead to a change in the habitat quality, reduce a species birth and survival rate to the extent that species can become locally extinct. Pollution may reduce and eliminate populations of sensitive species, for example, pesticide-linked decline of fish-eating birds and falcons. Lead poisoning is another major cause of mortality of many species, such as ducks, swans and cranes, as they ingest the spent shotgun pellets that fall into lakes and marshes. Eutrophication (nutrient enrichment) of water bodies drastically reduces species diversity.

Perhaps the most dramatic recent example of the potentially devastating effects of pollution on wild species relates to vultures (Oaks, 2004). In South Asia, vultures in the genus *Gyps* have declined by more than 95% in recent years owing to the toxic effects of the anti-inflammatory drug, Diclofenac, which causes renal failure and visceral gout in the birds feeding on carcasses of animals treated with the drug. Diclofenac is widely used in human medicine globally, but was introduced to the veterinary market on the Indian subcontinent during the early 1990s. Diclofenac has been found responsible for the decline of oriental white-backed vultures (Oaks, 2004).

Harvesting and Overexploitation

Exploitation, including hunting, collecting, fisheries and fisheries by-catch, and the impacts of trade in species and species' parts, constitutes a major threat for globally threatened birds (30% of all), mammals (33% of all), amphibians (6% of those assessed), reptiles and marine fishes (Baillie et al., 2004). Trade affects 13% of both threatened birds and mammals (IUCN, 2003).

Fishing and logging have a negative impact on biodiversity as well as non target species that directly depend on target species for food or shelter. Species of several trophic levels are tied to tree species and therefore clear-cut logging has a large impact on diversity. For example, approximately 300 species of insects and several bird species may depend on a single tree species. Thus tree species extinctions are conspicuously disastrous. Estimates indicate that about 80% of the world's 100,000 tree species are in the tropics, and 10% of all tree species are now threatened. Fishing often takes large incidental catches of non-target species, and trawling can destroy the whole marine community that lives on the forest floor.

Threats to marine Biodiversity

Over-fishing, habitat destruction, widespread marine pollution and human induced climate change threaten the survival of marine biodiversity. Pollution, oil and gas drilling and oil spills may increase the risks of extinction by increasing mortality of marine organisms. Massive inputs of sediments and other wastes from land severely threat biologically diverse and highly productive coastal ecosystems such as coral reefs, marshes and mangroves swamps. Many Marine fishes have been severely reduced in numbers by commercial fishing. Some of the major indicators of decline of marine biodiversity are: (i) drastic decline in fish stocks, (ii) decline in population of marine organisms, (iii) destruction of 50% mangrove vegetation, and (iv) local species extinctions.

26.12 EXTINCTION OF SPECIES

Extinction is a natural process. Species have disappeared and new ones have evolved to take their place over the long geological history of the earth. During the past 3.5 billion years, 4 billion species are estimated to have evolved, of which 99% have gone extinct. It is useful to distinguish three types of extinction processes.

Natural extinction: With the change in environmental conditions, some species disappear and others, which are more adapted to changed conditions, take their place. This loss of species which occurred in the geological past at a very slow rate is called natural or background extinction.

Mass extinction: There have been several periods in the earth's geological history when large number of species became extinct because of catastrophes. Mass extinctions occurred in millions of years. In geological times, large-scale species extinction was almost as common as species origination (Fig. 26.11). Average durations of species were generally less than 10 million years, and biological composition of earth at species level changed drastically several times. The five big mass extinctions occurred in the Late Ordovician, Late Devonian, Late Permian, Late Triassic, and end-Cretaceous (Fig. 26.11). The most serious extinction occurred near the end of the Permian period (250 million years ago), removing 60.9% of all life (Benton, 1995).

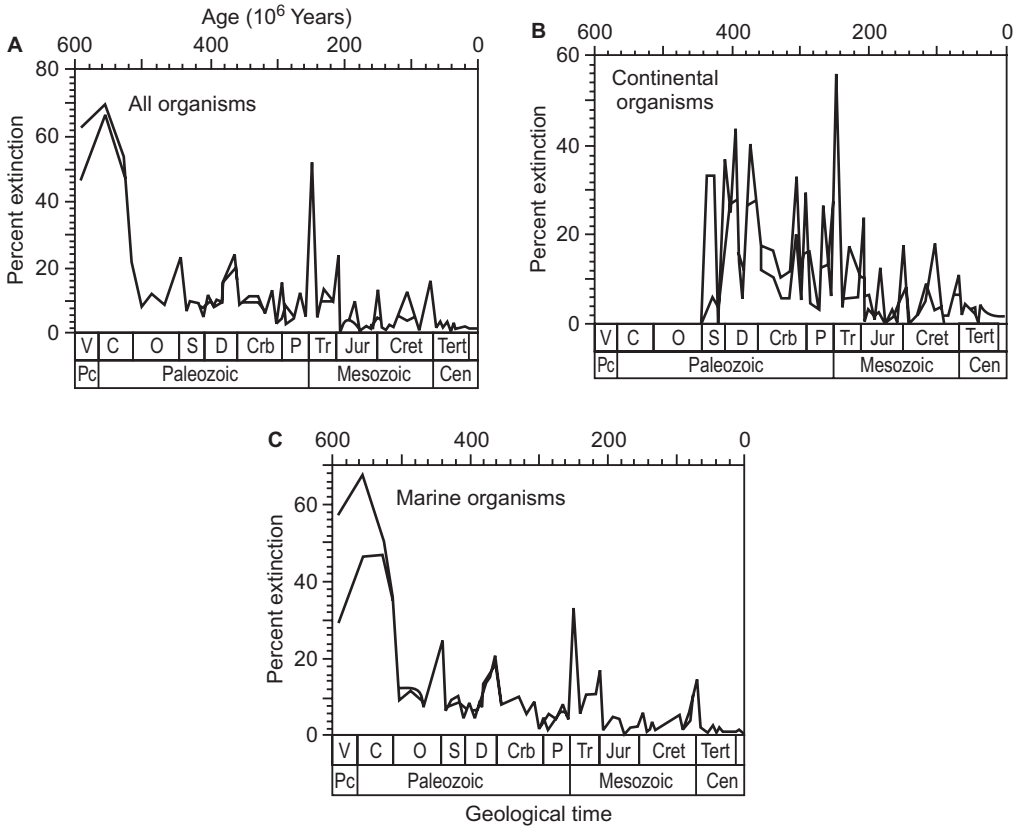


Fig. 26.11. The patterns of family extinctions over time (after Benton, 1995).

Anthropogenic extinction: An increasing number of species is disappearing from the face of the earth due to human activities. Each of the past mass extinctions was spread over million years or less, compared to only 200 years for the anticipated extinction due to human interference. Thus, in comparison to mass extinctions of geological past, the man-made mass extinction (the sixth mass extinction in Fig. 26.12) represents a very severe depletion of biodiversity. As many as 733 animals (mostly vertebrates and molluscs) and 110 plant species (flowering plants and lower group of plants) and one red algae (Bennett's Seaweed (*Vanvoorstia bennettiana*) have become Extinct or Extinct in the Wild since 1500A.D (Baillie et. al., 2004). *Euphorbia mayurnathanii* belonging to Euphorbiaceae family has become extinct only recently from India (Box 26.3). More species have gone extinct from the islands than from the mainland or the oceans.

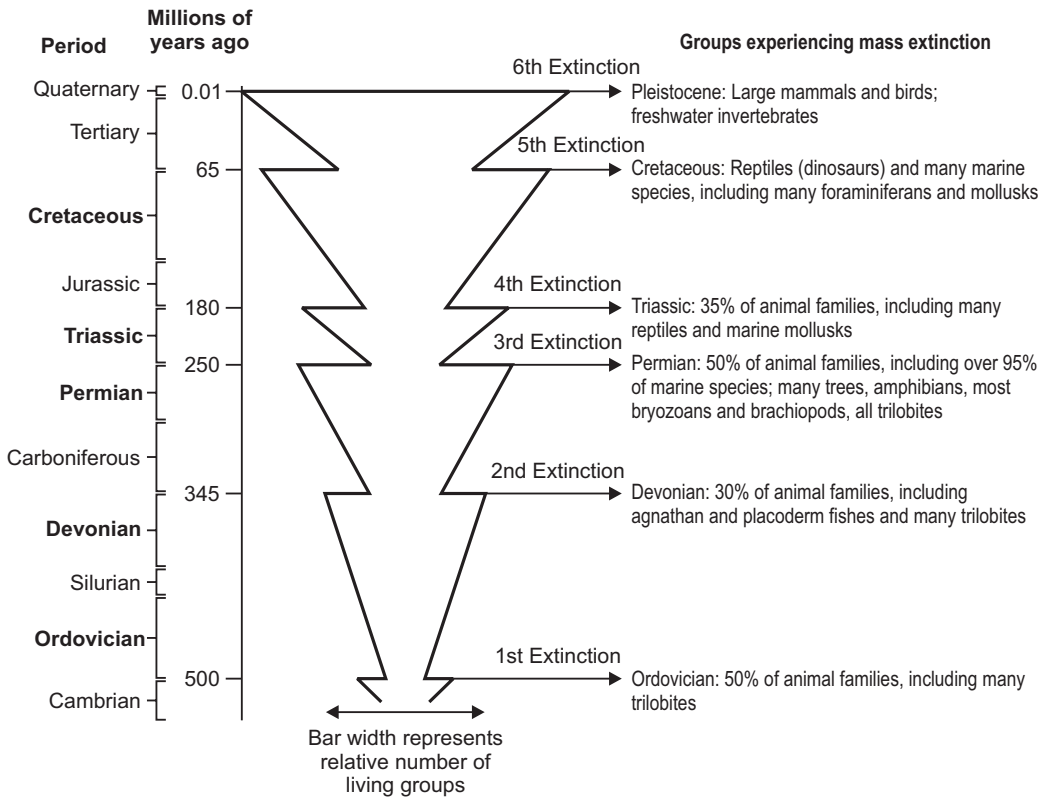


Fig. 26.12. The five major mass extinctions in the geological past and the sixth human driven mass extinction.

The current rate of extinction is 1000 to 10,000 times higher than the natural or background rate of extinction, and is in the order of 1000 species per decade per million years (Pimm and Raven, 2000). Some interesting observations about the current loss of species are:

1. Assuming that two-third of the estimated 14 million species exist in tropical forests, at the current rate of tropical deforestation (0.8% per year committing 0.1-0.3% species per year to extinction), the tropical forests alone are losing roughly 14000-40000 species per year (or 2-5 species per hour (Hughes et al., 1997).
2. The earth may lose up to 50% of the species by the end of the 21st century, if the current rate of loss continues.
3. Assuming that there are 220 Mendelian populations per species (i.e. 1.1-6.6 billion populations globally), and that the population extinction is a linear function of habitat loss, 16 million populations per year, or roughly 1800 per hour are being lost from tropical forests alone (Hughes et al., 1997).

Susceptibility to extinction

The characteristics of species particularly susceptible to extinction are: large body size (Bengal tiger, lion and elephant); small population size and low reproductive rate (Blue whale and Giant panda). Feeding at high trophic levels in the food chain (Bengal tiger and Bald eagle), fixed migratory routes and habitat (Blue whale and Whooping crane) and localized and narrow range of distribution (woodland caribou; many island species) also make the species susceptible to extinction.

Causes of species extinction

Natural factors that result in species extinction include small population size, deleterious environmental variations, periodic catastrophes, migration, overcrowding and inherent nonlinear population dynamics (Belovsky et al., 1999). Speciation reversal through introgressive hybridization can also contribute to species loss (Vonlanthen et al., 2012). The failure of organisms to (i) survive natural disasters, such as floods, droughts, ice ages, and impact of asteroids; (ii) coexist with one another as a result of competition, predation, etc., and (iii) live with man's activities of overexploitation, habitat fragmentation and destruction, or direct competition lead to extinction (Huston, 1994). In recent years, however, the third factor has become far more important than others, as discussed earlier.

The IUCN Red List Categories

The IUCN Red List is a catalogue of taxa that are facing the risk of extinction. It is important to understand that the Red List aims to impart information about the urgency and scale of conservation problems to the public and policy makers. The uses of the Red Lists are: developing awareness about the importance of threatened biodiversity; identification and documentation of endangered species; providing a global index of the decline of biodiversity; and defining conservation priorities at the local level and guiding conservation action.

The IUCN Red List System was initiated in 1963, and since then evaluation of the conservation status of species and subspecies is continuing on a global scale. It uses a set of criteria, relevant to all species and all regions of the world, to evaluate the extinction risk of species and subspecies. The Red List also provides information to international agreements such as the Convention on Biological Diversity and the Convention on International Trade in Endangered.

The World Conservation Union (IUCN, formerly known as International Union for the Conservation of Nature and Natural Resources) has recognized nine Red List Categories of species: *Extinct*, *Extinct in the Wild*, *Critically Endangered*, *Endangered*, *Vulnerable*, *Near Threatened*, *Least Concern*, *Data Deficient*, and *Not Evaluated*. These categories are briefly defined in Box 26.4. The species which are threatened with extinction are included in Vulnerable, Endangered, or Critically Endangered category. Critically endangered refers to a 50% probability of extinction in five years, endangered a 20% probability in 20 years and vulnerable a 10% probability in 100 years. Species with small world populations that are not at present endangered or vulnerable but are at risk are called **rare**. These species are usually localized within restricted geographical areas or habitats or are thinly scattered over a more extensive range.

Box 26.2. *Euphorbia mayurnathanii* extinct in the wild.

Euphorbia mayurnathanii is a species of plant in the Euphorbiaceae family. It was endemic to the Palghat Gap in India, but is now believed to be extinct in the wild. It still exists in cultivation, but international trade is controlled as it is included on Appendix II of CITES. The species was described in 1940 from three old specimens growing on a rocky ledge on the eastern spur of a hill in Pallasana situated in the Palghat Gap. The vegetation is monsoon forest and succulent plants are absent apart from this single species. It has been postulated that the climate in the past was drier and this small remnant of a previously xerophytic community survived because of its exposed position. Thorough searches have failed to locate any living wild specimens today.

Source : World Conservation Monitoring Centre 1998. *Euphorbia mayurnathanii*. In: IUCN 2013. IUCN Red List of Threatened Species. Version 2013.1. <www.iucnredlist.org>. Accessed 15 July 2013.

Box 26.3 The IUCN Threat categories		
Red list category	Abbreviation	Definition
Extinct	EX	Species is Extinct when there is no reasonable doubt that the last individual has died.
Extinct in the Wild	EW	Species is Extinct in the wild when exhaustive surveys in known and/or expected habitats, have failed to record an individual.
Critically Endangered	CR	Species is Critically Endangered when it is facing an extremely high risk of extinction in the wild in the immediate future.
Endangered	EN	Species is Endangered when it is not Critically Endangered but is facing a very high risk of extinction in the wild in the near future.
Vulnerable	VU	Species is Vulnerable when it is not Critically Endangered or Endangered, but is facing a high risk of extinction in the wild in the medium-term future.
Near Threatened	NT	Species that do not qualify for Critically Endangered, Endangered or vulnerable but are close to qualifying for a threatened category in the near future.
Least Concern	LC	Species is Lower Risk when it has been evaluated and does not satisfy the criteria for Critically Endangered, Endangered or Vulnerable or near threatened.
Data Deficient	DD	Species is Data Deficient when there is inadequate information to make a direct, or indirect, assessment of its risk of extinction.
Not Evaluated	NE	Species is Not Evaluated when it has not yet been assessed against the above criteria.

STATUS OF THREATENED SPECIES- 2004 to 2012

Status of threatened species: The 2004 IUCN Red List was the most comprehensive assessment of the global conservation status of threatened plant and animal species. Based on assessment of 2004 IUCN Red List, there are 15,589 species (7,266 animals, 8,321 plants and 2 lichens) listed as threatened (Critically Endangered, Endangered, or Vulnerable) (Baillie et al., 2004). Of these, 2,866 are listed as Critically Endangered (1,376 animals, and 1,490 plants). Two species of lichen, Perforate reindeer Lichen (*Cladonia perforate*) and Boreal Felt Lichen (*Erioderma pedicellatum*) were added to the IUCN 2003 Red List, the first additions from the Fungi kingdom.

The 2012 IUCN Red List shows a sharp increase in the threatened (Critically Endangered, Endangered, or Vulnerable) species of various groups vertebrates and plants (Table 26.9). However, it should be noted that many more species were assessed in 2012 than in 2004.

Table 26.9. The 2012 IUCN Red List shows a sharp increase in the threatened (Critically Endangered (CR), Endangered (End), or Vulnerable (Vul) species of selected groups.

Group	CR-2004	CR-2012	End 2004	End-2012	Vul 2004	Vul-2012
Mammals	162	196	352	446	587	497
Birds	179	197	345	389	688	727
Reptiles	64	144	79	296	161	367
Amphibian	413	509	728	767	628	657
Fishes	171	415	160	494	470	1149
Insects	47	119	120	207	392	503
Mollusks	265	549	221	480	488	828
Plants	1490	1820	2239	2655	4592	4914

In July, 2012, of the total 63,837 species globally assessed, the IUCN classified 3,947 as Critically Endangered, 81 as Extinct, 63 as Extinct in the Wild. In the lower risk categories, there were 5766 species in Endangered, 10,104 in Vulnerable and 4,467 in Near Threatened categories (Fig. 26.13). Scientific data regarding 10,497 species was not available and hence classified as Data Deficient (Fig. 26.13). At threat are 41 per cent of amphibian species, 33 per cent of reef-building corals, 25 per cent of mammals, 13% of birds, and 30% of conifers. Threatened species of Mammals, Birds and total fauna based on IUCN 2012 are shown in Fig. 26.14. Number of threatened species of total Plants and angiosperms worldwide in 2012 are shown in Fig. 26.15. The update to IUCN 2012 list has been released on 2 July, 2013. With this update, 4,807 species have been added to The IUCN Red List of 2012, bringing the total of assessed species to 70,294, of which 20,934 are threatened with extinction (<http://www.iucnredlist.org/news/worlds-oldest-and-largest-species-in-decline-iucn-red-list>. Accessed 17 July, 2013).

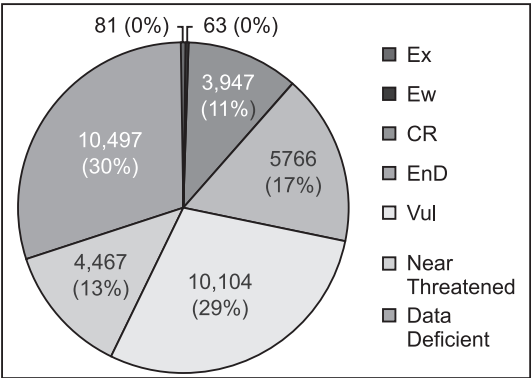


Fig. 26.13. Proportion of species in different threat categories based on 63,837 spp- IUCN 2012 based on data from http://www.iucn.org/about/work/programmes/species/our_work/the_iucn_red_list/ accessed 16 July, 2013.

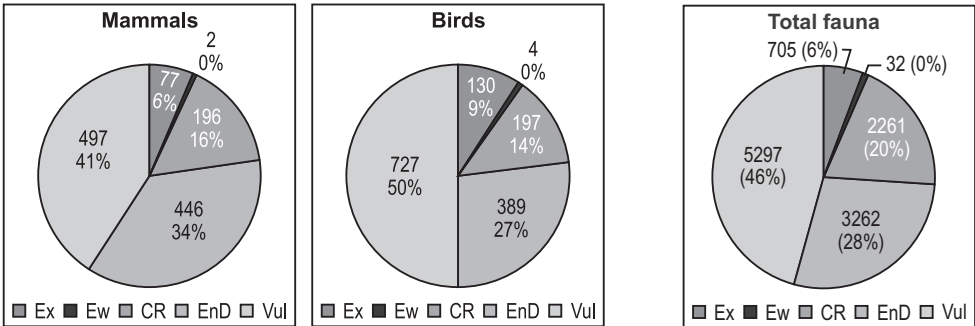


Fig. 26.14. (a) Threatened species of Mammals and Birds (based on IUCN, 2012).

Fig. 26.14. (b) Threatened species of Total Fauna (based on IUCN, 2012).

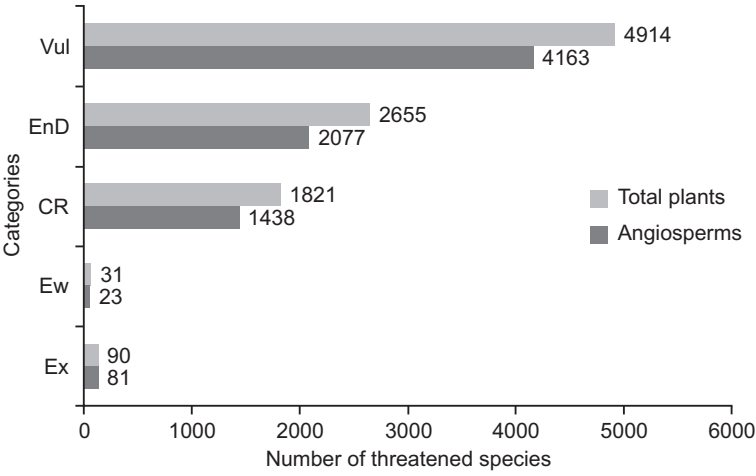


Fig. 26.15. Number of threatened species of total plants and angiosperms worldwide in 2012 (based on IUCN, 2012).

The update includes the first global reassessment of conifers. According to the results, 34% of the world’s cedars, cypresses, firs and other cone-bearing plants are now threatened with extinction – an increase by 4% since the last complete assessment in 1998 (<http://www.iucnredlist.org/news/worlds-oldest-and-largest-species-in-decline-iucn-red-list>. Accessed 17 July, 2013). Conifers are the oldest and largest species on the planet. The Bristlecone Pine (*Pinus longaeva*) for example, can live to an age of nearly 5,000 years and the Coastal Redwood (*Sequoia sempervirens*) grows to a height of 110 meters. The coniferous forests sequester more carbon than any other biome, three times the amount sequestered by temperate and tropical forests.

Global figures for the 2013 IUCN Red List of Threatened Species:

TOTAL SPECIES ASSESSED = 70,294

(Total threatened species = 20,934)

Extinct = 799

Extinct in the Wild = 61

Critically Endangered = 4,227

Endangered = 6,243

Vulnerable = 10,464

Near Threatened = 4,742

Lower Risk/conservation dependent = 241*

Least Concern = 31,846

Data Deficient = 11,671

(this is an old category that is gradually being phased out of the Red List)

(<http://www.iucn.org/?13243/Worlds-oldest-and-largest-species-in-decline--IUCN-Red-List>)

An overview of proportions of threatened species within each of the more comprehensively assessed groups shows that best estimates of percentage threatened species (with lower and upper estimates) for each group are: cycads 63% (63-64%); amphibians 41% (31-56%); conifers 34% (33-35%); reef-forming corals 33% (27-44%); sharks & rays 31% (17-63%); freshwater crabs 31%

(16-65%); mammals 25% (21-36%); groupers 18% (12-43%); birds 13% (13-14%); wrasses 4% (4-18%); lobsters <1% (0-35%) (http://www.iucnredlist.org/about/summary-statistics#How_many_threatened).

The Red List Index (RLI)

The **Red List Index (RLI)**, based on the IUCN Red List of Threatened Species, is an indicator of the changing state of global biodiversity. It defines the conservation status of major species groups, and measures trends in extinction risk over time. RLIs have been calculated using changes in threat status for species in each of the groups. Currently, the *Red List Index* (RLI) is available for four taxonomic groups only (those in which all species have been assessed at least twice): birds, mammals, amphibians, and corals (see Fig. 26.16) (IUCN and Global Biodiversity Outlook-3). The RLI clearly demonstrates that the status of these major groups is still declining. An increasing number of national RLIs are now being used by countries to monitor biodiversity loss (see BirdLife International, 2012). RLIs for reef-forming corals, birds, mammals, and amphibians. Coral species are moving towards increased extinction risk most rapidly, while amphibians are, on average, the most threatened group.

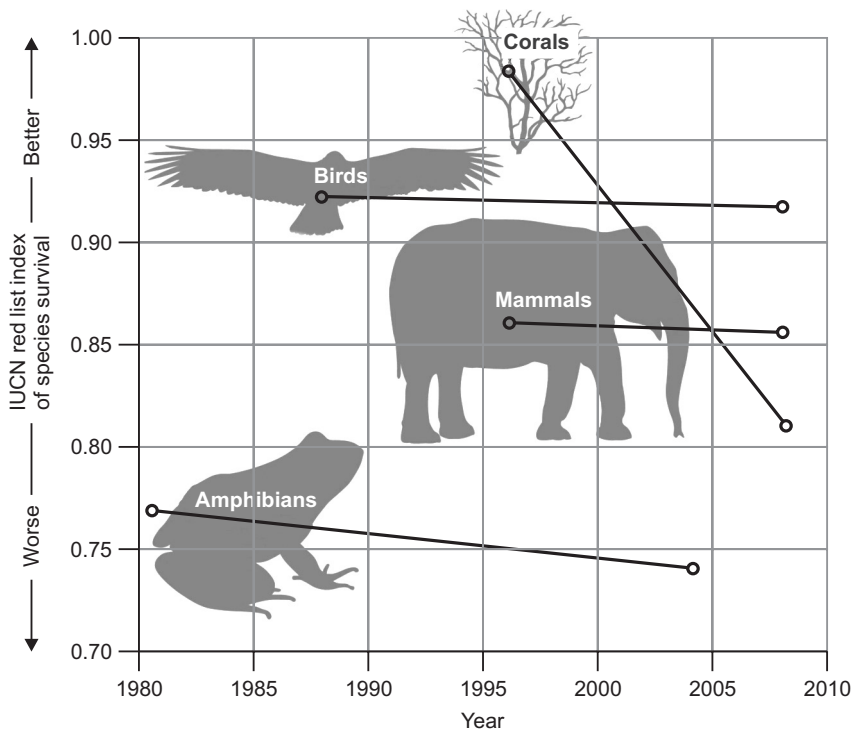


Fig. 26.16. The Red List Index (RLI) for four taxonomic groups (those in which all species have been assessed at least twice): birds, mammals, amphibians, and corals.

An RLI value of 1.0 equates to all species qualifying as Least Concern (i.e., not expected to become Extinct in the near future). An RLI value of 0 equates to all species having gone Extinct. A constant RLI value over time indicates that the overall extinction risk for the group is constant. If the rate of biodiversity loss were reducing, the RLI would show an upward trend. (adapted from IUCN http://www.iucnredlist.org/about/summary-statistics#How_many_threatened).

The World List of Threatened Trees (1998) indicates that more than 8000 tree species, 10 per cent of the world's total, are currently threatened with extinction. More than one in six mangrove species worldwide is on the IUCN Red List of Threatened Species. They are in danger of extinction because of such factors as coastal development, climate change, logging and agriculture. To promote the conservation of tree species, the United Nations Environment Programme World Conservation Monitoring Centre (UNEPWCMC) initiated a Global Tree Conservation Atlas in 2003, which presents profiles of threatened tree species around the world in a map-based format (<http://www.unep-wcmc.org>). The United Nations Food and Agriculture Organization is currently preparing the first-ever report *State of the World's Forest Genetic Resources*. Scheduled to be published in 2013, the report will be the most comprehensive effort undertaken to date to map the genetic diversity of trees and other forest resources (<http://www.fao.org/nr/cgrfa/cgrfa-global/cgrfa-globass/en/>).

26.13 BIODIVERSITY ASSESSMENT AND INVENTORY

The growing awareness of the benefits of biodiversity and its anticipated rapid decline has necessitated a speedy inventory and monitoring of biodiversity at all levels, although a complete inventory of the world's biodiversity would forever remain an elusive goal. The biodiversity can be assessed at genetic, population, species and ecosystem levels (Table 26.10). A variety of approaches and techniques have been proposed for this purpose (Table 26.11).

Table 26.10. Levels of inventory, and examples of techniques and information provided (see Singh 2002, Singh and Khurana, 2002).

Level	Techniques	Examples of information provided
Genetic	Molecular techniques, e.g. allozyme electrophoresis, RFLP (restriction fragment length polymorphism), DNA fingerprinting, PCR (polymerase chain reaction), etc.	Genetic variability among individuals within a population, subdivisions of populations, genetic differentiation among species, inbreeding and out breeding depression, morphologically cryptic species
Population	Point and Distance sampling, Population viability analyses	Rarity and threat status, sustainable harvest/yield, population size, minimum viable population
Species	Natural history collections, Taxonomic studies, Geographic information system, Point and Broad scale sampling, Indices for alpha, beta and gamma diversities	Identity and geographic distribution, new taxa, areas of endemism, phylogenetic relationships, possible uses, organisation of individuals in community, species turnover
Ecosystems and landscape	Broad scale sampling, Remote sensing, Forest inventories, Pattern diversity	Ecosystem inventories, fragmentation, habitat destruction, threatened ecosystems, ecosystem processes, earth cover and land use, compositional diversity

Table 26.11. Approaches for biodiversity assessment and inventory (see Singh, 2002).

Level/scale	Purpose	Sampling
Genetic	Genetic variation among individuals, within populations and among species.	Molecular techniques
Population	Size of populations for rarity and threat status.	Point sampling
Species	Identity, number, new taxa, geographical distribution, endemism, possible uses.	Point and broad-scale sampling
Single species inventories	• High impact species (e.g. keystone, exotic invasive). • Indicator species (pollution, disturbance, age indicators, etc.)	Point and broad-scale sampling
	• Conservation-focus species (umbrella species, flagship species, threatened species, etc.). • Species of value to humans. • Typical or representative species.	
Multiple taxa inventories	• All taxa biodiversity inventory (ATBI): site-specific, all species • All biota taxonomic inventories (ABTI): all species of selected taxa -global perspective. • Rapid biodiversity assessment (RBA): focus on certain taxa to provide estimate of biological richness of a site/region.	Point sampling Taxonomic survey Visual encounter survey (VES); point sampling, broad-scale sampling
Ecosystem and landscape	Type, geographical extent, land cover, unique habitat, threatened ecosystem, etc.	Broad-scale sampling; remote sensing
Spatial scale		
Local	Checklists of species, comprehensive inventories.	Point sampling
Regional	Checklists of species, can be assembled from local inventories.	Synthesis of point sampling data
Global	Extrapolation for global coverage.	Point sampling coupled with broad-scale sampling; remote sensing

The assessment of biodiversity is done both at spatial and temporal scales. Spatial scale includes local inventory (i.e. sampling within biotypes) and global inventory (i.e. sampling among biotypes). Local inventories can be combined to produce a regional inventory. At the temporal scale, monitoring would provide considerable insight into rhythmic versus stochastic patterns and successional events.

Careful inventory of selected single taxon may aid substantially in the conservation efforts. Such inventories also reflect the quality and changes in structure and function of ecosystems. Examples of such inventories are given in Table 26.12. Multitaxa inventory, comprised of several levels, provides a relatively complete and broader understanding of site, region or global biodiversity. Some approaches for multispecies inventory are given in Table 26.13. Sugden and Pennisi (2000) emphasized the importance of digitizing biodiversity data, sharing of information across databases, delivering the data to the desktop and continued monitoring of biodiversity.

Table 26.12. Assessment and monitoring of specific taxon (compiled from various sources, see Singh and Khurana, 2002).

Types of species	Categories	Importance	Selected examples
High impact species	Keystone species	Removal of species affects persistence of others, impact disproportionate to abundance	Palm & fig trees in Peruvian rain forest
Indicator species	Exotic invasives	Entry of species eliminates some native species	<i>Vespula germanica</i> from Europe to South Africa, Australia, New Zealand
	Sentinels	Introduced into atypical conditions as early warning devices	Canaries in coal mines
	Detectors	Show measurable response to environmental change	Heathland communities
	Exploiters	Presence indicate probability of disturbance or pollution	Tubificid worm in aquatic habitat
	Accumulators	Accumulate chemicals in tissues	Lichens, woodlice
	Bioassay organisms	Detect presence or/and concentration of pollutants	Trout
Conservation focus species	Umbrella species	Protection of its occupancy area encapsulates other species	Protected ranges for large mammals and birds become reserves for other species with smaller ranges
	Flagship species	Popular charismatic spp. to stimulate conservation awareness and action	Condors, pandas, rhinos, large cats, large primates, orchid, cacti
	Threatened species	Genetically impoverished, rare sp. prone to extinction	Red listed species
Species of value to humans		Species of direct human value (food/fodder/ medicines etc.)	The bark of rare tree, <i>Homolanthus nutans</i> in Solomon island used for its antiviral properties
Typical or representative species	Once widespread, change in rarity status		<i>Nicrophorus americanus</i> , once widespread in 32 states of USA, now only in two localities

Table 26.13. Important multispecies inventory approaches (compiled from various sources, see Singh and Khurana, 2002).

Inventory type	Proposed monitoring
All taxa biodiversity inventory (ATBI)	Presence, occupancy and natural history of all the species inhabiting a site
Rapid biodiversity assessment (RBA)	Selected focal taxa to measure biological richness and uniqueness of a site
Visual encounter surveys (VES)	Used for RBAs
All biota taxonomic inventory (ABTI)	Focus on specific taxa and inventory of all species in those groups on a global basis

26.14 CONSERVATION OF BIODIVERSITY

The fact that species are disappearing before even we have named them or determined their possible uses and role in the biosphere, suggests that it is wise to make serious attempts to conserve them. Most people are beginning to recognize that diversity at all levels - gene pool, species and biotic community is important and needs to be conserved for sustainable development. There is also an ethical imperative of stewardship of looking after our planet, so that we may pass it on in a good health to our future generations. This way we will not deprive the future generations from the ecological, economic and aesthetic benefits that they can derive from biodiversity. The decisions we make now, as individuals and as a society, will determine the diversity of genes, species and ecosystems that remain in future. We may appreciate the fact that the most effective and efficient mechanism for conserving biodiversity is to prevent further destruction or degradation of habitats by us. We require more knowledge to conserve biodiversity in reduced space and under increased pressure of human activities.

There are four complementary strategies for biodiversity conservation, (i) *in situ* strategy, (ii) *ex situ* strategy, (iii) reduction of anthropogenic pressure, and (iv) rehabilitation of endangered species (Table 26.14).

Table 26.14. Major strategies of biodiversity conservation.

Strategy	Work element
<i>In situ</i>	Establish protected area network, with appropriate management practices, corridors to link fragments; restore degraded habitats within and outside PAs.
<i>Ex situ</i>	Establish botanical and zoological gardens, conservation stands; banks of germplasm, pollen, seed, seedlings, tissue culture, gene and DNA, etc.
Reduction of biotic pressure	Reduce anthropogenic pressure on natural populations by cultivating them elsewhere.
Rehabilitation	Identify and rehabilitate threatened species; launch augmentation, reintroduction or introduction programmes.

***In situ* Conservation Strategies**

The *in situ* strategy emphasizes protection of total ecosystems for the conservation of overall biodiversity of genes, populations, species, communities and the ecological processes. The *in situ* approach includes protection of a group of typical ecosystems through a network of protected areas as recognized by the UNEP and the World Conservation Union (IUCN). Two international programmes, viz., Millennium Ecosystem Assessment (MEA) and Critical Ecosystems (CE) could play an important role in this direction. Myers et al. (2000) suggested that the biodiversity hotspots could be a focus for conservation as most species (particularly the endemics) could be protected at least cost. The Global Trees Campaign aims to save the world's most threatened tree species and their habitats through information, conservation and wise use (Newton et al., 2003). The campaign focuses on trees as flagship species for conservation of ecosystems and landscapes, and enables local people to carry out rescue and sustainable use operations (Newton et al., 2003).

Protected Areas

These are areas of land and/or sea especially dedicated to the protection and maintenance of biological diversity, and of natural and associated cultural resources. These are managed through legal or other effective means. The protected areas conserve the resident species as well as the

ecological processes and indirect ecosystem services, e.g. abatement of soil erosion, amelioration of climate and regulation of water discharge in spring and rivers. The UNEP has recognized six management categories of protected areas (Box 26.5).

Box 26.4. The IUCN Protected Area Management Categories

CATEGORY Ia : Strict Nature Reserve

The nature reserves represent the protected areas managed mainly for science. Area of land and/or sea possessing some outstanding or representative ecosystems, geological or physiological features and/or species, available primarily for scientific research and/or environmental monitoring.

CATEGORY Ib : Wilderness Area

These are protected areas managed mainly for wilderness protection. Large area of unmodified or slightly modified land, and/or sea, retaining its natural character and influence, without permanent or significant habitation, which is protected and managed so as to preserve its natural condition.

CATEGORY II : National Park

The protected area managed mainly for ecosystem protection and recreation. Natural area of land and/or sea, designated to (a) protect the ecological integrity of one or more ecosystems for present and future generations, (b) exclude exploitation or occupation inimical to the purposes of designation of the area and (c) provide a foundation for spiritual, scientific, educational, recreational and visitor opportunities, all of which must be environmentally and culturally compatible.

CATEGORY III : Natural Monument

These are protected areas managed mainly for conservation of specific natural features. Area containing one or more, specific natural or natural/cultural feature which is of outstanding or unique value because of its inherent rarity, representative or aesthetic qualities or cultural significance.

CATEGORY IV : Habitat/Species Management Area

These are protected areas managed mainly for conservation through management intervention. Area of land and/or sea subject to active intervention for management purposes so as to ensure the maintenance of habitats and/or to meet the requirements of specific species.

CATEGORY V : Protected Landscape/Seascape

The protected area managed mainly for landscape/ seascape conservation and recreation. Area of land, with coast and sea as appropriate, where the interaction of people and nature over time has produced an area of distinct character with significant aesthetic, ecological and/or cultural value, and often with high biological diversity. Safeguarding the integrity of this traditional interaction is vital to the protection, maintenance and evolution of such an area.

CATEGORY VI : Managed Resource Protected Area

In this case, the protected area is managed mainly for the sustainable use of natural ecosystems. Area containing predominantly unmodified natural systems, managed to ensure long-term protection and maintenance of biological diversity, while providing at the same time a sustainable flow of natural products and services to meet community needs.

Some of the main benefits of protected areas are: maintaining viable populations of all native species and subspecies; maintaining the number and distribution of communities and habitats, and conserving the genetic diversity of all the present species; preventing human-caused introductions of alien species; and making it possible for species/habitats to shift in response to environmental changes.

Terrestrial Protected areas: Examples of protected areas are National Parks, and Wildlife Sanctuaries. The earliest national parks, the Yellowstone in USA (established in 1872) and the Royal near Sydney, Australia, were chosen because of their scenic beauty and recreational values. Many similar areas throughout the world now protect rare species or wilderness areas.

Managing area-based conservation Protected areas are one of the core means of preventing ongoing losses of species and habitats. The number of protected areas has increased substantially from 9,214 in 1962 to 1,02,102 during 2003 (Box 26.6). The UN has recognized 1,02,102 protected areas covering more than 18.8 million km² covering 11.5 % of earth's land surface and 12.65 % including the marine areas During 2003 (Chape et al., 2003). Of these, 59,478 (58.25%) protected areas are less than 10 km². There are 41997 (41.13%) protected areas around the world which fulfill the norms of IUCN categories (Chape et al., 2003). Using the gap analysis approach by combining the distribution of species and protected areas, it has been found that the global network of protected areas is far from complete, even for terrestrial vertebrates and many species, especially plants and insects, exhibiting high level of endemism and species with smaller range sizes (Rodrigues et al., 2004).

Gibson et al. (2011) analysed 2,220 pairwise comparisons of biodiversity values in primary forests (with little or no human disturbance) and disturbed forests and found that biodiversity values were substantially lower in degraded forests, the authors concluded that primary forests are irreplaceable for sustaining tropical biodiversity. Similarly, Clark and Covey (2012) analysed published studies investigating 25 distinct paired human-disturbed and primary forests in the tropical and temperate biomes and concluded that a significant reduction in tree species richness following logging occurred; although logging may increase the richness of early successional tree species, it also tends to reduce the richness of late successional trees. It is important that the primary forests should be protected by enhancing of enforcement in existing protected areas, and the current network of reserves should be expanded. Protected areas have expanded over the past 50 years in both number and area and now cover 13 per cent of the world's land area (Box 26.6) (IUCN and UNEP-WCMC, 2011). However, coverage is uneven, and 6 of the 14 global biomes and half of the 821 terrestrial ecoregions do not meet the CBD target for 10 per cent of their area to be protected by 2010 (Jenkins and Joppa, 2009) (Fig. 26.17).

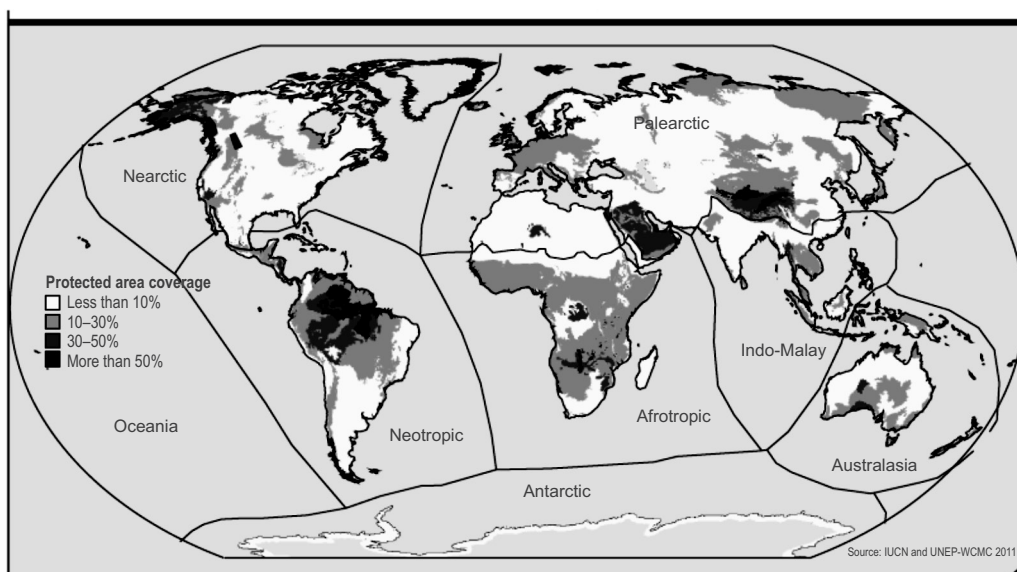


Fig. 26.17. Proportion of each terrestrial ecoregion covered by protected areas, 2011. (adapted from IUCN and UNEP-WCMC, 2011).

Some 51 per cent of the 587 sites identified by the Alliance for Zero Extinction as critical for the survival of hundreds of highly threatened species, and 49 per cent of the more than 10 000 important bird areas are still entirely outside the protected area network (Butchart et al., 2012). Even more importantly, the performance of protected areas in maintaining populations of their key species is poorly documented (UNEP, 2012). Laurance et al. (2012) conducted a systematic and uniquely comprehensive assessment of long-term changes within 60 protected areas stratified across the world's major tropical forest regions. They found that around half of the reserves are experiencing a severe loss of biodiversity. Crucially, the researchers also demonstrated the negative impact that environmental changes immediately outside the areas have on the health of the habitat inside the reserves. And they found that destructive activities such as forest clearance, fires and logging increasingly reach up to the edge of the protected areas. The results show that 85% of the reserves experienced declines in surrounding forest cover over the study period, whereas only 2% gained forest. The results underscore the importance of better management of areas around the reserves. About 15% of the global terrestrial carbon stock is stored in the world's protected area network (Campbell et al., 2008), and the role that this can play in climate mitigation is important.

Box 26.5. Expansion of Protected areas during the past 50 years

1962 : 9,214 protected areas

1981 : 40,000 protected areas covered 7,000,000 km²

2003 : 1,02,102 protected areas covering more than 18.8 million km²

2012 : 200,000+ protected areas covered 12% of the world

Target 11 of the Aichi Targets on protected areas coverage :

"By 2020, at least 17 per cent of terrestrial and inland water, and 10 per cent of coastal and marine areas, especially areas of particular importance for biodiversity and ecosystem services, are conserved through effectively and equitably managed, ecologically representative and well connected systems of protected areas and other effective area based conservation measures, and integrated into the wider landscapes and seascapes."

Source : UNEP (2012); (IUCN and UNEP-WCMC (2011); Chape et al. (2003)

Parts of protected areas which have already been degraded may require rehabilitative actions to restore them to a fully functional state or to their natural state. Fragmentation of natural ecosystems within and outside of protected areas due to agricultural development, forestry and urbanization has resulted into a matrix of modified ecosystems within which small remnants of natural ecosystems remain. Corridors of natural vegetation linking these small fragments are needed to enhance the probability of movement of the native species around the landscape to fulfill their resource and habitat requirements (Rodgers et al., 2000). The corridors should be wide enough to contain a significant area of the non-edge environment.

Marine protected areas: Since 1986, the World Conservation Union (IUCN) has been promoting establishment of global system of marine protected areas (MPAs). These are areas of inter tidal and sub tidal region taken together with their overlying water and associated flora and fauna, which have been reserved by law or other effective means to protect it (IUCN, 1992). The main objectives of marine protected areas are: Protection and restoration of depleted population of marine organisms; Protection of endangered species and critical habitats; Conserving and restoring marine ecosystem health, for effective fishing management, to maintain bio-diversity and ecological processes of marine and coastal ecosystems, to use marine resource in sustainable and equitable way, for restoration of degraded marine and coastal areas, to devise sustainable fishery and tourism management and conservation of biodiversity to minimize extinction risk.

According to World Database on Protected Areas records, 4,116 protected areas in the UN list contain marine and coastal elements, covering 4.3 million km². However, only part of this total figure is marine as the sites also contain land areas. Approximately 1.64 million km² comprise marine protected areas, which is less than one-tenth of the overall extent of protected areas worldwide or 8.7% of the 2003 global total of 18.8 million km² (Chape et al., 2003). This figure represents only 0.5% of the surface area of the oceans.

In 1995, IUCN released the first global assessment report of MPAs. It shows that almost 50% of MPAs are located in Oceanic and Asia (258 are within Australia and New Zealand waters alone). South America has least number of protected areas. About 50% of the existing MPA's are less than 1000 hectares in size. This area is small to accommodate breeding nursery and feeding areas of many of the species found in them. Only 2.2 % of MPAs exceed 1 million hectares. The largest marine protected area is the Great Barrier Reef Marine Park (345,400 km²) in Australia.

Large marine ecosystems (LME's) approach is now being adopted for evaluating fish stocks, stocker's competitors and predators. There are 49 LMEs on the margins of world oceans which also cover the most productive marine fisheries. The LME bio-regional management can provide a means of integrated fisheries management.

Uneven protected area coverage of biomes is most evident in the marine realm, despite a CBD target to protect 10 per cent of the ocean by 2012 (UNEP, 2012). By the end of 2010 marine protected areas covered 1.6 per cent of the ocean area (IUCN and UNEPW CMC, 2011). Indeed, at the end of 2010, only 12 countries had designated more than 10 per cent of their waters, often through large areas, while 121 countries had yet to designate more than 0.5 per cent of their marine jurisdiction (Toropova et al., 2010). In response, the CBD has retained the 10 per cent target, with a revised achievement date of 2020. Marine protected areas can be designated at a variety of levels of protection, but those with complete protection provide the greatest biodiversity benefits (UNEP, 2012; IUCN and UNEP-WCMC, 2011). The World Summit on Sustainable Development, the World Parks Congress, and the Convention on Biological Diversity have all committed to a goal of establishing a global network of marine protected areas by 2020, including on the high seas.

Biosphere Reserves

Biosphere reserves are a special category of protected areas of land and/or coastal environments, wherein people are an integral component of the system. These are representative examples of natural biomes and contain unique biological communities. The concept of Biosphere Reserves was launched in 1975 as a part of the UNESCO's Man and Biosphere Programme dealing with the conservation of ecosystems and the genetic resources contained therein. A Biosphere Reserve consists of core, buffer and transition zones (Fig. 26.18). *The natural or core zone* comprises an undisturbed and legally protected ecosystem. *The buffer zone* surrounds the core area, and is managed to accommodate a greater variety of resource use strategies, and research and educational activities. *The transition zone*, the outermost part of the Biosphere Reserve, is an area of active cooperation between reserve management and the local people, wherein activities like settlements, cropping, forestry and recreation and other economic uses continue in harmony with conservation goals.

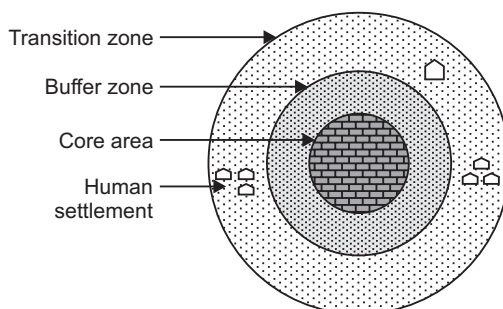


Fig. 26.18. The zonation in a terrestrial Biosphere Reserve.

The main functions of biosphere reserves are:

- (i) Conservation: to ensure the conservation of landscapes, ecosystems, species and genetic resources. It also encourages traditional resource use.
- (ii) Development: to promote economic development, which is culturally, socially and ecologically sustainable.
- (iii) Scientific research, monitoring and education: the aim is to provide support for research, monitoring, education and information exchange related to local, national and global issues of conservation and development.

The International Coordinating Council of UNESCO's Man and the Biosphere Programme (MAB), in May, 2013 has added 12 sites to the World Network of Biosphere Reserves to the list of 2012, which was 610 (UNESCO, 2012). Till May, 2013, there are 621 biosphere reserves located in 117 countries around the world, including 12 transboundary sites (http://www.unesco.org/new/en/natural-sciences/about-us/single-view/news/twelve_sites_added_to_unescos_world_biosphere_reserve_network/).

Marine Biosphere Reserves

One of the approaches to conserve marine biodiversity is to identify biodiversity hotspots that are worth protection (Roberts et al, 2002; Hooker and Gerber, 2004). Marine Reserves are those areas which are closed to all fishing and have a great potential conservation tool. Protection from fishing leads to rapid increase in biomass, abundance and average size of the exploited organisms and also increases species diversity. Life stages of some marine predators (e.g. baleen whales, prinnipeds and seabirds) are separated spatially into discrete feeding and breeding areas, with migration between them. Marine reserves can be placed in feeding, breeding or migratory habitats (Fig. 26.19).

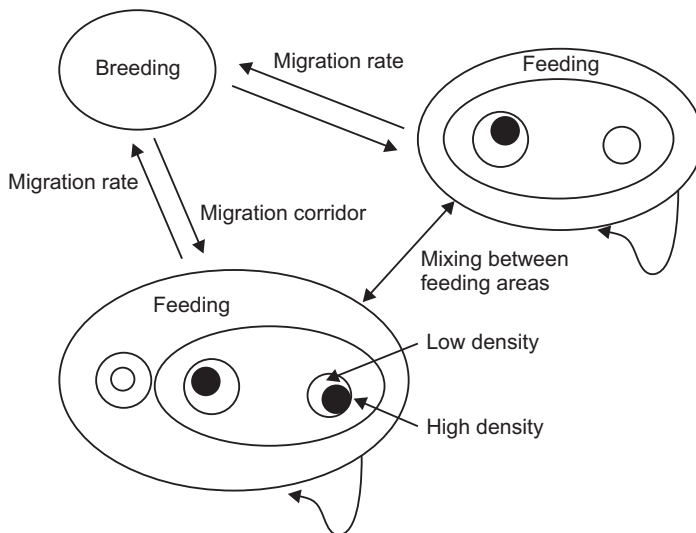


Fig. 26.19. Life stages of some marine predators (e.g., baleen whales, prinnipeds, seabirds) are separated spatially into discrete feeding and breeding areas, with migration between them. Marine Reserves can be placed in feeding, breeding, or migratory habitats (redrawn from Hooker and Gerber, 2004).

Ex-situ Conservation Strategies

The *ex-situ* conservation strategies include botanical gardens, zoos, conservation stands, and field gene banks as well as pollen, seed, seedling, tissue culture and DNA banks, etc. All these approaches help conserve species and population diversity outside the natural habitat.

Seed gene banks: Four crops – rice, maize, potatoes and wheat – provide about 60% of the dietary energy of humanity, out of an estimated total of 30,000 edible plant species (FAO, 1996). The crop species diversity has declined with the onset of modern agriculture techniques, which will have severe implication to the food security of the planet given environmental degradation, pests, epidemics and climate change. Seed gene banks are the easiest way to store germplasm of wild and cultivated plants at low temperature in cold rooms. Preservation of genetic resources is carried out in field gene banks under normal growing conditions in the case of plants, which do not produce seeds (e.g., bananas and plantains).

A Russian scientist N.I. Vavilov pioneered plant genetic collections (1887-1943). He undertook extraordinary expeditions during 1916-1933. Vavilov and his many students collected more than 250,000 plant accessions from around the world. The N.I. Vavilov Institute of Plant Industry (VIR) in St. Petersburg, Russia houses one of the most important gene banks in the world.

The ancestral genes could convey desirable characteristics such as resistance to new strains of disease or tolerance to novel growing conditions. About 90% of the accessions in gene banks are maintained as seeds (FAO, 1996).

In 2010, there are some 1,750 individual genebanks worldwide with total holdings of about 7.4 million accessions, of which about 6.6 million are held in national genebanks (FAO 2010 c). In India alone, the National Genebank (NGB) holds around 0.39 million accessions as its base collection. (http://www.nbgr.ernet.in/monthly_progress/cons_dec11.pdf. Accessed 16 July 2013).

Kew Botanical Gardens in London has stored plant material since the 1850s and has developed a project to establish the millennium seed banks, which will store seeds from the semiarid tropic and subtropics and contribute to cataloging of global biodiversity. The International Plant Genetic Resources Institute (IPGRI, Rome, Italy) is the world's largest organization and devoted uniquely to the conservation and use of agricultural biodiversity. Other organizations involved in conservation of crop plant genetic resources are:

- International Maize and Wheat Improvement Centre. (CIMMYT, Mexico).
- International Centre for Tropical Agriculture (CIAT, Colombia).
- International Centre for Agricultural Research in the Dry Areas (ICARDA, Syria).
- International Crops Research Institute for the Semi-arid Tropics (ICRISAT, India).
- International Rice Research Institute (IRRI, Philippines)

Orthodox seeds tolerate drying and low moisture and can be stored for long period of time at low temperatures. Recalcitrant seeds of tropical tree species (e.g., family Dipterocarpaceae) are damaged by desiccation as well as by low temperature. *Hopea odorata*, a tropical recalcitrant species from humid tropics is very desiccation sensitive, can not tolerate temperatures below 15°C. *Quercus robur* is a tropical-temperate recalcitrant species, which is desiccation sensitive but can tolerate storage temperature of -2°C.

For storing the seeds in seed gene banks, the seeds are cleaned and dehydrated prior to storage. Then cooled to 20°C (cooling lowers the metabolic rate and allows the seeds to be stored for a longer period of time). The seeds are germinated periodically to obtain fresh seeds.

***In-vitro* gene banks** These are short and medium term storage for a range of crops woody species, fruit trees and horticultural species using tissue culture techniques. Tissue culture systems allow the propagation of plants with high multiplication rates in an aseptic environment. The cells are grown on a gel and fed with suitable nutrients and hormones to give rise to entire plants. 38,000 accessions are conserved worldwide using the *in vitro* culture techniques (FAO, 1996). *In vitro* conservation, especially by cryopreservation in liquid nitrogen at a temperature of -196°C, is particularly useful for conserving vegetatively propagated crops like potato. Cryopreservation is the storage of material at ultra-low temperature either by very rapid cooling (used for storing seeds) or by gradual cooling and simultaneous dehydration at low temperature (used for tissue culture). The material can be stored for a long period of time in compact, low maintenance refrigeration units. It is an expensive technique; therefore limited amount of genetic diversity can be maintained *in vitro*.

DNA storage: DNA from nucleus, mitochondria and chloroplasts are now routinely extracted and immobilized into nitro-cellulose sheets. DNA can be probed with numerous cloned genes. PCR (Polymerase Chain Reaction) can amplify Oligonucleotides or genes from entire mixtures of genomic DNA International network of DNA repositories could be efficient and simple technique for the storage of Genomic DNA. However, there are problems with gene isolation, cloning and transfer which do not allow regeneration of the whole plants.

The DNA Bank Network was established in spring 2007 and is currently funded by the German Science Foundation (DFG) (www.dnabank-network.org). The network was initiated by GBIF Germany (Global Biodiversity Information Facility). It offers a worldwide unique concept. DNA bank databases of all partners are linked and are accessible via a central web portal, providing DNA samples of complementary collections (microorganisms, protists, plants, algae, fungi and animals).

The mission of the Frozen Ark Project is to collect, preserve and store DNA and viable cells from animals in danger of extinction. The Project will focus on the thousands of animals that are expected to disappear within the next few decades. Samples will be taken from captive breeding programmes, zoos and wild populations.

Animal Genetic Resources: On the basis of Commission On Genetic Resources For Food And Agriculture report published in April, 2013 (FAO, 2013), the Global Databank for Animal Genetic Resources currently contains data from 182 countries and 37 species. The total number of national breed populations recorded in the Global Databank has increased during the reporting period. The total number of mammalian national breed populations recorded in June 2012 was 10 712 as compared to 10 507 in October 2010. The total number of avian national breed populations recorded in 2012 was 3 482, compared 3 414 in 2010.

Botanical gardens and Zoos: Conservation of biological diversity in botanical gardens is already in practice. There are more than 3,000 botanic gardens and arboreta (botanical gardens where specific tree and shrub species are cultivated) in existence in 180 countries around the world according to Botanic Gardens Conservation International (BGCI, 2012). Collectively they cultivate over 300,000 taxa, representing around 100,000 species (almost one third of all known plant species). This also includes over 12,000 species which are under threat of extinction in the

wild. BGCI is presently working on recovery programmes for over 500 threatened plant species. Many of these now have seed banks, tissue culture facilities and other *ex situ* technologies. The Updated list indicates that there are currently 3156 botanical gardens and arboreta globally (<http://www.bgci.org>).

More than 250 zoos and aquariums are institutional members of the *World Association of Zoos and Aquariums* (WAZA). About 1,300 zoos and aquariums are linked to WAZA. Many of these zoos have well-developed captive breeding programmes. The IUCN list contains species, which were extinct in the wild in the past, but have “survived in human care” and several of them have been reintroduced by zoos and wildlife parks to their original habitat (e.g., Californian condor CR; Red wolf CR; Black-footed ferret EN; Arabian oryx EN; European bison VU; American bison NT).

The conservation of wild relatives of crop plants and the off-site conservation of crop varieties or cultures of microorganisms provides breeders and genetic engineers with a ready source of genetic material. Plants and animals conserved in botanical gardens, arboreta, zoos and aquaria can be used to restore degraded land, reintroduce species into wild, and restock depleted populations.

Restoration of Endangered Species

Restoration or rehabilitation of endangered species is extremely difficult and expensive process. This requires species specific plans and knowledge of ecology and reproductive biology of species. The species restoration plans should include diagnosis of factors responsible for the decline of species, habitat conservation, captive breeding and restriction of harvesting etc. The strategies for restoration include: (i) reintroduction programmes(release of captive bred or wild collected individuals in areas where they existed earlier but are currently absent), (ii) augmentation programmes (release of individuals to increase the existing population size and genetic diversity of species), and (iii) introduction programmes(establishment of populations outside their historic range (Miller et al., 1995;see Singh and Khurana, 2002).

Sustainable use and public participation

The long term conservation of the biodiversity can be ensured by creating public awareness about the value of biodiversity and allocating them a greater share of benefits from conservation. Ecotourism, recreation and education can also help in sustainable use and management of biodiversity. Creating public awareness on various aspects of biodiversity conservation at village, district and state levels is necessary for the involvement of people in effective conservation of biodiversity. Educating the local people about the economic and ecological values of biodiversity and threatened species could be effective to protect the threatened areas. For better management, it is essential to involve people at grass root level.

Ecotourism may be used to promote conservation of the natural and cultural heritage of an area, and simultaneously could improve the living standards of the inhabitants of host regions (Lindberg and Hawkins, 1993). A contingent valuation survey of Khangchendzonga National Park of Sikkim Himalaya and the sacred Khecheopalri lake has revealed the value of ecotourism for management and conservation of biodiversity hotspots and protected areas as well as for improving the living standards of the local inhabitants in India (Maharana et al., 2000).

Singh (2012) has discussed the role that biodiversity banking can play in conservation by attracting peoples participation. The biodiversity banking process, which includes attachment of value to the components of biodiversity and the services they provide for the humankind. Biodiversity banking allows accumulation of ‘tradable biodiversity credits’ to derive monetary benefits by the stakeholders if they commit to enhance the biodiversity value of the land (Kumaraswamy Udayakumar, 2011).

26.15 INDIA'S BIODIVERSITY AND ITS CONSERVATION

Status of India's Biodiversity

India, one of the 17 megadiverse countries, has 2.4 % of land area of the world but contributes 8.6% of all mammalian, 13.7% of all avian, 7.9% of all reptilian, 6% of all amphibian, 12.2% of all piscine, and 6.0% of all flowering plant species to the global diversity. It harbours 91,212 species of animals (7.43 % of the World's species), 17450 angiosperms, 65 Gymnosperms, 1200 pteridophytes, 2426 Bryophytes, 2021 Lichens, 14500 Fungi, and 6500 algae. A large number of animals and plants are yet to be explored, identified and described (MoEF, 2009 b). About 33% of the flowering plants recorded in India are endemic to our country. Indian region is also notable for endemic fauna. For example, out of the recorded vertebrates, 53% freshwater fishes, 60% amphibians, 36% reptiles and 10% mammalian fauna are endemic. The endemics are concentrated mainly in north-east, Western Ghats, north-west Himalaya and Andaman and Nicobar Islands. However, roughly 40% geographical area has remained under explored with a potential to yield hundreds of new species. For example, as many as 250 new species of flowering plants have been described only from Kerala alone during the past quarter century (Manilal, 2001). Continuous surveys and explorations have added 41 new plant species in 2007 alone by Botanical Survey of India (BSI).

Threatened Plants and Animals

As per the International Union for Conservation of Nature (IUCN) Red List 2012, India has 614 globally threatened faunal species, which is approximately 4.9 % of the world's total number of threatened faunal species and over 321 globally threatened plant species (Fig. 26.20). The Red List has listed 132 species of plants and animals as Critically Endangered from India. Plants seemed to be the most threatened life form with 60 species being listed as Critically Endangered and 145 as Endangered. The Critically Endangered list included 18 species of amphibians, 14 fishes and 10 mammals. There are also 15 bird species in the category. The agency listed 310 species as Endangered, including 69 fishes, 38 mammals and 32 amphibians. Two plant species were reported to be extinct in the wild, including *Euphorbia mayuranthanii* of Kerala. A leaf frog species and six plants were recorded as extinct, according to the latest assessment.

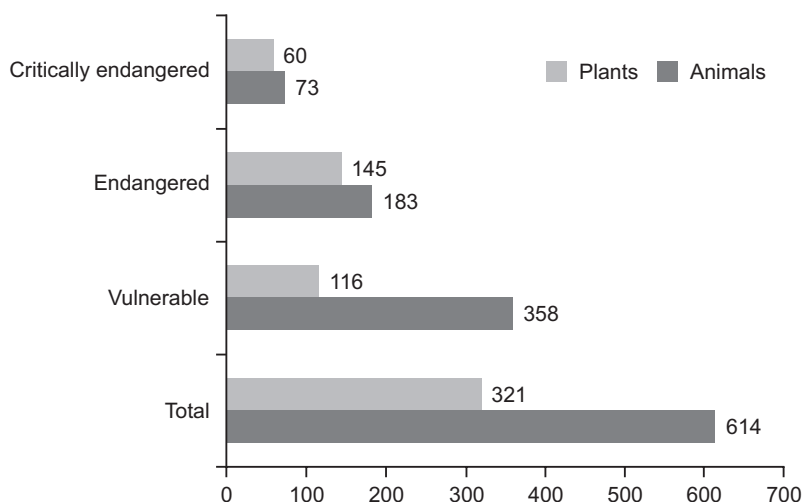


Fig. 26.20. The number of plant and animal species of various threat categories in India (Based on IUCN, Red Data List of Threatened Species, 2012).

Some examples of threatened species in India are given in Tables 26.15 and 26.16.

Table 26.15. Some threatened plant species of India (based on Nayar and Sastry, 1990 and Ramesh and Pascal, 1997; see IUCN, 2003).

Critical Endangered	Endangered	Vulnerable
<i>Ilex Khasiana</i>	<i>Kingiodendron pinnatum</i>	<i>Myristica malabarica</i>
<i>Syzygium travancoricum</i>	<i>Madhuca diplostemon</i>	<i>Ochreinauclea missionis</i>
<i>Dipterocarpus grandiflorus</i>	<i>Pterocarpus santalinus</i>	<i>Saraca asoca</i>
<i>Berberis nilghiriensis</i>	<i>Capparis pachyphylla</i>	<i>Cupressus cashmeriana</i>

Table 26.16. Some threatened animal species of India (based on IUCN, 2003).

Critical Endangered	Endangered	Vulnerable
Malabar Large Spotted Civet (<i>Viverra civettina</i>)	Asian Elephant (<i>Elephas maximus</i>)	Asiatic Black Bear (<i>Ursus thibetanus</i>)
Namdapha Flying Squirrel (<i>Biswamayopterus biswasi</i>)	Lion-tailed Macaque (<i>Macaca silenus</i>)	Asiatic Wild Ass (<i>Equus hemionus</i>)
Pygmy Hog (<i>Sus salvanius</i>)	Red Panda (<i>Ailurus fulgens</i>)	Blackbuck (<i>Antelope cervicapra</i>)
Sumatran Rhinoceros (<i>Dicerorhinus sumatrensis</i>)	Indian Rhinoceros (<i>Rhinoceros unicornis</i>)	Wild Yak (<i>Bos grunniens</i>)

The country is also rich in plants of medicinal importance. Nearly 3000 species have been mentioned in books dealing with medicinal plants, of which about 200 are used in bulk quantities for traditional systems of medicine, and plant products go into allopathic system from about 50 species (Jain and DeFilipps, 1991). Even species-poor localities may harbour a significant number of medicinal plants. Exploration of five sites in the species-poor dry deciduous forest revealed 125 plants from 57 families, which are therapeutically used by the tribal populations (Singh et al., 2002). Indian region has contributed to the global biodiversity species such as zebu, mithun, chicken, water buffalo, camel, rice, sugarcane, banana, tea, mango, millets, Asiatic vignas, brassicas, rice bean, tree cotton, jackfruit, edible diascoraeas, alocasia, colocasia, cardamom, black pepper, ginger, turmeric, cucurbits, bamboos, etc., and has been a secondary centre of domestication for horse, goat, sheep, cattle, yak, donkey, tobacco, potato, maize, soybean, etc. (Khoshoo, 1995).

Biodiversity Hotspots

Among the 34 global terrestrial biodiversity hotspots, four (viz., Himalaya, Indo-Burma, Western Ghats and Sri Lanka, and Sundaland) are partly located in India (Fig. 26.21). These areas are rich in flowering plants, also in reptiles, amphibians, swallow-tailed butterflies and some mammals; and also show a high degree of endemism. Besides being an active centre of evolution and rich diversity of flowering plants, the numerous primitive angiosperm families (e.g. Magnoliaceae and Winteraceae) and primitive genera of plants like *Magnolia* and *Betula* are found in eastern Himalaya. The Western Ghats region holds important population of Asian elephants, Indian tigers and the endangered lion-tailed macaque. About 14,000 km² area, 9% of the hotspot area, is protected in eight national parks and 39 sanctuaries. The Agasthyamalai hills and the Silent Valley, the new Amambalam Reserve, are the two main centres of diversity.

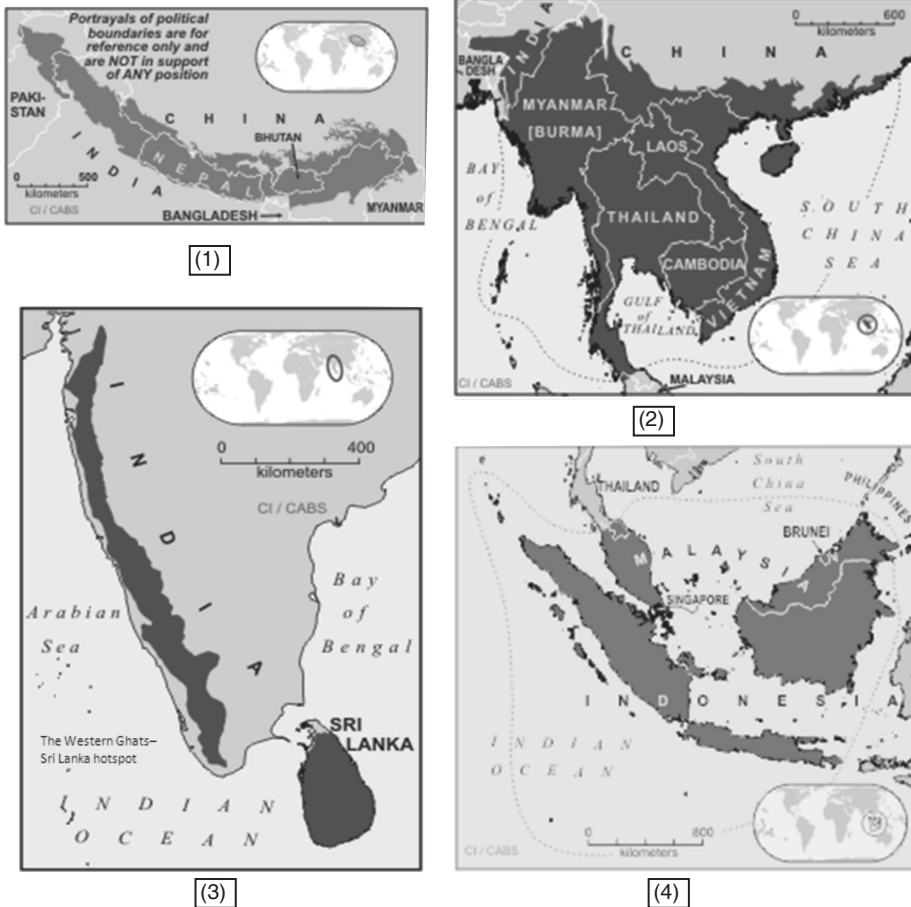


Fig. 26.21. The four global Hotspots of biodiversity in India (1) Himalaya; (2) Indo-Burma; (3) Western Ghats and Sri Lanka, (4) Sundaland. Note that only Nicobar Islands are located within Sundaland; the Andaman Island is in Indo Burma. (adapted from Conservation International).

Indian Scenario of Biodiversity Conservation

The country has taken significant steps for biodiversity conservation. Apart from establishing protected areas, a *National Biodiversity Act* was passed in 2002 which got the assent of President on 5 Feb 2003, and *Biodiversity Rules* were framed in 2004. A National Biodiversity Action Plan, 2008, was released on 24 Feb 2009.

Protected areas in India

As of February, 2013, India has 683 protected areas (102 National Parks and 520 Wildlife Sanctuaries http://wiienvis.nic.in/Database/Gis_1267.aspx) covering an area of 164980.75 km², representing 5.02 % of the land surface as against 10% internationally suggested norm.

National Parks of India

There are 102 existing national parks in India covering an area of 40,075 km², which is 1.22% of the geographical area of the country (National Wildlife Database, Feb. 2013), (Table 26.17). In addition to the above, 75 National Parks covering an area of 16,608 km² are proposed in the Protected Area Network Report (Rodgers and Panwar, 1988). The network of parks will go up 177 after full implementation of the above report.

Table 26.17. Table Number and extent of designated protected areas in India in 2013.Source: (http://wiienviis.nic.in/Database/Gis_1267.aspx).

Type	Status	Number	Area (km ²)	% of Geographical Area of India
National Parks	Declared	102	40,075	1.22 %
Wildlife Sanctuaries	Declared	520	121,980	3.74%
Conservation Reserves	Declared	57	2017.94 km ²	0.06 %
Community Reserves	Declared	4	20.69	0.0 %
Protected Areas (PAs)		683	164980.75 km ²	5.02 %

The Jim Corbett National Park in Uttranchal was the first National Park established in India. Under the World Heritage Convention, five natural sites have been declared as “World Heritage Sites”, which are Kaziranga National Park (Assam), Keoladeo Ghana National Park (Rajasthan), Manas Wildlife Sanctuary (Assam), Nanda Devi National Park (Uttar Pradesh), and Sunderbans National Park (West Bengal).

Some national parks and sanctuaries are briefly described below :

Kaziranga National Park, Assam : A World Heritage Site, the park hosts two-thirds of the world’s Great One-horned Rhinoceroses. Kaziranga has the highest density of tigers among protected areas in the world and was declared a Tiger Reserve in 2006.

The park is home to large breeding populations of elephants, wild water buffalo, and swamp deer.

Kaziranga is recognized as an *Important Bird Area* by Birdlife International for conservation of avifaunal species. Located on the edge of the Eastern Himalaya biodiversity hotspot, the park combines high species diversity and visibility.

Manas National Park, Assam : A UNESCO World Heritage Centre in India, Manas Tiger Reserve in Assam is the only Project Tiger in Assam located in the foothills of the Bhutan hills, far from human habitation. Manas National Park also houses approximately 20 highly endangered species, such as Hispid Hare and Pygmy Hog.

Valley of Flowers National Park: Is an Indian national park, Nestled high in West Himalaya, is renowned for its meadows of endemic alpine flowers and outstanding natural beauty. It is located in Uttarakhand state. This richly diverse area is also home to rare and endangered animals, including the Asiatic black bear, snow leopard, brown bear and blue sheep. The gentle landscape of the Valley of Flowers National Park complements the rugged mountain wilderness of Nanda Devi National Park. The National Park encompasses a unique transition zone between the mountain ranges of the Zaskar and Great Himalaya. The park stretches over an expanse of 87.50 km².

Gibbon Wildlife Sanctuary: Gibbon Wildlife Sanctuary has one of the most endangered primates called Hoolock Gibbon. This sanctuary is more than 22 sq kms and has the highest concentration of these animal species in the world. The Gibbon Wildlife Sanctuary was established in 1997. Gibbon Wildlife Sanctuary in Assam derives its name from the Gibbons (*Bunopithecus hoolock*), which are one of the rarely found primate species in India.

The sanctuary was originally a part of Hollongapar Reserve Forest rich in various plant species: Holong (*Dipterocarpus macrocarpus*) and Agar (*Aquilaria agallocha*). *Garcinia* spp., *Terminalia myocarpha*, *Artocarpus chaplasha*, *Mesua ferrea*, *Lagerstromia speciosa* and several species of bamboo, *Michelia champaca*, ferns and orchids.

Gene Sanctuary in India for Citrus in Garo Hills : The Tura Range in Goro Hills of Meghalaya is a gene sanctuary for conserving rich diversity of wild Citrus and Musa species. National Citrus Gene Sanctuary-cum-Biosphere Reserve, spread over an area of over 47 sq km, was established with the aim to protect the mother germo plasm of Citrus-indica. It is located in West Garo Hills District of Meghalaya. The sanctuary is covered with evergreen and semi-evergreen forests but has a rocky terrain and is nourished by numerous rivers and streams. The sanctuary is a natural habitat for rare and endangered species of animals, birds, butterflies and reptiles. It is also home to rare orchids.

Orchid Sanctuary : Sanctuaries for rhododendrons and orchids have been established in Sikkim. Sprawling over an area of 100 sq km, Sessa Orchid Sanctuary, West Kameng district, Arunachal Pradesh, is a protected area of India. The sanctuary is a home to beautiful and colourful orchids.

The project Tiger and Project Elephant

The project Tiger was launched in 1973 and there are 41 Tiger reserves spreading in 31 states (National Wildlife Database, Feb. 2013). A total area of 49,112 km² is covered by these project tiger areas, which is 1.49 % of the total geographical area of the country.

Project Elephant was launched in 1991-1992 for long-term survival of identifiable viable populations of elephants in their natural habitats. There are 11 elephant reserves in India (National Wildlife Database, Feb. 2013). Rhinos have been given special attention in selected sanctuaries and national parks in the North East and North-west India. In India sacred forests are located in several parts, e.g., Karnataka, Maharashtra, Kerala, Meghalaya, etc., and are serving as refugia (a safe place) for a number of rare, endangered and endemic taxa. Similarly, several water bodies (e.g., Khecheopalri lake in Sikkim) have been declared sacred by the people leading to protection of aquatic flora and fauna.

Biosphere Reserves

The Ministry of Environment and Forests has notified 19 biosphere reserves in India (Table 26.18), which are also notified as National Parks. Nine of the 19 biosphere reserves are a part of the World Network of Biosphere Reserves, based on the UNESCO Man and the Biosphere (MAB) Programme list. The Gulf of Mannar, Nilgiri and Sunderban biosphere reserves in India included in the list of the Man and Biosphere programme are briefly described (see Box 26.8 and 26.9).

Table 26.18. Biosphere reserves for in situ conservation of biodiversity in India (based on MoEF, 2009; UNESCO, 2012), <http://www.unesco.org/2013/India>.

Year	Name	Location	State	Type	Key Fauna	Area (km ²)
2008	Kuchchh	Part of Kutch, Rajkot, Surendranagar and Patan Districts	Gujarat	Desert	Indian Wild Ass	12,454
1989	Gulf of Mannar	Indian part of Gulf of Mannar extending from Rameswaram island in the North to Kanyakumari in the South of Tamil Nadu	Tamil Nadu	Coasts	Dugong or Sea Cow	10,500
1989	Sundarban	Part of delta of Ganges and Barahmaputra river system	West Bengal	Gangetic Delta	Royal Bengal Tiger	9630

Year	Name	Location	State	Type	Key Fauna	Area (km ²)
2009	Cold Desert	Pin Valley National Park and surroundings; Chandratal and Sarchu & Kibber Wildlife Sanctuary	Himachal Pradesh	Trans Himalayas	Snow Leopard	7,770
1988	Nandadevi	Parts of Chamoli District, Pithoragarh District & Bageshwar District	Uttarakhand	Western Himalayas	Snow Leopard Humalayan black bear Musk deer	5,860
1986	Nilgiri	Part of Waynad, Nagarhole, Bandipur and Mudumalai, Nilambur, Silent Valley and Siruvani Hills	Tamil Nadu, Kerala and Karnataka	Western Ghats	Nilgiri Tahr, Lion-tailed macaque	5,520
1998	Dehang-Dibang	Part of Siang and Dibang Valley	Arunachal Pradesh	Eastern Himalaya		5,112
1999	Pachmarhi	Parts of Betul District, Hoshangabad District and Chhindwara District	Madhya Pradesh	Semi-Arid	Giant Squirrel, Flying Squirrel	4,982
2010	Seshachalam	Seshachalam Hill Ranges covering parts of Chittoor and Kadapa districts	Andhra Pradesh	Eastern Ghats		4,756
1994	Simlipal	Part of Mayurbhanj district	Odisha	Deccan Peninsula	Gaur, Royal Bengal Tiger, Wild elephant	5,569
2012	Achanakamar - Amarkantak	Part of Anuppur, Dindori and Bilaspur districts	Madhya Pradesh, Chhattisgarh	Maikala hill Range	Four horned antelope, Indian wild dog Saras Crane	3,835
1989	Manas	Part of Kokrajhar, Bongaigaon, Barpeta, Nalbari, Kamrup and Darrang Districts	Assam	North East	Golden Langur, Red Panda	2,837
2000	Khangchen-dzonga	Parts of Kanchanjunga Hills	Sikkim	Himalayas (Central)	Snow Leopard, Red Panda	2,931
2001	Agasthyamalai	Neyyar, Peppara and Shenduruny Wildlife Sanctuary and their adjoining areas	Kerala, Tamil Nadu	Western Ghats	Nilgiri Tahr, Elephants	3,500
1989	Great Nicobar	Parts Andaman and Nicobar Islands	Andaman and Nicobar Islands	Islands	Saltwater Crocodile	885
1988	Nokrek	Part of Garo Hills	Meghalaya	North East Hills	Red Panda	820
1997	Dibru-Saikhowa	Part of Dibrugarh and Tinsukia districts	Assam	North East	Golden Langur	765
2011	Panna	Part of Panna and Chattarpur Districts	Madhya Pradesh	Catchment area of the Ken River	Tiger	2,999
2013	Great Nicobar	Southern most Andaman and Nicobar Islands archipelago	Andaman and Nicobar Islands	Islands	Crab eating Macaque Dugong Serpent eagle	1,039

BOX 26.6. Indian biosphere reserves as part of the World Network of Biosphere Reserves (UNESCO, 2012)

Based on the UNESCO Man and the Biosphere (MAB) Programme, these are : Nilgiri Biosphere Reserve, Gulf of Mannar Biosphere Reserve, Sundarbans national park, Nanda Devi Biosphere Reserve, Nokrek Biosphere Reserve, Pachmarhi Biosphere Reserve, Simlipal Biosphere Reserve, Achanakmar-Amarkantak Biosphere Reserve, Nicobar Islands

Source : UNESCO 2012

Box 26.7. Indian biosphere reserves recognized by the Man and Biosphere Programme (UNESCO, 2012)**Gulf of Mannar Biosphere Reserve**

The Gulf of Mannar Biosphere Reserve covers an area of 1,050,000 hectares on the south-east coast of India across from Sri Lanka (08°47' to 09°15'N; 78°12' to 79°14'E). It is one of the world's richest regions of marine biodiversity comprising 21 islands with estuaries, beaches, forests of the near shore environment. This biosphere reserve also includes the marine component with algal communities, sea grasses, coral reefs, salt marshes and mangroves. There are 3,600 plant and animal species, out of which sea cow (*Dugong dugon*) is endangered and six mangrove species endemic to peninsular India. The major ecosystems are Islands including coastal/marine component; coral reefs and mangrove. The mangroves are composed of dominant species such as *Rhizophora conjugata*, *Avicennia alba*, *Bruguiera gymnorrhiza*, *Ceriops tagal*, *Lumnitzera racemosa* etc.

Nilgiri Biosphere Reserve

This biosphere reserve established in the year 2000 is a unique and threatened ecosystem in the tropics. It is located within the Western Ghats Mountain system and represents one of the world's biodiversity hotspots. It provides habitat for the probably largest South Indian populations of tiger (*Panthera tigris*), elephant (*Elephas maximus*) and other large mammals. Major ecosystem types are (i) Tropical forest; Mixed mountain and highland systems; (ii) Tropical humid evergreen forests characterized by *Dipterocarpus indicus*, *Mesua ferrea*, *Palaquium ellipticum* etc.; (iii) tropical montane shola grasslands with *Cinnamomum wightii*, *Elaeocarpus serratus*, *Syzygium arnotianum* etc.; (iv) tropical moist deciduous forests with *Terminalia tomentosa*, *Dalbergia latifolia*, *Schleichera oleosa* etc.

Sunderban Biosphere Reserve

This biosphere reserve is located in the vast Delta of the Ganges, south of Calcutta and bordering Bangladesh in the east. Sunderban is the largest contiguous mangrove area (along with Bangladesh) in the world. The major ecosystems are tropical humid forest and mangroves. Coastal/marine area including mangrove such as *Avicennia alba*, *Bruguiera gymnorrhiza*, *Ceriops tagal*, *Rhizophora apiculata* etc.; tropical semi-evergreen forest; agroecosystems; silviculture, pisciculture; prawn culture. It represents the largest mangal diversity in the world with 81 mangal plant species and it provides habitat for the threatened Royal Bengal tiger (*Panthera tigris tigris*). The core area (Sunderban National Park) has been designated as a World Heritage site.

Nanda Devi biosphere reserve

It includes part of Chamoli, Pithoragarh, and Almora districts and valley of flowers, (Uttarakhand)

Conservation of cultivated plants and their wild relatives

Conservation of cultivated plants and their wild relatives has been mandated to the National Bureau of Plant Genetic Resources (NBPGR), which has established India's first National gene bank. Also, the National Facility for Plant Tissue Culture Repository (NFPTCR) has planned programme

on cryopreservation of seeds, pollens, embryos as well as in-vitro culture of economic and scientific value (Chandel, 1996). Botanical gardens and Zoos abound in various parts of the country, aiding in the ex-situ conservation.

26.16 INTERNATIONAL EFFORTS FOR CONSERVING BIODIVERSITY

IUCN – The World Conservation Union

The World Conservation Union (founded in 1948), brings together States, government agencies and a diverse range of non-governmental organizations in a unique world partnership covering some 81 countries, having its headquarters in Gland, Switzerland. IUCN seeks to influence, encourage and assist societies throughout the world to conserve the integrity and diversity of nature and sustainable use of natural resources. The Union has helped many countries to prepare National Conservation Strategies, and demonstrates the application of its knowledge through the field projects it supervises. The World Conservation Union builds on the strengths of its members, networks and partners to enhance their capacity and to support global alliances to safeguard natural resources at local, regional and global levels. The IUCN Species Survival Commission (SSC) represents the world's most complete source of scientific and management expertise on species, and their conservation.

CITES

The Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) was signed in Washington, DC, on 3 March 1973. Currently there are 178 Parties to this convention. The aim of CITES is to put a ban on the international trade in wildlife. The Convention establishes a world-wide system of controls on international trade in specified threatened animals and plants as well as products derived from them. Roughly 5,000 species of animals and 29,000 species of plants are protected by CITES against over-exploitation through international trade by the Convention. The Secretariat of the Convention is located in Switzerland which helps the Parties to interpret the provisions of CITES and advice on its practical implementation.

The Convention on Migratory Species

The Convention on the Conservation of Migratory Species of Wild Animals (also known as CMS or Bonn Convention) aims to conserve terrestrial, aquatic and avian migratory species throughout their range. It is an intergovernmental treaty, concluded under the aegis of the United Nations Environment Programme. The convention is concerned with the conservation of wildlife and habitats on a global scale. As of 01 April 2013, there are 119 Parties from Africa, Central and South America, Asia, Europe and Oceania (http://www.cms.int/about/map/world_english.jpg).

As the only global convention specializing in the conservation of migratory species, their habitats and migration routes, CMS complements and co-operates with a number of other international organizations, NGOs and partners in the media as well as in the corporate sector. CMS Parties strive towards strictly protecting these animals, conserving or restoring the places where they live, mitigating obstacles to migration and controlling other factors that might endanger them. Besides establishing obligations for each State joining the Convention, CMS promotes concerted action among the Range States of many of these species.

The World Heritage Convention

The most significant feature of the 1972 World Heritage Convention is that it links together in a single document the concepts of nature conservation and the preservation of cultural properties. The Convention recognizes the way in which people interact with nature, and the fundamental need to preserve the balance between the two. There are 190 States parties to the Convention as of September 19, 2012. States Parties are also expected to protect the World Heritage values of the properties inscribed and are encouraged to report periodically on their condition. It also encourages

States Parties to strengthen the appreciation of the public for World Heritage properties and to enhance their protection through educational and information programmes.

Convention on Biodiversity (CBD)

The Earth Summit held in 1992 at Rio de Janeiro resulted into a Convention on Biodiversity, which came into force on 29 December, 1993, and has been ratified by 193 countries as of 2013. Article 15 of CBD is very important as it states that access to genetic resources is the sovereign rights of states and that the authority to determine access to genetic resources rests with the National Governments, subject to National Legislation.

The convention on Biodiversity has three key objectives:

- (i) Conservation of biological diversity
- (ii) Sustainable use of biodiversity
- (iii) Fair and equitable sharing of benefits arising out of the utilization of genetic resources

This convention has stimulated many countries to develop National Biodiversity Strategy and Action Plans.

10th COP (Conference of Parties) at Nagoya (30 October 2010)

Highlights of the outcome: The United Nations General Assembly at its 65th session declared 2011-2020 as decade on biodiversity with following 5 goals (and 20 targets, the Aichi or SMART Targets that is Specific, Measurable, Ambitious, Realistic and Time bound).

- A. Address the underlying causes of biodiversity loss by mainstreaming biodiversity across government and society
- B. Reduce the direct pressures on biodiversity and promote sustainable use
- C. Improve the status of biodiversity by safeguarding ecosystems, species and genetic diversity
- D. Enhance the benefits to all from biodiversity and ecosystem Services
- E. Enhance implementation through participatory planning, knowledge management and capacity building

Nations agreed to set aside 17% land area of the planet (now 12.5%) and 10% of the marine area (now 1%) as protected.

The above goals will (i) address the underlying causes of biodiversity loss by mainstreaming biodiversity across government and society, (ii) reduce the direct pressures on biodiversity and promote sustainable use, (iii) improve the status of biodiversity by safeguarding ecosystems, species and genetic diversity, (iv) enhance the benefits to all from biodiversity and ecosystem Services, and (v) implementation through participatory planning, knowledge management and capacity building.

The 11th Conference of Parties to the UN Convention on Biological Diversity (CBD) was held in Hyderabad, from 8-19 October 2012. The Conference of Parties agreed that poverty eradication and biodiversity conservation are linked. The Government of India has earmarked US\$ 50 million to strengthen the institutional mechanism for biodiversity conservation in India and developing countries. At COP 11, India and the CBD Executive Secretary made strong calls to Parties, partners and other stakeholders to take urgent action towards achieving the Aichi Biodiversity Targets.

International Treaty on Plant Genetic Resources

The International Treaty on Plant Genetic Resources for Food and Agriculture ("IT") was adopted in Rome in November 2001 to create a legally binding framework for the protection and sustainable use of all plant genetic resources for food and agriculture.

On 29 June 2004, the International Treaty on Plant Genetic Resources for Food and Agriculture (popularly known as the International Seed Treaty) came into force. The Treaty will ensure that plant genetic resources for food and agriculture, which are vital for human survival, are conserved and sustainably used. The benefits from the use of genetic resources are equitably and fairly distributed. Some 164 countries negotiated the Treaty under the auspices of the United Nations' Food and Agriculture Organisation (FAO). To date 54 countries have ratified the Treaty representing a broad

range of both developing and industrialised countries. The Treaty now opens the way for a similar safeguard on livestock breeds under the FAO. Currently livestock breeds are being lost at twice the rate of seed varieties – 5 per cent annually.

Convention on wetlands of International Importance (Ramsar Convention)

This treaty, effective since 1971, provides the framework for international collaboration on wetland conservation which includes mangroves and coral reefs. There are 168 Contracting Parties to the convention. There are 2,1431 Wetlands of international importance worldwide.

Under the Ramsar Convention, 26 wetlands in India have been declared (up to January, 2014) as Ramsar sites (Table 26.19)

Table 26.19. List of wetlands of international importance in India identified under Ramsar Convention (source: Ministry of Environment and Forests, Government of India; Ramsar Handbooks series on the Ramsar website at www.ramsar.org/pdf/lib/hbk4-17.pdf)

Sl. No.	Name of Site	State Location	Date of Declaration	Area (in sq.km.)
1.	Asthmudi Wetland	Kerala	19.8.2002	61,400
2.	Bhitarkanika Mangroves	Orissa	19.8.2002	65,000
3.	Bhoj Wetlands	Madhya Pradesh	19.8.2002	3,201
4.	Chandertal Wetland	Himachal Pradesh	8.11.2005	49
5.	Chilika Lake	Orissa	1.10.1981	1,16,500
6.	Deepor Beel	Assam	19.8.2002	4,000
7.	East Calcutta Wetlands	West Bengal	19.8.2002	12,500
8.	Harike Lake	Punjab	23.3.1990	4,100
9.	Hokera Wetland	Jammu and Kashmir	8.11.2005	1,375
10.	Kanjli Lake	Punjab	22.1.2002	183
11.	Keoladeo National Park ^{MR}	Rajasthan	1.10.1981	2,873
12.	Kolleru Lake	Andhra Pradesh	19.8.2002	90,100
13.	Loktak Lake	Manipur	23.3.1990	26,600
14.	Point Calimere	Tamil Nadu	19.8.2002	38,500
15.	Pong Dam Lake	Himachal Pradesh	19.8.2002	15,662
16.	Renuka Wetland	Himachal Pradesh	8.11.2005	20
17.	Ropar Lake	Punjab	22.1.2002	1,365
18.	Rudrasagar Lake	Tripura	8.11.2005	240
19.	Sambhar Lake	Rajasthan	23.3.1990	24,000
20.	Sasthamkotta Lake	Kerala	19.8.2002	373
21.	Surinsar-Mansar Lakes	Jammu and Kashmir	8.11.2005	350
22.	Tsomoriri	Jammu and Kashmir	19.8.2002	1,200
23.	Vembanad Kol Wetland	Kerala	19.8.2002	151,250
24.	Upper Ganga River (Brijghat to Narora Stretch)	Uttar Pradesh	8.11.2005	26,590
25.	Wular Lake	Jammu & Kashmir	23.3.1990	18,900
26.	Nalsarovar Bird Sanctuary	Gujarat	24.09.12	12,000

MR: Sites included in the Nontreux Record a record of Ramsar Sites where changes in ecological character have occurred are occurring or are likely to occur maintained by the Secretariat in consultation with the Contracting Party concerned (Recommendation 4.8) www.ramsar.org/montreux.record.

The Convention on Conservation of Antarctic Marine Living Resources (CCAMLR)

The CCAMLR has been established to protect the southern ocean (see Hooker and Gerber, 2004). The aim is to regulate human activities so as to restrict changes in the Antarctic ecosystem. It applies to all areas south of 60°S and to waters between these latitudes and the Antarctic Convergence. The main goals of the CCAMLR are (i) to undertake research and studies of Antarctic marine living resources and the marine ecosystem, (ii) to prepare data base on the status and changes in population of Antarctic marine living resources, (iii) and to identify conservation needs and to analyze the effectiveness of conservation measures.

International Agreements for conserving marine Biodiversity

By International Law, a country's off shore fishing zone extends up to 370 kilometers (200 nautical miles) from its shore. This is known as the exclusive economic zone (EEZ). Fishing is allowed only in the EEZ zone with following restrictions:

1. Fishing techniques such as dynamiting and poisoning are outlawed.
2. Blue Whales have not been hunted commercially since 1964 and have been classified as endangered.

United Nations convention on the Law of the Sea was adopted in 1995. High seas are outside 200-miles exclusive economic zones (EEZ's) and are under the United Nations convention on the Law of the Sea. Areas closed to all types of fishing are Bermuda, Belize, Great Barrier Marine Park in Australia, and the Florida Keys National marine sanctuary in USA with an objective of (i) preservation and protection of marine environment (ii) conservation of marine living resources and (iii) limited mineral extraction from sea. Protection of coral reefs off the coast of Thailand and the Caribbean island enables ecotourism and at the same time gives protection to the threatened systems.

International Whaling Commission (IWC) has a mandate to establish whale sanctuary in different oceans of the world. There are now two important sanctuaries for protecting the whales, the Indian Ocean sanctuary and the southern ocean sanctuary.

Intergovernmental Science-Policy Platform for Biodiversity and Ecosystem Services (IPBES)

The lack of a global platform for biodiversity policy support, comparable with the IPCC for climate change, has hampered attempts for coordinated global response to the biodiversity loss. Therefore, the recognition and support for the process of establishing the IPBES by the UN general assembly is a first step in this direction. IPBES is an independent, international science panel (like IPCC) which would coordinate and highlight research on pressing topics (Perrings et al 2011). IPBES was set-up to provide scientific support for policy-making to protect the planet's biodiversity, its ecosystems and the services they provide to humanity.

IPBES was established in April 2012 in Panama City, Panama, and currently has 105 Member States. The Platform is an independent intergovernmental body open to all member countries of the United Nations. Its members are committed to building IPBES as the leading intergovernmental body for assessing the state of the planet's biodiversity, its ecosystems and the essential services they provide to society.

IPBES is unique in that it will aim to strengthen capacity for the effective use of science in decision-making at all levels. IPBES will also aim to address the needs of Multilateral Environmental Agreements that are related to biodiversity and ecosystem services, and build on existing processes ensuring synergy and complementarities in each other's work.

Over 500 delegates, including from most of the 105 Member States of the Intergovernmental Platform on Biodiversity and Ecosystem Services (IPBES), attended the first plenary meeting of the platform (IPBES-1), held in Bonn, Germany in January, 2013.

Remote Sensing and Geographic Information System

The natural environment represents a complex set of inter-relationships among flora, fauna, landforms, geological structure and atmospheric composition. These biophysical components interact to form ecosystems at various scales that define the surface characters of the earth. They also define renewable and non-renewable finite source bases for human activity. Maps have traditionally been used to explore the Earth's environment and to exploit its natural resources. Remote sensing has been used for acquiring large volume of spatial data for inventorying, monitoring and gaining new insight into the complexity of natural environment. Spatial information is related to geographical features such as topography, land use-land cover, climate and other such features on a large scale. Remote sensing has been widely used in many fields like ecosystem analysis and management, resource conservation, mineral exploration, biodiversity monitoring and conservation, meteorology, water resource management and mineral exploration and atmospheric chemistry.

Geographical Information System (GIS) is a computer based data processing tool for gathering, manipulating and analyzing the geographic information from different origins. It is a science of spatial analysis using spatial data and spatial information. The theory and practice of designing and using GIS is known as geographic information science, which is relevant to GIS, remote sensing, cartography and spatial statistics. The integration of remote sensing with GIS has become necessary for utilizing remotely sensed data. The most important feature of GIS is its analytical function which has been put to use by various disciplines as a tool for handling the geoinformatics data. GIS technology makes it possible to integrate information from direct surveys (using cartographic techniques) and remote sensing, enhancing the efficiency and analytic power of traditional mapping. Computer aided mapping and database management support the data analysis of GIS. Therefore, it is of interest to learn about the remote sensing processes, application of remote sensing, basic concepts, components and tools of GIS and its use in habitat analysis, biodiversity monitoring and natural resource management.

27.1 REMOTE SENSING

Remote Sensing of earth resources is a system for collecting information on biosphere from a distance. Remote sensing has evolved from aerial photography, which has been in use since 1920 for aerial surveys and geology. It has now emerged as a distinct field of study with the opening of satellite era during 1960 (Colwell, 1983). During the last 40-50 years, the field of remote sensing has undergone rapid growth because of its increasing use in exploring, mapping, and monitoring the resources of earth from local to global scale. In particular, the US space programme developed under the US National Aeronautical and Space Administration (NASA) has played a significant role to make remote sensing data readily available to scientific community throughout the world. The success of Landsat series of satellites launched by NASA led to the establishment of European Space Agency and National space agencies in many other countries like Canada, India, Japan and

China. The first Indian remote sensing satellite was launched in 1988; some of the best satellite remote sensing data are available from the remote sensing satellite series- IRS-1A, IRS-1B, IRS-1C, IRS-1D besides the P-2 and P-3 satellites. Advancement in computational capability and image processing by using computers has enabled wide use of remote sensing in diverse fields. According to Dahlberg and Jensen (1986) and Fischer and Lindenbrg (1989), there is often a three-way interaction between remote sensing, cartography, and GIS, which are used in physical, biological, and social science research (Fig. 27.1).

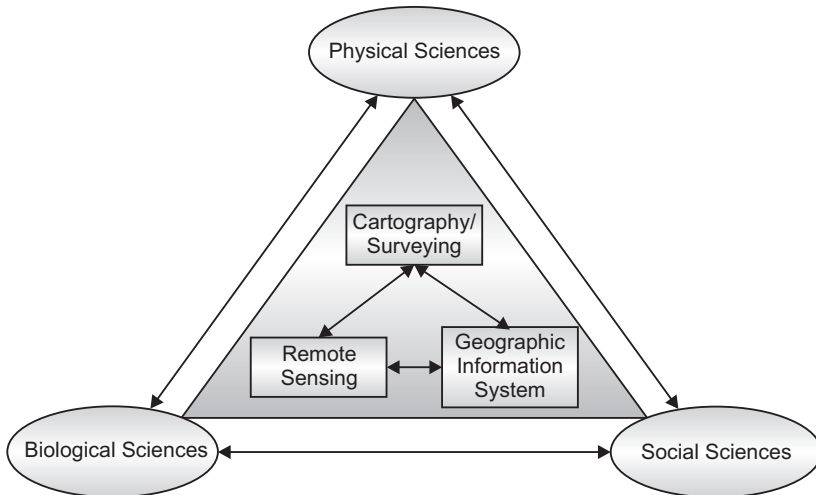


Fig. 27.1. A three-way model of interaction between the mapping sciences of remote sensing, geographic information, systems, and cartography surveying as they are used in the physical, biological, and social sciences.

Remote sensing has been variously defined depending upon the type of sensor used and utility of remote sensing data. Broadly speaking, “remote sensing is a non contact technique of acquiring information from the ultraviolet, visible, and infrared and microwave regions of the electromagnetic spectrum, using various scanners such as cameras, lasers, linear arrays and area arrays located on platforms such as aircraft and spacecraft and the analysis of the acquired information by means of visual and digital image processing” (Jensen, 2000). Remote sensing is a science of deriving information about an object or geographical area on the basis of measurements made from a distance.

Physical Basis for Remote Sensing

In remote sensing, observations of the targets (the physical object or geographical area) are made from a distance, using specialized instruments called sensors mounted on suitable platforms (Fig. 27.2). Reflected radiations from the objects are recorded by sensors such as photographic cameras, scanners or radiometers which are mounted on suitable platforms. Satellite mounted sensors have the unique capability to gather a wide range of information in different windows of the electromagnetic spectrum.

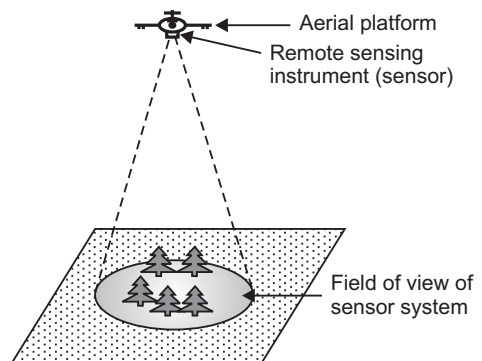


Fig. 27.2. A remote sensing sensor, mounted on an aerial platform, acquires information about an object, phenomenon in the field of view without being in physical contact.

The most important physical links between objects of measurement and remote sensing devices involve the electromagnetic energy, acoustic energy and force fields. Electromagnetic energy, in different wavelengths of the electromagnetic spectrum, is the most important medium for remote sensing of the environment. The majority of remote sensing instruments record electromagnetic radiation (EMR) that travels at a velocity of $3 \times 10^8 \text{ ms}^{-1}$ from the source, directly through the vacuum of space or indirectly by reflection or radiation to the sensor. The EMR represents an extremely efficient high-speed communications link between the sensor and the remotely located object (Table 27.1). The electromagnetic radiation in visible, infrared and ultraviolet and microwave length is either reflected or scattered by the surface (Fig. 27.3). The sensor detects solar radiation that has been reflected or emitted from the earth's surface. The EMR upon interaction with atmosphere and surface of land and water, experience changes in magnitude, direction and wavelength. Changes in the amount and properties of the EMR become, upon detection by the sensor, a valuable source of data for interpreting important properties of the environment. In the microwave region of the electromagnetic spectrum, for example, the sensor is radar, which is an active sensor, as it provides its own source of energy for observation and obtaining useful data. The sensor emits radiation, which is directed towards the target to be investigated, and the radiation reflected from the target is detected and measured by the sensor (Fig. 27.3). The active sensors are independent of solar and terrestrial radiation.

Table 27.1. The principal wavelengths of the Electromagnetic Spectrum used in remote sensing.

Wavelength	Type of sensor used
Visible Spectrum 0.4 μm – 0.75 μm	Aerial cameras.
0.3 to 1.1 μm	Multispectral scanners Vidicon / R.B.V.
Infrared (IR) Spectrum Near and middle infrared 0.72 μm – 15 μm	Thermal scanners Reflected IR (0.72 μm – 15 μm) is used for remote sensing.
Microwave Region 1 mm – 1 meter	This is the longest wavelengths used in remote sensing used mostly in RADAR.
Radio Waves (> 30 cm)	This is the longest portion of the spectrum mostly used for commercial broadcast and meteorology.

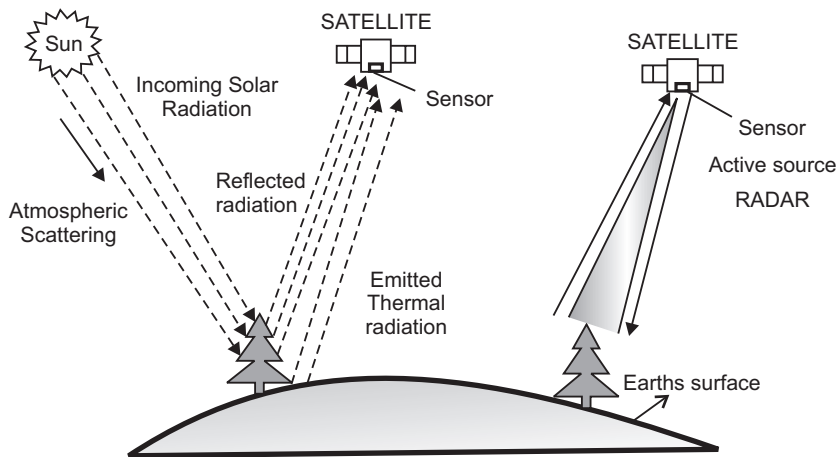


Fig. 27.3. Source of radiation from sun and active remote sensing sensors and its interaction in the atmosphere and at the earth surface.

Atmospheric scattering, also known as Rayleigh scattering, occurs due to multiple reflection of EM radiation by gas molecules and suspended particles in the atmosphere. This is responsible for blue colour of the sky and for the brilliant red and orange colour often seen at sunset. For remote sensing, Rayleigh scattering causes high path radiance in the blue end of the spectrum. Some of the incident solar radiations are absorbed by atmospheric water vapour, CO₂ and Ozone. Carbon dioxide is important in remote sensing because of its efficiency in absorbing radiation in mid-infrared region (13 to 17.5 µm). Water vapour is several times more effective in absorbing radiation than atmospheric gases.

Atmospheric windows are of much significance in remote sensing because these are put to use for viewing the ground surface phenomena from aerial or space platforms across the atmosphere (Table 27.2). The spectral regions of least absorption of EMR or those wavelengths that are easily transmitted through the atmosphere are called atmospheric windows (Fig. 27.4). In the visible part of the spectrum the atmospheric windows provide high energy conditions. In thermal IR region, two important windows occur at 8.0 to 9.2 µm and 10.5 to 12.5 µm. The latter is especially important because most emissions from the earth’s surface occur in these wavelengths. The atmosphere is essentially opaque in the region of 20 µm to 1 mm wavelength. The microwaves greater than 20 mm are propagated through the atmosphere with least attenuation. The largest wavelengths used in remote sensing are those about 1 mm to 1 m.

Table 27.2. Major atmospheric windows (Campbell, 2002).

Region	Wavelength
Ultraviolet and visible	0.30-0.75 µm 0.77-0.91 µm
Near-Infrared	1.55 – 1.75 µm 2.05 – 2.4 µm
Thermal Infrared	8.0 – 9.2 µm 10.2 – 12.4 µm
Microwave	7.5 – 11.5 mm 20.0 + mm

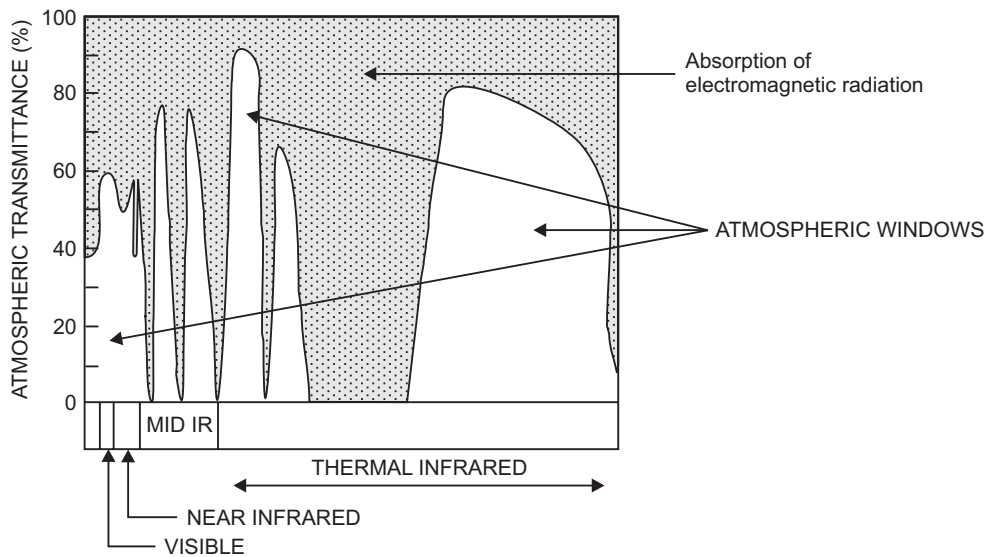


Fig. 27.4. Schematic representation of Atmospheric windows (redrawn from Campbell, 2002).

Thus electromagnetic radiation transfers the information from an object to a sensor. Electromagnetic energy can be detected physically or electronically. By using information from one or more wavelengths, it is possible to differentiate between different types of objects, e.g., soil, vegetation and water (Fig. 27.5). The spectral characteristics that differentiate an object from its surroundings constitute the spectral signature of the object. The curves showing different intensity of light reflected by objects at different wavelengths are called *spectral response curves*. The spectral data acquired from the remote sensors are processed, enhanced and interpreted for different applications.

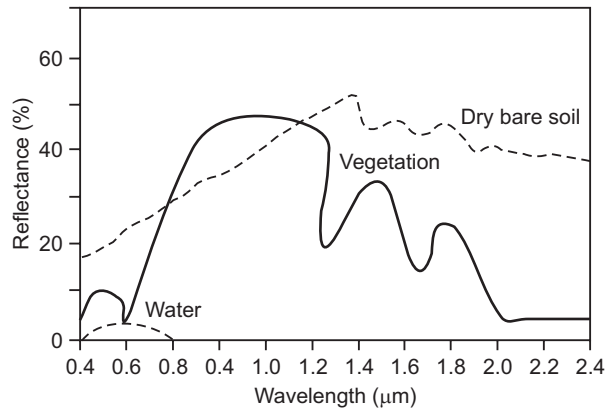


Fig. 27.5. Typical Spectral reflectance curves for vegetation, soil and water (redrawn from Jensen, 2000).

The acoustic sounding technique is used in atmospheric studies. The operational range is low and speed of propagation is slow. The most important force fields are gravity and magnetism and are used by the earth scientists.

Remote Sensing Process

The remote sensing process involves data collection and analysis in a systematic way which is summarized in Fig. 27.6 and described below.

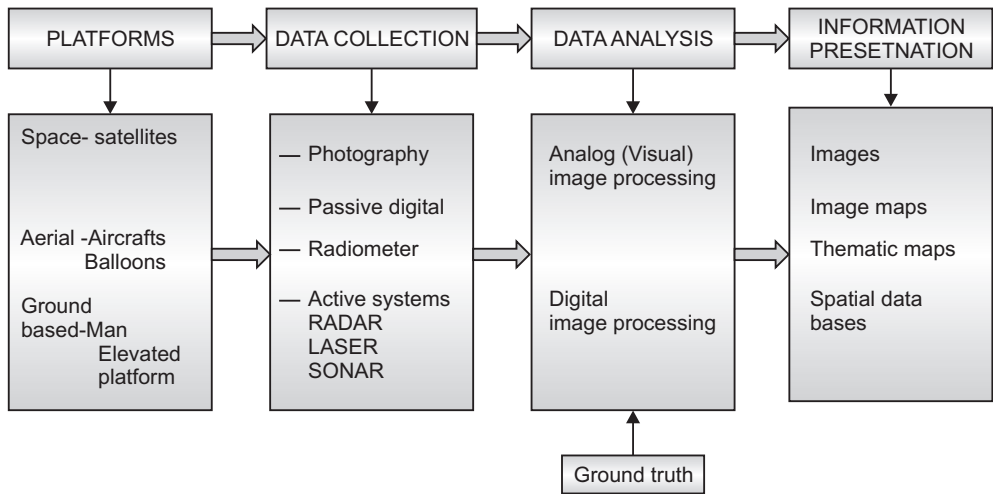


Fig. 27.6. Schematic representation of the remote sensing processes (modified from Gupta, 1991; Jensen, 2000).

Platforms and Sensors for Remote Sensing

For remote sensing data acquisition, the platforms could be ground based (e.g. hydraulic lift), aerial (balloon, helicopter and aircraft) and space borne satellites. Scanners are most commonly used in satellites to overcome the problem of returning photographic film on earth. The two most

important components of sensors are their spatial and spectral resolution (Table 27.3). The spatial resolution is defined by the smallest feature that can be “seen” by the sensors. The spectral resolution is the range and number of bands within the spectrum to which a sensor is sensitive. Remote sensing systems may be configured to collect data in just one band of the electromagnetic spectrum. In multispectral remote sensing, radiant energy is recorded in multiple bands of electromagnetic spectrum. In hyperspectral remote sensing, instruments acquire data in hundreds of contiguous spectral bands. For example, the Airborne Visible and Infrared Imaging Spectrometer (AVIRIS) has 224 bands in the region from 0.4 to 2.5 μm , which are just 10nm apart (Clark, 1999). Temporal resolution refers to the repetitivity of satellite to obtain spatial and spectral data at certain time intervals. It is the capability of the satellite to image exactly the same area at the same viewing angle at different periods of time.

Table 27.3. Some currently used remote sensing systems and their major characteristics.

Satellite platform	Sensor	Spectral Resolution (μm)						Resolution	
		Blue	Green	Red	Near IR	Mid IR	Thermal IR	Spatial (meters)	Temporal (days)
Landsat 1-5	Multispectral Scanner (MSS)	—	1	1	2	—	—	80	16-18
Landsat 4-5	Thematic Mapper (TM)	1	1	1	1	2	2	30 and 120	16
Landsat 7	Enhanced TM (ETM)	1	1	1	1	2	2	30 and 60	16
SPOT	HRV-Multispectral	—	1	1	1	—	—	20	Pointable
	Panchromatic	0.51 — 0.73 μm						10	Pointable
TERRA	AVIRIS	0.41 μm — 224 bands — 2.5 μm						2.5 or 20 m	Variable
	MODIS	0.405 μm — 36 bands — 14.38 μm						250.500.1000	1-2

Remote Sensing Satellites

A brief description of some remote sensing satellites is given below.

Landsat

NASA launched a series of LANDSAT satellites over the period of more than three decades. The **Landsat** missions have used various devices for collecting the information like the Return Beam Vidicon Camera (RBVC) and a four channel Multispectral Scanner (MSS) and Thematic Mapper-TM (used on Landsat –4 and –5). For the purpose of remote sensing, the satellites are placed in polar orbits, at 800-1000 km above the earth. These satellites can image all parts of the globe except the Polar Regions every nine days. The two Landsats (–5 and –7) that remain operational are equipped with Thematic Mapper (TM) as a principal sensor which provides high resolution images from spectral bands of visible and infra red wavelengths.

The Landsat satellites equipped with multispectral or multiband scanner image the earth surface in more than one region of the electromagnetic spectrum simultaneously from precisely the same location every repeat time (Table 27.3).

Landsat 4, 5 and 7 gather information in seven bands ranging from 0.45 to 12.5 mm spectral region (Table 27.4). The primary applications of the data collected in the various spectral bands are also shown in this table.

Table 27.4. Applications of the Landsat TM bands.

Band	Wavelength (μm)	Principal Application
1	0.45-0.52	Coastal water mapping, forest mapping, soil-vegetation discrimination
2	0.52-0.62	Vegetation discrimination and vigour assessment
3	0.63-0.69	Plant species differentiation
4	0.76-0.90	Vegetation types, plant biomass, water bodies
5	1.55-1.75	Soil moisture, snow
6	10.4-12.5	Vegetation stress, thermal mapping
7	2.08-2.35	Mineral & rock types, vegetation moisture

SPOT

The earth resource satellite of France was launched in February 1986. It has highly sensitive sensors, which can collect information from different direction. It can detect details as small as 10 m and resolve heights with an aggregation less than 5-km. It is a polar orbit satellite located at an altitude is 832 km. It carries a different type of sensor known as High Resolution Visible (HRV) sensing system. Spot images have diverse applications including agricultural and crop monitoring, terrain analysis, changes in ecosystems overtime and studies of river systems.

In March 1998, SPOT-4 was launched which has HRVIR sensor with an additional band in mid-IR region, and the panchromatic band of SPOT-1, was replaced by 0.61 – 0.68 μm. Spatial resolution was further increased to 5 m in PAN b and 10 m in multi spectral band in SPOT-5.

IKONOS

IKONOS was the first commercial high resolution satellite launched in 1999, with optical sensor assembly (OSA) having four spectral bands (1-4) and 1 m (PAN). The data are used for small to medium scale topographic mapping and also for updating the existing topographic maps.

Indian Remote Sensing Satellites

Indian Remote sensing satellites (IRS-1A/1B) have been of immense use in hydrology, forestry, urban and land use planning, agriculture, forestry, and geology and to identify and monitor various coastal zone features. IRS-1C and IRS-1D (launched in 1995 and 1997) have helped to study the natural resource management. These satellites carry three sensors; wide field sensor (WiFS) designed for regional vegetation mapping, Linear Imaging Self-Scanning Sensor 3 (LISS3), which yields multispectral data in four bands with a spatial resolution of 23 m, and the panchromatic camera (PAN, 0.5 – 0.75 μm) with spatial resolution of 6 m. Evolution of Indian Earth Observation Satellites with different spatial resolution(SR) and spectral bands is summarized in Box 27.1. Kalpana-1, INSAT-2E/3A are geo-stationary satellites, the others are polar orbiting satellites.

SAMIR, MSMR, MADRAS, SAPHIR are multifrequency passive microwave radio-meters.

SAR, Altika and Scatterometer are active microwave sensors. (Box 27.1)

Box 27.1. Evolution of Indian Earth Observation Satellites with different spatial resolution(SR) and spectral bands (courtesy: Raghavendra P. Singh, SAC)

Sl.	No Satellite	Sensors	Launch Date	Remarks
1	Bhaskara-1	TV camera, SAMIR	7 June, 1979	TV Camera (SR: 1 km) SAMIR (SR: 125 km) is passive microwave radiometer
2	Bhaskara-2	TV camera, SAMIR	20 Nov., 1981	Continuity mission with improvements in SAMIR
3	IRS-1A	LISS-I, LISS-II	17 Mar., 1988	LISS-I (SR: 72.5 m) and LISS-II (SR: 36.25 m) are multi spectral sensors with Blue, Green Red, NIR bands
4	IRS-1B	LISS-I, LISS-II	29 Aug., 1991	Continuity mission of IRS-1A
5	IRS-P2	LISS-II	15 Oct., 1994	Continuity mission
6	IRS-1C	LISS-III, PAN, WiFS	28 Dec., 1995	LISS-III (SR: 23.5 m) has Green, Red, NIR and SWIR bands, PAN (SR: 5.8m), WiFS (SR: 188 m) with red and NIR bands)
7	IRS-P3	MOS A, B, C, WiFS	21 Mar., 1996	MOS-B (SR: 520 m)
8	IRS-1D	LISS-III, PAN, WiFS	29 Sep., 1997	Continuity mission of IRS-1C
9	INSAT-2E	CCD, VHRR	03 Apr., 1999	CCD (SR: 1 km) multi spectral optical sensor first time put on Geostationary platform
10	IRS-P4 (Oceansat-1)	OCM, MSMR	26 May, 1999	OCM (SR: 360 m) 8 band sensor for ocean applications. MSMR is passive microwave radiometer
11	Kalpana-1	VHRR	12 Sept, 2002	VHRR (SR: 2 km in Visible and 8 km in TIR and MWIR (Water vapour)
12	INSAT-3A	CCD, VHRR	10 April, 2003	Continuity mission of INSAT series with CCD sensor
13	IRS-P6 (Resourcesat-1)	LISS-III, LISS-IV, AWiFS	17 Oct., 2003	LISS-IV (SR: 5.8 m) has Green, Red and NIR bands. AWiFS is advanced WiFS (SR: 56 m) with Green, Red, NIR and SWIR bands.
14	IRS-P5 (Cartosat-1)	PAN (Fore, Aft)	05 May, 2005	PAN (SR: 2.5 m)
15	Cartosat-2	PAN	10 Jan. 2007	PAN (SR: 0.8 m)
16	Cartosat-2A	PAN	28 April, 2008	Continuity Mission of Cartosat-2 series
17	IMS-1	Hysi, Mx	28 April, 2008	MX (SR: 36.25) is similar to LISS-II sensor. HySi (SR: 505.6 m) is 64 band hyper spectral imager
18	Oceansat-2	Scatterometer, OCM, ROSA	23 Sept. 2009	13.4 GHz, Scatterometer (SR: 50 km). Continuity of OCM with improvements
19	Cartosat-2B	PAN	12 July 2010	Continuity Mission of Cartosat-2 series

19	Cartosat-2B	PAN	12 July 2010	Continuity Mission of Cartosat-2 series
20	Resourcesat-2	LISS-III, LISS-IV, AWiFS	20 April 2011	Continuity Mission of IResourcesat-1 with improvements.
21	Megha Tropiques	MADRAS, SAPHIR, ScaRaB,	12. Oct. 2011	Indo French Cooperation Mission
21	RISAT	SAR	26. April 2012	Synthetic Aperture Radar (SAR) in C-band (5.35 GHz)
23	SARAL	Altika, ARGOS	25. Feb. 2013	Ka band (35.75 GHz). Altimeter

NOAA

The National Oceanic and Atmospheric Administration (NOAA) polar orbiting satellites (NOAA 10 and NOAA 11) carry the Advanced Very High Resolution Radiometer (AVHRR), a passive system that is sensitive to visible and infrared wavelengths. Today two NOAA satellites (NOAA 14 and 15) are operational. AVHRR data have been extensively used for large area vegetation mapping, surface water temperature mapping, flood monitoring, etc.

Nimbus-7

Marine biologists can gain information on the concentration of Phytoplankton from the colour of ocean. The coastal zone colour scanner of Nimbus-7 satellite provides data on primary production of the ocean, fisheries and marine food chain. Nimbus-7 satellite is operating since 1978.

TOPEX/POSEIDON satellite

The **TOPEX/POSEIDON satellite**, launched in 1992, uses radar altimeter to measure sea-surface height over 90 percent of the world's ice free oceans. The system acquires global maps of ocean topography (barely perceptible hills and valleys of the sea surface), which are used to calculate the speed and direction of ocean currents (Jones, 1992).

GOES

The Geostationary Operational Environmental Satellites known as GOES-8 and GOES-9 provide the daily infrared and visible images of the weather.

Terra Satellite

A series of multipurpose polar orbiting platforms known as Earth Observing System (EOS) have been initiated in 1988. These are carrying advanced multidisciplinary instruments to acquire 15 years of continuous Earth observations. EOS AM-1 (now referred to as the Terra satellite launched in 1999) has five remote sensing instruments, specifically designed to acquire global coverage to monitor changes in the nature and extent of earth's ecosystems. The spatial, spectral, and temporal characteristics of the main Terra sensor systems are MODIS, ASTER, and MISR (NASA, 1998; Jensen, 2000).

The Moderate Resolution Imaging Spectrometer (**MODIS**) has 36 bands from 0.405 to 14.385 μm that collects data at 250 and 500 m and 1-km spatial resolutions. MODIS views the entire surface of the Earth every 1 to 2 days, making observation at moderate resolution (0.25 to 1 km), of land and ocean surface temperature, primary productivity, land surface cover, clouds, aerosols, water vapor, temperature profiles, and fires.

The Advanced Space borne Thermal Emission and Reflection Radiometer (**ASTER**) has the highest spatial resolution sensor system on the EOS Terra platform. It provides information on surface temperature that can be used to model evapotranspiration.

The Multiangle Imaging Spectroradiometer (**MISR**) provides data on clouds, atmospheric aerosols, and ice cover. The Clouds and the Earth's Radiant Energy System (**CERES**) consists of two scanning radiometers that measure the Earth's radiation balance and provide cloud property estimates to assess their role in radiative fluxes from the surface of the earth to the top of the atmosphere.

MOPITT (Measurements of Pollution in the Troposphere) is a scanning radiometer that provides information on the distribution, transport, sources, and sinks of carbon monoxide and methane in the troposphere.

Ground stations

Worldwide networks of sophisticated ground station now receive the data from satellites.

Systems for Data Collection

Remote sensing data are collected using either passive or active remote systems. Passive sensors (e.g., cameras and video cameras) record naturally occurring radiation that is reflected or emitted from a terrain. A multispectral scanner may be used to record the amount of thermal radiant flux emitted from the terrain.

Remote sensing systems collect analog (e.g., hard-copy aerial photograph or video) digital data (e.g., a matrix (raster) of brightness values using a scanner, linear array or area array). The satellite/aerial scanner data are preprocessed on board, and relayed down to the ground receiving station. The data are stored temporarily on high density data tapes (HDDT, preprocessed). The two types of system for collecting information are described below.

Passive Systems

Sun is the major source of energy. A passive system consists of an array of small sensors or detectors, which record the amount of electromagnetic radiation reflected and/or emitted from the earth surface in the visible, near infrared and far red regions. For receiving the emission in the visible light, various types of camera and vidicon cameras are used. A photographic system consists of camera, filters and the film and the working principle of an aerial/space camera system is simple (Fig. 27.7). If the object is far away, the image is formed at infinity at the focal plane. Therefore, cameras used for remote sensing process are of fixed focus type. They carry a photosensitive film for registering the image in the field of view.

Various types of photographic films used in the cameras are: panchromatic, black and white – topography; infrared: Land and water distribution; false colour: Vegetation; thermal infrared: microwave emission and X and gamma rays.

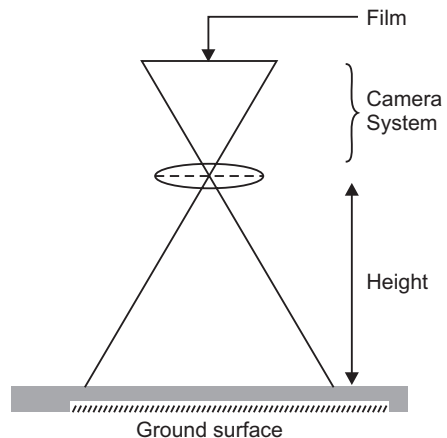


Fig. 27.7. The principle of a photographic system for taking the photographs in passive system.

The first three Landsat satellites carried the Return Beam Vidicon (RBV) camera. A multiple scanner is an example of the passive system. The microwave radiometry senses radiation in the 100 mm to 1-meter range. These systems have low spatial resolution; these are particularly useful for showing surface characteristics such as moisture.

There are also thermal infrared sensors, which measure thermal infrared radiation emitted from the objects. The photograph like images produced are useful for analyzing the surface temperature of land and water, and for detecting forest fires. Meteorological satellites have made greatest use of the thermal IR scanning systems. The spatial resolution (the size of the smallest feature that can be identified) is only of the order of a few kilometers. It produces sufficient details to make and complete weather forecasting.

A simplified representation of the events that take place for the conversion of digital data (collected from a satellite) into digital image is shown in Fig. 27.8. Briefly explained, the satellite image is digital and displayed as a matrix of numbers. Each number represents a single value for the specific pixel and bandwidth. These data are stored in the satellite and received on the earth stations. Using a computer, the data are numerically manipulated to produce a coloured imagery on the computer screen or can be stored as a hard copy (Fig. 27.8).

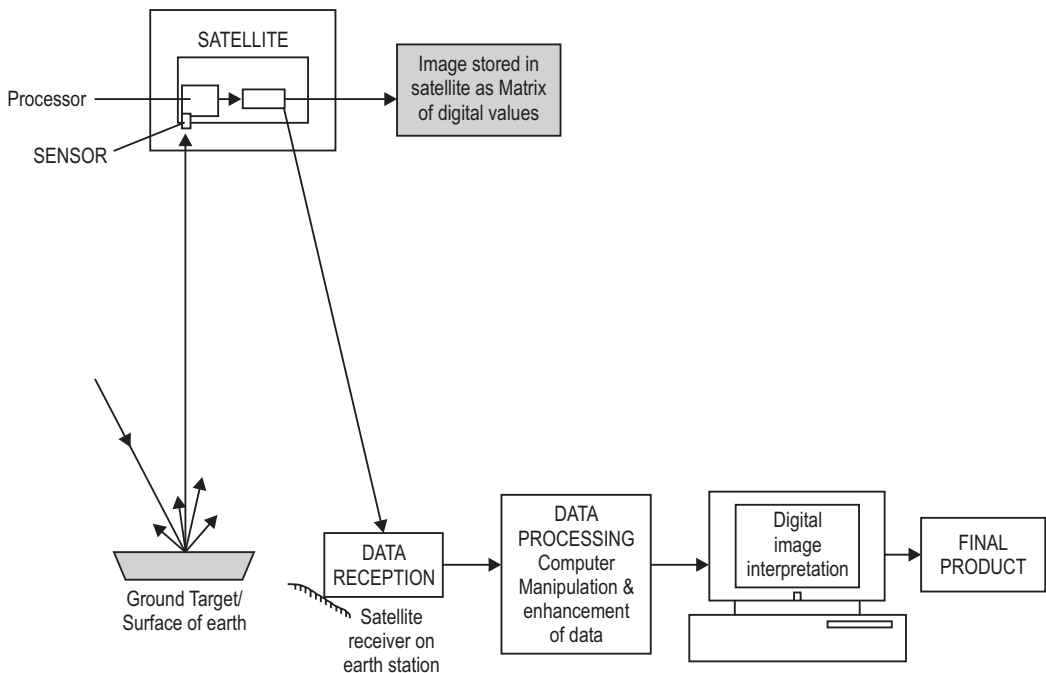


Fig. 27.8. A schematic representation of satellite imagery used in passive remote sensing (modified from Gupta, 1991).

Active Systems

An active system uses its own source of electromagnetic radiation and measures the intensity of the return signals as digital numbers. These systems include radar pulse, photo flash beams and laser beams and use the artificial source of energy resources for remote sensing. Electromagnetic energy is propagated near the sensor and bounced towards the sensor system (Fig. 27.9).

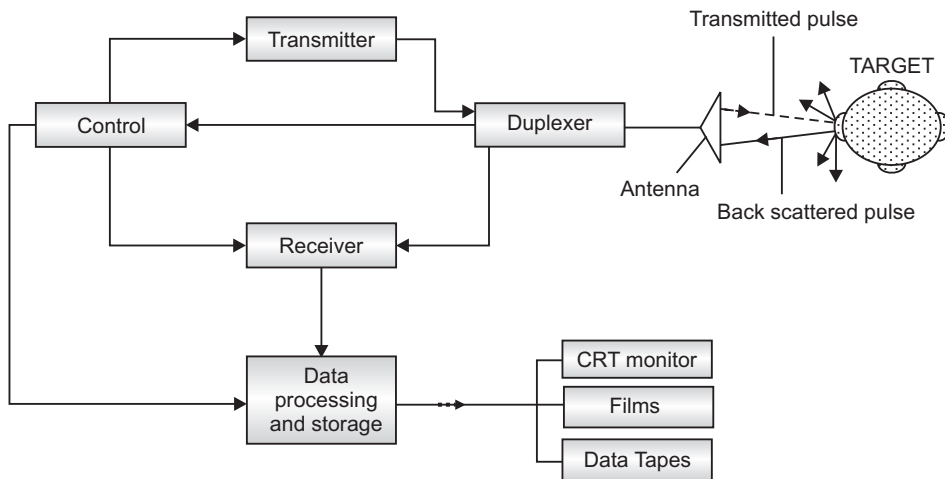


Fig. 27.9. Basic structure of a radar system used in active remote sensing (modified from Gupta, 1991).

RADAR is Radio Detection and Ranging. Some wet tropical areas have been imaged by using radar. Radar is an active microwave remote sensing system, which operates in 1 mm to 1 meter wavelength. The radar imaging is interpreted in a similar way as the aerial photographs using the elements like tone, texture, pattern and shape. Radar imagery is particularly useful for terrain analysis.

Synthetic Aperture Radar (SAR) and Side Looking Radar (SLAR) are examples of an active system, which can produce high resolution images of the earth. Rain mapping radar is used for measuring rainfall in the tropical regions.

Another active remote sensing system is **SONAR** (Sound Navigation and Ranging). Sonar permits under water imaging and determines the depth in aquatic systems. With the help of sonar systems (altimeter type), it is possible to determine the position of the earth crust on the ocean floor.

LiDAR (Light Detection and Ranging) is an active system. It consists of a laser which emits pulses of radiation in a continuous mode toward the ground. The system collects the radiation returned and focuses on the detector. The return time for each pulse back to the sensor is processed to calculate the variable distances between the sensor and the surfaces present on or above the ground. This is used in terrain and surface mapping and for measurement of water depth.

Microwave Remote Sensing

Analysis of the information that operates in the microwave portion of the electromagnetic spectrum is called microwave remote sensing. The microwave portion of the spectrum extends from 1 mm to 1 m. Microwave remote sensing is time and weather independent and sensitive to moisture in soil, vegetation and snow. It can penetrate soil and vegetation cover and has the ability to collect data far away from flight path. Microwave sensors are both passive and active type. Passive sensors are radiometers which are non-imaging sensors. They measure the intensity of electromagnetic radiation falling in their field of view. These operate at a fixed wavelength. Multiband spectroradiometers measure radiation intensity in more than one wavelength band. A multiband spectroradiometer simulating the four bands of the landsat-MSS, has been widely used for ground truth data collection.

Data Analysis

The analysis of remotely sensed data includes a variety of image processing techniques that include analog (visual) image processing of image data, and digital image processing of digital data (Fig. 27.10). The remote sensing data interpretation can be applied for a number of purposes in geology, forestry, agriculture, soil mapping, ecology, water resources and wild-life management.

Analog (visual) image processing

The principles of photo interpretation were developed for aerial photography and now applicable to all remote sensing visual data products (see Gupta, 1991; Jensen, 2000). With the availability of higher resolution imagery, visual photo interpretation is gaining popularity. It is based on man's ability to relate colours and patterns in an image on the computer screen or a printed hard-copy, to real world features. The visual image analysis includes image interpretation and photogrammetric measurement on the basis of the size, shape, shadow, colour, pattern, texture and site and precise measurement of objects (Fig. 27.10). The visual interpretation depends on the ability of a person to draw conclusions either by direct and spontaneous recognition of objects or by using the reasoning process. The photographs or images can be studied in terms of seven interpretation elements: tone or colour, texture, shape, size, shadow, site or association and pattern. Visual interpretation has been extensively used for soil mapping (panchromatic data) and land cover mapping (multispectral data). IRS-1C panchromatic data (5.8×5.8 m) are often photo interpreted and used as a base map in the GIS projects.

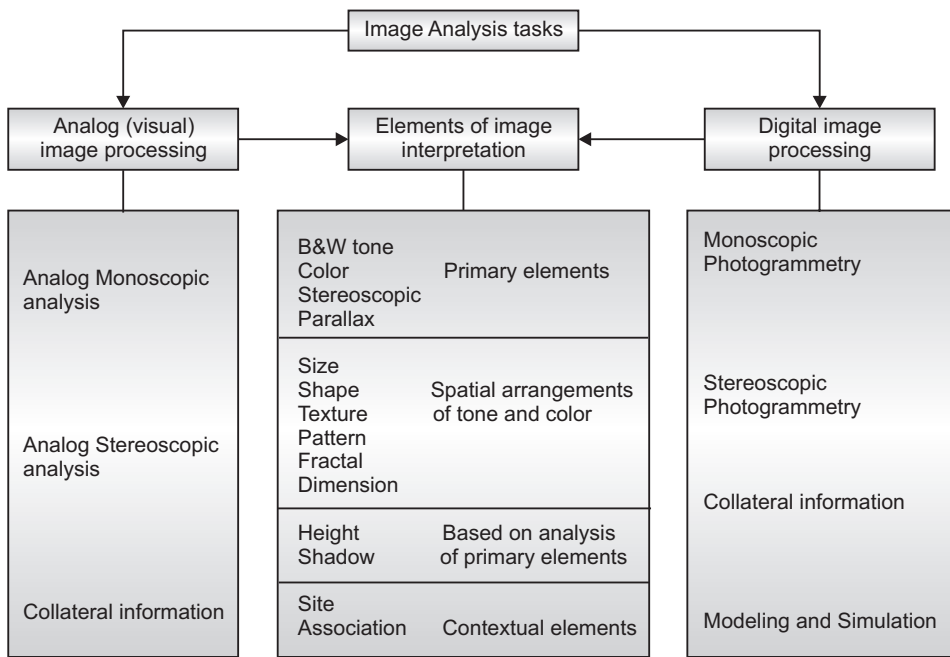


Fig. 27.10. The Schematic representation of analogue (visual) and computer associated digital image processing (modified from Jensen, 2000).

Digital image processing of digital data

The major type of digital image processing involves statistical and syntactical pattern recognition, photogrammetric image processing of the stereoscopic imagery, hyperspectral data analysis, expert system and neural network image analysis (for details see Jensen, 1996). In digital

image processing, instead of human vision, computer is instructed to perform an interpretation according to certain conditions. The goal of image classification is to convert image data into thematic data (thematic characteristics include land use, land cover, soil type, etc.). The classification is of two types, supervised and unsupervised. In the supervised approach, useful information categories are first defined (based on the familiarity with the area) and then their spectral separability is examined. In the unsupervised classification, the spectrally separable classes are first determined and then their informational utility is defined. Several classification algorithms are available. The classification results must be validated by comparing with samples from real world.

Ground Truth

In remote sensing, reference data (also called ground truths) are necessary in both research and operational applications. This involves data collection with statistically appropriate sampling design. Ground truth aids in analyzing and interpreting remotely sensed data, calibrating a sensor, and verifying information obtained from remote sensing data.

One form of ground truth is to measure the reflectance and emittance of various materials using spectrometer and radiometer. The spectrometer measures the spectral distribution of electromagnetic radiation. Radiometer quantitatively measures energy content of radiation in band of wavelengths. Spectrometers and spectroradiometers are used to prepare spectral reflectance for various objects.

In recent years, Global Position System (GPS) technology provides a convenient, inexpensive and accurate measurement of location of an object. GPS has enhanced the utility of remote sensing data when it is necessary to integrate image data with field data.

The ground truth accuracies have been improved by the increasing use of Global Positioning Systems (GPS). It is possible to get a high degree of accuracy in the case of the ground data using the GPS technology. The GPS is a satellite based system for determining accurate position of an object on or near the Earth's surface. GPS has been widely used in earthquake prediction, ocean floor mapping, volcano monitoring and a variety of mapping projects in different countries.

Information Presentation

The remotely sensed data are digitally processed with appropriate rectification and enhancement and combined with ground truth data. A variety of image data products are made available from scanner data. These can be categorized into graphic products, tabular data and digital data products. From the remote sensing data, much useful information is derived in the form of enhanced image, image map, thematic map, or graphs. Computers can easily generate graphic output which can be printed in black and white or in colour. Tables can easily list summary statistics on the areal extent of the cover types, and digital data files can be stored in the computer and can be conveniently input to a GIS for merger with other geographic data files.

Advantages of Remote Sensing

Remote sensing can provide fundamental scientific information concerning biophysical parameters like location, elevation, depth, vegetation biomass and moisture content of soil and atmosphere. The remotely sensed data have been used for modeling of numerous natural and cultural of processes. The main advantages of remote sensing are:

Synoptic view: A large area can be covered by a single image as produced from different sensors and satellites, enabling us to see all the features of the area together.

Repetitivity: The data for any area can be obtained repeatedly at regular intervals of time. This helps to monitor changes over the time across the revisit periods.

Mapping of difficult terrain: **Accurate** data can be collected for the inaccessible areas like mountains, swampy areas and thick forests.

Management planning: The digital data in different spectral regions can be processed and analyzed using a computer to get baseline information for a suitable management of an area.

Cost effective and time effective: If used efficiently, remote sensing could be cost effective and less time consuming for monitoring large areas as well as analyzing the structure and function of ecosystems.

Overcoming Limitations of Satellite Remote Sensing

Remote sensing has wide applications in diverse areas but it has certain limitations also. The earlier satellite missions were not equipped with sensors to collect data under all weather conditions. At the micro level, ground resolution is generally inadequate for detailed studies. Precision and geographic accuracy may not be sufficient in most cases. Stereo viewing and three dimensional viewing is only possible in the case of data from SPOT satellite (Narayan, 1999).

In the recently launched satellite missions, limitations are being overcome by using sensors that collect information in the microwave region of the electromagnetic spectrum. Highly improved and high resolution sensors have been developed to get ground resolution of less than 5 m.

Applications of Remote Sensing

Remote sensing data have been used for numerous applications for studying earth resources such as vegetation, soil, minerals, rocks and atmospheric characteristics. Such information is useful for modeling global carbon cycle, vegetation characteristics, ecosystem dynamics and atmospheric chemistry (Paylor et al., 1999). Remote sensing has immense potential in forest resource management for the assessment of forest cover, forest classification as well as for monitoring afforestation programmes. There are important applications in monitoring of water resources, town planning, fire detection, weather forecasting and communication. Some important applications of remote sensing data in ecosystem analysis, vegetation composition, biomass estimation and resource management are described below.

1. Analysis of ecosystem structure

The graphic outputs from the imagery taken in different spectral bands make it possible to determine the composition of different taxonomic levels and plant associations, soil properties and soil types. The individual properties of ecosystem such as above ground biomass, crop yield, soil fertility, population of large animals and habitat and feeding areas of wildlife can be studied. The multispectral aerial and space images make it possible to make good maps of plants, soil and wildlife distribution from regional to large scales. The remote sensing data have also been used for monitoring the threatened vegetation resources. A comprehensive inventory of the vegetation at the plant community level of the Everglades in Florida has been completed using a combination of satellite imagery, aerial photography, photogrammetry, GPS, and GIS technology (Doren et al., 1999; Welch et al., 1999).

Comparison of time lapse aerial and space imagery/photograph of an ecosystem taken at short time intervals can be used for quantitative tracing of plant growth and phenological development. Quantitative data on natural succession and man made change in land cover have been studied through change detection by using multitemporal data sets to discriminate areas of change between imaging dates.

Thermal infrared and passive microwave remote sensing have provided valuable information

on plant canopy evapotranspiration. The most practical application of plant moisture information is the regional assessment of crop water conditions for irrigation scheduling, stress assessment, and yield modeling for agriculture, rangeland, and forestry management. Remote sensing in the middle infrared, thermal infrared and passive microwave portions of the electromagnetic spectrum can provide information on plant water relations.

2. Ecosystem Functions

Biomass and productivity estimation: The relationship between red and near infrared canopy reflectance has facilitated the development of numerous remote sensing vegetation indices and biomass-estimating techniques (Roy, 1989). For this, multiple measurements in the visible and near-infrared region are utilized. Cohen (1991) suggests that the first true vegetation index was the Simple Ratio (SR), which is the near infrared (NIR) to red reflectance ratio as described by Birth and McVey (1968):

$$SR = NIR/Red$$

The Normalized Difference Vegetation Index (NDVI) has been developed on the basis of near infrared (NIR) to Red reflectance ratio:

$$NDVI = (NIR - Red) / (NIR + Red)$$

The radiations in 0.4 to 0.7 μm are strongly absorbed by green leaves. Radiation in the 0.7 to 1.1 μm are strongly back scattered or reflected. Reflected radiation from red and infra red spectral bands can be used to estimate the green leaf biomass of plant canopies.

NDVI values are derived from AVHRR (Advanced Very High Resolution Radiometer), Landsat MSS, Landsat TM and SPOT HRV data. The NDVI values can be related to *in situ* measurements of leaf area index (LAI), percent cover and biomass. Vegetated areas yield higher NDVI values. AVHRR sensors collect global radiation at a 4 km spatial resolution daily. A combination of channel 1 (0.55 to 0.68 μm) and channel 2 (0.75 to 1.1 μm) have been used for studying land cover, global vegetation phenology and net primary productivity. In Bilgiri Rangaswamy hills in Western Ghats, Bawa et al. (2002), have analyzed the areas of low and high tree diversity on the basis of species richness and NDVI.

The NDVI has been shown to be useful in estimating vegetation properties but some external and internal factors influence its global utility. Heute and Justice (1999) have proposed the Enhanced Vegetation Index (EVI), which is a modified NDVI with a soil adjustment factor. It is important to develop more rigorous remote sensing derived vegetation indices that can be used to accurately inventory the global distribution of vegetation types, leaf area index, and biomass and percent canopy closure. This will be invaluable to understand the functioning of ecosystems using remote sensing technology.

Nutrient cycling in forest ecosystems: Analytical spectroscopy shows that the infra red region from 0.7 to 2.5 μm contains information on various organic constituents. The high resolution imaging spectrometer (HIRIS) can sense the range of 0.4 to 2.45 μm in more than 200 bands. In temperate forest ecosystems, canopy chemistry and carbon and nitrogen cycling have been analyzed from the remotely sensed data (Martin and Aber, 1997). There is a big potential to apply canopy chemistry estimates in ecosystem models for estimating carbon balance and nutrient cycling in the forest ecosystems.

3. Oceanography and Coastal zone management

Remote sensing data have several applications in the field of oceanography and coastal zone management. Coastal zones are important because of large concentration of population in coastal zones and the presence of highly productive ecosystems. Some major applications of remote sensing include estimating fishing zones, identifying coral reef ecosystems, selecting sites for agriculture,

detecting sea level changes and monitoring coastal habitats like mangroves and wetlands. The IRS-P4, also called Oceansat-1, was launched in May 1999 by the Indian Space research Organization. It is equipped with Ocean Colour Monitor (OCFM) and Multifrequency Scanning Microwave Radiometer (MSMR). The Thematic Mapper provides data in four spectral bands of 0.45-0.52 μm , 0.52-0.60 μm , 0.63-0.69 μm and 0.76-0.86 μm . The aim is to study ocean dynamics towards a better planning and management of coastal zones and understanding the meteorological conditions in the Indian Ocean region.

4. Atmospheric pollution

Conventionally, atmospheric pollution has been investigated by sensors on balloons and aircrafts. Now satellite data have been found to be very useful to map the dispersion patterns and regional spread of atmospheric pollutants. Remote sensing of atmospheric gases can be done in various electromagnetic regions ranging from optical to microwave. Broadly two types of technique (Passive detection and active sensing) are used to detect the trace gases. Active detection requires illuminating the scene with a radiation source, usually a laser that is absorbed by the target gas. Passive system does not employ its own source for illumination and relies on measuring either reflected solar light or emitted thermal radiation from earth environment in very narrow spectral width channels. Three types of observational geometry (Nadir, Limb and Occultation) are used for remote sensing of gases. Differential Optical Absorption Spectroscopic (DOAS) technique is used to estimate the trace gases amount by detecting the reflected or emitted signal in very high spectral resolution in the absorption wavelength of the trace gases. NIMBUS-TOMS (1978) and continuity missions, ADEOS-IMG (1996), ENVISAT-SCIAMACHY (2002), METOP-IASI (2006), EOS-TERRA-MOPITT (1999), EOS-AURA-OMI (2004), EOS-AURA-TES (2004), JAXA-GOSAT (2009) are some of the important space missions which provided the valuable information on global distribution of trace gases such as Ozone, Carbon Monoxide, Carbon dioxide, Methane etc. Orbiting Carbon Observatory (OCO) of NASA is an important future mission to study the variability of Carbon dioxide.

Information on Aerosols is important in many air pollution studies. Satellite based measurement of aerosol optical thickness (AOT) on global scale is being regularly carried out by EOS-MODIS, EOS-MISR instruments. Space based detection of aerosols over land in comparison to ocean is a challenging task as it is difficult to separate the scattering from aerosols and surface. Dark Dense Vegetation (DDV) techniques are used to estimate aerosols over vegetated areas. Polarized multi angular measurements (as experimented on POLDER-1/2, PARASOL) are needed in future for improved characterization of aerosols.

27.2 GEOGRAPHIC INFORMATION SYSTEM (GIS)

Geographic information techniques in map visualization and data management have led to the development of GIS. Many of the basic concepts of GIS originated in the early 1960s, but recent advances in computer based GIS have been in use since 1970s at the interface between earth sciences and information science. GIS have evolved side by side with remote sensing systems because there are many close relationships between GIS and remote sensing. Remote sensing systems contribute data to GIS at a low cost and compatible with the requirements of a GIS. Besides this, both GIS and digital remote sensing systems use similar equipments and similar computer programmes for analysis and display. During 1970 and 1980, the developments in the field provided a strong theoretical basis of GIS in terms of software advances, database availability and geoinformatics. During 1990s, the GIS involved applications in various disciplines such as geography, computer science, and remote sensing concerned with collection, storage and analysis of spatial data. GIS uses the approaches and techniques from these disciplines to combine spatial and non spatial data from different sources. Current trends in the GIS use include a better integration between raster and

vector based systems, using remote sensing data and integration with video technology for better visualization and presentation. The aim is to provide useful information for planning, management and decision making through visual display and modeling. GIS is being increasingly used for varied applications in diverse fields of resource and environmental management, city planning, economic decision making and trade, remote sensing, survey engineering, forestry and land use planning as well as atmospheric science. Now GIS has grown into a large industry because of its utility in the socio-economic sector and applications in public and private sectors. In India, major developments in the use of GIS have occurred during the last ten years with significant contribution from the Department of Space in the fields of natural resource management, integrated mission for sustainable development and biodiversity characterization at the national level.

There are several definitions of GIS according to the capability and the purpose for which it is used. According to Clarke (1986), GIS is a computer assisted system for the capture, storage, retrieval, analysis and display of large amounts of spatial data. Geographic information systems include computer hardware, software, and procedures used for capturing, storing, retrieval, analysis and display of spatially referenced data from real world (Burrough, 1986). GIS allows data management using the link between tabular data and map. GIS has been put to use by various disciplines as tool for the handling of spatial data in a geoinformatics environment. Depending upon specific applications, GIS can be considered a data storage (using spatial data), a tool, a technology and liveware. Liveware are the people responsible for digitizing and implementation using the GIS.

GIS operational procedures and analytical tasks that are clearly useful for spatial analysis include: single layer operation, multiplayer operation, geometric modeling, network analysis, raster/grid analysis. Geoinformatics has multiple applications in environmental monitoring and natural resource management and in several other areas. Many internet protocols such as Hyper Text Transfer Protocol (HTTP), browsers and many tools and languages are based on GIS applications. The web based GIS are the systems of future as part of large organization or network of organizations dealing with transportation, law enforcement and environmental quality, etc.

Basic Concepts

Geographic information comprises data about the surface, subsurface, and atmosphere of the earth, interpretations and explanations applied to these data, and an organizational framework for understanding the information. Geographic information may be acquired through direct measurement, remote sensors, *in situ* observations; defined through survey or legal description; interpreted from data analysis or simulation. Geographic information obtains meaning from both spatial and non-spatial aspects of the data, i.e. “where” and “what”.

The GIS can be considered a computer-based tool for mapping and analyzing things that exist and events that happen on earth. GIS has the advantage of separation of data storage and presentation; therefore, the data may be viewed and presented in various ways for decision making (Figs 27.11 and 27.12). It is a computerized technology that permits the digital storage, processing and display of data simultaneously with their geographical location (Fig. 27.12). The geographic data could be in the form of point, line or area. For example, an oil well is represented by a point, a road by a series of coordinates, whereas a forest by an area. GIS organizes spatial data into layers representing information belonging to a particular class. The layer can be combined in various ways to create new layers.

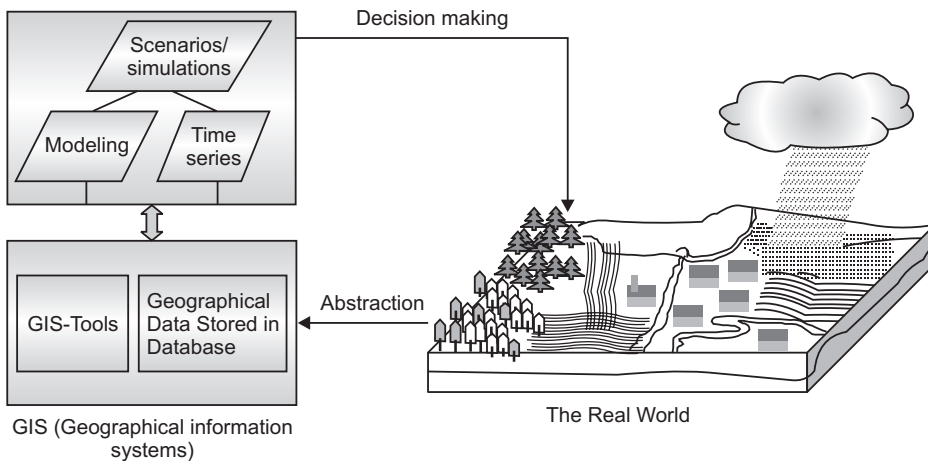


Fig. 27.11. Schematic representation of GIS as an abstraction of the real world and its utility in decision making.

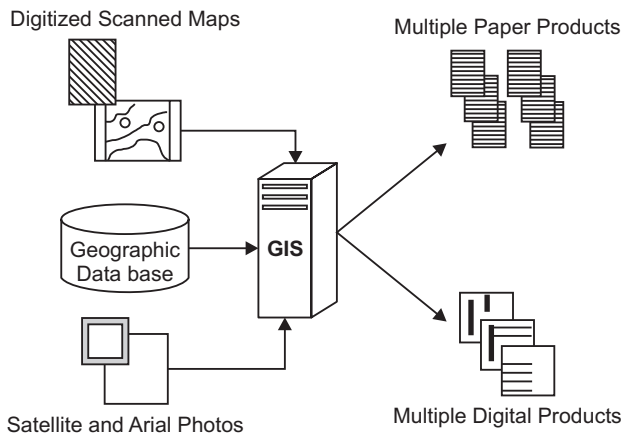


Fig. 27.12. Data input systems and output for GIS Application.

GIS technology integrates common database operations such as query and statistical analysis with the unique visualization and geographic analysis benefits offered by maps. These abilities distinguish GIS from other information systems. As a result, it is available to a wide range of public and private enterprises for explaining events, predicting outcomes, and planning strategies.

GIS Tools and Components

From a structural perspective, GIS comprises four subsystems, each having its specific function. These subsystems are input/data capture, data management, manipulation and analysis, output/display, (Fig 27.13). The data capture subsystem provides operational functions for acquiring data. The data management subsystem stores and retrieves the data elements. The manipulation and analysis subsystems handle the transformation of data from one form to another. The data output and display systems provide a way for the user to see the data (information) in the form of diagrams, maps or tables, etc.

The data management subsystem is connected directly to the data/capture manipulation/analysis and output/display subsystems (Fig. 27.13). It facilitates the passage of data between various subsystems. GIS has been considered as a database, graphic and analytic engine because it provides a mechanism to manage large volumes of spatial data and graphic information in a systematic way.

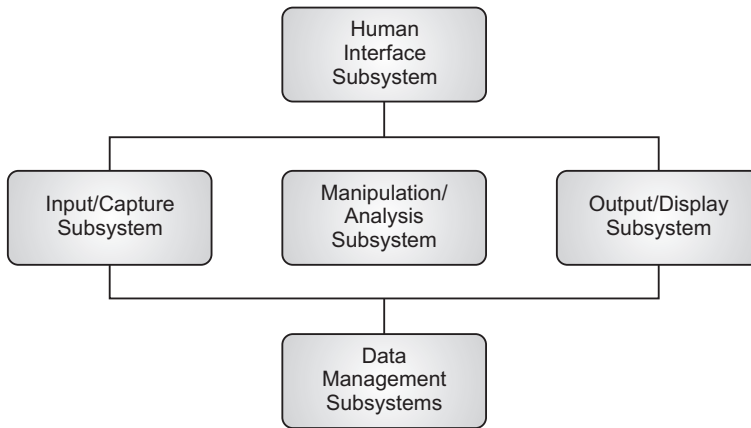


Fig. 27.13. GIS generic subsystems from a structural perspective reflecting the use and processing activities in a GIS (after Nyerges, 1993).

As shown in Fig. 27.14, GIS has five primary main components, namely,

1. **People**, computer staff, experts from various science disciplines, GIS operators, GIS experts, applications developers;
2. **Data**, which may be of type *spatial*, *temporal*, or *attribute*;
3. **Engines** that perform various data storage, retrieval, analysis, reporting, and communication functions;
4. **Computer Software** such X-Windows or MOTIF, Microsoft Windows; and those concerned with digitizing, database management, map/graphic editing. GIS Software can integrate multiple sources of data (maps, databases, satellite imagery), answer queries, conduct analysis and modeling, and produce maps.
5. **Hardware**, including workstations and networks, disk and tape storage, digitizers, plotters, scanner, printer and communications devices.

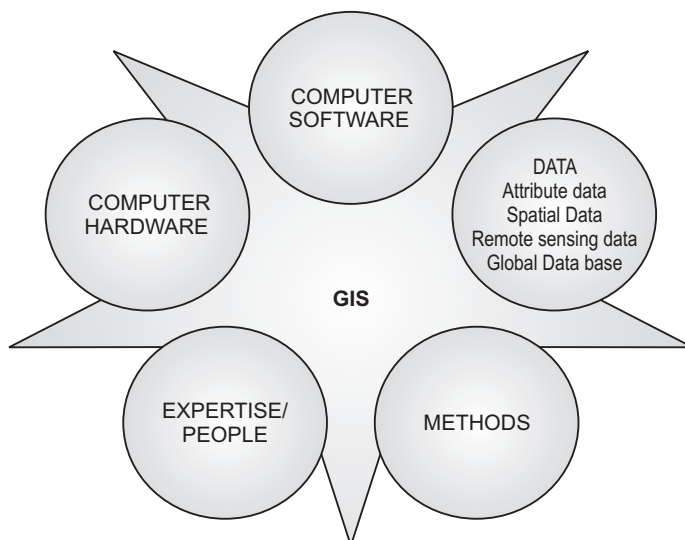


Fig. 27.14. Components in the GIS, It is important to give equal attention to all the links while planning GIS.

Data for GIS

GIS often depends on data that may have been gathered from other organizations. Therefore, it is important for the GIS to have access to mass storage (e.g., CD ROM drives) and to computer network communications for data transfer from one computer to other computer systems. A digitizing table or scanning system forms an important method for data entry. Scanning facilitates data entry recording patterns depicted on images or maps.

Data has to be of a type that the software is equipped to deal with. In order to represent the spatial information and their attributes, a data model (formalized descriptions of real world phenomena) is adopted, which establishes the linkage between the real world domain of geographical data and the computer and the GIS output of these features. The data storage models help to organize the real world geographic system into a systematic storage and retrieval mechanism.

Spatial analysis is one of the most important uses of GIS which can be either vector based or raster based. Data essentially have two kinds of data structures, i.e., vector or raster (in grid system) (Fig. 27.15). Vector data are integrated as pairs of X-Y coordinates to identify locations. This is the type of data necessary to translate linear features such as roads or geographical borders, or discrete information, such as the location of specific points in the form of digital information. Area can be described by polygons (set of X-Y coordinates) A vector structure is most appropriate for spatial management and cartographic production. The vector based system indicates the thematic maps precisely and needs less storage space, and at the same time the output is appreciable. However, the vector model is less useful for describing continuously varying features, for example soil type.

The raster (Cellular) data structure expresses data by assigning a value to grid cells analogous to the pixels of an MSS scene. A (grid) raster system used to display satellite imagery is composed of picture elements called pixels, essentially representing the grids cells (Fig. 27.15). It is a cell based system in which a map is represented as an array of rectangular or square cells. One set of cells and the associated value is a LAYER. Each cell contains a single value and every location corresponds to a cell. This is used to express geographically varying data like soil type, or precipitation. A raster structure is most appropriate for analysis and modeling. Satellite imagery and Aerial photos can be best represented by the raster based system. For example, different types of information on natural resources such as water resources, ground water potential, soil survey information, land use, vegetation composition can be extracted from the same satellite data for the GIS (Fig. 27.15). All the factors affecting the use and exploitation of natural resources are taken into consideration in the form of digital values along with the spatial location as reference to the true geographic position on the surface of the earth.

After this, the data can be manipulated with the help of suitable software to simulate various alternatives. The output of the model can give an idea of the consequences of decisions or enables to take effective decisions to obtain optimum results.

Procedural Perspective for GIS

Data management

GIS is an information system used for management decisions as well as to create databases of information. The databases are spatially oriented with respect to the position on the earth surface. The spatial data are generally

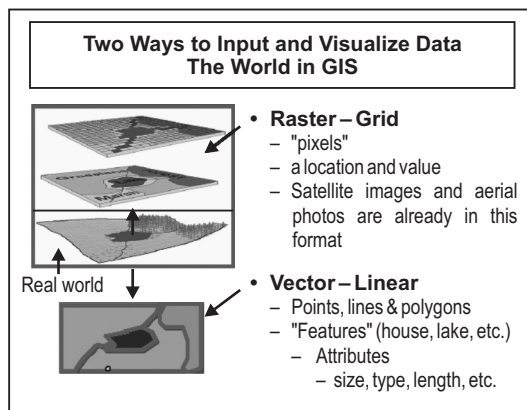


Fig. 27.15. The raster and Grid data used in GIS (reproduced from the UNESCO, 1999).

in the form of maps to show topography, geology, soil types, forest and vegetation, land use/land cover, water resources, census data regarding population and economic status. All these are stored in the digital form in the computer. The attributes like statistics, texts and tables can also be used besides the spatial data. Technically, GIS organizes and exploits digital map data stored in databases. The real world is described using digital map data that defines positions in space. Geographic information on objects may be entered in terms of points (no dimensions), lines (two dimensions) and area (two dimensions). Every single point, and every line and area are presented by coordinates. The straight line between two data constructs in a digital map is a vector. Modern GIS can process data from dissimilar sources, including digital map data, digital CAO and various computer based registers.

Databases are vital in all GIS. A database is a comprehensive collection of related data stored in logical files and collectively processed usually in a tabular form. A database must allow concurrent use by multiple users and should support storage optimization, must ensure data integrity; and should offer query facility and query optimization. A database management system (DBMS) is software that manipulates data in a data base so that data are maintained at one place, and can be easily accessed and updated. Remote sensing has become one of the most important data sources in the GIS. Spectral data are acquired using aerial and space platforms as well as using ground instruments. The various methods of spectral data acquisition have been discussed earlier in this chapter.

Steps to Using GIS

There are 6 main steps to using a GIS tool, i.e., data input, data manipulation, management, query and analysis, overlay analysis and visualization. The various steps are briefly described as follows:

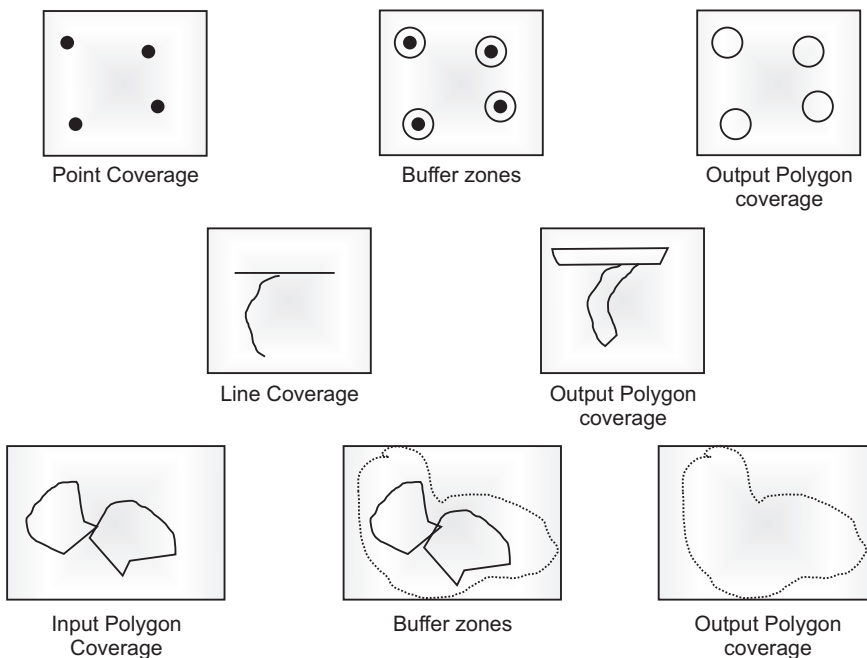


Fig. 27.16. Buffer zones can be established around points, lines and polygons. The buffer zones provide the basis for the creation of new polygons that the attributes of the original object and can be used in further analysis (redrawn from the CSSTEAP notes).

Data Input : This is the entry of the data necessary to create the map using the computer. The data could be derived from multiple resources such as reports, maps, laboratory instruments and measurements from ground, aerial and space borne sensors. The collateral geographical data could be in the form of point, line or area (Fig. 27.16). For example, the location of a thermal power station is a point data. Roads and canals are typical examples of linear data. Topographic maps are examples of area data.

Buffer Zones : Buffer zones are used to define spatial proximity which comprises one or more polygons of a prescribed extent around points, lines or areas (Fig. 27.16). Many GIS support automatic compilation of buffer zones.

Data manipulation : This is the manipulation of data into a format that will make it most suitable for visualization and interpretation. For example, maps can be converted into digital format by tracing them with an electronic cursor (digitizer). Once a map is digitized, its alteration is possible in any desired format for the purpose of analysis. The data encoding can be done either in the grid or raster format.

Management : This is necessary when dealing with very large data sets, in which case a Database Management System (DBMS) is necessary.

Query and analysis : Data extraction from the database without altering it is called query. This allows the user to obtain information directly from the map. This could range from obtaining the coordinates of a certain point, or finding out the mean annual temperature in an area or the geographical area of a parcel of land.

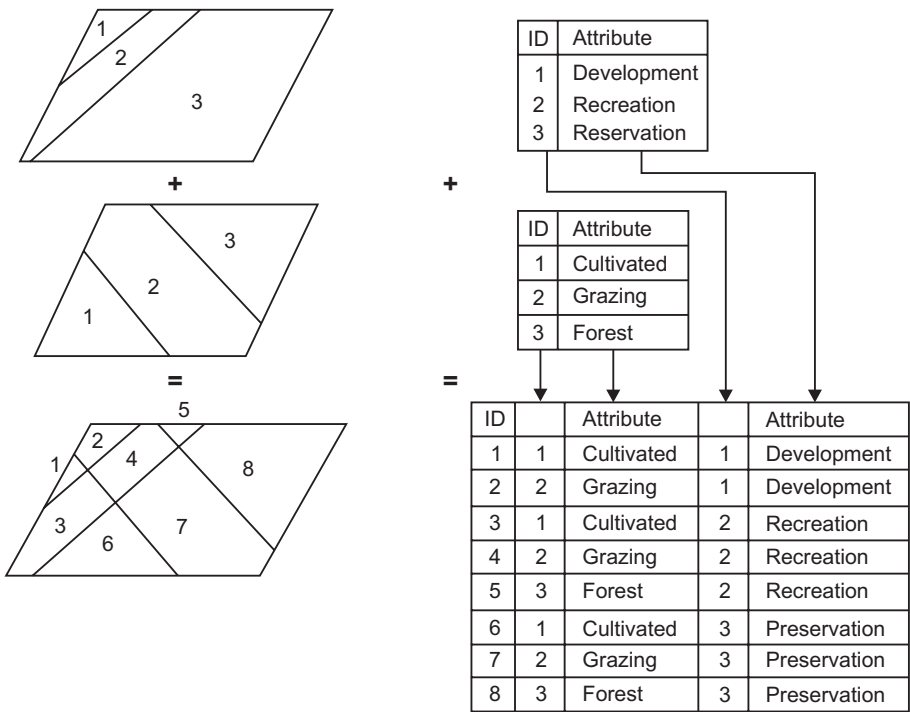


Fig. 27.17. A computerized thematic map overlays that can be used in multiple ways (adapted from Bernhardsen, 1997).

Overlay analysis : GIS is used to perform a variety of spatial analyses including the overlay analysis in which two or more data planes are combined for analyzing flows and for proximity analysis (Fig. 27.17). The layers are combined in various ways to create a new layer called a composite overlay. The input data is combined from diverse elements such as vegetation, land use, land ownership and land survey. The details of various components are converted into digital data. The final synthesis produces a reference map or data set.

Visualization : The end product is useful in decision making for solving complex problems. This can have visual implications, i.e. showing information about geology, soil type, precipitation and elevation on the same map. This is the end result (in a map or a graph) of the use of a GIS. Information presented in such a format is often much more effective than words and numbers (Fig. 27.18).

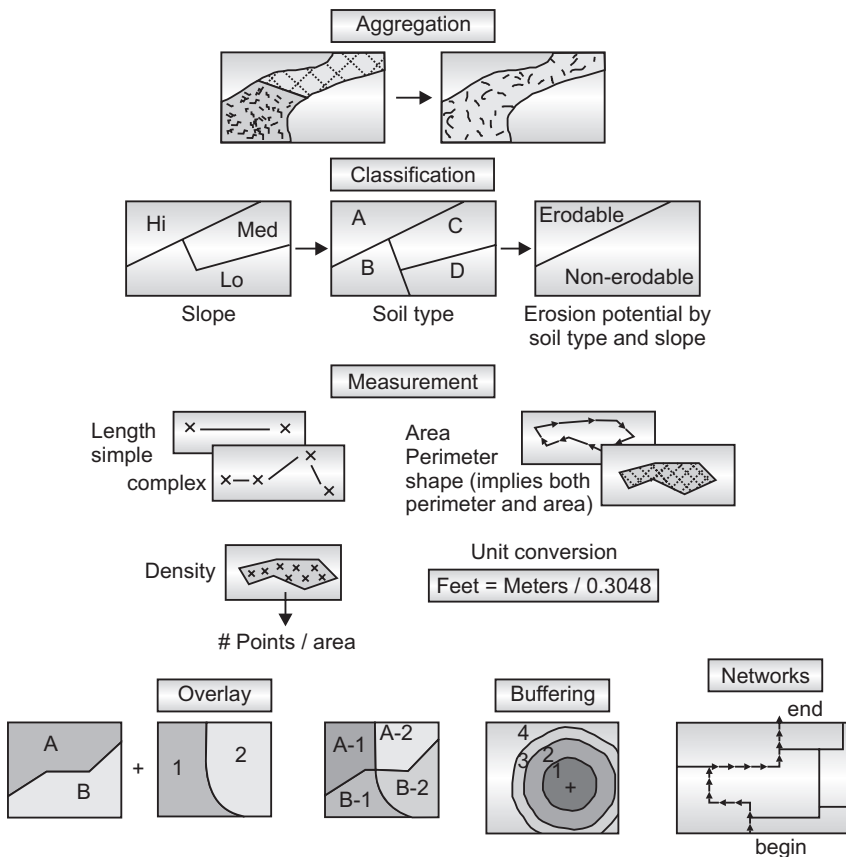


Fig. 27.18. Various data analysis tools used in GIS (redrawn from the UNESCO, 1999).

Main software used for GIS data analysis are summarized in Table 27.5. New maps are generated by integrating the multiple layers of maps using computer software, which show the relationship between different attributes and spatial data. A good background in land surveying is essential for any GIS programme, as GIS is structured on horizontal control survey networks, digital to geographic data bases and land survey data bases. It is also important to study user needs besides ensuring reliability, consistency, accuracy and easy to use approach for all types of users.

Table 27.5. Popular software used for GIS.

IDRISI	This software is designed to be easy to use and it provides a professional-level GIS. Image Processing and Spatial Statistics analysis can be done on Windows-based personal computers. This software is modular in design and has strong support for the manipulation and display of raster images. Manipulation and input of vector data is supported.
ARC GIS	This GIS software is based on tool box concept and mainly serves to visualize data to better understand its geographic distribution as well as produce maps. Each function can be visualized as a tool and has specific activity. This software combines the visualization functions of Arc View with advanced modeling and analysis capabilities.
MAP INFO	This is professional mapping software, which helps to see patterns and trends in data and helps in management of spatial data.
Auto Desk	A desktop application to integrate, analyze and display spatial data and provides seamless access to existing GIS data, in many different formats.
ER Mapper	ER Mapper allows combining raster and vectoring data and visualizing it, by performing enhancements (filters, etc.), combining and integrating all the data being used to make maps.

Integration of Related Systems in GIS

The GIS is made up of characteristics of several systems, i.e., digital cartography, computer aided design (CAD), remote sensing and database management, to obtain new information that can be used as a decision making tool (Fig. 27.19). From this figure it is evident that data analysis tools include aggregation, classification, measurement, overlay, buffering, networks and map algebra. Network functions examine the movement of objects along an interconnected pathway. Map algebra utility allows the user to specify mathematical relationships between map layers.

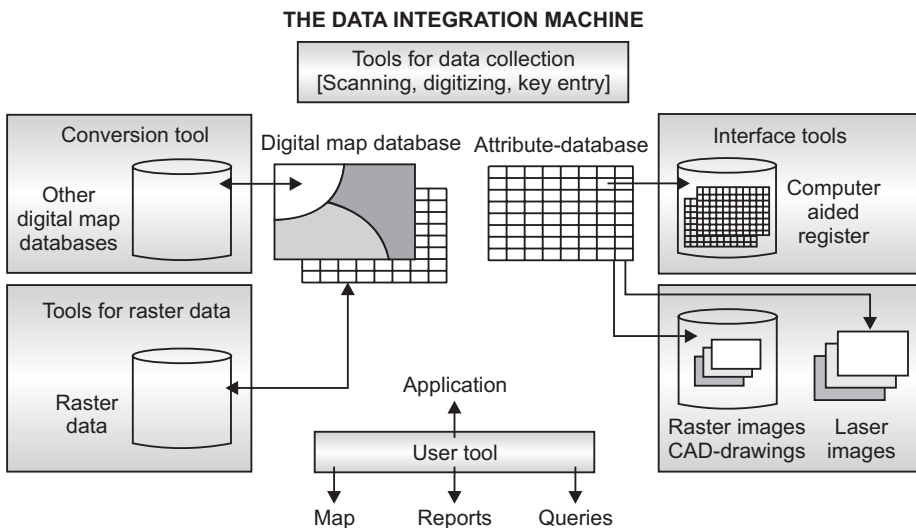


Fig. 27.19. Modern GIS is a typical “data integration machine” (redrawn from the UNESCO, 1999).

GIS Applications

Geographic information system (GIS) technology can be used for scientific investigations, resource management, and development planning (Figs. 27.20 and 27.21). GIS technology has useful applications for the analysis of natural and human caused environmental problems. It is possible to map ecosystems and to keep track of the number of species, or the population of a species, in a certain area. This can help in monitoring endangered populations and protected areas. The condition of the Earth's surface, atmosphere, and subsurface can be examined by feeding satellite data into a GIS. GIS has been widely used in water management, coastal zone management and biodiversity studies. Based on satellite data, areas that suffer most from deforestation may be identified and analyzed on the basis of soil types, the species composition and the impact of regulatory measures applicable to the area. It has been widely used to understand global change process, ozone depletion, and deforestation and soil erosion. Mainly, GIS has applications in the following areas:

- Land use change and farmland location.
- Forest inventory, mapping, monitoring and management.
- Soil inventories and surface water detection.
- Environmental impact assessment.
- Display of location and thematic data.
- Tracking of endangered species and habitat restoration.
- Watershed management.
- Geological resource mapping and exploration.
- Biodiversity and global change studies.

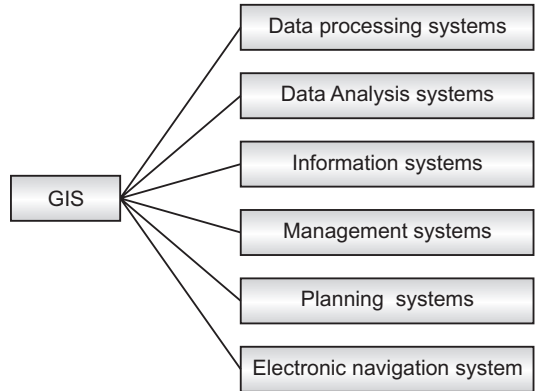


Fig. 27.20. GIS diversity on the basis of applications.

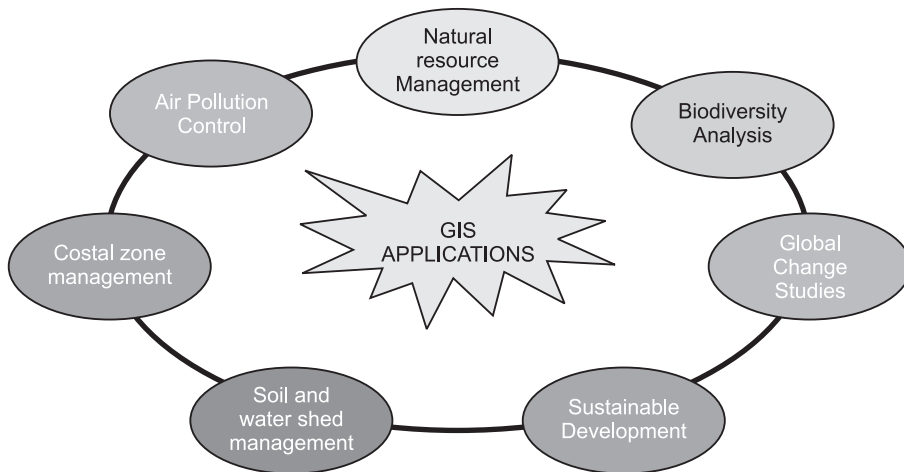


Fig. 26.21. GIS applications in different fields.

GIS has proved to be a useful tool in the analysis of species distributions and the prioritization of areas for their conservation. This includes studies on the richness and distribution of organisms in different parts of the world. It has also been used for understanding the impact of land use changes on the habitat of target species and how these contribute to the instability of such populations.

1. Species distribution and habitat conservation

GIS can contribute significantly to spatial modeling and can be used to identify and delineate a buffer zone in protected area taking into consideration several complex biophysical and socio-economic criteria for example, the establishment of a buffer zone around the Royal Chitwan National Park in Nepal by involving local people and integrating local knowledge to the management strategies.

Natural community and land cover GIS databases were used to identify shoreline areas within the highly fragmented Indian River Lagoon Ecosystem. In a GIS analysis, it was found that 40-100% of the areas of all existing nature reserves in Brazilian Amazon are directly accessible via navigable rivers and/or functional roads. This greatly facilitates the illegal harvest and conversion of forest resources in a region where each guard is responsible for protecting a huge area. Cost-effective defense of large areas can be achieved through appropriate delimitation of reserves along watershed divides and by efficient deployment of limited infrastructure and personnel (Peres and Terborgh, 1995). This study showed how access to even remote reserves must be taken into consideration when management plans are prepared.

2. Resource management

The management of natural resources involves topics such as the creation of inventories, characterization of areas, developing simulation models to predict the future condition of such resources, and decision making and the creation of management plans. GIS has been very important in each one of these steps. Remote sensing and GIS have been used in UNESCO sponsored international research programmes such as the Man and the Biosphere (MAB), for biosphere reserve development and ecosystem monitoring, management and conservation; the International Hydrological Programme (IHP) for surface and ground water resource management; the International Geological Correlation Programme and in several projects concerning ocean investigations and climate observation. In India, the IGCNC, New Delhi is developing Geographic Information Systems (GIS) data-bases for Protected Areas in the country so that the requisite spatial information becomes available to resource managers and planners. Similar databases are planned to be developed for species distribution, wetlands, degraded forests and other important ecosystems. These databases will provide useful inputs for effective eco-development and management planning. Presently IGCNC is executing the task of preparing base maps for three major landscapes which hold the potential habitats for large mammals such as elephants, rhinoceros and tigers covering the states of Assam, Arunachal Pradesh, Maharastra, Madhya Pradesh, Chhatisgarh, Uttar Pradesh, and Uttaranchal.

3. Global environmental database (GRID)

The United Nations Environmental Programme (UNEP) has set up the Global environmental database (GRID) within the framework of the Global Environmental Monitoring system (GEMS). The main functions of GRID are environmental data management and to establish a global network on environment using GIS and image processing technology. GRID is useful to examine interactions between different environmental datasets, and to provide models and scenarios. The International Geosphere and Biosphere programme uses grid as a useful data management system for studying

global change. The activities of grid also focus on problems of land degradation, forest depletion and loss of biological diversity.

4. Biodiversity and Gap Analysis

The conventional approach to maintaining biodiversity generally has been to protect endangered and threatened species. The species approach could be inefficient, expensive, and biased toward mega fauna. One approach for assessing the current status of biodiversity at all levels and not just endangered species is called Gap Analysis. It is heavily based on the use of remote sensing and geographic information systems (GIS) to identify gaps in biodiversity protection that may be filled by the establishment of new preserves, or changes in land use practices (Scott et al., 1993). Roy and Tomar (2000) have developed a geospatial model to identify the biodiversity distribution and gaps for the conservation prioritization.

Gap Analysis is a method to identify gaps in representation of biological diversity in protected (and less protected) areas and is a course-filter approach to conservation evaluation (Scott et al. 1993). Gap analysis uses the computerized geographic information systems to map the distribution of the various taxa in relation to landownership. This technique uses overlays maps to identify gaps in biodiversity protection. The aim is to identify the protected areas and to ascertain concentration of endemic species, species richness and the number of rare or endangered species. Gap analysis provides a tool for prioritizing and focusing habitat protection efforts for the conservation of scarce resources, conservation planning and to set priorities for biodiversity monitoring and analysis (Scott et al., 1993).

A typical Gap Analysis includes the creation and analysis of four primary GIS layers:

- The distribution of actual vegetation cover types delineated from satellite and aircraft remotely sensed data and ancillary information
- Land ownership
- Land-management status
- Distribution of terrestrial vertebrates as predicted from the distribution of vegetation and *in situ* observations

Gap Analysis provides an overview of the distribution and conservation of several components of biodiversity. It uses the distribution of actual vegetation types and terrestrial vertebrate and when available, invertebrate taxa as indicators for biodiversity. Digital map polygon overlays in a GIS are used to identify individual species, species rich areas and cover types that are not represented or under represented in existing management areas.

The main benefits of Gap Analysis are: consolidating diverse data sets using geographic information systems (GIS); standardizing habitat and land use classification systems for the area of study and developing predictive models based on habitat affinities for each species.

The main limitation of gap analysis is that it maps large minimum mapping units (typically 100 ha to 1 km²). The small habitat patches are often missed. It also fails to distinguish between most seral stages, and to detect gradual ecotones and other important landscape features (e.g., small wetlands and riparian zones). Under natural conditions, most of the land management units are small. Therefore large-scale Gap Analysis must be complemented with a down-scaled approach and systematic resource inventories at finer spatial resolution to be useful to land managers.

5. Coastal Zone Management

Indian remote sensing satellite data has been used to identify and monitor various coastal zone features. The satellite imagery has helped to collect information about tidal wetlands, mangrove forests, salt marshes, tidal flats, shoreline changes; dynamics of erosion, coastal land use/land cover and suspended sediments (Desai et al., 2000). GIS is being used for integrating thematic information from satellite data with socio-economic and cultural data for identifying suitable areas for coastal water aquaculture and for developing a sustainable coastal zone management plan (Table 27.6).

Table 27.6. Status of remote sensing based coastal zone studies in India (from Desai et al., 2000).

Project	Achievements
Wetland Landform	Maps of mudflats, Beach, mangroves and coastal
Shoreline changes	Maps of erosion/depositional areas for the entire coastline.
Coral reef	Maps of major four coral gulf reef areas, viz., Gulf of Kachchh, Lakshadweep, Bay and gulf of Mannar, Andaman and Nicobar.
Suspended sediment	Qualitative sediment mapping of pre-and post monsoon period for Kerala and Tamil Nadu coast.
Coastal regulation zone	Maps showing wetland areas between high and low water lines and land use features.
Coastal Zone information system	Pilot scale study mainly based on remote sensing inputs is under development for providing decision support for coastal zone planners.

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Part – VII

ENVIRONMENT AND SOCIETY

- ♦ ECOLOGICAL RESTORATION
- ♦ ENVIRONMENTAL IMPACT ASSESSMENT
- ♦ SUSTAINABLE DEVELOPMENT AND ECOLOGICAL ECONOMICS
- ♦ HUMAN POPULATION AND URBANIZATION

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CHAPTER

28

Ecological Restoration

Earth's ecosystems are under threat. The over exploitation of land and aquatic ecosystems by humans is evident at local to regional, national and global scales with marked negative impacts on biological diversity and the livelihood of the people. The United Nations Environmental Programme Yearbook 2009 reports that 21% Earth's land cover has been significantly degraded by human activity and 60% of the planet's assessed ecosystems are now damaged or threatened because of the unprecedented natural resource overexploitation and creating more waste than ecosystems can process (UNEP, 2009). New approaches are needed to decouple productivity from environmental degradation (MEA, 2005c). In many deforested, degraded and fragmented forest habitats investments in restoration and rehabilitation forests can yield high biodiversity conservation and livelihood benefits (Sayer et al., 2004; Chazdon, 2008).

In India, more than half of the land mass is already degraded because of increasing rate of deforestation, overgrazing of grasslands, mining and quarrying activities, degradation of agricultural lands, and dryland expansion. Perhaps the most significant human alteration of land cover is the creation of large areas of agricultural land. The damaging consequences of this degradation are directly associated with losses of ecological services and the subsistence of the people living in the forests. Aquatic ecosystems including lakes, streams, rivers and wetlands are critical resources for the people but suffer from degradation due to wide ranging human activities. The land-use changes, agriculture and urban development and other multiple uses caused a loss of about 40% mangrove systems in India (Upadhyay et al., 2002).

28.1 HISTORICAL PERSPECTIVE

The acts of humans repairing damaged ecosystems have remained a part and parcel of human societies. A conscious effort to restore degraded areas was practiced with the development of interest in conservation activities during the 1800s and 1900s. In North America, the Ecological Society of America (ESA) produced the Nature's Guide to the America in 1925 under the guidance of Victor Shelford. The Nature conservancy was established in 1950, which played an active role for the restoration of predator species and the American grasslands. The ecological restoration at the Arboretum of University of Wisconsin - Madison to its previous tall grass prairie was taken up by the civilian conservation corps in 1935 under the direction of Aldo-Leopold, a pioneer in conservation movement (Jordan et al., 1987; MacMahon, 1998). The Wisconsin Curtis Prairie Restoration is an example of the successful efforts of restoration based on ecological principles.

The Society for Ecological Restoration was formed in 1982 with an aim to implementing restoration activities based on ecological principles and socio-economic consideration of the people being affected. Now, restoration is a growing enterprise both in public and private sector covering many large and small regional programmes. Restoration has been viewed as a type of ecosystem

management including preservation; conservation and stewardship (see MacMahon 1998). Ecological restoration can be a valuable tool for analyzing community organization and ecosystem function (Dobson et al., 1997).

In different regions of the world, efforts are underway in forest, grassland, freshwater ecosystems, and wetlands for restoring biodiversity, ecosystem services and mitigating the risks of global change including exotic species invasions (see McCarty and Zedler, 2002). Ecological restoration is a fundamental element of ecosystem management for conserving biodiversity, sustainable human livelihoods, empowerment of local people and maintaining ecosystem productivity (SER, 2002; IUCN, 2002). With increasing interest in conserving biodiversity, ecological restoration is now being used as a tool to conserve species, communities and ecosystems. The IUCN Commission on Ecosystem Management (CEM) has adopted ecosystem approach for promoting ecosystem restoration for sustainable development.

A number of studies have indicated that degraded communities are often resilient to restoration efforts because of unpredictable expression in biotic composition, trophic interactions, invasion by exotic species and changes in biogeochemical processes (Suding et al., 2004). Keeping in view the unpredictability of ecological communities in the changing environment, the restoration paradigm has to change from historic to futuristic restoration by setting realistic and dynamic goals, assuming multiple trajectories, adopting an ecosystem and landscape approach and maintaining long-term monitoring of restoration outcomes (Choi, 2004). According to Choi (2007), a future-oriented restoration should (1) establish the ecosystems that are able to sustain in the future, not the past, environment; (2) have multiple alternative goals and trajectories for unpredictable endpoints; (3) focus on rehabilitation of ecosystem functions rather than recomposition of species or cosmetics of landscape surface.

Hobbs and Cramer (2008) have compiled information on major restoration projects being successfully implemented in different regions of the world. In different projects, ecological restoration has improved the biological diversity on degraded landscapes, increased the populations and distribution of rare and threatened species, enhanced the availability of environmental goods and services, and has contributed to the improvement of human well-being in various large scale projects in different parts of the world.

A systematic meta-analysis of 89 restoration assessments has shown that restoration actions in a wide range of ecosystem types are effective in improving ecosystem services, particularly in the tropical regions (Rey Benayas et al., 2009). The Economics of Ecosystems and Biodiversity (TEEB, 2010) study concluded that restoration activities can bring high rates of return across a range of biomes, particularly when the value of nature's goods and services are properly accounted for. New methods of ecosystem service valuation are suggesting that the economic benefits of restoration can outweigh costs. Payment for Ecosystem Service schemes could therefore provide incentives for restoration (Bullock et al., 2011).

28.2 WHAT IS ECOLOGICAL RESTORATION?

Humans have done more to alter the earth's ecosystems in the past 50 years than any other time in history. The degradation of ecosystem occurs when a system loses its biodiversity, structures and processes of production resulting in decline of ecosystem services. Ecological restoration is the process of assisting the recovery of degraded, damaged or destroyed ecosystems based on ecological principles and socio-economical consideration of the people being affected (SER, 2004). Ecological restoration also encompasses terms like rehabilitation, reclamation, recreation, remediation and ecological engineering. Healthy ecosystems, and the goods and services they provide, are the foundations of survival for all societies and human well being.

Keeping in view the broad perspectives in ecological restoration, some terms are in common use, such as ecosystem degradation, reference ecosystem, restoration, rehabilitation; remediation and reclamation (Bradshaw, 2002, SER, 2002) that are briefly described below.

Ecosystem degradation: Commonly used terms in ecological restoration defining the deviations from normal or desired state of an intact ecosystem are degradation, damage, destruction and transformations (SER, 2002). Ecosystem destruction results in the removal of macroscopic life and the damage to physical environment. Transformation is the conversion of an ecosystem to a modified or altogether different kind of ecosystem or land use type. Damaged land is incapable of normal recovery but can still produce goods and services. Degradation refers to subtle or gradual changes that reduce ecological integrity and health. Land becomes degraded when its soil conditions and biotic richness have been reduced such that their capacity to satisfy particular uses has declined, and derelict land is of very little use to people. So the transformation and overuse can successively result in damaged, degraded and derelict land.

Reference Ecosystem: A reference ecosystem is the actual pre-disturbance site (having written or oral description) and provides a dynamic model for planning a restoration project and its evolution. The reference or the target ecosystem along with its processes and conditions serve as the benchmark for evaluation and monitoring. The construction of target ecosystem depends on the available ecological and historical knowledge. The reference can be a reference landscape or a reference landscape unit (SER, 2002). The designated ecosystem or landscape unit can simply be called reference. An ecological trajectory describes the development pathway of an ecosystem through time, which begins with the unrestored ecosystem and progresses towards the desired state of recovery. The trajectory includes both abiotic and biotic components of ecosystems.

Restoration: The word restoration literally means “to bring back into a former or original state”. The relevant definition of restoration is “the act of restoring to a former state or position or to an impaired or perfect condition.” The activities necessary to bring a disturbed site into former or original state involves manipulation of nature to re-create species composition and ecosystem processes close to the state that existed before disturbance. It re-establishes the structure, productivity and species diversity of the original community (Figs. 28.1 to 28.3). In other words, it recovers the ecological processes and functions of the original community.

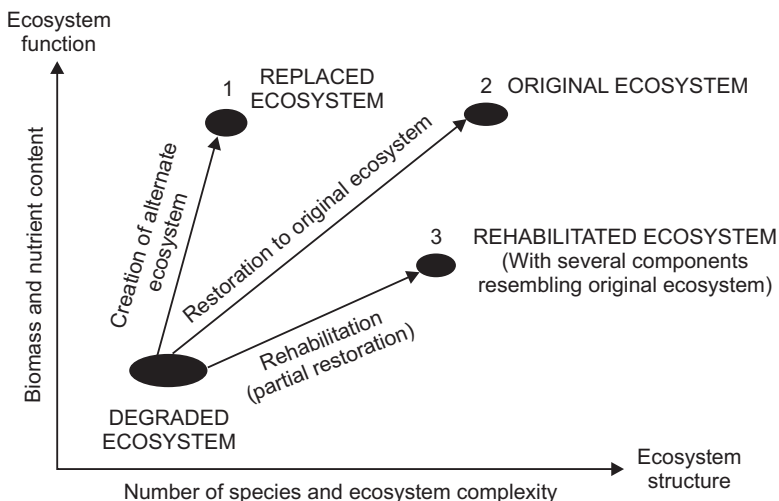


Fig. 28.1. Relationship between restoration, rehabilitation and creation of an alternate system. (based on Bradshaw, 1997 a).

Rehabilitation: Return of a damaged or degraded land to a fully functional ecosystem irrespective of its original state, but in conformity to a prior landuse plan is called rehabilitation. It is also called partial restoration where only selected ecological attributes of the site have been restored, e.g. partial recovery of species diversity and ecological complexity (Fig. 28.1) and medium human-well being benefits (Fig. 28.2). Rehabilitation projects involve structural measures and assisted recovery. Assisted recovery aims to remove a basic perturbation or disturbance such as excluding livestock grazing from the riparian zones, allowing natural growth of vegetation or restoring fluvial processes, allowing natural processes to operate.

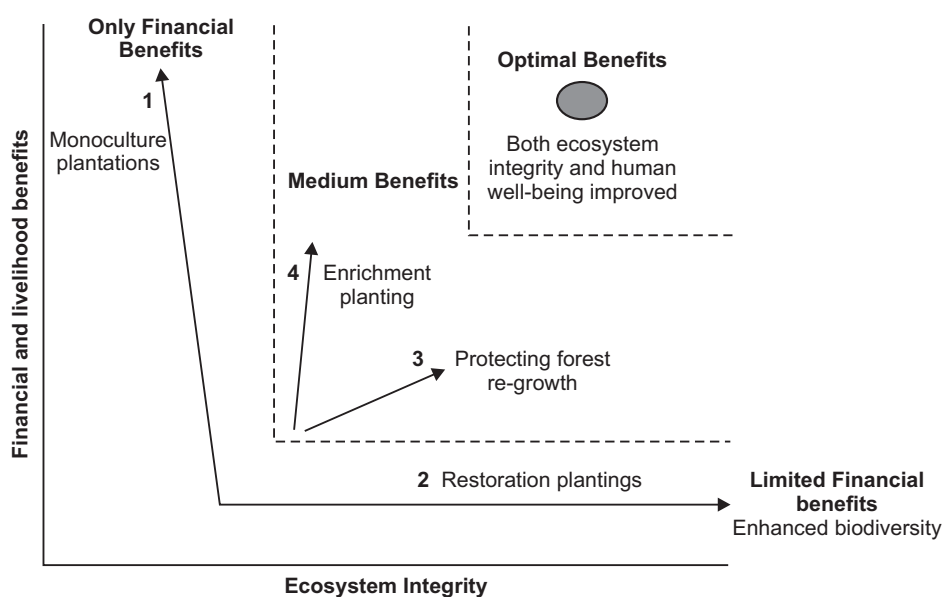


Fig. 28.2. The level of financial and livelihood benefits and ecosystem integrity varies with the type of ecological restoration within the landscape mosaic. The ideal restoration objective, shown with a dot, is often difficult to achieve for optimizing financial and livelihood benefits along with ecosystem integrity. (1) Traditional monoculture or exotic species plantations improves only the financial benefits. (2) Restoration plantings or direct seeding with many species maximize ecosystem integrity with few financial benefits to landowners in the short-term. (3) Protecting forest re-growth, (4) enrichment planting of secondary forest, etc., provide ecosystem integrity and livelihood benefits at medium level. (based on Lamb and Gilmour, 2003; Lamb et al., 2005).

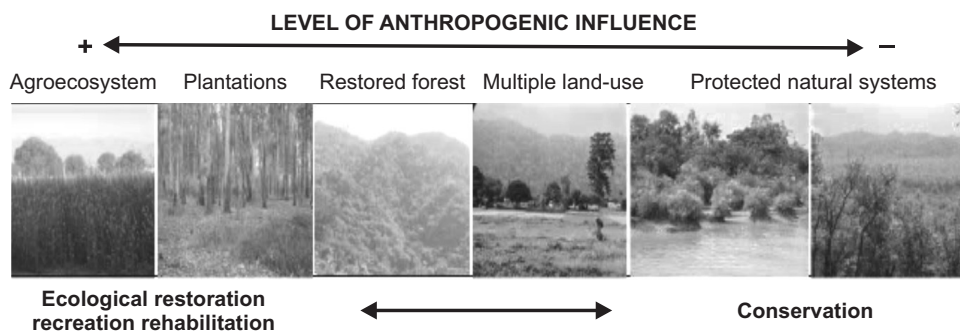


Fig. 28.3. The level of anthropogenic influence on ecological restoration.

Remediation, Reclamation and Mitigation: To remedy is to “to rectify, to make good”. Remediation is the process of cleaning chemical contaminants from a polluted area by physical and biological methods for protecting human and ecosystem health. Here the emphasis is on the process rather than the end point reached.

Reclamation is the process by which some measure of biotic function and productivity of a derelict or badly damaged land is restored, e.g., by liming of acidic soil, etc. This is not an ecological process. It involves a series of activities intended to change the biophysical capacity of an ecosystem. The resulting ecosystem is different from the historical ecosystem with minimum ecosystem integrity, the human well being benefits are maximum (Fig. 28.2). Mitigation has been defined by Bradshaw (1997a) as any restoration, rehabilitation or reclamation process to moderate the effects of a degrading action.

Replacement or Creation of an alternate ecosystem: On severely disturbed sites, there is virtually nothing left to restore. Replacement is an attempt to construct a new biological community on a severely disturbed site. The new system may be modelled on a system located outside the range of historical system or under different conditions that naturally occur. This is an interesting option since it may allow optimization of a given function e.g., productivity to a higher level than existed previously.

28.3 PRINCIPLES OF GOOD ECOLOGICAL RESTORATION

Ecological restoration is a well-established practice for the conservation of biological diversity and ecosystem management at various spatial scales. The goal is to maintain ecosystems that are resilient and self-sustaining as well as supporting sustainable livelihoods of people. The following attributes provide basis for good ecological restoration (SER, 2004):

Effective in restoring and maintaining ecological integrity.

Efficient in using practical and economical methods to achieve functional success.

Engaging through implementing inclusive processes and by recognizing and embracing interrelationships between culture and restoration.

Ecological restoration is a powerful tool in facilitating the inherent recovery of ecosystem integrity in terms of biodiversity characteristics and ecosystem processes. Many healthy ecosystems are the product of human endeavours over very long periods of time.

28.4 LINKING RESTORATION WITH HUMAN WELL-BEING AND SERVICES

The terms ecosystem integrity and ecosystem health are commonly used to describe the desired state of a restored ecosystem based on Lamb and Gilmour (2003), Fig. 28.4. Ecosystem integrity is the state and condition of an ecosystem that displays the biodiversity characteristics in terms of species composition, structure and function. Given that many people depend on the degraded ecosystems to sustain their livelihoods, ecological restoration needs to address biodiversity conservation, human livelihood, empowerment of local people and ecosystem productivity (IUCN, 2002). Ecological restoration has as its goal an ecosystem that is resilient and self

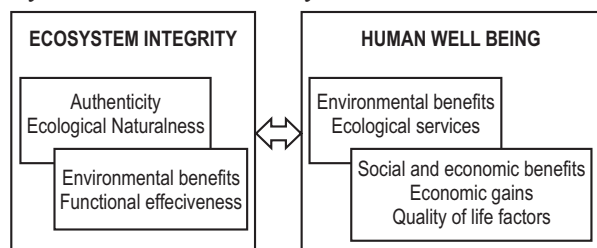


Fig. 28.4. The ecological restoration practices focus on balancing ecological integrity and human well-being. Ecological restoration has as its goal an ecosystem that is resilient and self sustaining with respect to structure and functional effectiveness, integrated with the natural landscape. It also aims to support sustainable livelihoods by improving environmental benefits and ecological services and providing social and economic benefits to people.

sustaining with respect to structure and functional effectiveness, integrated with the natural landscape. It also aims to support sustainable livelihoods by improving environmental benefits and ecological services and providing social and economic benefits to people.

According to Palmer and Filoso (2009), restoration of ecosystem services that can be bought, sold, or traded requires actions (use of “tools”) that directly influence biophysical processes. These processes interact with ecosystem features and are collectively called ecosystem functions or ecological processes (Fig. 28.5). To ensure the value of an ecosystem service credit, direct measurements of the service or ecosystem functions that support the service must be made. For some services and landscape settings, researchers are only now identifying the most relevant ecological processes (Palmer and Filoso, 2009).

Restoration ecology provides opportunities to link environmental science, technology, and engineering within a socially relevant context. Ecological restoration is fast becoming a critical component of nature. In common with ecological restoration, natural capital restoration is intended to improve the health, integrity, and self-sustainability of ecosystems for all living things according to the Society for Ecological Restoration International. The restoration of natural capital is an activity that integrates investment in and replenishment of natural capital stocks to improve the flows of ecosystem goods and services, while enhancing all aspects of human well-being.

28.5 STRATEGIES OF RESTORATION

According to Hobbs (2007), in order to restore an ecosystem, we need to (i) understand how it worked before it was modified or degraded, and then use this understanding to reassemble it and reinstate essential processes, (ii) identify the type and extent of damage, for, the type of intervention required in restoration depends heavily on the type and extent of damage to the ecosystem. In some cases, relatively small changes to management or manipulation of the species composition are required (e.g., removal of harmful invasive species or replacement of missing species). In others, substantial alteration of the physical and/or chemical environment, such as hydrology and nutrient dynamics, may be needed to restore ecosystem or landscape processes (Hobbs, 2007).

There are basically three strategies for ecological restoration, depending upon the degree of degradation.

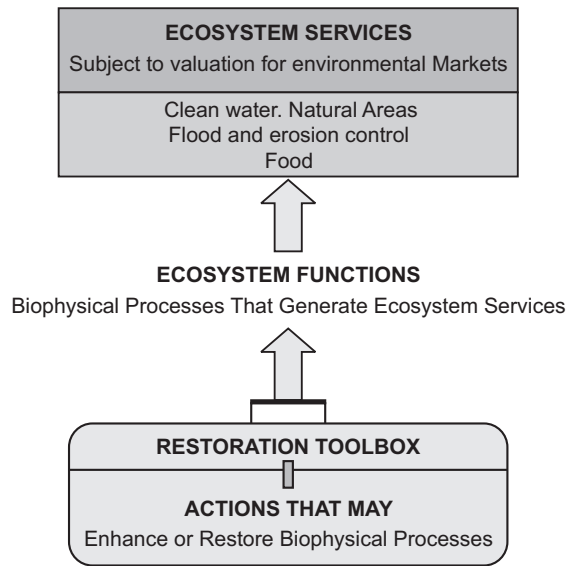


Fig. 28.5. Restoration of ecosystem services that can be bought, sold, or traded requires use of restoration “tools” that directly influence biophysical processes. These processes interact with ecosystem functions or ecological processes. To ensure the value of an ecosystem service credit, direct measurements of the service or ecosystem functions that support the service is needed. (modified from Palmer and Filoso, 2009).

1. Passive restoration or Natural Recovery

When only vegetation has been destroyed and soil is in place, natural recovery along the pathway of ecological succession is easy. The disturbed area is abandoned in conditions designed to be habitable to organisms that were originally present. The damaging factor is withdrawn and in time the original vegetation re-establishes (Fig. 28.6). For example, a mature forest can be converted to a secondary forest or to a temporary cropland (such as in slash and burn agriculture), without irreversible damage to the soil. When such transformed land is abandoned, ecological succession moves back the site to a mature forest stage.

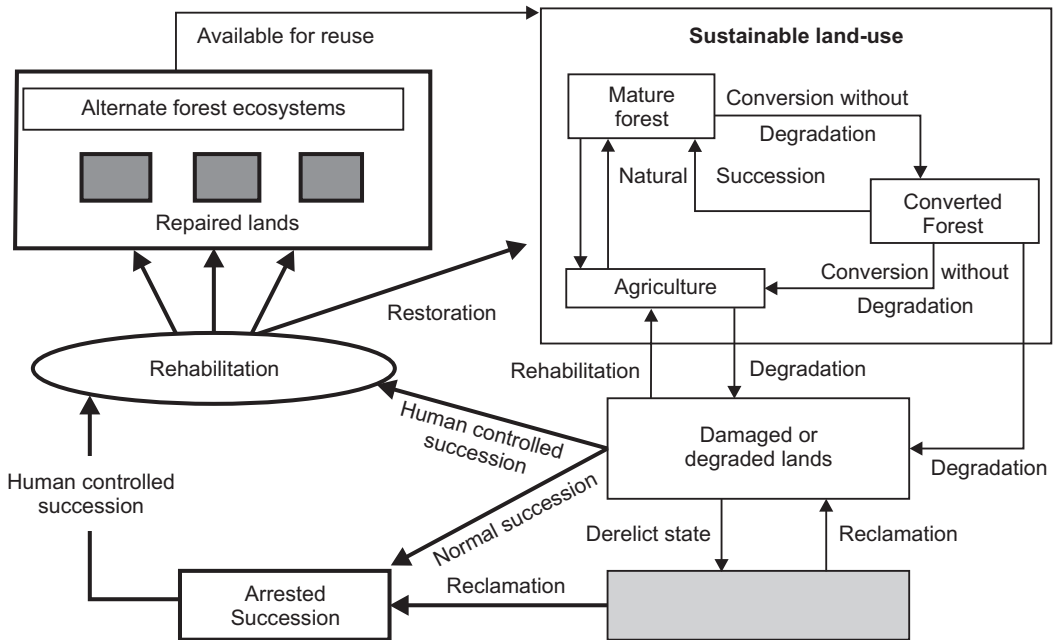


Fig. 28.6. Conversion of mature forest showing pathways that lead to sustainable land-use, damaged, degraded or derelict lands, arrested succession, and repaired lands (based on Brown and Lugo, 1994).

The restoration of natural flooding regimes of degraded prairie pothole communities and prescribed burning in degraded grasslands have been found to be successful to promote restoration of native plant assemblies (Suding et al., 2004). This successional-based approach has a focus to reestablish the historical physical environment to promote natural succession processes so as to return the biotic system to its original condition. In other cases, restoration based on successional recovery has been found to be unpredictable as the degraded systems shift to new state that cannot be restored to historical conditions through reestablishing physical condition or disturbance regimes (see Suding et al., 2004). Other limitations like climate change, fragmentation, exotic species, altered hydrology and other biophysical conditions need to be addressed.

In the case of wetlands, the passive restoration involves the removal of the factors that cause wetland degradation or loss, rest of the work is done by nature itself. For example, if cattle grazing are harming the wetland, then remove the cattle. It will allow natural regeneration of wetland plant communities, natural re-colonization by animals and re-establishment of wetland hydrology. Passive approach is appropriate if degraded sites retain basic wetland characteristics. In the case of aquatic ecosystems, the rate of establishment of population can be rapid if the disturbing factor is removed. But on land, natural succession is a rather slow process. The most restorable sites are those with some remnant of natural topography, hydrology and species composition.

2. Full or active restoration

This is accurate recreation of site conditions existing prior to the disturbance activity. The process is knowledge, money and time intensive. This is a laudable goal but is beset with several difficulties : (i) inadequate information about the original system, (ii) lack of adequate source of species if the original system was unique, (iii) difficulty in putting a halt to factors causing the damage (as acid rain may originate in a far away place or in another country), (iv) original community may be a result of processes which are no longer operative, (v) difficulty in duplicating the sequence of species introductions and disappearances. Practically speaking, recreation of historic community may not be always feasible (Cairns, 1991). Full restoration, therefore, is almost impossible to achieve in a majority of cases.

3. Rehabilitation

Disturbed ecosystem is restored to a form and productivity conforming to a pre-decided use plan defining ecosystem components of value. This involves repair of the damaged ecosystem and restores selected ecological attributes of the site, making the system functional and stable (Singh and Jha, 1993). Once the damaged ecosystem is rehabilitated, it can be further directed towards; full restoration, or can be maintained as alternate ecosystem depending upon the societal needs. If the land is derelict or badly degraded, reclamation may be necessary before rehabilitation task can be taken up taking in to consideration the human benefits in terms of productivity (Fig. 28.6).

Rehabilitation has several advantages : (i) It converts unproductive land to self-perpetuating ecosystem, (ii) it prevents further damage to adjoining ecosystems (e.g. downstream aquatic systems), (iii) it reverses the negative trend of land degradation, and (iv) it removes pressure on undisturbed lands (Brown and Lugo, 1994).

In this book we shall use the terms restoration and rehabilitation as synonyms.

28.6 RESTORATION PLAN AND REHABILITATIVE MEASURES

The process of restoration towards its particular target or its former state depends on the level of knowledge of the former system, availability of biota for restoration, genetic variation of biota, the level of alteration of hydrology, soil and geomorphology, cost and availability of funding, public policy and sociocultural values. A general framework for the restoration is shown in Fig. 28.7.

It should be recognized that the natural resources available for restoration at and in the vicinity of a disturbed site are climate, soil and biota. A complete data set should be collected on these resources. The restoration plan needs to address the following questions:

1. What was the ecosystem / land use prior to degradation?
2. What has caused the degradation in the first place and what components of the ecosystem have been affected, and to what extent?
3. How the causative factor (s) can be removed during and in the post-restoration period?
4. What is the biological potential of the site, and

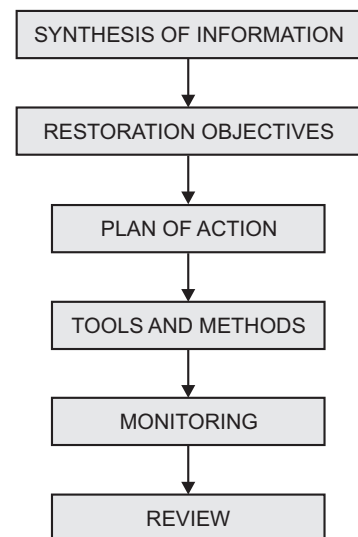


Fig. 28.7. Generalized Framework For Restoration of Degraded Ecosystem.

what type of vegetation cover will give the most productive use for each unit of land?

5. What factors are likely to limit the establishment and growth of recovering vegetation and how can these be removed?
6. What is the general successional sequence leading to natural recovery, and how can this rate be accelerated?
7. What after care will be required?

Depending upon the societal and ecological needs the goal and the end point for restoration is set i.e. whether the site should be restored to wilderness, pastoral land, forestry or agriculture or a combination of these. Frequently, if the area is large, multiple end points comprising a mosaic of ecological communities (natural, near-natural, and semi natural) are more desirable. Montaya et al., (2012) argue that not all species contribute equally to ecosystem function; rather, some species are functionally more 'important' than others; it is these functionally important species that should be prioritized in restoration programs. Further, restoration projects should consider functional traits of species and should aim to reintroduce the dominant species of key functional groups first, rather than the number of species, and then restore ecological redundancy by the addition of species that vary in functional traits. Different restoration aims require different assembly histories, and that different sequences of species seeding, planting, and removal, will be critical when designing restoration programs.

The following rehabilitative measures are built up in the restoration plan to speed up the process of regeneration :

1. Assist the natural immigration process by (a) spreading top soil from adjoining intact ecosystem to augment the seed bank, (b) hand seeding or seed broadcasting or (c) planting of nursery raised seedlings. The choice of species is important and will depend on the goal of restoration and ecological conditions of the site. When in doubt regarding the suitability of species, use mixed seeding, the site will ultimately select the species complement. Introduce keystone species.
2. Assist establishment of species by drilling, cultivation or dibbling (to cover the seeds), spreading mulch to create microsites for germination, and sow nurse crop to ameliorate excessive exposure. Sometimes hydro seeding (seeds applied to surface suspended in liquid mixture containing organic mulch, a stabilizer and nutrients) helps in better establishment.
3. Assist the accumulation of fine particles by (a) spreading top soil or fine material, and (b) introducing nurse plants which can capture wind-blown loose material and prevent loss of existing fine particles through wind or runoff.
4. Assist the accumulation of mineral nutrients (e.g., phosphorus) by adding the deficient elements as chemical fertilizers or sewage sludge.
5. Assist the accumulation of nitrogen by repeated application of N-fertilizers, or digested sewage sludge. Seeding or planting the site with N-fixing legumes is a more sustainable measure.
6. Assist the immigration of soil organisms by (a) inoculating seeds / seedlings with rhizobia and mycorrhizal fungi or scattering appropriate soil as inoculum, and (b) introducing earthworms.
7. Assist in improving water holding capacity by adding top soil or/and organic mulch, by using farmyard manure in tree-planting pits or by using hydrophilic polymers.
8. Assist in detoxification, for example reduce acidity by application of lime, and select acid-

tolerant species for seeding or planting. Metal decontamination can be achieved through phytoremediation (phytoaccumulation, phytoextraction, phytostabilization, phytovolatilization, and rhizofiltration) (see Prasad, 2004). Adding organic matter or covering the contaminated material with uncontaminated soil can be of temporary relief.

9. Regulate the speed of ecosystem processes, e.g. nutrient release and decomposition can be regulated by managing the quality and quantity of organic matter inputs by selecting appropriate species.
10. Control weeds during the early stages of revegetation, to keep the succession on desired trajectory.
11. Permit invasion / introduction of desired consumers while keeping the site free from undesired consumers.
12. Remove high impact stressors, for example, hydrology can be changed by draining or damming, and topography can be modified by terracing to control erosion.

While the restoration effort is in progress or even afterwards, stressors such as high fire frequency, over grazing, biomass extraction, etc. need to be alienated or controlled. The physical and chemical problems that can be found in the soils of degraded terrestrial ecosystems and their short and long term treatments are briefly listed in Table 28.1; the long- term treatments are closer to the natural ecological processes and are most likely to be self-sustaining.

Table 28.1. The physical and chemical problems that can be found in the soils of degraded terrestrial ecosystems and their short and long term treatments; the long- term treatment are closer to the natural ecological processes and are most likely to be self-sustaining. (adapted from Dobson et al., 1997).

Category	Problem	Immediate Treatment	Long-term treatment
Physical			
Texture	Coarse Fine	Organic matter or fines Organic matter	Vegetation Vegetation
Structure	Compact Loose	Rip or scarify compact	Vegetation Vegetation
Stability	Unstable	Stabilizer/ nurse	Regrade/ vegetation
Moisture	Wet Dry	Drain Irrigate or mulch	Drain Tolerant vegetation
Nutrition			
Macronutrients	Nitrogen Others	Fertilizer Fertilizer and lime	N-fixing species Fertilizer and lime
Micronutrients	Deficient	Fertilizer	—
Toxicity			
pH	Low High	Lime Pyretic waste or organic matter	Lime or tolerant species Weathering
Heavy metals	High	Organic matter or tolerant plants	Inert covering or bioremediation
Salinity	High	Weathering or irrigate	Weathering or tolerant species

Monitoring and Review

Keeping in view the multiplicity of ecological variables, objectives are evaluated on the basis of *performance Standards*, also known as design criteria or success criteria based on an understanding of the reference ecosystem (SER, 2002).

Objectives, performance standards, and protocols for monitoring and for data assessment are incorporated into restoration plans prior to the start of a project. If interpretation of the data collected during monitoring shows that performance standards have been met, then the project objectives indicate that the restored ecosystem has sufficient resilience and requires little or no further assistance.

Three strategies exist for conducting an evaluation: direct comparison, attribute analysis and trajectory analysis (see SER, 2002).

Direct comparison: selected parameters are determined or measured in the reference and restoration sites in terms of both the biota and the abiotic environment on the basis of sufficient variables to describe an ecosystem fully.

Attribute analysis: It is based on quantitative and semi-quantitative data from scheduled monitoring and other inventories that are useful in judging the degree to which each goal has been achieved.

Trajectory analysis: is a promising strategy, still under development, for interpreting large sets of comparative data. In this strategy, data collected periodically at the restoration site are plotted to establish trends pertaining to ecological variables as well as cultural and socio-economic goals.

According to Montoya et al. (2012), for monitoring species interactions, some relatively quick and cost-efficient ways include sampling during peak seasons, focusing on network-stability properties that change little with sampling effort, and, for pollination networks, sampling using transects instead of timed observations.

Plant and animal reintroductions

During full or partial replacement of biological populations, the emphasis is on species of plants or animals that have a social, economic or ecological value. Some successful examples of returning species to the wild are:

- (a) Returning peregrine falcons to suitable habitats in the Eastern United States.
- (b) Returning Arabian Oryx to Saudi Arabia.
- (c) The return of California condor to southern California.

Animal reintroductions to the wild have been successfully implemented in many cases. Griffith et al. (1989) have compiled information about re-introduction attempts of 198 bird and mammal species. Some successful cases have included the following categories.

- (a) The game species have their tolerance curves and acclimation abilities generally within the range of variation of their environment into which they are being introduced. Therefore, it is easier to introduce game species compared to the threatened species.
- (b) Introductions within the normal geographical range are more successful as compared to outside the range.
- (c) Wild caught animals are reintroduced better than captive raised animals. In other words, wild animals have a greater range of genetic variation than those of captive animals.
- (d) Herbivores have greater availabilities of food and a smaller home range. This results in greater successes with reintroduction of herbivores as compared to carnivores.
- (e) The number of animals also determines the success of reintroduction. The greater the number of animals, the better the chances of survival in wild.

Animal control

By removing exotic species, it is possible to rebuild a community for protecting native and endangered species. Some examples are:

- (a) Salt cedars dry out riparian habitats and crowd out native vegetation. By removing salt cedars has helped to protect the endangered Coachella valley fringe – toed lizard in California.
- (b) In Hawaii, feral pigs root out native plant species as well as eat native birds. Hunting and removing of feral pigs can help protect the native plants.
- (c) In Galapagos Islands, feral goats and rats are being removed to protect native species.

Creating artificial ecosystems

On highly disturbed sites, ‘designer’ ecosystems’ could serve the specific purpose for humans. The development of designer ecosystems may prove to be useful to halting a trend of degradation or habitat/ species loss (MacMahon, 1998). Some natural principles of ecosystem functioning have been used in creating designer ecosystems. A few examples are given below:

1. In Arcata, California, an artificial marsh has been created. Waste water from conventional treatment processes flows through oxidation ponds, where sunlight and pathogenic bacteria can kill pathogenic microbes. It is a low cost ecologically based treatment system for sewage effluents.
2. On a minor scale, the naturalist animal displays in zoological gardens and the displays of various plants by vegetation type in botanical gardens are a kind of designer ecosystems.
3. Coffee plantation of the traditional shade type is designed to meet the productive goals for human. But they also are harboring variety of threatened species. These sites are often rich in native species as they respond to artificially layered structure (Greenberg et al., 1997).
4. A number of forest sites have been ecologically engineered to hide industrial complexes, control pollution and stabilize soil (Miyawaki and Golley, 1993).
5. Created or constructed wetlands have been developed on site previously supporting natural vegetation.
6. Large scale recreation of forest, prairies and riparian ecosystem.

Revegetation in restoration

The major purposes of revegetating a restoration site are: 1) to control sediment erosion and transport, 2) to enhance habitat, and 3) to provide viable habitats for species that are in decline. Vegetation is critical in stabilizing soil materials. Plantations can play a critical role in restoring productivity, ecosystem stability and biological diversity of degraded areas (Singh et al., 2002).

Before revegetation, it is important to ensure that the physical characteristics of the landscape – e.g., slope, aspect, soil texture, organic material content, and depth of water table – are appropriate for the site location and will serve as a suitable planting medium for the desired vegetative community.

Disturbed sites are prime invasion locations for exotic species, which generally are faster and more adept at colonizing bare areas than native species. In addition, restored disturbed areas are also prime invasion locations, especially if the restoration activities involve significant earth disturbance. Therefore, it is important to decide about planting native plant species.

It is useful to use the locally adapted plant populations (e.g., ecotypes) in revegetation efforts. Revegetating with plants raised in a slightly different environment than that of the restoration site might lead to lower plant survival rates. Therefore, plants must be raised with organic and fertilizer

amendments at the restoration sites. The maintenance of the gene pools of natural areas near the restoration site is especially relevant when restoring sensitive habitats, reintroducing rare plant species and when working with plants which hybridize easily.

Ecological engineering

Ecotechnology or ecological engineering can be used for planning and prudently using nature. Restoration of ecosystems is thus an important component of ecological engineering. The aim is to recover or rehabilitate natural ecosystems that have already been damaged, based on a reasonable understanding of ecological principles. According to Mitsch and Jorgensen (1989), some important considerations in ecological engineering are as follows:

1. Ecosystems are used to solve or reduce pollution problems that would otherwise be harmful to other ecosystems. Examples are waste water recycling in terrestrial ecosystems or wild and sludge disposal on agricultural land.
2. Ecosystems are imitated to solve a pollution problem, such as, integrated fish ponds and creation of artificial wetlands.
3. Incineration is a cost effective method of cleaning oil contaminated soils.
4. Pumping of ground water and aeration helps to remove volatile organic compounds from the aquifers.
5. Phytoremediation: plants can be highly effective agents for cleaning many contaminants. Water hyacinth can absorb heavy metals and other toxins from polluted water. Surface pollution can be ameliorated by growing suitable plants and microbes. Constructed wetlands could be utilized to treat water pollution without consideration of species composition or overall ecological integrity.
6. Removal of chlorinated hydrocarbons: Microbes can be used or engineered in laboratory to destroy many chlorinated hydrocarbons.
7. Removal of chemicals from industrial effluents by using chopped up horsetails can be quite effective.

28.7 CENTRAL CONCERNS IN ECOLOGICAL RESTORATION

The National Research Council (NRC, 1992) has developed checklist of questions for restoration projects during the project planning and design as well as post restoration phase. Some basic concerns common to all types of restoration projects (NRC, 1992; MacMahon, 1998) include (i) the product, (ii) feasibility and authenticity (iii) scales and (iv) costs, which are briefly described below:

The Product

The basic goals in restoration are to return a particular habitat or ecosystem to conditions as similar as possible to its pre-degraded state. In a degraded ecosystem, the structure and function can be shown as the two axes (Fig. 28.2). Both the components of the ecosystems that are affected need to be restored. The restoration can be practiced at several levels depending upon the view point and objectives:

Habitat restoration: There are two distinct usages of the term habitat, one that is organism specific and another that is land based. The land-based approach believes that if habitat (substrate, hydrology, etc.) is restored, then a floral and faunal community appropriate to the restored habitat will automatically colonise the site. Therefore, habitat restoration should be the first step in every restoration programme. However, the organism specific approach considers that the first step in the process of restoration is identifying a focal or target species or group of species. The next step is to

identify the biotic and abiotic resources that are required by the species to persist. Then one has to ensure availability of resources through time (Miller and Hobbs, 2007).

Species / Community restoration: The emphasis is to reintroduce or augment the plant or animal species, whose population may have declined. Enrichment planting of a degraded forest or seeding of a degraded pasture are examples. However, just bringing together species may not result in a persistent biological community with a particular composition. The natural process of dispersal and colonization is a more successful technique for reintroducing a variety of species in wild rather than traditional seeding methods. It is useful to take in to account “safe site” for a successful reestablishment keeping in view the year-to-year variability of the natural ecosystems dynamics. The establishment phase is important for a successful restoration, which is consistent with the succession theory and life history strategies (Chambers and McMahon, 1994).

Quality restoration: Emphasis is placed on the restoration of the quality of the system. Examples are reducing the BOD level of a polluted water body, decontamination of soil or enrichment of biodiversity.

Restoration of ecosystem and ecosystem services: In this emphasis is on the restoration of the fundamental processes which permit the ecosystems to function. It may be possible to restore the ecological functions fairly completely. However, to achieve an original structure may be more difficult. For example, in the forest ecosystem, it may not be possible to achieve the original structure in less than 500 years or soil profile in less than 5000 years. It is possible to restore biological functions of soil in a short span of time of only ten years (Bradshaw, 1997b).

CBD, the Millennium Development Goals, and the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, link the conservation of biodiversity with the provision of ecosystem services to support sustainable development and poverty reduction (Sachs et al., 2009). Ecological restoration is assuming an increasingly central role in global environmental policy. For example, the 2020 Targets developed recently by the CBD aim for the restoration of ecosystems that provide essential services (Target 14), and the enhancement of the contribution of biodiversity to carbon stocks through restoration of at least 15% of degraded ecosystems (Target 15). In fact The CBD declared recently that the ‘restoration of terrestrial, inland water and marine ecosystems will be needed to re-establish ecosystem functioning and the provision of valuable ecosystem services’ (Normile, 2010). MEA (2005d) suggests that biodiversity and the provision of ecosystem services are positively related, thus management to enhance one should increase the other. A meta-analysis of 89 restoration projects indicated correlated increases in biodiversity and ecosystem services in restored compared with degraded systems (Rey Benayas, et al., 2009). The restoration of populations and, more generally, of the ecosystem services provided by species and their interactions, is a critical step for the recovery of important services, such as seed dispersal, pollination, and pest control (Montoya et al., 2012). However, analysis of empirical evidence shows that the relationship between biodiversity and ecosystem services is complex and not always positive (Bullock et al., 2011). Further, the meta-analysis of Rey Benayas et al., (2009) that included a range of biomes found that restoration was only partially successful in achieving reference conditions, as restored systems had median measures of 86% of the biodiversity and 80% of the services associated with reference ecosystems.

Feasibility and authenticity

The main concern is how feasible it is to restore an authentic ecosystem? The feasibility of restoration differs with ecosystem type as well as the nature and extent of degradation involved. The goal is how closely a restored ecosystem resembles the model or reference system dynamics. Authenticity in a larger context addresses to its historical and aesthetic value. The overall aim is to identify those aspects of ecosystems that are most important.

The feasibility of restoration varies markedly from system to system. Under favourable conditions, restoration of ecosystems has been achieved successfully, such as in the case of the Curtis Prairie at the University of Wisconsin Arboretum, Madison and the restoration of the estuary of the Salmon River in Oregon. The tidal wetlands having a few species of plants and simple structure can be restored quite easily under favourable conditions. However, ecosystems like peat bogs, where peat has been removed or disturbed are difficult to restore.

Scale

It relates to both temporal and spatial scales of the project, which determines the quality and authenticity of restored ecosystems (MacMahon, 1998). Some important considerations concerning the scale are as follows:

- Large project size to minimize the effect of boundary conditions and interval dynamics.
- To control necessary disturbances to the system.
- Monitoring to assess the project success.
- Affordable size.

On the basis of planning and implementation, three different scales of the projects have been recognized (NRC, 1992; MacMahon, 1998), which are:

- (i) Small scale (e.g., the introduction of a few species in the wild),
- (ii) Middle scale (e.g., to reclaim the mine spoils), and
- (iii) Large scale (e.g., an entire river system and associated wetlands)

The ideal situation is to work with a high degree of detail at large scale for a long time (NRC, 1992). On a large scale, it may be possible to restore ecosystems of high ecological quality based on the principle of island biogeography (MacMahon and Jordan, 1997). A large size has a greater functional infrastructure with respect to soil heterogeneity, exposures, and hydrological regimes. Large sites can support more species and have greater capacity of self repair than do the smaller sites. The abiotic system is also fairly easily repairable. Large scale projects can also improve gradually through “subsidized succession” (MacMahon and Jordan, 1997).

The restoration projects undertaken on large scale (hundreds to thousands of hectares) typically involve the reintroduction, removal or manipulation of ecosystem processes such as burning, grazing and hydrological processes. The restoration of a tropical dry forest at GuanaCaste National Park in Costa Rica involved control of grazing and burning to reverse the effects of centuries of these practices. In the case of large scale wetland projects, suitable hydrological regimes have been restored to ensure gradual recovery of the community (Larsen, 1996; Collier et al., 1997).

Cost and Benefits

Cost of the restoration activities places a price on the ecosystem. It also links ecology with human activity and determines the realistic limits on the scope of the project. The cost varies widely depending upon the scale of the restoration project, available land use options and objective of the project. It may be mentioned that the cost is defined realistically within the proper financial bounds. Restoration process can be costly. Recovery of contaminated wetlands in California exceeded US \$500,000 per ha, more commonly however, the costs are up to US \$ 20,000 per ha depending upon the degree of damage or contamination (Edwards and Abivardi, 1997).

There are several ways in which economic benefits arising out of restoration of a given ecosystem can be assessed (Filion and Adamowicz, 1994). Some of these methods are briefly described below:

- 1. Market price output:** The actual value of goods (e.g., fish, timber products) is calculated.

2. **Actual expenditure:** The actual expenditure on goods and services related to use of the area (e.g., ecotourism) is computed.
3. **Replacement cost:** Economic benefit for a given good or service (e.g., construction of a dike for flood control as replacement value of wetland) is calculated using a tradable substitute.
4. **Travel cost:** The value that visitors place on a site is estimated by the time and money spent by them to visit the area.
5. **Hedonic methods:** Value of an area is estimated by comparing the prices of surrogate goods, e.g., increased property price close to the restored area.
6. **Contingent valuation:** Value placed by the people upon a resource such as a restored ecosystem in terms of willingness to pay is assessed through questionnaires.

28.8 DEFINING RESTORATION SUCCESS AND EFFECTIVENESS

While defining standards for ecologically successful river restoration, Palmer et al., (2005) emphasized that stakeholder success has a focus on human satisfaction, whereas learning success focuses on scientific knowledge and management practices that will benefit restoration action. Ideally, implementation of national and international programmes to evaluate success in restoration would not only advance our understanding of how best to restore streams and rivers, but would also influence the expectations and goals of stakeholders (Palmer et al., 2005). For restoration to be effective, it is important to understand the interaction between ecological success, learning success and stakeholder success (Fig. 28.8). Recently, Sayer et al. (2013) have provided 10 summary principles to support implementation of a landscape approach for restoration, emphasizing adaptive management, stakeholder involvement, and multiple objectives. These principles can address landscape challenges more effectively and include the following : (i) Continual learning and adaptive management, (ii) Common concern entry point, (iii) Multiple scales, (iv) Multifunctionality, (v) Multiple stakeholders, (vi) Negotiated and transparent change logic, (vii) Clarification of rights and responsibilities, (viii) Participatory and user-friendly monitoring, (ix) Resilience, (x) Strengthened stakeholder capacity. These principles of the landscape approach are an innovation that should help address the challenge of increasing agricultural production while minimizing negative impacts on the environment (Sayer et al., 2013).

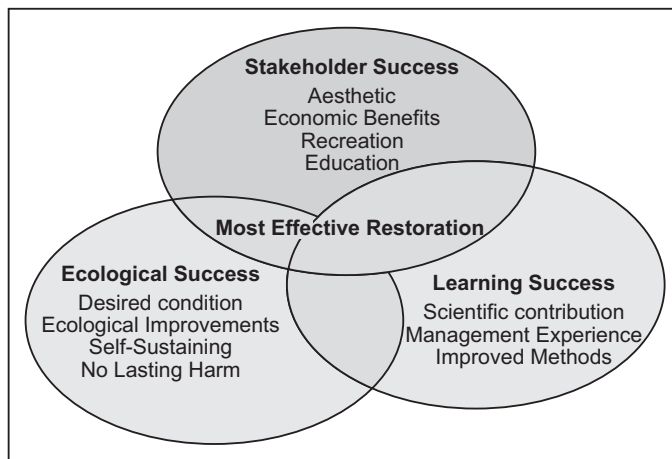


Fig. 28.8. The most effective river restoration projects is determined by the intersection of the three primary axes of success, i.e., ecological success, stakeholder success, learning success. (redrawn from Palmer et al., 2005; Journal of Applied Ecology 42 : 208-217).

28.9 SCOPE OF GLOBAL RESTORATION ACTIVITIES

Restoration projects in different regions of World

The areas of degraded land now present in various parts of the world are large and some of the ecosystems are severely degraded and will be costly to repair. We may not succeed in fully eradicating the causes of degradation in these circumstances but there is sufficient evidence from a variety of case studies to be optimistic about eradicating the causes of degradation and improve the livelihood opportunities for the poor (Hobbs and Cramer, 2008). In different regions of the world, restoration efforts are underway in forest, grassland, freshwater ecosystems, and wetlands for restoring biodiversity, ecosystem services and mitigating the risk of global change including exotic species invasions. The various restoration projects range in size from local to regional scales using site specific abiotic and biotic interventions. Hobbs and Cramer (2008) have listed the following restoration projects as being successfully implemented in different regions of the world:

- Restoring fire and grazing regimes to prairie remnants in the Midwest of North America.
- Restoring surface-mined areas in forests in southwestern Australia to return a forest ecosystem.
- Restoring rainforest ecosystems in areas formerly deforested in Costa Rica.
- Restoring woodland cover to large areas of Scotland.
- Restoring water flows to the Mesopotamian Marshes in southern Iraq.
- Restoring plant cover in arid lands in Africa and elsewhere those have been degraded through overgrazing, overuse.
- Restoring rivers and water flows in southern Florida and the internationally important Everglades National Park.

In different projects, ecological restoration has improved the biological diversity on degraded landscapes, increased the populations and distribution of rare and threatened species, enhanced the availability of environmental goods and services, and has contributed to the improvement of human well-being in various large scale projects in different parts of the world. Thus, ecological restoration will be a key element not only of conservation but also for sustainable development worldwide

Forest landscape restoration

Forest landscape restoration as: *“a planned process that aims to regain ecological integrity and enhance human wellbeing in deforested or degraded landscapes”*.

Forest landscape restoration brings people together to identify, negotiate and implement practices that restore an agreed optimal balance of the ecological, social and economic benefits of forests and trees within a broader pattern of land uses (Box 28.1).

More than 400 million hectares of southern and eastern Asia's deforested and degraded landscapes offer potential for restoration for alleviating poverty, improve food security, reduce climate change, and conserve biodiversity. The plan of action for Harapan Rainforest in Indonesia for ecosystem restoration is summarized in Fig. 28.9 (www.birdlife.org/forests). The salient features of the Harapan Rainforest project are:

- Business models that make forest conservation economically viable
- Development and implementation plan
- Demonstrate impact for biodiversity, people and climate
- Legislative frameworks that support sustainable forest conservation
- New models that allow effective management, local participation and benefit sharing

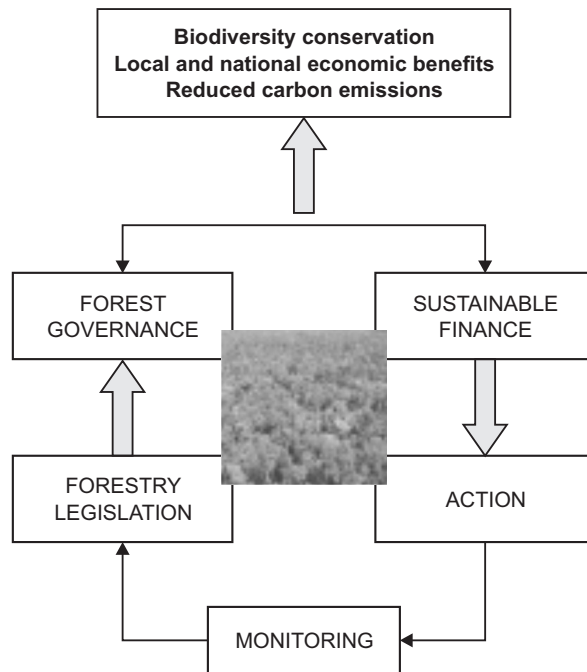


Fig. 28.9. Harapan Rainforest for Ecosystem Restoration in Indonesia (adapted from The BirdLife Forests of Hope Programme : www.birdlife.org/forests).

Box 28.1. Forest Landscape Restoration

The Global Restoration Initiative and the Bonn Challenge: Restoring 150 million hectares by 2020 (www.wri.org/restoring-forests). This is as a practical, action-orientated platform to facilitate the implementation of several existing international commitments that require restoration, including the CBD Aichi Target 15, the UNFCCC REDD+ goal and the Rio+20 land degradation target. IUCN estimates that the annual net benefit to national and local economies of restoring 150 million hectares is approximately US\$ 85 billion/year. This could also sequester an additional 1 Gt CO₂e per year, reducing the current emissions gap by 11-17% (The Bonn Challenge on forests, climate and biodiversity (www.ideastransformlandscapes.org)).

28.10 EXAMPLES OF TERRESTRIAL ECOSYSTEM RESTORATION

Restoration of Curtis Prairie

Prairies once covered 24% of total land cover worldwide (Howell and Kline, 1987). The tall grass prairie occurs under high rainfall conditions and has greater species richness and dominant species than the short grass or mixed grass prairie. Currently, the tall grass prairie ecosystem is one of the most threatened ecosystems in the mid west United States. Most of fertile prairie soils have been converted to agriculture. In the remaining prairies, most are dry prairies found on steep slopes and the mesic prairies. Most of the remnant prairies are severely degraded due to invasion by woody species, overgrazing, and poor management practices having negative impact on biodiversity.

The best restored prairies at the University of Wisconsin, Arboretum have been planted using the native grasses and annual plants manually, which ultimately became the University of Wisconsin's restored Curtis Prairie (Primack, 1996).

Habitat Restoration

For providing habitat for the endangered Bermuda Cahow (hook billed petrel), a nearly vanished ecosystem was created by replanting forests and digging ponds. The Cahow is a seabird endemic to Bermuda and adjacent islands off the Coast of North America. It is ground nesting and never experienced predation. Also, the bird had no defense against the exotic species. Over hunting and habitat destruction decimated the population of cahow. By 1600s, the bird disappeared from Bermuda and it was assumed to be extinct. In 1951, scientists found a few living Cahows in Bermuda harbour. A protection and recovery programme was started on a 6-hectare sanctuary on Nonsuch Island. Reintroduction of native vegetation and recreation of habitat resulted in the establishment of the natural community. In a few years, this became the most successful example of environmental restoration. There has been considerable recovery of indigenous species and the recovery of the Cahow populations amounted to nearly fifty nesting sites by 1990.

Restoration of Longleaf Pine Forest Landscape

Longleaf pine (*Pinus palustris*) forest ecosystems provide a good case history for restoration of forest landscapes. Longleaf pine forests now covering only less than 10% of the original forests once covered 60% of the area in Southeastern US coastal plain, stretching from North Carolina to Texas (Johnson and Gjerstad, 1998, see McCarty and Zedler, 2002). Longleaf pine forests are characterized by unique diversity of plant and animal species, including a large number of threatened and endangered species. These ecosystems are maintained by periodic fires carried by understory plants which prevent invasion by deciduous oaks. This valuable ecosystem has been restored by reintroducing fire and establishing populations of indigenous plants and animals. The US Department of Defense military bases alone manage approximately 400000 ha of longleaf-type vegetation (Johnson and Gjerstad, 1998).

Restoring a Dry Tropical forest: Guanacaste National Park

In Central America, dry tropical forests are one of the most threatened ecosystems. Due to the loss of forest cover, many of the unique plant and animal species have become rare or threatened. In Costa Rica, a new national park (now an international Biosphere Reserve) has been created. A restoration project is under way in a dry tropical forest under the leadership of David Janzen (professor of Biology at the University of Pennsylvania). Both scientists and local residents are participating to restore about 700 km² of dry tropical forest to its approximate original condition and relinking it to the rain forest. The Guanacaste National Park has been named, after the Costa Rican National tree (*Enterolobium cyclocarpum*, Fabaceae), that once grew in this forest. Studies have been undertaken to understand the ecosystem and to introduce native plants and animals for the restoration of the forest (see Janzen, 1988 b). The process of restoration involves the following steps:

- (a) To find indigenous species of the forest which are scattered across the countryside.
- (b) Creating habitat for the growth of plants.
- (c) Creating breaks to control the spread of fires with the help of local residents.
- (d) Livestock grazing is allowed in the National Park for efficient seed dispersal.
- (e) The project involves local people as native people are an invaluable resource in the restoration process through their knowledge of the forest and their skills as land stewards.
- (f) Providing jobs to local residents will help preserve the local culture along with saving biodiversity and cultural heritage.

It is estimated that it will take about 20 years to convert unused grassland to a closed canopy forest in an area of approximately 70,000 ha of ancient fields and pastures (see Janzen 1988 b, 2002). In about 200 years, it will look like the seasonal tropical forest that was once there due to elimination of fires. Within one half of a millennium the area should be indistinguishable from the original forest, except for those changes that are inevitable through climate change. This project is based on biocultural restoration as local residents are a part of the restoration project. The local people will get educational, economic and environmental benefits. This can serve as a model project, which can inspire restoration in many other tropical forest ecosystems as well as train scientists all over the world.

The Atlantic Forest Restoration Pact (AFRP): A Social Movement

The Brazilian Atlantic Forest Restoration Pact (AFRP) is one of the largest and most ambitious forest restoration programs in the world (Calmon et al., 2011). Its goals are to: restore 15 million hectares of degraded lands during the next four decades; protect and augment critical ecosystem services such as clean water for more than 120 million people ; generate more than 6 million 'green' jobs ; protect and revitalize one of the most threatened biodiversity hotspots in the world ;sustainable forestry industry based on native species and community livelihoods; Improving water security for urban areas and removing 200 million tons of atmospheric CO₂ per year.

28.11 RESTORATION OF AQUATIC ECOSYSTEMS

Aquatic ecosystems-lakes, streams, rivers and wetlands - are critical resources in any region. The ecological and societal benefits of river corridors and wetland restoration are substantial. Rivers transport water, sediment and nutrients from the land to sea, play an important role in building deltas and beaches. Rivers also regulate salinity and fertility of estuaries and coastal zones as well as provide corridors for many unique species of plants and animals (threatened and endangered species). Wetlands are a habitat for fish and wildlife species besides having recreational, commercial, ecological and economic value. Wetlands also remove pollutants from over the land flows before they reach lakes, rivers and bays.

Human activities such as damming, channelization, groundwater pumping, diversion and use as open sewers have many negative impacts on aquatic systems. Many of these problems are reversible, by removing the particular agent of destruction. For the restoration of San Juan River in South Western Colorado, understanding the natural and predictable riverine flows was the key to restoration of a system without use of heavy technology and creation of artificial state (NRC, 1992). Other degraded aquatic ecosystems may require greater restoration efforts. The conservation and management of wetlands in India need to be based on understanding the relationship of biota to variable hydrological regime and cultural eutrophication (Gopal and Chauhan, 2001).

Process of Lake Restoration

Over loading of lakes by nutrients from external source has caused eutrophication problems. In lake ecosystems, nutrients accumulate in the sediments both from organic and inorganic sources. This extra supply of nutrients causes internal loading, increased production and change in ecosystem processes at the water sediment interface. Restoration of lakes refers to the activities carried out in a lake to adjust and steer the structure and function of the ecosystem (Bjork, 1988). The lake is a dynamic system and it tends to change from a juvenile to an aging state depending on the lake size, altitude, geology and the climate. In aquatic ecosystems restoration can be achieved by removing the disturbance factor such as excessive nutrients, eliminating the source of excess phosphorus,

restarting a natural hydrological pattern or restructuring a stream channel. Restoration measures are applied by taking into consideration the interests of the local community. The goal is to achieve lasting results with minimum management cost. In the degraded lake ecosystem restorative activities are based on the following strategies.

- (a) Raising of water levels in lakes where it has been lowered.
- (b) Artificial aeration of hypolimnion through air bubbling.
- (c) Regulation of pH.
- (d) Neutralization of toxic substances.
- (e) Diversion of discharge of sewage and industrial waste water.
- (f) Sediment dredging.
- (g) Creating a mosaic of open water and emergent/submergent vegetation.

Restoration of Lake Trummen

Lake Trummen (1 km², maximum depth 1.2 m before restoration, and 4 months theoretical water retentive time), is located in southern central Sweden. The restoration of lake was based on the diversion of sewage to a control treatment plant (see Bjork, 1988). Originally, it was an oligotrophic lake. This became a humic lake due to pollution by sewage and industrial waste water. In the humic lake, winter fish kills were frequent and extremely heavy blooms of blue green algae occurred during the favourable period. Suction dredging was used to remove the soft bottom sediments causing internal nutrient loading. After a period of restoration, selective fishing was found to be useful for decreasing the concentration of phosphorus and nitrogen and phytoplankton biomass.

Bioengineering for restoring river banks

In the last five years, bioengineering has grown into one of the most popular approaches to tackling stream bank erosion. Bioengineering refers to the stabilization of soil or surface materials through the use of organic materials. Bioengineering controls erosion by establishing sustainable and native plant communities on destabilized surface materials. Vegetative cover is established by: 1) creating geomorphically stable landforms that can support vegetation, and 2) installing a combination of live, dormant, or decaying plant materials on the re-formed slopes.

The advantages to bioengineering are: (i) it is usually considerably less expensive than traditional “engineered” methods of erosion control, (ii) it is designed to be maintenance-free in the long run, (iii) it creates habitat and it enhances habitat (e.g., food and cover sources and temperature control for aquatic and terrestrial animals), (iv) it can stabilize slopes effectively in a short period of time (e.g., one growing season), (v) it is self-repairing, (vi) it filters overland runoff, increases infiltration, and attenuates flood peaks, and (vii) it creates aesthetically pleasing landscapes.

It is important to note that bioengineering, at times, may take longer to control erosion (longer establishment period, etc.) and can be labour intensive.

28.12 RESTORING DEGRADED ECOSYSTEMS IN INDIA

Restoration of surface mined lands

Throughout the world, surface mined lands cover large areas. The mining of metals and minerals represents a large range of activities from the extraction to milling, processing, refining and waste disposal. Mine spoils are drastically disturbed, nutritionally impoverished habitats. The process of natural succession on mine spoils is slow due to removal of top soil resulting in loss of

soil seed bank and root stock and due to soil profile disturbance. The rehabilitation of mine lands can be speeded up by planting suitable indigenous species with ground seeding of herbaceous flora. A desired species for planting on mine spoils should possess the ability (i) to grow on poor and dry soils, (ii) to develop the vegetation cover in a short time and to accumulate biomass rapidly, (iii) to bind soil for arresting soil erosion and checking nutrient loss, and (iv) to improve soil organic matter status and soil microbial biomass, thereby enriching the supply of plant available nutrients (Singh and Singh, 1999). On mine spoils of the Singrauli coal field in Madhya Pradesh, India, *Albizia lebbek* showed greater restoration potential than *Albizia procera*, although both are nitrogen fixing tree species indicating that choice of species is important in restoration efforts (Singh et al., 2004).

Restoring Lake Chilika

Chilika lagoon, a unique assemblage of marine, brackish and fresh water ecosystem, situated along the east coast of India is a Ramsar Site. It represents a hotspot of biodiversity, home of Irrawady dolphin. Tidal and salinity influx in to the lagoon was considerably reduced because of shoal formation (accumulation of sand and sediment along the channel). Orissa government constituted the Chilika Development Authority, which recommended opening an inlet close to the lagoon. The gradual reduction in the salinity from the mouth to the lagoon interior has provided desirable environment for the euhaline forms. The hydrological interventions for the restoration of the lagoon. The desiltation of the channel connecting the lake to the sea and opening of a new mouth to restore the natural flows of water and salinity levels by the Chilika Development Authority, Orissa. This restoration programme has resulted in improvement in fishery resources, water quality, the biodiversity and protection of the critically threatened Irrawady dolphin. A number of threatened and economic species of fish have re-appeared.

Restoring Rainforest Fragments in the Western Ghats, India

Western Ghats is a biodiversity hotspot. There is a chronic problem of human extraction of fuel wood, forest fragmentation and degradation; the current annual deforestation rate being 1.2%. Mature forest species are suitable for restoration of sites ranging from open-weedy sites to those with a fairly dense native or exotic canopy (Raman et al., 2009). Restoration affected by proximity to seed sources, presence of animal dispersal agents, control of invasive weeds.

Over the last ten years, restoration of degraded rainforest fragments in the Valparai plateau of the Anamalai hills is in progress in partnership with tea and coffee plantation companies to restore degraded sites in twelve rainforest fragments lying within their estates (Mudappa and Raman, 2010), Fig. 28.10. More than 160 native tree and liana species were raised in a rainforest plant nursery, planted out and monitored in this restoration effort.

Restoration of the mangrove wetlands on the east coast of India

Restoration of the mangrove wetlands has been implemented in Godavari and Krishna river basin in Eastern Ghats (Ramasubramanian and Ravishankar, 2004). Changes in the biophysical condition of the coupe felled area. A fishbone design was utilized in order to facilitate easy inflow and outflow of tidal water. A total area of 520 ha of degraded mangroves was restored with community participation. The species *Avicennia marina*, *Avicennia officinalis* and *Excoecaria agallocha* were selected for planting, as these species tolerate a wide range of salinity. *Aegiceras corniculatum*, *Bruguiera gymnorrhiza*, *Rhizophora apiculata*, *Rhizophora mucronata* and *Xylocarpus moluccensis*

were also planted to ensure genetic diversity. Mangroves are now naturally regenerating, and the canopy cover has become denser.

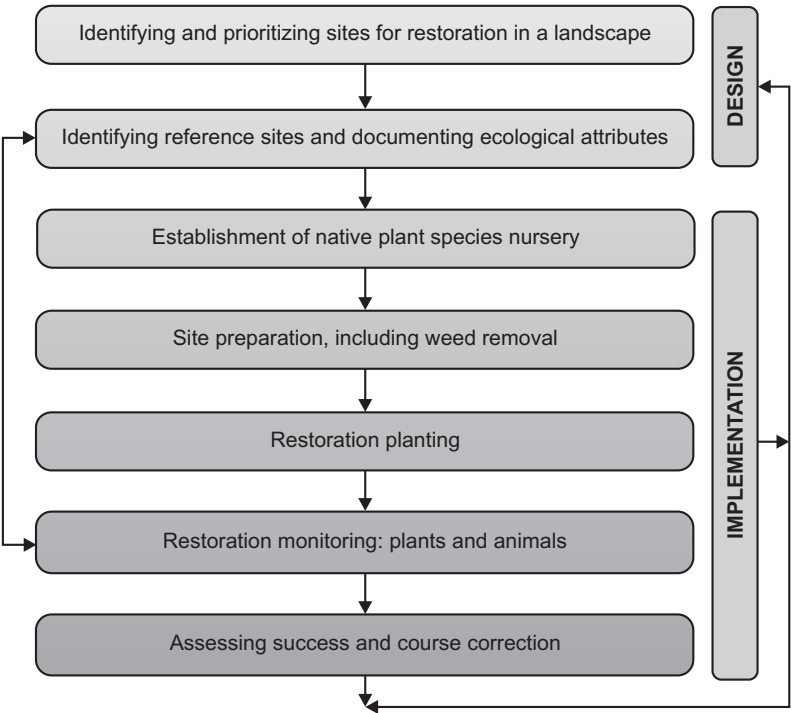


Fig. 28.10. Rainforest Restoration Project in Anamalai Hills (adapted from Mudappa and Raman, 2010).

CHAPTER

29

Environmental Impact Assessment

The rapid pace of social and economic development necessitates implementing and enforcing appropriate environmental policies and laws for resource conservation and ecological sustainability. Environmental awareness has created a sense of urgency and resolve for political and legislative actions for conserving natural resources. Environmental legislation has become one of the major tools for formulating environmental policy and implementing it for various development activities. In the developed world, growing environmental awareness in sixties focused attention on the environmental consequences of developmental actions. This led public to demand that environmental issues be incorporated into the decision making process. Now this concern is felt worldwide both in developed as well as developing countries. In recent times, several management tools are being applied for sustainable developmental actions. EIA (Environment impact assessment) and SEA (strategic environmental assessment) are among the available tools for environmental assessment. EIA is relatively a specific and simple exercise and is a project oriented planning tool. EIA links environment with development, and is an appraisal of the environmental health and social implications of a planned development. The impact statements are prepared to assess the effects of a project on human beings, plant and animal life, soil, water, air, climate, material assets as well as cultural heritage.

Environmental Impact Assessment is an anticipatory and participatory management tool for informed decision making on optimal use of natural resources and sustainable development. In EIA, the science of environmental analysis is integrated with policies of resource management and is an effective tool for achieving sustainability. EIA is useful for almost all developmental activities so as to ensure economic activity on a long-term basis because it takes into consideration risk, social, economic, culture and health and other relevant factors in different regions depending upon the requirements. It contributes to the transparency of the decision-making process during the development, implementation, monitoring and followup stages of a development project.

29.1 EIA-ORIGIN AND DEVELOPMENT

Environmental impact assessments have been carried out on ad hoc basis for decades. These were formalized under the United States Environmental Policy Act 1969. In the US, the first EIA law- National Environmental protection Act (NEPA) came into force on January 1, 1970, which required the developers to publish an environmental impact statement (EIS) before implementing any plans. Since then, a number of countries have established EIA as an element of environmental policy and legal requirement for the proposed development projects and activities. The role for EIA was formally recognized at the earth summit held at Rio conference in 1992. Principle 17 of the Rio

declaration states that – “EIA as a national instrument shall be undertaken for the proposed activities that are likely to have significant adverse impact on the environment and are subject to a decision of a competent national authority”.

Environmental Impact assessment, according to Munn (1979), refers to the need “to identify and predict the impact on the environment and on man’s health and wellbeing of legislative proposals, policies, programmes, projects and operational procedures, and to interpret and communicate information about the impacts.” MoEF (1994) gives a formal definition as, “a process by which information about the environmental effects of a project is collected, both by developed and from other sources, and taken into account by the relevant decision making body before a decision is given on whether a development should go ahead”. It is a process of identifying, estimating and evaluating the environmental consequences of current and proposed actions (Vanclay and Bronstein, 1995). Apart from the decision on whether a project should proceed, an EIA considers various project alternatives and mitigation measures to be taken into consideration. Thus, EIA is the process of identifying, predicting, evaluating and mitigating the biophysical, social, and other relevant effects of development proposals prior to major decisions being taken and commitments made (UNEP, 2002 c).

The United Nations Environment Programme (UNEP) has defined the goals and principles of EIA for the analysis and assessment of planned activities to ensure environmentally safe and sustainable development (UNEP, 1980). Besides, UNEP addresses several regional agreements such as United Nations/ European Commission Convention on Environmental Impacts in a Transboundary Context and Regional Strategic Environmental Assessment (SEAs) to assess the impact of major public and private projects on the environment. During 1985, the European Community Directive on EIA was introduced for all the EC member states (CEC, 1985). In the context of the Brundtland report, EIA constitutes a basis for planning for a common future (WCED, 1987). The nature of the EIA systems (e.g., mandatory or discretionary, level of public participation, types of actions requiring EIA and their implementation in practice) vary widely in different countries. However, the concept of EIA has been accepted due to its central role in environmental protection and its validity as a proactive planning tool. Now, more than 150 developing countries have incorporated EIA either within the legislature mechanism or as guidelines in decision making process. The basic aspects, applications and definitions of environmental impact assessment are summarised in Table 29.1.

The Rio Declaration on Environment and Development in its article 5 of the legal principles for environmental protection for sustainable development stresses the need for a state to conduct the environmental impact assessment of the human activities likely to affect the environment. EIAs are now becoming increasingly important for the biodiversity projects in different countries, particularly in the developing countries. The article 14 of the Biodiversity Convention emphasizes EIA as a potential implementation mechanism for proposed projects likely to have significant adverse impacts on biodiversity, particularly on threatened plant and animal species. EIA has a potential role in ensuring that the aims of the Convention on Biological diversity are promoted through planning and development control, along with conservation of ecosystem viability.

Table 29.1. Aspects, applications and definitions of environmental impact assessment (based on Treweek, 1999).

Aspects of process	Application	EIA category
Application	General Global, strategic (to policies, plans, programmes) Local, Specific (to individual projects/sites)	Impact Assessment (IA) Environmental Assessment (EA) Environmental Impact Assessment (EIA) Strategic Environmental Assessment (SEA) Project EA or EIA
Source of impact (actions)	Specific projects Policies Development aid New policy Change in Policy	Project EA or EIA SEA Policy assessment
Impact receptors	The whole environment Ecosystems and their components People (Individuals) Populations Communities	EA or EIA Ecological impact assessment Demographic impact assessment Health Impact Assessment
Impact categories (general)	Economic Ecological Cumulative Social and culture Health Climate Noise	Economic Impact Assessment Ecological Impact Assessment Cumulative Effects Assessment Social Impact Assessment Health Impact Assessment Climate Impact Assessment Noise Impact Assessment

29.2 EIA -PURPOSE AND AIM

The aim of the EIA is to provide an accurate prediction of the effects of a development before it is carried out so that new plan may be modified accordingly. It is an enabling tool for informed decision making and has a flexible approach for environmental management. The purpose of EIA is to facilitate sound and integrated decision making with due environmental considerations to support the goals of environmental protection and sustainable development (Caldwell, 1982). The World Bank Operational directive 4.01, 1991 states that “the purpose of EIA is to improve decision making and to ensure that the project options under consideration are environmentally sound and sustainable” (World Bank, 1991).

EIA is expected to supply decision makers with an indication of the likely outcomes of their decisions, and help them to select between alternative methods and sites of construction and avoiding unacceptable options. The main purpose of EIA is to give the environment its due place in the decision making process by clearly evaluating the consequences of a proposed activity before an action is taken (Gilpin, 1995). For sustainable development, EIA has a potential role in the following areas (Sadler, 1996; UNEP, 2002 c). Safeguarding valued ecological processes and heritage sites.

1. Protecting traditional ecological knowledge.
2. Maintaining natural resource base.
3. Optimizing the use of natural resources.
4. Protecting human health, community well being and traditional lifestyles of people.

5. Conserving natural resources and ecosystem integrity.

The supporting and secondary aims of EIA are as follows:

1. Improving coordination among participating agencies and actions.
2. Evolving better-designed and planned development projects.
3. Empowering community development and building local capacity building through public participation.
4. Instilling environmental values and accountability across a range of institutions.
5. Internalizing environmental cost and damages in industry.

29.3 CORE VALUES AND PRINCIPLES

A linked set of general and operational principles can be focussed on the core values and guiding principles that impart purpose and direction to EI actions and approaches (Sadler, 1996; Sadler and Bailey, 1997). The basic principles are identified in law, procedures and guidelines. The principles can be keyed to three core values that provide purpose and direction to EIA action and approach.

Core Values

1. **Integrity:** the process will be in agreement with accepted standards and principles of good practice.
2. **Utility:** The process will provide balanced and credible information for decision making.
3. **Sustainability:** The process will promote environmentally sound development, i.e., within the assimilative and regenerative capacity of the natural systems.

Guiding Principles

There are guiding principles for an effective EIA process. These principles elaborate the core values; identify fundamental legal and policy requirements. In a way, these serve as benchmarks for an effective performance of EIA systems. Some important principles for the process design and development are given as:

1. Should serve as a primary environmental tool, with a clear mandate and provisions.
2. Should have uniform and consistent application to all proposals and actions with potential environmental effects and consequences.
3. Should be carried out throughout the project cycle using best practice science and mitigation strategy.
4. Should address all factors including cumulative, long term and large-scale effects.
5. Should embody sustainable considerations including assimilative capacity, carrying capacity and biological diversity.
6. Should have flexible and problem solving approach as well as opportunities for public involvement.
7. Should be efficiently implementable to minimize delays and burden on proponents.
8. Should have follow-up and feedback in-built mechanism for checking on compliance with conditions, monitoring effects, managing impacts, auditing and evaluative performance.
9. Should provide cost effective outcomes to promote actions to ensure environmental protection at least cost to society.

29.4 EIA PROCESS

EIA is a process that examines the environmental consequences of development actions in advance with an emphasis on environmental protection (Fig. 29.1). The various steps of the EIA process are shown in Fig. 29.2. It may be mentioned that EIA is a cyclic activity, with interactions and feedback occurring between various steps. The order of the steps in the process may also vary depending upon the main objectives of the process. Thus, EIA can provide a basis for environmentally sound decision making, the designing and planning of developmental projects, monitoring, management and audit, mitigation and application of EIA process.

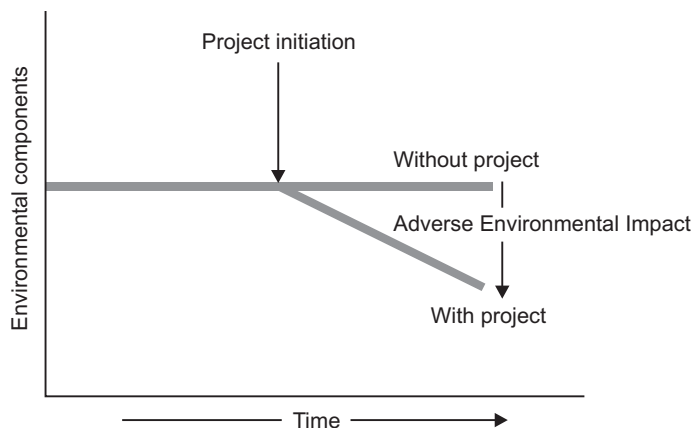


Fig. 29.1. Environmental impact of a development project (adapted from UNEP, 2002 c).

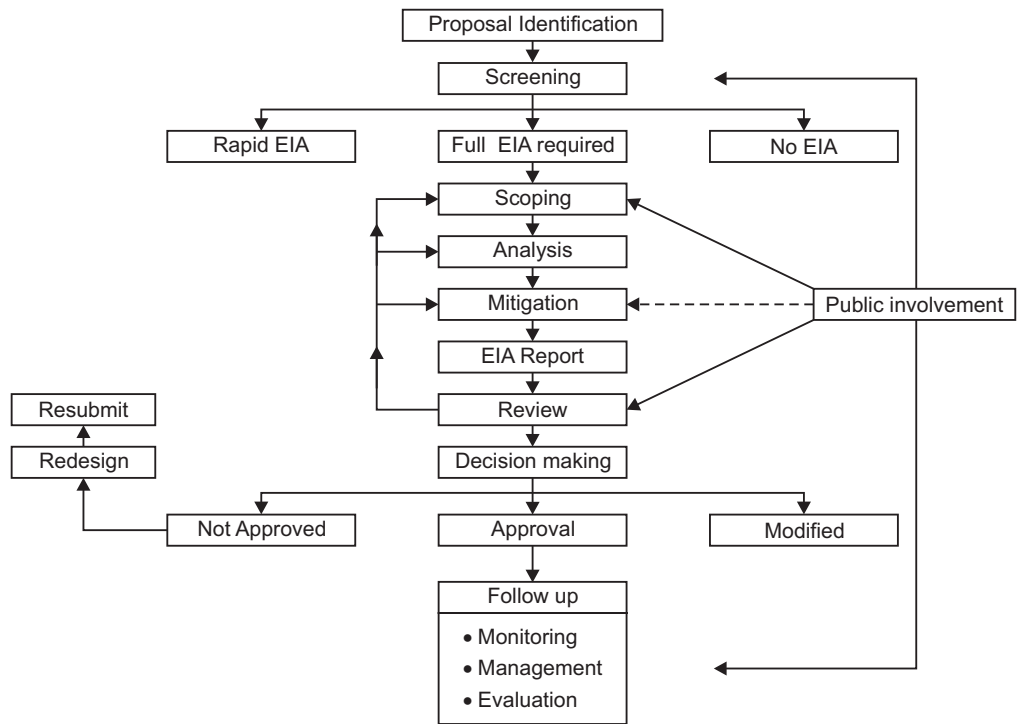


Fig. 29.2. Generalized Environmental Impact Assessment process flow chart EIA is a cyclic process with interaction between the various steps. EIA can be useful at most stages of the process (adapted from UNEP, 2002 c).

In the EIA process, the structure and logical approach accompanied by fact finding, gaining public input, analysis and tests of information and organisation of findings in a report are important. The EIA process is meant to be applied purposely (fitted to function), flexibly (not all steps are needed) and relatively (having focus on key issues). Projects involving large public investment in areas such as infrastructure require impact assessment linked to cost-benefit analysis. EIA can be viewed as an integral part of planning process and initiated at the project level from the very beginning. The main steps in the EIA include impact identification, base line study of the project area, impact evaluation (for working out potential mitigative measures) and cost-benefit analysis of environmental impacts.

The main steps of an idealized EIA process as shown in Figure 29.2 are briefly described in Table 29.2, and outlined below (see MoEF, 2001; UNEP, 2002 c):

Table 29.2. The main steps of an idealized EIA process.

Inception	Framing of a proposal.
Screening	To determine the need for EIA for the projects that may have significant impacts. Screening may partly be determined by the EIA regulations operating in a country at the time of the assessment.
Scoping	To identify key issues and impacts to establish study limits.
Assessment	Survey, measure and predict likely impacts on environment components at scoping phase.
Evaluation	To determine the significance and social importance of measured impacts, particularly adverse impacts.
Mitigation	To specify the measures to prevent, minimize or offset or otherwise compensate for environmental loss and damage.
Report Results	Environmental statement to document findings of EIA.
Decision-making	This is related to approval, modification or rejection of the proposal and involves a consideration of the EIS by the relevant authority together with other material considerations.
ES Review	Systematic appraisal of the quality of the EIS and compliance with legislation or modification required.
Monitoring	To comply with terms and conditions and to monitor environmental impacts.
Follow up management	The aim is to define contingency plan to be in place to deal with breakdown in mitigation.
Audit	This concerns with documentation of longer-term outcomes of the project and to check performance against environmental quality standards. It provides a vital step in the EIA learning.

Screening: Screening is done to see whether a project requires environmental clearance as per the statutory notifications. Screening Criteria are based upon scales of investment, type of development, and location of development. A Project requires statutory environmental clearance only if the provisions of EIA notification and/or one or more statutory notification cover it.

Scoping: It is a process of detailing the terms of reference of EIA. and preparing ground plan for subsequent steps by making preliminary assessment of (i) the project's potential impacts, (ii) impact area/zone, (iii) mitigation measures, (iv) need for monitoring, and (v) the methods and level of study. It has to be done in consultation with the project proponent and guidance, if need be, from Impact Assessment Agency. The Ministry of Environment and Forests, Government of India has published guidelines for different sectors, which outline the significant issues to be addressed in the EIA studies.

Baseline Data: Baseline data describes the existing environmental status of the study area. The site-specific primary data should be monitored for the identified parameters and supplemented by secondary data if available.

Impact Prediction: Impact prediction and assessment are the most important parts of the technical process (Erickson, 1994). The inputs and outputs related to evaluating the impacts are given in Figure 29.3. Impact prediction should include assessment of (i) direct primary impacts, (ii) indirect/secondary impacts, and (iii) cumulative impacts (that accrue over time and space from a number of projects of the area). Impacts may be positive or negative, reversible or irreversible and of various duration (Therivel and Morris, 2001). An impact prediction requires a good understanding of the nature of the proposed project, knowledge of outcomes of similar projects undertaken in the past, knowledge of past, existing and approved projects likely to cause interactive or cumulative impacts with the project being considered for evaluation, and information about the relevant receptors.

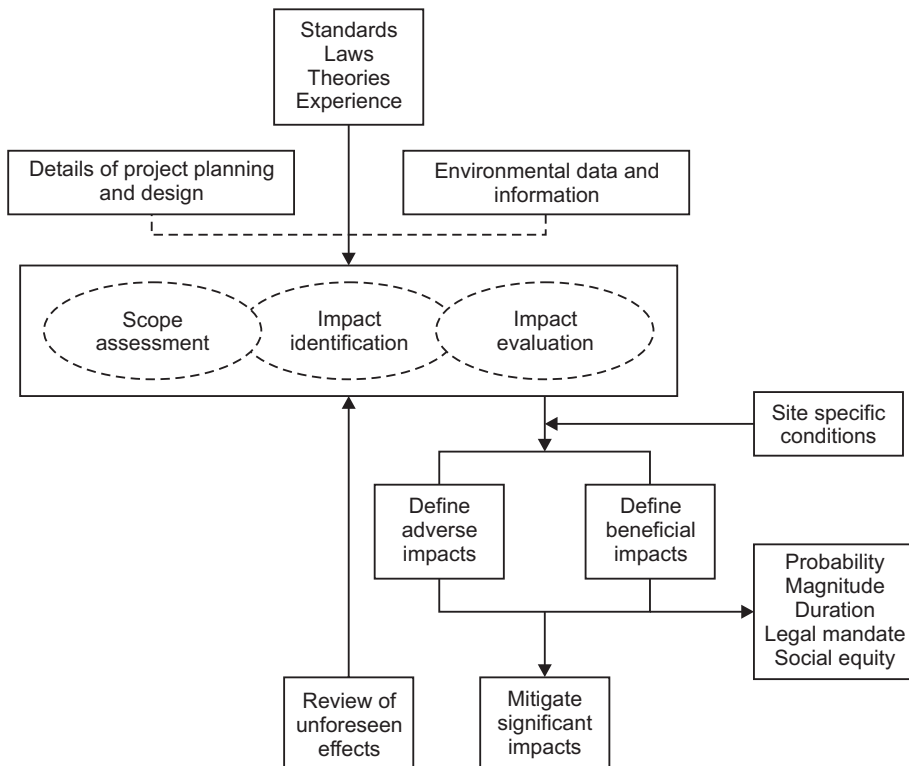


Fig. 29.3. Inputs and outputs related to evaluation of impacts (based on Erickson, 1994).

Mitigation Measures and Environmental Impact Assessment Report: For every project, possible alternatives should be identified and environmental attributes compared. Alternatives should cover both project location and process technologies. The EIA report should provide clear information to the decision-maker on the different environmental scenarios without the project, with the project and with project alternatives. Uncertainties should be clearly reflected in the EIA report.

Public Hearing Law: This requires that the public must be informed and consulted on a proposed development after the completion of EIA report by local associations, environmental groups active in the area or any other person located at the project site/ sites of displacement.

Decision Making: Decision making process involves consultation between the project proponent (assisted by a consultant) and the impact assessment authority (assisted by an expert group, if necessary). The decision on environmental clearance is arrived at through a number of

steps including evaluation of EIA and EMP. Decisions are seldom 'yes' or 'no', as any decision making involves trade-offs.

Monitoring the Clearance Conditions: Monitoring should be done during both construction and operation phases of a project. Monitoring will enable the regulatory agency to review the validity of predictions and the conditions of implementation of the Environmental Management Plan (EMP).

Risk Assessment: Hazard identification taking recourse to hazard indices, inventory analysis, dam break probability, natural hazard probability etc.

1. Maximum Credible Accident (MCA) analysis to identify potential hazardous scenarios
2. Consequence analysis of failures and accidents resulting in fire, explosion, hazardous releases and dam breaks, etc.
3. Hazard & Operability (HAZOP) studies
4. Assessment of risk on the basis of the above evaluations
5. Preparation of an onsite and offsite (project affected area) Disaster Management Plan

29.5 ENVIRONMENTAL COMPONENTS OF EIA

Both comprehensive EIA and Rapid EIA are followed depending on the availability of time. Rapid EIA is for speedier appraisal process. While both types of EIA require inclusion/ coverage of all significant environmental impacts and their mitigation, Rapid EIA achieves this through the collection of 'one season' (other than monsoon) data only to reduce the time required. The review of Rapid EIA submissions will show whether a comprehensive EIA is required or not. The components of the EIA have been described on the basis of MoEF manual (2001):

Air environment

- Determination of impact zone (through a screening model) and developing a monitoring network
- Monitoring the existing status of ambient air quality within the impacted region (7-10 km from the periphery) of the proposed project site
- Monitoring the site-specific meteorological data, viz. wind speed and direction, humidity, ambient temperature and environmental lapse rate
- Estimation of quantities of air emissions including fugitive emissions from the proposed project
- Identification, quantification and evaluation of other potential emissions (including those of vehicular traffic) within the impact zone and estimation of cumulative emissions/impacts
- Evaluation of the adequacy of the proposed pollution control devices to meet gaseous emission and ambient air quality standards

Noise environment

- Monitoring the present status of noise levels within the impact zone, and prediction of future noise levels resulting from the proposed project and related activities including increase in vehicular movement
- Recommendations on mitigation measures for noise pollution

Water environment

- Study of existing ground and surface water resources with respect to quantity and quality within the impact zone of the proposed project
- Prediction of impacts on water resources due to the proposed water use/pumping on account of the project

Biological environment

- Assessment of flora and fauna present within the impact zone of the project
- Assessment of potential damage to terrestrial and aquatic flora and fauna due to discharge of effluents and gaseous emissions from the project
- Assessment of damage to terrestrial flora and fauna due to air pollution, and land use and landscape changes
- Prediction of biological stresses within the impact zone of the proposed project

Land environment

- Studies on soil characteristics, existing land use and topography, landscape and drainage patterns within the impact zone
- Estimation of impacts of project on land use, landscape, topography, drainage and hydrology
- Identification of potential utility of treated effluent in land application and subsequent impacts
- Projection of anticipated changes in the socio-economic and health due to the project and related activities including traffic congestion and delineation of measures to minimize adverse impacts
- Assessment of impact on significant historical, cultural and archaeological sites/places in the area
- Assessment of economic benefits arising out of the project

29.6 MAIN PARTICIPANTS IN EIA PROCESS

The various types of developmental projects have a specific configuration of underlying interests, strategies and perspectives. There are four key players involved in the planning and development process; these can be categorized according to their role or degree of power (e.g., statutory, advisory), level of operation (e.g., international, national and local) or emphasis (e.g., environmental and economic). The key players are:

1. Developers
2. Affected parties
2. Regulators
3. Facilitators

Regulators are represented by Government Departments at various levels (e.g., Department of Environment, State Governments to local authorities). Facilitators include the various consultants, advocates and Advisers who participate in the EIA process and planning and development process. These agents are employed by the developers or by the local groups or the regulatory bodies to help them.

The interactive role among the key players is illustrated in Figure 29.4.

EIA applies to both public and private sectors. In accordance with the EIA Manual of MoEF (2001), the following have significant roles in the EIA Process :

1. The Project Proponent
2. The Environmental Consultants
3. The State Pollution Control Boards / Pollution Control Committees (PCCs)
4. The Public
5. The Impact Assessment Agency

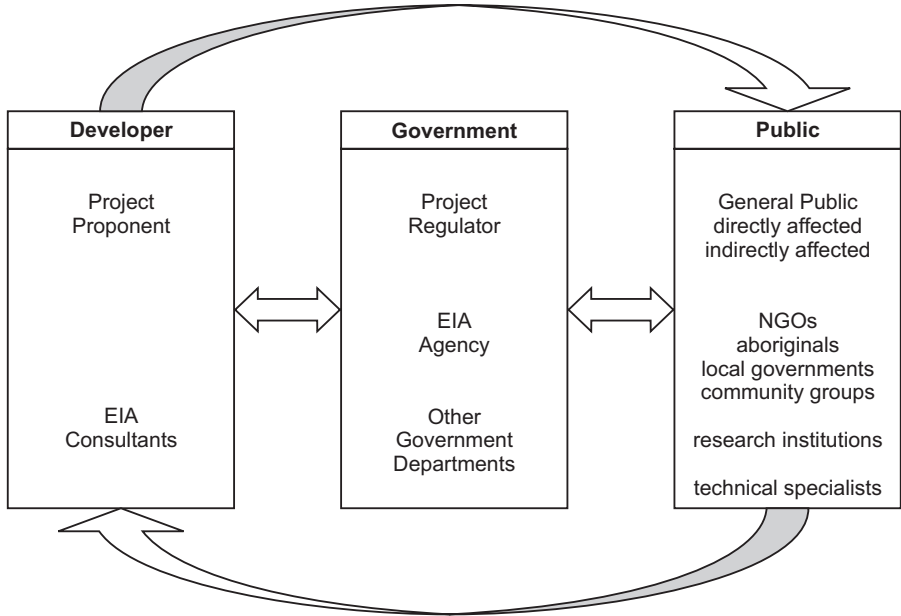


Fig. 29.4. Participants in the EIA process.

Role of the Project Proponent

The project proponent during the project planning stage decides the type of projects i.e. new establishment, expansion or modernization. The project proponent needs to prepare the detailed project report/feasibility report and submit the executive summary. The executive summary contains project details, and findings of EIA study, which is to be made available to concerned public. After the public hearing the proponent submits application to Impact Assessment Agency (IAA) for environmental clearance. The Environmental Management Plan is a crucial input to monitoring the clearance conditions and therefore details of monitoring should be included in the EMP.

The flow chart showing the responsibilities of participants in EIA is given in Figure 29.5 and explained below.

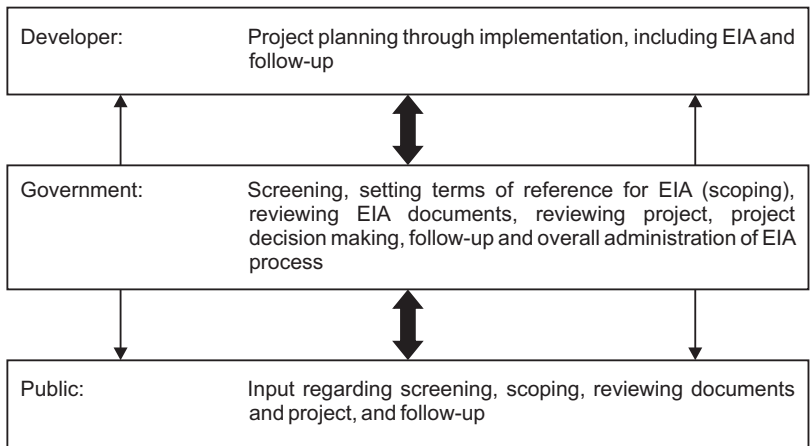


Fig. 29.5. Flow chart showing the responsibilities of participants in EIA.

Role of Environment Consultant

Environmental consultant should be conversant with the existing legal and procedural requirements of obtaining environmental clearance for proposed project. The consultant should guide the proponent through initial screening of the project and establish whether EIA studies are required to be conducted and if so finalise the scope of such study.

Role of the State Pollution Control Boards (PCBs) / Pollution Control Committees (PCCs)

The State PCBs/PCCs are responsible for assessing the compatibility of a proposed development with current operational and prescribed standards. If the development is in compliance, the PCB will then issue its no objection certificate. They shall also hold the public hearing as per the provisions of EIA notification. The details of public hearing shall be forwarded to IAA.

Role of the Public

The public also has an important role to play in EIA. The concerned persons will be invited through press advertisement to review information and provide their views on the proposed development requiring environmental clearance.

Role of the Impact Assessment Agency (IAA)

Where a proponent is required to obtain environmental clearance, the IAA will evaluate and assess the EIA report. During the implementation and operation of the project, the IAA will also be responsible for the environmental monitoring process.

29.7 IMPACT IDENTIFICATION METHODS

It is important to recognize the impacts of developmental activities on the environment. The various stages in the EIA relate to baseline information, impacted environment, policies and processes. A number of impact identification methods are in use ranging from simple checklists and matrices to complex computerized model and network (Table 29.3). The environmental impact matrix provides a simple inventory and display of these impacts relating to project actions and environmental components (Fig. 29.6). The pioneering work in this area was done by Leopold et al. (1971). The Leopold matrix is based on a horizontal list of 100 project actions and a vertical list of 88 environmental components. If a project action is likely to make an impact on environmental components, the appropriate cell is scored for magnitude and importance of the impact. A positive and negative sign is provided for harmful or beneficial nature of the impact. Row totals of the matrix reflect the total impact of all project actions on one environmental component, while the column totals reflect the impact of one project action on all components of the environment. The matrix total gives the total environmental impact. The simple methods such as Leopold matrices are easy to use, more effective and consistent to present information in the EIA. However, the simple methods have limitations, as these may not reveal impact significance, indirect impacts and alternatives in the EIA.

Table 29.3. Comparison of impact identification methods (based on Glasson et al., 1994).

Criterion											
	1	2	3	4	5	6	7	8	9	10	11
Checklists											
Simple/descriptive/ question	+	+	–	–	–	–	–	+	+	+	+
Threshold	+	+	+	–	+	+	+	–	+	+	+
Matrices											
Simple	+	+	–	–	–	–	–	+	+	+	+
Magnitude	+	+	+	–	–	–	–	+	+	+	+
Leopold	+	+	+	–	+	–	–	+	+	–	+
Weighted	+	+	–	–	+	+	–	+	–	–	+
Quantitative											
ESS	+		+		+	+	+	–	–	–	–
Network											
Sorensen	+	–	–	+	–	+	–	+	–	+	–
Overlay maps	–	+	+	–	+	+	–	+	+	+	+

1. Compliance with regulations; 2. Comprehensive coverage (social, economic and physical impacts); 3. Positive v. negative, reversible v. irreversible impacts, etc.; 4. Secondary, indirect, cumulative impacts; 5. Significant v. insignificant impact; 6. Compare alternative options; 7. Compare against carrying capacity; 8. Uses qualitative and quantitative information; 9. Easy to use; 10. Unbiased, consistent; 11. Summarizes impact for use in ES.

The Battelle Environmental evaluation system (BEES) is one of the best quantitative methods that have been used to assess water resource development, highways, nuclear power plants and other projects (Dee et al., 1973). It consists of a checklist of 78 measurable environmental parameters relating to ecology, environmental pollution, and social and economic variables that are likely to be affected by the proposal.

The Network methods such as Sorensen Network recognise that environmental systems consist of complex relationships (Sorensen, 1971). The overlay maps and geographical information systems are used to show the spatial distribution of impacts. The complex methods (quantitative methods, networks and overlays maps) include various aspects of the impact analysis and provide a better approach to impact identification. But these methods make communication and effective participation of the target groups somewhat difficult. The Geographical Information System (GIS) can use greater amount of information and enable a sensitivity analysis to be carried out for the decision making in the EIA.

Mitigation

Mitigation measures are taken to avoid, minimize, remedy or compensate for the predicted adverse impact, of the project. The measures may include: selection of alternative production technique, modification of methods and design features, minimisation of generation of pollutants, and wastes during operation and measures to compensate for losses. The construction phase is most damaging to the environment, therefore needs to be attended most effectively.

Modified Leopold Matrix		SOCIAL ENVIRONMENT							
Development	Environmental Effects	Recreation	Landscape/visual	Historical/cultural	Personal and social values	Risks and anxieties	Existing land uses	Land value	Settlement
		Public participation	Employment	Settlement	Land value	Existing land uses	Risks and anxieties	Personal and social values	Historical/cultural
Treatment Comminution Sedimentation Milliscreening Oxidation ponds Activated sludge Trickling filter Nutrient removal Chlorination Further treatment offsite									
Disposal – Land Rapid infiltration Surface flooding Spray irrigation Disposal – Inland Water River Lake Disposal – Marine Water Estuary Inshore marine Offshore marine Deep well injection									

Fig. 29.6. Modified Leopold matrix to study environmental impact. (adapted from UNEP, 2002 c).

29.8 DIVERSIFICATION OF EIA

Application and continued diversification of the environmental impact assessment, includes the following sub-components.

- Strategic Environmental Impact Assessment (SEA).
- Legislation for International and Trans-boundary Effects.
- Ecological impact assessment (EcIA).

Strategic Environmental Assessment (SEA)

SEA can be defined as “the formalized, systematic and comprehensive process of evaluating the environmental impacts of a policy, programme, plan or its alternatives as well as using the findings in publicly accountable decision making” (Therivel et al., 1992). It is an environmental appraisal system with a focus on policy and planning system. In SEA, the need for a more flexible approach is emphasised. The projects have to be fully open to even a fundamental change if necessary. It is more resource-led because it is emerging in response to the challenges of sustainable development.

SEA pertains to concepts rather than particular activities and gives importance to cross-cutting environmental and sustainability objectives to be achieved. SEA generally has a longer time frame than EIA. SEA may not be directed by a specific purpose and may essentially evolve. It helps to plan development to be compatible with the conservation of natural resources. The resource-based approach of the SEA has a focus on valued, sensitive and threatened ecosystems and services. It has been found to have important applications in activities such as exploiting natural resources directly such as forestry, fishing, and mining and mineral extraction activities. The impacts on large areas like transport can also be evaluated with the help of SEA. Current practices focus mainly on plan and programme level and at regional level. Primary areas for SEA are land use, natural resources, extraction of raw materials, and waste and pollution generating activities. The framing of transport policy for protected areas and suggesting alternative approaches for reducing biodiversity losses have also worked well with the SEA.

Legislation for International and Trans-boundary Effects

Some actions may cause environmental impacts in more than one country or in international waters. Examples are: discharge of industrial and municipal wastes into rivers that border or pass through several countries, the long range acid deposition problem, food import and export, the global warming, international whaling, desertification, the environmental management of Antarctica and the stratospheric ozone. Therefore, there is a movement towards environmental protection on a worldwide scale through conventions, treaties and international agreements. Many successful international treaties and conventions have been passed. A few of them relating to assessment of Trans-boundary effects include the following.

- Ramsar Convention- this is the Convention on wetlands of International importance especially as waterfowl habitat. The convention was adopted in Ramsar, Iran, 1971 and is the oldest international conservation treaty. In recent years, the criteria of wetland importance have been expanded to include its value to a wide range of plants and animals as well as its sustainable economic value to local human communities.
- Convention on Long-Range Trans-Boundary Air Pollution, 1979.
- CITES - Convention on International Trade in Endangered species of Wild Flora and Fauna. Concluded in 1973 and implemented from 1975. The convention bans the imports and exports of those species “threatened with extinction that are or may be threatened”.
- United Nations Convention on the Law of Sea, 1982.
- Convention of Biological Diversity, 1992.
- Framework Convention on Global Climate Change.

Ecological Impact Assessment (EcIA)

The ecological impact assessment deals with ecological consequences of defined development actions in localised, individual projects or regional developmental plans, which have impacts on ecosystem or their components (Treweek, 1999). It has a resource-based approach with wide applications at a number of scales from individual species to whole ecosystems. It may be practiced as an integral part of environmental impact assessment or as a part of land-use planning. EIA and related legislation is the main driving force for the development of ecological impact assessment. The various procedural steps for ecological impact analysis are summarised in Table 29.4.

Table 29.4. Common procedural steps in ecological impact analysis process.

Scoping	Interpretation of the proposal and its associated sources of ecological stress or disturbance Description, characterization, inventory of potential ecological receptors Impact screening to establish the range of possible interactions between potential stressors and receptors Defining study limits Identifying valued ecosystem components Establishing the range and focus of study.
Focusing	Identifying valued ecosystem components, processes and functions Refining the scope of the study
Impact Assessment	Baseline studies To identify and predict impacts on ecosystem components
Impact Mitigation	To reduce significant adverse effects
Impact Evaluation	To determine the significance and importance of predicted, residual, ecological impacts Measuring against internal and external criteria
Monitoring and feedback	To strengthen the knowledge base To provide opportunities for corrective action in the light of unforeseen outcomes Feedback to assess proposal implementation and compliance

29.9 BENEFITS AND IMPROVING EFFECTIVENESS OF EIA PROCESS

For an effective environmental assessment, it is important to make the assessment simple, focussed, time bound and cost effective. The environmental assessment reports need to be precise and goal oriented with a clear objective of highlighting opportunities for environmental enhancement as well as assessing the role of the target community. It is also important to explain various gaps and uncertainties in information and a plan of action for mitigative measures. The objective is to bridge the gap between environmental conservation and socioeconomic considerations.

Main Benefits

The main benefits of environmental assessment are:

- To reduce the cost and time of project implementation.
- To save cost from modifications in project design.
- To improve the acceptability of the project.
- Avoiding the impacts and violation of laws concerning environmental safeguards.
- Improved project performance.
- To minimize treatment or clean-up costs.

Benefits to Local Community

The environmental assessment is a useful component of good environmental management. The benefits to local community taking part in environmental assessment include:

- A healthier local environment.
- Improved human health.
- Maintenance of biological diversity.

- Optimization of resource use.
- To avoid conflicts over natural resource use.
- Increased community skills and knowledge.

Improving Effectiveness of an EIA

There are differences in scope of EIA projects in terms of actions and impacts covered by legislation and methods used to assess and evaluate the impacts. Various problems with environmental impact assessment at the project level may include.

- To find the site boundary or project-driven limits on spatial and temporal scales.
- Inability to assess cross boundary impacts.
- Lack of long-term data bases and base line information.
- Failure to identify mitigative measures.
- Project design and feasibility studies may be expensive.
- The predictions of environmental impact statements could be uncertain.

The regulatory and economic controls are not enough to correct environmental impacts of developmental projects. Public awareness or of those responsible for environmental protection and enforcement is also important. This could prove to be the most effective means of protecting environmental quality. The EIA needs a shift in focus from project proponents to people affected both beneficially and adversely. The EIA scheme needs adequate clarity along with reliability of information and data. There should be sufficient provisions to assess negative impacts in relation to developmental factors and resource capabilities.

Assessment can also be aided by the recent developments of environmental auditing for various organizations including both private and public sector industries. The information from environmental auditing could bring a significant change in the quality and quantity of baseline data for EIA. Education, training and availability of information to administrative and professional participants as well as public at large could play an important role in making the environmental impact assessment more effective and useful.

The broader context of EIA methods

A project does not operate in isolation; therefore, there is a need to broaden EIA's limit to include other projects, plans. Evidently, this would allow trade-offs between impacts and projects. In the global context environmental policy is moving towards a broader proportion of sustainability. For example, the suggestion that sustainable development can be achieved by mutually enforcing economic, social, and environmental objectives is being challenged by others who believe that the environmental life-support systems are preconditions for economic and social development. Thus the life-support system should be given unconditional protection. The globalisation of economy is posing new threats to environment as some countries are trying to achieve environmental sustainability at the cost of other countries. In a way environmental sustainability is being imported. Maybe we need to expand the scope of EIA to consider impacts such as these.

EIA is expected to help in achieving sustainable development, which is based on principles like: anticipate and prevent; maintain natural capital, avoid intensive uses of land, make the polluter to pay and exercise precautions. EIA can contribute to the achievement of these principles, provided it is linked effectively with other policy and planning instruments.

29.10 EIA IN INDIA

In India, Ministry of Environment and Forests (MoEF) has taken several policy initiatives and enacted environmental and pollution control legislation to promote integration of environmental concerns in developmental projects. MoEF initiated the EIA with the river valley projects during 1978-79. Its scope has subsequently been enhanced to cover other developmental sectors such as industries, thermal power projects, mining schemes, etc. The Government of India enacted the Environment (Protection) Act on 23rd May 1986. The notification of Environmental Impact Assessment (EIA) of developmental projects was issued on 27 January 1994 under the provisions of Environment (Protection) Act, 1986, and subsequently amended on 4th May 1994, 10th April 1997 and 27th January 2000. The EIA has now been made mandatory for 30 categories of developmental projects.

As per The Schedule-I of the EIA notification, the list of projects requiring environmental clearance from the Central Government is given in Table 29.5. The Schedule II of the notification deals with the Application form and the project report. The composition of the expert committee for the EIA in the various disciplines has been specified in The Schedule-III of the notification. The committees consist of experts in the disciplines of ecosystem management, air/water pollution, water resource management, flora/fauna conservation and management, land use planning, social science/rehabilitation and project appraisal. The Schedule IV deals with the procedure for Public hearing.

Appropriate guidelines have been evolved and circulated to the concerned central and state government departments for the collection of environmental data and preparation of management plans. EIA now covers all major developmental projects- e.g. oil refineries, river valley projects, power stations, and national highways. A compendium of the procedures and questionnaires entitled “Application Form and Questionnaire for Environmental Clearance” was published in September, 1999 in association with the Confederation of Indian Industry. Recently, the ministry has brought out an EIA Manual covering the various issues like regulatory requirements, the EIA methodology including baseline studies, identification of key issues and consideration of alternatives, impact analysis and remedial measures in a systematic way (MoEF, 2001). It also deals with the process of reviewing of EIA and Environmental Management Plan reports and post project management.

The EIA Notification 2006 has notified 39 development sectors, which require prior environmental clearance. Based on the capacity and on their potential environmental impacts. The Projects have been categorised into Category A or B, the latter has been further categorised as B₁ (project requires to submit EIA report) or B₂ (project does not require to submit EIA report). Category A projects (including expansion and modernization of existing projects) require clearance from the Ministry of Environment and Forest (MoEF), Govt. of India (GoI) and category B from State Environmental Impact Assessment Authority (SEIAA), constituted by Government of India. A State Level Environment Impact Assessment Authority (SEIAA) is constituted by the Central Government under the Environment (Protection) Act, 1986 for the purpose of the EIA notification. This is the ‘Environmental Clearance’ issuing authority, which scrutinizes the projects appraised by SEAC and gives the final decision on issuing ‘Environmental Clearance’ or rejecting the application. All categories ‘A’ and ‘B₁’ projects necessarily have to carry out EIA studies along with the “public Hearing” as per the procedure stipulated in the notification.

Table 29.5. Categories of development projects in which EIA is mandatory (MoEF, 1994; MoEF, 2001).

1. Nuclear Power and related projects such as Heavy Water Plants, nuclear fuel complex, Rare Earths.
2. River Valley projects including hydel power, major Irrigation and their combination including flood control.
3. Ports, Harbours, Airports (except minor ports and harbours).
4. Petroleum Refineries including crude and product pipelines.
5. Chemical Fertilizers (Nitrogenous and Phosphatic other than single superphosphate).
6. Pesticides (Technical).
7. Petrochemical complexes (Both Olefin and Aromatic) and Petro-chemical intermediates such as DMT, Caprolactam, LAB etc. and production of basic plastics such as LLDPE, HDPE, PP, PVC.
8. Bulk drugs and pharmaceuticals.
9. Exploration for oil and gas and their production, transportation and storage.
10. Synthetic Rubber.
11. Asbestos and Asbestos products.
12. Hydrocyanic acid and its derivatives.
- 13 (a) Primary metallurgical industries (such as production of Iron and Steel, Aluminium, Copper, Zinc, Lead and Ferro Alloys).
13. Electric arc furnaces (Mini Steel Plants).
14. Chlor alkali industry.
15. Integrated paint complex including manufacture of resins and basic raw materials required in the manufacture of paints.
16. Viscose Staple fibre and filament yarn.
17. Storage batteries integrated with manufacture of oxides of lead and lead antimony alloys.
18. All tourism projects between 200 m—500 metres of High Water Line and at locations with an elevation of more than 1000 metres with investment of more than ₹ 5 crores.
19. Thermal Power Plants.
20. Mining projects (major minerals) with leases more than 5 hectares.
21. Highway Projects except projects relating to improvement work including widening and strengthening of roads with marginal land acquisition along the existing alignments provided it does not pass through ecologically sensitive areas such as National Parks, Sanctuaries, Tiger Reserves, Reserve Forests.
22. Tarred Roads in the Himalayas and or Forest areas.
23. Distilleries.
24. Raw Skins and Hides
25. Pulp, paper and newsprint.
26. Dyes.; 27. Cement; 28 Foundaries (individual)
29. Electroplating; 30. Meta amino phenol

The existing manual on Environmental Impact Assessment (EIA) of MoEF, is common for all the sectors requiring prior environmental clearance. However, considering the diversity in all sectors related to infrastructure and industrial development projects, MoEF launched a program for development of sector specific technical EIA guidance manuals. The EIA guidance manual will help the project proponent/consultant in the preparation of the EIA report. It also helps the regulatory authority to review the report as well as the public to become aware of the related environmental issues.

Environmental Impact Notification S.O. 1533 (E), dt. 14th September 2006 has made it mandatory to obtain environmental clearance for scheduled development projects.

We have heavily drawn on the EIA manual of MoEf (2001) in describing the EIA process, environmental components, and participants and their role in the EIA process earlier in the chapter.

Environmental appraisal procedure

With a view to ensure multi-disciplinary input required for environmental appraisal of development projects, Expert Committees (Environmental Appraisal Committees) have been constituted by the Government of India for the following sectors:

- Mining Projects
- Industrial Projects
- Thermal Power Projects
- River Valley, Multipurpose, Irrigation and Hydroelectric Projects
- Infrastructure Development and Miscellaneous Projects
- Nuclear Power Projects

The Appraisal Committees evaluate the impact of the project based on the data furnished by the project authorities. If necessary, site visits or on-the-spot assessment of various environmental aspects are also undertaken. In the case of projects where the project proponents have submitted complete information, a decision is taken within 90 days. After considering all the facets of a project, environmental clearance is accorded subject to implementation of the stipulated environmental safeguards. The six regional offices of the Ministry functioning at Shillong, Bhubaneswar, Chandigarh, Bangalore, Lucknow and Bhopal undertake monitoring of cleared projects. The primary objective of such a procedure is to ensure adequacy of the suggested safeguards and also to undertake mid-course corrections required, if any. Any changes in the scope of project are identified to check whether review of earlier decision is called for or not.

Coastal States/Union Territories are required to prepare Coastal Zone Management Plans (CZMPs) as per the provisions of the Coastal Regulation Zone (CRZ) Notification 1991, identifying and categorising the coastal areas for different activities, and submit these to the Ministry for approval. The Ministry constitutes a Task Force for examining their plans. It has often been observed that one or more natural resource (s) becomes a limiting resource in a given region thereby restricting the scope of development portfolios. The Ministry of Environment and Forests has been sponsoring Carrying Capacity Studies for different regions. The studies involve:

1. Inventory of the natural resources available;
2. Preparation of the existing environmental settings;
3. Perspective plans and their impact on natural resources through creation of “Business As Usual Scenario”;
4. Identification of “Hot Spots” requiring immediate remedial action to overcome air, water or land pollution;
5. Formulation of alternative development scenarios includes Preferred Scenarios. A comparison between “Business As Usual” and the “Preferred Scenario” would indicate the future course of action to be adopted for development of the region after the package has been discussed with the local people as well as the planners.

CHAPTER

30

Sustainable Development and Ecological Economics

The term, sustainable development, was popularized in *Our Common Future*, a report published by the World Commission on Environment and Development in 1987. Also known as the Brundtland report, *Our Common Future* included the “classic” definition of sustainable development: “development which meets the needs of the present without compromising the ability of future generations to meet their own needs” (WCED, 1987). It is generally accepted that sustainable development calls for a convergence between the three pillars of economic development, social equity, and environmental protection. Sustainable development is a noble and necessary aspiration. It is a visionary development paradigm; and over the past 20 years governments, businesses, and civil society have committed to sustainable development goals (UN, 2012a). The United Nations Rio+20 summit held in Brazil in 2012 committed governments to create a set of sustainable development goals (SDGs) that would be integrated into the follow-up to the Millennium Development Goals (MEA, 2005a) after 2015 (Griggs et al., 2013).

Sustainable development emphasizes that rate of consumption and use of natural resources must approximate the rate at which these resources can be substituted or replaced. At the same time, it requires that society is able to satisfy social, economic and other needs without affecting the interests of future generations. The principles of sustainability emphasize on respect and care for community life, quality of human life and conservation of biodiversity. The aim is to achieve compatibility between ecology and economics and between local and global issues of development, and the management of the human use of the resources. The sustainability science has a major focus on management strategies for the global sustainability, appropriate methodologies and institutional needs, and political agenda of sustainable development and the nature society interactions (Kates et al., 2001). There are interactions between many of the social, economic and environmental challenges, which necessitate developing a strategic approach to sustainable development depending upon country’s own needs, priorities and resources (OECD/ DAC, 2001a; Dalal-Clayton and Bass, 2002). Thus sustainability has become one of the core global concepts keeping in view the fate of the earth system and the diversity of organisms that inhabit it.

30.1 DRIVERS OF CHANGE

In developing countries, extreme poverty still afflicts the lives of millions of people. In 1993, more than 1.3 billion people were living on less than US\$1 per day out of which 1 billion people were in Asia and Pacific regions (UNEP, 1999). The populations of the 47 least developed countries (i.e., 10% of the world population) exist on less than 0.5% of the world’s income, more than 2 billion people do not have access to adequate sanitation, 1 billion people do not have access to clean water, and 1.3 billion people are breathing air below the quality set by WHO (Serageldin, 2002). Poverty and socio-economic inequality raise serious issues of sustainability concerning peace and security, equity and solidarity, and the environment protection at the national, regional

and global levels. High rates of poverty are associated with various social evils like disease, family breakdown, endemic crime and use of narcotic drugs. The question that arises is ‘How can the quality of life of the poor be improved without undermining the possibility of realizing a decent quality of life later on?’.

Environmental pollution (see chapters 19-21), Global climate change (chapter 23), natural resource deterioration (chapter 25), loss of biological diversity (chapter 26) have complex interactions with socio-economic systems. Climate change poses numerous and stark challenges for sustainable development and its effects will be felt in all regions of the globe, although the intensity of exposure will vary. Coastal communities, notably in small islands and megadeltas, mountain settlements and urban communities in megacities of developing countries are particularly vulnerable (Intergovernmental Panel on Climate Change, 2012a).

The International Energy Agency (2012) has estimated that water consumption for energy production will increase by 85 % between now and 2035. Overall, global energy demand is projected to increase by about one third in this time period.

The Human activities significantly affect the sustainability of biosphere (Lubchenco et al. 1991). The various human activities intended to improve the quality of life are usually accompanied by environmental degradation (Fig. 30.1). Such activities as over fishing, agriculture, over use of fresh water supply, deforestation and industrialization cause environmental degradation and social disintegration on account of altered ecosystem biogeochemistry and declining resources (Fig. 30.2).

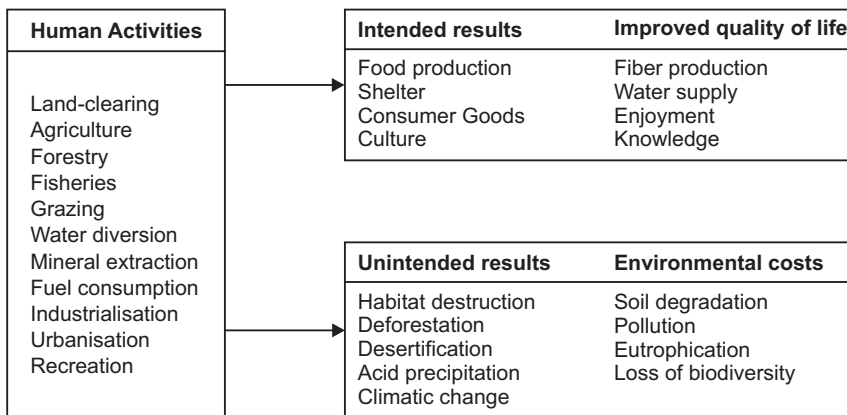


Fig. 30.1. Human activities affecting sustainability of the biosphere (Lubchenco et al., 1991).

The major global environmental changes that are being realized and are occurring at ever-increasing pace have been discussed at various international forums (see UNDP 1999, 2001; WRI/UNDP/UNEP/World Bank, 2000; OECD/DAC, 2001a,b). Global Environment Outlook 2000 and Global Environment Outlook 3 and 5 (UNEP GEO-3; UNEP GEO-5) give a comprehensive assessment of the global environment and present a range of scenarios and their possible consequences (UNEP, 1999; UNEP GEO-3, 2002; UNEP GEO-5, 2012). It is evident that global environmental changes are accelerating and they interact with each other and with other social and economic changes. Nearly two thirds of the services provided by nature to humankind are found to be on decline worldwide (UN, 2012a).

Most of the assessments of the human impact indicate that human activities are moving several of Earth's sub-systems outside the range of natural variability typical for the previous more than 500,000 years and may indeed be passing the ‘tipping points’ in the Earth system which may lead to rapid and irreversible change. The work of the Stockholm Resilience Centre on planetary boundaries (see Box 30.1) is an important example of the work in this field.

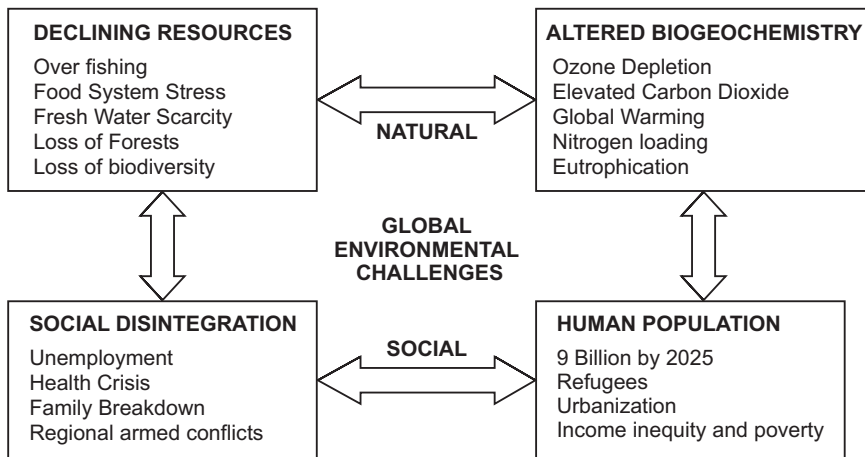


Fig. 30.2. The Global Challenges of sustainable development include unsustainable use of natural resources, altered biogeochemistry, social disintegration and human population increase. There is cause and effect relationship among the natural and social factors of unsustainability.

Box 30.1. A Safe Operating Space for Humanity

In 2009, a group of scientists led by Johan Rockström from the Stockholm Resilience Centre proposed a framework of “planetary boundaries”. These boundaries define the safe operating space for humanity with respect to the Earth system and are associated with the planet’s biophysical subsystems or processes. Crossing certain biophysical thresholds could have disastrous consequences for humanity and outside of these safe limits the Earth system cannot continue to function in a stable, Holocene-like state (Rockström et al., 2009a). This framework is based on scientific research that indicates that since the Industrial Revolution, human actions have gradually become the main driver of global environmental change. A total of nine boundaries are identified: climate change, rate of biodiversity loss, biogeochemical flows (both nitrogen and phosphorus), stratospheric ozone depletion, ocean acidification, global freshwater use, change in land use, atmospheric aerosol loading and chemical pollution. The scientists estimate that human activity appears to have already transgressed the boundaries associated with climate change, rate of biodiversity loss and changes to the global nitrogen cycle. For example, Rockström et al. (2009a,b) propose that human changes to atmospheric CO₂ concentrations should not exceed 350 parts per million by volume, and that radiative forcing should not exceed 1 watt per square metre above pre-industrial levels; but we have crossed 400 ppm CO₂ concentration and 1.5 watt/m² radiative forcing. Similarly against the planetary boundary of biodiversity loss (rate of extinction) of 10 species per million species per year, we are losing > 100 spp/million spp/yr, and against the planetary boundary of N₂ removal from atmosphere for human use of 35 million tons/yr, we are removing 121 million tons/yr. *source: The Stockholm Resilience Centre. More information is available at www.stockholmresilience.org.*

30.2 TOWARDS SUSTAINABLE DEVELOPMENT

The challenges for achieving sustainable development are numerous, the main threats are: economic disparity and political instability, extreme poverty, under nourishment and disease, population growth, high consumerism in developed economies and increasing global energy use. A number of steps have been and are being taken, however, to move the society on a path of sustainable development.

World Conservation Strategy

In 1980, the International Union for Conservation of Nature and Natural Resources (IUCN), the United Nations Environment Programme (UNEP), and the World Wide Fund for Nature (WWF) developed the world conservation strategy, a long range plan for conserving the world's biological resources (IUCN/ UNEP/WWF, 1980). This plan was expanded and followed by "*Caring for the Earth: a strategy for Sustainable Living*" (IUCN/ UNEP/WWF, 1991). This report enlisted a set of principles and strategies of a sustainable society based on practical integration of environmental, social and economic concerns. Its primary goals were to maintain essential ecological processes and life support systems, to preserve species and genetic diversity, to ensure the use of species and ecosystems in a sustainable manner and to improve the quality of human life. The ecological society of America published the "Sustainable Biosphere Initiative: an ecological research Agenda" (Lubchenco et al., 1991) and recognised sustainability as a unifying idea in applied ecology.

World Commission on the Environment and Development

Though the term sustainability has been in use for a long time, it became more widely used in the 1980s. The United Nations (UN) established a commission called the World Commission on the Environment and Development in 1983. This commission is frequently referred to as the Brundtland Commission, after Gro Harlem Brundtland, the Head of the commission and former the Prime Minister of Norway. The commission proposed a global agenda to address the world's environmental problems and the peoples' concerns relating to living conditions, resources, population pressures, international trade, education, and health.

The Brundtland Commission Report, *Our Common Future*, defined sustainable development as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" (WCED, 1987). The Brundtland report provided the momentum for the landmark 1992 Rio Summit that laid the foundations for the global institutionalization of sustainable development. The idea of sustainability figures as one of the leading models for societal development by indicating the direction in which societies ought to develop and is based on the premise that every human being has the right to live a decent life; sustainability is to be conceptualised as the attempt to accomplish intragenerational justice under the condition of intergenerational justice (Christen and Schmidt, 2012). Since the stable functioning of Earth systems including the atmosphere, oceans, forests, waterways, biodiversity and biogeochemical cycles is a prerequisite for a thriving global society, definition of sustainable development must be revised to include the security of people and the planet. The definition of sustainable development should therefore be "development that meets the needs of the present while safeguarding Earth's life-support system, on which the welfare of current and future generations depends" (Griggs et al., 2013).

The concept of sustainable development has three principal dimensions (pillars): economic, social and environment (Fig 30.3). Human well-being rests on the successful integration of these pillars of sustainability.

Underlying the economic dimension is the principle that society's well being would have to be maximized and poverty eradicated through the optimal and efficient use of natural resources. The concept of "needs" emphasized in the Brundtland commission's definition refers, in particular, to the essential needs of the world's poor, to which overriding priority should be given. The social aspect refers to a socially sustainable system based on distributional equity, uplifting the welfare of people, improving access to basic health and education services, gender equity, political accountability and participation. It also refers to the development of various cultures, diversity, pluralism and effective grass roots participation in decision-making. The issue of equity, in terms of the distribution of benefits and access to resources remains an essential component of both the economic and social dimensions of sustainable development.

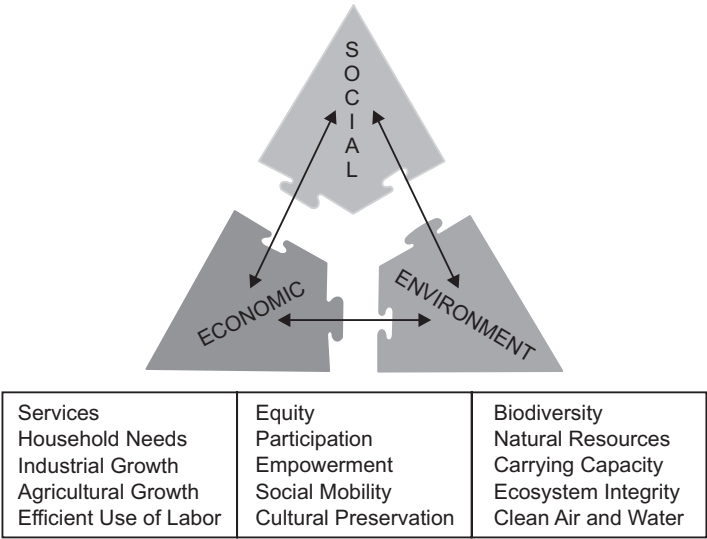


Fig. 30.3. The complex interactions among the different dimensions of sustainable development. (Source : <http://www.worldbank.org>)

The environmental dimension, on the other hand, is concerned with the conservation and enhancement of the physical and biological resource base, avoiding over exploitation of renewable resources and depletion of non-renewable resources. Environmental sustainability has a main focus on maintenance of biological diversity, atmospheric stability and ecosystem functions and services.

For the various development planning purposes, sustainability also takes into account the spatial and cultural aspects (Fig. 30.4). Spatial sustainability refers to the balanced rural-urban configuration, preventing destruction of fragile ecosystems, regenerative agriculture and establishment of network of natural protected areas. Cultural sustainability recognizes the indigenous cultures as well as the concept of eco-development.

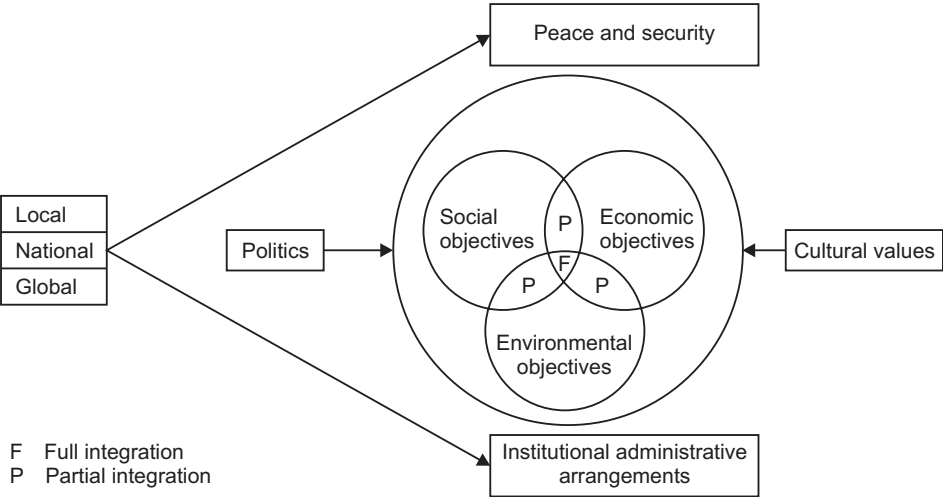


Fig. 30.4. The Systems of sustainable development (Dalal-Clayton, B., Bass, S., *Sustainable Development Strategies: A Resource Book*, © OECD 2002).

The United Nations Commission on Sustainable Development (CSD) was created in December 1992 to ensure effective follow-up of United Nations Conference on Environment and Development

(UNCED) held at Rio; to monitor and report on implementation of the Earth Summit agreements at the local, national, regional and international levels. The CSD is a functional commission of the UN Economic and Social Council (ECOSOC), with 53 members. The United Nations World Summit on Sustainable Development (WSSD) was held in Johannesburg, South Africa, 26 August– 4 September 2002. The WSSD urged in its Plan of Implementation that: “States should take immediate steps to make progress in the formulation and elaboration of national strategies for sustainable development and begin their implementation by 2005”.

The major challenge has been to understand the resilience of ecosystems so as to find ways to fulfill the demand of people with the tolerance range of nature. In this direction, the *Millennium Ecosystems Assessment* is an important effort that was launched in June 2001 by the UN. The Millennium Ecosystems Assessment was followed by PAGE (Pilot Analysis of Global Ecosystems), which assessed five major ecosystem types including agro ecosystems, forests, freshwater, grasslands and coastal and marine ecosystems. The results have been reported in the World Resources Report 2000–2001 (WRI/UNDP/UNEP/World Bank, 2000; Gewin, 2002). There have been regional assessments of the state of environment and of challenges to sustainable development. For example, “India; the State of environment–2001” has been published by the UNEP in association with the Ministry of Environment and Forests and The Energy Research Institute (TERI). The report has examined the most serious environmental issues and discussed the impact of pollution, hazardous wastes on human health (UNEP, 2001).

30.3 HAS SUSTAINABLE DEVELOPMENT FAILED?

Even after 20 years of Rio 92, unsustainable trends continue. The background paper developed by the United Nations Headquarters in 2010 lists the following trends: (i) Economic growth since the Rio Summit of 1992 has been fed by unprecedented resource and material consumption and related environmental impacts. (ii) Large portions of the natural world have been converted to human use. Prompting concerns about the ability of the world’s natural resource base to sustain such growth. (iii) Assessment of incomes and wealth reveal a large and growing gap between the rich and poor. Rising income inequality has been observed in most countries over the past two decades. Roughly 80 percent of the natural resources used each year are consumed by about 20 percent of the world’s population. Overall the world remains extremely unequal, with the richest 1 percent of the world’s population owning 40 percent of the world’s wealth; while the poorest 50 percent have barely 1 percent of world wealth. (iv) Increasing consumption, combined with population growth, mean that humanity’s demands on the planet have more than doubled over the past 45 years. (v) Global biodiversity continues to decline, and species in all groups with known trends are, on average, being driven closer to extinction. (vi) Provision of many critical ecosystem services—such as water, biodiversity, fibre, and food—is being compromised due to the impact of human development. (vii) Scarcity of and competition over freshwater is a growing concern for many regions in the world, with around 50 countries currently facing moderate to severe water stress.

The continuing unsustainability has prompted Brand (2012) to state “Looking back, it becomes clear that the concept of sustainable development was a political strategy of global environment and resource management, of ecological modernization and – at least at the beginning – an attempt to reconcile environmental problems with those of development. It was, right after 1989, part of a prevailing optimism that global problems could be solved cooperatively. However, Sustainable development has failed because of the absence of relevant socio-economic actors needed to significantly push this strategy”. The background paper also felt that the implementation of sustainable development has been largely unsuccessful. Notwithstanding the belief that sustainable development calls for a convergence between the three pillars of economic development, social equity, and environmental protection, over the past 20 years it has often been compartmentalized as

an environmental issue. Added to this is the reigning orientation of development as purely economic growth. Climate change has emerged as the de facto proxy for addressing sustainable development issues and the economic agenda and economic institutions have had greater influence than those governing environmental and social issues, at both the national and international level.

The Background paper suggests the following reasons for unsuccessful implementation of Sustainable Development: (i) Sustainable development is part of the international lexicon, but the concept remains too amorphous to be clearly defined, and hence implemented. (ii) Sustainable development remains fundamentally an environmental issue. (iii) Sustainable development has been subject to competing agendas. (iv) Development as economic growth continues to be the dominant paradigm. (v) Developed countries have not met commitments to developing countries, generating an atmosphere of distrust. (vi) Sustainable development has not been able to find the political entry points to make real progress.

Nevertheless, it can be said that Rio 1992 did not fail the world—it provided the vision and important pieces of the multilateral machinery to achieve a sustainable future. The following snapshots of progress or lack of progress on the basis of data from several sources, give an overview of a range of key areas, in each case looking at changes that took place between 1990 and 2010 (UN, 2012b), (BOX 30.2).

Box 30.2. The global track record on sustainable development

Poverty eradication

27 % of the world's population live in absolute poverty, down from 46 per cent in 1990.

Economic growth and inequality

Global economic growth is up 75 per cent since 1992 but inequality is still high

Hunger and undernourishment

An increase of 20 million undernourished people since 2000

Forests

5.2 million hectares net forest loss per year

The ozone layer

Ozone layer will recover to pre-1980 levels in 50 years plus

Biodiversity and ecosystems

Two thirds of the services provided by nature to humankind are in decline

The oceans

85 % of all fish stocks are overexploited, depleted, recovering or fully exploited

Climate change

38% increase in annual global carbon dioxide emissions between 1990 and 2009

Energy

20 % of the world's population lack access to electricity

Water and sanitation

884 million people lack access to clean water

2.6 billion people are without access to basic sanitation

Gender

43% of those in the agriculture workforce in developing countries are women

Education

67 million children of primary school age are out of school

Health

3.5-year increase in life expectancy between 1990 and 2010

Source : On the basis of data from various sources; Resilient people, resilient planet: a future worth choosing (UN, 2012b).

30.4 GUIDING PRINCIPLES OF SUSTAINABLE DEVELOPMENT

Sustainable development is a global concept and the preconditions for sustainable development are equity and social justice, economic efficiency and ecological harmony. The main Agenda for sustainable development has a focus on carrying capacity based sustainable development; structural changes in economic sector and preservative environmental planning. Conservation based development needs to protect the structure and function as well as diversity of natural ecosystems with a focus on the following aspects:

- To conserve life-support-systems and ecological processes.
- To conserve biological diversity in terrestrial, freshwater and marine systems.
- Sustainable use of resources
- Minimizing depletion of natural resources
- To care for the environment and the poor in south
- No to exceed the earth's carrying capacity
- To improve the quality of life including social and economic concerns
- To develop a national framework code to integrate development and conservation.

According to Daly (1994), the concept of sustainable development can be operationalised in terms of the conservation of natural capital both for renewable and non renewable resources. Natural capital consists of all the natural resources and the ecosystem services. In ecological perspectives, sustainability must involve limits on population and consumption levels, as humans are now using, co-opting or eliminating some 40% of the net primary productivity of all terrestrial ecosystems (Vitousek et al., 1986). The nature's natural capital is seen as a finite quantity whose use must be spread over time in order to be utilized by as many generations of people as possible. As summarized by Ramakrishnan (2001) on the basis of Hare et al. (1990), the guiding principles apply to ecological, social and cultural dimensions, which relate to the following:

- Intergenerational equity for today and tomorrow
- Conservation of cultural and biological diversity
- Reliance on nature's capital without depleting it.
- Precautionary policy approach to use resources
- Stable world population with low birth and death rates.
- Limits on natural resource use within the capacity to supply renewable resources and assimilate wastes.
- Pricing environmental values and natural resources so as to cover full environmental and social costs.
- Global perspective on environmental issues.
- Efficiency of resource use by all societies.
- Strong community participation in policy and practice.
- Sharing of benefits between rich and poor.
- An ethical transition in attitudes.

30.5 SUSTAINABILITY INDICATORS

Chapter 40 of Agenda 21 called for the development of indicators for sustainable development.

Indicators are defined as something that reveal or detect and are also a simple statistical figure that helps compress information into an easily understandable form that can facilitate the decision making process. They can help to measure the progress made towards sustainable development as well as provide an effective means of communicating ideas, thoughts and values. The United

Nations Division for Sustainable Development approved indicators of sustainable development. CSD Indicators of Sustainable Development – 3rd edition are summarized in Table 30.1. New set of 50 core indicators has been created. Core indicators are part of larger set of 96 indicators of sustainable development. Core indicators fulfill the following three criteria: (i) They cover issues that are relevant for sustainable development in most countries. (ii) They provide critical information not available from other core indicators. (iii) They can be calculated by most countries with data that is readily available or could be made available within reasonable time and costs.

Table 30.1. CSD Indicators of Sustainable Development – 3rd edition (The United Nations Division for Sustainable Development, 2007).

Theme	Sub-theme	Core indicator
Poverty	Income poverty	Proportion of population living below national poverty line
	Income inequality	Ratio of share in national income of highest to lowest quintile
	Sanitation	Proportion of population using an improved sanitation facility
	Drinking water	Proportion of population using an improved water source
	Access to energy	Share of households without electricity or other modern energy services
	Living conditions	Proportion of urban population living in slums
Governance	Corruption	Percentage of population having paid bribes
	Crime	Number of intentional homicides per 100,000 population
Health	Mortality	Under-five mortality rate Life expectancy at birth
	Health care delivery	Percent of population with access to primary health care facilities Immunization against infectious childhood diseases
	Nutritional status	Nutritional status of children
	Health status and risks	Morbidity of major diseases such as HIV/AIDS, malaria, tuberculosis
Education	Education level	Gross intake ratio to last grade of primary education Net enrolment rate in primary education Adult secondary (tertiary) schooling attainment level
	Literacy	Adult literacy rate
Demographics	Population	Population growth rate Dependency ratio
	Tourism	
Natural hazards	Vulnerability to natural hazards	Percentage of population living in hazard prone areas
	Disaster preparedness and response	
Atmosphere	Climate change	Carbon dioxide emissions
	Ozone layer depletion	Consumption of ozone depleting substances
	Air quality	Ambient concentration of air pollutants in urban areas

Land	Land use and status	
	Desertification	
	Agriculture	Arable and permanent cropland area
	Forests	Proportion of land area covered by forests
Oceans, seas and coasts	Coastal zone	Percentage of total population living in coastal areas
	Fisheries	Proportion of fish stocks within safe biological limits
	Marine environment	Proportion of marine area protected
Freshwater	Water quantity	Proportion of total water resources used Water use intensity by economic activity
	Water quality	Presence of fecal coliforms in freshwater
Biodiversity	Ecosystem	Proportion of terrestrial area protected, total and by ecological region
	Species	Change in threat status of species
Economic Development	Macroeconomic performance	Gross domestic product (GDP) per capita
		Investment share in GDP
	Sustainable public finance	Debt to GNI ratio
	Employment	Employment-population ratio Labor productivity and unit labor costs Share of women in wage employment in the non-agricultural sector
	Information and communication technologies	Internet users per 100 population
	Research and development	
	Tourism	Tourism contribution to GDP
Global economic partnership	Trade	Current account deficit as percentage of GDP
	External financing	Net Official Development Assistance (ODA) given or received as a percentage of GNI
Consumption and production patterns	Material consumption	Material intensity of the economy
	Energy use	Annual energy consumption, total and by main user category Intensity of energy use, total and by economic activity
	Waste generation and management	Generation of hazardous waste Waste treatment and disposal
	Transportation	Modal split of passenger transportation

The environmental indicators relating to atmosphere, land, oceans, seas, coasts, freshwater and biodiversity are briefly described as under:

Atmosphere

Agenda 21 suggests an integrated approach to protection of the atmosphere coordinated with social and economic development, which focuses on:

- improving the scientific basis for addressing uncertainties;
- preventing stratospheric ozone depletion;

- addressing transboundary air pollution; and
- promoting more sustainable and efficient energy use, transportation, consumption, industrial development, and land and marine resource use.

Land

The indicators under the land theme in the CSD framework focus on the key sub-themes of agriculture, forests, desertification, and urbanization. The selected indicators are relevant for assessing sustainable development at the national level, and are generally supported by appropriate data sets. The indicators from other environmental sub-themes complement these measures of land sustainability, for example, ecosystems, water quality, climate change, and coastal zones. In addition, pressures on land and land use impacts are reflected in social, economic, and institutional themes, such as poverty, drinking water, population change, energy use, and natural disaster preparedness and response.

Oceans, seas and coasts

The CSD core set includes three indicators under the sub-themes of coastal zone and fisheries that are national in scope and within the capabilities of most countries to develop. The total population in coastal areas and the algae concentration in coastal waters provide measures of the overall pressure on the coastal resource particularly from land-based activities. The annual catch by major species provides a core indicator where data is generally available to measure the intensity of fishery activity. In terms of key marine ecosystems such as coral, mangrove, and sea grass, countries may wish to consider trends in the extent of these areas under the biodiversity ecosystem sub-theme.

Freshwater

The freshwater indicators in the core set capture the two essential dimensions of quantity and quality. The withdrawal of available water measures a country's demand for water and reveals its vulnerability to water shortages. The measurement of biological oxygen demand and fecal coli form concentration reflect, respectively, the two significant aspects of ecosystem health and human health. These three indicators are policy-relevant and generally measurable at the national level.

Biodiversity

The selected biodiversity indicators measure the area of selected key ecosystems, the abundance of selected key species, and the protected area as a percent of total area. They are relatively easy to calculate, meaningful to decision makers, and allow countries flexibility in determining the most important ecosystems and species from a national perspective. Area of key ecosystems, for example, was suggested as a valuable indicator by testing countries and could include those areas subject to greatest change or those with special biodiversity value. It may also be possible in some countries to disaggregate the protected area indicator by ecosystem type to provide information on ecosystem representation. The Ministry of Environment and Forests, India, also included some human development indicators such as infant mortality rate, life expectancy at birth and adult literacy rate better suited to Indian conditions in addition to those of the UN list.

30.6 NATIONAL SUSTAINABLE DEVELOPMENT STRATEGIES

Since UNCED, extensive efforts have been made by governments and international organizations to integrate environmental, economic and social objectives into decision-making through new policies and strategies for sustainable development. For example, the IUCN is making

considerable efforts to develop the concepts of sustainable use of the components of biological diversity, particularly sustainable use of species. Sustainable use of wild species has a potential to provide developmental benefits; ensuring long-term supply of valuable resources to people; enabling the recovery of species and population depleted by overuse; conservation of species and conservation of ecosystems.

Agenda 21 promotes National Sustainable Development Strategies (NSDS) as mechanisms for translating a country's goals and aspirations of sustainable development into concrete policies and actions. NSDS is a coordinated, participatory interactive and cyclical process of thoughts and actions to achieve economic, environmental and social objectives in a balanced and integrated manner (Dalal-Clayton et al., 1994; IIED, 2000)). The process encompasses situation analysis, formulation of policies and action plans, implementation, monitoring and regular review. It is a process of planning, participation and action in which the emphasis is on managing progress towards sustainability goals rather than producing a "plan" as an end product. Agenda 21 states that countries should adopt national strategies for sustainable development, which "should build upon and harmonize the various sectors of economic, social and environmental policies and plans that are operating in the country".

The UN international forum on national strategies has contributed to the principles of sustainable development that are applicable to both developed and developing countries (UN DESA, 2002a; UN DESA, 2002b). These principles are summarized in Table 30.2. The rationale for a systematic sustainable development strategy is illustrated in Fig 30.5 with an emphasis on participation, communication, capacity strengthening and monitoring. The country dialogue also identifies specific mechanisms and processes so as to strengthen the effectiveness of national development strategies. It may be mentioned that development plans, poverty reduction strategies and the approaches provide a basis for achieving the goals of effective NSDS.



Fig. 30.5. Constellation of mechanisms contributing to a sustainable development strategy, Dalal-Clayton, B., Bass, S., *Sustainable Development Strategies: A Resource Book*, © OECD 2002).

Table 30.2. Elements of a national sustainable development strategy (based on UN DESA, 2002b).

A	Integration of economic, social and environmental objectives, and balance across sectors, territories and generations.	• Linking local, national, regional and global priorities and actions; • Linking the short term to the medium and long term; • Linking the national, regional and global levels; • Linking different sectors; • Coherence between budgets and strategy priorities.
B	Broad participation and effective partnerships	• Institutionalized channels for communication; • Access to information for all stakeholders and effective networking; • Transparency and accountability; • Trust and mutual respect; • Partnerships among government, civil society, private sector and external intuitions.
C	Country ownership and commitment	• Strong political and stakeholder commitment; • Sound leadership and good governance; • Shared strategic and pragmatic vision; • Strong institution of group of institutions spearheading the process; • Continuity of the national sustainable development strategy process.
D	Developing capacity and enabling environment	• Building on existing knowledge and expertise; • Building on existing mechanism and strategies.
E	Focus on outcomes and means of implementation	• The means to assess and agree priority issues in place; • Coherence between budge, capacity and strategy priorities; Realistic, flexible targets; • Linked to private sector investment; • Anchored in sound technical and economic analysis; • Integrated mechanisms for assessment follow up, evaluation and feedback.

The negotiations at the WSSD in 2002 demonstrated a major shift in the perception of sustainable development away from environmental issues toward social and economic development. This shift, which was driven by the needs of the developing countries and strongly influenced by the Millennium Development Goals (MDGs) (UN, 2010). The Fourth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC, 2007, chapter 12.1.1) pointed out the iterative relationship between climate change and sustainable development, and that the two can be mutually reinforcing. In many respects, the UNFCCC has become an international proxy for discussions around sustainable development, and a potential means to channel required funding and technology from developed to developing countries.

30.7 SUSTAINABLE DEVELOPMENT GOALS

One of the main outcomes of the Rio+20 Conference was the agreement by member States to launch a process to develop a set of Sustainable Development Goals (SDGs), which need to be combined with the the Millennium Development Goals and converge with the post 2015 development agenda (UN, 2012b). It was decided to establish an *“inclusive and transparent intergovernmental process open to all stakeholders, with a view to developing global sustainable development goals to be agreed by the General Assembly”*.

In the Rio+20 outcome document, member States agreed that sustainable development goals (SDGs) must address the following :

1. Be based on Agenda 21 and the Johannesburg Plan of Implementation.
2. Fully respect all the Rio Principles.
3. Be consistent with international law.
4. Build upon commitments already made.
5. Contribute to the full implementation of the outcomes of all major summits in the economic, social and environmental fields.
6. Focus on priority areas for the achievement of sustainable development, being guided by the outcome document.
7. Address and incorporate in a balanced way all three dimensions of sustainable development and their interlinkages.
8. Be coherent with and integrated into the United Nations development agenda beyond 2015.
9. Not divert focus or effort from the achievement of the Millennium Development Goals.
10. Include active involvement of all relevant stakeholders, as appropriate, in the process.

The Rio+20 outcome documents *The Future We Want* resolved to establish an inclusive and transparent intergovernmental process on SDGs that is open to all stakeholders with a view to developing global sustainable development goals. The sustainable development goals have been combined with the MDG targets, updated and extended for 2030, to produce six SDGs, i.e., thriving lives and livelihoods, sustainable food security, sustainable water security, universal clean energy, healthy and productive ecosystems, and governance for sustainable societies (Griggs et al., 2013), (Box 30.3).

30.8 MILLENNIUM DEVELOPMENT GOALS

The UN Millennium Declaration was adopted in 2000 and committed countries (189 countries) to reach eight Millennium Development Goals by 2015. The Millennium Development Goals Indicators currently consist of 48 indicators linked to the eight goals derived from the United Nations Millennium Declaration. However, the revised MDG monitoring framework presented by the Secretary-General of the United Nations in 2007 contains 58 indicators, as four new targets have been included to reflect commitments made at the 2005 World Summit. The eight goals are: (1) halving extreme poverty, (2) halting the spread of HIV/AIDS, (3) providing universal primary education, (4) eliminating gender disparity in education, (5) reducing the under-five mortality rate, (6) reducing the maternal mortality rate and achieving universal access to reproductive health, (7) developing a global partnership (to address the needs of the poorest countries, to further an open non-discriminatory trade system, and to deal with developing country debt); and (8) ensuring environmental sustainability (by integrating sustainable development into country policies and programs, reducing biodiversity loss, improving access to safe drinking water and sanitation, and improving the lives of slum dwellers) (UN, 2010).

30.9 RIO+20 (2012) AND GREEN ECONOMY

The United Nations Rio+20 summit in Brazil in 2012 laid the foundations for the green economy for an effective transition to a low carbon economy, efficient in the use of resources, with the primary objectives of creating jobs, increasing social inclusion and, ultimately, eradicating poverty. *Towards a Green Economy* is among UNEP's key contributions to the Rio+20 process and the overall goal of addressing poverty and delivering a sustainable 21st century (UNEP, 2011). UNEP defines a green economy as one that results in *improved human well-being and social equity, while significantly reducing environmental risks and ecological scarcities*. The concept of a "green

economy” does not *replace* sustainable development, but there is now a growing recognition that achieving sustainability rests almost entirely on getting the economy right. For example, reducing deforestation can therefore be a good investment: the climate regulation benefits of halving global deforestation alone have been estimated to exceed the costs by a factor of three. Similarly Green agriculture is characterized by shifting both industrial and subsistence farming towards ecologically sound farming practices such as efficient use of water, extensive use of organic and natural soil nutrients, optimal tillage, and integrated pest control. The gap between global supply and water withdrawals can only be addressed by investments in infrastructure and water policy reform – i.e., greening the water sector. Further details on green economy are provided below under section 30.15.

Box 30.3. Six Sustainable Development Goals: Some Provisional Targets for 2030

Goal 1: Thriving lives and livelihoods. End poverty and improve well-being through access to education, employment and information, better health and housing, and reduced inequality, besides moving towards sustainable consumption and production.

Goal 2: Sustainable food security. End hunger and achieve long-term food security including better nutrition through sustainable systems of production, distribution and consumption.

Goal 3: Sustainable water security. Achieve universal access to clean water and basic sanitation, and ensure efficient allocation through integrated water-resource management.

Goal 4: Universal clean energy. Improve universal, affordable access to clean energy that minimizes local pollution and health impacts and mitigates global warming.

Goal 5: Healthy and productive ecosystems. Sustain biodiversity and ecosystem services through better management, valuation, measurement, conservation and restoration.

Goal 6: Governance for sustainable societies. Transform governance and institutions at all levels to address the.

Source : (Griggs et al., 2013)

30.10 ENVIRONMENTAL PERFORMANCE INDEX (EPI)

The Environmental Sustainability index (ESI) and Environmental Performance Index (EPI) have been found to be useful for formulating a more functional concept of environmental sustainable development and for environmental decision making (see Brown, 2001; Esty and Cornelius, 2002; World Economic Forum 2002 a and 2002b). The 2000 Environmental Sustainability Index (ESI), the predecessor to the Environmental Performance Index (EPI), first responded to the growing need for rigorous, data driven environmental performance measurement. The 2012 EPI, the seventh iteration of this environmental measurement project, adds to the foundation of empirical support for sound policymaking and breaks further ground, establishing for the first time a basis for tracking changes in performance over time (Emerson et al., 2012). The EPI and the Pilot Trend Environmental Performance Index (Trend EPI) rank countries on 22 performance indicators spanning ten policy categories reflecting facets of both environmental public health and ecosystem vitality (Emerson et al., 2012), Fig. 30.6.

The methodology facilitates country comparisons and provides a way to assess the global community's performance over time with respect to established environmental policy goals (Emerson et al., 2012). The 2012 EPI ranks 132 countries on 22 performance indicators in the following ten policy categories: (1) Environmental Burden of Disease (2) Water (effects on human health) (3) Air Pollution (effects on human health) (4) Air Pollution (ecosystem effects) (5) Water Resources (ecosystem effects) (6) Biodiversity and Habitat (7) Forestry (8) Fisheries (9) Agriculture (10) Climate change.

Switzerland (with an EPI score of 76.69) leads the world in addressing pollution control and

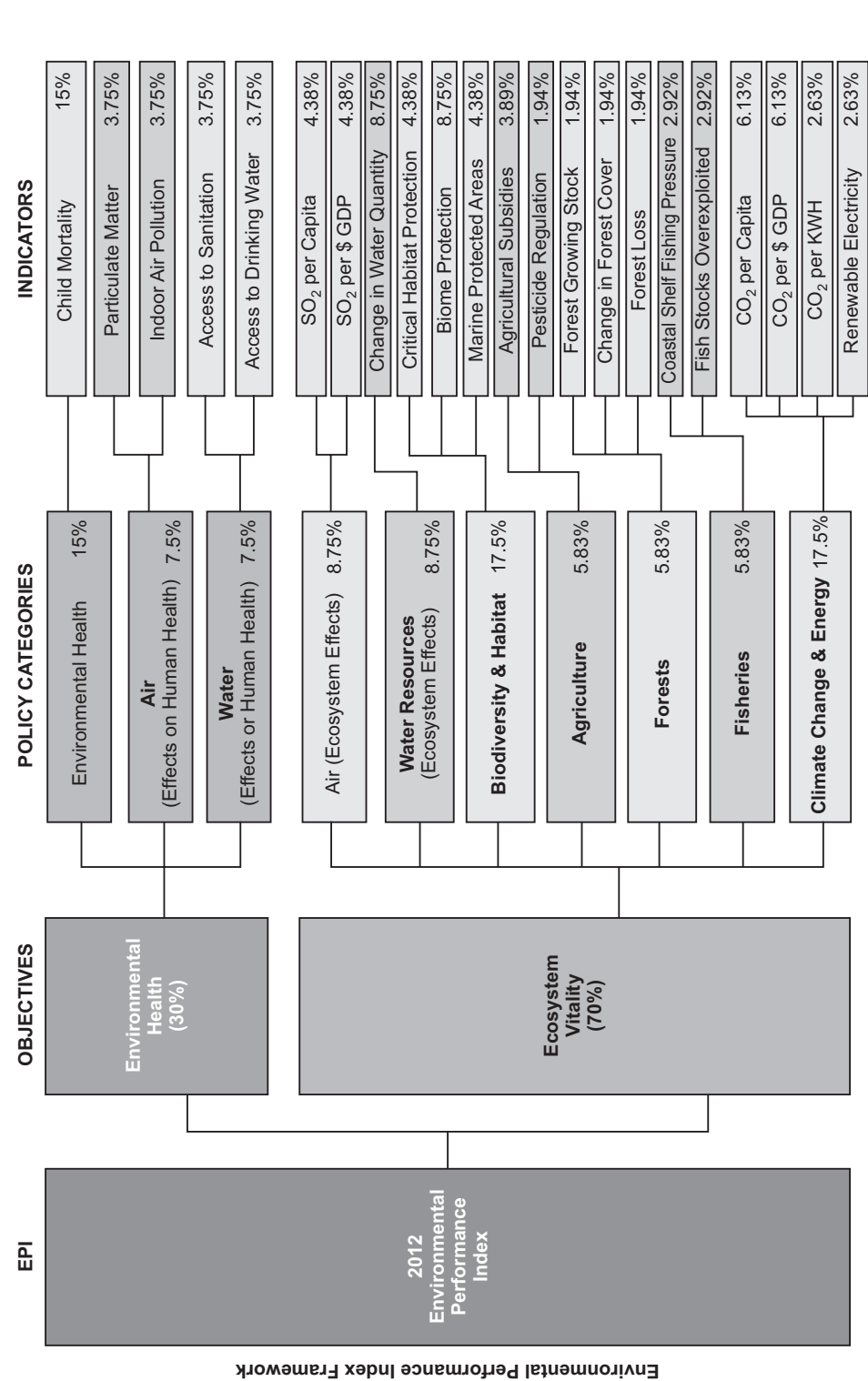


Fig. 30.6. The objectives, policy categories and indicators of the 2012 Environmental Performance Index. The percentages indicate the weightings used for aggregation (based on Emersen et al., 2012).

natural resource management challenges. Its top ranking on the 2012 EPI is in large part due to its high performance in air pollution control. Switzerland ranks first in the categories Air Pollution (effects on human health) and Air Pollution (ecosystem effects). It also has high marks for access to drinking water and the biodiversity and habitat indicators. Latvia (70.37), Norway (69.92), Luxembourg (69.2), and Costa Rica (69.03) round out the top five positions in the 2012 EPI. At the low end of the 2012 EPI rankings are South Africa (34.55), Kazakhstan (32.94), Uzbekistan (32.24), Turkmenistan (31.75), and Iraq (25.32). These countries are water scarce and face significant sustainability challenges; the last three are also known for weak governance. India (with an EPI score of 95.0) occupies 125 position among the 135 countries assessed for EPI.

Average Environmental Health and Ecosystem Vitality: Global Trends

By comparing results at the level of the Environmental Health and Ecosystem Vitality Objectives, differences in global performance can be revealed through trends over the last decade (Emerson et al., 2012). In Figure 30.7, it is clear that an imbalance between how global policy-making is organized with respect to environmental matters that affect human health directly (labeled “Environmental Health Objective”) and those that do not (labeled “Ecosystem Vitality Objective”). In 2000, we see that environmental health, averaged for the world, is significantly better than ecosystem vitality. And over the course of the next 11 years the gap has widened, environmental health has improved faster than ecosystem vitality. This imbalance reflects a failure to match policy-making capabilities to environmental objectives (Emerson et al., 2012). The 2012 EPI and Pilot Trend EPI provide a quantitative basis for comparing, analyzing, and understanding environmental performance for 132 countries. The two indexes rank these countries on their environmental performance using the most recent year of data available (the 2012 EPI) as well as performance over the last decade (the Trend EPI).

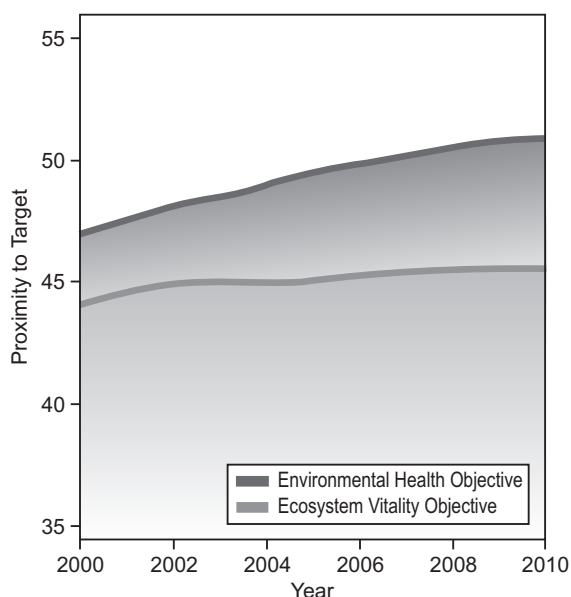


Fig. 30.7. Global trend in average Environmental Health and Ecosystem Vitality objective. (based on Emerson et al., 2012).

30.11 GLOBAL SUSTAINABILITY

A significant amount of progress has been made following the Brundtland commission’s report in meeting the challenges to the environment and sustainability. Improvements have been made in reusing materials and reducing wastes, increasing efficiency in the energy use and development of alternatives to fossil fuel. The seven billion people on Earth now have about 15% more food per capita than the population of four billion about 25 years ago. However, it is also true that the earth system is being pushed by human – driven changes beyond its natural operating domain and therefore stewardship of the Earth system as a whole has become a compelling issue. Several strategies of global sustainability have been suggested, a few are summarized below (see Steffen et al., 2004 for details).

- (1) **Adaptation:** Use a global change scenario to estimate the impact of global change on the system of interest, and then undertake such strategies that adapt the system to them. For

example, farmers can vary the planting date or switch to different crop types or develop new crop types.

- (2) **Technological solutions:** Employ geo-engineering techniques at the planetary scale, e.g. orbiting giant mirrors around the earth to deflect incoming solar radiation, injection of aerosols into the atmosphere to counteract greenhouse gases, injection of propane into the stratosphere to neutralize chlorine atoms, injection of CO₂ from industrial effluents into the deep ocean or massive reforestation to reduce the amount of CO₂ in the atmosphere. While technology has a key role to play in transiting to sustainable consumption, developed countries need to make conscious shifts from material to non-material consumption, from a private to a public mode of consumption, from one-time to multiple use of products, and from use of non-biodegradable to use of biodegradable material.
- (3) **Mitigation:** Lessen the intensity of the drivers of global change. For example, make the motor cars, aeroplanes and household appliances more efficient, increase the water use efficiency of agricultural crops.
- (4) **Standardisation:** Maintain the evolution of Earth System within a safe range of sustainability indicators. For example, maintain minimum air quality standards for urban air sheds by maintaining the global load of air pollutants within a certain range.
- (5) **Optimization:** Maximize utility at every point as the Earth System evolves into the future. For example, heavy exploitation of resources including fossil fuels by the developing countries in the first half of the 21st century in order to obtain an overall larger utility function for the earth system at the end of the century.
- (6) **Pessimisation:** Place the primary emphasis on the precautionary principle of preventing the worst from happening.
- (7) **Equity:** Reduce inequities amongst groups of people who currently reside on the planet, and also between current and future generations. For example, achieve a balance amongst the various stakeholders in global management.
- (8) **Equilibrium:** There are three options (i) preservation of a perceived idealized natural / human global environment, (ii) restoration of previously altered environment, and (iii) maintenance of an appropriate mix of natural and protected areas, and areas which are exploited for human purpose.
- (9) **Adaptive management:** Make tentative environment and development decisions on the basis of available knowledge and revise the decisions as new knowledge becomes available. The overall strategy must be to avoid catastrophic changes in the state of the Earth System, but within this overall strategy; a number of other approaches may be applied.

However, as Steffen et al. (2004) have discussed, none of the above strategies for global sustainability can be adopted by human societies in its pure form, and none is sufficient by itself to maintain global sustainability. Much remains to be known about the Earth System. Steffen et al. (2004) pose the basic question : “How can a large group of independent nations with differing cultures, values, wealth, social organization and world views come together to manage their own single, connected life support system in a coherent and effective way?”

30.12 SUSTAINABLE DEVELOPMENT IN A MORE INTERDEPENDENT WORLD

The Rio+20 outcome document, *The Future We Want*, resolved to establish an inclusive and transparent intergovernmental process on SDGs that is open to all stakeholders with a view to developing global sustainable development goals. The outcome document specifies that the process leading to the SDGs needs to be coordinated and coherent with the processes considering the post 2015 development agenda. Focus areas for priority attention at Rio+20 as articulated by member

states and major groups are summarized as:

Focus areas for priority attention at Rio+20
1. Green jobs, youth employment and social inclusion
2. Energy access, efficiency, sustainability
3. Food security and sustainable agriculture
4. Water
5. Sustainable cities
6. Management of the oceans, fisheries and other marine resources
7. Improved resilience and disaster preparedness

Achieving sustainable development post-2015 will entail progress in its four dimensions (i) inclusive economic development, (ii) inclusive social development, (iii) environmental sustainability and, (iv) effective governance and peace and security (United Nations System Task Team on the Post-2015 UN Development Agenda, 2012).

Sustainable improvements in human wellbeing is the ultimate purpose of all development effort, which substantially depends on a foundation that binds together and gives balanced weight to economic progress, social equity, a healthy environment and democratic governance (Geoghegan, 2013). The Independent Research Forum (IRF2015) is a grouping of international and regional research and knowledge organizations working in support of sustainable development. IRF2015's vision is a post-2015 development agenda based on the principles of sustainable development, which recognizes the mutual dependency of economic, social and environmental outcomes; is grounded in local experience and needs; and is adaptable to diverse contexts and capacities (Geoghegan, 2013), Fig. 30. 8.

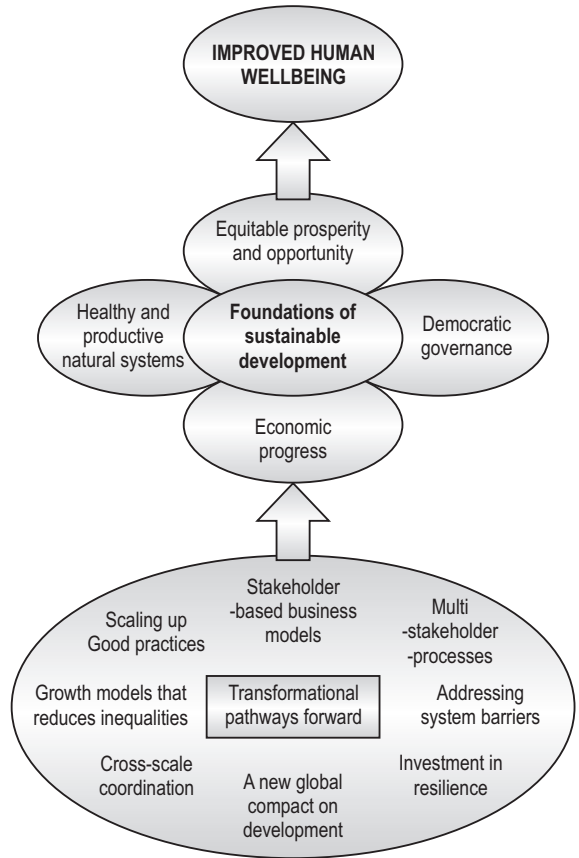


Fig. 30.8. The opportunities for progress that will secure the foundations for sustainable development (adapted from IRF 2015; Geoghegan, 2013).

30.13 SUSTAINABLE DEVELOPMENT IN INDIA: PERSPECTIVES AND STRATEGIES

Environmental sustainability considerations have been an integral part of the Indian culture. The need for conservation and sustainable use of natural resources has been expressed in Indian scriptures more than three thousand years old and is reflected in our constitutional, legislative and policy framework along with international commitments. The issues of increasing air and water

pollution, degradation of land and forests and loss of biodiversity have come into focus. Specific measures were initiated since the Stockholm Declaration way back in 1972. Sustainability concerns have become an intrinsic component of the planning process. The Ninth Five-Year Plan (1997-2002) and Tenth Five-Year Plan explicitly recognized a close relationship between environment, health and development and recognized the need for ensuring environmental sustainability.

Setting up of a Department of Environment in 1980 and its further up-gradation as a full fledged Ministry of Environment and Forests (MoEF) in 1985 strengthened country's progress in this direction. The Environment (Protection) Act 1986 was an 'umbrella legislation', which empowered the central government to take appropriate measures for the purpose of protecting and improving the environment. Subsequently, a number of legislations, policies and programmes were initiated to promote protection, conservation and sustainable use of country's natural resources, although their implementation has remained a problem. Indian judiciary has played a laudable role through pro-environment judgments. The National Conservation Strategy and Policy Statement on Environment and Development, 1992 outlined the specific means through which environmental considerations could become a significant part of the developmental process. India is a signatory to Convention on Biological Diversity (CBD), and has initiated the country wide process of developing National Biodiversity Strategy and Action Plan (NBSAP). A Biological Diversity Act, 2002, has been notified to provide for conservation of biodiversity, sustainable use of its components and equitable sharing of its benefits.

India acceded to the Montreal Protocol along with its London Amendment in June 1992. The MoEF has established an Ozone Cell and the Steering Committee on the Montreal Protocol to facilitate the implementation of programme. Also, the Government of India ratified Framework Convention on Climate Change (FCCC). MoEF, the nodal agency in the country, has constituted Working Groups on the FCCC and Kyoto Protocol and a Task Group on the pilot phase of activity implemented jointly to look at specific issues under the Climate Change Convention.

Keeping in view the realities of socio-economic system and the governmental support system, the main agenda of sustainable development in India has a focus on combating poverty, devolution of power to people, utilization of scientific and technical expertise, strengthening of institutional support systems (MoEF, 2002a, b). The sustainability of livelihood of the poor is a pre-condition for sustainable development. Some programmes like micro-crediting, participation in resource management and investment in poor people are already showing some encouraging results. For example, joint forest management and integrated watershed development have shown encouraging results of empowering people to manage their own resources for sustainable livelihoods. In resource rich areas, the community based natural resource management strategies by the involvement of local people in eco-development projects in Nagaland have been successful. In the tree-based Jhum system; Nepalese Alder (*Alnus nepalensis*) has been introduced to improve the biological fertility of soil. The traditional rain-harvesting systems and erosion control measures in arid regions of Rajasthan are helping to regenerate the systems. Controlling the exotic weeds (e.g., *Imperata cylindrica*) and improving the yield from home gardens is proving to be useful. Biodiversity conservation at all levels is essential and many ongoing projects are showing encouraging results. Similarly, biological management of soil fertility in tea plantations using traditional vermicomposting has provided dividends to the tea growers.

India still needs to evolve suitable strategy and implementable action plan wherein the issues of environmental protection and sustainable development are addressed in an integrated, holistic manner (Singh, 2004). Towards evolving such a strategy there is a need to prioritize the issues and subsequently develop an agenda of action to address these issues. The strategy has to be multisectoral and interdisciplinary, and incorporate possibilities of linking local specificity with global generalization. The strong considerations of disciplines, sub-disciplines and specializations

have hampered the communication across the boundaries. While this compartmentalization is inevitable, understanding of interfaces of disciplines or sub-disciplines may have great value. In this context, understanding of the interface between environment and economic development should be effectively attended to (see the section on ecological economics later in this chapter).

From both economic as well as ecological point of view, maintenance of ecosystem health emerges as the most important issue to be addressed. Considering the diversity of ecosystems and their sensitivity to human or climate-induced perturbations, we need to develop capabilities to maintain the integrity of these ecosystems. In this context following points deserve priority (Singh, 2004):

- Thorough investigation of native and endemic biotic components of the ecosystems must attract highest attention. This information is desired not only to address ecological issues such as, conservation priorities (species/communities/ecosystems) but also emerging international economic scenario. An endemic taxon possesses multiple range of attributes, such as small population size, high economic value (e.g., novel biomolecules) and low genetic diversity, which call for a thorough investigation of their life strategies and biology.
- Monitoring the changes in ecosystems, including those due to invasion of alien species, is yet another aspect which needs immediate attention. Changes in natural patterns and processes across disturbance gradients would provide a means for predicting the future structure and dynamics of ecosystems. Although impacts of global warming on distribution of species and communities are expected, the database on changes for scenario building is particularly missing. Specifically for sensitive situations, like the mountains, such data bases are urgently required.
- Ecosystem health has a direct bearing on well-being of the human societies. Unfortunately, the services rendered by natural ecosystems are being impaired and destroyed by a wide variety of human activities. Among others, the responsible drivers for creating this scenario are the frequent miss-match between short-term individual economic incentives and long-term societal well-being. In the wake of globalization, growing interest for market-valued goods and services have increased threats to natural services of the ecosystems (<http://esa.sdsc.edu/daily.htm>). There is a critical need for policy measures that address these issues. The value of ecosystem services needs to be embedded into national decision making frameworks. More research and development inputs must be diverted towards understanding, valuation and maintenance of ecosystem services.
- Maintaining ecological integrity involves protecting total native biodiversity at all levels, as well as the ecological patterns and processes that maintain that diversity. Biodiversity outside of the protected area network also needs to be conserved, and for this, innovative, intensive management techniques are required to be developed.
- ‘Anticipate and prevent’, instead of ‘react and cure’ strategy must be firmly established for dealing with environmental pollution. Use of plants and natural /constructed ecosystems in pollution abatement needs be thoroughly studied and applied. Adequate steps must be taken to promote clean and environment friendly technologies for all kinds of industry and automobiles. At the same time, the pollution monitoring and reporting system needs to be strengthened. Controlling water pollution, particularly from non-point sources, and programmes on rainwater harvesting, conserving forests and planting trees to ensure effective infiltration of rain water, and recycling industrial wastewater are urgent for ecological sustainability of water resources.

There is a need for extensive information generation and its wide dissemination across the country. This would require innovative changes in our traditional approach. If the underpinnings of

ecology and environment are to be integrated with improving the lot of underprivileged societies in India, what better option can one think of than involving the most vibrant human resource, that is, the youth of the country. The coordinated country wide effort of the Indian Academy of Sciences to deploy student power for monitoring India's lifescape diversity (Gadgil, 1996) is an initiative wherein the involvement of young students in environmental education is focused. Likewise, the ongoing programmes for promoting outreach through conservation education in the Himalayan region (Dhar et al., 2002) have shown the tremendous potential of involving school children in problem solving areas of ecology and environment in the country.

30.14 ECOLOGICAL ECONOMICS

Economics and sustainability have an intimate relationship. To some extent, understanding of both the issues is embedded in the emerging concept of ecological economics (EE). EE integrates the study and management of nature and economics, so that the cost of damage done to the air, water, soil and biota, and that of restoring this damage can be articulated in redefining development goals and in recalculating gross national product. However, the functioning of interdependent ecological-economics system is complex wherein the transformation of energy and materials between human economies and their natural ecosystems are subject to the laws of thermodynamics. The simple assumptions of conventional economics about nature as an unlimited source and sink lacked ecological perspectives and ignored the deeper implications. The concept of EE, in a way, focuses on issues of economic and ecological sustainability. But the sustainable management of both systems is tricky, especially in present scenario when incentives to maximize short-term economic growth or gain are strong. Further, the unified approach of integrating ecology and economics also raises the question of time scale in development process and the preservation of ecosystem. The ecosystems operate on a geological time scale, whereas the economic processes in terms of sustainability are viewed in human time scale. These realizations call for convergence of options and approaches of both the disciplines, and need thorough study.

In its modern form ecological economics is only a 15-20 years old discipline. It sees the economy, as a subsystem of environment or a larger natural ecosystem (Fig. 30.9 *a*), that is of a limited nature (Daly, 1999). This stand point is contrary to that of neoclassical economics according to which, economy is the total system, and nature is merely one of its sectors. According to this view, the ecosystem sector is required only for extracting resources and disposal of waste (Fig. 30.9 *b*). The supplies of nature are substitutable with the use of technology, and technology can keep economy ever-growing. The neoclassical economics stresses that the economy is whole, therefore, "the growth of economy is at the expense of nothing" (Daly, 1999); there is nothing like uneconomic growth.

Ecological economics emphasises that economic growth occurs as a result of the transformation of natural capital (environment and resources) into man-made capital. This transformation occurs within a "total environment", which is finite in several ways. The natural capital and man-made capital are complements; therefore, there is no point in extending the latter beyond the capacity of the former. For example, the fish catch in the sea was earlier limited by number of fishing boats (man-made capital), but is now limited by the reduced population of fish in the sea (natural capital). The fishing boat (the man-made capital) is of no use when fish population (natural capital) becomes the limiting factor.

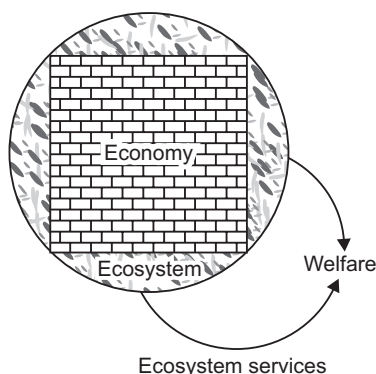


Fig. 30.9. (a) A representation of ecological economics view point, according to which economy is a part of ecosystem, and depends on ecosystem services (adapted from Daly, 1999).

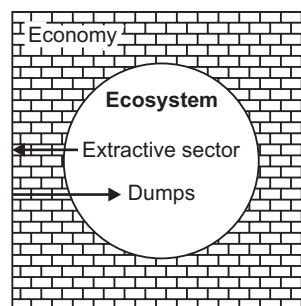


Fig. 30.9. (b) A representation of the neoclassical view of economics according to which ecosystem is one of the sectors of economy (adapted from Daly, 1999) Ecosystem required only for extracting resources and dumping wastes.

The ecological economics points out that growth would be uneconomic, when its cost is more than its worth. Economic indicators like gross national product (GNP) can be flawed measure of economic progress, because they do not explicitly account for the degradation in ecosystems and the loss of services flowing from them. Industries, in general, use ecosystem services and often damage them without bearing costs. For example, the GNP considers only the marketable timber while valuing forests, and conveniently ignores the indirect costs of soil erosion, nutrient depletion, increased flooding, reduced carbon sequestration and others which occur when forests are cut for timber.

Ecosystem services

We all are familiar with goods derived from natural ecosystems, such as fodder from grass lands, fish from oceans, fuel wood and timber from forests and pharmaceutical products from various natural ecosystems. Many of them are traded widely. What we less appreciate, is that natural ecosystems also perform fundamental life supporting services without which human civilization would come to a halt, but they remain outside market supply. In a broad sense, the benefits that come to humans from natural processes can be called ecosystem services. According to the Millennium Ecosystem Assessment (MEA, 2005 a) ecosystem services are “benefits people obtain from ecosystems”. These definitions include not only services of natural ecosystems but also agricultural ecosystems. Provision of clean air and water, carbon sequestration, generation of soil fertility and nutrient cycle, waste dissipation, pollination of plants including crops, recreation, biological control of pests and diseases, climate control, and provision of genetic resources apart from production of goods like food and fibre are some of the examples of ecosystems services. We can define ecosystem services as services which are generated as a result of interaction and exchange between biotic and abiotic components of ecosystems (Fig. 30.10). In a strict sense, products derived from ecosystems such as food and fibres need not be included, but for the sake of convenience often both the products and services are included while giving values to ecosystem services. According to the Millennium Ecosystem Assessment (MEA, 2005a), ecosystem services can be divided into four broad categories: provision service (e.g., timber from a forest), regulation service (e.g., CO_2 fixation), support service (e.g., soil conservation by forest vegetation), social service (e.g., ecotourism) (Table 30.3).

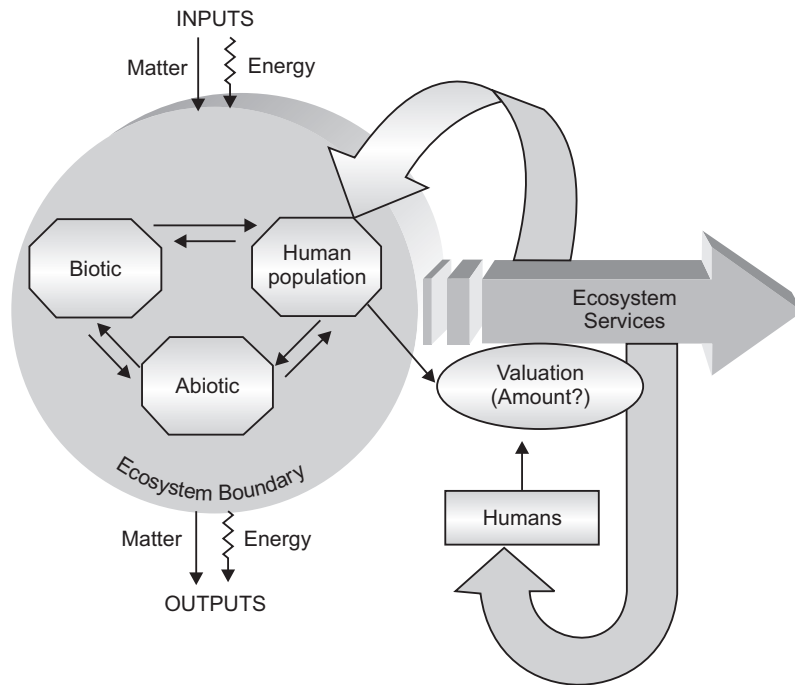


Fig. 30.10. A representation of connections among the components of an ecosystem and between ecosystem services and humans. Within the ecosystem, human population is separated from the biotic component to indicate its extraordinary impact. An ecosystem is an open system with inputs and outputs of energy and matter. Outputs such as cleansed water and air are part of ecosystems services. Ecosystems services are always in flow; their use and valuation depends on humans living both inside and outside the ecosystem. Valuation, however, is greatly influenced by human perception, which in turn is influenced by education and institutional context (from Singh, 2002).

Table 30.3. A summary of four principal types of ecosystem services and their measurement (derived from MEA, 2005a).

Type of service	Sub type	What to measure
1. Provision Service	Primary product Water Supply	Fruit, Timber Water Supply
2. Regulation service	Hydrological regulation	Rainfall interception, soil retention
	Environ Purification	SO ₂ absorption
	Gas regulation	CO ₂ fixation
3. Support Service	Soil conservation, wind protection and sand fixation	Reduction of soil nutrient loss, actual area protected / sand fixed
4. Social service	Recreation	Description of forest with recreation value
	Science & Education	Description of forest with function of science and education

The ecosystem services have been in place long before humans appeared on the earth, but their recognition and valuation are in context to humans. Valuation of ecosystem services depends on several factors, such as the size of territories and human population which receive them, and cultural

traits of societies. For example, the Gangetic Plains receive much of the supplies of ecosystem services from the western Himalayan forests because of the river connections (Fig. 30.11). Though the eastern Himalayan region is richer in forest biodiversity, the region has comparatively much smaller area of plains in its connections. Consequently, the value of ecosystem services flowing from the western Himalayan forest ecosystems to the lowland people would be much greater than those from the eastern Himalayan forests. The Gangetic plains which the rivers of western Himalayan region have nursed for millions of years support nearly 500 million people.

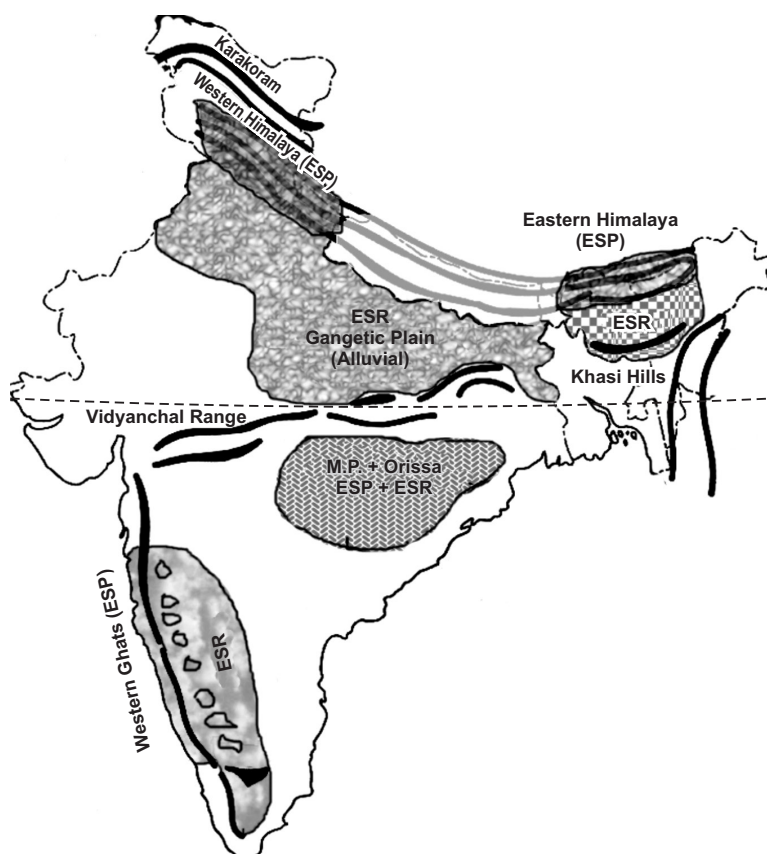


Fig. 30.11. A representation of principal ecosystem service providing (ESP) regions of India and adjoining service receiving regions (ESR). Areas outlined are approximate. These ESPs are also important in regard to terrestrial biodiversity. Western Ghats and Eastern Himalaya are among the top 22 hotspots. Though not equally important in terms of biodiversity, the Western Himalaya exceeds considerably the other regions in terms of ecosystem services, largely because of its connection with vast plains through river connections (Singh, S.P. unpublished).

Value of ecosystem services and natural capital at a global level

Costanza et al. (1997; a group of 13 researchers) made an estimate of the value of the world's ecosystem services and natural capital, which contribute to human welfare. They estimated the economic values of 17 ecosystem services for 16 major biome types, such as major forest types, estuaries, marine and freshwater based biomes (Tables 30.4 and 30.5). For the entire world, the estimated value was in the range of US \$ 16-54 trillion annually, with an average of US \$ 33 trillion/yr, which was about twice as much as the then global gross national product, US \$ 18 trillion/yr.

yr. Since most of the ecosystem services are not part of commercial market, they are often given little weight in policy decisions. For example, in Amazon basin, water equivalent to 63-73% of the annual rainfall evapotranspires from the tropical rain forests (Marengo and Nobre, 2001). Extensive deforestation is therefore likely to lead to a regional decline in rainfall and global climatic change. This kind of neglect is going to adversely affect the sustainability of humans on the planet.

Table 30.4. Total annual value of ecosystem services of major ecosystem types (from Costanza et al., 1997).

Ecosystem type	Value per unit area (US\$ ha ⁻¹)	Total (rate × area) (Trillion US\$)
Aquatic		
Open ocean	252	8.38
Coastal marine	4052	12.57
Wetland	14785	4.88
Lakes and rivers	8498	1.70
Total	27587	27.53
Terrestrial - natural and semi-natural		
Forest	969	4.71
Grassland	232	0.91
Total	1201	5.62
Manmade/intensely managed		
Crop land	92	0.13
Urban	0	0
Total	92	0.13
Grand Total	28880	33.28

Table 30.5. Total annual value of ecosystem services which include regulation of climate and gases, hazard protection, water regulation, erosion control, nutrient cycling, waste treatment, biological control resource production, and recreation/ culture. (based on Costanza et al., 1997).

Ecosystem service	Value in Trillion US\$
Soil formation	17.10
Recreation	3.00
Nutrient cycling	2.30
Water regulation and supply	2.03
Climatic regulation	1.80
Habitat	1.40
Flood and storm protection	1.10
Food and raw materials production	0.80
Genetic resources	0.80
Atmospheric gas balance	0.70
Pollination	0.40
Others	1.60
Total Value	33.03

Costanza et al. (1997) acknowledge that their estimate represents first approximation, and is a minimum value, which is likely to increase with improvement in valuation techniques and as ecosystem services become depleted and scarce in the future.

How are ecosystem functions and ecosystem services related? Their relationships are not straight ones. In some cases a single ecosystem service originates from two or more ecosystem functions, while in other cases a single ecosystem function contributes to two or more ecosystem services. Then, many ecosystem functions are interdependent. For example, in a forest a low proportional resorption of nutrients from senescing leaves of trees may lead to the deposition of leaf litter on forest floor with a low C:N ratio. This nutrient-rich litter, in turn is likely to promote microbial activity and hasten litter decomposition and increase nutrient supply to plants. These interdependent functions are likely to result in the generation of interrelated services, such as the generation of soil fertility and formation of deep soil that in turn holds more water for a longer period. Since ecosystems differ in their functions, they vary in their characteristic ecosystem services and values. Table 30.6 summaries ecosystem attributes and characteristic services of major forest types of Indian central Himalaya, and indicates the drop in services as a consequence of replacement of forest by a *Lantana camara* an exotic invasive species. An oak forest is certainly more important than others in terms of ecosystem services, but all forests may be required at a regional level. For example, a pine forest is able to sequester carbon in a sizeable amount even on rock outcrops and alder helps restoring a landslide site with little soil through nitrogen fixation and related facilitative influences.

Table 30.6. A summary of ecosystem attributes and services of major forest types in central Himalaya and of *Lantana camara* bushland that is established after deforestation (from Singh, 2003).

Forest	Attributes	Ecosystem services or their depletion
Banj oak (<i>Quercus leucotrichophora</i>)	● Large biomass (about 400-500 t ha⁻¹) ● Fine roots and carbon deposition up to 1.50 m or more ● High investment of photosynthate in ectomycorrhizae, massive yearly nutrient return to soil through litter fall	● Deep soil formation ● High soil fertility ● Effective carbon sequestration (also in deep soil) ● Considerable nutrient and water retention.
Chir pine (<i>Pinus roxburghii</i>)	● Small biomass (about 200 t ha⁻¹) ● High productivity (about 20 t ha⁻¹ yr⁻¹) ● High nutrient use efficiency, great ability to tolerate stress, ● Effective coloniser by trees and control over nutrients in soil	● Supply of ecosystem services under harsh conditions and on steep slopes with little soil. ● Retention of nutrients on steep and rocky slopes, associative nitrogen fixation.
Alder (<i>Alnus nepalensis</i>)	● Small biomass (about 100 t ha⁻¹) ● Very high productivity (20-30 t ha⁻¹ yr⁻¹) ● A great coloniser of landslips, high rate of N-fixation (up to 200 kg ha⁻¹ yr⁻¹)	● Development of tree cover without any cost ● Facilitation of other species ● Nutrient supply to other ecosystems.
Bushland <i>Lantana camara</i>	● Productivity similar to forests but low biomass, very shallow rooted ● Low biodiversity ● Flammable by cool fire	● Depletion of several ecosystem services ● Low soil C storage ● Hydrologic cycle adversely affected ● Persistent fire regime ● Loss of biodiversity

Changes in natural capital (capital represents a stock of materials or information present at a given point of time) and ecosystem functions have impact on human welfare. In a large natural capital like atmosphere even a small change can cause profound change at the global level, drastically affecting human welfare. The recent increase in greenhouse gases and expected global warming is an example of such changes. Drop in services flowing from a lake as a result of eutrophication represents a big change in a small natural capital. Conversion of oak forest into *Lantana camara* bush-land in central Himalaya results in a marked depletion in welfare of local population (Table 30.6) as they are deprived of firewood, spring water during dry seasons and leaf litter that is used to prepare compost for crop fields.

The immediate practical use of the estimates of ecosystem services is to help modify existing accounting at regional and national levels to better reflect values of ecosystem services and natural capital. Furthermore, while making appraisal of projects, the value of ecosystem services should be weighed against the benefits that they generate. We cite two well known examples to explain this point. In Hadejia-Jama'are flood plain region in Northern Nigeria, the net benefits of diversion of water from wetlands for irrigating crop fields was estimated at US\$ 29 per hectare. This is very small compared to the benefits already available from the wetlands to the local people, which were determined to be US\$ 167 per hectare. The wetland benefits included fishing, grazing livestock, gathering fuelwood and others. Needless to say, the proposed irrigation project would be greatly diminishing the benefits available. Recently, New York City found out that it could save US\$ 6-8 billion by taking suitable measures in protecting upstream watershed and improving flow of various ecosystem services, particularly purification of water by forests. The New York people had to choose between constructing an expensive new water filtration plant for the supply of drinking water or to improve water quality by investing a small amount of money in watershed protective measures around its reservoirs. The people chose the second option, which not only led to the supply of pure water at a low cost, but also to the enhancement of recreation, wildlife habitat and other ecological benefits.

Payment for Ecosystem Services (PES) using market-based mechanisms

The previous section dealing with the value of global ecosystem services indicated how important they are for the economy. It did not give any idea about ways of getting payment for providing ecosystem services. The market-based mechanism for PES is still in its initial stage of development. Getting payment from a standing forest represents a major advancement in conservation. All these years forest managers have remained glued to working out the sustainable harvest rate of forests, whereas getting payments for forest ecosystem services represents making money without extracting forest biomass. This approach of conservation is different from the one based on "polluter pays principle", which says that those who pollute environment should bear the costs. For example, owners of factories which dump their effluents in rivers should bear the costs of treating water and restoring rivers. The costs, however, are passed on to consumers of goods in the form of their increased prices. The ecosystem service-oriented approach holds that those who provide an environmental benefit should be rewarded for doing so, and this can be referred to as "provider gets principle" (Pagiola et al., 2002). This principle can promote conservation if applied successfully; failure of its application is one of the principal causes of ecosystem degradation and loss of ecosystem services. This can be explained by an example of forest services that flow from a mountain forest area (upstream area) to adjacent plains (downstream areas). There can be three levels of beneficiaries of ecosystem services flowing from forests of upstream areas: local forest managers, downstream developers and global community. Local people who manage forests get fuel wood, timber and non-timber forest products; the downstream water users are benefited by services, such as water purification and sediment reduction; and global community is benefited by carbon sequestration and consequent climatic mitigation (Fig. 30.12). The upstream farmers may decide to

clear their forests for use as agricultural land as they expect to derive benefits from increased food grain production. This decision would be taken when farmers expect that the value of increased food grain production would be greater than products they now get from forests. Deforestation in upstream areas might increase downstream flooding and sedimentation, for example, but these costs will not be borne by the upstream farmers, they would be borne by downstream people. Obviously, the upstream farmers would ignore these costs while taking decision on clearing forests. Since the benefits of clearing forests in terms of agricultural production is fully valued, but watershed services of maintaining forest (in terms of flood control and reduced sedimentation) are not, the upstream farmers decide to cut down forests. However, if downstream farmers take decision to make payment for the watershed services of forests, the upstream farmers may find the maintenance of forests more profitable. Therefore, there is a need to take steps for forest conservation. The strategy needed is to increase the benefits of conservation perceived by upstream farmers so that they exceed the benefits perceived from deforestation. However, there are several problems in getting to this stage because often it is difficult to identify all beneficiaries of various ecosystems services, making them to pay, and making the allotted money available to the forest managers. As shown in Fig 30.13 the market based mechanism may need to consider not only selling sediment reduction and other forest services to downstream users, but also selling carbon sequestration services to buyers needing to meet emission reduction under Kyoto Protocol, and other possible ones. The basic issue is that these ecosystem services are primarily used by people living away from the forests that generate them. The people in and near forests (because of whom deforestation is avoided) do not get any monetary gain for the services provided to the users living in far downstream areas. The local people, in fact, feel that they are unable to use their right to choose more profitable land use. Consequently, they do not give due importance to forest management, and forests degrade.

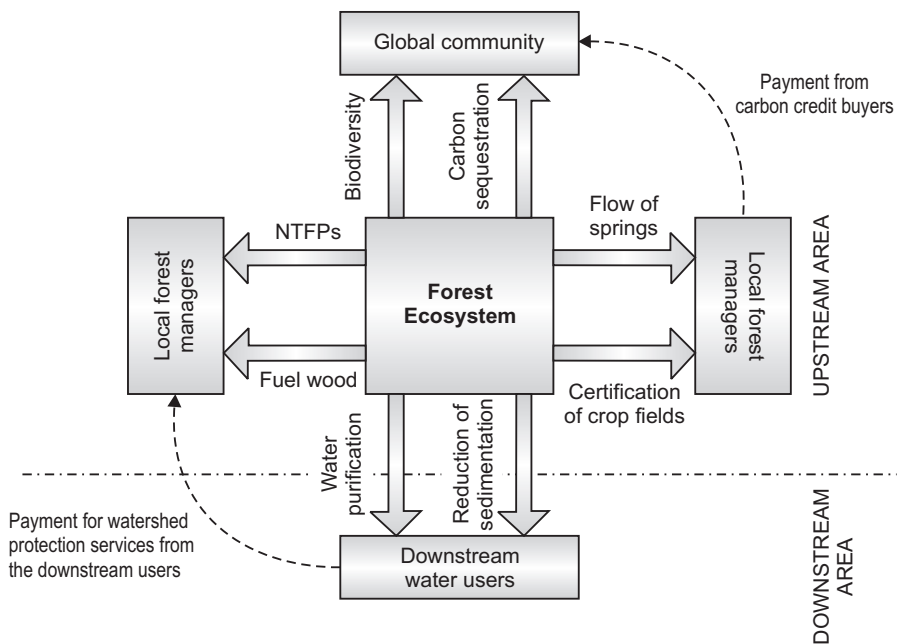


Fig. 30.12. A schematic representation of the beneficiaries of ecosystem services flowing from the forest of a mountain area (upstream). Beneficiaries are divisible into three groups: (i) local forest managers such as Van Panchayats (forest communities) of Uttaranchal, Himalaya, (ii) downstream communities (such as farmers of Gangetic Plains), and (iii) the Global community. The broken lines indicate payments for ecosystem services that may contribute to the conservation of the upstream forests.

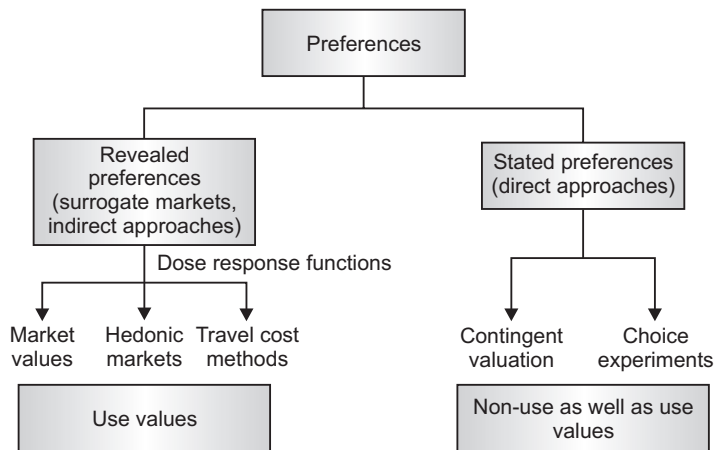


Fig. 30.13. Various preference methods for environmental valuation (adapted from Garrod and Hills, 1999).

Learning from PES

A formal definition of PES is “a voluntary, conditional agreement between at least one seller and one buyer over a well defined environmental service, or a land use presumed to produce that service” (Wunder, 2007). In last 15 years or so numerous PES initiatives have been taken, and Costa Rica, Mexico and China are particularly known for big PES initiatives. A PES project is expected to generate a set of environmental services in biophysical terms. It should be cost effective, and costs involved are: implementation costs, transaction costs and opportunity costs (costs of forgone alternative productive uses of the resource). There are several transaction costs, namely expense of negotiating contracts, performing baseline studies, and monitoring and enforcement (Jack et al., 2008).

With regard to services, the PES projects are more or less evenly distributed across three services, namely water (also includes watershed protection and hydrological services), biodiversity and carbon sequestration. However, a PES project generally offers more than one ecosystem service. Service providers are varied, such as landowners in upstream areas, private groups and individual who purchase wild areas to conserve as protected areas, coffee growers, municipalities, small farmers generally in groups, and investors in carbon projects both Kyoto eligible carbon and non-Kyoto eligible carbon credits GEF (Global Environmental Facility) and others. The range of service buyers too is quite wide – e.g., hydroelectricity power generators who require water with low silt, downstream users of lake and domestic water users, consumers of coffee who value the existence of biodiversity, protected area owners who enjoy biodiversity of parks, buyers of Kyoto eligible carbon credits, and governments. Governments are not only major buyers, but they also often play a key role in establishing payment mechanism and enabling small farmers to generate ecosystem services. The scales of schemes are both sub-national, involving small areas.

In the case of many ecosystem services payments are made for proxies, not for the actual services that are desired and negotiated. For example, downstream buyers pay to upstream sellers for maintaining a forest in a healthy state whether the healthy forest improves water quality or dry season water availability is not monitored. Only in the case of carbon, the payment is made for actual service, the amount of carbon sequestered. The relationship between forest conservation and water yield is uncertain. While evapotranspiration from forest canopy has a negative impact on water yield, the capacity of forest soil and litter to retain water has a positive impact on it. However, forests role in hydrology is always considered positive in PES schemes. Thus, payment for water service

corresponds to forest cover in a watershed. Compared to it, the relationship between the landuses being sold and biodiversity is more certain. For example, a piece of forest land has generally more species than a piece of non-forest land of same size. Therefore, the protection of a forest is a good proxy for biodiversity service. Furthermore, several ecosystem services flow simultaneously. For example, forests raised to control flood also improve water quality, increase regional biodiversity and sequester carbon as well. However, there are also possibilities of trade off. A monoculture of eucalypt may sequester far more carbon than a native forest, but at a high biodiversity cost.

Now we briefly describe a few case studies from different parts of the world in which payment for ecosystem services is in place (largely based on examples given in Pagiola et al. (2002) and Tacconi et al. (2010). Needless to say, a market-based mechanism has to address several issues relating to market structures, the process of market development and its impact on environmental, social and economic conditions.

PES based on upstream forest conservation in Costa Rica

One of the most elaborate systems of payment for ecosystem services in developing countries is of Costa Rica. Referred to as payments for environmental services (PSA), it successfully convinced several hydroelectric power generators and an industrial water user to pay for conservation activities in upstream region. Costa Ricans have developed a countrywide payment system for ecosystem services. In this payment system the landowners are directly paid for the land use decisions that promote the flow of ecosystem services, which include reducing sediment load in water bodies of downstream areas, regulation of timing of water flows so as to minimize the flood risk in rainy season and water shortages in dry periods, and improvement in water quality of available water for domestic use. In order to making payments for water services following issues were addressed (Pagiola, 2002). (i) Identification and quantification of water services generated from a land use, including their economic values. (ii) Identification of downstream beneficiaries of water services and level of charges to be imposed. (iii) Development of a payment mechanism that works - the payment needed to be enough to enable people to conserve forests and have desired land use sustainability. (iv) Addressing institutional, political, economic issues - creating favourable condition for establishing an institution of payment. There were five major types of water services beneficiaries in Costa Rica: hydroelectric power generators, municipal power supply system, irrigation systems, industrial users and people living in flood prone areas.

The countrywide system of environment service payments became very popular with landowners, with requests to participate far outstripping available financing. By the year 2000, over 200,000 ha of forest had been admitted to the project, and there were applications covering additional 800,000 ha. The programme is being adopted in adjoining countries.

Wetland banking in the United States

Wetlands are known for their several ecosystem services: trapping nutrients and sediments, water purification, ground water recharge, flood control, support of bird, fish and mammal populations. The American public does recognize the value of these services. To check the depletion of wetlands in the USA, the government developed market mechanisms, called wetlands mitigation banking, which is a kind of habitat trading project. In this, real estate developers can use a wetland area only by restoring or enhancing wetlands elsewhere. Thus, the total area of wetland in the USA is not allowed to decline. A market has been established that provides such services to developers. Therefore, the developers themselves have not to create wetlands to compensate for the loss caused by their projects.

Selling biodiversity along with shade-grown coffee in Meso-America

Mesoamerican countries e.g., Costa Rica, El Salvador, Mexico, Nicaragua, Panama and Guatemala have very rich biodiversity, as a result of their unique locations (also one of the megahotspots); they form a land bridge between the two continents, north and south Americas. An attempt has recently been made to harness consumers' willingness to pay for conservation by inducing them to pay a premium for shade-grown coffee that also permits the conservation of biodiversity (Pagiola and Rutherford, 2002). Known as traditional "rustic" form of the coffee cultivation, in this, coffee is grown under an essentially intact native forest canopy. Though production of shade-grown, rustic coffee cultivation is lower than sun-grown coffee (which is essentially a monoculture), it (i) conserves diversity of plants and birds, many of which are migratory, (ii) contributes to watershed protection services, and (iii) represents an organic and eco-friendly form of cultivation. In advertisements on websites, importance was given to the positive role of shade-grown coffee in conserving birds biodiversity. In order to assure consumers that the premium has desired effects, a certification of shade-grown coffee was developed.

Payment for Carbon sequestration by forests

Concerns over global warming have led to the possibility of selling carbon-sequestration services of forest ecosystem in international market. Assimilation of CO₂ and storage of carbon in vegetation through activities such as reforestation, agroforestry and improved management of existing forests or reduced impact on them can be referred to as carbon sequestration. In forests, long-term accumulation of carbon occurs in stems and large branches of trees and in soil. Under Kyoto Protocol finance from developed countries could be used to support certain forestry activities (under Kyoto Protocol) as such countries are required to cap their national CO₂ emissions at levels equal to 5.2% below those of 1990 for an initial five year period 2008-2012. That could be also achieved by developing carbon sinks in developing countries in the form of afforestation (raising forests on a land with no history of the presence of forest) and reforestation (raising forests on a land where forest did not exist since 1990).

To be eligible to participate in carbon trading, forestry projects will be required to monitor carbon stock over time, and to get the estimates certified. One needs to develop simple but reliable and valid methods to measure carbon sequestration. As for carbon stock in biomass, the assumption that the 50% biomass on dry weight basis is carbon is well accepted. Belowground biomass cannot be ignored as it represents up to 40% of total biomass, but it can be quite time taking and expensive to measure. Therefore, to simplify the estimation a constant root: shoot ratio is applied. To avoid measuring roots, a conservative approach is to estimate root biomass at no less than 10% or 15% of aboveground biomass. Soil carbon should be measured at least to 1 m depth, and a measurement of soil carbon and bulk density be taken from the same sample (see in Cacho et al., 2002).

Carbon trade through forestry had already been initiated, before the Kyoto Protocol becomes operational. For example, in the *Scolec Te'* project in southern Mexico farmer's communities are selling carbon at US \$ 12 per t C, however 40% of the sale price is used to cover the costs of technical support for farmers, administration of individual's carbon accounts, liaison with purchasers, and monitoring and reporting (Tipper, 2002). These costs can be reduced substantially by enabling farmer's community to take up many of above activities. Pioneering efforts have been made in Australia to support commercialization of carbon and related ecological services of forest. For example, the New South Wales (NSW) state has made a carbon rights legislation that allows investors to record ownership of the rights to carbon sequestration in forest on land titles. In brief, there is worldwide enthusiasm that carbon sequestration may be the first of the international environmental services to be traded in international market. Reduced emission from deforestation and degradation (REDD) is likely to be new agreement with regard to forest carbon trade (Box 30.4)

Box 30.4. REDD+ (Reduced Emission Deforestation and Forest Degradation')

REDD is acronym of 'Reduced Emission Deforestation and Forest Degradation', however originally it was used for 'Reduced Emission from Deforestation in Developing Countries'. The replacement of the second 'D' from the original REDD with 'forest degradation' emphasizes its recognition as a major source of emission in developing countries. In the Copenhagen Accord of the UNFCCC (2009) countries of the world created an international policy instrument by which reduction in green house gas emissions from deforestation and forest degradation, and increases in CO₂ sequestration by forests in developing countries may be rewarded financially. This arrangement is expected to address the problem of deforestation and forest degradation in developing countries which is estimated to account about 10% of total global green house gas emission annually (the percentage declined with time partly because of increase in green house gas emission from fossil fuel combustion). Unlike Clean Development Mechanism (CDM) of Kyoto Protocol in which carbon sequestration by afforestation and reforestation (only new forest carbon sink) is given credit, REDD+ gives value to carbon stock of a standing forest, thus emphasizes on the conservation of existing forests. In this, both the reduction in the rate of CO₂ emission from deforestation and forest degradation and enhancement in carbon sink of existing forests are eligible for payment. Initially REDD included only decrease in degradation, later on forest enhancement, sustainable management of forests (e.g., enrichment plantation) and conservation were included, and which is why it is now referred to as REDD+. Thus, it gives an opportunity to local communities dependent on forests to conserve their forests and get carbon credit (Skutsch, 2011). While in deforestation forest area decreases, in forest degradation carbon stock decreases without decrease in forest area. Forest degradation is widespread in Himalayas, generally because of selective logging of trees, cutting of tree branches for fodder and firewood, collection of forest floor litter for preparing manure and small but frequent man-made fires (Singh and Singh, 1992). Shifting cultivation or, cut and burn agriculture a common cause of forest degradation in tropics is common in the eastern part of Himalayas.

In well managed community forests of Himalayas (Uttarakhand) the rate of carbon sequestration ranges between 2-7 t ha⁻¹yr⁻¹ (Singh et al., 2011). These values are substantial seeing that local communities also collect firewood, fodder and litter from their forests. REDD+ is being proposed to be considered at national level, i.e, for payment for carbon credits estimates of the reduction in forest carbon emissions or carbon enhancements would be made at a country level.

Economic valuation of environment and ecosystem services

It is being increasingly recognized now that there is a scope for market based approaches to environmental regulation, largely because of the worldwide adoption of market philosophy and development of suitable market mechanisms (Garrod and Willis, 1999) A number of techniques have been developed to value environmental goods and services, of which some are interrelated (Fig. 30.14). The environmental valuation techniques which are based on a demand curves are generally divisible into two broad categories: (i) *revealed preference methods*, and (ii) *stated preference methods*.

The revealed preference methods depend on finding out actual purchase of goods (in the private market place) which are related to environmental goods for which demand is to be determined. In expressed or stated preference methods, the demand for environmental goods is not measured on the basis of actual purchase in the market place, but by examining individual's stated or expressed preference for environmental goods relative to their demands for other goods and services.

There are a number of revealed preference methods (Fig. 30.14). The *travel cost method* (TCM) is generally used to estimate the demand curve or marginal valuation curve for recreation sites, such as hill resorts and bird sanctuaries. Individuals have to spend money on transport to reach

a recreation site. Therefore, it is possible to estimate demand for the recreation site by observing how the number of visits to the site varies according to the price of the private good, the transport. The number of visits per year to recreation sites decreases with increasing cost of visiting them. For example, people living in Chennai make fewer visits to the hill resort, Nainital than people living in Delhi because of greater distance and costs on transport. The relationship between the cost of visiting the recreation site and number of visits per year or during a period provides the characteristic downward sloping demand curve, which makes the basis for estimating the recreational value of sites. The TCM can be expanded to include gross expenditure incurred to enjoy an ecosystem service. For example, total expenditure on travel, equipment and supplies for a fishing trip can be called fishing value of a stream. TCM can also be expanded to include entire economic impact. For example, tourists to a park impact regional economy through spending on food, travel, lodging etc. There would also be included income, such as spending by workers and jobs supported by various spending.

The **hedonic price method (HPM)** is another technique of revealed preference approach. It is based on the theory that every good provides a bundle of characteristics or attributes to consumers. For example, a house yields access to different quantities and qualities of public services, including environmental goods, such as space, scenic view and peace associated with woodlands. HPM depends on assessing demands for those characteristics. Suppose there are two houses of identical structures, e.g., number of rooms, plot size, garage, etc., but one is located near a beautiful lake and the other away from it. The difference in their costs could be attributed to the lake view.

Expressed preference techniques (EPT) depend on asking individuals explicitly how much they value environmental goods and services. In this, there is no requirement of complementary good (travel or house location) or a substitute good (compensating wage rate) to derive a demand curve. The two basic types of EPT's are: contingent valuation method (CVM) and stated **preference (SP) or choice experiment techniques**.

In CVM, first, the good to be valued is defined in detail, giving its attributes and describing what will be protected (Garrod and Willis, 1999). For example, in case of oil spill in an area, we may need to describe (i) the special birds and mammals that occur in the area, (ii) the estimated number of deaths of different birds and mammals due to the oil spill and time it would take to recover, and (iii) a feasible programme which could be used to prevent such a disaster occurring again. Respondents then, as an example, could be asked as following: The programme would cost you a total of ₹ 400, which you would pay as a special onetime charge in addition to regular taxes. If the programme costs you ₹ 400, would you vote for the programme or against it? In this the respondent has to choose between two choices (to pay ₹ 400 as one time charge or not), other CV question formats may elicit individual's marginal valuation in different ways. Conventional CV techniques value the goods in totality, the value of different attributes that comprise the good are not separately considered.

Environmental goods are not different from the numerous private goods that are part of our market place. For example, both show the same downward sloping demand curve, as price decreases demand increases. However, environmental goods differ from others in having "public good" elements. The consumption of public goods by one person does not detract from its availability to others (zero opportunity cost of consumption), thereby its consumers cannot be excluded. In other words producers of good have no property rights over it, it is not excludable.

In **benefit transfer method** estimates of ecosystem services in one setting are used to estimate value in another setting. The **commercial value approach** can be explained by the following example. The current commercial value of commercial fish production of a coastal wetland can be used as the value of future fish yield that would result from the protection of a given wetland. However, since it ignores other values of the wetland, this method captures only a part of its ecosystem service value.

30.15 GREEN ECONOMY

What is green economy?

Sustainable development involves both economic and environmental pillars. Green economy or green growth seeks to fuse them into a single intellectual and policy planning process (Samans, 2013). Green economy, thus seeks to promote economic growth as well as sustainable use of natural assets so that they continue to provide environmental services on which growth and well being rely (OECD, 2011). In this, the growth gives due importance to efficient use of natural resources, it is clean (e.g., keeps pollution at minimal levels) and is resilient in that it gives due considerations to natural hazards (World Bank, 2012). Green growth also fosters social equity and human well being (UNEP, 2011), and thus duly involves all the components of sustainable development lying in economic, environmental and social regimes.

Going beyond GDP, and Greening growth (GG) or making progress on green economy (GE) is highly challenging and multidimensional. It entails (I) valuing natural assets for their long-term services and attaching prices to externalities, (II) innovative strategies of breaking with growth that is unsustainable; and, (III) the creation and dissemination of environmentally sustainable technologies (GGKP, 2013). In this natural assets are not limited to natural ecosystems, they also include minerals metals and fossil fuels.

In GG and GE processes environment is regarded as natural capital, which delivers necessary inputs into production and consumption, including sink services for pollution. Reducing the natural asset base does not always against green growth. For example, by providing fossil fuel as a cooking energy to forest dependent people in Himalayas, firewood collection from forests can be drastically reduced. This in turn might enhance the carbon sink value of forests, more than carbon emission from fossil fuel combustion for cooking. Reduction in women drudgery and exposure to kitchen pollution and health improvement could be additional advantages.

GE/GG Indicators

Green house gas emission, change in natural assets including biodiversity, access to clean water and several other themes are relevant to GG/GE. They provide criteria for selecting suitable GG/GE indicators. However, even the best indicators have weaknesses, that is why convergence on a set of indicators is difficult to achieve.

It is important to have background information based on a large number of indicators of GG/GE to effectively communicate on the need for progress in the direction of GG/GE small number of key indicators are required. Constructing a single composite indicator is very challenging because (I) it is very difficult to combine very different measures with varying relevance across countries. Social valuations are adversely affected by poor information and awareness. There are knowledge gaps surrounding the interactions between the economy, the environment, and policies, and the long term nature of the many effects which are difficult to be compared across different times.

In view of these difficulties, OECD (2013) has worked out a small set of headline indicators to facilitate communication about comparing countries keeping in view main GG/GE concerns. The proposal (Table 30.7) however, is still preliminary in nature.

The headline indicators are from four areas: (I) natural asset base which includes ecosystems as well as raw material like minerals as an example; (II) environmental and resource productivity/intensity, which as an example considers GDP/CO₂ emitted and similar other indicators; (III) environmental quality of life, which considers indicators such as proportion of human population in a country exposed to particulate matter (PM_{2.5}); (IV) policies and opportunities, which range widely and may not be comparable across countries.

Table 30.7. A set of headline indicators, as proposed by OECD (2013).

Proposed Headline Indicator	Definition	Strength	Weakness
Natural asset base			
Index of natural resource use	Aggregated index of the changes in stocks of resources	– In principle, easy to communicate (index)	– Work in progress – Data availability problems
Change in land use and coverage	Land use by category as share of total	– Potential use of satellite imagery, can proxy for biodiversity	– Communication- currently no single index – Interpretation in light of different levels of development, geography and population density.
Environmental and resource productivity/intensity			
Carbon productivity	GDP/CO ₂ emitted & Income/CO ₂ in consumption	– Widely used and accepted. – Data availability	– Global interactions – displacement/leakage issue
Non-energy material productivity	GDP/Domestic material Consumption & GDP/Raw Material Consumption	– Policy-maker interest	– Currently environmentally meaningless – Cyclical
“Green” MFP measure	MFP adjusted for natural resource inputs and env. “bads”	– Promising way to incorporate the omitted environmental aspects into looking at productivity/ efficiency	– Questions on interpretation – Data availability problems – Work in progress
Environmental quality of life			
Population exposure to air pollution	Share of population exposed to health-threatening levels of PM _{2.5}		
Policies and opportunities			
Indicator of environmental policies	Placeholder-not yet selected	– Increasing amount of data on policies available	– Data collection on comparative policies (ongoing) is challenge.

Natural asset indicators pertain both to renewable natural resource stocks (e.g., fresh water, forest, fish and biodiversity) as well as non-renewable natural stock (e.g., mineral resources, metals fossil fuels), and consider varied indicators such as proportion of marine fish stock which is non-fully exploited, forest area, mineral reserves, land cover changes, degree of top soil losses on agricultural land or land area under sustainable management practices, and trends in abundance and extinction risk of selected species.

Examples of *environmental and resource productivity/intensity indicators* are : innovations with regard to GG which could be measured as R&D expenditure of importance to GG, productivity per unit energy use, GDP per unit of energy-related CO₂ emitted (or the inverse) and waste recycling and reuse. There are some limitations in use of these criteria. The underlying production function generally omit vital inputs like natural capital and ecosystem services, while income generated through the use of natural capital (e.g., timber, and fossil fuels) is captured in gross domestic product. GDP hardly reflects the benefits of investing in reducing pollution, while total costs in terms of inputs (labour and capital) are captured.

Examples of commonly used indicators of the *environmental quality of life* are access to clean water, sanitation, exposure to air pollutants like PM_{2.5} and noise, and trends in benefits that people derive from ecosystem services.

Policies and economic opportunities combine indicators such as number of persons trained in green job skills, level of environmentally related tax revenues, water pricing and cost recovery, renewable energy incentives, and international cooperation indicating parameters, such as financial flows of importance to GG (e.g., carbon market financing). According to the Dual citizen report (2012) in 2010-11 Germany and Denmark were given top rank as per GGEL. The other countries which were ranked high were Sweden, Huly, USA, UK, Norway, New Zealand, Spain, Brazil, China and Australia.

Some organizations and individuals warn that the use of price mechanisms to protect nature may extend corporate control into new areas such as forestry and water. It may unleash a massive resources grab. They argue that capacities of existing political systems are inadequate to promote green activities. Environmental economists like Clive Spash criticize the use of economic growth to address the depletion of environmental assets (see Spash, 2007).

Green economy industries

According to a study commissioned by American Solar Energy Society in 2007, the renewable energy and energy efficiency sectors generated \$ 1,045 billion in sales and created over 9 million jobs in 2008, with about 1.5 million hectares crop fields and pastures certified organic. The growth of organically grown food is about 20% in the USA from about \$1 billion in 1990 to \$ 20 billion in 2007. Green economy includes a variety of industries which work to reduce the consumption of material and services which is less damaging to environment.

In 2012, investment in renewable energy in China was US \$67.7 billion, which was more than in any other country. China's efforts towards energy efficiency are impressive, resulting in about 19% drop in energy intensity between 2006 and 2010.

CHAPTER

31

Human Population and Urbanization

Homo sapiens is a highly successful invasive species, a keystone species which can modify the surroundings to suit itself, and a species that consumes alone more than half of the total net production of the earth. Human population (P) is growing and so is the affluence (A), that is accelerating per capita consumption, and technologies and socio-political-economic arrangements to service that consumption (T). The human impact on the environment and resources as measured by $P \cdot A \cdot T$ is therefore, accelerating.

The urban expansion is due to population growth as well as the migration from rural to urban areas. Human population and urbanization are getting increasingly connected. Urbanization, in turn has improved access to education and health services. That is why fertility rate is clearly lower in urban areas than in rural areas, and now humans live longer in cities than in villages. Since urban areas are going to realize most of the human population growth, their sustainability is crucial for human welfare. This chapter is divided into two parts, one dealing with human population and the other with urbanization, but their linkages keep on surfacing up repeatedly in both parts.

31.1 HUMAN POPULATION

Salient Features of global population

The world population passed 7 billion in 2011, and is projected to be 9.3 billion in 2050 and ~10 billion 2100 (Fig 31.1). It took several million years to reach the first 1 billion, while it took only about 12 years to increase from 6 to 7 billion (Table 31.1). Here on, the rate of population increase is projected to slow down, and it is likely to take more than 70 years for global population to increase from 9 billion to 10 billion. In 100 years from 1900 to 2000 the number increased from 1 billion people to 6.1 billion.

Table 31.1. Changes in number of years taken to add each billion human population on earth (historical as well as projected). Historically the time kept on getting shorter and shorter until 7 billion populations. But the next 1 billion (from 7 to 8 billion) is likely to take more time than the last 1 billion (from 6 to 7 billion).

Population (billion)	Years		Number of years taken
	From *	To 1800	
0-1			
1-2	1800	1930	130
2-3	1930	1960	30
3-4	1960	1974	14
4-5	1974	1987	13
5-6	1987	1999	12
6-7	1999	2011	12
7-8	2011	2024	13

* Since the arrival of *Homo sapiens* on the planet.

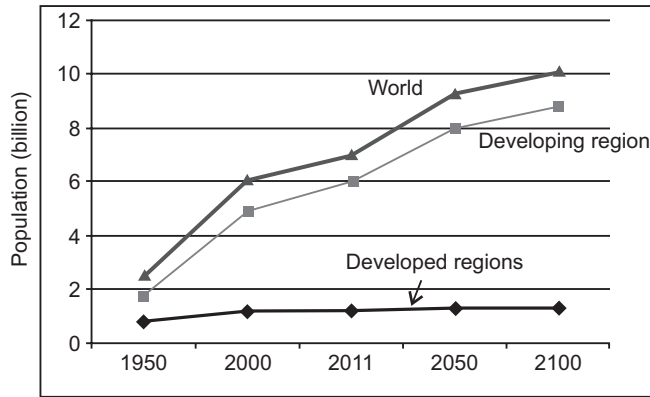


Fig. 31.1. Change in human population size in the world, in developing and developed regions, both historical and in future. Developed regions comprise Europe, North America, Australia, New Zealand and Japan, and developing regions comprise Africa, Asia, Latin America, and the Caribbeans, etc. The bottom curve is for developed regions

Much of the increase in global population during last 50 years or so has been because of the increase in life expectancy at birth in developing countries (Fig. 31. 2); in developed countries life expectancy was already high in 1950. For example, from 1950 to 2010 in North America it increased by only 9.5 years, while in Asia (generally developing category) the increase was by more than 26 years. (Table 31. 2). The period of peak rate of global population growth was during 1965-70 (just above 2% per year). Since then, the rate has decreased by about half (Fig. 31.3). The fertility rate at global level dropped from 5 to 2.5 per woman during last 50 years. In developing countries (China not included) the fertility rate has come down from 6 to 3 during the same period. However, the world is still sharply divided. The total fertility rate even in 2010 is high in some African countries, like Niger and Uganda (5 or more births per women). Many people believe that rather than family planning, it is the number of children which women desire, or say regard “ideal”, decides the change in population. For example , the ideal number of children (recorded in 2006/2007) per woman is 9.1 in Niger, 5.3 in Uganda, 3.2 in Jordan, 3.0 Egypt, 2.4 in UK and 1.6 in Australia (source: Demographic and Health services Euro barometers). A survey of 50 developing countries indicates that desired family size explains 90% of actual family size. A woman’s desired family size depends on income, her education, and infant’s survival (see in Kaiser, 2011). The fertility rate is negatively related to education (percentages of girls enrolled in secondary schools) and positively related to poverty (percentage of population living on less than 2 dollars per day). In other words, education and economic growth reduce fertility rate.

Table 31.2. Change in life expectancy at birth from 1950 to 2010 in world and different continents.

	Life expectancy (Years)	
	In 1950	In 2010
World	47.7	67.9
North America	68.7	78.2
Europe	65.6	75.4
Latin America & Caribbeans	51.3	73.4
Asia	42.9	69.0
Africa	38.2	55.2

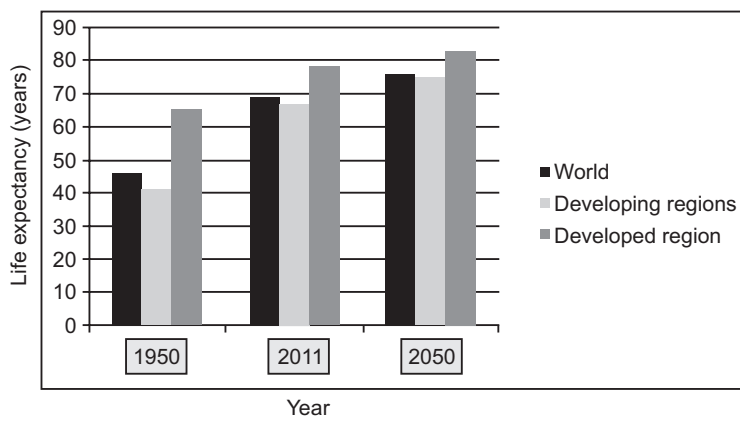


Fig. 31.2. Change in life expectancy in time. Difference between developing and developed regions is decreasing with time(values of 2050 are based on projections)

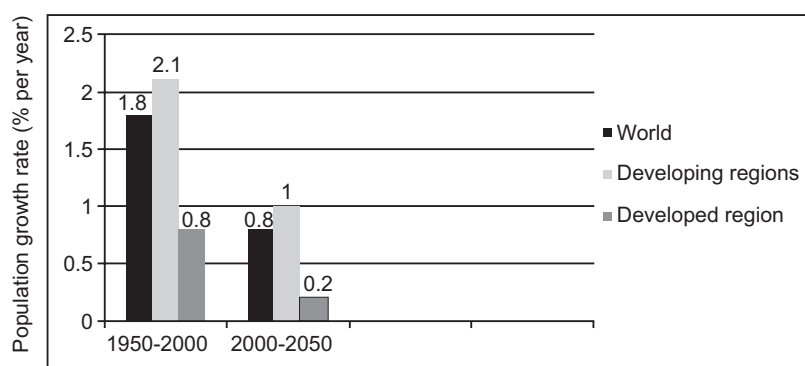


Fig. 31.3. Change in population growth rate of world, developing and developed regions, both historical and projected.

Here on until 2050 almost entire population growth is to occur in developing countries (Fig. 31. 1). Sub-Saharan Africa, the poorest part of the world will have the fastest growth. However, the number of people to be added also depends on the existing population size and its age structure, apart from population growth rate. Consequently, even with slower growth rates Asia (largely due to India and China) will have bulk of the word population in 2050 (Table 31.3). By the numbers the ratio of population of present developing countries to developed countries is projected to shift from 2.1 in 1950 to 6.1 in 2050.

Table 31.3. Sheer numbers of people in different continents of the world in 2010 and 2050(projected). Note, in Europe 2050 fewer people would be living than in 2010.

Continent	Number of people (in millions)	
	In 2010	In 2050
Africa	1022.2	2191.6
Asia	4164.0	5142.2
Europe	738.2	719.2
Latin America	590.1	751.0
North America	344.5	446.9

The contrasts in demographic features between a developing and poor country and developed and a rich country are indicated by citing examples of Nigeria and Japan (Fig. 31. 4): Nigeria has comparatively young and rapidly growing population despite, much higher mortality rate largely because of much higher fertility rate than Japan.

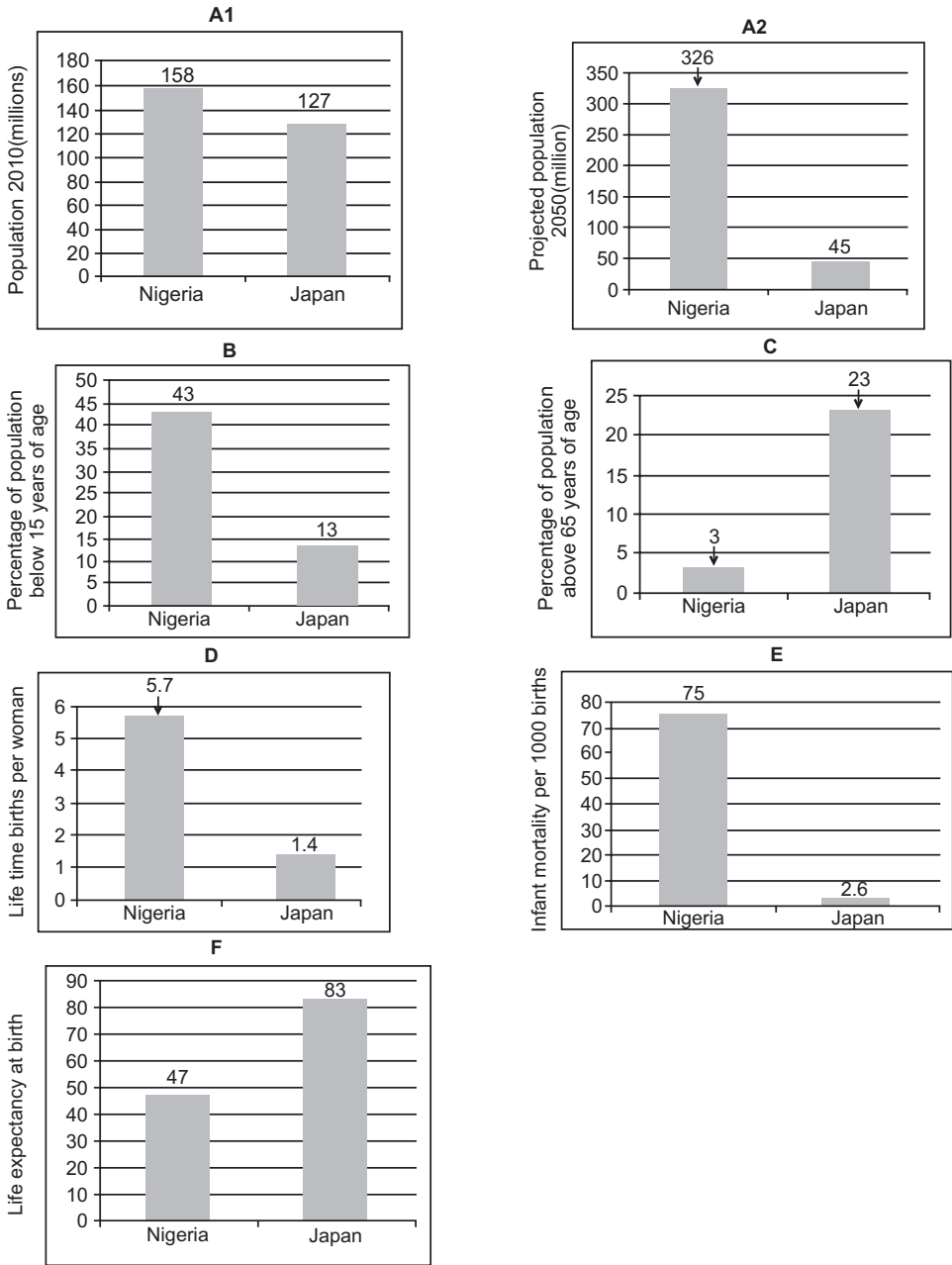


Fig. 31.4. The demographic divide as exemplified by Nigeria and Japan having similar population sizes (A1). Nigeria, a less developed country, has young (B) and growing population (population likely to more than double in 2050) (A2), and Japan, a more developed country has aged (C) population expected to decline (A2). differences are also large in birth per women (D), mortality (E) and life expectancy (F) (sources CHART PRB, 2010);

Demographic transition

Demographic transition refers to a framework in which both mortality and fertility decline with economic development, resulting in a high population growth during the transition but no growth afterwards (Lee, 2011). It refers to the transition from high **birth** and **death rates** to low birth and death rates as a country develops from a pre-industrial to an industrialized **economic system**. In this, mortality reduction precedes the fertility reduction, resulting in a transition phase of population growth (Fig. 31.5). Human population dynamics is considered to be tightly linked to its transition pattern. Projections about stable global population size, such as ~10 billion at the end of the 21st century (UN, 2011) assume that all countries would pass through demographic transition.

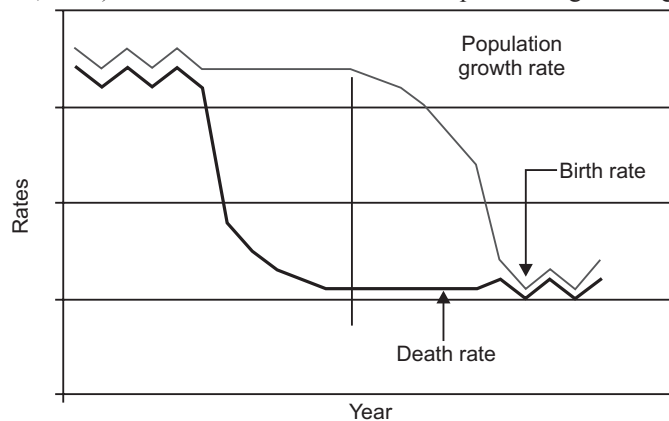


Fig. 31.5. Demographic transition - a typical pattern

The decline in mortality has generally occurred because of access to vaccines and antibiotics and other medical advances, improvement in diet, and health conditions, such as improved sanitation, safe drinking water, and vector control (Bloom, 2011). Women education is considered the principal contributor to fertility declines. Access to family planning devices also reduces fertility, and lower fertility leads to more child care and survival, which in turn further lowers fertility.

Mortality and Life expectancy

Historically, infant mortality rate (IMR) has been the principal driver of life expectancy. IMR is calculated as the number of children who die before their first birthday in a given year per thousand live births in that year (Bloom, 2011). Worldwide, it declined from 139 in 1950 to 43 at present. However, IMR values differ, being much lower in developed regions (68 in 1950 to 6 in 2020) than less developed regions (156 in 1950 and 47 in 2010). It is far lower in urban areas than rural areas because of higher income and education, and better access to emergency care and skilled health workers.

Worldwide life expectancy has been rising at the rate of 2.4 years per decade (or 24 years in a century) from 1840 to 2000. (Oppen and Vaupel, 2002). Life expectancy remained low, around 30 during most of the human history, and then rose by more than 20 years since 1950 for the world as whole. In developed countries it increased from 65 to 78 and in developing countries from 41 to 67. Medical and public health innovations account for such an increase in life expectancy. However, some people think that nutrition played a major role in it. In most countries life expectancy has been rising since 1950, the only exception being Russia after the separation from other countries of the Soviet republics. Russia's life expectancy dropped from close to 70 to 65 during 1990-2000, the period that saw about 15 million premature deaths, largely in working class. Several factors associated with the fall of the Soviet Union might explain the sharp decrease in life expectancy: physical, economic and psychological dislocations, weakening of health infrastructure, environmental deterioration and

increased alcoholism. This stresses that drop in life expectancy indicates overall failure of that country among men.

Life expectancy can differ between developed and developing regions, within developing countries, among continents and within states of a country (Fig. 31.6)

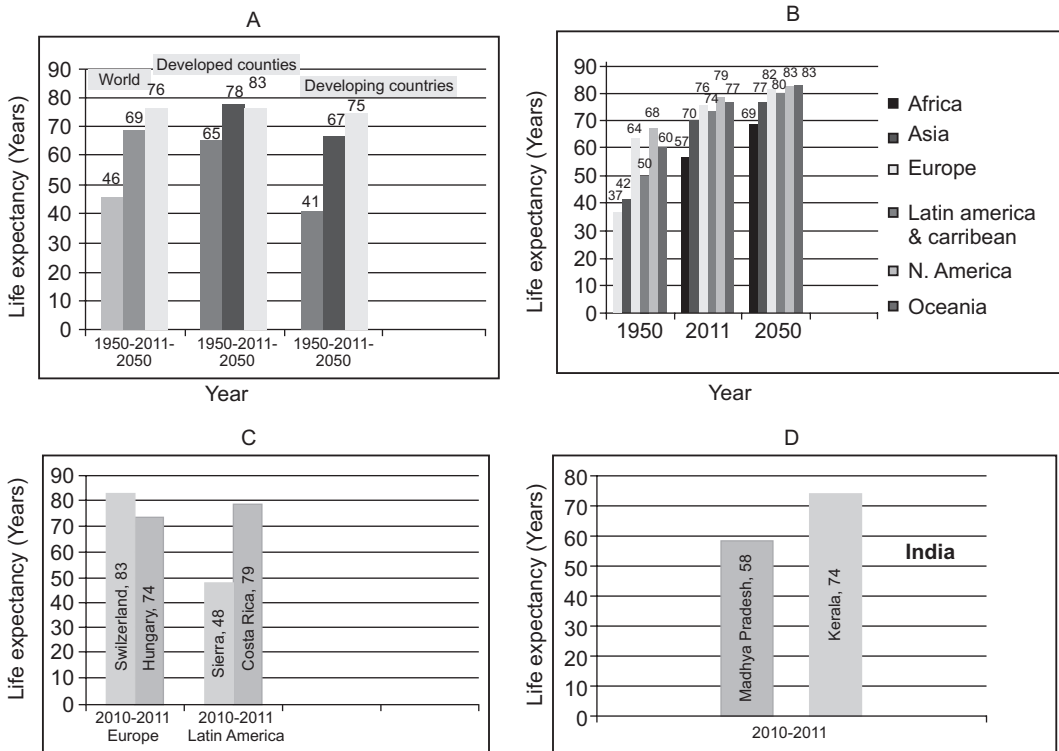


Fig. 31.6. Changes in life expectancy in time: (A) in world, developed and developing countries; (B) across continents; (C) contrast within a continent; and (D) between the states of India.

Two views about future of longevity

How far longevity will increase? From about 30 years during most of human history to over 80 years in several developed countries at present is a huge gain in longevity. Experts are clearly divided. One group argues that further increase in human longevity cannot be more than minimal (Fries, 1980; Olshansky et al., 2001) as infant and child mortality are already very low, and there are several new threats such as climate change, war, non-infections diseases like obesity and heart problem. Exceeding life expectancy of 85 years is unlikely. The other group (led by Oppen and Vaupel, 2002) emphasizes that the scope of further increase in life expectancy is considerable as present human lifestyle and health are far from “perfectly healthy” one, which includes receiving all available vaccinations, low fat and low-calorie diets, less abuse of alcohol, less use of tobacco and more exercises.

On average women live longer, average women life expectancy being 4.5 years greater (71.2 vs 66.7 years) than of men. The difference is more in developed regions (6.8 years) than developing regions (3.7 years). The average difference in women and men life expectancy has increased over time (from 2.1 in 1950 to 4.5 years in 2010). Both biological and life style (and environmental) differences account for this gender difference.

Impact of education and urbanization on fertility rate

World's total fertility rate (TFR) was halved between 1950 and 2010, from 5 to 2.5, largely because of its drop in less developed regions (from 6.2 to 2.6). Continent and country-level rates differ widely. In Asia, for example it ranges from 1.4 in Japan to 6.2 in Afghanistan and in Africa the range is from 2.0 in Tunisia to 7.0 in Niger. In Europe which is more developed region, TFR values are low and limited in range (1.2-2.2).

Fertility rate is significantly reduced by women education all over the world, presumably because educated women have desire for fewer children and have more access to birth control. (Fig. 31.7). Improved education is also associated with better health and lower mortality (Fig. 31.7) (Lutz and Samir, 2011). That is why there is a need to integrate education and population. Population far smaller than projected for future can be achieved by taking extra efforts in women education.

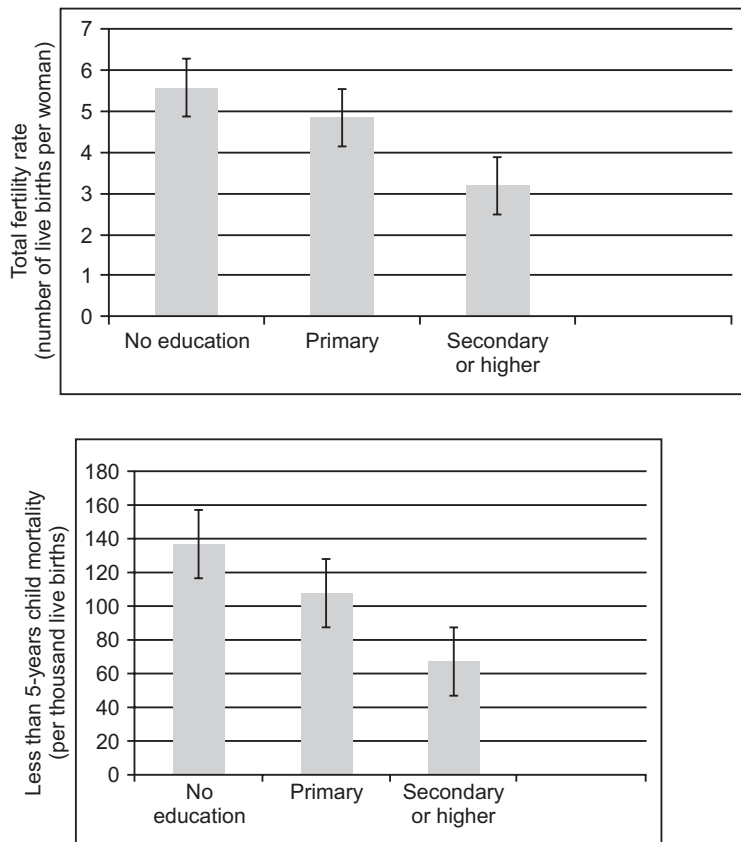


Fig. 31.7. Average mortality rate of less than 5-year old children (lower) and total fertility rate per woman (upper) in relation to level of education of mothers. Averages are for values of 29 developing countries. Data are of period 2005-2009 (includes India) (derived from Lutz and Samir, 2008).

For example, more than 1 billion smaller global population size can be achieved in 2050 by making the high level of progress in education than no education improvement scenario (Lutz and Samir, 2011).

In the case of India the population size in 2050 would be about 200 million less under the scenario of highest progress in education than under the lowest progress scenario (Fig. 31. 6).

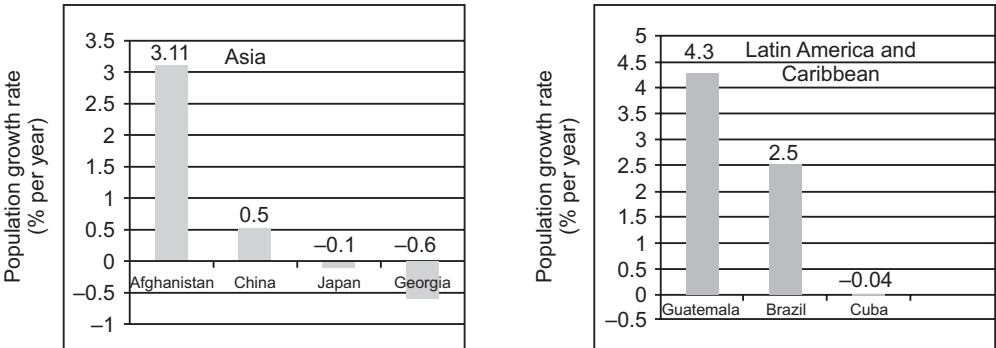
The world is experiencing a major shift to urban living. In 1900 urban population was only 10% of global population, in 2008 it exceeded 50%. Since developed countries have generally been urbanized, most of the net increase in global population will occur in developing countries (Grimm et al., 2008). Almost all mega-cities (>10 million population) in future are to be located in developing countries.

Cities now are known to have lower fertility and mortality rates and higher life expectancies than rural areas presumably because of education, and health infrastructure. For example, in Shanghai which is largely urban has total fertility rate 0.9 compared to 1.63 in whole of China, and life expectancy 79.3 years compared to 73.7 years for China?

Things were not like this in past, up to the early 20th century infant mortality was higher in cities (Mace, 2008). The curb in fertility and reduction in infant mortality occurred only with industrial revolution in Britain. Sanitation condition in Paris of the 19th century was poor. Parisian men were shorter than men elsewhere in France (Dye, 2008). By the start of the 20th century, however, health was improving faster in cities than their rural areas. At present fertility rate and percentage of population living in urban areas across the countries of world are negatively correlated. The effect urbanization on fertility reduction can be seen even Africa. In Addis Ababa the fertility rate is higher by four children than surrounding rural areas, possibly showing the greatest urban-rural divide (Mace, 2008).

Aging Population

Population growth rates of countries vary considerably (Fig. 31. 8), but a trend of slowing down has begun to affect humans. Population is a major consequence of decrease in fertility and mortality, and increase in life expectancy. Globally the population of those aged 60 or more at present represents 11% of total population (800 million) and is predicted to be 22% (about 1 billion) in 2050 (the UN population Division). The aged population is projected to reach 42% by 2050. Just imagine by 2050, 42 countries are likely to have a higher percentage of people aged 60 and over than Japan has at present (Bloom, 2011). The old-age-dependency ratio (number of individuals 65 years or older per 100 in the 15-to64 years old woking population) in Japan is already 38 per 100 and is to rise to 70 per 100 by 2050. However, recent experiences indicate that seniors in Japan continue to work, and pay taxes well beyond 65 years age. The work place is being made friendlier to senior citizens. However, ageing of population, lack of desire to make babies (Box 31.1) and resultant decline in population are likely to affect many developed countries.



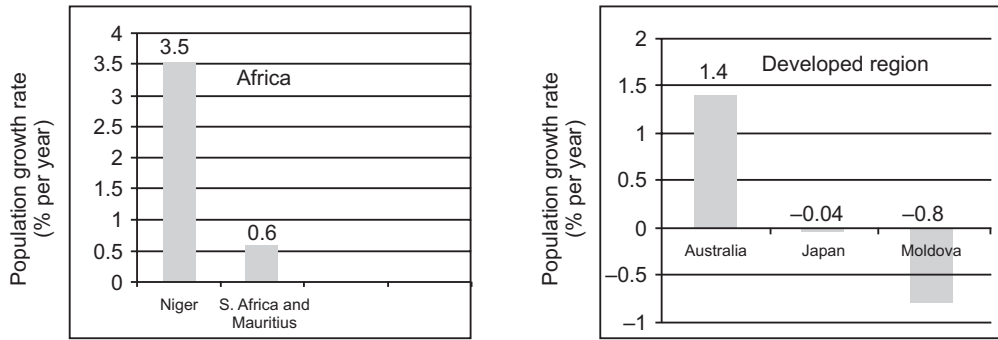


Fig. 31.8. Contrasts in population growth rates (2000-2010) of countries within continents and developed regions.

In the middle of historic changes in human population – fewer babies

The fertility rate of nearly half of the world has reached below the replenishable level (people are not interested in making babies). Since it takes another 30 years or so (about a generation) after the fertility falls for population to decline, the global population might stop growing in little more than a generation. It is perhaps going to be the greatest shift in the known history during which “biology and scriptures urged us to be fruitful and to multiply” (words in inverted comma are of Seccombe, 2013). Italy has a fairly low fertility rate (1.38 per women for last 5-7 years), though the Catholics are against the fertility control. Japan’s fertility rate has been below the replaceable level for last four decades, and if its fertility rate does not change, in 2100 Japan would have only 55 million people, that is less than half of present population (125 million in 2010). With a fertility rate 0.78 per women, Singapore represents an extreme case; its population will be down by over 90% by 2100 (from 5.2 million to below 0.5 million) if migration is stopped. A population falls by half in a generation (about 30 years), if its fertility rate is 1 per woman. It is reduced to about 6-7% of its original size in four generations. Thus, unless people take corrective measures, population of at least some developed countries will approach extinction in about a hundred years. Interestingly, fertility rate is below replaceable level in China, the most populous country. India’s population is still growing rapidly, but its fertility rate has decreased from 3.3 a decade ago to 2.6 per woman in 2011. In contrast, population is to increase rapidly in Nigeria with fertility rate of 7.6 per woman and some other African countries.

The two most populous countries of the world

India, the country of a high demographic heterogeneity. with a population of 1210 million as of 1 March 2011), India accounts for about 17% of the global population. Indian states are as populated as or more populated than many countries of the world: Some comparable values (in millions) are, UP 183, Brazil 187, Maharashtra 104, Mexico 104, Bihar 90, Germany 83, West Bengal 85, Vietnam 85, Andhra Pradesh 80, Philippines 84, MP 66, Thailand 63, Tamil Nadu 65, France 61, Rajasthan 62, Italy 59, Gujarat 55, South Africa 47, Orissa 39, Argentina 39, Kerala 33, Canada 32. India’s population growth rate was 1.6% per year from 2001 to 2011 (clearly lower than 1970s) which resulted in addition of about 181 million people which is only marginally lower than the present population of Brazil, the 5th most populous country of the world. India’s population will be more than that of China in less than a decade (by 2020) and by 2050 it would be 400 million more than of China at that time (Fig. 31.9). It is projected to stabilize at about 1.72 billion around 2060 (UNDP). But there are extreme uncertainties associated with all these projections, as India’s demographic heterogeneities are very high, more than most other countries. In several parts of India

demographic changes have occurred without substantial socioeconomic changes which are known to accompany these. Regional variation is very high. For example, in Andhra Pradesh total fertility rate (TFR) in 2008 was 1.7, compared to 3.9 in Bihar and infant mortality ranged from 12 per 1000 births in Kerala to 70 in Uttar Pradesh.

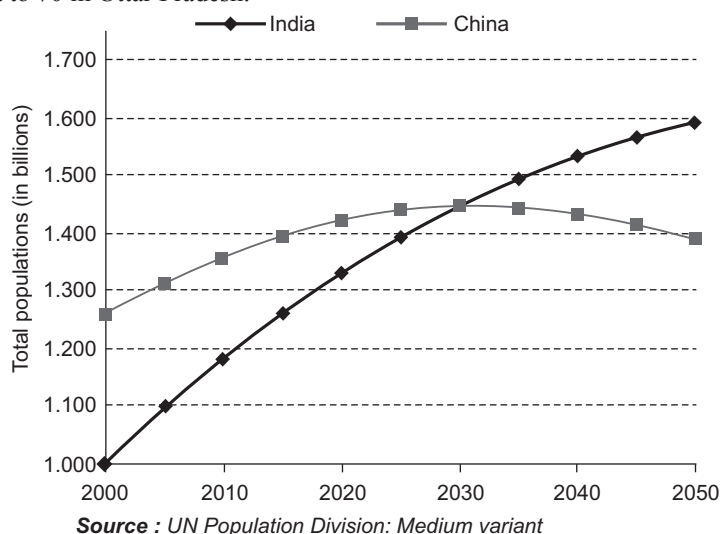


Fig. 31.9. India's population is predicted to overtake China's population in less than two decades (in 2030)

Kerala is the most progressive state as far as demographic achievements are concerned. Here fertility started declining as early as the 1960s, but at that time Kerala was relatively poor with a lower per capita income than national average and people were undernourished (James 2011). Similarly, Jammu & Kashmir, despite the lack of peace and economic growth has a healthy fertility rate, 1.8 per women. But it had relatively higher women literacy rates. However, the rapid decline in fertility in the late 1980s and 1990s in Andhra Pradesh and Tamil Nadu was not accompanied by any major socio-economic improvements. They had low female literacy. Even in 2011 female literacy rate in Andhra Pradesh was 59.7%, which was clearly lower than all country average of 65.5%. Orissa attained a net reproductive rate (NRR) of 1 with very high TFR (2.4) and IMR (69), while Kerala achieved it at 12 IMR and Maharashtra, West Bengal and Delhi at about 30 IMR.

In India the stage is set for the final stages of demographic transition. In spite of a marked progress in female literacy rates of over 12% in the past decade (2001-2011), India's gender ratio declined from 945 to 914 females per 1000 males during 1991-2011.

Experts generally agree that the demographic change in India, is creating new economic opportunities. A larger group is moving into a working age, compared to dependent group. Whether the country would be able to harness this demographic dividend, depends on several factors, such as suitable policy environment, good governance, carefully planned trade and sound macroeconomic management. The country is still deficient in quality education and skilled persons. The large working group (Figs. 31. 10 and 31.11) can become a liability if it is not suitably employable. In contrast, population in China is rapidly aging (Figs. 31. 10 and 11). To what extent China's single child policy has played role in this rapid demographic change is hard to know. The aging trend is still more evident in the population of a developed country (Fig. 31. 11, France).

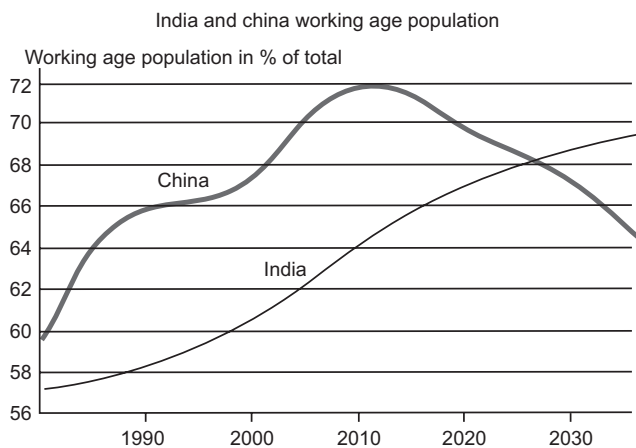


Fig. 31.10. India's population would remain young for quite some period (lower graph), while China's population is aging rapidly (upper graph). source UN Population Division)

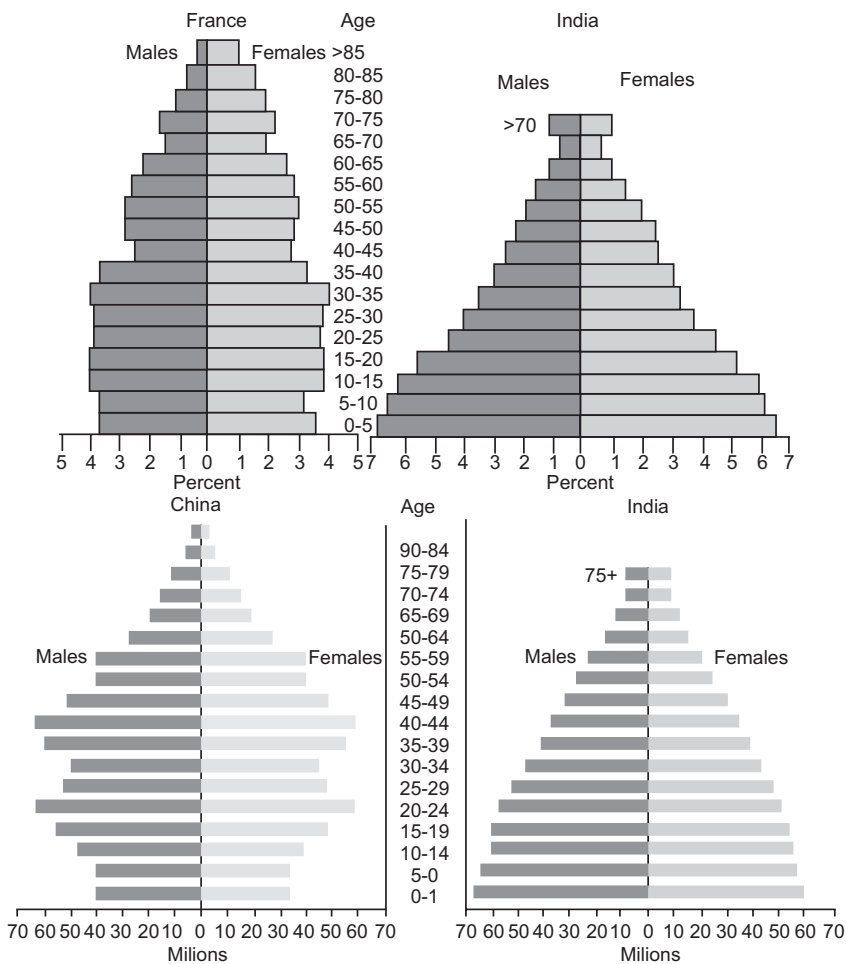


Fig. 31.11. Population structure of India and China, and a developed country, France (2010, 2011). While India's population still has a pyramidal shape with a broad base, China's has become urn-shaped. In France, the population has almost stabilized, and would soon start decreasing.

31.2 URBANIZATION

In the year 2008 for the first time ever in the history, more humans lived in urban areas than in rural areas. Urbanization, which amounts to increase in the proportion of cities or towns is an ongoing phenomenon. However, large-scale urbanization is a very recent phenomenon; if humans sojourn on the planet were taken as 24-hour, the urbanization began only a minute ago. In the next 25 years, about 1.7 billion new people will enter cities, and close to one million km² of new urban areas would be created (UNDP, 2005). The future world is going to be increasingly urban (Fig. 31.12).

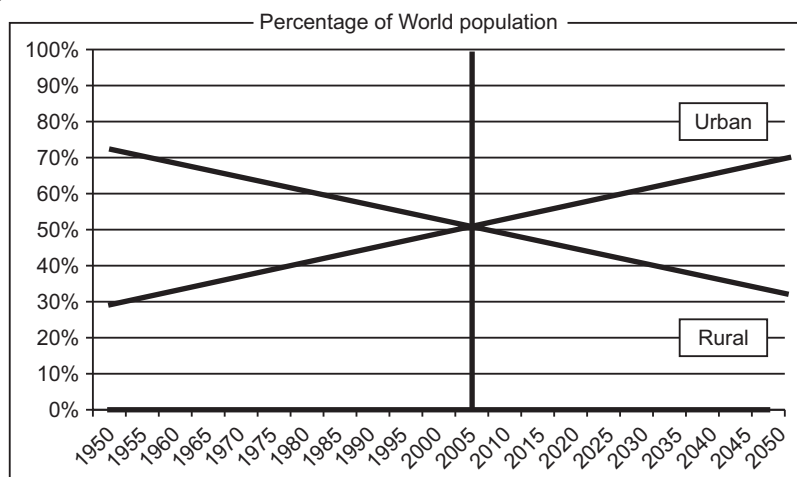


Fig. 31.12. The shift towards urban population in the world.

Though ecologists have begun to take interest in urbanization only recently, it is an important historic transformation of human social roots on a global scale, a shift from rural to urban culture with huge environmental implications. Cities are centres of concentrated production, consumption and waste generation and disposal that drive a host of global environmental changes. They, in fact are laboratories for studying the global environmental change.

This shift from a primarily rural to a more urban world (see Box 31.1) is the second major change in the human settlement patterns, the first being the shift from hunter-gatherer life to village based life several thousand years ago. Cities came into existence when there were agricultural surpluses, and consequently not all people were required to till the land. Several socio-economic factors, viz, specialization in labour, economic opportunities and economics of scale, institutional build up (e.g., government, religion and education), and the need for sophisticated political systems contributed to the development of cities which, in turn promoted and shaped those processes. There were also major changes in social connections. In villages people of common bloodline lived together with strong intimacy, and communal behavior. The urban culture lacks these, and is driven by competitive traits. Alienation, stress, more per capita consumption and waste generation characterize the urban culture.

Percentage of people living in urban areas and per capita GDP across the countries of world are correlated, 10% increase in urbanization being associated with 61% increase in per capita GDP. However, the causality of correlation is questionable and some analyses indicate that urbanization is not the direct cause of more wealth (Bloom et al., 2008). Urbanization occurs for many reasons, including strife in the hinterland and dictatorship that divert most public spending in a capital city. That may result in a big and wealthy city even amidst vast poverty.

Box 31. 1

Scale of urbanization : Some observations

- In 1900 only 10% of global population was urban, this percentage crossed 50% somewhere in the middle of 2007, and by the year 2050, about 64% and 86% of the developing and developed regions, respectively would be urbanized.
- In the next 50 years, more than 95% of the net increase in the planet's population will be in urban centres of developing countries (UNDP, 2005), and that too in small cities. The urban population in developing countries was 1.97 billion in 2000, it is to become 3.9 billion (out of the total urban population of 5 billion) in 2030 and 5.26 billion in 2050.
- Urban areas occupy only about 1% of the planet's area. But by 2030 about 270 million new housing units would be required in a developing country alone, which is similar to existing house units, 291 million in Europe.
- China has (in 2008) 670 cities, whereas it had only 69 in 1947 and 223 in 1980. It has 89 cities with a population of a million or more; US has 37 and India 32 such cities. China's real urbanization rate is already 50%.
- Changing population (millions) sizes and ranks of big cities of the world

	1975	Rank		2015	Rank
Tokyo	26.6	1	Tokyo	37.1	1
New York	15.9	2	Delhi	28.6	2
Mexico city	10.7	3	Mumbai	25.8	3
Osaka-Kobe	9.8	4	Sao Paulo	23.6	4
Sao-Paulo	9.6	5	Dhaka	20.9	5
LA-Long Beach	8.9	6	Mexico city	20.7	6
Buenos Airas	8.7	7	NY-New York	20.6	7
Paris	8.6	8	Kolkata	20.1	8
Kolkata	7.9	9	Shanghai	20.0	9
Moscow	7.6	10	Karachi	18.7	10
Rio de Janerio	7.5	11	Lagos	15.81	11
London	7.55	12	Kinshasa	15.0	12

- Individual cities are growing to unusually large sizes; most of mega cities (>10 million) are going to develop in developing world. (see the box table). In 1975 of the 12 biggest cities 6 were in developed countries, while in 2025 only two would be from them. Some Indian and Asian cities have unusually high population densities (in city proper, per km²): Chennai 26,903; Kolkata, 24,252; Mumbai, 20,694; Dhaka, 19,447; Seol 17,473; Cairo, 17,190; Jakarta, 15,400; Tokyo, 14,400; Surat, 13,666; Bengaluru, 11,876; New York, 10,637 and Tehran, 10,359.

Cities are hotspots of production, consumption and waste generation, including GHGs (Grimm et al., 2008). For example, one of the reasons why China has quickly surpassed the USA as the largest CO₂ emitter was its rapid urbanization, based on energy from coal -powered plants and tail pipe supply. Cities now consume 75% of global energy and emit 80% of GHGs. In their immediate vicinity, cities have several impacts, such as fragmenting, isolating and degrading natural habitats, simplifying and homogenizing species composition, disrupting hydrological systems and modifying energy flow and nutrient cycling. They thus appropriate a large share of Earth's carrying capacity

from other regions in terms of resources input and waste sinks. Thus urban areas do not have only local environmental impacts but also large ecological footprints. The ecological footprint of any specified population is the total area of productive land and water required on a continuous basis to produce all the resources consumed, and to assimilate all the wastes produced, by that population. Many cities appropriate an ecosystem area 200 times larger than the cities themselves.

Ecological aspects

Urban areas are not self sustaining, they survive only by importing food, water, energy, minerals, and other resources from somewhere else often using up agricultural land. They produce vast quantities of wastes, not only affect the health of their inhabitants but also the environmental health of rural areas and the health of the planet. However, cities can become more sustainable by modeling urban processes on ecological principles by which natural ecosystems operate; the characteristics of ecosystems of interest include diversity, adaptiveness, interconnectedness, resilience, regenerative capacity and symbiosis. These characteristics can be incorporated in the development of strategies to make them more productive and regenerative, resulting in ecological, social and economic benefits. Developing as a new discipline, urban ecology “integrates the theory and methods of both natural and social sciences to study the patterns and processes of urban ecosystems” (Grimm et al., 2008). Here a city is considered a heterogeneous, dynamic landscape and a complex adaptive, socioecological system in which ecosystem services link society and ecosystems at various scales (Grimm et al., 2000; Pickett et al., 2005).

Interaction between Global environmental change and urban areas

Global environmental changes that affect and are affected by urban areas can be described as following (Fig. 31.13).

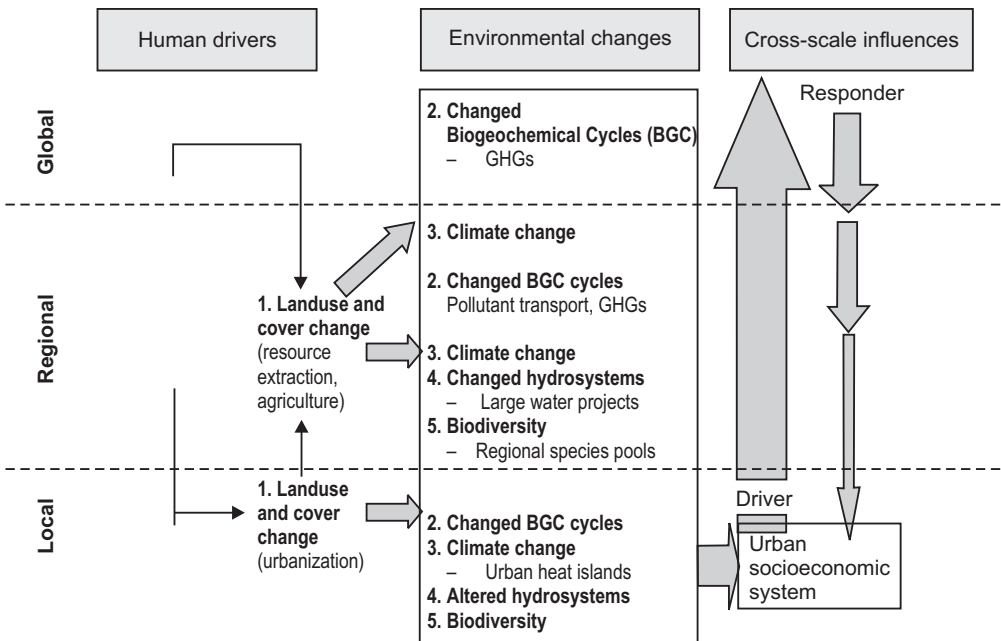


Fig. 31.13. Framework representing various interacting components in relation to an urban system. The emphasis is on major global changes (e.g., climate change and landuse change) that affect and are affected by an urban ecosystem. The socioeconomic system (bottom right) is the principal driver of (upward arrow) and responder (downward and horizontal arrow) to environmental change (derived from Grimm et al., 2008)

Urbanization- induced land use deterioration

The ecological foot prints of cities could be seen tens to hundreds of times of their total areas (<1% globally). The urban dwellers, primarily exert impact through their consumptive demands and waste assimilative needs. Even in the beginning of human history, city dwellers degraded areas beyond city boundaries. Salinization of land in Mesopotamia between rivers Euphrates and Tigris, way back in the third millennium BCE is a well documented story.

Demand for beef from European countries is now converting tropical rain forests of South America into grazing lands. In almost all developing countries can be seen the expansion of the edge of the city into surrounding rural landscapes, resulting in changes in soils, built structures, housing. All these affect fringe ecosystems. Urban construction in developed countries and rapidly growing economics like China is a major cause of tropical deforestation.

Regional to global effects of altered urban biogeochemical cycles

Urban transportation and industry are major point sources of CO₂ and other GHGs, and trace gases such as NO, NO₂, O₃, SO₂, HNO₃ and several organic acids. Because cities are generally located along rivers and coastal lines, they are major sources of their pollution and eutrophication. City wastes and pollutants have local to regional impacts. For example, the 20 largest US cities emit more CO₂ to the global atmosphere than the total area of the country can absorb. In Beijing, the carbon emitted from solid waste treatment rose by a factor of 2.8 from 1990 to 2003 (see in Grimm et al., 2008). Lack of sewage system in developing counties has markedly affected water bodies and nutrient cycling.

Cities themselves are affected by their impacts on biogeochemical cycles. High acid and N deposition and high concentration of CO₂, CH₄ and O₃ are common in cities. These chemicals have both growth- inhibiting and growth- promoting roles. Attempts are being made to convert waste water into resources by using innovative methods (see chapter on pollution). Urban aerosols, particularly Black Carbon, not only affect local air, but also global climate change in various ways.

Relationships between Urbanization and Climate Change

Cities account for about 80% CO₂ emission from fossil fuels and thus are greatly responsible for global climate change. However, urbanization is a matter of concern also because of its intense local environmental impact, which includes increased minimum temperatures and often lower maximum temperature, reduced wind flow and raised humidity, and altered precipitation. More known, however is the urban heat island (UHI) impact. Generally, cities have higher temperature than surrounding rural areas. The factors that could create UHI are suggested to be size of the city, increased impervious surface, decreased albedo, reduced vegetation cover and water area, and canyon like heat trapping morphology of high buildings. It may be pointed out that UHI is an entirely local phenomenon with no climate change implication.

UHI also affects water resources, air quality, human health and ecosystem functioning. It may also induce photochemical smog and retard air circulation, thus increasing residence time of pollutants in the atmosphere. Since even cold countries are known to spend energy on cooling because of UHI, this is likely to become a serious problem in cities of tropical countries with increasing wealth. The presence of impervious cover over a large fraction of city area causes several-fold increase in runoff and the transport of accumulated pollutants from buildings into streams and lakes. In Nainital, a lake city in Indian Himalaya, a centimeter of rainfall raises the lake water level by 3 Cm. The lake is surrounded by houses with tin roofs on surrounding mountains. Houses are the biggest contributor to the creation of impervious surface. Each new occupant on average adds 355m² in a high income country, 125m² in middle income and 85m² in a low income country. With weighted mean household size of 154m², worldwide some 90,000 km² new urban area would be required by 2030 (McDonald, 2008).

Urban biodiversity

Urbanization increases the biomass of birds, but reduces their richness and evenness, and causes a shift from insectivorous to grainivorous birds (Chace and Walsh, 2006). Species diversity of arthropods also decreases, and they become more generalists. Cities contribute to homogenization of organisms, in which edge species are major contributors. Because of an easy availability of food from human wastages, even birds like sea gulls which are essentially sea-based birds (“conceived by oceans and sky”), have begun to live in interior cities over a hundred miles from the coastal lines.

Fragmentation of vegetation and habitat is a characteristic feature of urban areas. Often individual trees, rather than stands are unit of vegetation. In stands, there are more of edges than interior component.

Effect of the forms of urban areas

As mentioned earlier the form of cities is important with regard to their environmental impact. The transportation energy increases when density of houses drops below 30 people/ha.

The heating energy use for a family is clearly higher in a detached home (about 10000 KW h⁻¹ per person per year) than in an apartment of a big building (7000 KW hr⁻¹ per person per year) (Holden and Norland, 2005). The form of cities and form of transport would also determine transport energy use, which can be huge in future, as urbanization in developing countries is going to produce millions of new consumers each year. For example, creation of walking lanes and bicycle lanes can markedly curtail CO₂ emission in small cities where distances are short.

Globally about one billion urban people are categorized as poor, generally occupying slums with inadequate sanitation facilities, and houses with poor ventilation, and surroundings with no space for recreation and physical exercises. Many workers use their home as work place, thus have little physical activities. Obesity among urban poor is getting common (Dye, 2008). With increasing GDP the slum populations have proportionately decreased. For example, in Indonesia while total urban population between 1990 and 2001 increased at the rate of 4.4% annually, the slum population growth was only 1.4% annually. The percentage of overweight people is increasing at a higher rate in urban areas than rural areas. The cost of obesity and related diseases, such as heart disease, diabetes and certain cancers is very high in cities, and is increasing rapidly in low income groups, partly because they cannot afford to eat “health food” (with low fat and carbohydrate).

Payment for sustainable urbanization can be used as mechanism to reduce energy consumption and CO₂ emission in cities of developing countries. The vastly wealthier countries would need to take a lead in supporting payment for ecosystem services like mechanisms to promote ecologically sound urban decisions (McDonald, 2008). They could include allocation to enough city area for walking and cycling lanes, mass transportation, compulsion of going to the nearest schools (but then huge finances would be required to have good quality schools throughout a city), use of sprinklers to reduce dust, planting shade trees, energy generation from waste water, maintenance of water bodies, etc. Most cities in developing countries have only 8-10% of area for roads, against 20-22% cities of developed countries.

The wealthy countries have to play a role in sustainable development, and for that ecologists would need to find out ecosystem services relevant to urban areas of developing countries for which payments are possible from developed countries. Developed countries can contribute to sustainable development by paying to promote responsible urbanization in developing countries where most of the future urbanization is to take place. Local governments in developing countries generally lack money to maintain parklands and to treat watersheds from which they get water. Developed countries can decide to support such initiatives. As an example, Citizens of Palampur (a small mountain town in Himachal Pradesh State of India) pay (though only a small amount) to an upstream village for conserving forests. But to make payment for clean water or maintain a park from distance is very difficult because of the involvement of a large numbers of institutions, and consensual nature of international treaties.

Following conditions are necessary to make the payment for ecosystem services feasible: (I) it is feasible to implement in many cities to achieve necessary scales. (II) It is flexible enough to be able to adapt to myriad local circumstances. (III) Involve costs within the capacity of developing counties (McDonald, 2008). Here the concept of greening at various levels might not only sensitize citizens but also bring about major reduction in energy use and generation of wastes. For example, some school owners in Gujarat are rewarded for their 'greening efforts', which may include recycling of paper, water harvesting, maintenance of trees, use of bicycle to go to school, holding classes in morning hours to reduce cooling energy temperatures are mild, use of solar energy and others. The concept of "greening buildings and life style" is still in infancy. The schemes of certification and reward and payment are either non-existent or weak. However, the approach of payment has a great scope in urban areas, as there people have more capacity to pay.

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